From Cantor to Ito

Combined all-volume build

Contents

Part I

Volume I: Logic, Sets, and Proof

Mathematical Logic and Proof

Chapter 1

Propositional Logic

1.1 Model-Theoretic View

Toolkit

This section presents classical propositional logic as a syntax-plus-semantics model, with the syntactic language connected to the two-valued semantics by the shared free-algebra evaluation machinery.

Theory (Propositional Logic)

Depends on: `Propositional language; formula; bridge:consequence;` $\mathrm{Cn}_{\vdash}$

Language

$$\mathcal{L}_{\mathsf{P}} = (\mathsf{P}, \neg, \wedge, \vee, \rightarrow, \leftrightarrow), \qquad \mathsf{Form}_{\mathsf{P}}.$$

Presented Theory

$$T := \mathrm{Cn}_{\vdash}(\Sigma), \qquad \Sigma \subseteq \mathsf{Form}_{\mathsf{P}}, \qquad \mathrm{CPL} := \mathrm{Cn}_{\vdash}(\varnothing).$$

Theorems and Consequences

1. $$\Gamma \vdash_{\mathrm{CPL}} \varphi \iff \Gamma \models_{\mathrm{CPL}} \varphi \qquad \text{(completeness)}.$$

Model (Classical Propositional Logic)

Model 1.1.1 (Classical Propositional Logic). **Depends on:** `Propositional language; valuation; bridge:free-sigma-algebra-evaluation; bridge:consequence`

Note

The seed is the model datum itself.

$$v : \mathsf{P} \to |\mathbf{2}|, \qquad |\mathbf{2}| = \{0, 1\}.$$

Binding

$$\Sigma := \Omega, \qquad X := \mathsf{P}, \qquad \text{target} := \mathbf{2}, \qquad \sigma := v, \qquad \widehat{v} := \text{eval}(\mathbf{2})(v).$$

Syntax–Semantics Bridge

1. $$v \models \varphi \quad :\Longleftrightarrow \quad \widehat{v}(\varphi) = 1.$$

Theorems and Consequences

1. Connective recursion is inherited from *Free Σ-Algebra and Evaluation*.
2. Semantic consequence, validity, and $\mathrm{Th}(v)$ are inherited from *Consequence*.

Definitional Extension (Propositional Logic)

Depends on: `theory:propositional-logic; truth-function; conservative extension`

Admissibility

A connective $\star$ of arity n is admissible by a defining equivalence:

$$\star(\bar{\varphi}) \leftrightarrow \theta(\bar{\varphi}), \qquad \widehat{v}(\star(\bar{\varphi})) = F_\star(\widehat{v}(\varphi_1), \dots, \widehat{v}(\varphi_n)).$$

There is no $\exists!$ obligation because formulas already evaluate to truth values.

Instances

$$\top :\equiv (p_0 \to p_0), \qquad \bot :\equiv \neg\top,$$
$$\varphi \oplus \psi :\equiv ((\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)).$$

Theorems and Consequences

1. $$\mathrm{CPL}^+ \vdash \sigma \quad \Longleftrightarrow \quad \mathrm{CPL} \vdash \sigma \qquad (\sigma \in \mathcal{L}_{\mathsf{P}}).$$
2. Every extended formula has a base-language equivalent.

1.2 Axiom Schemas

Propositional Axiom Schemas

The propositional-logic development uses the following classical Hilbert-style axiom schemas as logical starting points. They are schemas: each substitution of formulas for the displayed schematic letters gives an axiom instance.

Axiom Schema (Propositional Weakening)

Axiom 1.2.1 (Propositional Weakening)**.** For all formulas P and Q,

$$P \implies (Q \implies P).$$

Remark (Interpretation). If P is available, then it remains available under any additional assumption Q.

Axiom Schema (Implication Distribution)

Axiom 1.2.2 (Implication Distribution)**.** For all formulas P, Q, and R,

$$\big(P \implies (Q \implies R)\big) \implies \big((P \implies Q) \implies (P \implies R)\big).$$

Remark (Interpretation). Implications may be composed under a common hypothesis.

Axiom Schema (Classical Contraposition)

Axiom 1.2.3 (Classical Contraposition)**.** For all formulas P and Q,

$$(\neg Q \implies \neg P) \implies (P \implies Q).$$

Remark (Interpretation). This is the classical contrapositive principle. It is the axiom schema that marks the background logic as classical rather than intuitionistic.

1.3 Notation and Conventions

Toolkit

This notation ledger records the propositional symbols, metavariables, semantic notation, and normal-form abbreviations used in the chapter.

Remark (Exposition). The notation section fixes the symbolic vocabulary used throughout the chapter. It identifies which symbols are global conventions, which are local to propositional logic, and where each item is introduced by the formal development.

Symbol	Name	Meaning	Introduced by
$\mathbb{B}$	truth-value set	the set of truth values (global, canonical)	Truth Value
Prop	propositional variable set	the set of propositional variables (global, canonical)	Propositional Language
$\neg$	negation	unary propositional connective (global, canonical)	Propositional Language
$\wedge$	conjunction	binary propositional connective (global, canonical)	Propositional Language
$\vee$	disjunction	binary propositional connective (global, canonical)	Propositional Language
$\to$	conditional	binary propositional connective (global, canonical)	Propositional Language
$\leftrightarrow$	biconditional	binary propositional connective (global, canonical)	Propositional Language
P, Q, R	propositional variables	schematic propositional variables (chapter, canonical)	Propositional Variable
$P_1, \ldots, P_n$	finite list of propositional variables	a finite schematic list of propositional variables (chapter, local)	Propositional Variable
$\circ$	generic binary connective	an arbitrary member of $\{\wedge, \vee, \to, \leftrightarrow\}$ (global, canonical)	Logical Connective
WFF	well-formed formula set	the set of propositional well-formed formulas (chapter, canonical)	Well-Formed Formula
$\varphi, \psi, \chi, \theta$	formula metavariables	metavariables ranging over propositional well-formed formulas (global, canonical)	Well-Formed Formula
α, β	additional formula metavariables	formula metavariables used in comparison statements (chapter, local)	Constructor Injectivity for Propositional Formulas
$\mathrm{Tree}(\varphi)$	syntax tree	parse tree or formation tree of a formula (chapter, canonical)	Parse Tree
$\mathrm{Var}(\varphi)$	variable set of a formula	the set of propositional variables occurring in a formula (chapter, canonical)	Variable Set of a Formula
$\mathrm{Sub}(\varphi)$	subformula set	the set of subformulas of a formula (chapter, canonical)	Subformula
$\mathrm{depth}(\varphi)$	formula depth	height, rank, or complexity of the formula formation tree (chapter, canonical)	Formula Depth

$\mathrm{conn}(\varphi)$	connective count	the number of connective occurrences in a formula (chapter, canonical)	Connective Count
v	truth assignment	a truth assignment on propositional variables (chapter, canonical)	Truth Assignment
w	second truth assignment	truth assignment on propositional variables (global, canonical)	Truth Assignment
$\widehat{v}$	extended truth assignment	unique extension of a truth assignment to formulas (chapter, canonical)	Unique Extension of a Truth Assignment
$\neg, \wedge, \vee, \to, \leftrightarrow$	standard connectives	standard propositional connectives (global, canonical)	Unique Extension of a Truth Assignment
$v \models \varphi$	satisfaction of a formula	truth assignment satisfies a formula (chapter, canonical)	Satisfaction of a Formula
$\models \varphi$	tautology notation	formula valid under every truth assignment (global, canonical)	Satisfaction of a Formula
$v \models \Gamma$	satisfaction of a set	truth assignment satisfies each formula in a set (chapter, canonical)	Satisfaction of a Set of Formulas
Γ	set of premise formulas	a set of propositional formulas (chapter, canonical)	Satisfaction of a Set of Formulas
$\Gamma \models \varphi$	semantic consequence	semantic consequence from premises to conclusion (chapter, canonical)	Logical Consequence
$\varphi \equiv \psi$	logical equivalence	same truth value under every truth assignment (chapter, canonical)	Logical Equivalence
$f : \mathbb{B}^n \to \mathbb{B}$	n-ary Boolean function	function from truth-value tuples to a truth value (chapter, canonical)	Boolean Function
f_φ	truth function induced by a formula	Boolean function represented by a formula (chapter, canonical)	Truth Function Induced by a Formula
$\equiv$	logical equivalence symbol	logical equivalence relation between formulas (global, canonical)	Propositional Equivalence Law
$\top$	tautological formula	a formula chosen as a true constant (chapter, canonical)	Identity and Domination Laws

$\perp$	contradictory formula	a formula chosen as a false constant (chapter, canonical)	Identity and Domination Laws
$C[-]$	formula context	a one-hole propositional formula context (chapter, canonical)	Formula Context
$C[\varphi]$	context substitution	formula obtained by filling a context with a formula (chapter, canonical)	Formula Context

1.4 Syntax

Toolkit

This section uses truth values, propositional variables, connectives, well-formed formulas, formation trees, and subformulas to build syntax.

Definitional Root (Truth Value)

Definition 1.4.1 (Truth Value)**.** The set of truth values is

$$\mathbb{B} := \{\text{False}, \text{True}\}.$$

Some sources identify this set with $\{0, 1\}$, with 0 read as false and 1 read as true.

Remark (Standard quantified statement).

$$\mathbb{B} = \{\text{False}, \text{True}\}.$$

Remark (Interpretation). Truth values are the semantic targets of propositional formulas. Syntax determines which strings are formulas; semantics later assigns elements of $\mathbb{B}$ to formulas.

Definitional Root (Propositional Language)

Definition 1.4.2 (Propositional Language)**.** A propositional language consists of:

1. a set Prop of propositional variables;
2. logical connectives;
3. punctuation symbols such as parentheses;
4. recursive formation rules specifying the well-formed formulas.

Remark (Standard quantified statement).

$$\mathcal{L}_{\text{prop}} = (\mathsf{Prop}, \{\neg, \wedge, \vee, \to, \leftrightarrow\}, \text{punctuation}, \text{formation rules}).$$

Remark (Interpretation). The language supplies symbols and grammar. It does not yet assign truth values, truth conditions, or logical consequence.

Definition (Propositional Variable)

Definition 1.4.3 (Propositional Variable)**.** A propositional variable is an atomic symbol intended to stand for a proposition. A standard supply of variables is

$$\mathsf{Prop} = \{P_1, P_2, P_3, \ldots\}.$$

Informally, variables are often written $P, Q, R, S, \ldots$.

Remark (Standard quantified statement).

$$\text{PropositionalVariable}(P) \iff P \in \mathsf{Prop}.$$

Remark (Definition predicate reading). $\text{PropositionalVariable}(P)$ means $P \in \mathsf{Prop}$.

Remark (Interpretation). A propositional variable has no internal syntactic structure in propositional logic. It is a basic symbol from which more complex formulas are formed.

Remark (Examples). P, Q, R, P_1, P_2 are propositional variables when they belong to Prop.

Remark (Non-examples). $\neg P$, $P \wedge Q$, and $(P \to Q)$ are not propositional variables; they are compound formulas.

Remark (Dependencies).

- Propositional Language

Definition (Logical Connective)

Definition 1.4.4 (Logical Connective)**.** The standard primitive propositional connectives are

$$\neg, \qquad \wedge, \qquad \vee, \qquad \to, \qquad \leftrightarrow.$$

The connective $\neg$ is unary. The connectives $\wedge, \vee, \to, \leftrightarrow$ are binary.

Symbol	Name	Arity	Formation pattern	Reading
$\neg$	negation	unary	$\neg\varphi$	not φ
$\wedge$	conjunction	binary	$(\varphi \wedge \psi)$	φ and ψ
$\vee$	disjunction	binary	$(\varphi \vee \psi)$	φ or ψ
$\to$	conditional	binary	$(\varphi \to \psi)$	if φ, then ψ
$\leftrightarrow$	biconditional	binary	$(\varphi \leftrightarrow \psi)$	φ iff ψ

Remark (Standard quantified statement).

$$\text{UnaryConnective}(\neg) \quad \text{and} \quad \text{BinaryConnective}(\wedge) \wedge \text{BinaryConnective}(\vee) \wedge \text{BinaryConnective}(\to) \wedge \text{Binary}$$

Remark (Interpretation). Arity records how many formulas a connective uses to form a new formula. The truth behavior of these connectives belongs to semantics, not to the syntactic definition.

Remark (Dependencies).

- Propositional Language

Definition (Well-Formed Formula)

Definition 1.4.5 (Well-Formed Formula)**.** The set WFF of well-formed formulas is the smallest set of strings satisfying the following formation rules:

1. Every propositional variable $P \in \mathsf{Prop}$ belongs to WFF.
2. If $\varphi \in \mathsf{WFF}$, then $\neg\varphi \in \mathsf{WFF}$.
3. If $\varphi, \psi \in \mathsf{WFF}$ and $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$, then $(\varphi \circ \psi) \in \mathsf{WFF}$.
4. No string is a well-formed formula unless it is obtained by finitely many applications of the previous clauses.

Remark (Standard quantified statement).

$$\mathsf{WFF} = \bigcap \left\{ S : \mathsf{Prop} \subseteq S,\ (\forall\varphi \in S)(\neg\varphi \in S),\ (\forall\varphi, \psi \in S)(\forall\circ \in \{\wedge, \vee, \to, \leftrightarrow\})\big((\varphi \circ \psi) \in S\big) \right\}.$$

Remark (Definition predicate reading)*.* WellFormedFormula(φ) means $\varphi \in \mathsf{WFF}$, read as "φ is a well-formed formula."

Remark (Negated quantified statement).

$$\sigma \notin \mathsf{WFF} \iff \exists S\, [\mathsf{Prop} \subseteq S \wedge (\forall\varphi \in S)(\neg\varphi \in S) \wedge (\forall\varphi, \psi \in S)(\forall\circ \in \{\wedge, \vee, \to, \leftrightarrow\})((\varphi \circ \psi) \in S) \wedge \sigma \notin S]$$

Remark (Negation predicate reading)*.* $\neg$ WellFormedFormula(σ) means the string σ is excluded by at least one formation-closed set containing Prop; equivalently, it is not generated by finitely many applications of the formation rules.

Remark (Failure modes)*.* A string fails to be well formed when it violates the formation grammar: an operand may be missing, a connective may be missing, parentheses may be unmatched, or the string may not be generated by finitely many formation steps.

Remark (Failure mode decomposition).

$$\underbrace{P\wedge}_{\text{missing right operand}} \qquad \underbrace{(P\ Q)}_{\text{missing binary connective}} \qquad \underbrace{\neg\vee P}_{\text{negation applied to a non-formula}} \qquad .$$

Remark (Interpretation)*.* Well-formed formulas are exactly the strings to which the semantic and proof-theoretic machinery of propositional logic applies.

Remark (Examples)*.* P, $\neg P$, $(P \to Q)$, and $(P \to (Q \vee R))$ are well-formed formulas.

Remark (Non-examples)*.* $P\wedge$, (PQ), and $\neg \vee P$ are not well-formed formulas.

Remark (Dependencies).

- Propositional Variable

- Logical Connective

Lemma (Closure Under Formation)

Lemma 1.4.6 (Closure Under Formation)**.** *The set* WFF *is closed under the propositional formation rules.*
That is, if $\varphi \in \mathsf{WFF}$*, then*

$$\neg\varphi \in \mathsf{WFF}.$$

If $\varphi, \psi \in \mathsf{WFF}$ *and*

$$\circ \in \{\wedge, \vee, \to, \leftrightarrow\},$$

then

$$(\varphi \circ \psi) \in \mathsf{WFF}.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall\varphi \in \mathsf{WFF})(\neg\varphi \in \mathsf{WFF})$$

and

$$(\forall\varphi, \psi \in \mathsf{WFF})(\forall\circ \in \{\wedge, \vee, \to, \leftrightarrow\})((\varphi \circ \psi) \in \mathsf{WFF}).$$

Remark (Interpretation). Closure says that every legal use of a formation rule produces another well-formed formula. It is the forward direction of the recursive grammar.

Remark (Dependencies).

- Well-Formed Formula

Theorem (Minimality of Well-Formed Formulas)

Theorem 1.4.7 (Minimality of Well-Formed Formulas)**.** *Let* S *be a set of strings over the propositional alphabet. Suppose:*

1. $\mathsf{Prop} \subseteq S$*;*
2. *if* $\varphi \in S$*, then* $\neg\varphi \in S$*;*
3. *if* $\varphi, \psi \in S$ *and* $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$*, then* $(\varphi \circ \psi) \in S$*.*

Then

$$\mathsf{WFF} \subseteq S.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall S)\Big[\mathsf{Prop} \subseteq S \wedge (\forall\varphi \in S)(\neg\varphi \in S) \wedge (\forall\varphi, \psi \in S)(\forall\circ \in \{\wedge, \vee, \to, \leftrightarrow\})((\varphi \circ \psi) \in S)\Big] \to \mathsf{WFF} \subseteq S.$$

Remark (Interpretation)*.* Minimality is the formal content of the clause "nothing else is a wff." It says that WFF is contained in every set of strings that contains the propositional variables and is closed under the formation rules.

Remark (Dependencies)*.*

- Well-Formed Formula

Definition (Atomic Formula)

Definition 1.4.8 (Atomic Formula)**.** An atomic formula is a well-formed formula that is a propositional variable.

Remark (Standard quantified statement)*.*

$$\mathrm{AtomicFormula}(\varphi) \iff \varphi \in \mathsf{WFF} \wedge \varphi \in \mathsf{Prop}.$$

Remark (Definition predicate reading)*.* $\mathrm{AtomicFormula}(\varphi)$ means that φ is a propositional variable viewed as a well-formed formula.

Remark (Interpretation)*.* Atomic formulas are the base cases of the recursive grammar. They contain no connective structure.

Remark (Dependencies)*.*

- Well-Formed Formula
- Propositional Variable

Definition (Molecular Formula)

Definition 1.4.9 (Molecular Formula)**.** A molecular formula is a well-formed formula that is not atomic. Equivalently, it is a well-formed formula formed using at least one logical connective.

Remark (Standard quantified statement)*.*

$$\mathrm{MolecularFormula}(\varphi) \iff \varphi \in \mathsf{WFF} \wedge \neg\, \mathrm{AtomicFormula}(\varphi).$$

Remark (Definition predicate reading)*.* $\mathrm{MolecularFormula}(\varphi)$ means that φ has nontrivial connective structure.

Remark (Interpretation)*.* Molecular formulas are exactly the formulas obtained by applying negation or a binary connective at least once.

Remark (Dependencies)*.*

- Well-Formed Formula
- Atomic Formula
- Logical Connective

Definition (Main Connective)

Definition 1.4.10 (Main Connective)**.** The main connective of a molecular formula is the connective introduced at the final formation step:

1. for $\neg\varphi$, the main connective is $\neg$;
2. for $(\varphi \circ \psi)$, the main connective is $\circ$.

Atomic formulas have no main connective.

Remark (Standard quantified statement).

$$\mathrm{MainConnective}(\neg\varphi, \neg) \quad \text{and} \quad \mathrm{MainConnective}((\varphi \circ \psi), \circ).$$

Remark (Definition predicate reading). $\mathrm{MainConnective}(\varphi, c)$ means that c is the outermost connective introduced in the final formation step of φ.

Remark (Interpretation). The main connective identifies the outermost syntactic operation. It is the first connective used when analyzing a formula from the outside inward.

Remark (Dependencies).

- Well-Formed Formula
- Molecular Formula
- Logical Connective

Theorem (Structural Induction for Propositional Formulas)

Theorem 1.4.11 (Structural Induction for Propositional Formulas)**.** *Let $A(\varphi)$ be a property of well-formed formulas. Suppose:*

1. *$A(P)$ holds for every $P \in \mathsf{Prop}$;*
2. *if $A(\varphi)$ holds, then $A(\neg\varphi)$ holds;*
3. *if $A(\varphi)$ and $A(\psi)$ hold, then $A((\varphi \circ \psi))$ holds for every $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$.*

Then $A(\theta)$ holds for every $\theta \in \mathsf{WFF}$.
Go to proof.

Remark (Standard quantified statement).

$$\big[(\forall P \in \mathsf{Prop}) A(P) \wedge (\forall \varphi \in \mathsf{WFF})(A(\varphi) \to A(\neg\varphi)) \wedge (\forall \varphi, \psi \in \mathsf{WFF})(\forall \circ \in \{\wedge, \vee, \to, \leftrightarrow\})\big((A(\varphi) \wedge A(\psi)) -$$

Remark (Interpretation). To prove a statement for all formulas, it suffices to prove it for propositional variables and show it is preserved by each formula-forming operation.

Remark (Dependencies).

- Well-Formed Formula

Theorem (Structural Recursion for Propositional Formulas)

Theorem 1.4.12 (Structural Recursion for Propositional Formulas)**.** *To define a function F on* WFF*, it suffices to specify:*

1. *$F(P)$ for every $P \in \mathsf{Prop}$;*
2. *$F(\neg\varphi)$ in terms of $F(\varphi)$;*
3. *$F((\varphi \circ \psi))$ in terms of $F(\varphi)$, $F(\psi)$, and $\circ$.*

When these clauses respect the formation rules, they determine a unique function on WFF*.*
Go to proof.

Remark (Standard quantified statement).

recursive data on the formation clauses of $\mathsf{WFF} \Longrightarrow \exists! F : \mathsf{WFF} \to X$ satisfying those clauses. ∎

Remark (Interpretation). Structural recursion justifies recursive definitions such as variables of a formula, subformulas, depth, connective count, syntax tree, and truth evaluation.

Remark (Dependencies).

- Well-Formed Formula
- Structural Induction for Propositional Formulas

Lemma (Constructor Disjointness for Propositional Formulas)

Lemma 1.4.13 (Constructor Disjointness for Propositional Formulas)**.** *The three outer formation cases for propositional formulas are disjoint:*

1. *no propositional variable is a negation formula;*
2. *no propositional variable is a binary formula;*
3. *no negation formula is a binary formula.*

Go to proof.

Remark (Standard quantified statement).

$$P \in \mathsf{Prop} \Rightarrow [(\forall \varphi \in \mathsf{WFF})(P \neq \neg\varphi) \wedge (\forall \varphi, \psi \in \mathsf{WFF})(\forall \circ)(P \neq (\varphi \circ \psi))]$$

and

$$(\forall \varphi, \psi, \chi \in \mathsf{WFF})(\forall \circ)\big(\neg\varphi \neq (\psi \circ \chi)\big).$$

Remark (Predicate reading).

$$\underbrace{P}_{\text{atomic}} \qquad \underbrace{\neg\varphi}_{\text{negation case}} \qquad \underbrace{(\varphi \circ \psi)}_{\text{binary case}}$$

belong to three distinct outer-form classes.

Remark (Interpretation). A formula cannot be parsed as two different outer formation types.

Remark (Dependencies).

- Well-Formed Formula
- Atomic Formula
- Main Connective

Lemma (Constructor Injectivity for Propositional Formulas)

Lemma 1.4.14 (Constructor Injectivity for Propositional Formulas)**.** *The propositional formula constructors are injective:*

1. *if* $\neg\varphi = \neg\psi$, *then* $\varphi = \psi$;
2. *if* $(\varphi \circ \psi) = (\alpha \star \beta)$, *then* $\varphi = \alpha$, $\psi = \beta$, *and* $\circ = \star$.

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF})(\neg\varphi = \neg\psi \to \varphi = \psi)$$

and

$$(\forall \varphi, \psi, \alpha, \beta \in \mathsf{WFF})(\forall \circ, \star \in \{\wedge, \vee, \to, \leftrightarrow\})\left[(\varphi \circ \psi) = (\alpha \star \beta) \to (\varphi = \alpha \wedge \psi = \beta \wedge \circ = \star)\right].$$

Remark (Interpretation). If two formulas are built by the same outer constructor and are equal as strings, then their immediate constituents are equal.

Remark (Dependencies).

- Well-Formed Formula
- Logical Connective

Theorem (Unique Decomposition for Propositional Formulas)

Theorem 1.4.15 (Unique Decomposition for Propositional Formulas)**.** *Every* $\varphi \in \mathsf{WFF}$ *has exactly one outermost form:*

1. $\varphi = P$ *for a unique* $P \in \mathsf{Prop}$*;*
2. $\varphi = \neg\psi$ *for a unique* $\psi \in \mathsf{WFF}$*;*
3. $\varphi = (\psi \circ \chi)$ *for unique* $\psi, \chi \in \mathsf{WFF}$ *and a unique binary connective* $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$.

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall\varphi \in \mathsf{WFF}, \quad \text{exactly one of the following holds:}$$

$$\varphi \in \mathsf{Prop}, \qquad \exists!\psi \in \mathsf{WFF}\,(\varphi = \neg\psi), \qquad \exists!\psi, \chi \in \mathsf{WFF}\ \exists!\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}\,(\varphi = (\psi \circ \chi)).$$

Remark (Predicate reading)*.*

$$\varphi \in \underbrace{\mathsf{Prop}}_{\text{atomic case}} \quad \text{or} \quad \varphi = \underbrace{\neg\psi}_{\text{negation case}} \quad \text{or} \quad \varphi = \underbrace{(\psi \circ \chi)}_{\text{binary case}},$$

and exactly one of these alternatives holds.

Remark (Interpretation)*.* Unique decomposition makes recursive definitions and induction on formulas unambiguous.

Remark (Dependencies)*.*

- Well-Formed Formula
- Constructor Disjointness for Propositional Formulas
- Constructor Injectivity for Propositional Formulas

Definition (Parse Tree)

Definition 1.4.16 (Parse Tree)**.** The parse tree, or syntax tree, of a well-formed formula is the tree representation of its recursive construction:

1. a propositional variable is represented by a single leaf;
2. a formula $\neg\varphi$ is represented by a root labeled $\neg$ with one child, the parse tree of φ;
3. a formula $(\varphi \circ \psi)$ is represented by a root labeled $\circ$ with two children, the parse trees of φ and ψ.

Remark (Standard quantified statement).

$$\text{Tree} : \mathsf{WFF} \to \{\text{finite rooted labeled trees}\}$$

is defined recursively by the three formation clauses for propositional formulas.

Remark (Interpretation). The parse tree records the same structure described by unique decomposition. It turns the recursive construction of a formula into a visible tree.

Remark (Dependencies).

- Well-Formed Formula
- Unique Decomposition for Propositional Formulas

Theorem (Unique Syntax Tree for Propositional Formulas)

Theorem 1.4.17 (Unique Syntax Tree for Propositional Formulas)**.** *Every propositional well-formed formula has a unique parse tree.*
Go to proof.

Remark (Standard quantified statement).

$$\forall \varphi \in \mathsf{WFF}, \quad \exists! T \ (T = \text{Tree}(\varphi)).$$

Remark (Interpretation). A formula has exactly one tree representation of its recursive construction. This is the tree form of unique decomposition.

Remark (Dependencies).

- Parse Tree
- Unique Decomposition for Propositional Formulas

Definition (Variable Set of a Formula)

Definition 1.4.18 (Variable Set of a Formula)**.** The variable set $\text{Var}(\varphi)$ of a formula φ is defined recursively:

1. if $\varphi = P \in \mathsf{Prop}$, then $\text{Var}(\varphi) = \{P\}$;
2. if $\varphi = \neg\psi$, then $\text{Var}(\varphi) = \text{Var}(\psi)$;
3. if $\varphi = (\psi \circ \chi)$, then
$$\text{Var}(\varphi) = \text{Var}(\psi) \cup \text{Var}(\chi).$$

Remark (Standard quantified statement).

$$\text{Var}(P) = \{P\}, \qquad \text{Var}(\neg\psi) = \text{Var}(\psi), \qquad \text{Var}((\psi \circ \chi)) = \text{Var}(\psi) \cup \text{Var}(\chi).$$

Remark (Interpretation). The variable set records exactly which propositional variables occur in a formula.

Remark (Dependencies).

- Well-Formed Formula
- Structural Recursion for Propositional Formulas

Definition (Subformula)

Definition 1.4.19 (Subformula)**.** The set $\mathrm{Sub}(\varphi)$ of subformulas of a formula φ is defined recursively:

1. if φ is atomic, then $\mathrm{Sub}(\varphi) = \{\varphi\}$;
2. if $\varphi = \neg\psi$, then
$$\mathrm{Sub}(\varphi) = \{\varphi\} \cup \mathrm{Sub}(\psi);$$
3. if $\varphi = (\psi \circ \chi)$, then
$$\mathrm{Sub}(\varphi) = \{\varphi\} \cup \mathrm{Sub}(\psi) \cup \mathrm{Sub}(\chi).$$

Remark (Standard quantified statement).

$$\mathrm{Sub}(P) = \{P\}, \qquad \mathrm{Sub}(\neg\psi) = \{\neg\psi\} \cup \mathrm{Sub}(\psi),$$
$$\mathrm{Sub}((\psi \circ \chi)) = \{(\psi \circ \chi)\} \cup \mathrm{Sub}(\psi) \cup \mathrm{Sub}(\chi).$$

Remark (Interpretation)*.* Subformulas are the formulas that occur at nodes of the parse tree.

Remark (Dependencies).

- Well-Formed Formula
- Unique Decomposition for Propositional Formulas

Definition (Proper Subformula)

Definition 1.4.20 (Proper Subformula)**.** Let $\varphi \in \mathsf{WFF}$. A *proper subformula* of φ is a formula ψ such that
$$\psi \in \mathrm{Sub}(\varphi) \quad \text{and} \quad \psi \neq \varphi.$$

Remark (Standard quantified statement).

$$\mathrm{ProperSubformula}(\psi, \varphi) \iff \psi \in \mathrm{Sub}(\varphi) \land \psi \neq \varphi.$$

Remark (Interpretation)*.* Proper subformulas are the strict syntactic parts of a formula. They exclude the whole formula itself.

Remark (Dependencies).

- Subformula

- Well-Formed Formula

Definition (Formula Depth)

Definition 1.4.21 (Formula Depth)**.** The depth of a formula φ, denoted $\operatorname{depth}(\varphi)$, is defined recursively:

1. if φ is atomic, then $\operatorname{depth}(\varphi) = 0$;
2. if $\varphi = \neg\psi$, then $\operatorname{depth}(\varphi) = \operatorname{depth}(\psi) + 1$;
3. if $\varphi = (\psi \circ \chi)$, then

$$\operatorname{depth}(\varphi) = \max\{\operatorname{depth}(\psi), \operatorname{depth}(\chi)\} + 1.$$

Remark (Standard quantified statement)*.*

$$\operatorname{depth}(P) = 0, \qquad \operatorname{depth}(\neg\psi) = \operatorname{depth}(\psi) + 1,$$

$$\operatorname{depth}((\psi \circ \chi)) = \max\{\operatorname{depth}(\psi), \operatorname{depth}(\chi)\} + 1.$$

Remark (Interpretation)*.* Formula depth measures the height of the syntax tree. Some authors call this the rank, complexity, or formation depth of the formula.

Remark (Dependencies)*.*

- Well-Formed Formula
- Parse Tree

Definition (Connective Count)

Definition 1.4.22 (Connective Count)**.** The connective count of a formula φ, denoted $\operatorname{conn}(\varphi)$, is defined recursively:

1. if φ is atomic, then $\operatorname{conn}(\varphi) = 0$;
2. if $\varphi = \neg\psi$, then $\operatorname{conn}(\varphi) = \operatorname{conn}(\psi) + 1$;
3. if $\varphi = (\psi \circ \chi)$, then

$$\operatorname{conn}(\varphi) = \operatorname{conn}(\psi) + \operatorname{conn}(\chi) + 1.$$

Remark (Standard quantified statement)*.*

$$\operatorname{conn}(P) = 0, \qquad \operatorname{conn}(\neg\psi) = \operatorname{conn}(\psi) + 1,$$

$$\operatorname{conn}((\psi \circ \chi)) = \operatorname{conn}(\psi) + \operatorname{conn}(\chi) + 1.$$

Remark (Interpretation)*.* Formula depth measures the height of the syntax tree. Connective count measures the number of connective nodes in the syntax tree.

Remark (Dependencies).

- Well-Formed Formula
- Parse Tree

Lemma (Proper Subformula Depth)

Lemma 1.4.23 (Proper Subformula Depth)**.** *If ψ is a proper subformula of φ, then*

$$\operatorname{depth}(\psi) < \operatorname{depth}(\varphi).$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF})\left[\psi \in \operatorname{Sub}(\varphi) \wedge \psi \neq \varphi \rightarrow \operatorname{depth}(\psi) < \operatorname{depth}(\varphi)\right].$$

Remark (Interpretation). Every descent into a proper subformula lowers formula depth. This justifies induction on formula depth and recursive proofs over subformulas.

Remark (Dependencies).

- Subformula
- Formula Depth

Lemma (Finiteness of Variables in a Formula)

Lemma 1.4.24 (Finiteness of Variables in a Formula)**.** *For every $\varphi \in \mathsf{WFF}$, the set* $\operatorname{Var}(\varphi)$ *is finite.*
Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi \in \mathsf{WFF}) \quad \operatorname{Var}(\varphi) \text{ is finite.}$$

Remark (Interpretation). Every formula depends on only finitely many propositional variables, even though the language may contain infinitely many variables.

Remark (Dependencies).

- Variable Set of a Formula
- Structural Induction for Propositional Formulas

Lemma (Finiteness of Subformulas)

Lemma 1.4.25 (Finiteness of Subformulas)**.** *For every $\varphi \in \mathsf{WFF}$, the set* $\operatorname{Sub}(\varphi)$ *is finite.*
Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi \in \mathsf{WFF}) \quad \mathrm{Sub}(\varphi) \text{ is finite.}$$

Remark (Interpretation). Every formula has only finitely many syntactic parts.

Remark (Dependencies).

- Subformula
- Structural Induction for Propositional Formulas

1.5 Semantics

Toolkit

This section uses truth assignments, recursive extensions, satisfaction, truth tables, and logical equivalence to assign meanings to formulas.

Remark (Exposition). The semantics section assigns truth conditions to the syntactic formulas introduced earlier. Truth assignments, recursive extensions, satisfaction, and truth tables provide the machinery for validity, consequence, equivalence, and semantic classification.

Truth Assignments

Definition (Truth Assignment)

Definition 1.5.1 (Truth Assignment)**.** A *truth assignment* is a function

$$v : \mathsf{Prop} \to \mathbb{B}.$$

For each propositional variable $P \in \mathsf{Prop}$, the value $v(P)$ is the truth value assigned to P.

Remark (Standard quantified statement).

$$\mathrm{TruthAssignment}(v) \Longleftrightarrow v : \mathsf{Prop} \to \mathbb{B}.$$

Remark (Definition predicate reading).

$$\mathrm{TruthAssignment}(v)$$

means that v assigns exactly one truth value to each propositional variable.

Remark (Interpretation). A truth assignment gives semantic values to the atomic formulas. The recursive truth conditions for the connectives then determine the truth value of every well-formed formula.

Remark (Examples). If $\mathsf{Prop} = \{P_1, P_2, P_3, \ldots\}$, then a truth assignment may satisfy

$$v(P_1) = \mathsf{True}, \qquad v(P_2) = \mathsf{False}, \qquad v(P_3) = \mathsf{True},$$

with values assigned to every remaining propositional variable.

Remark (Non-Examples). A partial assignment that gives values only to P_1 and P_2 is not a truth assignment on Prop. It may be useful for local computations, but it is not a function $\mathsf{Prop} \to \mathbb{B}$.

Remark (Dependencies).

- Truth Value
- Propositional Variable
- Well-Formed Formula

Definition (Domain of a Truth Assignment)

Definition 1.5.2 (Domain of a Truth Assignment)**.** The *domain* of a truth assignment

$$v : \mathsf{Prop} \to \mathbb{B}$$

is the set Prop of propositional variables.

Remark (Standard quantified statement).

$$\mathrm{TruthAssignment}(v) \Longrightarrow \mathrm{dom}(v) = \mathsf{Prop}.$$

Remark (Interpretation). A truth assignment is global: it assigns truth values to all propositional variables in the language, even when a particular formula uses only finitely many of them.

Remark (Dependencies).

- Truth Assignment

Extension of a Truth Assignment

Definition (Extension of a Truth Assignment to Formulas)

Definition 1.5.3 (Extension of a Truth Assignment to Formulas)**.** Let

$$v : \mathsf{Prop} \to \mathbb{B}$$

be a truth assignment. The *extension* of v to well-formed formulas is the function

$$\widehat{v} : \mathsf{WFF} \to \mathbb{B}$$

defined recursively as follows:

$$\widehat{v}(P) = v(P) \quad \text{for every } P \in \mathsf{Prop},$$

$$\widehat{v}(\neg\varphi) = \mathsf{True} \iff \widehat{v}(\varphi) = \mathsf{False},$$

$$\widehat{v}(\varphi \wedge \psi) = \mathsf{True} \iff \widehat{v}(\varphi) = \mathsf{True} \text{ and } \widehat{v}(\psi) = \mathsf{True},$$

$$\widehat{v}(\varphi \vee \psi) = \mathsf{True} \iff \widehat{v}(\varphi) = \mathsf{True} \text{ or } \widehat{v}(\psi) = \mathsf{True},$$

$$\widehat{v}(\varphi \to \psi) = \mathsf{False} \iff \widehat{v}(\varphi) = \mathsf{True} \text{ and } \widehat{v}(\psi) = \mathsf{False},$$

and

$$\widehat{v}(\varphi \leftrightarrow \psi) = \mathsf{True} \iff \widehat{v}(\varphi) = \widehat{v}(\psi).$$

Remark (Standard quantified statement)*.* For every truth assignment $v : \mathsf{Prop} \to \mathbb{B}$, its extension is a function

$$\widehat{v} : \mathsf{WFF} \to \mathbb{B}$$

satisfying the recursive truth clauses for propositional variables, negation, conjunction, disjunction, material implication, and biconditional.

Remark (Definition predicate reading)*.*

$$\text{ExtendedTruthAssignment}(\widehat{v}, v)$$

means that $\widehat{v}$ is the recursively defined truth-value function on WFF induced by the atomic truth assignment v.

Remark (Interpretation)*.* The original assignment v gives truth values only to propositional variables. The extension $\widehat{v}$ evaluates every formula by following the formula's syntactic construction.

Remark (Examples)*.*

φ	$\neg\varphi$
True	False
False	True

φ	ψ	$\varphi \wedge \psi$
True	True	True
True	False	False
False	True	False
False	False	False

φ	ψ	$\varphi \vee \psi$
True	True	True
True	False	True
False	True	True
False	False	False

φ	ψ	$\varphi \to \psi$
True	True	True
True	False	False
False	True	True
False	False	True

φ	ψ	$\varphi \leftrightarrow \psi$
True	True	True
True	False	False
False	True	False
False	False	True

Remark (Examples). The truth tables are not additional formation rules. They specify how each primitive connective acts on truth values once the formula has already been formed syntactically.

Remark (Dependencies).

- Truth Assignment
- Well-Formed Formula
- Structural Recursion for Propositional Formulas
- Unique Decomposition for Propositional Formulas

Theorem (Unique Extension of a Truth Assignment)

Theorem 1.5.4 (Unique Extension of a Truth Assignment)**.** *Every truth assignment*

$$v : \mathsf{Prop} \to \mathbb{B}$$

has a unique extension

$$\widehat{v} : \mathsf{WFF} \to \mathbb{B}$$

satisfying the recursive truth clauses for propositional variables and the primitive connectives.
Go to proof.

Remark (Standard quantified statement).

$$(\forall v : \mathsf{Prop} \to \mathbb{B})(\exists! \widehat{v} : \mathsf{WFF} \to \mathbb{B})$$

such that $\widehat{v}$ agrees with v on Prop and satisfies the recursive truth clauses for

$$\neg, \wedge, \vee, \to, \leftrightarrow .$$

Remark (Predicate reading).

$$\text{TruthAssignment}(v) \implies \exists! \widehat{v}\ \text{ExtendedTruthAssignment}(\widehat{v}, v).$$

Remark (Interpretation). This theorem guarantees that semantic evaluation is well-defined. Once the truth values of the propositional variables are fixed, every well-formed formula has exactly one truth value.

Remark (Dependencies).

- Truth Assignment
- Extension of a Truth Assignment to Formulas
- Structural Recursion for Propositional Formulas
- Unique Decomposition for Propositional Formulas

Definition (Satisfaction of a Formula)

Definition 1.5.5 (Satisfaction of a Formula)**.** Let v be a truth assignment and let $\varphi \in \mathsf{WFF}$. We say that v satisfies φ, written

$$v \models \varphi,$$

if

$$\widehat{v}(\varphi) = \text{True}.$$

Remark (Standard quantified statement).

$$v \models \varphi \iff \text{TruthAssignment}(v) \wedge \varphi \in \mathsf{WFF} \wedge \widehat{v}(\varphi) = \text{True}.$$

Remark (Definition predicate reading).

$$\text{SatisfiesFormula}(v, \varphi)$$

means $v \models \varphi$.

Remark (Negated quantified statement).

$$v \not\models \varphi \iff \widehat{v}(\varphi) = \text{False}.$$

Remark (Interpretation). The notation $v \models \varphi$ means that the formula φ is true under the truth assignment v.

Remark (Dependencies).

- Truth Assignment
- Unique Extension of a Truth Assignment
- Well-Formed Formula

Definition (Satisfaction of a Set of Formulas)

Definition 1.5.6 (Satisfaction of a Set of Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A truth assignment v satisfies Γ, written

$$v \models \Gamma,$$

if $v \models \gamma$ for every $\gamma \in \Gamma$.

Remark (Standard quantified statement).

$$v \models \Gamma \iff \text{TruthAssignment}(v) \wedge \Gamma \subseteq \mathsf{WFF} \wedge (\forall \gamma \in \Gamma)(v \models \gamma).$$

Remark (Definition predicate reading).

$$\text{SatisfiesSet}(v, \Gamma)$$

means v satisfies every formula in Γ.

Remark (Negated quantified statement).

$$v \not\models \Gamma \iff \exists \gamma \in \Gamma \text{ such that } v \not\models \gamma.$$

Remark (Interpretation). Satisfaction of a set means simultaneous satisfaction of every formula in the set.

Remark (Dependencies).

- Satisfaction of a Formula
- Truth Assignment

Definition (Truth Table)

Definition 1.5.7 (Truth Table)**.** Let $\varphi \in \mathsf{WFF}$ and suppose the variables occurring in φ are among $P_1, \ldots, P_n$. A truth table for φ is a finite table with one row for each assignment

$$a : \{P_1, \ldots, P_n\} \to \mathbb{B}$$

and a final column recording the value of $\widehat{v}(\varphi)$ for any truth assignment v extending a.

Remark (Standard quantified statement).

$$\text{TruthTable}(\varphi) \iff \varphi \in \mathsf{WFF} \wedge \text{Var}(\varphi) = \{P_1, \ldots, P_n\} \wedge \text{the table has } 2^n \text{ assignment rows.}$$

Remark (Interpretation). A truth table is finite because every formula contains only finitely many variables. It records the complete truth behavior of a formula.

Remark (Dependencies).

- Variable Set of a Formula
- Finiteness of Variables in a Formula

- Unique Extension of a Truth Assignment

Definition (Tautology)

Definition 1.5.8 (Tautology)**.** A formula $\varphi \in \mathsf{WFF}$ is a tautology, written

$$\models \varphi,$$

if every truth assignment satisfies φ.

Remark (Standard quantified statement)*.*

$$\models \varphi \iff \varphi \in \mathsf{WFF} \land (\forall v : \mathsf{Prop} \to \mathbb{B})(v \models \varphi).$$

Remark (Definition predicate reading)*.*

$$\text{Tautology}(\varphi)$$

means $\models \varphi$.

Remark (Interpretation)*.* A tautology is true under every possible assignment of truth values to propositional variables.

Remark (Dependencies)*.*

- Satisfaction of a Formula
- Truth Assignment

Definition (Contradiction)

Definition 1.5.9 (Contradiction)**.** A formula $\varphi \in \mathsf{WFF}$ is a contradiction if no truth assignment satisfies φ.

Remark (Standard quantified statement)*.*

$$\text{Contradiction}(\varphi) \iff \varphi \in \mathsf{WFF} \land (\forall v : \mathsf{Prop} \to \mathbb{B})(v \not\models \varphi).$$

Remark (Definition predicate reading)*.*

$$\text{Contradiction}(\varphi)$$

means that φ is false under every truth assignment.

Remark (Interpretation)*.* A contradiction has no true row in its truth table.

Remark (Dependencies)*.*

- Satisfaction of a Formula
- Truth Assignment

Definition (Satisfiable Formula)

Definition 1.5.10 (Satisfiable Formula)**.** A formula $\varphi \in \mathsf{WFF}$ is satisfiable if at least one truth assignment satisfies it.

Remark (Standard quantified statement).

$$\mathrm{SatisfiableFormula}(\varphi) \iff \varphi \in \mathsf{WFF} \land \exists v : \mathsf{Prop} \to \mathbb{B} \text{ such that } v \models \varphi.$$

Remark (Definition predicate reading).

$$\mathrm{SatisfiableFormula}(\varphi)$$

means that φ is true under at least one truth assignment.

Remark (Interpretation)*.* A satisfiable formula has at least one true row in its truth table.

Remark (Dependencies).

- Satisfaction of a Formula
- Truth Assignment

Definition (Contingency)

Definition 1.5.11 (Contingency)**.** A formula $\varphi \in \mathsf{WFF}$ is a contingency if it is satisfiable but not a tautology.

Remark (Standard quantified statement).

$$\mathrm{Contingency}(\varphi) \iff \mathrm{SatisfiableFormula}(\varphi) \land \neg\, \mathrm{Tautology}(\varphi).$$

Remark (Definition predicate reading).

$$\mathrm{Contingency}(\varphi)$$

means that φ is true under some truth assignments and false under others.

Remark (Interpretation)*.* A contingency has at least one true row and at least one false row in its truth table.

Remark (Dependencies).

- Satisfiable Formula
- Tautology

Definition (Logical Consequence)

Definition 1.5.12 (Logical Consequence)**.** Let $\Gamma \subseteq \mathsf{WFF}$ and $\varphi \in \mathsf{WFF}$. The formula φ is a logical consequence of Γ, written

$$\Gamma \models \varphi,$$

if every truth assignment satisfying Γ also satisfies φ.

Remark (Standard quantified statement).

$$\Gamma \models \varphi \iff \Gamma \subseteq \mathsf{WFF} \wedge \varphi \in \mathsf{WFF} \wedge (\forall v : \mathsf{Prop} \to \mathbb{B}) \left[v \models \Gamma \to v \models \varphi \right].$$

Remark (Definition predicate reading).

$$\text{LogicalConsequence}(\Gamma, \varphi)$$

means $\Gamma \models \varphi$.

Remark (Negated quantified statement).

$$\Gamma \not\models \varphi \iff \exists v : \mathsf{Prop} \to \mathbb{B} \left[v \models \Gamma \wedge v \not\models \varphi \right].$$

Remark (Interpretation). Logical consequence means truth preservation from premises to conclusion. There is no truth assignment under which all premises are true and the conclusion is false.

Remark (Dependencies).

- Satisfaction of a Set of Formulas
- Satisfaction of a Formula

Definition (Logical Equivalence)

Definition 1.5.13 (Logical Equivalence)**.** Formulas $\varphi, \psi \in \mathsf{WFF}$ are logically equivalent, written

$$\varphi \equiv \psi,$$

if they have the same truth value under every truth assignment.

Remark (Standard quantified statement).

$$\varphi \equiv \psi \iff \varphi, \psi \in \mathsf{WFF} \wedge (\forall v : \mathsf{Prop} \to \mathbb{B}) \left[\widehat{v}(\varphi) = \widehat{v}(\psi) \right].$$

Remark (Definition predicate reading).

$$\text{LogicalEquivalence}(\varphi, \psi)$$

means $\varphi \equiv \psi$.

Remark (Interpretation). Logically equivalent formulas have identical truth-table columns. They are interchangeable for semantic purposes.

Remark (Dependencies).

- Unique Extension of a Truth Assignment
- Truth Assignment

Lemma (Coincidence Lemma for Propositional Logic)

Lemma 1.5.14 (Coincidence Lemma for Propositional Logic)**.** *Let $v, w : \mathsf{Prop} \to \mathbb{B}$ be truth assignments. If v and w agree on every propositional variable occurring in φ, then they assign the same truth value to φ:*

$$v|_{\mathrm{Var}(\varphi)} = w|_{\mathrm{Var}(\varphi)} \Longrightarrow \widehat{v}(\varphi) = \widehat{w}(\varphi).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi \in \mathsf{WFF})(\forall v, w : \mathsf{Prop} \to \mathbb{B}) \left[(\forall P \in \mathrm{Var}(\varphi))(v(P) = w(P)) \to \widehat{v}(\varphi) = \widehat{w}(\varphi)\right].$$

Remark (Interpretation)*.* The truth value of a formula depends only on the variables actually occurring in that formula.

Remark (Dependencies)*.*

- Variable Set of a Formula
- Unique Extension of a Truth Assignment
- Structural Induction for Propositional Formulas

Theorem (Finite Truth-Table Theorem)

Theorem 1.5.15 (Finite Truth-Table Theorem)**.** *Every propositional formula has a finite truth table. More precisely, if $\mathrm{Var}(\varphi)$ has n elements, then φ has a truth table with 2^n rows.*
Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi \in \mathsf{WFF}) \left[|\,\mathrm{Var}(\varphi)| = n \to \mathrm{TruthTable}(\varphi) \text{ has } 2^n \text{ rows}\right].$$

Remark (Interpretation)*.* Truth tables are finite because every formula contains only finitely many propositional variables.

Remark (Dependencies)*.*

- Truth Table
- Finiteness of Variables in a Formula

Definition (Unary Boolean Function)

Definition 1.5.16 (Unary Boolean Function)**.** A unary Boolean function is a function

$$f : \mathbb{B} \to \mathbb{B}.$$

Remark (Definition predicate reading).

$$UnaryBooleanFunction(f)$$

means that f takes one truth value as input and returns one truth value.

Remark (Interpretation). A unary Boolean function has one input row coordinate. It is the semantic shape of a one-place truth operation such as negation.

Remark (Dependencies).

- Truth Value

Definition (Binary Boolean Function)

Definition 1.5.17 (Binary Boolean Function)**.** A binary Boolean function is a function

$$f : \mathbb{B}^2 \to \mathbb{B},$$

equivalently a function

$$f : \mathbb{B} \times \mathbb{B} \to \mathbb{B}.$$

Remark (Definition predicate reading).

$$BinaryBooleanFunction(f)$$

means that f takes an ordered pair of truth values as input and returns one truth value.

Remark (Interpretation). A binary Boolean function assigns one output truth value to each of the four input rows

$$(T, T), \quad (T, F), \quad (F, T), \quad (F, F).$$

Remark (Dependencies).

- Truth Value
- Unary Boolean Function

Definition (Boolean Function)

Definition 1.5.18 (Boolean Function)**.** An n-ary Boolean function is a function

$$f : \mathbb{B}^n \to \mathbb{B}.$$

Remark (Standard quantified statement).

$$BooleanFunction_n(f) \iff f : \mathbb{B}^n \to \mathbb{B}.$$

Remark (Definition predicate reading).

$$BooleanFunction_n(f)$$

means that f takes n truth values as input and returns one truth value.

Remark (Interpretation). Boolean functions are the abstract truth-functions that formulas may represent.

Remark (Rows versus Boolean functions). An input to an n-ary Boolean function is a tuple $(b_1, \ldots, b_n) \in \mathbb{B}^n$. These tuples are the rows of the truth table, and there are 2^n of them. The Boolean function is not a single row; it is the full rule assigning exactly one truth value in $\mathbb{B}$ to every row.

Remark (Dependencies).

- Truth Value
- Unary Boolean Function
- Binary Boolean Function

Definition (Truth Function Induced by a Formula)

Definition 1.5.19 (Truth Function Induced by a Formula)**.** Let $\varphi \in \mathsf{WFF}$, and suppose $\mathrm{Var}(\varphi) \subseteq \{P_1, \ldots, P_n\}$. The truth function induced by φ is the Boolean function

$$f_\varphi : \mathbb{B}^n \to \mathbb{B}$$

defined by

$$f_\varphi(b_1, \ldots, b_n) = \widehat{v}(\varphi),$$

where $v(P_i) = b_i$ for each i.

Remark (Standard quantified statement).

$$f_\varphi(b_1, \ldots, b_n) = \widehat{v}(\varphi) \quad \text{whenever } v(P_i) = b_i \text{ for } 1 \le i \le n.$$

Remark (Definition predicate reading).

$$\mathrm{TruthFunctionOfFormula}(f_\varphi, \varphi)$$

means that f_φ records the truth behavior of φ.

Remark (Interpretation). The induced truth function packages the truth table of a formula as a Boolean function.

Remark (Dependencies).

- Boolean Function
- Variable Set of a Formula
- Unique Extension of a Truth Assignment
- Coincidence Lemma for Propositional Logic

Theorem (Counting Boolean Functions)

Theorem 1.5.20 (Counting Boolean Functions)**.** *There are exactly*

$$2^{2^n}$$

Boolean functions $\mathbb{B}^n \to \mathbb{B}$.
Go to proof.

Remark (Standard quantified statement)*.*

$$|\{f : \mathbb{B}^n \to \mathbb{B}\}| = 2^{2^n}.$$

Remark (Interpretation)*.* The domain $\mathbb{B}^n$ has 2^n elements. A Boolean function chooses one of two truth values for each input row.

Remark (Dependencies)*.*

- Boolean Function

1.6 Algebra of Propositions

Toolkit

This section uses logical equivalence, equivalence laws, formula contexts, and replacement of logical equivalents to treat propositions algebraically.

Remark (Exposition)*.* The algebra section treats logically equivalent formulas as interchangeable for calculation. Equivalence laws and replacement principles turn semantic agreement under every truth assignment into a controlled rewrite calculus for propositional formulas.

Definition (Propositional Equivalence Law)

Definition 1.6.1 (Propositional Equivalence Law)**.** A propositional equivalence law is a valid schema of the form

$$\varphi \equiv \psi,$$

with $\varphi, \psi \in \mathsf{WFF}$. Each instance of the schema says that the two formulas have the same truth value under every truth assignment.

Remark (Standard quantified statement)*.*

$$\mathrm{PropositionalEquivalenceLaw}(\varphi, \psi) \iff \varphi, \psi \in \mathsf{WFF} \land \varphi \equiv \psi.$$

Remark (Definition predicate reading)*.*

$$\mathrm{PropositionalEquivalenceLaw}(\varphi, \psi)$$

means that $\varphi \equiv \psi$ is a valid logical-equivalence schema.

Remark (Interpretation). Equivalence laws are the algebraic rewrite rules of propositional logic. They allow formulas to be transformed without changing truth conditions.

Remark (Dependencies).

- Logical Equivalence
- Well-Formed Formula

Theorem (Logical Equivalence is an Equivalence Relation)

Theorem 1.6.2 (Logical Equivalence is an Equivalence Relation)**.** *Logical equivalence is reflexive, symmetric, and transitive on* WFF*:*

$$\varphi \equiv \varphi, \qquad \varphi \equiv \psi \Longrightarrow \psi \equiv \varphi,$$

and

$$(\varphi \equiv \psi \wedge \psi \equiv \chi) \Longrightarrow \varphi \equiv \chi.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&(\forall \varphi \in \mathsf{WFF})(\varphi \equiv \varphi),\\&(\forall \varphi, \psi \in \mathsf{WFF})(\varphi \equiv \psi \to \psi \equiv \varphi),\\&(\forall \varphi, \psi, \chi \in \mathsf{WFF})((\varphi \equiv \psi \wedge \psi \equiv \chi) \to \varphi \equiv \chi).\end{aligned}$$

Remark (Interpretation). Logical equivalence behaves like equality at the semantic level.

Remark (Dependencies).

- Logical Equivalence

Theorem (Basic Boolean Equivalence Laws)

Theorem 1.6.3 (Basic Boolean Equivalence Laws)**.** *For all formulas* $\varphi, \psi, \chi \in \mathsf{WFF}$*, the following logical equivalences hold.*
Commutativity:

$$\varphi \wedge \psi \equiv \psi \wedge \varphi, \qquad \varphi \vee \psi \equiv \psi \vee \varphi.$$

Associativity:

$$(\varphi \wedge \psi) \wedge \chi \equiv \varphi \wedge (\psi \wedge \chi), \qquad (\varphi \vee \psi) \vee \chi \equiv \varphi \vee (\psi \vee \chi).$$

Idempotence:

$$\varphi \wedge \varphi \equiv \varphi, \qquad \varphi \vee \varphi \equiv \varphi.$$

Double negation:

$$\neg\neg\varphi \equiv \varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi, \chi \in \mathsf{WFF}) \begin{bmatrix} \varphi \wedge \psi \equiv \psi \wedge \varphi, \quad \varphi \vee \psi \equiv \psi \vee \varphi, \\ (\varphi \wedge \psi) \wedge \chi \equiv \varphi \wedge (\psi \wedge \chi), \quad (\varphi \vee \psi) \vee \chi \equiv \varphi \vee (\psi \vee \chi), \\ \varphi \wedge \varphi \equiv \varphi, \quad \varphi \vee \varphi \equiv \varphi, \quad \neg\neg\varphi \equiv \varphi \end{bmatrix}.$$

Remark (Interpretation). These are the basic algebraic laws for conjunction, disjunction, and negation.

Remark (Dependencies).

- Logical Equivalence
- Unique Extension of a Truth Assignment

Theorem (Identity and Domination Laws)

Theorem 1.6.4 (Identity and Domination Laws)**.** *Let* $\top$ *be any tautological formula and let* $\perp$ *be any contradictory formula. Then for every* $\varphi \in \mathsf{WFF}$*:*

$$\varphi \wedge \top \equiv \varphi, \qquad \varphi \vee \perp \equiv \varphi,$$

and

$$\varphi \wedge \perp \equiv \perp, \qquad \varphi \vee \top \equiv \top.$$

Go to proof.

Remark (Standard quantified statement).

$$(\mathrm{Tautology}(\top) \wedge \mathrm{Contradiction}(\perp)) \to (\forall \varphi \in \mathsf{WFF}) \\ [(\varphi \wedge \top \equiv \varphi) \wedge (\varphi \vee \perp \equiv \varphi) \wedge (\varphi \wedge \perp \equiv \perp) \wedge (\varphi \vee \top \equiv \top)].$$

Remark (Interpretation). Tautologies act like true constants; contradictions act like false constants.

Remark (Dependencies).

- Tautology
- Contradiction
- Logical Equivalence

Theorem (De Morgan Laws for Propositional Logic)

Theorem 1.6.5 (De Morgan Laws for Propositional Logic)**.** *For all* $\varphi, \psi \in \mathsf{WFF}$*,*

$$\neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi,$$

and

$$\neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF})\left[\neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi\right],$$
$$(\forall \varphi, \psi \in \mathsf{WFF})\left[\neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi\right].$$

Remark (Predicate reading).

$$\underbrace{\neg(\varphi \wedge \psi)}_{\text{not both}} \equiv \underbrace{\neg\varphi \vee \neg\psi}_{\text{at least one fails}}, \qquad \underbrace{\neg(\varphi \vee \psi)}_{\text{neither}} \equiv \underbrace{\neg\varphi \wedge \neg\psi}_{\text{both fail}}.$$

Remark (Interpretation). Negation swaps conjunction with disjunction when pushed inward.

Remark (Dependencies).

- Logical Equivalence
- Unique Extension of a Truth Assignment

Theorem (Distributive Laws for Propositional Logic)

Theorem 1.6.6 (Distributive Laws for Propositional Logic)**.** *For all $\varphi, \psi, \chi \in \mathsf{WFF}$,*

$$\varphi \wedge (\psi \vee \chi) \equiv (\varphi \wedge \psi) \vee (\varphi \wedge \chi),$$

and

$$\varphi \vee (\psi \wedge \chi) \equiv (\varphi \vee \psi) \wedge (\varphi \vee \chi).$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi, \chi \in \mathsf{WFF})\left[\varphi \wedge (\psi \vee \chi) \equiv (\varphi \wedge \psi) \vee (\varphi \wedge \chi)\right],$$
$$(\forall \varphi, \psi, \chi \in \mathsf{WFF})\left[\varphi \vee (\psi \wedge \chi) \equiv (\varphi \vee \psi) \wedge (\varphi \vee \chi)\right].$$

Remark (Interpretation). Conjunction distributes over disjunction, and disjunction distributes over conjunction. These laws are the main algebraic engine behind CNF and DNF conversion.

Remark (Dependencies).

- Logical Equivalence
- Unique Extension of a Truth Assignment

Theorem (Absorption Laws)

Theorem 1.6.7 (Absorption Laws)**.** *For all $\varphi, \psi \in \mathsf{WFF}$,*

$$\varphi \wedge (\varphi \vee \psi) \equiv \varphi,$$

and

$$\varphi \vee (\varphi \wedge \psi) \equiv \varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF})\,[\varphi \wedge (\varphi \vee \psi) \equiv \varphi]\,,$$
$$(\forall \varphi, \psi \in \mathsf{WFF})\,[\varphi \vee (\varphi \wedge \psi) \equiv \varphi]\,.$$

Remark (Interpretation). Once φ is already required, adding $\varphi \vee \psi$ contributes no additional constraint. Dually, once φ is already allowed, adding $\varphi \wedge \psi$ contributes no additional possibility.

Remark (Dependencies).

- Logical Equivalence
- Basic Boolean Equivalence Laws
- Distributive Laws for Propositional Logic

Theorem (Conditional Equivalence Laws)

Theorem 1.6.8 (Conditional Equivalence Laws)**.** *For all $\varphi, \psi \in \mathsf{WFF}$,*

$$\varphi \to \psi \equiv \neg\varphi \vee \psi,$$

and

$$\neg(\varphi \to \psi) \equiv \varphi \wedge \neg\psi.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF})\,[\varphi \to \psi \equiv \neg\varphi \vee \psi]\,,$$
$$(\forall \varphi, \psi \in \mathsf{WFF})\,[\neg(\varphi \to \psi) \equiv \varphi \wedge \neg\psi]\,.$$

Remark (Predicate reading).

$$\underbrace{\varphi \to \psi}_{\text{no counterexample}} \equiv \underbrace{\neg\varphi \vee \psi}_{\text{premise false or conclusion true}}\,, \qquad \underbrace{\neg(\varphi \to \psi)}_{\text{counterexample}} \equiv \underbrace{\varphi \wedge \neg\psi}_{\text{premise true and conclusion false}}\,.$$

Remark (Interpretation). A conditional fails exactly when the antecedent is true and the consequent is false.

Remark (Dependencies).

- Logical Equivalence
- Unique Extension of a Truth Assignment

Theorem (Biconditional Equivalence Laws)

Theorem 1.6.9 (Biconditional Equivalence Laws)**.** *For all* $\varphi, \psi \in \mathsf{WFF}$*,*

$$\varphi \leftrightarrow \psi \equiv (\varphi \to \psi) \wedge (\psi \to \varphi),$$

and

$$\varphi \leftrightarrow \psi \equiv (\varphi \wedge \psi) \vee (\neg\varphi \wedge \neg\psi).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi, \psi \in \mathsf{WFF})\left[\varphi \leftrightarrow \psi \equiv (\varphi \to \psi) \wedge (\psi \to \varphi)\right],$$
$$(\forall \varphi, \psi \in \mathsf{WFF})\left[\varphi \leftrightarrow \psi \equiv (\varphi \wedge \psi) \vee (\neg\varphi \wedge \neg\psi)\right].$$

Remark (Predicate reading)*.*

$$\varphi \leftrightarrow \psi \quad \iff \quad (\varphi \to \psi) \wedge (\psi \to \varphi)$$

and also

$$\varphi \leftrightarrow \psi \quad \iff \quad (\varphi \wedge \psi) \vee (\neg\varphi \wedge \neg\psi).$$

The biconditional says that the two formulas have the same truth value.

Remark (Interpretation)*.* A biconditional says that two formulas have the same truth value: both true or both false.

Remark (Dependencies)*.*

- Logical Equivalence
- Conditional Equivalence Laws

Definition (Formula Context)

Definition 1.6.10 (Formula Context)**.** A formula context $C[-]$ is a well-formed formula with one distinguished placeholder occurrence $[-]$. For each well-formed formula φ, the expression $C[\varphi]$ denotes the formula obtained by replacing the placeholder with φ.

Remark (Standard quantified statement)*.*

$$\text{FormulaContext}(C[-]) \implies (\forall \varphi \in \mathsf{WFF})(C[\varphi] \in \mathsf{WFF}).$$

Remark (Definition predicate reading)*.*

$$\text{FormulaContext}(C[-])$$

means that $C[-]$ is a one-hole propositional formula context.

Remark (Interpretation). A formula context is a syntactic environment into which a formula can be inserted. Contexts make precise the idea of replacing a subformula inside a larger formula.

Remark (Examples). If

$$C[-] := ([-] \wedge R),$$

then

$$C[P] = (P \wedge R)$$

and

$$C[Q \vee S] = ((Q \vee S) \wedge R).$$

Remark (Dependencies).

- Well-Formed Formula
- Subformula

Theorem (Replacement of Logical Equivalents)

Theorem 1.6.11 (Replacement of Logical Equivalents)**.** *Replacing logically equivalent formulas inside a formula context preserves logical equivalence:*

$$\varphi \equiv \psi \Longrightarrow C[\varphi] \equiv C[\psi].$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF})(\forall C[-])\left[\text{FormulaContext}(C[-]) \wedge \varphi \equiv \psi \to C[\varphi] \equiv C[\psi]\right].$$

Remark (Predicate reading).

$$\varphi \equiv \psi \quad \Longrightarrow \quad C[\varphi] \equiv C[\psi].$$

A logical equivalence may be used inside a larger formula context without changing the truth behavior of the whole formula.

Remark (Interpretation). Logical equivalence is substitutive: equivalent formulas may be replaced inside larger formulas without changing truth behavior.

Remark (Exposition). The equivalence laws in this section may be used as replacement rules inside larger formulas. If a formula contains a subformula matching one side of a listed equivalence law, that subformula may be replaced by the other side.

Common replacement rules include:

$$\neg\neg\varphi \equiv \varphi,$$
$$\neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi, \qquad \neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi,$$
$$\varphi \to \psi \equiv \neg\varphi \vee \psi,$$
$$\varphi \leftrightarrow \psi \equiv (\varphi \to \psi) \wedge (\psi \to \varphi),$$

and

$$\varphi \leftrightarrow \psi \equiv (\varphi \wedge \psi) \vee (\neg\varphi \wedge \neg\psi).$$

Remark (Exposition). The laws in this section allow formulas to be manipulated algebraically. They are not new formation rules; they are semantic equivalences. Later sections use these laws to transform formulas into normal forms and to reduce the set of primitive connectives.

Remark (Exposition). The algebra of propositions is the bridge from semantics to computation: once formulas are equivalent, they can be rewritten without changing truth behavior.

Remark (Dependencies).

- Formula Context
- Logical Equivalence
- Coincidence Lemma for Propositional Logic
- Unique Extension of a Truth Assignment

1.7 Normal Forms

Toolkit

This section uses logical equivalence, truth assignments, literals, clauses, and algebraic rewrite laws to motivate normal forms.

Remark (Exposition). The normal-forms section studies controlled syntactic shapes for formulas. Logical equivalence is the governing invariant: conversion may change a formula's shape, but it must preserve truth under every assignment while exposing regular literal, clause, DNF, CNF, and canonical structure.

Orientation

Remark (Section map). This section uses the preceding syntax, semantics, and algebra of propositions as its starting point. It first isolates literals and clauses, then introduces negation normal form, disjunctive normal form, conjunctive normal form, canonical normal forms, and semantic classification uses. Later arguments will use these forms to organize satisfiability questions, proof search, functional completeness, and resolution-style methods.

Remark (Why normal forms matter). Normal forms trade arbitrary expression for regular structure. In a truth-table argument, regular structure helps classify formulas as tautologies, contradictions, satisfiable formulas, or contingencies. In proof search and satisfiability procedures, regular structure turns a formula into a finite combinatorial object that can be searched. In functional completeness, normal forms show how complicated truth functions can be built from simpler connectives.

Definition (Literal)

Definition 1.7.1 (Literal). A *literal* is either a propositional variable P or the negation $\neg P$ of a propositional variable P.

Remark (Standard quantified statement)*.* For a formula $\ell \in \mathsf{WFF}$,

$$\ell \text{ is a literal} \iff (\exists P \in \mathsf{Prop})(\ell = P \text{ or } \ell = \neg P).$$

Remark (Definition predicate reading)*.*

$$\mathrm{Literal}(\ell)$$

means that ℓ is an atomic propositional variable or the negation of one atomic propositional variable.

Remark (Interpretation)*.* Literals are the indivisible signed atoms used in normal forms. They remember which propositional variable is being tested and whether it appears positively or negatively.

Remark (Examples)*.* The formulas P, Q, $\neg P$, and $\neg Q$ are literals.

Remark (Non-Examples)*.* The formulas $P \wedge Q$, $P \vee Q$, and $\neg(P \wedge Q)$ are not literals. The last formula is negated, but it negates a compound formula rather than a single propositional variable.

Remark (Dependencies)*.*

- Propositional Variable
- Logical Connective

Definition (Positive Literal)

Definition 1.7.2 (Positive Literal)**.** A *positive literal* is a literal of the form P, where P is a propositional variable.

Remark (Standard quantified statement)*.* For a formula $\ell \in \mathsf{WFF}$,

$$\ell \text{ is a positive literal} \iff (\exists P \in \mathsf{Prop})(\ell = P).$$

Remark (Definition predicate reading)*.*

$$\mathrm{PositiveLiteral}(\ell)$$

means that ℓ is a literal whose sign is positive.

Remark (Interpretation)*.* A positive literal asserts the propositional variable directly. It carries no leading negation.

Remark (Examples)*.* The formulas P and Q are positive literals.

Remark (Dependencies)*.*

- Literal

Definition (Negative Literal)

Definition 1.7.3 (Negative Literal)**.** A *negative literal* is a literal of the form $\neg P$, where P is a propositional variable.

Remark (Standard quantified statement). For a formula $\ell \in \mathsf{WFF}$,

$$\ell \text{ is a negative literal} \iff (\exists P \in \mathsf{Prop})(\ell = \neg P).$$

Remark (Definition predicate reading).

$$\mathrm{NegativeLiteral}(\ell)$$

means that ℓ is a literal whose sign is negative.

Remark (Interpretation). A negative literal denies exactly one propositional variable. In normal forms, negation is pushed down until it appears only at this literal level.

Remark (Examples). The formulas $\neg P$ and $\neg Q$ are negative literals.

Remark (Dependencies).

- Literal

Definition (Complementary Literals)

Definition 1.7.4 (Complementary Literals)**.** Two literals are *complementary* if they are P and $\neg P$ for the same propositional variable P.

Remark (Standard quantified statement). For literals ℓ_1, ℓ_2,

$$\ell_1 \text{ and } \ell_2 \text{ are complementary} \iff (\exists P \in \mathsf{Prop})\big((\ell_1 = P \wedge \ell_2 = \neg P) \vee (\ell_1 = \neg P \wedge \ell_2 = P)\big).$$

Remark (Definition predicate reading).

$$\mathrm{ComplementaryLiterals}(\ell_1, \ell_2)$$

means that the two literals name the same propositional variable with opposite signs.

Remark (Interpretation). Complementary literals are the local contradiction pattern inside clauses and terms. The pair $P, \neg P$ is tied to one variable, not merely to two formulas with opposite truth values under a particular assignment.

Remark (Examples). The literals P and $\neg P$ are complementary, as are Q and $\neg Q$.

Remark (Non-Examples). The literals P and $\neg Q$ are not complementary unless $P = Q$.

Remark (Dependencies).

- Literal

Definition (Clause)

Definition 1.7.5 (Clause)**.** A *clause* is a finite disjunction of literals.

Remark (Standard quantified statement). A formula $C \in \mathsf{WFF}$ is a clause if there are literals $\ell_1, \dots, \ell_n$ with $n \geq 1$ such that

$$C = \ell_1 \vee \cdots \vee \ell_n.$$

The empty clause is treated separately as the empty disjunction.

Remark (Definition predicate reading).

$$\mathrm{Clause}(C)$$

means that C is built by finitely disjoining literals.

Remark (Interpretation). A clause records a finite list of literal-level alternatives. It is satisfied when at least one of its literals is true.

Remark (Examples). The formulas $P \vee Q$, $P \vee \neg Q$, and $\neg P \vee Q \vee R$ are clauses.

Remark (Non-Examples). The formula $P \wedge Q$ is not a clause, because its top-level connective is a conjunction. The formula $\neg(P \vee Q)$ is not a clause, because its negation has not been pushed down to the literal level.

Remark (Dependencies).

- Literal

Definition (Term)

Definition 1.7.6 (Term)**.** A *term*, also called a *conjunction term*, is a finite conjunction of literals.

Remark (Standard quantified statement). A formula $T \in \mathsf{WFF}$ is a term if there are literals $\ell_1, \ldots, \ell_n$ with $n \geq 1$ such that

$$T = \ell_1 \wedge \cdots \wedge \ell_n.$$

The empty conjunction is treated separately.

Remark (Definition predicate reading).

$$\mathrm{Term}(T)$$

means that T is built by finitely conjoining literals.

Remark (Interpretation). A term records a finite list of literal-level requirements. It is satisfied when all of its literals are true.

Remark (Examples). The formulas $P \wedge Q$, $P \wedge \neg Q$, and $\neg P \wedge Q \wedge R$ are terms.

Remark (Non-Examples). The formula $P \vee Q$ is not a term, because its top-level connective is a disjunction. The formula $\neg(P \wedge Q)$ is not a term, because its negation has not been pushed down to the literal level.

Remark (Dependencies).

- Literal

Definition (Empty Clause)

Definition 1.7.7 (Empty Clause)**.** The *empty clause* is the empty disjunction of literals. Semantically, it is read as false.

Remark (Standard quantified statement). The empty clause is the disjunction over no literals and has truth value False under every truth assignment.

Remark (Definition predicate reading).

$$\text{EmptyClause}(C)$$

means that C is the clause with no disjuncts.

Remark (Interpretation). A nonempty clause is true when at least one listed literal is true. With no literals listed, there is no witness to make the disjunction true, so the empty clause is the false boundary case for clauses.

Remark (Dependencies).

- Clause
- Contradiction

Definition (Empty Conjunction)

Definition 1.7.8 (Empty Conjunction)**.** The *empty conjunction* is the conjunction over no literals. Semantically, it is read as true.

Remark (Standard quantified statement). The empty conjunction is the conjunction over no literals and has truth value True under every truth assignment.

Remark (Definition predicate reading).

$$\text{EmptyConjunction}(T)$$

means that T is the term with no conjuncts.

Remark (Interpretation). A nonempty term is true when every listed literal is true. With no literals listed, there is no requirement to violate, so the empty conjunction is the true boundary case for terms.

Remark (Dependencies).

- Term
- Tautology

Definition (Negation Normal Form)

Definition 1.7.9 (Negation Normal Form)**.** A formula is in *negation normal form* if it uses only propositional variables, negated propositional variables, conjunction, and disjunction, and every negation occurs immediately before a propositional variable.

Remark (Standard quantified statement). For a formula $\varphi \in \mathsf{WFF}$,

$$\varphi \text{ is in negation normal form} \iff \varphi \text{ is built from literals using only } \wedge \text{ and } \vee.$$

Remark (Definition predicate reading).

$$\text{NegationNormalForm}(\varphi)$$

means that φ has no conditionals, no biconditionals, and no negation outside the literal level.

Remark (Interpretation). Negation normal form is the first cleanup stage for propositional formulas. It does not force a formula to be a disjunction of terms or a conjunction of clauses; it only restricts the available connectives and pushes negation down to propositional variables.

Remark (Examples). The formulas P, $\neg P$, $P \wedge (\neg Q \vee R)$, and $(P \vee Q) \wedge (\neg R \vee S)$ are in negation normal form.

Remark (Non-Examples). The formulas $P \to Q$, $P \leftrightarrow Q$, and $\neg(P \wedge Q)$ are not in negation normal form. The first two use connectives not allowed in negation normal form, and the last has a negation applied to a compound formula.

Remark (Dependencies).

- Literal
- Well-Formed Formula
- Logical Equivalence

Definition (NNF Conversion Procedure)

Definition 1.7.10 (NNF Conversion Procedure)**.** The *NNF conversion procedure* converts a propositional formula to an equivalent formula in negation normal form by the following steps:

1. eliminate biconditionals using biconditional equivalence laws;
2. eliminate conditionals using conditional equivalence laws;
3. eliminate double negations;
4. push negations inward using De Morgan laws.

Remark (Standard quantified statement). Starting with $\varphi \in \mathsf{WFF}$, repeatedly apply the equivalences

$$\alpha \leftrightarrow \beta \equiv (\alpha \to \beta) \wedge (\beta \to \alpha), \qquad \alpha \to \beta \equiv \neg\alpha \vee \beta,$$
$$\neg\neg\alpha \equiv \alpha, \qquad \neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta, \qquad \neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta,$$

until negations occur only immediately before propositional variables and only $\wedge$ and $\vee$ remain above the literal level.

Remark (Definition predicate reading).

$$\text{NNFConversionProcedure}(\varphi, \psi)$$

means that ψ is obtained from φ by eliminating biconditionals and conditionals, removing double negations, and pushing remaining negations inward.

Remark (Interpretation). The procedure is a syntax-directed rewrite process. Each rewrite step is backed by a logical equivalence law, so the output formula has the same truth conditions as the input formula.

Remark (Dependencies).

- Conditional Equivalence Laws
- Biconditional Equivalence Laws
- De Morgan Laws for Propositional Logic
- Basic Boolean Equivalence Laws

Theorem (Existence of Negation Normal Form)

Theorem 1.7.11 (Existence of Negation Normal Form)**.** *For every formula* $\varphi \in \mathsf{WFF}$*, there exists a formula* $\psi \in \mathsf{WFF}$ *such that* ψ *is in negation normal form and*

$$\varphi \equiv \psi.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi \in \mathsf{WFF})(\exists \psi \in \mathsf{WFF})\big(\psi \text{ is in negation normal form} \land \varphi \equiv \psi\big).$$

Remark (Predicate reading).

$$\text{NegationNormalFormExists}(\varphi)$$

means that φ has a logically equivalent representative in negation normal form.

Remark (Interpretation). Every propositional formula can be rewritten so that negation appears only at the variable level. This makes negation normal form a semantic normalizing step: it changes shape while preserving logical equivalence.

Remark (Dependencies).

- Negation Normal Form
- NNF Conversion Procedure
- Replacement of Logical Equivalents
- Structural Induction for Propositional Formulas

Definition (Elementary Conjunction)

Definition 1.7.12 (Elementary Conjunction)**.** An *elementary conjunction* is a finite conjunction of literals. That is, it is a term of the form

$$L_1 \land L_2 \land \cdots \land L_k,$$

where each L_i is a literal. The one-literal case is permitted.

Remark (Standard quantified statement).

$$L_1 \wedge L_2 \wedge \cdots \wedge L_k, \qquad L_i \text{ is a literal for each } i.$$

Remark (Interpretation). Elementary conjunctions are the conjunction terms used as the building blocks of disjunctive normal form. They record simultaneous literal requirements.

Example (Examples).

$$P, \qquad P \wedge \neg Q, \qquad \neg P \wedge R \wedge S$$

are elementary conjunctions.

Remark (Dependencies).

- Term
- Literal

Definition (Minterm)

Definition 1.7.13 (Minterm)**.** Let $P_1, \ldots, P_n$ be a finite list of propositional variables. A *minterm* relative to $P_1, \ldots, P_n$ is an elementary conjunction that contains exactly one of P_i or $\neg P_i$ for each $i = 1, \ldots, n$.

Remark (Standard quantified statement).

$$M = L_1 \wedge \cdots \wedge L_n,$$

where for each i, exactly one of P_i or $\neg P_i$ appears among the literals $L_1, \ldots, L_n$.

Remark (Interpretation). A minterm relative to $P_1, \ldots, P_n$ specifies one complete truth-value assignment on those variables: choosing P_i requires P_i to be true, and choosing $\neg P_i$ requires P_i to be false.

Example (Examples). Relative to P, Q, R, the formula

$$P \wedge \neg Q \wedge R$$

is a minterm. Relative to P, Q, R, the formula $P \wedge R$ is not a minterm because it omits both Q and $\neg Q$.

Remark (Dependencies).

- Elementary Conjunction
- Variable Set of a Formula

Definition (Disjunctive Normal Form)

Definition 1.7.14 (Disjunctive Normal Form)**.** A formula is in *disjunctive normal form*, abbreviated *DNF*, when it is a finite disjunction of elementary conjunctions. Equivalently, it has the shape

$$C_1 \vee C_2 \vee \cdots \vee C_m,$$

where each C_j is an elementary conjunction.

Remark (Standard quantified statement).

$$\varphi \text{ is in DNF} \iff \varphi = C_1 \vee C_2 \vee \cdots \vee C_m$$

for finitely many elementary conjunctions $C_1, \ldots, C_m$.

Remark (Interpretation). DNF is a syntactic form, but it is used semantically: once a formula is replaced by a logically equivalent DNF formula, questions about satisfiability reduce to questions about individual conjunction terms.

Example (Examples).

$$(P \wedge Q) \vee (\neg P \wedge R) \vee S$$

is in DNF because each disjunct is an elementary conjunction.

Remark (Dependencies).

- Elementary Conjunction
- Logical Equivalence
- Satisfiable Formula

Theorem (Existence of Disjunctive Normal Form)

Theorem 1.7.15 (Existence of Disjunctive Normal Form). *For every formula $\varphi \in \mathsf{WFF}$, there exists a formula $\psi \in \mathsf{WFF}$ such that ψ is in disjunctive normal form and*

$$\varphi \equiv \psi.$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi \in \mathsf{WFF})(\exists \psi \in \mathsf{WFF})\big(\psi \text{ is in DNF} \wedge \varphi \equiv \psi\big).$$

Remark (Interpretation). One first converts φ to negation normal form, then repeatedly applies distributive laws to move conjunctions below disjunctions. Replacement of logical equivalents preserves equivalence at each step.

Remark (Dependencies).

- Disjunctive Normal Form
- Negation Normal Form
- Distributive Laws for Propositional Logic
- Replacement of Logical Equivalents
- Existence of Negation Normal Form

Theorem (DNF Satisfiability Criterion)

Theorem 1.7.16 (DNF Satisfiability Criterion)**.** *Let ψ be a formula in disjunctive normal form. Then ψ is satisfiable if and only if at least one conjunction term in ψ contains no complementary pair of literals.* *Go to proof.*

Remark (Standard quantified statement).

ψ is a satisfiable DNF formula $\iff$ some conjunction term of ψ has no complementary literals. ∎

Remark (Interpretation). A DNF formula is true exactly when at least one of its conjunction terms is true. A conjunction term can be made true exactly when it does not require both a variable and its negation.

Remark (Dependencies).

- Disjunctive Normal Form
- Complementary Literals
- Satisfiable Formula

Definition (Elementary Disjunction)

Definition 1.7.17 (Elementary Disjunction)**.** An *elementary disjunction* is a finite disjunction of literals. That is, it is a clause of the form

$$L_1 \vee L_2 \vee \cdots \vee L_k,$$

where each L_i is a literal. The one-literal case is permitted.

Remark (Standard quantified statement).

$$L_1 \vee L_2 \vee \cdots \vee L_k, \qquad L_i \text{ is a literal for each } i.$$

Remark (Interpretation). Elementary disjunctions are the clause-level building blocks used in conjunctive normal form. They record finite literal alternatives.

Example (Examples).

$$P, \qquad P \vee \neg Q, \qquad \neg P \vee R \vee S$$

are elementary disjunctions.

Remark (Dependencies).

- Clause
- Literal

Definition (Maxterm)

Definition 1.7.18 (Maxterm)**.** Let $P_1, \ldots, P_n$ be a finite list of propositional variables. A *maxterm* relative to $P_1, \ldots, P_n$ is an elementary disjunction that contains exactly one of P_i or $\neg P_i$ for each $i = 1, \ldots, n$.

Remark (Standard quantified statement).

$$M = L_1 \vee \cdots \vee L_n,$$

where for each i, exactly one of P_i or $\neg P_i$ appears among the literals $L_1, \ldots, L_n$.

Remark (Interpretation). A maxterm relative to $P_1, \ldots, P_n$ specifies one literal-level alternative for every variable in the list. It is false under exactly the assignment that makes each listed literal false.

Example (Examples). Relative to P, Q, R, the formula

$$P \vee \neg Q \vee R$$

is a maxterm. Relative to P, Q, R, the formula $P \vee R$ is not a maxterm because it omits both Q and $\neg Q$.

Remark (Dependencies).

- Elementary Disjunction
- Variable Set of a Formula

Definition (Conjunctive Normal Form)

Definition 1.7.19 (Conjunctive Normal Form)**.** A formula is in *conjunctive normal form*, abbreviated *CNF*, when it is a finite conjunction of clauses. Equivalently, it has the shape

$$C_1 \wedge C_2 \wedge \cdots \wedge C_m,$$

where each C_j is a clause.

Remark (Standard quantified statement).

$$\varphi \text{ is in CNF} \iff \varphi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

for finitely many clauses $C_1, \ldots, C_m$.

Remark (Interpretation). CNF is a syntactic form used to expose clause-by-clause constraints. Replacing a formula by a logically equivalent CNF formula turns global validity and contradiction questions into questions about the interaction of its clauses.

Example (Examples).

$$(P \vee Q) \wedge (\neg P \vee R) \wedge S$$

is in CNF because each conjunct is a clause.

Remark (Dependencies).

- Clause
- Logical Equivalence

- Contradiction

Theorem (Existence of Conjunctive Normal Form)

Theorem 1.7.20 (Existence of Conjunctive Normal Form)**.** *For every formula $\varphi \in \mathsf{WFF}$, there exists a formula $\psi \in \mathsf{WFF}$ such that ψ is in conjunctive normal form and*

$$\varphi \equiv \psi.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi \in \mathsf{WFF})(\exists \psi \in \mathsf{WFF})\big(\psi \text{ is in CNF} \wedge \varphi \equiv \psi\big).$$

Remark (Interpretation)*.* One first converts φ to negation normal form, then repeatedly applies distributive laws to move disjunctions below conjunctions. Replacement of logical equivalents preserves equivalence through every local rewrite.

Remark (Dependencies)*.*

- Conjunctive Normal Form
- Negation Normal Form
- Distributive Laws for Propositional Logic
- Replacement of Logical Equivalents
- Existence of Negation Normal Form

Remark (Equivalence-preserving conversion)*.* Normal form conversion is a rewrite process that replaces a formula by a logically equivalent formula with a prescribed syntactic shape. Each local rewrite is justified by an equivalence law, and each replacement inside a larger formula is justified by Replacement of Logical Equivalents.

The goal is not to preserve the exact parse tree of the original formula. The goal is to preserve truth value under every valuation while moving the formula toward negation normal form, disjunctive normal form, or conjunctive normal form.

Remark (Step 1: eliminate biconditionals)*.* Replace every biconditional by an equivalent formula using the Biconditional Equivalence Laws. A standard replacement is

$$\alpha \leftrightarrow \beta \quad \rightsquigarrow \quad (\alpha \to \beta) \wedge (\beta \to \alpha).$$

After this step, no occurrence of $\leftrightarrow$ remains.

Remark (Step 2: eliminate conditionals)*.* Replace every conditional by an equivalent disjunction using the Conditional Equivalence Laws. The standard replacement is

$$\alpha \to \beta \quad \rightsquigarrow \quad \neg\alpha \vee \beta.$$

After this step, the only connectives still present are $\neg$, $\wedge$, and $\vee$.

Remark (Step 3: push negations inward). Use De Morgan Laws for Propositional Logic and double-negation simplification to move every negation down to the literal level:

$$\neg(\alpha \wedge \beta) \rightsquigarrow \neg\alpha \vee \neg\beta, \qquad \neg(\alpha \vee \beta) \rightsquigarrow \neg\alpha \wedge \neg\beta, \qquad \neg\neg\alpha \rightsquigarrow \alpha.$$

When this step terminates, the result is in negation normal form: negations occur only immediately in front of propositional variables.

Remark (Step 4A: distribute for DNF). To obtain DNF from negation normal form, distribute conjunction over disjunction using the Distributive Laws for Propositional Logic:

$$\alpha \wedge (\beta \vee \gamma) \quad \rightsquigarrow \quad (\alpha \wedge \beta) \vee (\alpha \wedge \gamma),$$

and similarly

$$(\beta \vee \gamma) \wedge \alpha \quad \rightsquigarrow \quad (\beta \wedge \alpha) \vee (\gamma \wedge \alpha).$$

Repeatedly applying these rewrites moves disjunctions outward until the formula is a finite disjunction of conjunction terms.

Remark (Step 4B: distribute for CNF). To obtain CNF from negation normal form, distribute disjunction over conjunction:

$$\alpha \vee (\beta \wedge \gamma) \quad \rightsquigarrow \quad (\alpha \vee \beta) \wedge (\alpha \vee \gamma),$$

and similarly

$$(\beta \wedge \gamma) \vee \alpha \quad \rightsquigarrow \quad (\beta \vee \alpha) \wedge (\gamma \vee \alpha).$$

Repeatedly applying these rewrites moves conjunctions outward until the formula is a finite conjunction of clauses.

Remark (DNF conversion workflow). To convert a formula to DNF:

1. eliminate biconditionals;
2. eliminate conditionals;
3. push negations inward to obtain negation normal form;
4. distribute conjunction over disjunction until the result is a finite disjunction of elementary conjunctions.

The resulting formula is logically equivalent to the original formula, but it need not be the shortest or only possible DNF representation.

Remark (CNF conversion workflow). To convert a formula to CNF:

1. eliminate biconditionals;
2. eliminate conditionals;
3. push negations inward to obtain negation normal form;
4. distribute disjunction over conjunction until the result is a finite conjunction of clauses.

The resulting formula is logically equivalent to the original formula, but it need not be the shortest or only possible CNF representation.

Remark (Termination). The first three stages terminate because they remove or strictly simplify specific connective patterns. Eliminating biconditionals reduces the number of $\leftrightarrow$ symbols. Eliminating conditionals reduces the number of $\rightarrow$ symbols. Pushing negations inward reduces the distance between each negation and the propositional variables it governs, while double negations shorten the formula.

The distribution stages terminate when guided by a target measure: for DNF, keep applying distributions that move $\vee$ outward past $\wedge$; for CNF, keep applying distributions that move $\wedge$ outward past $\vee$. Each application lowers the number of target-obstructing connective patterns, even though it may duplicate subformulas.

Remark (Correctness). Correctness is by equivalence preservation. Each rewrite instance is one of the listed logical equivalence laws: biconditional laws, conditional laws, De Morgan laws, or distributive laws. Replacement of logical equivalents permits the rewrite to occur inside an arbitrary larger formula. Therefore the final normal form is logically equivalent to the original formula.

Remark (Non-uniqueness). Ordinary DNF and CNF are not unique. Reordering terms or clauses, reordering literals inside a term or clause, duplicating a term, deleting a redundant term, or applying absorption-style simplifications can produce different formulas with the same truth table and the same normal-form type.

Canonical normal forms impose extra conventions to remove this freedom. The ordinary conversion procedures in this topic do not impose those conventions.

Remark (Size blowup). Distribution can duplicate subformulas. As a result, converting a compact formula to DNF or CNF can produce a formula whose size is much larger than the original. This size blowup is a syntactic cost of forcing a formula into a flat disjunction-of-terms or conjunction-of-clauses shape.

Remark (Comparison).

Target form	Final outer connective pattern	Distribution used
NNF	negations only at literals	none
DNF	disjunction of conjunction terms	$\wedge$ over $\vee$
CNF	conjunction of clauses	$\vee$ over $\wedge$

All three workflows use equivalence-preserving rewrites. DNF and CNF share the same first three stages and differ only in the final distribution direction.

Definition (Truth-Table Row Conjunction)

Definition 1.7.21 (Truth-Table Row Conjunction)**.** Fix a finite variable list $P_1, \ldots, P_n$ and a row r of the truth table for those variables. The *truth-table row conjunction* for r is the conjunction

$$L_1 \wedge L_2 \wedge \cdots \wedge L_n,$$

where

$$L_i = \begin{cases} P_i, & \text{if row } r \text{ makes } P_i \text{ true,} \\ \neg P_i, & \text{if row } r \text{ makes } P_i \text{ false.} \end{cases}$$

Remark (Standard quantified statement). For a fixed row r on variables $P_1, \ldots, P_n$,

$$\bigwedge_{i=1}^{n} L_i, \qquad L_i = \begin{cases} P_i, & r(P_i) = \mathsf{True}, \\ \neg P_i, & r(P_i) = \mathsf{False}. \end{cases}$$

Remark (Interpretation). The row conjunction names exactly one truth-table row. It is true on that row and false on every row that disagrees with it on at least one listed variable.

Remark (Dependencies).

- Truth Table
- Literal
- Variable Set of a Formula

Definition (Truth-Table Row Clause)

Definition 1.7.22 (Truth-Table Row Clause)**.** Fix a finite variable list $P_1, \ldots, P_n$ and a row r of the truth table for those variables. The *truth-table row clause* for r is the clause

$$M_1 \vee M_2 \vee \cdots \vee M_n,$$

where

$$M_i = \begin{cases} P_i, & \text{if row } r \text{ makes } P_i \text{ false,} \\ \neg P_i, & \text{if row } r \text{ makes } P_i \text{ true.} \end{cases}$$

Remark (Standard quantified statement). For a fixed row r on variables $P_1, \ldots, P_n$,

$$\bigvee_{i=1}^{n} M_i, \qquad M_i = \begin{cases} P_i, & r(P_i) = \mathsf{False}, \\ \neg P_i, & r(P_i) = \mathsf{True}. \end{cases}$$

Remark (Interpretation). The row clause excludes exactly one truth-table row. It is false on row r and true on every row that disagrees with r on at least one listed variable.

Remark (Dependencies).

- Truth Table

- Clause
- Variable Set of a Formula

Definition (Canonical Disjunctive Normal Form)

Definition 1.7.23 (Canonical Disjunctive Normal Form)**.** Let $\varphi \in \mathsf{WFF}$, and fix an ordering $P_1, \ldots, P_n$ of the finite variable set of φ. The *canonical disjunctive normal form* of φ relative to this ordering is the disjunction of the truth-table row conjunctions corresponding to the rows on which φ is true.

Remark (Standard quantified statement). For $\mathrm{Var}(\varphi) = \{P_1, \ldots, P_n\}$, the canonical disjunctive normal form is

$$\bigvee_{\widehat{v}(\varphi)=\mathsf{True}} \big(L_1 \wedge \cdots \wedge L_n\big),$$

where each row conjunction is determined by the truth-table row of v.

Remark (Interpretation). Canonical DNF records exactly the truth-table rows on which the formula is true, using one row conjunction for each true row.

Remark (Examples). If φ is a contradiction, then it has no true rows. Its canonical DNF is the empty disjunction, represented by the empty clause.

Remark (Dependencies).

- Disjunctive Normal Form
- Truth-Table Row Conjunction
- Truth Table
- Empty Clause

Definition (Canonical Conjunctive Normal Form)

Definition 1.7.24 (Canonical Conjunctive Normal Form)**.** Let $\varphi \in \mathsf{WFF}$, and fix an ordering $P_1, \ldots, P_n$ of the finite variable set of φ. The *canonical conjunctive normal form* of φ relative to this ordering is the conjunction of the truth-table row clauses corresponding to the rows on which φ is false.

Remark (Standard quantified statement). For $\mathrm{Var}(\varphi) = \{P_1, \ldots, P_n\}$, the canonical conjunctive normal form is

$$\bigwedge_{\widehat{v}(\varphi)=\mathsf{False}} \big(M_1 \vee \cdots \vee M_n\big),$$

where each row clause is determined by the truth-table row of v.

Remark (Interpretation). Canonical CNF records exactly the truth-table rows on which the formula is false, using one row clause for each false row.

Remark (Examples). If φ is a tautology, then it has no false rows. Its canonical CNF is the empty conjunction, represented by the empty conjunction.

Remark (Dependencies).

- Conjunctive Normal Form
- Truth-Table Row Clause
- Truth Table
- Empty Conjunction

Theorem (Canonical Disjunctive Normal Form)

Theorem 1.7.25 (Canonical Disjunctive Normal Form Theorem). *Every formula $\varphi \in \mathsf{WFF}$ is logically equivalent to its canonical disjunctive normal form relative to a fixed ordering of its finitely many variables.*

$$\varphi \equiv \mathrm{CDNF}_{P_1,\dots,P_n}(\varphi).$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall\varphi \in \mathsf{WFF})\, \varphi \equiv \mathrm{CDNF}_{P_1,\dots,P_n}(\varphi),$$

where $P_1, \dots, P_n$ is a fixed ordering of the finitely many variables of φ.

Remark (Interpretation). Canonical DNF rebuilds the formula from exactly the truth-table rows where the formula is true. Each row conjunction recognizes one true row, and the final disjunction accepts any of those rows.

Remark (Dependencies).

- Canonical Disjunctive Normal Form
- Finite Truth-Table Theorem
- Coincidence Lemma for Propositional Logic
- Logical Equivalence

Theorem (Canonical Conjunctive Normal Form)

Theorem 1.7.26 (Canonical Conjunctive Normal Form Theorem). *Every formula $\varphi \in \mathsf{WFF}$ is logically equivalent to its canonical conjunctive normal form relative to a fixed ordering of its finitely many variables.*

$$\varphi \equiv \mathrm{CCNF}_{P_1,\dots,P_n}(\varphi).$$

Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi \in \mathsf{WFF})\, \varphi \equiv \mathrm{CCNF}_{P_1,\ldots,P_n}(\varphi),$$

where $P_1, \ldots, P_n$ is a fixed ordering of the finitely many variables of φ.

Remark (Interpretation). Canonical CNF rebuilds the formula by excluding exactly the truth-table rows where the formula is false. Each row clause rejects one false row, and the final conjunction requires all false rows to be rejected.

Remark (Dependencies).

- Canonical Conjunctive Normal Form
- Finite Truth-Table Theorem
- Coincidence Lemma for Propositional Logic
- Logical Equivalence

Theorem (Semantic Classification by Normal Forms)

Theorem 1.7.27 (Semantic Classification by Normal Forms)**.** *Normal forms give finite certificates for semantic classification. For any formula $\varphi \in \mathsf{WFF}$:*

1. *φ is satisfiable if and only if some DNF formula equivalent to φ has a satisfiable conjunction term.*
2. *φ is a contradiction if and only if some DNF formula equivalent to φ has no satisfiable conjunction term.*
3. *φ is a tautology if and only if some CNF formula equivalent to φ has no falsifiable clause.*
4. *φ is a contingency if and only if it is satisfiable and not a tautology.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} \varphi \text{ is satisfiable} &\iff \text{some equivalent DNF term is satisfiable,} \\ \varphi \text{ is a contradiction} &\iff \text{no equivalent DNF term is satisfiable,} \\ \varphi \text{ is a tautology} &\iff \text{no equivalent CNF clause is falsifiable.} \end{aligned}$$

Remark (Interpretation). The truth-table method classifies φ by evaluating every row directly. The normal-form method first rewrites φ into an equivalent DNF or CNF formula and then checks the finite terms or clauses. Canonical normal forms connect the two approaches: canonical DNF is built from true rows, and canonical CNF is built from false rows.

Remark (Exposition). DNF is naturally witness-producing. A satisfiable term gives an explicit way to make the whole disjunction true. CNF is naturally suited for clause checking: each clause is a local obstruction to falsifying the whole conjunction. This clause-centered view is also the starting point for later resolution methods.

Remark (Dependencies).

- Disjunctive Normal Form
- Conjunctive Normal Form
- Satisfiable Formula
- Contradiction
- Tautology
- Contingency
- DNF Satisfiability Criterion
- Canonical Disjunctive Normal Form Theorem
- Canonical Conjunctive Normal Form Theorem

Summary

Remark (Literal vocabulary). The vocabulary begins at the literal level. A literal is a propositional variable or the negation of one propositional variable. Positive literals and negative literals record the sign of that occurrence. Complementary literals are the opposed pair P and $\neg P$. Clauses are finite disjunctions of literals, while terms are finite conjunctions of literals. The empty clause is the false boundary case, and the empty conjunction is the true boundary case.

Remark (Negation normal form). Negation normal form is the first normalization target. It permits only $\neg$, $\wedge$, and $\vee$, with every negation pushed down to the literal level. Its purpose is to remove conditionals and biconditionals and to make negation local. Once a formula is in NNF, the remaining work is a matter of distributing $\wedge$ and $\vee$ in the direction required by the target normal form.

Remark (Disjunctive normal form). Disjunctive normal form writes a formula as a finite disjunction of terms. It is naturally witness-oriented: to make a DNF formula true, it is enough to make one term true, and a satisfiable term directly records compatible literal requirements. This is why DNF is especially useful for satisfiability checks and for building formulas from true rows of a truth table.

Remark (Conjunctive normal form). Conjunctive normal form writes a formula as a finite conjunction of clauses. It is naturally obstruction-oriented: to falsify a CNF formula, one must falsify a clause, and each clause gives a local disjunctive condition. This clause structure makes CNF the preferred shape for later clause checking, resolution-style arguments, and systematic proof search.

Remark (Canonical normal forms). Ordinary DNF and CNF are not unique. Reordering, duplicating, or simplifying terms and clauses can produce different formulas with the same truth table. Canonical normal forms add truth-table discipline. Canonical DNF records the true rows by minterm-style conjunctions, and canonical CNF records the false rows by maxterm-style clauses. These canonical forms show that every finite truth behavior can be represented syntactically.

Remark (Conversion workflow). The conversion workflow has a common beginning. First eliminate biconditionals. Then eliminate conditionals. Then push negations inward until the formula is in negation normal form. From there, distribute conjunction over disjunction to obtain DNF, or distribute disjunction over conjunction to obtain CNF. Each stage is justified by logical equivalence laws and by replacement of logical equivalents inside larger formulas.

Remark (Back to the algebra of propositions). Normal forms are an application of the algebra of propositions. The conversion rules are not new semantic principles; they are uses of earlier equivalence laws such as De Morgan laws, conditional and biconditional equivalences, distributivity, and replacement of equivalents. The normal-form viewpoint therefore shows how algebraic manipulation and semantic preservation work together.

Remark (Forward to functional completeness). Canonical normal forms point forward to functional completeness. If a truth function can be described by its true rows, then it can be represented by a disjunction of conjunctions of literals. If it can be described by its false rows, then it can be represented by a conjunction of disjunctions of literals. This is the bridge from truth tables to the question of which connectives are enough to express every truth function.

Remark (Forward to arguments and proof systems). Normal forms also prepare the transition from single formulas to arguments and proof systems. DNF gives explicit witnesses for satisfiability, while CNF turns formulas into collections of clauses that can be checked and transformed. The same syntactic regularity used here to classify formulas will later support validity tests, derivation systems, resolution methods, and comparisons between semantic consequence and formal proof.

1.8 Functional Completeness

Toolkit

This section uses Boolean functions, truth functions induced by formulas, logical equivalence, canonical normal forms, and connective rewrite laws.

Remark (Exposition). The functional-completeness section studies expressive power. A connective set is complete exactly when formulas built from it can represent every Boolean function, so normal forms and connective reductions become tests for the adequacy of propositional vocabularies.

Orientation

Remark (Section map). This section begins with Boolean functions and formula representation, then introduces connective bases and definability of connectives. It uses normal forms to prove completeness of the standard connectives and then reduces the primitive vocabulary further. The final topics show that NAND and NOR each form a one-connective complete basis, while some familiar connective sets fail to be functionally complete.

Remark (Connection with normal forms). Normal forms explain why functional completeness is possible. Canonical DNF and canonical CNF show that the truth behavior of a formula can be reconstructed from finitely many truth-table rows. Once those canonical constructions are available, proving that a connective set is functionally complete amounts to showing that the connectives needed for those normal forms can be defined from that set.

Boolean Functions and Representation

Definition (Representation of a Boolean Function)

Definition 1.8.1 (Representation of a Boolean Function)**.** Let $f : \mathbb{B}^n \to \mathbb{B}$ be a Boolean function. A formula $\varphi \in \mathsf{WFF}$ with variables among $P_1, \ldots, P_n$ *represents* f if for every truth assignment v,

$$\widehat{v}(\varphi) = f(v(P_1), \ldots, v(P_n)).$$

Remark (Standard quantified statement).

$$\text{Represents}(\varphi, f; P_1, \ldots, P_n) \iff (\forall v : \mathsf{Prop} \to \mathbb{B})\big(\widehat{v}(\varphi) = f(v(P_1), \ldots, v(P_n))\big).$$

Remark (Definition predicate reading). Represents$(\varphi, f; P_1, \ldots, P_n)$ means that φ has exactly the same truth table as the Boolean function f on the listed variables.

Remark (Interpretation). A formula represents a Boolean function when the formula computes the same truth value as the function for every input row. This turns a semantic table of values into a syntactic object inside the propositional language.

Remark (Dependencies).

- Boolean Function
- Truth Function Induced by a Formula
- Truth Assignment

Theorem (DNF Representation of Boolean Functions)

Theorem 1.8.2 (DNF Representation of Boolean Functions)**.** *Every Boolean function $f : \mathbb{B}^n \to \mathbb{B}$ is represented by a formula in disjunctive normal form using variables $P_1, \ldots, P_n$.*
Go to proof.

Remark (Standard quantified statement).

$$(\forall f : \mathbb{B}^n \to \mathbb{B})(\exists \varphi \in \mathsf{WFF})\big(\varphi \text{ is in DNF} \land \text{Represents}(\varphi, f; P_1, \ldots, P_n)\big).$$

Remark (Predicate reading). Every n-ary Boolean function has a propositional formula representative, and one may choose the representative to be in disjunctive normal form.

Remark (Interpretation). A truth table specifies which input rows make the Boolean function true. For each true row, form the corresponding conjunction of literals; the disjunction of those row conjunctions represents the entire function.

Remark (Dependencies).

- Representation of a Boolean Function
- Canonical Disjunctive Normal Form
- Canonical Disjunctive Normal Form Theorem
- Disjunctive Normal Form

Connective Bases

Definition (Connective Basis)

Definition 1.8.3 (Connective Basis)**.** A *connective basis* is a specified set B of propositional connectives allowed as primitive connectives for building formulas.

Remark (Standard quantified statement).

$$B \text{ is a connective basis} \iff B \subseteq \{\neg, \land, \lor, \to, \leftrightarrow, \ldots\}$$

and formulas over B are built using only connectives from B.

Remark (Definition predicate reading). ConnectiveBasis(B) means that B is the chosen primitive connective vocabulary for a propositional language or fragment.

Remark (Interpretation). A connective basis controls which connectives are primitive, not which truth functions are expressible. Different bases may have the same expressive power.

Remark (Dependencies).

- Logical Connective
- Propositional Language

Definition (Definable Connective)

Definition 1.8.4 (Definable Connective)**.** Let B be a connective basis. A connective $\circ$ is *definable from* B if there is a formula built using only connectives from B whose truth function agrees with the truth function of $\circ$.

Remark (Standard quantified statement).

$$\circ \text{ is definable from } B \iff \exists\theta \in \mathsf{WFF}_B \ \big(f_\theta = f_\circ\big),$$

where WFF_B denotes the formulas built using only connectives from B.

Remark (Definition predicate reading). DefinableFrom$(\circ, B)$ means that $\circ$ can be treated as an abbreviation for a formula pattern using only connectives from B.

Remark (Interpretation). Definability separates notation from expressive power. A connective can be omitted from the primitive syntax without losing its use, provided it is recoverable as a defined connective.

Remark (Dependencies).

- Connective Basis
- Truth Function Induced by a Formula
- Boolean Function

Definition (Functionally Complete Connective Set)

Definition 1.8.5 (Functionally Complete Connective Set)**.** A connective basis B is *functionally complete* if every Boolean function $f : \mathbb{B}^n \to \mathbb{B}$, for every $n \geq 1$, is represented by some formula using only connectives from B.

Remark (Standard quantified statement).

$$\text{FunctionallyComplete}(B) \iff (\forall n \geq 1)(\forall f : \mathbb{B}^n \to \mathbb{B})(\exists \varphi \in \mathsf{WFF}_B)\,\text{Represents}(\varphi, f; P_1, \ldots, P_n).$$

Remark (Definition predicate reading). FunctionallyComplete(B) means that formulas over B can represent every finite Boolean input-output table.

Remark (Interpretation). Functional completeness is the central expressive-power property for connective sets. It says that the basis can express all possible truth-functional behavior, not merely the connectives originally listed in the language.

Remark (Dependencies).

- Connective Basis
- Representation of a Boolean Function
- Boolean Function

Definition (Expressively Equivalent Connective Sets)

Definition 1.8.6 (Expressively Equivalent Connective Sets)**.** Two connective bases B and C are *expressively equivalent* if every connective in B is definable from C, and every connective in C is definable from B.

Remark (Standard quantified statement).

$$B \equiv_{\text{expr}} C \iff \big((\forall \circ \in B)\, \text{DefinableFrom}(\circ, C)\big) \wedge \big((\forall \star \in C)\, \text{DefinableFrom}(\star, B)\big).$$

Remark (Definition predicate reading). $B \equiv_{\text{expr}} C$ means that the two bases can define exactly the same connective behavior.

Remark (Interpretation). Expressive equivalence lets a proof replace one primitive vocabulary with another. This is why a language can be simplified syntactically without changing which truth functions can be expressed.

Remark (Dependencies).

- Connective Basis
- Definable Connective

Completeness of the Standard Connectives

Theorem (Standard Connectives are Functionally Complete)

Theorem 1.8.7 (Standard Connectives are Functionally Complete)**.** *The connective basis*

$$\{\neg, \wedge, \vee\}$$

is functionally complete.
Go to proof.

Remark (Standard quantified statement).

$$\text{FunctionallyComplete}(\{\neg, \wedge, \vee\}).$$

Remark (Predicate reading). Every Boolean function is represented by a formula using only negation, conjunction, and disjunction.

Remark (Interpretation). Canonical disjunctive normal form uses only literals, conjunctions, and disjunctions. Since literals use variables and negation, the normal-form theorem shows that $\{\neg, \wedge, \vee\}$ can express every Boolean function.

Remark (Dependencies).

- Functionally Complete Connective Set
- DNF Representation of Boolean Functions
- Disjunctive Normal Form

Theorem (Negation and Conjunction are Functionally Complete)

Theorem 1.8.8 (Negation and Conjunction are Functionally Complete)**.** *The connective basis*

$$\{\neg, \wedge\}$$

is functionally complete.
Go to proof.

Remark (Standard quantified statement).

$$\text{FunctionallyComplete}(\{\neg, \wedge\}).$$

Remark (Predicate reading). Every Boolean function is represented by a formula using only negation and conjunction.

Remark (Interpretation). Disjunction is definable from negation and conjunction by De Morgan's law:

$$\varphi \vee \psi \equiv \neg(\neg\varphi \wedge \neg\psi).$$

Thus every formula using $\neg, \wedge, \vee$ can be translated into an equivalent formula using only $\neg$ and $\wedge$.

Remark (Dependencies).

- Standard Connectives are Functionally Complete
- De Morgan Laws for Propositional Logic
- Replacement of Logical Equivalents

Theorem (Negation and Disjunction are Functionally Complete)

Theorem 1.8.9 (Negation and Disjunction are Functionally Complete)**.** *The connective basis*

$$\{\neg, \vee\}$$

is functionally complete.
Go to proof.

Remark (Standard quantified statement).

$$\text{FunctionallyComplete}(\{\neg, \vee\}).$$

Remark (Predicate reading). Every Boolean function is represented by a formula using only negation and disjunction.

Remark (Interpretation). Conjunction is definable from negation and disjunction by De Morgan's law:

$$\varphi \wedge \psi \equiv \neg(\neg\varphi \vee \neg\psi).$$

Thus every formula using $\neg, \wedge, \vee$ can be translated into an equivalent formula using only $\neg$ and $\vee$.

Remark (Dependencies).

- Standard Connectives are Functionally Complete
- De Morgan Laws for Propositional Logic
- Replacement of Logical Equivalents

Reduction of Connective Sets

Definition (Connective Reduction)

Definition 1.8.10 (Connective Reduction)**.** A *connective reduction* replaces a formula using one connective basis by a logically equivalent formula using a smaller or different connective basis.

Remark (Standard quantified statement)*.*

$$\mathrm{Reduction}_{B\to C}(\varphi,\psi) \iff \varphi \equiv \psi \quad \text{and} \quad \psi \in \mathsf{WFF}_C.$$

Remark (Definition predicate reading)*.* $\mathrm{Reduction}_{B\to C}(\varphi,\psi)$ means that ψ is a translation of φ into the connective basis C without changing truth behavior.

Remark (Interpretation)*.* Reduction shows that some primitive connectives are optional. If a connective is definable from the remaining basis, then formulas using that connective can be translated away.

Remark (Exposition)*.* The standard reductions are:

$$\varphi \to \psi \equiv \neg\varphi \lor \psi,$$

$$\varphi \leftrightarrow \psi \equiv (\varphi \to \psi) \land (\psi \to \varphi),$$

$$\varphi \lor \psi \equiv \neg(\neg\varphi \land \neg\psi), \qquad \varphi \land \psi \equiv \neg(\neg\varphi \lor \neg\psi).$$

Each replacement preserves logical equivalence and may be performed inside a larger formula context.

Remark (Exposition)*.* Reducing a connective basis can make formulas longer. Functional completeness is therefore an expressive result, not an efficiency result. A small basis may be powerful enough to express every Boolean function while still being awkward for ordinary human-readable formulas.

Remark (Dependencies)*.*

- Connective Basis
- Definable Connective
- Logical Equivalence

NAND

Definition (NAND Connective)

Definition 1.8.11 (NAND Connective)**.** The *NAND* connective, written $\uparrow$, is the binary connective defined by

$$\varphi \uparrow \psi \equiv \neg(\varphi \land \psi).$$

Remark (Standard quantified statement).

$$\widehat{v}(\varphi \uparrow \psi) = \mathsf{True} \Longleftrightarrow \text{not both } \widehat{v}(\varphi) = \mathsf{True} \text{ and } \widehat{v}(\psi) = \mathsf{True}.$$

Remark (Definition predicate reading). NAND(φ, ψ) means the negation of the conjunction of φ and ψ.

Remark (Interpretation). NAND is read as "not both." It is false only when both inputs are true. This single connective can reproduce negation, conjunction, and disjunction.

Remark (Dependencies).

- Logical Connective
- Truth Assignment

Definition (NAND-Only Formula)

Definition 1.8.12 (NAND-Only Formula)**.** A *NAND-only formula* is a well-formed formula whose only connective is $\uparrow$.

Remark (Standard quantified statement).

$$\varphi \text{ is NAND-only} \Longleftrightarrow \varphi \in \mathsf{WFF}_{\{\uparrow\}}.$$

Remark (Definition predicate reading). NANDOnly(φ) means that every connective occurrence in φ is an occurrence of $\uparrow$.

Remark (Interpretation). A NAND-only formula is syntactically restricted but expressively powerful. The restriction is severe at the level of primitive symbols, but not at the level of truth functions.

Remark (Dependencies).

- NAND Connective
- Well-Formed Formula

Theorem (NAND is Functionally Complete)

Theorem 1.8.13 (NAND is Functionally Complete)**.** *The one-connective basis* $\{\uparrow\}$ *is functionally complete.*
Go to proof.

Remark (Standard quantified statement).

$$\text{FunctionallyComplete}(\{\uparrow\}).$$

Remark (Predicate reading). Every Boolean function can be represented by a NAND-only formula.

Remark (Interpretation). NAND can define negation and conjunction:

$$\neg\varphi \equiv \varphi \uparrow \varphi, \qquad \varphi \wedge \psi \equiv (\varphi \uparrow \psi) \uparrow (\varphi \uparrow \psi).$$

Once negation and conjunction are definable, functional completeness follows from the functional completeness of $\{\neg, \wedge\}$.

Remark (Dependencies).

- NAND Connective
- NAND-Only Formula
- Functionally Complete Connective Set
- Negation and Conjunction are Functionally Complete

NOR

Definition (NOR Connective)

Definition 1.8.14 (NOR Connective)**.** The *NOR* connective, written $\downarrow$, is the binary connective defined by

$$\varphi \downarrow \psi \equiv \neg(\varphi \vee \psi).$$

Remark (Standard quantified statement).

$$\widehat{v}(\varphi \downarrow \psi) = \mathsf{True} \iff \widehat{v}(\varphi) = \mathsf{False} \text{ and } \widehat{v}(\psi) = \mathsf{False}.$$

Remark (Definition predicate reading). NOR(φ, ψ) means the negation of the disjunction of φ and ψ.

Remark (Interpretation). NOR is read as "neither." It is true exactly when both inputs are false. Like NAND, this single connective can reproduce the standard functionally complete basis.

Remark (Dependencies).

- Logical Connective
- Truth Assignment

Definition (NOR-Only Formula)

Definition 1.8.15 (NOR-Only Formula)**.** A *NOR-only formula* is a well-formed formula whose only connective is $\downarrow$.

Remark (Standard quantified statement).

$$\varphi \text{ is NOR-only} \iff \varphi \in \mathsf{WFF}_{\{\downarrow\}}.$$

Remark (Definition predicate reading). NOROnly(φ) means that every connective occurrence in φ is an occurrence of $\downarrow$.

Remark (Interpretation). A NOR-only formula uses a single primitive connective. Its expressive power comes from the ability of NOR to define negation and disjunction.

Remark (Dependencies).

- NOR Connective
- Well-Formed Formula

Theorem (NOR is Functionally Complete)

Theorem 1.8.16 (NOR is Functionally Complete). *The one-connective basis* $\{\downarrow\}$ *is functionally complete.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{FunctionallyComplete}(\{\downarrow\}).$$

Remark (Predicate reading). Every Boolean function can be represented by a NOR-only formula.

Remark (Interpretation). NOR can define negation and disjunction:

$$\neg\varphi \equiv \varphi \downarrow \varphi, \qquad \varphi \vee \psi \equiv (\varphi \downarrow \psi) \downarrow (\varphi \downarrow \psi).$$

Once negation and disjunction are definable, functional completeness follows from the functional completeness of $\{\neg, \vee\}$.

Remark (Dependencies).

- NOR Connective
- NOR-Only Formula
- Functionally Complete Connective Set
- Negation and Disjunction are Functionally Complete

Incomplete Connective Sets

Definition (Incomplete Connective Set)

Definition 1.8.17 (Incomplete Connective Set). A connective basis is *incomplete* if it is not functionally complete.

Remark (Standard quantified statement).

$$\mathrm{Incomplete}(B) \iff \neg\,\mathrm{FunctionallyComplete}(B).$$

Remark (Definition predicate reading). Incomplete(B) means that some Boolean function cannot be represented by any formula using only connectives from B.

Remark (Interpretation). Incompleteness is usually proved by finding an invariant preserved by every formula built from the basis. If some Boolean function fails that invariant, the basis cannot express every Boolean function.

Remark (Dependencies).

- Functionally Complete Connective Set
- Boolean Function

Definition (Monotone Boolean Function)

Definition 1.8.18 (Monotone Boolean Function)**.** A Boolean function $f : \mathbb{B}^n \to \mathbb{B}$ is *monotone* if changing input coordinates from False to True never changes the output from True to False.

Remark (Standard quantified statement).

$$\mathrm{Monotone}(f) \iff (\forall a, b \in \mathbb{B}^n)\big(a \leq b \to f(a) \leq f(b)\big),$$

where False $\leq$ True coordinatewise.

Remark (Definition predicate reading). Monotone(f) means that increasing truth inputs cannot make the output decrease from true to false.

Remark (Interpretation). The connectives $\wedge$ and $\vee$ are monotone. Negation is not monotone, because changing P from false to true changes $\neg P$ from true to false.

Remark (Dependencies).

- Boolean Function

Theorem (Conjunction and Disjunction are not Functionally Complete)

Theorem 1.8.19 (Conjunction and Disjunction are not Functionally Complete)**.** *The connective basis $\{\wedge, \vee\}$ is not functionally complete.*
Go to proof.

Remark (Standard quantified statement).

$$\neg\,\mathrm{FunctionallyComplete}(\{\wedge, \vee\}).$$

Remark (Predicate reading). There is at least one Boolean function that cannot be represented by any formula using only conjunction and disjunction.

Remark (Interpretation). Every formula built using only $\wedge$ and $\vee$ represents a monotone Boolean function. Since negation is not monotone, no formula using only $\wedge$ and $\vee$ can represent negation. Hence the basis is incomplete.

Remark (Dependencies).

- Incomplete Connective Set
- Monotone Boolean Function
- Functionally Complete Connective Set

Minimal Complete Bases

Definition (Minimal Complete Basis)

Definition 1.8.20 (Minimal Complete Basis)**.** A functionally complete connective basis B is *minimal complete* if no proper subset of B is functionally complete.

Remark (Standard quantified statement).

$$\mathrm{MinimalComplete}(B) \iff \mathrm{FunctionallyComplete}(B) \wedge (\forall C \subsetneq B) \neg\, \mathrm{FunctionallyComplete}(C).$$ ∎

Remark (Definition predicate reading). $\mathrm{MinimalComplete}(B)$ means that B is complete, but removing any primitive connective from B destroys completeness.

Remark (Interpretation). Minimality is relative to deletion of connectives from the basis. It does not mean that formulas using the basis are short or easy to read.

Remark (Dependencies).

- Functionally Complete Connective Set
- Connective Basis

Theorem (NAND and NOR are Minimal Complete Bases)

Theorem 1.8.21 (NAND and NOR are Minimal Complete Bases)**.** *The one-connective bases $\{\uparrow\}$ and $\{\downarrow\}$ are minimal complete bases.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{MinimalComplete}(\{\uparrow\}) \wedge \mathrm{MinimalComplete}(\{\downarrow\}).$$

Remark (Predicate reading). NAND alone is complete, NOR alone is complete, and deleting the sole connective from either basis leaves no functionally complete basis.

Remark (Interpretation). A one-connective complete basis is automatically minimal once the empty basis is not functionally complete. NAND and NOR are therefore minimal complete bases because each is complete by itself.

Remark (Dependencies).

- Minimal Complete Basis
- NAND is Functionally Complete
- NOR is Functionally Complete
- Incomplete Connective Set

Summary and Transition

Remark (Summary). Functional completeness measures the expressive strength of a connective basis. A functionally complete basis can represent every finite Boolean function by a formula using only the connectives in that basis. Normal forms supply the main construction: truth-table rows can be converted into formulas, and those formulas can be assembled into a representative for the desired Boolean function.

Remark (Section map). The standard connectives $\neg$, $\wedge$, and $\vee$ are functionally complete because they can build disjunctive normal forms. De Morgan laws reduce this basis to $\{\neg, \wedge\}$ or $\{\neg, \vee\}$. NAND and NOR go further: each single connective can define a complete two-connective basis and therefore can represent every Boolean function.

Remark (Transition). The next section moves from expressive power to argument forms and informal proof patterns. Functional completeness explains which truth functions can be expressed. Argument forms ask which patterns of inference preserve truth from premises to conclusions.

1.9 Argument Forms and Informal Proof Patterns

Toolkit

This section uses argument forms, inference patterns, replacement rules, informal proof patterns, and semantic validity.

Remark (Exposition). Argument forms organize recurring patterns of propositional reasoning. This section connects valid and invalid forms with replacement principles, informal proof practice, and the semantic tests that justify those patterns.

Orientation

Remark (Working vocabulary). This orientation uses truth assignments, logical consequence, tautology, logical equivalence, formula contexts, and counterexample assignments to justify common argument forms.

Remark (Exposition). Argument forms are reusable patterns of inference. The purpose of this section is to justify the basic inference tools that will be used throughout the rest of the project. The justification is semantic: an argument form is valid when no truth assignment makes every premise true while making the conclusion false.

Remark (Section map). This section introduces argument forms, substitution instances, validity, invalidity, and counterexample assignments. It then proves the validity of the standard inference patterns, isolates common invalid forms, explains rules of replacement, and connects informal proof patterns with semantic consequence. Formal proof systems are previewed but not developed here.

Remark (Boundary). The goal is rigorous senior-undergraduate propositional logic. The section proves that the proof moves used later are truth-preserving, but it does not build Hilbert systems, sequent calculi, cut elimination, or metatheoretic completeness machinery.

Argument Forms

Remark (Working vocabulary). This topic introduces argument forms, premises, conclusions, substitution instances, valid argument forms, invalid argument forms, and counterexample truth assignments.

Definition (Argument Form)

Definition 1.9.1 (Argument Form)**.** An *argument form* is a schematic pattern consisting of a finite list of premise formulas and one conclusion formula.

Remark (Standard quantified statement).

$$\mathrm{ArgumentForm}(\Gamma, \varphi) \iff \Gamma \subseteq \mathsf{WFF} \text{ is finite and } \varphi \in \mathsf{WFF}.$$

Remark (Definition predicate reading). $\mathrm{ArgumentForm}(\Gamma, \varphi)$ means that Γ is the set or finite list of premises and φ is the conclusion.

Remark (Interpretation). An argument form records a pattern of inference independently of the particular subject matter. Its validity depends on truth preservation, not on whether the individual formulas happen to be true in one example.

Remark (Dependencies).

- Well-Formed Formula

Definition (Valid Argument Form)

Definition 1.9.2 (Valid Argument Form)**.** An argument form with premises Γ and conclusion φ is *valid* if every truth assignment that satisfies every formula in Γ also satisfies φ.

Remark (Standard quantified statement).

$$\mathrm{ValidArgumentForm}(\Gamma, \varphi) \iff (\forall v)\big((\forall \gamma \in \Gamma)(v \models \gamma) \to v \models \varphi\big).$$

Remark (Definition predicate reading). $\mathrm{ValidArgumentForm}(\Gamma, \varphi)$ means that truth of all premises forces truth of the conclusion under every truth assignment.

Remark (Interpretation). Validity is a preservation property. A valid argument form may have false premises in a particular instance, but it cannot have true premises and a false conclusion under the same truth assignment.

Remark (Dependencies).

- Argument Form
- Satisfaction of a Formula
- Logical Consequence

Definition (Counterexample Assignment)

Definition 1.9.3 (Counterexample Assignment)**.** A *counterexample assignment* to an argument form (Γ, φ) is a truth assignment that satisfies every premise in Γ and does not satisfy φ.

Remark (Standard quantified statement)*.*

$$\mathrm{CounterexampleAssignment}(v, \Gamma, \varphi) \iff (\forall \gamma \in \Gamma)(v \models \gamma) \land v \not\models \varphi.$$

Remark (Definition predicate reading)*.* $\mathrm{CounterexampleAssignment}(v, \Gamma, \varphi)$ means that v shows the premises can all be true while the conclusion is false.

Remark (Interpretation)*.* A counterexample assignment is the semantic witness to invalidity. Finding one settles the question: the argument form is not truth-preserving.

Remark (Dependencies)*.*

- Valid Argument Form
- Truth Assignment

Theorem (Truth-Table Validity Test for Argument Forms)

Theorem 1.9.4 (Truth-Table Validity Test for Argument Forms)**.** *An argument form (Γ, φ) is valid if and only if no row of its truth table makes every formula in Γ true and φ false.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\mathrm{ValidArgumentForm}(\Gamma, \varphi) \iff \neg(\exists v)\, \mathrm{CounterexampleAssignment}(v, \Gamma, \varphi).$$

Remark (Predicate reading)*.* The argument form is valid exactly when there is no counterexample truth assignment.

Remark (Interpretation)*.* The truth-table test is finite because only the variables appearing in the premises and conclusion matter. A failed row is a counterexample assignment; the absence of such rows proves semantic validity.

Remark (Dependencies)*.*

- Valid Argument Form
- Counterexample Assignment
- Truth Table
- Finite Truth-Table Theorem

Core Valid Inference Patterns

Remark (Working vocabulary). This topic proves the semantic validity of the standard propositional inference patterns used throughout later proof work.

Theorem (Modus Ponens is Valid)

Theorem 1.9.5 (Modus Ponens is Valid)**.** *The argument form*

$$\varphi, \qquad \varphi \to \psi \quad \therefore \quad \psi$$

is valid.
Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF}) \;\; \text{ValidArgumentForm}(\{\varphi, \varphi \to \psi\}, \psi).$$

Remark (Predicate reading). Whenever φ and $\varphi \to \psi$ are both true, ψ must be true.

Remark (Interpretation). Modus ponens is the basic rule for using a conditional whose antecedent has already been established.

Remark (Dependencies).

- Valid Argument Form
- Truth Assignment

Theorem (Modus Tollens is Valid)

Theorem 1.9.6 (Modus Tollens is Valid)**.** *The argument form*

$$\varphi \to \psi, \qquad \neg\psi \quad \therefore \quad \neg\varphi$$

is valid.
Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF}) \;\; \text{ValidArgumentForm}(\{\varphi \to \psi, \neg\psi\}, \neg\varphi).$$

Remark (Predicate reading). If φ would imply ψ, and ψ is false, then φ must be false.

Remark (Interpretation). Modus tollens is the semantic form of denying an antecedent by denying the conditional's consequent.

Remark (Dependencies).

- Valid Argument Form
- Conditional Equivalence Laws

Theorem (Hypothetical Syllogism is Valid)

Theorem 1.9.7 (Hypothetical Syllogism is Valid)**.** *The argument form*

$$\varphi \to \psi, \qquad \psi \to \chi \quad \therefore \quad \varphi \to \chi$$

is valid.
Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi, \psi, \chi \in \mathsf{WFF})\ \text{ValidArgumentForm}(\{\varphi \to \psi, \psi \to \chi\}, \varphi \to \chi).$$

Remark (Predicate reading)*.* Conditional implication composes: if φ forces ψ, and ψ forces χ, then φ forces χ.

Remark (Interpretation)*.* Hypothetical syllogism justifies chaining conditional statements.

Remark (Dependencies)*.*

- Valid Argument Form
- Logical Consequence

Theorem (Disjunctive Syllogism is Valid)

Theorem 1.9.8 (Disjunctive Syllogism is Valid)**.** *The argument form*

$$\varphi \vee \psi, \qquad \neg\varphi \quad \therefore \quad \psi$$

is valid.
Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi, \psi \in \mathsf{WFF})\ \text{ValidArgumentForm}(\{\varphi \vee \psi, \neg\varphi\}, \psi).$$

Remark (Predicate reading)*.* If at least one of φ or ψ is true and φ is false, then ψ must be true.

Remark (Interpretation)*.* Disjunctive syllogism eliminates one disjunct after its negation is known.

Remark (Dependencies)*.*

- Valid Argument Form
- Truth Assignment

Theorem (Addition is Valid)

Theorem 1.9.9 (Addition is Valid)**.** *The argument form*

$$\varphi \quad \therefore \quad \varphi \vee \psi$$

is valid.
Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi, \psi \in \mathsf{WFF}) \ \ \text{ValidArgumentForm}(\{\varphi\}, \varphi \vee \psi).$$

Remark (Predicate reading)*.* A true formula remains true when placed as one disjunct of a disjunction.

Remark (Interpretation)*.* Addition introduces a disjunction from a known true disjunct.

Remark (Dependencies)*.*

- Valid Argument Form

Theorem (Simplification is Valid)

Theorem 1.9.10 (Simplification is Valid)**.** *The argument form*

$$\varphi \wedge \psi \quad \therefore \quad \varphi$$

is valid.
Go to proof.

Remark (Standard quantified statement)*.*

$$(\forall \varphi, \psi \in \mathsf{WFF}) \ \ \text{ValidArgumentForm}(\{\varphi \wedge \psi\}, \varphi).$$

Remark (Predicate reading)*.* If a conjunction is true, then each conjunct is true.

Remark (Interpretation)*.* Simplification extracts a component from a true conjunction.

Remark (Dependencies)*.*

- Valid Argument Form

Theorem (Conjunction Introduction is Valid)

Theorem 1.9.11 (Conjunction Introduction is Valid)**.** *The argument form*

$$\varphi, \qquad \psi \quad \therefore \quad \varphi \wedge \psi$$

is valid.
Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi \in \mathsf{WFF})\ \ \mathrm{ValidArgumentForm}(\{\varphi, \psi\}, \varphi \wedge \psi).$$

Remark (Predicate reading). If two formulas are both true, then their conjunction is true.

Remark (Interpretation). Conjunction introduction combines independently established formulas into one conjunction.

Remark (Dependencies).

- Valid Argument Form

Theorem (Constructive Dilemma is Valid)

Theorem 1.9.12 (Constructive Dilemma is Valid)**.** *The argument form*

$$\varphi \to \chi, \qquad \psi \to \theta, \qquad \varphi \vee \psi \quad \therefore \quad \chi \vee \theta$$

is valid.
Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi, \chi, \theta \in \mathsf{WFF})\ \mathrm{ValidArgumentForm}(\{\varphi \to \chi, \psi \to \theta, \varphi \vee \psi\}, \chi \vee \theta).$$

Remark (Predicate reading). If one of two antecedents holds, and each antecedent has its own consequence, then at least one corresponding consequence holds.

Remark (Interpretation). Constructive dilemma is a proof-by-cases pattern with conditionals attached to both cases.

Remark (Dependencies).

- Valid Argument Form
- Disjunctive Syllogism is Valid

Theorem (Destructive Dilemma is Valid)

Theorem 1.9.13 (Destructive Dilemma is Valid)**.** *The argument form*

$$\varphi \to \chi, \qquad \psi \to \theta, \qquad \neg\chi \vee \neg\theta \quad \therefore \quad \neg\varphi \vee \neg\psi$$

is valid.
Go to proof.

Remark (Standard quantified statement).

$$(\forall \varphi, \psi, \chi, \theta \in \mathsf{WFF})\ \mathrm{ValidArgumentForm}(\{\varphi \to \chi, \psi \to \theta, \neg\chi \vee \neg\theta\}, \neg\varphi \vee \neg\psi).$$

Remark (Predicate reading). If at least one consequent fails, and each antecedent would force its consequent, then at least one antecedent must fail.

Remark (Interpretation). Destructive dilemma combines disjunction with modus tollens-style reasoning.

Remark (Dependencies).

- Valid Argument Form
- Modus Tollens is Valid

Invalid Argument Forms

Remark (Working vocabulary). This topic uses counterexample assignments to show why familiar invalid forms fail to preserve truth from premises to conclusion.

Theorem (Affirming the Consequent is Invalid)

Theorem 1.9.14 (Affirming the Consequent is Invalid). *The argument form*

$$\varphi \to \psi, \qquad \psi \quad \therefore \quad \varphi$$

is invalid.
Go to proof.

Remark (Standard quantified statement).

$$\neg\,\mathrm{ValidArgumentForm}(\{\varphi \to \psi, \psi\}, \varphi).$$

Remark (Predicate reading). There is a truth assignment under which $\varphi \to \psi$ and ψ are true while φ is false.

Remark (Interpretation). A conditional can be true because both antecedent and consequent are true, but also because the antecedent is false. Therefore truth of the consequent does not force truth of the antecedent.

Remark (Dependencies).

- Counterexample Assignment
- Valid Argument Form

Theorem (Denying the Antecedent is Invalid)

Theorem 1.9.15 (Denying the Antecedent is Invalid). *The argument form*

$$\varphi \to \psi, \qquad \neg\varphi \quad \therefore \quad \neg\psi$$

is invalid.
Go to proof.

Remark (Standard quantified statement).

$$\neg\,\mathrm{ValidArgumentForm}(\{\varphi \to \psi, \neg\varphi\}, \neg\psi).$$

Remark (Predicate reading). There is a truth assignment under which $\varphi \to \psi$ and $\neg\varphi$ are true while $\neg\psi$ is false.

Remark (Interpretation). A conditional does not say that the antecedent is the only way for the consequent to be true. The antecedent can fail while the consequent still holds.

Remark (Exposition). For both invalid forms, take a truth assignment with φ false and ψ true. Then $\varphi \to \psi$ is true. This assignment makes the premises true and the conclusion false in each corresponding invalid pattern.

Remark (Dependencies).

- Counterexample Assignment
- Valid Argument Form

Rules of Replacement

Remark (Working vocabulary). This topic uses formula contexts, logical equivalence, and replacement of logical equivalents to justify algebraic rewriting inside larger formulas.

Definition (Rule of Replacement)

Definition 1.9.16 (Rule of Replacement)**.** A *rule of replacement* is a schematic permission to replace a subformula by a logically equivalent formula inside a formula context.

Remark (Standard quantified statement).

$$\varphi \equiv \psi \Longrightarrow C[\varphi] \equiv C[\psi].$$

Remark (Definition predicate reading). A rule of replacement licenses an equivalence-preserving rewrite inside a larger formula.

Remark (Interpretation). Rules of replacement differ from rules of inference. Inference rules move from premises to conclusions; replacement rules rewrite formulas without changing truth behavior.

Remark (Dependencies).

- Formula Context
- Logical Equivalence
- Replacement of Logical Equivalents

Theorem (Replacement Rules Preserve Logical Equivalence)

Theorem 1.9.17 (Replacement Rules Preserve Logical Equivalence)**.** *If a formula ψ is obtained from a formula φ by finitely many applications of rules of replacement, then $\varphi \equiv \psi$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{ReplacementSequence}(\varphi, \psi) \Longrightarrow \varphi \equiv \psi.$$

Remark (Predicate reading). A finite sequence of equivalence-preserving replacements preserves logical equivalence from the first formula to the last formula.

Remark (Interpretation). Replacement rules justify algebraic manipulation of formulas. Each local rewrite preserves truth under every truth assignment, so the entire finite rewrite chain also preserves truth under every truth assignment.

Remark (Exposition). Common replacement rules include double negation, De Morgan replacement, commutativity, associativity, distributivity, material implication, biconditional replacement, and contraposition. Each is justified by a logical equivalence law.

Remark (Dependencies).

- Rule of Replacement
- Replacement of Logical Equivalents
- Logical Equivalence is an Equivalence Relation

Informal Proof Patterns

Remark (Working vocabulary). This topic connects valid argument forms with common informal proof patterns: direct proof, conditional proof, proof by cases, contradiction, contrapositive proof, implication chains, and equivalence chains.

Definition (Informal Proof Pattern)

Definition 1.9.18 (Informal Proof Pattern)**.** An *informal proof pattern* is a reusable truth-preserving strategy for organizing an ordinary mathematical proof.

Remark (Standard quantified statement).

$$\text{InformalProofPattern}(\Pi) \Longrightarrow \Pi \text{ is justified by one or more valid argument forms.}$$

Remark (Definition predicate reading). InformalProofPattern(Π) means that Π is a recognized proof strategy whose correctness rests on truth-preserving inference.

Remark (Interpretation). Informal proof patterns are not a substitute for proof. They are templates for arranging valid moves so that a proof has a clear logical shape.

Remark (Exposition). A direct proof establishes a target conclusion by moving forward from known premises through valid argument forms. A conditional proof establishes $\varphi \to \psi$ by assuming φ and deriving ψ inside that assumption context. Proof by cases uses a disjunction of cases and shows that each case leads to the desired conclusion.

Remark (Exposition). Proof by contradiction assumes the negation of the desired conclusion and derives an impossible situation. Contrapositive proof establishes $\varphi \to \psi$ by showing $\neg\psi \to \neg\varphi$. Chain-of-implications proofs use repeated hypothetical syllogism, while equivalence-chain proofs use replacement and logical equivalence.

Remark (Exposition). If the target has the form $\varphi \to \psi$, a conditional proof is often natural. If the available information has the form $\varphi \vee \psi$, proof by cases is often natural. If the target is an equivalence, an equivalence chain or two conditional proofs are often natural.

Remark (Dependencies).

- Valid Argument Form
- Logical Consequence

Semantic Justification of Argument Forms

Remark (Working vocabulary). This topic uses tautology, logical consequence, valid argument forms, and truth-table reasoning to connect named inference patterns with semantic validity.

Theorem (Tautological Implication Gives a Valid Argument Form)

Theorem 1.9.19 (Tautological Implication Gives a Valid Argument Form)**.** *Let* $\Gamma = \{\gamma_1, \dots, \gamma_n\} \subseteq \mathsf{WFF}$ *and let* $\varphi \in \mathsf{WFF}$. *If*

$$(\gamma_1 \wedge \cdots \wedge \gamma_n) \to \varphi$$

is a tautology, then the argument form (Γ, φ) *is valid.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Tautology}((\gamma_1 \wedge \cdots \wedge \gamma_n) \to \varphi) \Longrightarrow \text{ValidArgumentForm}(\{\gamma_1, \dots, \gamma_n\}, \varphi).$$

Remark (Predicate reading). If the conditional from the conjunction of all premises to the conclusion is always true, then the premises cannot all be true while the conclusion is false.

Remark (Interpretation). This theorem converts a tautology check into an argument-validity check. It is one reason truth tables can validate named inference rules.

Remark (Exposition). The semantic proof pattern for a valid argument form is always the same. Let an arbitrary truth assignment make all premises true. Then show that the conclusion must also be true. If this can be done for an arbitrary truth assignment, then no counterexample assignment exists.

Remark (Exposition). This section does not yet define formal derivations. The word valid here is semantic: it refers to preservation of truth under truth assignments. Later proof systems will turn selected valid patterns into formal rules of inference.

Remark (Dependencies).

- Tautology
- Valid Argument Form
- Logical Consequence

Argument Forms as a Bridge to Formal Systems

Remark (Working vocabulary). This topic distinguishes semantic validity from formal derivability and explains why formal proof systems are deferred.

Remark (Exposition). A valid argument form is a semantic object: it preserves truth under all truth assignments. A formal rule of inference is a syntactic object inside a specified proof system. Many formal systems include rules corresponding to the valid argument forms justified in this section, but the semantic justification and the formal rule are not the same object.

Remark (Exposition). Formal proof systems require additional machinery: formal derivations, assumptions, discharge rules, axioms or axiom schemata, and precise proof-checking rules. Natural deduction, Hilbert systems, tableaux, resolution, and sequent calculus are examples of such systems. They are important, but they are not the main purpose of this introductory propositional-logic section.

Remark (Transition). The tools validated here are safe to use in ordinary mathematical proof because they are truth-preserving. Later, when formal proof systems are introduced, these same patterns can be represented as formal rules, derived rules, or admissible moves depending on the chosen system.

Summary and Transition

Remark (Working vocabulary). This summary reviews argument forms, validity, counterexamples, valid inference patterns, invalid forms, replacement rules, and the semantic foundation for ordinary proof moves.

Remark (Exposition). An argument form is valid when truth of the premises under an arbitrary truth assignment forces truth of the conclusion. The central negative test is a counterexample assignment: one assignment making all premises true and the conclusion false. This section justified both sides of the method by proving standard valid forms and refuting common invalid forms.

Remark (Exposition). The valid tools now grounded include modus ponens, modus tollens, hypothetical syllogism, disjunctive syllogism, addition, simplification, conjunction introduction, constructive dilemma, and destructive dilemma. The invalid forms of affirming the consequent and denying the antecedent were separated by explicit counterexample assignments.

Remark (Transition). The next section turns to compactness and finite satisfiability as a preview of a deeper theme: infinite-looking logical questions can sometimes be controlled by finite witnesses. The semantic foundation developed here remains the base for that discussion.

1.10 Compactness and Finite Satisfiability Preview

Toolkit

This section uses truth assignments, satisfiable sets of formulas, finite subsets, finite satisfiability, finite unsatisfiability witnesses, and propositional compactness.

Remark (Exposition). The compactness preview records the finite-information pattern behind propositional satisfiability: when a set of formulas is unsatisfiable, the failure is already visible in some finite subset.

Orientation

Remark (Orientation). The semantic ideas developed earlier apply formula by formula and also to sets of formulas. This section isolates the finite-information pattern behind that viewpoint. A truth assignment can satisfy a whole set Γ only when it makes every member of Γ true. Failure will be shown, in propositional logic, to be detected by some finite part of Γ. This is the content of compactness.

The goal is not to replace truth tables. It is to record the principle that truth tables suggest: whenever inconsistency appears in propositional logic, it appears after only finitely many formulas have been selected.

Remark (Guiding question). For a possibly infinite set $\Gamma \subseteq \mathsf{WFF}$, when can the existence of truth assignments for every finite piece of Γ be promoted to a truth assignment for all of Γ?

Remark (Scope of the preview). Everything in this section is stated using truth assignments, formulas, sets of formulas, and finite subsets. Later chapters will need richer semantic data, but this preview stays inside propositional logic.

Satisfiable Sets

Remark (Recall). By satisfaction of a set of formulas, $v \models \Gamma$ means that $v \models \gamma$ for every $\gamma \in \Gamma$.

Definition (Satisfiable Set of Propositional Formulas)

Definition 1.10.1 (Satisfiable Set of Propositional Formulas). Let $\Gamma \subseteq \mathsf{WFF}$. The set Γ is satisfiable if there exists a truth assignment v such that $v \models \Gamma$.

$$\mathrm{SatisfiableSet}(\Gamma) \iff \exists v \left(\mathrm{TruthAssignment}(v) \land v \models \Gamma\right).$$

Remark (Standard quantified statement).

$$\Gamma \text{ is satisfiable} \iff \exists v \Big(\mathrm{TruthAssignment}(v) \wedge \forall \varphi \in \Gamma \, (\widehat{v}(\varphi) = \mathrm{True})\Big).$$

Remark (Definition predicate reading).

$$\mathrm{SatisfiableSet}(\Gamma)$$

means that one truth assignment makes every formula in Γ true.

Remark (Negated quantified statement).

$$\neg\, \mathrm{SatisfiableSet}(\Gamma) \iff \forall v \Big(\mathrm{TruthAssignment}(v) \Rightarrow \exists \varphi \in \Gamma \, (\widehat{v}(\varphi) = \mathrm{False})\Big).$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{SatisfiableSet}(\Gamma)$$

means that every truth assignment fails on at least one formula in Γ.

Remark (Failure modes). Satisfiability fails when every truth assignment is blocked by at least one formula in Γ.

Remark (Failure mode decomposition).

$$\neg\, \mathrm{SatisfiableSet}(\Gamma) \iff \forall v \Big(\mathrm{TruthAssignment}(v) \Rightarrow \exists \varphi \in \Gamma \, (\widehat{v}(\varphi) = \mathrm{False})\Big).$$

Thus Γ is not satisfiable exactly when every truth assignment is defeated by at least one formula from Γ.

Remark (Interpretation). Satisfiability of a set is a compatibility condition. The formulas in Γ may impose many requirements, but they are satisfiable only if all of those requirements can be met by the same truth assignment.

Remark (Dependencies).

- Truth Assignment
- Satisfaction of a Set of Formulas

Definition (Unsatisfiable Set of Propositional Formulas)

Definition 1.10.2 (Unsatisfiable Set of Propositional Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. The set Γ is unsatisfiable if it is not satisfiable.

$$\mathrm{UnsatisfiableSet}(\Gamma) \iff \neg\, \mathrm{SatisfiableSet}(\Gamma).$$

Remark (Standard quantified statement).

$$\Gamma \text{ is unsatisfiable} \iff \forall v \Big(\mathrm{TruthAssignment}(v) \Rightarrow \exists \varphi \in \Gamma \, (\widehat{v}(\varphi) = \mathrm{False})\Big).$$

Remark (Definition predicate reading).

$$\mathrm{UnsatisfiableSet}(\Gamma)$$

means that no truth assignment makes every formula in Γ true.

Remark (Negated quantified statement).

$$\neg\,\mathrm{UnsatisfiableSet}(\Gamma) \iff \mathrm{SatisfiableSet}(\Gamma).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{UnsatisfiableSet}(\Gamma)$$

means that Γ has at least one satisfying truth assignment.

Remark (Interpretation). Unsatisfiability is global failure of joint truth. It does not say that every formula in Γ is false; it says that no single truth assignment can make them all true at once.

Remark (Dependencies).

- Satisfiable Set of Propositional Formulas

Finitely Satisfiable Sets

Definition (Finite Subset of a Set of Propositional Formulas)

Definition 1.10.3 (Finite Subset of a Set of Propositional Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A finite subset of Γ is a set $\Delta \subseteq \Gamma$ containing only finitely many formulas.

Remark (Standard quantified statement).

$$\Delta \text{ is a finite subset of } \Gamma \iff \Delta \subseteq \Gamma \wedge \Delta \text{ is finite}.$$

Remark (Definition predicate reading).

$$\Delta \subseteq \Gamma \quad \text{and} \quad \Delta \text{ is finite}$$

means that Δ is a finite selection of formulas from Γ.

Remark (Interpretation). A finite subset is a finite test case cut out of a larger set of formulas.

Remark (Dependencies).

- Well-Formed Formula

Definition (Finitely Satisfiable Set of Propositional Formulas)

Definition 1.10.4 (Finitely Satisfiable Set of Propositional Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. The set Γ is finitely satisfiable if every finite subset of Γ is satisfiable.

$$\mathrm{FinitelySatisfiableSet}(\Gamma) \iff \forall \Delta \Big((\Delta \subseteq \Gamma \wedge \Delta \text{ is finite}) \Rightarrow \mathrm{SatisfiableSet}(\Delta) \Big).$$

Remark (Standard quantified statement).

$$\Gamma \text{ is finitely satisfiable} \iff \forall\Delta \Big((\Delta \subseteq \Gamma \wedge \Delta \text{ is finite}) \Rightarrow \exists v \,(\text{TruthAssignment}(v) \wedge v \models \Delta)\Big).$$

Remark (Definition predicate reading).

$$\text{FinitelySatisfiableSet}(\Gamma)$$

means that every finite portion of Γ has at least one satisfying truth assignment.

Remark (Negated quantified statement).

$$\neg\,\text{FinitelySatisfiableSet}(\Gamma) \iff \exists\Delta \Big(\Delta \subseteq \Gamma \wedge \Delta \text{ is finite} \wedge \text{UnsatisfiableSet}(\Delta)\Big).$$

Remark (Negation predicate reading).

$$\neg\,\text{FinitelySatisfiableSet}(\Gamma)$$

means that some finite subset of Γ already has no satisfying truth assignment.

Remark (Failure modes). Finite satisfiability fails when at least one finite subset of Γ is unsatisfiable.

Remark (Failure mode decomposition).

$$\neg\,\text{FinitelySatisfiableSet}(\Gamma) \iff \exists\Delta \Big(\Delta \subseteq \Gamma \wedge \Delta \text{ is finite} \wedge \text{UnsatisfiableSet}(\Delta)\Big).$$

So failure of finite satisfiability is witnessed by one finite unsatisfiable subcollection.

Remark (Interpretation). Finite satisfiability says that no finite test has found a contradiction. It does not name a single truth assignment for all of Γ; it only says that each finite test can be passed.

Remark (Dependencies).

- Finite Subset of a Set of Propositional Formulas
- Satisfiable Set of Propositional Formulas
- Truth Assignment

Finite Unsatisfiability Witnesses

Definition (Unsatisfiability Witness)

Definition 1.10.5 (Unsatisfiability Witness)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A subset $\Delta \subseteq \Gamma$ is an unsatisfiability witness for Γ if Δ is unsatisfiable.

Remark (Standard quantified statement).

$$\Delta \text{ is an unsatisfiability witness for } \Gamma \iff \Delta \subseteq \Gamma \wedge \text{UnsatisfiableSet}(\Delta).$$

Remark (Definition predicate reading).

$$\Delta \subseteq \Gamma \quad \text{and} \quad \mathrm{UnsatisfiableSet}(\Delta)$$

means that the formulas in Δ already explain why the larger set Γ cannot be satisfied, provided Δ is part of Γ.

Remark (Interpretation). An unsatisfiability witness is a subcollection that already fails to be jointly satisfiable.

Remark (Dependencies).

- Unsatisfiable Set of Propositional Formulas

Definition (Finite Unsatisfiability Witness)

Definition 1.10.6 (Finite Unsatisfiability Witness)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A subset $\Delta \subseteq \Gamma$ is a finite unsatisfiability witness for Γ if Δ is finite and unsatisfiable.

$$\mathrm{FiniteUnsatisfiabilityWitness}(\Delta, \Gamma) \iff \Delta \subseteq \Gamma \wedge \Delta \text{ is finite} \wedge \mathrm{UnsatisfiableSet}(\Delta).$$

Remark (Standard quantified statement).

$$\Delta \text{ is a finite unsatisfiability witness for } \Gamma \iff \Delta \subseteq \Gamma \wedge \Delta \text{ is finite} \wedge \mathrm{UnsatisfiableSet}(\Delta).$$

Remark (Definition predicate reading).

$$\mathrm{FiniteUnsatisfiabilityWitness}(\Delta, \Gamma)$$

means that Δ is a finite subset of Γ and that Δ already has no satisfying truth assignment.

Remark (Interpretation). A finite unsatisfiability witness is a finite obstruction. It says that the failure of Γ is already visible before one inspects all formulas in Γ.

Remark (Dependencies).

- Finite Subset of a Set of Propositional Formulas
- Unsatisfiable Set of Propositional Formulas

Theorem (Finite Unsatisfiability Witness)

Theorem 1.10.7 (Finite Unsatisfiability Witness)**.** *Let $\Gamma \subseteq \mathsf{WFF}$. If there exists a finite $\Delta \subseteq \Gamma$ such that Δ is unsatisfiable, then Γ is unsatisfiable.*

Go to proof.

Remark (Standard quantified statement).

$$\exists \Delta \ \mathrm{FiniteUnsatisfiabilityWitness}(\Delta, \Gamma) \Rightarrow \mathrm{UnsatisfiableSet}(\Gamma).$$

Remark (Predicate reading).

$$\exists\Delta\ \mathrm{FiniteUnsatisfiabilityWitness}(\Delta,\Gamma) \Rightarrow \mathrm{UnsatisfiableSet}(\Gamma)$$

means that a single finite unsatisfiable subcollection forces the whole set Γ to be unsatisfiable.

Remark (Failure modes). The theorem can fail only if a truth assignment satisfied Γ while a finite subset $\Delta \subseteq \Gamma$ remained unsatisfiable.

Remark (Failure mode decomposition).

$$\mathrm{SatisfiableSet}(\Gamma) \Rightarrow \forall\Delta\left(\Delta \subseteq \Gamma \Rightarrow \mathrm{SatisfiableSet}(\Delta)\right).$$

If the whole set has a satisfying truth assignment, each subset inherits one.

Remark (Contrapositive quantified statement).

$$\mathrm{SatisfiableSet}(\Gamma) \Rightarrow \forall\Delta\left(\Delta \subseteq \Gamma \Rightarrow \mathrm{SatisfiableSet}(\Delta)\right).$$

Remark (Contrapositive predicate reading).

$$\mathrm{SatisfiableSet}(\Gamma) \Rightarrow \mathrm{FinitelySatisfiableSet}(\Gamma)$$

means that one truth assignment for all of Γ also satisfies every finite subset of Γ.

Remark (Interpretation). An unsatisfiable finite subset is decisive. Adding more formulas cannot restore a truth assignment that already fails on the finite subset.

Remark (Dependencies).

- Finite Unsatisfiability Witness
- Unsatisfiable Set of Propositional Formulas
- Satisfaction of a Set of Formulas

Corollary (Finite Witness Contrapositive)

Corollary 1.10.8 (Finite Witness Contrapositive)**.** *Let* $\Gamma \subseteq \mathsf{WFF}$. *If* Γ *is satisfiable, then every finite subset of* Γ *is satisfiable.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{SatisfiableSet}(\Gamma) \Rightarrow \mathrm{FinitelySatisfiableSet}(\Gamma).$$

Remark (Predicate reading).

$$\mathrm{SatisfiableSet}(\Gamma) \Rightarrow \mathrm{FinitelySatisfiableSet}(\Gamma)$$

means that full satisfiability automatically gives finite satisfiability.

Remark (Interpretation). This is the easy direction connecting full satisfiability to finite satisfiability: one truth assignment that works for all formulas also works for any finite selection.

Remark (Dependencies).

- Finite Unsatisfiability Witness
- Finitely Satisfiable Set of Propositional Formulas
- Satisfiable Set of Propositional Formulas

Propositional Compactness

Theorem (Propositional Compactness)

Theorem 1.10.9 (Propositional Compactness)**.** *Let* $\Gamma \subseteq \mathsf{WFF}$*. If every finite subset of* Γ *is satisfiable, then* Γ *is satisfiable.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{FinitelySatisfiableSet}(\Gamma) \Rightarrow \mathrm{SatisfiableSet}(\Gamma).$$

Remark (Predicate reading).

$$\mathrm{FinitelySatisfiableSet}(\Gamma) \Rightarrow \mathrm{SatisfiableSet}(\Gamma)$$

means that passing every finite satisfiability test is enough to produce a truth assignment for all of Γ.

Remark (Negated quantified statement).

$$\mathrm{FinitelySatisfiableSet}(\Gamma) \wedge \neg\,\mathrm{SatisfiableSet}(\Gamma)$$

is the direct negation of the compactness implication.

Remark (Negation predicate reading).

$$\mathrm{FinitelySatisfiableSet}(\Gamma) \wedge \neg\,\mathrm{SatisfiableSet}(\Gamma)$$

would mean that all finite tests pass but the full set has no satisfying truth assignment.

Remark (Failure modes).

$$\neg\,\mathrm{SatisfiableSet}(\Gamma) \Rightarrow \exists\Delta\,\Big(\Delta \subseteq \Gamma \wedge \Delta \text{ is finite} \wedge \neg\,\mathrm{SatisfiableSet}(\Delta)\Big).$$

Compactness says that unsatisfiability cannot first appear only at the infinite level.

Remark (Failure mode decomposition).

$$\mathrm{UnsatisfiableSet}(\Gamma) \Rightarrow \exists\Delta\ \mathrm{FiniteUnsatisfiabilityWitness}(\Delta, \Gamma).$$

Remark (Contrapositive quantified statement).

$$\text{UnsatisfiableSet}(\Gamma) \Rightarrow \exists\Delta\ \text{FiniteUnsatisfiabilityWitness}(\Delta, \Gamma).$$

Remark (Contrapositive predicate reading). The contrapositive says that if Γ is unsatisfiable, then the failure is already witnessed by some finite subset of Γ.

Remark (Interpretation). Propositional compactness turns infinitely many finite checks into one global truth assignment. Equivalently, every propositional inconsistency has a finite witness.

Remark (Dependencies).

- Finitely Satisfiable Set of Propositional Formulas
- Satisfiable Set of Propositional Formulas
- Finite Unsatisfiability Witness

Finite Truth Table Intuition

Remark (Finite truth table intuition). Truth tables are finite because a single formula contains only finitely many variables. More generally, a finite set of formulas contains only finitely many variables, so its satisfiability can be checked by a finite truth table over those variables.

Remark (Finite test). For a finite $\Delta \subseteq \mathsf{WFF}$, the variables appearing in Δ are finite in number. A truth table over those variables decides whether Δ is satisfiable.

Remark (Why compactness is surprising). Finite truth tables directly decide finite sets. Compactness says that, for propositional logic, passing all finite tests is enough to pass the entire possibly infinite test.

Remark (Dependencies).

- Finiteness of Variables
- Finite Truth Table
- Finitely Satisfiable Set of Propositional Formulas

Summary

Remark (Section summary). The compactness preview records three related ideas.

- A set of propositional formulas is satisfiable when one truth assignment makes all of its formulas true.
- A set is finitely satisfiable when every finite subset is satisfiable.
- Propositional compactness says that finite satisfiability is enough for satisfiability.

Remark (Conceptual takeaway). In propositional logic, an inconsistency always has a finite witness. If no finite witness exists, then the whole set of formulas can be satisfied.

Chapter 2

Predicate Logic

SERIES
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Logic, Sets, and Proof

BOOK I
Mathematical Logic and Proof

CHAPTER 2
Predicate Logic

2.1 Model-Theoretic View

Toolkit

This section presents first-order logic as a syntax-plus-semantics model, with the language interpreted by structures and formulas evaluated by the shared free-algebra machinery.

Theory (First-Order Logic)

Depends on: `Signature; term; formula; sentence; bridge:consequence;` $\mathrm{Cn}_{\vdash}$

Language

$$\mathcal{L} = (\mathsf{Const}_{\mathcal{L}}, \mathsf{Func}_{\mathcal{L}}, \mathsf{Rel}_{\mathcal{L}}, =).$$

$$\mathsf{Term}_{\mathcal{L}}, \qquad \mathsf{Form}_{\mathcal{L}}, \qquad \mathsf{Sent}_{\mathcal{L}}.$$

Presented Theory

$$T := \mathrm{Cn}_{\vdash}(\Sigma), \qquad \Sigma \subseteq \mathsf{Sent}_{\mathcal{L}}.$$

Theorems and Consequences

1. $$\Gamma \vdash_{\mathrm{FOL}} \varphi \quad \Longleftrightarrow \quad \Gamma \models_{\mathrm{FOL}} \varphi \qquad \text{(completeness)}.$$

Model (First-Order Logic)

Model 2.1.1 (First-Order Logic). **Depends on:** `ℒ-structure; interpretation; assignment; bridge:free-sigma-algebra-evaluation; bridge:consequence`

Note

FOL runs the shared machine twice: once for terms, then once for quantifier-free formulas. Quantifiers sit above that fixed-seed machine.

Structure

$$\mathcal{A} = \langle A, I \rangle, \qquad A \neq \varnothing.$$

$$I(c) \in A, \qquad I(f) : A^n \to A, \qquad I(R) \subseteq A^n, \qquad = \text{ is logical equality on } A.$$

Binding

$$s : \mathsf{Var} \to A \xrightarrow{\text{eval}(\mathcal{A}_{\text{term}})} [\![\cdot]\!]_{\mathcal{A},s} \xrightarrow{\text{relation/equality tests}} \sigma_s \xrightarrow{\text{eval}(\mathbf{2})} \models_{\text{qf}} .$$

Binding

$$\Sigma_{\text{term}} := \mathsf{Const}_{\mathcal{L}} \cup \mathsf{Func}_{\mathcal{L}}, \qquad \mathcal{A}_{\text{term}} := (A, (If)_{f \in \mathsf{Func}}, (Ic)_{c \in \mathsf{Const}}).$$

$$[\![\cdot]\!]_{\mathcal{A},s} := \text{eval}(\mathcal{A}_{\text{term}})(s).$$

Binding

$$\sigma_s\big(R(\bar{t})\big) := \big[([\![t_1]\!]_{\mathcal{A},s}, \ldots, [\![t_n]\!]_{\mathcal{A},s}) \in I(R)\big],$$

$$\sigma_s(t_1 = t_2) := \big[[\![t_1]\!]_{\mathcal{A},s} = [\![t_2]\!]_{\mathcal{A},s}\big].$$

Syntax–Semantics Bridge

$$\mathcal{A}, s \models \varphi \quad :\Longleftrightarrow \quad \text{eval}(\mathbf{2})(\sigma_s)(\varphi) = 1 \qquad (\varphi \text{ quantifier-free}).$$

Syntax–Semantics Bridge

$$\mathcal{A}, s \models \exists x\, \varphi \quad \Longleftrightarrow \quad \exists a \in A,\ \mathcal{A}, s[x \mapsto a] \models \varphi.$$

$$\mathcal{A}, s \models \forall x\, \varphi \quad \Longleftrightarrow \quad \forall a \in A,\ \mathcal{A}, s[x \mapsto a] \models \varphi.$$

Syntax–Semantics Bridge

For $\sigma \in \mathsf{Sent}_{\mathcal{L}}$, $\mathcal{A}, s \models \sigma$ is independent of s by the coincidence lemma, so

$$\mathcal{A} \models \sigma := \mathcal{A}, s \models \sigma \qquad (s \text{ arbitrary}).$$

Theorems and Consequences

1. Semantic consequence, $\mathcal{A} \models T$, and $\mathrm{Th}(\mathcal{A})$ are inherited from *Consequence*.
2. Prop is the base case: seed $= v$. FOL is the inductive case: s is evaluated into term values, then relation tests compute σ_s.

Definitional Extension (First-Order Logic)

Depends on: `theory:first-order-logic; expansion; reduct; conservativity; unique existence`

Note

$$\exists! y\, \varphi(y) \;:\equiv\; \exists y \big(\varphi(y) \wedge \forall y' (\varphi(y') \to y' = y)\big).$$

Admissibility

Relation symbol R, arity n:

$$\forall \bar{x}\, (R(\bar{x}) \leftrightarrow \varphi(\bar{x})).$$

Function symbol f, arity n:

$$T \vdash \forall \bar{x}\, \exists! y\, \psi(\bar{x}, y), \qquad \forall \bar{x}\, \forall y\, (f(\bar{x}) = y \leftrightarrow \psi(\bar{x}, y)).$$

Constant symbol c:

$$T \vdash \exists! y\, \psi(y), \qquad \forall y\, (c = y \leftrightarrow \psi(y)).$$

Theorems and Consequences

1. $$T^+ \vdash \sigma \quad \Longleftrightarrow \quad T \vdash \sigma \qquad (\sigma \in \mathcal{L}).$$
2. $$\mathcal{A} \models T \Longrightarrow \exists! \mathcal{A}^+\, (\mathcal{A}^+ \restriction \mathcal{L} = \mathcal{A}).$$

2.2 Axiom Schemas

Predicate-Logic Axiom Schemas

The predicate-logic development extends propositional logic with the following quantifier axiom schemas. The side conditions prevent variable capture and ensure that the displayed instances are well formed.

Axiom Schema (Predicate Logic 1)

Axiom 2.2.1 (Universal Instantiation)**.** For every formula $\varphi(x)$ and every term t that is free for x in φ,

$$\forall x\, \varphi(x) \implies \varphi(t).$$

Remark (Interpretation). A universally quantified statement may be specialized to any admissible instance.

Axiom Schema (Predicate Logic 2)

Axiom 2.2.2 (Quantifier Distribution)**.** For formulas φ and $\psi(x)$, if x is not free in φ, then

$$\forall x\, (\varphi \implies \psi(x)) \implies \big(\varphi \implies \forall x\, \psi(x)\big).$$

Remark (Interpretation). When a hypothesis does not depend on the quantified variable, a universal conclusion may be drawn uniformly under that hypothesis.

2.3 Syntax

2.3.1 Syntax of First-Order Logic: Terms and Formulas

Remark (Section toolkit: Syntax ledger). This section fixes the syntactic inventory for first-order logic before any semantic machinery is introduced. The ledger proceeds from symbols to terms, from terms to atomic formulas, and from atomic formulas to all well-formed formulas.

Layer	Item	Role
Symbol stock	Variable	reusable placeholder symbol
Symbol stock	Constant Symbol	nullary name-forming symbol
Symbol stock	Function Symbol	operation-forming symbol
Symbol stock	Predicate Symbol	assertion-forming symbol
Object syntax	Term	expression that names an object syntactically
Formula syntax	Atomic Formula	predicate symbol applied to terms
Formula syntax	Well-Formed Formula	formula generated by the grammar
Formula syntax	Molecular Formula	non-atomic well-formed formula

Remark (Exposition). The grammar has two tracks. Terms are built first, because atomic formulas need terms as their inputs. Formulas are built second, because they begin with atomic formulas and then close under logical connectives and quantifiers. Semantic notions such as structures, assignments, satisfaction, truth, and models enter later; the present section records only which strings count as grammatical expressions.

Remark (Grammar boundary). Terms name objects. Formulas make assertions. A term is not a formula, and a formula is not a term. For example, $f(x)$ is a term, while $P(f(x))$ is a formula.

Remark (Predicate-symbol convention). Object-language predicate symbols, such as P, Q, R, are symbols that occur inside first-order formulas. Project metalanguage predicate readings, such as $\mathrm{Term}(t)$ or $\mathrm{AtomicFormula}(\varphi)$, are external classification predicates used by these notes to describe the grammar. The two levels should not be confused.

Definition (Variable)

Definition 2.3.1 (Variable)**.** For a first-order language $\mathcal{L}$, a *variable* is a symbol drawn from the distinguished variable stock of $\mathcal{L}$. Variables are usually written

$$x, y, z, x_0, x_1, x_2, \ldots.$$

Remark (Standard quantified statement).

$$\mathsf{Var}_{\mathcal{L}} = \{x, y, z, x_0, x_1, x_2, \ldots\}.$$

Remark (Definition predicate reading). $\mathrm{Variable}(x)$ means $x \in \mathsf{Var}_{\mathcal{L}}$.

Remark (Negated quantified statement).

$$x \notin \mathsf{Var}_{\mathcal{L}}.$$

Remark (Negation predicate reading). $\neg\,\mathrm{Variable}(x)$ means that x is not one of the variable symbols of $\mathcal{L}$.

Remark (Failure modes). A symbol fails to be a variable when it belongs to a different syntactic role, such as a constant symbol, function symbol, predicate symbol, connective, or punctuation mark.

Remark (Failure mode decomposition).

$$\underbrace{c}_{\text{constant symbol}} \qquad \underbrace{f}_{\text{function symbol}} \qquad \underbrace{P}_{\text{predicate symbol}} \qquad \underbrace{\forall}_{\text{logical symbol}} \quad .$$

Remark (Interpretation). A variable is a basic syntactic placeholder. It can occur inside terms and formulas, and later sections explain how quantifiers bind occurrences of variables.

Remark (Examples). x, y, z, x_0, and x_1 are standard variable symbols.

Remark (Non-Examples). c, 0, f, P, $\forall$, and $\wedge$ are not variables merely because they are symbols of the language.

Remark (Dependencies).

- Language

Definition (Constant Symbol)

Definition 2.3.2 (Constant Symbol)**.** For a first-order language $\mathcal{L}$, a *constant symbol* is a non-logical symbol of arity 0. A constant symbol may occur as a term without taking input terms.

Remark (Standard quantified statement).

$$\mathsf{Const}_{\mathcal{L}} = \{c \in \mathcal{L} : \text{arity}(c) = 0\}.$$

Remark (Definition predicate reading). ConstantSymbol(c) means $c \in \mathsf{Const}_{\mathcal{L}}$.

Remark (Negated quantified statement).

$$c \notin \mathsf{Const}_{\mathcal{L}}.$$

Remark (Negation predicate reading). $\neg$ ConstantSymbol(c) means that c is not a nullary non-logical symbol of $\mathcal{L}$.

Remark (Failure modes). A symbol fails to be a constant symbol when it is not in the language, is logical punctuation, or has positive arity.

Remark (Failure mode decomposition).

$$\underbrace{x}_{\text{variable}} \qquad \underbrace{f}_{\text{positive arity}} \qquad \underbrace{(}_{\text{punctuation}} \quad .$$

Remark (Interpretation). A constant symbol is a simple name-forming symbol in the grammar. It supplies a base case for terms alongside variables.

Remark (Examples). In an arithmetic language, 0 and 1 are commonly used as constant symbols.

Remark (Non-Examples). $f(x)$ is not a constant symbol; it is a term built from a function symbol and an input term.

Remark (Dependencies).

- Variable

Definition (Function Symbol)

Definition 2.3.3 (Function Symbol)**.** For a first-order language $\mathcal{L}$, an n-ary *function symbol* is a non-logical symbol of arity $n \geq 1$. If f has arity n, then it forms terms by the pattern

$$f(t_1, \dots, t_n).$$

Remark (Standard quantified statement).

$$\mathsf{Func}_{\mathcal{L}}^{(n)} = \{f \in \mathcal{L} : \text{arity}(f) = n \text{ and } n \geq 1\}.$$

Remark (Definition predicate reading). FunctionSymbol(f) means $f \in \mathsf{Func}_{\mathcal{L}}^{(n)}$ for some $n \geq 1$.

Remark (Negated quantified statement).

$$\forall n \geq 1\ (f \notin \mathsf{Func}_{\mathcal{L}}^{(n)}).$$

Remark (Negation predicate reading). $\neg\,\text{FunctionSymbol}(f)$ means that f is not a positive-arity function symbol of $\mathcal{L}$.

Remark (Failure modes). A symbol fails to be a function symbol when it is a variable, constant symbol, predicate symbol, logical symbol, punctuation mark, or has no assigned positive arity.

Remark (Failure mode decomposition).

$$\underbrace{c}_{\text{arity }0} \qquad \underbrace{P}_{\text{predicate symbol}} \qquad \underbrace{\neg}_{\text{logical symbol}}.$$

Remark (Interpretation). A function symbol is an operation-forming symbol in the grammar. It is not a term by itself; it becomes part of a term only after it is supplied the required number of input terms.

Remark (Examples). In an arithmetic language, S may be unary, while $+$ and $\cdot$ may be binary function symbols.

Remark (Non-Examples). $f(x)$ is not a function symbol. It is a term formed by applying the function symbol f to the term x.

Remark (Dependencies).

- Constant Symbol

Definition (Predicate Symbol)

Definition 2.3.4 (Predicate Symbol)**.** For a first-order language $\mathcal{L}$, an n-ary *predicate symbol* is a non-logical symbol of arity $n \geq 1$. If P has arity n, then it forms atomic formulas by the pattern

$$P(t_1, \ldots, t_n).$$

Remark (Standard quantified statement).

$$\mathsf{Pred}_{\mathcal{L}}^{(n)} = \{P \in \mathcal{L} : \text{arity}(P) = n \text{ and } n \geq 1\}.$$

Remark (Definition predicate reading). $\text{PredicateSymbol}(P)$ means $P \in \mathsf{Pred}_{\mathcal{L}}^{(n)}$ for some $n \geq 1$.

Remark (Negated quantified statement).

$$\forall n \geq 1\ (P \notin \mathsf{Pred}_{\mathcal{L}}^{(n)}).$$

Remark (Negation predicate reading). $\neg\,\text{PredicateSymbol}(P)$ means that P is not a positive-arity predicate symbol of $\mathcal{L}$.

Remark (Failure modes). A symbol fails to be a predicate symbol when it belongs to the term-forming side of the language, is a logical symbol, is punctuation, or has no assigned positive arity as a predicate symbol.

Remark (Failure mode decomposition).

$$\underbrace{x}_{\text{variable}} \qquad \underbrace{c}_{\text{constant symbol}} \qquad \underbrace{f}_{\text{function symbol}} \qquad \underbrace{\exists}_{\text{logical symbol}} .$$

Remark (Interpretation). Predicate symbols are the atomic-formula-forming vocabulary of the language. They are syntactic symbols, not project metalanguage predicates.

Remark (Examples). P, Q, R, $<$, and $=$ may be predicate symbols when the language declares the corresponding arities.

Remark (Non-Examples). $P(x)$ is not a predicate symbol. It is an atomic formula formed from the predicate symbol P and the term x.

Remark (Dependencies).

- Function Symbol

Definition (Term)

Definition 2.3.5 (Term)**.** The set of *terms* of a first-order language $\mathcal{L}$, denoted $\mathsf{Term}_{\mathcal{L}}$, is the smallest set of strings satisfying:

1. every variable is a term;
2. every constant symbol is a term;
3. if f is an n-ary function symbol and $t_1, \ldots, t_n$ are terms, then $f(t_1, \ldots, t_n)$ is a term;
4. no string is a term unless it is generated by finitely many applications of the previous clauses.

Remark (Standard quantified statement).

$$\mathsf{Term}_{\mathcal{L}} = \bigcap \left\{ S : \mathsf{Var}_{\mathcal{L}} \subseteq S,\ \mathsf{Const}_{\mathcal{L}} \subseteq S,\ (\forall n \geq 1)(\forall f \in \mathsf{Func}_{\mathcal{L}}^{(n)})(\forall t_1, \ldots, t_n \in S) f(t_1, \ldots, t_n) \in S \right\}.$$

Remark (Definition predicate reading). $\mathrm{Term}(t)$ means $t \in \mathsf{Term}_{\mathcal{L}}$.

Remark (Negated quantified statement).

$$\sigma \notin \mathsf{Term}_{\mathcal{L}} \iff \exists S \begin{bmatrix} \mathsf{Var}_{\mathcal{L}} \subseteq S \wedge \mathsf{Const}_{\mathcal{L}} \subseteq S \\ \wedge\, (\forall n \geq 1)(\forall f \in \mathsf{Func}_{\mathcal{L}}^{(n)})(\forall t_1, \ldots, t_n \in S) f(t_1, \ldots, t_n) \in S \\ \wedge\ \ \sigma \notin S \end{bmatrix}.$$

Remark (Negation predicate reading). $\neg\,\mathrm{Term}(\sigma)$ means that σ is not generated by the term-formation clauses.

Remark (Failure modes). A string fails to be a term when it uses a predicate symbol, connective, or quantifier at the outer syntactic level, when a function symbol has too few or too many input terms, or when punctuation does not match the arity pattern.

Remark (Failure mode decomposition).

$$\underbrace{P(x)}_{\text{atomic formula, not a term}} \qquad \underbrace{f(x,y)}_{\text{wrong arity if } f \text{ is unary}} \qquad \underbrace{\forall x}_{\text{quantifier fragment}} \quad .$$

Remark (Interpretation). Terms are the noun-like expressions of first-order syntax. They provide the inputs to predicate symbols but do not themselves make assertions.

Remark (Examples). x, c, $f(x)$, $g(c,x)$, and $h(f(x),c)$ are terms when the language assigns the displayed arities.

Remark (Non-Examples). $P(x)$, $\neg P(x)$, $\forall x\, P(x)$, and $x \wedge y$ are not terms.

Remark (Dependencies).

- Language

Definition (Atomic Formula)

Definition 2.3.6 (Atomic Formula)**.** An *atomic formula* of a first-order language $\mathcal{L}$ is a string of the form

$$P(t_1,\ldots,t_n),$$

where P is an n-ary predicate symbol and $t_1,\ldots,t_n$ are terms.

Remark (Standard quantified statement).

$$\mathsf{Atom}_{\mathcal{L}} = \left\{P(t_1,\ldots,t_n) : P \in \mathsf{Pred}_{\mathcal{L}}^{(n)} \text{ and } t_1,\ldots,t_n \in \mathsf{Term}_{\mathcal{L}}\right\}.$$

Remark (Definition predicate reading). AtomicFormula(φ) means $\varphi \in \mathsf{Atom}_{\mathcal{L}}$.

Remark (Negated quantified statement).

$$\varphi \notin \mathsf{Atom}_{\mathcal{L}}.$$

Remark (Negation predicate reading). $\neg$ AtomicFormula(φ) means that φ is not a predicate symbol applied to the required number of terms.

Remark (Failure modes). A string fails to be atomic when it does not begin with a predicate symbol, when the predicate symbol receives the wrong number of term inputs, when an input is not a term, or when a logical connective or quantifier has already been applied.

Remark (Failure mode decomposition).

$$\underbrace{f(x)}_{\text{term, not formula}} \qquad \underbrace{P(x,y)}_{\text{wrong arity if } P \text{ is unary}} \qquad \underbrace{\neg P(x)}_{\text{molecular formula}} \quad .$$

Remark (Interpretation). Atomic formulas are the base cases of formula formation. They are the first strings in the grammar that count as formulas.

Remark (Examples). $P(x)$, $R(x,c)$, $x = y$, and $x < y$ are atomic formulas when the language supplies the relevant predicate symbols and arities.

Remark (Non-Examples). x, $f(x)$, $\neg P(x)$, and $(P(x) \wedge Q(x))$ are not atomic formulas.

Remark (Dependencies).

- Predicate Symbol
- Term

Definition (Formula)

Definition 2.3.7 (Formula)**.** For a first-order language $\mathcal{L}$, a *formula* is an expression formed by the formula grammar of $\mathcal{L}$. The grammar starts from atomic formulas and closes under the logical connectives and quantifiers of the language.

Remark (Interpretation). The language supplies the symbols and formation rules; the formula notion is the syntactic category governed by those rules.

Remark (Dependencies).

- Language

Definition (Well-Formed Formula)

Definition 2.3.8 (Well-Formed Formula)**.** The set of *well-formed formulas* of a first-order language $\mathcal{L}$, denoted $\mathsf{Form}_{\mathcal{L}}$, is the smallest set of strings satisfying:

1. every atomic formula is a well-formed formula;
2. if φ is a well-formed formula, then $\neg\varphi$ is a well-formed formula;
3. if φ and ψ are well-formed formulas and $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$, then $(\varphi \circ \psi)$ is a well-formed formula;
4. if φ is a well-formed formula and x is a variable, then $\forall x\, \varphi$ and $\exists x\, \varphi$ are well-formed formulas;
5. no string is a well-formed formula unless it is generated by finitely many applications of the previous clauses.

Remark (Standard quantified statement).

$$\begin{aligned}\mathsf{Form}_{\mathcal{L}} = \bigcap\{S : \ & \mathsf{Atom}_{\mathcal{L}} \subseteq S,\\ & (\forall \varphi \in S)(\neg\varphi \in S),\\ & (\forall \varphi, \psi \in S)(\forall \circ \in \{\wedge, \vee, \to, \leftrightarrow\})((\varphi \circ \psi) \in S),\\ & (\forall x \in \mathsf{Var}_{\mathcal{L}})(\forall \varphi \in S)(\forall x\, \varphi \in S \wedge \exists x\, \varphi \in S)\}.\end{aligned}$$

Remark (Definition predicate reading). WellFormedFormula(φ) means $\varphi \in \mathsf{Form}_{\mathcal{L}}$.

Remark (Negated quantified statement).

$$\varphi \notin \mathsf{Form}_{\mathcal{L}} \iff \exists S \begin{bmatrix} \mathsf{Atom}_{\mathcal{L}} \subseteq S \wedge (\forall \psi \in S)(\neg \psi \in S) \\ \wedge (\forall \psi, \chi \in S)(\forall \circ \in \{\wedge, \vee, \to, \leftrightarrow\})((\psi \circ \chi) \in S) \\ \wedge (\forall x \in \mathsf{Var}_{\mathcal{L}})(\forall \psi \in S)(\forall x\, \psi \in S \wedge \exists x\, \psi \in S) \\ \wedge\ \ \varphi \notin S \end{bmatrix}.$$

Remark (Negation predicate reading). $\neg\,\mathrm{WellFormedFormula}(\varphi)$ means that φ is not generated by the formula-formation clauses.

Remark (Failure modes). A string fails to be well formed when it is a bare term, has a missing operand, uses a quantifier without a variable and formula, has unmatched punctuation, or cannot be produced by finitely many formation steps.

Remark (Failure mode decomposition).

$$\underbrace{f(x)}_{\text{term only}} \qquad \underbrace{P(x)\wedge}_{\text{missing right formula}} \qquad \underbrace{\forall P(x)}_{\text{missing bound variable}} \qquad \underbrace{(P(x)}_{\text{unmatched punctuation}}.$$

Remark (Interpretation). Well-formed formulas are exactly the grammatical formula strings of the language. Later semantic and proof-theoretic sections apply only to these strings.

Remark (Examples). $P(x)$, $\neg P(x)$, $(P(x) \to Q(x))$, and $\forall x\,(P(x) \to Q(x))$ are well-formed formulas.

Remark (Non-Examples). x, $f(x)$, $P(x)\wedge$, and $\forall P(x)$ are not well-formed formulas.

Remark (Dependencies).

- Formula

Definition (Molecular Formula)

Definition 2.3.9 (Molecular Formula)**.** A *molecular formula* is a well-formed formula that is not atomic. Equivalently, it is a well-formed formula whose final formation step uses a logical connective or a quantifier.

Remark (Standard quantified statement).

$$\varphi \in \mathsf{Form}_{\mathcal{L}} \setminus \mathsf{Atom}_{\mathcal{L}}.$$

Remark (Definition predicate reading).

$$\mathrm{MolecularFormula}(\varphi) \coloneqq \mathrm{WellFormedFormula}(\varphi) \wedge \neg\,\mathrm{AtomicFormula}(\varphi).$$

Remark (Negated quantified statement).

$$\varphi \notin \mathsf{Form}_{\mathcal{L}} \quad \text{or} \quad \varphi \in \mathsf{Atom}_{\mathcal{L}}.$$

Remark (Negation predicate reading). $\neg\,\mathrm{MolecularFormula}(\varphi)$ means that φ is either not a well-formed formula or is atomic.

Remark (Failure modes). A string fails to be a molecular formula either because it is not a well-formed formula at all or because it is an atomic formula with no outer logical construction.

Remark (Failure mode decomposition).

$$\underbrace{x}_{\text{not a formula}} \qquad \underbrace{P(x)}_{\text{atomic formula}} \qquad \underbrace{P(x)\wedge}_{\text{not well formed}} \quad .$$

Remark (Interpretation). Molecular formulas are the compound formulas of first-order syntax. They are formed after the atomic stage by applying a connective or quantifier.

Remark (Examples). $\neg P(x)$, $(P(x) \wedge Q(x))$, and $\forall x\, P(x)$ are molecular formulas.

Remark (Non-Examples). $P(x)$ is atomic rather than molecular, and $f(x)$ is a term rather than a formula.

Remark (Dependencies).

- Well-Formed Formula
- Atomic Formula

Variables and Substitution — Quick Reference

Concept	Meaning
Scope of a quantifier	Formula governed by the quantifier
Bound/free occurrence	Whether x is captured by a quantifier
Free variable $\mathrm{FV}(\varphi)$	Variables whose values affect truth
Sentence	Formula with $\mathrm{FV}(\varphi) = \varnothing$
Substitution $\varphi[t/x]$	Replace free x by term t
Free for substitution	No variable in t becomes bound
Alpha-equivalence	Renaming bound variables

2.3.2 Syntax: Free Variables, Bound Variables, and Substitution

Definition (Scope of a Quantifier)

Definition 2.3.10 (Scope of a Quantifier)**.** Let φ be a formula of a first-order language. If φ is of the form $\forall x\, \psi$ or $\exists x\, \psi$, then the formula ψ is called the *scope* of the quantifier. The quantifier is said to *bind* all occurrences of the variable x that appear within its scope.

Remark (Dependencies).

- Formula
- Variable
- Universal Quantifier

- Existential Quantifier

Remark (English reading). The scope is the reach of a quantifier — the subformula it governs. Scope is syntactically determined by the parenthesisation of the formula, not by proximity to the quantifier symbol.

Example. In $(\forall x\, P(x)) \wedge Q(x)$, the scope of $\forall x$ is $P(x)$ only. The occurrence of x in $Q(x)$ falls outside the scope and is free.

Definition (Bound and Free Occurrences)

Definition 2.3.11 (Bound and Free Occurrences)**.** An occurrence of a variable x in a formula φ is *bound* if it lies within the scope of a quantifier $\forall x$ or $\exists x$.
An occurrence of x is *free* if it is not bound by any quantifier in φ.

Remark (Dependencies).

- Formula
- Variable
- Universal Quantifier
- Existential Quantifier
- Scope of a Quantifier

Remark (English reading). The same variable can have both bound and free occurrences in a single formula. Each occurrence is classified independently by checking whether it falls within the scope of a binding quantifier.

Definition (Free Variables of a Formula)

Definition 2.3.12 (Free Variables of a Formula)**.** The set $\mathrm{FV}(\varphi)$ of *free variables* of a formula φ is defined recursively:

1. If $\varphi = P(t_1, \ldots, t_n)$ is atomic, then $\mathrm{FV}(\varphi) = \bigcup_{i=1}^{n} \mathrm{Var}(t_i)$, where $\mathrm{Var}(t_i)$ is the set of variables in term t_i.
2. $\mathrm{FV}(\neg\varphi) = \mathrm{FV}(\varphi)$.
3. $\mathrm{FV}(\varphi \circ \psi) = \mathrm{FV}(\varphi) \cup \mathrm{FV}(\psi)$ for any binary connective $\circ$.
4. $\mathrm{FV}(\forall x\, \varphi) = \mathrm{FV}(\varphi) \setminus \{x\}$.
5. $\mathrm{FV}(\exists x\, \varphi) = \mathrm{FV}(\varphi) \setminus \{x\}$.

Remark (Dependencies).

- Bound and Free Occurrences

- Atomic Formula
- Formula
- Term
- Variable
- Union
- Set Difference

Remark (English reading). $\mathrm{FV}(\varphi)$ collects exactly those variables whose values can influence the truth of φ under a structure. Quantifying over x removes x from the free set because the quantifier takes responsibility for ranging over all (or some) values of x.

Remark (Common error). Terms themselves contain only free variables. Variables become bound only through quantification in formulas. The interpretation of a formula depends exactly on the values assigned to its free variables.

Definition (Sentence)

Definition 2.3.13 (Sentence)**.** A *sentence* (or *closed formula*) is a formula with no free variables.
Formally, φ is a sentence if and only if $\mathrm{FV}(\varphi) = \varnothing$.

Remark (Dependencies).

- Formula
- Free Variables of a Formula
- Empty Set

Remark (Consequence). For sentences, truth depends only on the structure, not on the variable assignment. This makes sentences the natural objects to be called true or false in a model, without qualification.

Definition (Substitution Notation)

Definition 2.3.14 (Substitution Notation)**.** Let φ be a formula, x a variable, and t a term.
The expression $\varphi[t/x]$ denotes the formula obtained from φ by replacing every *free* occurrence of x with the term t, leaving bound occurrences of x unchanged.

Remark (Dependencies).

- Variable
- Term
- Formula
- Bound and Free Occurrences

Definition (Free for Substitution)

Definition 2.3.15 (Free for Substitution)**.** A term t is *free for* x in φ if no free occurrence of x in φ lies within the scope of a quantifier $\forall y$ or $\exists y$ where y is a variable occurring in t.
Equivalently, t is free for x in φ if the substitution $\varphi[t/x]$ does not result in any variable in t becoming bound.

Remark (Standard quantified statement). The relation "t is free for x in φ" is defined by structural induction on formulas. It is automatic for atomic formulas. It is preserved through Boolean connectives:

$$t \text{ free for } x \text{ in } \neg\varphi \iff t \text{ free for } x \text{ in } \varphi,$$

and similarly componentwise for $\varphi \wedge \psi$, $\varphi \vee \psi$, and $\varphi \to \psi$. For a quantified formula, the capture-avoidance condition is

$$t \text{ free for } x \text{ in } Qy\,\varphi \iff \big(x \text{ is not free in } Qy\,\varphi\big) \text{ or } \big(y \notin \mathrm{Var}(t) \text{ and } t \text{ is free for } x \text{ in } \varphi\big),$$

where Q is either $\forall$ or $\exists$.

Remark (Dependencies).

- Variable
- Term
- Formula
- Bound and Free Occurrences
- Free Variables of a Formula

Remark (Capture-avoiding substitution). A substitution $\varphi[t/x]$ is admissible only if t is free for x in φ. If this condition is violated, a variable occurring in t may become bound after substitution, silently changing the meaning of the formula. This is called *variable capture*.

Example (Variable capture). In the formula $\exists y\,(x < y)$, the term $y + 1$ is not free for x. A blind substitution would produce $\exists y\,(y + 1 < y)$, where the y introduced by the term has been captured by the existing quantifier.

Remark (Common error). Variable capture is one of the most common syntactic mistakes in predicate logic. Always check that the term being substituted introduces no variables that fall inside a binding quantifier in the target formula.

Definition (Alpha-Equivalence)

Definition 2.3.16 (Alpha-Equivalence)**.** Formulas that differ only by the names of bound variables are logically equivalent. This is called *alpha-equivalence* (or *alphabetic variance*):

$$\forall x\,\varphi \;\equiv\; \forall y\,\varphi[y/x] \quad \text{(provided } y \text{ is not free in } \varphi\text{).}$$

Remark (Dependencies).

- Formula
- Bound and Free Occurrences
- Substitution Notation
- Free for Substitution

Remark (Consequence). Alpha-equivalence is the formal basis for the bound variable renaming used in prenex normal form conversion and in any proof where variable capture must be avoided. Two alpha-equivalent formulas are interchangeable in all contexts.

Formula Depth — Quick Reference

Concept	Meaning
Formula depth	Max formation steps from atomic formulas

2.3.3 Syntax: Formula Depth

Definition (Formula Depth)

Definition 2.3.17 (Formula Depth)**.** The *depth* (or *complexity*) of a formula φ, denoted $\mathrm{depth}(\varphi)$, is a natural number defined recursively:

1. **Atomic case.** If φ is atomic, then $\mathrm{depth}(\varphi) = 0$.
2. **Negation.** If $\varphi = \neg\psi$, then $\mathrm{depth}(\varphi) = \mathrm{depth}(\psi) + 1$.
3. **Binary connectives.** If $\varphi = (\psi \circ \chi)$ for $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then
$$\mathrm{depth}(\varphi) = \max\{\mathrm{depth}(\psi), \mathrm{depth}(\chi)\} + 1.$$
4. **Quantifiers.** If $\varphi = \forall x\, \psi$ or $\varphi = \exists x\, \psi$, then $\mathrm{depth}(\varphi) = \mathrm{depth}(\psi) + 1$.

Remark (Dependencies).

- Formula
- Natural Numbers
- Universal Quantifier
- Existential Quantifier

Remark (English reading)*.* The depth of a formula measures the maximum number of logical formation steps required to build the formula from atomic formulas. It corresponds to the height of the formula's syntactic parse tree.

Remark (Proof strategy)*.* Formula depth is the standard measure used in structural induction proofs over first-order formulas. The base case is depth 0 (atomic formulas); the inductive step handles each formation rule and assumes the result holds for sub-formulas of strictly smaller depth.

2.4 Semantics

2.4.1 Semantics: Structures and Variable Assignments

Remark (Section toolkit: Semantic setup)*.* This section is the first point at which semantic objects enter the Predicate Logic chapter. A language supplies the symbols; a structure supplies a domain and interpretations for the non-logical symbols; an assignment supplies values for variables; term evaluation turns terms into domain elements.

Item	Notation	Role
First-order language	$\mathcal{L}$	symbol stock with arities
Structure	$\mathcal{M} = \langle D, I \rangle$	semantic object for $\mathcal{L}$
Variable assignment	$s : \mathsf{Var}_{\mathcal{L}} \to D$	values for variables
Modified assignment	$s[x \mapsto d]$	update at one variable
Term evaluation	$[\![t]\!]_{\mathcal{M},s}$	value of a term

Definition (First-Order Language)

Definition 2.4.1 (First-Order Language)**.** A *first-order language* $\mathcal{L}$ consists of variables, logical symbols, and a signature of non-logical symbols: constant symbols, function symbols with assigned arities, and predicate symbols with assigned arities. Equality may be included as a distinguished logical predicate.

Remark (Standard quantified statement)*.*

$$\text{A first-order language } \mathcal{L} \text{ has symbol classes}$$
$$\mathsf{Var}_{\mathcal{L}}, \quad \mathsf{Const}_{\mathcal{L}}, \quad \mathsf{Func}^{(n)}_{\mathcal{L}}, \quad \mathsf{Pred}^{(n)}_{\mathcal{L}} \quad (n \geq 1),$$

together with the usual logical connectives, quantifiers, and punctuation.

Remark (Definition predicate reading)*.* $\mathrm{FirstOrderLanguage}(L)$ means that L supplies the variable stock, logical symbols, and arity-indexed non-logical symbols used to form first-order terms and formulas.

Remark (Negated quantified statement)*.*

$$L \text{ is not equipped with the required first-order symbol classes and arities.}$$

Remark (Negation predicate reading)*.* $\neg\,\mathrm{FirstOrderLanguage}(L)$ means that L lacks some required syntactic stock or arity assignment.

Remark (Failure modes)*.* A proposed language fails when its non-logical symbols are not sorted by role, when positive-arity symbols have no arity, or when the logical grammar is not specified.

Remark (Failure mode decomposition)*.*

$$\underbrace{\mathsf{Const}_L}_{\text{constant symbols}} \qquad \underbrace{\mathsf{Func}^{(n)}_L}_{\text{function symbols with arity}} \qquad \underbrace{\mathsf{Pred}^{(n)}_L}_{\text{predicate symbols with arity}} \qquad .$$

Remark (Interpretation)*.* A first-order language is still syntax. It says which expressions can be formed, but it does not assign objects, functions, or relations to the non-logical symbols.

Remark (Examples)*.* The arithmetic language with constant symbols $0, 1$, function symbols $S, +, \cdot$, and predicate symbol $<$ is a first-order language once the arities are fixed.

Remark (Dependencies)*.*

- Variable

- Constant Symbol
- Function Symbol
- Predicate Symbol

Definition (Structure)

Definition 2.4.2 (Structure)**.** Let $\mathcal{L}$ be a first-order language. A *structure* for $\mathcal{L}$ is a pair

$$\mathcal{M} = \langle D, I \rangle$$

where D is a nonempty set and I assigns each constant symbol an element of D, each n-ary function symbol a function $D^n \to D$, and each n-ary predicate symbol a relation on D^n.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } \mathcal{L} \text{ be a first-order language.}\\
&\mathcal{M} = \langle D, I \rangle \text{ is a structure for } \mathcal{L}\\
&\iff D \neq \varnothing,\ I(c) \in D,\ I(f) : D^n \to D,\ I(P) \subseteq D^n
\end{aligned}$$

for each $c \in \mathsf{Const}_{\mathcal{L}}$, $f \in \mathsf{Func}_{\mathcal{L}}^{(n)}$, and $P \in \mathsf{Pred}_{\mathcal{L}}^{(n)}$.

Remark (Definition predicate reading). Structure(M, L) means that M is a structure for the first-order language L.

Remark (Negated quantified statement).

$$D = \varnothing \quad \text{or} \quad I \text{ fails to assign the required object, function, or relation to some symbol.}$$

Remark (Negation predicate reading). $\neg$ Structure(M, L) means that M is not a well-formed structure for L.

Remark (Failure modes). A proposed structure fails if the domain is empty, if a symbol is left uninterpreted, or if the assigned object has the wrong arity or type.

Remark (Failure mode decomposition).

$$\underbrace{D = \varnothing}_{\text{empty domain}} \quad \vee \quad \underbrace{I(c) \notin D}_{\text{bad constant value}} \quad \vee \quad \underbrace{I(f) : D^n \not\to D}_{\text{bad function arity}} \quad \vee \quad \underbrace{I(P) \not\subseteq D^n}_{\text{bad predicate relation}}.$$

Remark (Interpretation). A structure is the minimal semantic object needed in Volume I. It gives meaning to the non-logical symbols of a language. Systematic topics such as substructures, expansions, reducts, and elementary classes belong to Volume VIII Model Theory.

Remark (Examples). The natural numbers with their usual interpretations of $0, S, +, \cdot, <$ form a structure for the arithmetic language.

Remark (Dependencies).

- First-Order Language

Definition (Variable Assignment)

Definition 2.4.3 (Variable Assignment)**.** Let $\mathcal{M} = \langle D, I\rangle$ be a structure for $\mathcal{L}$. A *variable assignment* for $\mathcal{M}$ is a function

$$s : \mathsf{Var}_{\mathcal{L}} \to D.$$

Remark (Standard quantified statement)*.*

$$s \text{ is a variable assignment for } \mathcal{M} \iff \operatorname{dom}(s) = \mathsf{Var}_{\mathcal{L}} \text{ and } \operatorname{ran}(s) \subseteq D.$$

Remark (Definition predicate reading)*.* VariableAssignment(s, M) means that s assigns a domain element of M to each variable of the language of M.

Remark (Negated quantified statement)*.*

$$\operatorname{dom}(s) \neq \mathsf{Var}_{\mathcal{L}} \quad \text{or} \quad \exists x \in \mathsf{Var}_{\mathcal{L}}\ (s(x) \notin D).$$

Remark (Negation predicate reading)*.* $\neg$ VariableAssignment(s, M) means that s is not a function assigning each variable a domain element of M.

Remark (Failure modes)*.* An assignment fails when it leaves a variable without a value or assigns a value outside the domain.

Remark (Failure mode decomposition)*.*

$$\underbrace{\operatorname{dom}(s) \neq \mathsf{Var}_{\mathcal{L}}}_{\text{missing variable value}} \quad \vee \quad \underbrace{s(x) \notin D}_{\text{value outside domain}}.$$

Remark (Interpretation)*.* Structures interpret non-logical symbols; assignments interpret variables. Open formulas require assignments, while sentences eventually do not depend on the particular assignment.

Remark (Dependencies)*.*

- Structure

Definition (Modified Assignment)

Definition 2.4.4 (Modified Assignment)**.** Let $s : \mathsf{Var}_{\mathcal{L}} \to D$, let $x \in \mathsf{Var}_{\mathcal{L}}$, and let $d \in D$. The *modified assignment* $s[x \mapsto d]$ is the assignment defined by

$$s[x \mapsto d](y) = \begin{cases} d, & y = x, \\ s(y), & y \neq x. \end{cases}$$

Remark (Standard quantified statement)*.*

$$\forall y \in \mathsf{Var}_{\mathcal{L}} \quad s[x \mapsto d](y) = d \iff y = x, \qquad s[x \mapsto d](y) = s(y) \iff y \neq x.$$

Remark (Definition predicate reading)*.* ModifiedAssignment(s, x, d, s') means that s' is the assignment obtained from s by sending x to d and leaving every other variable unchanged.

Remark (Negated quantified statement).

$$s'(x) \neq d \quad \text{or} \quad \exists y \in \mathsf{Var}_{\mathcal{L}} \ (y \neq x \wedge s'(y) \neq s(y)).$$

Remark (Negation predicate reading). $\neg$ ModifiedAssignment(s, x, d, s') means that s' is not the one-variable update of s at x with value d.

Remark (Failure modes). A proposed modified assignment fails if it does not change x to d, or if it changes some variable other than x.

Remark (Failure mode decomposition).

$$\underbrace{s'(x) \neq d}_{\text{target variable not updated}} \quad \vee \quad \underbrace{\exists y \neq x \ (s'(y) \neq s(y))}_{\text{unwanted extra change}}.$$

Remark (Interpretation). Modified assignments are the semantic mechanism behind quantifiers. To test a quantified formula, the assignment is varied at the bound variable while all other variable values remain fixed.

Remark (Dependencies).

- Variable Assignment

Definition (Term Evaluation)

Definition 2.4.5 (Term Evaluation)**.** Let $\mathcal{M} = \langle D, I \rangle$ be a structure and s a variable assignment. The *value* of a term t, denoted $[\![t]\!]_{\mathcal{M},s}$, is defined recursively by

$$[\![x]\!]_{\mathcal{M},s} = s(x), \qquad [\![c]\!]_{\mathcal{M},s} = I(c),$$

and

$$[\![f(t_1, \ldots, t_n)]\!]_{\mathcal{M},s} = I(f)\big([\![t_1]\!]_{\mathcal{M},s}, \ldots, [\![t_n]\!]_{\mathcal{M},s}\big).$$

Remark (Standard quantified statement).

$$\begin{aligned} &[\![x]\!]_{\mathcal{M},s} = s(x), \\ &[\![c]\!]_{\mathcal{M},s} = I(c), \\ &[\![f(t_1, \ldots, t_n)]\!]_{\mathcal{M},s} = I(f)([\![t_1]\!]_{\mathcal{M},s}, \ldots, [\![t_n]\!]_{\mathcal{M},s}). \end{aligned}$$

Remark (Definition predicate reading). TermEvaluation(M, s, t, a) means that the term t evaluates to the domain element a in M under s.

Remark (Negated quantified statement).

$$[\![t]\!]_{\mathcal{M},s} \neq a.$$

Remark (Negation predicate reading). $\neg$ TermEvaluation(M, s, t, a) means that a is not the value of t in M under s.

Remark (Failure modes). Term evaluation fails to have the proposed value when a variable, constant, or function-application clause computes a different domain element.

Remark (Failure mode decomposition).

$$\underbrace{s(x) \neq a}_{\text{variable case}} \quad \vee \quad \underbrace{I(c) \neq a}_{\text{constant case}} \quad \vee \quad \underbrace{I(f)(a_1, \ldots, a_n) \neq a}_{\text{function case}}.$$

Remark (Interpretation). Term evaluation is the semantic counterpart of term formation. It assigns a domain element to each grammatical term once a structure and assignment are fixed.

Remark (Dependencies).

- Term
- Structure
- Variable Assignment

2.4.2 Semantics: Satisfaction and Truth

Remark (Section toolkit: Satisfaction ledger). Satisfaction is the recursive semantic relation for formulas. Formulas with free variables are evaluated relative to an assignment. Sentences have no free variables, so their truth in a structure does not depend on which assignment is chosen.

Definition (Satisfaction)

Definition 2.4.6 (Satisfaction)**.** Let $\mathcal{M} = \langle D, I \rangle$ be a structure, let s be a variable assignment, and let φ be a well-formed formula. The relation $\mathcal{M}, s \models \varphi$ is defined recursively by the usual clauses: atomic formulas are evaluated by term values and interpreted predicate relations; equality compares term values; connectives follow the truth conditions for $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$; and

$$\mathcal{M}, s \models \forall x\, \psi \iff \forall d \in D\ (\mathcal{M}, s[x \mapsto d] \models \psi),$$

$$\mathcal{M}, s \models \exists x\, \psi \iff \exists d \in D\ (\mathcal{M}, s[x \mapsto d] \models \psi).$$

Remark (Standard quantified statement).

$$\begin{aligned}
&\mathcal{M}, s \models P(t_1, \ldots, t_n) \iff ([\![t_1]\!]_{\mathcal{M},s}, \ldots, [\![t_n]\!]_{\mathcal{M},s}) \in I(P),\\
&\mathcal{M}, s \models t_1 = t_2 \iff [\![t_1]\!]_{\mathcal{M},s} = [\![t_2]\!]_{\mathcal{M},s},\\
&\mathcal{M}, s \models \neg\psi \iff \mathcal{M}, s \not\models \psi,\\
&\mathcal{M}, s \models (\psi \wedge \chi) \iff \mathcal{M}, s \models \psi \text{ and } \mathcal{M}, s \models \chi,\\
&\mathcal{M}, s \models \forall x\, \psi \iff \forall d \in D\ (\mathcal{M}, s[x \mapsto d] \models \psi),\\
&\mathcal{M}, s \models \exists x\, \psi \iff \exists d \in D\ (\mathcal{M}, s[x \mapsto d] \models \psi).
\end{aligned}$$

Remark (Definition predicate reading). Satisfaction(M, s, φ) means $\mathcal{M}, s \models \varphi$.

Remark (Negated quantified statement).

$$\mathcal{M}, s \not\models \varphi.$$

Remark (Negation predicate reading). $\neg\,\text{Satisfaction}(M, s, \varphi)$ means that φ is not satisfied in M under s.

Remark (Failure modes). Satisfaction fails according to the outermost formation rule of the formula: an atomic tuple can miss the interpreted relation, equality can compare different term values, a connective can have the wrong truth pattern, or a quantifier can have a counterexample or lack a witness.

Remark (Failure mode decomposition).

$$\underbrace{(\llbracket t_i \rrbracket) \notin I(P)}_{\text{atomic failure}} \quad \underbrace{\llbracket t_1 \rrbracket \neq \llbracket t_2 \rrbracket}_{\text{equality failure}} \quad \underbrace{\exists d \in D\ (\mathcal{M}, s[x \mapsto d] \not\models \psi)}_{\text{universal failure}}.$$

Remark (Interpretation). Satisfaction is the bridge from grammar to semantic use. It is recursive because formulas are recursive. The assignment is essential exactly where free variables can affect the value of the formula.

Remark (Dependencies).

- Structure
- Variable Assignment
- Modified Assignment
- Term Evaluation
- Well-Formed Formula

Definition (Truth in a Structure)

Definition 2.4.7 (Truth in a Structure)**.** Let φ be a sentence and let $\mathcal{M}$ be a structure. The sentence φ is *true in* $\mathcal{M}$, written $\mathcal{M} \models \varphi$, exactly when $\mathcal{M}, s \models \varphi$ for every variable assignment s for $\mathcal{M}$.

Remark (Standard quantified statement).

$$\mathcal{M} \models \varphi \iff \forall s : \mathsf{Var}_{\mathcal{L}} \to D\ (\mathcal{M}, s \models \varphi), \qquad \varphi \text{ a sentence.}$$

Remark (Definition predicate reading). $\text{TruthInStructure}(M, \varphi)$ means $\mathcal{M} \models \varphi$.

Remark (Negated quantified statement).

$$\mathcal{M} \not\models \varphi \iff \exists s : \mathsf{Var}_{\mathcal{L}} \to D\ (\mathcal{M}, s \not\models \varphi).$$

Remark (Negation predicate reading). $\neg\,\text{TruthInStructure}(M, \varphi)$ means that φ is false in M.

Remark (Failure modes). Truth in a structure fails when some assignment makes the sentence fail. For sentences, this is independent of assignment, but the quantified form keeps the definition uniform with satisfaction.

Remark (Failure mode decomposition).

$$\underbrace{\exists s}_{\text{assignment witness}} \quad \underbrace{\mathcal{M}, s \not\models \varphi}_{\text{satisfaction failure}} .$$

Remark (Interpretation). Sentences have no free variables. Therefore the particular assignment does not carry mathematical information once the formula is a sentence; the notation $\mathcal{M} \models \varphi$ suppresses it.

Remark (Dependencies).

- Satisfaction

Definition (Validity)

Definition 2.4.8 (Validity)**.** A formula φ is *valid*, or *logically true*, exactly when $\mathcal{M}, s \models \varphi$ for every structure $\mathcal{M}$ for the language of φ and every variable assignment s for $\mathcal{M}$.

Remark (Standard quantified statement).

$$\models \varphi \iff \forall \mathcal{M}\ \forall s\ (\mathcal{M}, s \models \varphi).$$

Remark (Definition predicate reading). ValidFormula(φ) means that φ is satisfied in every structure under every assignment.

Remark (Negated quantified statement).

$$\not\models \varphi \iff \exists \mathcal{M}\ \exists s\ (\mathcal{M}, s \not\models \varphi).$$

Remark (Negation predicate reading). $\neg$ ValidFormula(φ) means that some structure and assignment make φ fail.

Remark (Failure modes). Validity fails by a counterexample structure together with an assignment.

Remark (Failure mode decomposition).

$$\underbrace{\mathcal{M}}_{\text{counterexample structure}} \quad \underbrace{s}_{\text{counterexample assignment}} \quad \underbrace{\mathcal{M}, s \not\models \varphi}_{\text{failure of satisfaction}} .$$

Remark (Interpretation). Validity is truth by logic alone. It does not depend on a special structure, special domain, or special interpretation of the non-logical symbols.

Remark (Dependencies).

- Satisfaction

Definition (Satisfiability)

Definition 2.4.9 (Satisfiability)**.** A formula φ is *satisfiable* exactly when there are a structure $\mathcal{M}$ and a variable assignment s such that $\mathcal{M}, s \models \varphi$.

Remark (Standard quantified statement)*.*

$$\varphi \text{ is satisfiable} \iff \exists \mathcal{M}\ \exists s\ (\mathcal{M}, s \models \varphi).$$

Remark (Definition predicate reading)*.* SatisfiableFormula(φ) means that φ is satisfied in at least one structure under at least one assignment.

Remark (Negated quantified statement)*.*

$$\varphi \text{ is unsatisfiable} \iff \forall \mathcal{M}\ \forall s\ (\mathcal{M}, s \not\models \varphi).$$

Remark (Negation predicate reading)*.* $\neg$SatisfiableFormula(φ) means that no structure and assignment satisfy φ.

Remark (Failure modes)*.* Satisfiability fails only globally: every attempted structure and assignment fails to satisfy the formula.

Remark (Failure mode decomposition)*.*

$$\underbrace{\forall \mathcal{M}}_{\text{all structures}} \quad \underbrace{\forall s}_{\text{all assignments}} \quad \underbrace{\mathcal{M}, s \not\models \varphi}_{\text{no satisfying instance}} .$$

Remark (Interpretation)*.* Satisfiability asks whether a formula can be made true somewhere. This is the semantic counterpart of consistency only in the minimal Volume I sense; deeper proof-theoretic distinctions are deferred.

Remark (Dependencies)*.*

- Satisfaction

2.4.3 Semantics: Models, Theories, and Logical Consequence

Remark (Section toolkit: Minimal model notation)*.* Volume I uses theories only as sets of sentences so that the notation $\mathcal{M} \models T$ and $T \models \varphi$ can be read. The systematic study of theories and their classes of models belongs to Volume VIII Model Theory.

Definition (Model of a Sentence)

Definition 2.4.10 (Model of a Sentence)**.** Let φ be a sentence and let $\mathcal{M}$ be a structure for the language of φ. The structure $\mathcal{M}$ is a *model of* φ exactly when $\mathcal{M} \models \varphi$.

Remark (Standard quantified statement)*.*

$$\mathcal{M} \text{ is a model of } \varphi \iff \mathcal{M} \models \varphi.$$

Remark (Definition predicate reading). ModelOfSentence(M, φ) means that M is a model of the sentence φ.

Remark (Negated quantified statement).

$$\mathcal{M} \text{ is not a model of } \varphi \iff \mathcal{M} \not\models \varphi.$$

Remark (Negation predicate reading). $\neg$ ModelOfSentence(M, φ) means that φ is not true in M.

Remark (Failure modes). The only failure mode is semantic: the sentence is false in the proposed structure.

Remark (Failure mode decomposition).

$$\underbrace{\mathcal{M}}_{\text{proposed structure}} \qquad \underbrace{\mathcal{M} \not\models \varphi}_{\text{sentence false in the structure}}.$$

Remark (Interpretation). A model is not an extra kind of object. It is a structure viewed relative to a sentence that it makes true.

Remark (Dependencies).

- Truth in a Structure

Definition (Model of a Set of Sentences)

Definition 2.4.11 (Model of a Set of Sentences)**.** Let T be a set of sentences and let $\mathcal{M}$ be a structure for their common language. The structure $\mathcal{M}$ is a *model of* T, written $\mathcal{M} \models T$, exactly when $\mathcal{M} \models \theta$ for every $\theta \in T$.

Remark (Standard quantified statement).

$$\mathcal{M} \models T \iff \forall \theta \in T \ (\mathcal{M} \models \theta).$$

Remark (Definition predicate reading). ModelOfTheory(M, T) means that M is a model of every sentence in T.

Remark (Negated quantified statement).

$$\mathcal{M} \not\models T \iff \exists \theta \in T \ (\mathcal{M} \not\models \theta).$$

Remark (Negation predicate reading). $\neg$ ModelOfTheory(M, T) means that at least one sentence of T is false in M.

Remark (Failure modes). A proposed model of a set of sentences fails by a single sentence of the set that is false in the structure.

Remark (Failure mode decomposition).

$$\underbrace{\exists \theta \in T}_{\text{failed sentence}} \qquad \underbrace{\mathcal{M} \not\models \theta}_{\text{not true in the structure}}.$$

Remark (Interpretation). The notation $\mathcal{M} \models T$ abbreviates a universal check over the sentences in T. No additional theory of model classes is developed here.

Remark (Dependencies).

- Model of a Sentence

Definition (Theory)

Definition 2.4.12 (Theory)**.** In Volume I, a *theory* is a set of sentences in a fixed first-order language.

Remark (Standard quantified statement).

$$T \text{ is a theory} \iff \forall\theta \in T\ (\theta \text{ is a sentence of the fixed language}).$$

Remark (Definition predicate reading). Theory(T) means that T is a set of sentences in a fixed first-order language.

Remark (Negated quantified statement).

$$\exists\theta \in T\ (\theta \text{ is not a sentence of the fixed language}).$$

Remark (Negation predicate reading). $\neg$ Theory(T) means that T is not a set consisting only of sentences of the fixed language.

Remark (Failure modes). A proposed theory fails in this minimal sense when it contains something other than a sentence of the fixed language.

Remark (Failure mode decomposition).

$$\underbrace{\theta \in T}_{\text{member of the proposed theory}} \qquad \underbrace{\theta \text{ is not a sentence}}_{\text{bad member}}.$$

Remark (Interpretation). This is intentionally modest. Complete theories, elementary classes, categoricity, saturation, types, diagrams, and elementary extensions are Volume VIII topics. A theory may be called satisfiable in Volume I when it has at least one model.

Remark (Dependencies).

- Truth in a Structure

Definition (Logical Consequence)

Definition 2.4.13 (Logical Consequence)**.** Let T be a set of sentences and let φ be a sentence in the same language. The sentence φ is a *logical consequence* of T, written $T \models \varphi$, exactly when every model of T is a model of φ.

Remark (Standard quantified statement).

$$T \models \varphi \iff \forall\mathcal{M}\ (\mathcal{M} \models T \Rightarrow \mathcal{M} \models \varphi).$$

Remark (Definition predicate reading). LogicalConsequence(T, φ) means that every model of T is a model of φ.

Remark (Negated quantified statement).

$$T \not\models \varphi \iff \exists \mathcal{M}\ (\mathcal{M} \models T \wedge \mathcal{M} \not\models \varphi).$$

Remark (Negation predicate reading). $\neg$ LogicalConsequence(T, φ) means that some structure models T while failing φ.

Remark (Failure modes). Logical consequence fails by a countermodel: a structure satisfying all sentences of T but not satisfying φ.

Remark (Failure mode decomposition).

$$\underbrace{\mathcal{M} \models T}_{\text{premises true}} \qquad \underbrace{\mathcal{M} \not\models \varphi}_{\text{conclusion false}}.$$

Remark (Interpretation). Logical consequence is semantic entailment. The premises in T leave no room, across structures, for φ to fail.

Remark (Dependencies).

- Model of a Set of Sentences
- Theory

Definition (Logical Equivalence)

Definition 2.4.14 (Logical Equivalence)**.** Two sentences φ and ψ are *logically equivalent*, written $\varphi \equiv \psi$, exactly when each is a logical consequence of the other:

$$\varphi \models \psi \quad \text{and} \quad \psi \models \varphi.$$

Remark (Standard quantified statement).

$$\varphi \equiv \psi \iff \forall \mathcal{M}\ (\mathcal{M} \models \varphi \iff \mathcal{M} \models \psi).$$

Remark (Definition predicate reading). LogicalEquivalence(φ, ψ) means that φ and ψ have the same truth value in every structure.

Remark (Negated quantified statement).

$$\varphi \not\equiv \psi \iff \exists \mathcal{M}\ \big((\mathcal{M} \models \varphi \wedge \mathcal{M} \not\models \psi) \vee (\mathcal{M} \models \psi \wedge \mathcal{M} \not\models \varphi)\big).$$

Remark (Negation predicate reading). $\neg$ LogicalEquivalence(φ, ψ) means that some structure separates the two sentences.

Remark (Failure modes). Logical equivalence fails when one sentence is true and the other false in the same structure.

Remark (Failure mode decomposition).

$$\underbrace{\mathcal{M} \models \varphi \wedge \mathcal{M} \not\models \psi}_{\text{first separation}} \quad \vee \quad \underbrace{\mathcal{M} \models \psi \wedge \mathcal{M} \not\models \varphi}_{\text{second separation}}.$$

Remark (Interpretation). Logical equivalence is semantic interchangeability at the sentence level. It records sameness of truth across all structures in the relevant language.

Remark (Dependencies).

- Logical Consequence

2.4.4 Semantics: Open Formulas, Definable Relations, and Substitution Lemmas

Remark (Section toolkit: Predicate vocabulary). This section separates three uses that are often blurred. A predicate symbol is object-language syntax from the signature. An open formula is a formula with free variables. A definable relation is the set-theoretic relation in a structure determined by an open formula.

Definition (Open Formula)

Definition 2.4.15 (Open Formula). An *open formula* is a well-formed formula with at least one free variable. When the free variables are among $x_1, \ldots, x_n$, it may be displayed as $\varphi(x_1, \ldots, x_n)$.

Remark (Standard quantified statement).

$$\varphi \text{ is open} \iff \varphi \in \mathsf{Form}_{\mathcal{L}} \text{ and } \mathsf{FV}(\varphi) \neq \varnothing.$$

Remark (Definition predicate reading). OpenFormula(φ) means that φ is a well-formed formula with at least one free variable.

Remark (Negated quantified statement).

$$\varphi \notin \mathsf{Form}_{\mathcal{L}} \quad \text{or} \quad \mathsf{FV}(\varphi) = \varnothing.$$

Remark (Negation predicate reading). $\neg$ OpenFormula(φ) means that φ is either not well formed or is a sentence.

Remark (Failure modes). An expression fails to be an open formula when it is not a well-formed formula or when all of its variables are bound.

Remark (Failure mode decomposition).

$$\underbrace{\varphi \notin \mathsf{Form}_{\mathcal{L}}}_{\text{not well formed}} \quad \vee \quad \underbrace{\mathsf{FV}(\varphi) = \varnothing}_{\text{sentence, not open}}.$$

Remark (Interpretation). Open formulas are syntactic predicates in the lightest Volume I sense. They do not by themselves name a set-theoretic relation until a structure is fixed.

Remark (Examples). $P(x)$, $x < y$, and $\exists z\, R(x, z)$ are open formulas when the displayed variables remain free as indicated.

Remark (Non-Examples). $\forall x\, P(x)$ is a sentence, not an open formula. A predicate symbol P alone is a symbol, not a formula.

Remark (Dependencies).

- Predicate Symbol
- Well-Formed Formula

Definition (Definable Relation)

Definition 2.4.16 (Definable Relation)**.** Let $\mathcal{M} = \langle D, I\rangle$ be a structure and let $\varphi(x_1, \ldots, x_n)$ be an open formula. The *relation defined by* φ in $\mathcal{M}$ is the set

$$R_\varphi^{\mathcal{M}} = \{(a_1, \ldots, a_n) \in D^n : \mathcal{M}, s[x_1 \mapsto a_1, \ldots, x_n \mapsto a_n] \models \varphi\}.$$

Remark (Standard quantified statement).

$$(a_1, \ldots, a_n) \in R_\varphi^{\mathcal{M}} \iff \mathcal{M}, s[x_1 \mapsto a_1, \ldots, x_n \mapsto a_n] \models \varphi.$$

Remark (Definition predicate reading). DefinableRelation(R, φ, M) means that R is the relation on the domain of M determined by φ in M.

Remark (Negated quantified statement).

$$R \neq R_\varphi^{\mathcal{M}}.$$

Remark (Negation predicate reading). $\neg$ DefinableRelation(R, φ, M) means that R is not the relation determined by φ in M.

Remark (Failure modes). A proposed definable relation fails when at least one tuple is included or excluded differently from the satisfaction condition imposed by the formula.

Remark (Failure mode decomposition).

$$\underbrace{(a_1, \ldots, a_n) \in R \wedge \mathcal{M}, s[\vec{x} \mapsto \vec{a}] \not\models \varphi}_{\text{extra tuple}} \quad \vee \quad \underbrace{(a_1, \ldots, a_n) \notin R \wedge \mathcal{M}, s[\vec{x} \mapsto \vec{a}] \models \varphi}_{\text{missing tuple}}.$$

Remark (Interpretation). A relation is a set-theoretic subset of D^n. A definable relation is one obtained from a formula after a structure supplies semantic meaning.

Remark (Dependencies).

- Open Formula
- Structure
- Satisfaction

Lemma (Substitution Lemma for Terms)

Lemma 2.4.17 (Substitution Lemma for Terms)**.** *Let $\mathcal{M}$ be a structure, let s be a variable assignment, let x be a variable, and let t and u be terms. Then*

$$[\![t[u/x]]\!]_{\mathcal{M},s} = [\![t]\!]_{\mathcal{M},\, s[x\mapsto [\![u]\!]_{\mathcal{M},s}]}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall t\, \forall u \quad [\![t[u/x]]\!]_{\mathcal{M},s} = [\![t]\!]_{\mathcal{M},\, s[x\mapsto [\![u]\!]_{\mathcal{M},s}]}.$$

Remark (Predicate reading)*.* Term substitution commutes with semantic term evaluation in a fixed structure and assignment.

Remark (Interpretation)*.* The lemma says that syntactically substituting a term before evaluation gives the same domain element as first evaluating the substituting term and then updating the assignment.

Remark (Dependencies)*.*

- Substitution Notation
- Free for Substitution
- Term Evaluation

Lemma (Substitution Lemma for Formulas)

Lemma 2.4.18 (Substitution Lemma for Formulas)**.** *Let φ be a formula and let u be a term free for x in φ. For every structure $\mathcal{M}$ and assignment s,*

$$\mathcal{M}, s \models \varphi[u/x] \iff \mathcal{M}, s[x \mapsto [\![u]\!]_{\mathcal{M},s}] \models \varphi.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall \varphi\, \forall u \quad u \text{ free for } x \text{ in } \varphi \Rightarrow \big(\mathcal{M}, s \models \varphi[u/x] \iff \mathcal{M}, s[x \mapsto [\![u]\!]_{\mathcal{M},s}] \models \varphi\big).$$

Remark (Predicate reading)*.* Formula substitution commutes with satisfaction when the substituted term is free for the variable being replaced.

Remark (Interpretation)*.* This lemma connects the syntactic substitution operation from the syntax section with the semantic modified-assignment operation from the semantics section. The free-for condition prevents variable capture.

Remark (Dependencies)*.*

- Substitution Notation

- Free for Substitution
- Satisfaction
- Substitution Lemma for Terms

2.5 Quantifiers

Toolkit: Quantifiers

Topic	Role
Basic syntax and semantics	universal, existential, bounded, sentence, open formula, scope
Quantifier laws	equivalence laws for negation, distribution, commutation, and renaming
Prenex normal form	syntactic normalization with quantifiers pulled forward
Logical strength	quantifier order, dependency, and non-equivalence examples

Remark (Exposition). Quantifiers extend propositional logic by allowing statements about all domain elements or about witnesses. The main new issue is dependency: later witnesses may depend on earlier universal choices.

Basic Syntax and Semantics

Remark (Section toolkit: Quantifier grammar). This section records the basic quantifier forms and the boundary between open formulas and sentences. Quantifiers bind variables. A formula with free variables requires an assignment for semantic evaluation; a sentence has no free variables and has a truth value in a structure.

Remark (Exposition). Sentences have no free variables and have truth values in structures. Open formulas may contain free variables and require assignments. Quantifiers turn some free occurrences into bound occurrences by placing them inside a scope.

Definition (Universal Formula)

Definition 2.5.1 (Universal Formula)**.** A *universal formula* is a well-formed formula of the form

$$\forall x\, \varphi,$$

where x is a variable and φ is a well-formed formula.

Remark (Standard quantified statement).

$$\forall x\, \varphi \in \mathsf{Form}_{\mathcal{L}} \iff x \in \mathsf{Var}_{\mathcal{L}} \text{ and } \varphi \in \mathsf{Form}_{\mathcal{L}}.$$

Remark (Definition predicate reading). UniversalFormula(Φ) means that Φ is a formula whose outermost constructor is a universal quantifier.

Remark (Negated quantified statement).

$$\Phi \text{ is not of the form } \forall x\, \varphi.$$

Remark (Interpretation). A universal formula asserts that its scope holds for every value assigned to the bound variable.

Remark (Examples). $\forall x\, P(x)$ and $\forall x\, \exists y\, R(x, y)$ are universal formulas.

Remark (Non-Examples). $P(x)$ and $\exists x\, P(x)$ are not universal formulas.

Remark (Dependencies).

- Variable

Definition (Existential Formula)

Definition 2.5.2 (Existential Formula)**.** An *existential formula* is a well-formed formula of the form

$$\exists x\, \varphi,$$

where x is a variable and φ is a well-formed formula.

Remark (Standard quantified statement).

$$\exists x\, \varphi \in \mathsf{Form}_{\mathcal{L}} \iff x \in \mathsf{Var}_{\mathcal{L}} \text{ and } \varphi \in \mathsf{Form}_{\mathcal{L}}.$$

Remark (Definition predicate reading). ExistentialFormula(Φ) means that Φ is a formula whose outermost constructor is an existential quantifier.

Remark (Negated quantified statement).

$$\Phi \text{ is not of the form } \exists x\, \varphi.$$

Remark (Interpretation). An existential formula asserts that its scope holds for at least one value of the bound variable.

Remark (Examples). $\exists x\, P(x)$ and $\exists x\, \forall y\, R(x, y)$ are existential formulas.

Remark (Non-Examples). $P(x)$ and $\forall x\, P(x)$ are not existential formulas.

Remark (Dependencies).

- Variable

Definition (Bounded Quantifier)

Definition 2.5.3 (Bounded Quantifier)**.** A *bounded quantifier* is an abbreviation that restricts a quantifier to a specified set or predicate:

$$\begin{aligned} \forall x \in A\, \varphi \quad &:= \quad \forall x\, (x \in A \to \varphi), \\ \exists x \in A\, \varphi \quad &:= \quad \exists x\, (x \in A \land \varphi). \end{aligned}$$

Remark (Standard quantified statement).

$$\forall x \in A\, \varphi \equiv \forall x\, (x \in A \to \varphi), \qquad \exists x \in A\, \varphi \equiv \exists x\, (x \in A \land \varphi).$$

Remark (Definition predicate reading). BoundedQuantifier(Φ) means that Φ uses a restricted universal or existential quantifier.

Remark (Negated quantified statement).

$$\Phi \text{ is not an abbreviation of either displayed bounded form.}$$

Remark (Interpretation). The restricted universal uses implication, while the restricted existential uses conjunction. This preserves the usual behavior on empty sets.

Remark (Examples). $\forall x \in A\, P(x)$ and $\exists x \in A\, P(x)$ are bounded-quantifier formulas.

Remark (Non-Examples). $\forall x\, (x \in A \wedge P(x))$ is not the correct expansion of the bounded universal form.

Remark (Dependencies).

- Universal Formula
- Existential Formula

Definition (Sentence)

Definition 2.5.4 (Sentence)**.** A *sentence* is a well-formed formula with no free variables.

Remark (Standard quantified statement).

$$\varphi \text{ is a sentence} \iff \varphi \in \mathsf{Form}_{\mathcal{L}} \text{ and } \mathsf{FV}(\varphi) = \varnothing.$$

Remark (Definition predicate reading). Sentence(φ) means that φ is a well-formed formula with no free variables.

Remark (Negated quantified statement).

$$\varphi \notin \mathsf{Form}_{\mathcal{L}} \quad \text{or} \quad \mathsf{FV}(\varphi) \neq \varnothing.$$

Remark (Interpretation). Sentences have truth values in structures without reference to a particular assignment.

Remark (Examples). $\forall x\, P(x)$ and $\exists x\, \forall y\, R(x, y)$ are sentences when all variables shown are bound.

Remark (Non-Examples). $P(x)$ and $\exists y\, R(x, y)$ are not sentences when x remains free.

Remark (Dependencies).

- Well-Formed Formula

Definition (Open Formula)

Definition 2.5.5 (Open Formula)**.** An *open formula* is a well-formed formula with at least one free variable.

Remark (Standard quantified statement).

$$\varphi \text{ is open} \iff \varphi \in \mathsf{Form}_{\mathcal{L}} \text{ and } \mathsf{FV}(\varphi) \neq \varnothing.$$

Remark (Definition predicate reading). OpenFormula(φ) means that φ has at least one free variable.

Remark (Negated quantified statement).

$$\varphi \notin \mathsf{Form}_{\mathcal{L}} \quad \text{or} \quad \mathsf{FV}(\varphi) = \varnothing.$$

Remark (Interpretation). Open formulas may become true or false only after a structure and assignment are fixed.

Remark (Examples). $P(x)$ and $R(x, y)$ are open formulas.

Remark (Non-Examples). $\forall x\, P(x)$ is a sentence rather than an open formula.

Remark (Dependencies).

- Well-Formed Formula
- Open Formula

Definition (Quantifier Scope)

Definition 2.5.6 (Quantifier Scope)**.** In a formula $Qx\,\varphi$, where $Q \in \{\forall, \exists\}$, the *scope* of the quantifier Qx is the formula φ. Occurrences of x inside that scope are bound by the quantifier unless an inner quantifier for x intervenes.

Remark (Standard quantified statement).

$$Qx\,\varphi \text{ has scope } \varphi, \qquad Q \in \{\forall, \exists\}.$$

Remark (Definition predicate reading). Scope(x, φ) means that φ is the scope governed by the quantifier binding x.

Remark (Negated quantified statement).

$$\varphi \text{ is not the formula governed by the displayed quantifier on } x.$$

Remark (Interpretation). Scope determines which variable occurrences a quantifier controls. Scope errors are the main source of incorrect translations from English into predicate logic.

Remark (Examples). In $\forall x\,(P(x) \to Q(x))$, the scope is $(P(x) \to Q(x))$. In $(\forall x\, P(x)) \to Q(x)$, the final occurrence of x is outside the scope of the quantifier.

Remark (Dependencies).

- Universal Formula
- Existential Formula

Theorem (Negation of Bounded Quantifiers)

Theorem 2.5.7 (Negation of Bounded Quantifiers)**.** *Let A be a set and let φ be a formula. Then*

$$\neg(\forall x \in A\, \varphi) \equiv \exists x \in A\, \neg\varphi, \qquad \neg(\exists x \in A\, \varphi) \equiv \forall x \in A\, \neg\varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$\neg\forall x\, (x \in A \to \varphi) \equiv \exists x\, (x \in A \wedge \neg\varphi), \qquad \neg\exists x\, (x \in A \wedge \varphi) \equiv \forall x\, (x \in A \to \neg\varphi).$$

Remark (Predicate reading). The theorem records the negation law for bounded universal and bounded existential formulas.

Remark (Interpretation). The domain restriction is preserved under negation. What changes is the quantifier and the inner assertion.

Remark (Dependencies).

- Bounded Quantifier
- Quantifier Negation Laws

Quantifier Laws

Remark (Section toolkit: Quantifier algebra). Quantifier algebra is the predicate-logic analogue of propositional equivalence laws. These laws justify rewriting quantified formulas while preserving logical equivalence.

Theorem (Quantifier Negation Laws)

Theorem 2.5.8 (Quantifier Negation Laws)**.** *For every formula φ,*

$$\neg\forall x\, \varphi \equiv \exists x\, \neg\varphi, \qquad \neg\exists x\, \varphi \equiv \forall x\, \neg\varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varphi \in \mathsf{Form}_{\mathcal{L}} \quad \neg\forall x\, \varphi \equiv \exists x\, \neg\varphi \quad \text{and} \quad \neg\exists x\, \varphi \equiv \forall x\, \neg\varphi.$$

Remark (Predicate reading). LogicalEquivalence$(\neg\forall x\, \varphi, \exists x\, \neg\varphi)$ and LogicalEquivalence$(\neg\exists x\, \varphi, \forall x\, \neg\varphi)$

Remark (Interpretation). Negating a quantified statement flips the quantifier and pushes the negation into the scope.

Remark (Dependencies).

- Universal Formula

- Existential Formula
- Logical Equivalence

Theorem (Double Negation and Quantifiers)

Theorem 2.5.9 (Double Negation and Quantifiers)**.** *For every formula φ,*

$$\neg\neg\forall x\,\varphi \equiv \forall x\,\varphi, \qquad \neg\neg\exists x\,\varphi \equiv \exists x\,\varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varphi \in \mathsf{Form}_{\mathcal{L}} \quad \neg\neg\forall x\,\varphi \equiv \forall x\,\varphi \quad \text{and} \quad \neg\neg\exists x\,\varphi \equiv \exists x\,\varphi.$$

Remark (Predicate reading)*.* LogicalEquivalence$(\neg\neg Qx\,\varphi, Qx\,\varphi)$ for $Q \in \{\forall, \exists\}$.

Remark (Interpretation)*.* Double negation does not change the quantifier form once the two negations are cancelled.

Remark (Dependencies).

- Quantifier Negation Laws

Theorem (Same-Type Quantifier Commutation)

Theorem 2.5.10 (Same-Type Quantifier Commutation)**.** *Quantifiers of the same type commute:*

$$\forall x\,\forall y\,\varphi \equiv \forall y\,\forall x\,\varphi, \qquad \exists x\,\exists y\,\varphi \equiv \exists y\,\exists x\,\varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varphi \in \mathsf{Form}_{\mathcal{L}} \quad \forall x\,\forall y\,\varphi \equiv \forall y\,\forall x\,\varphi \quad \text{and} \quad \exists x\,\exists y\,\varphi \equiv \exists y\,\exists x\,\varphi.$$

Remark (Predicate reading)*.* The theorem asserts logical equivalence after swapping adjacent quantifiers of the same type.

Remark (Interpretation)*.* Two universal choices can be checked in either order, and two existential witnesses can be selected in either order.

Remark (Dependencies).

- Universal Formula
- Existential Formula
- Logical Equivalence

Theorem (Mixed Quantifiers Do Not Commute)

Theorem 2.5.11 (Mixed Quantifiers Do Not Commute)**.** *In general,*

$$\forall x\, \exists y\, \varphi \not\equiv \exists y\, \forall x\, \varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$\exists \varphi \in \mathsf{Form}_{\mathcal{L}} \quad \forall x\, \exists y\, \varphi \not\equiv \exists y\, \forall x\, \varphi.$$

Remark (Predicate reading). $\neg\,\mathrm{LogicalEquivalence}(\forall x\, \exists y\, \varphi, \exists y\, \forall x\, \varphi)$ in general.

Remark (Interpretation). The first form permits the witness y to depend on x. The second form requires one uniform witness y that works for every x.

Remark (Dependencies).

- Same-Type Quantifier Commutation

Theorem (Valid Quantifier Distribution)

Theorem 2.5.12 (Valid Quantifier Distribution)**.** *For all formulas φ, ψ,*

$$\forall x\, (\varphi \wedge \psi) \equiv (\forall x\, \varphi) \wedge (\forall x\, \psi),$$

$$\exists x\, (\varphi \vee \psi) \equiv (\exists x\, \varphi) \vee (\exists x\, \psi).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varphi, \psi \in \mathsf{Form}_{\mathcal{L}} \quad \forall x\, (\varphi \wedge \psi) \equiv (\forall x\, \varphi) \wedge (\forall x\, \psi)$$

and

$$\exists x\, (\varphi \vee \psi) \equiv (\exists x\, \varphi) \vee (\exists x\, \psi).$$

Remark (Predicate reading). The displayed universal-conjunction and existential-disjunction distributions are logical equivalences.

Remark (Interpretation). Universal quantification distributes over conjunction because both conjuncts must hold for every value. Existential quantification distributes over disjunction because a witness for either disjunct witnesses the disjunction.

Remark (Dependencies).

- Logical Equivalence
- Universal Formula
- Existential Formula

Theorem (Invalid Quantifier Distribution)

Theorem 2.5.13 (Invalid Quantifier Distribution)**.** *In general,*

$$\forall x\,(\varphi \vee \psi) \not\equiv (\forall x\,\varphi) \vee (\forall x\,\psi),$$

$$\exists x\,(\varphi \wedge \psi) \not\equiv (\exists x\,\varphi) \wedge (\exists x\,\psi).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists \varphi, \psi \in \mathsf{Form}_{\mathcal{L}} \quad \forall x\,(\varphi \vee \psi) \not\equiv (\forall x\,\varphi) \vee (\forall x\,\psi),$$

$$\exists \varphi, \psi \in \mathsf{Form}_{\mathcal{L}} \quad \exists x\,(\varphi \wedge \psi) \not\equiv (\exists x\,\varphi) \wedge (\exists x\,\psi).$$

Remark (Predicate reading). The displayed distribution patterns are not logical equivalences in general.

Remark (Interpretation). The invalid laws fail because witnesses or counterexamples may vary between the two subformulas.

Remark (Dependencies).

- Valid Quantifier Distribution

Theorem (Vacuous Quantification)

Theorem 2.5.14 (Vacuous Quantification)**.** *If x does not occur free in φ, then*

$$\forall x\,\varphi \equiv \varphi, \qquad \exists x\,\varphi \equiv \varphi.$$

Go to proof.

Remark (Standard quantified statement).

$$x \notin \mathsf{FV}(\varphi) \Rightarrow (\forall x\,\varphi \equiv \varphi) \wedge (\exists x\,\varphi \equiv \varphi).$$

Remark (Predicate reading). Vacuous quantification preserves logical equivalence when the quantified variable is not free in the scope.

Remark (Interpretation). A quantifier that binds no occurrence has no semantic effect.

Remark (Dependencies).

- Quantifier Scope

Theorem (Renaming Bound Variables)

Theorem 2.5.15 (Renaming Bound Variables)**.** *If y is free for x in φ and no capture is introduced, then*

$$\forall x\,\varphi \equiv \forall y\,\varphi[y/x], \qquad \exists x\,\varphi \equiv \exists y\,\varphi[y/x].$$

Go to proof.

Remark (Standard quantified statement).

$$y \text{ free for } x \text{ in } \varphi \Rightarrow (\forall x\, \varphi \equiv \forall y\, \varphi[y/x]) \wedge (\exists x\, \varphi \equiv \exists y\, \varphi[y/x]).$$

Remark (Predicate reading). Renaming bound variables preserves logical equivalence when the renaming is capture-avoiding.

Remark (Interpretation). Bound variable names are placeholders. Renaming them changes notation, not meaning, provided no free variable becomes captured.

Remark (Dependencies).

- Free for Substitution
- Substitution Notation

Prenex Normal Form

Remark (Section toolkit: Prenex normalization). Prenex normal form is the predicate-logic analogue of the normal-form procedures from propositional logic. It is a syntactic normalization procedure, not a semantic change: each step preserves logical equivalence.

Definition (Quantifier Prefix)

Definition 2.5.16 (Quantifier Prefix). A *quantifier prefix* is a finite string

$$Q_1 x_1\, Q_2 x_2 \cdots Q_n x_n,$$

where each $Q_i \in \{\forall, \exists\}$ and each x_i is a variable.

Remark (Standard quantified statement).

$$\Pi = Q_1 x_1 \cdots Q_n x_n \iff \forall i \le n\ (Q_i \in \{\forall, \exists\} \text{ and } x_i \in \mathsf{Var}_{\mathcal{L}}).$$

Remark (Definition predicate reading). QuantifierPrefix(Π) means that Π is a finite string of quantifier-variable pairs.

Remark (Interpretation). A prefix records all quantifier choices before the quantifier-free part of a prenex formula is read.

Remark (Dependencies).

- Universal Formula
- Existential Formula

Definition (Matrix)

Definition 2.5.17 (Matrix). In a formula $Q_1 x_1 \cdots Q_n x_n\, \psi$, the *matrix* is the trailing formula ψ. For prenex normal form, the matrix is required to be quantifier-free.

Remark (Standard quantified statement).

$$Q_1x_1 \cdots Q_nx_n\,\psi \text{ has matrix } \psi, \qquad \psi \text{ quantifier-free in prenex normal form.}$$

Remark (Definition predicate reading). $\mathrm{Matrix}(\psi, \Phi)$ means that ψ is the matrix of the formula Φ.

Remark (Interpretation). The matrix is the part of a prenex formula where the propositional connectives and atomic formulas remain after all quantifiers have been pulled forward.

Remark (Dependencies).

- Well-Formed Formula

Definition (Prenex Formula)

Definition 2.5.18 (Prenex Formula)**.** A *prenex formula* is a formula of the form

$$Q_1x_1\,Q_2x_2 \cdots Q_nx_n\,\psi,$$

where $Q_1x_1 \cdots Q_nx_n$ is a quantifier prefix and ψ is quantifier-free.

Remark (Standard quantified statement).

$$\Phi \text{ is prenex} \iff \Phi = Q_1x_1 \cdots Q_nx_n\,\psi \text{ with } \psi \text{ quantifier-free.}$$

Remark (Definition predicate reading). $\mathrm{PrenexFormula}(\Phi)$ means that all quantifiers of Φ occur in an initial prefix.

Remark (Negated quantified statement).

$$\Phi \text{ is not of the form } Q_1x_1 \cdots Q_nx_n\,\psi \text{ with } \psi \text{ quantifier-free.}$$

Remark (Interpretation). Prenex formulas isolate quantifier order from the quantifier-free matrix.

Remark (Examples). $\forall x\, \exists y\, (P(x) \to R(x, y))$ is prenex.

Remark (Non-Examples). $\forall x\, (P(x) \to \exists y\, R(x, y))$ is not prenex because a quantifier remains inside the matrix.

Remark (Dependencies).

- Quantifier Prefix
- Matrix

Definition (Prenex Normal Form)

Definition 2.5.19 (Prenex Normal Form)**.** A formula ψ is a *prenex normal form* of a formula φ exactly when ψ is prenex and $\varphi \equiv \psi$.

Remark (Standard quantified statement).

$$\psi \text{ is a prenex normal form of } \varphi \iff \psi \text{ is prenex and } \varphi \equiv \psi.$$

Remark (Definition predicate reading). PrenexNormalForm(ψ, φ) means that ψ is a prenex formula logically equivalent to φ.

Remark (Interpretation). Prenex normal form changes the placement of quantifiers but not the meaning of the formula.

Remark (Exposition). Prenex normal form is not unique. Different choices of bound-variable names and different orders of valid pulling steps can produce different but logically equivalent prenex formulas.

Remark (Exposition). A prenex conversion proceeds in stages: eliminate implications and biconditionals, push negations inward, rename bound variables as needed, pull quantifiers forward, and leave a quantifier-free matrix.

Remark (Examples). Starting with

$$\neg\big(\forall x\, P(x) \to \exists y\, Q(y)\big),$$

eliminate the implication, push the negation inward, and pull the quantifiers forward to obtain

$$\forall x\, \forall y\, (P(x) \wedge \neg Q(y)).$$

Remark (Dependencies).

- Prenex Formula
- Logical Equivalence

Theorem (Existence of Prenex Normal Form)

Theorem 2.5.20 (Existence of Prenex Normal Form). *Every first-order formula is logically equivalent to some formula in prenex normal form. Go to proof.*

Remark (Standard quantified statement).

$$\forall \varphi \in \mathsf{Form}_{\mathcal{L}}\, \exists \psi \in \mathsf{Form}_{\mathcal{L}}\, (\psi \text{ is prenex} \wedge \varphi \equiv \psi).$$

Remark (Predicate reading). $\forall \varphi\, \exists \psi$ PrenexNormalForm(ψ, φ).

Remark (Interpretation). Every formula can be converted into an equivalent one with all quantifiers at the front and a quantifier-free matrix at the end.

Remark (Exposition). The proof proceeds by structural rewriting: remove implication and biconditional connectives, push negations inward, rename bound variables to avoid capture, and then pull quantifiers forward using equivalences whose side conditions have been made true by renaming.

Remark (Dependencies).

- Prenex Normal Form
- Quantifier Negation Laws
- Renaming Bound Variables
- Valid Quantifier Distribution

Logical Strength and Quantifier Order

Remark (Section toolkit: Strength and order). Quantifier order is one of the main new phenomena in predicate logic. In propositional logic, changing the order of independent connectives often keeps truth conditions recognizable. In predicate logic, moving $\forall$ and $\exists$ can change whether witnesses may depend on earlier choices.

Definition (Stronger Formula)

Definition 2.5.21 (Stronger Formula)**.** A sentence A is *stronger than* a sentence B exactly when

$$A \models B \quad \text{and} \quad B \not\models A.$$

Remark (Standard quantified statement).

$$A \text{ is stronger than } B \iff (\forall \mathcal{M}\ (\mathcal{M} \models A \Rightarrow \mathcal{M} \models B)) \wedge (\exists \mathcal{N}\ (\mathcal{N} \models B \wedge \mathcal{N} \not\models A)).$$

Remark (Definition predicate reading). StrongerFormula(A, B) means that A entails B, but B does not entail A.

Remark (Negated quantified statement).

$$A \not\models B \quad \text{or} \quad B \models A.$$

Remark (Interpretation). The stronger sentence has fewer possible structures in which it is true. It rules out more cases.

Remark (Examples). $\forall x\, P(x)$ is stronger than $\exists x\, P(x)$ over nonempty domains.

Remark (Dependencies).

- Logical Consequence

Definition (Weaker Formula)

Definition 2.5.22 (Weaker Formula)**.** A sentence B is *weaker than* a sentence A exactly when A is stronger than B.

Remark (Standard quantified statement).

$$B \text{ is weaker than } A \iff A \models B \text{ and } B \not\models A.$$

Remark (Definition predicate reading). WeakerFormula(B, A) means that B is entailed by A but does not entail A.

Remark (Interpretation). The weaker sentence permits more structures. It follows from the stronger sentence but carries less information.

Remark (Dependencies).

- Stronger Formula

Definition (Quantifier Order)

Definition 2.5.23 (Quantifier Order)**.** The *quantifier order* of a prenex formula is the ordered list of quantifiers in its prefix. A change of order is significant when it changes which witnesses may depend on earlier variables.

Remark (Standard quantified statement).

$$Q_1x_1 \cdots Q_nx_n\,\psi \text{ has quantifier order } (Q_1, \ldots, Q_n).$$

Remark (Definition predicate reading). QuantifierOrder$(\Phi, (Q_1, \ldots, Q_n))$ records the order of the quantifiers in the prefix of Φ.

Remark (Interpretation). Quantifier order controls dependency. In $\forall x\,\exists y$, the witness y may depend on x. In $\exists y\,\forall x$, the same y must work for all x.

Remark (Dependencies).

- Quantifier Prefix
- Prenex Formula

Theorem (Universal Implies Existential)

Theorem 2.5.24 (Universal Implies Existential)**.** *Over nonempty domains,*

$$\forall x\,\varphi(x) \models \exists x\,\varphi(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \mathcal{M}\,\big(\mathcal{M} \models \forall x\,\varphi(x) \Rightarrow \mathcal{M} \models \exists x\,\varphi(x)\big).$$

Remark (Predicate reading). LogicalConsequence$(\{\forall x\,\varphi(x)\}, \exists x\,\varphi(x))$.

Remark (Interpretation). If every domain element satisfies φ, then at least one does because first-order structures have nonempty domains.

Remark (Dependencies).

- Structure

- Logical Consequence

Theorem (Existential Does Not Imply Universal)

Theorem 2.5.25 (Existential Does Not Imply Universal)**.** *In general,*

$$\exists x\, \varphi(x) \not\models \forall x\, \varphi(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists \mathcal{M}\ \big(\mathcal{M} \models \exists x\, \varphi(x) \wedge \mathcal{M} \not\models \forall x\, \varphi(x)\big).$$

Remark (Predicate reading). $\neg\,\text{LogicalConsequence}(\{\exists x\, \varphi(x)\}, \forall x\, \varphi(x))$.

Remark (Interpretation). One witness does not make a universal claim true.

Remark (Exposition). In the real numbers, $\exists x\,(x = 0)$ is true, but $\forall x\,(x = 0)$ is false.

Remark (Dependencies).

- Universal Implies Existential

Theorem (Uniform Witness Implies Dependent Witness)

Theorem 2.5.26 (Uniform Witness Implies Dependent Witness)**.** *For every formula* $\varphi(x, y)$*,*

$$\exists y\, \forall x\, \varphi(x, y) \models \forall x\, \exists y\, \varphi(x, y).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \mathcal{M}\ \big(\mathcal{M} \models \exists y\, \forall x\, \varphi(x, y) \Rightarrow \mathcal{M} \models \forall x\, \exists y\, \varphi(x, y)\big).$$

Remark (Predicate reading). $\text{LogicalConsequence}(\{\exists y\, \forall x\, \varphi(x, y)\}, \forall x\, \exists y\, \varphi(x, y))$.

Remark (Interpretation). A single witness y that works for every x can be reused as the witness required for each separate x.

Remark (Dependencies).

- Quantifier Order
- Logical Consequence

Theorem (Dependent Witness Need Not Be Uniform)

Theorem 2.5.27 (Dependent Witness Need Not Be Uniform)**.** *In general,*

$$\forall x\, \exists y\, \varphi(x, y) \not\models \exists y\, \forall x\, \varphi(x, y).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists \mathcal{M}\ \big(\mathcal{M} \models \forall x\, \exists y\, \varphi(x,y) \wedge \mathcal{M} \not\models \exists y\, \forall x\, \varphi(x,y)\big).$$

Remark (Predicate reading). $\neg\,\mathrm{LogicalConsequence}(\{\forall x\, \exists y\, \varphi(x,y)\}, \exists y\, \forall x\, \varphi(x,y))$.

Remark (Interpretation). The dependent-witness statement allows y to vary with x. The uniform statement requires one y to handle all x at once.

Remark (Exposition). Over $\mathbb{R}$, let $\varphi(x,y)$ be $y > x$. Then $\forall x\, \exists y\, (y > x)$ is true, since $y = x + 1$ works for each x. But $\exists y\, \forall x\, (y > x)$ is false, since no real number is larger than every real number.

Remark (Exposition). Quantifier order matters because it records dependency. The phrase "for every x, there exists y" permits a different y after x is known. The phrase "there exists y such that for every x" chooses y before seeing x. This is a genuine expressive jump beyond propositional logic.

Remark (Dependencies).

- Uniform Witness Implies Dependent Witness

2.6 Proof Theory

Quantifier Inference Rules — Quick Reference

Rule	Schema	Key restriction
UI	$\forall x\, \varphi \Rightarrow \varphi[t/x]$	t free for x
UG	$\varphi \Rightarrow \forall x\, \varphi$	x not free in assumptions
EI	$\exists x\, \varphi \Rightarrow \varphi[c/x]$	c fresh witness
EG	$\varphi[t/x] \Rightarrow \exists x\, \varphi$	none

2.6.1 Proof Theory: Quantifier Inference Rules

Definition (Universal Instantiation — UI)

Definition 2.6.1 (Universal Instantiation — UI)**.** From a universally quantified statement, infer any instance obtained by substituting a term for the bound variable:

$$\forall x\, \varphi \;\Rightarrow\; \varphi[t/x].$$

Remark (Dependencies).

- Universal Quantifier
- Substitution Notation
- Free for Substitution

Remark (Side condition). The term t must be free for x in φ. If t contains a variable that would become bound after substitution, the inference is invalid.

Definition (Universal Generalization — UG)

Definition 2.6.2 (Universal Generalization — UG)**.** From a formula, infer a universally quantified statement:

$$\varphi \;\Rightarrow\; \forall x\, \varphi.$$

Remark (Dependencies).

- Universal Quantifier
- Formula
- Bound and Free Occurrences

Remark (Restriction on UG). The variable x must not occur free in any undischarged assumption on which φ depends. This restriction ensures that x is truly arbitrary — not tied to a specific hypothesis about x.

Remark (Arbitrary element argument). To prove $\forall x\, \varphi(x)$, fix an arbitrary element x (introducing no special assumptions about x), prove $\varphi(x)$, and then apply UG. The variable x must remain unconstrained throughout.

Definition (Existential Instantiation — EI)

Definition 2.6.3 (Existential Instantiation — EI)**.** From an existentially quantified statement, infer an instance with a fresh constant (witness):

$$\exists x\, \varphi \;\Rightarrow\; \varphi[c/x].$$

Remark (Dependencies).

- Existential Quantifier
- Constant Symbol
- Substitution Notation
- Free for Substitution

Remark (Witness discipline for EI). The constant c must be fresh: it must not appear in φ, in any undischarged assumption, or in the final conclusion of the proof. The constant represents an arbitrary but fixed witness to the existential claim, not a specific known object.

Remark (Common error). Using a witness constant introduced by EI outside its allowed scope, or choosing c before establishing the existential premise, invalidates the proof.

Definition (Existential Generalization — EG)

Definition 2.6.4 (Existential Generalization — EG)**.** From a formula containing a term, infer an existential statement:

$$\varphi[t/x] \;\Rightarrow\; \exists x\, \varphi.$$

Remark (Dependencies).

- Existential Quantifier
- Term
- Substitution Notation

Remark (No side conditions). EG has no side conditions: any term t may be replaced by an existentially quantified variable. To prove $\exists x\, \varphi(x)$, explicitly exhibit a term t such that $\varphi(t)$ holds, then apply EG.

Remark (Counterexample argument). To refute $\forall x\, \varphi(x)$: produce a single term t such that $\neg\varphi(t)$ holds. To refute $\exists x\, \varphi(x)$: show that $\varphi(t)$ fails for every term t in the domain.

Quantifier Proof Strategies — Quick Reference

Goal shape	**Strategy**	**Key rule**
Prove $\forall x\, \varphi(x)$	Fix an arbitrary x; prove $\varphi(x)$ for that x	UG
Prove $\exists x\, \varphi(x)$	Produce a specific witness t; verify $\varphi(t)$	EG
Use $\forall x\, \varphi(x)$	Instantiate at any term t to get $\varphi(t)$	UI
Use $\exists x\, \varphi(x)$	Introduce a fresh name c for the witness	EI
Negate $\forall x\, \varphi$	Becomes $\exists x\, \neg\varphi$	Neg. law
Negate $\exists x\, \varphi$	Becomes $\forall x\, \neg\varphi$	Neg. law
Prove $\forall x\, \exists y\, \varphi(x, y)$	Fix x; then construct y depending on x	UG then EG
Prove $\exists y\, \forall x\, \varphi(x, y)$	Find y first; then verify for all x	EG then UG

2.6.2 Proof Strategies for Quantified Statements

The inference rules UI, UG, EI, and EG were defined in the previous section. That section answered the question: what does each rule license? This section answers a different question: when the goal of a proof has a particular quantifier structure, which rule should be used, and what does the proof skeleton look like? The two questions are distinct, and both must be understood clearly before reading analysis proofs.

The single most important organizing principle is this: *the shape of the goal determines the proof structure, and the shape of the hypothesis determines how to use it.* A $\forall$ goal calls for a different move than a $\exists$ goal; a $\forall$ hypothesis is used differently from a $\exists$ hypothesis. The four combinations are the four strategies below.

Strategy 1: Proving a Universal $\forall x\, \varphi(x)$

To prove $\forall x\, \varphi(x)$: introduce a fresh variable x_0 representing an *arbitrary, unspecified* element of the domain. Prove $\varphi(x_0)$ without making any assumption about x_0 beyond its being in the domain. Then apply Universal Generalization (UG) to conclude $\forall x\, \varphi(x)$.

> **Let** x_0 be an arbitrary element of the domain.
> [**prove** $\varphi(x_0)$ **using only the hypotheses, not any special property of** x_0]
> **Therefore:** $\forall x\, \varphi(x)$. (by UG)

The critical constraint is that x_0 must be *arbitrary*: no hypothesis in the proof may impose a condition on x_0 that is not universally true. Violating this constraint is the UG fallacy ("illicit generalization").

Remark (Why "let x be arbitrary" works)*.* Fixing an arbitrary element x_0 is the formal counterpart of the English phrase "let x be any element." The word "arbitrary" means: the proof uses no information about x_0 that is specific to x_0 — only information that applies to every element of the domain. If the argument goes through for such an unspecified x_0, it goes through for every x, so the universal follows.

Remark (The UG side condition in practice)*.* UG requires that x_0 not appear free in any undischarged assumption. In practice: do not use any hypothesis of the form "x_0 has property P" unless P holds of every domain element. If a hypothesis says "$x_0 > 0$," then x_0 is not arbitrary, and UG to $\forall x$ is invalid.

Strategy 2: Proving an Existential $\exists x\, \varphi(x)$

To prove $\exists x\, \varphi(x)$: exhibit a specific term t and verify that $\varphi(t)$ holds. Then apply Existential Generalization (EG) to conclude $\exists x\, \varphi(x)$.

> **Claim:** $\exists x\, \varphi(x)$.
> **Take:** $t :=$ [specific term].
> **Verify:** $\varphi(t)$ holds. [proof that t works]
> **Therefore:** $\exists x\, \varphi(x)$. (by EG)

The entire burden of an existential proof is finding the right t. Once t is identified and verified, the logic is trivial: one application of EG.

Remark (Where the difficulty lies)*.* In existential proofs, the logical structure is simple but the mathematical insight is often deep. Finding the witness requires understanding the problem; verifying the witness is usually straightforward. This is why analysis proofs often say "take $\delta = \min(\varepsilon/2, 1)$" without explaining where that choice came from — the work happened before the formal proof was written.

Strategy 3: Using a Universal $\forall x\, \varphi(x)$

If a hypothesis or established fact has the form $\forall x\, \varphi(x)$, one can instantiate it at *any* specific term t to obtain $\varphi(t)$, by Universal Instantiation (UI). There are no side conditions beyond t being free for x in φ.

> **Available:** $\forall x\, \varphi(x)$.
> **Choose any term t.**
> **Obtain:** $\varphi(t)$. (by UI)

UI can be applied repeatedly to the same universal statement with different choices of t.

Strategy 4: Using an Existential $\exists x\, \varphi(x)$

If a hypothesis has the form $\exists x\, \varphi(x)$, introduce a *fresh constant* c to name the (unknown, unspecified) witness, obtaining $\varphi(c)$ by Existential Instantiation (EI). The constant c must be new: it must not appear anywhere else in the proof.

> **Available:** $\exists x\, \varphi(x)$.
> **Let c be a witness.** (fresh constant, not used elsewhere)
> **Obtain:** $\varphi(c)$. (by EI)

The freshness requirement is not a technicality. Using a constant that already appears in the proof — say, a constant a introduced under a different assumption — would illegally identify the witness with that specific element. The witness c is only known to satisfy $\varphi(c)$; nothing else about it is known, and it cannot be equated with anything specific.

Remark (The scope of a witness constant). Once c is introduced by EI, it names the witness for the duration of the argument. Conclusions derived using c depend on the existence of such a witness; they are valid (they do not depend on which specific element c actually names). However, c cannot be exported outside the scope in which the existential was instantiated, and no universal generalization may be applied to c.

The Central Asymmetry: Universal vs. Existential Goals

The following comparison is the most important thing to understand about quantifier proof structure. Universal and existential goals require fundamentally different strategies, and confusing them is one of the most common serious errors in mathematical writing.

Universal vs. Existential Goals: The Fundamental Difference

	$\forall x\, \varphi(x)$	$\exists x\, \varphi(x)$
To prove:	Fix *arbitrary* x; prove $\varphi(x)$	*Choose* specific t; verify $\varphi(t)$
Who chooses x?	The adversary (you cannot choose)	You (the prover)
Main rule:	UG	EG
Burden:	Works for every x without assumptions	A single witness suffices
Error:	Assuming something special about x	Failing to exhibit a concrete t

Remark (Who controls the variable). For a universal goal, the variable x is chosen by the "opponent" of the proof: the proof must work no matter what x is. The prover has no control over x; hence the term "arbitrary." For an existential goal, the prover chooses the witness t; the power is entirely on the prover's side, but so is the obligation to produce something concrete and verifiable.

Nested Quantifiers: The ∀∃ Pattern in Analysis

Every ε-δ proof in analysis is a proof of a statement with the pattern $\forall\varepsilon\,\exists\delta\,\cdots$. Understanding the proof structure of such statements is the direct payoff of this section.

Proof Skeleton: $\forall x\,\exists y\,\varphi(x, y)$

Step 1. Fix arbitrary x. (Opening move for the outer ∀; x is unspecified. This is Strategy 1.)

Step 2. Define y in terms of x. (The witness for $\exists y$ may, and typically does, depend on x. This is Strategy 2 — choosing a witness.)

Step 3. Verify $\varphi(x, y)$ using the specific y from Step 2 and the arbitrary x from Step 1.

Step 4. Conclude: $\exists y\,\varphi(x, y)$. (by EG)

Step 5. Conclude: $\forall x\,\exists y\,\varphi(x, y)$. (by UG, since x was arbitrary in Step 1)

The key insight: y is chosen *after* x is fixed, so y is allowed to depend on x.

Proof Skeleton: $\exists y\,\forall x\,\varphi(x, y)$

Step 1. Define y. (The witness for $\exists y$ must be chosen *before* x is introduced. This y must work for *every* x.)

Step 2. Fix arbitrary x. (Now x is introduced.)

Step 3. Verify $\varphi(x, y)$ for the specific y from Step 1 and arbitrary x from Step 2.

Step 4. Conclude: $\forall x\,\varphi(x, y)$. (by UG)

Step 5. Conclude: $\exists y\,\forall x\,\varphi(x, y)$. (by EG)

The key contrast: y is chosen *before* x, so y cannot depend on x. This is why $\exists y\,\forall x$ is a much stronger claim.

Remark (Why order determines strength). In the ∀∃ skeleton, the witness y is constructed from x, so a different y is allowed for each x. In the ∃∀ skeleton, a single y must work for all x simultaneously. The first pattern appears in continuity (δ may depend on ε and x); the second in uniform continuity (δ may depend on ε but not on x). The difference between continuity and uniform continuity is exactly the difference between these two proof skeletons.

A Complete Worked Example

The following proof traces every UI/UG/EI/EG step explicitly. In practice, proofs are written more concisely, but the logical structure is always this one.

Example (A worked multi-quantifier proof). **Claim:** If $\forall x\, \exists y\, P(x,y)$ and $\forall x\, \forall y\, (P(x,y) \to Q(x,y))$, then $\forall x\, \exists y\, Q(x,y)$.

Proof:

1. Assume $H_1 := \forall x\, \exists y\, P(x,y)$ and $H_2 := \forall x\, \forall y\, (P(x,y) \to Q(x,y))$. (hypotheses)
2. Let a be an arbitrary element of the domain. (introduce arbitrary variable for $\forall x$ in the goal)
3. From H_1, instantiate at a: $\exists y\, P(a,y)$. (by UI applied to H_1 with $t = a$)
4. Let b be a witness: $P(a,b)$. (by EI applied to Step 3; b is fresh)
5. From H_2, instantiate at a: $\forall y\, (P(a,y) \to Q(a,y))$. (by UI applied to H_2 with $t = a$)
6. From Step 5, instantiate at b: $P(a,b) \to Q(a,b)$. (by UI applied to Step 5 with $t = b$)
7. From Step 4 ($P(a,b)$) and Step 6 ($P(a,b) \to Q(a,b)$): obtain $Q(a,b)$. (by Modus Ponens)
8. From Step 7: $\exists y\, Q(a,y)$. (by EG: b witnesses the existential)
9. Since a was arbitrary: $\forall x\, \exists y\, Q(x,y)$. (by UG: a was not constrained)

Annotation: The proof follows the $\forall\exists$ skeleton exactly. Step 2 fixes the arbitrary x. Steps 3–4 use the first hypothesis to obtain a witness b depending on a. Steps 5–7 apply the second hypothesis to get the desired conclusion about b. Steps 8–9 wrap everything back up with EG and UG.

Remark (Reading analysis proofs). A proof that begins "Let $\varepsilon > 0$" is executing Step 2 of the $\forall\exists$ skeleton with domain $\mathbb{R}_{>0}$. A sentence like "take $\delta = \varepsilon/2$" is executing Step 2 of the inner existential proof: it is the witness for $\exists\delta$. Everything between "let $\varepsilon > 0$" and "take $\delta = \ldots$" is scaffolding to find the right witness. Everything after "take $\delta = \ldots$" is the verification that the witness works. Once this structure is seen, every ε-δ proof becomes readable by template.

Equality Rules — Quick Reference

Rule	Schema	Status
Reflexivity (EqI)	$t = t$	Primitive rule
Equality Elim (EqE)	$t_1 = t_2,\ \varphi[t_1/x] \Rightarrow \varphi[t_2/x]$	Primitive rule
Symmetry	$t_1 = t_2 \Rightarrow t_2 = t_1$	Derived
Transitivity	$t_1 = t_2,\ t_2 = t_3 \Rightarrow t_1 = t_3$	Derived
Term substitution	$f(\dots, t_1, \dots) = f(\dots, t_2, \dots)$	Derived
Predicate substitution	$P(\dots, t_1, \dots) \Leftrightarrow P(\dots, t_2, \dots)$	Derived

2.6.3 Proof Theory: Equality Rules

Definition (Equality Introduction — Reflexivity)

Definition 2.6.5 (Equality Introduction — Reflexivity)**.** For any term t, one may infer $t = t$.

Remark (Dependencies).

- Term

Remark (English reading). Reflexivity requires no premises and holds for all terms under all interpretations. It is the foundational rule that allows equality reasoning to get off the ground.

Definition (Equality Elimination — Substitution of Equals)

Definition 2.6.6 (Equality Elimination — Substitution of Equals)**.** From $t_1 = t_2$ and $\varphi[t_1/x]$, one may infer $\varphi[t_2/x]$.

Remark (Dependencies).

- Term
- Formula
- Substitution Notation
- Free for Substitution

Remark (English reading). Equality elimination permits replacing a term by an equal term in any formula position, provided the substitution is capture-avoiding. This formalizes the intuition that equal things can be substituted for each other.

Theorem (Symmetry of Equality)

Theorem 2.6.7 (Symmetry of Equality)**.** *From $t_1 = t_2$, one may infer $t_2 = t_1$.* *Go to proof.*

Remark (Standard quantified statement). $\forall t_1, t_2 : \; t_1 = t_2 \to t_2 = t_1$.

Remark (Dependencies).

- Equality Introduction
- Equality Elimination

Theorem (Transitivity of Equality)

Theorem 2.6.8 (Transitivity of Equality)**.** *From $t_1 = t_2$ and $t_2 = t_3$, one may infer $t_1 = t_3$. Go to proof.*

Remark (Standard quantified statement). $\forall t_1, t_2, t_3 : \; t_1 = t_2 \wedge t_2 = t_3 \to t_1 = t_3$.

Remark (Dependencies).

- Equality Introduction
- Equality Elimination

Theorem (Term Substitution under Equality)

Theorem 2.6.9 (Term Substitution under Equality)**.** *If $t_1 = t_2$, then for any function symbol f, $f(\dots, t_1, \dots) = f(\dots, t_2, \dots)$. Go to proof.*

Remark (Standard quantified statement). $\forall t_1, t_2 : \; t_1 = t_2 \to \forall$ function symbol f: $f(\dots, t_1, \dots) = f(\dots, t_2, \dots)$.

Remark (Dependencies).

- Equality Elimination

Theorem (Predicate Substitution under Equality)

Theorem 2.6.10 (Predicate Substitution under Equality)**.** *If $t_1 = t_2$ and P is an n-ary predicate symbol, then $P(\dots, t_1, \dots) \;\Leftrightarrow\; P(\dots, t_2, \dots)$. Go to proof.*

Remark (Standard quantified statement). $\forall t_1, t_2 : \; t_1 = t_2 \to \forall$ predicate symbol P: $P(\dots, t_1, \dots) \leftrightarrow P(\dots, t_2, \dots)$.

Remark (Dependencies).

- Equality Elimination

Remark (Summary). These derived rules collectively express that functions and predicates *respect equality*: equal inputs produce equal outputs (for functions) or equivalent propositions (for predicates). Together they amount to the principle of Leibniz substitutivity.

Remark (Common error). When substituting equals for equals, the substitution must be made consistently. Replacing a term by an equal in one occurrence but not another in the same formula can produce incorrect conclusions.

Soundness and Completeness — Quick Reference

Property	Direction
Soundness	$\Gamma \vdash \varphi \Rightarrow \Gamma \models \varphi$
Completeness	$\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi$
Soundness Theorem (FOL)	All standard FOL rules are sound
Gödel's Completeness Theorem	FOL is complete

2.6.4 Proof Theory: Soundness and Completeness

Definition (Soundness)

Definition 2.6.11 (Soundness)**.** A proof system is *sound* if every provable formula is valid:

$$\Gamma \vdash \varphi \implies \Gamma \models \varphi.$$

Remark (Dependencies).

- Satisfaction
- Logical Consequence

Definition (Completeness)

Definition 2.6.12 (Completeness)**.** A proof system is *complete* if every valid formula is provable:

$$\Gamma \models \varphi \implies \Gamma \vdash \varphi.$$

Remark (Dependencies).

- Satisfaction
- Logical Consequence

Theorem (Soundness of First-Order Logic)

Theorem 2.6.13 (Soundness of First-Order Logic)**.** *The standard inference rules for first-order logic (UI, UG, EI, EG, and the propositional rules) are sound: they preserve truth in all structures.*
If $\Gamma \vdash \varphi$, then $\Gamma \models \varphi$. Go to proof.

Remark (Standard quantified statement). $\forall \Gamma, \varphi : \ \Gamma \vdash \varphi \ \rightarrow \ \forall \mathcal{M}, \forall s, \ (\forall \gamma \in \Gamma, \ \mathcal{M}, s \models \gamma) \rightarrow \mathcal{M}, s \models \varphi.$

Remark (Dependencies).

- Soundness

- Satisfaction
- Logical Consequence
- Universal Instantiation
- Quantifier Distribution

Theorem (Gödel's Completeness Theorem)

First-order logic is complete: every logically valid formula is provable.

$$\Gamma \models \varphi \implies \Gamma \vdash \varphi.$$

Equivalently: if a set of sentences Γ is consistent (has no proof of contradiction), then Γ has a model.

Remark (English reading). Soundness ensures proofs do not lead us astray: we cannot prove false statements from true premises. Completeness ensures proofs are powerful enough: every statement that is true in all models can be established by a formal proof.

Together, soundness and completeness show that syntactic provability ($\vdash$) and semantic entailment ($\models$) coincide for first-order logic.

Remark (Completeness vs. incompleteness). Gödel's completeness theorem (1930) should not be confused with his incompleteness theorems (1931). The completeness theorem concerns first-order logic as a proof system. The incompleteness theorems concern the limitations of formal theories capable of expressing arithmetic, and are a separate result.

2.7 Translation

Toolkit: Translation

Task	**Question**	**Typical output**
Translation key	What does each symbol mean?	Domain, constants, predicates, relations
Universal claim	Which objects are restricted?	$\forall x\,(A(x) \to B(x))$
Existential claim	Which properties are jointly required?	$\exists x\,(A(x) \wedge B(x))$
Scope analysis	Which quantifier is outermost?	$\forall x \exists y\, R(x,y)$ or $\exists y \forall x\, R(x,y)$

Remark (Exposition). This section records the applied translation layer for predicate logic: keys, domains, restricted quantifiers, scope ambiguity, and the modern first-order reading of categorical forms.

English to Logic and Scope Ambiguity

Definition (Translation Key)

Definition 2.7.1 (Translation Key)**.** A *translation key* is a finite record that fixes a domain of discourse and assigns intended English readings to the constant symbols, predicate symbols, and relation symbols used in a translation.

Remark (Standard quantified statement)*.*

$$\begin{aligned} &K \text{ is a translation key} \\ &\iff K \text{ specifies a domain and readings for each non-logical symbol used.} \end{aligned}$$

Remark (Definition predicate reading)*.* $\mathrm{TranslationKey}(K)$ means that K supplies the domain and symbol readings needed to parse a translated formula.

Remark (Interpretation)*.* A translation key prevents hidden changes of meaning. The same symbol must keep the same reading throughout one exercise unless the key is explicitly replaced.

Remark (Examples)*.* For the English sentence "Every applicant submitted some form," a key may take the domain to be people and forms, with $A(x)$ for "x is an applicant," $F(y)$ for "y is a form," and $S(x, y)$ for "x submitted y."

Remark (Non-Examples)*.* Using $P(x)$ once for "x is a person" and later for "x passed" is not a translation key; it is a change of vocabulary inside one translation.

Remark (Dependencies)*.*

- Constant Symbol
- Predicate Symbol

Definition (Domain of Discourse)

Definition 2.7.2 (Domain of Discourse)**.** The *domain of discourse* for a translation is the collection of objects over which the variables of the translated formulas range.

Remark (Standard quantified statement)*.*

$$D \text{ is the domain of discourse} \iff \text{each variable in the translation ranges over objects in } D.$$

Remark (Definition predicate reading)*.* $\mathrm{DomainOfDiscourse}(D, K)$ means that D is the domain fixed by the translation key K.

Remark (Interpretation)*.* The domain decides what the quantifiers are allowed to talk about. If the domain is people, then an exam must either be named by a separate sort-like predicate or represented by a different translation choice.

Remark (Examples)*.* For "All cats are mammals," one may use the domain of all animals and write $\forall x\,(C(x) \to M(x))$. With the smaller domain of cats, the same English claim may be written $\forall x\, M(x)$.

Remark (Non-Examples)*.* Writing $\forall x\, M(x)$ while silently changing the domain from animals to cats halfway through the exercise is not a valid translation.

Remark (Dependencies)*.*

- Translation Key
- Variable

Definition (Predicate and Relation Symbols in a Translation)

Definition 2.7.3 (Predicate and Relation Symbols in a Translation)**.** In a translation key, a unary predicate symbol names a property of one object, and an n-ary relation symbol names a relation among n objects.

Remark (Standard quantified statement)*.*

$$\begin{aligned} P \text{ is unary in } K \quad &\iff P(x) \text{ has one argument,} \\ R \text{ is } n\text{-ary in } K \quad &\iff R(x_1, \dots, x_n) \text{ has } n \text{ arguments.} \end{aligned}$$

Remark (Definition predicate reading)*.* $\mathrm{TranslationPredicateSymbol}(P, K)$ records that P has a fixed predicate reading in K, and $\mathrm{TranslationRelationSymbol}(R, n, K)$ records that R is used with arity n in K.

Remark (Interpretation)*.* Object-language predicate symbols such as P, Q, R are symbols inside the formal language. Metalinguistic predicate readings such as $\mathrm{Term}(t)$ or $\mathrm{AtomicFormula}(\varphi)$ are repository descriptions of syntax; they are not formulas being translated.

Remark (Examples)*.* $A(x)$ may mean "x is an athlete." $L(x, y)$ may mean "x likes y." The arity is part of the key: $L(x)$ and $L(x, y)$ are not the same symbol use.

Remark (Non-Examples)*.* Writing $L(x, y, z)$ after declaring $L(x, y)$ as "x likes y" is an arity error.

Remark (Dependencies)*.*

- Predicate Symbol
- Atomic Formula

Definition (Constants and Variables in a Translation)

Definition 2.7.4 (Constants and Variables in a Translation)**.** A *constant* names a fixed object from the translation key, while a *variable* marks a place bound by a quantifier or left open for later binding.

Remark (Standard quantified statement)*.*

$$\begin{aligned} c \text{ is a translation constant} \quad &\iff c \text{ names one fixed object,} \\ x \text{ is a translation variable} \quad &\iff x \text{ is available for quantification or free occurrence.} \end{aligned}$$

Remark (Definition predicate reading). TranslationConstant(c, K) means that c names a fixed object in K, and TranslationVariable(x, K) means that x is used as a variable in translations governed by K.

Remark (Interpretation). Constants do not range. Variables do range when bound by quantifiers. Confusing the two usually leaves a formula with the wrong dependency pattern.

Remark (Examples). If a names Alice, then $P(a)$ says Alice is a participant. The formula $\forall x\,(S(x) \to P(x))$ uses x as a bound variable.

Remark (Non-Examples). The expression $\forall a\, S(a)$ is not appropriate if a has already been declared as a constant naming Alice.

Remark (Dependencies).

- Constant Symbol
- Variable

Proposition (Universal Translation Pattern)

Proposition 2.7.5 (Universal Translation Pattern)**.** *In a broad domain, an English statement of the form "All A are B" is translated by*

$$\forall x\,(A(x) \to B(x)).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x\,(A(x) \to B(x)).$$

Remark (Predicate reading). UniversalTranslationPattern(A, B) records the implication schema for translating universal restricted claims.

Remark (Interpretation). The antecedent $A(x)$ restricts attention to the relevant objects. Objects outside A do not have to satisfy B.

Remark (Exposition). "All cats are mammals" becomes $\forall x\,(C(x) \to M(x))$. "No cats are reptiles" becomes $\forall x\,(C(x) \to \neg R(x))$. The form $\forall x\,(C(x) \wedge M(x))$ says every object in the domain is both a cat and a mammal, which is far stronger than the intended restricted claim. In a broad domain, "all," "every," and "any" usually translate restricted claims with implication.

Remark (Dependencies).

- Universal Formula
- Molecular Formula

Proposition (Existential Translation Pattern)

Proposition 2.7.6 (Existential Translation Pattern)**.** *In a broad domain, an English statement of the form "Some A are B" is translated by*

$$\exists x\,(A(x) \wedge B(x)).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists x\,(A(x) \wedge B(x)).$$

Remark (Predicate reading). ExistentialTranslationPattern(A, B) records the conjunction schema for translating existential restricted claims.

Remark (Interpretation). The witness must satisfy both restrictions at once. Existence is not merely a conditional promise about what would happen if an A existed.

Remark (Exposition). "Some cats are black" becomes $\exists x\,(C(x) \wedge B(x))$. "Some cats are not black" becomes $\exists x\,(C(x) \wedge \neg B(x))$. The form $\exists x\,(C(x) \to B(x))$ may be true because the witness is not a cat, so it does not translate "Some cats are black." In a broad domain, "some," "there is," and "at least one" usually translate restricted claims with conjunction.

Remark (Dependencies).

- Existential Formula
- Molecular Formula

Proposition (Conditional Translation Pattern)

Proposition 2.7.7 (Conditional Translation Pattern)**.** *An English statement of the form "If A, then B" is translated by*

$$A \to B,$$

with quantifiers placed according to the variables occurring in A and B.
Go to proof.

Remark (Standard quantified statement).

$$A \to B.$$

Remark (Predicate reading). ConditionalTranslationPattern(A, B) records the implication schema for conditional English.

Remark (Interpretation). Conditionals are especially common inside universal translations. The conditional separates the restricting condition from the asserted consequence.

Remark (Exposition). "Every applicant who is eligible is interviewed" becomes $\forall x\,((A(x) \wedge E(x)) \to I(x))$. The form $\forall x\,(A(x) \wedge E(x) \wedge I(x))$ says every object is an eligible applicant who is interviewed, which is not the intended conditional claim.

Remark (Dependencies).

- Universal Translation Pattern

Proposition (Negation Translation Pattern)

Proposition 2.7.8 (Negation Translation Pattern)**.** *The negation of a translated quantified statement is obtained by negating the whole formula and then pushing the negation inward using the quantifier negation laws.*
Go to proof.

Remark (Standard quantified statement).

$$\neg\forall x\,\varphi \iff \exists x\,\neg\varphi,$$
$$\neg\exists x\,\varphi \iff \forall x\,\neg\varphi.$$

Remark (Predicate reading). $\mathrm{NegationTranslationPattern}(\varphi, \psi)$ records that ψ is obtained from $\neg\varphi$ by pushing negation through the quantifiers and connectives.

Remark (Interpretation). The negation of "all" is "some not." The negation of "some" is "none." For restricted universal claims, negating the implication produces a witness that satisfies the restriction and fails the consequent.

Remark (Exposition). The negation of $\forall x\,(A(x) \to B(x))$ is $\exists x\,(A(x) \wedge \neg B(x))$. The negation of $\exists x\,(A(x) \wedge B(x))$ is $\forall x\,(A(x) \to \neg B(x))$. The negation of $\forall x\,(A(x) \to B(x))$ is not $\forall x\,(A(x) \to \neg B(x))$; that latter formula says every A fails to be B, not merely that some counterexample exists.

Remark (Dependencies).

- Quantifier Negation Laws
- Universal Translation Pattern
- Existential Translation Pattern

Definition (Scope Ambiguity)

Definition 2.7.9 (Scope Ambiguity)**.** A sentence has *scope ambiguity* when the same ordinary-language wording admits two or more translations with different quantifier nesting or different connective scope.

Remark (Standard quantified statement).

$$\begin{aligned}&E \text{ has scope ambiguity}\\ \iff\ &\text{there exist distinct candidate translations } \varphi, \psi \text{ of } E \text{ with different scope.}\end{aligned}$$

Remark (Definition predicate reading). $\mathrm{ScopeAmbiguity}(E, \varphi, \psi)$ means that the English sentence E can reasonably be translated by φ or by ψ, and the difference is a scope difference.

Remark (Interpretation). Scope is not cosmetic. Changing the outer quantifier can change whether a single witness works for everyone or whether each object may have its own witness.

Remark (Exposition). The formulas $\forall x \exists y\, R(x,y)$ and $\exists y \forall x\, R(x,y)$ are not interchangeable. The first allows the witness y to depend on x; the second requires one y that works for every x. Symbols such as $A(x)$, $B(x)$, and $R(x,y)$ are object-language formula patterns used in translations. Expressions such as $\mathrm{TranslationKey}(K)$, $\mathrm{Term}(t)$, and $\mathrm{AtomicFormula}(\varphi)$ are metalinguistic predicate readings used by the project to describe mathematical objects.

$$\begin{array}{lll} \text{All } A \text{ are } B & \rightsquigarrow & \forall x\,(A(x) \to B(x)) \\ \text{Some } A \text{ are } B & \rightsquigarrow & \exists x\,(A(x) \wedge B(x)) \\ \text{No } A \text{ are } B & \rightsquigarrow & \forall x\,(A(x) \to \neg B(x)) \\ \text{Some } A \text{ are not } B & \rightsquigarrow & \exists x\,(A(x) \wedge \neg B(x)) \\ \text{Every } A \text{ has a } B & \rightsquigarrow & \forall x\,(A(x) \to \exists y\,(B(y) \wedge R(x,y))) \\ \text{There is a } B \text{ for every } A & \rightsquigarrow & \exists y\,(B(y) \wedge \forall x\,(A(x) \to R(x,y))). \end{array}$$

Remark (Examples). "Every A has a B" is usually $\forall x\,(A(x) \to \exists y\,(B(y) \wedge R(x,y)))$. "There is a B for every A" is usually $\exists y\,(B(y) \wedge \forall x\,(A(x) \to R(x,y)))$. The bare patterns $\forall x \exists y\, R(x,y)$ and $\exists y \forall x\, R(x,y)$ make the same scope contrast visible.

Remark (Non-Examples). Renaming a bound variable from x to z does not create a scope ambiguity. The scope is unchanged.

Remark (Dependencies).

- Quantifier Scope
- Quantifier Order

Square of Opposition

Definition (Universal Affirmative Form)

Definition 2.7.10 (Universal Affirmative Form)**.** The *universal affirmative* categorical form, called A, translates "All A are B" as

$$\forall x\,(A(x) \to B(x)).$$

Remark (Standard quantified statement).

$$\forall x\,(A(x) \to B(x)).$$

Remark (Definition predicate reading). $\mathrm{CategoricalAForm}(A, B, \varphi)$ means that φ is the universal affirmative translation from A to B.

Remark (Interpretation). The A-form says that every object satisfying the subject predicate also satisfies the predicate predicate. It does not by itself say that anything satisfies the subject predicate.

Remark (Dependencies).

- Universal Translation Pattern

Definition (Universal Negative Form)

Definition 2.7.11 (Universal Negative Form)**.** The *universal negative* categorical form, called E, translates "No A are B" as

$$\forall x\,(A(x) \to \neg B(x)).$$

Remark (Standard quantified statement)*.*

$$\forall x\,(A(x) \to \neg B(x)).$$

Remark (Definition predicate reading)*.* $\mathrm{CategoricalEForm}(A, B, \varphi)$ means that φ is the universal negative translation from A to B.

Remark (Interpretation)*.* The E-form says that anything satisfying A fails to satisfy B. It is still universal, so it can be true when no object satisfies A.

Remark (Dependencies)*.*

- Universal Translation Pattern
- Negation Translation Pattern

Definition (Particular Affirmative Form)

Definition 2.7.12 (Particular Affirmative Form)**.** The *particular affirmative* categorical form, called I, translates "Some A are B" as

$$\exists x\,(A(x) \land B(x)).$$

Remark (Standard quantified statement)*.*

$$\exists x\,(A(x) \land B(x)).$$

Remark (Definition predicate reading)*.* $\mathrm{CategoricalIForm}(A, B, \varphi)$ means that φ is the particular affirmative translation from A to B.

Remark (Interpretation)*.* The I-form requires a witness satisfying both predicates. It carries existential force.

Remark (Dependencies)*.*

- Existential Translation Pattern

Definition (Particular Negative Form)

Definition 2.7.13 (Particular Negative Form)**.** The *particular negative* categorical form, called O, translates "Some A are not B" as

$$\exists x\,(A(x) \land \neg B(x)).$$

Remark (Standard quantified statement).

$$\exists x\,(A(x) \land \neg B(x)).$$

Remark (Definition predicate reading). CategoricalOForm(A, B, φ) means that φ is the particular negative translation from A to B.

Remark (Interpretation). The O-form requires a witness satisfying the subject predicate and failing the predicate predicate.

Remark (Dependencies).

- Existential Translation Pattern
- Negation Translation Pattern

Proposition (A-O Contradiction)

Proposition 2.7.14 (A-O Contradiction)**.** *The A-form $\forall x\,(A(x) \to B(x))$ and the O-form $\exists x\,(A(x) \land \neg B(x))$ are contradictory: exactly one of them is true.*
Go to proof.

Remark (Standard quantified statement).

$$\neg\forall x\,(A(x) \to B(x)) \iff \exists x\,(A(x) \land \neg B(x)).$$

Remark (Predicate reading). ContradictoryPair(φ_A, φ_O) records that the A-form and O-form cannot share a truth value.

Remark (Interpretation). The O-form is exactly the counterexample to the A-form: an object that is A but not B.

Remark (Dependencies).

- Universal Affirmative Form
- Particular Negative Form
- Quantifier Negation Laws

Proposition (E-I Contradiction)

Proposition 2.7.15 (E-I Contradiction)**.** *The E-form $\forall x\,(A(x) \to \neg B(x))$ and the I-form $\exists x\,(A(x) \land B(x))$ are contradictory: exactly one of them is true.*
Go to proof.

Remark (Standard quantified statement).

$$\neg\forall x\,(A(x) \to \neg B(x)) \iff \exists x\,(A(x) \land B(x)).$$

Remark (Predicate reading). ContradictoryPair(φ_E, φ_I) records that the E-form and I-form cannot share a truth value.

Remark (Interpretation). The I-form is exactly the counterexample to the E-form: an object that is both A and B.

Remark (Dependencies).

- Universal Negative Form
- Particular Affirmative Form
- Quantifier Negation Laws

Proposition (Contrariety Requires Existential Import)

Proposition 2.7.16 (Contrariety Requires Existential Import). *If $\exists x\, A(x)$, then the A-form and E-form cannot both be true.*
Go to proof.

Remark (Standard quantified statement).

$$\exists x\, A(x) \to \neg\big(\forall x\, (A(x) \to B(x)) \wedge \forall x\, (A(x) \to \neg B(x))\big).$$

Remark (Predicate reading). $\mathrm{ContraryWithExistentialImport}(\varphi_A, \varphi_E)$ records that contrariety holds only after the subject predicate is known to have a witness.

Remark (Interpretation). With an A-object available, the two universal claims force both B and $\neg B$ of that object. Without such an object, both universals may be vacuously true. The traditional square of opposition assumes existential import; modern first-order logic does not automatically assume that a universal statement implies existence.

Remark (Dependencies).

- Universal Affirmative Form
- Universal Negative Form
- Existential Formula

2.8 Reference

Toolkit: Predicate Logic Reference

Reference	Use
Common errors	Diagnose translation, scope, and inference mistakes.
Summary tables	Compare translation patterns, quantifier laws, categorical forms, and proof cues.

Remark (Exposition). This section collects compact reference material for translation, quantifier negation, common inference errors, and proof strategy cues.

Common Errors and Fallacies

Remark (Translation and proof errors).

Error	Incorrect form	Corrected for
Confusing $\forall$ and $\exists$	$\exists x\, P(x)$ for "every x has P"	$\forall x\, P(x)$
Universal conjunction	$\forall x\,(A(x) \land B(x))$	$\forall x\,(A(x) \to B($
Existential implication	$\exists x\,(A(x) \to B(x))$	$\exists x\,(A(x) \land B(x$
Scope error	$\exists y \forall x\, R(x, y)$	$\forall x \exists y\, R(x, y)$, each x may h own y
Variable capture	$\varphi[t/x]$ without checking bound variables	Rename boun ables before s tion
Free variable left in a sentence	$\forall x\, R(x, y)$ as a closed claim	$\forall x \forall y\, R(x, y)$, o as intended
Affirming the consequent	$P \to Q,\ Q\ \therefore P$	Invalid; use a terexample va or structure
Denying the antecedent	$P \to Q,\ \neg P\ \therefore \neg Q$	Invalid; use a terexample va or structure
Invalid quantifier reversal	$\forall x \exists y\, R(x, y) \Rightarrow \exists y \forall x\, R(x, y)$	Invalid withou hypotheses

Remark (Failure-mode checklist). Before accepting a translation or inference, check the domain, each symbol reading, every bound-variable scope, and whether the final formula is intended to be a sentence. Then test suspicious implications by trying to build one case where the premise is true and the conclusion false.

Summary Tables

Remark (English phrase to logical pattern).

English phrase	Logical pattern
All A are B	$\forall x\,(A(x) \to B(x))$
Some A are B	$\exists x\,(A(x) \land B(x))$
No A are B	$\forall x\,(A(x) \to \neg B(x))$
Some A are not B	$\exists x\,(A(x) \land \neg B(x))$
Every A has a B	$\forall x\,(A(x) \to \exists y\,(B(y) \land R(x, y$
There is a B for every A	$\exists y\,(B(y) \land \forall x\,(A(x) \to R(x, y$

Remark (Quantifier negation laws).

$$\begin{aligned}
\neg\forall x\,\varphi &\iff \exists x\,\neg\varphi,\\
\neg\exists x\,\varphi &\iff \forall x\,\neg\varphi,\\
\neg\forall x\exists y\,\varphi &\iff \exists x\forall y\,\neg\varphi,\\
\neg\exists x\forall y\,\varphi &\iff \forall x\exists y\,\neg\varphi.
\end{aligned}$$

Remark (Categorical form translations).

Form	Name	Translation
A	Universal affirmative	$\forall x\,(A(x) \to B(x))$
E	Universal negative	$\forall x\,(A(x) \to \neg B(x))$
I	Particular affirmative	$\exists x\,(A(x) \land B(x))$
O	Particular negative	$\exists x\,(A(x) \land \neg B(x))$

Remark (Proof strategy cues).

Goal or premise	Useful cue
Prove $\forall x\,\varphi(x)$	Let x be arbitrary, then prove $\varphi(x)$.
Use $\forall x\,\varphi(x)$	Instantiate it with a term that is free for x.
Prove $\exists x\,\varphi(x)$	Produce a witness and verify φ.
Use $\exists x\,\varphi(x)$	Introduce a fresh witness and keep its scop trolled.
Disprove $\forall x\,\varphi(x)$	Find a counterexample satisfying $\neg\varphi$.
Disprove an implication	Make the hypothesis true and the conclusion f

Chapter 3

Many-Sorted Logic

SERIES	VOLUME I	BOOK I	CHAPTER 3
From Cantor to Ito	Logic, Sets, and Proof	Mathematical Logic and Proof	Many-Sorted Logic

3.1 Introduction

3.1.1 Examples

3.1.2 Reduction to and comparison with first-order logic

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3.1.4 Many-sorted logic as a unifier logic

3.2 Structures

3.2.1 Definition (signature)

3.2.2 Definition (structure)

3.3 Formal many-sorted language

3.3.1 Alphabet

3.3.2 Expressions: formulas and terms

3.3.3 Remarks on notation

3.3.4 Abbreviations

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3.4 Semantics

3.4.1 Definitions

3.4.2 Satisfiability, validity, consequence and logical equivalence

3.5 Substitution of a term for a variable

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Chapter 4

Standard Second-Order Logic

SERIES	VOLUME I	BOOK I	CHAPTER 4
From Cantor to Ito	Logic, Sets, and Proof	Mathematical Logic and Proof	Standard Second-Order Logic

Chapter 5

Languages, Axioms, and Models

Languages, Axioms, and Models Toolkit - Planned

This section is a rebuild placeholder.

This section is planned for the governance rebuild. No mathematical content has been restored here yet.

5.1 Signatures

Signatures Toolkit — Quick Reference

Concept	Role
Signature	non-logical vocabulary
Language	signature together with logical syntax
Constant symbol	names a distinguished object syntactically
Function symbol	names an operation syntactically
Relation symbol	names a relation syntactically
Arity	records number of input places

Signatures

Signature Presentation: A Simple Arithmetic Language

Let

$$\Sigma_{\text{arith}} = \big(\mathcal{C}_{\text{arith}}, \mathcal{F}_{\text{arith}}, \mathcal{R}_{\text{arith}}, \text{ar}\big)$$

where

$$\mathcal{C}_{\text{arith}} = \{0, 1\}, \qquad \mathcal{F}_{\text{arith}} = \{+, \cdot, S\}, \qquad \mathcal{R}_{\text{arith}} = \{<\}.$$

The arity function is specified by

$$\operatorname{ar}(0) = \operatorname{ar}(1) = 0, \qquad \operatorname{ar}(S) = 1, \qquad \operatorname{ar}(+) = \operatorname{ar}(\cdot) = 2, \qquad \operatorname{ar}(<) = 2.$$

Symbol	Kind	Arity	Intended Reading
$0, 1$	constants	0	distinguished constant names
S	function symbol	1	successor operation symbol
$+, \cdot$	function symbols	2	binary operation symbols
$<$	relation symbol	2	binary order relation symbol

The signature records only the available non-logical symbols and their arities. It does not assert any axioms about those symbols.

Definitional Root (First-Order Signature)

Definition 5.1.1 (First-Order Signature)**.** A *first-order signature* is a tuple

$$\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \operatorname{ar}),$$

where $\mathcal{C}$ is a set of constant symbols, $\mathcal{F}$ is a set of function symbols, $\mathcal{R}$ is a set of relation symbols, and

$$\operatorname{ar} : \mathcal{F} \cup \mathcal{R} \to \mathbb{Z}_{>0}$$

assigns to each function or relation symbol its number of arguments.

Remark (Standard quantified statement). For every $s \in \mathcal{F} \cup \mathcal{R}$, the value $\operatorname{ar}(s)$ is a positive integer. If $s \in \mathcal{F}$, then s forms terms with $\operatorname{ar}(s)$ inputs; if $s \in \mathcal{R}$, then s forms atomic formulas with $\operatorname{ar}(s)$ inputs.

Remark (Interpretation). A signature is the formal vocabulary available before any axioms are chosen. It says which symbols may occur and how many input places each symbol has.

Definition (Arity)

Definition 5.1.2 (Arity)**.** Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \operatorname{ar})$ be a first-order signature. The *arity function* is the map

$$\operatorname{ar} : \mathcal{F} \cup \mathcal{R} \to \mathbb{Z}_{>0}$$

that assigns to each function or relation symbol the number of arguments it requires. When constants are treated as 0-ary function symbols, the codomain is enlarged to include 0.

Remark (Standard quantified statement). For every $s \in \mathcal{F} \cup \mathcal{R}$, there is a unique positive integer n such that $\mathrm{ar}(s) = n$. If $f \in \mathcal{F}$ and $\mathrm{ar}(f) = n$, then an expression $f(t_1, \ldots, t_k)$ is a well-formed term exactly when $k = n$. If $R \in \mathcal{R}$ and $\mathrm{ar}(R) = n$, then an expression $R(t_1, \ldots, t_k)$ is a well-formed atomic formula exactly when $k = n$.

Remark (Interpretation). Arity is bookkeeping for grammatical well-formedness. It determines whether a symbol has been supplied with the correct number of arguments.

Remark (Dependencies).

- Signature

Definitional Framework (Language)

Definition 5.1.3 (Language)**.** A *first-order language* has two standard readings. As vocabulary, it is the same data as a signature,

$$\mathcal{L} = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \mathrm{ar}).$$

As a formal language generated by a signature Σ, it is the set of well-formed strings built from the alphabet

$$A = \Sigma \cup V \cup \Lambda,$$

where V is a countable set of variables and Λ is the fixed stock of logical symbols and punctuation. In this second reading,

$$\mathcal{L}(\Sigma) = \{s \in A^* \mid s \in \mathrm{Terms}(\Sigma) \text{ or } s \in \mathrm{WFF}(\Sigma)\}.$$

Remark (Standard quantified statement). The vocabulary reading records which non-logical symbols may occur. The generated-language reading then selects from all finite strings over A only the terms and well-formed formulas admitted by the formation rules determined by Σ.

Remark (Interpretation). The signature supplies the subject-specific vocabulary. The language adds the logical grammar needed to form expressions from that vocabulary.

Remark (Dependencies).

- Signature

Definition (Constant Symbol)

Definition 5.1.4 (Constant Symbol)**.** A *constant symbol* is a non-logical symbol that behaves syntactically as a 0-ary function symbol. In the streamlined convention where constants are included among function symbols, this is the condition

$$\mathrm{Constant}(c) \iff c \in \mathcal{F} \text{ and } \mathrm{ar}(c) = 0.$$

Because its arity is 0, the symbol c is already a well-formed term and takes no input terms.

Remark (Standard quantified statement). If $c \in \mathcal{F}$ and $\mathrm{ar}(c) = 0$, then $c \in \mathrm{Terms}(\mathcal{L})$. In a structure $\mathfrak{M} = (M, \mathcal{I})$, the interpretation of a constant is a unique element of the domain:

$$\forall c \in \mathcal{C},\ \exists! e \in M,\ \mathcal{I}(c) = e.$$

Remark (Interpretation). A constant symbol is syntactically rigid: unlike a variable, it does not vary with an assignment. In the arithmetic signature, the symbol 0 is a piece of formal syntax; under the standard interpretation, it denotes the number zero.

Remark (Dependencies).

- Signature
- Arity

Definition (Function Symbol)

Definition 5.1.5 (Function Symbol)**.** A *function symbol* is a non-logical symbol $f \in \mathcal{F}$ whose assigned arity determines how many terms it must take as inputs. If $\mathrm{ar}(f) = n$, then f is an n-ary term constructor: it takes exactly n terms and produces a new term, not a truth value.

Remark (Standard quantified statement). For every $f \in \mathcal{F}$, if $\mathrm{ar}(f) = n$, then

$$f(t_1, \ldots, t_k) \in \mathrm{Terms}(\mathcal{L}) \iff k = n$$

whenever $t_1, \ldots, t_k$ are terms. In a structure $\mathfrak{M} = (M, \mathcal{I})$, the interpretation of f is a total operation closed on the domain:

$$\mathcal{I}(f) : M^n \to M.$$

Remark (Interpretation). A function symbol is operation-forming syntax. In the arithmetic signature, S is unary, while $+$ and $\cdot$ are binary. Under an interpretation, each becomes an actual operation on the chosen domain.

Remark (Dependencies).

- Signature
- Arity

Definition (Relation Symbol)

Definition 5.1.6 (Relation Symbol)**.** A *relation symbol*, or predicate symbol, is a non-logical symbol $R \in \mathcal{R}$ whose assigned arity determines how many terms it must take as inputs. If $\mathrm{ar}(R) = n$, then R is an n-ary formula constructor: it takes exactly n terms and produces an atomic formula, hence something with a truth value rather than an object.

Remark (Standard quantified statement). For every $R \in \mathcal{R}$, if $\operatorname{ar}(R) = n$, then

$$R(t_1, \ldots, t_k) \in \mathrm{WFF}(\mathcal{L}) \iff k = n$$

whenever $t_1, \ldots, t_k$ are terms. In a structure $\mathfrak{M} = (M, \mathcal{I})$, the interpretation of R is a subset of n-tuples from the domain:

$$\mathcal{I}(R) \subseteq M^n.$$

Thus $R(t_1, \ldots, t_n)$ is true exactly when the tuple of interpreted terms lies in $\mathcal{I}(R)$.

Remark (Interpretation). A relation symbol is a statement-forming symbol once it is applied to the right number of terms. In the arithmetic signature, $+$ is a binary function symbol because it returns a number, while $<$ is a binary relation symbol because it returns a truth value. Under the standard interpretation, $<$ denotes the set of ordered pairs (a, b) such that a is less than b.

Remark (Dependencies).

- Signature
- Arity

Remark (Structural remarks). The same signature may support different choices of axioms later in the chapter. For example, a single arithmetic vocabulary can be used with stronger or weaker axiom sets.

The same signature may also be read over different mathematical universes once interpretations are introduced. At this stage, the signature itself records only the common vocabulary; it does not choose any particular interpretation.

5.2 Axioms

Axioms Toolkit — Quick Reference

Concept	Role
Axiom	formal assertion adopted without proof
Axiom system	collection of assertions in one language
Assumption	temporary assertion for reasoning
Consistency	absence of contradiction
Theory	deductive closure of axioms

Axioms

Axiom Presentation: A Simple Arithmetic Axiom System

Reuse the arithmetic signature

$$\Sigma_{\text{arith}} = \big(\{0, 1\}, \{+, \cdot, S\}, \{<\}, \text{ar}\big)$$

from the Signatures section. The signature supplies the symbols. An axiom system supplies assertions formed with those symbols.

For example, the following are familiar arithmetic-style axioms:

$$\forall x\ (x + 0 = x),$$

$$\forall x\ (S(x) \neq 0),$$

$$\forall x \forall y\ (S(x) = S(y) \to x = y).$$

These examples are illustrative. The signature says that 0, $+$, and S are available symbols. The axioms assert formal relationships involving those symbols.

Remark (Structural remarks). **Vocabulary vs assertion.** A signature is vocabulary: it specifies what may be said. An axiom is an assertion: it specifies what is adopted as true within that vocabulary. The signature supplies symbols; the axioms supply assertions involving those symbols.

Same signature, different axiom systems. A single signature may support multiple collections of axioms. For example, geometric languages can be used for Euclidean or non-Euclidean axiom systems. The usual group signature can be used for group theory, while adding the commutativity axiom gives the stronger axiom system for abelian groups. The vocabulary can remain fixed while the assertions change.

Forward conceptual roadmap. The chapter proceeds from collections of adopted assertions to their deductive and semantic roles:

$$\begin{array}{c}\text{Axiom System}\\ \Downarrow \\ \text{Theory}\\ \Downarrow \\ \text{Models}\end{array}$$

The present section introduces the assertion layer. Later sections formalize the theory generated by axioms and the models in which those assertions hold.

Definition (Axiom)

Definition 5.2.1 (Axiom)**.** Let $\mathcal{L}$ be a first-order language, and let $\mathrm{Sent}(\mathcal{L})$ denote the set of sentences of $\mathcal{L}$. If $T \subseteq \mathrm{Sent}(\mathcal{L})$ is a theory, then an *axiom of T* is an element $\alpha \in T$.

Remark (Standard quantified statement). For every first-order language $\mathcal{L}$ and every theory $T \subseteq \mathrm{Sent}(\mathcal{L})$, a sentence α is an axiom of T exactly when $\alpha \in T$.

Remark (Interpretation). An axiom is not part of the vocabulary itself. It is an assertion made using the vocabulary of a language. The signature determines which symbols may appear; the axiom states something with those symbols.

Remark (Proof-theoretic reading). In a deductive presentation, an axiom may be used as a derivation with no undischarged premises:

$$\frac{}{\vdash \alpha}.$$

Thus axioms are the starting assertions of the proof system.

Remark (Dependencies).

- Language

Definition (Axiom System)

Definition 5.2.2 (Axiom System)**.** An *axiom system* is a collection of axioms expressed in a common language.

Remark (Interpretation). An axiom system records the assertions chosen for a mathematical setting. Two axiom systems may use the same language while adopting different assertions. The shared language fixes the vocabulary; the axiom system fixes the adopted formal claims.

Remark (Dependencies).

- Axiom
- Language

Definition (Assumption)

Definition 5.2.3 (Assumption)**.** An *assumption* is a statement temporarily accepted for purposes of reasoning.

Remark (Interpretation). An axiom is adopted as part of the formal background of a system. An assumption is local to an argument: it may be introduced to prove a conditional, to reason within a case, or to derive a contradiction. Axioms belong to the system; assumptions belong to the proof context.

Remark (Dependencies).

- Axiom

5.3 Theories

Theories Toolkit — Quick Reference

Concept	Role
Theory	all consequences generated by axioms
Theorem	statement derivable from axioms
Consequence	statement following from a theory
Derivation	chain of justified steps
Deductive closure	all derivable statements

Theories

Theory Presentation: Arithmetic Consequences

Start with the arithmetic-style axiom system from the previous section. It uses the arithmetic signature and includes assertions such as

$$\forall x\ (x + 0 = x),$$

$$\forall x\ (S(x) \neq 0),$$

$$\forall x \forall y\ (S(x) = S(y) \to x = y).$$

Those axioms are the starting assertions. From them, together with other arithmetic axioms and ordinary logical reasoning, additional statements may be established. For instance, one may later prove new equations, cancellation facts, uniqueness facts, and order facts that were not listed as axioms.

A theory contains the original axioms. A theory also contains the statements derivable from those axioms. Thus the axioms are the generators, while the theory is the full collection of what those generators imply.

Remark (Structural remarks). **Axioms vs theory.** Axioms are starting assertions. A theory includes all logical consequences of those assertions. The distinction is the next step after the vocabulary and assertion distinction: a signature supplies vocabulary, axioms supply adopted assertions, and a theory contains everything derivable from those assertions.

Finite axioms, infinite theory. A small axiom system can generate an enormous collection of consequences. The listed axioms may be finite, but the statements derivable from them need not be. This is why a theory is usually larger than the axiom system that presents it.

Forward roadmap. The next conceptual step is semantic:

$$\begin{array}{c}\text{Theory}\\ \Downarrow\\ \text{Models}\end{array}$$

A model will later be introduced as a mathematical structure in which the statements of a theory hold.

Definition (Theory)

Definition 5.3.1 (Theory)**.** A *theory* over a first-order language $\mathcal{L}$ is a collection of closed sentences in that language:

$$T \subseteq \text{Sentences}(\mathcal{L}).$$

If Γ is a chosen set of axioms, the theory generated by Γ is the deductive closure

$$T = \{\varphi \in \text{Sentences}(\mathcal{L}) \mid \Gamma \vdash \varphi\}.$$

Remark (Standard quantified statement)*.* Syntactically, a sentence φ belongs to the theory generated by Γ exactly when φ is derivable from Γ. Semantically, for an $\mathcal{L}$-structure $\mathfrak{M}$, the complete theory of $\mathfrak{M}$ is

$$\text{Th}(\mathfrak{M}) = \{\varphi \in \text{Sentences}(\mathcal{L}) \mid \mathfrak{M} \models \varphi\}.$$

For every sentence φ, either $\mathfrak{M} \models \varphi$ or $\mathfrak{M} \models \neg\varphi$, so $\text{Th}(\mathfrak{M})$ decides every sentence of the language.

Remark (Interpretation)*.* A theory is not merely the list of axioms. It is the closure of those axioms under derivability: every statement that follows from the axioms belongs to the theory. The axioms start the system; the theory records everything the system proves.

Remark (Dependencies)*.*

- Language
- Axiom System

Definition (Theorem)

Definition 5.3.2 (Theorem)**.** A *theorem* of a theory T is a sentence φ for which there is a finite formal derivation from T, written

$$T \vdash \varphi.$$

Equivalently, in proof-theoretic terms, φ is a theorem of T exactly when some derivation has leaves drawn from logical axioms or sentences of T, uses valid inference rules, and has φ as its root.

Remark (Standard quantified statement)*.* Syntactically,

$$\text{Theorem}(T, \varphi) \iff \exists d\ (d : T \vdash \varphi).$$

Semantically, φ is entailed by T when it is true in every model of T:

$$T \models \varphi \iff \forall \mathfrak{M}\, \big(\mathfrak{M} \models T \implies \mathfrak{M} \models \varphi\big).$$

For first-order logic, the completeness theorem connects these readings: $T \vdash \varphi$ exactly when $T \models \varphi$.

Remark (Interpretation). Within an axiom system, a theorem is not an additional starting assumption. It is a statement reached from the adopted axioms by reasoning. Axioms are given; theorems are obtained.

Remark (Dependencies).

- Theory

Definition (Consequence)

Definition 5.3.3 (Consequence)**.** A sentence φ is a *semantic consequence* of a set of premises Γ, written $\Gamma \vDash \varphi$, if every model of Γ is also a model of φ. Equivalently, there is no structure in which all sentences of Γ are true while φ is false.

Remark (Standard quantified statement). Semantic consequence is the universal condition

$$\Gamma \vDash \varphi \iff \forall \mathfrak{M} \Big((\forall \gamma \in \Gamma,\ \mathfrak{M} \vDash \gamma) \implies \mathfrak{M} \vDash \varphi \Big).$$

Syntactic consequence, written $\Gamma \vdash \varphi$, means that there exists a finite valid deduction of φ from Γ. For first-order logic, the completeness theorem says that $\Gamma \vDash \varphi$ exactly when $\Gamma \vdash \varphi$.

Remark (Interpretation). Consequence is the relation that connects a theory to what it implies. If the axioms generate a theory, then consequences are the statements carried along by those axioms and by the statements already obtained from them.

Remark (Dependencies).

- Theory

Definition (Deductive Closure)

Definition 5.3.4 (Deductive Closure)**.** Let $\Gamma \subseteq \mathrm{Sentences}(\mathcal{L})$ be a set of premises. The *deductive closure* of Γ, denoted $\mathrm{Cn}(\Gamma)$, is the set of all sentences derivable from Γ:

$$\mathrm{Cn}(\Gamma) = \{\varphi \in \mathrm{Sentences}(\mathcal{L}) \mid \Gamma \vdash \varphi\}.$$

Remark (Standard quantified statement). A set of sentences T is deductively closed exactly when

$$T = \mathrm{Cn}(T),$$

or equivalently, for every sentence φ,

$$T \vdash \varphi \implies \varphi \in T.$$

The closure operator satisfies the standard Tarskian closure laws:

$$\Gamma \subseteq \mathrm{Cn}(\Gamma), \qquad \Gamma \subseteq \Delta \implies \mathrm{Cn}(\Gamma) \subseteq \mathrm{Cn}(\Delta), \qquad \mathrm{Cn}(\mathrm{Cn}(\Gamma)) = \mathrm{Cn}(\Gamma).$$

By completeness for first-order logic, this syntactic closure agrees with the semantic closure

$$\{\varphi \in \mathrm{Sentences}(\mathcal{L}) \mid \Gamma \vDash \varphi\}.$$

Remark (Interpretation). Deductive closure is the operation that turns an axiom system into the full theory it generates. Starting with the axioms, close under derivability; the result is the theory determined by those starting assertions.

Remark (Dependencies).

- Axiom System

5.4 Models

Models Toolkit — Quick Reference

Concept	Role
Interpretation	assigns meanings to symbols
Structure	domain with interpreted symbols
Model	structure satisfying the theory
Satisfaction	truth of a statement in a structure
Countermodel	model where a statement fails

Free Σ-Algebra and Evaluation

Depends on: `Signature; inductively generated set; unique readability`

Parameter

Shared bridge machinery. Parameter: a signature Σ. This card is instantiated by connective evaluation, term evaluation, parsing, and later by initial-algebra presentations such as $\mathbb{N}$.

Structure

$$\Sigma \text{ is a set of operation symbols,} \qquad \mathrm{ar} : \Sigma \to \mathbb{N}.$$

A Σ-algebra is a carrier $|\mathcal{A}|$ with operations

$$f^{\mathcal{A}} : |\mathcal{A}|^{\mathrm{ar}(f)} \to |\mathcal{A}| \qquad (f \in \Sigma).$$

Seeds land in the carrier; homomorphisms are maps of algebras.

Structure

$$T_\Sigma(X), \qquad f^T(t_1, \ldots, t_n) := f(t_1, \ldots, t_n), \qquad \eta : X \to T_\Sigma(X).$$

Syntax–Semantics Bridge

$$\mathrm{Hom}(h, \mathcal{A}, \mathcal{B}) \quad :\Longleftrightarrow \quad h(f^{\mathcal{A}}(\bar{a})) = f^{\mathcal{B}}(h(a_1), \ldots, h(a_n))$$

for every $f \in \Sigma$ and $\bar{a} \in |\mathcal{A}|^n$.

Axioms and Laws

No junk.

$$t \in T_\Sigma(X) \Longrightarrow t \in X \;\vee\; \exists f \in \Sigma\, \exists \bar{t}\, \big(t = f(\bar{t})\big).$$

No confusion.

$$x \neq f(\bar{t}), \qquad f(\bar{s}) = g(\bar{t}) \Longrightarrow f = g \wedge \bar{s} = \bar{t}.$$

Theorems and Consequences

1. **Universal property.**

$$\forall \mathcal{A}\, \forall \sigma : X \to |\mathcal{A}|\; \exists! \widehat{\sigma} : T_\Sigma(X) \to |\mathcal{A}|$$

$$\mathrm{Hom}(\widehat{\sigma}, T_\Sigma(X), \mathcal{A}) \wedge \widehat{\sigma} \circ \eta = \sigma.$$

2.

$$\mathrm{eval}(\mathcal{A})(\sigma) := \widehat{\sigma}.$$

Instances

use	Σ	X	target $\mathcal{A}$	seed σ
formula eval	Ω	atoms	$\mathbf{2}$	atom truth values
parse	Ω	X	$T_\Omega(X)$	η
term eval	Σ_{term}	Var	$\mathcal{A}_{\text{term}}$	assignment s
$\mathbb{N}$ preview	$\{0^{(0)}, S^{(1)}\}$	$\varnothing$	(C, c_0, succ)	none

Instances

$$\Omega = \{\neg^{(1)}, \wedge^{(2)}, \vee^{(2)}, \to^{(2)}, \leftrightarrow^{(2)}\},$$

$$\mathbf{2} = (\{0,1\}, \neg^{\mathbf{2}}, \wedge^{\mathbf{2}}, \vee^{\mathbf{2}}, \to^{\mathbf{2}}, \leftrightarrow^{\mathbf{2}}).$$

Boundary

The machine fixes one seed. Connective and term evaluation are exact instances. FOL quantifiers range over the assignment-indexed family $(\sigma_s)_s$:

$$\mathcal{A}, s \models \exists x\, \varphi \Longleftrightarrow \exists a \in A,\; \mathcal{A}, s[x \mapsto a] \models \varphi.$$

That is the downstream cylindric/polyadic layer. The absolutely free $T_\Sigma(X)$ is also distinct from the Lindenbaum–Tarski quotient by $\vdash$-equivalence.

Consequence

Depends on: `Satisfaction relation` $\models \subseteq \mathsf{Mod} \times \mathsf{Form}$

Parameter

A class of models Mod and a satisfaction relation

$$m \models \varphi \qquad (m \in \mathsf{Mod}).$$

Syntax–Semantics Bridge

$$\Gamma \models \varphi \quad :\Longleftrightarrow \quad \forall m \in \mathsf{Mod}\,\big((\forall \gamma \in \Gamma,\ m \models \gamma) \to m \models \varphi\big).$$

$$\mathrm{Cn}_{\models}(\Gamma) := \{\varphi : \Gamma \models \varphi\}, \qquad \models \varphi :\Longleftrightarrow \varnothing \models \varphi.$$

$$\mathrm{Th}(m) := \{\varphi : m \models \varphi\}.$$

Binding

logic	Mod	$m \models \varphi$
prop	valuations v	$\widehat{v}(\varphi) = 1$
FOL open	pairs $(\mathcal{A}, s)$	$\mathcal{A}, s \models \varphi$
FOL sentences	structures $\mathcal{A}$	$\mathcal{A} \models \sigma$

Models

Model Presentation: Arithmetic as a Model

Reuse the arithmetic signature and arithmetic-style axioms from the previous sections. A model begins by interpreting the symbols of that signature in an actual mathematical structure.

$$\begin{aligned} \text{Domain} &= \text{the natural numbers},\\ 0 &\mapsto \text{the number zero},\\ S &\mapsto \text{the successor operation},\\ + &\mapsto \text{addition}. \end{aligned}$$

In this structure, the statement

$$\forall x\,(x + 0 = x)$$

holds because adding zero to any natural number gives back that same natural number. The formal symbol 0 is only a symbol until it is interpreted as the number zero. The formal symbol + is only a symbol until it is interpreted as addition. The symbols acquire meaning only after interpretation.

Remark (Structural remarks). **Syntax vs meaning.** Signatures, axioms, and theories are formal objects. They belong to syntax: they specify vocabulary, adopted assertions, and consequences. Models provide semantic meaning by assigning mathematical content to the symbols and by determining which statements hold.

One theory, many models. The same theory may have multiple models. Euclidean geometry may be realized in different coordinate systems. The group axioms have many models, including different concrete groups. The theory fixes the assertions; a model is one mathematical realization in which those assertions hold.

Countermodels. A statement fails to follow from a theory if there is a model of the theory in which that statement is false. Countermodels are therefore evidence that a claim is not forced by the axioms.

Forward roadmap. The next sections use models to explain consistency and independence:

$$\begin{array}{c}\text{Models}\\ \Downarrow\\ \text{Consistency}\\ \Downarrow\\ \text{Independence}\end{array}$$

Definition (Interpretation)

Definition 5.4.1 (Interpretation)**.** Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \mathrm{ar})$ be a first-order signature, and let M be a nonempty set. An *interpretation of* Σ *on* M is an assignment $\mathcal{I}$ that maps each symbol of Σ to concrete data over M: constants are assigned elements of M, function symbols are assigned operations on M, and relation symbols are assigned relations on M.

Remark (Standard quantified statement). For every $c \in \mathcal{C}$, $\mathcal{I}(c) \in M$. For every $f \in \mathcal{F}$, if $\mathrm{ar}(f) = n$, then

$$\mathcal{I}(f) : M^n \to M.$$

For every $R \in \mathcal{R}$, if $\mathrm{ar}(R) = n$, then

$$\mathcal{I}(R) \subseteq M^n.$$

Remark (Interpretation). An interpretation turns formal vocabulary into mathematical content. It tells what each constant, function symbol, and relation symbol means in a particular setting.

Remark (Dependencies).

- Signature

Definition ($\mathcal{L}$-Structure)

Definition 5.4.2 ($\mathcal{L}$-Structure)**.** Let $\mathcal{L}$ be a first-order signature. An $\mathcal{L}$-*structure* is a pair

$$\mathfrak{M} = (M, \mathcal{I}),$$

where M is a nonempty domain and $\mathcal{I}$ is an interpretation of the symbols of $\mathcal{L}$ on M.

Remark (Standard quantified statement). For every first-order signature $\mathcal{L}$, an $\mathcal{L}$-structure consists of a nonempty domain M and an interpretation $\mathcal{I}$ such that every constant symbol is assigned an element of M, every n-ary function symbol is assigned a function $M^n \to M$, and every n-ary relation symbol is assigned a relation on M^n.

Remark (Interpretation). A structure is the environment in which formal symbols receive meaning. The domain supplies the objects, and the interpretation explains how the symbols refer to objects, operations, and relations on that domain.

Remark (Type-theoretic reading). In dependent type theory, this bundled data can be viewed schematically as

$$\sum_{M:\mathrm{Type}} \left(\prod_{c\in\mathcal{C}} M\right) \times \left(\prod_{f\in\mathcal{F}} (M^{\mathrm{ar}(f)} \to M)\right) \times \left(\prod_{R\in\mathcal{R}} (M^{\mathrm{ar}(R)} \to \mathrm{Prop})\right).$$

This is the same semantic package represented by a Lean structure carrying a carrier type, interpretations of constants and functions, and predicates for relations.

Remark (Dependencies).

- Signature
- Interpretation

Definition (Model)

Definition 5.4.3 (Model)**.** Let $\mathcal{L}$ be a first-order signature, let T be a theory over $\mathcal{L}$, and let $\mathfrak{M} = (M, \mathcal{I})$ be an $\mathcal{L}$-structure. The structure $\mathfrak{M}$ is a *model of* T, written $\mathfrak{M} \vDash T$, if every sentence in T is true in $\mathfrak{M}$.

Remark (Standard quantified statement). For every theory T over $\mathcal{L}$ and every $\mathcal{L}$-structure $\mathfrak{M}$,

$$\mathfrak{M} \vDash T \iff \forall \varphi \in T,\ \mathfrak{M} \vDash \varphi.$$

Remark (Interpretation). A model is a structure that makes the theory true. It is not just an interpretation of the vocabulary; it is an interpretation in which the adopted assertions and their consequences hold.

Remark (Dependencies).

- Structure
- Theory

Definition (Satisfaction)

Definition 5.4.4 (Satisfaction)**.** Let $\mathfrak{M} = (M, \mathcal{I})$ be an $\mathcal{L}$-structure. A *variable assignment* is a function $s : \mathrm{Var} \to M$. The assignment extends recursively to a term-evaluation map $\bar{s}$ by

$$\bar{s}(x) = s(x), \qquad \bar{s}(c) = \mathcal{I}(c),$$

and

$$\bar{s}\big(f(t_1, \ldots, t_n)\big) = \mathcal{I}(f)\big(\bar{s}(t_1), \ldots, \bar{s}(t_n)\big).$$

The satisfaction relation $\mathfrak{M} \vDash \varphi[s]$ is then defined by induction on formulas: atomic relations are tested inside the interpreted relation, equality is interpreted as equality in M, Boolean connectives are interpreted by the corresponding truth operations, and quantifiers range over all elements of M.

Remark (Standard quantified statement). For terms $t_1, \ldots, t_n$ and an n-ary relation symbol R,

$$\mathfrak{M} \vDash R(t_1, \ldots, t_n)[s] \iff \big(\bar{s}(t_1), \ldots, \bar{s}(t_n)\big) \in \mathcal{I}(R).$$

For equality,

$$\mathfrak{M} \vDash (t_1 = t_2)[s] \iff \bar{s}(t_1) = \bar{s}(t_2).$$

For quantifiers,

$$\mathfrak{M} \vDash (\forall x\, \varphi)[s] \iff \forall a \in M,\ \mathfrak{M} \vDash \varphi[s[x \mapsto a]],$$

and

$$\mathfrak{M} \vDash (\exists x\, \varphi)[s] \iff \exists a \in M,\ \mathfrak{M} \vDash \varphi[s[x \mapsto a]].$$

Remark (Interpretation). Satisfaction is the semantic test for a statement. Once the symbols have been interpreted in a structure, the statement can be checked as true or false in that setting.

Remark (Sentences). If φ is a sentence, then it has no free variables, so its truth does not depend on the initial assignment s. In that case the notation is shortened from $\mathfrak{M} \vDash \varphi[s]$ to $\mathfrak{M} \vDash \varphi$. This is the bridge from satisfaction of a single sentence to the definition of a model of a theory.

Remark (Dependencies).

- Structure

Definition (Countermodel)

Definition 5.4.5 (Countermodel)**.** Let $\Gamma \subseteq \mathrm{Sent}(\mathcal{L})$ be a set of premises, let $\varphi \in \mathrm{Sent}(\mathcal{L})$, and let $\mathfrak{M}$ be an $\mathcal{L}$-structure. A *countermodel to the entailment* $\Gamma \vDash \varphi$ is a structure $\mathfrak{M}$ that satisfies every sentence in Γ but does not satisfy φ.

Remark (Standard quantified statement). For $\Gamma \subseteq \mathrm{Sent}(\mathcal{L})$ and $\varphi \in \mathrm{Sent}(\mathcal{L})$,

$$\mathrm{CounterModel}(\mathfrak{M}, \Gamma, \varphi) \iff \big(\forall \gamma \in \Gamma,\ \mathfrak{M} \vDash \gamma\big) \wedge \big(\mathfrak{M} \nvDash \varphi\big).$$

Equivalently, for classical closed sentences,

$$\mathrm{CounterModel}(\mathfrak{M}, \Gamma, \varphi) \iff \big(\forall \gamma \in \Gamma,\ \mathfrak{M} \vDash \gamma\big) \wedge \big(\mathfrak{M} \vDash \neg\varphi\big).$$

Remark (Interpretation). A countermodel shows that the theory does not force the statement. It keeps the axioms true while making the proposed statement false.

Remark (Special cases). If $\Gamma = \emptyset$, then a countermodel to validity of φ is a structure $\mathfrak{M}$ such that $\mathfrak{M} \vDash \neg\varphi$. For a theory T, models of T satisfying φ and models of T satisfying $\neg\varphi$ witness the independence pattern: the theory alone does not settle φ.

Remark (Lean reading). In a mechanized setting, invalidity is witnessed by constructing semantic data:

$$\exists \mathfrak{M}\ \big((\forall \gamma \in \Gamma,\ \mathfrak{M} \vDash \gamma) \wedge \mathfrak{M} \vDash \neg\varphi\big).$$

The constructed structure and satisfaction proofs are the certificate that the entailment cannot be valid.

Remark (Dependencies).

- Model
- Satisfaction

5.5 Consistency

Consistency Toolkit — Quick Reference

Concept	Role
Contradiction	assertion that cannot be true in an interpretation
Consistency	absence of contradiction
Inconsistency	contradiction in or from axioms
Model existence	evidence that axioms fit together
Relative consistency	consistency transferred from another theory

Consistency

Consistency Presentation: A Simple Example

Consider two informal axiom systems.

Example A.

- All objects are red.
- Some object is not red.

These assertions cannot simultaneously hold. The first assertion says every object has the property; the second says at least one object lacks it.

Example B.

- All objects are red.
- Object a is red.

These assertions can hold simultaneously. The second assertion agrees with what the first assertion already requires.

These examples are only for intuition. A consistent axiom system is one whose assertions fit together without contradiction. An inconsistent axiom system is one whose assertions cannot be held together coherently.

Remark (Structural remarks). **Why contradictions matter.** Contradictory systems are unusable because their assertions do not fit together. If a system both requires and forbids the same situation, then it no longer gives a coherent basis for reasoning.

Models as evidence. If a model exists, the axioms hold together coherently in at least one interpretation. A model is therefore evidence that the axioms are mutually compatible.

Absolute vs relative consistency. Most consistency results in mathematics are relative consistency results. They show that one theory is consistent provided some other background theory is consistent. This section records the idea only; the technical development belongs elsewhere.

Forward roadmap. The next section uses models and consistency to introduce independence:

$$\begin{array}{c}\text{Consistency}\\ \Downarrow \\ \text{Independence}\end{array}$$

Definition (Contradiction)

Definition 5.5.1 (Contradiction)**.** A *contradiction* is a statement that cannot be true under a given interpretation.

Remark (Interpretation). A contradiction marks a breakdown in compatibility. It identifies an assertion that cannot be made true in the interpretation under consideration.

Remark (Dependencies).

- Satisfaction

Definition (Consistent Axiom System)

Definition 5.5.2 (Consistent Axiom System)**.** A *consistent axiom system* is an axiom system that does not lead to contradiction.

Remark (Interpretation). Consistency says that the adopted assertions fit together. A consistent axiom system avoids forcing an impossible situation.

Remark (Dependencies).

- Axiom System
- Contradiction

Definition (Inconsistent Axiom System)

Definition 5.5.3 (Inconsistent Axiom System)**.** An *inconsistent axiom system* is an axiom system containing or producing contradictory assertions.

Remark (Interpretation). Inconsistency means that the adopted assertions cannot all be maintained coherently. The system asks for incompatible requirements at once.

Remark (Dependencies).

- Axiom System

- Contradiction

Definition (Model Existence Criterion (Informal))

Definition 5.5.4 (Model Existence Criterion (Informal))**.** Informally, if an axiom system has a model, then the axioms are mutually compatible.

Remark (Interpretation). This is an informal criterion in this section, not a formal theorem. The idea is that a model provides a setting in which all the axioms hold together, so the axioms have at least one coherent realization.

Remark (Dependencies).

- Model

Definition (Relative Consistency)

Definition 5.5.5 (Relative Consistency)**.** Let T_0 and T_1 be theories. The theory T_1 is *consistent relative to* T_0 if

$$\mathrm{Con}(T_0) \implies \mathrm{Con}(T_1).$$

Equivalently, every contradiction in T_1 would imply a contradiction in T_0.

Remark (Standard quantified statement). For theories T_0 and T_1,

$$\mathrm{RelCon}(T_1, T_0) \iff \big(\neg\exists d_0\,(d_0 : T_0 \vdash \bot)\big) \implies \big(\neg\exists d_1\,(d_1 : T_1 \vdash \bot)\big).$$

Equivalently, in contrapositive form,

$$\exists d_1\,(d_1 : T_1 \vdash \bot) \implies \exists d_0\,(d_0 : T_0 \vdash \bot).$$

Remark (Interpretation). Relative consistency does not establish consistency from nothing. It says that the target theory introduces no new contradiction unless the base theory was already inconsistent. In practice, a relative consistency proof often gives a translation that turns any contradiction derived in T_1 into a contradiction derived in T_0.

Remark (Dependencies).

- Consistent Axiom System

5.6 Independence

Independence Toolkit - Planned

This section is a rebuild placeholder.

This section is planned for the governance rebuild. No mathematical content has been restored here yet.

5.7 Comparing Theories

Comparing Theories Toolkit - Planned

This section is a rebuild placeholder.

This section is planned for the governance rebuild. No mathematical content has been restored here yet.

5.8 Examples And Previews

Examples And Previews Toolkit - Planned

This section is a rebuild placeholder.

This section is planned for the governance rebuild. No mathematical content has been restored here yet.

5.9 Summary

Summary Toolkit - Planned

This section is a rebuild placeholder.

This section is planned for the governance rebuild. No mathematical content has been restored here yet.

Chapter 6

Proof Techniques

6.1 Proof Architecture

The Two Questions Before You Write Anything. Before writing a single symbol, ask two questions in order.

Question 1. *What is the logical form of the statement?*
Read the claim and identify its outermost logical structure: universal, conditional, biconditional, existential, uniqueness, equality, or structural assertion.

Question 2. *What proof strategy does this form demand?*
The form determines the skeleton of the proof almost mechanically, before any mathematics is done.

Remark 6.1.1 (Why order matters — and why most students get it backwards). Question 2 cannot be answered before Question 1, but almost every beginner tries to answer it first. The default instinct is to reach for a familiar technique — proof by contradiction is the usual reflex — and start writing. This is the wrong order.

Reading the statement *is* the first mathematical act of the proof. The logical form of the statement is not preliminary bookkeeping; it is the information that determines the entire structure of the argument. A proof whose structure mismatches the statement's form is not just inelegant — it is often invalid or at best needlessly circular.

Until you are fluent enough to read the form automatically, slow down deliberately before every proof. Write the statement out. Identify its outermost connective. Then and only then choose a strategy.

Example 6.1.2 (Applying the two questions). **Statement.** *The identity element of a group is unique.*

Q1. Logical form. Strip the natural-language phrasing down to its logical skeleton. "The identity is unique" means: for every group G and for any two elements $e, e' \in G$ that both satisfy the identity axiom, we have $e = e'$. The outermost structure is a universal statement with a uniqueness conclusion: $\forall G,\ \forall e, e' \in G$, [both satisfy identity axiom] $\Rightarrow e = e'$.

Q2. Strategy. A universally quantified uniqueness claim demands the *assume-two-and-compare* pattern: introduce an arbitrary group G, assume two objects e and e' both satisfying the definition, and derive $e = e'$.

The strategy is forced entirely by the form. No creative insight is required to select it. The selection is a consequence of reading.

Remark 6.1.3 (What "logical form" means in practice). You do not need to write out the full first-order formula every time. You need to identify one thing: the *outermost* structure.

Is the statement "for all x, ..."? Then introduce an arbitrary x. Is it "P implies Q"? Then assume P. Is it "there exists x"? Then construct a witness. Is it "there exists a unique x"? Then split into existence and uniqueness. Is it "$A = B$ as sets"? Then prove double inclusion. Is it "$a = b$ as elements in an algebraic structure"? Then either do algebra or use satisfy-and-cite.

The statement map in the next section makes this correspondence explicit.

Statement Forms and Their Proof Strategies. The following table maps every common statement form to the proof strategy it demands. This is not a list of suggestions — it is a mechanical lookup. Once you have identified the outermost logical form of the statement you are trying to prove, the table tells you how to open your proof.

Statement Form	**Proof Strategy**	**Opening Move**
$\forall x,\ P(x)$	Introduce arbitrary element	"Let x be arbitrary."
$P \Rightarrow Q$	Direct proof	"Assume P."
$P \Rightarrow Q$	Contrapositive	"Assume $\neg Q$."
$P \Rightarrow Q$ (when negation more useful)	Contradiction	"Assume P and $\neg Q$."
$P \Leftrightarrow Q$	Two separate directions	Prove $P \Rightarrow Q$, then $Q \Rightarrow P$.
$\exists x,\ P(x)$	Construct a witness	"Define $x := \ldots$ and verify $P(x)$."
$\exists!\, x,\ P(x)$	Existence then uniqueness	Construct witness; then assume-two-and-compare.
$A = B$ (sets)	Double inclusion	Prove $A \subseteq B$ and $B \subseteq A$.
$a = b$ (algebraic elements)	Satisfy-and-cite or algebra	Show a satisfies the defining property of b.
$P(n)$ for all $n \in \mathbb{N}$	Induction	Base case; inductive step.
"X is unique"	Assume-two-and-compare	"Let x, y both satisfy the definition."

Remark 6.1.4 (Reading the table: defaults and alternatives). Some statement forms admit more than one strategy, and the table reflects this with multiple rows for the same form.

For $P \Rightarrow Q$, the default is direct proof: assume P and derive Q by forward reasoning. Contrapositive is reached for when $\neg Q$ is more useful than P as an assumption — this happens when the negation of the conclusion is a strong, concrete condition (for example, "n is odd" is concrete; "n is not even" is the same thing but sounds weaker). Contradiction is reached for when neither P alone nor $\neg Q$ alone gets you far, but combining them produces a quick absurdity.

The hierarchy is: try direct proof first. If the hypothesis is hard to use forward, try contrapositive. If neither works, try contradiction. Do not reach for contradiction as a default just because the proof feels difficult.

Remark 6.1.5 (Set equality versus element equality are genuinely different). The two equality rows in the table look similar but demand completely different techniques, and conflating them is a common error.

$A = B$ as *sets* is proved by double inclusion because sets are defined extensionally: two sets are equal if and only if they have the same elements. Proving $A = B$ by any other means — saying they are "clearly the same" or "follow from the definition" — is not a proof. The double inclusion must be made explicit.

$a = b$ as *elements in an algebraic structure* is different. Here b is typically a uniquely-determined object: an inverse, an identity, a limit, a fixed point. The strategy is to show a satisfies the same defining property as b, and then invoke the previously proved uniqueness theorem to conclude $a = b$. This is the satisfy-and-cite pattern (Section 6.1).

The mistake is trying to use double inclusion on algebraic elements, or trying to use algebraic manipulation on sets. The form of the objects determines which technique applies.

Remark 6.1.6 (The biconditional is always two separate proofs). A biconditional $P \Leftrightarrow Q$ is *never* proved in a single argument that runs from P to Q and "loops back." It is always proved in two distinct proofs:

($\Rightarrow$) Assume P. Prove Q.
($\Leftarrow$) Assume Q. Prove P.

Label the two directions explicitly. The two directions often require completely different techniques and different amounts of work. In characterisation theorems in analysis, one direction is typically a two-line unraveling of a definition while the other requires a nontrivial inequality argument. Treating them as one argument is how gaps get hidden.

The Five Proof Archetypes. Every mathematical proof, at the highest level of compression, is an instance of one of five archetypes. The statement map tells you the specific opening move. The archetypes tell you the *global shape* of the argument — what kind of reasoning is being done, and why it constitutes a valid proof of the claim.

The Five Proof Archetypes

1. **Construct something.** Exhibit a concrete object and verify it has every required property. This is the proof structure for all existence claims.

2. **Assume two and compare.** Assume two objects both satisfy a definition; derive that they are equal. This is the proof structure for all uniqueness claims.

3. **Take a minimal element.** Apply the well-ordering principle to a nonempty set of counterexamples; exploit the minimality of the least element to derive a contradiction. Used in number theory, descent arguments, and equivalence with induction.

4. **Induct.** Verify a base case; show the property propagates to successors. Used whenever the claim concerns all natural numbers or any recursively defined structure.

5. **Chase an arbitrary element.** Take an arbitrary element satisfying one condition; track it through definitions and established facts until it satisfies the target condition. Used for set inclusions, function properties, and direct proofs where the logic flows in a straight line from hypothesis to conclusion.

Remark 6.1.7 (Why five archetypes, not more?). The five archetypes are not a classification chosen for convenience. They correspond to the five fundamental proof obligations that mathematical statements impose:

To prove something *exists*, you must produce it. To prove something is *unique*, you must show that any two instances are equal. To prove a statement holds *for all natural numbers*, the inductive structure of $\mathbb{N}$ provides the only available tool. To prove a *conditional* claim

$P \Rightarrow Q$, you must trace an arbitrary element or object satisfying P to a situation where Q holds. The minimal element archetype is the well-ordering counterpart to induction, available whenever you need to reason about the smallest counterexample.

Everything else in proof writing is refinement of these five. The specific proof structures in Section 6.3 (direct proof, contrapositive, contradiction, cases, existence-uniqueness) are instantiations of these archetypes adapted to specific logical forms. The tactics in Section 6.5 are the move-level tools used to execute whichever archetype applies.

Remark 6.1.8 (Archetypes can nest, and this is a feature). A single proof often uses more than one archetype. The most common nested pattern in abstract algebra is: construct a witness (archetype 1), then invoke a uniqueness theorem (archetype 2) to conclude that the constructed object equals a previously named one.

This is the satisfy-and-cite tactic (Section 6.1). It appears in almost every proof of the form "$a = b$" where b is an algebraically defined object. The nesting is not a complication — it is the standard pattern for proving equalities in algebraic structures, and once recognized, it becomes mechanical.

Another common nesting: induct (archetype 4), and inside the inductive step, use a direct proof argument (archetype 5) to move from $P(n)$ to $P(n+1)$. The outer structure is induction; the inner argument is element-chasing.

Remark 6.1.9 (The archetype identifies the proof's centre of gravity). When you read a finished proof in a textbook, the archetype is usually visible in the first line or two. "Suppose x, y both satisfy the definition" — that is archetype 2. "Define $f : X \to Y$ by $f(x) = \dots$" — that is archetype 1. "Let $n \in \mathbb{N}$ be arbitrary" followed by induction markup — that is archetype 4. "Let $x \in A$" followed by a chain of implications — that is archetype 5.

Reading proofs actively means identifying the archetype at the start and then watching how the author executes it. This is far more efficient than trying to absorb a proof line by line without a sense of its overall shape.

The Satisfy-and-Cite Pattern. There is a proof pattern that does not appear in any of the standard proof archetypes by name, yet appears constantly — in every chapter of abstract algebra and analysis. Most undergraduates have never been taught it explicitly, which means they either reinvent it poorly each time or miss it entirely and write longer, less clear proofs.

The pattern is called **satisfy-and-cite**. Here is its logic.

The Satisfy-and-Cite Pattern

Setup. You want to prove $a = b$, where b is a *uniquely determined* object: the unique identity, the unique inverse, the unique limit, the unique fixed point, the unique element satisfying some algebraic condition.

Step 1. Show that a satisfies the defining property of b.
Step 2. Cite the previously proved uniqueness theorem: there is exactly one object satisfying that property.
Step 3. Conclude that $a = b$, since b is the unique such object and a is one too.

Remark 6.1.10 (Why this is not the same as assume-two-and-compare). The confusion between these two patterns is extremely common, so it is worth being precise about what each one does.

The **assume-two-and-compare** pattern *proves* a uniqueness theorem. You use it when the goal is to establish that some object is unique. You say: let x and y both satisfy the definition; then show $x = y$.

The **satisfy-and-cite** pattern *uses* an already proved uniqueness theorem. You use it when you already know some object b is unique and you want to identify a with b. You show a satisfies b's defining property, then invoke uniqueness to collapse a to b.

Assume-two-and-compare appears in the proof that the identity is unique. Satisfy-and-cite appears in every proof that uses that fact afterward. The first creates the tool; the second uses it. Conflating them means writing a new uniqueness proof every time you want to identify two equal objects, which is circular at best and wrong at worst.

Example 6.1.11 (Socks-shoes property: $(ab)^{-1} = b^{-1}a^{-1}$). We want to show $(ab)^{-1} = b^{-1}a^{-1}$ in a group.

The inverse of ab is, by definition, the unique element x satisfying $(ab)x = e$ and $x(ab) = e$.

We compute directly, using only associativity and the inverse axiom:

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e.$$

Similarly $(b^{-1}a^{-1})(ab) = e$.

So $b^{-1}a^{-1}$ *satisfies the defining property of* $(ab)^{-1}$. By the uniqueness of inverses (which we have already proved), we conclude $(ab)^{-1} = b^{-1}a^{-1}$. □

Notice what did not happen: we never assumed two inverses of ab existed and compared them. We showed one specific element satisfies the inverse condition, then cited uniqueness to identify it with the named inverse.

Example 6.1.12 (Ring theorem: $a \cdot (-b) = -(ab)$). We want to show $a(-b) = -(ab)$ in a ring.

The additive inverse $-(ab)$ is, by definition, the unique element x satisfying $ab + x = 0$.

We compute using distributivity:

$$ab + a(-b) = a(b + (-b)) = a \cdot 0 = 0.$$

So $a(-b)$ *satisfies the defining property of* $-(ab)$. By the uniqueness of additive inverses (previously proved), $a(-b) = -(ab)$. □

Same pattern: compute, satisfy, cite.

Remark 6.1.13 (The pattern across mathematics). Once you see satisfy-and-cite, you will see it everywhere. It appears wherever uniqueness theorems have been proved:

In **algebra**: every proof of the form "this element equals the inverse/identity/zero" uses it. The four examples above are $(ab)^{-1} = b^{-1}a^{-1}$, $a \cdot (-b) = -(ab)$, $-(-a) = a$, and $a \cdot 0 = 0$.
In **analysis**: the uniqueness of limits means that to show $\lim_{n\to\infty} x_n = L$, you verify x_n converges to L by the definition; uniqueness then implies L equals any other expression you have shown to be the limit of the same sequence.

In **linear algebra**: the zero vector is unique; the additive inverse is unique; a linear transformation is uniquely determined by its values on a basis. Each uniqueness theorem creates a new application of satisfy-and-cite.

Each time a uniqueness theorem is proved, it creates a new proof tool that can be activated by satisfy-and-cite for every subsequent argument in that structure. The theorem is not just a fact — it is a lever.

6.2 The Proof Construction Algorithm

Remark 6.2.1 (What this section is for). Architecture (Section 6.1) tells you what kind of proof to write. This section tells you how to write it, line by line, once the architecture is chosen.

The procedure below is intentionally explicit and more detailed than anything you will find in a standard textbook. That is on purpose. With practice, these steps will become internalized and automatic. But until they are — until you have written enough proofs that the process feels natural — working through them consciously will prevent the most common errors: undefined objects, unjustified steps, and forgotten stop conditions.

A longer proof with every step explicit is always better than a shorter proof with implicit gaps.

Step 0: Classify the Statement. Before anything else, identify the logical form of the claim. Common forms:

- Universal: $\forall x\ P(x)$
- Conditional: $P \to Q$
- Biconditional: $P \leftrightarrow Q$
- Existential: $\exists x\ P(x)$
- Existence-and-uniqueness: $\exists!\, x\ P(x)$
- Set equality: $A = B$
- Algebraic equality: $a = b$
- Structural assertion: "f is injective," "R is an equivalence relation"

The logical form determines the skeleton of the proof. Consulting the statement map (Section 6.1) is the direct link from this classification to an opening strategy.

Step 1: Restate the Givens. Write out explicitly what is given. Introduce every set, relation, function, and algebraic structure, and record any properties assumed about them.

> Let G be a group. Let $a \in G$.

This is not a formality. By recording the givens, you license their use in the proof — you can now invoke associativity, the existence of inverses, and the identity element without reintroducing them each time. What is not in the givens cannot be used.

Step 2: Identify Objects and Their Types. Before reasoning begins, verify the *type* of each object you will work with: is it an element, a set, a function, or a relation? What is its ambient universe? What operations apply to it?

$$x \in A, \quad f : A \to B, \quad R \subseteq A \times A.$$

This step prevents the most common logical errors in undergraduate proofs: writing $x \in f$ when you mean $x \in \mathrm{dom}(f)$; writing $A = x$ when you mean $x \in A$; confusing membership ($\in$) with inclusion ($\subseteq$). Every object used in a proof must have a declared type. If you cannot say what type something is, you do not yet understand what you are working with.

Step 3: Introduce Arbitrary Elements. If the claim is universal ($\forall x\ P(x)$), immediately introduce an arbitrary element:

$$\text{Let } x \in A \text{ be arbitrary.}$$

The word "arbitrary" is not decorative. It means the element is not given any properties beyond those of the set A. Whatever you prove about this x will hold for all $x \in A$, precisely because you have assumed nothing about it beyond membership. The moment you specialize — "let $x = 0$" or "let x be such that ... " — you have lost the ability to conclude the universal statement.

Step 4: Expand the Goal. Rewrite the conclusion using definitions rather than named concepts.

- To prove f is injective: expand injectivity as "$f(a) = f(b) \Rightarrow a = b$" and take that as the new goal.
- To prove $A = B$ (sets): take as new goal "$A \subseteq B$ and $B \subseteq A$."
- To prove $x \in A \cup B$: rewrite the goal as "$x \in A$ or $x \in B$."
- To prove $\{x_n\}$ converges: expand to "$\forall \varepsilon > 0\ \exists N$ s.t. $n \geq N \Rightarrow d(x_n, L) < \varepsilon$."

This step is the one most students skip, and skipping it is the single most reliable predictor of a stuck proof. You cannot make progress toward a goal you have not named in operative terms. "Prove f is injective" is not an operative goal. "Prove that $f(a) = f(b)$ implies $a = b$" is.

Step 5: Introduce Helper Objects. Introduce any auxiliary objects needed for the argument: witnesses for existential claims, intermediate elements for indirect arguments, bounds for inequality proofs, function values.

$$\text{Let } N_1 \in \mathbb{N} \text{ s.t. } n \geq N_1 \Rightarrow d(x_n, L) < \varepsilon/2.$$

No object should appear in a proof without being explicitly introduced. If you write "let c be the cancellation element" without first establishing that such an element exists, your proof has an undefined object. Every name carries an obligation: the obligation to have said what the name refers to.

Step 6: Apply One Rule at a Time. Each line of a proof follows from exactly one of four sources:

1. A definition (unpacking what a term means),
2. A hypothesis (something assumed in the current proof),
3. A previously proved theorem (cited by name and reference),
4. A basic logical rule (modus ponens, universal instantiation, etc.).

Lines that are justified by "clearly" or "obviously" or no reason at all are gaps. At the learning stage, every step should have an explicit justification. When a proof stalls, the right question is always: which of the four sources has not been consulted? Almost always, the answer is that some definition has not been unpacked.

Step 7: Use Forward and Backward Reasoning. Proofs alternate between two modes of reasoning, and recognizing which mode is productive at each stage is a significant skill.

Forward reasoning: deduce consequences from what you have established so far. If you know $a^2 = e$ in a group, derive that $a(ae) = a \cdot a \cdot e = a^2 \cdot e = e$. You are moving away from the hypotheses, toward the goal.

Backward reasoning: examine the goal and ask what would suffice to establish it. If the goal is "$a = a^{-1}$," ask: by what definition would that follow? It follows if $aa = e$, since that is the condition for a to be its own inverse. Now your new goal is to prove $aa = e$, which may be more directly accessible.

In practice, most proofs proceed in both directions simultaneously: you derive forward from the hypotheses while simplifying the goal backward until the two sides meet. When you are stuck, try the direction you have not been working in.

Step 8: Handle Cases Explicitly. If a statement splits into mutually exclusive and exhaustive cases, enumerate all of them.

Either n is even, or n is odd.

Handle each case separately. Close each case before opening the next. Verify, explicitly, that the cases together cover every possibility. A proof with unlabeled cases — where it is unclear which case is being argued or whether all cases have been covered — is not a proof.

Step 9: Close the Argument. Once the desired conclusion has been established, state it explicitly:

- "Thus $x \in B$, and so $A \subseteq B$."
- "Hence $f(a) = f(b)$ implies $a = b$, so f is injective."
- "In both cases $P(n + 1)$ holds."

The explicit closing step serves two functions. First, it tells the reader (and you) that the proof goal has been achieved. Second, it forces you to check that what you have proved actually *is* the goal. Students who reach the end of a proof and write □without stating a conclusion often discover, on re-reading, that they proved something adjacent to the goal but not the goal itself.

Step 10: Signal Completion. Conclude with □, ■, or "This completes the proof."

Legal Moves and Stop Conditions

What you may never do:

- Choose an existential witness before proving it exists.
- Introduce an "arbitrary" element and then give it special properties.
- Assume the conclusion at any point in the proof.
- Apply a definition partially (all conditions must be checked).

A proof is complete when:

- A universal claim has been proved for a genuinely arbitrary element.
- An existential claim has produced a verified witness.
- A set equality has established both inclusions explicitly.
- Each direction of a biconditional has been proved separately.
- An existence-uniqueness claim has done both parts.

Line-by-Line Discipline. Before writing each line, ask three silent questions:

(i) Has every object mentioned in this line been defined or introduced earlier?

(ii) Which definition, hypothesis, or theorem justifies this step?

(iii) Does this step move the argument toward the stated goal?

If any answer is no, stop and resolve it before continuing.

6.3 Proof Structures

Direct Proof. A direct proof of $P \Rightarrow Q$ assumes P and derives Q through a chain of valid inferences.

Template: Direct Proof

Given. P.
Goal. Q.

Assume P.
$\vdots$ [expand definitions; apply lemmas; derive consequences]
Therefore Q. □

Remark 6.3.1 (Direct proof is the default — always try it first). Direct proof is not one option among equals. It is the default. Every time you sit down to prove $P \Rightarrow Q$, the first thing you write is "Assume P" and the first thing you do is unpack what P says. You use what P gives you, apply definitions and previously proved results, and derive Q.

Reach for contrapositive or contradiction only when direct proof has genuinely stalled — when you have unpacked the hypothesis and the goal and there is no visible path between them. Do not reach for contradiction just because the statement feels hard. Most statements that feel hard are hard only because a definition has not been unpacked.

Remark 6.3.2 (The anatomy of a direct proof). Every direct proof has three phases that correspond to the first four steps of the construction algorithm:

Phase 1 — Unpack the hypothesis. P is a named concept or a logical statement. Expand it into its operative form. If P says "G is a group," this licenses associativity, the existence of an identity, and the existence of inverses. Write those down as available tools.

Phase 2 — Expand the goal. Q is also a named concept. Expand it into what you need to establish. If Q says "$a = a^{-1}$," the operative target is showing a satisfies the defining property of an inverse: $aa = e$.

Phase 3 — Bridge the gap. Apply definitions, lemmas, and axioms to move from the expanded hypothesis to the expanded conclusion.

When Phase 3 stalls, the cause is almost always that Phase 1 or Phase 2 was not completed. The most common error is having a vague goal ("show $a = a^{-1}$") without having written out what that means in terms of the group axioms.

Example 6.3.3 (Direct proof in a group). **Claim.** In a group G, if $a^2 = e$ then $a = a^{-1}$.

Proof. Assume $a^2 = e$, meaning $aa = e$.

We want to show $a = a^{-1}$, which by definition of inverse means we need $aa = e$ and $ea = a$ (or equivalently, a is its own inverse).

Multiply both sides of $aa = e$ on the right by a^{-1}:

$$(aa)a^{-1} = e \cdot a^{-1}.$$

By associativity: $a(aa^{-1}) = a^{-1}$. By the inverse axiom: $ae = a^{-1}$. By the identity axiom: $a = a^{-1}$.

Hence $a^2 = e$ implies $a = a^{-1}$. □

Each step is justified by exactly one of: the hypothesis ($aa = e$), the inverse axiom ($aa^{-1} = e$), the identity axiom ($ae = a$), or associativity. No step is unjustified.

Proof by Contrapositive. The contrapositive of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$. These two statements are logically equivalent: one is true if and only if the other is. A proof by contrapositive establishes $\neg Q \Rightarrow \neg P$ by a direct argument, and since this is equivalent to $P \Rightarrow Q$, the original claim follows.

Template: Proof by Contrapositive

Goal. $P \Rightarrow Q$.
Equivalent goal. $\neg Q \Rightarrow \neg P$.

Assume $\neg Q$.

$\vdots$ [derive $\neg P$ from $\neg Q$]
Therefore $\neg P$.
Hence $P \Rightarrow Q$. □

Remark 6.3.4 (When contrapositive is the right tool). Contrapositive is useful when the negation of the conclusion is more concrete and easier to work with than the hypothesis.

The canonical example is injectivity. To prove $P \Rightarrow Q$ where P says "$f(a) = f(b)$" and Q says "$a = b$": the contrapositive is $a \neq b \Rightarrow f(a) \neq f(b)$, which may be directly accessible if the function is order-preserving or explicitly defined.

Another signal: when Q is the statement that some object exists or some set is non-empty. The negation $\neg Q$ says no such object exists, which may be a very restrictive and workable condition.

The general diagnostic: after writing $\neg Q$, look at what it says. If it gives you a strong, concrete assumption — something you can actually manipulate — contrapositive is likely the right tool. If $\neg Q$ is vague or complicated, try a different approach.

Remark 6.3.5 (Contrapositive versus contradiction: the essential difference). Students routinely conflate contrapositive and contradiction. They are not the same, and knowing the difference matters for clarity.

In a **contrapositive** proof, you assume *only* $\neg Q$ and derive $\neg P$. The hypothesis P is never mentioned. The logical engine is: you proved $\neg Q \Rightarrow \neg P$, which by equivalence gives $P \Rightarrow Q$.

In a **contradiction** proof, you assume *both* P and $\neg Q$ simultaneously and derive *any* absurdity — not necessarily $\neg P$. The logical engine is: the assumption $P \wedge \neg Q$ leads to a false statement, so $P \wedge \neg Q$ must itself be false, so $P \Rightarrow Q$.

When contrapositive works, it is cleaner: you need one assumption, not two, and the conclusion you derive ($\neg P$) is specific rather than "any contradiction." Always try contrapositive before reaching for contradiction.

Example 6.3.6 (Contrapositive for divisibility). **Claim.** If n^2 is even then n is even.

Proof (contrapositive). We prove the contrapositive: if n is odd then n^2 is odd.

Assume n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$.

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which has the form $2m + 1$ and is therefore odd.

Hence n odd $\Rightarrow n^2$ odd, which is the contrapositive of n^2 even $\Rightarrow n$ even. □

Notice: the hypothesis "n^2 is even" was never used. We proved the contrapositive directly, and the original implication follows by logical equivalence. There is no need to loop back and mention the original statement; the contrapositive suffices.

Proof by Contradiction. To prove P, assume $\neg P$ and derive a statement that contradicts a known truth: a hypothesis, an axiom, or a previously proved theorem. Since the assumption $\neg P$ led to a false statement, $\neg P$ is false, and therefore P is true.

Template: Proof by Contradiction

Goal. P.

Suppose for contradiction that $\neg P$.

$\vdots$ [derive consequences; identify the absurdity]
This contradicts [name the violated fact].
Therefore P. $\square$

Remark 6.3.7 (The logic: why it works). Contradiction relies on the law of the excluded middle: every statement is either true or false. To prove P, it suffices to show $\neg P$ is false. You show $\neg P$ is false by deriving from it a statement that is known to be false — a contradiction.

The contradiction can be anything: $0 = 1$, a set that is both empty and non-empty, a number that is simultaneously even and odd, an inequality $x < x$. The only requirement is that the contradicted fact was genuinely established before the current proof began.

This is the move that students most often get wrong: they think they have derived a contradiction when they have merely derived something surprising. A statement being counterintuitive is not a contradiction. The contradicted fact must be something you can point to: a hypothesis, an axiom, a definition, or a theorem with a name.

Remark 6.3.8 (When contradiction is the right tool). Contradiction is best reserved for claims whose negation is a strong, productive assumption. The archetypes where contradiction excels are:

Irrationality proofs. Assuming $\sqrt{2} \in \mathbb{Q}$ gives you a fraction in lowest terms; arithmetic then forces the numerator and denominator to be simultaneously even, contradicting the lowest-terms assumption.

Infinitude arguments. Assuming there are finitely many primes lets you list them; constructing a number not divisible by any of them contradicts the assumption that the list was complete.

Nonexistence results. If you want to show no object satisfies some property, assume one exists and derive an absurdity.

Uniqueness (via assume-two-and-compare). Assuming two distinct objects both satisfy a uniqueness condition produces a contradiction from the definitions.

Avoid contradiction as a default. When a direct proof or contrapositive works, prefer it: the argument is shorter and the logical structure is more transparent. Contradiction is a sign that the natural direction of inference does not work, and that you need to reason from an impossible assumption instead.

Remark 6.3.9 (Always name the contradiction explicitly). The closing line of a contradiction proof must name what has been contradicted. "This is a contradiction" with no further specification is a red flag: it suggests the student either does not know what was contradicted, or is hoping the reader will not check.

Write: "This contradicts the assumption that $\gcd(p, q) = 1$." Write: "This contradicts Theorem X, which asserts ..." Write: "This contradicts $m \in S$ being the least element."

Naming the violated fact performs two functions: it confirms that the contradiction is genuine, and it locates the logical break in the argument so that anyone reading the proof can verify it.

Example 6.3.10 ($\sqrt{2}$ is irrational). *Proof.* Suppose for contradiction that $\sqrt{2} \in \mathbb{Q}$. Write $\sqrt{2} = p/q$ where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$ (in lowest terms; this is possible by the definition of $\mathbb{Q}$).

Squaring: $2q^2 = p^2$. So p^2 is even, which forces p to be even (by the contrapositive example above). Write $p = 2k$.

Then $2q^2 = (2k)^2 = 4k^2$, so $q^2 = 2k^2$. So q^2 is even, which forces q to be even.

But then both p and q are even, so $\gcd(p, q) \geq 2$. This contradicts $\gcd(p, q) = 1$.

Therefore $\sqrt{2} \notin \mathbb{Q}$. □

The structure is: assume $\neg P$ (i.e., $\sqrt{2} \in \mathbb{Q}$); extract information from this assumption (lowest-terms representation); derive a consequence (both p and q even); identify the contradiction (conflicts with lowest terms); conclude P.

Proof by Cases. If the hypotheses or conclusion naturally divide into a finite number of mutually exclusive and collectively exhaustive alternatives, handle each case separately.

Template: Proof by Cases

Goal: Prove Q, given that exactly one of $P_1, P_2, \ldots, P_k$ holds (and one of them always holds).

Verify exhaustiveness: Every possibility falls under some P_i.
Verify mutual exclusivity (when needed).

Case 1: Assume P_1. [Argue.] Hence Q.

Case 2: Assume P_2. [Argue.] Hence Q.

$\vdots$

Case k: Assume P_k. [Argue.] Hence Q.

Since $P_1, \ldots, P_k$ are exhaustive and Q holds in each case, Q holds unconditionally. □

Remark 6.3.11 (The exhaustiveness obligation is non-negotiable). A case proof is only valid when the cases together cover every possibility. Always verify exhaustiveness explicitly, even when it feels obvious.

The most common exhaustive splits:

- n is even, or n is odd.

- $a > 0$, $a = 0$, or $a < 0$.
- $a \mid b$, or $a \nmid b$.
- $x \in A$, or $x \notin A$.
- $x = y$, or $x \neq y$.

Writing "we consider two cases" without naming what they are, or naming cases that do not cover all possibilities, is a structural error. The proof is not valid until exhaustiveness is established.

Remark 6.3.12 (Each case is a self-contained argument). Inside each case, you have an additional assumption: P_i is true. That assumption is local to the case. You may use it within Case i, but you may not carry it into Case j.

When you close a case — "hence Q, completing Case 1" — you are signalling that the local assumption is no longer in scope. Case 2 starts fresh, with only the original given plus the assumption P_2.

This discipline prevents the most common case-analysis error: drifting from one case into another by accidentally retaining a local assumption beyond its scope.

Remark 6.3.13 (Symmetric cases). When two cases are exactly symmetric — when the argument for Case 2 is obtained from the argument for Case 1 by renaming variables, with no other change — it is acceptable to write "Case 2 is symmetric to Case 1 by swapping a and b." But be careful: only write this when the symmetry is complete. A partial symmetry that requires a different step at one point must be written out in full.

Example 6.3.14 (Parity of $n(n+1)$). **Claim.** For every $n \in \mathbb{Z}$, $n(n+1)$ is even.

Proof. Every integer is either even or odd (these cases are exhaustive and mutually exclusive).

Case 1: n is even. Write $n = 2k$ for some $k \in \mathbb{Z}$. Then $n(n+1) = 2k(n+1)$. Since this is 2 times an integer, $n(n+1)$ is even.

Case 2: n is odd. Write $n = 2k+1$ for some $k \in \mathbb{Z}$. Then $n+1 = 2k+2 = 2(k+1)$, so $n(n+1) = (2k+1) \cdot 2(k+1)$. This is 2 times an integer, so $n(n+1)$ is even.

In both cases $n(n+1)$ is even. Since every integer falls into one of these cases, the proof is complete. □

Notice that the cases were named before the argument, the exhaustiveness was verified ("every integer is even or odd"), and each case was closed explicitly before moving to the next.

Example 6.3.15 (Case split induced by a definition). **Claim.** The discrete metric satisfies the triangle inequality.

Proof. Let $x, y, z \in X$. We must show $d(x,z) \leq d(x,y) + d(y,z)$.

Case 1: $x = z$. Then $d(x,z) = 0$. Since $d(x,y) \geq 0$ and $d(y,z) \geq 0$, we have $0 \leq d(x,y) + d(y,z)$.

Case 2: $x \neq z$. Then $d(x,z) = 1$. We cannot have both $x = y$ and $y = z$, for that would force $x = z$. So at least one of $d(x,y), d(y,z)$ equals 1, hence $d(x,y) + d(y,z) \geq 1 = d(x,z)$.

The cases $x = z$ and $x \neq z$ are exhaustive. The triangle inequality holds in both cases. □

The case split here is induced by the two-value structure of the discrete metric: distances are 0 or 1. The right case split to use is the one naturally suggested by the definition you are working with.

Existence and Uniqueness. The statement $\exists!\, x\ P(x)$ asserts two things simultaneously: that some object satisfying P exists, and that it is the *only* such object. These two obligations are proved separately, in order.

Template: Existence and Uniqueness ($\exists!\, x\ P(x)$)

Step 1 — Existence.
Define $x :=$ [explicit construction].
Verify that $P(x)$ holds. (Check every condition in P.)

Step 2 — Uniqueness (assume-two-and-compare).
Let x and y be arbitrary objects both satisfying P.
Apply the definition of x with y as input; apply the definition of y with x as input. The two applications collapse.
Conclude $x = y$.

Step 3 — Conclude.
By Steps 1 and 2, there exists exactly one object satisfying P. $\square$

Remark 6.3.16 (Existence must be proved before uniqueness). The uniqueness argument has the form: "let x and y both satisfy P." This assumption is only meaningful if we already know that at least one object satisfying P exists. Without Step 1, the uniqueness argument is vacuous: we are asserting something about objects that might not exist.

This is not pedantry. There are statements where the existence fails and the uniqueness argument would appear to go through — but the whole proof would then be establishing uniqueness for an empty set of objects, which is trivially true and useless. Existence first is not just a convention; it is logically necessary.

Remark 6.3.17 (The key move in assume-two-and-compare). The assume-two-and-compare argument has a characteristic structure that appears across mathematics. The critical move is this: if x and y both satisfy the defining property P, apply x's property with y as the argument, and simultaneously apply y's property with x as the argument. The two applications collapse to $x = y$.

Here is the pattern in the group identity example: if e satisfies the identity axiom, then $e \cdot e' = e'$ (using e as identity with argument e'). If e' satisfies the identity axiom, then $e \cdot e' = e$ (using e' as identity with argument e). Combining: $e' = e$.

The same move works for: uniqueness of inverses, uniqueness of the zero vector, uniqueness of limits, uniqueness of the supremum. In every case, the two assumed objects are "played against each other": each one's defining property is applied with the other as input.

Example 6.3.18 (Uniqueness of the group identity). *Proof.* Let G be a group. Suppose e and e' both satisfy the identity axiom:

$$e \cdot g = g \cdot e = g \quad \text{for all } g \in G,$$

$$e' \cdot g = g \cdot e' = g \quad \text{for all } g \in G.$$

Apply e''s identity property with $g = e$:

$$e' \cdot e = e.$$

Apply e's identity property with $g = e'$:

$$e \cdot e' = e'.$$

But $e' \cdot e = e \cdot e'$ (both equal the product $e \cdot e'$ in G by commutativity — actually, both equal $e \cdot e'$ directly). Actually, more directly: $e = e' \cdot e = e'$ where the first step uses e' as identity and the second uses e as identity. Hence $e = e'$. □

The single line is: $e = e \cdot e' = e'$. First equality: e acts as identity on e'. Second equality: e' acts as identity on e. Both identities applied to the same product; both sides collapse to the same expression; $e = e'$.

Remark 6.3.19 (Why proving uniqueness creates a proof tool). Every uniqueness theorem, once proved, creates a new application of the satisfy-and-cite tactic. This is the connection that most students miss.

Proving uniqueness says: there is exactly one object satisfying P. This means: if you can ever show that some element a satisfies P, you automatically get $a = b$, where b is the canonical name for the unique object satisfying P.

The group inverse provides the clearest example. Once you have proved that inverses are unique, the pattern for proving $(ab)^{-1} = b^{-1}a^{-1}$ is immediate: show $b^{-1}a^{-1}$ satisfies the defining property of the inverse of ab, then cite uniqueness. You never need to set up another assume-two-and-compare argument. The uniqueness theorem absorbs all future comparisons.

This is why uniqueness results are not just facts — they are *tools*. Prove them early; use them everywhere afterward.

6.4 Mathematical Induction

Why Induction Works. Mathematical induction is often presented as a technique: base case, inductive step, done. That presentation is useful but incomplete. Induction *feels* like a procedure, but it is actually a theorem, and it is worth understanding why it is true before using it.

Remark 6.4.1 (The Peano axiom behind induction). The natural numbers are characterized by the Peano axioms. The critical one for induction is Axiom P5:

If $A \subseteq \mathbb{N}$ *satisfies (i)* $0 \in A$ *and (ii)* $n \in A \Rightarrow S(n) \in A$*, then* $A = \mathbb{N}$.

This axiom says that $\mathbb{N}$ is the *smallest* set containing 0 and closed under the successor function S. There is no proper subset of $\mathbb{N}$ with these two properties.

The induction principle is a direct translation. Let $P(n)$ be any statement. Define $A = \{n \in \mathbb{N} : P(n)\}$. If $P(0)$ holds, then $0 \in A$. If $P(n) \Rightarrow P(S(n))$ for every n, then A is closed under S. By P5, $A = \mathbb{N}$. Therefore $P(n)$ holds for every $n \in \mathbb{N}$.

The full development of this from the Peano axioms is in the Axiom Systems chapter. Here, the takeaway is: induction is not a trick. It is a theorem derivable from the definition of $\mathbb{N}$.

Remark 6.4.2 (Why the base case is not optional). The inductive step alone says: *if* $P(n)$ holds for some n, then $P(n+1)$ holds. This is a conditional, not an assertion. Without a base case, there is no n for which $P(n)$ is known to hold. The chain of conditionals has no starting point, and the proof produces nothing.

The most famous illustration is the false theorem "all horses are the same colour." A flawed argument runs: base case trivial ($n = 1$), inductive step [fatally flawed reasoning]. The lesson is not just that the step is wrong — it is that even a correct base case does not rescue a bad step, and even a correct step does not rescue a missing base case. Each part is independent; both are necessary.

In terms of P5: the base case establishes $0 \in A$. The inductive step establishes closure under S. Both conditions of P5 are required for the conclusion $A = \mathbb{N}$.

Remark 6.4.3 (The domino picture and its limitations). The common picture of induction as an infinite row of falling dominoes is a useful mnemonic but a misleading explanation. It suggests that induction requires each domino to knock over the next, which captures the inductive step. But it does not explain why the argument is valid — the validity comes from P5, not from a physical metaphor.

More importantly, the domino picture suggests that the steps happen *in time*: first domino 0 falls, then 1, then 2, and so on. But induction proves a statement about *all* natural numbers simultaneously, not by a sequential process. The proof is instantaneous: once you have established the base case and the inductive step, the conclusion $P(n)$ for all n follows at once from P5.

Use the domino picture as memory aid. For understanding, use P5.

Remark 6.4.4 (Induction and well-ordering are equivalent). The induction principle and the well-ordering principle for $\mathbb{N}$ (every nonempty subset has a least element) are logically equivalent. This means every induction argument can be rephrased as a minimal-counterexample argument, and vice versa. Both are valid proof tools. The well-ordering approach is sometimes more natural when you want to reason about the *first* failure rather than climbing from a base case. The equivalence is proved in Section 6.4.

Theorem: Weak (Standard) Induction

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. Suppose:

1. **Base case.** $P(n_0)$ is true for some starting value n_0.
2. **Inductive step.** For all $n \geq n_0$, $P(n) \Rightarrow P(n+1)$.

Then $P(n)$ is true for all $n \geq n_0$.

Template: Weak Induction Proof

Let $P(n)$ denote the statement: [*write it out completely as a sentence*].

Base case ($n = [n_0]$). [*Verify $P(n_0)$ directly.*] Hence $P(n_0)$ holds.

Inductive step. Let $n \geq n_0$ and assume $P(n)$ holds. [*State explicitly what $P(n)$ says.*] We show $P(n+1)$ holds.
[*Rewrite $P(n+1)$ to expose $P(n)$ inside it. Substitute the inductive hypothesis. Simplify.*]
Hence $P(n+1)$ holds.

By induction, $P(n)$ holds for all $n \geq n_0$. $\square$

Remark 6.4.5 (The most important discipline: write $P(n)$ as a sentence). The single most impactful habit in induction proofs is writing the inductive hypothesis as a *complete mathematical sentence* before the inductive step begins.

"Assume $P(n)$ holds" is not sufficient if you have not first written out what $P(n)$ says. Write: "Assume $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$." That sentence is your inductive hypothesis. It is the precise tool you have available in the inductive step.

Without this sentence, students typically make one of two errors: they try to use $P(n)$ without knowing what it says, guessing at it from context; or they prove the inductive step using information beyond what $P(n)$ supplies, making the proof circular or invalid. Write the sentence. It takes five seconds and eliminates the most common induction error.

Remark 6.4.6 (The core skill of the inductive step). Every inductive step involves the same core action: *rewrite $P(n+1)$ to expose $P(n)$ inside it, then substitute the inductive hypothesis.*

For sum formulas, this means: split off the last term.

$$\sum_{k=1}^{n+1} k = \left(\sum_{k=1}^{n} k\right) + (n+1).$$

Now substitute $P(n)$: replace $\sum_{k=1}^{n} k$ with $\frac{n(n+1)}{2}$. Simplify to get the $P(n+1)$ form.

For divisibility, this means: write $4^{n+1} - 1 = 4(4^n - 1) + 3$ to expose $4^n - 1$ inside $4^{n+1} - 1$.

In both cases, the step is: factor out the $P(n)$ term, substitute, and verify the arithmetic produces $P(n+1)$.

When the rewriting step is not obvious, it usually means $P(n)$ needs to be strengthened (see Section 6.4).

Example 6.4.7 (Sum formula). **Proposition.** For all $n \geq 1$: $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$.

Proof. Let $P(n)$ denote: $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$.

Base case ($n = 1$). $\sum_{k=1}^{1} k = 1 = \frac{1 \cdot 2}{2}$. Hence $P(1)$ holds.

Inductive step. Let $n \geq 1$ and assume $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. *(Inductive hypothesis.)*

We show $\sum_{k=1}^{n+1} k = \frac{(n+1)(n+2)}{2}$.

Split off the last term:

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \left(\sum_{k=1}^{n} k\right) + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) && \text{(inductive hypothesis)} \\ &= (n+1)\left(\frac{n}{2}+1\right) = \frac{(n+1)(n+2)}{2}. \end{aligned}$$

Hence $P(n+1)$ holds. By induction, $P(n)$ holds for all $n \geq 1$. □

Example 6.4.8 (Divisibility). **Proposition.** For all $n \geq 0$, $3 \mid (4^n - 1)$.

Proof. Let $P(n)$ denote: $3 \mid (4^n - 1)$.

Base case ($n = 0$). $4^0 - 1 = 0 = 3 \cdot 0$. Hence $3 \mid 0$ and $P(0)$ holds.

Inductive step. Assume $3 \mid (4^n - 1)$, i.e., $4^n - 1 = 3m$ for some $m \in \mathbb{Z}$.

Expose $4^n - 1$ inside $4^{n+1} - 1$:

$$4^{n+1} - 1 = 4 \cdot 4^n - 1 = 4(4^n - 1) + 4 - 1 = 4(4^n - 1) + 3 = 4 \cdot 3m + 3 = 3(4m + 1).$$

Hence $3 \mid (4^{n+1} - 1)$ and $P(n+1)$ holds.

By induction, $P(n)$ holds for all $n \geq 0$. □

The rewriting step $4^{n+1} - 1 = 4(4^n - 1) + 3$ is the key move: it exposes the inductive hypothesis $3 \mid (4^n - 1)$ inside the goal $3 \mid (4^{n+1} - 1)$.

Choosing $P(n)$: The Hard Part. In textbook problems, the statement $P(n)$ is usually handed to you: the claim is explicitly "prove that $\sum_{k=1}^{n} k = n(n+1)/2$", and $P(n)$ is that equality. In original proofs, you must construct $P(n)$ from the statement of the goal. And sometimes even when $P(n)$ is given, the straightforward choice fails and must be strengthened.

Remark 6.4.9 (Rule 1: $P(n)$ mirrors the logical form of the goal). Every claim to be proved by induction has the form $\forall n,\ \Phi(n)$ for some predicate Φ. Strip the universal quantifier: what remains is $P(n)$.

Goal form	**Shape of $P(n)$**
$\forall n,\ f(n) = g(n)$	Equality: $P(n) :\equiv f(n) = g(n)$
$\forall n,\ A(n) \Rightarrow B(n)$	Conditional: $P(n) :\equiv A(n) \Rightarrow B(n)$
$\forall n,\ A(n) \Leftrightarrow B(n)$	Biconditional: $P(n) :\equiv A(n) \Leftrightarrow B(n)$

Do not simplify. Do not add conditions. Transcribe the logical form of $\Phi(n)$ exactly into $P(n)$.

Remark 6.4.10 (Rule 2: induct on the recursive variable). When the goal involves a recursively defined operation, identify which variable the definition recurses on. Induct on that variable and hold all others fixed as arbitrary.

To find the recursive variable: locate the definition of the operation and identify which variable is reduced in the recursive clause.

Operation	**Recursive clause**	**Induct on**
Addition (Peano)	$(n{+}{+}) + m := (n+m){+}{+}$	n
Multiplication (Peano)	$(n{+}{+}) \times m := (n \times m) + m$	n
Exponentiation	$m^{n{+}{+}} := m^n \times m$	n

Remark 6.4.11 (The strengthening principle: when $P(n)$ is too weak). Sometimes the natural $P(n)$ fails not because the statement is false but because the inductive hypothesis is insufficient to prove the step. The fix is counterintuitive: *add more to* $P(n)$, making the claim stronger.

This works because a stronger $P(n)$ gives you more in the inductive hypothesis, which is what you actually need to prove $P(n+1)$. The stronger $P(n)$ is harder to prove overall but easier to propagate.

The signal that strengthening is needed: you write the inductive step, you need to use some fact about the sequence that $P(n)$ alone does not provide, and you find yourself thinking "if only I also knew $P(n-1)$." That thought is the signal to switch to strong induction (which implicitly strengthens $P(n)$ to "$P(k)$ for all $k \leq n$").

Example 6.4.12 (Strengthening: Fibonacci bound requires two prior values). **Goal.** Show that the Fibonacci sequence satisfies $F_n < 2^n$ for all $n \geq 1$.

Naive attempt. Let $P(n)$: "$F_n < 2^n$."

Inductive step: assume $F_n < 2^n$; show $F_{n+1} < 2^{n+1}$.

We have $F_{n+1} = F_n + F_{n-1}$. We know $F_n < 2^n$ by the inductive hypothesis. But what do we know about F_{n-1}? Nothing: $P(n)$ says nothing about F_{n-1}. The step is stuck.

Fix: strong induction gives two terms. Use the strong inductive hypothesis: $F_k < 2^k$ for all $k \leq n$. Then:

$$F_{n+1} = F_n + F_{n-1} < 2^n + 2^{n-1} < 2^n + 2^n = 2^{n+1}.$$

The step completes because strong induction provides $F_{n-1} < 2^{n-1}$. This is the canonical signal that strong induction is needed: the recursive step requires knowing $P(k)$ for some $k < n$ other than $k = n-1$.

Remark 6.4.13 (Vacuous base cases in conditional statements). When $P(n)$ is a conditional and its hypothesis is false at the base value, the base case is vacuously true. This is expected and requires no further argument.

For example, if the claim is "for all $n \geq 1$, if $n > 0$ and $m > 0$ then $nm > 0$" and you induct starting at $n = 0$, then $P(0)$: "if $0 > 0$ and $m > 0$ then $0 \times m > 0$" has a false hypothesis ($0 > 0$ is false), so $P(0)$ is vacuously true.

This signals that the claim only has content for values of n where the hypothesis holds. The induction then establishes the claim for all n where the hypothesis is satisfied.

Remark 6.4.14 (Diagnostic questions when stuck choosing $P(n)$). 1. What exactly do I need to prove for $n = 0, 1, 2, 3$? Write it out. The pattern of $P(n)$ will appear.

2. Does the inductive step require knowing $P(n-1)$ as well as $P(n)$? If yes, switch to strong induction.

3. Does the inductive step produce a bound or equation slightly weaker than $P(n+1)$? If yes, strengthen $P(n)$.

4. After writing $P(n{+}{+})$, can I apply the recursive definition to expose $P(n)$? If not, the wrong variable may have been chosen.

Theorem: Strong Induction

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. Suppose that for every $n \in \mathbb{N}$:

$$\Big(\forall k < n,\ P(k)\Big) \Rightarrow P(n).$$

Then $P(n)$ holds for all $n \in \mathbb{N}$.

Remark 6.4.15 (Strong and weak induction prove the same statements). Strong induction is sometimes described as "more powerful" than weak induction, but this is imprecise. Both principles are logically equivalent (they both follow from P5 and each implies the other). Strong induction is not more powerful in the sense of proving statements that weak induction cannot — every statement provable by one is provable by the other.

What strong induction offers is convenience: when the proof of $P(n+1)$ genuinely requires knowing $P(k)$ for some $k < n$ that is not $n - 1$, strong induction supplies that information directly. Rephrasing the proof as weak induction would require artificially strengthening $P(n)$ to "$P(k)$ for all $k \leq n$", which is precisely the hypothesis that strong induction builds in automatically.

Use strong induction when the step needs information from more than one earlier case. Use weak induction when $P(n)$ alone suffices.

Remark 6.4.16 (The base case in strong induction). When $n = 0$, the hypothesis $\forall k < 0,\ P(k)$ is vacuously true: there are no natural numbers less than 0. So the step for $n = 0$ requires proving $P(0)$ with no inductive hypothesis at all — this is the base case, and it must be proved directly.

In practice, always write the base case explicitly, even though it is implicit in the universal statement. Students who skip the base case in strong induction often write "by the inductive hypothesis applied to $n - 1$" when $n = 0$, which is an error: $n - 1 = -1$ is not a natural number and has no inductive hypothesis.

Remark 6.4.17 (When to use strong induction: the diagnostic). The diagnostic for strong induction is simple. Write out the inductive step: $P(n+1) = f(P(?), P(?), \ldots)$, and ask what earlier values of P you need. If the answer includes anything other than exactly $P(n)$, use strong induction.

Canonical situations where strong induction is needed:

- Recursive sequences with two prior terms: $a_{n+1} = a_n + a_{n-1}$ requires both $P(n)$ and $P(n-1)$.
- Prime factorization: writing $n = ab$ with $a, b < n$ requires $P(a)$ and $P(b)$ where a and b are arbitrary.
- Any argument that splits $n+1$ into parts: you need P at the part values, which are not predetermined.

Template: Strong Induction Proof

Let $P(n)$ denote: [*write it out*].

Base case ($n = [\text{start}]$). [*Verify directly, with no inductive hypothesis.*]

Inductive step. Let $n > [\text{start}]$ and assume $P(k)$ holds for all $[\text{start}] \leq k < n$. *(Strong inductive hypothesis.)* We show $P(n)$ holds.
[*Use $P(k)$ for some specific $k < n$.*]
Hence $P(n)$ holds.
By strong induction, $P(n)$ holds for all $n \geq [\text{start}]$. □

Example 6.4.18 (Every integer $n \geq 2$ has a prime factor). *Proof.* Let $P(n)$: "n has a prime factor."

Base case ($n = 2$). 2 is prime, so 2 is a prime factor of itself. $P(2)$ holds.

Inductive step. Let $n > 2$ and assume $P(k)$ holds for all $2 \leq k < n$.
Case 1: n is prime. Then n is its own prime factor, and $P(n)$ holds.
Case 2: n is composite. Write $n = ab$ with $2 \leq a, b < n$. By the strong inductive hypothesis applied to a, a has a prime factor p. Then $p \mid a$ and $a \mid n$, so $p \mid n$. Hence $P(n)$ holds.
In both cases $P(n)$ holds. By strong induction, $P(n)$ holds for all $n \geq 2$. □

Why weak induction cannot work directly: in Case 2, the value a satisfies $2 \leq a < n$ but is otherwise arbitrary. It is not necessarily $n-1$. Weak induction supplies $P(n-1)$, which is useless when $a \neq n-1$. Strong induction supplies $P(k)$ for all $k < n$, covering whatever value a takes.

Well-Ordering Principle

Every nonempty subset of $\mathbb{N}$ has a least element.

Remark 6.4.19 (The minimal counterexample argument). Well-ordering drives a proof strategy that is the structural dual of induction: the **minimal counterexample argument**.

Instead of climbing up from a base case, you assume the claim fails somewhere, take the smallest failure, and derive a contradiction from its minimality. The logic:

1. Suppose $P(n)$ fails for some n. Let $S = \{n \in \mathbb{N} : \neg P(n)\}$. By assumption $S \neq \emptyset$.
2. By well-ordering, S has a least element m.
3. Derive a contradiction: either show $P(m)$ holds (contradicting $m \in S$), or find some $k < m$ with $\neg P(k)$ (contradicting the minimality of m).

The key tool is minimality: the fact that m is the *smallest* failure means all smaller values are successes, which is a powerful assumption. It is equivalent to the strong inductive hypothesis.

Proposition (Induction and well-ordering are equivalent)

Proposition 6.4.20 (Induction and well-ordering are equivalent)**.** *Over the axioms of* $\mathbb{N}$ *without P5, the following three principles are logically equivalent:*

1. *The (weak) induction principle.*
2. *The well-ordering principle.*
3. *The strong induction principle.*

Remark 6.4.21 (Proof sketch of the equivalence). **Induction $\Rightarrow$ Well-ordering.** Suppose $S \subseteq \mathbb{N}$ has no least element. Let $P(n)$: "$n \notin S$." Then $P(0)$ holds (otherwise $0 \in S$ would be the least element). If $P(k)$ holds for all $k \leq n$, then $n+1 \notin S$ (otherwise $n+1$ would be the least element of S). By induction, $P(n)$ holds for all n, so $S = \emptyset$.

Well-ordering $\Rightarrow$ Induction. Suppose the base case $P(n_0)$ holds and the inductive step holds. Let $S = \{n \geq n_0 : \neg P(n)\}$. If $S \neq \emptyset$, let m be its least element. Since $P(n_0)$ holds, $m \neq n_0$, so $m > n_0$ and $m - 1 \geq n_0$. By minimality of m, $P(m-1)$ holds. But by the inductive step, $P(m-1) \Rightarrow P(m)$, contradicting $m \in S$. Hence $S = \emptyset$.

Template: Minimal Counterexample

Suppose for contradiction that $P(n)$ fails for some $n \in \mathbb{N}$.
Let $S = \{n \in \mathbb{N} : \neg P(n)\}$. By assumption $S \neq \emptyset$. By well-ordering, S has a least element m.
[*Derive a contradiction using minimality of m: either show $P(m)$ holds, or construct a smaller counterexample.*]
This contradicts [minimality of m / the base case]. Therefore $P(n)$ holds for all $n \in \mathbb{N}$. $\square$

Example 6.4.22 (Minimal counterexample: exponential bound). **Claim.** For all $n \geq 1$, $2^n > n$.

Proof (minimal counterexample). Suppose the claim fails. Let $S = \{n \geq 1 : 2^n \leq n\}$. By assumption $S \neq \emptyset$; let m be its least element.

Since $2^1 = 2 > 1$, we have $m \neq 1$, so $m \geq 2$. Then $m - 1 \geq 1$ and $m - 1 < m$, so $m - 1 \notin S$ (by minimality of m). Thus $2^{m-1} > m - 1$.

Then:

$$2^m = 2 \cdot 2^{m-1} > 2(m-1) = 2m - 2 \geq m$$

(using $m \geq 2$ for the last inequality).

This contradicts $m \in S$ (which requires $2^m \leq m$). Therefore $S = \emptyset$ and the claim holds for all $n \geq 1$. $\square$

The minimal counterexample argument proceeds by taking the smallest failure and showing it cannot be the smallest failure. The contradiction is always of the same form: either the "failure" was not a failure, or there is a smaller one.

Structural Induction (Preview). Induction on $\mathbb{N}$ is a special case of a general principle that applies to any recursively defined structure. You do not need $\mathbb{N}$ to do induction; you need a recursive definition.

Remark 6.4.23 (The general principle). Let X be a set defined by:

1. A collection of *base cases* (atomic elements not built from other elements), and
2. A collection of *constructor rules* (ways of building larger elements from smaller ones).

Structural induction on X: to prove $P(x)$ for all $x \in X$, prove P for all base cases, then prove that if P holds for the parts, it holds for the whole.

This is exactly standard induction where the "less than" ordering on $\mathbb{N}$ is replaced by the "proper sub-element" ordering on X.

Example 6.4.24 (Induction on algebraic terms). Define an *algebraic term* over a ring R by:

- Base case: any element $r \in R$ is a term.
- Constructor: if s, t are terms, so are $s + t$ and $s \cdot t$.

To prove a property P holds for all terms:

1. Prove $P(r)$ for all $r \in R$ (base cases).
2. Prove: if $P(s)$ and $P(t)$ hold, then $P(s + t)$ holds.
3. Prove: if $P(s)$ and $P(t)$ hold, then $P(s \cdot t)$ holds.

The recursive structure of terms determines the structure of the proof.

Remark (Exposition). Structural induction is the proof principle paired with structural recursion. Both follow the constructors of the object being studied. Structural recursion defines a function by giving its value on the base cases and then giving its value on each constructed object in terms of the values already defined on the immediate parts. Structural induction proves a predicate or property by showing that it holds for the base cases and is preserved by every constructor.

Thus structural induction is like ordinary mathematical induction: it proves a statement $P(x)$ for every object x in a recursively generated class. The difference is that the next step is not always the Peano successor $n \mapsto n{+}{+}$. The next step is whatever constructor builds the object. For natural numbers the constructors are 0 and successor. For formulas the constructors are atomic formulas, negation, binary connectives, and quantifiers. For trees the constructors are leaves and nodes.

A useful way to see this is to attach a syntax tree to each well-formed formula. Define trees for propositional formulas recursively:

- if $P \in \mathsf{Prop}$, then $\mathrm{leaf}(P)$ is a tree;
- if T is the tree for φ, then $\mathrm{node}(\neg, T)$ is the tree for $\neg\varphi$;

- if T_φ and T_ψ are the trees for φ and ψ, then $\text{node}(\circ, T_\varphi, T_\psi)$ is the tree for $(\varphi \circ \psi)$.

Operations on this tree are recursive in the same sense. The depth of a leaf is 0; the depth of a negation node is one plus the depth of its child; the depth of a binary node is one plus the maximum of the depths of its two children. Structural induction then proves statements about all such trees, and hence about all formulas, by checking the leaf case, the negation-node case, and the binary-node case. The proof follows the same recursive blueprint as the definition.

Remark 6.4.25 (Where structural induction appears in this project). Induction on $\mathbb{N}$ is structural induction on the Peano structure $(\mathbb{N}, 0, S)$: the base case is $n = 0$, the constructor is $n \mapsto n$++.

In the Abstract Algebra chapter, polynomial properties are proved by induction on degree, which is structural induction on the recursive definition of polynomials over a ring.

In computer science, correctness of recursive algorithms is proved by structural induction on the input data structure (lists, trees, expressions). The full treatment of structural induction appears in the Abstract Algebra chapter.

Common Induction Mistakes.

Remark 6.4.26 (Mistake 1: Missing base case). A proof that establishes the inductive step but not the base case proves nothing. The step says: *if* $P(n)$ holds for some n, *then* $P(n+1)$ holds. Without a base case, there is no n for which $P(n)$ is established, so the chain never starts.

The classic illustration: the false theorem "all horses are the same colour." The base case for $n = 1$ is trivially true (one horse has one colour). The inductive step has a subtle flaw in the overlap argument. The lesson is that both parts are essential and independent. Always verify the base case, even when it seems trivial. A trivial base case takes one line; write the one line.

Remark 6.4.27 (Mistake 2: Circular inductive step). The inductive step proves $P(n+1)$ using only $P(n)$ and previously established facts. It is a fatal error to assume $P(n+1)$ at any point in the proof of $P(n+1)$.

This error most often appears when a student "derives" both sides of an equation and meets in the middle, inadvertently assuming what they are trying to prove. The symptom: the final few lines of the step are not deductions from $P(n)$ but rather manipulations of $P(n+1)$ itself.

The cure: always work in one direction. Start with the left side of $P(n+1)$ (or whatever the goal is), apply the inductive hypothesis once, and algebraically arrive at the right side. Never write the target as an intermediate line and work toward it from both sides.

Remark 6.4.28 (Mistake 3: Unstated inductive hypothesis). The most common error in practice is writing "assume $P(n)$" without saying what $P(n)$ actually says. This is a sign that $P(n)$ has not been written out as a full sentence.

The effect is that the inductive step becomes vague at the crucial moment: the student needs to use the inductive hypothesis but cannot cite it precisely, so they either paraphrase it loosely (which may introduce errors) or implicitly use a stronger version than what was stated.

Write $P(n)$ as a complete sentence at the top of the proof. Then write "assume $P(n)$: [full sentence]." This forces precision and prevents the most common errors in the body of the proof.

Remark 6.4.29 (Mistake 4: Wrong base case)*.* Induction proves $P(n)$ for all $n \geq n_0$, where n_0 is whatever value you use as the base case. If the claim is only true for $n \geq 2$, the base case $n = 0$ will fail, and the proof is invalid.

Always check: what is the smallest n for which the claim is true? That is your base case. Do not default to $n = 0$ or $n = 1$ without verifying that the claim holds there. For divisibility statements, $n = 0$ often gives $0 = 0 \cdot k$, which works. For growth bounds, $n = 1$ or $n = 2$ may be needed. For Fibonacci identities, specific small values may require individual checking.

Remark 6.4.30 (Mistake 5: Using strong induction where weak suffices, and vice versa)*.* Strong induction is not automatically better than weak. A proof that invokes the full strong hypothesis "$P(k)$ for all $k < n$" when only $P(n-1)$ is actually used is unnecessarily complicated and obscures the proof's logic.

Conversely, a proof that tries to use weak induction on a recursion requiring two prior terms will stall at the step.

The diagnostic: write the step and check explicitly which prior values you use. If only $P(n)$, use weak. If $P(k)$ for some $k < n$ that is not $n - 1$, use strong.

Remark 6.4.31 (Mistake 6: Forgetting vacuity in strong induction)*.* In strong induction, the case $n = 0$ has a vacuously true hypothesis $\forall k < 0,\ P(k)$, since no natural number is less than 0. This means $P(0)$ must be proved *directly*, not from the inductive hypothesis. Many students write "by the inductive hypothesis applied to $n - 1$" for $n = 0$, which refers to $n - 1 = -1$, a non-existent value.

The base case in strong induction is always proved from scratch. The strong hypothesis is only available for $n \geq 1$.

6.5 Algebraic Tactics

The DU/TA/AM Framework. Every line of an algebraic proof is justified by exactly one of three sources. This framework makes that obligation explicit and provides the tagging system used throughout these notes.

The Three Line Tags

- DU **(Definition / Unpack).** The step applies a definition or unpacks a named concept into its explicit conditions. *Example*: expanding "e is an identity" into "$ea = ae = a$ for all $a \in G$".
- TA **(Theorem / Axiom).** The step cites a previously proved proposition or a structural axiom (group axiom, ring axiom, field axiom, Peano axiom). *Example*: citing G1 (associativity) to regroup $(ab)c = a(bc)$.
- TA **(Algebraic Manipulation, abbreviated AM).** The step performs a computation whose justification is itself a combination of DU and TA moves, compressed for readability. *Example*: simplifying $a \cdot e = a$ in one step after the identity axiom has already been cited.

Remark 6.5.1 (Why the framework matters: finding where proofs break). The DU/TA/AM system serves a diagnostic purpose. Every unjustified step in a proof is either a DU step that has not been taken (some definition has not been unpacked) or a TA step that has not been cited (some axiom or theorem is being used silently).

When an algebraic proof stalls, the right question is: which definition have I not yet unpacked? Which axiom am I using but not citing? The tags make these omissions visible.

In practice, stalling almost always traces to a definition that has been left in its named form ("G is a group") when it needed to be expanded into operative conditions ("associativity holds; an identity exists; every element has an inverse"). Expand the definition and the stuck proof typically becomes unstuck.

Remark 6.5.2 (The hierarchy: DU and TA are atomic; AM is shorthand). DU and TA are the atomic justifications. There is nothing more basic. AM is not a separate category of reasoning — it is a compressed sequence of DU and TA steps that have been verified and labeled as a single unit for concision.

A fully formal proof contains only DU and TA steps, explicitly tagged. The three-column proof format used in this project — tag in the left column, statement in the middle, commentary in the right — enforces explicit justification of every line and is the recommended format while you are learning.

As fluency develops, AM steps can absorb more and more of the lower-level manipulations. But the underlying DU/TA structure is always present, even when not written. The ability to decompose any AM step into its DU and TA components is the hallmark of genuine understanding.

Remark 6.5.3 (A proof is a chain of citations). The deepest way to understand the DU/TA/AM system is this: a mathematical proof is a sequence of statements, each of which is justified by citing something that is already known (a definition, an axiom, or a previously proved theorem). The proof is valid if and only if every step has a valid citation.

When you read a proof and feel uncertain whether it is correct, the right move is to label every step. If you can assign a DU or TA justification to every line, the proof is valid. If any line has no justification, there is a gap.

This is not just a learning technique. Working mathematicians who find a suspicious step in a proof do exactly this: they ask "what justifies this?" and try to answer. If no answer is available, the step is an error.

Catalog of Algebraic Tactics. The following catalog lists the specific algebraic moves available once a proof is in progress and the architecture has been chosen. Each entry states the tactic, its precondition (what must already be established or assumed), and its effect (what it achieves in the proof).

Tactic	Precondition	Effect
Multiply by inverse	$a \neq 0$ in a field; or $a \in G$ in a group	Multiply both sides by a^{-1}; use associativity to collapse $a^{-1}a = e$; isolates the target variable.
Cancellation	$ac = bc$ (or $ca = cb$) in a group or integral domain	Conclude $a = b$. In a group: multiply by c^{-1}. In an integral domain: cite no-zero-divisors.
Distributivity bridge	Multiplication by 0, -1, or a negative element	Rewrite the product as a sum using distributivity, then use additive group structure to collapse. Crosses from $\times$ to $+$.
Add zero	Need to introduce a term without changing the expression	Write $a = a+0 = a+(b+(-b))$; regroup to expose the needed structure.
Satisfy-and-cite	A uniqueness theorem for some named object b is in scope	Show a satisfies b's defining property; cite uniqueness; conclude $a = b$. Avoids assume-two-and-compare entirely.
Double negation	$-(-a)$ appears or is the goal	$-a$ is the unique element x with $a + x = 0$; $-(-a)$ satisfies this; by uniqueness, $-(-a) = a$.
Absorb into axiom	A subexpression matches the LHS of an axiom	Rewrite using the axiom. Tag as TA.

Remark 6.5.4 (The distributivity bridge in detail). The distributivity bridge is the central tactic for ring and field proofs involving zero and negatives. Every instance of "$a \cdot 0 = 0$" and "$a(-b) = -(ab)$" uses it. Once you have seen the pattern, it is immediately recognizable.

The pattern always has four steps:

1. Introduce $a \cdot 0$ or $a \cdot (-b)$ into the argument.
2. Expand: write $0 = x + (-x)$ or $-b = b + (-b) + (-b)$ using the additive inverse definition.

3. Apply distributivity: $a(x + y) = ax + ay$.
4. Use uniqueness of the additive inverse to identify the result with $-(ab)$ or 0.

The reason this requires distributivity as a bridge is that the additive structure (with its identity 0 and its inverses $-a$) and the multiplicative structure (with its products ab) are separate. The only connection between them is distributivity. To get from the multiplicative side $(a \cdot 0)$ to the additive side $(0 = 0 + 0)$, you must cross the bridge.

Remark 6.5.5 (Cancellation: two theorems that must not be conflated). "Cancellation" names two distinct results with different preconditions.

Group cancellation. In any group, $ac = bc \Rightarrow a = b$. Proved by multiplying both sides on the right by c^{-1}, which always exists in a group. The hypothesis $c \neq 0$ is not needed because inverses always exist in a group.

Domain cancellation (or integral domain cancellation). In an integral domain, $ac = bc$ and $c \neq 0$ imply $a = b$. Proved by rewriting as $(a - b)c = 0$ and invoking the no-zero-divisors property. The hypothesis $c \neq 0$ is essential: without it, cancellation can fail (e.g., in $\mathbb{Z}_6$, $2 \cdot 3 = 4 \cdot 3$ but $2 \neq 4$).

The error to avoid: using group cancellation in an integral domain argument (it works but uses stronger machinery than needed and may not be available), or trying to use domain cancellation without verifying $c \neq 0$.

Remark 6.5.6 (Add zero: the simplest but most overlooked tactic). The add-zero tactic sounds trivial but appears in almost every algebraic proof where an intermediate term needs to be introduced without changing the value of an expression. The template is:

$$a = a + 0 = a + (b + (-b)) = (a + b) + (-b).$$

This exposes $a + b$ as a subexpression of a, which can then be worked with. It is the algebraic equivalent of adding and subtracting the same term in a real analysis inequality argument.

When you need a term to appear in an expression and cannot see how to introduce it, ask: can I write some zero as $x + (-x)$ and distribute?

6.6 Proof Strategies Reference

Lookup Table: Stuck Situations and Strategies.

Remark 6.6.1 (How to use this table). When progress on a proof stalls, locate the row that describes your situation. The middle column names the strategy. The right column gives the specific first move and a cross-reference to the section where that strategy is developed in detail.

This table is not a substitute for understanding the strategies; it is a navigation aid for when you know what you are trying to do but have forgotten the precise machinery. The full explanations and worked examples are in the sections cited.

Stuck because...	Strategy	First move / See
I need to prove P but don't know where to start	Two-question check	Write the logical form of P. Consult the statement map. §6.1
I need to show two things are equal and one is uniquely defined	Satisfy-and-cite	Show the first satisfies the definition of the second. Cite the uniqueness theorem. §6.1
I need to show something is the only object of its kind	Assume-two-and-compare	"Let x, y both satisfy the definition. Show $x = y$." §6.3
I have a conditional $P \Rightarrow Q$ and P is hard to use forward	Contrapositive	"Assume $\neg Q$." Derive $\neg P$. §6.3
I have a conditional and the conclusion seems impossible to reach directly	Contradiction	"Assume P and $\neg Q$." Derive any contradiction. §6.3
The claim splits naturally into a finite number of cases	Case analysis	List all cases explicitly. Verify exhaustiveness first. §6.3
The claim is about all $n \in \mathbb{N}$	Weak induction	Write $P(n)$ as a full sentence. Prove the base case directly. §6.4
The inductive step needs $P(k)$ for some $k < n$ (not just $n - 1$)	Strong induction	Replace "assume $P(n)$" with "assume $P(k)$ for all $k < n$." §6.4
The claim is about all $n \geq n_0$ and feels easier to negate	Minimal counterexample	"Let $S = \{n : \neg P(n)\} \neq \emptyset$. Let $m = \min S$." §6.4
I need to prove $a \cdot 0 = 0$ or $a(-b) = -(ab)$ in a ring	Distributivity bridge	Write $0 = x + (-x)$; apply distributivity; use uniqueness of additive inverse. §6.5
I have $ac = bc$ and need $a = b$	Cancellation	Group: multiply by c^{-1}. Domain: write $(a - b)c = 0$; cite no zero divisors. §6.5
I need to prove two sets are equal	Double inclusion	Prove $A \subseteq B$: let $x \in A$ arbitrary, derive $x \in B$. Then prove $B \subseteq A$. §6.1
A definition has not been unpacked	DU move	Identify the named concept and write out its definition. Tag the line DU. §6.5
A step uses an axiom or	TA move	Cite the axiom or theorem explicitly

Remark 6.6.2 (Cross-references to primary worked examples). The following table maps each strategy to a worked example in these notes where it is the primary tool.

Strategy	**Primary worked examples**
Assume-two-and-compare	Uniqueness of group identity (this section)
Satisfy-and-cite	$(ab)^{-1} = b^{-1}a^{-1}$; $a(-b) = -(ab)$ (this section)
Distributivity bridge	$a \cdot 0 = 0$; $a(-b) = -(ab)$ (Tactics section)
Cancellation (group)	Group left-cancellation (Tactics section)
Cancellation (domain)	Integral domain cancellation (Tactics section)
Weak induction	Sum formula $\sum k = n(n+1)/2$; $3 \mid 4^n - 1$
Strong induction	Fibonacci bound; prime factor theorem
Minimal counterexample	$2^n > n$
Contrapositive	n^2 even $\Rightarrow n$ even
Contradiction	$\sqrt{2} \notin \mathbb{Q}$

6.7 Analysis: What Changes and Why

Why Analysis Proofs Feel Different. Everything in the preceding sections applies to mathematics broadly: direct proof, induction, contradiction, and the rest are the universal grammar of mathematical argument. But real analysis — sequences, metric spaces, limits, continuity — introduces a particular *flavour* of proof work that surprises many students when they first encounter it.

This section prepares you for that surprise before you arrive at Volume III. The goal is not to teach you analysis; it is to give you the framework so that when you get there, the proofs do not feel alien.

Remark 6.7.1 (The shift from computation to verification). In calculus, most tasks are *computational*: differentiate this function, evaluate this integral, find this limit by l'Hôpital's rule. You apply known procedures and the answer is a number or a function.

In analysis, most tasks are *verificational*: prove that this sequence converges, show that this space is complete, establish that this function is continuous. The answer is a logical argument. There is no procedure to apply; there is a structure to find.

This shift is not just psychological. It changes what "progress" means. In a computation, you know you are making progress when the expressions simplify. In a proof, you know you are making progress when the definitions are unpacked, the hypotheses are applied, and the conclusion comes into reach. The habits of mind required are different.

Remark 6.7.2 (Definitions are not background; they are the machinery). The single most important habit in analysis is **returning to definitions**. Every concept — convergence, Cauchy, compactness, continuity — has a precise definition in terms of quantifiers and inequalities. That definition is not background knowledge. It is the working machinery of every proof involving that concept.

Here is the canonical example. To prove that $1/n \to 0$, a calculus student says: "obviously it approaches zero." An analysis student unpacks the definition:

For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|1/n - 0| < \varepsilon$.

Then they find N: choose $N > 1/\varepsilon$ (possible by the Archimedean property). For $n \geq N$: $1/n \leq 1/N < \varepsilon$.

The proof is three lines, and every line is either the definition (unpacked), the Archimedean property (a theorem), or arithmetic. There is no mystery. The entire proof is the definition plus two lines of calculation. This is the prototype of an analysis proof.

The lesson: when a proof stalls, the first question to ask is always: *which definition have I not yet unpacked?*

Remark 6.7.3 (Analysis proofs are structured, not creative). Here is perhaps the most liberating thing to understand about analysis: **proofs are far less creative than they look.**

Experienced analysts do not invent proofs from scratch. They recognize the type of exercise, recall the standard move for that type, and apply it. The vocabulary of standard moves is small — twelve moves cover virtually all of the proofs in a first course in metric spaces. The creativity, if any, is in recognizing which move to apply.

This is the same observation that governs the Architecture section of this chapter (Section 6.1), scaled down to the specific landscape of analysis. The statement map tells you the global structure. In analysis, a more fine-grained map is available: there are specific named moves for specific recurring situations. Learning to name them is the same skill as learning to read a statement and immediately identify its proof type.

The rest of this section makes that vocabulary explicit.

The Six Exercise Types. Every exercise in sequences and metric spaces belongs to one of six structural types. Identifying the type is the first step before writing anything. The type determines the shape of the solution — not the details, but the overall structure.

Quick Reference: The Six Types

Type	Signal words	Shape	First move
Proof	*Prove, show, establish*	Hypotheses to conclusion	Unpack all definitions
Verification	*Verify that . . . is a metric, show . . . satisfies*	Check each condition	List every condition
Prove/Counterexample	*True? Prove or find a counterexample*	Decide, then justify	Decide first
Construction	*Find, construct, exhibit, give an example of*	Name the object; verify it	Name the object
Computation	*Compute, calculate, find the diameter of*	Evaluate from a definition	Write the definition
Characterization	*If and only if, characterize all . . . that*	Two separate directions	Split two directions

Remark 6.7.4 (Type 1 (Proof): you are told the statement is true). In a proof exercise, the statement is true and your job is to trace the logical path from the hypotheses to the conclusion. There is no choice to make about truth value.

The opening move is always the same: unpack every definition in the hypotheses and in the conclusion. The definitions turn abstract concepts into concrete quantifier-and-inequality statements that can be manipulated. Without this step, you have nothing to work with.

Example. *Prove that every convergent sequence is Cauchy.* Unpack "convergent": $\forall \varepsilon > 0\ \exists N$ s.t. $n \geq N \Rightarrow d(x_n, p) < \varepsilon$. Unpack "Cauchy": $\forall \varepsilon > 0\ \exists M$ s.t. $m, n \geq M \Rightarrow d(x_m, x_n) < \varepsilon$. Now the proof goal is visible: use the convergence condition (which controls $d(x_n, p)$) to establish the Cauchy condition (which asks for control over $d(x_m, x_n)$). The triangle inequality bridges them.

Remark 6.7.5 (Type 2 (Verification): check every condition). In a verification exercise, a specific named object is given and you must confirm it satisfies every condition of a definition. The definition has a finite list of conditions. Your job is to work through every one, explicitly and separately.

The error students make: checking four out of five conditions and writing QED. A verification is complete only when every condition is checked. When you begin a verification, write the complete list of conditions first. Then work through them in order.

Example. *Verify that the discrete metric is a metric.* A metric has four conditions: non-negativity, identity of indiscernibles, symmetry, triangle inequality. List all four. Check each one. The triangle inequality requires a case split (on whether $x = z$ or $x \neq z$), which is itself a standard move in analysis.

Remark 6.7.6 (Type 3 (Prove or counterexample): decide before you argue). This is the most dangerous type for students who rush. The exercise gives you a statement and asks you to determine its truth value, then justify. If you skip the determination stage and try to prove a false statement, you will waste effort and grow confused.

Stage 1 is always: decide. Test the statement on simple cases — $\mathbb{R}$ with the standard metric first, then discrete spaces, then other special cases. One counterexample falsifies; failure to find a counterexample after testing several cases suggests the statement is true.

Stage 2: justify. If true, write a proof. If false, exhibit a specific counterexample and verify it fails.

Example. *Is the open ball always equal to the closed ball?* Test in $\mathbb{R}$: $B(0,1) = (-1,1)$ and $C(0,1) = [-1,1]$. The point -1 is in $C(0,1)$ but not $B(0,1)$. False. Exhibit this specific counterexample in the solution, with the witness (-1) and the calculation ($d(-1,0) = 1$, so $-1 \in C(0,1)$ and $-1 \notin B(0,1)$).

Remark 6.7.7 (Type 4 (Construction): name first, verify second). A construction exercise asks you to produce an object with given properties. The solution has two parts: stating the object, and verifying it has every required property. Both parts are required.

State the object completely and unambiguously before any verification. If you cannot state it clearly, you do not yet have a solution. Then verify every required property from scratch — do not assume properties that were not established.

Example. *Find a metric on* $\mathbb{N}$ *such that* $\mathbb{N}$ *is bounded.* The idea: embed $\mathbb{N}$ into $(0,1] \subset \mathbb{R}$ via $n \mapsto 1/n$ and pull back the metric. Define $d(m,n) = |1/m - 1/n|$. Then verify: (i) it is a metric (four conditions), and (ii) it is bounded ($d(m,n) \leq 1$ for all m,n).

Remark 6.7.8 (Type 5 (Computation): the definition is the answer). In a computation exercise, a specific numerical or set-theoretic value is asked for. The answer always comes from writing out the relevant definition as a supremum, infimum, or set, and then evaluating it.

Do not skip straight to the answer. Write the definition first. This is not just for clarity — it prevents errors that arise from misremembering what is being computed.

Example. *Compute* $\operatorname{diam}([a,b])$. Write: $\operatorname{diam}([a,b]) = \sup\{|x-y| : x,y \in [a,b]\}$. Upper bound: $|x-y| \leq b-a$ for all $x,y \in [a,b]$. Attained: $|b-a| = b-a$. Therefore $\operatorname{diam}([a,b]) = b-a$.

Remark 6.7.9 (Type 6 (Characterization): two proofs, not one). A characterization exercise asks for an equivalence or an if-and-only-if. The solution is always two separate proofs, labeled ($\Rightarrow$) and ($\Leftarrow$). Label them before writing a single line.

The two directions often require completely different techniques and different amounts of work. The ($\Rightarrow$) direction may be three lines while the ($\Leftarrow$) direction requires a sequence argument. Writing them separately prevents the error of running arguments in both directions simultaneously and losing track of which assumptions are in play.

Example. *Show that a sequence converges if and only if it is eventually constant (in the discrete metric).* ($\Leftarrow$): Suppose eventually constant; then convergence holds trivially. ($\Rightarrow$): Suppose it converges; apply the definition with $\varepsilon = 1/2$ and use the fact that distances in the discrete metric are either 0 or 1.

The Twelve Canonical Proof Moves. A *proof move* is a recurring logical or computational technique that appears across many different proofs in analysis. Once you learn to name these moves, reading and writing proofs becomes a matter of recognizing which move applies — not inventing an argument from scratch.

The moves below are not tricks. Each one encodes a genuine mathematical idea. But once the idea is understood, the move can be deployed almost mechanically in the right situations.

The Twelve Moves at a Glance

#	Move	Use when ...	Level
1	Unpack a definition	starting any proof	L1
2	Triangle inequality	need to bound $d(x, z)$ via y	L2
3	ε-splitting	triangle ineq. produces a sum	L2
4	Tail argument	combining two N's	L3
5	Case splitting	discrete or binary structure	L2
6	Choose N explicitly	ε-N arguments	L3
7	Contradiction	uniqueness or non-existence	L4
8	Subsequence extraction	sequence misbehaves globally	L3
9	Squeeze (sandwich)	bound $d(x_n, p)$ from above	L2
10	Inherited metric / embedding	constructing a new metric	L4
11	Open cover / finite subcover	compactness arguments	L4
12	Diagonal argument	simultaneous subsequence conditions	L3/4

Remark 6.7.10 (Move 1: Unpack a definition)*.* This is always the first move. Every analysis concept — convergence, Cauchy, compactness, closure, diameter — is a compressed form of a precise statement with quantifiers and inequalities. Unpacking the definition turns the abstract concept into something you can work with.

Without this step, you have named objects but no machinery. With it, you have the raw material for every subsequent move.

Template. Write out the definition of every term in the hypothesis and in the goal. The hypotheses become your assumptions; the goal becomes your target inequality or quantifier statement.

Remark 6.7.11 (Moves 2 and 3: Triangle inequality and ε-splitting)*.* These two moves almost always appear together. The triangle inequality lets you insert an intermediate point y and split $d(x, z)$ into $d(x, y) + d(y, z)$. ε-splitting means choosing the bounds on each piece so they add up to ε.

The intermediate point y is not arbitrary: it is chosen to be something you already have control over (a limit, a fixed center, a sequence element). The fractions $\varepsilon/2$, $\varepsilon/3$ are chosen *before* picking N, so that the final bound comes out to ε exactly.

Template. When you need $d(x_m, x_n) < \varepsilon$ and you know each term is close to a limit p: insert p via the triangle inequality, budget $\varepsilon/2$ for each piece, and choose N so both pieces are bounded.

Remark 6.7.12 (Moves 4 and 6: Tail argument and choosing N)*.* The ε-N definition of convergence requires you to *produce* an N. This is Move 6: work backwards from the target inequality to determine what N must satisfy, then assert its existence (typically by the Archimedean property: for any $\varepsilon > 0$ there is an integer greater than $1/\varepsilon$).

Move 4 (the tail argument) handles the situation where two separate conditions each give their own N_i. The resolution: take $N = \max(N_1, N_2, \ldots)$. For $n \geq N$, all conditions hold simultaneously.

These two moves are the engine of every ε-N argument. They appear in virtually every proof about convergent sequences.

Remark 6.7.13 (Move 5: Case splitting in analysis). Case splitting is the same structural technique described in Section 6.3, now applied to the specific context of metric spaces. The most common case split in metric space proofs is: $x = y$ (distance is 0) versus $x \neq y$ (distance is 1 in the discrete metric, or some positive value in other spaces).

Case splitting is particularly important in verification exercises: the triangle inequality almost always requires splitting on whether the two outer points coincide.

Remark 6.7.14 (Move 7: Contradiction in analysis). In analysis, contradiction is used most characteristically for *uniqueness* results: suppose two limits exist; assume they are different; derive a contradiction from the triangle inequality and the convergence hypothesis.

The template: suppose the goal fails (e.g., two limits exist with $d(x, y) > 0$). Choose $\varepsilon = d(x, y)/2$. Then for large enough n, both $d(x_n, x) < \varepsilon$ and $d(x_n, y) < \varepsilon$. The triangle inequality gives $d(x, y) \leq 2\varepsilon = d(x, y)$, a contradiction.

Notice: contradiction here uses the triangle inequality (Move 2) and the tail argument (Move 4) internally. Proof moves nest and combine.

Remark 6.7.15 (Moves 8 and 12: Subsequence extraction and diagonal argument). Subsequence extraction is the operation of selecting indices $n_1 < n_2 < \cdots$ so that the restricted sequence $\{x_{n_k}\}$ has some desired property (being monotone, being close to a given point, etc.). It is a Level 3 operation: you are selecting from an infinite sequence, which requires genuine sequence machinery.

The diagonal argument applies subsequence extraction infinitely many times and diagonalizes the result: the k-th term of the diagonal sequence is $x_k^{(k)}$, where $\{x_n^{(k)}\}$ is the k-th extracted subsequence. The diagonal sequence satisfies all conditions simultaneously. This is the technique behind Arzelà–Ascoli and related compactness results.

Remark 6.7.16 (Move 9: The squeeze (sandwich)). If you can show $0 \leq d(x_n, p) \leq c_n$ where $c_n \to 0$, then $x_n \to p$ follows immediately: any ε that works for c_n also works for $d(x_n, p)$.

The squeeze is particularly useful when $d(x_n, p)$ is hard to bound directly but can be dominated by a simpler sequence whose convergence is known.

Remark 6.7.17 (Move 10: Inherited metric / embedding). To construct a metric on a new space X, it is often easiest to find an injective function $f : X \to Y$ into a space Y whose metric is known, and define $d_X(x, z) := d_Y(f(x), f(z))$. All four metric axioms are then inherited from d_Y via f. The only things to verify are injectivity of f (to ensure the identity axiom) and whatever additional properties are required.

This is a Level 4 move because it operates on the global structure of the space: you are choosing an ambient space and an embedding, not just manipulating inequalities.

Remark 6.7.18 (Move 11: Open cover / finite subcover). Compactness says: every open cover of a compact set K has a finite subcover. The power of this is that it converts infinite problems into finite ones. If you need a single bound δ that works everywhere on K, you find a δ_x at each x (infinitely many), then extract a finite subcover and take the minimum of finitely many δ_x. A finite minimum exists; a minimum over infinitely many values might not.

This move appears in all compactness arguments: uniform continuity on compact sets, compactness implies boundedness, compactness implies completeness in complete metric spaces.

The Four-Layer Hierarchy. The twelve moves are not all at the same level. They form a four-layer hierarchy that reflects the logical structure of metric space proofs: each layer depends on the one below it, and together the four layers describe a progression from the most basic (unpacking definitions) to the most structural (global topological reasoning).

Understanding the hierarchy is not an organizational convenience. It tells you something deep: *why* metric spaces are defined the way they are, and why nearly every proof in the subject follows the same basic upward shape. Once you see this, metric space theory stops looking like a collection of unrelated tricks and starts looking like a small, elegant machine built from a few core mechanisms.

The Four Levels

L	Name	Moves	What it does
L1	Definition Expansion	Move 1	Turns abstract concepts into inequalities
L2	Inequality Mechanics	Moves 2, 3, 5, 9	Combines and bounds distances
L3	Sequence Control	Moves 4, 6, 8, 12	Controls asymptotic and tail behavior
L4	Structural / Topological	Moves 7, 10, 11	Reasons about the space as a whole

Remark 6.7.19 (Level 1: Definition Expansion — the foundation of everything). Every other move operates on inequalities. Those inequalities do not appear until you unpack a definition. Before Level 1, you have a statement about abstract objects. After Level 1, you have quantifiers and specific inequality conditions that can be manipulated.

This is why Level 1 is always the first step in any analysis proof, regardless of how complex the argument becomes. The inequality $d(x_n, p) < \varepsilon$ does not exist in the proof until you write it down. You write it down by unpacking the definition of convergence.

What Level 1 produces:

- "$x_n \to p$" $\mapsto \forall\, \varepsilon > 0\; \exists\, N$ s.t. $n \geq N \Rightarrow d(x_n, p) < \varepsilon$
- "$\{x_n\}$ is Cauchy" $\mapsto \forall\, \varepsilon > 0\; \exists\, M$ s.t. $m, n \geq M \Rightarrow d(x_m, x_n) < \varepsilon$
- "K is compact" $\mapsto$ every open cover of K has a finite subcover
- "E is closed" $\mapsto$ every convergent sequence in E converges to a point of E

The abstract word is a compressed form of the inequality. Unpacking it is not simplification; it is *activation*.

Remark 6.7.20 (Level 2: Inequality Mechanics — the algebra of metric spaces). Once Level 1 has produced inequalities, Level 2 manipulates them. This is the algebra of metric spaces: you combine, bound, split, and chain inequalities to move from what you know toward what you need.

The triangle inequality is a *metric axiom* for a reason: it was included in the definition of a metric precisely because it is the tool needed to chain distance bounds. Without it, you cannot get from control over $d(x, y)$ and $d(y, z)$ to control over $d(x, z)$. The axiom is there because it enables Level 2.

Notice that Level 2 does not involve sequences, indices, or tails. It is purely about bounding distances between specific points. The sequence machinery comes at the next level.

Remark 6.7.21 (Level 3: Sequence Control — the engine of limit arguments). Analysis is fundamentally about sequences approaching limits, distances becoming arbitrarily small, behavior stabilizing as an index grows large. Level 3 controls this asymptotic behavior.

The key idea: sequence control is about managing the word "eventually." The definition of convergence says that for large enough n, the distance $d(x_n, p)$ is small. The tail argument combines multiple "large enough n" conditions by taking the maximum index. Choosing N explicitly produces the specific threshold. Subsequence extraction selects which terms of the sequence to keep.

These moves require genuine sequence machinery — you are reasoning about indexed infinite processes, not just about individual distances between points. That is what places them at Level 3 rather than Level 2.

Remark 6.7.22 (Level 4: Structural / Topological — global reasoning). Level 4 moves operate on the global structure of the metric space, not on individual sequences or inequalities. Contradiction reasons about the logical structure of the whole proof. Embedding reasons about how the space sits inside a larger ambient space. Compactness reasons about the global covering properties of the space.

Compactness is the flagship Level 4 property. It converts infinite problems into finite ones: an infinite family of open sets reduces to a finite subcover, and a finite collection has a minimum, a maximum, a single controlling constant. This is the most powerful tool in metric space theory.

Level 4 and Level 3 interact bidirectionally. Sequential compactness (every sequence has a convergent subsequence) is a Level 3 characterization of a Level 4 property. Many deep results in analysis are precisely about these interactions.

Remark 6.7.23 (Reading a proof using the hierarchy). When you read a proof in the textbook, label each step by level. This is not a mechanical exercise; it reveals the logical architecture and tells you why each step appears where it does.

Here is the proof that convergent implies Cauchy, labeled:

> Let $\varepsilon > 0$. *[L1: introduce ε from the Cauchy definition]*
>
> Since $x_n \to p$, there exists N such that $n \geq N \Rightarrow d(x_n, p) < \varepsilon/2$. *[L3: choose N from the convergence hypothesis]*

For $m, n \geq N$: *[L3: tail condition]*

$d(x_m, x_n) \leq d(x_m, p) + d(p, x_n)$ *[L2: triangle inequality]*

$< \varepsilon/2 + \varepsilon/2 = \varepsilon.$ *[L2: ε-splitting]*

Hence $\{x_n\}$ is Cauchy. *[L1: conclusion matches Cauchy definition]*

This proof uses Levels 1, 2, and 3, but not Level 4. It is a local argument about sequences, not a global structural argument. To reach Level 4, you would need compactness or an embedding argument.

The levels a proof reaches tell you its depth and what tools it uses. A verification exercise stays at Levels 1–2. A limit argument reaches Level 3. A compactness argument reaches Level 4.

Remark 6.7.24 (Why the metric axioms are the axioms they are). The metric axioms — non-negativity, identity of indiscernibles, symmetry, triangle inequality — are not arbitrary. They are exactly the conditions needed to make Level 2 possible.

Without the triangle inequality, you cannot chain distance bounds. Without identity of indiscernibles, the notion of a limit point degenerates. Without symmetry, the definitions of balls and convergence become asymmetric and unworkable.

The axioms are the *minimal conditions on a distance function that support the full four-level proof hierarchy.* When you study a new property of metric spaces, ask: which level does it live at? The answer tells you what tools to expect and what kind of proofs will be required.

How to Read an Analysis Exercise Before Writing Anything. The preceding sections have given you the vocabulary: six exercise types, twelve proof moves, four levels. This section shows how to put them together in the five minutes before you write your first symbol.

Pre-Writing Checklist

1. **Identify the type.** Which of the six types is this: Proof, Verification, Prove-or-Counterexample, Construction, Computation, Characterization? Write the type at the top of your scratch paper before anything else.

2. **State the hypotheses and the goal.** In plain English, what are you assuming? What are you trying to show? Do not proceed until both are stated explicitly.

3. **Unpack every definition in the goal.** If the goal involves a named concept — convergence, compactness, closed, bounded — write out its definition in full. The unpacked goal is your actual target.

4. **Identify the key move.** Which of the twelve moves is most likely to bridge the hypothesis to the unpacked goal? Which level does that move belong to? Knowing the level tells you roughly how deep the proof will go and what other moves will likely appear alongside it.

5. **Sketch the shape in words.** Write in plain language: "I will choose $\varepsilon > 0$. I will use convergence of $\{x_n\}$ to find N_1. I will use convergence of $\{y_n\}$ to find N_2. I will take $N = \max(N_1, N_2)$. Then I will apply the triangle inequality." Only after this sketch do you fill in the symbols.

Remark 6.7.25 (The shape tells you what you need before you need it). Step 5 is the most important and the most skipped. Students who jump directly to symbols frequently paint themselves into corners: they choose N for one condition and then need it to satisfy a second condition that their choice does not support.

If you write the shape of the proof in words first, you see *how many conditions N must satisfy* before you commit to a specific value. Then you choose N to satisfy all of them simultaneously. The tail argument tells you how: take the maximum of however many separate N_i's the conditions require.

The shape also tells you how many terms the triangle inequality will produce and therefore what the right ε-splitting fractions are. These are not things you discover mid-proof; they are visible from the sketch.

Remark 6.7.26 (Common proofs and their shapes). The following table summarizes the shape of the most common proof patterns in a first analysis course, listed by the move sequence and the levels they traverse.

Goal	Move sequence	Levels
Convergent $\Rightarrow$ Cauchy	Unpack both definitions; triangle inequality; ε-splitting; tail argument (max of two N's).	L1–L3
Uniqueness of limits	Contradiction (assume two limits); triangle inequality; choose $\varepsilon = d(x, y)/2$; tail argument.	L2–L4
Convergent $\Rightarrow$ bounded	Unpack convergence; choose $\varepsilon = 1$; tail argument (bound $n \geq N$); take max of first N terms.	L1–L3
Verify d is a metric	List four conditions; check non-negativity, identity, symmetry directly; case split for triangle inequality.	L1–L2
Compute diameter of $[a, b]$	Write diameter as supremum; give upper bound; exhibit achieving pair.	L1–L2

Remark 6.7.27 (A note on templates). For each of the six exercise types, there is a standard template structure. These templates live in Appendix A of the companion *Primer for Real Analysis Exercises*, which you will receive at the start of Volume III. The templates are starting-point structures, not scripts to be memorized. Their purpose is to ensure that every solution has all required parts and that no structural element is accidentally omitted.

What matters at this stage is the vocabulary: six types, twelve moves, four levels. You now have the map. Volume III will fill in the terrain.

Set Theory

Chapter 7

Set Theory

SERIES
From Cantor to Ito

VOLUME I
Logic, Sets, and Proof

BOOK II
Set Theory

CHAPTER 5
Set Theory

7.1 Model of Set Theory

Theory (ZF, ZFC)

Depends on: `theory:first-order-logic`

Language

$$\mathcal{L}_{\in} = \{\in\}$$

First-order language with equality. The symbol $\in$ is binary. Equality is logical.

Presented Theory

$$\mathrm{ZF} := \mathrm{Cn}_{\vdash}(\Sigma_{\mathrm{ZF}}), \qquad \mathrm{ZFC} := \mathrm{Cn}_{\vdash}\big(\Sigma_{\mathrm{ZF}} \cup \{\mathrm{AC}\}\big).$$

Axioms and Laws

1. **1. Extensionality**

$$\forall x\, \forall y\, \big[\forall z\, (z \in x \leftrightarrow z \in y) \to x = y\big].$$

2. **2. Empty Set**

$$\exists x\, \forall y\, \neg(y \in x).$$

3. **3. Pairing**

$$\forall x\, \forall y\, \exists z\, \forall w\, \big(w \in z \leftrightarrow (w = x \lor w = y)\big).$$

4. **4. Union**

$$\forall x\, \exists y\, \forall z\, \big[z \in y \leftrightarrow \exists w\, (w \in x \land z \in w)\big].$$

5. **5. Power Set**

$$\forall x\, \exists y\, \forall z\, \big[z \in y \leftrightarrow \forall w\, (w \in z \to w \in x)\big].$$

6. **6. Infinity**

$$\begin{aligned}\exists x\, \Big[&\exists y \big(y \in x \land \forall w\, \neg(w \in y)\big)\\ &\land \forall y \big(y \in x \to \exists z\, (z \in x \land \forall w\, (w \in z \leftrightarrow (w \in y \lor w = y)))\big)\Big].\end{aligned}$$

7. **7. Separation Schema** For each formula $\varphi(z, \bar{w})$ in which y is not free:

$$\forall \bar{w}\, \forall x\, \exists y\, \forall z\, \big[z \in y \leftrightarrow (z \in x \land \varphi(z, \bar{w}))\big].$$

8. **8. Replacement Schema** For each formula $\varphi(x, y, \bar{w})$ in which B is not free:

$$\begin{aligned}\forall \bar{w}\, \forall A\, \Big[&\forall x \big(x \in A \to \exists! y\, \varphi(x, y, \bar{w})\big)\\ &\to \exists B\, \forall y \big(y \in B \leftrightarrow \exists x\, (x \in A \land \varphi(x, y, \bar{w}))\big)\Big].\end{aligned}$$

9. **9. Foundation**

$$\forall x\, \Big[\exists y\, (y \in x) \to \exists y \big(y \in x \land \neg \exists z\, (z \in x \land z \in y)\big)\Big].$$

Layer Delta

AC

$$\begin{aligned}\forall x\, \Big[&\big(\forall y\, (y \in x \to \exists z\, z \in y)\\ &\land \forall y\, \forall y'\, ((y \in x \land y' \in x \land y \neq y') \to \neg \exists z\, (z \in y \land z \in y'))\big)\\ &\to \exists c\, \forall y\, (y \in x \to \exists! z\, (z \in y \land z \in c))\Big].\end{aligned}$$

Model (Set Theory)

Model 7.1.1 (Set Theory Model). **Depends on:** `model:first-order-logic`; `bridge:free-sigma-algebra-evaluation`; `theory:set-theory`; `bridge:consequence`

Structure

$$\mathcal{A} = (A, E), \qquad A \neq \varnothing, \qquad E \subseteq A \times A.$$
$$a \; E \; b \quad :\Longleftrightarrow \quad (a, b) \in E.$$
$$\mathcal{A} = \langle A, I \rangle, \qquad I(\in) = E.$$

Sorts and Domains

A	universe

Interpretation

Constants. None.

Interpretation

Functions. None.

Relations

E	$E \subseteq A \times A$ Interpretation of $\in$.

Layer Delta

No constant or function symbols occur in $\mathcal{L}_{\in}$, so every term is a variable and FOL term evaluation reduces to $[\![x]\!]_{\mathcal{A},s} = s(x)$.

Syntax–Semantics Bridge

$$\mathcal{A}, s \models x \in y \quad \Longleftrightarrow \quad (s(x), s(y)) \in E.$$

Models of the Theory

$$\mathcal{A} \models \mathrm{ZF} \quad :\Longleftrightarrow \quad \forall \varphi \, (\varphi \in \mathrm{ZF} \to \mathcal{A} \models \varphi).$$
$$\mathcal{A} \models \mathrm{ZFC} \quad :\Longleftrightarrow \quad \mathcal{A} \models \mathrm{ZF} \wedge \mathcal{A} \models \mathrm{AC}.$$

Theorems and Consequences

1. One nonlogical symbol is interpreted: $I(\in) = E$.
2. $\mathrm{Th}(\mathcal{A})$ is inherited from *Consequence*.

Definitional Extension (Set Theory)

Depends on: `defext:first-order-logic; theory:set-theory; model:set-theory`

Layer Delta

$$\begin{aligned}\text{existence} &\leftarrow \text{Empty Set, Pairing, Union, Power Set,}\\ &\qquad \text{Infinity, Separation, Replacement;}\\ \text{uniqueness} &\leftarrow \text{Extensionality;}\\ \text{default value} &\leftarrow \varnothing.\end{aligned}$$

Representative-independence is delegated to quotient-specific extensions.

Instances

$\subseteq$ relation

$$\forall x\,\forall y\,\bigl(x \subseteq y \leftrightarrow \forall w\,(w \in x \to w \in y)\bigr).$$

No unique-existence obligation.

$\varnothing$ constant

$$\psi_\varnothing(y) :\equiv \forall w\,\neg(w \in y), \qquad \exists!\ \text{receipt: Empty Set + Extensionality.}$$

$$\forall y\,\bigl(\varnothing = y \leftrightarrow \forall w\,\neg(w \in y)\bigr).$$

$\{\cdot,\cdot\}$ binary function

$$\psi(x_1, x_2, y) :\equiv \forall w\,(w \in y \leftrightarrow (w = x_1 \lor w = x_2)).$$

$$\exists!\ \text{receipt: Pairing + Extensionality.}$$

$$\forall x_1\,\forall x_2\,\forall y\,\bigl(\{x_1, x_2\} = y \leftrightarrow \psi(x_1, x_2, y)\bigr).$$

$\bigcup$ unary function

$$\psi(x, y) :\equiv \forall z\,(z \in y \leftrightarrow \exists w\,(w \in x \land z \in w)).$$

$$\exists!\ \text{receipt: Union + Extensionality.}$$

$$\forall x\,\forall y\,\bigl(\bigcup x = y \leftrightarrow \psi(x, y)\bigr).$$

$\mathcal{P}$ unary function

$$\psi(x, y) :\equiv \forall z\,(z \in y \leftrightarrow \forall w\,(w \in z \to w \in x)).$$

$$\exists!\ \text{receipt: Power Set + Extensionality.}$$

$$\forall x\,\forall y\,\bigl(\mathcal{P}(x) = y \leftrightarrow \psi(x, y)\bigr).$$

7.2 Sets

7.2.1 Sets, Membership, and ZFC Axioms

Axiom (Extensionality)

Axiom 7.2.1 (Axiom of Extensionality)**.** Two sets are equal iff they have the same elements.

$$\forall A \forall B \Big(A = B \iff \forall x \,(x \in A \leftrightarrow x \in B) \Big).$$

Axiom (Empty Set)

Axiom 7.2.2 (Axiom of Empty Set)**.** There exists a set with no elements.

$$\exists A \, \forall x \,(x \notin A).$$

Axiom (Pairing)

Axiom 7.2.3 (Axiom of Pairing)**.** For any two sets, there exists a set containing exactly those two sets.

$$\forall A \, \forall B \, \exists C \, \forall x \,(x \in C \leftrightarrow (x = A \lor x = B)).$$

Axiom (Union)

Axiom 7.2.4 (Axiom of Union)**.** For any family of sets, there exists a set containing exactly the elements of those sets.

$$\forall A \, \exists U \, \forall x \,\big(x \in U \leftrightarrow \exists B \,(B \in A \land x \in B)\big).$$

Axiom (Power Set)

Axiom 7.2.5 (Axiom of Power Set)**.** For any set, there exists a set of all its subsets.

$$\forall A \, \exists P \, \forall x \,(x \in P \leftrightarrow x \subseteq A).$$

Axiom (Infinity)

Axiom 7.2.6 (Axiom of Infinity)**.** There exists an infinite set.

$$\exists A \Big(\varnothing \in A \land \forall x \,(x \in A \to x \cup \{x\} \in A) \Big).$$

Axiom Schema (Separation)

Axiom 7.2.7 (Axiom Schema of Separation)**.** Given a set and a property, there exists a subset containing exactly the elements satisfying that property. For any formula $\varphi(x)$,

$$\forall A\, \exists B\, \forall x\, (x \in B \leftrightarrow (x \in A \wedge \varphi(x))).$$

Axiom Schema (Replacement)

Axiom 7.2.8 (Axiom Schema of Replacement)**.** If $\varphi(x, y)$ defines a functional relation on a set A, then the image of A under φ is a set. For any formula $\varphi(x, y)$,

$$\forall A \left((\forall x \in A\, \exists! y\, \varphi(x, y)) \rightarrow \exists B\, \forall y\, \big(y \in B \leftrightarrow \exists x \in A\, \varphi(x, y)\big) \right).$$

Axiom (Foundation)

Axiom 7.2.9 (Axiom of Foundation)**.** Every nonempty set has an $\in$-minimal element.

$$\forall A \left(A \neq \varnothing \rightarrow \exists x \in A\, (x \cap A = \varnothing) \right).$$

Axiom (Choice)

Axiom 7.2.10 (Axiom of Choice)**.** For any family of nonempty sets, there exists a selection function.

$$\forall A \left((\forall B \in A\, (B \neq \varnothing)) \rightarrow \exists f\, \forall B \in A\, (f(B) \in B) \right).$$

Definition (Set and Membership)

Definition 7.2.11 (Set and Membership)**.** In axiomatic set theory, the notions of *set* and *membership* are *primitive*.

- A *set* is an object.
- *Membership* is a binary relation, denoted by $\in$, between objects.

If x is an object and A is a set, the statement $x \in A$ is read as "x is an element of A." No definition of "set" or "$\in$" is given in more basic terms. Their meaning is determined entirely by the axioms governing them.

Remark (English reading)*.* Primitive means we do not define these in terms of simpler concepts. Rather, we fix their meaning implicitly by stating the axioms they must obey. This is the standard approach in formal mathematics: rules of behaviour replace informal definitions.

Remark (Consequence)*.* All subsequent notions—subsets, ordered pairs, relations, functions, number systems—are *definitions* introduced within this axiomatic framework. Every proved result is a *theorem* derived logically from the axioms via the rules of inference.

Remark (Role of the axioms). The axioms are not statements to be proved, but rules specifying how $\in$ behaves and which sets are permitted to exist. From this point onward, set theory functions as the underlying language of mathematics: all reasoning about mathematical objects is ultimately grounded in these axioms, even when they are not cited explicitly.

Remark (Separation vs. Replacement). Separation is a schema: one axiom instance per definable property φ. It carves out subsets of an already-existing set. Replacement is strictly stronger: it can produce sets whose elements are not already contained in any pre-existing set, allowing the construction of large stages of the cumulative hierarchy.

Remark (Axiom of Choice and right inverses). The Axiom of Choice is equivalent to many statements used later in analysis, including Zorn's Lemma, the well-ordering theorem, and the statement that every surjective function has a right inverse. It is independent of the other ZFC axioms.

Foundations — Quick Reference

Concept	**Meaning**
Set, membership	Primitive notions; meaning fixed by axioms
Extensionality	Sets equal iff same elements
Empty Set	Exists a set with no elements
Pairing	Any two sets can be collected
Union	Elements of sets in a family
Power Set	All subsets form a set
Infinity	An infinite set exists
Separation	Subsets by definable property
Replacement	Functional image of a set is a set
Foundation	No infinite descending $\in$-chains
Choice	Selection function exists

Set Operations — Quick Reference

Concept	**Notation**	**Membership condition**
Empty set	$\varnothing$	$x \in \varnothing$ never
Subset	$A \subseteq B$	$x \in A \Rightarrow x \in B$
Proper subset	$A \subsetneq B$	$A \subseteq B$ and $A \neq B$
Set equality	$A = B$	same elements both ways
Union	$A \cup B$	in A or in B
Intersection	$A \cap B$	in A and in B
Set difference	$A \setminus B$	in A but not B
Symmetric diff.	$A \triangle B$	in exactly one of A, B
Complement	$A^c = U \setminus A$	in U but not A
Cartesian product	$A \times B$	all ordered pairs (a, b)
Power set	$\mathcal{P}(A)$	all subsets of A
De Morgan	$(A \cup B)^c = A^c \cap B^c$	complement of union

7.2.2 Set Constructions and Operations

Definition (Empty Set)

Definition 7.2.12 (Empty Set)**.** The unique set with no elements is called the *empty set* and is denoted $\varnothing$.

Remark (Dependencies).

- Axiom of Empty Set

Remark (Vacuous truth). Statements of the form $\forall x \in \varnothing,\ P(x)$ are *vacuously true*: there is no element $x \in \varnothing$ for which $P(x)$ could fail. This is not a special convention but a consequence of how universal quantification is defined.

Definition (Subset)

Definition 7.2.13 (Subset)**.** Let A and B be sets. We say A is a *subset* of B, written $A \subseteq B$, if every element of A is also an element of B:

$$A \subseteq B \iff \forall x\,(x \in A \to x \in B).$$

Remark (Dependencies).

- Axiom Schema of Separation
- Universal Formula

Definition (Proper Subset)

Definition 7.2.14 (Proper Subset)**.** A is a *proper subset* of B, written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Remark (Dependencies).

- Subset

Remark (Notation convention). Some authors write $A \subset B$ for a proper subset; others use it to mean $A \subseteq B$. In these notes, $\subseteq$ always denotes subset (possibly equal) and $\subsetneq$ always denotes proper subset.

Definition (Set Equality)

Definition 7.2.15 (Set Equality)**.** Two sets A and B are *equal*, written $A = B$, if they have the same elements:

$$A = B \iff \forall x\,(x \in A \leftrightarrow x \in B).$$

Equivalently, $A = B \iff (A \subseteq B \wedge B \subseteq A)$.

Remark (Dependencies).

- Subset
- Axiom of Extensionality

Remark (Proof strategy). In practice, to prove $A = B$ one shows mutual inclusion: first $A \subseteq B$ (take arbitrary $x \in A$ and deduce $x \in B$), then $B \subseteq A$. This two-step structure appears throughout set-theoretic proofs.

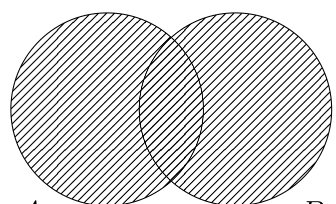
Definition (Union)

Definition 7.2.16 (Union)**.** The *union* of A and B, denoted $A \cup B$, is the set of all elements belonging to at least one of the two sets:

$$A \cup B \;=\; \{\, x \mid x \in A \vee x \in B \,\}.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation



A B

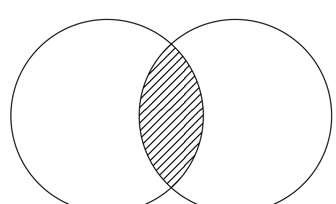
Definition (Intersection)

Definition 7.2.17 (Intersection)**.** The *intersection* of A and B, denoted $A \cap B$, is the set of all elements common to both:

$$A \cap B \;=\; \{\, x \mid x \in A \wedge x \in B \,\}.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation



A B

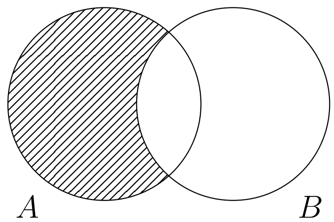
Definition (Set Difference)

Definition 7.2.18 (Set Difference)**.** The *set difference* $A \setminus B$ is the set of elements in A but not in B:

$$A \setminus B \;=\; \{\, x \mid x \in A \wedge x \notin B \,\}.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation



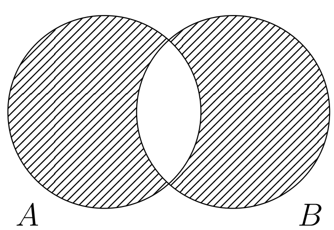
Definition (Symmetric Difference)

Definition 7.2.19 (Symmetric Difference)**.** The *symmetric difference* $A \triangle B$ is the set of elements belonging to exactly one of A and B:

$$A \triangle B \;=\; (A \setminus B) \cup (B \setminus A) \;=\; \{\, x \mid (x \in A \vee x \in B) \wedge x \notin A \cap B \,\}.$$

Remark (Dependencies).

- Set Difference
- Union
- Intersection



Remark (Algebraic structure of $\triangle$)*.* The symmetric difference is commutative and associative. Under $\triangle$, the power set $\mathcal{P}(U)$ forms an abelian group with identity $\varnothing$ and every element its own inverse ($A \triangle A = \varnothing$).

Definition (Complement)

Definition 7.2.20 (Complement)**.** Let U be a fixed universe and $A \subseteq U$. The *complement* of A relative to U, denoted A^c, is:

$$A^c = U \setminus A.$$

Remark (Dependencies).

- Set and Membership
- Axiom Schema of Separation

Definition (Cartesian Product)

Definition 7.2.21 (Cartesian Product)**.** The *Cartesian product* of sets A and B is:

$$A \times B = \{ (a, b) \mid a \in A \text{ and } b \in B \}.$$

Remark (Dependencies).

- Ordered Pair
- Axiom of Pairing
- Axiom of Power Set
- Axiom Schema of Separation

Remark (Ordered pairs). The ordered pair (a, b) is formally defined as $\{\{a\}, \{a, b\}\}$. This Kuratowski encoding ensures the key property: $(a, b) = (c, d)$ iff $a = c$ and $b = d$.

Remark (Non-commutativity). In general, $A \times B \neq B \times A$. The product is however associative up to canonical isomorphism: $(A \times B) \times C \cong A \times (B \times C)$.

Definition (Power Set)

Definition 7.2.22 (Power Set)**.** The *power set* of A, denoted $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) := \{ S \mid S \subseteq A \}.$$

Remark (Dependencies).

- Axiom of Power Set

Remark (Size). If A has n elements, then $|\mathcal{P}(A)| = 2^n$. For infinite A, Cantor's theorem shows $|\mathcal{P}(A)| > |A|$ strictly.

Theorem (De Morgan's Laws)

Theorem 7.2.23 (De Morgan's Laws)**.** *Let U be a universe and $A, B \subseteq U$. Then*

$$(A \cup B)^c = A^c \cap B^c \quad \textit{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B \subseteq U : \forall x \left(x \notin A \cup B \leftrightarrow (x \notin A \wedge x \notin B)\right)$, and dually $\forall x \left(x \notin A \cap B \leftrightarrow (x \notin A \vee x \notin B)\right)$.

Remark (Dependencies).

- Union
- Intersection
- Complement

Remark (Connection to propositional logic). De Morgan's laws mirror the propositional logic identities $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ and $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ under the correspondence $\cap \leftrightarrow \wedge$, $\cup \leftrightarrow \vee$, $A^c \leftrightarrow \neg A$.

Definition (Set Duality)

Definition 7.2.24 (Set Duality)**.** Two set-theoretic expressions over a fixed universe U are *dual* if one is obtained from the other by simultaneously replacing $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$, with complements unchanged.

Remark (Standard quantified statement). $\forall$ expressions E, E' over U: E and E' are dual iff E' is obtained from E by the substitution $\cup \mapsto \cap$, $\cap \mapsto \cup$, $\varnothing \mapsto U$, $U \mapsto \varnothing$.

Remark (Dependencies).

- Union
- Intersection
- Empty Set
- Complement

Corollary

Corollary 7.2.25 (Principle of Set Duality)**.** *Any identity involving $\cup$, $\cap$, $\varnothing$, and U that holds for all subsets of a universe remains valid when each operation and constant is replaced by its dual.*

Remark (Standard quantified statement). For any identity $\forall A_1, \ldots, A_n \subseteq U : E(A_i, \cup, \cap, \varnothing, U) = E'(A_i, \cup, \cap, \varnothing, U)$, the dual identity $\forall A_1, \ldots, A_n \subseteq U : E^d = E'^d$ holds, where E^d replaces $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$.

Remark (Dependencies).

- Set Duality
- Axiom of Extensionality

Remark (Using duality in proofs). To prove a statement involving unions and intersections, it often suffices to prove one version; the dual statement follows immediately.

Example (Distributive law via duality). The two distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad \text{and} \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

are duals of each other. Proving either one and applying the Principle of Set Duality yields the other immediately.

Remark (Transition to relations). The Cartesian product provides a way to encode ordered information. Relations and functions will be defined as special subsets of Cartesian products, so they require no new foundational objects.

Set Algebra Laws — Quick Reference

Law	Union form	Intersection form
Commutativity	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$	analogous
Distributivity	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	dual
Identity	$A \cup \varnothing = A$	$A \cap U = A$
Absorption	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$
Involution	$(A^c)^c = A$	—

7.2.3 Algebraic Laws of Set Operations

Remark (Algebraic structure of set operations). The operations $\cup$, $\cap$, and complement satisfy algebraic laws analogous to those of logical connectives. Together they endow $\mathcal{P}(U)$ with the structure of a *Boolean algebra*. These laws justify manipulation of set expressions in later proofs, particularly those involving equivalence classes, partitions, and functions.

Theorem (Commutativity of Union and Intersection)

Theorem 7.2.26 (Commutativity of Union and Intersection)**.** *Let A, B be sets. Then*

$$A \cup B = B \cup A \quad \text{and} \quad A \cap B = B \cap A.$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B : \forall x\,(x \in A \cup B \leftrightarrow x \in B \cup A)$ and $\forall x\,(x \in A \cap B \leftrightarrow x \in B \cap A)$.

Remark (Dependencies).

- Axiom of Extensionality
- Union
- Intersection

Theorem (Associativity of Union and Intersection)

Theorem 7.2.27 (Associativity of Union and Intersection)**.** *Let A, B, C be sets. Then*

$$(A \cup B) \cup C = A \cup (B \cup C) \quad \text{and} \quad (A \cap B) \cap C = A \cap (B \cap C).$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B, C : \forall x \big((x \in (A \cup B) \cup C) \leftrightarrow (x \in A \cup (B \cup C))\big)$, and analogously for $\cap$.

Remark (Dependencies).

- Axiom of Extensionality
- Union
- Intersection

Theorem (Distributive Laws)

Theorem 7.2.28 (Distributive Laws)**.** *Let A, B, C be sets. Then*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Go to proof.

Remark (Standard quantified statement). $\forall A, B, C : \forall x \big(x \in A \cap (B \cup C) \leftrightarrow x \in (A \cap B) \cup (A \cap C)\big)$, and dually for $\cup$ over $\cap$.

Remark (Dependencies).

- Axiom of Extensionality
- Union
- Intersection

Theorem (Identity and Absorption Laws)

Theorem 7.2.29 (Identity and Absorption Laws)**.** *Let A, B be sets and U a universe with $A \subseteq U$. Then*

$$A \cup \varnothing = A, \qquad A \cap U = A,$$

$$A \cup (A \cap B) = A, \qquad A \cap (A \cup B) = A.$$

Go to proof.

Remark (Standard quantified statement). $\forall A \subseteq U: \forall x\,(x \in A \cup \varnothing \leftrightarrow x \in A)$ and $\forall x\,(x \in A \cap U \leftrightarrow x \in A)$. Absorption: $\forall A, B: \forall x\,(x \in A \cup (A \cap B) \leftrightarrow x \in A)$, and dually.

Remark (Dependencies).

- Axiom of Extensionality
- Empty Set
- Union
- Intersection
- Subset

Theorem (Involution of Complement)

Theorem 7.2.30 (Involution of Complement)**.** *For any $A \subseteq U$,*

$$(A^c)^c = A.$$

Go to proof.

Remark (Standard quantified statement). $\forall A \subseteq U: \forall x\,(x \in (A^c)^c \leftrightarrow x \in A)$.

Remark (Dependencies).

- Axiom of Extensionality
- Complement

Remark (Non-commutative and non-associative operations). Set difference $\setminus$ is neither commutative nor associative:

$$A \setminus B \neq B \setminus A, \qquad (A \setminus B) \setminus C \neq A \setminus (B \setminus C) \quad \text{in general.}$$

It interacts with union and intersection via:

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C), \qquad A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

These follow from $A \setminus B = A \cap B^c$ together with De Morgan's laws.

Remark (Cartesian product). The Cartesian product $\times$ is not commutative, is associative only up to canonical isomorphism, and distributes over union in each coordinate:

$$A \times (B \cup C) = (A \times B) \cup (A \times C).$$

7.3 Families

Indexed Families — Quick Reference

Concept	Meaning
Indexed family	Function $F : I \to \mathcal{P}(U)$; written $\{A_i\}_{i\in I}$
Indexed union	$\bigcup_{i\in I} A_i$: in at least one A_i
Indexed intersection	$\bigcap_{i\in I} A_i$: in every A_i
Pairwise disjoint	$i \neq j \Rightarrow A_i \cap A_j = \varnothing$
Cover	$\bigcup_{C\in\mathcal{C}} C = A$
Indexed De Morgan	Complements exchange indexed unions and intersections
Arbitrary product	$\prod_{i\in I} A_i$: choice functions $f : I \to \bigcup A_i$

7.3.1 Indexed Families of Sets

Remark (Motivation). Many set-theoretic constructions require not just pairs of sets but infinite families. Indexed families provide the formal language for partitions, equivalence classes, unions over countably or uncountably many sets, and products indexed by arbitrary index sets.

Definition (Indexed Family of Sets)

Definition 7.3.1 (Indexed Family of Sets)**.** Let I be a set (the *index set*) and U a universe. An *indexed family of sets* is a function

$$F : I \to \mathcal{P}(U),$$

with $F(i)$ typically written A_i. The family is denoted $\{A_i\}_{i\in I}$.

Remark (Dependencies).

- Function
- Power Set

Remark (Family vs. set of sets). An indexed family is formally a *function*, not a set of sets. Different indices may correspond to the same set: $i \neq j$ does not imply $A_i \neq A_j$. This distinction matters for equivalence class constructions.

Definition (Indexed Union)

Definition 7.3.2 (Indexed Union)**.** The *union* of the indexed family $\{A_i\}_{i\in I}$ is:

$$\bigcup_{i\in I} A_i \;:=\; \{\, x \mid \exists i \in I \text{ such that } x \in A_i \,\}.$$

Remark (Dependencies).

- Indexed Family of Sets
- Union
- Existential Formula

Definition (Indexed Intersection)

Definition 7.3.3 (Indexed Intersection)**.** For $I \neq \varnothing$, the *intersection* of $\{A_i\}_{i\in I}$ is:

$$\bigcap_{i\in I} A_i \;:=\; \{\, x \mid \forall i \in I,\; x \in A_i \,\}.$$

Remark (Dependencies).

- Indexed Family of Sets
- Intersection
- Universal Formula

Remark (Why $I \neq \varnothing$ is required). If $I = \varnothing$, then $\forall i \in I, x \in A_i$ holds vacuously for every x, making the intersection the entire universe U. This is normally left undefined or requires specifying a background universe explicitly.

Theorem (De Morgan's Laws for Indexed Families)

Theorem 7.3.4 (De Morgan's Laws for Indexed Families)**.** *Let $\{A_i\}_{i\in I}$ be an indexed family of subsets of a universe U. Then*

$$U \setminus \bigcup_{i\in I} A_i = \bigcap_{i\in I} (U \setminus A_i).$$

If $I \neq \varnothing$, then

$$U \setminus \bigcap_{i\in I} A_i = \bigcup_{i\in I} (U \setminus A_i).$$

Go to proof.

Remark (Dependencies).

- Indexed Family of Sets
- Indexed Union
- Indexed Intersection
- Complement

Definition (Pairwise Disjoint Family)

Definition 7.3.5 (Pairwise Disjoint Family)**.** The family $\{A_i\}_{i\in I}$ is *pairwise disjoint* if

$$\forall i, j \in I,\ i \neq j \to A_i \cap A_j = \varnothing.$$

Remark (Dependencies).

- Indexed Family of Sets
- Intersection
- Empty Set
- Set Equality

Definition (Cover)

Definition 7.3.6 (Cover)**.** A collection $\mathcal{C} \subseteq \mathcal{P}(A)$ is a *cover* of A if

$$\bigcup_{C\in\mathcal{C}} C = A.$$

Remark (Standard quantified statement)*.* $\mathcal{C}$ covers $A \leftrightarrow \forall x \in A,\ \exists C \in \mathcal{C},\ x \in C.$

Arbitrary Cartesian Products

Definition (Arbitrary Cartesian Product)

Definition 7.3.7 (Arbitrary Cartesian Product)**.** Let $\{A_i\}_{i\in I}$ be an indexed family of sets. The *Cartesian product* is:

$$\prod_{i\in I} A_i := \left\{ f : I \to \bigcup_{i\in I} A_i \;\middle|\; \forall i \in I,\ f(i) \in A_i \right\}.$$

Remark (Dependencies).

- Indexed Family of Sets
- Function
- Indexed Union

Remark (Elements as choice functions)*.* An element of $\prod_{i\in I} A_i$ is a *choice function*: a function that assigns to each index i an element of the corresponding set A_i. For finite $I = \{1, \ldots, n\}$, this reduces to the familiar n-tuple $(a_1, \ldots, a_n)$.

Remark (Axiom of Choice connection)**.** For infinite I, the product $\prod_{i \in I} A_i$ is nonempty iff a choice function exists. This existence is not guaranteed by the other ZFC axioms: it is equivalent to the Axiom of Choice. Thus the Axiom of Choice is precisely the assertion that arbitrary products of nonempty sets are nonempty.

Covers and FIP — Quick Reference

Concept	Meaning
Cover of A	$\bigcup_{C \in \mathcal{C}} C \supseteq A$
Subcover	Sub-collection of a cover that is still a cover
Finite cover	Cover with finitely many members
Finite subcover	Finite subcollection that is still a cover
Open cover	Cover whose members are all open sets
Finite Intersection Property	Every finite subcollection has nonempty intersection
FIP–cover duality	$\mathcal{C}$ covers $A \iff$ closed complements fail FIP

7.3.2 Covers, Subcovers, and the Finite Intersection Property

Compactness is one of the central concepts of analysis, and its definition is stated in terms of covers: a space is compact if every open cover has a finite subcover. This is a purely set-theoretic condition — it makes no reference to distance or continuity. The work of understanding compactness therefore begins here, with the vocabulary of covers. Once these definitions are in place, the definition of compactness will be a single sentence, and the difficulty will lie entirely in the geometric and analytic content, not the set-theoretic framework.

Definition (Cover)

Definition 7.3.8 (Cover)**.** Let A be a set and $\mathcal{C}$ a collection of sets. $\mathcal{C}$ is a *cover* of A if every element of A belongs to at least one member of $\mathcal{C}$:

$$A \subseteq \bigcup_{C \in \mathcal{C}} C.$$

Remark (Dependencies).

- Subset
- Union

Definition (Subcover)

Definition 7.3.9 (Subcover)**.** $\mathcal{C}'$ is a *subcover* of $\mathcal{C}$ if $\mathcal{C}' \subseteq \mathcal{C}$ and $\mathcal{C}'$ is still a cover of A.

Remark (Dependencies).

- Cover

- Subset

Definition (Finite Cover)

Definition 7.3.10 (Finite Cover)**.** A cover is *finite* if it has finitely many members. A *finite subcover* is a subcover $\mathcal{C}' \subseteq \mathcal{C}$ with $|\mathcal{C}'| < \infty$.

Remark (Dependencies).

- Cover
- Finite and Infinite Sets
- Subset

Remark (The key question). The condition of being a cover is easy to satisfy: take $\mathcal{C} = \mathcal{P}(A)$ and it is trivially a cover. The interesting question is whether large covers can be replaced by smaller ones — specifically, finite ones. An infinite cover has infinitely many sets, which is hard to work with. A finite subcover is just as good for covering purposes, but is finite and therefore tractable. When a finite subcover always exists, no matter which cover one starts with, the space is called compact.

Definition (Open Cover)

Definition 7.3.11 (Open Cover)**.** In a topological space (X, τ), an *open cover* of a subset $A \subseteq X$ is a cover $\mathcal{C}$ of A all of whose members are open sets: $\mathcal{C} \subseteq \tau$.

Remark (Standard quantified statement). $\mathcal{C}$ is an open cover of $A \leftrightarrow (\forall C \in \mathcal{C},\ C \in \tau) \wedge (\forall x \in A,\ \exists C \in \mathcal{C},\ x \in C)$.

Remark (Dependencies).

- Cover
- Finite Intersection Property
- Complement
- Intersection
- Subset

Remark (Why open covers). The emphasis on open covers in compactness is because open sets are the sets one controls in a topological space. An open cover $\{U_\alpha\}$ of A means: every point of A has at least one U_α containing it. A finite subcover then gives a list of finitely many open sets whose union already contains all of A. Compactness says this reduction is always possible, which is a very strong structural property.

Definition (Finite Intersection Property)

Definition 7.3.12 (Finite Intersection Property)**.** A collection $\mathcal{F}$ of sets has the *finite intersection property* (FIP) if every finite subcollection has nonempty intersection: for all finite $\mathcal{F}' \subseteq \mathcal{F}$,

$$\bigcap_{F \in \mathcal{F}'} F \neq \varnothing.$$

Remark (Dependencies).

- Intersection
- Empty Set
- Subset
- Finite and Infinite Sets

Remark (How to read the FIP). The FIP is a consistency condition: no finite list of sets from $\mathcal{F}$ is mutually exclusive. It does *not* say the entire intersection $\bigcap_{F \in \mathcal{F}} F$ is nonempty — only that every *finite* sub-intersection is. The difference matters when $\mathcal{F}$ is infinite.

Example: In $\mathbb{R}$, the family $\mathcal{F} = \{(0, 1/n) : n \in \mathbb{N}\}$ has the FIP (any finite subcollection has nonempty intersection, since $(0, 1/n) \supseteq (0, 1/N)$ for the largest N in the subcollection). But $\bigcap_{n=1}^{\infty} (0, 1/n) = \varnothing$. Compact spaces are characterized by the property that the FIP implies nonempty total intersection.

Proposition (FIP–Cover Duality)

Proposition 7.3.13 (FIP–Cover Duality)**.** *Let X be a set, $A \subseteq X$, and $\{U_\alpha\}_{\alpha \in I}$ a family of subsets of X. Let $F_\alpha = U_\alpha^c = X \setminus U_\alpha$. Then:*

$$\{U_\alpha\} \text{ covers } A \iff \{F_\alpha \cap A\}_{\alpha \in I} \text{ does not have the FIP.}$$

More precisely: the family $\{U_\alpha\}$ does not *cover A if and only if the family of closed complements $\{F_\alpha\}$ has the FIP relative to A.* *Go to proof.*

Remark (Standard quantified statement). $A \subseteq \bigcup_{\alpha \in I} U_\alpha \;\leftrightarrow\; \neg(\forall$ finite $I_0 \subseteq I,\ \bigcap_{\alpha \in I_0} F_\alpha \cap A \neq \varnothing)$. Reading pointwise: $\forall x \in A,\ \exists \alpha \in I,\ x \in U_\alpha \;\leftrightarrow\; \{F_\alpha \cap A\}$ fails the FIP.

Remark (Dependencies).

- Cover
- Finite Intersection Property
- Complement
- Intersection
- Subset

Remark (Proof by De Morgan). The duality is a direct consequence of De Morgan's law for indexed families (Theorem 7.3.4). The family $\{U_\alpha\}$ fails to cover A means:

$$A \not\subseteq \bigcup_\alpha U_\alpha \iff A \cap \Big(\bigcup_\alpha U_\alpha\Big)^c \neq \varnothing \iff A \cap \bigcap_\alpha U_\alpha^c \neq \varnothing \iff A \cap \bigcap_\alpha F_\alpha \neq \varnothing.$$

Applying this to every finite subcollection gives the equivalence with the FIP.

Remark (Why this duality matters for compactness). Compactness has two equivalent formulations:

(i) Every open cover has a finite subcover.

(ii) Every family of closed sets with the FIP has nonempty intersection.

These formulations are equivalent by FIP–Cover Duality: take complements and apply De Morgan. Formulation (i) is the standard definition; formulation (ii) is often easier to apply when working with closed sets (e.g., nested compact sets). Both are purely set-theoretic conditions, and the equivalence is entirely a matter of set algebra. When compactness is encountered in topology and metric spaces, the only new content is the geometric meaning — the set-theoretic equivalence is established here.

Example (Nested closed sets and FIP). In $\mathbb{R}$, the family $\{[n,\infty) : n \in \mathbb{N}\}$ has the FIP: any finite subcollection $\{[n_1,\infty),\dots,[n_k,\infty)\}$ has intersection $[\max(n_i),\infty) \neq \varnothing$. But $\bigcap_{n=1}^\infty [n,\infty) = \varnothing$. This shows $\mathbb{R}$ is not compact (the family of closed sets has the FIP but empty total intersection). A compact space would not allow this failure.

Chapter 8

Relations

SERIES	VOLUME I	BOOK II	CHAPTER 6
From Cantor to Ito	Logic, Sets, and Proof	Set Theory	**Relations**

Remark (Source orientation). This chapter follows Tarski's treatment of relations as objects that can be formed, compared, reversed, and specialized into functional and one-one relations Tarski [1941].

8.1 Relations — Quick Reference

Relations — Quick Reference

Concept	**Meaning**
Ordered pair	$(a, b) := \{\{a\}, \{a, b\}\}$; identity iff both coords equal
Cartesian product	$A \times B$: all ordered pairs from A and B
Relation	Any subset $R \subseteq A \times B$
Domain of R	Inputs that occur in R
Range of R	Outputs that occur in R
Converse	Reverse every ordered pair
Identity relation	Diagonal relation Δ_A
Relation composition	Compose through an intermediate witness

Relation Algebra and Classification — Quick Reference

Concept	Meaning
Null relation	$\varnothing$
Universal relation	$A \times B$
Relation inclusion	$R \subseteq S$ as set inclusion
Relation equality	Mutual relation inclusion
Union of relations	Pair belongs to at least one relation
Intersection	Pair belongs to both relations
Difference	Pair belongs to first relation but not second
Complement	Pairs in $A \times B$ not in R
Converse laws	Converse reverses composition and preserves Boolean operations
Composition laws	Associative; distributes over unions, not all Boolean operations
One-to-many	One input has two distinct outputs
Many-to-one	Two distinct inputs share one output
Many-to-many	One-to-many and many-to-one phenomena both occur
Ternary relation	Subset of $A \times B \times C$

8.1.1 Ordered Pairs and Relations

Remark (Geometric picture). A binary relation from A to B can be pictured as directed edges from the A-side to the B-side of a bipartite diagram. The converse reverses every edge, and composition follows two edges in series through an intermediate set.

Definition (Ordered Pair)

Definition 8.1.1 (Ordered Pair)**.** Let a and b be sets. The *ordered pair* (a, b) is defined (Kuratowski) as

$$(a, b) \ := \ \{\{a\}, \{a, b\}\}.$$

Remark (Dependencies).

- Axiom of Pairing

Theorem (Uniqueness of Ordered Pairs)

Theorem 8.1.2 (Uniqueness of Ordered Pairs)**.** *For any sets a, b, c, d,*

$$(a, b) = (c, d) \iff (a = c \wedge b = d).$$

Go to proof.

Remark (Standard quantified statement). $\forall a, b, c, d : \ (a, b) = (c, d) \ \leftrightarrow \ (a = c \wedge b = d)$.

Remark (Dependencies).

- Ordered Pair

- Axiom of Pairing
- Axiom of Extensionality

Remark (Kuratowski encoding). The purpose of the Kuratowski definition is purely to guarantee the uniqueness theorem above. Once $(a, b) = (c, d) \iff a = c \wedge b = d$ is established, we may treat ordered pairs as a primitive with this property and forget the encoding.

Definition (Cartesian Product — as foundation for relations)

Definition 8.1.3 (Cartesian Product — as foundation for relations)**.** The *Cartesian product* of sets A and B is:

$$A \times B \;:=\; \{\,(a, b) \mid a \in A \text{ and } b \in B\,\}.$$

Remark (Dependencies).

- Ordered Pair
- Axiom of Pairing
- Axiom of Power Set
- Axiom Schema of Separation

Definition (Relation)

Definition 8.1.4 (Relation)**.** A *relation* from A to B is any subset $R \subseteq A \times B$. If $(a, b) \in R$ we write $a\,R\,b$ and say "a is related to b." When $A = B$ we say R is a *relation on* A.

Remark (Dependencies).

- Cartesian Product
- Subset

Remark (Relations as structured sets). Relations introduce no new foundational objects: they are simply sets of ordered pairs, constructed using operations already available. Properties of relations can therefore be studied with ordinary set-theoretic reasoning.

Definition (Domain of a Relation)

Definition 8.1.5 (Domain of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *domain* of R is

$$\mathrm{Dom}(R) \;:=\; \{\, a \in A \mid \exists b \in B,\ (a, b) \in R \,\}.$$

Remark (Dependencies).

- Relation

- Axiom Schema of Separation

Definition (Range of a Relation)

Definition 8.1.6 (Range of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *range* of R is

$$\mathrm{Ran}(R) := \{ b \in B \mid \exists a \in A,\ (a, b) \in R \}.$$

Tarski also calls this class the *counter-domain* or *converse domain*.

Remark (Dependencies).

- Relation
- Axiom Schema of Separation

Remark (Domain and range). The domain records the first coordinates that occur in a relation; the range records the second coordinates that occur.

Definition (Converse Relation)

Definition 8.1.7 (Converse Relation)**.** If $R \subseteq A \times B$, the *converse* of R is the relation $R^{-1} \subseteq B \times A$ defined by

$$(b, a) \in R^{-1} \iff (a, b) \in R.$$

Equivalently, $b\, R^{-1}\, a$ iff aRb.

Remark (Dependencies).

- Relation
- Ordered Pair

Definition (Identity Relation)

Definition 8.1.8 (Identity Relation)**.** For a set A, the *identity relation* on A is

$$\Delta_A := \{ (a, a) \mid a \in A \}.$$

Remark (Dependencies).

- Relation
- Cartesian Product

Definition (Composition of Relations)

Definition 8.1.9 (Composition of Relations)**.** Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The *composition* $S \circ R \subseteq A \times C$ is defined by

$$(a, c) \in S \circ R \iff \exists b \in B,\ (a, b) \in R \text{ and } (b, c) \in S.$$

Remark (Dependencies).

- Relation
- Cartesian Product

Theorem (Associativity of Relation Composition)

Theorem 8.1.10 (Associativity of Relation Composition)**.** *Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Then*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Go to proof.

Remark (Standard quantified statement). For all $a \in A$ and $d \in D$, $(a, d) \in T \circ (S \circ R)$ iff there are $b \in B$ and $c \in C$ with $(a, b) \in R$, $(b, c) \in S$, and $(c, d) \in T$, which is exactly the membership condition for $(a, d) \in (T \circ S) \circ R$.

Remark (Dependencies).

- Composition of Relations
- Relation Equality

Theorem (Identity Laws for Relation Composition)

Theorem 8.1.11 (Identity Laws for Relation Composition)**.** *Let $R \subseteq A \times B$. Then*

$$R \circ \Delta_A = R \quad \text{and} \quad \Delta_B \circ R = R.$$

Go to proof.

Remark (Dependencies).

- Identity Relation
- Composition of Relations

Definition (Null Relation)

Definition 8.1.12 (Null Relation)**.** For sets A and B, the *null relation* from A to B is $\varnothing \subseteq A \times B$. No element of A is related to any element of B.

Remark (Dependencies).

- Relation
- Empty Set
- Cartesian Product

Definition (Universal Relation)

Definition 8.1.13 (Universal Relation)**.** For sets A and B, the *universal relation* from A to B is $A \times B$ itself. Every element of A is related to every element of B.

Remark (Dependencies).

- Relation
- Cartesian Product

Remark (Extreme relations). The null relation and universal relation are the two extreme binary relations from A to B.

Definition (Relation Inclusion)

Definition 8.1.14 (Relation Inclusion)**.** Let $R, S \subseteq A \times B$. Relation inclusion is ordinary set inclusion:

$$R \subseteq S \iff \forall a \in A \forall b \in B,\ ((a,b) \in R \Rightarrow (a,b) \in S),$$

Remark (Dependencies).

- Relation
- Subset

Definition (Relation Equality)

Definition 8.1.15 (Relation Equality)**.** Let $R, S \subseteq A \times B$. Relation equality is ordinary set equality:

$$R = S \iff R \subseteq S \text{ and } S \subseteq R.$$

Remark (Dependencies).

- Relation
- Relation Inclusion
- Set Equality

Definition (Union of Relations)

Definition 8.1.16 (Union of Relations)**.** Let $R, S \subseteq A \times B$. The *union* $R \cup S$ is the relation with

$$(a,b) \in R \cup S \iff (a,b) \in R \text{ or } (a,b) \in S.$$

Remark (Dependencies).

- Relation

- Union

Definition (Intersection of Relations)

Definition 8.1.17 (Intersection of Relations)**.** Let $R, S \subseteq A \times B$. The *intersection* $R \cap S$ is the relation with

$$(a, b) \in R \cap S \iff (a, b) \in R \text{ and } (a, b) \in S.$$

Remark (Dependencies).

- Relation
- Intersection

Definition (Difference of Relations)

Definition 8.1.18 (Difference of Relations)**.** Let $R, S \subseteq A \times B$. The *difference* $R \setminus S$ is the relation with

$$(a, b) \in R \setminus S \iff (a, b) \in R \text{ and } (a, b) \notin S.$$

Remark (Dependencies).

- Relation
- Set Difference

Definition (Complement of a Relation)

Definition 8.1.19 (Complement of a Relation)**.** Let $R \subseteq A \times B$. The *complement* of R relative to $A \times B$ is

$$(A \times B) \setminus R.$$

Remark (Dependencies).

- Relation
- Set Difference

Remark (Relation algebra operations). Relation inclusion, equality, union, intersection, difference, and complement are inherited from the corresponding set operations on subsets of $A \times B$.

Theorem (Converse Laws for Relation Operations)

Theorem 8.1.20 (Converse Laws for Relation Operations)**.** *Let $R, S \subseteq A \times B$, let $T \subseteq B \times C$, and take complements relative to the indicated Cartesian products. Then*

$$(R^{-1})^{-1} = R, \qquad (T \circ R)^{-1} = R^{-1} \circ T^{-1},$$

$$(R \cup S)^{-1} = R^{-1} \cup S^{-1}, \qquad (R \cap S)^{-1} = R^{-1} \cap S^{-1},$$
$$(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}, \qquad ((A \times B) \setminus R)^{-1} = (B \times A) \setminus R^{-1}.$$

Go to proof.

Remark (Dependencies).

- Converse Relation
- Composition of Relations
- Union of Relations
- Intersection of Relations
- Difference of Relations
- Complement of a Relation

Theorem (Relation Composition and Boolean Operations)

Theorem 8.1.21 (Relation Composition and Boolean Operations)**.** *Let $P, Q \subseteq A \times B$, $R \subseteq B \times C$, and $S, T \subseteq B \times C$. Then*

$$R \circ (P \cup Q) = (R \circ P) \cup (R \circ Q), \qquad (S \cup T) \circ P = (S \circ P) \cup (T \circ P).$$

Also,

$$R \circ (P \cap Q) \subseteq (R \circ P) \cap (R \circ Q), \qquad (S \cap T) \circ P \subseteq (S \circ P) \cap (T \circ P).$$

The analogous formulas for difference and complement hold only as inclusions or under extra hypotheses; they are not general distributive laws. *Go to proof.*

Remark (Dependencies).

- Composition of Relations
- Union of Relations
- Intersection of Relations

Remark (Warning on relation composition). Relation composition distributes over unions on both sides. It does not generally distribute over intersections, differences, or complements. The reason is existential: a pair can lie in both composed relations using two different middle witnesses, even when no single witness satisfies both conditions at once.

Remark (Relation algebra interaction table).

Operation	**Converse**	**Composition**
Union	$(R \cup S)^{-1} = R^{-1} \cup S^{-1}$	Distributes exactly on both sides
Intersection	$(R \cap S)^{-1} = R^{-1} \cap S^{-1}$	Gives $\subseteq$ in general; equality needs extra hypotheses
Difference	$(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}$	No general distributive equality
Complement	$((A \times B) \setminus R)^{-1} = (B \times A) \setminus R^{-1}$	No general distributive equality
Composition	$(T \circ R)^{-1} = R^{-1} \circ T^{-1}$	Associative with identity relation Δ_A

Definition (Transitive Closure)

Definition 8.1.22 (Transitive Closure)**.** Let R be a relation on A. The *transitive closure* of R is the smallest transitive relation on A containing R.

Remark (Dependencies).

- Relation
- Transitive Relation
- Relation Inclusion

Definition (Equivalence Closure)

Definition 8.1.23 (Equivalence Closure)**.** Let R be a relation on A. The *equivalence closure* of R is the smallest equivalence relation on A containing R.

Remark (Dependencies).

- Relation
- Equivalence Relation
- Relation Inclusion

Definition (One-to-Many Relation)

Definition 8.1.24 (One-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *one-to-many* if some input is related to two distinct outputs:

$$\exists a \in A \, \exists b_1, b_2 \in B, \quad (a, b_1) \in R \wedge (a, b_2) \in R \wedge b_1 \neq b_2.$$

Remark (Dependencies).

- Relation

Definition (Many-to-One Relation)

Definition 8.1.25 (Many-to-One Relation)**.** Let $R \subseteq A \times B$. R is *many-to-one* if two distinct inputs are related to a common output:

$$\exists a_1, a_2 \in A \, \exists b \in B, \quad (a_1, b) \in R \wedge (a_2, b) \in R \wedge a_1 \neq a_2.$$

Remark (Dependencies).

- Relation

Definition (Many-to-Many Relation)

Definition 8.1.26 (Many-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *many-to-many* if it is one-to-many and many-to-one.

Remark (Dependencies).

- One-to-Many Relation
- Many-to-One Relation

Remark (Relation cardinality patterns). The one-to-many, many-to-one, and many-to-many labels classify how many inputs and outputs can be connected by a relation.

Remark (Why one-to-many is not functional). A one-to-many relation fails the determinacy condition needed for a function: one input admits at least two possible outputs. A many-to-one relation may still be a function, because different inputs are allowed to share an output.

Definition (Many-Place Relation)

Definition 8.1.27 (Many-Place Relation)**.** A *ternary relation* among sets A, B, and C is a subset of $A \times B \times C$. More generally, an n-place relation on $A_1, \ldots, A_n$ is a subset of

$$A_1 \times \cdots \times A_n.$$

For example, betweenness in geometry is naturally a three-place relation, and the arithmetic statement $x + y = z$ is naturally a three-place relation among x, y, and z.

Remark (Dependencies).

- Relation
- Cartesian Product

Remark (Tarski-style hierarchy). The hierarchy used in the next chapter is:

$$\begin{aligned}\text{relation} &\longrightarrow \text{functional relation} \longrightarrow \text{function}\\ &\longrightarrow \text{one-one or onto function} \longrightarrow \text{bijection.}\end{aligned}$$

The first step adds determinacy; the second adds totality; the later steps control whether outputs are shared or missed.

Relational Properties — Quick Reference

Property	Formal definition	Canonical example
Reflexive	$\forall a,\ (a,a) \in R$	$=$ on any set
Irreflexive	$\forall a,\ (a,a) \notin R$	$<$ on $\mathbb{N}$
Symmetric	$(a,b) \in R \Rightarrow (b,a) \in R$	"same age as"
Antisymmetric	$(a,b),(b,a) \in R \Rightarrow a = b$	$\leq$ on $\mathbb{N}$
Asymmetric	$(a,b) \in R \Rightarrow (b,a) \notin R$	$<$ on $\mathbb{N}$
Transitive	$(a,b),(b,c) \in R \Rightarrow (a,c) \in R$	$\leq$ on $\mathbb{N}$
Total (Connex)	$(a,b) \in R \lor (b,a) \in R$	$\leq$ on $\mathbb{R}$
Structural classes formed by combining properties:		
Equivalence	Reflexive + Symmetric + Transitive	$=$ on any set
Preorder	Reflexive + Transitive	$\leq$ on $\mathbb{N}$
Partial order	Reflexive + Antisymmetric + Transitive	$\subseteq$ on $\mathcal{P}(A)$
Strict partial order	Irreflexive + Transitive ($\Rightarrow$ Asymmetric)	$<$ on $\mathbb{R}$
Total order	Partial order + Total	$\leq$ on $\mathbb{R}$

8.1.2 Properties of Relations

Definition (Reflexive Relation)

Definition 8.1.28 (Reflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *reflexive* on A if every element of the domain is related to itself:

$$\forall a \in A,\ (a,a) \in R.$$

In infix notation, this is the condition $\forall a \in A,\ aRa$.

Remark (Standard quantified statement). Reflexivity is a universal requirement over the whole domain:

$$\text{Reflexive}_A(R) \iff \forall a \in A,\ (a,a) \in R.$$

For first-order logic with equality, the corresponding logical baseline is the identity axiom

$$\forall x\,(x = x),$$

which forces equality to behave reflexively in every structure.

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Irreflexive Relation)

Definition 8.1.29 (Irreflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *irreflexive* on A if no element of the domain is related to itself:

$$\forall a \in A,\ (a, a) \notin R.$$

In infix notation, this is the condition $\forall a \in A,\ \neg(aRa)$.

Remark (Standard quantified statement). Irreflexivity is not the same as merely failing to be reflexive:

$$\mathrm{Irreflexive}_A(R) \iff \forall a \in A,\ \neg(aRa),$$

whereas

$$\neg\,\mathrm{Reflexive}_A(R) \iff \exists a \in A,\ \neg(aRa).$$

Thus a relation may be neither reflexive nor irreflexive. In first-order axiomatizations of strict order, irreflexivity is commonly imposed by the axiom

$$\forall x\, \neg(x < x).$$

Remark (Dependencies).

- Relation
- Universal Formula

Remark (Independence of reflexive and irreflexive). A relation may be neither reflexive nor irreflexive (if some but not all elements satisfy $(a, a) \in R$). However, a relation cannot be both reflexive and irreflexive unless $A = \varnothing$.

Definition (Symmetric Relation)

Definition 8.1.30 (Symmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *symmetric* on A if every related pair also appears in the reverse order:

$$\forall a, b \in A,\ (a, b) \in R \to (b, a) \in R.$$

In infix notation, this is $\forall a, b \in A,\ aRb \to bRa$.

Remark (Standard quantified statement). Symmetry is a conditional property:

$$\mathrm{Symmetric}_A(R) \iff \forall a, b \in A,\ (aRb \to bRa).$$

It does not require every pair to be related; it requires only that any existing directed relation can be reversed. For first-order logic with equality, symmetry is expressed by the identity axiom

$$\forall x \forall y\, (x = y \to y = x).$$

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Antisymmetric Relation)

Definition 8.1.31 (Antisymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *antisymmetric* on A if any two elements related in both directions must be equal:

$$\forall a, b \in A,\ \big((a,b) \in R \wedge (b,a) \in R\big) \to a = b.$$

In infix notation, this is $\forall a, b \in A,\ ((aRb \wedge bRa) \to a = b)$.

Remark (Standard quantified statement). Antisymmetry forbids two-way relation cycles between distinct elements:

$$\text{Antisymmetric}_A(R) \iff \forall a, b \in A,\ ((aRb \wedge bRa) \to a = b).$$

It is not the same as asymmetry. Antisymmetry permits aRa; asymmetry forbids it. Thus $\leq$ is antisymmetric, while $<$ is asymmetric. In first-order axiomatizations of partial orders, antisymmetry is usually written

$$\forall x \forall y\, ((x \leq y \wedge y \leq x) \to x = y).$$

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Asymmetric Relation)

Definition 8.1.32 (Asymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *asymmetric* on A if every related pair excludes the reverse pair:

$$\forall a, b \in A,\ (a,b) \in R \to (b,a) \notin R.$$

In infix notation, this is $\forall a, b \in A,\ (aRb \to \neg(bRa))$.

Remark (Standard quantified statement). Asymmetry is the strict one-way condition

$$\text{Asymmetric}_A(R) \iff \forall a, b \in A,\ (aRb \to \neg(bRa)).$$

It implies irreflexivity by taking $a = b$, and it implies antisymmetry because two-way relation between distinct elements is impossible. Conversely, a relation that is both irreflexive and antisymmetric is asymmetric. For strict orders, the typical first-order axiom is

$$\forall x \forall y\, (x < y \to \neg(y < x)).$$

Remark (Dependencies).

- Relation
- Universal Formula

Remark (Asymmetric vs. antisymmetric). Every asymmetric relation is antisymmetric, but not conversely. $\leq$ on $\mathbb{R}$ is antisymmetric but not asymmetric (since $1 \leq 1$). Strict order $<$ is both asymmetric and irreflexive.

Definition (Transitive Relation)

Definition 8.1.33 (Transitive Relation)**.** A binary relation $R \subseteq A \times A$ is *transitive* on A if any two composable related pairs collapse to a direct related pair:

$$\forall a, b, c \in A,\ \big((a, b) \in R \wedge (b, c) \in R\big) \rightarrow (a, c) \in R.$$

In infix notation, this is $\forall a, b, c \in A,\ ((aRb \wedge bRc) \rightarrow aRc)$.

Remark (Standard quantified statement). Transitivity is the formal chaining law:

$$\text{Transitive}_A(R) \iff \forall a, b, c \in A,\ ((aRb \wedge bRc) \rightarrow aRc).$$

For equality, the corresponding first-order axiom is

$$\forall x \forall y \forall z\, ((x = y \wedge y = z) \rightarrow x = z).$$

Together with reflexivity and symmetry, transitivity gives equivalence relations; together with reflexivity and antisymmetry, it gives partial orders.

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Total (Connex) Relation)

Definition 8.1.34 (Total (Connex) Relation)**.** R is *total* (or *connex*) if every pair is comparable:

$$\forall a, b \in A,\ (a, b) \in R \vee (b, a) \in R.$$

Remark (Dependencies).

- Relation
- Universal Formula

Definition (Equivalence Relation)

Definition 8.1.35 (Equivalence Relation)**.** R is an *equivalence relation* on A if it is reflexive, symmetric, and transitive.

Remark (Standard quantified statement). R is an equivalence relation on A iff:

$$(\forall a \in A,\ (a,a) \in R) \wedge (\forall a, b \in A,\ (a,b) \in R \to (b,a) \in R) \wedge (\forall a, b, c \in A,\ (a,b) \in R \wedge (b,c) \in R \to (a,c) \in$$

Remark (Role in mathematics). Equivalence relations axiomatize the idea of "sameness up to a chosen criterion." They partition sets into equivalence classes and are the basis for quotient constructions throughout algebra, topology, and analysis.

Definition (Preorder)

Definition 8.1.36 (Preorder)**.** R is a *preorder* on A if it is reflexive and transitive.

Remark (Standard quantified statement). R is a preorder on A iff:

$$(\forall a \in A,\ (a,a) \in R) \ \wedge\ (\forall a, b, c \in A,\ (a,b) \in R \wedge (b,c) \in R \to (a,c) \in R).$$

Remark (Dependencies).

- Reflexive Relation
- Transitive Relation

Definition (Partial Order)

Definition 8.1.37 (Partial Order)**.** R is a *partial order* on A if it is reflexive, antisymmetric, and transitive.

Remark (Standard quantified statement). R is a partial order on A iff:

$$(\forall a \in A,\ (a,a) \in R) \wedge (\forall a, b \in A,\ (a,b) \in R \wedge (b,a) \in R \to a = b) \wedge (\forall a, b, c \in A,\ (a,b) \in R \wedge (b,c) \in R \to$$

Remark (Dependencies).

- Reflexive Relation
- Antisymmetric Relation
- Transitive Relation

Definition (Strict Partial Order)

Definition 8.1.38 (Strict Partial Order)**.** R is a *strict partial order* on A if it is irreflexive and transitive.

Remark (Standard quantified statement). R is a strict partial order on A iff:

$$(\forall a \in A,\ (a,a) \notin R) \ \wedge\ (\forall a, b, c \in A,\ (a,b) \in R \wedge (b,c) \in R \to (a,c) \in R).$$

Remark (Dependencies).

- Irreflexive Relation

- Transitive Relation

Remark (Asymmetry is automatic). Irreflexivity and transitivity together force asymmetry: if $(a, b) \in R$ and $(b, a) \in R$, transitivity gives $(a, a) \in R$, contradicting irreflexivity. Hence every strict partial order is asymmetric, and a strict partial order can equivalently be characterised as an asymmetric, transitive relation.

Remark (Correspondence with partial orders). Strict and non-strict orders carry the same information. Given a partial order $\leq$, the relation $a < b \iff (a \leq b \wedge a \neq b)$ is a strict partial order; conversely, given a strict partial order $<$, the relation $a \leq b \iff (a < b \vee a = b)$ is a partial order. The two presentations are interchangeable, which is why an ordered structure may be specified by either.

Definition (Total Order)

Definition 8.1.39 (Total Order)**.** R is a *total order* on A if it is a partial order and is total.

Remark (Standard quantified statement). R is a total order on A iff R is a partial order on A and additionally:

$$\forall a, b \in A,\ (a, b) \in R \vee (b, a) \in R.$$

Remark (Dependencies).

- Partial Order
- Total Relation

Remark (Using structural properties in proofs). When R is asserted to be an equivalence relation or partial order, this is shorthand for the conjunction of its defining properties. In proofs, cite only the specific component needed: "since R is a partial order, antisymmetry implies ..." rather than restating all properties.

Example (Using a structural property in a proof). Let R be an equivalence relation on A. For any $a, b \in A$, $(a, b) \in R \Rightarrow [a] = [b]$, where $[a] = \{x \in A \mid (a, x) \in R\}$.

Remark (Properties are logically independent). No basic property implies another in general. A relation may be transitive without being reflexive, or symmetric without being transitive. The structural classes are distinguished precisely by requiring specific combinations.

8.2 Equivalence

Equivalence and Partitions — Quick Reference

Concept	Meaning
Equivalence class	$[a]_R = \{x \in A \mid (a,x) \in R\}$
Quotient set	A/R: all equivalence classes
Index of R	Cardinality of A/R
Canonical surjection	$\pi : A \to A/R$, $\pi(a) = [a]$
Partition	Nonempty, disjoint, covering collection of subsets
Rep. independence	$[a] = [b] \iff (a,b) \in R$
Equiv.–partition correspondence	Bijection between equiv. relations and partitions

8.2.1 Equivalence Classes and Partitions

Definition (Equivalence Class)

Definition 8.2.1 (Equivalence Class)**.** Let R be an equivalence relation on A. For $a \in A$, the *equivalence class* of a is:

$$[a]_R \;:=\; \{\, x \in A \mid (a,x) \in R \,\}.$$

When R is clear from context, we write $[a]$.

Remark (Dependencies).

- Equivalence Relation
- Relation

Remark (English reading). $[a]_R$ is the set of all elements that R declares "the same as a." Two elements lie in the same class iff they are related: this is precisely the Representative Independence Lemma below.

Definition (Quotient Set)

Definition 8.2.2 (Quotient Set)**.** The *quotient set* of A by R is the set of all equivalence classes:

$$A/R \;:=\; \{\, [a]_R \mid a \in A \,\}.$$

Remark (Dependencies).

- Equivalence Relation
- Equivalence Class

Definition (Index of an Equivalence Relation)

Definition 8.2.3 (Index of an Equivalence Relation)**.** The *index* of R on A is the cardinality $|A/R|$, i.e. the number of equivalence classes.

Remark (Standard quantified statement). $\operatorname{index}(R) = |A/R| = |\{[a]_R \mid a \in A\}|$.

Remark (Dependencies).

- Quotient Set
- Same Cardinality

Definition (Canonical Surjection)

Definition 8.2.4 (Canonical Surjection)**.** The *canonical surjection* (quotient map) is the function

$$\pi : A \to A/R, \qquad \pi(a) := [a].$$

Remark (Forward reference). This definition uses the function vocabulary developed in the next chapter. It is included here because quotient sets are naturally introduced with their canonical map; proofs using its function-theoretic properties may be revisited after the function chapter.

Remark (Dependencies).

- Function
- Quotient Set
- Equivalence Class
- Surjective Function

Remark (Properties of π). π is surjective by construction. Elements $a, b \in A$ satisfy $\pi(a) = \pi(b)$ iff $(a, b) \in R$. The canonical surjection is the prototype for all quotient constructions: it reappears as the quotient homomorphism in algebra and the quotient map in topology.

Definition (Partition)

Definition 8.2.5 (Partition)**.** A *partition* of A is a collection $\mathcal{P}$ of subsets of A such that:

(i) every block is nonempty: $\forall P \in \mathcal{P},\ P \neq \varnothing$;

(ii) distinct blocks are disjoint: $\forall P, Q \in \mathcal{P},\ P \neq Q \to P \cap Q = \varnothing$;

(iii) the blocks cover A: $\bigcup_{P \in \mathcal{P}} P = A$.

The sets $P \in \mathcal{P}$ are called the *blocks* of the partition.

Remark (Dependencies).

- Subset
- Empty Set
- Intersection
- Union
- Cover

Remark (Partition vs. cover). Every partition of A is a cover of A whose members are nonempty and pairwise disjoint. Partitions are precisely the covers satisfying the disjointness condition.

Lemma (Representative Independence Lemma)

Lemma 8.2.6 (Representative Independence Lemma). *Let R be an equivalence relation on A. For any $a, b \in A$,*

$$[a] = [b] \iff (a, b) \in R.$$

Go to proof.

Remark (Standard quantified statement). $\forall a, b \in A : \ [a]_R = [b]_R \leftrightarrow (a, b) \in R.$

Remark (Dependencies).

- Equivalence Relation
- Equivalence Class
- Partition
- Quotient Set

Remark (Well-definedness on quotient sets). This lemma is the key fact underlying well-definedness of functions on quotient sets: a function $f : A \to B$ defined by $f([a]) = \cdots$ is well-defined iff the formula gives the same output for all representatives of $[a]$.

Theorem (Equivalence Relations and Partitions)

Theorem 8.2.7 (Equivalence Relations and Partitions). *Let A be a set.*

(i) If R is an equivalence relation on A, then A/R is a partition of A.

(ii) If $\mathcal{P}$ is a partition of A, then the relation $R_{\mathcal{P}}$ defined by

$$(a, b) \in R_{\mathcal{P}} \iff \exists P \in \mathcal{P} \text{ with } a \in P \text{ and } b \in P$$

is an equivalence relation on A.

(iii) These constructions are inverse: $R_{A/R} = R$ and $A/R_{\mathcal{P}} = \mathcal{P}$.

Go to proof.

Remark (Standard quantified statement). (i) $\forall a \in A,\ [a]_R \neq \varnothing$; $\forall a, b \in A,\ [a]_R \neq [b]_R \to [a]_R \cap [b]_R = \varnothing$; $\bigcup_{a \in A} [a]_R = A$.
(ii) $\forall a, b \in A :\ (a, b) \in R_{\mathcal{P}} \ \leftrightarrow\ \exists P \in \mathcal{P},\ a \in P \wedge b \in P$.

Remark (Dependencies).

- Equivalence Relation
- Equivalence Class
- Quotient Set
- Canonical Surjection
- Function
- Composition

Remark (Duality of perspectives). This theorem establishes a bijection between equivalence relations on A and partitions of A. The two perspectives—"same block" (partition) and "related" (equivalence relation)—are interchangeable and each is more natural in different contexts.

Example (Extremal equivalence relations). On any set A:

1. The *equality relation* $(a, b) \in R \iff a = b$ gives singleton classes $[a] = \{a\}$ and the finest partition of A.
2. The *universal relation* $R = A \times A$ gives $[a] = A$ for all a, and the coarsest partition (one block).

All other equivalence relations on A lie strictly between these extremes.

Theorem (Universal Property of the Quotient Map)

Theorem 8.2.8 (Universal Property of the Quotient Map)**.** *Let R be an equivalence relation on A, let $\pi : A \to A/R$ be the canonical surjection, and let $f : A \to B$ be any function. Then f is* constant on equivalence classes *— meaning $(a, b) \in R$ implies $f(a) = f(b)$ — if and only if there exists a unique function $\bar{f} : A/R \to B$ such that*

$$f = \bar{f} \circ \pi.$$

When it exists, $\bar{f}$ is defined by $\bar{f}([a]) = f(a)$. Go to proof.

Remark (Forward reference). This universal property depends on functions, surjectivity, and composition. It is stated here to complete the quotient-set picture, but its dependency ceiling is intentionally forward-looking.

Remark (Standard quantified statement). $\big(\forall a, b \in A,\ (a, b) \in R \to f(a) = f(b)\big) \ \leftrightarrow\ \big(\exists!\, \bar{f} : A/R \to B,\ \forall a \in A,\ \bar{f}([a]) = f(a)\big)$.

Remark (Dependencies).

- Equivalence Relation
- Canonical Surjection
- Quotient Set
- Function
- Composition

Remark (Why this is called the universal property). The condition $f = \bar{f} \circ \pi$ says the diagram commutes: going from A to B directly via f gives the same result as going first to A/R via π and then to B via $\bar{f}$. The word "universal" refers to the fact that $\pi : A \to A/R$ is the *unique* quotient with this property: any other map that "respects the equivalence" factors through π.

The uniqueness of $\bar{f}$ is forced by π being surjective: every element of A/R is $[a]$ for some a, so $\bar{f}([a]) = f(a)$ is the only possible definition.

Remark (Well-definedness is the key verification). The heart of the theorem is well-definedness: the formula $\bar{f}([a]) = f(a)$ could be ambiguous if $[a] = [b]$ but $f(a) \neq f(b)$. The hypothesis that f is constant on equivalence classes is precisely what prevents this: $(a, b) \in R$ (i.e., $[a] = [b]$) implies $f(a) = f(b)$, so the output does not depend on which representative of the class is chosen.

This "well-definedness check" recurs throughout mathematics. Whenever a function is defined on equivalence classes by a formula involving representatives, the first obligation is to verify that the formula gives the same answer for all representatives. The universal property tells exactly what condition on f makes this work.

Remark (Where this appears downstream). This theorem is not an abstract curiosity. It is the engine behind:

- *Integer construction*: the integers $\mathbb{Z}$ are defined as equivalence classes of pairs of natural numbers, and arithmetic operations are defined by formulas on representatives. The universal property verifies the operations are well-defined.
- *Rational construction*: $\mathbb{Q}$ is defined as equivalence classes of pairs (p, q) with $q \neq 0$, and addition/multiplication on representatives must be checked.
- *Quotient groups in algebra*: the quotient group G/N is defined by the same factorization, and the quotient homomorphism is π.
- *Quotient topology*: the quotient topological space is defined as the finest topology on A/R making π continuous — a condition stated precisely via the universal property.

Chapter 9

Functions

Remark (Source orientation). This chapter follows Tarski's treatment of functions as special relations: first many-one relations, then one-one relations and bijective correspondences Tarski [1941].

9.1 Functions — Quick Reference

Functions — Quick Reference

Concept	Meaning
Functional relation	Each input has at most one output
Total relation on A	Each input has at least one output
Function	Left-total, right-unique relation
Partial function	Functional, not necessarily total
Function equality	Same domain, codomain, and pointwise values
Many-place function	Functional relation in the last coordinate
Domain	$\mathrm{dom}(f) = A$ for $f : A \to B$
Codomain	$\mathrm{cod}(f) = B$ for $f : A \to B$
Image of function	$\mathrm{im}(f) = \{f(a) \mid a \in A\}$
Image of set	$f(S) = \{f(a) \mid a \in S\}$ for $S \subseteq A$
Preimage	$f^{-1}(T) = \{a \in A \mid f(a) \in T\}$ for $T \subseteq B$
Fiber	$f^{-1}(\{b\})$: preimage of a singleton
Graph	$\{(a, b) \in A \times B \mid b = f(a)\}$
Injective	Distinct inputs $\Rightarrow$ distinct outputs
Surjective	Every codomain element is achieved
Bijective	Injective and surjective
Function test	Function iff total and functional
One-one test	Injective iff converse graph is functional
Bijective test	Unique correspondence in both directions
Identity	$\mathrm{id}_A(a) = a$
Inclusion map	$\iota : A \hookrightarrow B$, $\iota(a) = a$ for $A \subseteq B$
Composition	$(g \circ f)(a) = g(f(a))$
Inverse	Defined for bijections; $f^{-1} \circ f = \mathrm{id}_A$
Left inverse	$g \circ f = \mathrm{id}_A$
Right inverse	$f \circ h = \mathrm{id}_B$
Restriction	$f\|_C : C \to B$
Extension	$g : A' \to B$ with $g\|_A = f$
Empty function	Unique function $\varnothing \to B$

9.1.1 Functions

Remark (Geometric picture). A function is a deterministic machine: every input in the domain is accepted, and no input can fork into two different outputs. Fibers are the level sets sitting over points of the codomain; they collect all inputs that the machine sends to the same output.

Definition (Functional Relation)

Definition 9.1.1 (Functional Relation)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *functional from A to B* if each input has at most one output:

$$\forall a \in A \, \forall b_1, b_2 \in B, \quad \big((a, b_1) \in R \wedge (a, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

Remark (Dependencies).

- Relation
- Cartesian Product

Definition (Total Relation on a Domain)

Definition 9.1.2 (Total Relation on a Domain)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *total on A* if each input has at least one output:

$$\forall a \in A \, \exists b \in B, \quad (a, b) \in R.$$

Remark (Dependencies).

- Relation

Definition (Function)

Definition 9.1.3 (Function)**.** Let A and B be sets. A *function* from A to B is a relation $f \subseteq A \times B$, together with the declared domain A and codomain B, such that:

(i) (*Existence / left-total*) for every $a \in A$, there exists $b \in B$ with $(a, b) \in f$;

(ii) (*Uniqueness / right-unique*) if $(a, b_1) \in f$ and $(a, b_2) \in f$ then $b_1 = b_2$.

If $(a, b) \in f$ we write $f(a) = b$.

Remark (Dependencies).

- Relation
- Cartesian Product

Remark (English reading). A function is a rule assigning to each input exactly one output. The two conditions formalize “defined everywhere” (existence) and “single-valued” (uniqueness).

Remark (Function as relation). Every function is a relation, but not every relation is a function. The two conditions that distinguish functions are left-totality and right-uniqueness. The codomain is part of the function data, not something recoverable from the graph alone.

Theorem (Functional Relation Characterization of Functions)

Theorem 9.1.4 (Functional Relation Characterization of Functions)**.** *Let A and B be sets, and let $R \subseteq A \times B$. Then R is a function from A to B if and only if*

$$\forall a \in A\, \exists! b \in B, \quad (a, b) \in R.$$

Equivalently, R is a function from A to B if and only if R is total on A and functional from A to B. *Go to proof.*

Remark (Dependencies).

- Function
- Functional Relation
- Total Relation on a Domain

Definition (Partial Function)

Definition 9.1.5 (Partial Function)**.** Let A and B be sets. A *partial function* from A to B is a functional relation $R \subseteq A \times B$. It need not be total on A: some inputs may have no output.

Remark (Dependencies).

- Functional Relation

Proposition (Partial Functions Are Functional Relations)

Proposition 9.1.6 (Partial Functions Are Functional Relations)**.** *For sets A and B and a relation $R \subseteq A \times B$, R is a partial function from A to B if and only if R is functional from A to B.* *Go to proof.*

Remark (Dependencies).

- Partial Function
- Functional Relation

Definition (Function Equality)

Definition 9.1.7 (Function Equality)**.** For functions $f : A \to B$ and $g : C \to D$, we write $f = g$ if

$$A = C, \qquad B = D, \qquad \text{and} \qquad \forall a \in A,\ f(a) = g(a).$$

Remark (Dependencies).

- Function

- Set Equality

Proposition (One-to-Many Relations Are Not Functional)

Proposition 9.1.8 (One-to-Many Relations Are Not Functional)**.** *If $R \subseteq A \times B$ is one-to-many, then R is not functional from A to B. Consequently, R is not a function from A to B.* *Go to proof.*

Remark (Dependencies).

- One-to-Many Relation
- Functional Relation
- Function

Remark (Many-to-one is allowed). A function may send different inputs to the same output. That is the ordinary many-one case in Tarski's terminology. What is forbidden for a function is not shared output, but one input having two distinct outputs.

Definition (Domain of a Function)

Definition 9.1.9 (Domain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where A is the *domain* $\mathrm{dom}(f)$.

Remark (Dependencies).

- Function

Definition (Codomain of a Function)

Definition 9.1.10 (Codomain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where B is the *codomain* $\mathrm{cod}(f)$.

Remark (Dependencies).

- Function

Remark (Domain and codomain). The notation $f : A \to B$ records both the domain A and the codomain B. The same graph may define maps into larger codomains, but surjectivity and bijectivity depend on the declared codomain.

Definition (Image of a Function)

Definition 9.1.11 (Image of a Function)**.** The *image* (or *range*) of $f : A \to B$ is:

$$\mathrm{im}(f) \;:=\; \{\, b \in B \mid \exists a \in A,\; f(a) = b \,\}.$$

The image may be a proper subset of the codomain.

Remark (Dependencies).

- Function
- Set and Membership
- Axiom Schema of Separation

Remark (Range and image). For a function viewed as its graph, the relation range and the function image coincide. The codomain may be larger than this range/image.

Definition (Image of a Set)

Definition 9.1.12 (Image of a Set). For $S \subseteq A$, the *image of S under f* is:

$$f(S) \ := \ \{\, f(a) \mid a \in S \,\}.$$

Note $f(A) = \mathrm{im}(f)$.

Remark (Dependencies).

- Function
- Subset
- Axiom Schema of Separation

Definition (Preimage)

Definition 9.1.13 (Preimage). For $T \subseteq B$, the *preimage* (inverse image) of T under f is:

$$f^{-1}(T) \ := \ \{\, a \in A \mid f(a) \in T \,\}.$$

Remark (Dependencies).

- Function
- Subset
- Axiom Schema of Separation

Remark (Preimage notation warning). $f^{-1}(T)$ does *not* denote an inverse function. It is defined for any function f, regardless of whether f is injective or bijective.

Definition (Fiber)

Definition 9.1.14 (Fiber). The *fiber* of f over $b \in B$ is the preimage of the singleton:

$$f^{-1}(\{b\}) \ = \ \{\, a \in A \mid f(a) = b \,\}.$$

Remark (Dependencies).

- Function

- Preimage

Remark (Fibers partition the domain). The collection $\{f^{-1}(\{b\}) \mid b \in \mathrm{im}(f)\}$ is a partition of A. Every function thus induces an equivalence relation $a_1 \sim_f a_2 \iff f(a_1) = f(a_2)$, whose equivalence classes are precisely the fibers.

Definition (Graph of a Function)

Definition 9.1.15 (Graph of a Function)**.** The *graph* of $f : A \to B$ is:

$$\mathrm{Graph}(f) \;:=\; \{\,(a,b) \in A \times B \mid b = f(a)\,\}.$$

Remark (Dependencies).

- Function
- Ordered Pair
- Cartesian Product

Remark (Function as graph). A relation $G \subseteq A \times B$ is the graph of a function $f : A \to B$ iff G is left-total and right-unique. Functions may therefore be identified with their graphs only after the domain and codomain have been fixed. As bare sets of ordered pairs, graphs do not encode the codomain.

Definition (Injective Function)

Definition 9.1.16 (Injective Function)**.** $f : A \to B$ is *injective* (one-to-one) if distinct elements of A have distinct images:

$$\forall a_1, a_2 \in A,\; f(a_1) = f(a_2) \implies a_1 = a_2.$$

Remark (Dependencies).

- Function

Definition (Empty Function)

Definition 9.1.17 (Empty Function)**.** For every set B, the *empty function* from $\varnothing$ to B is the empty relation

$$\varnothing : \varnothing \to B.$$

It is the unique function with domain $\varnothing$ and codomain B.

Remark (Dependencies).

- Function
- Empty Set

- Codomain of a Function

Definition (Surjective Function)

Definition 9.1.18 (Surjective Function)**.** $f : A \to B$ is *surjective* (onto) if every element of B is achieved:

$$\forall b \in B,\ \exists a \in A \text{ such that } f(a) = b.$$

Equivalently, $\operatorname{im}(f) = B$.

Remark (Dependencies).

- Function

Definition (Bijective Function)

Definition 9.1.19 (Bijective Function)**.** $f : A \to B$ is *bijective* if it is both injective and surjective.

Remark (Dependencies).

- Function
- Injective Function
- Surjective Function

Remark (Fiber interpretation)*.* Injectivity: each fiber has at most one element. Surjectivity: every fiber is nonempty. Bijectivity: every fiber has exactly one element.

Proposition (One-One Relations and Converse Functionality)

Proposition 9.1.20 (One-One Relations and Converse Functionality)**.** *Let $f : A \to B$ be a function, identified with its graph $G_f \subseteq A \times B$. Then f is injective if and only if the converse relation $G_f^{-1} \subseteq B \times A$ is functional from B to A.* *Go to proof.*

Remark (Dependencies).

- Injective Function
- Converse Relation
- Functional Relation
- Graph of a Function

Proposition (Bijections Have Unique Correspondence Both Ways)

Proposition 9.1.21 (Bijections Have Unique Correspondence Both Ways)**.** *Let $f : A \to B$ be a function. Then f is bijective if and only if*

$$\forall a \in A\, \exists! b \in B, \quad f(a) = b,$$

and

$$\forall b \in B\, \exists! a \in A, \quad f(a) = b.$$

Go to proof.

Remark (Dependencies).

- Bijective Function
- Injective Function
- Surjective Function

Definition (Many-Place Functional Relation)

Definition 9.1.22 (Many-Place Functional Relation)**.** A relation $R \subseteq A_1 \times \cdots \times A_n \times B$ is *functional in the last coordinate* if

$$\forall a_1 \in A_1 \cdots \forall a_n \in A_n\, \forall b_1, b_2 \in B,$$

$$\big((a_1, \ldots, a_n, b_1) \in R \wedge (a_1, \ldots, a_n, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

When such a relation is also total in the first n coordinates, it determines a function of n variables, written $f(a_1, \ldots, a_n) = b$.

Remark (Dependencies).

- Many-Place Relation
- Functional Relation

Definition (Identity Function)

Definition 9.1.23 (Identity Function)**.** The *identity function* on A is $\mathrm{id}_A : A \to A$, $\mathrm{id}_A(a) = a$.

Remark (Dependencies).

- Function

Definition (Inclusion Map)

Definition 9.1.24 (Inclusion Map)**.** If $A \subseteq B$, the *inclusion map* is $\iota : A \hookrightarrow B$, $\iota(a) = a$.

Remark (Dependencies).

- Function
- Subset

Remark (Identity vs. inclusion). Both send each element to itself, but the inclusion map has a strictly larger codomain when $A \subsetneq B$. The inclusion map is always injective.

Definition (Constant Function)

Definition 9.1.25 (Constant Function). $f : A \to B$ is *constant* if there exists $b_0 \in B$ with $f(a) = b_0$ for all $a \in A$.

Remark (Standard quantified statement). f is constant $\leftrightarrow \exists b_0 \in B, \forall a \in A, f(a) = b_0$.

Remark (Dependencies).

- Function

Composition & Inverses — Quick Reference

Result	Statement	Proof method
Associativity of $\circ$	$h \circ (g \circ f) = (h \circ g) \circ f$	element-wise
Identity and $\circ$	$f \circ \mathrm{id}_A = \mathrm{id}_B \circ f = f$	element-wise
Inj/Surj under $\circ$	Closed under composition; partial converses	element-wise
Inverse characterization	$f^{-1} \circ f = \mathrm{id}_A$, $f \circ f^{-1} = \mathrm{id}_B$	bijectivity
Inverse of composition	$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$	verify both sides
One-sided inverses	Left inverse $\iff$ injective; right $\iff$ surjective	
Preimage preserves ops	f^{-1} commutes with $\cup, \cap, \setminus, c$	element-wise
Image and set ops	Images preserve $\cup$; contain $\subseteq$ for $\cap$	counterexample
Image/preimage adjunction	$S \subseteq f^{-1}(f(S))$, $f(f^{-1}(T)) \subseteq T$	element-wise

9.1.2 Composition, Inverses, and Set-Image Laws

Definition (Composition)

Definition 9.1.26 (Composition). Let $f : A \to B$ and $g : B \to C$. The *composition* $g \circ f : A \to C$ is:

$$(g \circ f)(a) := g\big(f(a)\big) \quad \text{for all } a \in A.$$

Remark (Dependencies).

- Function
- Cartesian Product

Remark (Non-commutativity). Composition is generally not commutative: $g \circ f \neq f \circ g$ even when both are defined.

Theorem (Associativity of Composition)

Theorem 9.1.27 (Associativity of Composition)**.** *For $f : A \to B$, $g : B \to C$, $h : C \to D$:*

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Go to proof.

Remark (Standard quantified statement). $\forall f : A \to B,\ g : B \to C,\ h : C \to D,\ \forall a \in A : [h \circ (g \circ f)](a) = [(h \circ g) \circ f](a)$.

Remark (Dependencies).

- Function
- Composition

Theorem (Identity and Composition)

Theorem 9.1.28 (Identity and Composition)**.** *For any $f : A \to B$:*

$$f \circ \mathrm{id}_A = f \quad \textit{and} \quad \mathrm{id}_B \circ f = f.$$

Go to proof.

Remark (Standard quantified statement). $\forall f : A \to B,\ \forall a \in A :\ (f \circ \mathrm{id}_A)(a) = f(a)$ and $(\mathrm{id}_B \circ f)(a) = f(a)$.

Remark (Dependencies).

- Function
- Composition
- Identity Function

Theorem (Injectivity and Surjectivity Under Composition)

Theorem 9.1.29 (Injectivity and Surjectivity Under Composition)**.** *Let $f : A \to B$ and $g : B \to C$.*

(i) If f and g are injective, then $g \circ f$ is injective.

(ii) If f and g are surjective, then $g \circ f$ is surjective.

(iii) If $g \circ f$ is injective, then f is injective.

(iv) If $g \circ f$ is surjective, then g is surjective.

Go to proof.

Remark (Standard quantified statement). (i) $(\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2) \wedge (\forall b_1, b_2 \in B,\ g(b_1) = g(b_2) \to b_1 = b_2) \ \to\ \forall a_1, a_2 \in A,\ g(f(a_1)) = g(f(a_2)) \to a_1 = a_2$.
(iii) $(\forall a_1, a_2 \in A,\ g(f(a_1)) = g(f(a_2)) \to a_1 = a_2) \ \to\ \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2$.

Remark (Dependencies).

- Function
- Composition
- Injective Function
- Surjective Function

Remark (Sharpness of the implications). In (iii), g need not be injective. In (iv), f need not be surjective. Bijectivity of $g \circ f$ does not imply bijectivity of either f or g individually.

Definition (Inverse Function)

Definition 9.1.30 (Inverse Function)**.** Let $f : A \to B$ be bijective. The *inverse function* is $f^{-1} : B \to A$ defined by: $f^{-1}(b) = a$ iff $f(a) = b$.

Remark (Dependencies).

- Function
- Bijective Function
- Identity Function

Theorem (Characterization of Inverse Functions)

Theorem 9.1.31 (Characterization of Inverse Functions)**.** *Let $f : A \to B$ be bijective. Then*

$$f^{-1} \circ f = \mathrm{id}_A \quad \text{and} \quad f \circ f^{-1} = \mathrm{id}_B\,.$$

Conversely, a function admits an inverse iff it is bijective. *Go to proof.*

Remark (Standard quantified statement). $\forall a \in A:\ f^{-1}(f(a)) = a$ and $\forall b \in B:\ f(f^{-1}(b)) = b$.
Converse: $(\exists g : B \to A,\ g \circ f = \mathrm{id}_A \wedge f \circ g = \mathrm{id}_B) \ \leftrightarrow\ f$ is bijective.

Remark (Dependencies).

- Function
- Composition
- Inverse Function

- Bijective Function
- Identity Function

Theorem (Inverse of a Composition)

Theorem 9.1.32 (Inverse of a Composition)**.** *Let $f : A \to B$ and $g : B \to C$ be bijective. Then $g \circ f$ is bijective and*

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Go to proof.

Remark (Standard quantified statement)*.* $\forall c \in C : \ (g \circ f)^{-1}(c) = f^{-1}(g^{-1}(c))$.

Remark (Dependencies)*.*

- Function
- Composition
- Inverse Function
- Bijective Function

Definition (Left Inverse)

Definition 9.1.33 (Left Inverse)**.** Let $f : A \to B$. A *left inverse* of f is a function $g : B \to A$ with $g \circ f = \mathrm{id}_A$.

Remark (Dependencies)*.*

- Function
- Composition
- Identity Function

Definition (Right Inverse)

Definition 9.1.34 (Right Inverse)**.** Let $f : A \to B$. A *right inverse* (or *section*) of f is a function $h : B \to A$ with $f \circ h = \mathrm{id}_B$.

Remark (Dependencies)*.*

- Function
- Composition
- Identity Function

Remark (One-sided inverses)*.* Left inverses and right inverses are the two one-sided inverse notions.

Theorem (One-Sided Inverses and Function Properties)

Theorem 9.1.35 (One-Sided Inverses and Function Properties)**.** *Let $f : A \to B$.*

(i) f has a left inverse $\iff$ f is injective (and $A \neq \varnothing$ or $A = B = \varnothing$).

(ii) f has a right inverse $\iff$ f is surjective.

(iii) If f has both a left inverse g and a right inverse h, then $g = h$ and f is bijective.

Go to proof.

Remark (Standard quantified statement). (i) $(\exists g : B \to A,\ \forall a \in A,\ g(f(a)) = a) \leftrightarrow \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2$.
(ii) $(\exists h : B \to A,\ \forall b \in B,\ f(h(b)) = b) \leftrightarrow \forall b \in B,\ \exists a \in A,\ f(a) = b$.

Remark (Dependencies).

- Function
- Composition
- Left Inverse
- Right Inverse
- Injective Function
- Surjective Function
- Bijective Function

Remark (Axiom of Choice). The existence of a right inverse for every surjective function is equivalent to the Axiom of Choice.

Definition (Restriction)

Definition 9.1.36 (Restriction)**.** Let $f : A \to B$ and $C \subseteq A$. The *restriction* of f to C is $f|_C : C \to B$, $f|_C(a) = f(a)$ for all $a \in C$.

Remark (Dependencies).

- Function
- Subset
- Domain

Definition (Extension)

Definition 9.1.37 (Extension)**.** Let $A \subseteq A'$, and let $f : A \to B$. A function $g : A' \to B$ is an *extension* of f if $g|_A = f$.

Remark (Dependencies).

- Function
- Subset
- Restriction
- Domain
- Codomain

Remark (Extensions not unique)*.* An extension agrees with f on all of A but may be defined on a larger domain A'. Extensions are generally not unique; uniqueness requires additional constraints (such as continuity in analysis or linearity in algebra), leading to important results like the Tietze Extension Theorem and the Hahn–Banach Theorem.

Theorem (Preimages Preserve Set Operations)

Theorem 9.1.38 (Preimages Preserve Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq B$. Then*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$,

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$,

(iii) $f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$,

(iv) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Remark (Standard quantified statement)*.* (i) $\forall a \in A : \ a \in f^{-1}(S \cup T) \leftrightarrow (f(a) \in S \vee f(a) \in T)$.
(ii) $\forall a \in A : \ a \in f^{-1}(S \cap T) \leftrightarrow (f(a) \in S \wedge f(a) \in T)$.
(iv) $\forall a \in A : \ a \in f^{-1}(S^c) \leftrightarrow f(a) \notin S$.

Remark (Dependencies)*.*

- Function
- Preimage
- Union
- Intersection
- Set Difference
- Complement

Theorem (Images and Set Operations)

Theorem 9.1.39 (Images and Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq A$. Then*

(i) $f(S \cup T) = f(S) \cup f(T)$,

(ii) $f(S \cap T) \subseteq f(S) \cap f(T)$,

(iii) $f(S \setminus T) \supseteq f(S) \setminus f(T)$.

Equality holds in (ii) and (iii) for all S, T iff f is injective. Go to proof.

Remark (Standard quantified statement). (i) $\forall b \in B :\ b \in f(S \cup T) \leftrightarrow (\exists a \in S,\ f(a) = b) \vee (\exists a \in T,\ f(a) = b)$.
(ii) $\forall b \in B :\ (\exists a \in S \cap T,\ f(a) = b) \to (\exists a \in S,\ f(a) = b) \wedge (\exists a \in T,\ f(a) = b)$; equality iff f injective.

Remark (Dependencies).

- Function
- Image of a Set
- Union
- Intersection
- Set Difference
- Injective Function

Theorem (Image–Preimage Adjunction)

Theorem 9.1.40 (Image–Preimage Adjunction)**.** *Let $f : A \to B$, let $S \subseteq A$, and let $T \subseteq B$. Then*

$$S \subseteq f^{-1}(f(S)), \qquad f(f^{-1}(T)) \subseteq T.$$

Moreover, $S = f^{-1}(f(S))$ for every $S \subseteq A$ iff f is injective, and $f(f^{-1}(T)) = T$ for every $T \subseteq B$ iff f is surjective. Go to proof.

Remark (Dependencies).

- Function
- Image of a Set
- Preimage
- Injective Function
- Surjective Function

Remark (Images, preimages, and Boolean operations).

Operation	**Preimage law, for $S, T \subseteq B$**	**Image law, for $C, D \subseteq A$**
Union	$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$	$f(C \cup D) = f(C) \cup f(D)$
Intersection	$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$	$f(C \cap D) \subseteq f(C) \cap f(D)$; equality for all C, D iff f is injective
Difference	$f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$	$f(C \setminus D) \supseteq f(C) \setminus f(D)$; equality for all C, D iff f is injective
Complement	$f^{-1}(B \setminus S) = A \setminus f^{-1}(S)$	In general, $f(A \setminus C)$ need not equal $B \setminus f(C)$

Remark (Composition and inverse interaction).

Construction	**Set-operation interaction**	**Reference**
Composition	If $f : A \to B$, $g : B \to C$, and $U \subseteq C$, then $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$.	Composition, Preimage
Inverse function	If $f : A \to B$ is bijective, inverse images along f^{-1} are ordinary preimages for the function $f^{-1} : B \to A$.	Inverse Function, Inverse Characterization
Inverse of composition	If f and g are bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$, so inverse order reverses composition.	Inverse of a Composition
One-sided inverse	A left inverse detects injectivity; a right inverse detects surjectivity. These properties control when image inclusions become equalities.	One-Sided Inverses

Remark (Asymmetry between images and preimages). Preimages preserve union, intersection, difference, and complement exactly. Forward images preserve unions exactly, but generally do not preserve intersections, differences, or complements. The inclusions for intersections and differences become equalities uniformly when the function is injective. This asymmetry is fundamental and explains why preimages appear more naturally in topology and measure theory.

Chapter 10

Orderings

SERIES
From Cantor to Ito

VOLUME I
Logic, Sets, and Proof

BOOK II
Set Theory

CHAPTER 8
Orderings

10.1 Order

Ordered Sets — Quick Reference

Concept	Meaning
Partially ordered set (poset)	$(A, \leq)$: set with a partial order
Strict order	$a < b \iff a \leq b \wedge a \neq b$
Comparable / incomparable	$a \leq b$ or $b \leq a$ / neither
Totally ordered set (toset/loset)	Partial order with all pairs comparable
Upper / lower bound	$u \geq s$ / $\ell \leq s$ for all $s \in S$
Minimal element	No smaller element in S
Maximal element	No larger element in S
Least element	Smaller than all elements of S
Greatest element	Larger than all elements of S
Order-preserving map	Function f with $a \leq b \Rightarrow f(a) \leq' f(b)$
Order isomorphism	Bijective map preserving and reflecting order
Well-ordered set (woset)	Every nonempty subset has a least element
Chain / antichain	All comparable / none comparable
Initial segment	Downward-closed subset

10.1.1 Ordered Sets

Definition (Ordered Set)

Definition 10.1.1 (Ordered Set)**.** An *ordered set* is a set equipped with an order relation. Unless explicitly qualified, "ordered set" means a partially ordered set, and its relation is written $\leq$.

Remark (Dependencies).

- Relation
- Partial Order

Remark (Genus and species). "Ordered set" is a genus term: a species is fixed by specifying which order relation the carrier carries — a partial order gives a poset, a total order gives a toset, and so on. The order relations themselves (preorder, partial order, total order) are defined once, in the Relations chapter; this chapter studies the *structures* built on them and never restates a relation's axioms, so the two chapters cannot drift apart.

Definition (Partially Ordered Set)

Definition 10.1.2 (Partially Ordered Set)**.** A *partially ordered set* is a pair $(A, \leq)$ where A is a set and $\leq$ is a partial order on A (a relation that is reflexive, antisymmetric, and transitive).

Remark (Dependencies).

- Partial Order

Remark (English reading). A partially ordered set is a set with a specified way to compare elements that need not make every pair comparable.

Definition (Order-Structure Synonyms)

Definition 10.1.3 (Order-Structure Synonyms)**.** The following short forms abbreviate their full names and carry no independent mathematical content; each resolves to the canonical structure it names:

- *poset* := partially ordered set.
- *toset*, *loset*, *linearly ordered set* := totally ordered set (total order and linear order coincide, so these name one structure).
- *woset* := well-ordered set.

Definition (Strict Order)

Definition 10.1.4 (Strict Order)**.** Let $(A, \leq)$ be an ordered set. The *strict order* $<$ is defined by:

$$a < b \iff (a \leq b \wedge a \neq b).$$

Remark (Dependencies).

- Ordered Set
- Partial Order

Remark (Strict and non-strict are equivalent). The relation $<$ is irreflexive and transitive. Conversely, given any strict partial order $<$, one recovers a non-strict order by setting $a \leq b \iff (a < b \vee a = b)$. The two formulations carry exactly the same information.

Definition (Comparable and Incomparable Elements)

Definition 10.1.5 (Comparable and Incomparable Elements)**.** In an ordered set $(A, \leq)$, elements $a, b \in A$ are *comparable* if $a \leq b$ or $b \leq a$; they are *incomparable* if neither holds.

Remark (Standard quantified statement). a, b comparable $\leftrightarrow$ $a \leq b \vee b \leq a$. a, b incomparable $\leftrightarrow$ $\neg(a \leq b) \wedge \neg(b \leq a)$.

Remark (Incomparable elements). Incomparable elements can only occur in partial orders. In a total order, every pair is comparable by definition.

Definition (Totally Ordered Set)

Remark (Dependencies).

- Partially Ordered Set
- Total Order
- Comparable and Incomparable Elements

Definition 10.1.6 (Totally Ordered Set)**.** A *totally ordered set* (or *linearly ordered set*) is a pair $(A, \leq)$ where $\leq$ is a total order on A — equivalently, a partially ordered set in which every pair of elements is comparable:

$$\forall a, b \in A,\ a \leq b \vee b \leq a.$$

Remark (Examples). $(\mathbb{R}, \leq)$ is a total order. $(\mathcal{P}(A), \subseteq)$ is a partial order that is generally not total.

Definition (Upper and Lower Bounds)

Definition 10.1.7 (Upper and Lower Bounds)**.** Let $(A, \leq)$ be an ordered set and $S \subseteq A$.
An element $u \in A$ is an *upper bound* of S if $\forall s \in S,\ s \leq u$.
An element $\ell \in A$ is a *lower bound* of S if $\forall s \in S,\ \ell \leq s$.

Remark (Dependencies).

- Ordered Set
- Subset

Remark (Consequence). Bounds need not exist and need not be unique. They need not lie in S itself. Bounds are central to the definition of completeness for ordered fields.

Definition (Minimal Element)

Definition 10.1.8 (Minimal Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $m \in S$ is a *minimal element* of S if there is no $s \in S$ with $s < m$.

Remark (Dependencies).

- Ordered Set
- Subset
- Strict Order

Definition (Maximal Element)

Definition 10.1.9 (Maximal Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $M \in S$ is a *maximal element* of S if there is no $s \in S$ with $M < s$.

Remark (Dependencies).

- Ordered Set
- Subset
- Strict Order

Definition (Least Element)

Definition 10.1.10 (Least Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $\ell \in S$ is the *least element* of S if

$$\forall s \in S,\ \ell \leq s.$$

Remark (Dependencies).

- Ordered Set
- Subset

Definition (Greatest Element)

Definition 10.1.11 (Greatest Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $g \in S$ is the *greatest element* of S if

$$\forall s \in S,\ s \leq g.$$

Remark (Dependencies).

- Ordered Set
- Subset

Remark (Minimal vs. least). A *least* element is below every element of S; it must be unique if it exists. A *minimal* element merely has nothing below it within S; there may be many. Every least element is minimal, but not conversely.

Definition (Order-Preserving Map)

Definition 10.1.12 (Order-Preserving Map)**.** Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \to M'$ is *order-preserving* (or *monotone*) if

$$\forall a, b \in M, \quad a \leq b \implies f(a) \leq' f(b).$$

Remark (Dependencies).

- Partially Ordered Set
- Function

Definition (Order Isomorphism)

Definition 10.1.13 (Order Isomorphism)**.** Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \to M'$ is an *order isomorphism* if it is bijective and

$$\forall a, b \in M, \quad a \leq b \iff f(a) \leq' f(b).$$

Remark (Dependencies).

- Partially Ordered Set
- Function
- Bijective Function

Remark (Order-isomorphic posets). An order isomorphism preserves and reflects the order structure exactly, including comparability, minimal and maximal elements, and bounds. Two posets are *order-isomorphic* if such a map exists; they are structurally identical from the order-theoretic viewpoint.

Definition (Well-Ordered Set)

Definition 10.1.14 (Well-Ordered Set)**.** A totally ordered set $(A, \leq)$ is *well-ordered* if every nonempty subset $S \subseteq A$ has a least element:

$$\forall S \subseteq A, \ \big(S \neq \varnothing \ \Rightarrow \ \exists m \in S \text{ s.t. } m \leq s \ \forall s \in S\big).$$

Remark (Dependencies).

- Totally Ordered Set
- Subset

Remark (Well-order implies total order). Every well-ordered set is totally ordered. The well-ordering condition is strictly stronger: it requires a least element in every nonempty subset, not just in A as a whole.

Example (Well-order examples). $(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not: the set $\{\ldots, -3, -2, -1\}$ has no least element. $(\mathbb{R}, \leq)$ is not: $(0, 1)$ has no least element.

Remark (Connection to ordinal numbers). An *ordinal number* is defined as an equivalence class of well-ordered sets under order isomorphism: it measures the *order type* of a well-ordered set, not merely its cardinality.

Definition (Chain and Antichain)

Definition 10.1.15 (Chain and Antichain)**.** A subset $C \subseteq A$ of a poset $(A, \leq)$ is a *chain* if every pair of elements in C is comparable. It is an *antichain* if no two distinct elements of C are comparable.

Remark (Standard quantified statement). C is a chain $\leftrightarrow \forall a, b \in C,\ a \leq b \vee b \leq a$.
C is an antichain $\leftrightarrow \forall a, b \in C,\ a \neq b \to \neg(a \leq b) \wedge \neg(b \leq a)$.

Definition (Initial Segment)

Definition 10.1.16 (Initial Segment)**.** A subset $I \subseteq A$ is an *initial segment* of $(A, \leq)$ if

$$a \in I \text{ and } b \leq a \;\Rightarrow\; b \in I.$$

Remark (Standard quantified statement). I is an initial segment $\leftrightarrow \forall a \in I,\ \forall b \in A,\ b \leq a \to b \in I$.

Remark (Transition). Ordered sets provide the abstract framework for order structures on the real numbers, function spaces, and metric spaces. In later sections, order interacts with topology and analysis through intervals, monotone functions, and the completeness axiom for $\mathbb{R}$.

Definition (Directed Set)

Definition 10.1.17 (Directed Set)**.** A preorder $(I, \leq)$ is *directed* (or a *directed set*) if every finite subset of I has an upper bound in I: for all $i, j \in I$ there exists $k \in I$ with $i \leq k$ and $j \leq k$.

Remark (Dependencies).

- Partial Order
- Upper and Lower Bounds
- Finite and Infinite Sets

Remark (What directed means). In a directed set, any two elements can always be "dominated" by a third. The condition does not require a maximum element to exist, only that no two elements are ever incomparable in a final way: there is always a common upper bound available. Every totally ordered set is directed (take $k = \max(i, j)$), but directed sets can be far more general.

Example (Canonical directed sets). (i) $(\mathbb{N}, \leq)$ is directed: given $m, n \in \mathbb{N}$, take $k = \max(m, n)$.

(ii) The collection of all finite subsets of a set X, ordered by inclusion, is directed: given finite sets $F_1, F_2 \subseteq X$, take $k = F_1 \cup F_2$.

(iii) The set of all open neighborhoods of a point x in a topological space, ordered by reverse inclusion ($U \leq V$ iff $U \supseteq V$), is directed: given neighborhoods U and V of x, take $k = U \cap V$, which is also a neighborhood of x.

(iv) A partially ordered set is directed if and only if every finite subset has an upper bound. Partially ordered sets that are *not* directed have "incomparable maximal elements" — elements with no common upper bound.

Definition (Net)

Definition 10.1.18 (Net)**.** Let $(I, \leq)$ be a directed set and X a set. A *net* in X is a function $x : I \to X$. The value at $i \in I$ is written x_i, and the net is written $(x_i)_{i \in I}$.

Remark (Dependencies).

- Directed Set
- Function

Remark (Nets generalize sequences). A sequence is a net with index set $(\mathbb{N}, \leq)$. Nets are the correct generalization of sequences to general topological spaces: in a metric space, sequences suffice to detect all topological properties (closure, continuity, compactness). In a general topological space, sequences can fail to detect these properties — one needs nets. Specifically, a subset A of a topological space is closed if and only if every convergent net in A has its limit in A; and a function is continuous if and only if it preserves limits of nets.

The definition of net is placed here because it is purely order-theoretic: a directed index set and a function. The analytic content — what it means for a net to converge — belongs to topology and functional analysis. By establishing the set-theoretic definition now, those chapters can immediately state convergence conditions without set-theoretic digression.

Supremum, Infimum, and Bounds — Quick Reference

Concept	**Meaning**
Upper bound	$u \geq s$ for all $s \in S$
Lower bound	$\ell \leq s$ for all $s \in S$
Supremum	Least upper bound of S
Infimum	Greatest lower bound of S
Uniqueness	Suprema and infima are unique when they exist
Max/min vs. sup/inf	Maximum $\Rightarrow$ supremum; not conversely
Duality	sup in $(A, \leq)$ = inf in $(A, \geq)$

10.1.2 Supremum and Infimum

Every student of analysis encounters the least upper bound property of $\mathbb{R}$ early and prominently. But this property is not a peculiarity of real numbers: it is an order-theoretic concept that can be formulated in any partially ordered set. Before we reach $\mathbb{R}$, it is essential to understand supremum and infimum in their natural habitat — abstract ordered sets — so that when the completeness axiom appears in Volume II, you will recognise exactly what it is asserting and why $\mathbb{Q}$ fails to satisfy it.

The definitions in this section are also the backbone of measure theory. The extended real line $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, equipped with the usual order, is the ambient poset in which measures take their values. Countable additivity, the monotone convergence theorem, and Fatou's lemma are all statements about suprema of sequences. Understanding supremum and infimum at the abstract level here means those later theorems will feel inevitable rather than arbitrary.

The definition of upper and lower bound was given in the Ordered Sets section. For convenience we recall: $u \in A$ is an *upper bound* of $S \subseteq A$ if $\forall s \in S,\ s \leq u$; a *lower bound* satisfies the reverse. We say S is *bounded above* (resp. *bounded below*, *bounded*) if it has at least one upper (resp. lower, both) bound.

Remark (Bounds depend on the ambient poset). Whether S is bounded above depends on the poset $(A, \leq)$, not on S alone. The set $\{r \in \mathbb{Q} : r^2 < 2\}$ is bounded above in $(\mathbb{Q}, \leq)$ — the rational number 2 is an upper bound — but it has *no* supremum inside $\mathbb{Q}$. The same set in $(\mathbb{R}, \leq)$ has supremum $\sqrt{2} \in \mathbb{R}$. Being bounded above is necessary but not sufficient for a supremum to exist; the completeness axiom for $\mathbb{R}$ is precisely the assertion that sufficiency holds in $\mathbb{R}$.

Definition (Supremum and Infimum)

Definition 10.1.19 (Supremum and Infimum)**.** Let $(A, \leq)$ be a poset and $S \subseteq A$. The *supremum* (or *least upper bound*) of S, written $\sup S$, is an element $u^* \in A$ satisfying:

(i) u^* is an upper bound of S: $\forall s \in S,\ s \leq u^*$;

(ii) u^* is the *least* upper bound: if u is *any* upper bound of S, then $u^* \leq u$.

The *infimum* (or *greatest lower bound*) of S, written $\inf S$, is an element $\ell^* \in A$ satisfying:

(i) ℓ^* is a lower bound of S: $\forall s \in S,\ \ell^* \leq s$;

(ii) ℓ^* is the *greatest* lower bound: if ℓ is *any* lower bound of S, then $\ell \leq \ell^*$.

Remark (Standard quantified statement). $u^* = \sup S \ \leftrightarrow\ (\forall s \in S,\ s \leq u^*) \ \wedge\ (\forall u \in A,\ (\forall s \in S,\ s \leq u) \ \to\ u^* \leq u)$.
$\ell^* = \inf S \ \leftrightarrow\ (\forall s \in S,\ \ell^* \leq s) \ \wedge\ (\forall \ell \in A,\ (\forall s \in S,\ \ell \leq s) \ \to\ \ell \leq \ell^*)$.

Remark (Dependencies).

- Upper and Lower Bounds

- Partial Order
- Subset

Remark (How to read these definitions). Condition (i) says u^* is a *candidate*: it stands above everything in S. Condition (ii) says u^* is the *best* candidate: it is at most as large as any other upper bound. Together they say: u^* is the unique element of A that is simultaneously an upper bound of S and no larger than any other upper bound.

A useful restatement of condition (ii) is its contrapositive: if $v < u^*$, then v is *not* an upper bound of S, meaning there exists some $s \in S$ with $v < s$. This is how the condition is typically used in proofs in analysis: to verify that $u^* = \sup S$, you show (i) u^* is an upper bound, and (ii) for every $\varepsilon > 0$, the value $u^* - \varepsilon$ is *not* an upper bound, i.e., there exists $s \in S$ with $s > u^* - \varepsilon$.

Proposition

Proposition 10.1.20 (Uniqueness of Supremum and Infimum)**.** *Let $(A, \leq)$ be a poset and $S \subseteq A$. If $\sup S$ exists, it is unique. If $\inf S$ exists, it is unique.*

Remark (Standard quantified statement). $\forall u^*, v^* \in A$: if both u^* and v^* satisfy the two conditions in the definition of $\sup S$, then $u^* = v^*$.

Remark (Dependencies).

- Supremum and Infimum
- Partial Order
- Subset

Remark (Proof idea). If u^* and v^* are both suprema, then u^* is an upper bound, so the minimality of v^* gives $v^* \leq u^*$; and symmetrically $u^* \leq v^*$. Antisymmetry of the partial order then forces $u^* = v^*$. This is a standard antisymmetry argument: when two elements are mutually $\leq$-related in a partial order, they must be equal.

Remark (Supremum need not belong to S). The set $S = (0, 1) \subseteq \mathbb{R}$ is open: it contains neither endpoint. Yet $\sup S = 1$ and $\inf S = 0$. Neither 0 nor 1 belongs to S.

When the supremum does happen to lie in S, it is called the *maximum* of S, written $\max S$. Thus $\max S = \sup S$ when $\sup S \in S$, and $\max S$ is undefined when $\sup S \notin S$. Every maximum is a supremum, but the converse fails: a supremum may exist without being a maximum. The same relationship holds between infimum and minimum.

This distinction is critical in analysis. A continuous function on a closed bounded interval always achieves its maximum (the Extreme Value Theorem), but on an open interval it may approach a supremum without ever attaining it.

Remark (Supremum need not exist). Being bounded above is necessary but not sufficient for a supremum to exist. Consider $S = \{r \in \mathbb{Q} : r^2 < 2\}$ in the poset $(\mathbb{Q}, \leq)$. Every element of S satisfies $r < 2$, so $2 \in \mathbb{Q}$ is an upper bound. But the candidate supremum would need to be $\sqrt{2}$, and $\sqrt{2} \notin \mathbb{Q}$. Any rational upper bound $q > \sqrt{2}$ is not the least upper bound, because $\frac{q+\sqrt{2}}{2}$ is a smaller upper bound (and is also irrational, which means we can always find a rational strictly between any rational upper bound and $\sqrt{2}$, and that rational is still an upper bound). In short: S is bounded above in $\mathbb{Q}$, but $\sup S$ does not exist in $\mathbb{Q}$.

This is the precise failure that the *completeness axiom* for $\mathbb{R}$ corrects: it asserts that every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. The rational numbers satisfy every other ordered field axiom but fail this one.

Example (Sup and inf in divisibility). Order the set $D = \{1, 2, 3, 4, 6, 12\}$ by divisibility: $a \leq b$ iff $a \mid b$. Take $S = \{4, 6\}$.

Upper bounds of S: an element $u \in D$ must satisfy $4 \mid u$ and $6 \mid u$, i.e., u must be a common multiple of 4 and 6 in D. The only such element is 12, so $\sup S = 12 = \operatorname{lcm}(4, 6)$.

Lower bounds of S: an element $\ell \in D$ must satisfy $\ell \mid 4$ and $\ell \mid 6$, i.e., ℓ must be a common divisor of 4 and 6 in D. The common divisors are $\{1, 2\}$; the greatest is 2, so $\inf S = 2 = \gcd(4, 6)$.

This example reveals a deep pattern: in any divisibility poset, sup computes the least common multiple and inf computes the greatest common divisor. Order theory and number theory are speaking the same language.

Example (Sup and inf for function spaces). Let $B(X)$ be the set of bounded real-valued functions on a set X, ordered pointwise: $f \leq g$ iff $f(x) \leq g(x)$ for all $x \in X$. For a collection $\{f_\alpha\}$ of bounded functions, the pointwise supremum is $(\sup_\alpha f_\alpha)(x) = \sup_\alpha f_\alpha(x)$. In measure theory and integration theory, one constantly takes suprema and infima of sequences of functions; this example shows that such operations are instances of the abstract sup/inf defined here.

Proposition

Proposition 10.1.21 (Characterisation of Supremum). *Let $(A, \leq)$ be a poset, $S \subseteq A$, and $u^* \in A$. Then $u^* = \sup S$ if and only if:*

(i) u^* is an upper bound of S; and

(ii) for every $v \in A$ with $v < u^*$, there exists $s \in S$ with $v < s$.

Remark (Standard quantified statement). $u^* = \sup S \;\leftrightarrow\; (\forall s \in S,\ s \leq u^*) \;\wedge\; (\forall v \in A,\ v < u^* \;\rightarrow\; \exists s \in S,\ v < s)$.

Remark (Dependencies).

- Supremum and Infimum
- Partial Order
- Upper and Lower Bounds
- Subset

Remark (Why this characterisation is useful). Condition (ii) here is the contrapositive of the minimality condition in the definition. In analysis this is the working form: to check that

u^* is the supremum, you verify that nothing strictly below u^* is an upper bound. In $\mathbb{R}$ this becomes: for every $\varepsilon > 0$, there exists $s \in S$ with $s > u^* - \varepsilon$. This is the exact form used in the proof that sup interacts correctly with limits, integrals, and measures.

Proposition

Proposition 10.1.22 (Duality of Supremum and Infimum)**.** *Let* $(A, \leq)$ *be a poset and* $S \subseteq A$*. Then* $\inf_{(A,\leq)} S = \sup_{(A,\geq)} S$*, where* $(A, \geq)$ *denotes the dual poset.*
In particular: if every nonempty bounded-above subset of $(A, \leq)$ *has a supremum, then every nonempty bounded-below subset of* $(A, \leq)$ *has an infimum.*

Remark (Standard quantified statement)*.* $\forall S \subseteq A$: ℓ^* is the infimum of S in $(A, \leq)$ $\leftrightarrow$ ℓ^* is the supremum of S in $(A, \geq)$.

Remark (Dependencies)*.*

- Supremum and Infimum
- Partial Order
- Duality Principle for Posets
- Subset

Remark (Consequence for $\mathbb{R}$)*.* This proposition explains why the completeness axiom for $\mathbb{R}$ is stated only for suprema: the infimum statement comes for free by duality. The axiom says every nonempty bounded-above subset has a supremum; the proposition immediately gives that every nonempty bounded-below subset has an infimum. You will use this constantly without comment in Volume II.

Remark (Sequences and sup/inf)*.* A sequence $(a_n)_{n=1}^{\infty}$ in $\mathbb{R}$ is a function $\mathbb{N} \to \mathbb{R}$, so its range $\{a_n : n \in \mathbb{N}\}$ is a subset of $\mathbb{R}$. The quantities $\sup_n a_n$ and $\inf_n a_n$ are the supremum and infimum of this range set in the poset $(\mathbb{R}, \leq)$.

More subtle are the *limit superior* and *limit inferior*:

$$\limsup_{n\to\infty} a_n = \inf_{n\geq 1} \sup_{k\geq n} a_k, \qquad \liminf_{n\to\infty} a_n = \sup_{n\geq 1} \inf_{k\geq n} a_k.$$

These are iterated suprema and infima. The fact that they always exist in $\overline{\mathbb{R}}$ (the extended reals, where $\pm\infty$ are adjoined to supply missing bounds) is a direct application of the completeness of $\mathbb{R}$. Mastering the abstract definitions here makes those formulas transparent rather than mysterious.

Remark (Transition)*.* Supremum and infimum are defined for arbitrary subsets of a poset, but they need not always exist. When *every* two-element subset $\{a, b\}$ has both a supremum and an infimum, the poset has enough structure to define binary operations $a \vee b = \sup\{a, b\}$ and $a \wedge b = \inf\{a, b\}$. This additional structure is called a *lattice*, and it is developed in the next section. Lattices are the precise abstract setting for the σ-algebras and topologies encountered in measure theory and topology.

Lattices — Quick Reference

Concept	Meaning
Lattice	Poset with $\sup\{a,b\}$ and $\inf\{a,b\}$ for all a,b
Join / Meet	$a \vee b = \sup\{a,b\}$; $a \wedge b = \inf\{a,b\}$
Complete lattice	Every subset has sup and inf
Distributive lattice	$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
Boolean lattice	Complemented distributive lattice
$\mathcal{P}(X)$	Complete Boolean lattice under $\subseteq$
σ-algebras	Sub-lattices of $\mathcal{P}(X)$ closed under countable ops
Knaster–Tarski	Every monotone map on a complete lattice has a fixed point

10.1.3 Lattices

The preceding section defined supremum and infimum for arbitrary subsets of a poset. In many posets that appear in practice, every pair of elements already has a supremum and an infimum. When this is the case, the binary operations $a \vee b = \sup\{a,b\}$ and $a \wedge b = \inf\{a,b\}$ are well-defined everywhere, giving the poset the algebraic structure of a *lattice*.

Lattices are ubiquitous:

- The power set $\mathcal{P}(X)$ ordered by $\subseteq$ is a complete lattice, with join = union and meet = intersection.
- Every σ-algebra is a sub-lattice of $\mathcal{P}(X)$ that is additionally closed under countable joins and meets and under complementation.
- Every topology is a sub-structure of $\mathcal{P}(X)$ that is closed under arbitrary joins and finite meets.
- In functional analysis, the lattice of closed subspaces of a Hilbert space is a complete lattice under inclusion.

Understanding lattices here means that when σ-algebras, topologies, and sub-σ-algebras are encountered downstream, their closure conditions will be recognisable instances of a single abstract pattern.

Definition (Lattice)

Definition 10.1.23 (Lattice)**.** A *lattice* is a poset $(L, \leq)$ in which every two-element subset $\{a,b\} \subseteq L$ has both a supremum and an infimum in L.
The supremum $\sup\{a,b\}$ is called the *join* of a and b, written $a \vee b$. The infimum $\inf\{a,b\}$ is called the *meet* of a and b, written $a \wedge b$.

Remark (Standard quantified statement)*.* $(L, \leq)$ is a lattice $\leftrightarrow$ $\forall a,b \in L,\ \exists\, a \vee b \in L$ and $\exists\, a \wedge b \in L$, where $a \vee b$ and $a \wedge b$ satisfy the sup and inf conditions of Definition (Supremum and Infimum).

Remark (Join and meet as operations). In a lattice, $\vee$ and $\wedge$ are binary operations $L \times L \to L$. They satisfy, for all $a, b, c \in L$:

(i) *Commutativity*: $a \vee b = b \vee a$ and $a \wedge b = b \wedge a$.

(ii) *Associativity*: $(a \vee b) \vee c = a \vee (b \vee c)$ and $(a \wedge b) \wedge c = a \wedge (b \wedge c)$.

(iii) *Idempotency*: $a \vee a = a$ and $a \wedge a = a$.

(iv) *Absorption*: $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$.

Properties (i)–(iv) follow directly from the order-theoretic definitions of join and meet. Conversely, any set with two binary operations satisfying (i)–(iv) can be given a partial order making it a lattice: define $a \leq b$ iff $a \wedge b = a$. The two approaches — order-theoretic and algebraic — are equivalent.

Example (Power set lattice). For any set X, the power set $(\mathcal{P}(X), \subseteq)$ is a lattice: $A \vee B = A \cup B$ and $A \wedge B = A \cap B$. The algebraic laws of the Set Algebra section (commutativity, associativity, distributivity, absorption) are exactly the lattice laws in this example. This is not a coincidence: Boolean algebra is a special kind of lattice.

Example (Divisibility lattice). The divisibility order on $\mathbb{N}^+ = \{1, 2, 3, \ldots\}$ makes it a lattice: $a \vee b = \operatorname{lcm}(a, b)$ and $a \wedge b = \gcd(a, b)$, as we noted in Example 10.1.2.

Example (Not every poset is a lattice). Consider the poset $(\{a, b, c, d\}, \leq)$ with $a < c$, $a < d$, $b < c$, $b < d$, and a, b incomparable. The subset $\{a, b\}$ has two minimal upper bounds, c and d, which are incomparable. Hence $\sup\{a, b\}$ does not exist, and this poset is not a lattice.

The power set $\mathcal{P}(\{1, 2, 3\})$ restricted to subsets of cardinality $\neq 1$ provides a cleaner example: $\{1, 2\}$ and $\{1, 3\}$ have no meet in this restricted collection.

Definition (Complete Lattice)

Remark (Dependencies).

- Partial Order
- Subset
- Supremum and Infimum

Definition 10.1.24 (Complete Lattice)**.** A lattice $(L, \leq)$ is a *complete lattice* if every subset $S \subseteq L$ (including the empty set and L itself) has both a supremum and an infimum in L:

$$\forall S \subseteq L, \quad \sup S \in L \text{ and } \inf S \in L.$$

Remark (Standard quantified statement). $(L, \leq)$ is a complete lattice $\leftrightarrow$ $\forall S \subseteq L$, $\exists u^* \in L$, $\exists \ell^* \in L$ satisfying the sup and inf conditions of Definition (Supremum and Infimum).

Remark (The empty set forces top and bottom). When $S = \varnothing$, the sup condition says: u^* is an upper bound of the empty set (which is vacuously satisfied by every element of L), and u^*

is at most every other upper bound. This forces $u^* = \inf L$, the *bottom element* of L (often written $\perp$). Symmetrically, $\inf \varnothing = \sup L$, the *top element* (often written $\top$).

Concretely: in $(\mathcal{P}(X), \subseteq)$, we have $\sup \varnothing = \varnothing$ (the empty union) and $\inf \varnothing = X$ (the empty intersection equals the whole universe, by convention). Every complete lattice therefore has a top element $\top = \sup L$ and a bottom element $\perp = \inf L$.

Example ($\mathcal{P}(X)$ is a complete lattice). For any set X, the power set $(\mathcal{P}(X), \subseteq)$ is a complete lattice: for any family $\{A_\alpha\}_{\alpha \in I} \subseteq \mathcal{P}(X)$,

$$\sup_{\alpha \in I} A_\alpha = \bigcup_{\alpha \in I} A_\alpha, \qquad \inf_{\alpha \in I} A_\alpha = \bigcap_{\alpha \in I} A_\alpha.$$

The bottom element is $\varnothing$ and the top element is X. This is the canonical example of a complete lattice, and every σ-algebra on X is a complete sub-lattice of $(\mathcal{P}(X), \subseteq)$.

Example ($[0, \infty]$ is a complete lattice). The extended positive half-line $[0, \infty] = [0, \infty) \cup \{+\infty\}$ with the usual order is a complete lattice. Every nonempty subset has a supremum (which may be $+\infty$) and an infimum (which may be 0). This is the value space for measures: a measure assigns to each measurable set a value in $[0, \infty]$, and the key properties of measures (countable additivity, monotone convergence) are stated in terms of sups and infs in this complete lattice.

Definition (Distributive Lattice)

Remark (Dependencies).

- Lattice
- Subset
- Supremum and Infimum

Definition 10.1.25 (Distributive Lattice)**.** A lattice $(L, \leq)$ is *distributive* if for all $a, b, c \in L$:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Remark (Dependencies).

- Lattice

Remark (One law implies the other). By the duality principle for lattices, distributivity of $\wedge$ over $\vee$ is equivalent to distributivity of $\vee$ over $\wedge$: $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$. Proving one gives the other for free.

Remark (Why distributivity matters). The distributive law is what makes a lattice behave like set algebra. In the power set $\mathcal{P}(X)$, the distributive laws are exactly the set identities $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ from Theorem 3.22. Distributivity is the condition that separates lattices that "look like sets" from those that do not (e.g., the lattice of closed subspaces of a Hilbert space is not distributive).

Definition (Complement and Boolean Lattice)

Definition 10.1.26 (Complement and Boolean Lattice)**.** Let $(L, \leq)$ be a bounded distributive lattice with top element $\top$ and bottom element $\bot$. An element $a' \in L$ is a *complement* of $a \in L$ if

$$a \vee a' = \top \quad \text{and} \quad a \wedge a' = \bot.$$

A *Boolean lattice* (or *Boolean algebra*) is a bounded distributive lattice in which every element has a complement.

Remark (The power set is the canonical Boolean lattice). In $\mathcal{P}(X)$, the complement of a set A is $A^c = X \setminus A$, since $A \cup A^c = X = \top$ and $A \cap A^c = \varnothing = \bot$. The power set is therefore a complete Boolean lattice. Every σ-algebra on X is a Boolean sub-algebra of $\mathcal{P}(X)$: it is closed under complements (by definition) and under finite meets and joins (since every σ-algebra is an algebra).

Remark (σ-algebras as sub-lattices). A σ-algebra $\mathcal{M}$ on X is, from the lattice-theoretic viewpoint, a sub-structure of $(\mathcal{P}(X), \subseteq)$ that is:

- a complete sub-lattice (closed under all countable joins and meets);
- a Boolean sub-algebra (closed under complementation).

The condition "closed under countable joins" is exactly the σ-algebra axiom that countable unions of measurable sets are measurable. The condition "closed under meets" for countable families follows from De Morgan together with closure under countable unions and complements.

This perspective makes the definition of σ-algebra feel natural: it is exactly a complete Boolean sub-algebra of the power set lattice, where "complete" is interpreted countably (countable joins and meets), not set-theoretically.

Remark (Topologies as join-sub-structures). A topology τ on X is *not* a sub-lattice of $\mathcal{P}(X)$ in general, because it is not closed under arbitrary meets (i.e., arbitrary intersections of open sets need not be open). However, τ is closed under arbitrary joins (arbitrary unions of open sets are open) and finite meets (finite intersections of open sets are open). This makes τ an *inf-semilattice* for finite meets and a complete *join-semilattice*. The failure of closure under infinite meets is fundamental: it is precisely what allows open sets to "separate" points, and it is the reason topology is structurally different from measure theory.

Theorem (Knaster–Tarski Fixed-Point Theorem)

Let $(L, \leq)$ be a complete lattice and $f : L \to L$ a monotone function (i.e., $a \leq b \Rightarrow f(a) \leq f(b)$). Then f has a least fixed point and a greatest fixed point in L. Specifically:

$$\mathrm{lfp}(f) = \inf\{\, a \in L : f(a) \leq a \,\}, \qquad \mathrm{gfp}(f) = \sup\{\, a \in L : a \leq f(a) \,\}.$$

Remark (Standard quantified statement). $\forall$ complete lattices $(L, \leq)$, $\forall$ monotone $f : L \to L$: $\exists p, q \in L$ such that $f(p) = p$, $f(q) = q$, and $\forall r \in L$ with $f(r) = r$: $p \leq r \leq q$.

Remark (Why this theorem appears here). The Knaster–Tarski theorem is not needed for most of Volume I, but it is placed here because its hypotheses — complete lattice, monotone function — are now in place and the statement is short. Its significance reaches further:

- In set theory, it is used to prove the Schröder–Bernstein theorem without the Axiom of Choice (the monotone map sends $A \mapsto X \setminus g(Y \setminus f(A))$ on $\mathcal{P}(X)$).
- In measure theory, it is used to construct the smallest σ-algebra containing a given class of sets (the generated σ-algebra is a fixed point of the "closure under countable operations" operator).
- In theoretical computer science and domain theory, it is the foundation for denotational semantics.

For now, note that $(\mathcal{P}(X), \subseteq)$ is a complete lattice and any set operation that is monotone (such as $A \mapsto \overline{A}$, closure in a topological space, or $A \mapsto \sigma(A)$, generated σ-algebra) has a fixed point by this theorem.

Remark (Transition). With the notion of lattice in hand, the Hasse diagram becomes a more powerful visualisation tool: the Hasse diagram of a lattice has a unique bottom element ($\inf L$) and a unique top element ($\sup L$), and the join and meet of any two elements can be read off as the unique path structure in the diagram. The next section introduces Hasse diagrams, the duality principle, and the connection between duality and the passage from sup to inf.

Remark (Dependencies).

- Distributive Lattice
- Complete Lattice

Hasse Diagrams and Duality — Quick Reference

Concept	Meaning
Cover relation	$a \lessdot b$: $a < b$ with nothing strictly between
Hasse diagram	Graph encoding a poset; edges only for cover relations
Dual poset	$(A, \geq)$ obtained by reversing the order on $(A, \leq)$
Duality principle	Every theorem about posets yields a dual theorem for free

10.1.4 Hasse Diagrams and the Duality Principle

A partial order can be an enormous set of pairs, impossible to visualise directly. The *Hasse diagram* is a compression: it retains only the pairs that are "immediate" in the order, discarding those implied by transitivity and reflexivity. From the compressed diagram the full order can be recovered by taking transitive closure. Hasse diagrams are the standard visual language for finite posets; knowing how to read and draw them is essential for understanding the order examples in topology and lattice theory.

Definition (Cover Relation)

Definition 10.1.27 (Cover Relation)**.** Let $(A, \leq)$ be a poset and $a, b \in A$. We say b *covers* a, written $a \lessdot b$, if $a < b$ and there is no $c \in A$ with $a < c < b$.

Remark (Standard quantified statement)*.* $a \lessdot b \ \leftrightarrow \ (a < b) \ \wedge \ \neg(\exists c \in A, \ a < c \ \wedge \ c < b)$.

Remark (Intuition)*.* The cover relation strips the partial order down to its "essential edges." If $a \lessdot b$, then b is the *immediate successor* of a: you cannot insert anything between them. In a finite poset, every strict relation $a < b$ factors as a finite chain of cover steps: $a = a_0 \lessdot a_1 \lessdot \cdots \lessdot a_k = b$. In an infinite poset, this factorisation may fail (e.g., in $(\mathbb{R}, \leq)$, no two distinct real numbers are in a cover relation because there is always a rational strictly between them).

Definition (Hasse Diagram)

Remark (Dependencies)*.*

- Partial Order
- Strict Order

Definition 10.1.28 (Hasse Diagram)**.** The *Hasse diagram* of a finite poset $(A, \leq)$ is the directed graph whose vertices are the elements of A, with an upward edge from a to b whenever $a \lessdot b$. Three graphical conventions apply:

(i) Reflexive loops $(a \leq a)$ are suppressed.

(ii) Edges implied by transitivity are suppressed: if $a \lessdot b$ and $b \lessdot c$, no direct edge from a to c is drawn.

(iii) Direction is encoded by height: $a \lessdot b$ is drawn with b strictly above a, so arrowheads are unnecessary.

Remark (Dependencies)*.*

- Partial Order
- Cover Relation
- Finite and Infinite Sets

Remark (How to recover the full order)*.* The full order can be read back from the Hasse diagram: $a \leq b$ if and only if either $a = b$ or there is an upward path in the diagram from a to b (a sequence of cover edges going strictly upward). The diagram is therefore a lossless encoding of the partial order, compressed by removing the redundant reflexive and transitive edges.

Example (Divisibility on $\{1, 2, 3, 4, 6, 12\}$). Order the set $D = \{1, 2, 3, 4, 6, 12\}$ by divisibility ($a \le b$ iff $a \mid b$). The covers are:

$$1 \lessdot 2, \quad 1 \lessdot 3, \quad 2 \lessdot 4, \quad 2 \lessdot 6, \quad 3 \lessdot 6, \quad 4 \lessdot 12, \quad 6 \lessdot 12.$$

The Hasse diagram places 1 at the bottom and 12 at the top. The relation $1 \le 4$ holds in the poset but the edge $1 \to 4$ is *not* drawn because $1 \lessdot 2 \lessdot 4$ provides an upward path. Drawing the direct edge would be redundant.

Notice also: 4 and 6 are incomparable, so there is no edge between them — they sit at the same level with no path connecting them. The join $4 \vee 6 = 12 = \mathrm{lcm}(4, 6)$ and the meet $4 \wedge 6 = 2 = \gcd(4, 6)$ are readable from the diagram as the lowest common ancestor and the highest common descendant, respectively.

Example (Power set $\mathcal{P}(\{a, b, c\})$ under inclusion). The Hasse diagram of $(\mathcal{P}(\{a, b, c\}), \subseteq)$ has four levels: $\varnothing$ at the bottom; the three singletons $\{a\}, \{b\}, \{c\}$ one level up; the three two-element sets $\{a, b\}, \{a, c\}, \{b, c\}$ above those; and $\{a, b, c\}$ at the top. Each singleton is covered by the two two-element sets containing it. This diagram is the Boolean lattice B_3: it has a bottom ($\varnothing$), a top ($\{a, b, c\}$), and every element has a complement ($A' = \{a, b, c\} \setminus A$).

Remark (Limitation to finite posets). For infinite posets such as $(\mathbb{N}, \le)$, the cover relation is well-defined ($n \lessdot n+1$) but the full diagram cannot be drawn. In $(\mathbb{R}, \le)$ or $(\mathbb{Q}, \le)$, the cover relation is empty: no two elements are in a cover relation. Hasse diagrams are a practical visualisation tool for finite or schematic examples and should not be used as a proof device in infinite settings.

Definition (Dual Poset)

Definition 10.1.29 (Dual Poset)**.** Let $(A, \le)$ be a poset. The *dual poset* is $(A, \ge)$, where $a \ge b$ is defined to mean $b \le a$. The dual is also written $(A, \le^{\mathrm{op}})$.

Remark (Dependencies).

- Partial Order
- Relation

Remark (Intuition). The dual poset is obtained by flipping the Hasse diagram upside down. Every structural feature is preserved but reflected: elements at the top are now at the bottom, minimal elements become maximal, upper bounds become lower bounds, and suprema become infima and vice versa (as established in Proposition 10.1.22).

Proposition

Proposition 10.1.30 (Duality Principle for Posets)**.** *Let Φ be any first-order statement about a poset $(A, \le)$ expressed using only the relation $\le$. Let Φ^* be the statement obtained from Φ by replacing every occurrence of $\le$ with $\ge$ (equivalently, working in the dual poset). Then:*

$$(A, \le) \models \Phi \iff (A, \ge) \models \Phi.$$

In particular, if Φ is a theorem about all posets, then so is Φ^*.

Remark (Standard quantified statement). $\forall$ posets $(A, \leq)$, $\forall$ first-order sentences Φ in the language of $\leq$: $(A, \leq) \models \Phi \leftrightarrow (A, \geq) \models \Phi^*$, where Φ^* substitutes $\geq$ for every $\leq$.

Remark (Dependencies).

- Partial Order
- Relation

Remark (Theorems come in pairs at no extra cost). The duality principle means every proved theorem about posets automatically yields a second theorem — its dual — without any additional work. You will encounter this repeatedly in analysis. Here are the most important instances:

- *Supremum $\leftrightarrow$ Infimum.* Every theorem about sup yields a theorem about inf by duality. In particular, the completeness axiom for $\mathbb{R}$ (every nonempty bounded-above subset has a supremum) immediately implies that every nonempty bounded-below subset has an infimum — not a separate axiom but a consequence by duality.
- *Minimal $\leftrightarrow$ Maximal.* A statement about minimal elements dualises to a statement about maximal elements. Zorn's Lemma (every poset in which every chain is bounded above has a maximal element) dualises to: every poset in which every chain is bounded below has a minimal element.
- *Well-ordering is not self-dual.* $(\mathbb{N}, \leq)$ is well-ordered (every nonempty subset has a least element); its dual $(\mathbb{N}, \geq)$ is not well-ordered because the whole set has no greatest element. Duality preserves the form of a definition but not whether a particular structure satisfies it.
- *Join $\leftrightarrow$ Meet in lattices.* Every lattice identity involving $\vee$ and $\wedge$ dualises by swapping the two operations. The distributive law $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ dualises to $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$: proving one gives the other for free.

Remark (Transition). The order vocabulary developed across the order sections — partial orders, total orders, well-orders, bounds, suprema, infima, lattices, directed sets, and duality — is the complete toolkit for order-theoretic reasoning in analysis. In Volume II, the construction of $\mathbb{R}$ requires only the completeness axiom (existence of suprema) and the Dedekind cut or Cauchy sequence constructions, both of which use only the vocabulary established here. In Volume III, metric space arguments, the definition of compactness, and the construction of Lebesgue measure all draw on this vocabulary repeatedly.

Order Extensions — Quick Reference

Concept	Meaning
Preorder	Reflexive, transitive (not necessarily antisymmetric)
Loset / toset	Synonym for totally ordered set (total = linear order)
Symmetric part	$xIy \iff xRy \wedge yRx$
Asymmetric part	$xPy \iff xRy \wedge \neg(yRx)$
Transitive closure	Smallest transitive relation containing R
Extension	$\succsim'$ agrees with $\succsim$ on strict and weak ranks
OWC	$xT(\succsim)y \Rightarrow \neg(y \succ x)$
Szpilrajn's Thm	Every partial order extends to a linear order

10.1.5 Order Extensions

Definition (Symmetric and Asymmetric Parts)

Definition 10.1.31 (Symmetric and Asymmetric Parts)**.** For any binary relation R on X, define:

- the *symmetric part*: $xIy \iff xRy$ and yRx
- the *asymmetric part*: $xPy \iff xRy$ but not yRx

Note that $R = P \cup I$.

Remark (Dependencies).

- Relation
- Union
- Set Difference

Remark (Preference-theoretic reading). In decision theory, R encodes a preference relation: xRy means "x is at least as good as y." Then I is the *indifference* relation (equally good) and P is the *strict preference* relation (strictly better). This decomposition is fundamental to utility representation theory.

Remark (Preorder and loset are defined elsewhere). The order relations this section uses are defined in the Relations chapter: a preorder is reflexive and transitive, and a linear order is a total (connex) partial order. The corresponding *structure* — the loset, i.e. the linearly/totally ordered set — is catalogued with the order structures in the Ordered Sets section. Nothing is restated here, so no definition can drift.

Remark (Preorder vs. partial order). A preorder need not be antisymmetric: two distinct elements may satisfy xRy and yRx simultaneously (they are "indifferent" but not equal). Adding antisymmetry promotes a preorder to a partial order. Every partial order is a preorder, but not conversely.

Definition (Maximal Elements and Upper Bounds)

Definition 10.1.32 (Maximal Elements and Upper Bounds)**.** Let $\succsim$ be a binary relation on X.

- The set of *maximal elements* of X is

$$\mathrm{Max}(X, \succsim) = \{x \in X \mid y \succsim x \text{ for no } y \in X \text{ with } y \neq x\}.$$

- The set of *upper bounds* of X is

$$M(X, \succsim) = \{x \in X \mid x \succsim y \text{ for all } y \in X\}.$$

Remark (Dependencies).

- Relation
- Upper and Lower Bounds
- Strict Order

Remark (Maximal vs. greatest)*.* A maximal element has nothing strictly above it; an upper bound (greatest element) is above everything. In a partial order, the greatest element is unique if it exists and is always maximal, but maximal elements need not be greatest. In a linear order, maximal and greatest coincide.

Remark (Transitive closure (defined in the Relations chapter))*.* The *transitive closure* $T(R)$ of a binary relation R on X — the smallest transitive relation containing R — is defined in the Relations chapter. We recall it here only to use it, and do not restate the definition, so the two chapters cannot drift apart. (For the explicit construction, see the description below.)

Remark (Dependencies).

- Transitive Closure (Relations)
- Transitive Relation

Remark (Constructive description)*.* Define $R^0 = R$ and $xR^m y$ if there exist $z_1, \ldots, z_m \in X$ such that $xRz_1R\cdots Rz_mRy$. Then

$$T(R) = R \cup \bigcup_{m \in \mathbb{N}} R^m.$$

Existence follows from the fact that the intersection of any collection of transitive relations is transitive, so the intersection of all transitive supersets of R is the smallest such relation.

Definition (Extension of a Preorder)

Definition 10.1.33 (Extension of a Preorder)**.** Let $\succsim$ be a preorder on X. A preorder $\succsim'$ is an *extension* of $\succsim$ if

$$x \succsim y \;\Rightarrow\; x \succsim' y, \qquad x \succ y \;\Rightarrow\; x \succ' y.$$

A *complete extension* is an extension that is also complete.

Remark (Dependencies).

- Preorder
- Relation

Remark (What extensions do). An extension of $\succsim$ "fills in" the comparisons left undecided by $\succsim$ without reversing any existing strict comparison. The goal is to promote a partial ranking to a total ranking while respecting the original judgements.

Theorem (Szpilrajn, 1930)

For any nonempty set X and partial order $\succsim$ on X, there exists a linear order that is an extension of $\succsim$.

Remark (Proof sketch via Hausdorff Maximum Principle). Let T_X be the set of all partial orders on X extending $\succsim$, ordered by inclusion. By the Hausdorff Maximum Principle (equivalent to AC), T_X has a maximal chain; its union $\succsim^*$ is a partial order extending $\succsim$. If $\succsim^*$ were not complete, one could enlarge it via transitive closure, contradicting maximality. Hence $\succsim^*$ is a linear order extending $\succsim$.

Dependence on AC. The theorem is equivalent to the Axiom of Choice for infinite sets; no proof avoiding AC is known.

Corollary (Complete preorder extension)

Corollary 10.1.34 (Complete preorder extension)**.** *For any nonempty set X and preorder $\succsim$ on X, there exists a complete preorder that is an extension of $\succsim$.*

Remark (Proof). Proof details.

Remark (Standard quantified statement). $\forall X \neq \varnothing$, $\forall \succsim$ preorder on X: $\exists \succsim'$ complete preorder on X s.t. $(x \succsim y \to x \succsim' y) \wedge (x \succ y \to x \succ' y)$.

Remark (Dependencies).

- Preorder
- Extension of a Preorder
- Quotient Set

- Symmetric and Asymmetric Parts

Remark (Proof). Pass to the quotient $X/\sim$ (where $\sim$ is the symmetric part), apply Szpilrajn's Theorem to obtain a linear order on $X/\sim$, then pull back to a complete preorder on X.

Definition and Proposition (Only Weak Cycles)

Definition 10.1.35 (and Proposition (Only Weak Cycles))**.** Let $\succsim$ be a binary relation on X with asymmetric part $\succ$ and transitive closure $T(\succsim)$. We say $\succsim$ satisfies *only weak cycles* (OWC) if

$$xT(\succsim)y \;\Rightarrow\; \neg(y \succ x).$$

Proposition. A binary relation $\succsim$ on a nonempty set X can be extended to a complete preorder if and only if it satisfies OWC.

Remark (Dependencies).

- Relation
- Symmetric and Asymmetric Parts
- Transitive Closure
- Extension of a Preorder

Remark (Intuition for OWC). OWC prohibits the following: there is a "chain" of weak preferences $x_1 \succsim x_2 \succsim \cdots \succsim x_n$ but a strict reversal $x_n \succ x_1$. Such a configuration cannot be extended to a complete preorder because the chain forces $x_1 \succsim' x_n$ in any extension, but strict preference $x_n \succ x_1$ in the extension would mean $x_n \succsim' x_1$ but not $x_1 \succsim' x_n$ — contradicting $x_1 \succsim' x_n$.

Remark (Transition to AC). Szpilrajn's Theorem relies on Zorn's Lemma / Hausdorff Maximum Principle, which are equivalent to the Axiom of Choice. The next section develops these equivalences directly.

Induced Orders and Order Embeddings — Quick Reference

Concept	**Meaning**
Induced preorder	$x \leq_f y \;\Leftrightarrow\; f(x) \leq' f(y)$; always a preorder
Induced partial order	Induced order is a partial order iff f is injective
f-indistinguishable	$x \sim_f y$: both $x \leq_f y$ and $y \leq_f x$
Order embedding	Injective, order-preserving, and order-reflecting
Embedding vs. isomorphism	Embedding is iso onto image; isomorphism is surjective too
Suborder	Restriction of $\leq'$ to a subset $S \subseteq B$

Key results:

$\leq_f$ preorder: always. Antisymmetry: iff f injective.
f order embedding $\Rightarrow (A, \leq_f) \cong (f(A), \leq')$.

10.1.6 Induced Orders and Order Embeddings

The simplest way to put an order on a set A is to compare elements of A *indirectly* through a function into an already-ordered set. This construction, called the *induced order* or *pullback order*, is ubiquitous: it explains how the usual order on $\mathbb{Z}$ restricts to $\mathbb{N}$, how functions on a common domain acquire a pointwise order, and how subsets inherit the order of the ambient space.

Definition (Induced Order)

Definition 10.1.36 (Induced Order)**.** Let $(B, \leq')$ be a partially ordered set and let $f : A \to B$ be a function. The *order induced by* f (or *pullback order along* f) is the relation $\leq_f$ on A defined by:

$$x \ \leq_f \ y \iff f(x) \ \leq' \ f(y).$$

Remark (Intuition)*.* Two elements of A are compared by sending them through f and comparing their images in $(B, \leq')$. The order on A is entirely inherited from B via f: we never compare elements of A directly, only their f-values.

Remark (Standard quantified statement)*.* $\forall x, y \in A, \quad x \leq_f y \iff f(x) \leq' f(y)$.

Remark (Dependencies)*.*

- Partial Order
- Function
- Relation

Proposition

Proposition 10.1.37 ($\leq_f$ is always a preorder)**.** *For any function $f : A \to B$ and partial order $(B, \leq')$, the relation $\leq_f$ is a preorder on A: it is reflexive and transitive.*

Remark (Standard quantified statement)*.* $\forall x \in A : \ f(x) \leq' f(x)$ (reflexivity); $\forall x, y, z \in A :$ $f(x) \leq' f(y) \wedge f(y) \leq' f(z) \to f(x) \leq' f(z)$ (transitivity).

Remark (Dependencies)*.*

- Induced Order
- Preorder
- Function
- Partial Order

Remark (Proof strategy)*.* Both properties are inherited directly from $(B, \leq')$ by pulling back through f: reflexivity at x uses reflexivity at $f(x)$, and transitivity along x, y, z uses transitivity along $f(x), f(y), f(z)$.

Definition (f-Indistinguishable Elements)

Definition 10.1.38 (f-Indistinguishable Elements)**.** Let $f : A \to B$ and $\leq_f$ be the induced order. Two elements $x, y \in A$ are *f-indistinguishable*, written $x \sim_f y$, if both $x \leq_f y$ and $y \leq_f x$, i.e. if

$$f(x) \leq' f(y) \quad \text{and} \quad f(y) \leq' f(x).$$

Remark (Dependencies).

- Induced Order
- Function
- Partial Order

Remark (Intuition). Two elements are f-indistinguishable when they map to the same position in the $\leq'$-order — not necessarily to the same *point*, but to mutually comparable points. When $\leq'$ is a partial order (antisymmetric), $x \sim_f y$ forces $f(x) = f(y)$, making f non-injective the only obstruction to antisymmetry.

Proposition

Proposition 10.1.39 ($\leq_f$ is a partial order iff f is injective)**.** *Let $(B, \leq')$ be a partially ordered set and $f : A \to B$. The induced order $\leq_f$ is a partial order on A if and only if f is injective.*

Remark (Standard quantified statement). $\leq_f$ is a partial order $\leftrightarrow$ $\forall x, y \in A,\ f(x) \leq' f(y) \wedge f(y) \leq' f(x) \to x = y$ $\leftrightarrow$ $\forall x, y \in A,\ f(x) = f(y) \to x = y$ (injectivity of f, since $\leq'$ is antisymmetric).

Remark (Dependencies).

- Induced Order
- Partial Order
- Injective Function

Remark (Common error). It is tempting to think the induced order is automatically a partial order "because $\leq'$ is one." It is not. The issue is antisymmetry: if f collapses two distinct points $x \neq y$ to the same or comparable images, we get $x \leq_f y$ and $y \leq_f x$ with $x \neq y$, violating antisymmetry. Reflexivity and transitivity lift freely; antisymmetry does not.

Example (Non-injective f destroys antisymmetry). Let $B = (\mathbb{N}, \leq)$ and let $f : \{a, b\} \to \mathbb{N}$ be the constant function $f(a) = f(b) = 0$. Then $a \leq_f b$ (since $0 \leq 0$) and $b \leq_f a$ (since $0 \leq 0$), but $a \neq b$. So $\leq_f$ is not antisymmetric, confirming that non-injectivity forces a failure.

Example (Injective f gives a genuine partial order). Let $B = (\mathcal{P}(\{1, 2, 3\}), \subseteq)$ and let $A = \{x, y, z\}$ with $f(x) = \{1\}$, $f(y) = \{2\}$, $f(z) = \{1, 2\}$. Then f is injective. The induced order has $x \leq_f z$ (since $\{1\} \subseteq \{1, 2\}$) and $y \leq_f z$ (since $\{2\} \subseteq \{1, 2\}$), but x and y are incomparable. This is a genuine partial order on A.

Definition (Suborder / Restriction)

Definition 10.1.40 (Suborder / Restriction)**.** Let $(B, \leq')$ be a partially ordered set and $S \subseteq B$. The *suborder* on S (or *induced order on* S) is the relation $\leq'_S$ defined by:

$$x \ \leq'_S \ y \iff x \leq' y, \quad \text{for } x, y \in S.$$

Remark (Dependencies).

- Partial Order
- Subset
- Relation

Remark (Connection to induced orders). The suborder on $S \subseteq B$ is exactly the order induced by the inclusion map $\iota : S \hookrightarrow B$, $\iota(x) = x$. Since ι is injective, the suborder is always a partial order whenever $(B, \leq')$ is. This justifies speaking of "S with the inherited order" without further verification.

Definition (Order Embedding)

Definition 10.1.41 (Order Embedding)**.** Let $(A, \leq)$ and $(B, \leq')$ be partially ordered sets. A function $f : A \to B$ is an *order embedding* if for all $x, y \in A$:

$$x \ \leq \ y \iff f(x) \ \leq' \ f(y).$$

Remark (Dependencies).

- Ordered Set
- Function

Remark (Two directions, one condition). The ($\Rightarrow$) direction says f is order-preserving (see Definition (Order-Preserving Map)). The ($\Leftarrow$) direction — $f(x) \leq' f(y) \Rightarrow x \leq y$ — is called *order-reflection*. An order embedding both preserves and reflects the order, so f is a perfect local copy of $(A, \leq)$ inside $(B, \leq')$.

Remark (Relation to the induced order). If $f : A \to B$ is an order embedding from $(A, \leq)$ to $(B, \leq')$, then $\leq$ coincides with the induced order $\leq_f$:

$$x \leq y \iff f(x) \leq' f(y) \iff x \leq_f y.$$

In other words, an order embedding is exactly an injective function whose induced order *agrees with the existing order on* A.

Proposition (Order embeddings are injective)

Proposition 10.1.42 (Order embeddings are injective)**.** *Every order embedding* $f : (A, \leq) \to (B, \leq')$ *is injective. Go to proof.*

Remark (Standard quantified statement). $\forall x, y \in A : \ f(x) = f(y) \to x = y.$

Remark (Dependencies).

- Order Embedding
- Injective Function
- Partial Order

Remark (Proof strategy). Injectivity falls out of the reflection direction: if two elements have the same image, reflection forces them to be mutually $\leq$-related, and antisymmetry collapses them to equality.

Proposition (Order embedding is isomorphism onto image)

Proposition 10.1.43 (Order embedding is isomorphism onto image)**.** *If $f : (A, \leq) \to (B, \leq')$ is an order embedding, then f is an order isomorphism from $(A, \leq)$ to the suborder $(f(A), \leq'_{f(A)})$. Go to proof.*

Remark (Standard quantified statement). $\forall x, y \in A : \ x \leq y \ \leftrightarrow \ f(x) \leq'_{f(A)} f(y)$, i.e. f is an order isomorphism $(A, \leq) \cong (f(A), \leq')$.

Remark (Dependencies).

- Order Embedding
- Image of a Set
- Suborder
- Partial Order

Remark (Consequence). This proposition is the order-theoretic analogue of the first isomorphism theorem: an order embedding always realises $(A, \leq)$ as an isomorphic copy of a subposet of $(B, \leq')$. When f is also surjective, the embedding becomes a full order isomorphism (see Definition (Order Isomorphism)).

Example (Standard embeddings). Each of the following is an order embedding:

(i) The inclusion $\mathbb{N} \hookrightarrow \mathbb{Z}$ with standard $\leq$ on both: $m \leq n \iff m \leq n$ trivially. The natural numbers sit inside the integers as an isomorphic copy of $(\mathbb{N}, \leq)$.

(ii) The function $f : \mathbb{N} \to \mathbb{N}$, $f(n) = 2n$, embeds $(\mathbb{N}, \leq)$ into itself: $m \leq n \iff 2m \leq 2n$. The image is the even numbers with the inherited order.

(iii) The map $A \mapsto A$ from $(\mathcal{P}(X), \subseteq)$ to itself is an order isomorphism (the identity), but any injective $g : \mathcal{P}(X) \to \mathcal{P}(X)$ that preserves and reflects $\subseteq$ is an order embedding.

Remark (Transition). Induced orders and embeddings are the order-theoretic counterpart of substructures in algebra. Just as a subgroup is a subset closed under the group operations, a sub-poset is a subset with the inherited order — and an order embedding is the morphism that witnesses this. In analysis, these ideas appear when restricting an order from $\mathbb{R}$ to subsets (intervals, sequences, function spaces), and when the completeness property is transferred between ordered structures.

10.2 Zorn

AC Equivalences — Quick Reference

Statement	**Role**
Axiom of Choice (AC)	Choice function from any collection of nonempty sets
Product form of AC	Cartesian product of nonempty sets is nonempty
Zorn's Lemma	Poset with every chain bounded $\Rightarrow$ maximal element
Hausdorff Maximum Principle	Every poset has a $\subseteq$-maximal chain
Well-Ordering Theorem	Every set admits a well-ordering

All five statements are equivalent (over ZF set theory).

10.2.1 Zorn's Lemma and the Axiom of Choice

Remark (Why AC is non-constructive). For finite collections, the axiom is provable without extra assumptions (see SRF-STO-C08-S8.1). For infinite collections, no construction can produce the choice function: AC asserts its existence without exhibiting it. This non-constructive character is both AC's power and the source of the foundational controversy surrounding it.

Zorn's Lemma

Let $(P, \leq)$ be a poset in which every chain (loset) has an upper bound in P. Then P has at least one maximal element.

Remark (Reading Zorn's Lemma). The hypothesis "every chain has an upper bound" does *not* require the bound to lie in the chain itself. The conclusion gives a maximal element: an element above which nothing in P sits. In a partial order, there may be many maximal elements; Zorn guarantees at least one.

Remark (Typical proof pattern using Zorn's Lemma). To show an object of type X exists with a maximal property:

1. Partially order candidate objects by inclusion (or extension).
2. Verify every chain of candidates has an upper bound (usually its union).
3. Conclude by Zorn that a maximal candidate exists.

4. Show the maximal candidate has the desired property (often: if it lacked it, it could be enlarged, contradicting maximality).

This pattern recurs throughout algebra and analysis: maximal ideals, algebraic bases (Hamel bases), maximal consistent sets, and Szpilrajn's Theorem.

Hausdorff Maximum Principle

In every poset $(P, \leq)$ there exists a $\subseteq$-maximal chain; that is, a chain $C \subseteq P$ such that no chain $C' \subseteq P$ strictly contains C.

Remark (Relation to Zorn). The Hausdorff Maximum Principle is an immediate corollary of Zorn's Lemma: take the poset of chains ordered by inclusion; every chain of chains is bounded by its union; Zorn gives a maximal chain. Conversely, Hausdorff implies Zorn. Both are equivalent to AC over ZF.

Remark (Application: Szpilrajn via Hausdorff). In the proof of Szpilrajn's Theorem (↑ Thm), let T_X be the set of all partial orders on X extending $\succsim$, ordered by $\subseteq$. By the Hausdorff Maximum Principle, T_X has a maximal chain A; its union $\succsim^* = \bigcup A$ is a partial order extending $\succsim$. If $\succsim^*$ were not total, there would exist incomparable x, y and one could extend $\succsim^*$ by adding (x, y) (closing under transitivity), contradicting maximality of A. Hence $\succsim^*$ is a linear order. $\square$

Remark (Transition). Zorn's Lemma is the workhorse for existence proofs throughout abstract algebra and analysis. It first appears in the project context here and at the proof stubs in §11 of the Chapter VIII exercises.

Chapter 11

Cardinality

SERIES	VOLUME I	BOOK II	CHAPTER 9
From Cantor to Ito	Logic, Sets, and Proof	Set Theory	Cardinality

11.1 Cardinality

Cardinality — Quick Reference

Concept	Meaning
Same cardinality	Bijection $f : A \to B$ exists; written $A \sim B$
Finite set	$A \sim \{1, \dots, n\}$ for some $n \in \mathbb{N}$, or $A = \varnothing$
Countably infinite	$A \sim \mathbb{N}$
Countable	Finite or countably infinite
Uncountable	Infinite but not countable
$\lvert A\rvert \le \lvert B\rvert$	Injection $A \hookrightarrow B$ exists
$\mathbb{Q}$ countable	Diagonal enumeration
$\mathbb{R}$ uncountable	Cantor diagonal argument
Schröder–Bernstein	$\lvert A\rvert \le \lvert B\rvert$ and $\lvert B\rvert \le \lvert A\rvert \Rightarrow A \sim B$
Cantor's theorem	$\lvert A\rvert < \lvert \mathcal{P}(A)\rvert$ for every set A
Countable union	Countable union of countable sets is countable

11.1.1 Cardinality and Infinite Sets

The intuitive idea of "size" works perfectly for finite sets: $\{a, b, c\}$ has three elements, $\{1, 2, 3, 4\}$ has four, and any two finite sets can be compared by counting. For infinite sets, counting breaks down, and a different foundation is needed. The key insight, due to Cantor, is that two sets have the "same size" if and only if their elements can be matched perfectly — one-to-one and onto — without any element left over on either side. This single idea, made precise through the notion of a bijection, gives a theory of size that applies uniformly to both finite and infinite sets, and produces genuinely surprising results: the integers and the rationals turn out to have the same size as the natural numbers, while the real numbers are provably larger.

Definition (Same Cardinality)

Definition 11.1.1 (Same Cardinality)**.** Two sets A and B have the *same cardinality*, written $A \sim B$, if there exists a bijection $f : A \to B$.

Remark (Standard quantified statement). $A \sim B \;\leftrightarrow\; \exists f : A \to B,\ (\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2) \wedge (\forall b \in B,\ \exists a \in A,\ f(a) = b)$.

Remark (Dependencies).

- Bijective Function

Remark (Same cardinality is an equivalence relation). The relation $\sim$ is an equivalence relation on the class of all sets:

(i) *Reflexive*: $A \sim A$ via id_A.

(ii) *Symmetric*: if $f : A \to B$ is a bijection then $f^{-1} : B \to A$ is a bijection, so $A \sim B$ implies $B \sim A$.

(iii) *Transitive*: if $f : A \to B$ and $g : B \to C$ are bijections then $g \circ f : A \to C$ is a bijection, so $A \sim B$ and $B \sim C$ imply $A \sim C$.

This is a genuine equivalence relation, so all the theory of equivalence classes developed in the previous section applies. The equivalence class of A under $\sim$ is called the *cardinality* of A, written $|A|$.

Example (Infinite sets can be equinumerous with proper subsets). Let $E = \{2, 4, 6, \dots\}$ be the set of positive even integers. The function $f : \mathbb{N} \to E$ defined by $f(n) = 2n$ is a bijection, so $\mathbb{N} \sim E$. This seems paradoxical: E is a proper subset of $\mathbb{N}$, yet the two sets have the same cardinality. This is not a contradiction but rather one of the defining characteristics of infinite sets — a phenomenon that simply cannot occur for finite sets.

Remark (Dedekind's characterization of infinite sets). The phenomenon in Example 11.1.1 is not an accident. An infinite set can always be matched bijectively with some proper subset of itself. This is sometimes taken as the *definition* of an infinite set (Dedekind's definition): a set is infinite if and only if it is equinumerous with a proper subset of itself.

Definition (Finite and Infinite Sets)

Definition 11.1.2 (Finite and Infinite Sets)**.** A set A is *finite* if $A = \varnothing$ or $A \sim \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. A set is *infinite* if it is not finite.

Remark (Standard quantified statement). A is finite $\;\leftrightarrow\;$ $A = \varnothing \vee \exists n \in \mathbb{N},\ \exists f : A \to \{1, \dots, n\},\ f$ bijective. A is infinite $\leftrightarrow \neg(A$ is finite$)$.

Remark (Dependencies).

- Empty Set
- Same Cardinality
- Natural Numbers

Definition (Countable and Uncountable Sets)

Definition 11.1.3 (Countable and Uncountable Sets)**.** A set A is *countably infinite* if $A \sim \mathbb{N}$. A set is *countable* if it is finite or countably infinite. A set is *uncountable* if it is infinite but not countable.

Remark (Standard quantified statement)*.* A is countably infinite $\leftrightarrow$ $\exists f : A \to \mathbb{N},\ f$ bijective. A is countable $\leftrightarrow$ A finite $\vee\ A \sim \mathbb{N}$. A is uncountable $\leftrightarrow$ A infinite $\wedge \neg(A \sim \mathbb{N})$.

Remark (Dependencies)*.*

- Finite and Infinite Sets
- Same Cardinality
- Natural Numbers

Remark (Countable means listable)*.* A set A is countably infinite if and only if its elements can be arranged into an infinite list $a_1, a_2, a_3, \ldots$ with no repetitions and every element of A appearing somewhere in the list. The bijection $f : \mathbb{N} \to A$ is precisely such a listing: $a_n = f(n)$. This is why countability is often informally described as "can be listed."

Definition (Comparing Cardinalities)

Definition 11.1.4 (Comparing Cardinalities)**.** Write $|A| \le |B|$ if there exists an injection $f : A \hookrightarrow B$. Write $|A| < |B|$ if $|A| \le |B|$ but $A \not\sim B$.

Remark (Standard quantified statement)*.* $|A| \le |B| \ \leftrightarrow\ \exists f : A \to B,\ \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \to a_1 = a_2$ (injection exists).

Remark (Dependencies)*.*

- Injective Function
- Same Cardinality

Remark (Why injections, not surjections)*.* An injection $A \hookrightarrow B$ places a "copy" of A inside B without overlap, which formalizes "A is no larger than B." The direction matters: an injection from A to B is not the same as an injection from B to A.

Theorem ($\mathbb{Q}$ is countable)

Theorem 11.1.5 ($\mathbb{Q}$ is countable)**.** *The set $\mathbb{Q}$ of rational numbers is countable.* *Go to proof.*

Remark (Dependencies)*.*

- Countable Set
- Natural Numbers
- Bijective Function

Remark (Proof strategy — diagonal enumeration). The argument arranges all positive rationals into an infinite two-dimensional array: place p/q (in lowest terms, $p, q > 0$) in row p, column q. Then traverse the array along successive diagonals — the anti-diagonal $p+q=k$ for $k = 2, 3, 4, \ldots$ — skipping any fraction already seen. This produces an explicit list of all positive rationals, and one then interleaves the negative rationals and zero. The precise details are left as a guided proof exercise. The core idea is: a doubly infinite list can be converted into a singly infinite list by reading diagonals.

Remark (What this means). $\mathbb{Q}$ is dense in $\mathbb{R}$ — between any two real numbers there is a rational — yet $\mathbb{Q}$ is countable. This means the rationals are in some precise sense "thin" even though they are everywhere. The uncountability of $\mathbb{R}$ that follows shows the reals are genuinely larger.

Theorem (Countable union of countable sets is countable)

Theorem 11.1.6 (Countable union of countable sets is countable)**.** *Let $\{A_n\}_{n\in\mathbb{N}}$ be a countable family of countable sets. Then $\bigcup_{n=1}^{\infty} A_n$ is countable.* *Go to proof.*

Remark (Dependencies).

- Countable Set
- Natural Numbers
- Indexed Family of Sets
- Union

Remark (Proof strategy). The argument again uses a diagonal enumeration. Arrange the elements of each A_n as a row in an infinite array (padding with repetitions if A_n is finite), then read the array along anti-diagonals. This produces a surjection from $\mathbb{N}$ onto $\bigcup A_n$, which is enough to establish countability. The formal proof is a standard exercise.

Remark (Why this theorem matters downstream). This result is used constantly. In topology: a countable union of nowhere-dense sets is called a *meager* set, and Baire's theorem says a complete metric space is not meager — the proof requires knowing that countable unions behave predictably. In measure theory: a countable union of sets of measure zero has measure zero, by countable additivity. Whenever an argument says "take the union over all $n \in \mathbb{N}$" and needs the result to remain manageable in size, this theorem is the justification.

Theorem ($\mathbb{R}$ is uncountable)

Theorem 11.1.7 ($\mathbb{R}$ is uncountable)**.** *The set $\mathbb{R}$ of real numbers is not countable.* *Go to proof.*

Remark (Dependencies).

- Countable Set
- Finite and Infinite Sets
- Countable Union of Countable Sets

Remark (Proof strategy — Cantor's diagonal argument). This is one of the most important proof techniques in all of mathematics. Suppose, for contradiction, that $\mathbb{R}$ were countable, so its elements could be listed: $r_1, r_2, r_3, \ldots$. Write each r_n in decimal expansion. Construct a new real number x by choosing the n-th decimal digit of x to differ from the n-th decimal digit of r_n (and avoiding 9 to prevent the $0.999\cdots = 1$ ambiguity). Then $x \neq r_n$ for every n, because x and r_n differ in the n-th decimal place. So x is not on the list — contradicting the assumption that the list contains every real number.

The key move is the diagonal construction: to defeat every row r_n simultaneously, make x disagree with row n at position n. This "diagonal sweep" is a technique that reappears in logic (Gödel's incompleteness theorem), computability theory (the halting problem), and the proof of Cantor's theorem below.

Remark (Consequence). Since $\mathbb{Q}$ is countable and $\mathbb{R}$ is uncountable, the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ must be uncountable: if they were countable, then $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be a countable union of countable sets, hence countable — contradiction.

Theorem (Cantor's Theorem)

Theorem 11.1.8 (Cantor's Theorem)**.** *For any set A, there is no surjection $A \to \mathcal{P}(A)$. In particular, $|A| < |\mathcal{P}(A)|$.* *Go to proof.*

Remark (Standard quantified statement). $\forall A : \neg\exists f : A \to \mathcal{P}(A),\ \forall S \in \mathcal{P}(A),\ \exists a \in A,\ f(a) = S$. Equivalently: $\forall f : A \to \mathcal{P}(A),\ \exists D \in \mathcal{P}(A),\ \forall a \in A,\ f(a) \neq D$.

Remark (Dependencies).

- Function
- Surjective Function
- Power Set
- Axiom Schema of Separation

Remark (Proof strategy — the same diagonal idea). Suppose $f : A \to \mathcal{P}(A)$ is any function. Construct the set

$$D = \{\, a \in A \mid a \notin f(a) \,\}.$$

D is a well-defined subset of A, so $D \in \mathcal{P}(A)$. Claim: D is not in the image of f. For suppose $D = f(a_0)$ for some $a_0 \in A$. Then:

- if $a_0 \in D$, then by definition $a_0 \notin f(a_0) = D$ — contradiction.
- if $a_0 \notin D$, then by definition $a_0 \in f(a_0) = D$ — contradiction.

Either case is impossible, so $D \neq f(a_0)$ for every $a_0 \in A$, and f is not surjective. Since f was arbitrary, no surjection $A \to \mathcal{P}(A)$ exists.

Notice the structure: D is defined by "$a \in D$ iff $a \notin f(a)$," which is a diagonal construction — the membership of a in D is decided by looking at position (a, a) in the "membership table" of f, and then disagreeing. This is the abstract form of the same move used in the proof that $\mathbb{R}$ is uncountable.

Remark (A hierarchy of infinities). Cantor's theorem implies there is no largest cardinality: for any set A, the power set $\mathcal{P}(A)$ is strictly larger. Starting from $\mathbb{N}$, the sets

$$\mathbb{N}, \quad \mathcal{P}(\mathbb{N}), \quad \mathcal{P}(\mathcal{P}(\mathbb{N})), \quad \dots$$

form a strictly increasing sequence of cardinalities. One can show $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$ via the characteristic function bijection (described below), so the second level is already the cardinality of the continuum.

Definition (Function Space B^A)

Definition 11.1.9 (Function Space B^A)**.** For sets A and B, the *function space* B^A denotes the set of all functions from A to B:

$$B^A \;:=\; \{\, f : A \to B \,\}.$$

Remark (Dependencies).

- Function
- Set and Membership

Remark (Why the exponential notation). For finite sets with $|A| = m$ and $|B| = n$, the number of functions $A \to B$ is n^m (there are n choices for each of the m inputs). The notation B^A reflects this count. In particular, $|\{0,1\}^A| = 2^{|A|}$ for finite A, which matches $|\mathcal{P}(A)| = 2^{|A|}$.

Remark (The characteristic function bijection). For any set A, the power set $\mathcal{P}(A)$ and the function space $\{0,1\}^A$ have the same cardinality via the *characteristic function* bijection:

$$\mathcal{P}(A) \;\to\; \{0,1\}^A, \qquad S \;\mapsto\; \mathbf{1}_S,$$

where $\mathbf{1}_S(a) = 1$ if $a \in S$ and $\mathbf{1}_S(a) = 0$ if $a \notin S$. This bijection is why Cantor's theorem is sometimes stated as: there is no surjection $A \to \{0,1\}^A$. The notation 2^A is used interchangeably with $\{0,1\}^A$, and $|\mathcal{P}(A)| = 2^{|A|}$ is the general statement.

In functional analysis, spaces of the form B^A — functions from A to B — are the central objects of study. Knowing they are sets (with a definite cardinality when A and B are concrete) is the set-theoretic foundation for everything that follows.

Theorem (Schröder–Bernstein Theorem)

Theorem 11.1.10 (Schröder–Bernstein Theorem)**.** *If* $|A| \leq |B|$ *and* $|B| \leq |A|$*, then* $A \sim B$*.*
Equivalently: if there exist injections $f : A \hookrightarrow B$ *and* $g : B \hookrightarrow A$*, then there exists a bijection* $h : A \to B$*. Go to proof.*

Remark (Standard quantified statement)*.* $\big(\exists f : A \hookrightarrow B\big) \wedge \big(\exists g : B \hookrightarrow A\big) \;\to\; \exists h : A \to B,\ h$ bijective.

Remark (Dependencies)*.*

- Comparing Cardinalities
- Injective Function
- Bijective Function

Remark (Why this theorem is indispensable)*.* In practice, proving $A \sim B$ by exhibiting an explicit bijection can be very difficult. Schröder–Bernstein gives a two-injection strategy: find any injection each way, and the theorem guarantees a bijection exists (even if you cannot describe it explicitly). This is a powerful tool for cardinality computations. For example, to show $|(0,1)| = |\mathbb{R}|$, find an injection $(0,1) \hookrightarrow \mathbb{R}$ (inclusion) and an injection $\mathbb{R} \hookrightarrow (0,1)$ (e.g., $x \mapsto \frac{1}{\pi}\arctan(x) + \frac{1}{2}$); the theorem does the rest.

Remark (Proof idea)*.* The proof constructs h by partitioning A into two pieces: those elements whose "ancestry" under repeated application of $g \circ f$ is finite (they go through f), and those whose ancestry is infinite (they go through g^{-1}). The formal argument is a standard exercise; the key insight is that the two injections together cover A in a structured enough way to stitch together a bijection.

Remark (Sequences as functions)*.* A *sequence* in a set X is a function $f : \mathbb{N} \to X$. The element $f(n)$ is written x_n, and the sequence is written $(x_n)_{n\in\mathbb{N}}$ or simply (x_n). This is not a new object: it is an element of the function space $X^{\mathbb{N}}$. When metric spaces and topology discuss convergence of sequences $x_1, x_2, x_3, \ldots$ in an arbitrary space X, the formal object is always a function $\mathbb{N} \to X$, and all the function machinery — injectivity, composition, preimages — applies to it directly.

Remark (Transition)*.* Cardinality distinguishes between the countable (manageable, listable) and the uncountable (genuinely larger). This distinction organizes much of what follows. In metric spaces: separable spaces have countable dense subsets, and separability is a strong structural property. In measure theory: σ-algebras are closed under *countable* unions — not arbitrary ones — precisely because countable operations preserve the kind of control that makes measure theory work. In functional analysis: L^2 spaces have countable orthonormal bases (Hilbert spaces are separable), while non-separable spaces behave very differently. The line between countable and uncountable is not a technicality; it is one of the organizing principles of analysis.

Foundational Geometry

Chapter 12

Euclidean Geometry

12.1 Foundations

Definitions

Remark (Source note). The following definitions are Euclid's Book I definitions in the Heath translation [Euclid, 1956, Book I, Definitions 1–23].

Definition

Definition 12.1.1 (Point)**.** A point is that which has no part.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.2 (Line)**.** A line is breadthless length.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.3 (Ends of a line)**.** The ends of a line are points.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.4 (Straight line)**.** A straight line is a line which lies evenly with the points on itself.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.5 (Surface)**.** A surface is that which has length and breadth only.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.6 (Edges of a surface)**.** The edges of a surface are lines.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.7 (Plane surface)**.** A plane surface is a surface which lies evenly with the straight lines on itself.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.8 (Plane angle)**.** A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.9 (Rectilinear angle)**.** And when the lines containing the angle are straight, the angle is called rectilinear.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.10 (Right angle and perpendicular)**.** When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.11 (Obtuse angle)**.** An obtuse angle is an angle greater than a right angle.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.12 (Acute angle)**.** An acute angle is an angle less than a right angle.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.13 (Boundary)**.** A boundary is that which is an extremity of anything.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.14 (Figure)**.** A figure is that which is contained by any boundary or boundaries.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.15 (Circle)**.** A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.16 (Center of a circle)**.** And the point is called the center of the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.17 (Diameter)**.** A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.18 (Semicircle)**.** A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.19 (Rectilinear figures)**.** Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.20 (Equilateral, isosceles, and scalene triangles)**.** Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.21 (Right-angled, obtuse-angled, and acute-angled triangles)**.** Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.22 (Quadrilateral classes)**.** Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 12.1.23 (Parallel straight lines)**.** Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Postulates

Remark (Source note). The following are Euclid's five postulates from Book I of the *Elements* in the Heath translation [Euclid, 1956, Book I, Postulates 1–5].

Remark (Exposition). Postulates are geometry-specific starting assumptions. They say which constructions are permitted and which basic geometric facts may be used without proof. In Euclid's system, these are not conclusions reached from earlier results; they are part of the foundation from which the theory begins.

Later propositions are proved from the definitions, the postulates, the common notions, and propositions already established. Thus the postulates are the rules of play for Euclidean geometry: they determine what can be constructed, what may be assumed about straight lines and circles, and where the theory's geometric content enters before proof begins.

Axiom

Axiom 12.1.24 (Euclid Postulate I: Drawing a straight line)**.** Let it have been postulated to draw a straight line from any point to any point.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.25 (Euclid Postulate II: Producing a finite straight line)**.** Let it have been postulated to produce a finite straight line continuously in a straight line.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.26 (Euclid Postulate III: Describing a circle)**.** Let it have been postulated to describe a circle with any center and distance.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.27 (Euclid Postulate IV: Equality of right angles)**.** Let it have been postulated that all right angles are equal to one another.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.28 (Euclid Postulate V: Parallel postulate)**.** Let it have been postulated that, if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axioms

Remark (Source note)*.* Euclid calls the following principles *common notions*. They function as general axioms governing equality and comparison in Book I of the *Elements* [Euclid, 1956, Book I, Common Notions 1–5].

Axiom

Axiom 12.1.29 (Euclid Common Notion I: Equality is transitive through a common equal)**.** Things which are equal to the same thing are also equal to one another.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.30 (Euclid Common Notion II: Adding equals to equals)**.** If equals be added to equals, the wholes are equal.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.31 (Euclid Common Notion III: Subtracting equals from equals)**.** If equals be subtracted from equals, the remainders are equal.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.32 (Euclid Common Notion IV: Coincidence implies equality)**.** Things which coincide with one another are equal to one another.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 12.1.33 (Euclid Common Notion V: Whole and part)**.** The whole is greater than the part.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Remark (Exposition). Common notions are general principles of reasoning rather than specifically geometric assumptions. They express basic behavior of equality, comparison, and logical transfer: equal things may be substituted for equal things, adding or subtracting equals preserves equality, and a whole exceeds its part.

Because these principles are not tied to points, lines, circles, or planes, they can be used throughout mathematics, not only in geometry. In modern language, they play the role of background axioms governing how equality and comparison may be used inside proofs.

12.2 Compass-and-Straightedge Constructions

Propositions

Remark (Source note). The following construction propositions are Euclid's Propositions I.1–I.3 in modern theorem form, following the Heath translation of the *Elements* [Euclid, 1956, Book I, Propositions 1–3].

Theorem

Theorem 12.2.1 (Euclid I.1: Constructing an equilateral triangle)**.** *Given a finite straight line segment AB, there exists an equilateral triangle ABC constructed on AB.*

Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 12.2.2 (Euclid I.2: Copying a segment from a given point)**.** *Given a point A and a finite straight line segment BC, there exists a straight line segment beginning at A and equal to BC.*

Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 12.2.3 (Euclid I.3: Cutting off an equal segment). *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Chapter 13

Trigonometry

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VOLUME I
Logic, Sets, and Proof

BOOK III
Foundational Geometry

CHAPTER 11
Trigonometry

Chapter 14

Analytical Geometry

14.1 Analytical Geometry

Remark (Status). Notes, proofs, and exercises will appear here in a future revision.

14.2 Lines

Definition

Definition 14.2.1 (Point-slope form). Let $P_1 = (x_1, y_1)$ be a point in the Euclidean plane and let $m \in \mathbb{R}$. The point-slope form of the line through P_1 with slope m is

$$y - y_1 = m(x - x_1).$$

Remark (Standard quantified statement). For every point $P_1 = (x_1, y_1) \in \mathbb{R}^2$ and every $m \in \mathbb{R}$, the corresponding line is the set of all $(x, y) \in \mathbb{R}^2$ satisfying $y - y_1 = m(x - x_1)$.

Remark (Derivation). The formula records the condition that the slope from P_1 to an arbitrary point (x, y) on the line is constant and equal to m.

Definition

Remark (Dependencies).

- Real Numbers
- Cartesian Product
- Subtraction on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition 14.2.2 (Two-point form)**.** Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points with $x_1 \neq x_2$. The two-point form of the line through P_1 and P_2 is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1).$$

Remark (Standard quantified statement). For all distinct points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ in $\mathbb{R}^2$ with $x_1 \neq x_2$, the line through P_1 and P_2 is the set of all $(x, y) \in \mathbb{R}^2$ satisfying the displayed two-point equation.

Definition

Remark (Dependencies).

- Point-Slope Form
- Subtraction on $\mathbb{R}$
- $\mathbb{R}$ Is a Field

Definition 14.2.3 (General linear form)**.** The general linear form of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where $A, B, C \in \mathbb{R}$ and $A^2 + B^2 \neq 0$.

Remark (Standard quantified statement). For all $A, B, C \in \mathbb{R}$ with $A^2 + B^2 \neq 0$, the equation $Ax + By + C = 0$ determines a nondegenerate line in $\mathbb{R}^2$.

Remark (Interpretation). The vector (A, B) is normal to the line. This form includes vertical lines and therefore avoids the exceptional cases present in slope-based forms.

Definition

Remark (Dependencies).

- Real Numbers
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- $\mathbb{R}$ Is a Field

Definition 14.2.4 (Vector parametric form)**.** A line in $\mathbb{R}^2$ may be represented as the image of $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where $\mathbf{r}_0 \in \mathbb{R}^2$ is a point on the line and $\mathbf{v} \in \mathbb{R}^2$ is a nonzero direction vector.

Remark (Standard quantified statement). For every $\mathbf{r}_0 \in \mathbb{R}^2$ and every nonzero $\mathbf{v} \in \mathbb{R}^2$, the set $\{\mathbf{r}_0 + t\mathbf{v} : t \in \mathbb{R}\}$ is a line in $\mathbb{R}^2$.

Definition

Remark (Dependencies).

- Real Numbers
- Cartesian Product
- Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition 14.2.5 (Intercept form)**.** If a line meets the x-axis at $(a, 0)$ and the y-axis at $(0, b)$ with $a, b \neq 0$, its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Remark (Standard quantified statement). For all nonzero $a, b \in \mathbb{R}$, the line with intercepts $(a, 0)$ and $(0, b)$ is the set of all $(x, y) \in \mathbb{R}^2$ satisfying $x/a + y/b = 1$.

Remark (Derivation). The intercept form follows by applying the two-point form to the points $(a, 0)$ and $(0, b)$ and then rearranging the resulting equation.

Definition

Remark (Dependencies).

- Two-Point Form
- $\mathbb{R}$ Is a Field

Definition 14.2.6 (Affine parametric line in $\mathbb{R}^n$)**.** Given distinct points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \qquad t \in \mathbb{R}.$$

Remark (Standard quantified statement)*.* For all distinct $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the set $\{\mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1) : t \in \mathbb{R}\}$ is the affine line through $\mathbf{x}_1$ and $\mathbf{x}_2$.

Remark (Interpretation)*.* For $0 \le t \le 1$, the same formula describes the line segment joining $\mathbf{x}_1$ and $\mathbf{x}_2$. For arbitrary real t, it describes the entire affine line.

Definition

Remark (Dependencies).

- Real Numbers
- Cartesian Product
- Subtraction on $\mathbb{R}$
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition 14.2.7 (Normal form)**.** A line in $\mathbb{R}^2$ may be written in normal form as

$$x \cos\alpha + y \sin\alpha - p = 0,$$

where $p \ge 0$ is the distance from the origin to the line.

Remark (Standard quantified statement)*.* For every $p \ge 0$ and every real angle α, the equation $x \cos\alpha + y \sin\alpha - p = 0$ describes a line whose unit normal direction is $(\cos\alpha, \sin\alpha)$.

Definition

Remark (Dependencies).

- General Linear Form
- Sine and Cosine
- Order on $\mathbb{R}$

Definition 14.2.8 (Symmetric form)**.** Given a point (x_1, y_1) and direction cosines $(\cos\theta, \sin\theta)$, the symmetric form of a line is

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r.$$

Remark (Standard quantified statement). For every point (x_1, y_1) and every direction with nonzero displayed denominators, the symmetric equation describes the same line as the parametric equations $x = x_1 + r\cos\theta$ and $y = y_1 + r\sin\theta$.

Remark (Dependencies).

- Vector Parametric Form
- Sine and Cosine
- $\mathbb{R}$ Is a Field

Part II

Volume II: Origins of Numbers

Discrete Number Systems

Chapter 15

Formalizing the Number Systems

Why Formalization Is Needed

Ordinary arithmetic is one of the most familiar parts of mathematics. We learn to count, add, subtract, multiply, and divide long before we ask what sort of objects numbers are or why the usual rules are justified. For foundational work, however, intuition is not enough. The basic number systems must be specified by precise axioms and constructions, so that their operations and order relations are not merely assumed but accounted for.

The Natural-Number Structure

The first place where this becomes visible is the natural numbers. A foundational treatment cannot simply point to the counting numbers and begin calculating. It has to identify a first element, a successor operation, and the induction principle that makes recursive definitions and proofs possible. Peano's role was to isolate this structure: a first element, a successor map, conditions saying that successor is injective and never returns to the first element, and an induction principle strong enough to characterize the system. This is the role of Peano's 1889 presentation of arithmetic [Peano, 1889].

Structural Characterization

Dedekind's work emphasized the same structural point of view. What matters is not the particular names attached to the elements, but the pattern determined by a distinguished first element, successor, recursion, and induction. This perspective underlies the treatment of Peano systems earlier in the volume, and Dedekind's name also sits behind the cut-based construction of the real numbers developed later in the usual foundational tradition [Dedekind, 1901].

Later Number Systems

Once the natural numbers have been fixed, the later systems require their own explicit constructions. The integers are built so that subtraction becomes an internal operation. The rational numbers are built so that division by nonzero quantities is represented inside the system. The real numbers are then introduced to address the completeness failures of the rationals. In each case, the construction supplies the operations, the order, and the structural properties needed for analysis, rather than treating them as background facts.

Chapter 16

Identity, Equality, and Equivalence

SERIES
From Cantor to Ito

VOLUME II
Origins of Numbers

BOOK I
Discrete Number Systems

CHAPTER 2
Identity, Equality, and Equivalence

16.1 Identity, Equality, and Equivalence

16.2 Equality

Definition (Identity)

Definition 16.2.1 (Identity)**.** Two expressions x and y are *identical* when they denote the same object.

Remark (Standard quantified statement).

$$x = y \iff x \text{ and } y \text{ denote the same object.}$$

Remark (Interpretation). Identity is sameness, not similarity. It is the relation used when no distinction remains between two denotations.

Remark (Dependencies).

- Relation

Axiom (Leibniz's Law)

Axiom 16.2.2 (Leibniz's Law)**.** Identical objects have exactly the same properties.

$$\forall x\, \forall y \left(x = y \implies \forall P \left(P(x) \iff P(y) \right) \right).$$

Remark (Standard quantified statement).

$$\text{Indiscernible}(x, y) \iff \forall P \left(P(x) \iff P(y)\right).$$

Remark (Interpretation). Leibniz's law says that equality permits no observable mathematical difference. Any property true of one equal object is true of the other.

Definition (Equality on a Domain)

Definition 16.2.3 (Equality on a Domain)**.** Let A be a domain. *Equality on A* is the relation, written $=$, that holds exactly when two terms or constructions denote the same element of A.

Remark (Standard quantified statement).

$$\text{Equality}_A(=) \iff \forall x, y \in A, \left(x = y \iff x \text{ and } y \text{ denote the same element of } A\right).$$

Remark (Interpretation). Equality is the identity relation restricted to the domain currently under discussion. Later equivalence relations may identify objects for a purpose, but equality records literal sameness.

Remark (Dependencies).

- Relation

Definition (Definitional and Propositional Equality)

Definition 16.2.4 (Definitional and Propositional Equality)**.** *Definitional equality* records an introduced meaning, written with $:=$. *Propositional equality* is a statement of sameness inside the theory, written with $=$.

Remark (Standard quantified statement).

$$a := b \quad \text{introduces notation, while} \quad a = b \quad \text{asserts equality.}$$

Remark (Interpretation). The distinction prevents definitions from being confused with theorems. Definitions create abbreviations; propositional equalities must be available as axioms, assumptions, or proved statements.

16.3 Properties of Equality

Axiom (E1, Reflexivity of Equality)

Axiom 16.3.1 (Reflexivity of Equality)**.**

$$\forall x,\ x = x.$$

Remark (Standard quantified statement).

$$\text{Reflexive}(=) \iff \forall x,\ x = x.$$

Remark (Interpretation). Every object is itself. This is the primitive equality axiom used before symmetry, transitivity, or arithmetic structure is derived.

Theorem (E2, Symmetry of Equality)

Theorem 16.3.2 (Symmetry of Equality)**.**

$$\forall x\, \forall y,\ (x = y \implies y = x).$$

Remark (Standard quantified statement).

$$\text{Symmetric}(=) \iff \forall x\, \forall y,\ (x = y \implies y = x).$$

Remark (Interpretation). If x is the same object as y, then y is the same object as x. Symmetry follows from Leibniz's law by substituting into the property $P(t) \equiv (t = x)$ and using reflexivity.

Remark (Dependencies).

- Leibniz's Law
- Reflexivity of Equality

Theorem (E3, Transitivity of Equality)

Theorem 16.3.3 (Transitivity of Equality)**.**

$$\forall x\, \forall y\, \forall z,\ \big(x = y \wedge y = z \implies x = z\big).$$

Remark (Standard quantified statement).

$$\text{Transitive}(=) \iff \forall x\, \forall y\, \forall z,\ \big(x = y \wedge y = z \implies x = z\big).$$

Remark (Interpretation). Sameness chains: if the first object is the second and the second is the third, then the first is the third.

Remark (Dependencies).

- Leibniz's Law
- Reflexivity of Equality
- Symmetry of Equality

Corollary (Equality Is an Equivalence Relation)

Corollary 16.3.4 (Equality Is an Equivalence Relation)**.** *The equality relation is reflexive, symmetric, and transitive.*

Remark (Standard quantified statement).

$$\text{EquivalenceRelation}(=) \iff \text{Reflexive}(=) \wedge \text{Symmetric}(=) \wedge \text{Transitive}(=).$$

Remark (Interpretation). Equality is the strictest equivalence relation: it groups only objects that are literally the same.

Remark (Dependencies).

- Reflexivity of Equality
- Symmetry of Equality
- Transitivity of Equality

16.4 Substitution

Axiom (Substitution of Equals)

Axiom 16.4.1 (Substitution of Equals)**.** For every formula $\varphi(z)$ whose substitution is admissible,

$$x = y \implies \big(\varphi(x) \iff \varphi(y)\big).$$

Remark (Standard quantified statement).

$$\forall x\, \forall y \left(x = y \implies \big(\varphi(x) \iff \varphi(y)\big)\right).$$

Remark (Interpretation). Substitution is the operational meaning of equality. Equal objects may replace one another in legitimate statements without changing truth.

Proposition (Substitution Preserves Predicates)

Proposition 16.4.2 (Substitution Preserves Predicates)**.**

$$x = y \implies \big(P(x) \iff P(y)\big).$$

Remark (Standard quantified statement).

$$\forall x\, \forall y,\ x = y \implies \big(P(x) \iff P(y)\big).$$

Remark (Interpretation). Predicates cannot distinguish equal arguments.

Remark (Dependencies).

- Substitution of Equals

Proposition (Substitution Preserves Relations on the Left)

Proposition 16.4.3 (Substitution Preserves Relations on the Left)**.**

$$x = y \implies \big(R(x, z) \iff R(y, z)\big).$$

Remark (Standard quantified statement).

$$\forall x\,\forall y\,\forall z,\ x = y \implies \big(R(x,z) \iff R(y,z)\big).$$

Remark (Interpretation). Relations respect equality in their first coordinate.

Remark (Dependencies).

- Substitution of Equals

Proposition (Substitution Preserves Relations on the Right)

Proposition 16.4.4 (Substitution Preserves Relations on the Right)**.**

$$x = y \implies \big(R(z,x) \iff R(z,y)\big).$$

Remark (Standard quantified statement).

$$\forall x\,\forall y\,\forall z,\ x = y \implies \big(R(z,x) \iff R(z,y)\big).$$

Remark (Interpretation). Relations respect equality in their second coordinate.

Remark (Dependencies).

- Substitution of Equals

Proposition (Substitution Preserves Relations)

Proposition 16.4.5 (Substitution Preserves Relations)**.**

$$x = x' \wedge y = y' \implies \big(R(x,y) \iff R(x',y')\big).$$

Remark (Standard quantified statement).

$$\forall x, x', y, y',\ \big(x = x' \wedge y = y'\big) \implies \big(R(x,y) \iff R(x',y')\big).$$

Remark (Interpretation). Relations are congruent with respect to equality in all displayed arguments.

Remark (Dependencies).

- Substitution Preserves Relations on the Left
- Substitution Preserves Relations on the Right

Proposition (Substitution Preserves Functions)

Proposition 16.4.6 (Substitution Preserves Functions)**.**

$$x = y \implies f(x) = f(y).$$

Remark (Standard quantified statement).

$$\forall x\,\forall y,\; x = y \Longrightarrow f(x) = f(y).$$

Remark (Interpretation). Functions send equal inputs to equal outputs.

Remark (Dependencies).

- Substitution of Equals

Proposition (Left Congruence for Operations)

Proposition 16.4.7 (Left Congruence for Operations)**.**

$$x = x' \Longrightarrow x\,O\,y = x'\,O\,y.$$

Remark (Standard quantified statement).

$$\forall x, x', y,\; x = x' \Longrightarrow x\,O\,y = x'\,O\,y.$$

Remark (Interpretation). A binary operation respects equality in its left argument.

Remark (Dependencies).

- Substitution Preserves Functions

Proposition (Right Congruence for Operations)

Proposition 16.4.8 (Right Congruence for Operations)**.**

$$y = y' \Longrightarrow x\,O\,y = x\,O\,y'.$$

Remark (Standard quantified statement).

$$\forall x, y, y',\; y = y' \Longrightarrow x\,O\,y = x\,O\,y'.$$

Remark (Interpretation). A binary operation respects equality in its right argument.

Remark (Dependencies).

- Substitution Preserves Functions

Proposition (Full Congruence for Operations)

Proposition 16.4.9 (Full Congruence for Operations)**.**

$$x = x' \wedge y = y' \Longrightarrow x\,O\,y = x'\,O\,y'.$$

Remark (Standard quantified statement).

$$\forall x, x', y, y',\; \big(x = x' \wedge y = y'\big) \Longrightarrow x\,O\,y = x'\,O\,y'.$$

Remark (Interpretation). Binary operations are congruent with respect to equality in both arguments.

Remark (Dependencies).

- Left Congruence for Operations
- Right Congruence for Operations

Proposition (Congruence with Respect to Equality Is Automatic)

Proposition 16.4.10 (Congruence with Respect to Equality Is Automatic)**.** *Every function, predicate, relation, and operation respects equality in its displayed arguments.*

Remark (Standard quantified statement).

$$\begin{aligned} x = y &\Longrightarrow f(x) = f(y), \\ x = y &\Longrightarrow \big(P(x) \iff P(y)\big), \\ x = x' \wedge y = y' &\Longrightarrow \big(R(x, y) \iff R(x', y')\big), \\ x = x' \wedge y = y' &\Longrightarrow x \, O \, y = x' \, O \, y'. \end{aligned}$$

Remark (Interpretation). Respecting equality is automatic. Respecting a nontrivial equivalence relation is not automatic; that is why quotient constructions must prove separate respect lemmas before operations, predicates, or relations descend to classes.

Remark (Dependencies).

- Substitution Preserves Predicates
- Substitution Preserves Relations
- Substitution Preserves Functions
- Full Congruence for Operations

16.5 Equivalence

Definition (Equivalence Relation)

Definition 16.5.1 (Equivalence Relation)**.** Let A be a set. A relation $\sim$ on A is an *equivalence relation* if it is reflexive, symmetric, and transitive.

Remark (Standard quantified statement)*.*

$$\mathrm{EquivalenceRelation}_A(\sim) \iff \Big(\forall x \in A,\ x \sim x\Big) \wedge \Big(\forall x, y \in A,\ x \sim y \Rightarrow y \sim x\Big) \wedge \Big(\forall x, y, z \in A,\ (x \sim y \wedge y$$

Remark (Interpretation)*.* An equivalence relation formalizes sameness for a chosen purpose.

Remark (Dependencies)*.*

- Relation
- Reflexive Relation
- Symmetric Relation
- Transitive Relation

Corollary (Equality Is the Finest Equivalence)

Corollary 16.5.2 (Equality Is the Finest Equivalence)**.** *Let $\sim$ be an equivalence relation on A. For all $a, b \in A$,*

$$a = b \Longrightarrow a \sim b.$$

Remark (Standard quantified statement)*.*

$$\forall a, b \in A,\ a = b \Longrightarrow a \sim b.$$

Remark (Interpretation)*.* Equality is the finest equivalence relation because it identifies only objects that are already the same. Any coarser equivalence relation must at least identify equal representatives.

Remark (Dependencies)*.*

- Equivalence Relation
- Reflexivity of Equality

Definition (Equivalence Class)

Definition 16.5.3 (Equivalence Class)**.** Let $\sim$ be an equivalence relation on A. The *equivalence class* of $a \in A$ is

$$[a] := \{x \in A \mid x \sim a\}.$$

Remark (Standard quantified statement)*.*

$$x \in [a] \iff x \sim a.$$

Remark (Interpretation)*.* The class $[a]$ collects all elements that are equivalent to the representative a.

Remark (Dependencies)*.*

- Equivalence Relation

Definition (Quotient Set)

Definition (Quotient Set)**.** The *quotient set* of A by $\sim$ is the set of equivalence classes

$$A/\!\sim \, := \{[a] \mid a \in A\}.$$

Theorem (Equivalence Classes Partition the Domain)

Theorem 16.5.4 (Equivalence Classes Partition the Domain)**.** *The equivalence classes of an equivalence relation on A are nonempty, cover A, and are pairwise disjoint.*

Remark (Standard quantified statement)*.*

$$\forall a \in A,\ a \in [a], \qquad A = \bigcup_{a \in A} [a], \qquad [a] \cap [b] \neq \emptyset \Longrightarrow [a] = [b].$$

Remark (Interpretation)*.* Equivalence classes divide the domain into non-overlapping blocks.

Remark (Dependencies)*.*

- Equivalence Relation
- Equivalence Class

Proposition (Representatives Related iff Classes Equal)

Proposition 16.5.5 (Representatives Related iff Classes Equal)**.**

$$a \sim b \iff [a] = [b].$$

Remark (Standard quantified statement)*.*

$$\forall a, b \in A,\ a \sim b \iff [a] = [b].$$

Remark (Interpretation). Changing representatives inside an equivalence class does not change the class.

Remark (Dependencies).

- Equivalence Classes Partition the Domain

Canonical Projection

Definition (Canonical Projection)

Definition 16.5.6 (Canonical Projection). Let $\sim$ be an equivalence relation on A. The *canonical projection* is the map

$$\pi : A \to A/{\sim}, \qquad \pi(a) := [a].$$

Remark (Dependencies).

- Quotient Set
- Equivalence Class

Proposition (Projection Is Surjective)

Proposition 16.5.7 (Projection Is Surjective). *The canonical projection $\pi : A \to A/{\sim}$ is surjective.*

Remark (Standard quantified statement).

$$\forall C \in A/{\sim}\, \exists a \in A,\ \pi(a) = C.$$

Remark (Dependencies).

- Canonical Projection
- Quotient Set

Proposition (Projection Identifies Exactly the Equivalent Elements)

Proposition 16.5.8 (Projection Identifies Exactly the Equivalent Elements). *For all $a, b \in A$,*

$$\pi(a) = \pi(b) \iff a \sim b.$$

Remark (Dependencies).

- Canonical Projection
- Representatives Related iff Classes Equal

Theorem (Existence of a Quotient)

Theorem 16.5.9 (Existence of a Quotient)**.** *For any equivalence relation $\sim$ on A, the quotient set $A/{\sim}$ exists as the set of equivalence classes, and the canonical projection*

$$\pi : A \to A/{\sim}$$

is well-defined.

Remark (Dependencies).

- Equivalence Relation
- Quotient Set
- Canonical Projection

Induced Structure on a Quotient

Definition (Operation Respects an Equivalence)

Definition 16.5.10 (Operation Respects an Equivalence)**.** Let $O : A \times A \to A$ be a binary operation and let $\sim$ be an equivalence relation on A. The operation O *respects* $\sim$ if

$$\forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Longrightarrow \big(a\,O\,b\big) \sim \big(a'\,O\,b'\big).$$

Remark (Dependencies).

- Equivalence Relation
- Binary Arithmetic Operation
- Cartesian Product

Definition (Left and Right Respect)

Definition 16.5.11 (Left and Right Respect)**.** The operation O *respects* $\sim$ on the left if

$$a \sim a' \Longrightarrow (a\,O\,b) \sim (a'\,O\,b),$$

and respects $\sim$ on the right if

$$b \sim b' \Longrightarrow (a\,O\,b) \sim (a\,O\,b').$$

Remark (Dependencies).

- Operation Respects an Equivalence
- Equivalence Relation

Lemma (Respect Splits)

Lemma 16.5.12 (Respect Splits)**.** *A binary operation O respects $\sim$ if and only if it respects $\sim$ on the left and on the right.*

Remark (Dependencies).

- Operation Respects an Equivalence
- Left and Right Respect

Theorem (Induced Operation on a Quotient)

Theorem 16.5.13 (Induced Operation on a Quotient)**.** *If $O : A \times A \to A$ respects $\sim$, then*

$$[a]\,\overline{O}\,[b] := [a\,O\,b]$$

defines a well-defined binary operation on $A/{\sim}$.

Remark (Dependencies).

- Operation Respects an Equivalence
- Representatives Related iff Classes Equal
- Existence of a Quotient

Theorem (Induced Unary Operation)

Theorem 16.5.14 (Induced Unary Operation)**.** *Let $f : A \to A$. If*

$$a \sim a' \implies f(a) \sim f(a'),$$

then

$$\overline{f}([a]) := [f(a)]$$

defines a well-defined unary operation on $A/{\sim}$.

Remark (Dependencies).

- Representatives Related iff Classes Equal
- Existence of a Quotient

Definition (Predicate Respects an Equivalence)

Definition 16.5.15 (Predicate Respects an Equivalence)**.** Let P be a predicate on A. The predicate P *respects* $\sim$ if

$$\forall a, a' \in A,\ a \sim a' \implies \big(P(a) \iff P(a')\big).$$

Remark (Dependencies).

- Equivalence Relation
- Predicate

Theorem (Induced Predicate on a Quotient)

Theorem 16.5.16 (Induced Predicate on a Quotient)**.** *If P respects $\sim$, then*

$$\overline{P}([a]) :\Longleftrightarrow P(a)$$

defines a well-defined predicate on $A/{\sim}$.

Remark (Dependencies).

- Predicate Respects an Equivalence
- Representatives Related iff Classes Equal

Definition (Relation Respects an Equivalence)

Definition 16.5.17 (Relation Respects an Equivalence)**.** Let R be a binary relation on A. The relation R *respects* $\sim$ if

$$\forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Longrightarrow \big(R(a,b) \Longleftrightarrow R(a',b')\big).$$

Remark (Dependencies).

- Equivalence Relation
- Relation

Theorem (Induced Relation on a Quotient)

Theorem 16.5.18 (Induced Relation on a Quotient)**.** *If R respects $\sim$, then*

$$\overline{R}([a],[b]) :\Longleftrightarrow R(a,b)$$

defines a well-defined binary relation on $A/{\sim}$.

Remark (Dependencies).

- Relation Respects an Equivalence
- Representatives Related iff Classes Equal

Theorem (Induced Map out of a Quotient)

Theorem 16.5.19 (Induced Map out of a Quotient). *Let $g : A \to B$. If*

$$a \sim a' \Longrightarrow g(a) = g(a'),$$

then there is a unique well-defined map

$$\overline{g} : A/\!\sim \to B$$

such that

$$\overline{g}([a]) = g(a) \qquad \text{and} \qquad \overline{g} \circ \pi = g,$$

where $\pi : A \to A/\!\sim$ is the canonical projection.

Remark (Dependencies).

- Canonical Projection
- Projection Identifies Exactly the Equivalent Elements
- Projection Is Surjective

Theorem (Induced Partial Operation on a Quotient)

Theorem 16.5.20 (Induced Partial Operation on a Quotient). *Let $D \subseteq A$ be saturated under $\sim$, meaning*

$$a \in D \wedge a \sim a' \Longrightarrow a' \in D.$$

If $f : D \to A$ respects $\sim$ on D, then

$$\overline{f}([a]) := [f(a)]$$

is a well-defined partial operation on $A/\!\sim$ with domain

$$D/\!\sim := \{[a] \in A/\!\sim : a \in D\}.$$

Remark (Dependencies).

- Induced Unary Operation
- Induced Predicate on a Quotient

Proposition (Representative Independence Is Necessary)

Proposition 16.5.21 (Representative Independence Is Necessary). *If a rule on classes*

$$[a]\,\overline{O}\,[b] := [a\,O\,b]$$

is well-defined on $A/\!\sim$, then O respects $\sim$.

Remark (Dependencies).

- Operation Respects an Equivalence
- Projection Identifies Exactly the Equivalent Elements
- Induced Operation on a Quotient

Lemma (Distinct Equivalence Classes Are Disjoint)

Lemma 16.5.22 (Distinct Equivalence Classes Are Disjoint)**.**

$$[a] \neq [b] \implies [a] \cap [b] = \emptyset.$$

Remark (Standard quantified statement)*.*

$$\forall a, b \in A,\ [a] \neq [b] \implies [a] \cap [b] = \emptyset.$$

Remark (Interpretation)*.* Two equivalence classes either coincide or do not overlap.

Remark (Dependencies)*.*

- Equivalence Classes Partition the Domain

Chapter 17

Arithmetic Operations and Relations

17.1 Arithmetic Operations and Relations

17.2 Closure

Definition (Unary Arithmetic Operation)

Definition 17.2.1 (Unary Arithmetic Operation)**.** Let K be a domain. A *unary arithmetic operation* on K is a function

$$O : K \to K.$$

Remark (Standard quantified statement)*.*

$$\mathrm{UnaryOperation}_K(O) \iff \forall x \in K,\ O(x) \in K.$$

Remark (Interpretation)*.* Unary closure is the abstract form behind successor, negation, and other single-input transformations.

Remark (Dependencies)*.*

- Function

Definition (Binary Arithmetic Operation)

Definition 17.2.2 (Binary Arithmetic Operation)**.** Let K be a domain. A *binary arithmetic operation* on K is a function

$$O : K \times K \to K.$$

Remark (Standard quantified statement)*.*

$$\mathrm{BinaryOperation}_K(O) \iff \forall x, y \in K,\ x\, O\, y \in K.$$

Remark (Interpretation)*.* Binary closure is the abstract form behind addition and multiplication.

Remark (Dependencies)*.*

- Function
- Cartesian Product

Definition (Unary Closure)

Definition 17.2.3 (Unary Closure)**.** A unary operation O is *closed on* K if applying it to an element of K produces an element of K.

Remark (Standard quantified statement)*.*

$$\forall x \in K,\ O(x) \in K.$$

Remark (Interpretation)*.* The operation does not leave the domain.

Remark (Dependencies)*.*

- Unary Arithmetic Operation

Definition (Binary Closure)

Definition 17.2.4 (Binary Closure)**.** A binary operation O is *closed on* K if combining two elements of K produces an element of K.

Remark (Standard quantified statement)*.*

$$\forall x, y \in K,\ x\, O\, y \in K.$$

Remark (Interpretation)*.* The operation is internal to the domain.

Remark (Dependencies)*.*

- Binary Arithmetic Operation

Definition (Operation Well-Definedness)

Definition 17.2.5 (Operation Well-Definedness)**.** Let $\sim$ be an equivalence relation on A. A binary operation O respects $\sim$ if

$$a \sim a' \wedge b \sim b' \Longrightarrow a\,O\,b \sim a'\,O\,b'.$$

Remark (Standard quantified statement)*.*

$$\forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Longrightarrow a\,O\,b \sim a'\,O\,b'.$$

Remark (Interpretation)*.* This is the condition that lets an operation descend to equivalence classes.

Remark (Dependencies)*.*

- Equivalence Relation

Lemma (Left Respect)

Lemma 17.2.6 (Left Respect)**.** *If a binary operation respects* $\sim$*, then*

$$a \sim a' \Longrightarrow a\,O\,b \sim a'\,O\,b.$$

Remark (Standard quantified statement)*.*

$$\forall a, a', b \in A,\ a \sim a' \Rightarrow a\,O\,b \sim a'\,O\,b.$$

Remark (Interpretation)*.* Full respect includes respect in the left coordinate.

Remark (Dependencies)*.*

- Operation Well-Definedness

Lemma (Right Respect)

Lemma 17.2.7 (Right Respect)**.** *If a binary operation respects* $\sim$*, then*

$$b \sim b' \Longrightarrow a\,O\,b \sim a\,O\,b'.$$

Remark (Standard quantified statement)*.*

$$\forall a, b, b' \in A,\ b \sim b' \Rightarrow a\,O\,b \sim a\,O\,b'.$$

Remark (Interpretation)*.* Full respect includes respect in the right coordinate.

Remark (Dependencies)*.*

- Operation Well-Definedness

Theorem (Induced Operation)

Theorem 17.2.8 (Induced Operation)**.** *The operation*

$$[a]\,\overline{O}\,[b] := [a\,O\,b]$$

on $A/{\sim}$ is well-defined if and only if O respects $\sim$.

Remark (Standard quantified statement)*.*

$$\mathrm{WellDefined}(\overline{O}) \iff \forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Rightarrow a\,O\,b \sim a'\,O\,b'.$$

Remark (Interpretation)*.* This theorem is the abstract engine behind quotient constructions such as integers and rationals.

Remark (Dependencies)*.*

- Quotient Set
- Operation Well-Definedness

17.3 Identity Elements

Definition (Left Identity)

Definition 17.3.1 (Left Identity)**.**

$$\forall x,\ e_L\,O\,x = x.$$

Remark (Dependencies)*.*

- Binary Arithmetic Operation

Definition (Right Identity)

Definition 17.3.2 (Right Identity)**.**

$$\forall x,\ x\,O\,e_R = x.$$

Remark (Dependencies)*.*

- Binary Arithmetic Operation

Definition (Two-Sided Identity)

Definition 17.3.3 (Two-Sided Identity)**.** An element e is a *two-sided identity* for O if it is both a left identity and a right identity.

Remark (Dependencies)*.*

- Left Identity
- Right Identity

Definition (Absorbing Element)

Definition 17.3.4 (Absorbing Element)**.** An element z is an *absorbing element* for a binary operation O if

$$z \, O \, x = z \quad \text{and} \quad x \, O \, z = z$$

for every element x in the domain of the operation.

Remark (Dependencies).

- Binary Arithmetic Operation

Lemma (Left and Right Identities Coincide)

Lemma 17.3.5 (Left and Right Identities Coincide)**.** *If e_L is a left identity and e_R is a right identity for the same operation, then $e_L = e_R$.*

Remark (Standard quantified statement).

$$(\forall x,\ e_L \, O \, x = x) \wedge (\forall x,\ x \, O \, e_R = x) \Longrightarrow e_L = e_R.$$

Remark (Dependencies).

- Left Identity
- Right Identity

Theorem (Uniqueness of Identity)

Theorem 17.3.6 (Uniqueness of Identity)**.** *A two-sided identity for a binary operation is unique.*

Remark (Standard quantified statement).

$$\text{Identity}(e) \wedge \text{Identity}(e') \Longrightarrow e = e'.$$

Remark (Dependencies).

- Left and Right Identities Coincide

17.4 Inverse Operations

Definition (Left Inverse)

Definition 17.4.1 (Left Inverse)**.** An element x' is a left inverse of x with respect to identity e if

$$x' \, O \, x = e.$$

Remark (Dependencies).

- Binary Arithmetic Operation
- Two-Sided Identity

Definition (Right Inverse)

Definition 17.4.2 (Right Inverse)**.** An element x'' is a right inverse of x with respect to identity e if

$$x \, O \, x'' = e.$$

Remark (Dependencies).

- Binary Arithmetic Operation
- Two-Sided Identity

Definition (Two-Sided Inverse)

Definition 17.4.3 (Two-Sided Inverse)**.** An element y is a two-sided inverse of x if

$$y \, O \, x = e \quad \text{and} \quad x \, O \, y = e.$$

Remark (Dependencies).

- Left Inverse
- Right Inverse

Lemma (Left and Right Inverses Coincide)

Lemma 17.4.4 (Left and Right Inverses Coincide)**.** *In an associative operation with identity, a left inverse and a right inverse of the same element are equal.*

Remark (Standard quantified statement).

$$(x' \, O \, x = e) \land (x \, O \, x'' = e) \implies x' = x''.$$

Remark (Dependencies).

- Associativity

- Two-Sided Identity
- Left Inverse
- Right Inverse

Theorem (Uniqueness of Inverse)

Theorem 17.4.5 (Uniqueness of Inverse)**.** *In an associative operation with identity, inverses are unique.*

Remark (Dependencies).

- Left and Right Inverses Coincide

Definition (Right Solvability)

Definition 17.4.6 (Right Solvability)**.** Right solvability means

$$\forall x, y\, \exists z,\ x = y\, O\, z,$$

Remark (Dependencies).

- Binary Arithmetic Operation

Definition (Left Solvability)

Definition 17.4.7 (Left Solvability)**.** Left solvability means

$$\forall x, y\, \exists z,\ x = z\, O\, y.$$

Remark (Dependencies).

- Binary Arithmetic Operation

Definition (Two-Sided Solvability)

Definition 17.4.8 (Two-Sided Solvability)**.** Two-sided solvability means both left solvability and right solvability hold.

Remark (Dependencies).

- Left Solvability
- Right Solvability

Proposition (Solvability and Inverses)

Proposition 17.4.9 (Solvability and Inverses)**.** *For an associative operation with a two-sided identity, solvability is equivalent to the existence of inverses.*

Remark (Dependencies).

- Associativity
- Two-Sided Identity
- Two-Sided Inverse
- Right Solvability
- Left Solvability
- Two-Sided Solvability

17.5 Algebraic Laws

Definition (Associativity)

Definition 17.5.1 (Associativity).

$$\forall x, y, z,\ x \, O \, (y \, O \, z) = (x \, O \, y) \, O \, z.$$

Remark (Dependencies).

- Binary Arithmetic Operation

Theorem (General Associativity)

Theorem 17.5.2 (General Associativity). *If O is associative, then every bracketing of $x_1 \, O \cdots O \, x_n$ has the same value.*

Remark (Dependencies).

- Associativity

Definition (Commutativity)

Definition 17.5.3 (Commutativity).

$$\forall x, y,\ x \, O \, y = y \, O \, x.$$

Remark (Dependencies).

- Binary Arithmetic Operation

Theorem (General Commutativity)

Theorem 17.5.4 (General Commutativity). *If O is associative and commutative, then any finite reordering of $x_1 \, O \cdots O \, x_n$ has the same value.*

Remark (Dependencies).

- Associativity
- Commutativity

Definition (Left Distributivity)

Definition 17.5.5 (Left Distributivity)**.** The operation $\otimes$ is left distributive over $\oplus$ if

$$x \otimes (y \oplus z) = (x \otimes y) \oplus (x \otimes z),$$

Remark (Dependencies).

- Binary Arithmetic Operation

Definition (Right Distributivity)

Definition 17.5.6 (Right Distributivity)**.** The operation $\otimes$ is right distributive over $\oplus$ if

$$(y \oplus z) \otimes x = (y \otimes x) \oplus (z \otimes x),$$

Remark (Dependencies).

- Binary Arithmetic Operation

Definition (Two-Sided Distributivity)

Definition 17.5.7 (Two-Sided Distributivity)**.** The operation $\otimes$ is two-sided distributive over $\oplus$ if it is both left distributive and right distributive.

Remark (Dependencies).

- Left Distributivity
- Right Distributivity

Definition (Left Cancellation)

Definition 17.5.8 (Left Cancellation)**.** An operation has left cancellation if

$$x \, O \, y = x \, O \, z \Longrightarrow y = z,$$

Remark (Dependencies).

- Binary Arithmetic Operation

Definition (Right Cancellation)

Definition 17.5.9 (Right Cancellation)**.** An operation has right cancellation if

$$y \, O \, x = z \, O \, x \Longrightarrow y = z.$$

Remark (Dependencies).

- Binary Arithmetic Operation

Definition (Two-Sided Cancellation)

Definition 17.5.10 (Two-Sided Cancellation)**.** An operation has two-sided cancellation if it has both left cancellation and right cancellation.

Remark (Dependencies).

- Left Cancellation
- Right Cancellation

Proposition (Invertibility Implies Left Cancellation)

Proposition 17.5.11 (Invertibility Implies Left Cancellation)**.** *In an associative operation with identity, invertible elements satisfy left cancellation.*

Remark (Dependencies).

- Associativity
- Two-Sided Identity
- Two-Sided Inverse
- Left Cancellation

Proposition (Invertibility Implies Cancellation)

Proposition 17.5.12 (Invertibility Implies Cancellation)**.** *In an associative operation with identity, invertible elements satisfy two-sided cancellation.*

Remark (Dependencies).

- Invertibility Implies Left Cancellation
- Invertibility Implies Right Cancellation
- Two-Sided Cancellation

Proposition (Invertibility Implies Right Cancellation)

Proposition 17.5.13 (Invertibility Implies Right Cancellation)**.** *In an associative operation with identity, invertible elements satisfy right cancellation.*

Remark (Dependencies).

- Associativity
- Two-Sided Identity
- Two-Sided Inverse
- Right Cancellation

17.6 Derived Laws

Proposition (Left Zero Absorption)

Proposition 17.6.1 (Left Zero Absorption)**.** *When multiplication distributes over addition and* 0 *is the additive identity,*

$$0 \cdot a = 0.$$

Remark (Dependencies).

- Two-Sided Distributivity
- Two-Sided Identity
- Absorbing Element

Proposition (Right Zero Absorption)

Proposition 17.6.2 (Right Zero Absorption)**.** *When multiplication distributes over addition and* 0 *is the additive identity,*

$$a \cdot 0 = 0.$$

Remark (Dependencies).

- Two-Sided Distributivity
- Two-Sided Identity
- Absorbing Element

Proposition (Zero Absorption)

Proposition 17.6.3 (Zero Absorption)**.** *When multiplication distributes over addition and* 0 *is the additive identity,*

$$0 \cdot a = 0 \qquad \textit{and} \qquad a \cdot 0 = 0.$$

Remark (Dependencies).

- Left Zero Absorption
- Right Zero Absorption

Proposition (Left Sign Rule)

Proposition 17.6.4 (Left Sign Rule)**.** *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot b = -(a \cdot b).$$

Remark (Dependencies).

- Two-Sided Distributivity
- Two-Sided Inverse
- Left Zero Absorption

Proposition (Right Sign Rule)

Proposition 17.6.5 (Right Sign Rule). *When multiplication distributes over addition and additive inverses exist,*

$$a \cdot (-b) = -(a \cdot b).$$

Remark (Dependencies).

- Two-Sided Distributivity
- Two-Sided Inverse
- Right Zero Absorption

Proposition (Sign Rule)

Proposition 17.6.6 (Sign Rule). *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot b = -(a \cdot b) \qquad \text{and} \qquad a \cdot (-b) = -(a \cdot b).$$

Remark (Dependencies).

- Left Sign Rule
- Right Sign Rule

Corollary (Negative One Rule)

Corollary 17.6.7 (Negative One Rule). *When multiplication distributes over addition and additive inverses exist,*

$$(-1) \cdot a = -a.$$

Remark (Dependencies).

- Left Sign Rule

Proposition (Product of Negatives)

Proposition 17.6.8 (Product of Negatives). *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot (-b) = a \cdot b.$$

Remark (Dependencies).

- Left Sign Rule
- Right Sign Rule

Definition (Zero Divisor)

Definition 17.6.9 (Zero Divisor)**.** Let K be an arithmetic system with multiplication and zero. A nonzero element $a \in K$ is a *zero divisor* if there exists a nonzero element $b \in K$ such that $ab = 0$.

Remark (Dependencies).

- Binary Arithmetic Operation

Theorem (No Zero Divisors)

Theorem 17.6.10 (No Zero Divisors)**.** *In an arithmetic system with no nonzero zero divisors,*

$$a \cdot b = 0 \implies a = 0 \vee b = 0.$$

Remark (Dependencies).

- Binary Arithmetic Operation

17.7 Order Relations

Definition (Reflexive Relation)

Definition 17.7.1 (Reflexive Relation)**.**

$$\forall x,\ xRx.$$

Remark (Dependencies).

- Relation
- Reflexive Relation

Definition (Irreflexive Relation)

Definition 17.7.2 (Irreflexive Relation)**.**

$$\forall x,\ \neg(xRx).$$

Remark (Dependencies).

- Relation
- Irreflexive Relation

Definition (Symmetric Relation)

Definition 17.7.3 (Symmetric Relation)**.**

$$\forall x, y,\ xRy \implies yRx.$$

Remark (Dependencies).

- Relation
- Symmetric Relation

Definition (Antisymmetric Relation)

Definition 17.7.4 (Antisymmetric Relation)**.**

$$\forall x, y,\ (xRy \wedge yRx) \implies x = y.$$

Remark (Dependencies).

- Relation
- Antisymmetric Relation

Definition (Asymmetric Relation)

Definition 17.7.5 (Asymmetric Relation)**.**

$$\forall x, y,\ xRy \implies \neg(yRx).$$

Remark (Dependencies).

- Relation
- Asymmetric Relation

Definition (Transitive Relation)

Definition 17.7.6 (Transitive Relation)**.**

$$\forall x, y, z,\ (xRy \wedge yRz) \implies xRz.$$

Remark (Dependencies).

- Relation
- Transitive Relation

Definition (Total Relation)

Definition 17.7.7 (Total Relation)**.**

$$\forall x, y,\ xRy \vee yRx.$$

Remark (Dependencies).

- Relation
- Total Relation

Definition (Trichotomy)

Definition 17.7.8 (Trichotomy)**.** The relation $<$ satisfies *trichotomy* if exactly one of

$$x < y, \qquad x = y, \qquad y < x$$

holds for every pair x, y.

Remark (Dependencies).

- Strict Order
- Total Relation

17.8 Order Compatibility

Definition (Right Monotony)

Definition 17.8.1 (Right Monotony)**.** Right monotony means

$$yRz \Longrightarrow (x\,O\,y)R(x\,O\,z),$$

Remark (Dependencies).

- Relation
- Binary Arithmetic Operation

Definition (Left Monotony)

Definition 17.8.2 (Left Monotony)**.** Left monotony means

$$yRz \Longrightarrow (y\,O\,x)R(z\,O\,x).$$

Remark (Dependencies).

- Relation

- Binary Arithmetic Operation

Definition (Two-Sided Monotony)

Definition 17.8.3 (Two-Sided Monotony)**.** Two-sided monotony means both left monotony and right monotony hold.

Remark (Dependencies).

- Left Monotony
- Right Monotony

Definition (Right Order Reflection)

Definition 17.8.4 (Right Order Reflection)**.** Right reflection means

$$(x \, O \, y) R (x \, O \, z) \implies y R z,$$

Remark (Dependencies).

- Relation
- Binary Arithmetic Operation

Definition (Left Order Reflection)

Definition 17.8.5 (Left Order Reflection)**.** Left reflection means

$$(y \, O \, x) R (z \, O \, x) \implies y R z.$$

Remark (Dependencies).

- Relation
- Binary Arithmetic Operation

Definition (Two-Sided Order Reflection)

Definition 17.8.6 (Two-Sided Order Reflection)**.** Two-sided order reflection means both left order reflection and right order reflection hold.

Remark (Dependencies).

- Left Order Reflection
- Right Order Reflection

Definition (Positive Cone)

Definition 17.8.7 (Positive cone)**.** Let K be an arithmetic system with addition, multiplication, and zero. A subset $P \subset K$ is called a *positive cone* if it satisfies:

1. if $a, b \in P$, then $a + b \in P$;
2. if $a, b \in P$, then $ab \in P$;
3. for every $a \in K$, exactly one of the following holds:

$$a \in P, \qquad a = 0, \qquad -a \in P.$$

The strict order associated to P is

$$a < b \iff b - a \in P.$$

Remark (Dependencies).

- Subset
- Trichotomy
- Binary Arithmetic Operation

Definition (Monotone Map)

Definition 17.8.8 (Monotone Map)**.**

$$x \leq y \Longrightarrow f(x) \leq f(y).$$

Remark (Dependencies).

- Function
- Relation

Definition (Strictly Monotone Map)

Definition 17.8.9 (Strictly Monotone Map)**.**

$$x < y \Longrightarrow f(x) < f(y).$$

Remark (Dependencies).

- Function
- Relation

Proposition (Addition Preserves Order on the Left)

Proposition 17.8.10 (Addition Preserves Order on the Left)**.**

$$y < z \Longrightarrow x + y < x + z$$

in ordered arithmetic systems where addition is strictly order-preserving.

Remark (Dependencies).

- Right Monotony

Proposition (Addition Preserves Order on the Right)

Proposition 17.8.11 (Addition Preserves Order on the Right)**.**

$$y < z \Longrightarrow y + x < z + x$$

in ordered arithmetic systems where addition is strictly order-preserving.

Remark (Dependencies).

- Left Monotony

Proposition (Addition Preserves Order)

Proposition 17.8.12 (Addition Preserves Order)**.** *In ordered arithmetic systems where addition is strictly order-preserving, addition preserves order on both sides.*

Remark (Dependencies).

- Addition Preserves Order on the Left
- Addition Preserves Order on the Right

Proposition (Multiplication Preserves Order on the Left)

Proposition 17.8.13 (Multiplication Preserves Order on the Left)**.**

$$0 < x \wedge y < z \Longrightarrow x \cdot y < x \cdot z$$

in ordered arithmetic systems where multiplication by positive elements is strictly order-preserving.

Remark (Dependencies).

- Right Monotony
- Strict Total Order

Proposition (Multiplication Preserves Order on the Right)

Proposition 17.8.14 (Multiplication Preserves Order on the Right)**.**

$$0 < x \wedge y < z \Longrightarrow y \cdot x < z \cdot x$$

in ordered arithmetic systems where multiplication by positive elements is strictly order-preserving.

Remark (Dependencies).

- Left Monotony
- Strict Total Order

Proposition (Multiplication Preserves Order)

Proposition 17.8.15 (Multiplication Preserves Order)**.** *In ordered arithmetic systems where multiplication by positive elements is strictly order-preserving, multiplication preserves order on both sides.*

Remark (Dependencies).

- Multiplication Preserves Order on the Left
- Multiplication Preserves Order on the Right

17.9 Composed Relations

Definition (Preorder)

Definition (Preorder)**.** A *preorder* is a reflexive and transitive relation.

Definition (Partial Order)

Definition (Partial Order)**.** A *partial order* is a preorder that is antisymmetric.

Definition (Strict Partial Order)

Definition 17.9.1 (Strict Partial Order)**.** A *strict partial order* is an irreflexive and transitive relation.

Remark (Dependencies).

- Irreflexive Relation
- Transitive Relation

Definition (Total Order)

Definition (Total Order)**.** A *total order* is a partial order whose relation is total.

Definition (Strict Total Order)

Definition 17.9.2 (Strict Total Order)**.** A *strict total order* is a strict partial order satisfying trichotomy.

Remark (Dependencies).

- Strict Partial Order
- Trichotomy

Definition (Well-Order)

Definition (Well-Order)**.** A *well-order* is a total order in which every nonempty subset has a least element.

17.10 Structure-Preserving Maps

Definition (Homomorphism)

Definition 17.10.1 (Homomorphism)**.** A map f is a *homomorphism* for an operation O when

$$f(x\,O\,y) = f(x)\,O'\,f(y).$$

For ring-like structures it preserves both addition and multiplication, and preserves 1 when the language includes a distinguished multiplicative identity.

Remark (Dependencies).

- Function
- Binary Arithmetic Operation

Definition (Injective Homomorphism)

Definition 17.10.2 (Injective Homomorphism)**.** An *injective homomorphism* is a homomorphism whose underlying function is injective.

Remark (Dependencies).

- Homomorphism
- Injective Function

Definition (Order Embedding)

Definition 17.10.3 (Order Embedding)**.**

$$xRy \iff f(x)R'f(y).$$

Remark (Dependencies).

- Function
- Relation
- Order Embedding

Definition (Structure Embedding)

Definition 17.10.4 (Structure Embedding)**.** A *structure embedding* is an injective homomorphism that also embeds the relevant order relation.

Remark (Dependencies).

- Injective Homomorphism
- Order Embedding

Proposition (Composition of Homomorphisms)

Proposition 17.10.5 (Composition of Homomorphisms)**.** *The composition of two homomorphisms is a homomorphism.*

Remark (Dependencies).

- Homomorphism

Corollary (Identity Map Is a Homomorphism)

Corollary 17.10.6 (Identity Map Is a Homomorphism)**.** *The identity map on a structure is a homomorphism from that structure to itself.*

Remark (Dependencies).

- Homomorphism

17.11 Algebraic Structures

17.12 Field and Ordered-Field Laws

Chapter 18

Constructing a Number System

18.1 Ground Rules

Toolkit: Construction Obligations

- **Carrier:** the underlying collection of objects.
- **Operations:** the arithmetic rules defined on the carrier.
- **Relations:** equality, equivalence, order, or congruence relations.
- **Verification:** closure, well-definedness, laws, and embeddings.

Construction Obligations

Remark (Exposition). A number-system construction is a structured promise. The construction must say what the objects are, how arithmetic is performed, which objects are identified, and which algebraic or order laws the resulting system satisfies.

Definition

Definition 18.1.1 (Number-System Construction Target)**.** A *number-system construction target* is the specified algebraic and order structure that a proposed system is meant to satisfy. The target fixes the required carrier, operations, relations, distinguished elements, and laws.

Remark (Standard quantified statement). A construction target specifies the structure and laws the new arithmetic system is intended to satisfy.

Remark (Interpretation). The target determines the proof obligations; without it, the phrase number system has no fixed mathematical content.

Remark (Dependencies).

Definition

Definition 18.1.2 (Construction Data)**.** The *construction data* for a number system consists of the raw carrier objects, the intended equality or equivalence relation, the distinguished elements, and the operation formulas from which the system is built.

Remark (Standard quantified statement). Construction data specifies the raw objects, identifications, distinguished elements, and operation formulas used in the construction.

Remark (Interpretation). The data is the material being tested against the target; it is not yet a verified number system.

Remark (Dependencies).

- Number-System Construction Target

Definition

Definition 18.1.3 (Construction Verification)**.** *Construction verification* is the proof that the construction data actually satisfies the construction target. It includes, as applicable, closure of operations, equivalence-relation proofs, well-definedness of operations on classes, algebraic laws, order laws, and compatibility of embeddings with earlier systems.

Remark (Standard quantified statement). Construction verification proves that the construction data satisfies the chosen construction target.

Remark (Interpretation). Verification is the step that turns proposed arithmetic into a justified structure.

Remark (Dependencies).

- Number-System Construction Target
- Construction Data

18.2 Finite Modular Number Systems

Toolkit: Residue Class Arithmetic

- **Modulus:** a positive integer n.
- **Congruence:** $a \equiv b \pmod{n}$ when $n \mid (a - b)$.
- **Elements:** equivalence classes of integers modulo n.
- **Operations:** class addition and multiplication inherited from $\mathbb{Z}$.

Congruence and Residue Classes

Remark (Exposition). Modular arithmetic is the quotient version of the two-element example. The carrier is obtained by identifying integers that have the same remainder modulo a fixed positive integer.

Definition

Definition 18.2.1 (Congruence Modulo an Integer). Let $n \in \mathbb{Z}$ be positive. For $a, b \in \mathbb{Z}$, define

$$a \equiv b \pmod{n} \quad :\Longleftrightarrow \quad n \mid (a - b).$$

Remark (Standard quantified statement). For positive integer n, two integers are congruent modulo n exactly when n divides their difference.

Remark (Interpretation). Congruence modulo n records sameness of remainder using divisibility in $\mathbb{Z}$.

Remark (Dependencies).

- Integers
- Positive Integer
- Divisibility in $\mathbb{Z}$
- Subtraction on $\mathbb{Z}$

Theorem

Theorem 18.2.2 (Congruence Modulo an Integer Is an Equivalence Relation). *For each positive integer n, congruence modulo n is an equivalence relation on $\mathbb{Z}$.*

Remark (Standard quantified statement). For each positive integer n, congruence modulo n is reflexive, symmetric, and transitive on $\mathbb{Z}$.

Remark (Interpretation). This permits the quotient set $\mathbb{Z}/n\mathbb{Z}$ to be formed from congruence classes.

Remark (Dependencies).

- Congruence Modulo an Integer
- Divisibility in $\mathbb{Z}$

Definition

Definition 18.2.3 (Integers Modulo n)**.** Let $n \in \mathbb{Z}$ be positive. The *integers modulo n* are the quotient set

$$\mathbb{Z}/n\mathbb{Z} := \mathbb{Z}/\equiv \pmod{n}.$$

Remark (Standard quantified statement)*.* For positive integer n, $\mathbb{Z}/n\mathbb{Z}$ is the quotient of $\mathbb{Z}$ by congruence modulo n.

Remark (Interpretation)*.* The elements are not single integers but equivalence classes of integers.

Remark (Dependencies).

- Quotient Set
- Integers
- Congruence Modulo an Integer

Definition

Definition 18.2.4 (Residue Class Modulo n)**.** For $a \in \mathbb{Z}$, the *residue class of a modulo n* is

$$[a]_n := \{b \in \mathbb{Z} : b \equiv a \pmod{n}\}.$$

Remark (Standard quantified statement)*.* The set $[a]_n$ contains exactly the integers congruent to a modulo n.

Remark (Interpretation)*.* Each such set is one element of the quotient number system.

Remark (Dependencies).

- Integers Modulo n
- Equivalence Class
- Congruence Modulo an Integer

Operations

Definition

Definition 18.2.5 (Addition Modulo n)**.** For residue classes modulo a positive integer n, define

$$[a]_n + [b]_n := [a + b]_n.$$

Remark (Standard quantified statement)*.* The sum $[a]_n + [b]_n$ is $[a + b]_n$.

Remark (Interpretation)*.* This formula becomes a valid operation on classes only after well-definedness is proved.

Remark (Dependencies).

- Residue Class Modulo n
- Addition on $\mathbb{Z}$-Classes

Definition

Definition 18.2.6 (Multiplication Modulo n)**.** For residue classes modulo a positive integer n, define

$$[a]_n \cdot [b]_n := [ab]_n.$$

Remark (Standard quantified statement)*.* The product $[a]_n \cdot [b]_n$ is $[ab]_n$.

Remark (Interpretation)*.* This formula inherits multiplication from $\mathbb{Z}$, subject to the same well-definedness obligation.

Remark (Dependencies).

- Residue Class Modulo n
- Multiplication on $\mathbb{Z}$-Classes

Theorem

Theorem 18.2.7 (Modular Addition and Multiplication Are Well-Defined)**.** *For each positive integer n, addition and multiplication modulo n are well-defined operations on $\mathbb{Z}/n\mathbb{Z}$.*

Remark (Standard quantified statement)*.* For each positive integer n, the representative formulas for modular addition and multiplication determine unique outputs.

Remark (Interpretation)*.* Changing representatives does not change the output, so the operations belong to $\mathbb{Z}/n\mathbb{Z}$ itself.

Remark (Dependencies).

- Addition Modulo n

- Multiplication Modulo n
- Congruence Modulo an Integer

Definition

Definition 18.2.8 (Finite Modular Number System)**.** For a positive integer n, the *finite modular number system modulo* n is

$$\mathbb{Z}/n\mathbb{Z}$$

with addition and multiplication modulo n, additive identity $[0]_n$, and multiplicative identity $[1]_n$.

Remark (Standard quantified statement)*.* For positive integer n, the finite modular number system modulo n is $\mathbb{Z}/n\mathbb{Z}$ with modular addition, modular multiplication, $[0]_n$, and $[1]_n$.

Remark (Interpretation)*.* This is the quotient arithmetic system obtained by reducing integer arithmetic modulo n.

Remark (Dependencies)*.*

- Integers Modulo n
- Addition Modulo n
- Multiplication Modulo n
- Zero in $\mathbb{Z}$
- One in $\mathbb{Z}$

Basic Example

Theorem

Theorem 18.2.9 ($\mathbb{Z}/4\mathbb{Z}$ Has Zero Divisors)**.** *In $\mathbb{Z}/4\mathbb{Z}$, the nonzero class $[2]_4$ satisfies*

$$[2]_4 \cdot [2]_4 = [0]_4.$$

Therefore $\mathbb{Z}/4\mathbb{Z}$ is not a field.

Remark (Standard quantified statement)*.* In $\mathbb{Z}/4\mathbb{Z}$, $[2]_4 \neq [0]_4$ and $[2]_4 \cdot [2]_4 = [0]_4$.

Remark (Interpretation)*.* This shows that finite modular number systems need not be fields.

Remark (Dependencies)*.*

- Finite Modular Number System
- Multiplication Modulo n

18.3 The $\{0,1\}$ Number System

Toolkit: The System with Two Elements

- **Carrier:** the set $\{0,1\}$.
- **Addition:** addition with $1+1=0$.
- **Multiplication:** multiplication with $1 \cdot 1 = 1$.
- **Purpose:** a complete finite model of the construction checklist.

The Carrier and Operations

Definition

Definition 18.3.1 (Two-Element Carrier)**.** The *two-element carrier* is the set

$$\mathbb{B}_2 := \{0,1\}.$$

Remark (Standard quantified statement). The carrier of the two-element system is the set containing exactly the integer elements 0 and 1.

Remark (Interpretation). This fixes the underlying finite set before any arithmetic is assigned to it.

Remark (Dependencies).

- Zero in $\mathbb{Z}$
- One in $\mathbb{Z}$

Definition

Definition 18.3.2 (Addition on the Two-Element System)**.** Addition on $\mathbb{B}_2$ is the binary operation $+_2$ determined by

$+_2$	0	1
0	0	1
1	1	0

Remark (Standard quantified statement). Addition on the two-element system is the binary operation whose values are given by the displayed table.

Remark (Interpretation). The table is the definition of addition; in particular, 1 plus 1 equals 0 in this system.

Remark (Dependencies).

- Two-Element Carrier

Definition

Definition 18.3.3 (Multiplication on the Two-Element System)**.** Multiplication on $\mathbb{B}_2$ is the binary operation $\cdot_2$ determined by

$$\begin{array}{c|cc} \cdot_2 & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}$$

Remark (Standard quantified statement). Multiplication on the two-element system is the binary operation whose values are given by the displayed table.

Remark (Interpretation). The table is the definition of multiplication; 1 is multiplicative identity and 0 is absorbing.

Remark (Dependencies).

- Two-Element Carrier

Definition

Definition 18.3.4 (Two-Element Number System)**.** The *two-element number system* is

$$\mathbb{F}_2 := (\mathbb{B}_2, +_2, \cdot_2, 0, 1).$$

Remark (Standard quantified statement). The two-element number system is the carrier $\mathbb{B}_2$ equipped with its addition, multiplication, zero, and one.

Remark (Interpretation). This packages the carrier and operations as the arithmetic structure to be verified.

Remark (Dependencies).

- Two-Element Carrier
- Addition on the Two-Element System
- Multiplication on the Two-Element System

Verification

Theorem

Theorem 18.3.5 (The Two-Element System Is a Field)**.** *The two-element number system $\mathbb{F}_2$ is a field.*

Remark (Standard quantified statement). The structure $\mathbb{F}_2$ satisfies the field axioms.

Remark (Interpretation). The two tables make the smallest field visible by direct finite verification.

Remark (Dependencies).

- Two-Element Number System

Theorem

Theorem 18.3.6 (The Two-Element System Is Not an Ordered Field)**.** *There is no order on $\mathbb{B}_2$ that makes $\mathbb{F}_2$ an ordered field.*

Remark (Standard quantified statement)*.* No order on $\mathbb{B}_2$ makes $\mathbb{F}_2$ satisfy the ordered-field axioms.

Remark (Interpretation)*.* The equation $1 + 1 = 0$ is compatible with field arithmetic but incompatible with ordered-field arithmetic.

Remark (Dependencies)*.*

- The Two-Element System Is a Field

As Modular Arithmetic

Theorem

Theorem 18.3.7 ($\mathbb{Z}/2\mathbb{Z}$ Is the Two-Element System)**.** *The finite modular number system $\mathbb{Z}/2\mathbb{Z}$ has exactly two classes, $[0]_2$ and $[1]_2$, and its addition and multiplication tables are the tables of $\mathbb{F}_2$.*

Remark (Standard quantified statement)*.* $\mathbb{Z}/2\mathbb{Z}$ has the same carrier size and operation tables as $\mathbb{F}_2$.

Remark (Interpretation)*.* The two-element table system is the first modular arithmetic system.

Remark (Dependencies)*.*

- Finite Modular Number System
- Two-Element Number System
- Modular Addition and Multiplication Are Well-Defined

Chapter 19

Peano Systems

SERIES
From Cantor to Ito

VOLUME II
Origins of Numbers

BOOK I
Discrete Number Systems

CHAPTER 5
Peano Systems

19.1 Presburger Arithmetic

Definition (Presburger Signature)

Definition 19.1.1 (Presburger Signature)**.** The *Presburger signature* is the first-order signature

$$\mathcal{L}_{\text{Pres}} = \{0, S, +, <, =\},$$

where 0 is a constant symbol, S is a unary function symbol, $+$ is a binary function symbol, and $<$ and $=$ are binary relation symbols.

Remark (Standard quantified statement)*.*

$$\text{PresburgerSignature}(\mathcal{L}) \iff \mathcal{L} = \{0, S, +, <, =\} \wedge \text{ar}(S) = 1 \wedge \text{ar}(+) = 2 \wedge \text{ar}(<) = 2 \wedge \text{ar}(=) = 2.$$

Remark (Interpretation)*.* The Presburger signature names addition and order, but it deliberately omits multiplication as a function symbol. This makes Presburger arithmetic a theory of the additive structure of the natural numbers rather than a theory of full arithmetic.

Definition (Presburger Arithmetic)

Definition 19.1.2 (Presburger Arithmetic)**.** *Presburger arithmetic* is the first-order theory of the natural numbers in the Presburger signature. Its intended structure is

$$(\mathbb{N}, 0, S, +, <, =),$$

and its axioms describe the behavior of zero, successor, addition, order, and induction for formulas written in the language $\mathcal{L}_{\text{Pres}}$.

Remark (Standard quantified statement)*.*

$$\text{PresburgerArithmetic}(T) \iff T \subseteq \text{Sentences}(\mathcal{L}_{\text{Pres}}) \wedge T \text{ axiomatizes } (\mathbb{N}, 0, S, +, <, =).$$

Remark (Interpretation)*.* Presburger arithmetic isolates the part of arithmetic governed by addition. It is strong enough to express linear equations, congruences, and order properties of natural numbers, but it cannot directly express multiplication as a primitive operation. The later Peano-system material supplies the more general recursive framework from which addition, multiplication, order, and induction are developed structurally.

19.2 Defining Peano Systems

defining-peano-systems

Peano Systems → **Peano Axioms** → Recursion on Peano Systems

The Peano Axioms

The Peano axioms make precise what it means for a system to behave like the natural numbers. They isolate the first element, the successor map, and the induction principle. These axioms do not define addition, multiplication, or order. They fix the structure that supports those later constructions. The formulation used here belongs to the Dedekind–Peano line of axiomatizing the natural-number structure [Dedekind, 1901, Peano, 1889].

Definition (Peano System)

Definition 19.2.1 (Peano System)**.** A triple $(P, S, 1)$ is called a *Peano system* exactly when P is a set, $1 \in P$, $S : P \to P$ is a function, and the following axioms hold:

P1. $1 \in P$.

P2. For every $n \in P$, $S(n) \in P$.

P3. For every $n \in P$, $S(n) \neq 1$.

P4. For all $m, n \in P$, if $S(m) = S(n)$ then $m = n$.

P5. If $A \subseteq P$, $1 \in A$, and

$$\forall n \in P \ (n \in A \Longrightarrow S(n) \in A),$$

then $A = P$.

Remark (Standard quantified statement).

$$\begin{aligned}
&(P, S, 1) \text{ is a Peano system} \\
\iff &\Big(P \text{ is a set} \wedge 1 \in P \wedge S : P \to P \\
&\wedge \forall n \in P \ (S(n) \in P) \\
&\wedge \forall n \in P \ (S(n) \neq 1) \\
&\wedge \forall m, n \in P \ (S(m) = S(n) \Longrightarrow m = n) \\
&\wedge \forall A \subseteq P \ \big((1 \in A \wedge \forall n \in P \ (n \in A \Longrightarrow S(n) \in A)) \Longrightarrow A = P\big)\Big).
\end{aligned}$$

Remark (Failure modes). A triple $(P, S, 1)$ fails to be a Peano system if the distinguished element does not lie in the underlying set, if the successor map leaves the set, if the successor hits 1, if the successor identifies two distinct elements, or if there exists a proper subset of P that contains 1 and is closed under successor.

Remark (Failure mode decomposition).

$$\begin{aligned}
\neg\,(P, S, 1) \text{ is a Peano system} \Longrightarrow\ & 1 \notin P \\
&\vee \exists n \in P \ (S(n) \notin P) \\
&\vee \exists n \in P \ (S(n) = 1) \\
&\vee \exists m, n \in P \ (m \neq n \wedge S(m) = S(n)) \\
&\vee \exists A \subseteq P \ \big(1 \in A \wedge \forall n \in P \ (n \in A \Longrightarrow S(n) \in A) \wedge A \neq P\big).
\end{aligned}$$

Remark (Interpretation). A Peano system is the abstract data of counting: an underlying set, a first element, and a successor operation satisfying the usual non-collision and induction principles. The purpose of the definition is to isolate the ambient object before the individual axioms are recorded and used one by one.

Remark (Source alignment). The five displayed clauses are the house-style Peano-system presentation of Peano's axiomatization of arithmetic [Peano, 1889]. They are stated here in structural form so that later definitions and proofs can refer directly to the successor and induction clauses.

Remark (Comparison with Feferman). This definition corresponds closely to Definition 3.1 in Feferman [1964], where Peano systems are introduced as the abstract structural framework underlying the positive integers. The present notes expand the definition into explicit quantified, failure-mode, and decomposition forms in order to support later logical analysis and dependency extraction.

Remark (Dependencies).

- Function
- Subset

Axiom (Base Element Lies in the Underlying Set)

Axiom 19.2.2 (Distinguished Element Lies in the Underlying Set)**.**

$$1 \in P.$$

Remark (Standard quantified statement).

$$1 \in P.$$

Remark (Interpretation). The counting system begins from a distinguished first element of the underlying set. This is the base point from which the remaining elements are generated by the successor operation.

Axiom (Closure Under Successor)

Axiom 19.2.3 (Closure Under Successor)**.** If $n \in P$, then $S(n) \in P$.

Remark (Standard quantified statement).

$$\forall n \in P \ (S(n) \in P).$$

Remark (Interpretation). Applying the successor operation to an element of the underlying set never leaves that set. Successor is therefore an internal operation on the counting system.

Axiom (Base Element Is Not a Successor)

Axiom 19.2.4 (Distinguished Element Is Not a Successor)**.** For every $n \in P$, $S(n) \neq 1$.

Remark (Standard quantified statement).

$$\forall n \in P \ (S(n) \neq 1).$$

Remark (Interpretation). The first natural number does not arise by taking the successor of an earlier natural number. This keeps the counting system from folding backward onto its starting point.

Axiom (Injectivity of Successor)

Axiom 19.2.5 (Injectivity of Successor)**.** If $m, n \in P$ and $S(m) = S(n)$, then $m = n$.

Remark (Standard quantified statement).

$$\forall m, n \in P \ (S(m) = S(n) \implies m = n).$$

Remark (Interpretation). Different natural numbers do not collapse to the same successor. The successor operation preserves distinctness and lets predecessor reasoning work whenever a successor is known.

Axiom (Induction Axiom)

Axiom 19.2.6 (Induction Axiom)**.** Let $A \subseteq P$. If $1 \in A$ and if

$$\forall n \in P \ (n \in A \implies S(n) \in A),$$

then $A = P$.

Remark (Standard quantified statement).

$$\forall A \subseteq P \ \Big(\big(1 \in A \land \forall n \in P \ (n \in A \implies S(n) \in A)\big) \implies A = P \Big).$$

Remark (Interpretation). Any subset of the underlying set that contains the distinguished element and is closed under successor must already contain the entire set. This axiom rules out rogue elements and is the logical source of mathematical induction.

Axiom (Existence of a Peano System)

Axiom 19.2.7 (Existence of a Peano System)**.** There exists a Peano system.

Remark (Standard quantified statement).

$$\exists P \ \exists S \ \big((P, S, 1) \text{ is a Peano system}\big).$$

Remark (Interpretation). The Peano axioms are not merely a formal pattern; at least one structure satisfies them. This axiom guarantees that the abstract counting system is available before we give it the conventional name $\mathbb{N}$.

Remark (Historical note). This axiom corresponds to Axiom 3.2 of Feferman [1964], where the existence of at least one Peano system is taken as a foundational assumption.

Remark (Dependencies).

- Peano System
- Language

First Theorems on Peano Systems

Induction as a Working Principle

The induction axiom for a Peano system is formulated in terms of inductive subsets of the underlying set. Arithmetic arguments are usually written in terms of predicates rather than subsets. This theorem identifies the predicate-form induction principle derived from the Peano axioms. It is the standard form of induction used in the rest of the chapter. Historically, this is the working proof principle attached to Peano's axiomatization of the natural numbers [Peano, 1889].

Theorem (Induction Principle for a Peano System)

Theorem 19.2.8 (Induction Principle for a Peano System)**.** *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\varphi(1) \qquad \textit{and} \qquad \forall n \in P\ (\varphi(n) \Longrightarrow \varphi(S(n))),$$

then

$$\forall n \in P\ \varphi(n).$$

Go to proof.

Lean 4 Verification Record

Source anchor: Peano, *Arithmetices principia*; Feferman, *The Number Systems*, Section 3.1.
Formalizes: Theorem 19.2.8.
Lean module: `LRA.VolumeII.PeanoSystems.Induction`.
Lean declaration: `induction_principle`.
Status: checked against `lra-lean`.

```
theorem induction_principle
    (ps : PeanoSystem)
    (predicate : ps.carrier -> Prop)
    (base_case : predicate ps.one)
    (successor_step :
      forall element : ps.carrier,
        predicate element ->
        predicate (ps.successor element)) :
    forall element : ps.carrier,
      predicate element :=
  ps.induction predicate base_case successor_step
```

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system and let φ be a predicate on P.

$$\Big(\varphi(1) \wedge \forall n \in P\ (\varphi(n) \Longrightarrow \varphi(S(n)))\Big) \Longrightarrow \forall n \in P\ \varphi(n).$$

Remark (Contrapositive quantified statement).

Let $(P, S, 1)$ be a Peano system and let φ be a predicate on P.

$$\neg\forall n \in P\ \varphi(n) \implies \neg\Big(\varphi(1) \wedge \forall n \in P\ (\varphi(n) \implies \varphi(S(n)))\Big).$$

Remark (Interpretation). This theorem converts the subset form of induction into the predicate form used in ordinary arithmetic arguments. A property propagates through the entire Peano system once it holds at the distinguished first element and is preserved by successor. The theorem therefore turns the structural induction axiom into a proof principle for arbitrary predicates on P.

Remark (Comparison with Feferman). Peano's axiomatization includes induction as part of the natural-number structure [Peano, 1889]. Feferman derives the working form of mathematical induction from the subset formulation of the Peano axioms immediately after Theorem 3.3 in Chapter 3 of Feferman [1964]. The present notes formalize that induction principle as an explicit reusable theorem.

Remark (Dependencies).

- Peano System
- Induction Axiom

Strong Induction

Ordinary induction propagates a property from each element to its immediate successor. Strong induction allows the inductive step to use the property at every predecessor, not just the immediate one. This broader hypothesis makes strong induction convenient when the successor step needs information from the entire earlier segment of the Peano system.

Theorem (Strong Induction on a Peano System)

Theorem 19.2.9 (Strong Induction on a Peano System). *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\forall n \in P\ \big(\forall k \in P\ (k < n \implies \varphi(k))\big) \implies \varphi(n),$$

then

$$\forall n \in P\ \varphi(n).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system and let φ be a predicate on P.

$$\Big(\forall n \in P\ \big(\forall k \in P\ (k < n \implies \varphi(k))\big) \implies \varphi(n)\Big) \implies \forall n \in P\ \varphi(n).$$

Remark (Interpretation). Strong induction grants the inductive step access to the entire history below n, not just the single predecessor. It is the proof form behind least counterexample arguments.

Remark (Dependencies).

- Induction Principle for a Peano System
- Strict Less Than on a Peano System
- Trichotomy on a Peano System

Inductive Subsets and Predecessors

Induction is expressed in terms of subsets of a Peano system. Two structural notions appear repeatedly in that setting: closure under successor and membership generated from the distinguished first element. Predecessor language is also convenient once it is known that every non-initial element arises as a successor.

Definition (Successor-Closed Subset)

Definition 19.2.10 (Successor-Closed Subset)**.** Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *successor-closed* exactly when

$$\forall n \in P\ (n \in A \Longrightarrow S(n) \in A).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P.\\ &A \text{ is successor-closed} \Longleftrightarrow \forall n \in P\ (n \in A \Longrightarrow S(n) \in A).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P.\\ &A \text{ is not successor-closed} \Longleftrightarrow \exists n \in P\ (n \in A \wedge S(n) \notin A).\end{aligned}$$

Remark (Failure modes). A subset fails to be successor-closed exactly when it contains some element of P but fails to contain that element's successor.

Remark (Failure mode decomposition).

$$A \text{ is not successor-closed} \Longleftrightarrow \exists n \in P\ \underbrace{(n \in A \wedge S(n) \notin A)}_{\text{closure fails at } n}.$$

Remark (Interpretation). A successor-closed subset is stable under one step of counting. Once an element of A is present, its successor remains in A as well.

Remark (Comparison with Feferman). The notion of a successor-closed subset is implicit in condition (iii) of Feferman's Definition 3.1 [Feferman, 1964]. The present notes isolate it as a separate definition so that induction arguments can refer to the closure condition directly.

Remark (Dependencies).

- Peano System

Definition (Inductive Subset of a Peano System)

Definition 19.2.11 (Inductive Subset of a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *inductive* exactly when $1 \in A$ and A is successor-closed.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P. \\ &A \text{ is inductive} \iff \big(1 \in A \land \forall n \in P\ (n \in A \implies S(n) \in A)\big). \end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned} &\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P. \\ &A \text{ is not inductive} \iff \big(1 \notin A \lor \exists n \in P\ (n \in A \land S(n) \notin A)\big). \end{aligned}$$

Remark (Failure modes). A subset fails to be inductive in exactly two ways: it may fail to contain the distinguished first element, or it may contain some element while failing to contain that element's successor.

Remark (Failure mode decomposition).

$$\begin{aligned} A \text{ is not inductive} \iff \quad & \underbrace{1 \notin A}_{\text{base point missing}} \\ & \lor \underbrace{\exists n \in P\ (n \in A \land S(n) \notin A)}_{\text{successor closure fails}}. \end{aligned}$$

Remark (Interpretation). An inductive subset contains the distinguished first element and is closed under successor. Such a subset contains every stage obtained by starting at 1 and applying successor repeatedly.

Remark (Comparison with Feferman). The notion of an inductive subset packages the two hypotheses appearing in the induction clause of Feferman's Definition 3.1 [Feferman, 1964]. The present notes name this package so that the internal structure of induction proofs remains explicit.

Remark (Dependencies).

- Peano System

- Successor-Closed Subset

Theorem (Minimality of a Peano System)

Theorem 19.2.12 (Minimality of a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. If $A \subseteq P$ contains 1 and is closed under successor, then $A = P$.*

$$\big(1 \in A \wedge \forall n \in P\ (n \in A \Longrightarrow S(n) \in A)\big) \Longrightarrow A = P.$$

Go to proof.

Remark (Interpretation)*.* A Peano system has no elements outside the successor chain generated by its base element. Any subset that contains the base element and is closed under successor already exhausts the whole carrier.

Remark (Dependencies)*.*

- Induction Axiom
- Inductive Subset of a Peano System

Definition (Predecessor in a Peano System)

Definition 19.2.13 (Predecessor in a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $x, u \in P$. The element u is called a *predecessor* of x exactly when

$$S(u) = x.$$

Remark (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } (P, S, 1) \text{ be a Peano system and let } x, u \in P.\\ u \text{ is a predecessor of } x \Longleftrightarrow S(u) = x.\end{gathered}$$

Remark (Negated quantified statement)*.*

$$\begin{gathered}\text{Let } (P, S, 1) \text{ be a Peano system and let } x, u \in P.\\ u \text{ is not a predecessor of } x \Longleftrightarrow S(u) \neq x.\end{gathered}$$

Remark (Interpretation)*.* A predecessor of x is an element whose successor is x. This language isolates the inverse-looking question attached to the successor map without assuming that successor is globally invertible.

Remark (Dependencies)*.*

- Peano System

Generation by Successor

The Peano axioms describe a counting system generated from a distinguished first element by the successor operation. The next theorems make that generation principle explicit: every element is either the first element or a successor, every non-initial element has a unique predecessor, and no element can be its own successor.

Theorem (Every Element Is Either 1 or a Successor)

Theorem 19.2.14 (Every Element Is Either 1 or a Successor)**.** *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, either $x = 1$ or there exists $u \in P$ such that $S(u) = x$.* *Go to proof.*

Lean 4 Verification Record

Source anchor: Landau, *Foundations of Analysis*, Section 1; Feferman, *The Number Systems*, Theorem 3.3.
Formalizes: Theorem 19.2.14.
Lean module: `LRA.VolumeII.PeanoSystems.BasicTheorems`.
Lean declaration: `every_element_is_one_or_successor`.
Status: checked against `lra-lean`.

```
theorem every_element_is_one_or_successor
    (ps : PeanoSystem) :
    forall element : ps.carrier,
      Or (element = ps.one)
        (Exists (fun predecessor : ps.carrier =>
          ps.successor predecessor = element)) := by
  let D : ps.carrier -> Prop :=
    fun candidate_element =>
      Or (candidate_element = ps.one)
        (Exists (fun predecessor : ps.carrier =>
          ps.successor predecessor = candidate_element))

  apply induction_principle ps D

  case base_case =>
    exact Or.inl rfl

  case successor_step =>
    intro element induction_hypothesis
    exact Or.inr (Exists.intro element rfl)
```

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall x \in P \; \big(x = 1 \lor \exists u \in P \; (S(u) = x)\big).$$

Remark (Interpretation). Every element of a Peano system is reached by starting at the distinguished first element and moving forward by successor. The only possible exception to having a predecessor is the element 1 itself.

Remark (Historical note). This theorem is one of the basic successor consequences of Peano's axiomatization [Peano, 1889]. In the present organization it corresponds directly to Theorem 3.3 in Feferman [1964]. Feferman uses this result to show that the distinguished initial element is the only possible non-successor in a Peano system.

Remark (Dependencies).

- Peano System
- Distinguished Element Lies in the Underlying Set
- Closure Under Successor
- Induction Principle for a Peano System

Theorem (Successor Inequality Reflection)

Theorem 19.2.15 (Successor Inequality Reflection)**.** *Let* $(P, S, 1)$ *be a Peano system. For all* $a, b \in P$*, if* $a \neq b$*, then* $S(a) \neq S(b)$*. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall a, b \in P\ (a \neq b \implies S(a) \neq S(b)).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\exists a, b \in P\ (a \neq b \wedge S(a) = S(b)).\end{aligned}$$

Remark (Failure modes). Successor inequality reflection fails exactly when two distinct elements have the same successor.

Remark (Failure mode decomposition).

$$\exists a, b \in P \qquad \underbrace{(a \neq b \wedge S(a) = S(b))}_{\text{distinct elements with a common successor}} \qquad .$$

Remark (Contrapositive quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall a, b \in P\ (S(a) = S(b) \implies a = b).\end{aligned}$$

Remark (Interpretation). The successor operation reflects inequality: distinct inputs remain distinct after one successor step. This is the contrapositive form of successor injectivity, and it is the form used when an induction step must preserve a negative equality statement.

Remark (Dependencies).

- Peano System

- Injectivity of Successor

Corollary (Non-One Elements Have a Predecessor)

Corollary 19.2.16 (Non-One Elements Have a Predecessor)**.** *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, if $x \neq 1$, then there exists $u \in P$ such that $S(u) = x$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\ &\forall x \in P\ \big(x \neq 1 \Longrightarrow \exists u \in P\ (S(u) = x)\big).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\ &\exists x \in P\ \big(x \neq 1 \land \forall u \in P\ (S(u) \neq x)\big).\end{aligned}$$

Remark (Failure modes). The statement fails exactly when some element distinct from 1 has no predecessor.

Remark (Failure mode decomposition).

$$\exists x \in P\ \underbrace{\big(x \neq 1 \land \forall u \in P\ (S(u) \neq x)\big)}_{\text{non-initial element with no predecessor}}.$$

Remark (Interpretation). Every non-initial element of a Peano system is reached by one successor step from an earlier element. This separates predecessor existence from predecessor uniqueness so that later arguments can cite only the part they need.

Remark (Comparison with Feferman). Feferman's Theorem 3.3 in Feferman [1964] contains this predecessor-existence fact as part of the structural analysis of Peano systems. The present notes record it separately before adding uniqueness.

Remark (Dependencies).

- Peano System
- Every Element Is Either 1 or a Successor

Corollary (Predecessor Existence and Uniqueness Away from 1)

Corollary 19.2.17 (Predecessor Existence and Uniqueness Away from 1)**.** *Let $(P, S, 1)$ be a Peano system, and let $x \in P$ with $x \neq 1$. Then there exists a unique $u \in P$ such that $S(u) = x$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\ &\forall x \in P\ \big(x \neq 1 \Longrightarrow \exists! u \in P\ (S(u) = x)\big).\end{aligned}$$

Remark (Interpretation). Every element distinct from 1 has exactly one predecessor. The successor map therefore organizes the non-initial part of a Peano system into one-step extensions of earlier elements.

Remark (Comparison with Feferman). Feferman proves predecessor existence and uniqueness together inside Theorem 3.3 of Feferman [1964]. The present notes separate the uniqueness statement into an independent corollary so that it can be cited directly in later proofs.

Remark (Dependencies).

- Peano System
- Non-One Elements Have a Predecessor
- Injectivity of Successor

Corollary (Unique Predecessor Characterization Away from 1)

Corollary 19.2.18 (Unique Predecessor Characterization Away from 1)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $x \in P$*,* $x \neq 1$ *if and only if there exists a unique* $u \in P$ *such that* $S(u) = x$*.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall x \in P\ \big(x \neq 1 \iff \exists! u \in P\ (S(u) = x)\big).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\exists x \in P\ \Big(\big(x \neq 1 \wedge (\forall u \in P\ (S(u) \neq x)\\&\qquad\qquad \vee \exists u, v \in P\ (S(u) = x \wedge S(v) = x \wedge u \neq v))\big)\\&\qquad \vee \big(x = 1 \wedge \exists u \in P\ (S(u) = x \wedge \forall v \in P\ (S(v) = x \implies v = u))\big)\Big).\end{aligned}$$

Remark (Failure modes). The biconditional can fail in two ways: a non-initial element can fail to have a unique predecessor, or the distinguished first element can have a unique predecessor.

Remark (Failure mode decomposition).

$$\begin{aligned}\exists x \in P\ \Big(&\underbrace{\big(x \neq 1 \wedge (\forall u \in P\ (S(u) \neq x) \vee \exists u, v \in P\ (S(u) = x \wedge S(v) = x \wedge u \neq v))\big)}_{\text{non-initial element lacks a unique predecessor}}\\&\vee \underbrace{\big(x = 1 \wedge \exists u \in P\ (S(u) = x \wedge \forall v \in P\ (S(v) = x \implies v = u))\big)}_{\text{the first element has a unique predecessor}}\Big).\end{aligned}$$

Remark (Interpretation)*.* Being different from the distinguished first element is exactly the same as having a unique predecessor. This turns the informal phrase "away from 1" into an internal characterization of the successor structure.

Remark (Comparison with Feferman)*.* This corollary packages the predecessor analysis surrounding Feferman's Theorem 3.3 in Feferman [1964]. The forward direction is the predecessor existence-and-uniqueness result, while the reverse direction uses the axiom that 1 is not a successor.

Remark (Dependencies)*.*

- Peano System
- Distinguished Element Is Not a Successor
- Predecessor Existence and Uniqueness Away from 1

Corollary (The Distinguished Element Is the Unique Non-Successor)

Corollary 19.2.19 (The Distinguished Element Is the Unique Non-Successor)**.** *Let $(P, S, 1)$ be a Peano system. An element $x \in P$ is not a successor of any element of P if and only if $x = 1$.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall x \in P \left(\neg\exists u \in P\ (S(u) = x) \iff x = 1\right).$$

Remark (Interpretation)*.* The distinguished first element is characterized internally as the unique element with no predecessor. This gives a structural description of 1 that does not depend on external notation.

Remark (Comparison with Feferman)*.* This corollary extracts a structural characterization of the distinguished initial element that is implicit in the discussion surrounding Theorem 3.3 of Feferman [1964].

Remark (Dependencies)*.*

- Distinguished Element Is Not a Successor
- Predecessor Existence and Uniqueness Away from 1

Theorem (No Object Is Its Own Successor)

Theorem 19.2.20 (No Object Is Its Own Successor)**.** *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$S(n) \neq n.$$

Go to proof.

Lean 4 Verification Record

Source anchor: Landau, *Foundations of Analysis*, Section 2, Theorem 2.
Formalizes: Theorem 19.2.20.
Lean module: `LRA.VolumeII.PeanoSystems.BasicTheorems`.
Lean declaration: `successor_not_self`.
Status: checked against `lra-lean`.

```
theorem successor_not_self
    (ps : PeanoSystem) :
    forall element : ps.carrier,
      Not (ps.successor element = element) := by
  apply induction_principle ps

  case base_case =>
    exact ps.one_not_successor ps.one

  case successor_step =>
    intro element induction_hypothesis
    exact
      successor_preserves_inequality
        ps
        (ps.successor element)
        element
        induction_hypothesis
```

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall n \in P\ (S(n) \neq n).$$

Remark (Interpretation). Successor always moves forward in the counting structure. No element can be recovered by applying successor to itself.

Remark (Historical note). This theorem is a standard consequence of the Peano axioms in the classical development of the natural numbers [Peano, 1889]. Feferman includes the corresponding result as an exercise in Chapter 3 of Feferman [1964].

Remark (Dependencies).

- Distinguished Element Is Not a Successor
- Injectivity of Successor
- Induction Principle for a Peano System

Peano System Figure Plates

The following figure plates collect the visual diagrams for the Peano system and iterator material. Each plate is presented separately so that the diagram can be read as a full-page reference.

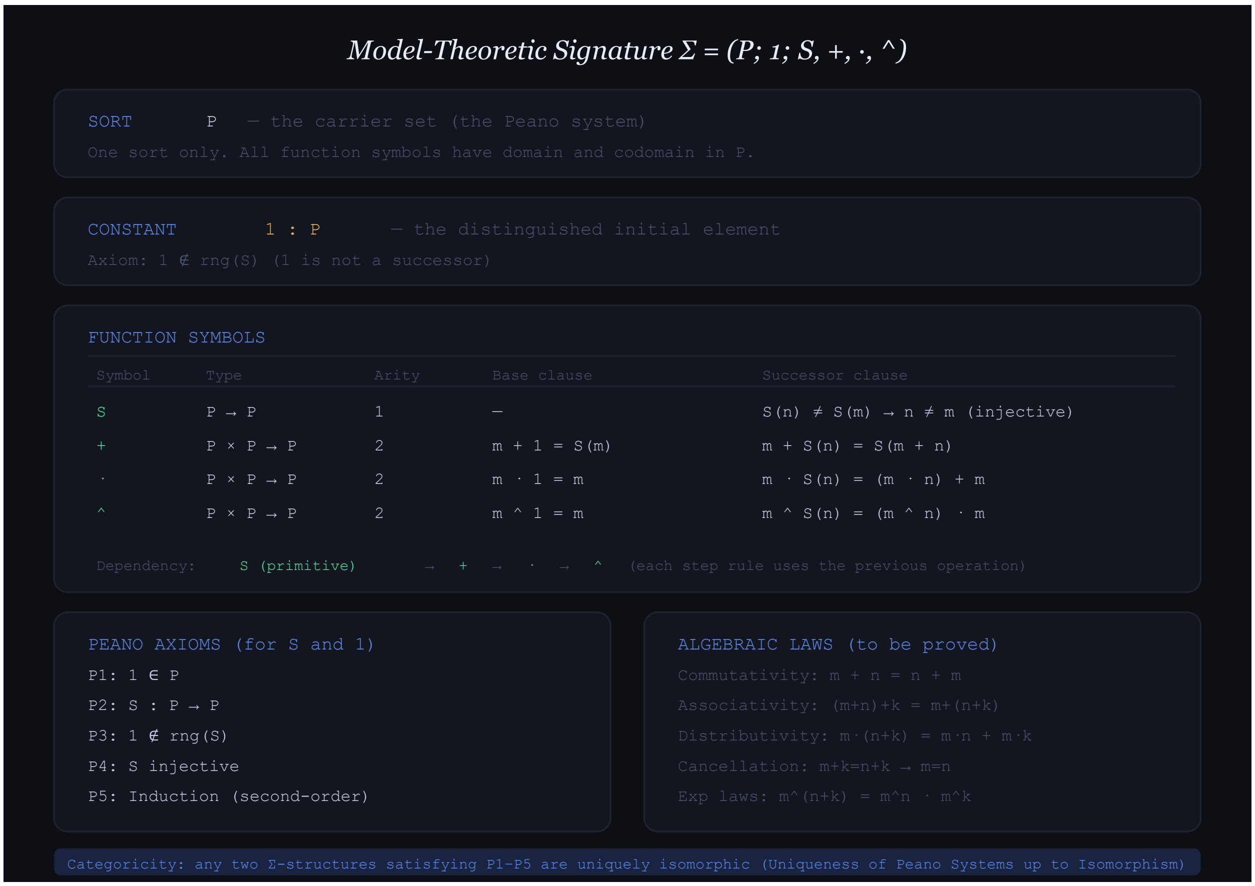

Figure 19.1: The model-theoretic signature for a Peano system and the later arithmetic operations generated from it.

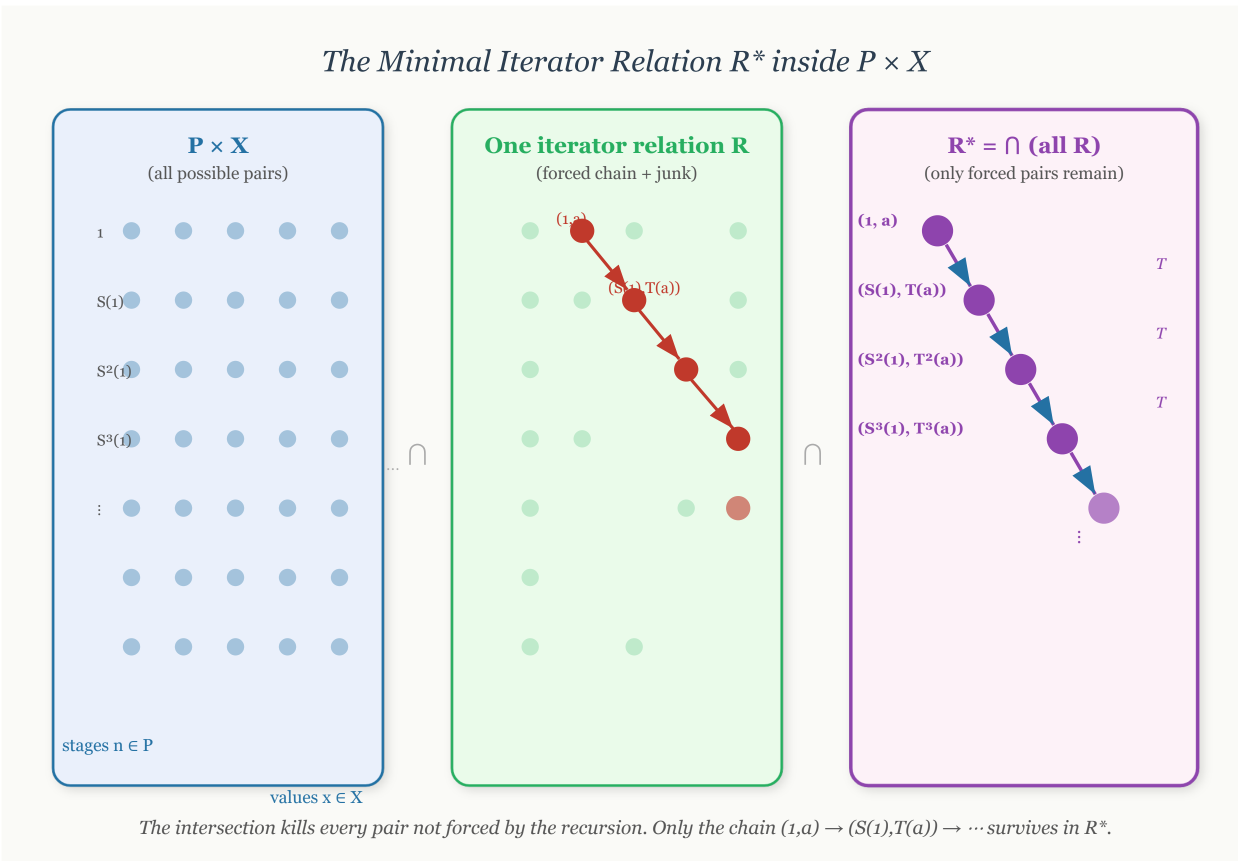

Figure 19.2: The intuitive construction of the minimal iterator relation R^* inside $P \times W$.

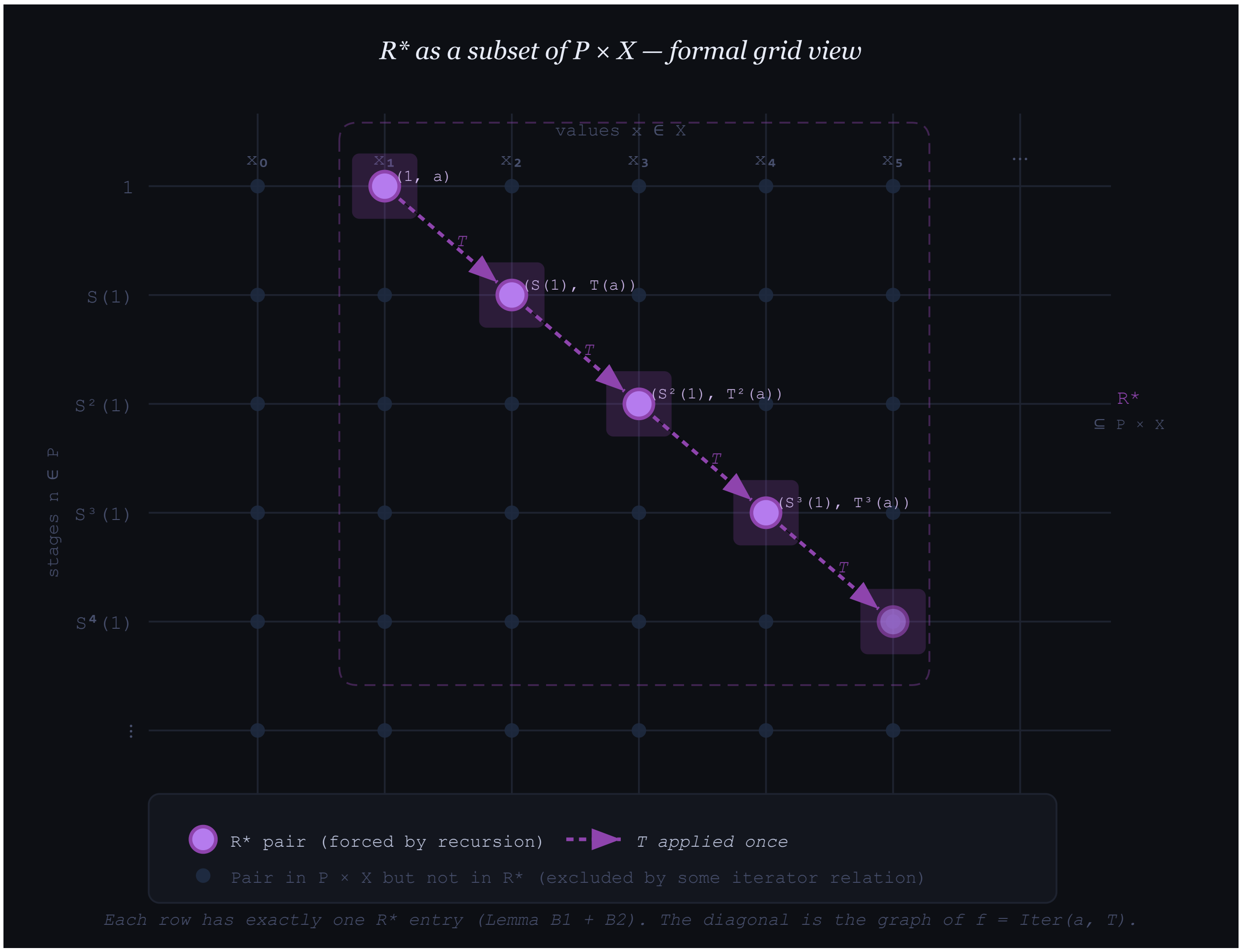

Figure 19.3: A formal grid view of R^* as the chain of forced ordered pairs in $P \times W$.

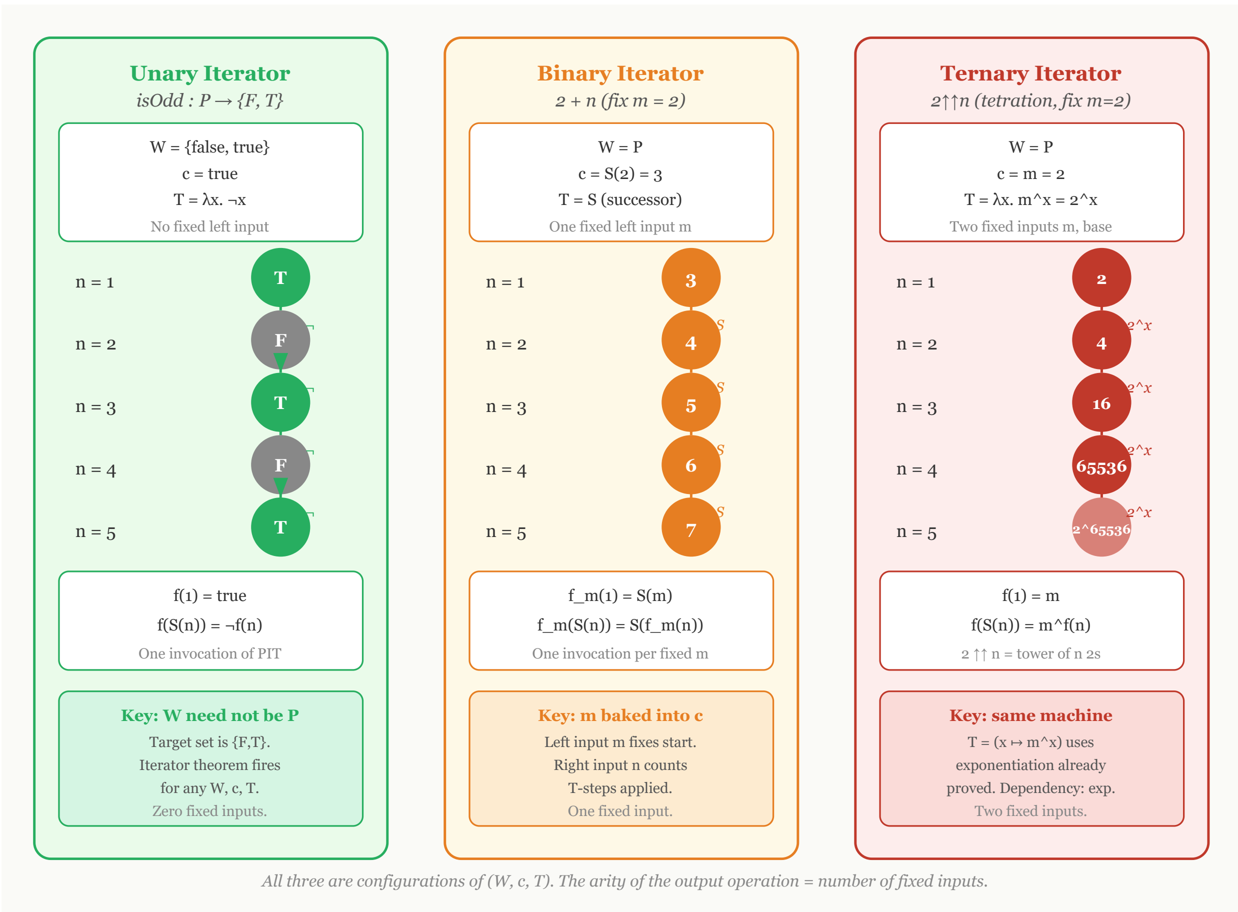

Figure 19.4: Unary, binary, and higher-arity iterator configurations after fixing the relevant parameters.

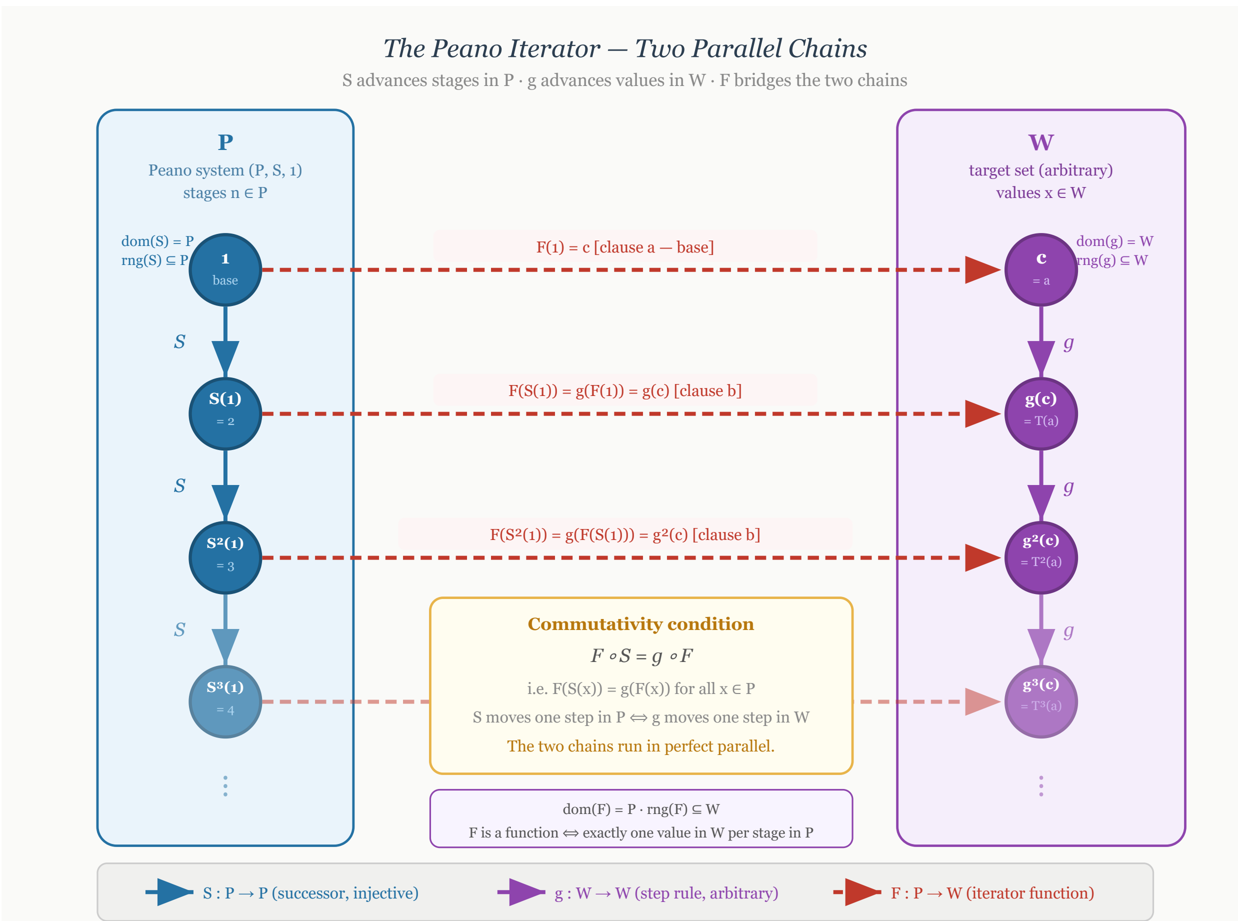

Figure 19.5: The Peano iterator as two parallel chains, one in P and one in W, connected by the iterator function.

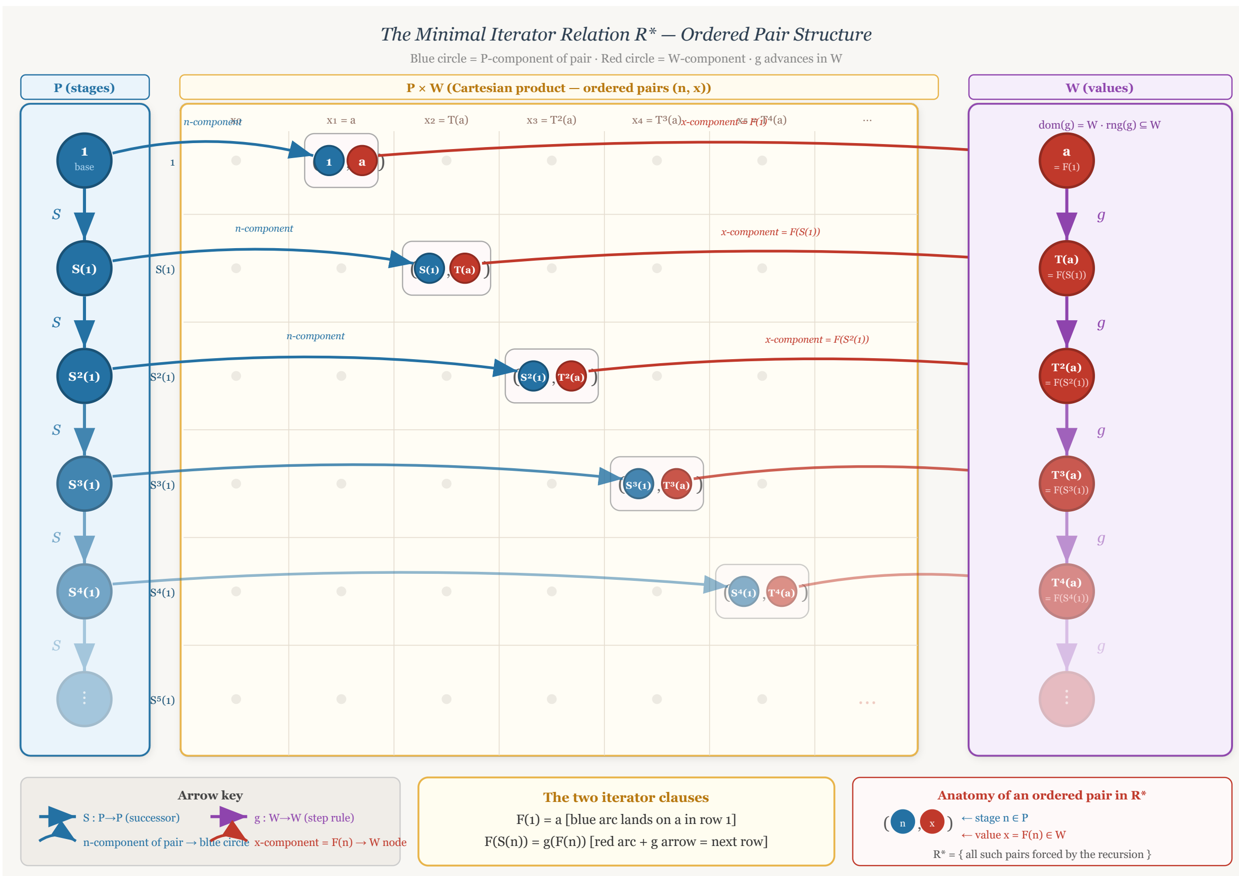

Figure 19.6: The three-zone view of R^*, separating stages in P, ordered pairs in $P \times W$, and values in W.

19.3 Examples of Peano Systems

examples-of-peano-systems

Peano Systems Defining Peano Systems → **Examples of Peano Systems** → Recursion on Peano Systems

Peano Systems as Abstract Structures

The examples in this section separate the abstract idea of a Peano system from any particular representation of the natural numbers. A Peano system is not a specific set such as $\mathbb{N}$. It is a structure consisting of a carrier set, a distinguished initial element, and a successor operation satisfying the Peano conditions. Different carriers can therefore realize the same abstract successor structure.

Remark (Formal reference)*.* The formal definition used throughout this section is Peano System. In each example, the data are displayed as a triple (P, S, e) consisting of a carrier P, a successor map $S : P \to P$, and a distinguished first element $e \in P$.

Remark (Interpretation)*.* The axioms say that the system has one starting point, the successor operation never merges two different elements, and induction reaches every element of the carrier. Thus the carrier is an infinite one-way chain generated from its initial element.

The Standard One-Based System

The Usual One-Based Carrier

The familiar natural-number model is the reference example, but it is only one realization of the Peano axioms. The carrier is the usual one-based set of natural numbers, the distinguished element is 1, and successor is the usual next-number operation.

Example (The Usual Natural Numbers)*.* Let

$$P = \mathbb{N}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S(n) = n + 1.$$

Then $(P, S, e) = (\mathbb{N}, S, 1)$ is the usual one-based Peano system.

Remark (Structure)*.* The Peano data are

$$\bigl(P, S, e\bigr) = \bigl(\mathbb{N},\, n \mapsto n + 1,\, 1\bigr).$$

Remark (Interpretation)*.* This is the familiar model, but it is not the only possible carrier for a Peano system. The remaining examples use different elements while preserving the same abstract successor pattern.

Von Neumann Ordinals

Set-Theoretic Carriers

Von Neumann ordinals realize natural-number structure inside set theory. In this representation, each stage is a set, and successor is formed by adjoining the current stage as a new element.

Example (The Zero-Based Von Neumann Chain). Let

$$0 = \varnothing, \qquad 1 = \{0\}, \qquad 2 = \{0,1\}, \qquad 3 = \{0,1,2\}, \quad \dots$$

Let $P = \omega$, let $e = 0$, and define

$$S : P \to P, \qquad S(n) = n \cup \{n\}.$$

Then $(P, S, e) = (\omega, S, 0)$ is a zero-based Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\omega,\, n \mapsto n \cup \{n\},\, 0\big).$$

This example uses the zero-based convention common in set-theoretic foundations. The house convention for arithmetic in this volume remains one-based.

Example (The One-Based Von Neumann Chain). Let

$$P = \omega \setminus \{0\}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S(n) = n \cup \{n\}.$$

Then (P, S, e) is a one-based Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\omega \setminus \{0\},\, n \mapsto n \cup \{n\},\, 1\big).$$

Remark (Interpretation). In the Von Neumann model, each natural number is literally the set of all smaller natural numbers. This is not merely notation; it is a concrete set-theoretic realization of the abstract Peano structure.

Nonnumeric Peano Systems

Different Carriers, Same Successor Shape

A Peano carrier need not be the usual natural-number set, and its successor operation need not agree with the ambient order or arithmetic of a larger structure. What matters is the internally generated successor chain.

Example (The Powers of Ten). Let

$$P = \{10^{n-1} : n \in \mathbb{N}\}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S(10^{n-1}) = 10^n \quad (n \in \mathbb{N}).$$

Equivalently, for every $x \in P$,

$$S(x) = 10x.$$

Then (P, S, e) is a Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\{1, 10, 10^2, 10^3, \ldots\},\ x \mapsto 10x,\ 1\big).$$

Remark (Interpretation). The elements are not all natural numbers. They are only powers of ten. Nevertheless, starting from 1 and repeatedly multiplying by 10 produces a successor chain with exactly the Peano shape:

$$1 \mapsto 10 \mapsto 10^2 \mapsto 10^3 \mapsto \cdots .$$

Example (Strings of Repeated Letters). Let $\Sigma = \{a\}$ and let

$$P = \{a^n : n \in \mathbb{N}\}, \qquad e = a,$$

where a^n denotes the string consisting of n copies of a. Define

$$S : P \to P, \qquad S(a^n) = a^{n+1} \quad (n \in \mathbb{N}).$$

Then (P, S, e) is a Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\{a, aa, aaa, aaaa, \ldots\},\ a^n \mapsto a^{n+1},\ a\big).$$

Remark (Interpretation). This example is useful because the elements are not numbers at all. They are finite strings. The Peano structure is carried by the successor relation, not by the surface appearance of the elements.

Example (The Reciprocal Chain). Let

$$P = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S\!\left(\frac{1}{n}\right) = \frac{1}{n+1} \quad (n \in \mathbb{N}).$$

Then (P, S, e) is a Peano system.

Remark (Structure)*.* The Peano data are

$$(P, S, e) = \left(\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}, \frac{1}{n} \mapsto \frac{1}{n+1}, 1\right).$$

Remark (Interpretation)*.* This is a good example precisely because the carrier is a decreasing sequence in the usual order on $\mathbb{R}$:

$$1 > \frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \cdots .$$

The Peano successor does not have to mean "larger than." It only means the next element in the generated successor chain. The order inherited from $\mathbb{R}$ is external structure, not part of the Peano-system data.

A Finite Alphabet Is Not a Peano System

A Failed Carrier

Finite chains help isolate what the Peano axioms forbid. A finite alphabet can support a partial next-letter operation, or it can support a cyclic successor operation, but neither choice gives a Peano system.

Example (The Finite Alphabet Fails)*.* Let

$$P = \{A, B, C, \dots, Z\}.$$

If one tries to define

$$S(A) = B, \quad S(B) = C, \quad \dots,$$

then there is no value of $S(Z)$ inside the same finite alphabet that extends the one-way chain. If instead one defines $S(Z) = A$, then $S : P \to P$ is total but the successor operation forms a cycle.

Remark (Failure modes)*.* The finite alphabet fails in one of two ways:

either S is not a total map $P \to P$,
or S loops back into a cycle.

The first failure violates closure of successor. The second failure violates the one-way successor structure forced by the Peano axioms.

Remark (Interpretation)*.* A finite alphabet cannot be a Peano system. Peano systems are necessarily infinite because induction and successor closure generate an unending chain from the distinguished first element.

Summary

The Core Lesson

The carrier of a Peano system can be numerical, set-theoretic, symbolic, or embedded in another ordered structure. The common content is the same abstract successor pattern.

Remark (Examples). The following are Peano systems once their successor operations are specified:

$$
\begin{aligned}
&(\mathbb{N},\, n \mapsto n+1,\, 1),\\
&(\omega,\, n \mapsto n \cup \{n\},\, 0),\\
&(\omega \setminus \{0\},\, n \mapsto n \cup \{n\},\, 1),\\
&(\{1, 10, 10^2, 10^3, \ldots\},\, x \mapsto 10x,\, 1),\\
&(\{a, aa, aaa, aaaa, \ldots\},\, a^n \mapsto a^{n+1},\, a),\\
&\left(\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\},\, \frac{1}{n} \mapsto \frac{1}{n+1},\, 1\right).
\end{aligned}
$$

They look different, but they have the same abstract Peano shape. Each is an infinite chain generated from a distinguished initial element by repeated successor.

19.4 Recursion on Peano Systems

Why Recursion Requires a Theorem

The Peano axioms describe the shape of a Peano system: there is a first element, a successor operation, no predecessor of the first element, no successor collisions, and an induction principle. They do not, by themselves, say that every recursive specification determines a function. A recursive display such as

$$f(1) = c \qquad \text{and} \qquad f(S(n)) = g(f(n))$$

is only a specification. It becomes a definition only after one proves that there exists a unique function satisfying it. This is the "specification is not existence" point emphasized in Feferman's treatment of recursive definition [Feferman, 1964, p. 67].
The proof used here follows the function-as-graph viewpoint. A function $f : P \to W$ is a subset of $P \times W$ whose elements are ordered pairs (n, w), one for each stage $n \in P$. The construction therefore moves through three zones: the stage set P, the Cartesian product grid $P \times W$, and the value set W. The recursion theorem is the result that identifies the right graph inside $P \times W$ and proves that it is the graph of exactly one function.

Remark (Prerequisite definition). An *inductive subset* of a Peano system is defined in Inductive Subset of a Peano System. In this section, induction is used only through the earlier Induction Principle for a Peano System.

Arity Examples Before the Theorem

Iterator Configurations

The iterator theorem has one formal shape but many uses. In every example below, the stage variable is the Peano stage $n \in P$, the value set is W, the first value is c, and the update rule is $g : W \to W$. These examples are motivating configurations only; the theorem establishing their legitimacy is proved afterward.

Remark (Example: isOdd).

component	choice
W	$\{\text{false}, \text{true}\}$
c	true
g	$x \mapsto \neg x$

n	1	2	3	4	5
$h(n)$	true	false	true	false	true

The defining clauses are

$$h(1) = \text{true} \qquad \text{and} \qquad h(S(n)) = \neg h(n).$$

This example shows that the value set W need not be the Peano system P. The stages are natural-number-like, but the values are truth values.

Remark (Example: addition with the left parameter fixed). Fix the left parameter at $m = 2$.

component	choice
W	P
c	$S(2) = 3$
g	S

n	1	2	3	4	5
$h(n)$	3	4	5	6	7

The defining clauses are

$$h(1) = S(2) \qquad \text{and} \qquad h(S(n)) = S(h(n)).$$

This is the iterator pattern behind $2 + n$ in the one-based convention. One parameter is fixed, and recursion runs in the remaining argument.

Remark (Example: tetration with the base fixed). Fix the base at $m = 2$.

component	choice
W	P
c	2
g	$x \mapsto 2^x$

n	1	2	3	4	5
$h(n)$	2	2^2	2^{2^2}	$2^{2^{2^2}}$	$2^{2^{2^{2^2}}}$

The defining clauses are

$$h(1) = 2 \qquad \text{and} \qquad h(S(n)) = 2^{h(n)}.$$

This example shows that the iterator theorem is not limited to the first arithmetic operations. Higher arithmetic still has the same formal configuration once the relevant operation on W is available.

Remark (Example: string repetition). Fix a string $s \in \Sigma^*$.

component	choice
W	Σ^*
c	s
g	$x \mapsto x \cdot s$

n	1	2	3	4	5
$h(n)$	s	$s \cdot s$	$s \cdot s \cdot s$	$s \cdot s \cdot s \cdot s$	$s \cdot s \cdot s \cdot s \cdot s$

The defining clauses are

$$h(1) = s \qquad \text{and} \qquad h(S(n)) = h(n) \cdot s.$$

This example is not arithmetic. It illustrates that recursion on P can generate values in any set equipped with an appropriate update rule.

Formal Development

Minimal Relations and Iterator Functions

The formal proof is organized in Feferman's graph style. First define the admissible relations in $P \times W$. Then take their intersection. The resulting minimal relation is shown to be consistent, deterministic, and complete, hence it is the graph of the required function. This follows the minimal-relation construction used in Feferman, Hrbacek–Jech, and related lecture-note presentations of the recursion theorem [Feferman, 1964, Hrbacek and Jech, 1999, Freiwald, 2009].

Definition (Iterator Data)

Definition 19.4.1 (Iterator Data)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$ be a function. The triple (W, c, g) is called *iterator data* for $(P, S, 1)$.

Remark (Standard quantified statement).

$$(W, c, g) \text{ is iterator data for } (P, S, 1) \iff \big(W \text{ is a set} \wedge c \in W \wedge g : W \to W\big).$$

Remark (Definition predicate reading).

$$\mathrm{IteratorData}(W, c, g) \iff \big(W \text{ is a set} \wedge c \in W \wedge g : W \to W\big).$$

Remark (Negated quantified statement).

$$(W, c, g) \text{ is not iterator data for } (P, S, 1) \iff \big(W \text{ is not a set} \vee c \notin W \vee g \text{ is not a function } W \to W\big).$$

Remark (Failure modes). Iterator data fails if the proposed value collection is not a set, if the initial value does not lie in the value set, or if the proposed update rule is not a self-map of the value set.

Remark (Failure mode decomposition).

$$
\begin{aligned}
&(W, c, g) \text{ is not iterator data for } (P, S, 1) \\
\iff &\underbrace{W \text{ is not a set}}_{\text{no value set}} \\
&\vee \quad \underbrace{c \notin W}_{\text{invalid initial value}} \\
&\vee \underbrace{g \text{ is not a function } W \to W}_{\text{invalid update rule}}.
\end{aligned}
$$

Remark (Interpretation). Iterator data records the value set, the first value, and the fixed rule used to pass from each value to the next value.

Remark (Dependencies).

- Peano System

Definition (Iterator Relation)

Definition 19.4.2 (Iterator Relation)**.** Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. A relation $R \subseteq P \times W$ is called an *iterator relation* for (W, c, g) exactly when

$$(1, c) \in R$$

and

$$\forall n \in P \; \forall w \in W \; \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in R\big).$$

Remark (Standard quantified statement).

$$
\begin{aligned}
R \text{ is an iterator relation for } (W, c, g) \iff &R \subseteq P \times W \wedge (1, c) \in R \\
&\wedge \forall n \in P \; \forall w \in W \; \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in R\big).
\end{aligned}
$$

Remark (Definition predicate reading).

$$
\begin{aligned}
\text{IteratorRelation}(R, (P, S, 1), W, c, g) \iff &\text{Relation}(R, P, W) \wedge (1, c) \in R \\
&\wedge \forall n \in P \; \forall w \in W \; \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in R\big).
\end{aligned}
$$

Remark (Negated quantified statement).

$$
\begin{aligned}
R \text{ is not an iterator relation for } (W, c, g) \iff &R \nsubseteq P \times W \vee (1, c) \notin R \\
&\vee \exists n \in P \; \exists w \in W \; \big((n, w) \in R \wedge (S(n), g(w)) \notin R\big).
\end{aligned}
$$

Remark (Failure modes). An iterator relation fails if it is not a relation inside $P \times W$, if it omits the required base pair, or if it is not closed under the iterator step.

Remark (Failure mode decomposition).

$$R \text{ is not an iterator relation for } (W,c,g) \iff \underbrace{R \not\subseteq P \times W}_{\text{wrong ambient relation}} \vee \underbrace{(1,c) \notin R}_{\text{base pair missing}} \vee \underbrace{\exists n \in P\ \exists w \in W\ \big((n,w) \in R \wedge (S(n), g(w)) \notin R\big)}_{\text{step closure fails}}.$$

Remark (Interpretation). An iterator relation is any graph-like set of possible stage-value pairs that contains the required first pair and is closed under the recursive step.

Remark (Dependencies).

- Iterator Data

Definition (Minimal Iterator Relation)

Definition 19.4.3 (Minimal Iterator Relation)**.** Let $(P,S,1)$ be a Peano system, and let (W,c,g) be iterator data for it. The *minimal iterator relation* for (W,c,g) is

$$R^* = \bigcap \{R \subseteq P \times W : R \text{ is an iterator relation for } (W,c,g)\}.$$

Lean 4 Verification Record

Source anchor: Feferman, *The Number Systems*, Section 3.4; Mendelson, *Number Systems and the Foundations of Analysis*, Section 2.2.
Formalizes: Definition 19.4.3.
Lean module: `LRA.VolumeII.PeanoSystems.Recursion`.
Lean declaration: `minimal_iterator_relation`.
Status: checked against `lra-lean`.

```
def minimal_iterator_relation
    (ps : PeanoSystem)
    (data : IteratorData ps)
    (element : ps.carrier)
    (value : data.target) : Prop :=
  forall relation : ps.carrier -> data.target -> Prop,
    iterator_relation ps data relation ->
    relation element value
```

Remark (Standard quantified statement).

$$(n,w) \in R^* \iff \forall R \subseteq P \times W\ \big(R \text{ is an iterator relation for } (W,c,g) \implies (n,w) \in R\big).$$

Remark (Definition predicate reading). Let $\mathcal{C}_{\mathrm{it}}$ be the closure condition

$$\mathcal{C}_{\mathrm{it}}(R) \iff \mathrm{IteratorRelation}(R, (P,S,1), W, c, g).$$

Then the definition says

$$\mathrm{MinimalRelation}(R^*, \mathcal{C}_{\mathrm{it}}).$$

Remark (Negated quantified statement).

$$(n, w) \notin R^* \iff \exists R \subseteq P \times W \ \big(R \text{ is an iterator relation for } (W, c, g) \wedge (n, w) \notin R\big).$$

Remark (Failure modes). A pair fails to belong to the minimal iterator relation exactly when some iterator relation omits that pair. Such a pair is not forced by the base and successor-closure clauses.

Remark (Interpretation). The minimal iterator relation consists exactly of the stage-value pairs forced by the base pair and the successor closure rule. This is the R^* construction used in Feferman's proof of recursive definition, and in the related set-theoretic presentations cited here [Feferman, 1964, Hrbacek and Jech, 1999, Freiwald, 2009].

Remark (Dependencies).

- Iterator Relation

Lemma (Consistency of the Minimal Iterator Relation)

Lemma 19.4.4 (Consistency of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation for (W, c, g). Then R^* is an iterator relation for (W, c, g).* *Go to proof.*

Remark (Standard quantified statement).

$$R^* \text{ is an iterator relation for } (W, c, g).$$

Remark (Predicate reading).

$$\text{MinimalRelation}(R^*, \mathcal{C}_{\text{it}}) \Longrightarrow \text{IteratorRelation}(R^*, (P, S, 1), W, c, g).$$

Remark (Negated quantified statement).

$$R^* \text{ is not an iterator relation for } (W, c, g).$$

Remark (Contrapositive quantified statement).

$$R^* \text{ is not an iterator relation for } (W, c, g) \Longrightarrow R^* \text{ is not the minimal iterator relation for } (W, c, g).$$

Remark (Interpretation). The intersection construction does not lose the base pair or the successor closure condition. The minimal relation is therefore still one of the admissible iterator relations.

Remark (Dependencies).

- Minimal Iterator Relation

Lemma (Forced Values Are Unique)

Lemma 19.4.5 (Forced Values Are Unique)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. If $(n, w_0) \in R^*$ and $(n, w_1) \in R^*$, then $w_0 = w_1$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall n \in P\ \forall w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \implies w_0 = w_1\big).$$

Remark (Predicate reading).

$$\mathrm{FunctionalRelation}(R^*, P, W).$$

Remark (Negated quantified statement).

$$\exists n \in P\ \exists w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \wedge w_0 \neq w_1\big).$$

Remark (Failure modes). Forced value uniqueness fails exactly when one stage is assigned two distinct values by the minimal iterator relation.

Remark (Contrapositive quantified statement).

$\exists n \in P\ \exists w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \wedge w_0 \neq w_1\big) \implies R^*$ is not functional at every stage. $\blacksquare$

Remark (Interpretation). No stage can be forced to carry two different values. This is the key determinacy step in the minimal-relation construction used to turn recursive specifications into functions [Feferman, 1964].

Remark (Dependencies).

- Consistency of the Minimal Iterator Relation

Lemma (Determinism of the Minimal Iterator Relation)

Lemma 19.4.6 (Determinism of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there is at most one $w \in W$ such that $(n, w) \in R^*$.* *Go to proof.*

Lean 4 Verification Record

Source anchor: Feferman, *The Number Systems*, Section 3.4; Mendelson, *Number Systems and the Foundations of Analysis*, Section 2.2.
Formalizes: Lemma 19.4.6.
Lean module: `LRA.VolumeII.PeanoSystems.Recursion`.
Lean declaration: `minimal_iterator_relation_deterministic`.
Status: checked against `lra-lean`.

```
theorem minimal_iterator_relation_deterministic
    (ps : PeanoSystem)
    (data : IteratorData ps) :
    forall element : ps.carrier,
      forall first_value second_value : data.target,
        minimal_iterator_relation ps data element first_value ->
        minimal_iterator_relation ps data element second_value ->
        first_value = second_value := by
  apply induction_principle ps

  case base_case =>
    intro first_value second_value first_forced second_forced
    exact
      Eq.trans
        (minimal_iterator_relation_base_unique ps data first_value first_forced)
        (minimal_iterator_relation_base_unique ps data second_value second_forced).symm

  case successor_step =>
    intro element induction_hypothesis
    intro first_successor_value second_successor_value
    intro first_forced second_forced
    cases minimal_iterator_relation_complete ps data element with
    | intro current_value current_forced =>
        exact
          Eq.trans
            (forced_successor_values_are_unique
              ps data element current_value first_successor_value
              induction_hypothesis current_forced first_forced)
            (forced_successor_values_are_unique
              ps data element current_value second_successor_value
              induction_hypothesis current_forced second_forced).symm
```

Remark (Standard quantified statement).

$$\forall n \in P\ \forall w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \Longrightarrow w_0 = w_1\big).$$

Remark (Predicate reading).

$$\text{FunctionalRelation}(R^*, P, W).$$

Remark (Negated quantified statement).

$$\exists n \in P\ \exists w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \wedge w_0 \neq w_1\big).$$

Remark (Failure modes). Determinism fails exactly when a single stage has two distinct values in the minimal iterator relation.

Remark (Dependencies).

- Forced Values Are Unique

Lemma (Completeness of the Minimal Iterator Relation)

Lemma 19.4.7 (Completeness of the Minimal Iterator Relation)**.** *Let* $(P, S, 1)$ *be a Peano system, let* (W, c, g) *be iterator data for it, and let* R^* *be the minimal iterator relation. For each* $n \in P$*, there exists* $w \in W$ *such that* $(n, w) \in R^*$*.* *Go to proof.*

Lean 4 Verification Record

Source anchor: Feferman, *The Number Systems*, Section 3.4; Mendelson, *Number Systems and the Foundations of Analysis*, Section 2.2.
Formalizes: Lemma 19.4.7.
Lean module: `LRA.VolumeII.PeanoSystems.Recursion`.
Lean declaration: `minimal_iterator_relation_complete`.
Status: checked against `lra-lean`.

```
theorem minimal_iterator_relation_complete
    (ps : PeanoSystem)
    (data : IteratorData ps) :
    forall element : ps.carrier,
      Exists (fun value : data.target =>
        minimal_iterator_relation ps data element value) := by
  apply induction_principle ps

  case base_case =>
    exact
      Exists.intro
        data.initial_value
        (minimal_iterator_relation_is_iterator_relation ps data).left

  case successor_step =>
    intro element induction_hypothesis
    cases induction_hypothesis with
    | intro value value_is_forced =>
        exact
          Exists.intro
            (data.step_rule value)
            ((minimal_iterator_relation_is_iterator_relation ps data).right
              element value value_is_forced)
```

Remark (Standard quantified statement).

$$\forall n \in P \; \exists w \in W \; ((n, w) \in R^*).$$

Remark (Predicate reading).

$$\mathrm{TotalRelation}(R^*, P, W).$$

Remark (Negated quantified statement).

$$\exists n \in P \; \forall w \in W \; ((n, w) \notin R^*).$$

Remark (Failure modes). Completeness fails exactly when some stage of P receives no value in the minimal iterator relation.

Remark (Source alignment). This is the totality half of Feferman's minimal-relation construction for recursive definition: after determinacy gives at most one value at each stage, completeness gives at least one value at each stage [Feferman, 1964].

Remark (Dependencies).

- Consistency of the Minimal Iterator Relation
- Induction Principle for a Peano System

Lemma (Uniqueness of Iterator Functions)

Lemma 19.4.8 (Uniqueness of Iterator Functions). *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = g(f(n))) \qquad \text{and} \qquad \forall n \in P\ (h(S(n)) = g(h(n))),$$

then $f = h$. Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}\forall f, h : P \to W\ \Big(& f(1) = c \land h(1) = c \\ & \land \forall n \in P\ (f(S(n)) = g(f(n))) \\ & \land \forall n \in P\ (h(S(n)) = g(h(n))) \implies f = h\Big).\end{aligned}$$

Remark (Predicate reading).

$$\forall f, h : P \to W\ \big(\mathrm{IteratorFunction}(f, (P, S, 1), W, c, g) \land \mathrm{IteratorFunction}(h, (P, S, 1), W, c, g) \implies f = h\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}\exists f, h : P \to W\ \Big(& f(1) = c \land h(1) = c \\ & \land \forall n \in P\ (f(S(n)) = g(f(n))) \\ & \land \forall n \in P\ (h(S(n)) = g(h(n))) \land f \neq h\Big).\end{aligned}$$

Remark (Failure modes). Uniqueness fails exactly when two distinct functions satisfy the same base clause and the same successor clause.

Remark (Contrapositive quantified statement).

$$\begin{aligned}f \neq h \implies & f(1) \neq c \lor h(1) \neq c \\ & \lor \exists n \in P\ (f(S(n)) \neq g(f(n))) \\ & \lor \exists n \in P\ (h(S(n)) \neq g(h(n))).\end{aligned}$$

Remark (Interpretation). The iterator clauses determine at most one function. Any difference between two candidate functions must show up as a failure of the base clause or a failure of one of the successor clauses.

Remark (Dependencies).

- Iterator Data
- Induction Principle for a Peano System

Lemma (Existence of an Iterator Function)

Lemma 19.4.9 (Existence of an Iterator Function)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a function $f : P \to W$ such that*

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n))).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists f : P \to W\ \big(f(1) = c \wedge \forall n \in P\ (f(S(n)) = g(f(n)))\big).$$

Remark (Predicate reading).

$$\exists f : P \to W\ \ \text{IteratorFunction}(f, (P, S, 1), W, c, g).$$

Remark (Negated quantified statement).

$$\forall f : P \to W\ \big(f(1) \neq c \vee \exists n \in P\ (f(S(n)) \neq g(f(n)))\big).$$

Remark (Failure modes). Existence fails exactly when every candidate function misses the base value or misses the recursive successor equation at some stage.

Remark (Contrapositive quantified statement).

$$\neg\exists f : P \to W\ \big(f(1) = c \wedge \forall n \in P\ (f(S(n)) = g(f(n)))\big) \implies R^* \text{ is not both deterministic and complete.}$$

Remark (Interpretation). Existence is the assertion that the minimal relation is not merely graph-like locally, but supplies a value at every stage and hence determines an actual function.

Remark (Dependencies).

- Determinism of the Minimal Iterator Relation
- Completeness of the Minimal Iterator Relation

Theorem (Peano Iterator Theorem)

Theorem 19.4.10 (Peano Iterator Theorem)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n))).$$

Go to proof.

Lean 4 Verification Record

Source anchor: Feferman, *The Number Systems*, Theorem 3.4; Mendelson, *Number Systems and the Foundations of Analysis*, Section 2.2.
Formalizes: Theorem 19.4.10.
Lean module: `LRA.VolumeII.PeanoSystems.Recursion`.
Lean declaration: `peano_iterator_theorem`.
Status: checked against `lra-lean`.

```
theorem peano_iterator_theorem
    (ps : PeanoSystem)
    (target : Type)
    (initial_value : target)
    (step_rule : target -> target) :
    Exists (fun iterator_function : ps.carrier -> target =>
      And
        (satisfies_iterator_clauses
          ps target initial_value step_rule iterator_function)
        (forall other_iterator : ps.carrier -> target,
          satisfies_iterator_clauses
            ps target initial_value step_rule other_iterator ->
          forall element : ps.carrier,
            other_iterator element = iterator_function element)) := by
  refine
    Exists.intro
      (iter ps target initial_value step_rule)
      ?_
  constructor
  exact iter_satisfies_iterator_clauses ps target initial_value step_rule
  intro other_iterator other_satisfies element
  exact
    Eq.symm
      (iterator_function_unique
        ps target initial_value step_rule
        (iter ps target initial_value step_rule)
        other_iterator
        (iter_satisfies_iterator_clauses ps target initial_value step_rule)
        other_satisfies
        element)
```

Remark (Standard quantified statement).

$$\exists! f : P \to W\ \big(f(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n)))\big).$$

Remark (Predicate reading).

$$\exists! f : P \to W\ \ \mathrm{IteratorFunction}(f, (P, S, 1), W, c, g).$$

Remark (Negated quantified statement).

$$\neg\exists! f : P \to W \ \big(f(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n)))\big).$$

Remark (Failure modes). The iterator theorem fails only if no function satisfies the iterator clauses or if more than one function satisfies them.

Remark (Failure mode decomposition).

$$\begin{aligned} &\neg\exists! f : P \to W \ \big(f(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n)))\big) \\ \iff &\underbrace{\neg\exists f : P \to W \ \big(f(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n)))\big)}_{\text{existence fails}} \\ &\lor \underbrace{\exists f, h : P \to W \ \big(f \neq h \land f(1) = c \land h(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n))) \land \forall n \in P\ (h(S(n)) = g}_{\text{uniqueness fails}} \end{aligned}$$

Remark (Interpretation). The theorem says that iterator data is sufficient to turn a recursive specification into a genuine definition: there is exactly one graph and hence exactly one function satisfying the base and successor clauses.

Remark (Historical note). This is the one-based Peano-system version of the iterator theorem in Feferman [1964], Mendelson [1982], Simpson [2009]. It is also the set-theoretic instance of the natural numbers object universal property [Lawvere, 1969]; Dedekind's recursion theorem is the classical ancestor [Dedekind, 1901].

Remark (Dependencies).

- Uniqueness of Iterator Functions
- Existence of an Iterator Function

Definition (Iterator-Generated Function)

Definition 19.4.11 (Iterator-Generated Function). Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. The unique function $f : P \to W$ satisfying

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n)))$$

is called the *iterator-generated function determined by* (W, c, g).

Remark (Standard quantified statement).

$$f \text{ is the iterator-generated function determined by } (W, c, g) \iff \big(f : P \to W \land f(1) = c \land \forall n \in P\ (f(S(n))$$

Remark (Definition predicate reading).

$$\text{IteratorGeneratedFunction}(f, (P, S, 1), W, c, g) \iff \text{IteratorFunction}(f, (P, S, 1), W, c, g) \land \exists! h : P \to W \ \ \text{It}$$

Remark (Negated quantified statement).

$$f \text{ is not the iterator-generated function determined by } (W, c, g) \iff f \text{ is not a function } P \to W \\ \vee f(1) \neq c \vee \exists n \in P \; (f(S(n)) \neq g($$

Remark (Failure modes). A candidate fails to be the iterator-generated function if it is not a function from P to W, if it has the wrong base value, or if it violates the successor clause.

Remark (Interpretation). This definition packages the unique function supplied by the Peano Iterator Theorem as a named object available for later constructions.

Remark (Dependencies).

- Peano Iterator Theorem

Theorem (Iterator Base Clause)

Theorem 19.4.12 (Iterator Base Clause)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$f(1) = c.$$

Go to proof.

Remark (Standard quantified statement).

$$f \text{ is the iterator-generated function determined by } (W, c, g) \implies f(1) = c.$$

Remark (Predicate reading).

$$\text{IteratorGeneratedFunction}(f, (P, S, 1), W, c, g) \implies f(1) = c.$$

Remark (Contrapositive quantified statement).

$$f(1) \neq c \implies f \text{ is not the iterator-generated function determined by } (W, c, g).$$

Remark (Interpretation). The base clause is not an additional theorem about the generated function; it is one of the clauses built into the unique recursive specification.

Remark (Dependencies).

- Iterator-Generated Function

Theorem (Iterator Successor Clause)

Theorem 19.4.13 (Iterator Successor Clause)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$\forall n \in P \; (f(S(n)) = g(f(n))).$$

Go to proof.

Remark (Standard quantified statement).

f is the iterator-generated function determined by $(W, c, g) \implies \forall n \in P\ (f(S(n)) = g(f(n)))$. $\blacksquare$

Remark (Predicate reading).

$$\text{IteratorGeneratedFunction}(f, (P, S, 1), W, c, g) \implies \forall n \in P\ (f(S(n)) = g(f(n))).$$

Remark (Contrapositive quantified statement).

$\exists n \in P\ (f(S(n)) \neq g(f(n))) \implies f$ is not the iterator-generated function determined by (W, c, g). $\blacksquare$

Remark (Interpretation). The successor clause says that the generated function advances by applying the update rule to the previous value at every Peano stage.

Remark (Dependencies).

- Iterator-Generated Function

Definition (Iteration of a Self-Map)

Definition 19.4.14 (Iteration of a Self-Map)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$. The iterator-generated function determined by (W, c, g) is called the *iteration of g from c along P*.

Remark (Standard quantified statement).

f is the iteration of g from c along $P \iff f$ is the iterator-generated function determined by (W, c, g). $\blacksquare$

Remark (Definition predicate reading).

f is the iteration of g from c along $P \iff \text{IteratorGeneratedFunction}(f, (P, S, 1), W, c, g)$.

Remark (Negated quantified statement).

f is not the iteration of g from c along $P \iff f$ is not the iterator-generated function determined by $(W$

Remark (Interpretation). Iteration is the special case of recursion in which the same self-map is used at every stage.

Remark (Dependencies).

- Iterator-Generated Function

Corollary (Uniqueness of Binary Iterator Operations)

Corollary 19.4.15 (Uniqueness of Binary Iterator Operations)**.** *Let $(P, S, 1)$ be a Peano system, let A be a set, let W be a set, and suppose each $a \in A$ determines iterator data (W, c_a, g_a) for $(P, S, 1)$. If $F, G : A \times P \to W$ satisfy*

$$F(a, 1) = c_a, \qquad G(a, 1) = c_a$$

and

$$F(a, S(n)) = g_a(F(a, n)), \qquad G(a, S(n)) = g_a(G(a, n))$$

for all $a \in A$ and all $n \in P$, then $F = G$. *Go to proof.*

Remark (Standard quantified statement).

$$\forall a \in A\, \forall n \in P\, \Big(F(a,1) = c_a \wedge G(a,1) = c_a \\ \wedge F(a,S(n)) = g_a(F(a,n)) \wedge G(a,S(n)) = g_a(G(a,n))\Big) \implies F = G.$$

Remark (Predicate reading). For each $a \in A$, write $F_a(n) = F(a,n)$ and $G_a(n) = G(a,n)$. Then

$$\forall a \in A\, \big(\mathrm{IteratorFunction}(F_a, (P,S,1), W, c_a, g_a) \wedge \mathrm{IteratorFunction}(G_a, (P,S,1), W, c_a, g_a) \implies F_a = G_a\big).$$

Remark (Negated quantified statement).

$$\exists a \in A\, \exists n \in P\, \Big(F(a,1) = c_a \wedge G(a,1) = c_a \\ \wedge F(a,S(n)) = g_a(F(a,n)) \wedge G(a,S(n)) = g_a(G(a,n)) \wedge F \neq G\Big).$$

Remark (Failure modes). Binary iterator uniqueness fails if two different binary operations satisfy the same one-parameter iterator specification for every fixed parameter.

Remark (Interpretation). For each fixed first argument, the binary operation is an ordinary iterator in the second argument. Uniqueness for the binary operation is therefore pointwise uniqueness of iterator functions.

Remark (Historical note). This isolates the binary-operation uniqueness pattern used by Thurston before existence is invoked [Thurston, 1956].

Remark (Dependencies).

- Uniqueness of Iterator Functions

General Recursion

Pure iteration uses a fixed self-map $g : W \to W$. General recursion allows the successor rule to depend on the current stage as well as on the current value. The state-encoding theorem below reduces that stage-dependent case to the Peano Iterator Theorem.

Definition (Stage-Dependent Step Rule)

Definition 19.4.16 (Stage-Dependent Step Rule)**.** Let $(P, S, 1)$ be a Peano system and let W be a set. A *stage-dependent step rule* on W over P is a function

$$\Phi : P \times W \to W.$$

Remark (Standard quantified statement).

$$\Phi \text{ is a stage-dependent step rule on } W \text{ over } P \iff \Phi : P \times W \to W.$$

Remark (Definition predicate reading).

$$\mathrm{StageDependentStepRule}(\Phi, P, W) \iff \Phi : P \times W \to W.$$

Remark (Negated quantified statement).

$$\Phi \text{ is not a stage-dependent step rule on } W \text{ over } P \iff \Phi \text{ is not a function } P \times W \to W.$$

Remark (Failure modes). A stage-dependent step rule fails precisely when it is not a well-defined function from stage-value pairs back into the value set.

Remark (Interpretation). A stage-dependent step rule allows the next value to depend on both the current Peano stage and the current value.

Remark (Dependencies).

- Peano System

Definition (General Recursive Function)

Definition 19.4.17 (General Recursive Function)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. A function $f : P \to W$ is called the *general recursive function determined by* (W, c, Φ) exactly when

$$f(1) = c \qquad \text{and} \qquad \forall n \in P \; (f(S(n)) = \Phi(n, f(n))).$$

Remark (Standard quantified statement).

$$\begin{aligned} f \text{ is the general recursive function determined by } (W, c, \Phi) \iff{} & f : P \to W \wedge f(1) = c \\ & \wedge \forall n \in P \; (f(S(n)) = \Phi(n, f(n))). \end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned} \mathrm{GeneralRecursiveFunction}(f, (P, S, 1), W, c, \Phi) \iff{} & f : P \to W \wedge f(1) = c \\ & \wedge \forall n \in P \; (f(S(n)) = \Phi(n, f(n))). \end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned} f \text{ is not the general recursive function determined by } (W, c, \Phi) \iff{} & f \text{ is not a function } P \to W \\ & \vee f(1) \neq c \vee \exists n \in P \; (f(S(n)) \neq \Phi(n \end{aligned}$$

Remark (Failure modes). A candidate fails to be the general recursive function if it has the wrong domain or codomain, the wrong base value, or a failed stage-dependent successor equation.

Remark (Interpretation). General recursive functions are the stage-dependent analogues of iterator-generated functions.

Remark (Dependencies).

- Stage-Dependent Step Rule

Lemma (Uniqueness of General Recursive Functions)

Lemma 19.4.18 (Uniqueness of General Recursive Functions)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \qquad \text{and} \qquad \forall n \in P\ (h(S(n)) = \Phi(n, h(n))),$$

then $f = h$. *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} \forall f, h : P \to W\ \Big(& f(1) = c \wedge h(1) = c \\ & \wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \\ & \wedge \forall n \in P\ (h(S(n)) = \Phi(n, h(n))) \Longrightarrow f = h\Big). \end{aligned}$$

Remark (Predicate reading).

$\forall f, h : P \to W\ \big(\mathrm{GeneralRecursiveFunction}(f, (P, S, 1), W, c, \Phi) \wedge \mathrm{GeneralRecursiveFunction}(h, (P, S, 1), W,$

Remark (Negated quantified statement).

$$\begin{aligned} \exists f, h : P \to W\ \Big(& f(1) = c \wedge h(1) = c \\ & \wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \\ & \wedge \forall n \in P\ (h(S(n)) = \Phi(n, h(n))) \wedge f \neq h\Big). \end{aligned}$$

Remark (Failure modes). Uniqueness of general recursive functions fails if two distinct functions satisfy the same base clause and the same stage-dependent successor clause.

Remark (Contrapositive quantified statement).

$$\begin{aligned} f \neq h \Longrightarrow\ & f(1) \neq c \vee h(1) \neq c \\ & \vee \exists n \in P\ (f(S(n)) \neq \Phi(n, f(n))) \\ & \vee \exists n \in P\ (h(S(n)) \neq \Phi(n, h(n))). \end{aligned}$$

Remark (Interpretation). Once the initial value and stage-dependent rule are fixed, two recursive candidates cannot diverge without violating one of the defining clauses.

Remark (Dependencies).

- General Recursive Function
- Induction Principle for a Peano System

Theorem (General Recursion by State Encoding)

Theorem 19.4.19 (General Recursion by State Encoding)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. There exists a function $H : P \to P \times W$ such that*

$$H(1) = (1, c)$$

and

$$\forall n \in P \ \big(H(S(n)) = (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\big).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists H : P \to P \times W \ \Big(H(1) = (1, c) \wedge \forall n \in P \ H(S(n)) = (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\Big).$$ ∎

Remark (Predicate reading). Let $\Gamma_\Phi : P \times W \to P \times W$ be defined by

$$\Gamma_\Phi(p, w) = (S(p), \Phi(p, w)).$$

Then state encoding is the iterator assertion

$$\exists H : P \to P \times W \ \ \mathrm{IteratorFunction}(H, (P, S, 1), P \times W, (1, c), \Gamma_\Phi).$$

Remark (Negated quantified statement).

$$\forall H : P \to P \times W \ \Big(H(1) \neq (1, c) \\ \vee \exists n \in P \ H(S(n)) \neq (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\Big).$$

Remark (Failure modes). State encoding fails if every candidate state function has the wrong initial state or violates the encoded successor step at some stage.

Remark (Interpretation). The theorem converts stage-dependent recursion into ordinary iteration by carrying the current stage together with the current value as the recursive state.

Remark (Dependencies).

- Peano Iterator Theorem

- Stage-Dependent Step Rule

Theorem (General Recursion Theorem)

Theorem 19.4.20 (General Recursion Theorem)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P\ (f(S(n)) = \Phi(n, f(n))).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists! f : P \to W\ \big(f(1) = c \wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n)))\big).$$

Remark (Predicate reading).

$$\exists! f : P \to W\ \ \mathrm{GeneralRecursiveFunction}(f, (P, S, 1), W, c, \Phi).$$

Remark (Negated quantified statement).

$$\neg\exists! f : P \to W\ \big(f(1) = c \wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n)))\big).$$

Remark (Failure modes). General recursion fails if no function satisfies the general recursive clauses or if two distinct functions satisfy them.

Remark (Failure mode decomposition).

$$\begin{aligned} &\neg\exists! f : P \to W\ \big(f(1) = c \wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n)))\big) \\ \iff &\underbrace{\neg\exists f : P \to W\ \big(f(1) = c \wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n)))\big)}_{\text{existence fails}} \\ &\vee \underbrace{\exists f, h : P \to W\ \big(f \neq h \wedge f(1) = c \wedge h(1) = c \wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \wedge \forall n \in P\ (h(S(n))}_{\text{uniqueness fails}} \end{aligned}$$

Remark (Interpretation). General recursion justifies recursive definitions whose successor rule depends on the current stage as well as the current value.

Remark (Historical note). This is the stage-dependent form corresponding to Feferman's Theorem 3.4' [Feferman, 1964]. It is the form used later for arithmetic definitions; those definitions are deferred to their own sections [Feferman, 1964, Mendelson, 1982].

Remark (Dependencies).

- Uniqueness of General Recursive Functions
- General Recursion by State Encoding

Remark (Deferred arithmetic definitions). Definitions of addition, multiplication, exponentiation, and later arithmetic operations are not part of this section. They begin after the recursion principles have been established.

Theorem (Uniqueness of Peano Systems up to Isomorphism)

Theorem 19.4.21 (Uniqueness of Peano Systems up to Isomorphism)**.** *Let* $(P, S_P, 1_P)$ *and* $(Q, S_Q, 1_Q)$ *be Peano systems. Then there exists a unique bijection* $\theta : P \to Q$ *such that*

$$\theta(1_P) = 1_Q$$

and

$$\forall n \in P\ (\theta(S_P(n)) = S_Q(\theta(n))).$$

Thus any two Peano systems are isomorphic. Go to proof.

Remark (Standard quantified statement).

$$\exists! \theta : P \to Q\ \big(\theta \text{ is bijective} \wedge \theta(1_P) = 1_Q \wedge \forall n \in P\ (\theta(S_P(n)) = S_Q(\theta(n)))\big).$$

Remark (Predicate reading). Let $A = (P, S_P, 1_P)$ and $B = (Q, S_Q, 1_Q)$. The theorem asserts

$$\text{Isomorphism}(A, B),$$

with unique witness $\theta : P \to Q$ satisfying the displayed base and successor-preservation clauses.

Remark (Negated quantified statement).

$$\neg\exists! \theta : P \to Q\ \big(\theta \text{ is bijective} \wedge \theta(1_P) = 1_Q \\ \wedge \forall n \in P\ (\theta(S_P(n)) = S_Q(\theta(n)))\big).$$

Remark (Failure modes). Categoricity fails if no structure-preserving bijection exists between the two Peano systems or if more than one such bijection exists.

Remark (Interpretation). All Peano systems have the same structure up to a unique isomorphism. The elements may be named differently, but the distinguished first element and successor structure determine the translation.

Remark (Historical note). This is the categoricity theorem for Peano systems, following Feferman's Theorem 3.5 and Simpson's Peano-system presentation [Feferman, 1964, Simpson, 2009].

Remark (Dependencies).

- Peano System
- General Recursion Theorem

Chapter 20

Natural Numbers (ℕ)

20.1 Axioms for ℕ

20.1.1 Axioms for ℕ

The Canonical Peano System

The natural numbers are now fixed as the canonical one-based Peano system. The preceding Peano chapter developed the abstract interface (X, e, S); in this chapter, $\mathbb{N}$ is the distinguished instance obtained by taking the base element to be 1. The symbol 0 is not part of this system. It is adjoined in the next chapter when the whole numbers are constructed.

Definition (The Natural Numbers)

Definition 20.1.1 (The Natural Numbers)**.** The natural numbers $\mathbb{N}$ are the canonical Peano system

$$(\mathbb{N}, 1, S),$$

where $1 \in \mathbb{N}$ is the distinguished base element and $S : \mathbb{N} \to \mathbb{N}$ is the successor operation.

Remark (Standing convention)*.* Throughout this chapter, variables m, n, k range over $\mathbb{N}$ unless a larger ambient set is explicitly named. The element 1 is the base natural number, and $S(n)$ is the successor of n.

Remark (Dependencies)*.*

- Peano System

Peano Axioms Specialized to $\mathbb{N}$

The following axioms are not new principles beyond the Peano-system interface; they are the abstract Peano axioms specialized to the canonical natural-number system. Naming them here makes later references inside the arithmetic chapter point directly to $\mathbb{N}$ rather than to an arbitrary Peano system.

Axiom (N1, Base)

Axiom 20.1.2 (N1, Base)**.**

$$1 \in \mathbb{N}.$$

Axiom (N2, Successor Closure)

Axiom 20.1.3 (N2, Successor Closure)**.**

$$\forall n \in \mathbb{N}\ (S(n) \in \mathbb{N}).$$

Axiom (N3, Base Has No Predecessor)

Axiom 20.1.4 (N3, Base Has No Predecessor)**.**

$$\forall n \in \mathbb{N}\ (S(n) \neq 1).$$

Axiom (N4, Successor Injectivity)

Axiom 20.1.5 (N4, Successor Injectivity)**.**

$$\forall m, n \in \mathbb{N}\ (S(m) = S(n) \implies m = n).$$

Axiom (N5, Induction)

Axiom 20.1.6 (N5, Induction)**.** For every predicate φ on $\mathbb{N}$,

$$\Big(\varphi(1) \land \forall n \in \mathbb{N}\ (\varphi(n) \implies \varphi(S(n)))\Big) \implies \forall n \in \mathbb{N}\ \varphi(n).$$

20.2 Canonical Natural Number System

Working in $\mathbb{N}$

The preceding chapter established the Peano-system facts needed for arithmetic: induction, successor analysis, recursion, and categoricity. From this point forward, $\mathbb{N}$ denotes the canonical natural-number Peano system, with distinguished first element 1 and successor map S. The arithmetic operations introduced in this chapter are defined inside that system by the recursion principles already proved for Peano systems.

Remark (Standing convention). Unless otherwise stated, variables m, n, k range over $\mathbb{N}$, the symbol 1 denotes the distinguished first natural number, and $S(n)$ denotes the successor of n.

20.3 Addition on N

Recursive Definition of Addition

Addition on a Peano system is defined from the successor primitive by recursion. For each fixed left input m, the map $n \mapsto m + n$ is generated from the initial value $S(m)$ and the successor operator on the same Peano system. This definition makes addition a derived operation rather than an independent axiom.

Definition (Addition on a Peano System)

Definition 20.3.1 (Addition on a Peano System)**.** Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$f_m : P \to P$$

to be the unique function satisfying

$$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P \ (f_m(S(n)) = S(f_m(n))).$$

For $m, n \in P$, define

$$m + n \coloneqq f_m(n).$$

Remark (Standard quantified statement).

> Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $f_m : P \to P$ satisfy
> $f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P \ (f_m(S(n)) = S(f_m(n)))$.
> Then for all $m, n \in P$, $\quad m + n \coloneqq f_m(n)$.

Remark (Negated quantified statement).

> Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $f_m : P \to P$ satisfy
> $f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P \ (f_m(S(n)) = S(f_m(n)))$.
> Then for all $m, n, x \in P$, $\quad x \neq m + n \iff x \neq f_m(n)$.

Remark (Failure modes). The definition fails only when the designated value does not agree with the unique recursively generated function attached to the fixed left input m. Once the recursive map f_m is fixed, every incorrect candidate for $m + n$ is simply a value of P different from $f_m(n)$.

Remark (Failure mode decomposition).

Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $f_m : P \to P$ satisfy

$$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

$$x \neq m + n \iff \underbrace{x \neq f_m(n)}_{\text{candidate value disagrees with the recursive output}}.$$

Remark (Interpretation). Addition is introduced by fixing the left input and recursively generating the right-input map. The value $m + 1$ is set to be the successor of m, and each further successor step in the right input advances the output by one more successor. This makes addition repeated succession.

Remark (Dependencies).

- Peano System
- Peano Iterator Theorem

Theorem (Addition Is Well-Defined on a Peano System)

Theorem 20.3.2 (Addition Is Well-Defined on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$f_m : P \to P$$

such that

$$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

Consequently the rule

$$(m, n) \longmapsto m + n \coloneqq f_m(n)$$

defines a unique binary operation on P. *Go to proof.*

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.

$$\forall m \in P\ \exists! f_m : P \to P\ \Big(f_m(1) = S(m) \land \forall n \in P\ (f_m(S(n)) = S(f_m(n)))\Big).$$

Remark (Interpretation). The theorem shows that the recursive clauses for addition determine exactly one right-input map for each fixed left input. The addition operation is therefore not assumed; it is constructed from the successor map by recursion.

Remark (Source alignment). This is the house-style well-definedness form of addition obtained by applying the Peano recursion theorem in the manner of Feferman's construction of arithmetic operations [Feferman, 1964].

Remark (Dependencies).

- Addition on a Peano System
- Peano Iterator Theorem

Recursive Clauses as Reusable Equalities

The recursive definition of addition is given by the base value and successor step for the right-input map $n \mapsto m + n$. Those clauses are part of the definition, but they become substantially more useful once isolated as named theorems. Later algebraic arguments repeatedly cite these equalities rather than reopening the recursive construction each time.

Theorem (Addition With 1)

Theorem 20.3.3 (Addition With 1). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m + 1 = S(m).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall m \in P\ (m + 1 = S(m)).\end{aligned}$$

Remark (Interpretation). This theorem records the base clause of the recursive definition in the operation notation itself. Adding the distinguished first element applies one successor step to the left input.

Remark (Dependencies).

- Addition on a Peano System

Theorem (Addition Successor on the Right)

Theorem 20.3.4 (Addition Successor on the Right). *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m + S(n) = S(m + n).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall m, n \in P\ (m + S(n) = S(m + n)).\end{aligned}$$

Remark (Interpretation). This theorem records the recursive step of addition in operation notation. Moving one successor step forward in the right input moves the sum one successor step forward as well.

Remark (Dependencies).

- Addition on a Peano System

Left Input Initialization

The recursive definition controls addition when the right input is advanced by successor. To use addition symmetrically, the first structural task is to understand what happens when the distinguished first element appears on the left. The next theorem shows that left addition by 1 also produces a single successor step.

Theorem (One Plus n)

Theorem 20.3.5 (One Plus n)**.** *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 + n = S(n).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{array}{l}\text{Let } (P, S, 1) \text{ be a Peano system.} \\ \forall n \in P \ (1 + n = S(n)).\end{array}$$

Remark (Interpretation)*.* The theorem shows that the distinguished first element acts on either side as a single successor step. This is the first sign that the recursively defined operation is not tied permanently to the asymmetry of its definition.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- Induction Principle for a Peano System

Associativity and Commutativity

Once the recursive clauses are available as reusable theorems, addition can be shown to satisfy the two fundamental algebraic symmetries of a binary operation. Associativity makes repeated addition unambiguous, and commutativity shows that the recursively defined operation does not depend on which input is treated as the evolving one.

Theorem (Addition Is Associative)

Theorem 20.3.6 (Addition Is Associative)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a + b) + c = a + (b + c).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P,S,1) \text{ be a Peano system.}\\&\forall a,b,c\in P\ ((a+b)+c=a+(b+c)).\end{aligned}$$

Remark (Interpretation). Associativity shows that successive additions can be regrouped without changing the value. This makes longer sums structurally coherent and prepares addition for later interaction with order and multiplication.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- Induction Principle for a Peano System

Theorem (Addition Is Commutative)

Theorem 20.3.7 (Addition Is Commutative)**.** *Let $(P,S,1)$ be a Peano system. For every $a,b\in P$,*

$$a+b=b+a.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P,S,1) \text{ be a Peano system.}\\&\forall a,b\in P\ (a+b=b+a).\end{aligned}$$

Remark (Interpretation). Although addition is defined by recursion on the right input, the resulting operation is symmetric in its two arguments. Commutativity shows that the apparent asymmetry of the defining clauses is an artifact of construction rather than a feature of the operation itself.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- One Plus n
- Addition Is Associative
- Induction Principle for a Peano System

Cancellation Laws

The next structural property shows that addition preserves enough information to allow one to remove a common summand from an equation. These cancellation laws are the first substitute for subtraction inside a Peano system and are indispensable when order is later defined through additive comparison.

Theorem (Left Cancellation for Addition)

Theorem 20.3.8 (Left Cancellation for Addition)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$a + b = a + c \Longrightarrow b = c.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } (P, S, 1) \text{ be a Peano system.}\\ \forall a, b, c \in P\ \big((a + b = a + c) \Longrightarrow b = c\big).\end{gathered}$$

Remark (Contrapositive quantified statement).

$$\begin{gathered}\text{Let } (P, S, 1) \text{ be a Peano system.}\\ \forall a, b, c \in P\ \big(b \neq c \Longrightarrow a + b \neq a + c\big).\end{gathered}$$

Remark (Interpretation). Left cancellation shows that adding the same element to two inputs cannot erase a difference between them. This is the algebraic core that later makes additive definitions of order behave well.

Remark (Dependencies).

- Injectivity of Successor
- Addition With 1
- Addition Successor on the Right
- One Plus n
- Addition Is Associative
- Addition Is Commutative
- Induction Principle for a Peano System

Theorem (Right Cancellation for Addition)

Theorem 20.3.9 (Right Cancellation for Addition)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$b + a = c + a \Longrightarrow b = c.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, c \in P\ \big((b + a = c + a) \Longrightarrow b = c\big).$$

Remark (Contrapositive quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, c \in P\ \big(b \neq c \Longrightarrow b + a \neq c + a\big).$$

Remark (Interpretation). Right cancellation is the companion to left cancellation. Once commutativity is known, the placement of the common summand no longer matters, so equality can be tested after adding the same element on either side.

Remark (Dependencies).

- Addition Is Commutative
- Left Cancellation for Addition

Theorem (Cancellation for Addition)

Theorem 20.3.10 (Cancellation for Addition)**.** *Let $(P, S, 1)$ be a Peano system. Addition on P satisfies cancellation on both sides:*

$$a + b = a + c \Longrightarrow b = c \qquad \textit{and} \qquad b + a = c + a \Longrightarrow b = c.$$

Go to proof.

Remark (Interpretation). The left and right cancellation laws together say that adding a common natural number never destroys equality information, regardless of which side contains the common summand.

Remark (Dependencies).

- Left Cancellation for Addition
- Right Cancellation for Addition

Additive Non-Return

Addition on $\mathbb{N}$ always moves forward by at least one successor step in its right input. The next theorem records the resulting non-return law: adding a natural number to m never returns the added natural number itself.

Theorem (No Natural Number Equals Itself Plus a Natural Number)

Theorem 20.3.11 (No Natural Number Equals Itself Plus a Natural Number)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $m, n \in P$*,*

$$n \neq m + n.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall m, n \in P\ (n \neq m + n).\end{aligned}$$

Remark (Interpretation)*.* Because every element of P is at least one successor step, the sum $m + n$ lies strictly beyond n. The theorem rules out additive cycles and is a standard tool for order arguments on the natural numbers.

Remark (Source alignment)*.* This is the house-style form of the additive non-return theorem in [Feferman, 1964].

Remark (Dependencies)*.*

- Addition With 1
- Addition Successor on the Right
- No Object Is Its Own Successor
- Induction Principle for a Peano System

20.4 Multiplication on N

Recursive Definition of Multiplication

Multiplication on a Peano system is defined from addition by recursion. For each fixed left input m, the map $n \mapsto m \cdot n$ is generated by starting at m and adding one further copy of m at each successor stage of the right input. This makes multiplication repeated addition rather than an independent primitive operation.

Definition (Multiplication on a Peano System)

Definition 20.4.1 (Multiplication on a Peano System)**.** Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$g_m : P \to P$$

to be the unique function satisfying

$$g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$$

For $m, n \in P$, define

$$m \cdot n \coloneqq g_m(n).$$

Remark (Standard quantified statement)*.*

> Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $g_m : P \to P$ satisfy
> $g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$
> Then for all $m, n \in P$, $\quad m \cdot n \coloneqq g_m(n)$.

Remark (Negated quantified statement)*.*

> Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $g_m : P \to P$ satisfy
> $g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$
> Then for all $m, n, x \in P$, $\quad x \neq m \cdot n \iff x \neq g_m(n)$.

Remark (Failure modes)*.* The definition fails only when a proposed value for $m \cdot n$ does not agree with the recursively generated function attached to the fixed left factor m. Once g_m is determined, every incorrect candidate product is simply a value different from $g_m(n)$.

Remark (Failure mode decomposition)*.*

> Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $g_m : P \to P$ satisfy
> $g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$
> $x \neq m \cdot n \iff \underbrace{x \neq g_m(n)}_{\text{candidate value disagrees with the recursive output}}.$

Remark (Interpretation)*.* Multiplication is introduced by fixing the left input and recursively generating the right-input map. The value $m \cdot 1$ is initialized at m, and each successor step in the right input adds one further copy of m. Multiplication is therefore repeated addition.

Remark (Dependencies)*.*

- Peano System
- General Recursion Theorem for a Peano System

- Addition on a Peano System

Theorem (Multiplication Is Well-Defined on a Peano System)

Theorem 20.4.2 (Multiplication Is Well-Defined on a Peano System)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $m \in P$*, there exists a unique function*

$$g_m : P \to P$$

such that

$$g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$$

Consequently the rule

$$(m, n) \longmapsto m \cdot n := g_m(n)$$

defines a unique binary operation on P*. Go to proof.*

Remark (Standard quantified statement).

> Let $(P, S, 1)$ be a Peano system.
> $\forall m \in P\ \exists! g_m : P \to P\ \Big(g_m(1) = m \wedge \forall n \in P\ (g_m(S(n)) = g_m(n) + m)\Big).$

Remark (Interpretation)*.* The recursive clauses for multiplication determine exactly one right-input map for each fixed left input. The multiplication operation is therefore constructed from recursion and addition rather than assumed at the outset.

Remark (Source alignment)*.* This is the house-style well-definedness form of multiplication obtained by applying the Peano recursion theorem in the manner of Feferman's construction of arithmetic operations [Feferman, 1964].

Remark (Dependencies).

- Multiplication on a Peano System
- General Recursion Theorem for a Peano System
- Addition on a Peano System

Recursive Clauses as Reusable Equalities

The recursive definition of multiplication is given by a base value and a successor step. As with addition, these clauses become more useful once they are isolated as named theorems that can be cited directly in later algebraic arguments.

Theorem (Multiplication With 1)

Theorem 20.4.3 (Multiplication With 1)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $m \in P$,

$$m \cdot 1 = m.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall m \in P\ (m \cdot 1 = m).$$

Remark (Interpretation). This theorem records the base clause of the recursive definition in operation notation. Multiplying by the distinguished first element leaves the left input unchanged.

Remark (Dependencies).

- Multiplication on a Peano System

Theorem (Multiplication Successor on the Right)

Theorem 20.4.4 (Multiplication Successor on the Right)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $m, n \in P$,

$$m \cdot S(n) = (m \cdot n) + m.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall m, n \in P\ (m \cdot S(n) = (m \cdot n) + m).$$

Remark (Interpretation). Moving one successor step forward in the right input adds one further copy of m to the product. This is the recursive step of multiplication written in operation notation.

Remark (Dependencies).

- Multiplication on a Peano System
- Addition on a Peano System

Left Identity

The recursive definition controls multiplication when the right input evolves. To use multiplication symmetrically, the first structural task is to understand what happens when the distinguished first element appears on the left. The next theorem identifies that left action explicitly.

Theorem (One Times n)

Theorem 20.4.5 (One Times n)**.** *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 \cdot n = n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{array}{l} \text{Let } (P, S, 1) \text{ be a Peano system.} \\ \forall n \in P\ (1 \cdot n = n). \end{array}$$

Remark (Interpretation). The distinguished first element acts as a multiplicative identity on the left. This is the multiplicative analogue of the theorem that $1 + n$ produces the successor of n in the additive theory.

Remark (Dependencies).

- Multiplication With 1
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Distributivity and Associativity

Multiplication is built from repeated addition, so its first major structural law is distributivity over addition. Once distributivity is available, the operation can be shown to satisfy associativity, which makes iterated products well-defined without ambiguity of grouping.

Theorem (Multiplication Distributes Over Addition)

Theorem 20.4.6 (Multiplication Distributes Over Addition)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, c \in P \ (a \cdot (b + c) = (a \cdot b) + (a \cdot c)).$$

Remark (Interpretation). Distributivity expresses formally that multiplying by a means repeatedly adding copies of a, so a sum in the second factor splits into two separate blocks of repeated addition. This is the bridge between the additive and multiplicative structures.

Remark (Dependencies).

- Multiplication Successor on the Right
- Addition Is Associative
- Addition Is Commutative
- Induction Principle for a Peano System

Theorem (Left Distributivity of Multiplication Over Addition)

Theorem 20.4.7 (Left Distributivity of Multiplication Over Addition)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$,

$$(a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, c \in P \ \big((a + b) \cdot c = (a \cdot c) + (b \cdot c)\big).$$

Remark (Interpretation). The existing distributive law expands a sum in the right factor. This theorem expands a sum in the left factor. It follows from right distributivity together with commutativity of multiplication.

Remark (Source alignment). This is the house-style form of Feferman's left distributive law for multiplication [Feferman, 1964].

Remark (Dependencies).

- Multiplication Distributes Over Addition
- Multiplication Is Commutative
- Addition Is Commutative

Theorem (Multiplication Distributes over Addition on Both Sides)

Theorem 20.4.8 (Multiplication Distributes over Addition on Both Sides)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \qquad \text{and} \qquad (a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Go to proof.

Remark (Dependencies).

- Multiplication Distributes Over Addition
- Left Distributivity of Multiplication Over Addition

Theorem (Multiplication Is Associative)

Theorem 20.4.9 (Multiplication Is Associative)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P \ ((a \cdot b) \cdot c = a \cdot (b \cdot c)).$

Remark (Interpretation)*.* Associativity shows that repeated multiplication can be regrouped without changing the value. Once this law is established, longer products can be used without parenthetical ambiguity.

Remark (Dependencies).

- Multiplication Distributes Over Addition
- Multiplication With 1
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Commutativity

The recursive definition of multiplication treats the right input as the evolving variable, but the resulting operation is not meant to privilege one factor over the other. The next theorem shows that the asymmetry of the construction disappears in the final operation.

Theorem (Multiplication Is Commutative)

Theorem 20.4.10 (Multiplication Is Commutative)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b \in P$,

$$a \cdot b = b \cdot a.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b \in P \ (a \cdot b = b \cdot a).$$

Remark (Interpretation). Although multiplication is defined recursively in one argument, the resulting operation is symmetric in the two factors. Commutativity shows that repeated addition of a to build $a \cdot b$ agrees exactly with repeated addition of b to build $b \cdot a$.

Remark (Dependencies).

- One Times n
- Multiplication Successor on the Right
- Multiplication Distributes Over Addition
- Addition Is Commutative
- Induction Principle for a Peano System

Cancellation Laws

Multiplication by a natural number preserves enough information to allow a common nonzero factor to be removed from an equality. Since every element of the one-based natural-number system is nonzero, no extra nonzero hypothesis is needed in $\mathbb{N}$.

Theorem (Left Cancellation for Multiplication)

Theorem 20.4.11 (Left Cancellation for Multiplication)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$,

$$a \cdot b = a \cdot c \Longrightarrow b = c.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.} \quad \forall a, b, c \in P \ \big(a \cdot b = a \cdot c \Longrightarrow b = c\big).$$

Remark (Interpretation). Left cancellation says that multiplying two values by the same natural number cannot collapse a genuine difference between them.

Remark (Dependencies).

- Multiplication Preserves and Reflects Strict Order
- Trichotomy on a Peano System

Theorem (Right Cancellation for Multiplication)

Theorem 20.4.12 (Right Cancellation for Multiplication)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$b \cdot a = c \cdot a \Longrightarrow b = c.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.} \quad \forall a, b, c \in P \; \big(b \cdot a = c \cdot a \Longrightarrow b = c\big).$$

Remark (Interpretation)*.* Right cancellation is the symmetric form of multiplicative cancellation. It follows from left cancellation together with commutativity of multiplication.

Remark (Dependencies).

- Left Cancellation for Multiplication
- Multiplication Is Commutative

Theorem (Cancellation for Multiplication)

Theorem 20.4.13 (Cancellation for Multiplication)**.** *Let* $(P, S, 1)$ *be a Peano system. Multiplication on* P *satisfies cancellation on both sides:*

$$a \cdot b = a \cdot c \Longrightarrow b = c \qquad \textit{and} \qquad b \cdot a = c \cdot a \Longrightarrow b = c.$$

Go to proof.

Remark (Dependencies).

- Left Cancellation for Multiplication
- Right Cancellation for Multiplication

20.5 Order on N

Strict and Non-Strict Order

Order on a Peano system is extracted from addition. Since the chapter is using the distinguished first element 1 rather than an additive zero, the strict order is the cleaner primitive notion: $a < b$ means that b is reached from a by adding a nontrivial amount. The non-strict order is then obtained by allowing equality in addition to strict increase.

Definition (Strict Less Than on a Peano System)

Definition 20.5.1 (Strict Less Than on a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a < b$$

exactly when there exists $c \in P$ such that

$$b = a + c.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.\\ &a < b \iff \exists c \in P\ (b = a + c).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.\\ &a \not< b \iff \forall c \in P\ (b \neq a + c).\end{aligned}$$

Remark (Failure modes). Strict inequality fails exactly when no additive witness in P carries a to b. In that case, every nontrivial additive step from a misses b.

Remark (Failure mode decomposition).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.\\ &a \not< b \iff \underbrace{\forall c \in P\ (b \neq a + c)}_{\text{every proposed additive witness fails}}.\end{aligned}$$

Remark (Interpretation). The strict order records whether b lies genuinely ahead of a in the counting structure. The witness c is the amount of additive displacement needed to move from a to b.

Remark (Dependencies).

- Addition on a Peano System

Definition (Less Than or Equal on a Peano System)

Definition 20.5.2 (Less Than or Equal on a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a \leq b$$

exactly when

$$a < b \qquad \text{or} \qquad a = b.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.\\ &a \leq b \iff \big(a < b \lor a = b\big).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.\\ &a \not\leq b \iff \big(a \not< b \land a \neq b\big).\end{aligned}$$

Remark (Failure modes). The non-strict order fails in exactly two simultaneous ways: a is not strictly below b, and a is not equal to b. Thus b neither lies ahead of a nor coincides with it.

Remark (Failure mode decomposition).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.\\ &a \not\leq b \iff \underbrace{a \not< b}_{\text{no strict additive witness}} \land \underbrace{a \neq b}_{\text{equality also fails}}.\end{aligned}$$

Remark (Interpretation). The relation $a \leq b$ allows either equality or a genuine forward step from a to b. It is the additive order relation derived from the strict witness definition.

Remark (Dependencies).

- Strict Less Than on a Peano System

Basic Order Laws

Once the order relations have been defined, the first task is to show that they satisfy the expected structural laws: reflexivity, transitivity, and antisymmetry. These are the properties that make the additive comparison relation behave as an honest order rather than a merely suggestive notation.

Theorem (Order Is Reflexive on a Peano System)

Theorem 20.5.3 (Order Is Reflexive on a Peano System)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a \in P$*,*

$$a \leq a.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall a \in P\ (a \leq a).\end{aligned}$$

Remark (Interpretation). Every element is comparable to itself. Reflexivity supplies the equality case inside the non-strict order.

Remark (Dependencies).

- Less Than or Equal on a Peano System

Theorem (Order Is Transitive on a Peano System)

Theorem 20.5.4 (Order Is Transitive on a Peano System)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$\big(a \leq b \wedge b \leq c\big) \implies a \leq c.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall a, b, c \in P\ \big((a \leq b \wedge b \leq c) \implies a \leq c\big).\end{aligned}$$

Remark (Interpretation). If b lies at or beyond a, and c lies at or beyond b, then c also lies at or beyond a. Transitivity expresses the consistency of additive comparison along a chain.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition Is Associative

Theorem (Order Is Antisymmetric on a Peano System)

Theorem 20.5.5 (Order Is Antisymmetric on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$\big(a \leq b \wedge b \leq a\big) \implies a = b.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } (P, S, 1) \text{ be a Peano system.}\\ \forall a, b \in P \ \big((a \leq b \wedge b \leq a) \implies a = b\big).\end{gathered}$$

Remark (Contrapositive quantified statement)*.*

$$\begin{gathered}\text{Let } (P, S, 1) \text{ be a Peano system.}\\ \forall a, b \in P \ \big(a \neq b \implies \neg(a \leq b \wedge b \leq a)\big).\end{gathered}$$

Remark (Interpretation)*.* Two distinct elements cannot each lie at or beyond the other. Antisymmetry separates genuine equality from bidirectional comparability.

Remark (Dependencies)*.*

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Left Cancellation for Addition
- Addition Is Associative
- No Object Is Its Own Successor

Compatibility with Addition

The order on a Peano system is built from addition, so it must interact coherently with addition. The next theorems show that adding the same element to both sides preserves the comparison relation, regardless of whether the common summand is placed on the right or on the left.

Theorem (Addition Preserves Order on the Right)

Theorem 20.5.6 (Addition Preserves Order on the Right)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$,

$$a \le b \Longrightarrow a + c \le b + c.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\ &\forall a, b, c \in P\ \big(a \le b \Longrightarrow a + c \le b + c\big).\end{aligned}$$

Remark (Interpretation). Adding the same quantity to two comparable elements does not disturb their order. The additive witness for $a \le b$ survives after both sides are shifted by the same summand.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition Is Associative

Theorem (Addition Preserves Order on the Left)

Theorem 20.5.7 (Addition Preserves Order on the Left)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$,

$$a \le b \Longrightarrow c + a \le c + b.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\ &\forall a, b, c \in P\ \big(a \le b \Longrightarrow c + a \le c + b\big).\end{aligned}$$

Remark (Interpretation). The same order preservation holds when the common summand is added on the left. This is the left-handed form of the additive compatibility law.

Remark (Dependencies).

- Addition Preserves Order on the Right
- Addition Is Commutative

Theorem (Addition Preserves Order)

Theorem 20.5.8 (Addition Preserves Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \Longrightarrow a + c \leq b + c \qquad \text{and} \qquad a \leq b \Longrightarrow c + a \leq c + b.$$

Go to proof.

Remark (Dependencies).

- Addition Preserves Order on the Right
- Addition Preserves Order on the Left

Compatibility with Multiplication

Multiplication by a natural number is repeated addition by a positive amount. It therefore preserves strict comparisons, and because every multiplier lies at or beyond 1, it also reflects strict comparisons.

Theorem (Multiplication Preserves and Reflects Strict Order)

Theorem 20.5.9 (Multiplication Preserves and Reflects Strict Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $k, m, n \in P$,*

$$m < n \Longleftrightarrow k \cdot m < k \cdot n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\ &\forall k, m, n \in P \ \big(m < n \Longleftrightarrow k \cdot m < k \cdot n\big).\end{aligned}$$

Remark (Interpretation)*.* Multiplication by a fixed natural number is an order embedding of the natural numbers into themselves. It shifts strict inequalities forward without collapsing distinct values.

Remark (Source alignment)*.* This is the house-style multiplication-order comparison theorem corresponding to Feferman's order laws for products [Feferman, 1964].

Remark (Dependencies).

- Strict Less Than on a Peano System
- Multiplication on a Peano System

- Multiplication Distributes Over Addition
- Multiplication Is Commutative
- Addition Preserves Order on the Right
- Left Cancellation for Addition

Successor Characterization and Trichotomy

Strict inequality can be recast in terms of stepping one successor beyond the smaller element. This successor reformulation sharpens the order relation and leads to trichotomy, which states that any two elements are linearly comparable.

Theorem (Strict Order Successor Characterization)

Theorem 20.5.10 (Strict Order Successor Characterization). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a < b \iff S(a) \leq b.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b \in P \ \big(a < b \iff S(a) \leq b\big).$$

Remark (Interpretation). To say that b is strictly larger than a is to say that b lies at or beyond the first successor step past a. This theorem converts strict order into a non-strict comparison involving successor.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition With 1
- Addition Successor on the Right
- Addition Is Associative
- Addition Is Commutative
- Every Element Is Either 1 or a Successor

Theorem (Trichotomy on a Peano System)

Theorem 20.5.11 (Trichotomy on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$, exactly one of the following holds:*

$$a < b, \qquad a = b, \qquad b < a.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } (P, S, 1) \text{ be a Peano system.} \\
&\forall a, b \in P \Big((a < b \vee a = b \vee b < a) \wedge \\
&\qquad \neg(a < b \wedge a = b) \wedge \neg(a < b \wedge b < a) \wedge \neg(a = b \wedge b < a) \Big).
\end{aligned}$$

Remark (Interpretation). Any two elements of a Peano system are linearly comparable: one lies below the other, or they coincide. Trichotomy is the theorem that upgrades the additive comparison relation from a partial-order structure to a total order.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Order Is Reflexive on a Peano System
- Order Is Transitive on a Peano System
- Order Is Antisymmetric on a Peano System
- Strict Order Successor Characterization
- Induction Principle for a Peano System

Theorem (Natural-Number Order Is Total)

Theorem 20.5.12 (Natural-Number Order Is Total)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a \leq b \qquad \text{or} \qquad b \leq a.$$

Consequently $\leq$ is a total order on P. Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.} \quad \forall a, b \in P \ (a \leq b \vee b \leq a).$$

Remark (Interpretation). Trichotomy says exactly one of $a < b$, $a = b$, or $b < a$ holds. Since the non-strict order allows either strict comparison or equality, trichotomy immediately yields total comparability for $\leq$.

Remark (Dependencies).

- Less Than or Equal on a Peano System
- Trichotomy on a Peano System
- Order Is Reflexive on a Peano System
- Order Is Transitive on a Peano System
- Order Is Antisymmetric on a Peano System

20.6 Powers and Growth

Exponentiation by Recursion

Exponentiation is defined on a Peano system by recursion on the exponent: $a^1 = a$ and $a^{S(n)} = a^n \cdot a$. This makes exponentiation a derived operation built from multiplication in exactly the way multiplication is built from addition. The recursive clauses are immediately isolated as named theorems so that later algebraic arguments can cite them directly.

Definition (Exponentiation on a Peano System)

Definition 20.6.1 (Exponentiation on a Peano System)**.** Let $(P, S, 1)$ be a Peano system. For each $a \in P$, define the function

$$h_a : P \to P$$

to be the unique function satisfying

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$$

For $a, n \in P$, define

$$a^n := h_a(n).$$

Remark (Standard quantified statement).

> Let $(P, S, 1)$ be a Peano system. For each $a \in P$, let $h_a : P \to P$ satisfy
> $h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$
> Then for all $a, n \in P$, $\quad a^n := h_a(n)$.

Remark (Negated quantified statement).

Let $(P, S, 1)$ be a Peano system. For each $a \in P$, let $h_a : P \to P$ satisfy
$h_a(1) = a \quad$ and $\quad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a)$.
Then for all $a, n, x \in P$, $\quad x \neq a^n \iff x \neq h_a(n)$.

Remark (Interpretation). Exponentiation is repeated multiplication in the same sense that multiplication is repeated addition. The base a is the fixed left factor; the exponent n counts how many times a is multiplied together. The recursive definition propagates rightward: $a^1 = a$, $a^2 = a^1 \cdot a = a \cdot a$, and so on.

Remark (Dependencies).

- Peano System
- General Recursion Theorem for a Peano System
- Multiplication on a Peano System

Theorem (Exponentiation Is Well-Defined on a Peano System)

Theorem 20.6.2 (Exponentiation Is Well-Defined on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a \in P$, there exists a unique function*

$$h_a : P \to P$$

such that

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$$

Consequently the rule

$$(a, n) \longmapsto a^n := h_a(n)$$

defines a unique binary operation on P. *Go to proof.*

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.

$\forall a \in P\ \exists! h_a : P \to P\ \Big(h_a(1) = a \wedge \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a)\Big)$.

Remark (Interpretation). The recursive clauses for exponentiation determine exactly one exponent map for each fixed base. Exponentiation is therefore constructed from recursion and multiplication rather than assumed as a primitive operation.

Remark (Source alignment). This is the house-style existence-and-uniqueness form of the recursive definition of exponentiation in [Feferman, 1964].

Remark (Dependencies).

- General Recursion Theorem for a Peano System
- Multiplication on a Peano System

Exponent Laws

The recursive definition immediately yields the standard exponent laws. These are not assumed; they are proved from the recursive clauses by induction. The three primary laws are: the sum law $a^{m+n} = a^m \cdot a^n$, the product law $(a \cdot b)^n = a^n \cdot b^n$, and the power law $(a^m)^n = a^{mn}$.

Theorem (Exponent Sum Law)

Theorem 20.6.3 (Exponent Sum Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$a^{m+n} = a^m \cdot a^n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.} \quad \forall a, m, n \in P \; (a^{m+n} = a^m \cdot a^n).$$

Remark (Interpretation). Adding exponents corresponds to concatenating the two blocks of repeated multiplication. The proof is induction on n using the recursive clause $a^{S(n)} = a^n \cdot a$ and associativity of multiplication.

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Is Associative
- Induction Principle for a Peano System

Theorem (Exponent Product Law)

Theorem 20.6.4 (Exponent Product Law). *Let $(P, S, 1)$ be a Peano system. For all $a, b, n \in P$,*

$$(a \cdot b)^n = a^n \cdot b^n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.} \quad \forall a, b, n \in P \; ((a \cdot b)^n = a^n \cdot b^n).$$

Remark (Interpretation). Raising a product to a power distributes over the factors. The proof is induction on n, using commutativity and associativity of multiplication to rearrange the interleaved factors at each successor step.

Remark (Dependencies).

- Exponentiation on a Peano System

- Multiplication Is Associative
- Multiplication Is Commutative
- Induction Principle for a Peano System

Theorem (Exponent Power Law)

Theorem 20.6.5 (Exponent Power Law)**.** *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$(a^m)^n = a^{m \cdot n}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{Let } (P, S, 1) \text{ be a Peano system.} \quad \forall a, m, n \in P\ ((a^m)^n = a^{m \cdot n}).$$

Remark (Interpretation)*.* Raising a power to a further power multiplies the exponents. The proof is induction on n using the exponent sum law and the recursive definition of multiplication.

Remark (Dependencies)*.*

- Exponentiation on a Peano System
- Exponent Sum Law
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Order Compatibility of Powers

Exponentiation is order-sensitive in both its base and its exponent. For a fixed positive exponent, taking powers preserves and reflects strict order of bases. For a fixed base beyond 1, increasing the exponent strictly increases the value.

Theorem (Powers with Common Exponent Preserve Strict Order)

Theorem 20.6.6 (Powers with Common Exponent Preserve Strict Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, n \in P$,*

$$a < b \Longleftrightarrow a^n < b^n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, n \in P \ \big(a < b \iff a^n < b^n\big).$$

Remark (Interpretation). A positive exponent does not collapse the strict order between bases. Since the exponent is a natural number, exponentiation repeatedly multiplies by positive factors and preserves the order information.

Remark (Source alignment). This is the house-style common-exponent comparison theorem corresponding to Feferman's natural-number exponent laws [Feferman, 1964].

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Preserves and Reflects Strict Order
- Induction Principle for a Peano System

Theorem (Powers with Common Base Preserve Strict Order)

Theorem 20.6.7 (Powers with Common Base Preserve Strict Order)**.** *Let* $(P, S, 1)$ *be a Peano system. For every* $a, m, n \in P$*,*

$$a^m < a^n \iff 1 < a \wedge m < n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, m, n \in P \ \big(a^m < a^n \iff (1 < a \wedge m < n)\big).$$

Remark (Interpretation). Strict growth in the exponent occurs exactly when the base is beyond the multiplicative identity. If the base is 1, all powers are 1; if the base is larger than 1, each successor step in the exponent multiplies by a factor that strictly increases the value.

Remark (Source alignment). This is the house-style common-base comparison theorem corresponding to Feferman's exponent-order theorem [Feferman, 1964].

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Preserves and Reflects Strict Order
- Exponent Sum Law
- Induction Principle for a Peano System

Exponential Growth: $n < 2^n$

The most important growth comparison on $\mathbb{N}$ is that exponential growth outpaces linear growth: $n < 2^n$ for all $n \in P$. This single inequality captures why exponential towers dwarf polynomial bounds and is the prototypical example of an induction argument whose inductive step uses both the hypothesis and a multiplicative amplification.

Theorem ($n < 2^n$ for All $n \in \mathbb{N}$)

Theorem 20.6.8 ($n < 2^n$ for All $n \in \mathbb{N}$)**.** *Let $(P, S, 1)$ be a Peano system, and let $2 := S(1)$. For every $n \in P$,*

$$n < 2^n.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system and let $2 := S(1)$.
$\forall n \in P\ (n < 2^n)$.

Remark (Interpretation). The base case $1 < 2 = 2^1$ holds by definition of 2 and the strict order. For the inductive step, assume $n < 2^n$. Then $2^{S(n)} = 2^n \cdot 2$. Since $2^n > n \geq 1$, we have $2^n \cdot 2 > n \cdot 2 = n + n > n + 1 = S(n)$, where the last step uses that $n \geq 1$ so $n + n > n + 1$. Hence $S(n) < 2^{S(n)}$. The inequality $n < 2^n$ is the threshold that distinguishes the growth regimes and reappears in every comparison between polynomial and exponential bounds.

Remark (Dependencies).

- Exponentiation on a Peano System
- Strict Less Than on a Peano System
- Multiplication Distributes Over Addition
- Addition Preserves Order on the Right
- Induction Principle for a Peano System

20.7 Well Ordering Principle

The Well-Ordering Principle

The well-ordering principle asserts that every nonempty subset of $\mathbb{N}$ contains a least element. This is not merely a consequence of the order axioms; it is a property that distinguishes $\mathbb{N}$ from dense ordered sets such as $\mathbb{Q}$, which have no smallest positive element. The well-ordering principle is equivalent to ordinary induction over $\mathbb{N}$, but it supports a distinct proof style: to establish a universal statement, assume it fails for some element and derive a contradiction by examining the least failure.

Theorem (Well-Ordering Principle)

Theorem 20.7.1 (Well-Ordering Principle)**.** *Let $(P, S, 1)$ be a Peano system. Every nonempty subset of P has a least element with respect to the order $\leq$ on P. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall A \subseteq P\ \Big(A \neq \varnothing \implies \exists m \in A\ \forall a \in A\ (m \leq a)\Big).$$

Remark (Predicate reading).

$$\text{WellOrderingPrinciple} \coloneqq \forall A\ \big((A \subseteq P \land \exists a\,(a \in A)) \Rightarrow \exists m\,(m \in A \land \forall a\,(a \in A \Rightarrow m \leq a))\big).$$

Remark (Negated quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\exists A \subseteq P\ \Big(A \neq \varnothing \land \forall m \in A\ \exists a \in A\ (a < m)\Big).$$

Remark (Failure modes). The well-ordering principle would fail if some nonempty subset of P had no least element. That means for every candidate minimum, a strictly smaller element of the subset exists, producing an infinite strictly descending chain in P. Induction rules this out.

Remark (Failure mode decomposition).

$$\neg\,\text{WellOrderingPrinciple} \iff \exists A \subseteq P\ \Big(\underbrace{A \neq \varnothing}_{\text{nonempty}} \land \underbrace{\forall m \in A\ \exists a \in A\ (a < m)}_{\text{no least element}}\Big).$$

Remark (Interpretation). The well-ordering principle is the order-theoretic complement to induction. Induction says one can build up: start at 1 and keep stepping forward. Well-ordering says one can always descend to a minimum: any nonempty set has a floor. Proof strategies that use well-ordering typically assume a property fails somewhere, form the set of failures, extract its minimum by well-ordering, and derive a contradiction from the minimality of that element.

Remark (Dependencies).

- Peano System
- Less Than or Equal on a Peano System
- Induction Principle for a Peano System
- Trichotomy on a Peano System

Equivalence of Induction, Strong Induction, and Well-Ordering

Ordinary induction, strong induction, and the well-ordering principle are logically equivalent over a Peano system. Each can be used as the primitive proof engine and the others recovered as theorems. This equivalence matters structurally: it means that induction proofs, strong-induction proofs, and least-counterexample proofs are different presentations of the same foundational mechanism.

Theorem (Induction, Strong Induction, and Well-Ordering Are Equivalent)

Theorem 20.7.2 (Induction, Strong Induction, and Well-Ordering Are Equivalent)**.** *Let $(P, S, 1)$ be a Peano system equipped with the order $\leq$. The following are equivalent:*

1. *The induction principle: for every predicate φ on P, if $\varphi(1)$ holds and the successor step holds, then $\varphi(n)$ holds for all $n \in P$.*
2. *The strong induction principle: for every predicate φ on P, if $\varphi(k)$ holds for every $k < n$ implies $\varphi(n)$, then $\varphi(n)$ holds for every $n \in P$.*
3. *The well-ordering principle: every nonempty subset of P has a least element with respect to $\leq$.*

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.

$$\mathrm{WellOrderingPrinciple} \iff \mathrm{StrongInduction} \iff \Big(\forall \varphi \ (\varphi(1) \wedge \forall n \in P \ (\varphi(n) \implies \varphi(S(n))) \implies \forall n \in P \ \varphi$$

Remark (Interpretation). Induction proves well-ordering by contrapositive: if a nonempty set A had no least element, the set of non-members of A would be inductive, forcing $A = \varnothing$, a contradiction. Well-ordering proves induction by forming the set of elements where φ fails and, if nonempty, extracting a least failure to derive a contradiction with the successor step.

Remark (Dependencies).

- Induction Principle for a Peano System
- Strong Induction on a Peano System
- Well-Ordering Principle

20.8 Induction

Induction from an Arbitrary Base

Ordinary and strong induction each begin at the distinguished first element 1. Many arguments require starting at a later element. Induction from an arbitrary base m restricts the claim to elements at or above m and starts the inductive process there.

Theorem (Induction from an Arbitrary Base)

Theorem 20.8.1 (Induction from an Arbitrary Base). *Let $(P, S, 1)$ be a Peano system, let $m \in P$, and let φ be a predicate on P. If*

$$\varphi(m) \qquad \textit{and} \qquad \forall n \in P \ \big(n \geq m \wedge \varphi(n) \implies \varphi(S(n))\big),$$

then

$$\forall n \in P \ \big(n \geq m \implies \varphi(n)\big).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system, $m \in P$, and φ a predicate on P.
$$\Big(\varphi(m) \wedge \forall n \in P \ \big(n \geq m \wedge \varphi(n) \implies \varphi(S(n))\big)\Big) \implies \forall n \in P \ \big(n \geq m \implies \varphi(n)\big).$$

Remark (Interpretation). Apply ordinary induction to the shifted predicate $\psi(n) :=(n \geq m \implies \varphi(n))$, which holds vacuously below m and reduces to φ above m. This theorem is the formal license for all arguments of the form "for all $n \geq m$, proceed by induction."

Remark (Dependencies).

- Induction Principle for a Peano System
- Less Than or Equal on a Peano System

20.9 Algebraic Structure of N

$\mathbb{N}$ as a Commutative Monoid Under Addition

The arithmetic established in the preceding sections can now be packaged into a named algebraic structure. Under addition, $\mathbb{N}$ is a commutative monoid with identity 0 — or, in the Peano-system setting that uses 1 as the distinguished element with no additive identity available internally, the correct statement is that $(\mathbb{N}, +)$ satisfies associativity and commutativity. This topic states the structural conclusion that follows from the individual arithmetic theorems proved above.

Theorem (ℕ Is Closed, Associative, and Commutative Under Addition)

Theorem 20.9.1 (ℕ Is Closed, Associative, and Commutative Under Addition)**.** *Let $(P, S, 1)$ be a Peano system. The binary operation $+$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a + b \in P$.*
2. ***Associativity.*** *For all $a, b, c \in P$, $(a + b) + c = a + (b + c)$.*
3. ***Commutativity.*** *For all $a, b \in P$, $a + b = b + a$.*
4. ***Left cancellation.*** *For all $a, b, c \in P$, if $a + b = a + c$ then $b = c$.*
5. ***Right cancellation.*** *For all $a, b, c \in P$, if $b + a = c + a$ then $b = c$.*

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P : \ (a + b) + c = a + (b + c) \ \wedge \ a + b = b + a \ \wedge \ (a + b = a + c \Longrightarrow b = c)$.

Remark (Interpretation). This theorem is a packaging result. It names the algebraic structure that has been proved piece by piece: the three individual theorems on associativity, commutativity, and cancellation together certify that $(P, +)$ is a cancellative commutative semigroup. There is no additive identity in P because P has no element 0 with $0 + n = n$ for all n; the smallest element under addition is 1, not an identity.

Remark (Dependencies).

- Addition Is Associative
- Addition Is Commutative
- Left Cancellation for Addition
- Right Cancellation for Addition

ℕ as a Commutative Monoid Under Multiplication

Multiplication on P has a two-sided multiplicative identity, namely 1, the distinguished first element. Under multiplication, P is therefore a commutative monoid. This topic states the structural packaging for multiplication that parallels the additive packaging above.

Theorem (ℕ Is a Commutative Monoid Under Multiplication)

Theorem 20.9.2 (ℕ Is a Commutative Monoid Under Multiplication)**.** *Let $(P, S, 1)$ be a Peano system. The binary operation $\cdot$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a \cdot b \in P$.*
2. ***Associativity.*** *For all $a, b, c \in P$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.*
3. ***Commutativity.*** *For all $a, b \in P$, $a \cdot b = b \cdot a$.*
4. ***Identity.*** *For all $a \in P$, $1 \cdot a = a = a \cdot 1$.*

Consequently $(P, \cdot, 1)$ is a commutative monoid. Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\ &\forall a, b, c \in P: \; (a \cdot b) \cdot c = a \cdot (b \cdot c) \;\wedge\; a \cdot b = b \cdot a \;\wedge\; 1 \cdot a = a.\end{aligned}$$

Remark (Interpretation)*.* The distinguished first element 1 is a multiplicative identity on both sides, so multiplication promotes P from a semigroup to a monoid. The absence of multiplicative inverses (other than $1^{-1} = 1$, which does not lie in P in the usual sense) is what prevents $(P, \cdot)$ from being a group. The structural limitation is intrinsic: no element of P other than 1 has a reciprocal in P.

Remark (Dependencies).

- Multiplication Is Associative
- Multiplication Is Commutative
- Multiplication With 1
- One Times n

ℕ as a Semiring and the Absence of Subtraction

Together, addition and multiplication on P form a semiring: addition is a commutative semigroup, multiplication is a commutative monoid, and multiplication distributes over addition on both sides. The critical structural limitation is that subtraction is not a total operation on P. Since P has no additive identity and no additive inverses, the equation $a + x = b$ has a solution in P only when $b > a$; it has no solution when $b \leq a$. This failure of internal subtraction is the reason ℤ is introduced immediately after ℕ.

Theorem (ℕ Is a Semiring)

Theorem 20.9.3 (ℕ Is a Semiring)**.** *Let $(P, S, 1)$ be a Peano system. Then $(P, +, \cdot)$ satisfies:*

1. *$(P, +)$ is a commutative semigroup.*
2. *$(P, \cdot, 1)$ is a commutative monoid.*
3. *Multiplication distributes over addition: for all $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P : \ a \cdot (b + c) = (a \cdot b) + (a \cdot c) \ \wedge \ (a + b) + c = a + (b + c) \ \wedge \ a + b = b + a \ \wedge \ 1 \cdot a = a.$

Remark (Interpretation). A semiring packages the interaction of two operations without requiring additive inverses. The natural numbers are the canonical example of a semiring that is not a ring: they satisfy every ring axiom except the existence of additive inverses, which fails because there is no element -1 in P. The integers are the smallest ring extending P that repairs this failure.

Remark (Dependencies).

- ℕ Is Closed, Associative, and Commutative Under Addition
- ℕ Is a Commutative Monoid Under Multiplication
- Multiplication Distributes Over Addition

Chapter 21

Constructing the Whole Numbers

21.1 Why Whole Numbers Are Introduced

Adjoining an Additive Identity

The natural-number system developed in the preceding chapter is one-based: its distinguished first element is 1. This is sufficient for counting and recursive arithmetic, but addition on $\mathbb{N}$ has no additive identity inside $\mathbb{N}$. The whole numbers are introduced by adjoining a new element before constructing the integers.

The Missing Identity

Remark (Exposition). In the one-based system, every element of $\mathbb{N}$ is positive. The operation $+$ is defined and behaves associatively and commutatively, but there is no element $e \in \mathbb{N}$ satisfying $e + n = n$ for every $n \in \mathbb{N}$. The element needed for that identity is not another successor-stage element; it is a new element placed before 1.

Remark (Interpretation). The passage from $\mathbb{N}$ to $\mathbb{W}$ is therefore not a change in the positive arithmetic already constructed. It is an extension that adds the missing additive identity and leaves the positive part intact.

Remark (Dependencies).

- Addition on a Peano System
- Addition Is Associative
- Addition Is Commutative

Remark (Exposition)*.* Once 0 is available, addition can be packaged as a genuine commutative monoid. That structure is the natural input for the integer construction, where ordered pairs are interpreted as formal differences and subtraction is made available by a quotient construction.

21.2 Definition of the Whole Numbers

The Positive Part with a New Zero

The whole numbers are obtained by adjoining a single new element to the one-based natural numbers. The notation records two facts at once: the positive natural numbers remain present by inclusion, and the new element 0 is not one of those positive natural numbers.

The Underlying Set

Definition (Whole Numbers)

Definition 21.2.1 (Whole Numbers)**.** Let $\mathbb{N}$ be the one-based natural-number system developed in the preceding chapter. Let 0 be a new element with $0 \notin \mathbb{N}$. Define

$$\mathbb{W} := \mathbb{N} \cup \{0\}.$$

Remark (Standard quantified statement)*.*

$$\mathbb{W} = \mathbb{N} \cup \{0\} \quad \text{and} \quad 0 \notin \mathbb{N}.$$

Remark (Interpretation)*.* Elements of $\mathbb{W}$ are either the new element 0 or positive natural numbers. The natural numbers embed into $\mathbb{W}$ by the inclusion map $n \mapsto n$. The element 0 is not a successor-stage element of the original one-based Peano system.

Remark (Dependencies)*.*

- Peano System

Theorem ($0 \notin \mathbb{N}$)

Theorem 21.2.2 ($0 \notin \mathbb{N}$)**.** *The adjoined element 0 is not an element of the one-based natural-number system:*

$$0 \notin \mathbb{N}.$$

Go to proof.

Remark (Interpretation)*.* This theorem records the disjointness condition built into the construction of $\mathbb{W}$. The element 0 is new; it is not a renamed natural number.

Remark (Dependencies)*.*

- Whole Numbers

21.3 Extending Successor

Successor After Adjoining Zero

The new successor map on $\mathbb{W}$ agrees with the old successor on the positive part and sends the new initial element 0 to 1. This makes $\mathbb{W}$ a zero-based successor chain while preserving the one-based successor behavior already proved on $\mathbb{N}$.

The Extended Successor

Definition (Successor on $\mathbb{W}$)

Definition 21.3.1 (Successor on $\mathbb{W}$)**.** Define $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$ by

$$S_{\mathbb{W}}(0) = 1 \qquad \text{and} \qquad S_{\mathbb{W}}(n) = S_{\mathbb{N}}(n) \quad (n \in \mathbb{N}).$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W},\\ &S_{\mathbb{W}}(0) = 1,\\ &\forall n \in \mathbb{N}\ \big(S_{\mathbb{W}}(n) = S_{\mathbb{N}}(n)\big).\end{aligned}$$

Remark (Interpretation)*.* The definition inserts 0 immediately before the positive chain. After the first step, successor is exactly the successor already defined on $\mathbb{N}$.

Remark (Dependencies)*.*

- Whole Numbers
- Peano System

Theorem (Successor on $\mathbb{W}$ Is Well-Defined)

Theorem 21.3.2 (Successor on $\mathbb{W}$ Is Well-Defined)**.** *The rule defining $S_{\mathbb{W}}$ determines a function $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$S_{\mathbb{W}} \text{ is a well-defined function from } \mathbb{W} \text{ to } \mathbb{W}.$$

Remark (Interpretation)*.* The cases are exclusive because $0 \notin \mathbb{N}$, and the positive case lands in $\mathbb{N} \subseteq \mathbb{W}$.

Remark (Dependencies)*.*

- Whole Numbers
- Successor on $\mathbb{W}$

Theorem (0 Is Not a Successor in $\mathbb{W}$)

Theorem 21.3.3 (0 Is Not a Successor in $\mathbb{W}$)**.** *There is no $a \in \mathbb{W}$ such that $S_{\mathbb{W}}(a) = 0$. Go to proof.*

Remark (Standard quantified statement).

$$\neg \exists a \in \mathbb{W} \ \big(S_{\mathbb{W}}(a) = 0\big).$$

Remark (Interpretation). The new element 0 is the first element of the extended chain. It is placed before 1, not obtained by applying successor to some earlier element.

Remark (Dependencies).

- Successor on $\mathbb{W}$
- 1 Is Not a Successor

Theorem (Every Nonzero Element of $\mathbb{W}$ Is a Successor)

Theorem 21.3.4 (Every Nonzero Element of $\mathbb{W}$ Is a Successor)**.** *If $a \in \mathbb{W}$ and $a \neq 0$, then there exists $b \in \mathbb{W}$ such that $a = S_{\mathbb{W}}(b)$. Go to proof.*

Remark (Standard quantified statement).

$$\forall a \in \mathbb{W} \ \big(a \neq 0 \implies \exists b \in \mathbb{W} \ (a = S_{\mathbb{W}}(b))\big).$$

Remark (Interpretation). The element 1 is the successor of 0, and every positive natural number beyond 1 remains a successor in the original Peano chain.

Remark (Dependencies).

- Whole Numbers
- Successor on $\mathbb{W}$
- Induction Axiom

Theorem (Successor on $\mathbb{W}$ Is Injective)

Theorem 21.3.5 (Successor on $\mathbb{W}$ Is Injective)**.** *For all $a, b \in \mathbb{W}$,*

$$S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{W} \ \big(S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b\big).$$

Remark (Interpretation). The extended successor still never merges two inputs. The only new case is the comparison between 0 and a positive natural number, and that case is ruled out by the original non-return condition for the first element.

Remark (Dependencies).

- Successor on $\mathbb{W}$
- Injectivity of Successor
- 1 Is Not a Successor

Theorem ($(\mathbb{W}, 0, S_{\mathbb{W}})$ Satisfies the Successor Properties)

Theorem 21.3.6 ($(\mathbb{W}, 0, S_{\mathbb{W}})$ Satisfies the Successor Properties)**.** *The triple* $(\mathbb{W}, 0, S_{\mathbb{W}})$ *satisfies the successor-side properties:*

$$0 \in \mathbb{W}, \qquad S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W},$$

$$\neg\exists a \in \mathbb{W}\ (S_{\mathbb{W}}(a) = 0), \qquad \forall a, b \in \mathbb{W}\ (S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b).$$

Go to proof.

Remark (Interpretation). Adjoining 0 shifts the base of the successor chain from 1 to 0. The resulting successor structure has a base element, closes under successor, has no predecessor for the base, and keeps successor injective.

Remark (Dependencies).

- Whole Numbers
- Successor on $\mathbb{W}$
- Successor on $\mathbb{W}$ Is Well-Defined
- 0 Is Not a Successor in $\mathbb{W}$
- Successor on $\mathbb{W}$ Is Injective

21.4 Extending Addition to $\mathbb{W}$

21.4.1 Extending Addition to $\mathbb{W}$

Addition with Zero

Addition on $\mathbb{W}$ keeps the old natural-number addition whenever both inputs are positive and adds identity clauses for the new element 0. The case split is exclusive because $0 \notin \mathbb{N}$.

The Extended Addition Operation

Definition (Addition on $\mathbb{W}$)

Definition 21.4.1 (Addition on $\mathbb{W}$)**.** Define $+_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 +_{\mathbb{W}} a = a, \qquad a +_{\mathbb{W}} 0 = a,$$

and, for $a, b \in \mathbb{N}$,

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Remark (Standard quantified statement).

$$\begin{aligned}
&+_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}, \\
&\forall a \in \mathbb{W}\ (0 +_{\mathbb{W}} a = a \ \wedge\ a +_{\mathbb{W}} 0 = a), \\
&\forall a, b \in \mathbb{N}\ (a +_{\mathbb{W}} b = a +_{\mathbb{N}} b).
\end{aligned}$$

Remark (Interpretation). If either input is 0, the new identity clause applies. If both inputs lie in $\mathbb{N}$, the old operation is used. Since $0 \notin \mathbb{N}$, the positive-positive clause does not overlap with the zero clauses.

Remark (Dependencies).

- Whole Numbers
- Addition on a Peano System

Theorem (Addition on $\mathbb{W}$ Is Well-Defined)

Theorem 21.4.2 (Addition on $\mathbb{W}$ Is Well-Defined)**.** *The case definition of $+_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$.* *Go to proof.*

Remark (Standard quantified statement).

$$+_{\mathbb{W}} \text{ is a well-defined binary operation on } \mathbb{W}.$$

Remark (Interpretation). Well-definedness is a case check. The zero cases are new, the positive-positive case is inherited from $\mathbb{N}$, and compatibility is automatic because the positive-positive case cannot contain 0.

Remark (Dependencies).

- Whole Numbers
- Addition on $\mathbb{W}$
- Addition Is Well-Defined on a Peano System

Theorem (Zero Is an Additive Identity on $\mathbb{W}$)

Theorem 21.4.3 (Zero Is an Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 +_{\mathbb{W}} a = a \qquad \textit{and} \qquad a +_{\mathbb{W}} 0 = a.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{W}\ \big(0 +_{\mathbb{W}} a = a\ \wedge\ a +_{\mathbb{W}} 0 = a\big).$$

Remark (Interpretation). This is the structural feature missing from the one-based natural numbers. After adjoining 0, addition has a genuine identity element.

Remark (Dependencies).

- Addition on $\mathbb{W}$

Theorem (Left Additive Identity on $\mathbb{W}$)

Theorem 21.4.4 (Left Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 +_{\mathbb{W}} a = a.$$

Go to proof.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$

Theorem (Right Additive Identity on $\mathbb{W}$)

Theorem 21.4.5 (Right Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$a +_{\mathbb{W}} 0 = a.$$

Go to proof.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$

Theorem (Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$)

Theorem 21.4.6 (Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$)**.** *For all $a, b \in \mathbb{N}$,*

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{N}\ \big(a +_{\mathbb{W}} b = a +_{\mathbb{N}} b\big).$$

Remark (Interpretation). The extension does not alter sums of positive natural numbers.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Addition on a Peano System

Theorem (Addition on $\mathbb{W}$ Is Associative)

Theorem 21.4.7 (Addition on $\mathbb{W}$ Is Associative)**.** *For all $a, b, c \in \mathbb{W}$,*

$$(a +_{\mathbb{W}} b) +_{\mathbb{W}} c = a +_{\mathbb{W}} (b +_{\mathbb{W}} c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{W}\ \big((a +_{\mathbb{W}} b) +_{\mathbb{W}} c = a +_{\mathbb{W}} (b +_{\mathbb{W}} c)\big).$$

Remark (Interpretation). Associativity follows by reducing zero cases to the identity law and reducing the positive-positive-positive case to associativity on $\mathbb{N}$.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Addition Is Associative

Theorem (Addition on $\mathbb{W}$ Is Commutative)

Theorem 21.4.8 (Addition on $\mathbb{W}$ Is Commutative)**.** *For all $a, b \in \mathbb{W}$,*

$$a +_{\mathbb{W}} b = b +_{\mathbb{W}} a.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{W}\ \big(a +_{\mathbb{W}} b = b +_{\mathbb{W}} a\big).$$

Remark (Interpretation). The zero cases are symmetric by definition, and the positive-positive case is the commutativity theorem already proved for $\mathbb{N}$.

Remark (Dependencies).

- Addition on $\mathbb{W}$

- Addition Is Commutative

Theorem (Addition on $\mathbb{W}$ Satisfies Cancellation)

Theorem 21.4.9 (Addition on $\mathbb{W}$ Satisfies Cancellation)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a +_{\mathbb{W}} b = a +_{\mathbb{W}} c \implies b = c.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{W} \left(a +_{\mathbb{W}} b = a +_{\mathbb{W}} c \implies b = c\right).$$

Remark (Interpretation). When $a = 0$, cancellation is the identity law. When all relevant entries are positive, it is the natural-number cancellation theorem.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Left Cancellation for Addition

21.5 Extending Multiplication to $\mathbb{W}$

21.5.1 Extending Multiplication to $\mathbb{W}$

Multiplication with Zero

Multiplication on $\mathbb{W}$ keeps the old natural-number multiplication on positive inputs and adds the absorbing behavior of the new element 0. The cases are compatible because $0 \notin \mathbb{N}$.

The Extended Multiplication Operation

Definition (Multiplication on $\mathbb{W}$)

Definition 21.5.1 (Multiplication on $\mathbb{W}$)**.** Define $\cdot_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 \cdot_{\mathbb{W}} a = 0, \qquad a \cdot_{\mathbb{W}} 0 = 0,$$

and, for $a, b \in \mathbb{N}$,

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Remark (Standard quantified statement).

$$\begin{aligned}
&\cdot_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}, \\
&\forall a \in \mathbb{W} \left(0 \cdot_{\mathbb{W}} a = 0 \ \wedge \ a \cdot_{\mathbb{W}} 0 = 0\right), \\
&\forall a, b \in \mathbb{N} \left(a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b\right).
\end{aligned}$$

Remark (Interpretation). The zero cases are new. The positive-positive case is inherited from $\mathbb{N}$. Since $0 \notin \mathbb{N}$, these clauses do not compete.

Remark (Dependencies).

- Whole Numbers
- Multiplication on a Peano System

Theorem (Multiplication on $\mathbb{W}$ Is Well-Defined)

Theorem 21.5.2 (Multiplication on $\mathbb{W}$ Is Well-Defined)**.** *The case definition of $\cdot_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$. Go to proof.*

Remark (Standard quantified statement).

$$\cdot_{\mathbb{W}} \text{ is a well-defined binary operation on } \mathbb{W}.$$

Remark (Interpretation). The inherited positive case is already well-defined on $\mathbb{N}$, and the new zero cases land in the distinguished element $0 \in \mathbb{W}$.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication Is Well-Defined on a Peano System

Theorem (Zero Is Absorbing for Multiplication on $\mathbb{W}$)

Theorem 21.5.3 (Zero Is Absorbing for Multiplication on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \cdot_{\mathbb{W}} a = 0 \qquad \textit{and} \qquad a \cdot_{\mathbb{W}} 0 = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{W} \left(0 \cdot_{\mathbb{W}} a = 0 \ \wedge \ a \cdot_{\mathbb{W}} 0 = 0\right).$$

Remark (Interpretation). The adjoined element 0 is absorbing for multiplication on both sides.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$

Theorem (Left Zero Absorption on $\mathbb{W}$)

Theorem 21.5.4 (Left Zero Absorption on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \cdot_{\mathbb{W}} a = 0.$$

Go to proof.

Remark (Dependencies).

- Zero Is Absorbing for Multiplication on $\mathbb{W}$

Theorem (Right Zero Absorption on $\mathbb{W}$)

Theorem 21.5.5 (Right Zero Absorption on $\mathbb{W}$). *For every $a \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} 0 = 0.$$

Go to proof.

Remark (Dependencies).

- Zero Is Absorbing for Multiplication on $\mathbb{W}$

Theorem (One Is a Multiplicative Identity on $\mathbb{W}$)

Theorem 21.5.6 (One Is a Multiplicative Identity on $\mathbb{W}$). *For every $a \in \mathbb{W}$,*

$$1 \cdot_{\mathbb{W}} a = a \qquad \text{and} \qquad a \cdot_{\mathbb{W}} 1 = a.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{W} \left(1 \cdot_{\mathbb{W}} a = a \ \wedge \ a \cdot_{\mathbb{W}} 1 = a\right).$$

Remark (Interpretation). The old multiplicative identity remains an identity after adjoining 0.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication With 1
- One Times n

Theorem (Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$)

Theorem 21.5.7 (Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$). *For all $a, b \in \mathbb{N}$,*

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{N} \left(a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b\right).$$

Remark (Interpretation). The extension does not alter products of positive natural numbers.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication on a Peano System

Theorem (Multiplication on $\mathbb{W}$ Is Associative)

Theorem 21.5.8 (Multiplication on $\mathbb{W}$ Is Associative)**.** *For all $a, b, c \in \mathbb{W}$,*

$$(a \cdot_{\mathbb{W}} b) \cdot_{\mathbb{W}} c = a \cdot_{\mathbb{W}} (b \cdot_{\mathbb{W}} c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{W} \ \big((a \cdot_{\mathbb{W}} b) \cdot_{\mathbb{W}} c = a \cdot_{\mathbb{W}} (b \cdot_{\mathbb{W}} c)\big).$$

Remark (Interpretation)*.* Associativity follows from the absorbing zero cases and the inherited associativity of multiplication on positive natural numbers.

Remark (Dependencies).

- Zero Is Absorbing for Multiplication on $\mathbb{W}$
- Multiplication Is Associative

Theorem (Multiplication on $\mathbb{W}$ Is Commutative)

Theorem 21.5.9 (Multiplication on $\mathbb{W}$ Is Commutative)**.** *For all $a, b \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} b = b \cdot_{\mathbb{W}} a.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{W} \ \big(a \cdot_{\mathbb{W}} b = b \cdot_{\mathbb{W}} a\big).$$

Remark (Interpretation)*.* The zero cases are symmetric, and the positive-positive case is the inherited commutativity theorem for $\mathbb{N}$.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication Is Commutative

Theorem (Multiplication Distributes over Addition on $\mathbb{W}$)

Theorem 21.5.10 (Multiplication Distributes over Addition on $\mathbb{W}$)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} (b +_{\mathbb{W}} c) = (a \cdot_{\mathbb{W}} b) +_{\mathbb{W}} (a \cdot_{\mathbb{W}} c).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{W} \left(a \cdot_{\mathbb{W}} (b +_{\mathbb{W}} c) = (a \cdot_{\mathbb{W}} b) +_{\mathbb{W}} (a \cdot_{\mathbb{W}} c)\right).$$

Remark (Interpretation)*.* Distributivity reduces zero cases to the absorbing law and positive cases to the distributive law already proved on $\mathbb{N}$.

Remark (Dependencies)*.*

- Addition on $\mathbb{W}$
- Multiplication on $\mathbb{W}$
- Multiplication Distributes Over Addition

21.6 Order on $\mathbb{W}$

21.6.1 Order on $\mathbb{W}$

Zero as the Least Element

The order on $\mathbb{W}$ extends the order on $\mathbb{N}$ and places the new element 0 below every whole number. Thus the positive part keeps its old order, while the extended system gains a least element.

The Extended Order

Definition (Order on $\mathbb{W}$)

Definition 21.6.1 (Order on $\mathbb{W}$)**.** Define $\leq_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 \leq_{\mathbb{W}} a \quad (a \in \mathbb{W}),$$

and, for $a, b \in \mathbb{N}$,

$$a \leq_{\mathbb{W}} b \quad \Longleftrightarrow \quad a \leq_{\mathbb{N}} b.$$

Remark (Standard quantified statement)*.*

$$\forall a, b \in \mathbb{W} \left(a \leq_{\mathbb{W}} b \Longleftrightarrow (a = 0 \ \vee \ (a, b \in \mathbb{N} \ \wedge \ a \leq_{\mathbb{N}} b))\right).$$

Remark (Interpretation). The non-strict order makes 0 least and otherwise preserves the order already defined on $\mathbb{N}$.

Remark (Dependencies).

- Whole Numbers
- Less Than or Equal on a Peano System

Definition (Strict Order on $\mathbb{W}$)

Definition 21.6.2 (Strict Order on $\mathbb{W}$)**.** Define $<_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 <_{\mathbb{W}} n \quad (n \in \mathbb{N}),$$

and, for $a, b \in \mathbb{N}$,

$$a <_{\mathbb{W}} b \quad \Longleftrightarrow \quad a <_{\mathbb{N}} b.$$

No element of $\mathbb{W}$ is strictly less than 0.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{W} \ \big(a <_{\mathbb{W}} b \Longleftrightarrow (a = 0 \ \wedge \ b \in \mathbb{N}) \vee (a, b \in \mathbb{N} \ \wedge \ a <_{\mathbb{N}} b)\big).$$

Remark (Interpretation). The non-strict order makes 0 least. The strict order says exactly that the positive natural numbers lie after 0 and keep their old strict ordering.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Strict Less Than on a Peano System

Theorem (Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$)

Theorem 21.6.3 (Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$)**.** *For all* $a, b \in \mathbb{N}$,

$$a \leq_{\mathbb{W}} b \Longleftrightarrow a \leq_{\mathbb{N}} b, \qquad a <_{\mathbb{W}} b \Longleftrightarrow a <_{\mathbb{N}} b.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{N} \ \big((a \leq_{\mathbb{W}} b \Longleftrightarrow a \leq_{\mathbb{N}} b) \wedge (a <_{\mathbb{W}} b \Longleftrightarrow a <_{\mathbb{N}} b)\big).$$

Remark (Interpretation). The extended order changes only comparisons involving the new element 0.

Remark (Dependencies).

- Order on $\mathbb{W}$

- Strict Order on $\mathbb{W}$

Theorem (Zero Is the Least Element of $\mathbb{W}$)

Theorem 21.6.4 (Zero Is the Least Element of $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \leq_{\mathbb{W}} a.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{W}\ (0 \leq_{\mathbb{W}} a).$$

Remark (Interpretation)*.* The adjoined element 0 is the first element of the whole-number order.

Remark (Dependencies)*.*

- Order on $\mathbb{W}$

Theorem (Order on $\mathbb{W}$ Is Reflexive)

Theorem 21.6.5 (Order on $\mathbb{W}$ Is Reflexive)**.** *For every $a \in \mathbb{W}$, $a \leq_{\mathbb{W}} a$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{W}\ (a \leq_{\mathbb{W}} a).$$

Remark (Interpretation)*.* Reflexivity is immediate for 0 and inherited from $\mathbb{N}$ for positive elements.

Remark (Dependencies)*.*

- Order on $\mathbb{W}$
- Order Is Reflexive on a Peano System

Theorem (Order on $\mathbb{W}$ Is Antisymmetric)

Theorem 21.6.6 (Order on $\mathbb{W}$ Is Antisymmetric)**.** *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} a \Longrightarrow a = b.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b \in \mathbb{W}\ \big(a \leq_{\mathbb{W}} b\ \wedge\ b \leq_{\mathbb{W}} a \Longrightarrow a = b\big).$$

Remark (Interpretation)*.* Antisymmetry follows from the least-element behavior of 0 and the inherited antisymmetry on positive natural numbers.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Antisymmetric on a Peano System

Theorem (Order on $\mathbb{W}$ Is Transitive)

Theorem 21.6.7 (Order on $\mathbb{W}$ Is Transitive). *For all $a, b, c \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} c \Longrightarrow a \leq_{\mathbb{W}} c.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{W} \ \big(a \leq_{\mathbb{W}} b \ \wedge \ b \leq_{\mathbb{W}} c \Longrightarrow a \leq_{\mathbb{W}} c\big).$$

Remark (Interpretation). Transitivity reduces to the least-element clause when 0 occurs and to the transitivity theorem on $\mathbb{N}$ otherwise.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Transitive on a Peano System

Theorem (Order on $\mathbb{W}$ Is Total)

Theorem 21.6.8 (Order on $\mathbb{W}$ Is Total). *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \quad \text{or} \quad b \leq_{\mathbb{W}} a.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{W} \ \big(a \leq_{\mathbb{W}} b \ \vee \ b \leq_{\mathbb{W}} a\big).$$

Remark (Interpretation). Any comparison involving 0 is decided by leastness, while comparisons between positive elements are decided by totality on $\mathbb{N}$.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Trichotomy on a Peano System

Theorem (Well-Ordering Principle for $\mathbb{W}$)

Theorem 21.6.9 (Well-Ordering Principle for $\mathbb{W}$). *Every nonempty subset of $\mathbb{W}$ has a least element with respect to $\leq_{\mathbb{W}}$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{W}\ \big(A \neq \varnothing \Longrightarrow \exists m \in A\ \forall a \in A\ (m \leq_{\mathbb{W}} a)\big).$$

Remark (Interpretation). A nonempty subset either contains 0, which is least, or lies in the positive part where the well-ordering principle for $\mathbb{N}$ applies.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Zero Is the Least Element of $\mathbb{W}$
- Well-Ordering Principle

21.7 Algebraic Structure of $\mathbb{W}$

21.7.1 Algebraic Structure of $\mathbb{W}$

Packaging the Extended Operations

After addition, multiplication, and order have been extended, the main point of $\mathbb{W}$ can be recorded structurally. The whole numbers form the zero-based arithmetic system used as the bridge from positive natural numbers to the integer construction.

Algebraic Packages

Theorem ($(\mathbb{W}, +, 0)$ Is a Commutative Monoid)

Theorem 21.7.1 ($(\mathbb{W}, +, 0)$ Is a Commutative Monoid)**.** *The structure* $(\mathbb{W}, +_{\mathbb{W}}, 0)$ *is a commutative monoid. Go to proof.*

Remark (Standard quantified statement).

$$(\mathbb{W}, +_{\mathbb{W}}, 0) \text{ is a commutative monoid.}$$

Remark (Interpretation). Addition on $\mathbb{W}$ has an identity element, is associative, and is commutative.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Addition on $\mathbb{W}$ Is Associative
- Addition on $\mathbb{W}$ Is Commutative

Theorem ($(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid)

Theorem 21.7.2 ($(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid)**.** *The structure* $(\mathbb{W}, \cdot_{\mathbb{W}}, 1)$ *is a commutative monoid. Go to proof.*

Remark (Standard quantified statement).

$$(\mathbb{W}, \cdot_{\mathbb{W}}, 1) \text{ is a commutative monoid.}$$

Remark (Interpretation). Multiplication on $\mathbb{W}$ has identity 1, is associative, and is commutative.

Remark (Dependencies).

- One Is a Multiplicative Identity on $\mathbb{W}$
- Multiplication on $\mathbb{W}$ Is Associative
- Multiplication on $\mathbb{W}$ Is Commutative

Theorem ($(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring)

Theorem 21.7.3 ($(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring)**.** *The structure* $(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1)$ *is a commutative semiring. Go to proof.*

Remark (Standard quantified statement).

$$(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1) \text{ is a commutative semiring.}$$

Remark (Interpretation). The whole numbers have the additive and multiplicative structure needed for ordinary arithmetic, but no subtraction operation is included.

Remark (Dependencies).

- $(\mathbb{W}, +, 0)$ Is a Commutative Monoid
- $(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid
- Multiplication Distributes over Addition on $\mathbb{W}$
- Zero Is Absorbing for Multiplication on $\mathbb{W}$

Theorem ($\mathbb{W}$ Is a Commutative Ordered Semiring)

Theorem 21.7.4 ($\mathbb{W}$ Is a Commutative Ordered Semiring)**.** *With the operations* $+_{\mathbb{W}}$ *and* $\cdot_{\mathbb{W}}$ *and the order* $\leq_{\mathbb{W}}$*, the whole numbers form a commutative ordered semiring. Go to proof.*

Remark (Standard quantified statement).

$$(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1, \leq_{\mathbb{W}}) \text{ is a commutative ordered semiring.}$$

Remark (Interpretation). This packages the algebraic semiring laws together with the order facts: $\mathbb{W}$ is totally ordered, 0 is least, and the inherited arithmetic is compatible with that order.

Remark (Dependencies).

- $(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring
- Order on $\mathbb{W}$ Is Total
- Zero Is the Least Element of $\mathbb{W}$
- Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$

Theorem ($\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part)

Theorem 21.7.5 ($\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part)**.** *The inclusion map identifies $\mathbb{N}$ with the positive part $\mathbb{W} \setminus \{0\}$ and preserves successor, 1, addition, multiplication, and order. Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{N} \hookrightarrow \mathbb{W}, \qquad \iota(n) = n,$$

identifies $\mathbb{N}$ with $\mathbb{W} \setminus \{0\}$ and preserves successor, 1, addition, multiplication, and order.

Remark (Interpretation). The positive part of $\mathbb{W}$ is exactly the previously constructed natural-number system, with the same arithmetic and order.

Remark (Dependencies).

- Whole Numbers
- Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$
- Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$
- Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$

Theorem ($\mathbb{W}$ Has No Additive Inverses Except 0)

Theorem 21.7.6 ($\mathbb{W}$ Has No Additive Inverses Except 0)**.** *If $a, b \in \mathbb{W}$ and*

$$a +_{\mathbb{W}} b = 0,$$

then $a = 0$ and $b = 0$. Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{W}\ \big(a +_{\mathbb{W}} b = 0 \Longrightarrow (a = 0\ \wedge\ b = 0)\big).$$

Remark (Interpretation)*.* The only way to sum to zero inside $\mathbb{W}$ is for both summands already to be zero.

Remark (Dependencies)*.*

- Addition on $\mathbb{W}$
- Zero Is the Least Element of $\mathbb{W}$
- No Natural Number Equals Itself Plus a Natural Number

Theorem ($\mathbb{W}$ Is Not a Ring)

Theorem 21.7.7 ($\mathbb{W}$ Is Not a Ring)**.** *With the operations $+_{\mathbb{W}}$ and $\cdot_{\mathbb{W}}$, the whole numbers do not form a ring.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1) \text{ is not a ring.}$$

Remark (Interpretation)*.* The obstruction is not associativity, commutativity, or distributivity. The obstruction is the absence of additive inverses for nonzero whole numbers.

Remark (Dependencies)*.*

- $(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring
- $\mathbb{W}$ Has No Additive Inverses Except 0

21.8 Transition to $\mathbb{Z}$

21.8.1 Transition to $\mathbb{Z}$

From Zero to Additive Inverses

The whole numbers fix the missing additive identity, but they do not make subtraction available in general. The integer construction repairs this by adjoining additive inverses formally.

The Next Construction

Remark (Exposition)*.* The passage from $\mathbb{N}$ to $\mathbb{W}$ adds the element needed for $0 + a = a$. It does not add solutions to equations such as $a + x = b$ when $b < a$. Thus $\mathbb{W}$ is still not closed under subtraction.

Remark (Exposition)*.* The construction of $\mathbb{Z}$ repairs this by adjoining additive inverses formally. One standard route uses ordered pairs from $\mathbb{W} \times \mathbb{W}$, with a pair (a, b) interpreted as a formal difference $a - b$. The equivalence relation then identifies different pairs that represent the same intended difference.

Remark (Dependencies).

- $(\mathbb{W}, +, 0)$ Is a Commutative Monoid
- $\mathbb{W}$ Has No Additive Inverses Except 0
- $\mathbb{W}$ Is Not a Ring

Chapter 22

Integers ($\mathbb{Z}$)

SERIES
From Cantor to Ito

VOLUME II
Origins of Numbers

BOOK I
Discrete Number Systems

CHAPTER 8
Integers ($\mathbb{Z}$)

22.1 Construction of $\mathbb{Z}$

Integer Prepairs

Definition

Definition 22.1.1 (Integer Prepair)**.** An *integer prepair* is an ordered pair $(a, b) \in \mathbb{W} \times \mathbb{W}$, read as the formal difference $a - b$.

Remark (Dependencies).

- Ordered Pair
- Whole Numbers

Definition

Definition 22.1.2 (Integer Prepair Set)**.** The integer prepair set is

$$\operatorname{Pre}\mathbb{Z} := \mathbb{W} \times \mathbb{W}.$$

Remark (Dependencies).

- Cartesian Product
- Whole Numbers

Formal-Difference Equivalence

Definition

Definition 22.1.3 (Formal-Difference Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Z}$, define

$$(a, b) \sim_{\mathbb{Z}} (c, d) \quad :\Longleftrightarrow \quad a + d = c + b.$$

Remark (Dependencies).

- Relation
- Integer Prepair Set
- Addition on $\mathbb{W}$

Lemma

Lemma 22.1.4 (Formal-Difference Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{Z}$*,*

$$(a, b) \sim_{\mathbb{Z}} (a, b).$$

Remark (Dependencies).

- Formal-Difference Equivalence
- Addition on $\mathbb{W}$

Lemma

Lemma 22.1.5 (Formal-Difference Equivalence Is Symmetric)**.** *For all prepairs* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{Z}} (c, d) \Longrightarrow (c, d) \sim_{\mathbb{Z}} (a, b).$$

Remark (Dependencies).

- Formal-Difference Equivalence
- Addition on $\mathbb{W}$

Lemma

Lemma 22.1.6 (Formal-Difference Equivalence Is Transitive)**.** *For all prepairs* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{Z}} (c, d) \wedge (c, d) \sim_{\mathbb{Z}} (e, f) \Longrightarrow (a, b) \sim_{\mathbb{Z}} (e, f).$$

Remark (Dependencies).

- Addition on $\mathbb{W}$ Satisfies Cancellation

Theorem

Theorem 22.1.7 (Formal-Difference Equivalence Is an Equivalence Relation)**.** *The relation $\sim_{\mathbb{Z}}$ is an equivalence relation on* $\operatorname{Pre}\mathbb{Z}$*.*

Remark (Dependencies).

- Formal-Difference Equivalence
- Formal-Difference Equivalence Is Reflexive
- Formal-Difference Equivalence Is Symmetric
- Formal-Difference Equivalence Is Transitive

Definition of $\mathbb{Z}$

Definition

Definition 22.1.8 (Integers)**.** The integers are the quotient

$$\mathbb{Z} := \operatorname{Pre}\mathbb{Z}/{\sim_{\mathbb{Z}}}.$$

Remark (Dependencies).

- Quotient Set
- Integer Prepair Set
- Formal-Difference Equivalence

Definition

Definition 22.1.9 (Integer Class Notation)**.** The notation $[a, b]$ denotes the $\sim_{\mathbb{Z}}$-equivalence class of the prepair (a, b).

Remark (Dependencies).

- Integers
- Formal-Difference Equivalence

Zero and One in $\mathbb{Z}$

Definition

Definition 22.1.10 (Zero in $\mathbb{Z}$)**.** Define

$$0_{\mathbb{Z}} := [0, 0].$$

Remark (Dependencies).

- Integer Class Notation
- Whole Numbers

Definition

Definition 22.1.11 (One in $\mathbb{Z}$)**.** Define

$$1_{\mathbb{Z}} := [1, 0].$$

Remark (Dependencies).

- Integer Class Notation
- Whole Numbers

Theorem

Theorem 22.1.12 (Zero Is an Integer)**.** *The class $0_{\mathbb{Z}}$ is an element of $\mathbb{Z}$.*

Remark (Dependencies).

- Integers
- Zero in $\mathbb{Z}$
- Integer Class Notation

Theorem

Theorem 22.1.13 (One Is an Integer)**.** *The class $1_{\mathbb{Z}}$ is an element of $\mathbb{Z}$.*

Remark (Dependencies).

- Integers
- One in $\mathbb{Z}$
- Integer Class Notation

Theorem

Theorem 22.1.14 (Zero Is Not One in $\mathbb{Z}$)**.** *In $\mathbb{Z}$,*

$$0_{\mathbb{Z}} \neq 1_{\mathbb{Z}}.$$

Remark (Dependencies).

- $0 \notin \mathbb{N}$

22.2 Operations on $\mathbb{Z}$-Classes

Addition on $\mathbb{Z}$-Classes

Definition

Definition 22.2.1 (Addition on $\mathbb{Z}$-Classes)**.** Define addition on representatives by

$$[a, b] + [c, d] := [a + c, b + d].$$

Remark (Dependencies).

- Integer Class Notation
- Addition on $\mathbb{W}$

Lemma

Lemma 22.2.2 (Addition Respects Formal-Difference Equivalence on the Left)**.** *If* $[a, b] = [a', b']$*, then*

$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Formal-Difference Equivalence
- Addition on $\mathbb{W}$ Satisfies Cancellation

Lemma

Lemma 22.2.3 (Addition Respects Formal-Difference Equivalence on the Right)**.** *If* $[c, d] = [c', d']$*, then*

$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Formal-Difference Equivalence
- Addition on $\mathbb{W}$ Satisfies Cancellation

Lemma

Lemma 22.2.4 (Addition Respects Formal-Difference Equivalence)**.** *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$*, then*

$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Addition Respects Formal-Difference Equivalence on the Left
- Addition Respects Formal-Difference Equivalence on the Right

Theorem

Theorem 22.2.5 (Addition on $\mathbb{Z}$ Is Well-Defined). *Addition determines a well-defined binary operation $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.*

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Addition Respects Formal-Difference Equivalence

Theorem

Theorem 22.2.6 (Addition on $\mathbb{Z}$ Is Associative). *For all $x, y, z \in \mathbb{Z}$,*

$$(x + y) + z = x + (y + z).$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Well-Defined
- Addition on $\mathbb{W}$ Is Associative

Theorem

Theorem 22.2.7 (Addition on $\mathbb{Z}$ Is Commutative). *For all $x, y \in \mathbb{Z}$,*

$$x + y = y + x.$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Well-Defined
- Addition on $\mathbb{W}$ Is Commutative

Theorem

Theorem 22.2.8 (Zero Is a Left Additive Identity on $\mathbb{Z}$). *For every $x \in \mathbb{Z}$,*

$$0_{\mathbb{Z}} + x = x.$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Zero in $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Well-Defined
- Zero Is an Additive Identity on $\mathbb{W}$

Corollary

Corollary 22.2.9 (Zero Is a Right Additive Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$x + 0_{\mathbb{Z}} = x.$$

Remark (Dependencies).

- Zero Is a Left Additive Identity on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.2.10 (Zero Is an Additive Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$0_{\mathbb{Z}} + x = x \qquad \textit{and} \qquad x + 0_{\mathbb{Z}} = x.$$

Remark (Dependencies).

- Zero Is a Left Additive Identity on $\mathbb{Z}$
- Zero Is a Right Additive Identity on $\mathbb{Z}$

Negation on $\mathbb{Z}$-Classes

Definition

Definition 22.2.11 (Negation on $\mathbb{Z}$-Classes)**.** Define negation on representatives by

$$-[a, b] := [b, a].$$

Remark (Dependencies).

- Integer Class Notation

Lemma

Lemma 22.2.12 (Negation Respects Formal-Difference Equivalence)**.** *If $[a, b] = [a', b']$, then*

$$-[a, b] = -[a', b'].$$

Remark (Dependencies).

- Negation on $\mathbb{Z}$
- Formal-Difference Equivalence

Theorem

Theorem 22.2.13 (Negation on $\mathbb{Z}$ Is Well-Defined). *Negation determines a well-defined unary operation $\mathbb{Z} \to \mathbb{Z}$.*

Remark (Dependencies).

- Negation on $\mathbb{Z}$
- Negation Respects Formal-Difference Equivalence

Theorem

Theorem 22.2.14 (Negation Gives a Left Additive Inverse on $\mathbb{Z}$). *For every $x \in \mathbb{Z}$,*

$$(-x) + x = 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Negation on $\mathbb{Z}$
- Addition on $\mathbb{Z}$
- Zero in $\mathbb{Z}$
- Negation on $\mathbb{Z}$ Is Well-Defined
- Addition on $\mathbb{Z}$ Is Well-Defined

Corollary

Corollary 22.2.15 (Negation Gives a Right Additive Inverse on $\mathbb{Z}$). *For every $x \in \mathbb{Z}$,*

$$x + (-x) = 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Negation Gives a Left Additive Inverse on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.2.16 (Every Integer Has an Additive Inverse). *For every $x \in \mathbb{Z}$ there exists $y \in \mathbb{Z}$ such that*

$$x + y = 0_{\mathbb{Z}} \qquad \text{and} \qquad y + x = 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Negation Gives a Left Additive Inverse on $\mathbb{Z}$
- Negation Gives a Right Additive Inverse on $\mathbb{Z}$

Subtraction on $\mathbb{Z}$-Classes

Definition

Definition 22.2.17 (Subtraction on $\mathbb{Z}$)**.** For $x, y \in \mathbb{Z}$, define

$$x - y := x + (-y).$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Negation on $\mathbb{Z}$

Theorem

Theorem 22.2.18 (Subtraction on $\mathbb{Z}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.*

Remark (Dependencies).

- Subtraction on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Well-Defined
- Negation on $\mathbb{Z}$ Is Well-Defined

Multiplication on $\mathbb{Z}$-Classes

Definition

Definition 22.2.19 (Multiplication on $\mathbb{Z}$-Classes)**.** Define multiplication on representatives by

$$[a, b] \cdot [c, d] := [ac + bd, ad + bc].$$

Remark (Dependencies).

- Integer Class Notation
- Addition on $\mathbb{W}$
- Multiplication on $\mathbb{W}$

Lemma

Lemma 22.2.20 (Multiplication Respects Formal-Difference Equivalence on the Left)**.** *If $[a, b] = [a', b']$, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c, d].$$

Remark (Dependencies).

- Addition on $\mathbb{W}$ Satisfies Cancellation
- Order on $\mathbb{W}$ Is Total

Lemma

Lemma 22.2.21 (Multiplication Respects Formal-Difference Equivalence on the Right). *If* $[c, d] = [c', d']$, *then*

$$[a, b] \cdot [c, d] = [a, b] \cdot [c', d'].$$

Remark (Dependencies).

- Addition on $\mathbb{W}$ Satisfies Cancellation
- Order on $\mathbb{W}$ Is Total

Lemma

Lemma 22.2.22 (Multiplication Respects Formal-Difference Equivalence). *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$, *then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c', d'].$$

Remark (Dependencies).

- Multiplication on $\mathbb{Z}$
- Multiplication Respects Formal-Difference Equivalence on the Left
- Multiplication Respects Formal-Difference Equivalence on the Right

Theorem

Theorem 22.2.23 (Multiplication on $\mathbb{Z}$ Is Well-Defined). *Multiplication determines a well-defined binary operation* $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.

Remark (Dependencies).

- Multiplication on $\mathbb{Z}$
- Multiplication Respects Formal-Difference Equivalence

Theorem

Theorem 22.2.24 (Multiplication on $\mathbb{Z}$ Is Associative). *For all* $x, y, z \in \mathbb{Z}$,

$$(x \cdot y) \cdot z = x \cdot (y \cdot z).$$

Remark (Dependencies).

- Multiplication on $\mathbb{Z}$

- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Multiplication on $\mathbb{W}$ Is Associative

Theorem

Theorem 22.2.25 (Multiplication on $\mathbb{Z}$ Is Commutative)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \cdot y = y \cdot x.$$

Remark (Dependencies).

- Multiplication on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Multiplication on $\mathbb{W}$ Is Commutative

Theorem

Theorem 22.2.26 (One Is a Left Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$1_{\mathbb{Z}} \cdot x = x.$$

Remark (Dependencies).

- Multiplication on $\mathbb{Z}$
- One in $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined
- One Is a Multiplicative Identity on $\mathbb{W}$

Corollary

Corollary 22.2.27 (One Is a Right Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$x \cdot 1_{\mathbb{Z}} = x.$$

Remark (Dependencies).

- One Is a Left Multiplicative Identity on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.2.28 (One Is a Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$1_{\mathbb{Z}} \cdot x = x \qquad \textit{and} \qquad x \cdot 1_{\mathbb{Z}} = x.$$

Remark (Dependencies).

- One Is a Left Multiplicative Identity on $\mathbb{Z}$
- One Is a Right Multiplicative Identity on $\mathbb{Z}$

Distributivity on $\mathbb{Z}$-Classes

Theorem

Theorem 22.2.29 (Left Distributivity on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$
- Multiplication Distributes over Addition on $\mathbb{W}$

Corollary

Corollary 22.2.30 (Right Distributivity on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Remark (Dependencies).

- Left Distributivity on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.2.31 (Multiplication Distributes over Addition on $\mathbb{Z}$)**.** *Multiplication distributes over addition on both sides in $\mathbb{Z}$.*

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$
- Left Distributivity on $\mathbb{Z}$
- Right Distributivity on $\mathbb{Z}$

Class-Level Ring Summary

Theorem

Theorem 22.2.32 ($\mathbb{Z}$ Is an Additive Abelian Group)**.** *The structure $(\mathbb{Z}, +, 0_{\mathbb{Z}})$ is an abelian group.*

Remark (Dependencies).

- Addition on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Associative
- Addition on $\mathbb{Z}$ Is Commutative
- Zero Is an Additive Identity on $\mathbb{Z}$
- Every Integer Has an Additive Inverse

Theorem

Theorem 22.2.33 ($\mathbb{Z}$ Is a Multiplicative Commutative Monoid)**.** *The structure* $(\mathbb{Z}, \cdot, 1_{\mathbb{Z}})$ *is a commutative monoid.*

Remark (Dependencies).

- Multiplication on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Associative
- Multiplication on $\mathbb{Z}$ Is Commutative
- One Is a Multiplicative Identity on $\mathbb{Z}$

Theorem

Theorem 22.2.34 ($\mathbb{Z}$ Is a Commutative Ring)**.** *The structure* $(\mathbb{Z}, +, \cdot, 0_{\mathbb{Z}}, 1_{\mathbb{Z}})$ *is a commutative ring.*

Remark (Dependencies).

- $\mathbb{Z}$ Is an Additive Abelian Group
- $\mathbb{Z}$ Is a Multiplicative Commutative Monoid
- Multiplication Distributes over Addition on $\mathbb{Z}$

22.3 Algebraic Structure of $\mathbb{Z}$

Integral Domain Structure

Theorem

Theorem 22.3.1 ($\mathbb{Z}$ Has No Zero Divisors)**.** *For all* $x, y \in \mathbb{Z}$*,*

$$x \cdot y = 0 \Longrightarrow x = 0 \text{ or } y = 0.$$

Remark (Dependencies).

- Order on $\mathbb{W}$ Is Total

- Cancellation for Multiplication

Theorem

Theorem 22.3.2 ($\mathbb{Z}$ Is an Integral Domain). *The integers form an integral domain: $\mathbb{Z}$ is a commutative ring, $0 \neq 1$, and $\mathbb{Z}$ has no zero divisors.*

Remark (Dependencies).

- $\mathbb{Z}$ Is a Commutative Ring
- Zero Is Not One in $\mathbb{Z}$
- $\mathbb{Z}$ Has No Zero Divisors

Corollary

Corollary 22.3.3 (Multiplicative Cancellation on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$z \neq 0 \wedge z \cdot x = z \cdot y \Longrightarrow x = y.$$

Remark (Dependencies).

- $\mathbb{Z}$ Has No Zero Divisors
- Multiplication on $\mathbb{Z}$ Is Commutative

Derived Laws on $\mathbb{Z}$

Corollary

Corollary 22.3.4 (Zero Absorption on $\mathbb{Z}$). *For every $x \in \mathbb{Z}$,*

$$0 \cdot x = 0 \qquad \text{and} \qquad x \cdot 0 = 0.$$

Remark (Dependencies).

- $\mathbb{Z}$ Is a Commutative Ring
- Zero in $\mathbb{Z}$

Corollary

Corollary 22.3.5 (Sign Rule on $\mathbb{Z}$). *For all $x, y \in \mathbb{Z}$,*

$$(-x) \cdot y = -(x \cdot y) \qquad \text{and} \qquad x \cdot (-y) = -(x \cdot y).$$

Remark (Dependencies).

- $\mathbb{Z}$ Is a Commutative Ring

- Negation on $\mathbb{Z}$ Is Well-Defined
- Multiplication Distributes over Addition on $\mathbb{Z}$

Corollary

Corollary 22.3.6 (Negative One Rule on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$(-1) \cdot x = -x.$$

Remark (Dependencies)*.*

- Sign Rule on $\mathbb{Z}$
- One in $\mathbb{Z}$

Corollary

Corollary 22.3.7 (Product of Negatives on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

Remark (Dependencies)*.*

- Sign Rule on $\mathbb{Z}$
- Negative One Rule on $\mathbb{Z}$

22.4 Order on $\mathbb{Z}$

Positivity on $\mathbb{Z}$

Definition

Definition 22.4.1 (Positive Integer)**.** An integer $x \in \mathbb{Z}$ is *positive* if $x = [a, b]$ for some $a, b \in \mathbb{W}$ with $b < a$.

Remark (Dependencies)*.*

- Integer Class Notation
- Whole Numbers
- Order on $\mathbb{W}$

Lemma

Lemma 22.4.2 (Positivity Respects Formal-Difference Equivalence)**.** *If $[a, b] = [a', b']$, then*

$$b < a \iff b' < a'.$$

Remark (Dependencies)*.*

- Addition on $\mathbb{W}$ Satisfies Cancellation
- Order on $\mathbb{W}$ Is Total

Theorem

Theorem 22.4.3 (Positivity on $\mathbb{Z}$ Is Well-Defined). *The predicate "x is positive" is well-defined on equivalence classes in $\mathbb{Z}$.*

Remark (Dependencies).

- Positive Integer
- Formal-Difference Equivalence
- Positivity Respects Formal-Difference Equivalence

Definition

Definition 22.4.4 (Strict Order on $\mathbb{Z}$). For $x, y \in \mathbb{Z}$, define

$$x < y \quad :\Longleftrightarrow \quad y - x \text{ is positive.}$$

Remark (Dependencies).

- Subtraction on $\mathbb{Z}$
- Positive Integer

Definition

Definition 22.4.5 (Order on $\mathbb{Z}$). For $x, y \in \mathbb{Z}$, define

$$x \leq y \quad :\Longleftrightarrow \quad x < y \text{ or } x = y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Z}$

Proposition

Proposition 22.4.6 (Product of Positives Is Positive). *For all $x, y \in \mathbb{Z}$,*

$$0 < x \wedge 0 < y \Longrightarrow 0 < x \cdot y.$$

Remark (Dependencies).

- Positive Integer
- Multiplication on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Order on $\mathbb{W}$

Canonical Form and Sign

Theorem

Theorem 22.4.7 (Canonical Form of Integers). *For every $x \in \mathbb{Z}$, exactly one of the following holds:*

$$x = [n, 0] \text{ for some nonzero } n \in \mathbb{W}, \qquad x = [0, 0], \qquad x = [0, n] \text{ for some nonzero } n \in \mathbb{W}. \blacksquare$$

Remark (Dependencies).

- Integers
- Integer Class Notation
- Formal-Difference Equivalence
- Whole Numbers
- Order on $\mathbb{W}$ Is Total

Corollary

Corollary 22.4.8 (Integers Split by Sign). *Every integer is nonnegative or the negative of a nonnegative integer, and the two descriptions overlap only at $0_{\mathbb{Z}}$.*

Remark (Dependencies).

- Canonical Form of Integers
- Positive Integer
- Zero in $\mathbb{Z}$

Corollary

Corollary 22.4.9 (Image of the Embedding Is the Nonnegative Integers). *The image of the canonical embedding $\mathbb{W} \to \mathbb{Z}$ is exactly the set of nonnegative integers.*

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- Integers Split by Sign
- Embedding $\mathbb{W}$ into $\mathbb{Z}$ Is Injective

Theorem

Theorem 22.4.10 (Sign Trichotomy on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$, exactly one of*

$$x < 0, \qquad x = 0, \qquad 0 < x$$

holds.

Remark (Dependencies).

- Canonical Form of Integers
- Order on $\mathbb{W}$ Is Total

Theorem

Theorem 22.4.11 (Trichotomy on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$, exactly one of*

$$x < y, \qquad x = y, \qquad y < x$$

holds.

Remark (Dependencies).

- Strict Order on $\mathbb{Z}$
- Subtraction on $\mathbb{Z}$
- Sign Trichotomy on $\mathbb{Z}$

Total Order on $\mathbb{Z}$

Theorem

Theorem 22.4.12 (Strict Order on $\mathbb{Z}$ Is Irreflexive)**.** *For every $x \in \mathbb{Z}$, not $x < x$.*

Remark (Dependencies).

- Strict Order on $\mathbb{Z}$
- Subtraction on $\mathbb{Z}$
- Zero Is an Integer

Theorem

Theorem 22.4.13 (Strict Order on $\mathbb{Z}$ Is Asymmetric)**.** *For all $x, y \in \mathbb{Z}$,*

$$x < y \Longrightarrow \neg(y < x).$$

Remark (Dependencies).

- Strict Order on $\mathbb{Z}$
- Strict Order on $\mathbb{Z}$ Is Irreflexive
- Trichotomy on $\mathbb{Z}$

Theorem

Theorem 22.4.14 (Strict Order on $\mathbb{Z}$ Is Transitive)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \wedge y < z \implies x < z.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Z}$
- Subtraction on $\mathbb{Z}$
- Positive Integer
- Addition Preserves Strict Order on $\mathbb{Z}$

Theorem

Theorem 22.4.15 (Order on $\mathbb{Z}$ Is Reflexive)**.** *For every $x \in \mathbb{Z}$, $x \leq x$.*

Remark (Dependencies).

- Order on $\mathbb{Z}$
- Strict Order on $\mathbb{Z}$

Theorem

Theorem 22.4.16 (Order on $\mathbb{Z}$ Is Antisymmetric)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \leq y \wedge y \leq x \implies x = y.$$

Remark (Dependencies).

- Order on $\mathbb{Z}$
- Strict Order on $\mathbb{Z}$
- Trichotomy on $\mathbb{Z}$

Theorem

Theorem 22.4.17 (Order on $\mathbb{Z}$ Is Transitive)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \wedge y \leq z \implies x \leq z.$$

Remark (Dependencies).

- Order on $\mathbb{Z}$
- Strict Order on $\mathbb{Z}$
- Strict Order on $\mathbb{Z}$ Is Transitive
- Trichotomy on $\mathbb{Z}$

Theorem

Theorem 22.4.18 (Order on $\mathbb{Z}$ Is Total)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \leq y \text{ or } y \leq x.$$

Remark (Dependencies).

- Order on $\mathbb{Z}$
- Trichotomy on $\mathbb{Z}$

Theorem

Theorem 22.4.19 ($\mathbb{Z}$ Is a Totally Ordered Set)**.** *The ordered pair $(\mathbb{Z}, \leq)$ is a totally ordered set.*

Remark (Dependencies).

- Ordered Set
- Total Order
- Order on $\mathbb{Z}$
- Order on $\mathbb{Z}$ Is Reflexive
- Order on $\mathbb{Z}$ Is Antisymmetric
- Order on $\mathbb{Z}$ Is Transitive
- Order on $\mathbb{Z}$ Is Total

Addition Order Compatibility on $\mathbb{Z}$

Theorem

Theorem 22.4.20 (Left Addition Preserves Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \Longrightarrow z + x < z + y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Z}$

- Addition on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Well-Defined

Corollary

Corollary 22.4.21 (Right Addition Preserves Strict Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \implies x + z < y + z.$$

Remark (Dependencies).

- Left Addition Preserves Strict Order on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.4.22 (Addition Preserves Strict Order on $\mathbb{Z}$). *Addition preserves strict order on both sides in $\mathbb{Z}$.*

Remark (Dependencies).

- Left Addition Preserves Strict Order on $\mathbb{Z}$
- Right Addition Preserves Strict Order on $\mathbb{Z}$

Theorem

Theorem 22.4.23 (Left Addition Preserves Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \implies z + x \leq z + y.$$

Remark (Dependencies).

- Order on $\mathbb{Z}$
- Left Addition Preserves Strict Order on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Well-Defined

Corollary

Corollary 22.4.24 (Right Addition Preserves Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \implies x + z \leq y + z.$$

Remark (Dependencies).

- Left Addition Preserves Order on $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.4.25 (Addition Preserves Order on $\mathbb{Z}$). *Addition preserves non-strict order on both sides in* $\mathbb{Z}$.

Remark (Dependencies).

- Left Addition Preserves Order on $\mathbb{Z}$
- Right Addition Preserves Order on $\mathbb{Z}$

Corollary

Corollary 22.4.26 (Addition Reflects Order on $\mathbb{Z}$). *Addition by a fixed integer reflects both strict and non-strict order.*

Remark (Dependencies).

- Addition Preserves Strict Order on $\mathbb{Z}$
- Addition Preserves Order on $\mathbb{Z}$
- Negation Is a Left Additive Inverse on $\mathbb{Z}$

Multiplication Order Compatibility on $\mathbb{Z}$

Theorem

Theorem 22.4.27 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$). *For all* $x, y, z \in \mathbb{Z}$,

$$0 < z \wedge x < y \Longrightarrow z \cdot x < z \cdot y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Z}$
- Product of Positives Is Positive
- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Multiplication Distributes over Addition on $\mathbb{Z}$

Corollary

Corollary 22.4.28 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$). *For all* $x, y, z \in \mathbb{Z}$,

$$0 < z \wedge x < y \Longrightarrow x \cdot z < y \cdot z.$$

Remark (Dependencies).

- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$

- Multiplication on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.4.29 (Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$)**.** *Multiplication by a positive integer preserves strict order on both sides.*

Remark (Dependencies).

- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$
- Right Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$

Theorem

Theorem 22.4.30 (Left Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x \leq y \Longrightarrow z \cdot x \leq z \cdot y.$$

Remark (Dependencies).

- Order on $\mathbb{Z}$
- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$

Corollary

Corollary 22.4.31 (Right Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x \leq y \Longrightarrow x \cdot z \leq y \cdot z.$$

Remark (Dependencies).

- Left Multiplication by Positives Preserves Order on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 22.4.32 (Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *Multiplication by a positive integer preserves non-strict order on both sides.*

Remark (Dependencies).

- Left Multiplication by Positives Preserves Order on $\mathbb{Z}$
- Right Multiplication by Positives Preserves Order on $\mathbb{Z}$

Theorem

Theorem 22.4.33 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z < 0 \land x < y \Longrightarrow z \cdot y < z \cdot x.$$

Remark (Dependencies).

- Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$
- Sign Rule on $\mathbb{Z}$
- Strict Order on $\mathbb{Z}$ Is Asymmetric

Corollary

Corollary 22.4.34 (Multiplication by Negatives Reverses Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z < 0 \land x \leq y \Longrightarrow z \cdot y \leq z \cdot x.$$

Remark (Dependencies).

- Multiplication by Negatives Reverses Strict Order on $\mathbb{Z}$
- Order on $\mathbb{Z}$

Theorem

Theorem 22.4.35 (Multiplication by Zero Collapses Order on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$0 \cdot x = 0 \cdot y.$$

Remark (Dependencies).

- Zero Absorption on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined

Absolute Value on $\mathbb{Z}$

Definition

Definition 22.4.36 (Absolute Value on $\mathbb{Z}$)**.** Define

$$|x| := \begin{cases} x, & 0 \leq x, \\ -x, & x < 0. \end{cases}$$

Remark (Dependencies).

- Order on $\mathbb{Z}$

- Trichotomy on $\mathbb{Z}$
- Negation on $\mathbb{Z}$ Is Well-Defined

Theorem

Theorem 22.4.37 (Absolute Value Is Nonnegative)**.** *For every $x \in \mathbb{Z}$,*

$$0 \leq |x|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Z}$
- Trichotomy on $\mathbb{Z}$

Theorem

Theorem 22.4.38 (Absolute Value Vanishes Only at Zero)**.** *For every $x \in \mathbb{Z}$,*

$$|x| = 0 \iff x = 0.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Z}$
- Trichotomy on $\mathbb{Z}$

Theorem

Theorem 22.4.39 (Absolute Value Is Symmetric under Negation)**.** *For every $x \in \mathbb{Z}$,*

$$|-x| = |x|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Z}$
- Trichotomy on $\mathbb{Z}$
- Sign Rule on $\mathbb{Z}$

Theorem

Theorem 22.4.40 (Absolute Value Is Multiplicative)**.** *For all $x, y \in \mathbb{Z}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Z}$

- Product of Positives Is Positive
- Product of Negatives on $\mathbb{Z}$
- Multiplication by Positives Preserves Order on $\mathbb{Z}$

Theorem

Theorem 22.4.41 (Bounding by Absolute Value). *For every $x \in \mathbb{Z}$,*

$$-|x| \leq x \leq |x|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Z}$
- Trichotomy on $\mathbb{Z}$
- Absolute Value Is Symmetric under Negation

Theorem

Theorem 22.4.42 (Triangle Inequality on $\mathbb{Z}$). *For all $x, y \in \mathbb{Z}$,*

$$|x + y| \leq |x| + |y|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Z}$
- Bounding by Absolute Value
- Addition Preserves Order on $\mathbb{Z}$

Well-Ordering Fails on $\mathbb{Z}$

Theorem

Theorem 22.4.43 ($\mathbb{Z}$ Is Not Well-Ordered). *The ordered set $(\mathbb{Z}, \leq)$ is not a well-order.*

Remark (Dependencies).

- Well-Ordered Set
- Order on $\mathbb{Z}$
- Zero in $\mathbb{Z}$
- One in $\mathbb{Z}$
- Trichotomy on $\mathbb{Z}$

22.5 Embedding $\mathbb{W}$ into $\mathbb{Z}$

Embedding $\mathbb{W}$ into $\mathbb{Z}$

Definition

Definition 22.5.1 (Embedding of $\mathbb{W}$ into $\mathbb{Z}$)**.** Define

$$\iota : \mathbb{W} \to \mathbb{Z}, \qquad \iota(n) := [n, 0].$$

Remark (Dependencies).

- Whole Numbers
- Integers
- Integer Class Notation
- Function

Theorem

Theorem 22.5.2 (Embedding Preserves Zero)**.** *The canonical embedding satisfies*

$$\iota(0_{\mathbb{W}}) = 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- Zero in $\mathbb{Z}$
- Whole Numbers

Theorem

Theorem 22.5.3 (Embedding Preserves One)**.** *The canonical embedding satisfies*

$$\iota(1_{\mathbb{W}}) = 1_{\mathbb{Z}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- One in $\mathbb{Z}$
- Whole Numbers

Theorem

Theorem 22.5.4 (Embedding $\mathbb{W}$ into $\mathbb{Z}$ Is Injective). *For all $a, b \in \mathbb{W}$,*

$$\iota(a) = \iota(b) \Longrightarrow a = b.$$

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- Injective Function
- Formal-Difference Equivalence
- Addition on $\mathbb{W}$ Satisfies Cancellation

Theorem

Theorem 22.5.5 (Embedding Preserves Addition). *For all $a, b \in \mathbb{W}$,*

$$\iota(a + b) = \iota(a) + \iota(b).$$

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- Addition on $\mathbb{W}$
- Addition on $\mathbb{Z}$ Is Well-Defined

Theorem

Theorem 22.5.6 (Embedding Preserves Multiplication). *For all $a, b \in \mathbb{W}$,*

$$\iota(a \cdot b) = \iota(a) \cdot \iota(b).$$

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- Multiplication on $\mathbb{W}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined

Theorem

Theorem 22.5.7 (Embedding Preserves Order). *For all $a, b \in \mathbb{W}$,*

$$a \leq b \Longleftrightarrow \iota(a) \leq \iota(b).$$

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- Order on $\mathbb{W}$
- Order on $\mathbb{Z}$

Theorem

Theorem 22.5.8 ($\mathbb{W}$ Embeds into $\mathbb{Z}$)**.** *The map $\iota : \mathbb{W} \to \mathbb{Z}$ is an injective order-preserving semiring homomorphism.*

Remark (Dependencies).

- Embedding of $\mathbb{W}$ into $\mathbb{Z}$
- Embedding $\mathbb{W}$ into $\mathbb{Z}$ Is Injective
- Embedding Preserves Addition
- Embedding Preserves Multiplication
- Embedding Preserves Order

22.6 Divisibility in $\mathbb{Z}$

Divisibility in $\mathbb{Z}$

Definition

Definition 22.6.1 (Divisibility in $\mathbb{Z}$)**.** For $a, b \in \mathbb{Z}$, define

$$a \mid b \quad :\Longleftrightarrow \quad \exists c \in \mathbb{Z}\,(b = a \cdot c).$$

Remark (Dependencies).

- Integers
- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Existential Formula

Basic Divisibility Laws

Proposition

Proposition 22.6.2 (Divisibility Is Reflexive on $\mathbb{Z}$)**.** *For every $a \in \mathbb{Z}$,*

$$a \mid a.$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- One in $\mathbb{Z}$
- $\mathbb{Z}$ Multiplicative Commutative Monoid

Proposition

Proposition 22.6.3 (Divisibility Is Transitive on $\mathbb{Z}$). *For all $a, b, c \in \mathbb{Z}$,*

$$a \mid b \wedge b \mid c \Longrightarrow a \mid c.$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Associative

Proposition

Proposition 22.6.4 (One and Negative One Divide Every Integer). *For every $a \in \mathbb{Z}$,*

$$1 \mid a \qquad \text{and} \qquad (-1) \mid a.$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- One in $\mathbb{Z}$
- Negative One Rule on $\mathbb{Z}$

Proposition

Proposition 22.6.5 (Every Integer Divides Zero). *For every $a \in \mathbb{Z}$,*

$$a \mid 0.$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- Zero Absorption on $\mathbb{Z}$

Proposition

Proposition 22.6.6 (Divisibility Is Compatible with Negation on the Divisor). *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \Longleftrightarrow (-a) \mid b.$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- Sign Rule on $\mathbb{Z}$

Proposition

Proposition 22.6.7 (Divisibility Is Compatible with Negation on the Dividend)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff a \mid (-b).$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- Sign Rule on $\mathbb{Z}$

Proposition

Proposition 22.6.8 (Divisibility Is Compatible with Negation on $\mathbb{Z}$)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff (-a) \mid b \iff a \mid (-b).$$

Remark (Dependencies).

- Divisibility Is Compatible with Negation on the Divisor
- Divisibility Is Compatible with Negation on the Dividend

Proposition

Proposition 22.6.9 (Divisibility Is Compatible with Addition on $\mathbb{Z}$)**.** *For all $a, b, c \in \mathbb{Z}$,*

$$a \mid b \wedge a \mid c \implies a \mid (b + c).$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- Addition on $\mathbb{Z}$ Is Well-Defined
- Multiplication Distributes over Addition on $\mathbb{Z}$

Proposition

Proposition 22.6.10 (Divisibility Respects Linear Combinations on $\mathbb{Z}$)**.** *For all $a, b, c, x, y \in \mathbb{Z}$,*

$$a \mid b \wedge a \mid c \implies a \mid (b \cdot x + c \cdot y).$$

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- Divisibility Is Compatible with Addition on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Associative
- Multiplication Distributes over Addition on $\mathbb{Z}$

Units in $\mathbb{Z}$

Definition

Definition 22.6.11 (Unit in $\mathbb{Z}$)**.** An integer $u \in \mathbb{Z}$ is a *unit* if

$$\exists v \in \mathbb{Z}\,(u \cdot v = 1).$$

Remark (Dependencies).

- Integers
- One in $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Existential Formula

Theorem

Theorem 22.6.12 (The Units of $\mathbb{Z}$ Are Exactly ± 1)**.** *An integer u is a unit if and only if*

$$u = 1 \quad or \quad u = -1.$$

Remark (Dependencies).

- Unit in $\mathbb{Z}$
- $\mathbb{Z}$ Is an Integral Domain
- Trichotomy on $\mathbb{Z}$
- Negative One Rule on $\mathbb{Z}$

Definition

Definition 22.6.13 (Associates in $\mathbb{Z}$)**.** Integers $a, b \in \mathbb{Z}$ are *associates* if $a = u \cdot b$ for some unit $u \in \mathbb{Z}$.

Remark (Dependencies).

- Unit in $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined

Division Algorithm

Theorem

Theorem 22.6.14 (Division Algorithm on $\mathbb{Z}$)**.** *If $a, b \in \mathbb{Z}$ and $b \neq 0$, then there exist unique $q, r \in \mathbb{Z}$ such that*

$$a = bq + r \qquad \text{and} \qquad 0 \leq r < |b|.$$

Remark (Dependencies).

- Well-Ordering Principle
- Multiplicative Cancellation on $\mathbb{Z}$
- $\mathbb{Z}$ Is Not Well-Ordered

Greatest Common Divisor

Definition

Definition 22.6.15 (Greatest Common Divisor in $\mathbb{Z}$)**.** A greatest common divisor $\gcd(a, b)$ is the positive common divisor of a and b divisible by every other common divisor.

Remark (Dependencies).

- Divisibility in $\mathbb{Z}$
- Positive Integer
- Order on $\mathbb{Z}$

Bezout Identity

Theorem

Theorem 22.6.16 (Bezout Identity on $\mathbb{Z}$)**.** *For $a, b \in \mathbb{Z}$, there exist $x, y \in \mathbb{Z}$ such that*

$$\gcd(a, b) = a \cdot x + b \cdot y.$$

Remark (Dependencies).

- Well-Ordering Principle

22.7 Transition to $\mathbb{Q}$

Remark (Why Rational Numbers Are Introduced). The integers have additive inverses, but they do not have multiplicative inverses for every nonzero element. The rational numbers are introduced by adjoining those missing inverses through equivalence classes of integer pairs with nonzero second coordinate.

The Continuum

Chapter 23

Rational Numbers ($\mathbb{Q}$)

23.1 Construction of $\mathbb{Q}$

Prefractions

Definition

Definition 23.1.1 (Nonzero Integers)**.** The set of nonzero integers is

$$\mathbb{Z}^* := \mathbb{Z} \setminus \{0_{\mathbb{Z}}\}.$$

Remark (Dependencies).

- Integers
- Zero in $\mathbb{Z}$

Definition

Definition 23.1.2 (Prefraction)**.** A *prefraction* is an ordered pair $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, read as the formal quotient a/b.

Remark (Dependencies).

- Ordered Pair
- Integers
- Nonzero Integers

Definition

Definition 23.1.3 (Prefraction Set)**.** The prefraction set is

$$\operatorname{Pre}\mathbb{Q} := \mathbb{Z} \times \mathbb{Z}^*.$$

Remark (Dependencies).

- Cartesian Product
- Integers
- Nonzero Integers

Fraction Equivalence

Definition

Definition 23.1.4 (Fraction Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$, define

$$(a, b) \sim_{\mathbb{Q}} (c, d) \quad :\Longleftrightarrow \quad a \cdot d = b \cdot c.$$

Remark (Dependencies).

- Relation
- Prefraction Set
- Multiplication on $\mathbb{Z}$

Lemma

Lemma 23.1.5 (Fraction Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{Q}$,

$$(a, b) \sim_{\mathbb{Q}} (a, b).$$

Remark (Dependencies).

- Fraction Equivalence
- Multiplication on $\mathbb{Z}$ Is Commutative

Lemma

Lemma 23.1.6 (Fraction Equivalence Is Symmetric)**.** *For all prefractions* $(a, b), (c, d)$,

$$(a, b) \sim_{\mathbb{Q}} (c, d) \Longrightarrow (c, d) \sim_{\mathbb{Q}} (a, b).$$

Remark (Dependencies).

- Fraction Equivalence

- Multiplication on $\mathbb{Z}$ Is Commutative

Lemma

Lemma 23.1.7 (Fraction Equivalence Is Transitive)**.** *For all prefractions* $(a, b), (c, d), (e, f)$,

$$(a, b) \sim_{\mathbb{Q}} (c, d) \wedge (c, d) \sim_{\mathbb{Q}} (e, f) \implies (a, b) \sim_{\mathbb{Q}} (e, f).$$

Remark (Dependencies).

- Multiplicative Cancellation on $\mathbb{Z}$

Theorem

Theorem 23.1.8 (Fraction Equivalence Is an Equivalence Relation)**.** *The relation* $\sim_{\mathbb{Q}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{Q}$.

Remark (Dependencies).

- Fraction Equivalence
- Fraction Equivalence Is Reflexive
- Fraction Equivalence Is Symmetric
- Fraction Equivalence Is Transitive

Definition of $\mathbb{Q}$

Definition

Definition 23.1.9 (Rational Numbers)**.** The rational numbers are the quotient

$$\mathbb{Q} := \operatorname{Pre}\mathbb{Q}/{\sim_{\mathbb{Q}}}.$$

Remark (Dependencies).

- Quotient Set
- Prefraction Set
- Fraction Equivalence

Definition

Definition 23.1.10 (Rational Class Notation)**.** The notation $[a, b]$ denotes the $\sim_{\mathbb{Q}}$-equivalence class of the prefraction (a, b), where $b \neq 0_{\mathbb{Z}}$.

Remark (Dependencies).

- Rational Numbers
- Fraction Equivalence

Proposition

Proposition 23.1.11 (Equality Criterion for Rational Classes). *For prefractions* $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$,

$$[a, b] = [c, d] \iff a \cdot d = b \cdot c.$$

Remark (Dependencies).

- Rational Class Notation
- Fraction Equivalence
- Quotient Set
- Equivalence Class

Zero and One in $\mathbb{Q}$

Definition

Definition 23.1.12 (Zero in $\mathbb{Q}$). Define

$$0_{\mathbb{Q}} := [0_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Remark (Dependencies).

- Rational Class Notation
- Zero in $\mathbb{Z}$
- One in $\mathbb{Z}$

Definition

Definition 23.1.13 (One in $\mathbb{Q}$). Define

$$1_{\mathbb{Q}} := [1_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Remark (Dependencies).

- Rational Class Notation
- One in $\mathbb{Z}$

Lemma

Lemma 23.1.14 (Vanishing Criterion in $\mathbb{Q}$)**.** *For every prefraction* $(a, b) \in \operatorname{Pre}\mathbb{Q}$*,*

$$[a, b] = 0_{\mathbb{Q}} \iff a = 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Rational Class Notation
- Fraction Equivalence
- Zero in $\mathbb{Q}$
- Zero in $\mathbb{Z}$

Theorem

Theorem 23.1.15 (Zero Is a Rational)**.** *The class* $0_{\mathbb{Q}}$ *is an element of* $\mathbb{Q}$*.*

Remark (Dependencies).

- Rational Numbers
- Zero in $\mathbb{Q}$
- Rational Class Notation

Theorem

Theorem 23.1.16 (One Is a Rational)**.** *The class* $1_{\mathbb{Q}}$ *is an element of* $\mathbb{Q}$*.*

Remark (Dependencies).

- Rational Numbers
- One in $\mathbb{Q}$
- Rational Class Notation

Theorem

Theorem 23.1.17 (Zero Is Not One in $\mathbb{Q}$)**.** *In* $\mathbb{Q}$*,*

$$0_{\mathbb{Q}} \neq 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Zero Is Not One in $\mathbb{Z}$

23.2 Operations on $\mathbb{Q}$-Classes

Addition on $\mathbb{Q}$-Classes

Definition

Definition 23.2.1 (Addition on $\mathbb{Q}$-Classes)**.** Define addition on representatives by

$$[a, b] + [c, d] := [a \cdot d + b \cdot c,\ b \cdot d].$$

Remark (Dependencies).

- Rational Class Notation
- Addition on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$

Lemma

Lemma 23.2.2 (Addition Lands in Pre $\mathbb{Q}$)**.** *If $b \neq 0_{\mathbb{Z}}$ and $d \neq 0_{\mathbb{Z}}$, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- $\mathbb{Z}$ Has No Zero Divisors

Lemma

Lemma 23.2.3 (Addition Respects Fraction Equivalence on the Left)**.** *If $[a, b] = [a', b']$, then*

$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 23.2.4 (Addition Respects Fraction Equivalence on the Right)**.** *If $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 23.2.5 (Addition Respects Fraction Equivalence). *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$, *then*

$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition Respects Fraction Equivalence on the Left
- Addition Respects Fraction Equivalence on the Right

Theorem

Theorem 23.2.6 (Addition on $\mathbb{Q}$ Is Well-Defined). *Addition determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition Respects Fraction Equivalence

Theorem

Theorem 23.2.7 (Addition on $\mathbb{Q}$ Is Associative). *For all* $x, y, z \in \mathbb{Q}$,

$$(x + y) + z = x + (y + z).$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Addition on $\mathbb{Z}$ Is Associative
- Multiplication on $\mathbb{Z}$ Is Associative
- Multiplication Distributes over Addition on $\mathbb{Z}$

Theorem

Theorem 23.2.8 (Addition on $\mathbb{Q}$ Is Commutative)**.** *For all $x, y \in \mathbb{Q}$,*

$$x + y = y + x.$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Addition on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 23.2.9 (Zero Is a Left Additive Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} + x = x.$$

Remark (Dependencies).

- Zero in $\mathbb{Q}$
- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Zero Is an Additive Identity on $\mathbb{Z}$
- One Is a Multiplicative Identity on $\mathbb{Z}$

Corollary

Corollary 23.2.10 (Zero Is a Right Additive Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$x + 0_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- Zero Is a Left Additive Identity on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 23.2.11 (Zero Is an Additive Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} + x = x \qquad \textit{and} \qquad x + 0_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- Zero Is a Left Additive Identity on $\mathbb{Q}$
- Zero Is a Right Additive Identity on $\mathbb{Q}$

Negation on $\mathbb{Q}$-Classes

Definition

Definition 23.2.12 (Negation on $\mathbb{Q}$-Classes)**.** Define negation on representatives by

$$-[a, b] := [-a, b].$$

Remark (Dependencies).

- Rational Class Notation
- Negation on $\mathbb{Z}$

Lemma

Lemma 23.2.13 (Negation Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$*, then*

$$-[a, b] = -[a', b'].$$

Remark (Dependencies).

- Negation on $\mathbb{Q}$
- Fraction Equivalence
- Sign Rule on $\mathbb{Z}$

Theorem

Theorem 23.2.14 (Negation on $\mathbb{Q}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation* $\mathbb{Q} \to \mathbb{Q}$*.*

Remark (Dependencies).

- Negation on $\mathbb{Q}$
- Negation Respects Fraction Equivalence

Theorem

Theorem 23.2.15 (Negation Gives a Left Additive Inverse on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*

$$(-x) + x = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Negation on $\mathbb{Q}$
- Addition on $\mathbb{Q}$

- Negation on $\mathbb{Q}$ Is Well-Defined
- Addition on $\mathbb{Q}$ Is Well-Defined
- Negation Gives a Left Additive Inverse on $\mathbb{Z}$

Corollary

Corollary 23.2.16 (Negation Gives a Right Additive Inverse on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$x + (-x) = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Negation Gives a Left Additive Inverse on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 23.2.17 (Every Rational Has an Additive Inverse)**.** *For every $x \in \mathbb{Q}$ there exists $y \in \mathbb{Q}$ such that*

$$x + y = 0_{\mathbb{Q}} \qquad \textit{and} \qquad y + x = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Negation on $\mathbb{Q}$ Is Well-Defined
- Negation Gives a Left Additive Inverse on $\mathbb{Q}$
- Negation Gives a Right Additive Inverse on $\mathbb{Q}$

Subtraction on $\mathbb{Q}$

Definition

Definition 23.2.18 (Subtraction on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x - y := x + (-y).$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Negation on $\mathbb{Q}$

Theorem

Theorem 23.2.19 (Subtraction on $\mathbb{Q}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.*

Remark (Dependencies).

- Subtraction on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Negation on $\mathbb{Q}$ Is Well-Defined

Multiplication on $\mathbb{Q}$-Classes

Definition

Definition 23.2.20 (Multiplication on $\mathbb{Q}$-Classes)**.** Define multiplication on representatives by

$$[a, b] \cdot [c, d] := [a \cdot c,\ \ b \cdot d].$$

Remark (Dependencies).

- Rational Class Notation
- Multiplication on $\mathbb{Z}$

Lemma

Lemma 23.2.21 (Multiplication Lands in Pre $\mathbb{Q}$)**.** *If $b \neq 0_{\mathbb{Z}}$ and $d \neq 0_{\mathbb{Z}}$, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- $\mathbb{Z}$ Has No Zero Divisors

Lemma

Lemma 23.2.22 (Multiplication Respects Fraction Equivalence on the Left)**.** *If $[a, b] = [a', b']$, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c, d].$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 23.2.23 (Multiplication Respects Fraction Equivalence on the Right)**.** *If $[c, d] = [c', d']$, then*

$$[a, b] \cdot [c, d] = [a, b] \cdot [c', d'].$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 23.2.24 (Multiplication Respects Fraction Equivalence). *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$, *then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c', d'].$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Multiplication Respects Fraction Equivalence on the Left
- Multiplication Respects Fraction Equivalence on the Right

Theorem

Theorem 23.2.25 (Multiplication on $\mathbb{Q}$ Is Well-Defined). *Multiplication determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Multiplication Respects Fraction Equivalence

Theorem

Theorem 23.2.26 (Multiplication on $\mathbb{Q}$ Is Associative). *For all* $x, y, z \in \mathbb{Q}$,

$$(x \cdot y) \cdot z = x \cdot (y \cdot z).$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Z}$ Is Associative

Theorem

Theorem 23.2.27 (Multiplication on $\mathbb{Q}$ Is Commutative). *For all* $x, y \in \mathbb{Q}$,

$$x \cdot y = y \cdot x.$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 23.2.28 (One Is a Left Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x.$$

Remark (Dependencies).

- One in $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- One Is a Multiplicative Identity on $\mathbb{Z}$

Corollary

Corollary 23.2.29 (One Is a Right Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$x \cdot 1_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- One Is a Left Multiplicative Identity on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 23.2.30 (One Is a Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x \qquad \textit{and} \qquad x \cdot 1_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- One Is a Left Multiplicative Identity on $\mathbb{Q}$
- One Is a Right Multiplicative Identity on $\mathbb{Q}$

Reciprocal on Nonzero $\mathbb{Q}$-Classes

Lemma

Lemma 23.2.31 (Nonzero Predicate Is Well-Defined on $\mathbb{Q}$). *For every prefraction* $(a, b) \in \operatorname{Pre}\mathbb{Q}$,

$$[a, b] \neq 0_{\mathbb{Q}} \iff a \neq 0_{\mathbb{Z}}.$$

Thus nonzeroness is independent of the representative.

Remark (Dependencies).

- Zero in $\mathbb{Q}$
- Vanishing Criterion in $\mathbb{Q}$

Definition

Definition 23.2.32 (Reciprocal on Nonzero $\mathbb{Q}$-Classes). For $[a, b] \neq 0_{\mathbb{Q}}$, define

$$[a, b]^{-1} := [b, a].$$

Remark (Dependencies).

- Rational Class Notation
- Vanishing Criterion in $\mathbb{Q}$

Lemma

Lemma 23.2.33 (Reciprocal Lands in $\operatorname{Pre}\mathbb{Q}$). *If* $[a, b] \neq 0_{\mathbb{Q}}$*, then* $a \neq 0_{\mathbb{Z}}$*, so* (b, a) *is a prefraction.*

Remark (Dependencies).

- Prefraction
- Prefraction Set
- Vanishing Criterion in $\mathbb{Q}$
- Zero in $\mathbb{Z}$

Lemma

Lemma 23.2.34 (Reciprocal Respects Fraction Equivalence). *If* $[a, b] = [a', b']$ *and* $[a, b] \neq 0_{\mathbb{Q}}$*, then*

$$[a, b]^{-1} = [a', b']^{-1}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{Q}$

- Fraction Equivalence
- Reciprocal Lands in Pre $\mathbb{Q}$

Theorem

Theorem 23.2.35 (Reciprocal on $\mathbb{Q}$ Is Well-Defined). *Reciprocal determines a well-defined map*

$$\mathbb{Q} \setminus \{0_{\mathbb{Q}}\} \to \mathbb{Q} \setminus \{0_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{Q}$
- Reciprocal Lands in Pre $\mathbb{Q}$
- Reciprocal Respects Fraction Equivalence

Theorem

Theorem 23.2.36 (Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$x^{-1} \cdot x = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- One in $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Q}$ Is Well-Defined

Corollary

Corollary 23.2.37 (Reciprocal Gives a Right Multiplicative Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$x \cdot x^{-1} = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 23.2.38 (Every Nonzero Rational Has a Multiplicative Inverse). *For every $x \in \mathbb{Q}$ with $x \neq 0_{\mathbb{Q}}$, there exists $y \in \mathbb{Q}$ such that*

$$x \cdot y = 1_{\mathbb{Q}} \qquad \text{and} \qquad y \cdot x = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$
- Reciprocal Gives a Right Multiplicative Inverse on $\mathbb{Q}$

Division on $\mathbb{Q}$

Definition

Definition 23.2.39 (Division on $\mathbb{Q}$). For $x, y \in \mathbb{Q}$ with $y \neq 0_{\mathbb{Q}}$, define

$$x/y := x \cdot y^{-1}.$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$

Theorem

Theorem 23.2.40 (Division on $\mathbb{Q}$ Is Well-Defined). *Division determines a well-defined map*

$$\mathbb{Q} \times (\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}) \to \mathbb{Q}.$$

Remark (Dependencies).

- Division on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Reciprocal on $\mathbb{Q}$ Is Well-Defined

Distributivity on $\mathbb{Q}$-Classes

Theorem

Theorem 23.2.41 (Left Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Multiplication Distributes over Addition on $\mathbb{Z}$

Corollary

Corollary 23.2.42 (Right Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Remark (Dependencies).

- Left Distributivity on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 23.2.43 (Multiplication Distributes over Addition on $\mathbb{Q}$). *Multiplication distributes over addition on both sides in $\mathbb{Q}$.*

Remark (Dependencies).

- Left Distributivity on $\mathbb{Q}$
- Right Distributivity on $\mathbb{Q}$

Class-Level Field Summary

Theorem

Theorem 23.2.44 ($\mathbb{Q}$ Is an Additive Abelian Group). *The structure $(\mathbb{Q}, +, 0_{\mathbb{Q}})$ is an abelian group.*

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Associative
- Addition on $\mathbb{Q}$ Is Commutative
- Zero Is an Additive Identity on $\mathbb{Q}$
- Every Rational Has an Additive Inverse

Theorem

Theorem 23.2.45 (Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group)**.** *The structure* $(\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}, \cdot, 1_{\mathbb{Q}})$ *is an abelian group.*

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Nonzero Predicate Is Well-Defined on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Associative
- Multiplication on $\mathbb{Q}$ Is Commutative
- One Is a Multiplicative Identity on $\mathbb{Q}$
- Every Nonzero Rational Has a Multiplicative Inverse

Theorem

Theorem 23.2.46 ($\mathbb{Q}$ Is a Field)**.** *The structure* $(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}})$ *is a field.*

Remark (Dependencies).

- $\mathbb{Q}$ Is an Additive Abelian Group
- Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group
- Multiplication Distributes over Addition on $\mathbb{Q}$
- Zero Is Not One in $\mathbb{Q}$

23.3 Field Structure of Q

Derived Field Laws on $\mathbb{Q}$

Corollary

Corollary 23.3.1 (Zero Product on $\mathbb{Q}$)**.** *For all* $x, y \in \mathbb{Q}$,

$$x \cdot y = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}} \text{ or } y = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field

Corollary

Corollary 23.3.2 (Double Negation on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$-(-x) = x.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field

Corollary

Corollary 23.3.3 (Negation of a Product on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot y = -(x \cdot y) \qquad \text{and} \qquad x \cdot (-y) = -(x \cdot y).$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field

Corollary

Corollary 23.3.4 (Product of Negatives on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

Remark (Dependencies).

- Negation of a Product on $\mathbb{Q}$
- Double Negation on $\mathbb{Q}$

Corollary

Corollary 23.3.5 (Additive Cancellation on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x + z = y + z \Longrightarrow x = y.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Additive Abelian Group

Corollary

Corollary 23.3.6 (Multiplicative Cancellation on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$z \neq 0_{\mathbb{Q}} \wedge x \cdot z = y \cdot z \Longrightarrow x = y.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field

Corollary

Corollary 23.3.7 (Uniqueness of Additive Inverses on $\mathbb{Q}$). *Every additive inverse in $\mathbb{Q}$ is unique.*

Remark (Dependencies).

- $\mathbb{Q}$ Is an Additive Abelian Group

Corollary

Corollary 23.3.8 (Uniqueness of Multiplicative Inverses on $\mathbb{Q}$). *Every multiplicative inverse of a nonzero rational number is unique.*

Remark (Dependencies).

- Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group

Corollary

Corollary 23.3.9 (The Inverse of One Is One on $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$1_{\mathbb{Q}}^{-1} = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- One in $\mathbb{Q}$
- Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$
- One Is a Multiplicative Identity on $\mathbb{Q}$

Corollary

Corollary 23.3.10 (Inverse of a Product on $\mathbb{Q}$). *For all nonzero $x, y \in \mathbb{Q}$,*

$$(x \cdot y)^{-1} = x^{-1} \cdot y^{-1}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Reciprocal on $\mathbb{Q}$ Is Well-Defined
- Uniqueness of Multiplicative Inverses on $\mathbb{Q}$

Corollary

Corollary 23.3.11 (Inverse of an Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$(x^{-1})^{-1} = x.$$

Remark (Dependencies).

Integer Powers on $\mathbb{Q}$

Definition

Definition 23.3.12 (Integer Power on $\mathbb{Q}$)**.** Let $x \in \mathbb{Q}$. For $n \in \mathbb{W}$, define x^n by the usual recursive power operation. If $x \neq 0_{\mathbb{Q}}$ and $n \in \mathbb{Z}$ is negative, define

$$x^n := (x^{-1})^{-n}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Every Nonzero Rational Has a Multiplicative Inverse

Corollary

Corollary 23.3.13 (Exponent Addition Law on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$x^{m+n} = x^m \cdot x^n.$$

Remark (Dependencies).

- Integer Power on $\mathbb{Q}$
- Integer Power on $\mathbb{Q}$

Corollary

Corollary 23.3.14 (Power of a Power on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$(x^m)^n = x^{m \cdot n}.$$

Remark (Dependencies).

- Integer Power on $\mathbb{Q}$
- Integer Power on $\mathbb{Q}$

Corollary

Corollary 23.3.15 (Power of a Product on $\mathbb{Q}$)**.** *For all nonzero $x, y \in \mathbb{Q}$ and every $n \in \mathbb{Z}$,*

$$(x \cdot y)^n = x^n \cdot y^n.$$

Remark (Dependencies).

- Integer Power on $\mathbb{Q}$
- Integer Power on $\mathbb{Q}$
- Inverse of a Product on $\mathbb{Q}$

23.4 Order on $\mathbb{Q}$

Positivity on $\mathbb{Q}$

Definition

Definition 23.4.1 (Positive Rational)**.** A rational number $[a, b] \in \mathbb{Q}$ is *positive* if

$$a \cdot b > 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Rational Class Notation
- Multiplication on $\mathbb{Z}$
- Order on $\mathbb{Z}$

Lemma

Lemma 23.4.2 (Positivity Respects Fraction Equivalence on $\mathbb{Q}$)**.** *If* $[a, b] = [c, d]$, *then*

$$a \cdot b > 0_{\mathbb{Z}} \iff c \cdot d > 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- $\mathbb{Z}$ Is a Totally Ordered Set
- $\mathbb{Z}$ Has No Zero Divisors

Theorem

Theorem 23.4.3 (Positivity on $\mathbb{Q}$ Is Well-Defined)**.** *The predicate "x is positive" is well-defined on equivalence classes in $\mathbb{Q}$.*

Remark (Dependencies).

- Positive Rational
- Positivity Respects Fraction Equivalence on $\mathbb{Q}$

Definition

Definition 23.4.4 (Strict Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x < y \quad :\iff \quad y - x \text{ is positive.}$$

Remark (Dependencies).

- Positive Rational
- Subtraction on $\mathbb{Q}$

Definition

Definition 23.4.5 (Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x \leq y \quad :\Longleftrightarrow \quad x < y \text{ or } x = y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$

Proposition

Proposition 23.4.6 (Product of Positives Is Positive on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \wedge 0_{\mathbb{Q}} < y \Longrightarrow 0_{\mathbb{Q}} < x \cdot y.$$

Remark (Dependencies).

- Positive Rational
- Multiplication on $\mathbb{Q}$
- $\mathbb{Z}$ Has No Zero Divisors

Trichotomy on $\mathbb{Q}$

Theorem

Theorem 23.4.7 (Sign Trichotomy on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$, exactly one of*

$$x < 0_{\mathbb{Q}}, \qquad x = 0_{\mathbb{Q}}, \qquad 0_{\mathbb{Q}} < x$$

holds.

Remark (Dependencies).

- Positive Rational
- Strict Order on $\mathbb{Q}$
- Zero in $\mathbb{Q}$

Theorem

Theorem 23.4.8 (Trichotomy on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$, exactly one of*

$$x < y, \qquad x = y, \qquad y < x$$

holds.

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Subtraction on $\mathbb{Q}$
- Sign Trichotomy on $\mathbb{Q}$

Total Order on $\mathbb{Q}$

Theorem

Theorem 23.4.9 (Strict Order on $\mathbb{Q}$ Is Irreflexive)**.** *For every $x \in \mathbb{Q}$, not $x < x$.*

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 23.4.10 (Strict Order on $\mathbb{Q}$ Is Asymmetric)**.** *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies \neg(y < x).$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 23.4.11 (Strict Order on $\mathbb{Q}$ Is Transitive)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \wedge y < z \implies x < z.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Addition Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 23.4.12 (Order on $\mathbb{Q}$ Is Reflexive)**.** *For every $x \in \mathbb{Q}$, $x \leq x$.*

Remark (Dependencies).

- Order on $\mathbb{Q}$

Theorem

Theorem 23.4.13 (Order on $\mathbb{Q}$ Is Antisymmetric). *For all* $x, y \in \mathbb{Q}$,

$$x \leq y \land y \leq x \implies x = y.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Strict Order on $\mathbb{Q}$ Is Asymmetric

Theorem

Theorem 23.4.14 (Order on $\mathbb{Q}$ Is Transitive). *For all* $x, y, z \in \mathbb{Q}$,

$$x \leq y \land y \leq z \implies x \leq z.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Strict Order on $\mathbb{Q}$ Is Transitive

Theorem

Theorem 23.4.15 (Order on $\mathbb{Q}$ Is Total). *For all* $x, y \in \mathbb{Q}$,

$$x \leq y \text{ or } y \leq x.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 23.4.16 ($\mathbb{Q}$ Is a Totally Ordered Set). *The ordered pair* $(\mathbb{Q}, \leq)$ *is a totally ordered set.*

Remark (Dependencies).

- Ordered Set
- Total Order
- Order on $\mathbb{Q}$
- Order on $\mathbb{Q}$ Is Reflexive
- Order on $\mathbb{Q}$ Is Antisymmetric
- Order on $\mathbb{Q}$ Is Transitive
- Order on $\mathbb{Q}$ Is Total

Addition Order Compatibility on $\mathbb{Q}$

Theorem

Theorem 23.4.17 (Left Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies z + x < z + y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Addition on $\mathbb{Q}$
- Positive Rational

Corollary

Corollary 23.4.18 (Right Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies x + z < y + z.$$

Remark (Dependencies).

- Left Addition Preserves Strict Order on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 23.4.19 (Addition Preserves Strict Order on $\mathbb{Q}$). *Addition preserves strict order on both sides in $\mathbb{Q}$.*

Remark (Dependencies).

- Left Addition Preserves Strict Order on $\mathbb{Q}$
- Right Addition Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 23.4.20 (Left Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies z + x \leq z + y.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Left Addition Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 23.4.21 (Right Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies x + z \leq y + z.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Right Addition Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 23.4.22 (Addition Preserves Order on $\mathbb{Q}$). *Addition preserves non-strict order on both sides in $\mathbb{Q}$.*

Remark (Dependencies).

- Left Addition Preserves Order on $\mathbb{Q}$
- Right Addition Preserves Order on $\mathbb{Q}$

Corollary

Corollary 23.4.23 (Addition Reflects Order on $\mathbb{Q}$). *Addition by a fixed rational reflects both strict and non-strict order.*

Remark (Dependencies).

- Addition Preserves Strict Order on $\mathbb{Q}$
- Addition Preserves Order on $\mathbb{Q}$
- Negation Gives a Left Additive Inverse on $\mathbb{Q}$

Multiplication Order Compatibility on $\mathbb{Q}$

Theorem

Theorem 23.4.24 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \implies z \cdot x < z \cdot y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Product of Positives Is Positive on $\mathbb{Q}$

- Multiplication on $\mathbb{Q}$

Corollary

Corollary 23.4.25 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \implies x \cdot z < y \cdot z.$$

Remark (Dependencies).

- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 23.4.26 (Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves strict order on both sides.*

Remark (Dependencies).

- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$
- Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 23.4.27 (Left Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \implies z \cdot x \leq z \cdot y.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 23.4.28 (Right Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \implies x \cdot z \leq y \cdot z.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 23.4.29 (Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves non-strict order on both sides.*

Remark (Dependencies).

- Left Multiplication by Positives Preserves Order on $\mathbb{Q}$
- Right Multiplication by Positives Preserves Order on $\mathbb{Q}$

Theorem

Theorem 23.4.30 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x < y \Longrightarrow z \cdot y < z \cdot x.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Negation of a Product on $\mathbb{Q}$
- Positive Multiplication Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 23.4.31 (Multiplication by Negatives Reverses Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x \leq y \Longrightarrow z \cdot y \leq z \cdot x.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Multiplication by Negatives Reverses Strict Order on $\mathbb{Q}$

Reciprocal Order Laws on $\mathbb{Q}$

Theorem

Theorem 23.4.32 (Sign of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}, \qquad x < 0_{\mathbb{Q}} \Longrightarrow x^{-1} < 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{Q}$
- Order on $\mathbb{Q}$

- Product of Positives Is Positive on $\mathbb{Q}$
- Reciprocal Is a Left Multiplicative Inverse on $\mathbb{Q}$

Theorem

Theorem 23.4.33 (Reciprocal Reverses Order on Positives in $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x < y \Longrightarrow 0_{\mathbb{Q}} < y^{-1} < x^{-1}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{Q}$
- Order on $\mathbb{Q}$
- Sign of a Reciprocal on $\mathbb{Q}$
- Positive Multiplication Preserves Strict Order on $\mathbb{Q}$

Absolute Value on $\mathbb{Q}$

Definition

Definition 23.4.34 (Absolute Value on $\mathbb{Q}$). For $x \in \mathbb{Q}$, define

$$|x| := \begin{cases} x, & 0_{\mathbb{Q}} \leq x, \\ -x, & x < 0_{\mathbb{Q}}. \end{cases}$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Zero in $\mathbb{Q}$
- Negation on $\mathbb{Q}$

Theorem

Theorem 23.4.35 (Absolute Value Is Nonnegative on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \leq |x|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Q}$
- Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 23.4.36 (Absolute Value Vanishes Only at Zero on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$|x| = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}}.$$

Remark (Dependencies)*.*

- Absolute Value on $\mathbb{Q}$
- Absolute Value Is Nonnegative on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 23.4.37 (Absolute Value Is Multiplicative on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Remark (Dependencies)*.*

- Absolute Value on $\mathbb{Q}$
- Positive Multiplication Preserves Order on $\mathbb{Q}$
- Product of Negatives on $\mathbb{Q}$

Theorem

Theorem 23.4.38 (Triangle Inequality on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x + y| \leq |x| + |y|.$$

Remark (Dependencies)*.*

- Absolute Value on $\mathbb{Q}$
- Addition on $\mathbb{Q}$
- Order on $\mathbb{Q}$
- $\mathbb{Q}$ Is a Totally Ordered Set
- Addition Preserves Order on $\mathbb{Q}$

Theorem

Theorem 23.4.39 (Absolute Value of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$|x^{-1}| = |x|^{-1}.$$

Remark (Dependencies)*.*

- Absolute Value on $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$ Is Well-Defined
- Sign of a Reciprocal on $\mathbb{Q}$
- Absolute Value Is Multiplicative on $\mathbb{Q}$

23.5 Ordered Field Structure of Q

Ordered Field Instantiation

Theorem

Theorem 23.5.1 ($\mathbb{Q}$ Is an Ordered Field)**.** *The structure*

$$(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}}, \leq)$$

is an ordered field.

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- $\mathbb{Q}$ Is a Totally Ordered Set
- Addition Preserves Order on $\mathbb{Q}$
- Multiplication by Positives Preserves Order on $\mathbb{Q}$

Corollary

Corollary 23.5.2 (Ordered Field Laws Hold on $\mathbb{Q}$)**.** *Every theorem that depends only on the ordered-field axioms applies to $\mathbb{Q}$ with its constructed operations and order.*

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field

Standard Ordered-Field Consequences

Corollary

Corollary 23.5.3 (Squares Are Nonnegative on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \leq x^2.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field

- Product of Positives Is Positive on $\mathbb{Q}$

Corollary

Corollary 23.5.4 (One Is Positive in $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$0_{\mathbb{Q}} < 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- One in $\mathbb{Q}$

Corollary

Corollary 23.5.5 (Natural Numbers Are Positive in $\mathbb{Q}$). *Every natural number embedded in $\mathbb{Q}$ is positive.*

Remark (Dependencies).

- One Is Positive in $\mathbb{Q}$

Corollary

Corollary 23.5.6 (Positive Rationals Have Positive Inverses). *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Reciprocal Reverses Order on Positives in $\mathbb{Q}$

Corollary

Corollary 23.5.7 (Fraction Comparison in $\mathbb{Q}$). *For rational classes with positive denominator product,*

$$[a, b] < [c, d] \quad \Longleftrightarrow \quad a \cdot d < c \cdot b.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Product of Positives Is Positive on $\mathbb{Q}$
- One in $\mathbb{Q}$
- Reciprocal Reverses Order on Positives in $\mathbb{Q}$
- Positive Multiplication Preserves Order on $\mathbb{Q}$
- Reciprocal Reverses Order on Positives in $\mathbb{Q}$

23.6 Arbitrary Closeness in $\mathbb{Q}$

Density and the Archimedean Property

Definition

Definition 23.6.1 (Two in $\mathbb{Q}$)**.** Define

$$2_{\mathbb{Q}} := 1_{\mathbb{Q}} + 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- One in $\mathbb{Q}$
- Addition on $\mathbb{Q}$

Lemma

Lemma 23.6.2 (Two Is Nonzero in $\mathbb{Q}$)**.** *In* $\mathbb{Q}$,

$$2_{\mathbb{Q}} \neq 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- One Is Positive in $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Lemma

Lemma 23.6.3 (Midpoint Lies Strictly Between Rationals)**.** *For all* $x, y \in \mathbb{Q}$,

$$x < y \Longrightarrow x < (x + y) \cdot 2_{\mathbb{Q}}^{-1} < y.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Two Is Nonzero in $\mathbb{Q}$

Theorem

Theorem 23.6.4 (Density of $\mathbb{Q}$ in Itself)**.** *For all* $x, y \in \mathbb{Q}$,

$$x < y \Longrightarrow \exists r \in \mathbb{Q}\ (x < r < y).$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Midpoint Lies Strictly Between Rationals

Theorem

Theorem 23.6.5 (Archimedean Property of $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \implies \exists n \in \mathbb{N}\ (y < n \cdot x).$$

Remark (Dependencies).

- Well-Ordering Principle
- Division Algorithm on $\mathbb{Z}$
- $\mathbb{Z}$ Is a Totally Ordered Set

From Discreteness to Arbitrary Closeness

Remark 23.6.6 (Discrete Systems Before $\mathbb{Q}$). The systems $\mathbb{N}$, $\mathbb{W}$, and $\mathbb{Z}$ are discrete in the order sense: consecutive integer points have no same-system point between them. The rational numbers are the first number system in this construction chain where intervals can be subdivided internally without leaving the system.

Definition

Definition 23.6.7 (Arbitrarily Close Distinct Points)**.** A number system has arbitrarily close distinct points if for every point x and every positive tolerance ε, there is a point $y \neq x$ such that

$$|x - y| < \varepsilon.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Absolute Value on $\mathbb{Q}$

Theorem

Theorem 23.6.8 ($\mathbb{Q}$ Has Arbitrarily Close Distinct Points)**.** *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \implies \exists y \in \mathbb{Q}\ (y \neq x \text{ and } |x - y| < \varepsilon).$$

Remark (Dependencies).

- Reciprocals Become Arbitrarily Small in $\mathbb{Q}$
- Addition Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 23.6.9 ($\mathbb{Q}$ Has No Adjacent Points)**.** *There do not exist rational numbers $x < y$ with no rational number strictly between them.*

Remark (Dependencies).

- Density of $\mathbb{Q}$ in Itself

Proposition

Proposition 23.6.10 ($\mathbb{Q}$ Has No Least Positive Rational)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \implies \exists y \in \mathbb{Q}\ (0_{\mathbb{Q}} < y < x).$$

Remark (Dependencies).

- Density of $\mathbb{Q}$ in Itself

Corollary

Corollary 23.6.11 (Reciprocals Become Arbitrarily Small in $\mathbb{Q}$)**.** *For every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \implies \exists n \in \mathbb{N}\ \left(0_{\mathbb{Q}} < n^{-1} < \varepsilon\right).$$

Remark (Dependencies).

- Archimedean Property of $\mathbb{Q}$
- Sign of a Reciprocal on $\mathbb{Q}$

Limit-Like Behavior Before Limits

Proposition

Proposition 23.6.12 (One-Sided Rational Approximation)**.** *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \implies \exists y, z \in \mathbb{Q}\ (x - \varepsilon < y < x < z < x + \varepsilon).$$

Remark (Dependencies).

- $\mathbb{Q}$ Has Arbitrarily Close Distinct Points
- Density of $\mathbb{Q}$ in Itself

Remark 23.6.13 (Refinement and Bounds). The Archimedean property supplies arbitrarily small rational increments, so a rational point can be approached from one side at any prescribed tolerance. Mediants supply a complementary order-refinement mechanism: once two rational bounds are known, their mediant is a new rational bound strictly between them. The following approximation topics make these two mechanisms explicit through dyadic grids, mediant refinement, Stern–Brocot paths, and rational Cauchy sequences.

23.7 Rational Gaps

No Rational Square Root of Two

Theorem

Theorem 23.7.1 (No Rational Square Root of Two). *There is no rational number $q \in \mathbb{Q}$ such that*

$$q \cdot q = 2_{\mathbb{Q}}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- $\mathbb{Z}$ Is an Integral Domain
- TODO: formalize the parity or infinite-descent facts in $\mathbb{Z}$.

The Square-Two Gap

Definition

Definition 23.7.2 (The Gap Set in $\mathbb{Q}$). Define

$$S := \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \text{ and } q \cdot q < 2_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rationals
- Order on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- Zero in $\mathbb{Q}$

Proposition

Proposition 23.7.3 (The Gap Set Is Bounded Above in $\mathbb{Q}$). *The set S is nonempty and bounded above in $\mathbb{Q}$.*

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Natural Numbers Are Positive in $\mathbb{Q}$

Theorem

Theorem 23.7.4 (The Square-Two Gap Has No Rational Supremum). *The set S has no least upper bound in $\mathbb{Q}$.*

Remark (Dependencies).

- No Rational Square Root of Two
- Density of $\mathbb{Q}$ in Itself
- $\mathbb{Q}$ Is an Ordered Field

Corollary

Corollary 23.7.5 ($\mathbb{Q}$ Is Order-Incomplete)**.** *The ordered field $\mathbb{Q}$ is not order-complete: there exists a nonempty bounded-above subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$.*

Remark (Dependencies).

- The Gap Set Is Bounded Above in $\mathbb{Q}$
- The Square-Two Gap Has No Rational Supremum

Cauchy Sequences in $\mathbb{Q}$

Definition

Definition 23.7.6 (Cauchy Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is Cauchy if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists N \in \mathbb{N}\, \forall m, n \geq N\ |x_m - x_n| < \varepsilon)\,.$$

Remark (Dependencies).

- Rationals
- Order on $\mathbb{Q}$
- Absolute Value on $\mathbb{Q}$
- Subtraction on $\mathbb{Q}$

Definition

Definition 23.7.7 (Null Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is null if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N\ |x_n| < \varepsilon)\,.$$

Remark (Dependencies).

- Rationals
- Order on $\mathbb{Q}$
- Absolute Value on $\mathbb{Q}$

Lemma

Lemma 23.7.8 (Rational Limits Are Unique). *A convergent sequence in $\mathbb{Q}$ has at most one rational limit.*

Remark (Dependencies).

- Triangle Inequality on $\mathbb{Q}$
- $\mathbb{Q}$ Is an Ordered Field

23.8 Embedding $\mathbb{Z}$ into $\mathbb{Q}$

The Integer Embedding

Definition

Definition 23.8.1 (Embedding of $\mathbb{Z}$ into $\mathbb{Q}$). Define

$$\iota : \mathbb{Z} \to \mathbb{Q}, \qquad \iota(n) := [n, 1_{\mathbb{Z}}].$$

Remark (Dependencies).

- Rational Numbers
- One in $\mathbb{Z}$
- Zero Is Not One in $\mathbb{Z}$

Theorem

Theorem 23.8.2 (Embedding Preserves Zero). *The embedding sends integer zero to rational zero:*

$$\iota(0_{\mathbb{Z}}) = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{Z}$ into $\mathbb{Q}$
- Zero in $\mathbb{Z}$
- Zero in $\mathbb{Q}$

Theorem

Theorem 23.8.3 (Embedding Preserves One). *The embedding sends integer one to rational one:*

$$\iota(1_{\mathbb{Z}}) = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{Z}$ into $\mathbb{Q}$
- One in $\mathbb{Z}$
- One in $\mathbb{Q}$

Theorem

Theorem 23.8.4 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$ Is Injective). *For all* $m, n \in \mathbb{Z}$,

$$\iota(m) = \iota(n) \implies m = n.$$

Remark (Dependencies).

- Equality Criterion for Rational Classes
- One Is a Multiplicative Identity on $\mathbb{Z}$

Preservation of Algebra and Order

Theorem

Theorem 23.8.5 (Embedding Preserves Addition). *For all* $m, n \in \mathbb{Z}$,

$$\iota(m + n) = \iota(m) + \iota(n).$$

Remark (Dependencies).

- Embedding $\mathbb{Z}$ into $\mathbb{Q}$
- Addition on $\mathbb{Z}$
- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Z}$ Is Well-Defined
- Addition on $\mathbb{Q}$ Is Well-Defined

Theorem

Theorem 23.8.6 (Embedding Preserves Multiplication). *For all* $m, n \in \mathbb{Z}$,

$$\iota(m \cdot n) = \iota(m) \cdot \iota(n).$$

Remark (Dependencies).

- Embedding $\mathbb{Z}$ into $\mathbb{Q}$
- Multiplication on $\mathbb{Z}$
- Multiplication on $\mathbb{Q}$

- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Multiplication on $\mathbb{Q}$ Is Well-Defined

Theorem

Theorem 23.8.7 (Embedding Preserves Order). *For all $m, n \in \mathbb{Z}$,*

$$m \leq n \iff \iota(m) \leq \iota(n).$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$-Classes
- Multiplication on $\mathbb{Q}$-Classes
- Order on $\mathbb{Q}$
- $\mathbb{Z}$ Is a Totally Ordered Set

Theorem

Theorem 23.8.8 ($\mathbb{Z}$ Embeds into $\mathbb{Q}$). *The map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an injective order-preserving ring homomorphism.*

Remark (Dependencies).

- Embedding Preserves Zero
- Embedding Preserves One
- Embedding $\mathbb{Z}$ into $\mathbb{Q}$ Is Injective
- Embedding Preserves Addition
- Embedding Preserves Multiplication
- Embedding Preserves Order

Field of Fractions

Theorem

Theorem 23.8.9 ($\mathbb{Q}$ Is the Field of Fractions of $\mathbb{Z}$). *For every $x \in \mathbb{Q}$, there exist $m, n \in \mathbb{Z}$ with $n \neq 0_{\mathbb{Z}}$ such that*

$$x = \iota(m) \cdot \iota(n)^{-1}.$$

Remark (Dependencies).

- Rational Class Notation

- Every Nonzero Rational Has a Multiplicative Inverse
- Vanishing Criterion

Corollary

Corollary 23.8.10 (The Embedding of $\mathbb{Z}$ into $\mathbb{Q}$ Is Not Surjective)**.** *The image of $\iota : \mathbb{Z} \to \mathbb{Q}$ is a proper subset of $\mathbb{Q}$.*

Remark (Dependencies).

- Density of $\mathbb{Q}$ in Itself
- Embedding Preserves Order

23.9 Transition to $\mathbb{R}$

Why the Real Numbers Are Introduced

Remark 23.9.1 (Why the Real Numbers Are Introduced). The rational numbers form an ordered field, are dense in themselves, and satisfy the Archimedean property. These facts make $\mathbb{Q}$ a rich arithmetic and order structure, but they do not make it complete.

The square-two gap shows the defect precisely: there is a nonempty bounded-above subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$. The real numbers are introduced as a completion of $\mathbb{Q}$, so that bounded-above sets acquire suprema and Cauchy sequences acquire limits.

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Density of $\mathbb{Q}$ in Itself
- Archimedean Property of $\mathbb{Q}$
- $\mathbb{Q}$ Is Order-Incomplete
- Cauchy Sequence in $\mathbb{Q}$

23.10 Canonical Rational Number Statement Plan

Architecture Rule for the Rational Numbers

The rational-number chapter is organized around a local-versus-generic separation. Local results open the quotient construction box: prefractions, fraction equivalence, well-defined operations, field axioms, order, density, gaps, and the embedding of $\mathbb{Z}$ into $\mathbb{Q}$. Generic field and ring consequences are not re-proved here. They are cited as corollaries from the general algebraic infrastructure developed earlier.

Remark (Partiality of reciprocal). The new structural issue relative to $\mathbb{Z}$ is that reciprocal is a partial operation. It is defined only on nonzero rational classes, so the nonzero predicate must be proved well-defined before reciprocal is introduced.

Construction of $\mathbb{Q}$

- Def `def:nonzero-integers`: $\mathbb{Z}^* := \mathbb{Z} \setminus \{0\}$.
- Def `def:prefraction`: a prefraction is $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, read as a/b.
- Def `def:prefraction-set`: $\mathrm{PreQ} := \mathbb{Z} \times \mathbb{Z}^*$.
- Def `def:fraction-equivalence`: $(a, b) \sim_Q (c, d)$ iff $ad = bc$.
- Lem `lem:fraction-equivalence-reflexive`: $(a, b) \sim_Q (a, b)$.
- Lem `lem:fraction-equivalence-symmetric`: $(a, b) \sim_Q (c, d) \Rightarrow (c, d) \sim_Q (a, b)$.
- Lem `lem:fraction-equivalence-transitive`: fraction equivalence is transitive; depends on nonzero multiplicative cancellation in $\mathbb{Z}$.
- Thm `thm:fraction-equivalence-relation`: $\sim_Q$ is an equivalence relation on PreQ.
- Def `def:rationals`: $\mathbb{Q} := \mathrm{PreQ} / \sim_Q$.
- Def `def:rational-class-notation`: $[a, b]$ denotes the $\sim_Q$-class of (a, b) with $b \neq 0$.
- Def `def:zero-in-q`: $0_Q := [0, 1]$.
- Def `def:one-in-q`: $1_Q := [1, 1]$.
- Lem `lem:vanishing-criterion-q`: $[a, b] = 0_Q$ iff $a = 0$.
- Thm `thm:zero-is-a-rational`: $0_Q \in \mathbb{Q}$.
- Thm `thm:one-is-a-rational`: $1_Q \in \mathbb{Q}$.
- Thm `thm:zero-not-one-in-q`: $0_Q \neq 1_Q$; depends on $0_Z \neq 1_Z$.

Operations on Rational Classes

- Def `def:addition-on-q-classes`: $[a, b] + [c, d] := [ad + bc, bd]$.
- Lem `lem:addition-denominator-nonzero-q`: $b \neq 0$ and $d \neq 0$ imply $bd \neq 0$.
- Lem `lem:addition-on-q-respects-left-equivalence`: addition respects fraction equivalence in the left argument.
- Lem `lem:addition-on-q-respects-right-equivalence`: addition respects fraction

equivalence in the right argument.

- Lem `lem:addition-on-q-respects-equivalence`: addition respects fraction equivalence in both arguments.
- Thm `thm:addition-on-q-well-defined`: addition is a well-defined operation $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.
- Thm `thm:addition-on-q-associative`: $(x + y) + z = x + (y + z)$.
- Thm `thm:addition-on-q-commutative`: $x + y = y + x$.
- Thm `thm:zero-left-additive-identity-on-q`: $0_Q + x = x$.
- Cor `cor:zero-right-additive-identity-on-q`: $x + 0_Q = x$.
- Def `def:negation-on-q-classes`: $-[a, b] := [-a, b]$.
- Lem `lem:negation-on-q-respects-equivalence`: negation respects fraction equivalence.
- Thm `thm:negation-on-q-well-defined`: negation is well-defined on $\mathbb{Q}$.
- Thm `thm:negation-left-additive-inverse-on-q`: $(-x) + x = 0_Q$.
- Cor `cor:negation-right-additive-inverse-on-q`: $x + (-x) = 0_Q$.
- Thm `thm:every-rational-has-additive-inverse`: every rational has a two-sided additive inverse.
- Def `def:subtraction-on-q`: $x - y := x + (-y)$.
- Thm `thm:subtraction-on-q-well-defined`: subtraction is well-defined.
- Def `def:multiplication-on-q-classes`: $[a, b] \cdot [c, d] := [ac, bd]$.
- Lem `lem:multiplication-denominator-nonzero-q`: $b \neq 0$ and $d \neq 0$ imply $bd \neq 0$.
- Lem `lem:multiplication-on-q-respects-left-equivalence`: multiplication respects equivalence in the left argument.
- Lem `lem:multiplication-on-q-respects-right-equivalence`: multiplication respects equivalence in the right argument.
- Lem `lem:multiplication-on-q-respects-equivalence`: multiplication respects equivalence in both arguments.
- Thm `thm:multiplication-on-q-well-defined`: multiplication is well-defined on $\mathbb{Q}$.
- Thm `thm:multiplication-on-q-associative`: $(xy)z = x(yz)$.

- Thm `thm:multiplication-on-q-commutative`: $xy = yx$.
- Thm `thm:one-left-multiplicative-identity-on-q`: $1_Q x = x$.
- Cor `cor:one-right-multiplicative-identity-on-q`: $x 1_Q = x$.
- Lem `lem:nonzero-predicate-well-defined-q`: $[a, b] \neq 0_Q$ iff $a \neq 0$.
- Def `def:reciprocal-on-q-classes`: for $a \neq 0$, $[a, b]^{-1} := [b, a]$.
- Lem `lem:reciprocal-denominator-nonzero-q`: $a \neq 0$ puts (b, a) in PreQ.
- Lem `lem:reciprocal-on-q-respects-equivalence`: reciprocal respects equivalence on nonzero classes.
- Thm `thm:reciprocal-on-q-well-defined`: reciprocal is a well-defined map $\mathbb{Q} \setminus \{0_Q\} \to \mathbb{Q} \setminus \{0_Q\}$.
- Thm `thm:reciprocal-left-multiplicative-inverse-on-q`: $x \neq 0_Q \Rightarrow x^{-1}x = 1_Q$.
- Cor `cor:reciprocal-right-multiplicative-inverse-on-q`: $x \neq 0_Q \Rightarrow xx^{-1} = 1_Q$.
- Thm `thm:every-nonzero-rational-has-multiplicative-inverse`: every nonzero rational has a two-sided multiplicative inverse.
- Def `def:division-on-q`: for $y \neq 0_Q$, $x/y := x \cdot y^{-1}$.
- Thm `thm:division-on-q-well-defined`: division is well-defined on $\mathbb{Q} \times (\mathbb{Q} \setminus \{0_Q\})$.
- Thm `thm:left-distributivity-on-q`: $x(y + z) = xy + xz$.
- Cor `cor:right-distributivity-on-q`: $(y + z)x = yx + zx$.

Field Structure and Generic Field Laws

- Thm `thm:q-additive-abelian-group`: $(\mathbb{Q}, +, 0_Q)$ is an abelian group.
- Thm `thm:q-nonzero-multiplicative-abelian-group`: $(\mathbb{Q} \setminus \{0_Q\}, \cdot, 1_Q)$ is an abelian group.
- Thm `thm:q-is-a-field`: $\mathbb{Q}$ is a field.
- Cor `cor:zero-product-on-q`: $xy = 0_Q$ iff $x = 0_Q$ or $y = 0_Q$; cites the generic field zero-product theorem.
- Cor `cor:double-negation-on-q`: $-(-x) = x$; cites the generic ring result.

- Cor `cor:negation-of-product-on-q`: $(-x)y = -(xy) = x(-y)$; cites the generic ring result.
- Cor `cor:product-of-negatives-on-q`: $(-x)(-y) = xy$; cites the generic ring result.
- Cor `cor:additive-cancellation-on-q`: $x + z = y + z \Rightarrow x = y$; cites generic group cancellation.
- Cor `cor:multiplicative-cancellation-on-q`: $z \neq 0_Q$ and $xz = yz$ imply $x = y$; cites generic field cancellation.
- Cor `cor:uniqueness-additive-inverse-on-q`: additive inverses are unique.
- Cor `cor:uniqueness-multiplicative-inverse-on-q`: multiplicative inverses of nonzero elements are unique.
- Cor `cor:inverse-of-one-is-one-on-q`: $1_Q^{-1} = 1_Q$.
- Cor `cor:inverse-of-product-on-q`: if $x, y \neq 0$, then $(xy)^{-1} = x^{-1}y^{-1}$.
- Cor `cor:inverse-of-inverse-on-q`: if $x \neq 0$, then $(x^{-1})^{-1} = x$.
- Def `def:integer-power-on-q`: x^n for $n \in \mathbb{Z}$, with $x \neq 0_Q$ when $n < 0$.
- Cor `cor:exponent-addition-on-q`: $x^{m+n} = x^m x^n$.
- Cor `cor:power-of-power-on-q`: $(x^m)^n = x^{mn}$.
- Cor `cor:power-of-product-on-q`: $(xy)^n = x^n y^n$.

Order on $\mathbb{Q}$

- Def `def:positive-rational`: $[a, b]$ is positive iff $ab > 0$ in $\mathbb{Z}$.
- Lem `lem:positivity-respects-equivalence-q`: positivity respects fraction equivalence.
- Thm `thm:positivity-on-q-well-defined`: positivity is well-defined on $\mathbb{Q}$.
- Def `def:strict-order-on-q`: $x < y$ iff $y - x$ is positive.
- Def `def:order-on-q`: $x \leq y$ iff $x < y$ or $x = y$.
- Prop `prop:product-of-positives-is-positive-q`: positive rationals have positive product.
- Thm `thm:sign-trichotomy-on-q`: exactly one of $x < 0_Q$, $x = 0_Q$, $0_Q < x$ holds.
- Thm `thm:trichotomy-on-q`: exactly one of $x < y$, $x = y$, $y < x$ holds.

- Thm `thm:strict-order-on-q-irreflexive`: not $x < x$.
- Thm `thm:strict-order-on-q-asymmetric`: $x < y \Rightarrow \neg(y < x)$.
- Thm `thm:strict-order-on-q-transitive`: $x < y$ and $y < z$ imply $x < z$.
- Thm `thm:order-on-q-reflexive`: $x \le x$.
- Thm `thm:order-on-q-antisymmetric`: $x \le y$ and $y \le x$ imply $x = y$.
- Thm `thm:order-on-q-transitive`: $x \le y$ and $y \le z$ imply $x \le z$.
- Thm `thm:order-on-q-total`: $x \le y$ or $y \le x$.
- Thm `thm:q-totally-ordered-set`: $(\mathbb{Q}, \le)$ is a total order.
- Thm `thm:left-addition-preserves-strict-order-on-q`: $x < y \Rightarrow z + x < z + y$.
- Cor `cor:right-addition-preserves-strict-order-on-q`: $x < y \Rightarrow x + z < y + z$.
- Thm `thm:left-addition-preserves-order-on-q`: $x \le y \Rightarrow z + x \le z + y$.
- Cor `cor:right-addition-preserves-order-on-q`: $x \le y \Rightarrow x + z \le y + z$.
- Cor `cor:addition-reflects-order-on-q`: addition by a fixed rational reflects strict and non-strict order.
- Thm `thm:left-positive-multiplication-preserves-strict-order-on-q`: positive left multiplication preserves strict order.
- Cor `cor:right-positive-multiplication-preserves-strict-order-on-q`: positive right multiplication preserves strict order.
- Thm `thm:positive-multiplication-preserves-order-on-q`: positive multiplication preserves non-strict order.
- Thm `thm:negative-multiplication-reverses-strict-order-on-q`: negative multiplication reverses strict order.
- Thm `thm:reciprocal-reverses-order-on-positives-q`: $0_Q < x < y \Rightarrow 0_Q < y^{-1} < x^{-1}$.
- Thm `thm:sign-of-reciprocal-q`: reciprocals preserve positive and negative sign.
- Def `def:absolute-value-on-q`: $|x| := x$ if $0_Q \le x$, and $|x| := -x$ otherwise.
- Thm `thm:absolute-value-nonnegative-on-q`: $0_Q \le |x|$.
- Thm `thm:absolute-value-zero-iff-on-q`: $|x| = 0_Q$ iff $x = 0_Q$.

- Thm `thm:absolute-value-multiplicative-on-q`: $|xy| = |x||y|$.
- Thm `thm:triangle-inequality-on-q`: $|x + y| \leq |x| + |y|$.
- Thm `thm:absolute-value-of-reciprocal-q`: $x \neq 0_Q \Rightarrow |x^{-1}| = |x|^{-1}$.
- Thm `thm:q-is-an-ordered-field`: $\mathbb{Q}$ is an ordered field.

Density, Gaps, and Embedding

- Lem `lem:midpoint-between-q`: $x < y \Rightarrow x < (x + y)2_Q^{-1} < y$.
- Thm `thm:density-of-q`: between any two rationals lies a rational.
- Prop `prop:no-least-positive-rational`: every positive rational has a smaller positive rational.
- Thm `thm:archimedean-property-of-q`: for $0_Q < x$, some $n \in \mathbb{N}$ has $y < nx$.
- Cor `cor:reciprocals-arbitrarily-small-q`: reciprocals of naturals become arbitrarily small.
- Thm `thm:no-rational-square-root-of-two`: no rational square is 2_Q.
- Def `def:gap-set-q`: $S := \{q \in \mathbb{Q} : 0_Q < q \wedge q^2 < 2_Q\}$.
- Prop `prop:gap-set-bounded-above-q`: the gap set is nonempty and bounded above.
- Thm `thm:gap-set-no-rational-supremum-q`: the gap set has no rational supremum.
- Cor `cor:q-order-incomplete`: $\mathbb{Q}$ is not order-complete.
- Def `def:cauchy-sequence-in-q`: rational Cauchy sequence.
- Def `def:null-sequence-in-q`: rational null sequence.
- Lem `lem:rational-limit-unique`: a convergent rational sequence has at most one rational limit.
- Def `def:embedding-z-into-q`: $\iota : \mathbb{Z} \to \mathbb{Q}$, $\iota(n) := [n, 1]$.
- Thm `thm:embedding-z-into-q-preserves-zero`: $\iota(0_Z) = 0_Q$.
- Thm `thm:embedding-z-into-q-preserves-one`: $\iota(1_Z) = 1_Q$.
- Thm `thm:embedding-z-into-q-injective`: $\iota(m) = \iota(n) \Rightarrow m = n$.
- Thm `thm:embedding-z-into-q-preserves-addition`: $\iota(m + n) = \iota(m) + \iota(n)$.

- Thm `thm:embedding-z-into-q-preserves-multiplication`: $\iota(mn) = \iota(m)\iota(n)$.
- Thm `thm:embedding-z-into-q-preserves-order`: $m \leq n$ iff $\iota(m) \leq \iota(n)$.
- Thm `thm:z-embeds-into-q`: ι is an injective order-preserving ring homomorphism.
- Thm `thm:q-is-field-of-fractions-of-z`: every rational is a quotient of integer images.
- Cor `cor:embedding-z-into-q-not-surjective`: $\iota(\mathbb{Z}) \subsetneq \mathbb{Q}$.
- Rem `rem:why-real-numbers-are-introduced`: $\mathbb{Q}$ is an ordered field but order-incomplete, motivating $\mathbb{R}$.

Chapter 24

Real Numbers ($\mathbb{R}$)

SERIES	VOLUME II	BOOK II	CHAPTER 11
From Cantor to Ito	Origins of Numbers	The Continuum	Real Numbers ($\mathbb{R}$)

24.1 Construction of $\mathbb{R}$

Dedekind Cuts

Definition

Definition 24.1.1 (Dedekind Cut)**.** A *Dedekind cut* is a set $\alpha \subseteq \mathbb{Q}$ satisfying:

$$\alpha \neq \emptyset, \qquad \alpha \neq \mathbb{Q},$$

$$p \in \alpha \wedge q < p \Longrightarrow q \in \alpha,$$

and

$$p \in \alpha \Longrightarrow \exists r \in \alpha\, (p < r).$$

Remark (Dependencies).

- Density of $\mathbb{Q}$

Definition

Definition 24.1.2 (The Real Numbers)**.** The real numbers are the set

$$\mathbb{R} := \{\alpha : \alpha \text{ is a Dedekind cut in } \mathbb{Q}\}.$$

Remark (Dependencies).

- Dedekind Cut

Rational Cuts

Definition

Definition 24.1.3 (Rational Cut)**.** For $q \in \mathbb{Q}$, define the rational cut

$$q^* := \{r \in \mathbb{Q} : r < q\}.$$

Remark (Dependencies).

- Rational Numbers
- Order on $\mathbb{Q}$

Lemma

Lemma 24.1.4 (Rational Cuts Are Cuts)**.** *For every $q \in \mathbb{Q}$, the set q^* is a Dedekind cut.*

Remark (Dependencies).

- Density of $\mathbb{Q}$
- No Least Positive Rational

Zero and One in $\mathbb{R}$

Definition

Definition 24.1.5 (Zero in $\mathbb{R}$)**.** Define

$$0_{\mathbb{R}} := (0_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rational Cut
- Zero in $\mathbb{Q}$

Definition

Definition 24.1.6 (One in $\mathbb{R}$)**.** Define

$$1_{\mathbb{R}} := (1_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 1_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rational Cut
- One in $\mathbb{Q}$

Theorem

Theorem 24.1.7 (Zero and One Are Reals)**.** *The cuts* $0_{\mathbb{R}}$ *and* $1_{\mathbb{R}}$ *are elements of* $\mathbb{R}$.

Remark (Dependencies).

- The Real Numbers
- Zero in $\mathbb{R}$
- One in $\mathbb{R}$
- Rational Cuts Are Cuts

Theorem

Theorem 24.1.8 (Zero Is Not One in $\mathbb{R}$)**.** *In* $\mathbb{R}$,

$$0_{\mathbb{R}} \neq 1_{\mathbb{R}}.$$

Remark (Dependencies).

- Zero in $\mathbb{R}$
- One in $\mathbb{R}$
- One Is Positive in $\mathbb{Q}$

24.2 Order on $\mathbb{R}$

Order by Inclusion

Definition

Definition 24.2.1 (Order on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \leq \beta \quad :\Longleftrightarrow \quad \alpha \subseteq \beta.$$

Remark (Dependencies).

- The Real Numbers
- Subset

Definition

Definition 24.2.2 (Strict Order on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha < \beta \quad :\Longleftrightarrow \quad \alpha \subsetneq \beta.$$

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Proper Subset

Theorem

Theorem 24.2.3 (Order on $\mathbb{R}$ Is Reflexive). *For every $\alpha \in \mathbb{R}$, $\alpha \leq \alpha$.*

Remark (Dependencies).

- Order on $\mathbb{R}$
- Subset

Theorem

Theorem 24.2.4 (Order on $\mathbb{R}$ Is Antisymmetric). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \alpha \implies \alpha = \beta.$$

Remark (Dependencies).

- Order on $\mathbb{R}$
- Set Equality
- Subset

Theorem

Theorem 24.2.5 (Order on $\mathbb{R}$ Is Transitive). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \gamma \implies \alpha \leq \gamma.$$

Remark (Dependencies).

- Order on $\mathbb{R}$
- Subset

Theorem

Theorem 24.2.6 (Comparability of Cuts). *For all cuts $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \subseteq \beta \vee \beta \subseteq \alpha.$$

Remark (Dependencies).

- Dedekind Cut

- $\mathbb{Q}$ Is a Totally Ordered Set

Theorem

Theorem 24.2.7 (Order on $\mathbb{R}$ Is Total)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \vee \beta \leq \alpha.$$

Remark (Dependencies).

- Order on $\mathbb{R}$
- Comparability of Cuts

Theorem

Theorem 24.2.8 (Trichotomy on $\mathbb{R}$)**.** *For all $\alpha, \beta \in \mathbb{R}$, exactly one of*

$$\alpha < \beta, \qquad \alpha = \beta, \qquad \beta < \alpha$$

holds.

Remark (Dependencies).

- Strict Order on $\mathbb{R}$
- Order on $\mathbb{R}$ Is Antisymmetric
- Order on $\mathbb{R}$ Is Total

Theorem

Theorem 24.2.9 ($\mathbb{R}$ Is a Totally Ordered Set)**.** *The ordered pair $(\mathbb{R}, \leq)$ is a total order.*

Remark (Dependencies).

- Ordered Set
- Total Order
- Order on $\mathbb{R}$
- Order on $\mathbb{R}$ Is Reflexive
- Order on $\mathbb{R}$ Is Antisymmetric
- Order on $\mathbb{R}$ Is Transitive
- Order on $\mathbb{R}$ Is Total

24.3 Completeness of $\mathbb{R}$

Definition

Definition 24.3.1 (Upper Bound in $\mathbb{R}$)**.** Let $S \subseteq \mathbb{R}$. A cut $\gamma \in \mathbb{R}$ is an upper bound of S if

$$\forall \alpha \in S\, (\alpha \leq \gamma).$$

The set S is bounded above if it has an upper bound in $\mathbb{R}$.

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Subset

Definition

Definition 24.3.2 (Supremum in $\mathbb{R}$)**.** For $S \subseteq \mathbb{R}$, a cut σ is the supremum of S if σ is an upper bound of S and every upper bound γ of S satisfies $\sigma \leq \gamma$.

Remark (Dependencies).

- Upper Bound in $\mathbb{R}$
- Order on $\mathbb{R}$

Definition

Definition 24.3.3 (Least-Upper-Bound Property on $\mathbb{R}$)**.** The ordered set $\mathbb{R}$ has the *least-upper-bound property* if every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Upper Bound in $\mathbb{R}$
- Supremum in $\mathbb{R}$

Lemma

Lemma 24.3.4 (Union of a Bounded Nonempty Family of Cuts Is a Cut)**.** *If $S \subseteq \mathbb{R}$ is nonempty and bounded above, then*

$$\bigcup_{\alpha \in S} \alpha$$

is a Dedekind cut.

Remark (Dependencies).

- Dedekind Cut
- Upper Bound in $\mathbb{R}$
- Union

Theorem

Theorem 24.3.5 (Least-Upper-Bound Property). *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded above has a least upper bound, namely*

$$\sup S = \bigcup_{\alpha \in S} \alpha.$$

Remark (Dependencies).

- Least-Upper-Bound Property on $\mathbb{R}$
- Union of a Bounded Nonempty Family of Cuts Is a Cut
- Supremum in $\mathbb{R}$

Corollary

Corollary 24.3.6 (Greatest-Lower-Bound Property). *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded below has a greatest lower bound.*

Remark (Dependencies).

- Least-Upper-Bound Property

24.4 Addition on $\mathbb{R}$

Definition

Definition 24.4.1 (Addition of Cuts). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha + \beta := \{p + q : p \in \alpha,\ q \in \beta\}.$$

Remark (Dependencies).

- The Real Numbers
- Addition on $\mathbb{Q}$

Lemma

Lemma 24.4.2 (Sum of Cuts Is a Cut). *If $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is a Dedekind cut.*

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Theorem

Theorem 24.4.3 (Addition on $\mathbb{R}$ Is Associative). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma).$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Sum of Cuts Is a Cut
- Addition on $\mathbb{Q}$ Is Associative

Theorem

Theorem 24.4.4 (Addition on $\mathbb{R}$ Is Commutative). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha + \beta = \beta + \alpha.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Sum of Cuts Is a Cut
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 24.4.5 (Zero Is an Additive Identity on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha + 0_{\mathbb{R}} = \alpha.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Zero in $\mathbb{R}$
- Zero Is an Additive Identity on $\mathbb{Q}$

Definition

Definition 24.4.6 (Negation of a Cut)**.** For $\alpha \in \mathbb{R}$, define

$$-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q}\,(r > 0 \land -p - r \notin \alpha)\}.$$

Remark (Dependencies).

- The Real Numbers
- Rational Numbers
- Positive Rational
- Order on $\mathbb{Q}$

Lemma

Lemma 24.4.7 (Negation of a Cut Is a Cut)**.** *If $\alpha \in \mathbb{R}$, then $-\alpha$ is a Dedekind cut.*

Remark (Dependencies).

- Negation on $\mathbb{R}$
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Theorem

Theorem 24.4.8 (Negation Is an Additive Inverse on $\mathbb{R}$)**.** *For every $\alpha \in \mathbb{R}$,*

$$\alpha + (-\alpha) = 0_{\mathbb{R}}.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Negation on $\mathbb{R}$
- Negation of a Cut Is a Cut
- Zero in $\mathbb{R}$

Theorem

Theorem 24.4.9 ($\mathbb{R}$ Is an Additive Abelian Group)**.** *The structure $(\mathbb{R}, +, 0_{\mathbb{R}})$ is an abelian group.*

Remark (Dependencies).

- Addition on $\mathbb{R}$

- Addition on $\mathbb{R}$ Is Associative
- Addition on $\mathbb{R}$ Is Commutative
- Zero Is an Additive Identity on $\mathbb{R}$
- Negation Is an Additive Inverse on $\mathbb{R}$

Definition

Definition 24.4.10 (Subtraction on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha - \beta := \alpha + (-\beta).$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Negation on $\mathbb{R}$

24.5 Multiplication on $\mathbb{R}$

Definition

Definition 24.5.1 (Nonnegative Cut)**.** A cut $\alpha \in \mathbb{R}$ is nonnegative if $0_{\mathbb{R}} \leq \alpha$ and positive if $0_{\mathbb{R}} < \alpha$.

Remark (Dependencies).

- The Real Numbers
- Zero in $\mathbb{R}$
- Order on $\mathbb{R}$

Definition

Definition 24.5.2 (Product of Nonnegative Cuts)**.** For nonnegative cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \cdot \beta := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\} \cup \{p \cdot q : p \in \alpha,\ q \in \beta,\ p \geq 0_{\mathbb{Q}},\ q \geq 0_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Nonnegative Cut
- Multiplication on $\mathbb{Q}$
- Order on $\mathbb{Q}$

Lemma

Lemma 24.5.3 (Product of Nonnegative Cuts Is a Cut)**.** *If* $\alpha, \beta \geq 0_{\mathbb{R}}$*, then* $\alpha \cdot \beta$ *is a Dedekind cut.*

Remark (Dependencies).

- Product of Nonnegative Cuts
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Definition

Definition 24.5.4 (Absolute Value on $\mathbb{R}$)**.** For $\alpha \in \mathbb{R}$, define

$$|\alpha| := \begin{cases} \alpha, & 0_{\mathbb{R}} \leq \alpha, \\ -\alpha, & \alpha < 0_{\mathbb{R}}. \end{cases}$$

Remark (Dependencies).

- Order on $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Zero in $\mathbb{R}$
- Negation on $\mathbb{R}$

Theorem

Theorem 24.5.5 (Triangle Inequality)**.** *For all* $\alpha, \beta \in \mathbb{R}$*,*

$$|\alpha + \beta| \leq |\alpha| + |\beta|.$$

Go to proof.

Remark (Dependencies).

- Absolute Value on $\mathbb{R}$
- $\mathbb{R}$ Is an Ordered Field

Definition

Definition 24.5.6 (Multiplication of Cuts)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define $\alpha \cdot \beta$ by sign cases: use $|\alpha| \cdot |\beta|$ when α and β have the same sign, use $-(|\alpha| \cdot |\beta|)$ when they have opposite signs, and use $0_{\mathbb{R}}$ when either factor is $0_{\mathbb{R}}$.

Remark (Dependencies).

- The Real Numbers
- Product of Nonnegative Cuts
- Absolute Value on $\mathbb{R}$
- Negation on $\mathbb{R}$
- Zero in $\mathbb{R}$

Lemma

Lemma 24.5.7 (Product of Cuts Is a Cut)**.** *If* $\alpha, \beta \in \mathbb{R}$*, then* $\alpha \cdot \beta \in \mathbb{R}$.

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Product of Nonnegative Cuts
- Product of Nonnegative Cuts Is a Cut
- Negation of a Cut Is a Cut

Theorem

Theorem 24.5.8 (Multiplication on $\mathbb{R}$ Is Associative)**.** *For all* $\alpha, \beta, \gamma \in \mathbb{R}$,

$$(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma).$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Product of Cuts Is a Cut
- Multiplication on $\mathbb{Q}$ Is Associative

Theorem

Theorem 24.5.9 (Multiplication on $\mathbb{R}$ Is Commutative)**.** *For all* $\alpha, \beta \in \mathbb{R}$,

$$\alpha \cdot \beta = \beta \cdot \alpha.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Product of Cuts Is a Cut
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 24.5.10 (One Is a Multiplicative Identity on $\mathbb{R}$)**.** *For every $\alpha \in \mathbb{R}$,*

$$\alpha \cdot 1_{\mathbb{R}} = \alpha.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- One in $\mathbb{R}$
- One Is a Multiplicative Identity on $\mathbb{Q}$

Definition

Definition 24.5.11 (Reciprocal of a Positive Cut)**.** For $\alpha > 0_{\mathbb{R}}$, define

$$\alpha^{-1} := \{r \in \mathbb{Q} : r \leq 0_{\mathbb{Q}}\} \cup \{r \in \mathbb{Q} : r > 0_{\mathbb{Q}} \wedge \exists s \in \mathbb{Q}\,(s > r \wedge s^{-1} \notin \alpha)\}.$$

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Order on $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$

Definition

Definition 24.5.12 (Reciprocal of a Nonzero Cut)**.** For $\alpha \neq 0_{\mathbb{R}}$, define α^{-1} to be $|\alpha|^{-1}$ if $\alpha > 0_{\mathbb{R}}$ and $-(|\alpha|^{-1})$ if $\alpha < 0_{\mathbb{R}}$.

Remark (Dependencies).

- Reciprocal of a Positive Cut
- Absolute Value on $\mathbb{R}$
- Negation on $\mathbb{R}$
- Zero in $\mathbb{R}$
- Order on $\mathbb{R}$

Lemma

Lemma 24.5.13 (Reciprocal of a Nonzero Cut Is a Cut)**.** *If $\alpha \neq 0_{\mathbb{R}}$, then $\alpha^{-1} \in \mathbb{R}$.*

Remark (Dependencies).

- Reciprocal on $\mathbb{R}$
- Reciprocal of a Positive Cut
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Theorem

Theorem 24.5.14 (Reciprocal Is a Multiplicative Inverse on $\mathbb{R}$)**.** *If $\alpha \neq 0_{\mathbb{R}}$, then*

$$\alpha \cdot \alpha^{-1} = 1_{\mathbb{R}}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- Reciprocal of a Nonzero Cut Is a Cut
- One in $\mathbb{R}$

Definition

Definition 24.5.15 (Division on $\mathbb{R}$)**.** For $\beta \neq 0_{\mathbb{R}}$, define

$$\alpha/\beta := \alpha \cdot \beta^{-1}.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Reciprocal on $\mathbb{R}$
- Zero in $\mathbb{R}$

Theorem

Theorem 24.5.16 (Distributivity on $\mathbb{R}$)**.** *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- Multiplication Distributes over Addition on $\mathbb{Q}$

24.6 Field and Ordered-Field Structure of $\mathbb{R}$

Theorem

Theorem 24.6.1 (Nonzero $\mathbb{R}$ Is a Multiplicative Abelian Group). *The structure* $(\mathbb{R} \setminus \{0_{\mathbb{R}}\}, \cdot, 1_{\mathbb{R}})$ *is an abelian group.*

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Multiplication on $\mathbb{R}$ Is Associative
- Multiplication on $\mathbb{R}$ Is Commutative
- One Is a Multiplicative Identity on $\mathbb{R}$
- Reciprocal Is a Multiplicative Inverse on $\mathbb{R}$

Theorem

Theorem 24.6.2 ($\mathbb{R}$ Is a Field). *The structure* $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}})$ *is a field.*

Remark (Dependencies).

- $\mathbb{R}$ Is an Additive Abelian Group
- Nonzero $\mathbb{R}$ Is a Multiplicative Abelian Group
- Distributivity on $\mathbb{R}$
- Zero Is Not One in $\mathbb{R}$

Corollary

Corollary 24.6.3 (Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for fields holds in* $\mathbb{R}$*, including zero product, double negation, sign rules, cancellation, uniqueness of inverses, inverse laws, and exponent laws.*

Remark (Dependencies).

Order Compatibility

Theorem

Theorem 24.6.4 (Addition Preserves Order on $\mathbb{R}$). *For all* $\alpha, \beta, \gamma \in \mathbb{R}$,

$$\alpha < \beta \implies \alpha + \gamma < \beta + \gamma,$$

and the corresponding non-strict statement also holds.

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Order on $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Addition Preserves Order on $\mathbb{Q}$

Theorem

Theorem 24.6.5 (Multiplication by Positives Preserves Order on $\mathbb{R}$)**.** *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \gamma \wedge \alpha < \beta \Longrightarrow \gamma \cdot \alpha < \gamma \cdot \beta.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Order on $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Positive Multiplication Preserves Order on $\mathbb{Q}$

Theorem

Theorem 24.6.6 ($\mathbb{R}$ Is an Ordered Field)**.** *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}}, \leq)$ is an ordered field.*

Remark (Dependencies).

- $\mathbb{R}$ Is a Field
- $\mathbb{R}$ Is a Totally Ordered Set
- Addition Preserves Order on $\mathbb{R}$
- Multiplication by Positives Preserves Order on $\mathbb{R}$

Corollary

Corollary 24.6.7 (Ordered-Field Laws Hold on $\mathbb{R}$)**.** *Every theorem proved generically for ordered fields holds in $\mathbb{R}$.*

Remark (Dependencies).

Theorem

Theorem 24.6.8 ($\mathbb{R}$ Is a Complete Ordered Field)**.** *The ordered field $\mathbb{R}$ satisfies the least-upper-bound property.*

Remark (Dependencies).

- Complete Ordered Field
- $\mathbb{R}$ Is an Ordered Field
- Least-Upper-Bound Property

24.7 Embedding $\mathbb{Q}$ into $\mathbb{R}$

Definition

Definition 24.7.1 (Embedding of $\mathbb{Q}$ into $\mathbb{R}$)**.** Define $\iota : \mathbb{Q} \to \mathbb{R}$ by

$$\iota(q) := q^*.$$

Remark (Dependencies).

- Rational Numbers
- The Real Numbers
- Rational Cut

Theorem

Theorem 24.7.2 (Embedding Preserves Zero and One)**.** *The embedding satisfies*

$$\iota(0_{\mathbb{Q}}) = 0_{\mathbb{R}} \qquad \textit{and} \qquad \iota(1_{\mathbb{Q}}) = 1_{\mathbb{R}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Zero in $\mathbb{Q}$
- One in $\mathbb{Q}$
- Zero in $\mathbb{R}$
- One in $\mathbb{R}$

Theorem

Theorem 24.7.3 (Embedding $\mathbb{Q}$ into $\mathbb{R}$ Is Injective)**.** *For all $p, q \in \mathbb{Q}$,*

$$\iota(p) = \iota(q) \Longrightarrow p = q.$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Rational Cut

- $\mathbb{Q}$ Is a Totally Ordered Set

Theorem

Theorem 24.7.4 (Embedding Preserves Addition and Multiplication)**.** *For all $p, q \in \mathbb{Q}$,*

$$\iota(p+q) = \iota(p) + \iota(q) \qquad \textit{and} \qquad \iota(p \cdot q) = \iota(p) \cdot \iota(q).$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$

Theorem

Theorem 24.7.5 (Embedding Preserves Order)**.** *For all $p, q \in \mathbb{Q}$,*

$$p \leq q \Longleftrightarrow \iota(p) \leq \iota(q).$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Order on $\mathbb{Q}$
- Order on $\mathbb{R}$
- Rational Cut

Theorem

Theorem 24.7.6 ($\mathbb{Q}$ Embeds into $\mathbb{R}$ as an Ordered Field)**.** *The map ι is an injective order-preserving field homomorphism.*

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Embedding Preserves Zero and One
- Embedding $\mathbb{Q}$ into $\mathbb{R}$ Is Injective
- Embedding Preserves Addition and Multiplication
- Embedding Preserves Order

Theorem

Theorem 24.7.7 (Density of $\mathbb{Q}$ in $\mathbb{R}$). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha < \beta \implies \exists q \in \mathbb{Q}\,(\alpha < \iota(q) < \beta).$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Density of $\mathbb{Q}$ in Itself
- Dedekind Cut

Theorem

Theorem 24.7.8 (Archimedean Property of $\mathbb{R}$). *For $\alpha, \beta \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \alpha \implies \exists n \in \mathbb{N}\,(\beta < \iota(n) \cdot \alpha).$$

Remark (Dependencies).

- Least-Upper-Bound Property

24.8 The Gap Filled

Definition

Definition 24.8.1 (The Square-Root-of-Two Cut). Define

$$\sqrt{2} := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}} \vee r \cdot r < 2_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rational Numbers
- Order on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- Two in $\mathbb{Q}$

Lemma

Lemma 24.8.2 (The Square-Root-of-Two Cut Is a Cut). *The set $\sqrt{2}$ is a Dedekind cut.*

Remark (Dependencies).

- Density of $\mathbb{Q}$

Theorem

Theorem 24.8.3 (Its Square Is Two)**.** *In* $\mathbb{R}$*,*

$$\sqrt{2} \cdot \sqrt{2} = 2_{\mathbb{R}},$$

where $2_{\mathbb{R}} := \iota(2_{\mathbb{Q}})$*.*

Remark (Dependencies).

- The Square-Root-of-Two Cut
- The Square-Root-of-Two Cut Is a Cut
- Multiplication on $\mathbb{R}$
- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Two in $\mathbb{Q}$

Corollary

Corollary 24.8.4 ($\mathbb{R}$ Contains the Supremum $\mathbb{Q}$ Was Missing)**.** *Let*

$$S = \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \wedge q \cdot q < 2_{\mathbb{Q}}\}.$$

Then

$$\sqrt{2} = \sup \iota(S).$$

Thus the bounded-above rational set with no rational supremum has a supremum in $\mathbb{R}$*.*

Remark (Dependencies).

- The Gap Set in $\mathbb{Q}$
- The Gap Set Has No Rational Supremum

24.9 Characterization of $\mathbb{R}$

Definition

Definition 24.9.1 (Complete Ordered Field)**.** A *complete ordered field* is an ordered field satisfying the least-upper-bound property.

Remark (Dependencies).

- Least-Upper-Bound Property on $\mathbb{R}$

Theorem

Theorem 24.9.2 (Uniqueness of the Complete Ordered Field)**.** *Any two complete ordered fields are isomorphic by a unique order-isomorphism.*

Remark (Dependencies).

- Complete Ordered Field
- $\mathbb{R}$ Is a Complete Ordered Field

Corollary

Corollary 24.9.3 ($\mathbb{R}$ Is the Real Numbers)**.** *The field* $\mathbb{R}$ *is determined up to unique order-isomorphism by being a complete ordered field.*

Remark (Dependencies).

- Complete Ordered Field
- Uniqueness of the Complete Ordered Field
- $\mathbb{R}$ Is a Complete Ordered Field

24.10 Transition and Forward

Remark 24.10.1 (What Completeness Buys)*.* The least-upper-bound property promotes the arbitrary closeness available in $\mathbb{Q}$ to guaranteed limits in $\mathbb{R}$: bounded monotone sequences converge, Cauchy sequences converge, and suprema exist. The general theory of these phenomena belongs to analysis; this chapter supplies the complete ordered field in which they live.

Remark 24.10.2 (Parallel Constructions)*.* Cauchy sequences, nested intervals, and dyadic expansions give alternative constructions of complete ordered fields. By Uniqueness of the Complete Ordered Field, each yields this same $\mathbb{R}$ up to unique isomorphism.

Chapter 25

Complex Numbers

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CHAPTER 12
Complex Numbers

25.1 Construction of $\mathbb{C}$

Complex Prepairs

The Last Number System Extension

The complex numbers extend the real numbers by adding a formal square root of -1. Unlike the construction of $\mathbb{Z}$ from $\mathbb{W}$ or $\mathbb{Q}$ from $\mathbb{Z}$, no nontrivial identification is needed: ordered pairs of real numbers already have the correct equality. We still use the same construction pattern: representatives, an equivalence relation, equivalence classes, then operations whose well-definedness is checked before they are used.

Definition

Definition 25.1.1 (Complex Prepair)**.** A *complex prepair* is an ordered pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, read informally as the expression $a + bi$.

Remark (Dependencies).

- Ordered Pair
- The Real Numbers

Definition

Definition 25.1.2 (Complex Prepair Set)**.** The complex prepair set is

$$\operatorname{Pre}\mathbb{C} := \mathbb{R} \times \mathbb{R}.$$

Remark (Dependencies).

- Cartesian Product
- The Real Numbers

Coordinate Equivalence

Definition

Definition 25.1.3 (Complex Coordinate Equivalence)**.** For $(a,b),(c,d) \in \operatorname{Pre}\mathbb{C}$, define

$$(a,b) \sim_{\mathbb{C}} (c,d) \quad :\Longleftrightarrow \quad a = c \ \text{ and } \ b = d.$$

Remark (Predicate reading).

$$(a,b) \sim_{\mathbb{C}} (c,d) \Longleftrightarrow (a = c) \wedge (b = d).$$

Thus the complex-number quotient is formally present, but the equivalence classes are singleton coordinate-equality classes.

Remark (Dependencies).

- Relation
- Complex Prepair Set

Lemma

Lemma 25.1.4 (Complex Coordinate Equivalence Is Reflexive)**.** *For every* $(a,b) \in \operatorname{Pre}\mathbb{C}$*,*

$$(a,b) \sim_{\mathbb{C}} (a,b).$$

Lemma

Lemma 25.1.5 (Complex Coordinate Equivalence Is Symmetric)**.** *For all prepairs* $(a,b),(c,d)$*,*

$$(a,b) \sim_{\mathbb{C}} (c,d) \Longrightarrow (c,d) \sim_{\mathbb{C}} (a,b).$$

Lemma

Lemma 25.1.6 (Complex Coordinate Equivalence Is Transitive)**.** *For all prepairs* $(a,b),(c,d),(e,f)$*,*

$$(a,b) \sim_{\mathbb{C}} (c,d) \wedge (c,d) \sim_{\mathbb{C}} (e,f) \Longrightarrow (a,b) \sim_{\mathbb{C}} (e,f).$$

Theorem

Theorem 25.1.7 (Complex Coordinate Equivalence Is an Equivalence Relation)**.** *The relation* $\sim_{\mathbb{C}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{C}$*.*

Remark (Proof sketch). Reflexivity, symmetry, and transitivity are checked coordinate by coordinate using equality in $\mathbb{R}$. Thus this quotient construction has the same formal shape as the constructions of $\mathbb{Z}$ and $\mathbb{Q}$, even though the equivalence classes are singleton classes.

Remark (Dependencies).

- Complex Coordinate Equivalence
- Complex Coordinate Equivalence Is Reflexive
- Complex Coordinate Equivalence Is Symmetric
- Complex Coordinate Equivalence Is Transitive

Definition of $\mathbb{C}$

Definition

Definition 25.1.8 (Complex Numbers). The complex numbers are the quotient

$$\mathbb{C} := \operatorname{Pre}\mathbb{C}/\sim_{\mathbb{C}}.$$

Remark (Dependencies).

- Quotient Set
- Complex Prepair Set
- Complex Coordinate Equivalence

Definition

Definition 25.1.9 (Complex Class Notation). The notation $[a, b]_{\mathbb{C}}$ denotes the $\sim_{\mathbb{C}}$-equivalence class of the prepair (a, b).

Remark (Standard form). Since coordinate equivalence is equality of coordinates, every complex number has a unique representative

$$[a, b]_{\mathbb{C}}.$$

We later write this same element as $a + bi$.

Definition

Definition 25.1.10 (Zero, One, and the Imaginary Unit in $\mathbb{C}$). Define

$$0_{\mathbb{C}} := [0, 0]_{\mathbb{C}}, \qquad 1_{\mathbb{C}} := [1, 0]_{\mathbb{C}}, \qquad i := [0, 1]_{\mathbb{C}}.$$

Remark (Dependencies).

- Equivalence Class
- Complex Class Notation
- The Real Numbers

25.2 Operations on $\mathbb{C}$-Classes

Addition on $\mathbb{C}$-Classes

Definition

Definition 25.2.1 (Addition on $\mathbb{C}$-Classes)**.** Define addition on representatives by

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} := [a + c, b + d]_{\mathbb{C}}.$$

Remark (Dependencies)*.*

- Complex Class Notation
- Addition on $\mathbb{R}$

Lemma

Lemma 25.2.2 (Addition Respects Complex Coordinate Equivalence)**.** *If* $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ *and* $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$*, then*

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} + [c', d']_{\mathbb{C}}.$$

Remark (Dependencies)*.*

- Addition on $\mathbb{C}$
- Complex Coordinate Equivalence

Theorem

Theorem 25.2.3 (Addition on $\mathbb{C}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Remark (Proof sketch)*.* Suppose the input classes are represented in two ways. Coordinate equivalence gives $a = a'$, $b = b'$, $c = c'$, and $d = d'$. Addition in $\mathbb{R}$ gives $a + c = a' + c'$ and $b + d = b' + d'$, so the resulting complex classes agree.

Remark (Dependencies)*.*

- Function
- Addition on $\mathbb{C}$
- Addition Respects Complex Coordinate Equivalence

Negation and Subtraction

Definition

Definition 25.2.4 (Negation on $\mathbb{C}$-Classes)**.** Define negation by

$$-[a, b]_{\mathbb{C}} := [-a, -b]_{\mathbb{C}}.$$

Lemma

Lemma 25.2.5 (Negation Respects Complex Coordinate Equivalence)**.** *If* $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$*, then*

$$-[a, b]_{\mathbb{C}} = -[a', b']_{\mathbb{C}}.$$

Theorem

Theorem 25.2.6 (Negation on $\mathbb{C}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation* $\mathbb{C} \to \mathbb{C}$*.*

Remark (Dependencies)*.*

- Function
- Negation on $\mathbb{C}$
- Negation Respects Complex Coordinate Equivalence

Remark (Proof sketch)*.* If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$, then $a = a'$ and $b = b'$. Hence $-a = -a'$ and $-b = -b'$, so the negated representatives determine the same complex class.

Definition

Definition 25.2.7 (Subtraction on $\mathbb{C}$)**.** For $z, w \in \mathbb{C}$, define

$$z - w := z + (-w).$$

Theorem

Theorem 25.2.8 (Subtraction on $\mathbb{C}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation* $\mathbb{C} \times \mathbb{C} \to \mathbb{C}$*.*

Remark (Dependencies)*.*

- Function
- Subtraction on $\mathbb{C}$
- Addition on $\mathbb{C}$ Is Well-Defined
- Negation on $\mathbb{C}$ Is Well-Defined

Multiplication on $\mathbb{C}$-Classes

Definition

Definition 25.2.9 (Multiplication on $\mathbb{C}$-Classes)**.** Define multiplication on representatives by

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} := [ac - bd, ad + bc]_{\mathbb{C}}.$$

Remark (Why this formula)*.* The rule is forced by the intended identity $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Remark (Dependencies)*.*

- Complex Class Notation
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{R}$

Lemma

Lemma 25.2.10 (Multiplication Respects Complex Coordinate Equivalence)**.** *If* $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ *and* $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$*, then*

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} \cdot [c', d']_{\mathbb{C}}.$$

Theorem

Theorem 25.2.11 (Multiplication on $\mathbb{C}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Remark (Proof sketch)*.* Changing representatives preserves each coordinate. Therefore the real expressions $ac - bd$ and $ad + bc$ are unchanged after replacing (a, b) by an equivalent prepair and (c, d) by an equivalent prepair. The product class is independent of the chosen representatives.

Remark (Dependencies)*.*

- Function
- Multiplication on $\mathbb{C}$
- Multiplication Respects Complex Coordinate Equivalence

Conjugation and Inverses

Definition

Definition 25.2.12 (Complex Conjugation)**.** For $z = [a,b]_{\mathbb{C}}$, define its *complex conjugate* by

$$\overline{z} := [a,-b]_{\mathbb{C}}.$$

Theorem

Theorem 25.2.13 (Complex Conjugation Is Well-Defined)**.** *Complex conjugation determines a well-defined unary operation $\mathbb{C} \to \mathbb{C}$.*

Remark (Dependencies).

- Function
- Complex Conjugation

Remark (Proof sketch). Equivalent prepairs have equal first coordinates and equal second coordinates. Keeping the first coordinate and negating the second preserves that equality, so conjugation does not depend on the representative.

Lemma

Lemma 25.2.14 (Product with the Conjugate Is Real and Nonnegative)**.** *For $z = [a,b]_{\mathbb{C}}$,*

$$z\overline{z} = [a^2 + b^2, 0]_{\mathbb{C}}.$$

Moreover, if $z \neq 0_{\mathbb{C}}$, then $a^2 + b^2 \neq 0$.

Definition

Definition 25.2.15 (Multiplicative Inverse in $\mathbb{C}$)**.** For $z = [a,b]_{\mathbb{C}} \neq 0_{\mathbb{C}}$, define

$$z^{-1} := \left[\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right]_{\mathbb{C}}.$$

Theorem

Theorem 25.2.16 (Inversion on $\mathbb{C}^{\times}$ Is Well-Defined)**.** *Inversion determines a well-defined operation on the nonzero complex numbers.*

Remark (Dependencies).

- Function
- Multiplicative Inverse in $\mathbb{C}$
- Product with the Conjugate Is Real and Nonnegative

Remark (Proof sketch). If $z = [a, b]_{\mathbb{C}} \neq 0_{\mathbb{C}}$, then a and b are not both zero, so $a^2 + b^2 \neq 0$ in $\mathbb{R}$. Equivalent representatives have the same coordinates, hence the same denominator and the same inverse coordinates.

25.3 Field Structure of $\mathbb{C}$

Additive Structure

Theorem

Theorem 25.3.1 (Addition on $\mathbb{C}$ Is Associative). *For all $z, w, u \in \mathbb{C}$,*

$$(z + w) + u = z + (w + u).$$

Theorem

Theorem 25.3.2 (Addition on $\mathbb{C}$ Is Commutative). *For all $z, w \in \mathbb{C}$,*

$$z + w = w + z.$$

Theorem

Theorem 25.3.3 (Zero Is an Additive Identity on $\mathbb{C}$). *For every $z \in \mathbb{C}$,*

$$0_{\mathbb{C}} + z = z \qquad \textit{and} \qquad z + 0_{\mathbb{C}} = z.$$

Theorem

Theorem 25.3.4 (Every Complex Number Has an Additive Inverse). *For every $z \in \mathbb{C}$,*

$$z + (-z) = 0_{\mathbb{C}} \qquad \textit{and} \qquad (-z) + z = 0_{\mathbb{C}}.$$

Multiplicative Structure

Theorem

Theorem 25.3.5 (Multiplication on $\mathbb{C}$ Is Associative). *For all $z, w, u \in \mathbb{C}$,*

$$(zw)u = z(wu).$$

Theorem

Theorem 25.3.6 (Multiplication on $\mathbb{C}$ Is Commutative). *For all $z, w \in \mathbb{C}$,*

$$zw = wz.$$

Theorem

Theorem 25.3.7 (One Is a Multiplicative Identity on $\mathbb{C}$). *For every $z \in \mathbb{C}$,*

$$1_{\mathbb{C}} z = z \qquad \text{and} \qquad z 1_{\mathbb{C}} = z.$$

Theorem

Theorem 25.3.8 (Every Nonzero Complex Number Has a Multiplicative Inverse). *For every $z \in \mathbb{C}$ with $z \neq 0_{\mathbb{C}}$,*

$$zz^{-1} = 1_{\mathbb{C}} \qquad \text{and} \qquad z^{-1}z = 1_{\mathbb{C}}.$$

Distributivity and the Field Theorem

Theorem

Theorem 25.3.9 (Multiplication Distributes over Addition on $\mathbb{C}$). *For all $z, w, u \in \mathbb{C}$,*

$$z(w+u) = zw + zu \qquad \text{and} \qquad (w+u)z = wz + uz.$$

Theorem

Theorem 25.3.10 ($\mathbb{C}$ Is a Field). *With the operations defined above,*

$$(\mathbb{C}, +, \cdot, 0_{\mathbb{C}}, 1_{\mathbb{C}})$$

is a field.

Remark (Dependencies).

- Addition on $\mathbb{C}$ Is Well-Defined
- Multiplication on $\mathbb{C}$ Is Well-Defined
- Every Nonzero Complex Number Has a Multiplicative Inverse

The Imaginary Unit

Theorem

Theorem 25.3.11 (The Imaginary Unit Squares to Negative One). *In $\mathbb{C}$,*

$$i^2 = -1_{\mathbb{C}}.$$

Remark (Interpretation). The equation $i^2 = -1$ is the structural reason for introducing $\mathbb{C}$. It makes the polynomial $x^2 + 1$ have a root, while the field operations remain compatible with the real arithmetic already constructed.

Theorem

Theorem 25.3.12 ($\mathbb{C}$ Is Not an Ordered Field)**.** *There is no order relation on $\mathbb{C}$ making $\mathbb{C}$ an ordered field and extending the usual order on $\mathbb{R}$.*

Remark (Reason). In any ordered field, every square is nonnegative. If $\mathbb{C}$ were an ordered field extending the real order, i^2 would be nonnegative. But $i^2 = -1$, and $-1 < 0$ in the real subfield.

25.4 Embedding $\mathbb{R}$ into $\mathbb{C}$

The Real Axis

Definition

Definition 25.4.1 (Canonical Embedding of $\mathbb{R}$ into $\mathbb{C}$)**.** Define

$$\iota : \mathbb{R} \to \mathbb{C}, \qquad \iota(a) := [a, 0]_{\mathbb{C}}.$$

Remark (Dependencies).

- The Real Numbers
- Complex Numbers
- Complex Class Notation

Theorem

Theorem 25.4.2 (The Real Embedding into $\mathbb{C}$ Is Injective)**.** *If $\iota(a) = \iota(b)$, then $a = b$.*

Theorem

Theorem 25.4.3 (The Real Embedding Preserves Addition and Multiplication)**.** *For all $a, b \in \mathbb{R}$,*

$$\iota(a + b) = \iota(a) + \iota(b), \qquad \iota(ab) = \iota(a)\iota(b).$$

Theorem

Theorem 25.4.4 (The Real Embedding Preserves Zero and One)**.**

$$\iota(0) = 0_{\mathbb{C}}, \qquad \iota(1) = 1_{\mathbb{C}}.$$

Remark (Identification). After these preservation results, it is standard to identify $a \in \mathbb{R}$ with $[a, 0]_{\mathbb{C}} \in \mathbb{C}$. Under this convention,

$$[a, b]_{\mathbb{C}} = [a, 0]_{\mathbb{C}} + [0, b]_{\mathbb{C}} = a + bi.$$

Real and Imaginary Parts

Definition

Definition 25.4.5 (Real and Imaginary Parts)**.** For $z = [a, b]_{\mathbb{C}}$, define

$$\mathrm{Re}(z) := a, \qquad \mathrm{Im}(z) := b.$$

Theorem

Theorem 25.4.6 (Real and Imaginary Parts Are Well-Defined)**.** *The functions*

$$\mathrm{Re}, \mathrm{Im} : \mathbb{C} \to \mathbb{R}$$

are well-defined.

Remark (The completed number-system ladder)*.* The construction of $\mathbb{C}$ completes the standard ladder of number systems in this volume:

$$\mathbb{N} \subseteq \mathbb{W} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

The final step sacrifices order compatibility in exchange for algebraic power: -1 now has a square root.

Chapter 26

Number Lines and Intervals

26.1 Number Lines

The Geometric Model of an Ordered System

A number line is the geometric realization of a linearly ordered system: points on a line arranged so that "left of" matches the order. The construction so far is algebraic; the number line is the picture that accompanies it, and it becomes exact for $\mathbb{R}$.

Definition (Number Line)

Definition 26.1.1 (Number Line)**.** A *number line* for a linearly ordered system $(A, <_A)$ is a linearly ordered set of points L together with an order isomorphism $\lambda : A \to L$, so that for all $a, a' \in A$,

$$a <_A a' \iff \lambda(a) \text{ lies to the left of } \lambda(a').$$

Remark (Standard quantified statement)*.*

$$\lambda : A \to L \text{ is a bijection with } \forall a, a' \in A \; (a <_A a' \Leftrightarrow \lambda(a) \prec \lambda(a')).$$

Remark (Interpretation)*.* The number line records only order, not arithmetic: it is the visual home of betweenness and length. For $\mathbb{R}$ the order isomorphism is onto a full continuum, which is why the real line has no gaps; for $\mathbb{Q}$ the same picture has holes at every irrational location.

Remark (Dependencies)*.*

- Order Embedding

26.2 Intervals on a Number Line

The Basic Pieces of the Line

Intervals are the natural named pieces of a number line. This section gives each basic interval shape its notation and spells out which endpoints are included.

Definition (Open Interval)

Definition 26.2.1 (Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \le b$. The *open interval* with endpoints a and b is

$$(a, b) = \{\, x \in \mathbb{R} : a < x < b \,\}.$$

Remark (Standard quantified statement).

$$(a, b) = \{\, x \in \mathbb{R} : a < x < b \,\}.$$

Remark (Interpretation). The open interval excludes both endpoints. It is one of the four bounded interval shapes.

Remark (Dependencies).

Definition (Closed Interval)

Definition 26.2.2 (Closed Interval)**.** Let $a, b \in \mathbb{R}$ with $a \le b$. The *closed interval* with endpoints a and b is

$$[a, b] = \{\, x \in \mathbb{R} : a \le x \le b \,\}.$$

Remark (Standard quantified statement).

$$[a, b] = \{\, x \in \mathbb{R} : a \le x \le b \,\}.$$

Remark (Interpretation). The closed interval includes both endpoints. It is one of the four bounded interval shapes.

Remark (Dependencies).

Definition (Left-Half-Open Interval)

Definition 26.2.3 (Left-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \le b$. The *left-half-open interval* with endpoints a and b is

$$(a, b] = \{\, x \in \mathbb{R} : a < x \le b \,\}.$$

Remark (Standard quantified statement).

$$(a, b] = \{\, x \in \mathbb{R} : a < x \le b \,\}.$$

Remark (Interpretation). The left-half-open interval excludes the left endpoint, includes the right. It is one of the four bounded interval shapes.

Remark (Dependencies).

Definition (Right-Half-Open Interval)

Definition 26.2.4 (Right-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *right-half-open interval* with endpoints a and b is

$$[a, b) = \{\, x \in \mathbb{R} : a \leq x < b \,\}.$$

Remark (Standard quantified statement).

$$[a, b) = \{\, x \in \mathbb{R} : a \leq x < b \,\}.$$

Remark (Interpretation). The right-half-open interval includes the left endpoint, excludes the right. It is one of the four bounded interval shapes.

Remark (Dependencies).

Definition (Ray)

Definition 26.2.5 (Ray)**.** Let $a \in \mathbb{R}$. The four *rays* determined by a are

$$(a, \infty) = \{x : x > a\},\ [a, \infty) = \{x : x \geq a\},\ (-\infty, a) = \{x : x < a\},\ (-\infty, a] = \{x : x \leq a\}.$$

Remark (Standard quantified statement).

$$\begin{aligned} (a, \infty) &= \{\, x \in \mathbb{R} : a < x \,\}, & [a, \infty) &= \{\, x \in \mathbb{R} : a \leq x \,\}, \\ (-\infty, a) &= \{\, x \in \mathbb{R} : x < a \,\}, & (-\infty, a] &= \{\, x \in \mathbb{R} : x \leq a \,\}. \end{aligned}$$

Remark (Interpretation). Rays are the unbounded one-sided intervals. Together with the bounded intervals and the whole line $(-\infty, \infty) = \mathbb{R}$, they exhaust the interval shapes.

Remark (Dependencies).

Chapter 27

Embedding the Number Systems

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27.1 Embeddings as Structure Preserving Maps

From Construction to Identification

The preceding chapters built $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ one at a time. In that construction-first development the symbol 2 denotes a different object in each system: a Peano element, a whole number, an integer difference object, a rational fraction object, and a real number are literally distinct sets. Nothing in the constructions has made them identical.

An *embedding* is the device that closes this gap without pretending the objects were the same to begin with. It is an injective map that preserves the relevant structure present in the source system — the distinguished constants, the operations, and the order being considered. Once such a map is fixed, the source system is identified with its image inside the target, and only then is the familiar chain $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ written down. The inclusion symbol records an identification along an embedding, not a literal subset relation.

Definition (Structure-Preserving Map)

Definition 27.1.1 (Structure-Preserving Map)**.** Let A and B be ordered arithmetic systems with a specified common source structure: selected constants, selected operations, and a strict order. A map $\varphi : A \to B$ is *structure-preserving* exactly when it preserves each selected constant and, for every selected binary operation $\star$ and all $a, a' \in A$,

$$\varphi(a \star_A a') = \varphi(a) \star_B \varphi(a'),$$

and $a <_A a' \implies \varphi(a) <_B \varphi(a')$.

Remark (Standard quantified statement).

$$\begin{aligned}
&\varphi : A \to B \text{ is structure-preserving}\\
&\iff \Big(\text{every selected constant is preserved}\\
&\qquad \land \ \text{every selected operation is preserved}\\
&\qquad \land \forall a, a' \in A \ (a <_A a' \Rightarrow \varphi(a) <_B \varphi(a'))\Big).
\end{aligned}$$

Remark (Interpretation). A structure-preserving map carries the chosen arithmetic and order from A into B without distortion. The choice of structure matters: the positive part of $\mathbb{W}$ carries the Peano successor from $\mathbb{N}$, while the later ring and field embeddings focus on 0, 1, addition, multiplication, and order. A structure-preserving map need not be injective; collapsing distinct elements is still allowed at this stage. Injectivity is added separately to obtain an embedding.

Remark (Dependencies).

- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part

Definition (Order Embedding)

Definition (Order Embedding)**.** Let A and B be ordered arithmetic systems and $\varphi : A \to B$ a map. Then φ is an *order embedding* exactly when, for all $a, a' \in A$,

$$a <_A a' \iff \varphi(a) <_B \varphi(a').$$

Definition (Embedding of Number Systems)

Definition 27.1.2 (Embedding of Number Systems)**.** Let A and B be ordered arithmetic systems. A map $\varphi : A \to B$ is an *embedding* exactly when φ is injective and structure-preserving and is an order embedding.

Remark (Standard quantified statement).

$$\begin{aligned}
&\varphi \text{ is an embedding}\\
&\iff \Big(\varphi \text{ is injective} \land \varphi \text{ is structure-preserving}\\
&\qquad \land \forall a, a' \in A \ (a <_A a' \Leftrightarrow \varphi(a) <_B \varphi(a'))\Big).
\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\varphi \text{ is not an embedding}\\
&\iff \Big(\exists a, a' \in A \ (a \neq a' \land \varphi(a) = \varphi(a'))\\
&\qquad \lor \ \varphi \text{ fails to preserve a constant, sum, or product}\\
&\qquad \lor \ \exists a, a' \in A \ \big((a <_A a' \land \neg(\varphi(a) <_B \varphi(a'))) \lor (\varphi(a) <_B \varphi(a') \land \neg(a <_A a'))\big)\Big).
\end{aligned}$$

Remark (Failure modes). An embedding can fail in three structurally distinct ways: it may collapse two distinct elements to the same image (injectivity fails), fail to transport one of the selected constants or operations (algebraic preservation fails), or fail to preserve or reflect a strict-order comparison (order embedding fails).

Remark (Interpretation). An embedding is the precise meaning of "A sits inside B." It is injective, so distinct elements stay distinct; it preserves the arithmetic and reflects the order, so the copy of A inside B behaves exactly as A did. This is what licenses treating $2 \in \mathbb{N}$ and $2 \in \mathbb{R}$ as "the same number."

Remark (Interpretation). Most embeddings after $\mathbb{W}$ transport the *ordered ring/field* structure: constants, the two operations, and order. The first bridge $\mathbb{N} \hookrightarrow \mathbb{W}$ is slightly different because $\mathbb{N}$ is one-based and carries Peano successor. That local theorem is proved in the whole-number chapter. In every case, induction is not transported to the larger target: the phrase "the naturals sit inside the integers" must not be read as carrying induction across all of $\mathbb{Z}$.

Remark (Dependencies).

- Structure-Preserving Map
- Order Embedding

Definition (Image of an Embedding)

Definition 27.1.3 (Image of an Embedding)**.** Let $\varphi : A \to B$ be an embedding. The *image* of φ is the set $\varphi(A) = \{\, \varphi(a) : a \in A \,\} \subseteq B$, equipped with the selected constants, operations, and order of B restricted to $\varphi(A)$.

Remark (Standard quantified statement).

$$\varphi(A) = \{\, b \in B : \exists a \in A \ (b = \varphi(a)) \,\}.$$

Remark (Interpretation). The image is the literal subset of B that plays the role of A. The identification theorem below states that φ, viewed as a map onto its image, is an isomorphism, so A and $\varphi(A)$ are the same ordered arithmetic system under two names.

Remark (Dependencies).

- Embedding of Number Systems

Theorem (Embeddings Identify a System with Its Image)

Theorem 27.1.4 (Embeddings Identify a System with Its Image)**.** *Let $\varphi : A \to B$ be an embedding of ordered arithmetic systems relative to a selected source structure. Then the corestriction $\varphi : A \to \varphi(A)$ is an isomorphism onto its image: it is a bijection that preserves and reflects the selected constants, operations, and order. Consequently A may be identified with the corresponding substructure $\varphi(A) \subseteq B$.* *Go to proof.*

Remark (Standard quantified statement).

$$\varphi : A \to B \text{ an embedding}$$
$$\implies \varphi : A \to \varphi(A) \text{ is a bijective isomorphism for the selected structure.}$$

Remark (Interpretation). This is the theorem that makes the inclusion notation legitimate. After identification, $A \subseteq B$ abbreviates "A embeds via φ as the substructure $\varphi(A)$." Every later use of $\mathbb{N} \subseteq \mathbb{R}$ is shorthand for this identification.

Remark (Dependencies).

- Embedding of Number Systems
- Image of an Embedding

27.2 Embedding N Into W

Embedding $\mathbb{N}$ into $\mathbb{W}$

This section records, from the global embedding viewpoint, the local bridge proved in the whole-number chapter: $\mathbb{N}$ sits inside $\mathbb{W}$ as the positive part. The map is the bridge that lets elements of $\mathbb{N}$ be regarded as nonzero elements of $\mathbb{W}$ after identification.

Definition (Embedding $\mathbb{N}$ into $\mathbb{W}$)

Definition 27.2.1 (Embedding $\mathbb{N}$ into $\mathbb{W}$)**.** The *canonical embedding* $\iota : \mathbb{N} \to \mathbb{W}$ is defined by

$$\iota(n) = n \quad \text{(the copy of each natural number inside the whole numbers).}$$

Remark (Standard quantified statement).

$$\iota : \mathbb{N} \to \mathbb{W}, \qquad \iota(n) = n \quad \text{(the copy of each natural number inside the whole numbers).}$$

Remark (Interpretation). The map sends each element of $\mathbb{N}$ to its natural representative in $\mathbb{W}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{N}$ may be identified with its image in $\mathbb{W}$.

Remark (Dependencies).

- Construction of $\mathbb{W}$ from $\mathbb{N}$ by adjoining 0
- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part

Theorem ($\mathbb{N}$ Embeds in $\mathbb{W}$)

Theorem 27.2.2 ($\mathbb{N}$ Embeds in $\mathbb{W}$)**.** *The canonical map $\iota : \mathbb{N} \to \mathbb{W}$ identifies $\mathbb{N}$ with the positive part $\mathbb{W} \setminus \{0\}$ and preserves successor, 1, addition, multiplication, and order.* *Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{N} \to \mathbb{W} \setminus \{0\} \text{ is bijective and preserves successor, arithmetic, and order.}$$

Remark (Interpretation). Once this theorem is in place, $\mathbb{N}$ may be treated as the positive substructure $\iota(\mathbb{N}) = \mathbb{W} \setminus \{0\}$ inside $\mathbb{W}$. The new element of $\mathbb{W}$ is exactly the adjoined zero.

Remark (Dependencies).

- Construction of $\mathbb{W}$ from $\mathbb{N}$ by adjoining 0
- Embedding $\mathbb{N}$ into $\mathbb{W}$
- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part

27.3 Embedding W Into Z

Embedding $\mathbb{W}$ into $\mathbb{Z}$

This section fixes the canonical embedding of $\mathbb{W}$ into $\mathbb{Z}$ and records that it is injective, structure-preserving, and order-reflecting. The map is the bridge that lets elements of $\mathbb{W}$ be regarded as elements of $\mathbb{Z}$ after identification.

Definition (Embedding $\mathbb{W}$ into $\mathbb{Z}$)

Definition 27.3.1 (Embedding $\mathbb{W}$ into $\mathbb{Z}$)**.** The *canonical embedding* $\iota : \mathbb{W} \to \mathbb{Z}$ is defined by

$$\iota(w) = [(w, 0)] \quad \text{(the integer represented by } w \text{ and } 0).$$

Remark (Standard quantified statement).

$$\iota : \mathbb{W} \to \mathbb{Z}, \qquad \iota(w) = [(w, 0)] \quad \text{(the integer represented by } w \text{ and } 0).$$

Remark (Interpretation). The map sends each element of $\mathbb{W}$ to its natural representative in $\mathbb{Z}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{W}$ may be identified with its image in $\mathbb{Z}$.

Remark (Dependencies).

- Formal-Difference Equivalence Is an Equivalence Relation
- Embedding of Number Systems

Theorem ($\mathbb{W}$ Embeds in $\mathbb{Z}$)

Theorem 27.3.2 ($\mathbb{W}$ Embeds in $\mathbb{Z}$)**.** *The canonical map $\iota : \mathbb{W} \to \mathbb{Z}$ is an embedding of ordered arithmetic systems: it is injective, preserves 0, 1, addition, and multiplication, and reflects the order.* *Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{W} \to \mathbb{Z} \text{ is injective, structure-preserving, and order-reflecting.}$$

Remark (Interpretation). Once this theorem is proved, the identification theorem of §27.1 applies, and $\mathbb{W}$ may be treated as the substructure $\iota(\mathbb{W}) \subseteq \mathbb{Z}$.

Remark (Dependencies).

- Formal-Difference Equivalence Is an Equivalence Relation
- Embedding $\mathbb{W}$ into $\mathbb{Z}$
- Embeddings Identify a System with Its Image

27.4 Embedding Z Into Q

Embedding $\mathbb{Z}$ into $\mathbb{Q}$

This section fixes the canonical embedding of $\mathbb{Z}$ into $\mathbb{Q}$ and records that it is injective, structure-preserving, and order-reflecting. The map is the bridge that lets elements of $\mathbb{Z}$ be regarded as elements of $\mathbb{Q}$ after identification.

Definition (Embedding $\mathbb{Z}$ into $\mathbb{Q}$)

Definition 27.4.1 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$)**.** The *canonical embedding* $\iota : \mathbb{Z} \to \mathbb{Q}$ is defined by

$$\iota(z) = [(z, 1)] \quad \text{(the rational represented by } z/1\text{)}.$$

Remark (Standard quantified statement).

$$\iota : \mathbb{Z} \to \mathbb{Q}, \qquad \iota(z) = [(z, 1)] \quad \text{(the rational represented by } z/1\text{)}.$$

Remark (Interpretation). The map sends each element of $\mathbb{Z}$ to its natural representative in $\mathbb{Q}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{Z}$ may be identified with its image in $\mathbb{Q}$.

Remark (Dependencies).

- Construction of $\mathbb{Q}$ as fraction classes of integers
- Embedding of Number Systems

Theorem ($\mathbb{Z}$ Embeds in $\mathbb{Q}$)

Theorem 27.4.2 ($\mathbb{Z}$ Embeds in $\mathbb{Q}$)**.** *The canonical map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order.* *Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{Z} \to \mathbb{Q} \text{ is injective, structure-preserving, and order-reflecting.}$$

Remark (Interpretation). Once this theorem is proved, the identification theorem of §27.1 applies, and $\mathbb{Z}$ may be treated as the substructure $\iota(\mathbb{Z}) \subseteq \mathbb{Q}$.

Remark (Dependencies).

- Construction of $\mathbb{Q}$ as fraction classes of integers
- Embedding $\mathbb{Z}$ into $\mathbb{Q}$
- Embeddings Identify a System with Its Image

27.5 Embedding Q Into R

Embedding $\mathbb{Q}$ into $\mathbb{R}$

This section fixes the canonical embedding of $\mathbb{Q}$ into $\mathbb{R}$ and records that it is injective, structure-preserving, and order-reflecting. The map is the bridge that lets elements of $\mathbb{Q}$ be regarded as elements of $\mathbb{R}$ after identification.

Definition (Embedding $\mathbb{Q}$ into $\mathbb{R}$)

Definition 27.5.1 (Embedding $\mathbb{Q}$ into $\mathbb{R}$)**.** The *canonical embedding* $\iota : \mathbb{Q} \to \mathbb{R}$ is defined by

$$\iota(q) = q^* \quad \text{(the real number determined by } q\text{)}.$$

Remark (Standard quantified statement).

$$\iota : \mathbb{Q} \to \mathbb{R}, \qquad \iota(q) = q^* \quad \text{(the real number determined by } q\text{)}.$$

Remark (Interpretation). The map sends each element of $\mathbb{Q}$ to its natural representative in $\mathbb{R}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{Q}$ may be identified with its image in $\mathbb{R}$.

Remark (Dependencies).

- Construction of $\mathbb{R}$ by Dedekind cuts
- Embedding of Number Systems

Theorem ($\mathbb{Q}$ Embeds in $\mathbb{R}$)

Theorem 27.5.2 ($\mathbb{Q}$ Embeds in $\mathbb{R}$)**.** *The canonical map $\iota : \mathbb{Q} \to \mathbb{R}$ is an embedding of ordered arithmetic systems: it is injective, preserves 0, 1, addition, and multiplication, and reflects the order. Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{Q} \to \mathbb{R} \text{ is injective, structure-preserving, and order-reflecting.}$$

Remark (Interpretation). Once this theorem is proved, the identification theorem of §27.1 applies, and $\mathbb{Q}$ may be treated as the substructure $\iota(\mathbb{Q}) \subseteq \mathbb{R}$.

Remark (Dependencies).

- Construction of $\mathbb{R}$ by Dedekind cuts
- Embedding $\mathbb{Q}$ into $\mathbb{R}$
- Embeddings Identify a System with Its Image

27.6 Compatibility of Arithmetic and Order

One Arithmetic Through the Chain

Each embedding preserves arithmetic and order individually. What remains is to confirm that the embeddings are mutually compatible: composing $\mathbb{N} \hookrightarrow \mathbb{W} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}$ agrees with the direct embedding of $\mathbb{N}$ into $\mathbb{R}$, so that "2" has one unambiguous meaning after all identifications.

Theorem (Compatibility of the Embedding Chain)

Theorem 27.6.1 (Compatibility of the Embedding Chain). *Let $\iota_1 : \mathbb{N} \to \mathbb{W}$, $\iota_2 : \mathbb{W} \to \mathbb{Z}$, $\iota_3 : \mathbb{Z} \to \mathbb{Q}$, and $\iota_4 : \mathbb{Q} \to \mathbb{R}$ be the canonical embeddings. For every $n \in \mathbb{N}$ the composite $(\iota_4 \circ \iota_3 \circ \iota_2 \circ \iota_1)(n)$ is the real number identified with n, and the composite preserves the arithmetic and order carried by the source system. Hence arithmetic computed in any system agrees with arithmetic on its image in $\mathbb{R}$. Go to proof.*

Remark (Standard quantified statement).

$$\forall n, m \in \mathbb{N} \; \big(\Phi(n{+}m) = \Phi(n){+}\Phi(m) \wedge \Phi(n{\cdot}m) = \Phi(n){\cdot}\Phi(m) \wedge (n < m \Leftrightarrow \Phi(n) < \Phi(m))\big), \quad \Phi = \iota_4{\circ}\iota_3{\circ}\iota_2{\circ}\iota$$

Remark (Interpretation). Compatibility is what makes the single symbol 2 well-defined across the whole chain: the answer to $2 + 2$ is the same whether computed in $\mathbb{N}$ or after embedding into $\mathbb{R}$. Without this coherence the inclusion notation would silently carry two different meanings for the same numeral.

Remark (Dependencies).

- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part
- $\mathbb{W}$ Embeds in $\mathbb{Z}$
- $\mathbb{Z}$ Embeds in $\mathbb{Q}$
- $\mathbb{Q}$ Embeds in $\mathbb{R}$

27.7 One Chain Many Structures

One Chain, Many Structures

The embeddings assemble into a single chain $\mathbb{N} \hookrightarrow \mathbb{W} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}$. Each arrow is injective, preserves the relevant arithmetic, and reflects order; each step also *adds* structure that the previous system lacked.

From $\mathbb{N}$ to $\mathbb{W}$ an additive identity 0 appears. From $\mathbb{W}$ to $\mathbb{Z}$ additive inverses appear, completing a commutative ring. From $\mathbb{Z}$ to $\mathbb{Q}$ multiplicative inverses of nonzero elements appear, completing an ordered field. From $\mathbb{Q}$ to $\mathbb{R}$ order-completeness appears: the gaps in the rational line are filled. The numerals carry through unchanged by compatibility, but the ambient structure grows at every step. This chain is the bridge from number construction to the analysis of the real line.

Remark (Exposition). Read as a ladder of structures: counting ($\mathbb{N}$) $\to$ counting with zero ($\mathbb{W}$) $\to$ signed counting / ring ($\mathbb{Z}$) $\to$ ordered field ($\mathbb{Q}$) $\to$ complete ordered field ($\mathbb{R}$). Only the last rung is order-complete, and that completeness is the property the structure-of-the-real-line chapter turns into metric topology.

Chapter 28

Summary: The Number Systems

28.1 Axiom Systems for $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$

Derived Properties — Quick Reference

Name	Statement	Proof sketch
Zero multiplication	$a \cdot 0 = 0$	$a \cdot 0 = a \cdot (0+0) = a \cdot 0 + a \cdot 0$; cancel $a \cdot 0$ via A4.
Negation rule	$(-a) \cdot b = -(ab)$	$ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$.
(-1) rule	$(-1) \cdot a = -a$	Special case: $b = a$, $-a = (-1)a$.
Double negation	$(-a)(-b) = ab$	$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.
No zero divisors	$ab = 0 \Rightarrow a = 0$ or $b = 0$	$\mathbb{Z}$ is an integral domain; proved via well-ordering + sign analysis.

28.2 Number Systems Summary

The Natural Numbers $\mathbb{N}$

One-Based Peano Axioms for $\mathbb{N}$

Let $\mathbb{N}$ be a set with distinguished element 1 and successor operation $n \mapsto n{+}{+}$.

P1 $1 \in \mathbb{N}$.

P2 $n \in \mathbb{N} \Rightarrow n{+}{+} \in \mathbb{N}$. *(successor closure)*

P3 $n{+}{+} \neq 1$ for all $n \in \mathbb{N}$. *(the first element has no predecessor)*

P4 $n{+}{+} = m{+}{+} \Rightarrow n = m$. *(successor is injective)*

P5 If $P(0)$ holds and $P(n) \Rightarrow P(n{+}{+})$, then $P(n)$ holds for all $n \in \mathbb{N}$. *(induction)*

Remark. P1–P4 rule out wrap-around, ceiling, and rogue elements. P5 eliminates anything unreachable from 1 by finite iteration of the successor. Addition, multiplication, and order are not axioms – they are *defined* recursively from the successor and then shown to satisfy familiar laws by induction.

Remark (Well-ordering and induction are equivalent over $\mathbb{N}$). The following are logically equivalent:

- Principle of Mathematical Induction (P5).
- Well-Ordering Principle: every nonempty $A \subseteq \mathbb{N}$ has a least element.
- Strong Induction: $(\forall k < n,\ P(k)) \Rightarrow P(n)$ for all n implies P holds everywhere.

One must assume at least one; the others follow. Well-ordering is the engine behind the Division Algorithm, Bézout's Identity, and unique prime factorization.

The Whole Numbers $\mathbb{W}$

$\mathbb{W}$ as $\mathbb{N}$ with an Adjoined Zero

The whole numbers are

$$\mathbb{W} = \mathbb{N} \cup \{0\}, \qquad 0 \notin \mathbb{N}.$$

The successor, addition, multiplication, and order on $\mathbb{W}$ extend the corresponding structures on $\mathbb{N}$:

- $S_{\mathbb{W}}(0) = 1$, and $S_{\mathbb{W}}$ agrees with $S_{\mathbb{N}}$ on $\mathbb{N}$;
- 0 is the additive identity;
- 0 is absorbing for multiplication;
- 0 is the least element of $\mathbb{W}$.

Remark (What $\mathbb{W}$ gains over $\mathbb{N}$, and what it lacks). **Gain:** an additive identity. This makes $(\mathbb{W}, +, 0)$ a commutative monoid and prepares the ordered-pair construction of the integers.

Lack: additive inverses for nonzero elements. Subtraction is still not an internal operation in general, so $\mathbb{W}$ is not a ring.

The Integers $\mathbb{Z}$

Axioms for $\mathbb{Z}$ — Ordered Commutative Ring

Addition

A1	$a + b = b + a$	*(commutativity)*
A2	$(a + b) + c = a + (b + c)$	*(associativity)*
A3	$\exists\, 0 \in \mathbb{Z}$ such that $a + 0 = a$	*(additive identity)*
A4	$\forall\, a,\ \exists\, -a \in \mathbb{Z}$ with $a + (-a) = 0$	*(additive inverses)*

Multiplication

M1	$a \cdot b = b \cdot a$	*(commutativity)*
M2	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$	*(associativity)*
M3	$\exists\, 1 \in \mathbb{Z},\ 1 \neq 0$, such that $a \cdot 1 = a$	*(multiplicative identity)*

Distributivity

D	$a \cdot (b + c) = a \cdot b + a \cdot c$	*(distributivity)*

Order

O1	$a \leq b$ or $b \leq a$	*(totality)*
O2	$a \leq b \wedge b \leq a \Rightarrow a = b$	*(antisymmetry)*
O3	$a \leq b \wedge b \leq c \Rightarrow a \leq c$	*(transitivity)*
O4	$b \leq c \Rightarrow a + b \leq a + c$	*(order + addition)*
O5	$a \geq 0 \wedge b \geq 0 \Rightarrow ab \geq 0$	*(order + multiplication)*

Remark (What $\mathbb{Z}$ gains over $\mathbb{W}$, and what it lacks). **Gain:** Axiom A4 – additive inverses. Subtraction is always defined in $\mathbb{Z}$; it is not in $\mathbb{W}$.

Lack: Multiplicative inverses. There is no $x \in \mathbb{Z}$ with $2x = 1$, so M4 (every nonzero element is invertible) fails. In the language of abstract algebra, $\mathbb{Z}$ is a *commutative ordered ring* but not a field. The passage to $\mathbb{Q}$ restores multiplicative inverses by formally adjoining fractions.

Derived Properties of $\mathbb{Z}$ (from the Ring Axioms)

Remark (Zero multiplication — the step most often looked up). The identity $a \cdot 0 = 0$ is not an axiom; it is derived. The proof uses only A3, A4, and D:

$$a \cdot 0 = a \cdot (0 + 0) \stackrel{\text{D}}{=} a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides (A4) gives $0 = a \cdot 0$, i.e. $a \cdot 0 = 0$. No order axioms, no well-ordering, no properties of $\mathbb{N}$ are needed — this holds in any ring.

Structural Comparison: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

Axiom Comparison Table

Group	**Axiom**	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$
Addition	A1 commutativity	✓	✓	✓	✓
	A2 associativity	✓	✓	✓	✓
	A3 identity (0)	×	✓	✓	✓
	A4 inverses $(-a)$	×	×	✓	✓
Multiplication	M1 commutativity	✓	✓	✓	✓
	M2 associativity	✓	✓	✓	✓
	M3 identity (1)	✓	✓	✓	✓
	M4 inverses $(1/a)$	×	×	×	✓
	D distributivity	✓	✓	✓	✓
Order	O1–O3 total order	✓	✓	✓	✓
	O4–O5 order + arithmetic	✓	✓	✓	✓
	Well-ordering	✓	✓	×	×
Completeness	Least upper bound	×	×	×	×
Structure		*positive arithmetic*	*comm. semiring*	*comm. ring*	*ordered field*

✓ = satisfied × = fails

Remark (The extension hierarchy). Each number system repairs exactly one deficiency of its predecessor:

$$\mathbb{N} \xrightarrow{+\,0} \mathbb{W} \xrightarrow{+\,\text{A4}} \mathbb{Z} \xrightarrow{+\,\text{M4}} \mathbb{Q} \xrightarrow{+\,\text{LUB}} \mathbb{R}.$$

$\mathbb{N} \to \mathbb{W}$: adjoin 0 so addition has an identity. $\mathbb{W} \to \mathbb{Z}$: adjoin additive inverses so subtraction is always defined. $\mathbb{Z} \to \mathbb{Q}$: adjoin multiplicative inverses so division by any nonzero element is always defined. $\mathbb{Q} \to \mathbb{R}$: adjoin least upper bounds so every bounded increasing sequence converges.

Note that $\mathbb{N}$ and $\mathbb{W}$ are the well-ordered systems in the chain. $\mathbb{Z}$ loses well-ordering the moment negative integers are introduced (the set of all negative integers has no least element).

From Extension to Embedding

The hierarchy above describes what each new system adds. The next chapter looks at a separate question: how the earlier systems are identified inside the later ones. That identification is not merely a matter of writing subset symbols. It is carried by structure-preserving embeddings that justify treating a number from an earlier construction as the corresponding number in a larger system.

Part III

Volume III: Classical Analysis

Bounds, Sequences, and Series

Chapter 29

Real Analysis

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 1
Real Analysis

29.1 Foundational Analysis Properties

29.1.1 Foundational Analysis Properties

Foundational Analysis Properties — Quick Reference

Concept	Goal	Proof mechanism / logical form
Well-definedness	Consistency	$x \sim y \Rightarrow F(x) = F(y)$
Existence	Feasibility	$\exists x\ P(x)$
Uniqueness	Singularity	$P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2$
Linearity	Decomposition	$L(f + g) = L(f) + L(g)$ and $L(cf) = cL(f)$
Uniformity	One choice works globally	$\forall \varepsilon > 0, \exists \delta > 0, \forall x, y,\ Q(\varepsilon, \delta, x, y)$
Equivalence	Interchangeable descriptions	$A \Rightarrow B$ and $B \Rightarrow A$
Characterisation	Usable test	$P(x) \iff C(x)$
Estimate	Control	$A \le B$, $\|A\| < \varepsilon$, or $d(A, B) < \varepsilon$
Boundedness	Two-sided control	$\exists l, u \in S\ \forall x \in A\ (l \le x \le u)$
Upper/lower control	One-sided domination	$\forall x \in A\ (x \le u)$ or $\forall x \in A\ (l \le x)$
Extremality	Realized optimality	$m \in A \wedge \mathrm{Bound}(m, A)$
Least/greatest character	Optimal bound	$\mathrm{Bound}(b, A) \wedge \forall c\,(\mathrm{Bound}(c, A) \Rightarrow b \preceq c)$
Monotonicity	Order preservation	Start with $x \le y$ and prove $f(x) \le f(y)$, or the reverse
Closure	Staying inside a class	$P(x) \wedge P(y) \Rightarrow P(x \star y)$
Approximation	Arbitrary precision	$\forall \varepsilon > 0\ \exists x\ d(x, a) < \varepsilon$

Many proofs in analysis are not isolated inventions. They prove recurring structural properties: a definition is independent of choices, an object exists, there is only one possible object, an operation respects addition and scalars, a single parameter works everywhere, or two definitions describe the same phenomenon. Naming the property first tells us what the proof must do.

Well-Definedness

Well-definedness appears whenever an object is defined using choices: representatives of equivalence classes, selected sequences, selected partitions, or selected approximations. The goal is to prove that different allowed choices give the same output.

$$x \sim x' \Rightarrow F(x) = F(x').$$

Remark 29.1.1 (Proof sketch).

1. Identify the choice built into the definition.
2. Take two allowed choices, usually x and x' with $x \sim x'$.
3. Compute or compare the two proposed outputs.

4. Show the outputs agree in the codomain.

Remark 29.1.2 (Crucial logic). The output must be independent of the chosen representative. If changing the representative changes the output, the proposed definition is not a function on the quotient or chosen domain.

Existence

Existence proves that at least one object satisfying a property is available:

$$\exists x \; P(x).$$

In analysis the object might be a limit, a supremum, a point in an interval, a subsequence, a bound, an index N, a radius δ, or a partition.

Remark 29.1.3 (Proof sketch).

1. Determine every requirement in the property P.
2. Produce a candidate explicitly, or invoke a theorem that guarantees one.
3. Verify the candidate belongs to the required domain.
4. Verify the candidate satisfies every part of P.

Remark 29.1.4 (Constructive and nonconstructive existence). A constructive proof names the object directly. A nonconstructive proof uses a theorem such as completeness, the intermediate value theorem, or Bolzano–Weierstrass. In both cases, the final burden is the same: the object must satisfy the stated property.

Uniqueness

Uniqueness proves that at most one object can satisfy a property:

$$P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2.$$

It does not by itself prove existence. It only proves that two successful candidates cannot be distinct.

Remark 29.1.5 (Proof sketch).

1. Assume x_1 and x_2 both satisfy P.
2. Use the defining property of x_1 and the defining property of x_2.
3. Derive two-sided comparison, zero distance, or identical defining data.
4. Conclude $x_1 = x_2$.

Remark 29.1.6 (Common analysis forms). For suprema, show $s \leq t$ and $t \leq s$. For limits, show $|L - M| < \varepsilon$ for every $\varepsilon > 0$, hence $L = M$. In metric spaces, show $d(x_1, x_2) = 0$.

Linearity

Linearity proves that an operation decomposes over addition and scalar multiplication:

$$L(f+g) = L(f) + L(g), \qquad L(cf) = cL(f).$$

The goal is not merely algebraic neatness. Linearity says that a complicated input can be analysed by splitting it into simpler pieces.

Remark 29.1.7 (Proof sketch).
1. Take arbitrary admissible inputs f, g and scalar c.
2. Expand $L(f+g)$ using the definition of L and the definition of addition.
3. Rearrange the expression until $L(f) + L(g)$ appears.
4. Repeat with $L(cf)$ and show it equals $cL(f)$.

Remark 29.1.8 (Typical analysis examples). Limits, derivatives, integrals, and many summation operators have linearity theorems. Each proof follows the same shape: expand the left side, split the expression, and identify the two pieces.

Uniformity

Uniformity means that one choice works for all relevant points at once. The prototype is not just uniform continuity, but any statement where a witness is chosen before the point is allowed to vary:

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y,\ Q(\varepsilon, \delta, x, y).$$

The key phrase is "one choice." The same N, δ, bound, or partition must work globally over the specified domain.

Remark 29.1.9 (Proof sketch).
1. Fix the tolerance or target requirement.
2. Choose a single witness that depends only on the allowed global data.
3. Let the point, pair of points, function, or index vary arbitrarily.
4. Verify that the same witness works in every permitted case.

Remark 29.1.10 (Contrast with pointwise statements). Pointwise statements allow the witness to depend on the point. Uniform statements do not. The order of quantifiers is the whole issue:

$$\forall x\, \exists \delta \quad \text{is pointwise, while} \quad \exists \delta\, \forall x \quad \text{is uniform.}$$

Equivalence

Equivalence proves that two statements are logically interchangeable:

$$A \iff B.$$

Such theorems are common because analysis often has several descriptions of the same phenomenon: epsilon-delta, sequential, order-theoretic, topological, or integral.

Remark 29.1.11 (Proof sketch).

1. Prove $A \Rightarrow B$.
2. Prove $B \Rightarrow A$.
3. Keep the assumptions for the two directions separate.
4. After both directions are proved, either description may be used.

Remark 29.1.12 (Typical analysis examples). Sequential and epsilon-delta definitions of continuity are equivalent. Riemann and Darboux integrability are equivalent under the appropriate hypotheses. A set is closed exactly when it contains the limits of all convergent sequences from the set.

Characterisation

A characterisation is an equivalence whose purpose is to turn a definition into a usable test:

$$P(x) \iff C(x).$$

The theorem does not merely say two statements agree. It supplies a criterion that can be used in later proofs.

Remark 29.1.13 (Proof sketch).

1. Identify the original definition.
2. Identify the proposed criterion.
3. Prove definition $\Rightarrow$ criterion.
4. Prove criterion $\Rightarrow$ definition.

Remark 29.1.14 (Typical analysis examples). The epsilon characterisation of the supremum turns least-upper-bound language into an approximation rule. Cauchy criteria turn convergence questions into internal distance estimates. Sequential criteria turn topological conditions into statements about sequences.

Estimates and Bounds

An estimate proves control:

$$A \leq B, \qquad |A| < \varepsilon, \qquad d(A, B) < \varepsilon.$$

The aim is not to compute the exact value. The aim is to put the target quantity inside a range that is strong enough for the theorem.

Remark 29.1.15 (Proof sketch).

1. Identify the quantity that must be controlled.
2. Replace it by a larger or simpler quantity using valid inequalities.
3. Use hypotheses to replace variable terms by constants or small errors.
4. End with the required bound.

Remark 29.1.16 (Typical analysis examples). Showing a sequence is bounded, proving an error term tends to zero, bounding a Riemann-sum error, or proving $|f(x) - L| < \varepsilon$ are all estimate problems. The proof is a chain of control.

Boundedness

Boundedness proves two-sided control. In an ordered setting, the usual form is that a set or sequence is bounded above and bounded below:

$$\exists l, u \in S \; \forall x \in A \; (l \le x \le u).$$

Remark 29.1.17 (Proof sketch).

1. Prove bounded above by finding a candidate upper bound.
2. Prove bounded below by finding a candidate lower bound.
3. Verify both candidates work for every element under consideration.
4. Conclude that the object is bounded.

Upper and Lower Control

Upper and lower control prove one-sided domination. The proof focuses on a candidate bound and checks that every element lies on the required side of it:

$$\forall x \in A \; (x \le u), \qquad \forall x \in A \; (l \le x).$$

Remark 29.1.18 (Proof sketch).

1. Name the candidate upper or lower bound.
2. Let an arbitrary element of the set be given.
3. Use the hypotheses or definition of the set to compare that element with the candidate.
4. Since the element was arbitrary, conclude the one-sided bound.

Extremality

Extremality proves that an optimal value is actually realized by the object being studied. A maximum is not merely an upper bound; it is an upper bound that belongs to the set. A minimum is the corresponding realized lower bound.

$$\mathrm{Maximum}(m, A) \iff (m \in A) \land \mathrm{UpperBound}(m, A),$$

$$\mathrm{Minimum}(m, A) \iff (m \in A) \land \mathrm{LowerBound}(m, A).$$

Remark 29.1.19 (Proof sketch).

1. Prove the candidate belongs to the set.
2. Prove the candidate is the appropriate one-sided bound.
3. Combine membership with the bound condition.
4. Conclude maximum or minimum.

Least and Greatest Character

Least and greatest character proves optimality among competitors. For a supremum, the candidate must first be an upper bound, and then it must be no larger than every other upper bound. Infimum proofs reverse the direction.

$$\text{Supremum}(s, A) \iff \text{UpperBound}(s, A) \wedge \forall u \in S \left(\text{UpperBound}(u, A) \Rightarrow s \leq u\right),$$

$$\text{Infimum}(i, A) \iff \text{LowerBound}(i, A) \wedge \forall l \in S \left(\text{LowerBound}(l, A) \Rightarrow l \leq i\right).$$

Remark 29.1.20 (Proof sketch).

1. Prove the candidate is a bound of the required kind.
2. Let an arbitrary competing bound be given.
3. Compare the candidate with that competing bound.
4. Conclude the candidate is the least upper bound or greatest lower bound.

Monotonicity

Monotonicity proves order preservation or order reversal:

$$x \leq y \Rightarrow f(x) \leq f(y), \qquad x \leq y \Rightarrow f(y) \leq f(x).$$

The proof begins with an arbitrary ordered pair and carries that order through the definitions.

Remark 29.1.21 (Proof sketch).

1. Take arbitrary points x and y with $x \leq y$.
2. Expand the definitions of the function or operation.
3. Use order rules and hypotheses to compare the outputs.
4. Conclude that the order is preserved or reversed.

Closure

Closure proves that a class is stable under an operation. One starts with objects already known to belong to the class, combines them, and then verifies that the result still satisfies the defining property.

$$P(x) \wedge P(y) \Rightarrow P(x \star y).$$

For closure under scalar operations, the same pattern becomes

$$P(x) \Rightarrow P(c \star x)$$

for every admissible scalar or parameter c.

Remark 29.1.22 (Proof sketch).

1. Take arbitrary objects from the class.
2. Form the object produced by the operation.
3. Check every condition in the definition of the class.
4. Conclude the result remains inside the class.

Approximation

Approximation proves arbitrary precision. The proof usually begins with a tolerance and constructs an object close enough to a target:

$$\forall \varepsilon > 0 \, \exists x \; d(x, a) < \varepsilon.$$

Remark 29.1.23 (Proof sketch).

1. Fix an arbitrary tolerance $\varepsilon > 0$.
2. Choose or construct an approximating object.
3. Estimate the error between the approximation and the target.
4. Show the error is below ε.

A Classification Pass

Before proving a theorem, classify its job:

- Does the statement depend on choices? Check well-definedness.
- Does it ask for an object? Prove existence.
- Does it say only one object can work? Prove uniqueness.
- Does it say an operation splits across sums and scalars? Prove linearity.
- Does one witness have to work for every point? Prove uniformity.
- Does it say two descriptions agree? Prove equivalence.
- Does it give a test for a definition? Prove a characterisation.
- Does it ask for smallness or control? Prove an estimate.
- Does it ask for two-sided order control? Prove boundedness.
- Does it ask for one-sided order control? Prove an upper or lower bound.
- Does it ask for a realized best element? Prove extremality.
- Does it ask for the best bound among all bounds? Prove least or greatest character.
- Does it preserve or reverse order? Prove monotonicity.
- Does it stay inside a class after an operation? Prove closure.
- Does it ask for arbitrary precision? Prove approximation.

Remark 29.1.24 (The point). The classification does not solve the proof, but it tells us what a completed proof must contain. For example, an existence proof must exhibit or obtain an object, while a uniqueness proof must compare two candidates. An equivalence proof must have two directions, while a uniformity proof must choose one witness before the arbitrary point is fixed.

29.1.2 Manipulating Inequalities and the Logic of Bounding

Inequality Toolkit — Quick Reference

Tool	Statement	Typical use
Triangle inequality	$\lvert a+b\rvert \le \lvert a\rvert + \lvert b\rvert$	Split a distance at an intermediate poin
Distance form	$\lvert a-b\rvert \le \lvert a-c\rvert + \lvert c-b\rvert$	Introduce pivot c to separate two estima
Reverse triangle ineq.	$\big\lvert \lvert a\rvert - \lvert b\rvert \big\rvert \le \lvert a-b\rvert$	Bound magnitudes using closeness
Absolute value interval	$\lvert x\rvert \le y \iff -y \le x \le y$	Convert $\lvert\cdot\rvert$ bound to two-sided inequali
Boundedness replacement	$\lvert b_n\rvert \le M$ (constant): replace $\lvert b_n\rvert$ by M	Eliminate variable factors in products
ε-budget split	$\lvert A\rvert < \frac{\varepsilon}{2}$ and $\lvert B\rvert < \frac{\varepsilon}{2} \Rightarrow \lvert A+B\rvert < \varepsilon$	Combine two independent estimates
$+1$ trick	Replace $\lvert L\rvert$ by $\lvert L\rvert + 1$ in denominators	Avoid division by zero when L may vani

The Fundamental Asymmetry Between Equality and Inequality

Equality is bilateral: $a = b$ licenses substitution in either direction and carries no information about magnitude. An inequality is directional: $a \le b$ locates a relative to b on the line, says nothing about the gap, and does not support backwards substitution.

This asymmetry is the source of a strategic shift. In an equality argument the goal is to show that two expressions are the same object. In a bounding argument the goal is to show that one expression is *controlled* by another. Control is inherently one-sided: a ceiling above $|f(x)|$ is useful regardless of how far below it $|f(x)|$ actually falls.

Remark 29.1.25 (Consequence for proof strategy). The question in analysis is rarely "what is $|a_n - L|$?" It is "is $|a_n - L|$ smaller than ε?" A valid crude bound that answers the second question is more useful than an exact answer to the first. Deliberate over-estimation — absent from algebra — is a required habit.

Remark 29.1.26 (Common error). Treating an inequality as if it were an equality by reversing direction. Given $a \le b$, negation gives $-a \ge -b$, not $-a \le -b$. Multiplication by a negative constant reverses direction. Failing to track sign changes when multiplying or negating is the most frequent algebraic error in ε-δ computations.

The Basic Rules of Inequality Arithmetic

Proposition 29.1.27 (Order arithmetic)**.** *Let $a, b, c, d \in \mathbb{R}$.*

(i) If $a \le b$ and $c \le d$, then $a + c \le b + d$.

(ii) If $a \le b$ and $c > 0$, then $ac \le bc$.

(iii) If $a \le b$ and $c < 0$, then $ac \ge bc$.

(iv) $|x| \le y$ (with $y \ge 0$) if and only if $-y \le x \le y$.

Go to proof.

Remark.

Remark 29.1.28 (Fully quantified form). (i) $\forall a, b, c, d \in \mathbb{R},\ (a \leq b) \wedge (c \leq d) \Rightarrow a + c \leq b + d$. (iv) $\forall x \in \mathbb{R},\ \forall y \geq 0,\ |x| \leq y\ \Leftrightarrow\ -y \leq x \leq y$.

Remark 29.1.29 (Inequalities add but do not subtract). Rule (i) licenses adding inequalities in the same direction. There is no subtraction rule: $a \leq b$ and $c \leq d$ yields no conclusion about $a - c$ versus $b - d$ in general. The direction of $a - c$ versus $b - d$ depends on the relative sizes of $(b - a)$ and $(d - c)$, which are not controlled.

Remark 29.1.30 (Consequence for absolute values). Rule (iv) is the bridge between $|\cdot|$ bounds and two-sided linear inequalities. It is invoked silently every time an ε-N argument converts $|a_n - L| < \varepsilon$ into positional information about a_n.

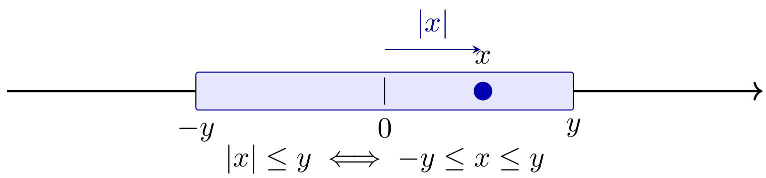


Figure 29.1: The absolute value characterisation as a symmetric interval. The shaded region is $[-y, y]$; the point x lies inside with $|x| \leq y$. The equivalence $|x| \leq y \Leftrightarrow -y \leq x \leq y$ (Proposition 29.1.27(iv)) converts an absolute value bound into a two-sided linear constraint.

The Triangle Inequality and the Pivot Move

Theorem (Triangle Inequality)

For all $a, b \in \mathbb{R},\ |a+b| \leq |a|+|b|$. Equivalently, for all $a, b, c \in \mathbb{R},\ |a-b| \leq |a-c|+|c-b|$.

Remark 29.1.31 (Fully quantified form). $\forall a, b \in \mathbb{R},\ |a + b| \leq |a| + |b|$.

Remark 29.1.32 (The pivot move). The distance form $|a-b| \leq |a-c|+|c-b|$ is the operative version in analysis. The element c is a *pivot*: an intermediate point that decomposes a hard distance into two pieces, each controllable by hypothesis. In the proof that $a_n + b_n \to L + M$, the pivot $c = L + b_n$ gives

$$|(a_n + b_n) - (L + M)| = |(a_n - L) + (b_n - M)| \leq |a_n - L| + |b_n - M|,$$

and the two pieces are controlled by the individual convergences.

Remark 29.1.33 (Reverse triangle inequality). $\big||a| - |b|\big| \leq |a - b|$. Proof: apply the standard inequality to $a = (a - b) + b$ to get $|a| - |b| \leq |a - b|$; symmetry gives the other side. Consequence: $|\cdot| : \mathbb{R} \to \mathbb{R}$ is Lipschitz with constant 1, so closeness of inputs implies closeness of magnitudes.

Remark 29.1.34 (Common error). Writing $|a| + |b| \leq |a + b|$. The triangle inequality always produces an *upper* bound on a sum, never a lower bound. The reverse $(|1| + |-1| = 2 > |1 + (-1)| = 0)$ is a one-line counterexample.

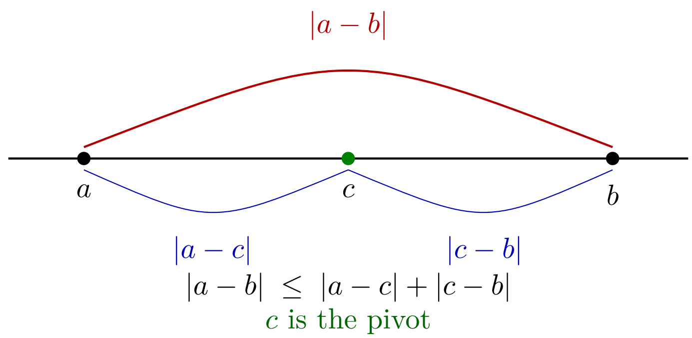


Figure 29.2: The triangle inequality in distance form. The direct distance $|a-b|$ (red) is bounded above by the sum of indirect distances $|a-c|$ and $|c-b|$ (blue) through pivot c. In analysis, c is chosen to split $|a-b|$ into two pieces each controlled by a separate hypothesis.

Bounding Is Not Solving

In algebra, a computation terminates when the exact value is found. In analysis, a bounding argument terminates when the expression is shown to fall within an acceptable range. These are structurally different goals.

Remark 29.1.35 (Deliberate crudeness is valid). To bound $|x^2-9x+1|$ on $[-1,5]$:

$$|x^2-9x+1| \leq |x|^2 + 9|x| + 1 \leq 25 + 45 + 1 = 71.$$

The actual supremum is far smaller than 71. That is irrelevant. The bound is valid, explicit, and was obtained cheaply. The test for a bound is not "is this sharp?" but "is this true and sufficient for the argument's purpose?"

Remark 29.1.36 (Consequence for proof reading). A reader who checks that each inequality is valid can trust the conclusion without caring whether the bounds are optimal. Optimality is irrelevant to logical correctness.

The ε-Budget Split

The prototype of all two-piece bounding arguments:

$$|A+B| \leq |A| + |B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The total tolerance ε is a budget allocated among the summands. With k summands, allocate ε/k each; with summands of unequal difficulty, split unevenly provided the shares sum to at most ε.

Remark 29.1.37 (The four-step procedure). Standard structure of an ε-N or ε-δ proof:

1. Apply the triangle inequality: express $|f-L|$ as a sum $|A_1| + \cdots + |A_k|$.
2. Assign budgets $\varepsilon_i > 0$ with $\sum_i \varepsilon_i = \varepsilon$.

3. For each piece, find N_i (or δ_i) so that $|A_i| < \varepsilon_i$ for all $n \geq N_i$ (or $|x-a| < \delta_i$).

4. Set $N = \max_i N_i$ (or $\delta = \min_i \delta_i$) to enforce all constraints simultaneously.

Step 4 reflects the logical structure of the conclusion: all conditions must hold at once, so the threshold is the most restrictive one.

Remark 29.1.38 (Common error). Taking $N = N_1$ and ignoring N_2. For $n \geq N_1$ the first piece is controlled; the second is not. Both pieces must be within budget simultaneously, which requires $n \geq \max\{N_1, N_2\}$.

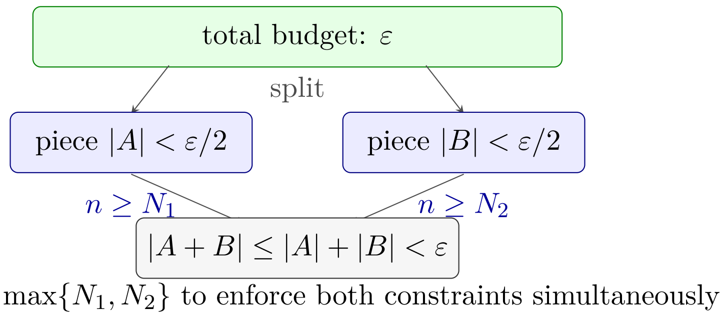


Figure 29.3: The ε-budget split. The total tolerance ε is distributed as shares $\varepsilon/2$ between two pieces. Each piece is controlled independently for $n \geq N_i$. Taking $N = \max\{N_1, N_2\}$ enforces both bounds simultaneously, giving $|A+B| < \varepsilon$ for all $n \geq N$. With k pieces, assign ε/k each and take $N = \max_i N_i$.

The Boundedness Replacement

Proposition 29.1.39 (Convergent sequences are bounded)**.** *If $(b_n) \to b$, then there exists $M > 0$ such that $|b_n| \leq M$ for all $n \in \mathbb{N}$.* *Go to proof.*

Remark.

Remark 29.1.40 (Fully quantified form). $\forall (b_n) \subset \mathbb{R},\ \lim_{n\to\infty} b_n = b \ \Rightarrow\ \exists M > 0,\ \forall n \in \mathbb{N},\ |b_n| \leq M.$

Remark 29.1.41 (The replacement move). Given $|b_n| \leq M$:

$$|b_n| \cdot |a_n - L| \leq M \cdot |a_n - L|.$$

The variable factor is gone. Choose N so that $|a_n - L| < \varepsilon/M$ for $n \geq N$; the whole product is then less than ε. The constant M need not be tight.

Remark 29.1.42 (Common error). Choosing $|a_n - L| < \varepsilon/|b_n|$. This is circular: N would depend on n, since $|b_n|$ changes with n. The threshold N must be a fixed number, independent of n. Replacing $|b_n|$ by the constant M breaks the circularity.

The Work-Backwards Principle

ε-N proofs are *stated* forwards but *discovered* backwards. The proof asserts: "Let $\varepsilon > 0$. Choose $N = \lceil C/\varepsilon \rceil$." The value N was found by scratchwork: "the expression simplifies to C/n; this is less than ε when $n > C/\varepsilon$; so set $N = \lceil C/\varepsilon \rceil$."

The scratchwork is a private computation that produces the *witness*. The proof states the witness and verifies it, working forwards.

Scratchwork (private):	Work backwards from $\lvert a_n - L\rvert < \varepsilon$ to find what N must satisfy.
Proof (public):	State N explicitly. Verify $n \geq N \Rightarrow \lvert a_n - L\rvert < \varepsilon$.

Remark 29.1.43 (Common error). Allowing scratchwork to appear in the proof: "we need $n > C/\varepsilon$, so choose n this large." The proof must assert a fixed N and verify the conclusion holds. A threshold stated as "n large enough" is an incomplete scratchwork note, not a proof.

Remark 29.1.44 (Connection to the geometric phase). The work-backwards process is the analytic counterpart of the geometric phase in limit proofs. The geometric phase identifies structure and chooses a pivot; the backwards computation determines the explicit threshold. The proof states both and verifies the chain forwards.

Chaining Inequalities

A standard proof move builds a chain $|f(x)| \leq A(x) \leq B \leq \varepsilon$, where each step uses a different tool: the triangle inequality, a domain bound, and the choice of δ.

Remark 29.1.45 (Replacing by a larger quantity). A quantity on the right-hand side of $\leq$ may be freely replaced by anything larger, since this only weakens the inequality. If $|x| \leq |x-1|+1$ and $|x-1| < \delta$, then $|x| < \delta + 1$. The replacement of $|x-1|$ by δ is valid because $|x-1|$ appears on the right. The same replacement on the *left* would be invalid.

Remark 29.1.46 (Common error). Including a step where the inequality points the wrong direction, or where a step holds only generically rather than pointwise. A chain is only as strong as its weakest link; every $\leq$ must be unconditionally valid.

The $+1$ Trick for Denominators

When $\varepsilon/|L|$ appears in an estimate and L may be zero, replace $|L|$ by $|L| + 1 \geq 1$:

$$\frac{\varepsilon}{|L|+1} \leq \varepsilon.$$

Remark 29.1.47 (Fully quantified form). $\forall L \in \mathbb{R},\ |L| + 1 \geq 1 > 0$. Hence $\varepsilon/(|L|+1)$ is well-defined and positive for all $\varepsilon > 0$ and all $L \in \mathbb{R}$.

Remark 29.1.48 (Generalisation). The $+1$ is an instance of a wider principle: any constant intended for a denominator must be guaranteed strictly positive. If the constant is a given quantity C that might be zero, replace it by $C+1$. The choice of 1 is conventional; any fixed positive number serves the same purpose.

Remark 29.1.49 (Common error). Using $|L|$ as a denominator without first verifying $L \neq 0$. The product limit proof $a_n b_n \to LM$ is the canonical location for this error: the bound $\varepsilon/(2|L|)$ fails when $L = 0$, and the $L = 0$ case requires separate treatment or the use of $|L| + 1$.

Summary: The Bounding Mindset

Algebra	**Analysis**
Find the exact value of x	Show $\lvert x - L\rvert < \varepsilon$
Equality: both sides are the same object	Inequality: one side controls the other
Work backwards and forwards symmetrically	Discover witness backwards; verify forwards
Exact answer is the goal	Any valid bound is sufficient
Fractions must be simplified	Crude over-estimates are acceptable
Multiply both sides freely	Track sign changes; direction is load-bearing

Remark 29.1.50 (The structure of a bounding proof). A bounding proof is a *chain of control*: inequalities connecting the target expression to ε, where each link uses one of the toolkit tools above. The proof is complete when the chain reaches ε. Every link must be unconditionally valid; every variable factor must be replaced by a constant before the chain can close.

29.1.3 Completeness as a Constructive Engine

Completeness Construction Toolkit — Quick Reference

Tool	**What it supplies**
ε-characterisation of sup	$\forall \varepsilon > 0,\ \exists x \in S,\ x > \sup S - \varepsilon$
Inductive selection	Monotone $(x_n) \subset S$ with $x_n \to \sup S$
Monotone Approximation	Such a sequence exists for any nonempty bounded S
LUB $\Leftrightarrow$ MCT	The two completeness statements are equivalent

The ε-Characterisation of Suprema and Infima

Proposition 29.1.51 (ε-characterisation of sup). *Let $S \subset \mathbb{R}$ be nonempty and bounded above, and let $s = \sup S$. Then for every $\varepsilon > 0$ there exists $x \in S$ with $x > s - \varepsilon$.* *Go to proof.*

Remark.

Remark 29.1.52 (Fully quantified form). $\forall \varepsilon > 0,\ \exists x \in S,\ s - \varepsilon < x \leq s$. Equivalently: any $M < s$ is not an upper bound for S.

Remark 29.1.53 (Proof strategy). If no such x existed for some $\varepsilon_0 > 0$, then $s - \varepsilon_0$ would be an upper bound for S smaller than s, contradicting $s = \sup S$.

Remark 29.1.54 (Operative role). The ε-characterisation is the *existential engine* of constructive completeness arguments. At each inductive step it is invoked with a specific ε value, discharging the universal quantifier and yielding a concrete existence claim. Existential instantiation then names a witness — call it x_{n+1} — and the rest of the argument uses only the single property $x_{n+1} > s - \varepsilon$. The witness is underspecified; the argument makes no commitment about which element of S is chosen, and no further properties of x_{n+1} are used.

Remark 29.1.55 (Symmetric form for inf). For $t = \inf S$: for every $\varepsilon > 0$ there exists $y \in S$ with $y < t + \varepsilon$. The proof is symmetric: any $M > t$ is not a lower bound.

Inductive Selection at Bounds

Theorem (Inductive Selection at sup)

Let $S \subset \mathbb{R}$ be nonempty and bounded above, and let $s = \sup S$. Then there exists a strictly increasing sequence $(x_n)_{n=1}^{\infty} \subset S$ with $x_n \to s$.

Remark 29.1.56 (Fully quantified form). $\exists (x_n) \subset S$: $\forall n,\ x_n < x_{n+1};\ \forall n,\ s - \frac{1}{n} < x_n \le s$; $\lim_{n\to\infty} x_n = s$.

Remark 29.1.57 (Why no algebraic formula is possible in general). The set S may have large gaps; any fixed formula might land outside S entirely. The construction must be inductive, selecting from S at each stage using the ε-characterisation rather than evaluating a closed expression.

Remark 29.1.58 (Proof strategy — the max device). The inductive step must simultaneously satisfy two lower bounds: $x_{n+1} > x_n$ (monotonicity) and $x_{n+1} > s - \frac{1}{n+1}$ (proximity). A single invocation of the existential can impose only one threshold. Setting

$$M_n := \max\Big(x_n,\ s - \tfrac{1}{n+1}\Big)$$

collapses both requirements into one: $x_{n+1} > M_n$ delivers both $x_{n+1} > x_n$ and $x_{n+1} > s - \frac{1}{n+1}$ simultaneously. Since $M_n < s$ (both entries are strictly below s whenever $s \notin S$, or the constant sequence $x_n = s$ handles the case $s \in S$), the ε-characterisation with $\varepsilon = s - M_n > 0$ supplies the required witness.

Base case. Set $\varepsilon = 1$. The ε-characterisation gives $x_1 \in S$ with $x_1 > s - 1$; since $x_1 \in S$, also $x_1 \le s$.

Inductive step. Given x_n with $s - \frac{1}{n} < x_n \le s$, form M_n as above and instantiate the ε-characterisation with $\varepsilon = s - M_n > 0$ to obtain $x_{n+1} \in S$ with $x_{n+1} > M_n$. Then:

$$\begin{aligned} x_{n+1} > M_n \ge x_n \quad &\Rightarrow \quad x_{n+1} > x_n, \\ x_{n+1} > M_n \ge s - \tfrac{1}{n+1} \quad &\Rightarrow \quad s - \tfrac{1}{n+1} < x_{n+1} \le s. \end{aligned}$$

Convergence. $0 \le s - x_n < \frac{1}{n}$; the squeeze theorem and the Archimedean Property give $x_n \to s$.

Remark 29.1.59 (What the construction requires). Three ingredients: (i) completeness of $\mathbb{R}$ ($\sup S$ exists); (ii) the ε-characterisation (Proposition 29.1.51), supplying the witness at each step; (iii) the Archimedean Property (Theorem 31.1.4), forcing $|s - x_n| \le \frac{1}{n} \to 0$. The internal structure of S — its cardinality, openness, density — plays no role.

Remark 29.1.60 (Common error). Instantiating the existential with only the proximity threshold $s - \frac{1}{n+1}$ and ignoring x_n. The resulting witness satisfies proximity but not monotonicity: one might obtain $x_1 = 0.9$, $x_2 = 0.4$, $x_3 = 0.85$, each close to s but not increasing. The max is not a convenience; it is the minimal device that encodes both requirements into one threshold.

Remark 29.1.61 (Infimum case). Replace max by min, $s - \frac{1}{n}$ by $t + \frac{1}{n}$, and reverse all strict inequalities. The resulting sequence $(y_n) \subset S$ is strictly decreasing with $y_n \to t = \inf S$.

Monotone Approximation of Bounds

Proposition 29.1.62 (Monotone Approximation of Bounds). *Let $S \subset \mathbb{R}$ be nonempty and bounded. Then there exist monotone sequences $(x_n), (y_n) \subset S$ with $x_n \to \sup S$ and $y_n \to \inf S$. Go to proof.*

Remark.

Remark 29.1.63 (Fully quantified form). $\forall S \neq \emptyset$ bounded: $\exists (x_n) \subset S$, $x_n \uparrow \sup S$; $\exists (y_n) \subset S$, $y_n \downarrow \inf S$.

Remark 29.1.64 (Proof strategy). The strictly increasing sequence from Section 29.1.3 is already monotone and converges to $\sup S$. Taking $X_n = \max\{x_1, \ldots, x_n\}$ makes it explicitly non-decreasing (useful when the inductive step does not guarantee strict increase). The infimum case is symmetric.

Remark 29.1.65 (Consequence). Every supremum is the limit of a sequence from the set. The sequential characterisation of sup and inf — that they are realised as limits of internal sequences — follows from this construction, making Proposition 29.1.62 the canonical proof of that characterisation.

Remark 29.1.66 (Forward connections). The inductive selection pattern reappears in:

- **Monotone Convergence Theorem** — every increasing bounded sequence converges to the supremum of its range; this construction exhibits that the supremum is always realised as such a limit.
- **Nested Interval Property** — the left and right endpoints of nested intervals form monotone sequences converging to a common point by exactly this mechanism.
- **Existence of square roots** — the standard proof sets $s = \sup\{x \in \mathbb{R} : x^2 < 2\}$ and extracts an approximating sequence from that set by inductive selection.

LUB $\iff$ MCT: The Completeness Equivalence

Definition (Least Upper Bound Property)

$\mathbb{R}$ has the **least upper bound property** if every nonempty set $A \subset \mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Theorem (LUB $\iff$ MCT)

The following are equivalent over the Archimedean ordered field axioms:

(i) $\mathbb{R}$ has the least upper bound property (LUB).

(ii) Every bounded monotone sequence of real numbers converges (MCT).

Remark 29.1.67 (Fully quantified form). (i) $\forall A \subset \mathbb{R},\ (A \neq \emptyset \wedge \exists M,\ \forall a \in A,\ a \leq M) \Rightarrow \exists \sup A \in \mathbb{R}$.

(ii) $\forall (x_n) \subset \mathbb{R},\ (\forall n,\ x_n \leq x_{n+1}) \wedge (\exists M,\ \forall n,\ x_n \leq M) \Rightarrow \exists L \in \mathbb{R},\ \lim_{n\to\infty} x_n = L$.

Remark 29.1.68 (Proof strategy — LUB $\Rightarrow$ MCT). Let (x_n) be increasing and bounded. The image set $A = \{x_n : n \in \mathbb{N}\}$ is bounded above, so LUB gives $\alpha = \sup A$. The ε-characterisation produces N with $x_N > \alpha - \varepsilon$. Monotonicity promotes this to a tail statement: $\alpha - \varepsilon < x_n \leq \alpha$ for all $n \geq N$. Hence $x_n \to \alpha$.

The quantifier structure is:

$$\alpha = \sup A \ \Rightarrow\ (\forall \varepsilon > 0)\ \exists N\ (x_N > \alpha - \varepsilon) \ \xrightarrow{\text{monotone}}\ (\forall \varepsilon > 0)\ \exists N\ (\forall n \geq N)\ (|x_n - \alpha| < \varepsilon).$$

Remark 29.1.69 (Proof strategy — MCT $\Rightarrow$ LUB). Given $A \neq \emptyset$ and bounded above by u_1, pick $l_1 \in A$. Build a bisection bracket: at each stage, let $m_n = (l_n + u_n)/2$; if m_n is an upper bound for A set $u_{n+1} = m_n$, otherwise set $l_{n+1} = m_n$ (and pick a witnessing $a \in A$ with $a > l_{n+1}$). The sequences (l_n) and (u_n) are monotone and bounded, so MCT gives limits $l_n \to s$ and $u_n \to s$ (since $u_n - l_n = (u_1 - l_1)/2^n \to 0$). The invariants — "$u_n$ is an upper bound for A" and "l_n is not" — force $s = \sup A$.

The construction maintains:

- a *universal* invariant: $(\forall a \in A)(a \leq u_n)$,
- a *failure* invariant: $(\exists a \in A)(a > l_n)$.

These two invariants, together with $u_n - l_n \to 0$, pin s as the least upper bound.

Remark 29.1.70 (Why this matters structurally). The chain LUB $\Leftrightarrow$ MCT $\Leftrightarrow$ NIP $\Leftrightarrow$ BW $\Leftrightarrow$ CC (Cauchy Criterion) is the backbone of the equivalence cluster at the heart of real analysis. All five statements assert that $\mathbb{R}$ has no gaps, each in its own language. The inductive selection technique — building sequences from sets using the ε-characterisation — is the constructive face of all five equivalences.

Remark 29.1.71 (Downstream reuse). The quantifier pattern

$$(\forall \varepsilon > 0)(\exists \text{witness}) \ \Rightarrow\ (\exists \text{sequence with controlled tail})$$

reappears in Bolzano–Weierstrass (extracting convergent subsequences), $\limsup/\liminf$ (tail suprema as a decreasing sequence), compactness (finite subcovers as finite witnesses), and approximation theorems throughout analysis.

29.1.4 Walking a Sequence with a Predicate

Predicate-Walking Toolkit — Quick Reference

Theorem	Predicate $P(n)$	Tool
Monotone Subseq.	n is a peak of (x_n)	Dichotomy on peak count
Bolzano–Weierstrass	interval has ∞ many terms	Bisection + pigeonhole
Arzelà–Ascoli	converges at $d_1, \dots, d_m$	Diagonal extraction

The Infinitely-Often / Eventually Dichotomy

Definition (Infinitely Often / Eventually)

Let (x_n) be a sequence and P a predicate on $\mathbb{N}$. P holds **infinitely often** if $\{n \in \mathbb{N} : P(n)\}$ is infinite; P holds **eventually** if there exists $N \in \mathbb{N}$ such that $P(n)$ holds for all $n \geq N$.

Remark 29.1.72 (Fully quantified form). P infinitely often: $\forall N \in \mathbb{N},\ \exists n \geq N,\ P(n)$.
P eventually: $\exists N \in \mathbb{N},\ \forall n \geq N,\ P(n)$.

Proposition 29.1.73 (The dichotomy)**.** *For any predicate P on $\mathbb{N}$, exactly one of the following holds:*

(i) P holds infinitely often.

(ii) $\neg P$ holds eventually.

Go to proof.

Remark.

Remark 29.1.74 (Fully quantified form). $\big(\forall N,\ \exists n \geq N,\ P(n)\big) \ \veebar\ \big(\exists N,\ \forall n \geq N,\ \neg P(n)\big)$. The two cases are exhaustive because $\mathbb{N}$ is infinite: if P fails for all but finitely many n, those indices have a maximum N, and $\neg P(n)$ holds for all $n > N$.

Remark 29.1.75 (Relation to Infinite Pigeonhole). The index set $\mathbb{N}$ is partitioned into $\{n : P(n)\}$ and $\{n : \neg P(n)\}$; an infinite set cannot have both parts finite. The "infinitely often" case provides an inexhaustible reservoir of indices from which to select subsequence terms inductively.

The Abstract Extraction Template

Remark 29.1.76 (Four-step pattern). Every predicate-walking argument follows the same skeleton:

1. *Define the predicate.* Identify the property $P(n)$ to propagate: peak, infinite-occupancy, pointwise convergence, etc.

2. *Apply the dichotomy.* Either P holds infinitely often, or $\neg P$ holds eventually. Handle each case.

3. *Extract the subsequence.* In the "infinitely often" case, enumerate indices satisfying P in increasing order: $n_1 < n_2 < \cdots$. In the "eventually" case, work in the $\neg P$-tail and build inductively, using $\neg P$ as the fuel.

4. *Verify the property.* Show (x_{n_k}) has the desired property, using the predicate and any additional hypotheses (boundedness, MCT, NIP, equicontinuity).

The critical structural requirement: at each inductive stage, one must find $n_{k+1} > n_k$ satisfying the relevant condition. The "infinitely often" guarantee ensures the reservoir is never exhausted.

The Peak Argument: Monotone Subsequence Theorem

Definition (Peak Index)

An index $m \in \mathbb{N}$ is a **peak** of (x_n) if $x_m \geq x_n$ for all $n \geq m$.

Remark 29.1.77 (Fully quantified form). m is a peak: $\forall n \in \mathbb{N},\ n \geq m \Rightarrow x_m \geq x_n$.

Theorem (Monotone Subsequence)

Every sequence (x_n) in $\mathbb{R}$ has a monotone subsequence.

Remark 29.1.78 (Fully quantified form). $\forall (x_n) \subset \mathbb{R},\ \exists (n_k)$ strictly increasing, (x_{n_k}) is monotone

Remark 29.1.79 (Proof strategy — peak dichotomy). Apply the dichotomy to $P(m) =$ "m is a peak."

Case 1: infinitely many peaks. Let $n_1 < n_2 < \cdots$ enumerate the peaks. Since n_k is a peak and $n_{k+1} > n_k$: $x_{n_k} \geq x_{n_{k+1}}$. The subsequence is decreasing.

Case 2: finitely many peaks. Let N be an index beyond all peaks. No $n \geq N$ is a peak, so for each such n there exists $m > n$ with $x_m > x_n$. Set $n_1 = N$. Since n_1 is not a peak, $\exists n_2 > n_1$ with $x_{n_2} > x_{n_1}$. Inductively, this produces a strictly increasing subsequence.

The no-peak condition $\neg P(n_k)$ is the fuel for each inductive step: it is precisely the statement "there exists a later term exceeding x_{n_k}," which supplies n_{k+1}.

Remark 29.1.80 (Consequence: Bolzano–Weierstrass). Every bounded sequence has a bounded monotone subsequence. By MCT, that subsequence converges. Bolzano–Weierstrass follows immediately.

Remark 29.1.81 (Common error). Assuming Case 2 produces an increasing subsequence "by default." It does, but only because $\neg P(n)$ supplies a strictly larger term. If the predicate were defined differently, $\neg P$ might provide no useful structure. The conclusion depends on the specific content of $\neg P$, not the abstract case structure.

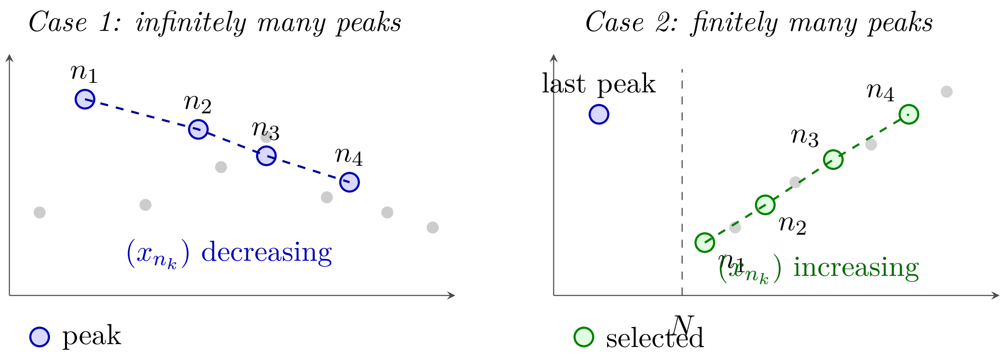


Figure 29.4: The two cases of the Monotone Subsequence Theorem (Section 29.1.4). *Left:* Infinitely many peaks (blue) yield a decreasing subsequence (dashed blue). *Right:* After the last peak, every $n \geq N$ is a non-peak, so a later term always exceeds the current one; this fuels the inductive construction of an increasing subsequence (dashed green).

The Infinite-Occupancy Predicate: Bolzano–Weierstrass via Bisection

Proposition 29.1.82 (Bolzano–Weierstrass via bisection)**.** *Every bounded sequence in* $\mathbb{R}$ *has a convergent subsequence. Go to proof.*

Remark.

Remark 29.1.83 (Proof strategy — infinite-occupancy predicate)*.* The predicate at each stage: $P(I) =$ "interval I contains infinitely many terms of (x_n)." Since $(x_n) \subseteq [a_0, b_0]$ for some bounded interval, $P([a_0, b_0])$ holds.

Inductive step. Given $I_k = [a_k, b_k]$ with $P(I_k)$, bisect at $m_k = (a_k + b_k)/2$. The two halves cover I_k; at least one contains infinitely many terms. Set I_{k+1} to be that half. Pick $x_{n_{k+1}} \in I_{k+1}$ with $n_{k+1} > n_k$. The predicate $P(I_{k+1})$ holds by construction.

Convergence. The intervals have $|I_k| = (b_0 - a_0)/2^k \to 0$. By NIP there exists a unique $L \in \bigcap I_k$. Since $a_k \leq x_{n_k} \leq b_k$ and both endpoints converge to L, the squeeze theorem gives $x_{n_k} \to L$.

Remark 29.1.84 (Comparison with the peak proof)*.* The peak proof is combinatorial: it operates on the values of the sequence directly and produces a monotone subsequence without requiring boundedness. Convergence follows from MCT.

The bisection proof is geometric: it operates on intervals containing the sequence's range and produces a subsequence squeezed to a limit directly. Boundedness is required from the outset to produce the initial interval.

Both proofs use the same dichotomy, but in the peak proof it is applied to an index predicate; in the bisection proof, to an interval predicate.

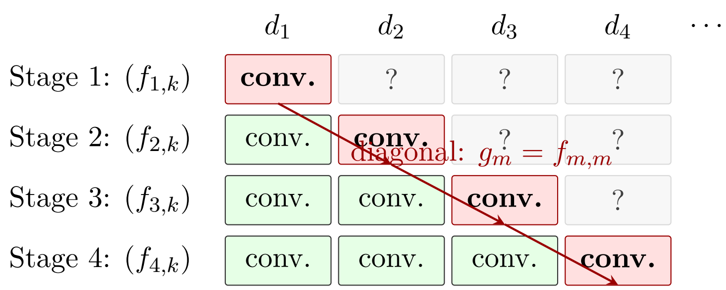
Diagonal Extraction: Arzelà–Ascoli

Remark 29.1.85 (Diagonal extraction pattern). The diagonal argument generalises predicate-walking to sequences of functions. Let (f_n) be bounded and equicontinuous in $C^0([a,b],\mathbb{R})$, and let $D = \{d_1, d_2, \dots\} = \mathbb{Q} \cap [a,b]$.

Stage 1. $(f_n(d_1))$ is bounded; extract by BW a subsequence $(f_{1,k})$ converging at d_1.
Stage 2. $(f_{1,k}(d_2))$ is bounded; extract a sub-subsequence $(f_{2,k})$ converging at d_2. As a subsequence of $(f_{1,k})$, it still converges at d_1.
Stage m. Extract $(f_{m,k})$ as a subsequence of $(f_{m-1,k})$ converging at d_m. By induction, $(f_{m,k})$ converges at $d_1, \dots, d_m$.
Diagonal step. Define $g_m = f_{m,m}$. For any fixed j and $m \geq j$, the term $g_m = f_{m,m}$ is a term of $(f_{j,k})$ with index $k = m \geq j$. Hence $g_m(d_j) \to y_j$. The sequence (g_m) converges pointwise on all of D.

Equicontinuity then propagates pointwise convergence on D to uniform convergence on $[a,b]$ via an $\varepsilon/3$ argument.

Remark 29.1.86 (Why the diagonal works). At stage m, the predicate "converges at $d_1, \dots, d_m$" holds for all terms of $(f_{m,k})$. The diagonal term $g_m = f_{m,m}$ is a tail term of every earlier row $f_{j,k}$ for $j \leq m$, so it satisfies the predicate for all $j \leq m$ simultaneously. Sending $m \to \infty$ gives convergence at every d_j.



(g_m) converges pointwise on all of $D = \{d_1, d_2, \dots\}$

Figure 29.5: Diagonal extraction in the Arzelà–Ascoli argument. Each row $(f_{m,k})$ converges at $d_1, \dots, d_m$ (green) and is a subsequence of the row above. The diagonal $g_m = f_{m,m}$ (red) is a tail of every row, so it converges at every d_j. Equicontinuity then extends pointwise convergence on the dense set D to uniform convergence on $[a,b]$.

Limitations of the Technique

Remark 29.1.87 (The predicate must be decidable at each stage). At each inductive step, one must determine which branch of the dichotomy applies. If the predicate requires knowledge of the limit — a quantity not yet known — the construction cannot proceed. Predicate-walking requires a finitary, combinatorial, or topological condition on the terms already seen.

Remark 29.1.88 (Multiple simultaneous predicates may exhaust the reservoir). The conjunction $P \wedge Q$ can hold only finitely often even when P and Q each hold infinitely often. The diagonal argument avoids this by handling one predicate at a time, composing via the diagonal.

Remark 29.1.89 (Subsequences, not sequences). Predicate-walking yields existence of a subsequence with the desired property, not a conclusion about the full sequence. The Monotone Subsequence Theorem does not assert (x_n) is monotone, or that the extracted subsequence is unique.

Remark 29.1.90 (Convergence requires a separate argument). The extracted subsequence has structural properties but not yet a limit. Converting structure into convergence requires a separate theorem: monotone subsequence → MCT (requires boundedness); bisection subsequence → NIP and squeeze; pointwise convergence on D → equicontinuity and $\varepsilon/3$.

Remark 29.1.91 (Quantitative control is absent). Predicate-walking proofs are pure existence arguments. They specify neither the rate of convergence nor which indices are selected. Explicit rates or constructive witnesses require a different approach.

Remark 29.1.92 (Countability is essential for the diagonal). The Arzelà–Ascoli diagonal requires a countable dense D so that points can be enumerated and handled one at a time. In non-separable metric spaces (those without a countable dense subset) the argument breaks down.

The Unifying Principle

Remark 29.1.93 (Infinite Pigeonhole as the common core). Every predicate-walking argument rests on the fact that an infinite set cannot be partitioned into two finite parts. Applied to $\mathbb{N}$: if P holds only finitely often, the P-indices have a maximum, and $\neg P$ holds from that point on. The method reduces a complex existence question to a sequence of binary choices, each justified by the Infinite Pigeonhole Principle. The construction is inductive, automatic, and never runs dry, provided the predicate is well-chosen and the reservoir is genuinely infinite at each stage.

Remark 29.1.94 (Dependency summary).

1. Monotone Subsequence Theorem — peak dichotomy alone; no completeness.
2. Bolzano–Weierstrass via MCT — Monotone Subseq. + MCT (requires completeness via AoC).
3. Bolzano–Weierstrass via bisection — NIP + squeeze (requires completeness via NIP $\equiv$ AoC).
4. Arzelà–Ascoli — BW at each diagonal stage + equicontinuity.

The Monotone Subsequence Theorem is the only result in the cluster requiring no completeness. All others depend on completeness via MCT or NIP.

29.1.5 Interval Bisection and Nested Intervals

Bisection and NIP Toolkit — Quick Reference

Label	Statement	Proof method
NIP	$\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ for nested closed bounded intervals	Supremum of left endpoints
IVT	f continuous, $f(a) < L < f(b) \Rightarrow \exists\, c \in (a,b),\ f(c) = L$	Bisection + NIP
UNC	$\mathbb{R}$ is uncountable	Nested interval diagonalisation

The Nested Interval Property

Theorem (Nested Interval Property)

For each $n \in \mathbb{N}$, let $I_n = [a_n, b_n]$ be a closed bounded interval with $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Remark 29.1.95 (Fully quantified form). $\forall n \in \mathbb{N},\ I_n = [a_n, b_n] \neq \emptyset,\ [a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \implies \exists\, x \in \mathbb{R},\ \forall n,\ x \in I_n$.

Remark 29.1.96 (Proof strategy). Let $A = \{a_n : n \in \mathbb{N}\}$. Nestedness implies every b_n is an upper bound for A. By AoC, $x = \sup A$ exists. Then $a_n \leq x \leq b_n$ for all n, so $x \in I_n$ for all n.

Remark 29.1.97 (Closedness is essential). The open intervals $(0, 1/n)$ are nested but $\bigcap (0, 1/n) = \emptyset$. Closedness ensures the endpoints can serve as witnesses in the AoC argument.

Remark 29.1.98 (Boundedness is essential). The unbounded closed intervals $[n, \infty)$ are nested but their intersection is empty. Both hypotheses are necessary.

Remark 29.1.99 (Singleton intersection). If additionally $b_n - a_n \to 0$, the intersection is a singleton $\{x\}$, since $a_n \leq x \leq b_n$ and the squeeze theorem forces $a_n \to x$ and $b_n \to x$. This strengthened form is used in most applications.

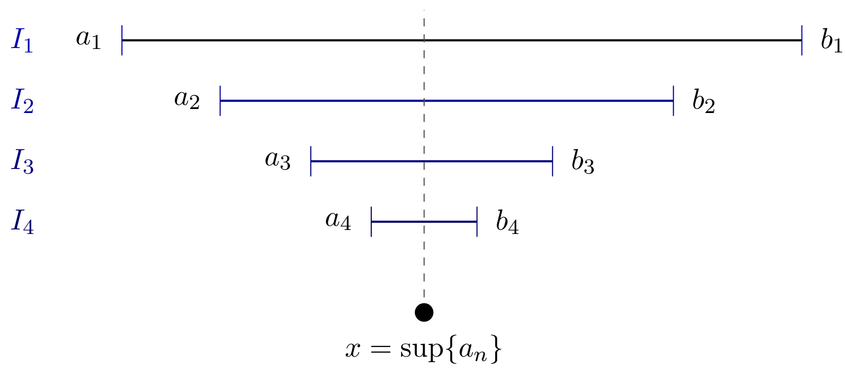


Figure 29.6: The Nested Interval Property (Section 29.1.5). Left endpoints a_n increase; right endpoints b_n decrease; each b_n is an upper bound for $\{a_n\}$. By the Axiom of Completeness, $x = \sup\{a_n\}$ exists and satisfies $a_n \leq x \leq b_n$ for all n, so $x \in \bigcap I_n$.

Remark 29.1.100 (Equivalence with completeness). NIP is equivalent to AoC over the Archimedean ordered field axioms. In particular, NIP fails over $\mathbb{Q}$: the intervals $[1, \sqrt{2}\,] \cap \mathbb{Q}$ shrink to a gap that $\mathbb{Q}$ cannot fill.

The Bisection Technique

Definition (Bisection Sequence)

Let $I_0 = [a_0, b_0]$ carry a property P. The **bisection sequence** (I_n) is defined inductively: given $I_n = [a_n, b_n]$, let $m_n = (a_n + b_n)/2$ and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } [a_n, m_n] \text{ inherits } P, \\ [m_n, b_n] & \text{otherwise.} \end{cases}$$

Remark 29.1.101 (Fully quantified form). $I_0 = [a_0, b_0]$; $\forall n,\ I_{n+1} \subseteq I_n$; $|I_n| = (b_0 - a_0)/2^n$.

Remark 29.1.102 (Length control). Bisection gives automatic length control: $|I_n| = (b_0 - a_0)/2^n \to 0$. No separate argument for the length condition of NIP is required.

Remark 29.1.103 (Abstract template). Every bisection proof has the same five-step shape:

1. *Initialize.* Identify P and starting interval I_0.
2. *Bisect.* Check which half inherits P.
3. *Extract.* Apply NIP to obtain $x \in \bigcap I_n$; note $a_n \to x$ and $b_n \to x$.
4. *Conclude.* Pass P through the limit, using continuity, MCT, or squeeze as appropriate.
5. *Verify hypotheses.* Confirm intervals are closed and nested; length control is free from bisection.

Remark 29.1.104 (Connection to predicate-walking). The bisection technique is a special case of predicate-walking: the predicate $P(I) =$ "interval I inherits the relevant property" is propagated at each stage by bisecting and keeping the half that satisfies P. The infinite-pigeonhole dichotomy appears in the Bolzano–Weierstrass proof (Proposition 29.1.82) but not in the IVT proof, where neither half is discarded based on cardinality — instead the sign-test determines the choice deterministically.

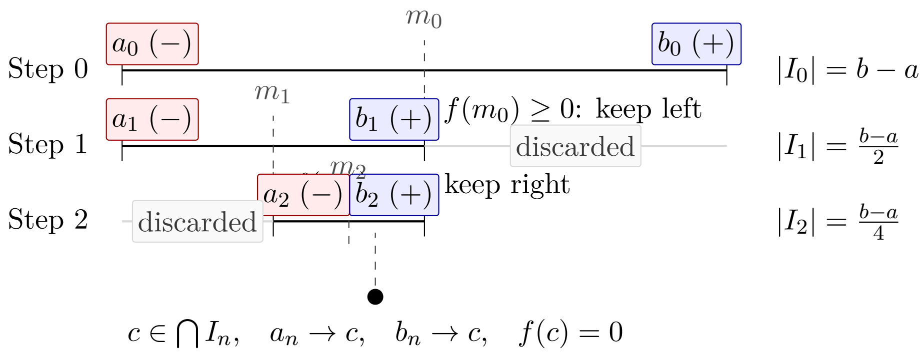


Figure 29.7: Three steps of the bisection proof of the Intermediate Value Theorem (Proposition 29.1.105) for $f(a) < 0 < f(b)$. At each step the half preserving a sign change is retained and the other discarded; lengths halve as $(b-a)/2^n \to 0$. By NIP, $c \in \bigcap I_n$ exists uniquely; continuity forces $f(c) = 0$.

The Intermediate Value Theorem via Bisection

Proposition 29.1.105 (Intermediate Value Theorem)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous. If $L \in \mathbb{R}$ satisfies $f(a) < L < f(b)$ or $f(a) > L > f(b)$, then there exists $c \in (a,b)$ with $f(c) = L$. Go to proof.*

Remark.

Remark 29.1.106 (Proof strategy — bisection approach)*.* Reduce to $L = 0$, $f(a) < 0 < f(b)$ (replace f by $f - L$).

Property to propagate. $f(a_n) < 0$ and $f(b_n) \geq 0$ — a sign change across the interval.
Branching rule. Let $m_n = (a_n + b_n)/2$: if $f(m_n) \geq 0$ set $[a_{n+1}, b_{n+1}] = [a_n, m_n]$; otherwise set $[a_{n+1}, b_{n+1}] = [m_n, b_n]$. The sign-change property is preserved in both cases.
Closing argument. NIP yields $c \in \bigcap I_n$ with $a_n \to c$ and $b_n \to c$. Since $f(a_n) < 0$ for all n: continuity gives $f(c) \leq 0$. Since $f(b_n) \geq 0$ for all n: continuity gives $f(c) \geq 0$. Therefore $f(c) = 0$.

Remark 29.1.107 (Two completeness paths)*.* IVT admits two proofs from completeness:

1. *AoC path.* Set $K = \{x \in [a,b] : f(x) \leq 0\}$, $c = \sup K$; rule out $f(c) > 0$ and $f(c) < 0$ using the LUB property.

2. *NIP path.* Bisection as above.

The two paths are equivalent over the Archimedean ordered field axioms.

Remark 29.1.108 (Applied interpretation)*.* The bisection construction is directly algorithmic: it locates a root to any desired precision in finitely many steps. After n bisections the root is known to within $(b-a)/2^n$. This is the bisection method of numerical analysis.

Uncountability of $\mathbb{R}$ via Nested Intervals

Proposition 29.1.109. *$\mathbb{R}$ is uncountable. Go to proof.*

Remark.

Remark 29.1.110 (Proof strategy — nested interval diagonalisation)*.* Suppose for contradiction $\mathbb{R} = \{x_1, x_2, \ldots\}$. Build nested intervals (I_n) inductively with $I_{n+1} \subseteq I_n$ and $x_n \notin I_{n+1}$: at each step I_n contains two disjoint closed subintervals; x_{n+1} lies in at most one, so choose the other.

By NIP, $\bigcap I_n \neq \emptyset$: pick $y \in \bigcap I_n$. Then $y \neq x_n$ for every n (since $y \in I_n$ but $x_n \notin I_n$), contradicting the assumption that $\{x_n\}$ enumerates $\mathbb{R}$.

Remark 29.1.111 (Relation to Cantor's diagonal argument)*.* This is the geometric form of Cantor's diagonalisation: instead of altering a decimal digit, the nth real is excluded from the nth nested interval. The witness y is the interval analogue of the "diagonal" real.

Summary: When to Use Each Technique

Remark 29.1.112 (Bisection vs. general nested intervals)*.*

Feature	Bisection
How built	Halving at midpoints
Length control	Automatic: $(b_0 - a_0)/2^n$
Typical use	Property propagation via si
Canonical examples	IVT, BW
Guarantor	NIP $\equiv$ AoC

Remark 29.1.113 (Completeness is indispensable)*.* Neither technique works over $\mathbb{Q}$. The bisection procedure for $f(x) = x^2 - 2$ on $[1, 2]$ runs mechanically over $\mathbb{Q}$, but the limiting point $\sqrt{2}$ is not in $\mathbb{Q}$. Every bisection proof is, at bottom, an appeal to the completeness of $\mathbb{R}$.

Remark 29.1.114 (Cross-section of the chapter)*.* The four sections of this chapter are not independent: each builds on the last. Inequality bounding (§29.1.2) requires only the ordered field axioms. Completeness as a constructive engine (§29.1.3) requires AoC and the Archimedean Property. Predicate-walking (§29.1.4) requires completeness via MCT or NIP, depending on the variant. Bisection and NIP (§29.1.5) is the geometric synthesis: it encapsulates completeness in a form that drives the IVT, BW, and uncountability proofs simultaneously, connecting back to both the inductive-selection constructions of §29.1.3 and the predicate-walking patterns of §29.1.4.

29.1.6 Subsequence Partition via Residue Classes

Residue Partition Toolkit — Quick Reference

Tool	Statement	Use
k-Periodicity Criterion	$(a_n) \to L \iff \forall r,\ (a_{kn+r}) \to L$	Prove convergence by analysing k c
Forward direction	Subseqs. of convergent seqs. converge to L	Reduce to subsequence analysis
Divergence corollary	Two classes $\to$ different limits $\Rightarrow (a_n)$ diverges	Detect divergence by comparing res
Index translation	$m = kn + r \Rightarrow n \geq N_r$ once $m \geq k \max_r N_r$	Core of the backward direction

The Residue-Class Partition

For any sequence (a_n) and any integer $k \geq 2$, the division algorithm partitions $\mathbb{N}$ into k disjoint residue classes:

$$\mathbb{N} = \bigsqcup_{r=0}^{k-1} \{kn + r : n \geq 0\}.$$

Each class defines a subsequence: $(a_{kn+r})_{n\geq 0}$ selects every kth term of (a_n) beginning at position r. These k subsequences partition (a_n) exactly — every term appears in precisely one class, order within each class is preserved, and their union recovers (a_n).

Remark 29.1.115 (Fully quantified form of the partition). $\forall n \in \mathbb{N},\ \exists!\, r \in \{0, \ldots, k-1\},\ \exists!\, q \in \mathbb{N},\ n = kq + r$. Hence $\{(a_{kn+r})_{n\geq 0} : r = 0, \ldots, k-1\}$ is a partition of (a_n) into k subsequences.

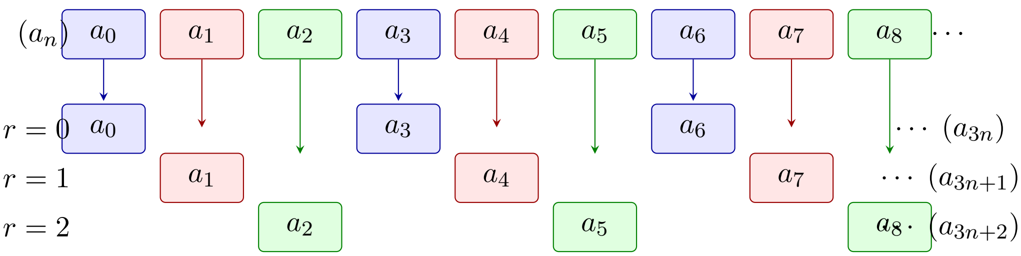


Figure 29.8: The residue-class partition of (a_n) with modulus $k = 3$. Each term is coloured by residue class: blue ($r = 0$), red ($r = 1$), green ($r = 2$). The partition is exact; the k-Periodicity Criterion (Section 29.1.6) states that $(a_n) \to L$ if and only if each coloured subsequence converges to the same L.

The k-Periodicity Convergence Criterion

Theorem (k-Periodicity Convergence Criterion)

Let (a_n) be a sequence, $k \geq 2$ an integer, and $L \in \mathbb{R}$. The following are equivalent:

(i) $(a_n) \to L$.

(ii) For every $r \in \{0, 1, \ldots, k-1\}$, the subsequence $(a_{kn+r}) \to L$.

Remark 29.1.116 (Fully quantified form). $(i) \Leftrightarrow (ii)$, where:

(i): $\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall m \geq N,\ |a_m - L| < \varepsilon$.

(ii): $\forall r \in \{0, \dots, k-1\},\ \forall \varepsilon > 0,\ \exists N_r \in \mathbb{N},\ \forall n \geq N_r,\ |a_{kn+r} - L| < \varepsilon$.

Remark 29.1.117 (Proof strategy). **Forward direction** $(i) \Rightarrow (ii)$. Each (a_{kn+r}) is a subsequence of (a_n). Subsequences of a convergent sequence converge to the same limit (Proposition 29.1.39).

Backward direction $(ii) \Rightarrow (i)$. The operative direction. Given $\varepsilon > 0$, for each residue r let N_r satisfy $|a_{kn+r} - L| < \varepsilon$ for all $n \geq N_r$. Set $N^* = k \cdot \max_r N_r$. For any $m \geq N^*$, write $m = kn + r$ via the division algorithm. Then:

$$n = \frac{m - r}{k} \geq \frac{N^* - (k-1)}{k} \geq \frac{k \max_r N_r - (k-1)}{k} \geq N_r,$$

so $|a_m - L| = |a_{kn+r} - L| < \varepsilon$.

Remark 29.1.118 (The index translation). The key step converts a global index m to a local index n within its residue class via $m = kn + r$. The factor k in the definition of N^* absorbs the offset $r \leq k-1$ from every class simultaneously, ensuring that $n \geq N_r$ holds for whichever class m lands in.

The Divergence Corollary

Proposition 29.1.119 (Residue divergence criterion)**.** *If there exist residues $r, s \in \{0, \dots, k-1\}$ such that $(a_{kn+r}) \to L$ and $(a_{kn+s}) \to M$ with $L \neq M$, then (a_n) diverges. Go to proof.*

Remark.

Remark 29.1.120 (Fully quantified form). $(a_{kn+r}) \to L \ \wedge\ (a_{kn+s}) \to M \ \wedge\ L \neq M \ \Rightarrow$ (a_n) diverges.

Remark 29.1.121 (Proof strategy). Contrapositively: if $(a_n) \to L$, then every subsequence converges to L (forward direction of Section 29.1.6). If two residue-class subsequences converge to different limits, (a_n) cannot converge.

Remark 29.1.122 (Canonical example). The sequence $((-1)^n)$ has $a_{2n} = 1 \to 1$ and $a_{2n+1} = -1 \to -1$. Since $1 \neq -1$, the sequence diverges — established immediately without any ε-N computation.

The Alternating Series Test via Even/Odd Partition

The $k = 2$ case of Section 29.1.6 drives the cleanest proof of the Alternating Series Test.

Theorem (Alternating Series Test)

Let (a_n) be a positive decreasing sequence with $a_n \to 0$. Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges.

Remark 29.1.123 (Fully quantified form). $(\forall n,\ a_n > 0) \ \wedge\ (\forall n,\ a_{n+1} \leq a_n) \ \wedge\ (a_n \to 0) \ \Rightarrow$ $\exists L,\ \sum_{n=1}^{\infty} (-1)^{n+1} a_n = L$.

Remark 29.1.124 (Proof strategy — even/odd partition). Let s_n denote the nth partial sum. Apply Section 29.1.6 with $k = 2$: it suffices to show $(s_{2k}) \to L$ and $(s_{2k+1}) \to L$ for the same L.

Even subsequence (s_{2k}): decreasing and bounded. Group consecutive pairs:

$$s_{2k} = \sum_{j=1}^{k} (-a_{2j-1} + a_{2j}).$$

Since (a_n) is decreasing, each summand $-a_{2j-1} + a_{2j} \le 0$, so (s_{2k}) is decreasing. Also:

$$s_{2k} = -a_1 + \sum_{j=1}^{k-1} (a_{2j} - a_{2j+1}) + a_{2k} \ge -a_1,$$

since each term in the sum is non-negative. Thus (s_{2k}) is decreasing and bounded below by $-a_1$. By MCT, $s_{2k} \to L$ for some $L \ge -a_1$.

Odd subsequence (s_{2k+1}): linked to the even. Observe $s_{2k+1} = s_{2k} + a_{2k+1}$. Since $s_{2k} \to L$ and $a_{2k+1} \to 0$, the Algebraic Limit Theorem gives $s_{2k+1} \to L$.

Applying the criterion. Both residue-class subsequences converge to the same L. By Section 29.1.6 with $k = 2$, $(s_n) \to L$.

Remark 29.1.125 (Why the partition is the right tool). The even and odd partial sums have structurally different characters: the even sums are monotone (enabling MCT), the odd sums are then controlled via the algebraic relationship $s_{2k+1} = s_{2k} + a_{2k+1}$. No single uniform argument handles both. The partition separates the two cases into independent analyses, then the criterion reassembles the conclusion.

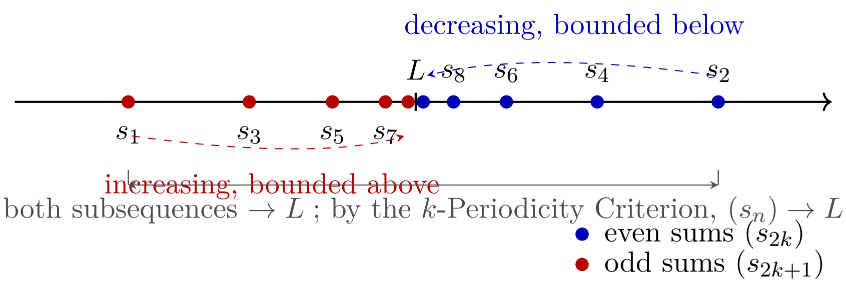


Figure 29.9: Even and odd partial sums of an alternating series $\sum(-1)^{n+1}a_n$ with (a_n) positive, decreasing, and tending to zero. Even sums (s_{2k}) (blue) decrease and are bounded below; by MCT they converge to L. Odd sums (s_{2k+1}) (red) satisfy $s_{2k+1} = s_{2k} + a_{2k+1} \to L$. Since both residue classes (mod $k = 2$) converge to L, the k-Periodicity Criterion (Section 29.1.6) gives $(s_n) \to L$.

Remark 29.1.126 (Alternative proofs). The Alternating Series Test admits three proofs from different completeness avatars:

1. *Cauchy criterion* — show (s_n) is eventually ε-steady directly.

2. *NIP* — the nested intervals $[s_{2k}, s_{2k-1}]$ shrink to a point by the Nested Interval Property.

3. *MCT + residue partition* — as above.

The partition proof is the most transparent structurally: the monotone behaviour and the limit identity are isolated into the two residue classes rather than handled simultaneously.

Higher k and Choosing the Right Period

For sequences with period $k > 2$, the criterion applies directly. The sequence $a_n = \cos(2\pi n/3)$ takes values $1, -\frac{1}{2}, -\frac{1}{2}, 1, -\frac{1}{2}, -\frac{1}{2}, \ldots$, cycling with period 3. Partitioning modulo 3:

$$\begin{aligned} a_{3n} &= 1 \to 1, \\ a_{3n+1} &= -\tfrac{1}{2} \to -\tfrac{1}{2}, \\ a_{3n+2} &= -\tfrac{1}{2} \to -\tfrac{1}{2}. \end{aligned}$$

The three limits differ ($1 \neq -\frac{1}{2}$), so the sequence diverges by Proposition 29.1.119.

Remark 29.1.127 (Choosing k to match the period)*.* The right choice of k is the natural period of the sequence's structure. Using $k = 2$ for a period-3 sequence produces residue classes that are themselves period-3 subsequences; convergence of each class then requires a further partition with $k = 3$. Choosing k to match the period compresses the argument to a single application of the criterion.

Remark 29.1.128 (Applying the criterion to prove convergence)*.* To prove $(a_n) \to L$ via the criterion, one must:

1. Select k matching the period of the sequence's behaviour.

2. Prove $(a_{kn+r}) \to L$ for each $r \in \{0, \ldots, k-1\}$, using whatever tool fits that residue class.

3. Invoke Section 29.1.6 to conclude $(a_n) \to L$.

The method does not produce L — it confirms a candidate. Identifying L requires separate work (direct computation, MCT applied to a class, etc.) before the criterion is invoked.

Contrast with Predicate-Walking

Remark 29.1.129 (Dual logical directions)*.* The residue partition technique and predicate-walking (§29.1.4) operate in opposite logical directions.

Feature	**Predicate-walking**	**Residue partition**
Flow	Whole $\to$ one part	All parts $\to$ whole
Goal	Property of extracted subseq.	Convergence of full (a_n)
Coverage	One "good" subsequence	All k residue classes
Divergence use	Two classes, different limits	Same: two classes, different limits
Completeness role	Via MCT or NIP	Via MCT or algebraic limit theorem

Predicate-walking extracts what is needed; the residue partition reassembles from all pieces. The divergence criterion is available in both frameworks: two subsequences with different limits immediately imply divergence.

Limitations

Remark 29.1.130 (The same-limit condition is non-negotiable). The backward direction of Section 29.1.6 requires all k residue-class subsequences to converge to the *same* L. There is no weakening: a convergent sequence has all its subsequences converging to the identical limit, so any two classes converging to different limits already certify divergence. The technique cannot produce convergence from mixed limits.

Remark 29.1.131 (The limit must be known in advance). The criterion confirms a candidate limit L; it does not produce L from scratch. In practice, L is identified by direct computation on one residue class (e.g., the even partial sums converge to L by MCT), then the criterion assembles global convergence.

Remark 29.1.132 (Exact periodicity is required). The technique applies when the sequence's qualitative behaviour repeats with exact period k in the index. For sequences whose oscillation decays without exact periodicity, the residue classes are not individually tractable, and a direct ε-N argument or a squeeze is more appropriate.

Remark 29.1.133 (Common error). Applying the criterion with k smaller than the true period, then concluding convergence after verifying only the even residue class. If the odd class is not verified, the conclusion does not follow. Every residue class must be checked.

Remark 29.1.134 (Forward connections). The residue partition technique reappears in:

- **Cesàro means** — the Cesàro average of a sequence (a_n) can be analysed by partitioning the partial sums of the average into residue classes.
- **Fourier series** — the convergence of partial sums of a Fourier series is often established by analysing even and odd partial sums (or partial sums along specific subsequences of the approximation order).
- **Recursively defined sequences** — when a recurrence has period k, the residue-class subsequences each satisfy a simpler recurrence, enabling separate analysis.

29.2 Asymptotic Notation

Asymptotic Notation — Quick Reference

Concept/Statement	Meaning/Role
Little-o at a point	One function is negligible relative to another near a point.
Little-o at infinity	Negligibility along an unbounded input range.
Increment little-o	The local error notation $r(h) = o(h)$ as $h \to 0$.
Quotient criterion	Little-o is detected by a quotient tending to 0.
Sum rule	Negligible errors add.
Bounded factor rule	Bounded multipliers preserve negligible errors.

Little-o Notation

Asymptotic notation separates the main term of an expression from the part that becomes negligible in a limiting process. The notation $f = o(g)$ means that f is smaller than every fixed multiple of g near the limiting point. It does not say that f is zero. It says that f is negligible relative to the chosen scale g.

Definition 29.2.1 (Little-o at a Point)**.** Let $a \in \mathbb{R}$, let f and g be real-valued functions defined on a punctured neighbourhood of a, and suppose $g(x) \neq 0$ for all x sufficiently close to a with $x \neq a$. We write

$$f(x) = o(g(x)) \qquad (x \to a)$$

if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x)| \leq \varepsilon |g(x)|.$$

Remark (Standard quantified statement)*.*

$$f(x) = o(g(x)) \ (x \to a) \Longleftrightarrow \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \ \big(0 < |x - a| < \delta \Rightarrow |f(x)| \leq \varepsilon |g(x)|\big).$$

Remark (Interpretation)*.* The scale g is the unit of comparison. The assertion $f = o(g)$ says that no matter how small a tolerance multiple εg is chosen, f eventually fits inside that multiple.

Definition 29.2.2 (Little-o at Infinity)**.** Let f and g be real-valued functions defined on some interval (M, ∞), and suppose $g(x) \neq 0$ for all sufficiently large x. We write

$$f(x) = o(g(x)) \qquad (x \to \infty)$$

if and only if for every $\varepsilon > 0$ there exists $R > 0$ such that

$$x > R \quad \Longrightarrow \quad |f(x)| \leq \varepsilon |g(x)|.$$

Remark (Standard quantified statement)*.*

$$f(x) = o(g(x)) \ (x \to \infty) \Longleftrightarrow \forall \varepsilon > 0 \ \exists R > 0 \ \forall x \ \big(x > R \Rightarrow |f(x)| \leq \varepsilon |g(x)|\big).$$

Remark (Interpretation). The point version and the infinity version have the same logical shape. Only the neighbourhood changes: small punctured intervals around a are replaced by tails (R, ∞).

Definition 29.2.3 (Increment Little-o)**.** Let r be a real-valued function defined for all sufficiently small nonzero h. We write

$$r(h) = o(h) \qquad (h \to 0)$$

if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |h| < \delta \quad \Longrightarrow \quad |r(h)| \leq \varepsilon |h|.$$

Remark (Standard quantified statement).

$$r(h) = o(h)\ (h \to 0) \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall h\ \big(0 < |h| < \delta \Rightarrow |r(h)| \leq \varepsilon |h|\big).$$

Remark (Interpretation). This is the form used in first-order approximation. A remainder $r(h)$ is smaller than the increment h itself, so the quotient $r(h)/h$ tends to zero.

Remark (Dependencies).

- Little-o at a Point

Proposition 29.2.4 (Little-o Quotient Characterisation)**.** *Let $a \in \mathbb{R}$, and let f and g be real-valued functions defined on a punctured neighbourhood of a, with $g(x) \neq 0$ for x sufficiently close to a and $x \neq a$. Then*

$$f(x) = o(g(x))\ (x \to a) \quad \iff \quad \frac{f(x)}{g(x)} \to 0\ (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$[\forall \varepsilon > 0\ \exists \delta > 0\ 0 < |x - a| < \delta \Rightarrow |f(x)| \leq \varepsilon |g(x)|] \iff \frac{f(x)}{g(x)} \to 0.$$

Remark (Interpretation). The quotient form is often the fastest test for little-o. The epsilon form is often the safest proof form because it avoids manipulating a quotient until the nonvanishing of the scale has been checked.

Remark (Dependencies).

- Little-o at a Point

Proposition 29.2.5 (Sum Rule for Little-o)**.** *If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$ as $x \to a$, then*

$$f_1(x) + f_2(x) = o(g(x)) \qquad (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$f_1 = o(g) \wedge f_2 = o(g) \Longrightarrow f_1 + f_2 = o(g).$$

Remark (Interpretation). Negligible errors can be added. The proof is the usual epsilon-budget split: make each error smaller than half the final tolerance.

Remark (Dependencies).

- Little-o at a Point

Proposition 29.2.6 (Bounded Factor Rule for Little-o). *If $f(x) = o(g(x))$ as $x \to a$ and m is bounded near a, then*

$$m(x)f(x) = o(g(x)) \qquad (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$f = o(g) \wedge \exists M > 0\ \exists \eta > 0\ \forall x \left(0 < |x - a| < \eta \Rightarrow |m(x)| \leq M\right) \Longrightarrow mf = o(g).$$

Remark (Interpretation). A bounded multiplier cannot turn a negligible error into a main-order term. It only changes the constant in the epsilon estimate.

Remark (Dependencies).

- Little-o at a Point

29.3 Inequality

Inequality — Quick Reference

Concept/Statement	Meaning/Role
Add both sides	Adding the same term preserves strict inequalities.
Nonstrict add both sides	Addition also preserves nonstrict inequalities.
Add inequalities	Pairwise strict inequalities add.
Mixed addition	Strict and nonstrict hypotheses combine predictably.
Multiply by positive	Positive factors preserve order.
Multiply by negative	Negative factors reverse order.
Squeeze theorem	Double bounding by one value forces equality.
Reciprocal order	Reciprocals reverse positive inequalities.
Square monotonicity	Squaring preserves order on nonnegative inputs.
AM–GM	Arithmetic mean dominates geometric mean.
Bernoulli inequality	Powers of $1 + x$ dominate the tangent line.

Inequality

Theorem 29.3.1 (Addition Preserves Strict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \;\Rightarrow\; a + c < b + c.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \;\Rightarrow\; a + c < b + c.$$

Remark (Dependencies)*.*
Ordered field, ordered field axioms, additive cancellation.

Theorem 29.3.2 (Addition Preserves Nonstrict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \;\Rightarrow\; a + c \leq b + c.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \;\Rightarrow\; a + c \leq b + c.$$

Remark (Dependencies)*.*
Ordered field, addition preserves strict inequality.

Theorem 29.3.3 (Addition of Strict Inequalities)**.** *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a < b \;\wedge\; c < d \;\Rightarrow\; a + c < b + d.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c, d \in \mathbb{R}, \quad a < b \;\wedge\; c < d \;\Rightarrow\; a + c < b + d.$$

Remark (Dependencies)*.*
addition preserves strict inequality, transitivity of strict inequality.

Theorem 29.3.4 (Addition of Nonstrict Inequalities)**.** *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a \leq b \;\wedge\; c \leq d \;\Rightarrow\; a + c \leq b + d.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \;\wedge\; c \leq d \;\Rightarrow\; a + c \leq b + d.$$

Remark (Dependencies)*.*
addition preserves nonstrict inequality.

Theorem 29.3.5 (Mixed Addition of Inequalities)**.** *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ c < d \ \Rightarrow \ a + c < b + d.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \ \wedge \ c < d \ \Rightarrow \ a + c < b + d.$$

Remark (Dependencies)*.*

addition preserves nonstrict inequality, addition preserves strict inequality, mixed transitivity.

Theorem 29.3.6 (Multiplication by a Positive Scalar Preserves Strict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge \ 0 < c \ \Rightarrow \ ac < bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ 0 < c \ \Rightarrow \ ac < bc.$$

Remark (Dependencies)*.*

Ordered field, multiply positive, positive cone.

Theorem 29.3.7 (Multiplication by a Negative Scalar Reverses Strict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge \ c < 0 \ \Rightarrow \ ac > bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ c < 0 \ \Rightarrow \ ac > bc.$$

Remark (Dependencies)*.*

Ordered field, multiply negative, positive cone.

Theorem 29.3.8 (Multiplication by a Positive Scalar Preserves Nonstrict Inequality)**.** *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ 0 < c \ \Rightarrow \ ac \leq bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \ \wedge \ 0 < c \ \Rightarrow \ ac \leq bc.$$

Remark (Dependencies)*.*

multiplication by positive preserves strict inequality.

Theorem 29.3.9 (Multiplication by a Nonneg Scalar Preserves Nonstrict Inequality)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a \leq b \ \wedge \ 0 \leq c \ \Rightarrow \ ac \leq bc.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \ \wedge \ 0 \leq c \ \Rightarrow \ ac \leq bc.$$

Remark (Interpretation)*.* This extends the positive-scalar result to include $c = 0$, where both sides become 0 and the inequality holds trivially.

Remark (Dependencies)*.*
multiplication by positive preserves nonstrict inequality, ordered field.

Theorem (Squeeze)

Theorem 29.3.10 (Squeeze)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a \leq b \leq c \ \wedge \ a = c \ \Rightarrow \ b = a.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \leq c \ \wedge \ a = c \ \Rightarrow \ b = a.$$

Remark (Interpretation)*.* The squeeze principle forces equality when a value is simultaneously bounded above and below by equal quantities. The continuous analogue (if $f(x) \leq g(x) \leq h(x)$ and $f(x) \to L$ and $h(x) \to L$ then $g(x) \to L$) is the central tool for computing limits. The algebraic version stated here is its foundation.

Remark (Dependencies)*.*
Ordered field, additive cancellation.

Theorem 29.3.11 (Transitivity of Strict Inequality)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a < b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Remark (Dependencies)*.*
Ordered field, positive cone.

Theorem 29.3.12 (Mixed Transitivity)**.** *For all $a, b, c \in \mathbb{R}$,*

$$a \leq b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \wedge b < c \Rightarrow a < c.$$

Remark (Dependencies)*.*

Ordered field, transitivity of strict inequality.

Theorem 29.3.13 (Reciprocal Reverses Strict Inequality for Positives)**.** *For all $a, b \in \mathbb{R}$,*

$$0 < a < b \Rightarrow 0 < \frac{1}{b} < \frac{1}{a}.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b \in \mathbb{R}, \quad 0 < a < b \Rightarrow 0 < \tfrac{1}{b} < \tfrac{1}{a}.$$

Remark (Dependencies)*.*

Ordered field, inverse positive, reciprocal order for positive elements, multiplication by positive preserves strict inequality.

Theorem 29.3.14 (Reciprocal Equivalence for Positives)**.** *For all $a, b \in \mathbb{R}$ with $a > 0$ and $b > 0$,*

$$a < b \iff \frac{1}{b} < \frac{1}{a}.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b \in \mathbb{R},\ a > 0 \wedge b > 0 \Rightarrow \big(a < b \iff \tfrac{1}{b} < \tfrac{1}{a}\big).$$

Remark (Dependencies)*.*

reciprocal reverses strict inequality for positives, reciprocal order for positive elements.

Theorem 29.3.15 (Square Is Strictly Monotone on Nonneg Reals)**.** *For all $a, b \in \mathbb{R}$,*

$$0 \leq a < b \Rightarrow a^2 < b^2.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a, b \in \mathbb{R}, \quad 0 \leq a < b \Rightarrow a^2 < b^2.$$

Remark (Dependencies)*.*

Ordered field, addition of strict inequalities, multiplication by positive preserves strict inequality, squares are nonnegative.

Theorem 29.3.16 (Square Root Is Monotone on Nonneg Reals). *For all $a, b \in \mathbb{R}$,*

$$0 \le a \le b \;\Rightarrow\; \sqrt{a} \le \sqrt{b}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 \le a \le b \;\Rightarrow\; \sqrt{a} \le \sqrt{b}.$$

Remark (Audit note). This theorem depends on `def:square-root`, which does not yet appear in the repository label inventory. That definition must be created and registered before this result can be fully formalised.

Remark (Dependencies).

`def:square-root` (not yet registered — see audit note above), square is strictly monotone on nonneg reals.

Theorem 29.3.17 (AM-GM Inequality for Two Terms). *For all $a, b \ge 0$,*

$$\sqrt{ab} \le \frac{a+b}{2}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R},\; a \ge 0 \;\wedge\; b \ge 0 \;\Rightarrow\; \sqrt{ab} \le \frac{a+b}{2}.$$

Remark (Interpretation). The arithmetic–geometric mean inequality for two terms is the simplest case of the AM-GM family. The standard proof proceeds by squaring both sides (using square monotonicity) and reducing to $(\sqrt{a} - \sqrt{b})^2 \ge 0$, which follows from squares being nonnegative.

Remark (Dependencies).

squares are nonnegative, square is strictly monotone on nonneg reals, `def:square-root` (not yet registered — see audit note in square root monotonicity).

Theorem (Bernoulli's Inequality)

Theorem 29.3.18 (Bernoulli's Inequality). *For all $x \in \mathbb{R}$ with $x \ge -1$ and all $n \in \mathbb{N}$,*

$$(1+x)^n \ge 1 + nx.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R},\; \forall n \in \mathbb{N},\; x \ge -1 \;\Rightarrow\; (1+x)^n \ge 1 + nx.$$

Remark (Interpretation). Bernoulli's inequality is proved by induction and is a key estimate in analysis: it provides a linear lower bound for exponential growth and is used to establish the divergence of the harmonic series, bounds on compound interest, and properties of the exponential function.

Remark (Dependencies).

Ordered field, multiplication by nonneg preserves nonstrict inequality, addition preserves nonstrict inequality.

29.4 Modulus

Modulus — Quick Reference

Concept/Statement	**Meaning/Role**
Absolute value	Magnitude stripped of sign.
Nonnegativity	Modulus is always nonnegative.
Zero characterisation	Only 0 has modulus 0.
Self or negation	$\lvert a\rvert$ is either a or $-a$.
Symmetry	Negation does not change magnitude.
Product rule	Modulus is multiplicative.
Quotient rule	Division respects modulus when the denominator is nonzero.
Bounds by modulus	a lies between $-\lvert a\rvert$ and $\lvert a\rvert$.
Nonstrict modulus inequality	$\lvert a\rvert \leq r$ is equivalent to a two-sided bound.
Strict modulus inequality	$\lvert a\rvert < r$ is equivalent to a strict two-sided bound.
Reverse triangle inequality	Magnitudes vary no faster than distance.
Finite triangle inequality	The modulus of a finite sum is bounded by summed moduli.

Modulus

Definition (Absolute Value)

Definition 29.4.1 (Absolute Value)**.** Let $a \in \mathbb{R}$. The *absolute value* (or *modulus*) of a is

$$|a| \;:=\; \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Equivalently, $|a| := \max(a, -a)$.

Remark (Standard quantified statement).

$$|a| \;=\; \max(a,\,-a) \;:=\; \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Remark (Interpretation). The absolute value measures magnitude independent of sign. On the real line it equals the distance from a to 0: $|a| = d(a, 0)$ under the standard metric. The two-branch definition partitions $\mathbb{R}$ at 0 and reflects negative values to positive ones.

Remark (Dependencies).
Ordered field, $\mathbb{R}$ is an ordered field.

Theorem 29.4.2 (Absolute Value Is Nonnegative)**.** *For all $a \in \mathbb{R}$,*

$$|a| \geq 0.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |a| \geq 0.$$

Remark (Dependencies).
Absolute value, squares are nonnegative, ordered field.

Theorem 29.4.3 (Absolute Value Zero Iff Zero)**.** *For all $a \in \mathbb{R}$,*

$$|a| = 0 \iff a = 0.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |a| = 0 \iff a = 0.$$

Remark (Dependencies).
Absolute value, absolute value is nonnegative, ordered field.

Theorem 29.4.4 (Absolute Value Equals Self or Negation)**.** *For all $a \in \mathbb{R}$, either $|a| = a$ or $|a| = -a$.*

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |a| = a \ \lor \ |a| = -a.$$

Remark (Dependencies).
Absolute value.

Theorem 29.4.5 (Absolute Value Is Symmetric)**.** *For all $a \in \mathbb{R}$,*

$$|-a| = |a|.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}, \quad |-a| = |a|.$$

Remark (Dependencies).

Absolute value, double negation.

Theorem 29.4.6 (Absolute Value of a Product)**.** *For all $a, b \in \mathbb{R}$,*

$$|ab| = |a|\,|b|.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad |ab| = |a|\,|b|.$$

Remark (Dependencies).

Absolute value, negation of product, product of negatives, squares are nonnegative.

Theorem 29.4.7 (Absolute Value of a Quotient)**.** *For all $a \in \mathbb{R}$ and $b \in \mathbb{R}$ with $b \neq 0$,*

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall b \in \mathbb{R} \setminus \{0\}, \quad \left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$

Remark (Dependencies).

Absolute value, absolute value of a product, absolute value zero iff zero, uniqueness of multiplicative inverse.

Theorem 29.4.8 (Bound by Modulus)**.** *For all $a \in \mathbb{R}$,*

$$-|a| \leq a \leq |a|.$$

Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad -|a| \leq a \leq |a|.$$

Remark (Dependencies).

Absolute value, absolute value is nonnegative, ordered field.

Theorem (Nonstrict Modulus Inequality)

Theorem 29.4.9 (Nonstrict Modulus Inequality)**.** *For all $a \in \mathbb{R}$ and $r \geq 0$,*

$$|a| \leq r \iff -r \leq a \leq r.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall r \in \mathbb{R},\ r \geq 0 \ \Rightarrow\ \big(|a| \leq r \iff -r \leq a \leq r\big).$$

Remark (Interpretation). This biconditional is the primary tool for replacing a modulus inequality with a pair of simultaneous linear inequalities. It underlies the ε-δ manipulations throughout analysis: the condition $|x - a| \leq \varepsilon$ is equivalent to $a - \varepsilon \leq x \leq a + \varepsilon$.

Remark (Dependencies).

Absolute value, bound by modulus, ordered field.

Theorem 29.4.10 (Strict Modulus Inequality)**.** *For all $a \in \mathbb{R}$ and $r > 0$,*

$$|a| < r \iff -r < a < r.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall r \in \mathbb{R},\ r > 0 \ \Rightarrow\ \big(|a| < r \iff -r < a < r\big).$$

Remark (Dependencies).

Absolute value, nonstrict modulus inequality, ordered field.

Theorem (Triangle Inequality)

Theorem (Triangle Inequality)**.** *For all $a, b \in \mathbb{R}$,*

$$|a + b| \leq |a| + |b|.$$

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad |a + b| \leq |a| + |b|.$$

Remark (Interpretation). The triangle inequality is the foundational estimate of analysis. Every convergence argument, continuity proof, and metric-space result that requires bounding a sum reduces, at some step, to this inequality. The name reflects its geometric meaning: the length of one side of a triangle does not exceed the sum of the other two.

Theorem 29.4.11 (Reverse Triangle Inequality)**.** *For all $a, b \in \mathbb{R}$,*

$$\big|\,|a| - |b|\,\big| \leq |a - b|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad \big|\,|a| - |b|\,\big| \leq |a - b|.$$

Remark (Interpretation). The reverse triangle inequality provides a lower bound on $|a-b|$ in terms of the difference of moduli. It is used to bound the modulus of a difference from below, particularly when proving that a function is not uniformly continuous or that a sequence does not converge.

Remark (Dependencies).

Absolute value, triangle inequality, absolute value is symmetric.

Theorem 29.4.12 (Generalised Triangle Inequality). *For all $n \in \mathbb{N}$ and $a_1, \ldots, a_n \in \mathbb{R}$,*

$$\left|\sum_{k=1}^{n} a_k\right| \leq \sum_{k=1}^{n} |a_k|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N},\ \forall a_1, \ldots, a_n \in \mathbb{R}, \quad \left|\sum_{k=1}^{n} a_k\right| \leq \sum_{k=1}^{n} |a_k|.$$

Remark (Dependencies).

Absolute value, triangle inequality.

Chapter 30

Structure of the Real Line

30.1 Real Line One Dimensional

The Line as Order Plus Distance

The real numbers carry two compatible structures at once: a linear order $<$ and a notion of distance $d(x, y) = |x - y|$. Together these make $\mathbb{R}$ a one-dimensional space — a line. This chapter develops only the metric topology of $\mathbb{R}$, not topology in general spaces. Every open set is built from intervals, and compactness is introduced only on $\mathbb{R}$, with Heine–Borel as the capstone.

The intuition is geometric: points are positions on a line, order is "left of," and distance is the length of the segment between two points. The caution to keep throughout is that $\mathbb{R}$ is special. Order and metric agree here in a way that fails on a general space, so definitions are stated in the form that survives to metric spaces and manifolds, with $\mathbb{R}$ as the first instance rather than the only one.

30.2 Order Distance Length

Distance on the Line

The absolute value of a difference turns $\mathbb{R}$ into a metric space. Length of an interval is the distance between its endpoints. These are the quantitative tools underneath every later definition of nearness.

Definition (Distance on the Real Line)

Definition 30.2.1 (Distance on the Real Line)**.** The *distance* on $\mathbb{R}$ is the map $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ given by

$$d(x, y) = |x - y|.$$

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}\ (d(x, y) = |x - y|).$$

Remark (Interpretation). The distance between two points is the length of the segment joining them. It is the metric that makes "close" precise and is the basis for balls, neighborhoods, and open sets.

Remark (Dependencies).

- Triangle Inequality

Definition (Length of a Bounded Interval)

Definition 30.2.2 (Length of a Bounded Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *length* of any bounded interval with endpoints a and b is

$$\ell = b - a.$$

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \big(a \leq b \Longrightarrow \ell([a, b]) = b - a = d(a, b)\big).$$

Remark (Interpretation). Length depends only on the endpoints, not on whether they are included. It is the distance $d(a, b)$ and measures the size of the basic pieces of the line.

Remark (Dependencies).

- Distance on the Real Line

Theorem (The Distance Function Is a Metric)

Theorem 30.2.3 (The Distance Function Is a Metric)**.** *The map $d(x, y) = |x - y|$ satisfies, for all $x, y, z \in \mathbb{R}$: (i) $d(x, y) \geq 0$ with $d(x, y) = 0$ iff $x = y$; (ii) $d(x, y) = d(y, x)$; and (iii) $d(x, z) \leq d(x, y) + d(y, z)$. Hence $(\mathbb{R}, d)$ is a metric space. Go to proof.*

Remark (Standard quantified statement).

$$\forall x, y, z \in \mathbb{R}\ \big(d(x, y) \geq 0 \land (d(x, y) = 0 \Leftrightarrow x = y) \land d(x, y) = d(y, x) \land d(x, z) \leq d(x, y) + d(y, z)\big).$$

Remark (Interpretation). The three metric axioms are exactly the properties of $|\cdot|$: nonnegativity with the zero-distance characterization, symmetry, and the triangle inequality. This is the precise sense in which $\mathbb{R}$ is a one-dimensional metric space.

Remark (Dependencies).

- Distance on the Real Line
- Triangle Inequality

30.3 Absolute Value as Distance

Absolute Value Is Distance to Zero

The metric d specializes to absolute value when one point is the origin. This is the geometric reading of $|a|$: how far a sits from 0 on the line.

Proposition (Absolute Value Is Distance to the Origin)

Proposition 30.3.1 (Absolute Value Is Distance to the Origin)**.** *For every* $a \in \mathbb{R}$*,* $|a| = d(a, 0)$*.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$\forall a \in \mathbb{R}\ (|a| = d(a, 0)).$$

Remark (Interpretation)*.* Absolute value is not a separate idea from distance; it is the distance to the origin. The two-sided bound $|x| \le r \Leftrightarrow -r \le x \le r$ is then exactly the statement that x lies within distance r of 0, i.e. in the interval $[-r, r]$.

Remark (Dependencies)*.*

- Distance on the Real Line

30.4 Intervals as Subsets

Intervals as the Basic Pieces

The interval shapes were introduced in §26.2. Here they are recast as subsets of the metric line and supplemented with boundedness, which is the size condition compactness will need.

Definition (Bounded Subset of the Real Line)

Definition 30.4.1 (Bounded Subset of the Real Line)**.** A set $E \subseteq \mathbb{R}$ is *bounded* exactly when there exists $M \ge 0$ such that $|x| \le M$ for all $x \in E$; equivalently, E is contained in some closed interval $[-M, M]$.

Remark (Standard quantified statement)*.*

$$E \text{ is bounded} \iff \exists M \ge 0\ \forall x \in E\ (|x| \le M).$$

Remark (Negated quantified statement)*.*

$$E \text{ is unbounded} \iff \forall M \ge 0\ \exists x \in E\ (|x| > M).$$

Remark (Interpretation)*.* Boundedness says E fits inside a single closed interval — it does not run off to infinity. Together with closedness it is one half of the Heine–Borel characterization of compact subsets of $\mathbb{R}$.

Remark (Dependencies).

- Closed Interval
- Triangle Inequality

Set Operations on Intervals

Intervals as Sets

Once intervals are regarded as subsets of $\mathbb{R}$, the usual set operations apply to them. This is the first place where intervals behave like the building blocks of topology rather than merely as pieces of notation on the number line. Intersections often remain intervals, while unions, complements, and differences may split into several interval pieces.

Definition (Union of Intervals)

Definition 30.4.2 (Union of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *union* is the set

$$I \cup J = \{x \in \mathbb{R} : x \in I \text{ or } x \in J\}.$$

Remark (Dependencies).

- Open Interval
- Closed Interval
- Union

Definition (Intersection of Intervals)

Definition 30.4.3 (Intersection of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *intersection* is the set

$$I \cap J = \{x \in \mathbb{R} : x \in I \text{ and } x \in J\}.$$

Remark (Dependencies).

- Open Interval
- Closed Interval
- Intersection

Definition (Complement of an Interval)

Definition 30.4.4 (Complement of an Interval)**.** Let $I \subseteq \mathbb{R}$ be an interval. Its *complement* in $\mathbb{R}$ is the set

$$I^c = \mathbb{R} \setminus I = \{x \in \mathbb{R} : x \notin I\}.$$

Remark (Dependencies).

- Open Interval
- Closed Interval
- Complement

Definition (Difference of Intervals)

Definition 30.4.5 (Difference of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. The *difference* of I by J is the set

$$I \setminus J = \{x \in \mathbb{R} : x \in I \text{ and } x \notin J\} = I \cap J^c.$$

Remark (Predicate readings)*.* For any $x \in \mathbb{R}$,

$$x \in I \cup J \iff (x \in I) \vee (x \in J),$$

$$x \in I \cap J \iff (x \in I) \wedge (x \in J),$$

$$x \in I^c \iff x \notin I, \qquad x \in I \setminus J \iff (x \in I) \wedge (x \notin J).$$

These are set-theoretic operations; no arithmetic operation is being performed on the endpoints.

Remark (Dependencies)*.*

- Open Interval
- Closed Interval
- Left-Half-Open Interval
- Right-Half-Open Interval
- Union of Intervals
- Intersection of Intervals
- Complement of an Interval

Remark (Intersections of intervals)*.* The intersection of two intervals is either empty or another interval. For example, if $a \leq b$ and $c \leq d$, then

$$[a, b] \cap [c, d] = \begin{cases} [\max\{a, c\}, \min\{b, d\}], & \max\{a, c\} \leq \min\{b, d\}, \\ \varnothing, & \max\{a, c\} > \min\{b, d\}. \end{cases}$$

Endpoint inclusion follows the endpoint rules of the original intervals.

Remark (Unions of intervals). The union of two intervals need not be an interval:

$$(0,1) \cup (2,3)$$

has a gap. If two intervals overlap or touch without a missing endpoint, their union is again an interval, for instance

$$(0,2] \cup [2,5) = (0,5).$$

The missing or included endpoints determine whether the joined set is truly one interval.

Remark (Complements of intervals). The complement of a bounded interval usually splits into two rays:

$$(a,b)^c = (-\infty, a] \cup [b, \infty), \qquad [a,b]^c = (-\infty, a) \cup (b, \infty).$$

Complements reverse endpoint inclusion: an endpoint excluded from the interval is included in the complement, and an endpoint included in the interval is excluded from the complement.

Remark (Differences of intervals). A difference is an intersection with a complement, so it can also split:

$$(0,5) \setminus [2,3] = (0,2) \cup (3,5).$$

If the removed interval only cuts off one side, the result may remain a single interval:

$$(0,5) \setminus (3,\infty) = (0,3].$$

The bracket at 3 appears because $3 \notin (3,\infty)$, so it is not removed.

Remark (Topological motivation). Open sets in $\mathbb{R}$ are built from unions of open intervals. Closed sets are understood through complements of open sets. Boundaries arise when a point is not cleanly inside a set or its complement. Thus interval set operations are the grammar behind the metric topology of the real line.

30.5 Neighborhoods and Balls

Balls Are Open Intervals

On the line, the ball of radius r about a point is an open interval centered at that point. Neighborhoods are sets that contain such a ball. These are the local objects from which open sets are assembled.

Definition (Open Ball on the Real Line)

Definition 30.5.1 (Open Ball on the Real Line). Let $x \in \mathbb{R}$ and $r > 0$. The *open ball* of radius r about x is

$$B_r(x) = \{\, y \in \mathbb{R} : |y - x| < r \,\} = (x - r,\, x + r).$$

Remark (Standard quantified statement).

$$B_r(x) = \{\, y \in \mathbb{R} : d(y, x) < r \,\} = (x - r, x + r).$$

Remark (Interpretation). A ball on the line is just a symmetric open interval: all points strictly within distance r of x. Writing balls as intervals keeps the chapter inside the metric topology of $\mathbb{R}$.

Remark (Dependencies).

- Distance on the Real Line
- Open Interval

Definition (Neighborhood of a Point)

Definition 30.5.2 (Neighborhood of a Point)**.** A set $N \subseteq \mathbb{R}$ is a *neighborhood* of $x \in \mathbb{R}$ exactly when there exists $r > 0$ with $B_r(x) \subseteq N$.

Remark (Standard quantified statement).

$$N \text{ is a neighborhood of } x \iff \exists r > 0\ (B_r(x) \subseteq N).$$

Remark (Interpretation). A neighborhood of x is any set with room to spare around x — it contains a whole ball, not just x itself. Neighborhoods are the language of "near x" used to define interior points and limits.

Remark (Dependencies).

- Open Ball on the Real Line

30.6 Open Sets

Sets Built from Balls

An open set is one that contains a ball around each of its points. On the line these are exactly the unions of open intervals. The closure properties recorded here are the metric-topology core.

Definition (Open Set in $\mathbb{R}$)

Definition 30.6.1 (Open Set in $\mathbb{R}$)**.** A set $U \subseteq \mathbb{R}$ is *open* exactly when

$$\forall x \in U\ \exists r > 0\ (B_r(x) \subseteq U).$$

Remark (Standard quantified statement).

$$U \text{ is open} \iff \forall x \in U\ \exists r > 0\ (B_r(x) \subseteq U).$$

Remark (Negated quantified statement).

$$U \text{ is not open} \iff \exists x \in U\ \forall r > 0\ (B_r(x) \not\subseteq U).$$

Remark (Failure modes). Openness fails exactly when some point of U has no ball contained in U — a boundary point trapped inside U, every ball around which pokes outside.

Remark (Interpretation). An open set has interior room around each of its points: a small step in either direction remains inside the set. The empty set and $\mathbb{R}$ are open, and open intervals are the model examples.

Remark (Dependencies).

- Open Ball on the Real Line

Theorem (Open Intervals Are Open)

Theorem 30.6.2 (Open Intervals Are Open)**.** *Every open interval* (a, b)*, every open ray* (a, ∞) *and* $(-\infty, b)$*, and* $\mathbb{R}$ *itself are open sets. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ (a < b \Longrightarrow (a, b) \text{ is open}),$$
$$\forall a \in \mathbb{R}\ \big((a, \infty) \text{ is open} \wedge (-\infty, a) \text{ is open}\big), \qquad \mathbb{R} \text{ is open.}$$

Remark (Interpretation). The name "open interval" is justified: each such interval satisfies the open-set condition, with the radius at x taken as the distance to the nearer endpoint.

Remark (Dependencies).

- Open Set in $\mathbb{R}$
- Open Interval

Theorem (Unions and Finite Intersections of Open Sets)

Theorem 30.6.3 (Unions and Finite Intersections of Open Sets)**.** *An arbitrary union of open subsets of* $\mathbb{R}$ *is open, and a finite intersection of open subsets of* $\mathbb{R}$ *is open. Go to proof.*

Remark (Standard quantified statement).

$$\Big(\forall i \in I\ U_i \text{ open}\Big) \Longrightarrow \Big(\bigcup_{i \in I} U_i\Big) \text{ open;} \qquad \Big(\bigwedge_{k=1}^{n} U_k \text{ open}\Big) \Longrightarrow \Big(\bigcap_{k=1}^{n} U_k\Big) \text{ open.}$$

Remark (Interpretation). Openness is preserved by arbitrary unions but only finite intersections. The infinite intersection $\bigcap_n (-1/n, 1/n) = \{0\}$ shows why "finite" cannot be dropped: a shrinking family of open intervals can close down onto a single point.

Remark (Dependencies).

- Open Set in $\mathbb{R}$

30.7 Closed Sets

Complements of Open Sets

Closed sets are the complements of open sets. They can equivalently be described as the sets that contain all of their limit points, the description used when reasoning about convergence.

Definition (Closed Set in $\mathbb{R}$)

Definition 30.7.1 (Closed Set in $\mathbb{R}$)**.** A set $F \subseteq \mathbb{R}$ is *closed* exactly when its complement $\mathbb{R} \setminus F$ is open.

Remark (Standard quantified statement).

$$F \text{ is closed} \iff (\mathbb{R} \setminus F) \text{ is open.}$$

Remark (Interpretation). Closed and open are complementary, not opposite: a set can be both (e.g. $\emptyset$, $\mathbb{R}$) or neither (e.g. $[0, 1)$). Closed intervals $[a, b]$ are closed, as are finite sets and $\mathbb{R}$ itself.

Remark (Dependencies).

- Open Set in $\mathbb{R}$

Theorem (Closed Sets Contain Their Limit Points)

Theorem 30.7.2 (Closed Sets Contain Their Limit Points)**.** *A set $F \subseteq \mathbb{R}$ is closed if and only if it contains every one of its limit points. Go to proof.*

Remark (Standard quantified statement).

$$F \text{ closed} \iff \forall x \in \mathbb{R}\ \big(x \text{ is a limit point of } F \Rightarrow x \in F\big).$$

Remark (Interpretation). This is the convergence-flavored description of closedness: a closed set cannot be approached from outside without already containing the approached point. It is the form used when showing limits of sequences in F stay in F.

Remark (Dependencies).

- Closed Set in $\mathbb{R}$
- Limit Point

30.8 Interior Exterior Boundary

Inside, Outside, and Edge

Relative to a set E, each point of the line is interior to E, exterior to E, or on its boundary. These three regions partition $\mathbb{R}$ and refine the open/closed distinction.

Definition (Interior Point)

Definition 30.8.1 (Interior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an interior point of E exactly when

$$\exists r > 0\ (B_r(x) \subseteq E).$$

Remark (Standard quantified statement).

$$x \text{ is an interior point of } E \iff \exists r > 0\ (B_r(x) \subseteq E).$$

Remark (Interpretation)*.* An interior point of E sits inside E with room to spare: a whole ball around it lies in E.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Exterior Point)

Definition 30.8.2 (Exterior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an exterior point of E exactly when

$$\exists r > 0\ (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Remark (Standard quantified statement).

$$x \text{ is an exterior point of } E \iff \exists r > 0\ (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Remark (Interpretation)*.* An exterior point of E is interior to the complement: a whole ball around it misses E.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Boundary Point)

Definition 30.8.3 (Boundary Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a boundary point of E exactly when

$$\forall r > 0\ (B_r(x) \cap E \neq \emptyset\ \wedge\ B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Remark (Standard quantified statement).

$$x \text{ is a boundary point of } E \iff \forall r > 0\ (B_r(x) \cap E \neq \emptyset\ \wedge\ B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Remark (Interpretation). A boundary point of E is approached from both sides: every ball around it meets both E and its complement.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Interior of a Set)

Definition 30.8.4 (Interior of a Set)**.** The *interior* of $E \subseteq \mathbb{R}$, written $\operatorname{int}(E)$, is the set of all interior points of E.

Remark (Standard quantified statement).

$$\operatorname{int}(E) = \{\, x \in \mathbb{R} : \exists r > 0\ (B_r(x) \subseteq E) \,\}.$$

Remark (Interpretation). The interior is the largest open set inside E; E is open exactly when $E = \operatorname{int}(E)$. The boundary is what E and its complement share, and a set is closed exactly when it contains its boundary.

Remark (Dependencies).

- Interior Point

30.9 Limit and Isolated Points

Approachable and Detached Points

A limit point of E is one that E crowds arbitrarily closely, even with the point itself removed. An isolated point of E belongs to E but has a ball meeting E only at itself. These notions drive the convergence description of closed sets.

Definition (Limit Point)

Definition 30.9.1 (Limit Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a *limit point* of E exactly when

$$\forall r > 0\ \exists y \in E\ \big(0 < |y - x| < r\big).$$

Remark (Standard quantified statement).

$$x \text{ is a limit point of } E \iff \forall r > 0\ \exists y \in E\ (0 < |y - x| < r).$$

Remark (Negated quantified statement).

$$x \text{ is not a limit point of } E \iff \exists r > 0\ \forall y \in E\ (y = x \vee |y - x| \geq r).$$

Remark (Interpretation). Every punctured ball around a limit point meets E: the set accumulates at x. The strict inequality $0 < |y - x|$ excludes x itself, so x need not belong to E to be a limit point. This is the exact condition under which a sequence in E can converge to x.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Isolated Point)

Definition 30.9.2 (Isolated Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in E$. Then x is an *isolated point* of E exactly when

$$\exists r > 0\ \forall y \in E\ \big(|y - x| < r \Rightarrow y = x\big).$$

Remark (Standard quantified statement).

$$x \text{ is an isolated point of } E \iff x \in E \wedge \exists r > 0\ \forall y \in E\ \big(|y - x| < r \Rightarrow y = x\big).$$

Remark (Interpretation). An isolated point sits in E but stands apart: some ball around it contains no other point of E. A point of E is either isolated or a limit point of E, never both.

Remark (Dependencies).

- Open Ball on the Real Line

30.10 Unions and Intersections

Closure Laws for Open and Closed Sets

The open closure laws of §30.6 dualize to closed sets by complementation. Collecting them gives the bookkeeping rules used throughout the rest of the development.

Theorem (Closure Laws for Closed Sets)

Theorem 30.10.1 (Closure Laws for Closed Sets)**.** *An arbitrary intersection of closed subsets of $\mathbb{R}$ is closed, and a finite union of closed subsets of $\mathbb{R}$ is closed. Go to proof.*

Remark (Standard quantified statement).

$$\Big(\forall i \in I\ F_i \text{ closed}\Big) \Longrightarrow \Big(\bigcap_{i\in I} F_i\Big) \text{ closed}; \qquad \Big(\bigwedge_{k=1}^{n} F_k \text{ closed}\Big) \Longrightarrow \Big(\bigcup_{k=1}^{n} F_k\Big) \text{ closed}.$$

Remark (Interpretation). The roles of union and intersection swap relative to open sets: closed sets are stable under arbitrary intersection but only finite union. The asymmetry mirrors the open case under De Morgan's laws.

Remark (Dependencies).

- Closed Set in $\mathbb{R}$
- Unions and Finite Intersections of Open Sets

30.11 Covers and Subcovers

Covering a Set with Open Pieces

A cover of E is a family of sets whose union contains E; an open cover uses open sets. A subcover keeps enough of the family to still cover, and a finite subcover keeps only finitely many. These are the exact ingredients of compactness, defined so as to survive beyond the line.

Definition (Open Cover)

Definition 30.11.1 (Open Cover)**.** Let $E \subseteq \mathbb{R}$. An *open cover* of E is a family $\{U_i\}_{i\in I}$ of open subsets of $\mathbb{R}$ with

$$E \subseteq \bigcup_{i\in I} U_i.$$

Remark (Standard quantified statement).

$$\{U_i\}_{i\in I} \text{ is an open cover of } E \iff \big(\forall i \in I\ U_i \text{ open}\big) \wedge E \subseteq \textstyle\bigcup_{i\in I} U_i.$$

Remark (Interpretation). An open cover is a way of surrounding every point of E with an open piece of the family. The index set I may be infinite; the question compactness asks is whether finitely many pieces already suffice.

Remark (Dependencies).

- Open Set in $\mathbb{R}$

Definition (Subcover and Finite Subcover)

Definition 30.11.2 (Subcover and Finite Subcover)**.** Let $\{U_i\}_{i\in I}$ be an open cover of E. A *subcover* is a subfamily $\{U_i\}_{i\in J}$ with $J \subseteq I$ that still covers E. It is a *finite subcover* when J is finite.

Remark (Standard quantified statement).

$$\{U_i\}_{i\in J} \text{ is a finite subcover of } E \iff \big(J \subseteq I \wedge J \text{ is finite} \wedge E \subseteq \textstyle\bigcup_{i\in J} U_i\big).$$

Remark (Interpretation). A subcover discards some pieces while still covering E; a finite subcover discards all but finitely many. The existence of a finite subcover for *every* open cover is the defining property of compactness.

Remark (Dependencies).

- Open Cover

30.12 Compactness on R

The Open-Cover Definition

Compactness is defined by the open-cover property, deliberately and not as "closed and bounded." The open-cover formulation is the one that survives to general metric spaces and to manifolds such as the torus, where "bounded" has no intrinsic meaning. On $\mathbb{R}$ the two descriptions coincide, and that coincidence is the Heine–Borel theorem of the next section, proved rather than assumed.

Definition (Compact Set in $\mathbb{R}$)

Definition 30.12.1 (Compact Set in $\mathbb{R}$)**.** A set $K \subseteq \mathbb{R}$ is *compact* exactly when every open cover of K admits a finite subcover:

$$\forall \{U_i\}_{i\in I} \text{ open cover of } K\ \exists \text{finite } J \subseteq I\ \Big(K \subseteq \bigcup_{i\in J} U_i\Big).$$

Remark (Standard quantified statement).

$$K \text{ compact} \iff \forall \{U_i\}_{i\in I}\Big(\big(\forall i \in I\ U_i \text{ open}\big) \wedge K \subseteq \textstyle\bigcup_{i\in I} U_i \Rightarrow \exists \text{finite } J \subseteq I\ K \subseteq \bigcup_{i\in J} U_i\Big).$$

Remark (Negated quantified statement).

$$K \text{ not compact} \iff \exists \{U_i\}_{i\in I}\Big(\big(\forall i \in I\ U_i \text{ open}\big) \wedge K \subseteq \textstyle\bigcup_{i\in I} U_i \wedge \forall \text{finite } J \subseteq I\ K \not\subseteq \bigcup_{i\in J} U_i\Big).$$

Remark (Failure modes). Compactness fails exactly when some open cover has no finite subcover. Two standard witnesses: $(0, 1]$ fails via the cover $\{(1/n, 2)\}_n$ (not closed), and $[0, \infty)$ fails via $\{(-1, n)\}_n$ (not bounded).

Remark (Interpretation). Compactness is a finiteness property phrased entirely in open sets: no matter how a set is covered by open pieces, finitely many already do the job. This is what later guarantees, on a compact domain, that continuous functions attain extrema and are uniformly continuous — the facts Analysis I relies on. Crucially, this definition does not mention order or boundedness, so it transfers unchanged to the torus.

Remark (Dependencies).

- Open Cover
- Subcover and Finite Subcover

Theorem (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded)

Theorem 30.12.2 (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded)**.** *If $K \subseteq \mathbb{R}$ is compact, then K is closed and bounded.* *Go to proof.*

Remark (Standard quantified statement).

$$K \text{ compact} \Longrightarrow (K \text{ closed} \wedge K \text{ bounded}).$$

Remark (Interpretation). This is the easy half of Heine–Borel and holds the same way in every metric space. Boundedness comes from covering K by growing balls about a fixed point; closedness comes from separating an outside point from K by disjoint balls. The converse — closed and bounded implies compact — is special to $\mathbb{R}^n$ and is the next theorem.

Remark (Dependencies).

- Compact Set in $\mathbb{R}$
- Closed Set in $\mathbb{R}$
- Bounded Subset of the Real Line

30.13 Heine Borel

The Capstone of the Real Line

Heine–Borel closes the chapter by proving the converse direction: on $\mathbb{R}$, closed and bounded is enough for compactness. The keystone is that every closed bounded interval $[a, b]$ is compact; the general statement follows because a closed subset of a compact set is compact.

Theorem (Closed Bounded Intervals Are Compact)

Theorem 30.13.1 (Closed Bounded Intervals Are Compact). *Every closed bounded interval $[a, b] \subseteq \mathbb{R}$ is compact. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \big(a \leq b \Longrightarrow [a, b] \text{ is compact}\big).$$

Remark (Interpretation). This is the hard half. The standard proof uses completeness directly: let S be the set of $x \in [a, b]$ such that $[a, x]$ has a finite subcover, and show $\sup S = b$ is attained. Bisection (the nested-interval method) gives the same result. Either route shows completeness is what makes the line compact in the large.

Remark (Dependencies).

- Compact Set in $\mathbb{R}$
- Closed Interval
- Least-Upper-Bound Property

Theorem (Heine–Borel Theorem for $\mathbb{R}$)

Theorem 30.13.2 (Heine–Borel Theorem for $\mathbb{R}$). *A set $K \subseteq \mathbb{R}$ is compact if and only if K is closed and bounded. Go to proof.*

Remark (Standard quantified statement).

$$K \text{ compact} \iff (K \text{ closed} \wedge K \text{ bounded}).$$

Remark (Interpretation). Heine–Borel is the bridge into Analysis I: it certifies that the closed bounded sets — the ones on which the Extreme Value and uniform-continuity theorems live — are exactly the compact ones. It is stated here as the equivalence *on* $\mathbb{R}$; the open-cover side is the definition that later carries to the torus, while "closed and bounded" does not.

Remark (Exposition). With Heine–Borel in hand the chain is complete: number construction $\to$ embedded number systems $\to$ the real line as an ordered metric space $\to$ Analysis I. Completeness, introduced as the last rung of the embedding ladder, is exactly the property that makes the line compact in the large, and compactness is the hypothesis Analysis I leans on.

Remark (Dependencies).

- Compact Subsets of $\mathbb{R}$ Are Closed and Bounded
- Closed Bounded Intervals Are Compact

Chapter 31

Bounds

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 3
Bounds

31.1 Completeness Axiom

Axiom of Completeness

Remark (Why completeness is needed). The ordered field axioms alone do not prevent "holes" in the number line. The rationals $\mathbb{Q}$ satisfy all field and order axioms, yet fail completeness. Consider the set

$$S = \{x \in \mathbb{Q} : x^2 < 2\}.$$

This set is nonempty (e.g. $1 \in S$) and bounded above in $\mathbb{Q}$ (e.g. 2 is an upper bound). Yet $\sup S$ does not exist as a rational number: the candidate $\sqrt{2}$ is irrational. The set S has no least upper bound in $\mathbb{Q}$ — a hole sits exactly where $\sqrt{2}$ should be.

The Completeness Axiom asserts that $\mathbb{R}$ has no such holes: every nonempty bounded-above set has a supremum *inside* $\mathbb{R}$. This single axiom is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$, and it underlies every major theorem in analysis.

Definition (Least Upper Bound Property)

Definition 31.1.1 (Least Upper Bound Property)**.** Let $(S, \leq)$ be an ordered set. The set S has the *Least Upper Bound Property* if and only if every nonempty subset of S that is bounded above in S has a supremum in S.

Remark (Standard quantified statement).

Let $(S, \leq)$ be an ordered set.
S has the Least Upper Bound Property

$$\iff \forall A \subseteq S \ \Big[\big(A \neq \varnothing \wedge \exists u \in S \ \forall x \in A \ (x \leq u)\big) \to \exists s \in S \ \text{Supremum}(s, A)\Big].$$

Remark (Definition predicate reading).

$$\text{LeastUpperBoundProperty}(S) :\equiv \forall A \subseteq S \left[\big(A \neq \varnothing \wedge \exists u \in S \; \forall x \in A \; (x \leq u) \big) \rightarrow \exists s \in S \; \text{Supremum}(s, A) \right]$$

Remark (Negated quantified statement).

Let $(S, \leq)$ be an ordered set.
S does not have the Least Upper Bound Property
$$\iff \exists A \subseteq S \left[A \neq \varnothing \wedge \exists u \in S \; \forall x \in A \; (x \leq u) \wedge \forall s \in S \; \neg \, \text{Supremum}(s, A) \right].$$

Remark (Negation predicate reading).

$$\neg \, \text{LeastUpperBoundProperty}(S) \iff \exists A \subseteq S \left[A \neq \varnothing \wedge \exists u \in S \; \forall x \in A \; (x \leq u) \wedge \forall s \in S \; \neg \, \text{Supremum}(s, A) \right.$$

Remark (Failure modes). The property fails when there is a nonempty subset of the ambient ordered set that has at least one upper bound but has no least upper bound inside that same ambient set.

Remark (Failure mode decomposition).

$$\neg \, \text{LeastUpperBoundProperty}(S) = \underbrace{\exists A \subseteq S \; (A \neq \varnothing)}_{\text{nonempty witness set}} \wedge \underbrace{\exists u \in S \; \forall x \in A \; (x \leq u)}_{\text{bounded above in } S} \wedge \underbrace{\forall s \in S \; \neg \, \text{Supremum}(s,}_{\text{no supremum in } S}$$

Remark (Interpretation). The Least Upper Bound Property says that upper-bounded nonempty subsets do not point toward missing ceiling elements. Whenever the order supplies at least one upper bound, it also supplies a smallest upper bound in the same ambient set. Failure therefore indicates a hole in the ordered structure rather than a defect in the subset alone.

Remark (Dependencies).

- Ordered Set
- Subset
- Upper Bound
- Supremum

Axiom (Completeness of $\mathbb{R}$)

Axiom 31.1.2 (Completeness of $\mathbb{R}$)**.** The ordered field $\mathbb{R}$ has the Least Upper Bound Property.

Remark (Standard quantified statement).

$$\text{LeastUpperBoundProperty}(\mathbb{R}).$$

Equivalently,

$$\forall A \subseteq \mathbb{R} \left[\big(A \neq \varnothing \wedge \exists u \in \mathbb{R} \; \forall x \in A \; (x \leq u) \big) \rightarrow \exists s \in \mathbb{R} \; \text{Supremum}(s, A) \right].$$

Remark (Axiom predicate reading).

$$\text{LeastUpperBoundProperty}(\mathbb{R}).$$

Remark (Negated quantified statement).

$$\begin{aligned} &\neg\,\text{LeastUpperBoundProperty}(\mathbb{R}) \\ &\iff \exists A \subseteq \mathbb{R}\,\Big[A \neq \varnothing \wedge \exists u \in \mathbb{R}\;\forall x \in A\;(x \leq u) \wedge \forall s \in \mathbb{R}\;\neg\,\text{Supremum}(s, A)\Big]. \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{LeastUpperBoundProperty}(\mathbb{R}) \iff \exists A \subseteq \mathbb{R}\,\Big[A \neq \varnothing \wedge \exists u \in \mathbb{R}\;\forall x \in A\;(x \leq u) \wedge \forall s \in \mathbb{R}\;\neg\,\text{Supremum}(s, A$$

Remark (Failure modes). Completeness would fail if some nonempty subset of $\mathbb{R}$ were bounded above in $\mathbb{R}$ but had no supremum in $\mathbb{R}$.

Remark (Failure mode decomposition).

$$\neg\,\text{LeastUpperBoundProperty}(\mathbb{R}) = \underbrace{\exists A \subseteq \mathbb{R}\;(A \neq \varnothing)}_{\text{nonempty witness set}} \wedge \underbrace{\exists u \in \mathbb{R}\;\forall x \in A\;(x \leq u)}_{\text{bounded above in } \mathbb{R}} \wedge \underbrace{\forall s \in \mathbb{R}\;\neg\,\text{Supremum}(s,}_{\text{missing real supremum}}$$

Remark (Interpretation). The completeness axiom asserts that the real line has no order gaps of the least-upper-bound kind. Nonempty sets of real numbers that are bounded above always determine a real number sitting exactly at the least possible ceiling. In Volume II, $\mathbb{R}$ is constructed as the complete ordered field arising from Dedekind cuts; in this volume, completeness is used as the standing least-upper-bound principle for real analysis. This is the structural property that distinguishes $\mathbb{R}$ from ordered fields such as $\mathbb{Q}$.

Remark (Dependencies).

- The Real Numbers
- Ordered Field
- Least Upper Bound Property

31.1.1 Nested Intervals

Nested Interval Property

Theorem (Nested Interval Property)

Theorem 31.1.3 (Nested Interval Property)**.** *Let (I_n) be a sequence of nonempty closed intervals in $\mathbb{R}$. If $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, then*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (a_n) \subseteq \mathbb{R}\ \forall (b_n) \subseteq \mathbb{R}\\ &\Big[\big(\forall n \in \mathbb{N}\ (a_n \le b_n)\big) \wedge \big(\forall n \in \mathbb{N}\ ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n])\big)\\ &\qquad \to \exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x \in [a_n, b_n])\Big].\end{aligned}$$

Remark (Predicate reading).

$$\text{NestedClosedIntervals}(I_n) \implies \text{NonemptyIntersection}\left(\bigcap_{n=1}^{\infty} I_n\right).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists (a_n) \subseteq \mathbb{R}\ \exists (b_n) \subseteq \mathbb{R}\\ &\Big[\big(\forall n \in \mathbb{N}\ (a_n \le b_n)\big) \wedge \big(\forall n \in \mathbb{N}\ ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n])\big)\\ &\qquad \wedge \forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x \notin [a_n, b_n])\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\text{NestedClosedIntervals}(I_n) \wedge \neg\,\text{NonemptyIntersection}\left(\bigcap_{n=1}^{\infty} I_n\right).$$

Remark (Failure modes). The theorem fails when a nested sequence of nonempty closed real intervals has no common real point.

Remark (Failure mode decomposition).

$$\underbrace{\text{NestedClosedIntervals}(I_n)}_{\text{closed intervals remain nested}} \wedge \underbrace{\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x \notin I_n)}_{\text{no real point lies in every interval}}.$$

Remark (Interpretation). Nested closed intervals cannot keep shrinking or relocating in a way that loses every real point. Nesting keeps the intervals aligned, nonemptiness prevents trivial collapse at each stage, and completeness supplies a real point that survives all stages at once.

Remark (Dependencies).

Axiom 31.1.2.

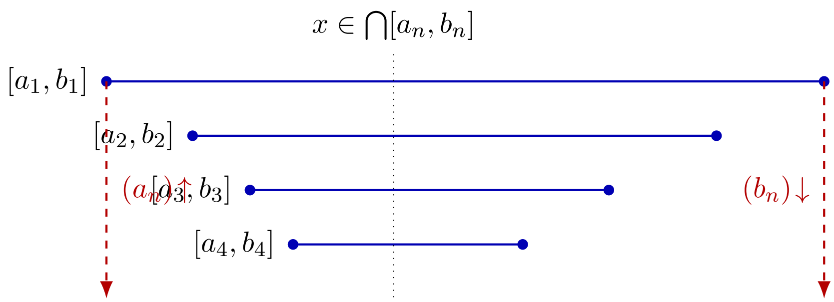


Figure 31.1: Nested intervals $[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$ pinch toward a common point. The sequence of left endpoints is increasing and bounded above; its supremum x lies in every interval.

31.1.2 Archimedean Property

Archimedean Property

Remark (Why this requires proof). The Archimedean Property states that the natural numbers are unbounded in $\mathbb{R}$. That conclusion is not a formal consequence of the ordered-field axioms alone. There exist ordered fields satisfying all ordered-field axioms in which the natural numbers are bounded above, so the statement must be justified from stronger structure.

The needed structure is completeness. If $\mathbb{N}$ were bounded above in $\mathbb{R}$, then the Least Upper Bound Property would produce a supremum $\alpha \in \mathbb{R}$ for $\mathbb{N}$. But then $\alpha - 1$ cannot be an upper bound, so there exists $n \in \mathbb{N}$ with $n > \alpha - 1$. Hence $n + 1 > \alpha$, contradicting the fact that α is an upper bound of $\mathbb{N}$.

Go to proof.

Theorem (Archimedean Property)

Theorem 31.1.4 (Archimedean Property)**.** *For every real number x, there exists a natural number n such that $n > x$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (n > x).$$

Remark (Predicate reading).

$$\text{ArchimedeanProperty}(\mathbb{R}) \iff \forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (n > x).$$

Remark (Negated quantified statement).

$$\exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (n \leq x).$$

Remark (Negation predicate reading).

$$\neg\, \text{ArchimedeanProperty}(\mathbb{R}) \iff \exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (n \leq x).$$

Remark (Failure modes). The theorem fails exactly when the natural numbers are bounded above by some real number.

Remark (Failure mode decomposition).

$$\neg\,\text{ArchimedeanProperty}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R}}_{\text{real upper bound}} \quad \underbrace{\forall n \in \mathbb{N}\,(n \leq x)}_{\text{all natural numbers remain below it}} .$$

Remark (Interpretation). The Archimedean Property says that the natural numbers do not stop below any real ceiling. No real number is so large that every natural number remains below it. In this chapter, the result is a consequence of completeness: a bounded copy of $\mathbb{N}$ would have a supremum, and that supremum is contradicted by shifting one natural number upward.

Remark (Dependencies).

Axiom 31.1.2, Definition 31.2.7.

Corollary (Archimedean Reciprocal Corollary)

Corollary 31.1.5 (Archimedean Reciprocal Corollary)**.** *For every $x > 0$ and every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $nx > y$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}\,\big(x > 0 \rightarrow \exists n \in \mathbb{N}\,(nx > y)\big).$$

Remark (Predicate reading).

$$\text{ArchimedeanReciprocal}(\mathbb{R}) \iff \forall x, y \in \mathbb{R}\,\big(x > 0 \rightarrow \exists n \in \mathbb{N}\,(nx > y)\big).$$

Remark (Negated quantified statement).

$$\exists x, y \in \mathbb{R}\,\big(x > 0 \wedge \forall n \in \mathbb{N}\,(nx \leq y)\big).$$

Remark (Negation predicate reading).

$$\neg\,\text{ArchimedeanReciprocal}(\mathbb{R}) \iff \exists x, y \in \mathbb{R}\,\big(x > 0 \wedge \forall n \in \mathbb{N}\,(nx \leq y)\big).$$

Remark (Failure modes). The corollary fails when a positive real step size has all of its natural multiples bounded above by some real threshold.

Remark (Failure mode decomposition).

$$\neg\,\text{ArchimedeanReciprocal}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R}\,(x > 0)}_{\text{positive step size}} \wedge \underbrace{\exists y \in \mathbb{R}\,\forall n \in \mathbb{N}\,(nx \leq y)}_{\text{all natural multiples stay bounded}} .$$

Remark (Interpretation). Any fixed positive real step eventually passes any real threshold when repeated enough times. The statement is the Archimedean Property applied after rescaling by the positive number x: growth by units becomes growth by steps of size x.

Remark (Dependencies).

Theorem 31.1.4.

31.1.3 Density

Density

Definition (Dense Subset)

Definition 31.1.6 (Dense Subset)**.** Let $D \subseteq S \subseteq \mathbb{R}$. The set D is *dense in* S if and only if for every $x \in S$ and every $\varepsilon > 0$ there exists $d \in D$ such that $|x - d| < \varepsilon$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } D \subseteq S \subseteq \mathbb{R}.\\ &D \text{ is dense in } S \iff \forall x \in S\ \forall \varepsilon > 0\ \exists d \in D\ \big(|x - d| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\operatorname{DenseSubset}(D, S) :\equiv \forall x \in S\ \forall \varepsilon > 0\ \exists d \in D\ \big(|x - d| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } D \subseteq S \subseteq \mathbb{R}.\\ &D \text{ is not dense in } S \iff \exists x \in S\ \exists \varepsilon_0 > 0\ \forall d \in D\ \big(|x - d| \geq \varepsilon_0\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \operatorname{DenseSubset}(D, S) \iff \exists x \in S\ \exists \varepsilon_0 > 0\ \forall d \in D\ \big(|x - d| \geq \varepsilon_0\big).$$

Remark (Failure modes). Density fails when some point of S has a positive-radius neighborhood that contains no point of D.

Remark (Failure mode decomposition).

$$\neg \operatorname{DenseSubset}(D, S) = \underbrace{\exists x \in S}_{\text{point not approximated}} \quad \underbrace{\exists \varepsilon_0 > 0}_{\text{positive exclusion radius}} \quad \underbrace{\forall d \in D\ \big(|x - d| \geq \varepsilon_0\big)}_{\text{no point of } D \text{ enters the neighborhood}}.$$

Remark (Interpretation). Density is an approximation property inside S. It does not say that every point of S already belongs to D; it says that points of D occur arbitrarily close to every point of S. Thus D can be a proper subset of S while still leaving no positive-width gap inside S.

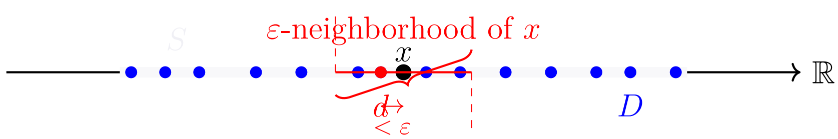


Figure. Illustration of a Dense Subset $D \subseteq S$. Given any $x \in S$ (black dot) and any $\varepsilon > 0$ (red interval width), there exists $d \in D$ (red-filled blue dot) such that $|x - d| < \varepsilon$.

Definition (Order Dense Subset)

Definition 31.1.7 (Order Dense Subset)**.** Let X be a linearly ordered set and let $A \subseteq X$. The subset A is *order dense in* X if and only if for every $a, b \in X$ with $a < b$ there exists $c \in A$ such that $a < c < b$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } X \text{ be a linearly ordered set and } A \subseteq X.\\ &A \text{ is order dense in } X \iff \forall a, b \in X \ \big(a < b \to \exists c \in A \ (a < c < b)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\operatorname{OrderDenseSubset}(A, X) :\!= \forall a, b \in X \ \big(a < b \to \exists c \in A \ (a < c < b)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } X \text{ be a linearly ordered set and } A \subseteq X.\\ &A \text{ is not order dense in } X \iff \exists a, b \in X \ \big(a < b \land \forall c \in A \ (c \leq a \lor b \leq c)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \operatorname{OrderDenseSubset}(A, X) \iff \exists a, b \in X \ \big(a < b \land \forall c \in A \ (c \leq a \lor b \leq c)\big).$$

Remark (Failure modes). Order density fails when there is a nonempty open interval of X that contains no point of A.

Remark (Failure mode decomposition).

$$\neg \operatorname{OrderDenseSubset}(A, X) = \underbrace{\exists a, b \in X \ (a < b)}_{\text{nonempty interval}} \land \underbrace{\forall c \in A \ (c \leq a \lor b \leq c)}_{\text{no point of } A \text{ lies between } a \text{ and } b}.$$

Remark (Interpretation). Order density is an interval-penetration property. It ignores metric distance and asks only whether every ordered gap contains a point of the subset. A subset can therefore be order dense in X precisely when no nonempty open interval of X is missed entirely by that subset.

Remark (Dependencies).

- Total Order
- Subset

Theorem (Density of the Rational Numbers in $\mathbb{R}$)

Theorem 31.1.8 (Density of the Rational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \ \big(a < b \to \exists q \in \mathbb{Q} \ (a < q < b)\big).$$

Remark (Predicate reading).

$$\text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R} \; \big(a < b \to \exists q \in \mathbb{Q} \; (a < q < b)\big).$$

Remark (Negated quantified statement).

$$\exists a, b \in \mathbb{R} \; \big(a < b \land \forall q \in \mathbb{Q} \; (q \leq a \lor b \leq q)\big).$$

Remark (Negation predicate reading).

$$\neg \, \text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R} \; \big(a < b \land \forall q \in \mathbb{Q} \; (q \leq a \lor b \leq q)\big).$$

Remark (Failure modes). The theorem would fail precisely if some nonempty real interval contained no rational number.

Remark (Failure mode decomposition).

$$\neg \, \text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) = \underbrace{\exists a, b \in \mathbb{R} \; (a < b)}_{\text{nonempty real interval}} \land \underbrace{\forall q \in \mathbb{Q} \; (q \leq a \lor b \leq q)}_{\text{no rational lies inside}}.$$

Remark (Interpretation). Rational numbers penetrate every real interval. The theorem does not say that every real number is rational; it says that rational numbers are never separated from a real interval by a positive order gap. This is the interval form of rational density in the real line.

Remark (Dependencies).

Definition 31.1.7, Theorem 31.1.4.

Theorem (Density of the Irrational Numbers in $\mathbb{R}$)

Theorem 31.1.9 (Density of the Irrational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \; \big(a < b \to \exists s \in \mathbb{R} \setminus \mathbb{Q} \; (a < s < b)\big).$$

Remark (Predicate reading).

$$\text{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R} \; \big(a < b \to \exists s \in \mathbb{R} \setminus \mathbb{Q} \; (a < s < b)\big).$$

Remark (Negated quantified statement).

$$\exists a, b \in \mathbb{R} \; \big(a < b \land \forall s \in \mathbb{R} \setminus \mathbb{Q} \; (s \leq a \lor b \leq s)\big).$$

Remark (Negation predicate reading).

$$\neg \, \text{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R} \; \big(a < b \land \forall s \in \mathbb{R} \setminus \mathbb{Q} \; (s \leq a \lor b \leq s)\big).$$

Remark (Failure modes). The theorem would fail precisely if some nonempty real interval contained no irrational number.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{R}\setminus\mathbb{Q},\mathbb{R}) = \underbrace{\exists a,b\in\mathbb{R}\ (a<b)}_{\text{nonempty real interval}} \ \wedge\ \underbrace{\forall s\in\mathbb{R}\setminus\mathbb{Q}\ (s\le a\vee b\le s)}_{\text{no irrational lies inside}}.$$

Remark (Interpretation). Irrational numbers also penetrate every real interval. One way to see the result is to insert a rational point into a translated and rescaled interval, then shift it by an irrational amount. The outcome is that the irrational numbers, like the rational numbers, leave no interval gap in $\mathbb{R}$.

Remark (Dependencies).

Definition 31.1.7, Theorem 31.1.8.

Corollary (Rational and Irrational Points in Every Open Interval)

Corollary 31.1.10 (Rational and Irrational Points in Every Open Interval)**.** *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number. Go to proof.*

Remark (Standard quantified statement).

$$\forall a,b\in\mathbb{R}\ \Big[a<b\to\big(\exists q\in\mathbb{Q}\ (a<q<b)\wedge\exists s\in\mathbb{R}\setminus\mathbb{Q}\ (a<s<b)\big)\Big].$$

Remark (Predicate reading).

$$\mathrm{OrderDenseSubset}(\mathbb{Q},\mathbb{R})\wedge\mathrm{OrderDenseSubset}(\mathbb{R}\setminus\mathbb{Q},\mathbb{R}).$$

Remark (Negated quantified statement).

$$\exists a,b\in\mathbb{R}\ \Big[a<b\wedge\big(\forall q\in\mathbb{Q}\ (q\le a\vee b\le q)\vee\forall s\in\mathbb{R}\setminus\mathbb{Q}\ (s\le a\vee b\le s)\big)\Big].$$

Remark (Negation predicate reading).

$$\neg\Big[\mathrm{OrderDenseSubset}(\mathbb{Q},\mathbb{R})\wedge\mathrm{OrderDenseSubset}(\mathbb{R}\setminus\mathbb{Q},\mathbb{R})\Big].$$

Remark (Failure modes). The corollary fails if a nonempty real interval misses all rational points or misses all irrational points.

Remark (Failure mode decomposition).

$$\exists a,b\in\mathbb{R}\ (a<b)\ \wedge\ \Big[\underbrace{\forall q\in\mathbb{Q}\ (q\le a\vee b\le q)}_{\text{no rational point}}\ \vee\ \underbrace{\forall s\in\mathbb{R}\setminus\mathbb{Q}\ (s\le a\vee b\le s)}_{\text{no irrational point}}\Big].$$

Remark (Interpretation). The corollary combines the two density theorems. Every real interval, no matter how small, contains representatives of both arithmetic types. Thus neither the rational numbers nor the irrational numbers occupy isolated regions of the real line.

Remark (Dependencies).

Theorem 31.1.8, Theorem 31.1.9.

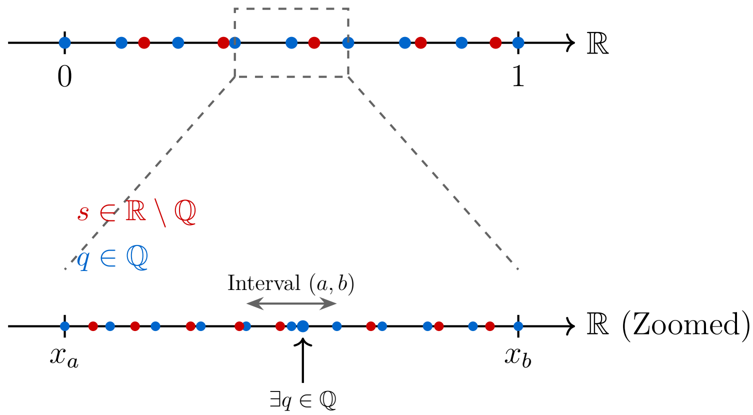


Figure 31.2: Visualization of the Density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$. Zooming into any interval reveals that both sets are always present.

Remark (Zoom invariance and structural consequences).
Density in $\mathbb{R}$ has a scale-invariant consequence: no matter how small a nonempty open interval becomes, it still contains both rational and irrational points. The interleaving of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ is therefore not a large-scale phenomenon. It persists at every interval scale.

This has three important consequences. First, the continuum does not become locally discrete under magnification. For every nonempty open interval (a, b), both

$$\mathbb{Q} \cap (a, b) \qquad \text{and} \qquad (\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)$$

remain nonempty, and in fact infinite. Density is therefore stable under repeated restriction to smaller intervals.

Second, this behavior sharply distinguishes the real line from discrete numerical models. Floating-point systems form a discrete set of representable values, so repeated magnification eventually reaches a scale at which adjacent representable numbers leave no further point strictly between them. At that stage, interval penetration fails in the discrete model even though it continues to hold in $\mathbb{R}$.

Third, the Archimedean Property supplies the scale mechanism behind this behavior. Given any positive scale ε, there exists $n \in \mathbb{N}$ with

$$\frac{1}{n} < \varepsilon.$$

This makes it possible to construct rational approximations at arbitrarily small interval scales, which is one of the basic engines behind density arguments.

The resulting contrast is structural. Smooth mathematical models are built in a setting where interval penetration persists at every scale, whereas finite computational models eventually encounter a smallest representable separation. The transition from continuum behavior to discrete behavior occurs exactly where nonempty intervals in the numerical model cease to contain further
representable points.

Density Dependency Map

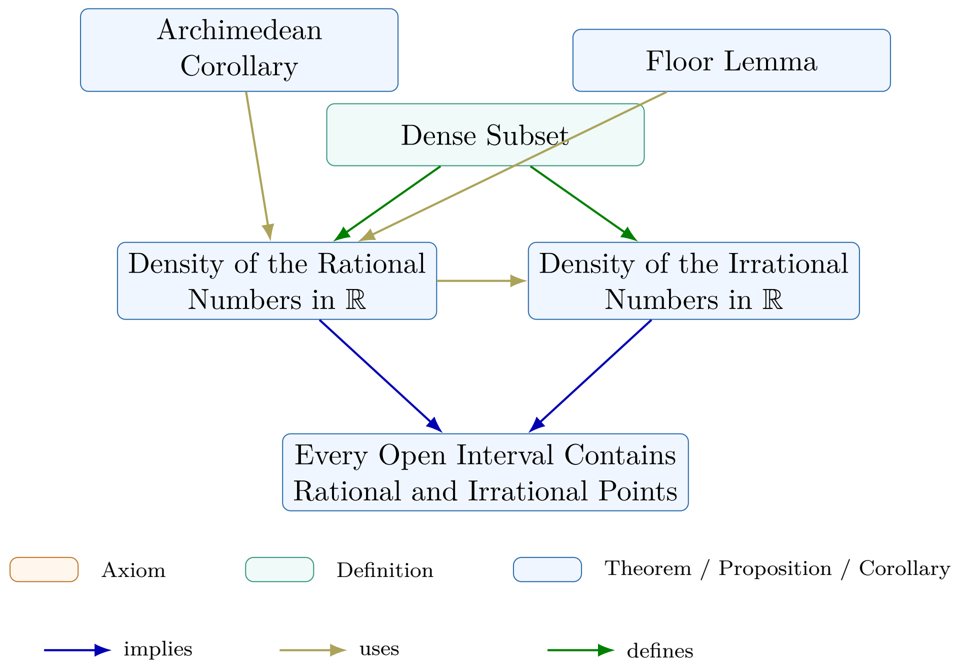


Figure 31.3: Dependency map for the density subchapter. The Floor Lemma and the Archimedean Corollary support the proof of the density of the rational numbers in $\mathbb{R}$. The irrational density theorem is then derived from the rational density result, and the final corollary combines both density theorems into a single interval statement. The definition records the local vocabulary of the subchapter.

31.1.4 Completeness Summary

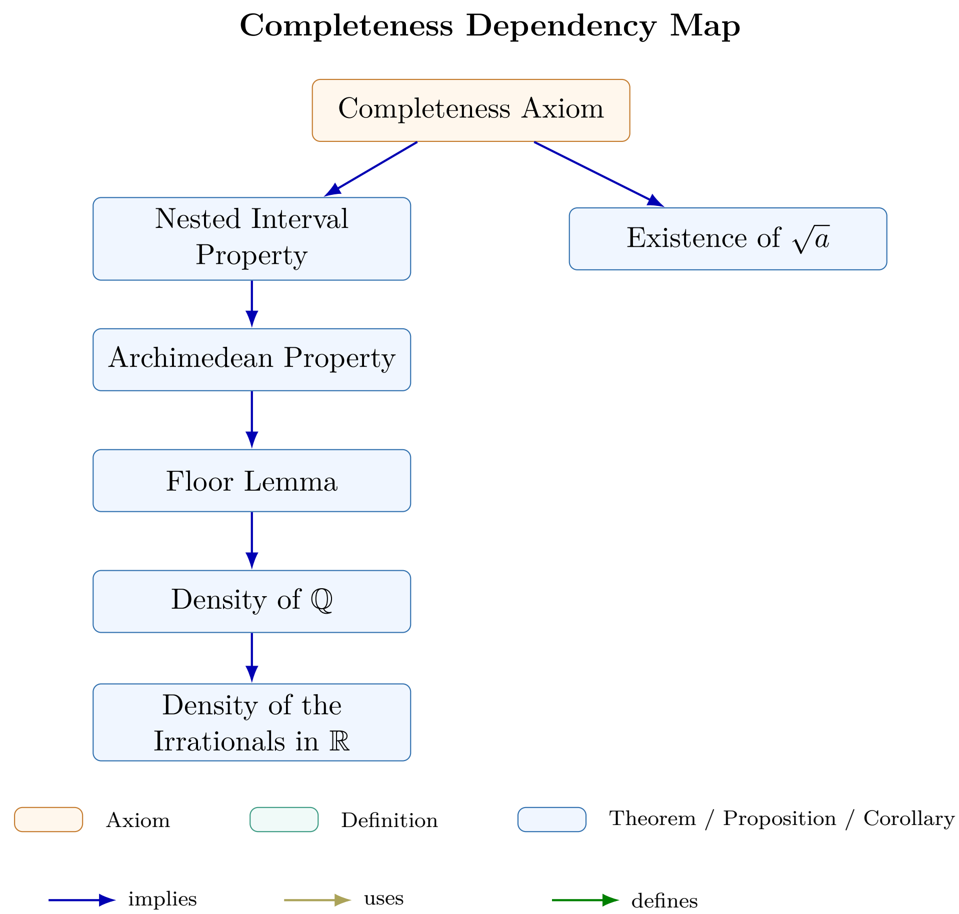


Figure 31.4: Dependency map for the completeness subchapter. The Completeness Axiom yields the Nested Interval Property and the existence of $\sqrt{a}$; the Nested Interval Property then supports the Archimedean Property, the Floor Lemma, and the density results for $\mathbb{Q}$ and for the irrational numbers.

Remark (Connections forward). Completeness appears in every major limit theorem in the notes. The Monotone Convergence Theorem (convergence chapter) constructs the limit of a bounded monotone sequence as $\sup\{a_n : n \in \mathbb{N}\}$; existence of this supremum requires completeness. The Bolzano–Weierstrass Theorem applies the Nested Interval Property directly to extract a convergent subsequence from any bounded sequence. The Cauchy Criterion identifies convergent sequences purely in terms of their own spacing, without naming the limit in advance; its proof constructs the limit as a supremum and again requires completeness. In metric spaces (Distance chapter), completeness in the sense of Cauchy sequences is the natural generalization of the same idea to abstract settings. Every ε-δ proof that locates a limit or extremum is drawing on this chapter's infrastructure.

31.2 Bounds and Extremal Values

31.2.1 Upper and Lower Bounds

Bounds are the first level of extremal information. They compare one candidate element with every element of a nonempty set, but they do not require the candidate to be sharp or to belong to the set.

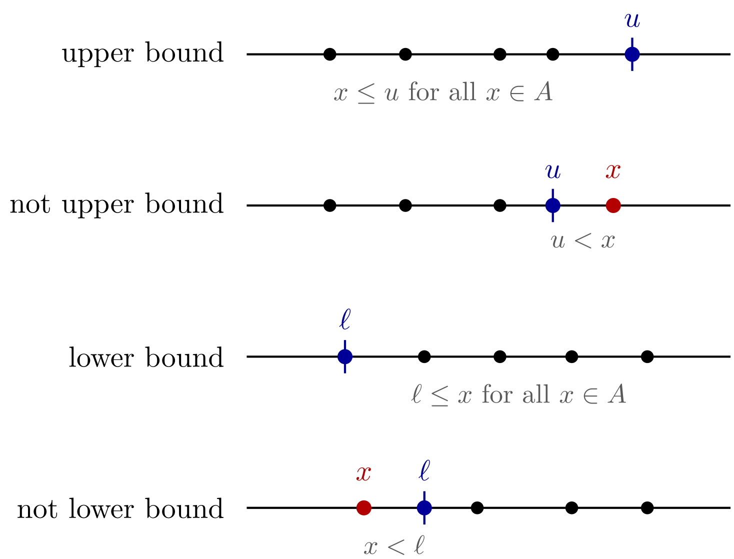


Figure 31.5: Upper and lower bound conditions on a nonempty set.

Bound

Definition 31.2.1 (Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $b \in S$. A *bound* of A means a one-sided comparison between b and every element of A: either $x \leq b$ for all $x \in A$, or $b \leq x$ for all $x \in A$.

Remark (Standard quantified statement)*.*

$$A \neq \varnothing \quad \text{and} \quad \big[(\forall x \in A)(x \leq b) \text{ or } (\forall x \in A)(b \leq x)\big].$$

Remark (Definition predicate reading)*.*

$$\mathrm{Bound}_{S,\leq}(b, A) := \mathrm{UB}_{S,\leq}(b, A) \vee \mathrm{LB}_{S,\leq}(b, A).$$

Remark (Negated quantified statement)*.*

$$A = \varnothing \quad \text{or} \quad \big[(\exists x \in A)(b < x) \text{ and } (\exists y \in A)(y < b)\big].$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{Bound}_{S,\leq}(b, A) \iff A = \varnothing \vee \big(\neg\,\mathrm{UB}_{S,\leq}(b, A) \wedge \neg\,\mathrm{LB}_{S,\leq}(b, A)\big).$$

Remark (Failure modes). The statement fails if the set is empty, or if b lies strictly between two elements of A.

Remark (Failure mode decomposition). Failure of the upper-bound side means some element of A is strictly larger than b. Failure of the lower-bound side means some element of A is strictly smaller than b. Failure to be any bound requires both witnesses.

Upper Bound

Definition 31.2.2 (Upper Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $u \in S$. The element u is an *upper bound* of A if every element of A is less than or equal to u.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad (\forall x \in A)(x \leq u).$$

Remark (Definition predicate reading).

$$\mathrm{UB}_{S,\leq}(u, A) := A \neq \varnothing \wedge (\forall x \in A)(x \leq u).$$

Remark (Negated quantified statement).

$$A = \varnothing \quad \text{or} \quad (\exists x \in A)(u < x).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{UB}_{S,\leq}(u, A) \iff A = \varnothing \vee (\exists x \in A)(u < x).$$

Remark (Failure modes). The set may be empty, or a counterexample element of A may sit strictly above the proposed bound.

Remark (Failure mode decomposition). To disprove that u is an upper bound, produce either the structural failure $A = \varnothing$ or a witness $x \in A$ with $u < x$.

Lower Bound

Definition 31.2.3 (Lower Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $\ell \in S$. The element ℓ is a *lower bound* of A if every element of A is greater than or equal to ℓ.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad (\forall x \in A)(\ell \leq x).$$

Remark (Definition predicate reading).

$$\mathrm{LB}_{S,\leq}(\ell, A) := A \neq \varnothing \wedge (\forall x \in A)(\ell \leq x).$$

Remark (Negated quantified statement).

$$A = \varnothing \quad \text{or} \quad (\exists x \in A)(x < \ell).$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{LB}_{S,\le}(\ell, A) \iff A = \varnothing \vee (\exists x \in A)(x < \ell).$$

Remark (Failure modes). The set may be empty, or a counterexample element of A may sit strictly below the proposed bound.

Remark (Failure mode decomposition). To disprove that ℓ is a lower bound, produce either the structural failure $A = \varnothing$ or a witness $x \in A$ with $x < \ell$.

Bounded Above

Definition 31.2.4 (Bounded Above)**.** Let $(S, \le)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded above* if it has at least one upper bound in S.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad (\exists u \in S)(\forall x \in A)(x \le u).$$

Remark (Definition predicate reading).

$$\mathrm{BddAbove}_{S,\le}(A) := (\exists u \in S)\, \mathrm{UB}_{S,\le}(u, A).$$

Remark (Negated quantified statement).

$$A = \varnothing \quad \text{or} \quad (\forall u \in S)(\exists x \in A)(u < x).$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{BddAbove}_{S,\le}(A) \iff A = \varnothing \vee (\forall u \in S) \neg\, \mathrm{UB}_{S,\le}(u, A).$$

Remark (Failure modes). The set may be empty, or every proposed upper bound may be defeated by some larger element of A.

Remark (Failure mode decomposition). Unboundedness above is not one failed candidate; it is a uniform procedure that defeats every candidate $u \in S$.

Bounded Below

Definition 31.2.5 (Bounded Below)**.** Let $(S, \le)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded below* if it has at least one lower bound in S.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad (\exists \ell \in S)(\forall x \in A)(\ell \le x).$$

Remark (Definition predicate reading).

$$\mathrm{BddBelow}_{S,\le}(A) := (\exists \ell \in S)\, \mathrm{LB}_{S,\le}(\ell, A).$$

Remark (Negated quantified statement).

$$A = \varnothing \quad \text{or} \quad (\forall \ell \in S)(\exists x \in A)(x < \ell).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BddBelow}_{S,\leq}(A) \iff A = \varnothing \lor (\forall \ell \in S)\neg\,\mathrm{LB}_{S,\leq}(\ell, A).$$

Remark (Failure modes). The set may be empty, or every proposed lower bound may be defeated by some smaller element of A.

Remark (Failure mode decomposition). Unboundedness below is a uniform defeat of all lower-bound candidates.

Bounded Set

Definition 31.2.6 (Bounded Set)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded* if it is bounded above and bounded below.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad (\exists \ell, u \in S)(\forall x \in A)(\ell \leq x \leq u).$$

Remark (Definition predicate reading).

$$\mathrm{Bdd}_{S,\leq}(A) := \mathrm{BddBelow}_{S,\leq}(A) \land \mathrm{BddAbove}_{S,\leq}(A).$$

Remark (Negated quantified statement).

$$A = \varnothing \quad \text{or} \quad (\forall \ell, u \in S)(\exists x \in A)(x < \ell \text{ or } u < x).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Bdd}_{S,\leq}(A) \iff A = \varnothing \lor \neg\,\mathrm{BddBelow}_{S,\leq}(A) \lor \neg\,\mathrm{BddAbove}_{S,\leq}(A).$$

Remark (Failure modes). The set may be empty, unbounded below, unbounded above, or both.

Remark (Failure mode decomposition). Boundedness has two independent one-sided requirements. A failure on either side is enough to destroy boundedness.

31.2.2 Suprema and Infima

Suprema and infima are sharp bounds. A supremum is the least upper bound; an infimum is the greatest lower bound. The set remains required to be nonempty throughout this section.

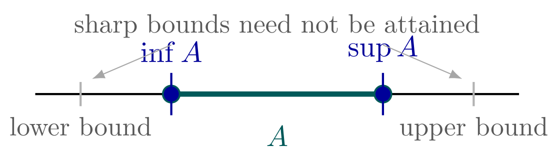


Figure 31.6: Suprema and infima are sharp bounds, whether or not they are elements of the set.

Supremum

Definition 31.2.7 (Supremum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $s \in S$. The element s is the *supremum* of A if s is an upper bound of A and no smaller element of S is an upper bound of A.

Remark (Standard quantified statement)*.*

$$A \neq \varnothing \quad \text{and} \quad (\forall x \in A)(x \leq s) \quad \text{and} \quad (\forall u \in S)\big[(\forall x \in A)(x \leq u) \Rightarrow s \leq u\big].$$

Remark (Definition predicate reading)*.*

$$\mathrm{Sup}_{S,\leq}(s, A) := \mathrm{UB}_{S,\leq}(s, A) \wedge (\forall u \in S)\big(\mathrm{UB}_{S,\leq}(u, A) \Rightarrow s \leq u\big).$$

Remark (Negated quantified statement)*.*

$$A = \varnothing \quad \text{or} \quad (\exists x \in A)(s < x) \quad \text{or} \quad (\exists u \in S)\big[(\forall x \in A)(x \leq u) \text{ and } u < s\big].$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{Sup}_{S,\leq}(s, A) \iff A = \varnothing \vee \neg\,\mathrm{UB}_{S,\leq}(s, A) \vee (\exists u \in S)\big(\mathrm{UB}_{S,\leq}(u, A) \wedge u < s\big).$$

Remark (Failure modes)*.* The set may be empty; s may fail to be an upper bound; or s may be an upper bound that is not least among upper bounds.

Remark (Failure mode decomposition)*.* The first clause is a domain failure. The second is a bound failure. The third is a sharpness failure witnessed by a strictly smaller upper bound.

Infimum

Definition 31.2.8 (Infimum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $i \in S$. The element i is the *infimum* of A if i is a lower bound of A and no larger element of S is a lower bound of A.

Remark (Standard quantified statement)*.*

$$A \neq \varnothing \quad \text{and} \quad (\forall x \in A)(i \leq x) \quad \text{and} \quad (\forall \ell \in S)\big[(\forall x \in A)(\ell \leq x) \Rightarrow \ell \leq i\big].$$

Remark (Definition predicate reading)*.*

$$\mathrm{Inf}_{S,\leq}(i, A) := \mathrm{LB}_{S,\leq}(i, A) \wedge (\forall \ell \in S)\big(\mathrm{LB}_{S,\leq}(\ell, A) \Rightarrow \ell \leq i\big).$$

Remark (Negated quantified statement)*.*

$$A = \varnothing \quad \text{or} \quad (\exists x \in A)(x < i) \quad \text{or} \quad (\exists \ell \in S)\big[(\forall x \in A)(\ell \leq x) \text{ and } i < \ell\big].$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{Inf}_{S,\leq}(i, A) \iff A = \varnothing \vee \neg\,\mathrm{LB}_{S,\leq}(i, A) \vee (\exists \ell \in S)\big(\mathrm{LB}_{S,\leq}(\ell, A) \wedge i < \ell\big).$$

Remark (Failure modes)*.* The set may be empty; i may fail to be a lower bound; or i may be a lower bound that is not greatest among lower bounds.

Remark (Failure mode decomposition). The first clause is a domain failure. The second is a bound failure. The third is a sharpness failure witnessed by a strictly larger lower bound.

Uniqueness of the Supremum

Proposition 31.2.9 (Uniqueness of the Supremum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s, t \in S$ are both suprema of A, then $s = t$.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Sup}_{S,\leq}(s, A) \quad \text{and} \quad \mathrm{Sup}_{S,\leq}(t, A) \quad \Longrightarrow \quad s = t.$$

Remark (Predicate reading).

$$\mathrm{Sup}_{S,\leq}(s, A) \wedge \mathrm{Sup}_{S,\leq}(t, A) \Longrightarrow s = t.$$

Remark (Negated quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Sup}_{S,\leq}(s, A) \quad \text{and} \quad \mathrm{Sup}_{S,\leq}(t, A) \quad \text{and} \quad s \neq t.$$

Remark (Negation predicate reading).

$$\mathrm{Sup}_{S,\leq}(s, A) \wedge \mathrm{Sup}_{S,\leq}(t, A) \wedge s \neq t.$$

Remark (Failure modes). Failure would require two distinct least upper bounds for the same nonempty set.

Remark (Failure mode decomposition). Each candidate is an upper bound and each lies below every upper bound; applying these minimality clauses to the other candidate gives both $s \leq t$ and $t \leq s$.

Uniqueness of the Infimum

Proposition 31.2.10 (Uniqueness of the Infimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i, j \in S$ are both infima of A, then $i = j$.*

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Inf}_{S,\leq}(i, A) \quad \text{and} \quad \mathrm{Inf}_{S,\leq}(j, A) \quad \Longrightarrow \quad i = j.$$

Remark (Predicate reading).

$$\mathrm{Inf}_{S,\leq}(i, A) \wedge \mathrm{Inf}_{S,\leq}(j, A) \Longrightarrow i = j.$$

Remark (Negated quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Inf}_{S,\leq}(i, A) \quad \text{and} \quad \mathrm{Inf}_{S,\leq}(j, A) \quad \text{and} \quad i \neq j.$$

Remark (Negation predicate reading).

$$\mathrm{Inf}_{S,\leq}(i, A) \wedge \mathrm{Inf}_{S,\leq}(j, A) \wedge i \neq j.$$

Remark (Failure modes). Failure would require two distinct greatest lower bounds for the same nonempty set.

Remark (Failure mode decomposition). Each candidate is a lower bound and each lies above every lower bound; applying these maximality clauses to the other candidate gives both $i \leq j$ and $j \leq i$.

Upper Bound Property of the Supremum

Proposition 31.2.11 (Upper Bound Property of the Supremum). *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $s = \sup A$ exists in S. Then every element of A lies below s:*

$$(\forall x \in A)(x \leq s).$$

Go to proof.

Remark (Predicate reading).

$$\mathrm{Sup}_{S,\leq}(s, A) \Longrightarrow \mathrm{UB}_{S,\leq}(s, A).$$

Remark (Negated quantified statement).

$$\mathrm{Sup}_{S,\leq}(s, A) \quad \text{and} \quad (\exists x \in A)(s < x).$$

Remark (Failure modes). Failure would mean that a supremum is not an upper bound.

Remark (Failure mode decomposition). This is exactly the first clause in the definition of supremum.

Lower Bound Property of the Infimum

Proposition 31.2.12 (Lower Bound Property of the Infimum). *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $i = \inf A$ exists in S. Then every element of A lies above i:*

$$(\forall x \in A)(i \leq x).$$

Remark (Predicate reading).

$$\mathrm{Inf}_{S,\leq}(i, A) \Longrightarrow \mathrm{LB}_{S,\leq}(i, A).$$

Remark (Negated quantified statement).

$$\mathrm{Inf}_{S,\leq}(i, A) \quad \text{and} \quad (\exists x \in A)(x < i).$$

Remark (Failure modes). Failure would mean that an infimum is not a lower bound.

Remark (Failure mode decomposition). This is exactly the first clause in the definition of infimum.

Subset Inclusion Preserves Upper Bounds

Lemma 31.2.13 (Subset Inclusion Preserves Upper Bounds). *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If u is an upper bound of B, then u is an upper bound of A.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad A \subseteq B, \quad (\forall y \in B)(y \leq u) \quad \Longrightarrow \quad (\forall x \in A)(x \leq u).$$

Remark (Predicate reading).

$$A \subseteq B \wedge \mathrm{UB}_{S,\leq}(u, B) \Longrightarrow \mathrm{UB}_{S,\leq}(u, A).$$

Remark (Negated quantified statement).

$$A \subseteq B, \quad \mathrm{UB}_{S,\leq}(u, B), \quad \text{and} \quad (\exists x \in A)(u < x).$$

Remark (Failure modes). Failure would require a counterexample in the smaller set even though the larger set is already bounded by u.

Remark (Failure mode decomposition). Any counterexample $x \in A$ is also an element of B, contradicting the upper-bound hypothesis for B.

Subset Inclusion Preserves Lower Bounds

Lemma 31.2.14 (Subset Inclusion Preserves Lower Bounds)**.** *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If ℓ is a lower bound of B, then ℓ is a lower bound of A.*

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad A \subseteq B, \quad (\forall y \in B)(\ell \leq y) \quad \Longrightarrow \quad (\forall x \in A)(\ell \leq x).$$

Remark (Predicate reading).

$$A \subseteq B \wedge \mathrm{LB}_{S,\leq}(\ell, B) \Longrightarrow \mathrm{LB}_{S,\leq}(\ell, A).$$

Remark (Negated quantified statement).

$$A \subseteq B, \quad \mathrm{LB}_{S,\leq}(\ell, B), \quad \text{and} \quad (\exists x \in A)(x < \ell).$$

Remark (Failure modes). Failure would require a smaller-set counterexample to a lower bound that already works on the larger set.

Remark (Failure mode decomposition). Any counterexample $x \in A$ lies in B, contradicting the lower-bound hypothesis for B.

Supremum Is Monotone Under Inclusion

Theorem 31.2.15 (Supremum Is Monotone Under Inclusion)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If $A \subseteq B$, then $\sup A \leq \sup B$.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad B \neq \varnothing, \quad A \subseteq B, \quad \mathrm{Sup}_{\mathbb{R},\leq}(s_A, A), \quad \mathrm{Sup}_{\mathbb{R},\leq}(s_B, B) \quad \Longrightarrow \quad s_A \leq s_B.$$

Remark (Predicate reading).

$$A \subseteq B \wedge \mathrm{Sup}(s_A, A) \wedge \mathrm{Sup}(s_B, B) \Longrightarrow s_A \leq s_B.$$

Remark (Negated quantified statement).

$$A \subseteq B, \quad \mathrm{Sup}(s_A, A), \quad \mathrm{Sup}(s_B, B), \quad s_B < s_A.$$

Remark (Failure modes). Failure would mean enlarging a nonempty set lowers its least upper bound.

Remark (Failure mode decomposition). The value $\sup B$ bounds B, hence bounds A; since $\sup A$ is least among upper bounds of A, $\sup A \leq \sup B$.

Infimum Is Monotone Under Inclusion

Theorem 31.2.16 (Infimum Is Monotone Under Inclusion)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. If $A \subseteq B$, then $\inf B \leq \inf A$.*

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad B \neq \varnothing, \quad A \subseteq B, \quad \mathrm{Inf}_{\mathbb{R},\leq}(i_A, A), \quad \mathrm{Inf}_{\mathbb{R},\leq}(i_B, B) \quad \Longrightarrow \quad i_B \leq i_A.$$

Remark (Predicate reading).

$$A \subseteq B \wedge \mathrm{Inf}(i_A, A) \wedge \mathrm{Inf}(i_B, B) \Longrightarrow i_B \leq i_A.$$

Remark (Negated quantified statement).

$$A \subseteq B, \quad \mathrm{Inf}(i_A, A), \quad \mathrm{Inf}(i_B, B), \quad i_A < i_B.$$

Remark (Failure modes). Failure would mean enlarging a nonempty set raises its greatest lower bound.

Remark (Failure mode decomposition). The value $\inf B$ bounds B below, hence bounds A below; since $\inf A$ is greatest among lower bounds of A, $\inf B \leq \inf A$.

Upper Bound Comparison with the Supremum

Proposition 31.2.17 (Upper Bound Comparison with the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. For $u \in \mathbb{R}$, the number u is an upper bound of A if and only if $s \leq u$.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad \mathrm{Sup}(s, A) \quad \Longrightarrow \quad (\forall u \in \mathbb{R})\big[\mathrm{UB}(u, A) \Longleftrightarrow s \leq u\big].$$

Remark (Predicate reading).

$$\mathrm{Sup}(s, A) \Longrightarrow \forall u\,(\mathrm{UB}(u, A) \leftrightarrow s \leq u).$$

Remark (Negated quantified statement).

$$\operatorname{Sup}(s, A) \quad \text{and} \quad (\exists u)\big[(\operatorname{UB}(u, A) \wedge u < s) \vee (s \leq u \wedge \neg \operatorname{UB}(u, A))\big].$$

Remark (Failure modes). Failure would mean an upper bound below the supremum, or a number above the supremum that fails to bound the set.

Remark (Failure mode decomposition). Leastness gives the forward implication. The fact that s bounds A gives the reverse implication by transitivity.

Lower Bound Comparison with the Infimum

Proposition 31.2.18 (Lower Bound Comparison with the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. For $\ell \in \mathbb{R}$, the number ℓ is a lower bound of A if and only if $\ell \leq i$.*

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad \operatorname{Inf}(i, A) \quad \Longrightarrow \quad (\forall \ell \in \mathbb{R})\big[\operatorname{LB}(\ell, A) \Longleftrightarrow \ell \leq i\big].$$

Remark (Predicate reading).

$$\operatorname{Inf}(i, A) \Longrightarrow \forall \ell\, (\operatorname{LB}(\ell, A) \leftrightarrow \ell \leq i).$$

Remark (Negated quantified statement).

$$\operatorname{Inf}(i, A) \quad \text{and} \quad (\exists \ell)\big[(\operatorname{LB}(\ell, A) \wedge i < \ell) \vee (\ell \leq i \wedge \neg \operatorname{LB}(\ell, A))\big].$$

Remark (Failure modes). Failure would mean a lower bound above the infimum, or a number below the infimum that fails to bound the set below.

Remark (Failure mode decomposition). Greatestness gives the forward implication. The fact that i bounds A below gives the reverse implication by transitivity.

Every Element Lies Below the Supremum

Proposition 31.2.19 (Every Element Lies Below the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. Then $x \leq s$ for every $x \in A$.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad \operatorname{Sup}(s, A) \quad \Longrightarrow \quad (\forall x \in A)(x \leq s).$$

Remark (Predicate reading).

$$\operatorname{Sup}(s, A) \Longrightarrow \operatorname{UB}(s, A).$$

Remark (Negated quantified statement).

$$\operatorname{Sup}(s, A) \quad \text{and} \quad (\exists x \in A)(s < x).$$

Remark (Failure modes). Failure would mean the supremum is not an upper bound.

Remark (Failure mode decomposition). This is the upper-bound clause in the definition of supremum.

Every Element Lies Above the Infimum

Proposition 31.2.20 (Every Element Lies Above the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. Then $i \leq x$ for every $x \in A$.*

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad \mathrm{Inf}(i, A) \quad \Longrightarrow \quad (\forall x \in A)(i \leq x).$$

Remark (Predicate reading).

$$\mathrm{Inf}(i, A) \Longrightarrow \mathrm{LB}(i, A).$$

Remark (Negated quantified statement).

$$\mathrm{Inf}(i, A) \quad \text{and} \quad (\exists x \in A)(x < i).$$

Remark (Failure modes). Failure would mean the infimum is not a lower bound.

Remark (Failure mode decomposition). This is the lower-bound clause in the definition of infimum.

Infimum Is Less Than or Equal to the Supremum

Theorem 31.2.21 (Infimum Is Less Than or Equal to the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above. Then* $\inf A \leq \sup A$.
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad \mathrm{Inf}(i, A), \quad \mathrm{Sup}(s, A) \quad \Longrightarrow \quad i \leq s.$$

Remark (Predicate reading).

$$\mathrm{Inf}(i, A) \wedge \mathrm{Sup}(s, A) \Longrightarrow i \leq s.$$

Remark (Negated quantified statement).

$$\mathrm{Inf}(i, A), \quad \mathrm{Sup}(s, A), \quad \text{and} \quad s < i.$$

Remark (Failure modes). Failure would put the greatest lower bound strictly above the least upper bound of the same nonempty set.

Remark (Failure mode decomposition). Choose $a \in A$. Then $i \leq a$ because i is a lower bound, and $a \leq s$ because s is an upper bound.

Order Comparison of Suprema

Proposition 31.2.22 (Order Comparison of Suprema)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If every $x \in A$ is below some $y \in B$, then $\sup A \le \sup B$.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad B \neq \varnothing, \quad (\forall x \in A)(\exists y \in B)(x \le y) \quad \Longrightarrow \quad \sup A \le \sup B.$$

Remark (Predicate reading).

$$[\forall x \in A\, \exists y \in B\, (x \le y)] \wedge \mathrm{Sup}(s_A, A) \wedge \mathrm{Sup}(s_B, B) \Longrightarrow s_A \le s_B.$$

Remark (Negated quantified statement).

$$[\forall x \in A\, \exists y \in B\, (x \le y)] \quad \text{and} \quad \sup B < \sup A.$$

Remark (Failure modes). Failure would mean A reaches higher in supremum than B even though each element of A is dominated by an element of B.

Remark (Failure mode decomposition). Every element of B lies below $\sup B$, so every element of A also lies below $\sup B$. Thus $\sup B$ is an upper bound for A.

Least Upper Bound Property Implies Existence of Suprema

Theorem 31.2.23 (Least Upper Bound Property Implies Existence of Suprema)**.** *Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. By the least upper bound property of the real numbers, there exists a real number s such that $s = \sup S$.*

Remark (Standard quantified statement).

$$S \neq \varnothing \quad \text{and} \quad (\exists u \in \mathbb{R})(\forall x \in S)(x \le u) \quad \Longrightarrow \quad (\exists s \in \mathbb{R})\, \mathrm{Sup}_{\mathbb{R},\le}(s, S).$$

Remark (Predicate reading).

$$S \neq \varnothing \wedge \mathrm{BddAbove}_{\mathbb{R},\le}(S) \Longrightarrow (\exists s \in \mathbb{R})\, \mathrm{Sup}_{\mathbb{R},\le}(s, S).$$

Remark (Negated quantified statement).

$$S \neq \varnothing, \quad \mathrm{BddAbove}_{\mathbb{R},\le}(S), \quad \text{and} \quad (\forall s \in \mathbb{R}) \neg\, \mathrm{Sup}_{\mathbb{R},\le}(s, S).$$

Remark (Failure modes). In $\mathbb{R}$ the theorem cannot fail once the set is nonempty and bounded above; outside complete ordered settings, the existence clause can fail.

Remark (Failure mode decomposition). The hypotheses separate domain nonemptiness from one-sided boundedness. The conclusion asserts existence of the sharp upper bound.

Greatest Lower Bound Property Implies Existence of Infima

Theorem 31.2.24 (Greatest Lower Bound Property Implies Existence of Infima)**.** *Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Then there exists a real number i such that $i = \inf S$.*

Remark (Standard quantified statement).

$$S \neq \varnothing \quad \text{and} \quad (\exists \ell \in \mathbb{R})(\forall x \in S)(\ell \leq x) \quad \Longrightarrow \quad (\exists i \in \mathbb{R})\, \mathrm{Inf}_{\mathbb{R},\leq}(i, S).$$

Remark (Predicate reading).

$$S \neq \varnothing \wedge \mathrm{BddBelow}_{\mathbb{R},\leq}(S) \Longrightarrow (\exists i \in \mathbb{R})\, \mathrm{Inf}_{\mathbb{R},\leq}(i, S).$$

Remark (Negated quantified statement).

$$S \neq \varnothing, \quad \mathrm{BddBelow}_{\mathbb{R},\leq}(S), \quad \text{and} \quad (\forall i \in \mathbb{R}) \neg\, \mathrm{Inf}_{\mathbb{R},\leq}(i, S).$$

Remark (Failure modes). In $\mathbb{R}$ the theorem cannot fail once the set is nonempty and bounded below.

Remark (Failure mode decomposition). Apply the least upper bound property to $-S$ or argue by the order-dual form of completeness.

31.2.3 Maxima and Minima

Maxima and minima are attained sharp bounds. They are stronger than suprema and infima because the extremal element must belong to the nonempty set itself.

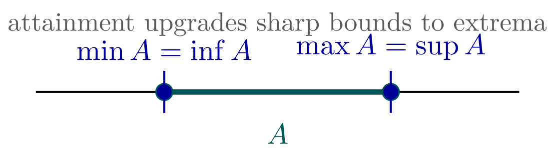


Figure 31.7: Maxima and minima are suprema and infima that are attained by the set.

Maximum

Definition 31.2.25 (Maximum)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *maximum* of A if $m \in A$ and every element of A is less than or equal to m.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad m \in A \quad \text{and} \quad (\forall x \in A)(x \leq m).$$

Remark (Definition predicate reading).

$$\mathrm{Max}_{S,\leq}(m, A) := m \in A \wedge \mathrm{UB}_{S,\leq}(m, A).$$

Remark (Negated quantified statement).

$$A = \varnothing \quad \text{or} \quad m \notin A \quad \text{or} \quad (\exists x \in A)(m < x).$$

Remark (Negation predicate reading).

$$\neg \operatorname{Max}_{S,\leq}(m, A) \iff A = \varnothing \vee m \notin A \vee \neg \operatorname{UB}_{S,\leq}(m, A).$$

Remark (Failure modes). The set may be empty; the candidate may not be a member of the set; or a larger element of the set may exist.

Remark (Failure mode decomposition). Maximum combines membership and an upper-bound condition. Failure of either component prevents attainment.

Minimum

Definition 31.2.26 (Minimum)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *minimum* of A if $m \in A$ and every element of A is greater than or equal to m.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad m \in A \quad \text{and} \quad (\forall x \in A)(m \leq x).$$

Remark (Definition predicate reading).

$$\operatorname{Min}_{S,\leq}(m, A) := m \in A \wedge \operatorname{LB}_{S,\leq}(m, A).$$

Remark (Negated quantified statement).

$$A = \varnothing \quad \text{or} \quad m \notin A \quad \text{or} \quad (\exists x \in A)(x < m).$$

Remark (Negation predicate reading).

$$\neg \operatorname{Min}_{S,\leq}(m, A) \iff A = \varnothing \vee m \notin A \vee \neg \operatorname{LB}_{S,\leq}(m, A).$$

Remark (Failure modes). The set may be empty; the candidate may not be a member of the set; or a smaller element of the set may exist.

Remark (Failure mode decomposition). Minimum combines membership and a lower-bound condition. Failure of either component prevents attainment.

Uniqueness of the Maximum

Proposition 31.2.27 (Uniqueness of the Maximum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both maxima of A, then $m = n$.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \operatorname{Max}_{S,\leq}(m, A) \quad \text{and} \quad \operatorname{Max}_{S,\leq}(n, A) \quad \Longrightarrow \quad m = n.$$

Remark (Predicate reading).

$$\operatorname{Max}_{S,\leq}(m, A) \wedge \operatorname{Max}_{S,\leq}(n, A) \Longrightarrow m = n.$$

Remark (Negated quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Max}_{S,\leq}(m, A) \quad \text{and} \quad \mathrm{Max}_{S,\leq}(n, A) \quad \text{and} \quad m \neq n.$$

Remark (Negation predicate reading).

$$\mathrm{Max}_{S,\leq}(m, A) \wedge \mathrm{Max}_{S,\leq}(n, A) \wedge m \neq n.$$

Remark (Failure modes). Failure would require two distinct elements of A each to dominate the entire set.

Remark (Failure mode decomposition). Since m is maximal over all elements and $n \in A$, we get $n \leq m$. Since n is maximal over all elements and $m \in A$, we get $m \leq n$.

Uniqueness of the Minimum

Proposition 31.2.28 (Uniqueness of the Minimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both minima of A, then $m = n$.*

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Min}_{S,\leq}(m, A) \quad \text{and} \quad \mathrm{Min}_{S,\leq}(n, A) \quad \Longrightarrow \quad m = n.$$

Remark (Predicate reading).

$$\mathrm{Min}_{S,\leq}(m, A) \wedge \mathrm{Min}_{S,\leq}(n, A) \Longrightarrow m = n.$$

Remark (Negated quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Min}_{S,\leq}(m, A) \quad \text{and} \quad \mathrm{Min}_{S,\leq}(n, A) \quad \text{and} \quad m \neq n.$$

Remark (Negation predicate reading).

$$\mathrm{Min}_{S,\leq}(m, A) \wedge \mathrm{Min}_{S,\leq}(n, A) \wedge m \neq n.$$

Remark (Failure modes). Failure would require two distinct elements of A each to lie below the entire set.

Remark (Failure mode decomposition). Since m is minimal and $n \in A$, we get $m \leq n$. Since n is minimal and $m \in A$, we get $n \leq m$.

Maximum Implies Supremum

Proposition 31.2.29 (Maximum Implies Supremum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \max A$, then $m = \sup A$.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Max}_{S,\leq}(m, A) \quad \Longrightarrow \quad \mathrm{Sup}_{S,\leq}(m, A).$$

Remark (Predicate reading).

$$\mathrm{Max}_{S,\leq}(m, A) \Longrightarrow \mathrm{Sup}_{S,\leq}(m, A).$$

Remark (Negated quantified statement).

$$\mathrm{Max}_{S,\leq}(m, A) \quad \text{and} \quad \neg\, \mathrm{Sup}_{S,\leq}(m, A).$$

Remark (Failure modes). Failure would require an attained upper bound that is not least among upper bounds.

Remark (Failure mode decomposition). Membership gives $m \in A$. Any upper bound of A must lie above every element of A, hence above m itself.

Minimum Implies Infimum

Proposition 31.2.30 (Minimum Implies Infimum)**.** *Let* $(S, \leq)$ *be an ordered set and let* $A \subseteq S$ *be nonempty. If* $m = \min A$*, then* $m = \inf A$*.*

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Min}_{S,\leq}(m, A) \quad \Longrightarrow \quad \mathrm{Inf}_{S,\leq}(m, A).$$

Remark (Predicate reading).

$$\mathrm{Min}_{S,\leq}(m, A) \Longrightarrow \mathrm{Inf}_{S,\leq}(m, A).$$

Remark (Negated quantified statement).

$$\mathrm{Min}_{S,\leq}(m, A) \quad \text{and} \quad \neg\, \mathrm{Inf}_{S,\leq}(m, A).$$

Remark (Failure modes). Failure would require an attained lower bound that is not greatest among lower bounds.

Remark (Failure mode decomposition). Membership gives $m \in A$. Any lower bound of A must lie below every element of A, hence below m itself.

Supremum in the Set is the Maximum

Proposition 31.2.31 (Supremum in the Set is the Maximum)**.** *Let* $(S, \leq)$ *be an ordered set and let* $A \subseteq S$ *be nonempty. If* $s = \sup A$ *and* $s \in A$*, then* $s = \max A$*.*
Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Sup}_{S,\leq}(s, A) \quad \text{and} \quad s \in A \quad \Longrightarrow \quad \mathrm{Max}_{S,\leq}(s, A).$$

Remark (Predicate reading).

$$\mathrm{Sup}_{S,\leq}(s, A) \wedge s \in A \Longrightarrow \mathrm{Max}_{S,\leq}(s, A).$$

Remark (Negated quantified statement).

$$\mathrm{Sup}_{S,\leq}(s, A) \quad \text{and} \quad s \in A \quad \text{and} \quad \neg\, \mathrm{Max}_{S,\leq}(s, A).$$

Remark (Failure modes). Failure would require a supremum that belongs to the set but is not its maximum.

Remark (Failure mode decomposition). The supremum condition supplies the upper-bound clause. Membership supplies attainment.

Infimum in the Set is the Minimum

Proposition 31.2.32 (Infimum in the Set is the Minimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i = \inf A$ and $i \in A$, then $i = \min A$.*

Remark (Standard quantified statement).

$$A \neq \varnothing \quad \text{and} \quad \mathrm{Inf}_{S,\leq}(i, A) \quad \text{and} \quad i \in A \quad \Longrightarrow \quad \mathrm{Min}_{S,\leq}(i, A).$$

Remark (Predicate reading).

$$\mathrm{Inf}_{S,\leq}(i, A) \wedge i \in A \Longrightarrow \mathrm{Min}_{S,\leq}(i, A).$$

Remark (Negated quantified statement).

$$\mathrm{Inf}_{S,\leq}(i, A) \quad \text{and} \quad i \in A \quad \text{and} \quad \neg\, \mathrm{Min}_{S,\leq}(i, A).$$

Remark (Failure modes). Failure would require an infimum that belongs to the set but is not its minimum.

Remark (Failure mode decomposition). The infimum condition supplies the lower-bound clause. Membership supplies attainment.

31.2.4 Epsilon Characterization

ε-Characterization of Supremum

Definition (Epsilon Characterization of Supremum)

Definition 31.2.33 (Epsilon Characterization of Supremum)**.** Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. The element s satisfies the *epsilon characterization of the supremum* of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $s - \varepsilon < a$.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}. \\ &s \text{ satisfies the epsilon characterization of } \sup A \\ &\Longleftrightarrow \Big[\forall x \in A\ (x \leq s)\Big] \wedge \Big[\forall \varepsilon > 0\ \exists a \in A\ (s - \varepsilon < a)\Big]. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{EpsilonSupremum}(s, A) :\equiv \mathrm{UpperBound}(s, A) \land \forall \varepsilon > 0 \; \exists a \in A \; (s - \varepsilon < a).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.\\ &s \text{ does not satisfy the epsilon characterization of } \sup A\\ &\iff \Big[\exists x \in A \; (s < x)\Big] \lor \Big[\exists \varepsilon_0 > 0 \; \forall a \in A \; (a \leq s - \varepsilon_0)\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{EpsilonSupremum}(s, A) \iff \Big[\exists x \in A \; (s < x)\Big] \lor \Big[\exists \varepsilon_0 > 0 \; \forall a \in A \; (a \leq s - \varepsilon_0)\Big].$$

Remark (Failure modes). The characterization fails either because s is not an upper bound of A, or because A remains at least some positive distance below s.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{EpsilonSupremum}(s, A) = \underbrace{\exists x \in A \; (s < x)}_{\text{upper-bound failure}} \lor \underbrace{\exists \varepsilon_0 > 0 \; \forall a \in A \; (a \leq s - \varepsilon_0)}_{\text{positive gap below } s}.$$

Remark (Interpretation). This condition says that s caps the set from above while points of A approach s from below at every positive scale. The upper-bound clause prevents A from crossing above s, and the epsilon clause prevents s from floating strictly above the set. Geometrically, s is a ceiling that the set presses against arbitrarily closely.

Remark (Dependencies).

Definition 31.2.2.

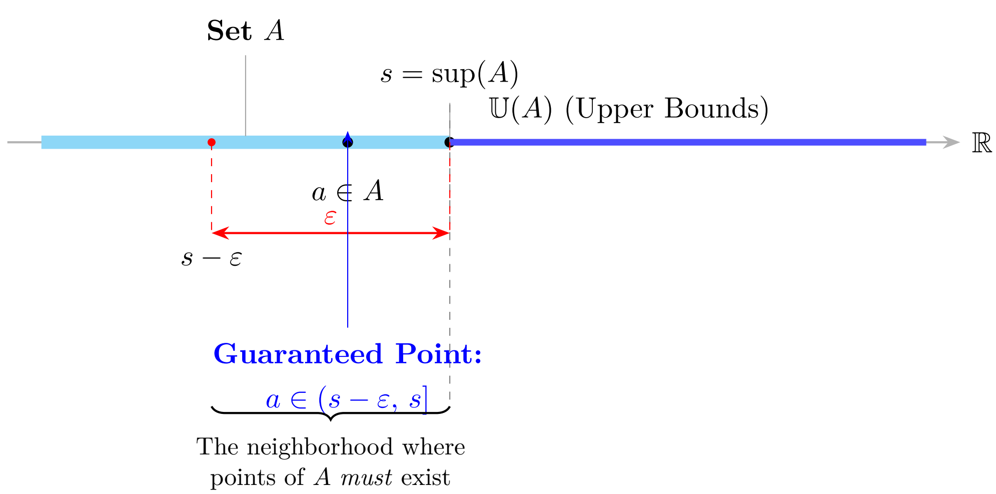


Figure 31.8: Refined visualization of the ε-characterization. Labels occupy staggered vertical lanes with leader lines to eliminate all horizontal collision, while the dashed spine preserves the geometric correspondence between the real line and the probe below it.

Definition (Epsilon Characterization of Infimum)

Definition 31.2.34 (Epsilon Characterization of Infimum)**.** Let $A \subseteq \mathbb{R}$ and let $t \in \mathbb{R}$. The element t satisfies the *epsilon characterization of the infimum* of A if and only if t is a lower bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $a < t + \varepsilon$.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } t \in \mathbb{R}.\\ &t \text{ satisfies the epsilon characterization of } \inf A\\ &\iff \Big[\forall x \in A\ (t \le x)\Big] \wedge \Big[\forall \varepsilon > 0\ \exists a \in A\ (a < t + \varepsilon)\Big].\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{EpsilonInfimum}(t, A) :\!= \mathrm{LowerBound}(t, A) \wedge \forall \varepsilon > 0\ \exists a \in A\ (a < t + \varepsilon).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } t \in \mathbb{R}.\\ &t \text{ does not satisfy the epsilon characterization of } \inf A\\ &\iff \Big[\exists x \in A\ (x < t)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)\Big].\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{EpsilonInfimum}(t, A) \iff \Big[\exists x \in A\ (x < t)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)\Big].$$

Remark (Failure modes)*.* The characterization fails either because t is not a lower bound of A, or because A remains at least some positive distance above t.

Remark (Failure mode decomposition)*.*

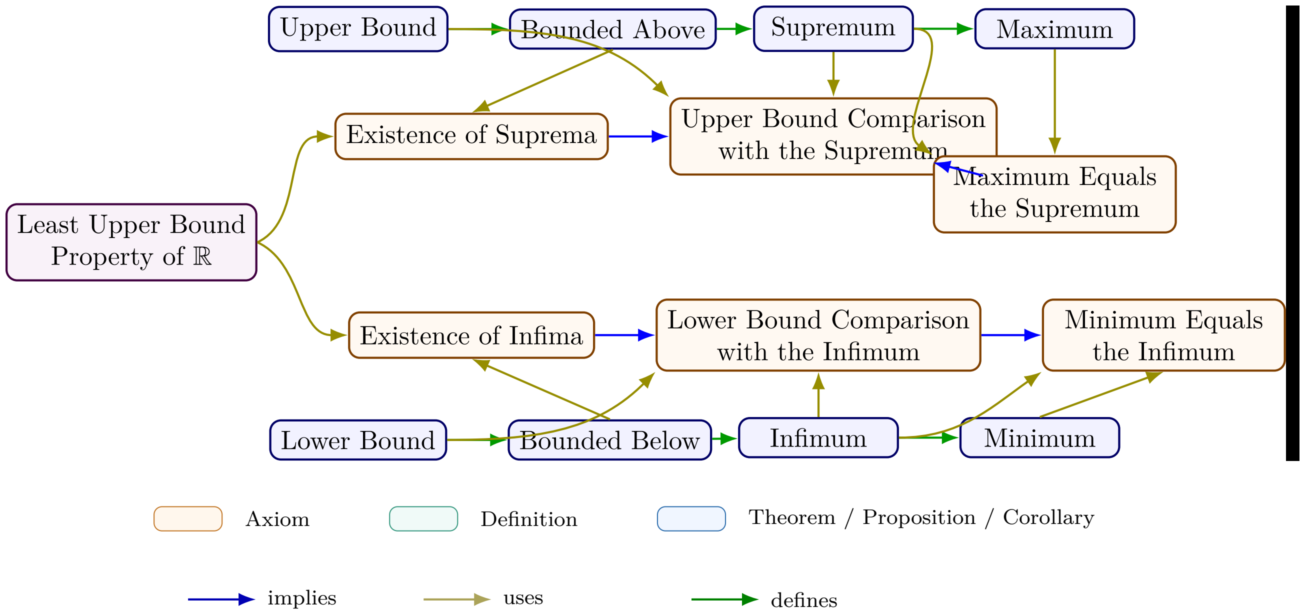
$$\neg\,\mathrm{EpsilonInfimum}(t, A) = \underbrace{\exists x \in A\ (x < t)}_{\text{lower-bound failure}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)}_{\text{positive gap above } t}.$$

Remark (Interpretation)*.* This condition says that t sits below the set while points of A approach t from above at every positive scale. The lower-bound clause prevents A from crossing below t, and the epsilon clause prevents t from sitting strictly below the set with a positive gap. Geometrically, t is a floor that the set presses against arbitrarily closely from above.

Remark (Dependencies)*.*

Definition 31.2.3.

31.2.5 Extremal Bounds Summary



Remark (Equivalent formulations).
Over the ordered-field axioms, completeness is equivalent to each of the following:

- Every nonempty set bounded below has an infimum.
- Every Cauchy sequence in $\mathbb{R}$ converges.
- *(Nested Interval Property)* Every nested sequence of nonempty closed bounded intervals has nonempty intersection.

31.3 Algebra of Bounds

31.3.1 Set Operations and Bounds

Remark (Orientation). Set operations change the collection of possible counterexamples to a proposed bound. Each result below isolates one operation and one direction of bound transport.

Lemma (Union Preserves Upper Bounds)

Lemma 31.3.1 (Union Preserves Upper Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. If u_A is an upper bound of A and u_B is an upper bound of B, then* $\max\{u_A, u_B\}$ *is an upper bound of $A \cup B$. Go to proof.*

Remark (Standard quantified statement).

$$(\forall a \in A)(a \leq u_A) \wedge (\forall b \in B)(b \leq u_B) \implies (\forall x \in A \cup B)(x \leq \max\{u_A, u_B\}).$$

Remark (Predicate reading).

$$\mathrm{UB}(u_A, A) \wedge \mathrm{UB}(u_B, B) \implies \mathrm{UB}(\max\{u_A, u_B\}, A \cup B).$$

Remark (Negated quantified statement).

$$(\forall a \in A)(a \leq u_A), \quad (\forall b \in B)(b \leq u_B), \quad (\exists x \in A \cup B)(\max\{u_A, u_B\} < x).$$

Remark (Negation predicate reading).

$$\mathrm{UB}(u_A, A) \wedge \mathrm{UB}(u_B, B) \wedge \neg\, \mathrm{UB}(\max\{u_A, u_B\}, A \cup B).$$

Remark (Failure modes). A failure would require a union element larger than both proposed upper bounds.

Remark (Failure mode decomposition). Every $x \in A \cup B$ lies in A or in B. In the first case $x \leq u_A \leq \max\{u_A, u_B\}$; in the second case $x \leq u_B \leq \max\{u_A, u_B\}$.

Remark (Dependencies).

Upper Bound.

Lemma (Union Preserves Lower Bounds)

Lemma 31.3.2 (Union Preserves Lower Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. If ℓ_A is a lower bound of A and ℓ_B is a lower bound of B, then $\min\{\ell_A, \ell_B\}$ is a lower bound of $A \cup B$.* *Go to proof.*

Remark (Standard quantified statement).

$$(\forall a \in A)(\ell_A \leq a) \wedge (\forall b \in B)(\ell_B \leq b) \implies (\forall x \in A \cup B)(\min\{\ell_A, \ell_B\} \leq x).$$

Remark (Predicate reading).

$$\mathrm{LB}(\ell_A, A) \wedge \mathrm{LB}(\ell_B, B) \implies \mathrm{LB}(\min\{\ell_A, \ell_B\}, A \cup B).$$

Remark (Negated quantified statement).

$$(\forall a \in A)(\ell_A \leq a), \quad (\forall b \in B)(\ell_B \leq b), \quad (\exists x \in A \cup B)(x < \min\{\ell_A, \ell_B\}).$$

Remark (Negation predicate reading).

$$\mathrm{LB}(\ell_A, A) \wedge \mathrm{LB}(\ell_B, B) \wedge \neg\, \mathrm{LB}(\min\{\ell_A, \ell_B\}, A \cup B).$$

Remark (Failure modes). A failure would require a union element smaller than both proposed lower bounds.

Remark (Failure mode decomposition). Every $x \in A \cup B$ lies in A or in B. In either case the appropriate lower bound lies below x, and $\min\{\ell_A, \ell_B\}$ lies below that lower bound.

Remark (Dependencies).

Lower Bound.

Proposition (Union Is Bounded Above Exactly When Both Pieces Are)

Proposition 31.3.3 (Union Is Bounded Above Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded above if and only if A is bounded above and B is bounded above.* *Go to proof.*

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad B \neq \varnothing \quad \Longrightarrow \quad \mathrm{BddAbove}(A \cup B) \iff \mathrm{BddAbove}(A) \wedge \mathrm{BddAbove}(B).$$

Remark (Predicate reading).

$$\mathrm{BddAbove}(A \cup B) \iff \mathrm{BddAbove}(A) \wedge \mathrm{BddAbove}(B).$$

Remark (Negated quantified statement). The equivalence fails precisely if one side holds and the other does not.

Remark (Negation predicate reading).

$$\mathrm{BddAbove}(A \cup B) \oplus \big(\mathrm{BddAbove}(A) \wedge \mathrm{BddAbove}(B)\big).$$

Remark (Failure modes). The forward direction could fail only if a bound for the union did not bound a piece. The reverse direction could fail only if bounds for both pieces did not combine into a bound for the union.

Remark (Failure mode decomposition). Use subset inheritance for the forward direction and Union Preserves Upper Bounds for the reverse direction.

Remark (Dependencies).

Bounded Above; Union Preserves Upper Bounds; Subsets Preserve Upper Bounds.

Proposition (Union Is Bounded Below Exactly When Both Pieces Are)

Proposition 31.3.4 (Union Is Bounded Below Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded below if and only if A is bounded below and B is bounded below.* *Go to proof.*

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad B \neq \varnothing \quad \Longrightarrow \quad \mathrm{BddBelow}(A \cup B) \iff \mathrm{BddBelow}(A) \wedge \mathrm{BddBelow}(B).$$

Remark (Predicate reading).

$$\mathrm{BddBelow}(A \cup B) \iff \mathrm{BddBelow}(A) \wedge \mathrm{BddBelow}(B).$$

Remark (Negated quantified statement). The equivalence fails precisely if one side holds and the other does not.

Remark (Negation predicate reading).

$$\mathrm{BddBelow}(A \cup B) \oplus \big(\mathrm{BddBelow}(A) \wedge \mathrm{BddBelow}(B)\big).$$

Remark (Failure modes). The forward direction could fail only if a lower bound for the union did not bound a piece. The reverse direction could fail only if lower bounds for both pieces did not combine into a lower bound for the union.

Remark (Failure mode decomposition). Use subset inheritance for the forward direction and Union Preserves Lower Bounds for the reverse direction.

Remark (Dependencies).

Bounded Below; Union Preserves Lower Bounds; Subsets Preserve Lower Bounds.

Corollary (Union Is Bounded Exactly When Both Pieces Are)

Corollary 31.3.5 (Union Is Bounded Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded if and only if A is bounded and B is bounded. Go to proof.*

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad B \neq \varnothing \quad \Longrightarrow \quad \mathrm{Bdd}(A \cup B) \Longleftrightarrow \mathrm{Bdd}(A) \wedge \mathrm{Bdd}(B).$$

Remark (Predicate reading).

$$\mathrm{Bdd}(A \cup B) \Longleftrightarrow \mathrm{Bdd}(A) \wedge \mathrm{Bdd}(B).$$

Remark (Negated quantified statement). The two-sided boundedness equivalence fails if either the upper or lower one-sided equivalence fails.

Remark (Negation predicate reading).

$$\mathrm{Bdd}(A \cup B) \oplus \big(\mathrm{Bdd}(A) \wedge \mathrm{Bdd}(B)\big).$$

Remark (Failure modes). Two-sided boundedness fails exactly through an upper-bound failure or a lower-bound failure.

Remark (Failure mode decomposition). Apply the bounded-above and bounded-below union equivalences separately.

Remark (Dependencies).

Bounded Set; Union Is Bounded Above Exactly When Both Pieces Are; Union Is Bounded Below Exactly When Both Pieces Are.

Lemma (Subsets Preserve Upper Bounds)

Lemma 31.3.6 (Subsets Preserve Upper Bounds)**.** *Let $C \subseteq A \subseteq \mathbb{R}$. If u is an upper bound of A, then u is an upper bound of C. Go to proof.*

Remark (Standard quantified statement).

$$C \subseteq A \wedge (\forall a \in A)(a \leq u) \Longrightarrow (\forall c \in C)(c \leq u).$$

Remark (Predicate reading).

$$C \subseteq A \wedge \mathrm{UB}(u, A) \Longrightarrow \mathrm{UB}(u, C).$$

Remark (Negated quantified statement).

$$C \subseteq A, \quad (\forall a \in A)(a \leq u), \quad (\exists c \in C)(u < c).$$

Remark (Negation predicate reading).

$$C \subseteq A \wedge \mathrm{UB}(u, A) \wedge \neg\, \mathrm{UB}(u, C).$$

Remark (Failure modes). A failure would require a counterexample in C that is not already a counterexample in A.

Remark (Failure mode decomposition). Since $C \subseteq A$, every $c \in C$ is an element of A. The upper-bound condition on A therefore applies directly to c.

Remark (Dependencies).

Upper Bound.

Lemma (Subsets Preserve Lower Bounds)

Lemma 31.3.7 (Subsets Preserve Lower Bounds)**.** *Let $C \subseteq A \subseteq \mathbb{R}$. If ℓ is a lower bound of A, then ℓ is a lower bound of C.* *Go to proof.*

Remark (Standard quantified statement).

$$C \subseteq A \wedge (\forall a \in A)(\ell \leq a) \Longrightarrow (\forall c \in C)(\ell \leq c).$$

Remark (Predicate reading).

$$C \subseteq A \wedge \mathrm{LB}(\ell, A) \Longrightarrow \mathrm{LB}(\ell, C).$$

Remark (Negated quantified statement).

$$C \subseteq A, \quad (\forall a \in A)(\ell \leq a), \quad (\exists c \in C)(c < \ell).$$

Remark (Negation predicate reading).

$$C \subseteq A \wedge \mathrm{LB}(\ell, A) \wedge \neg\, \mathrm{LB}(\ell, C).$$

Remark (Failure modes). A failure would require a point in C below ℓ, hence a point in A below ℓ.

Remark (Failure mode decomposition). Subset inheritance removes possible counterexamples; it cannot create a new one.

Remark (Dependencies).

Lower Bound.

Corollary (Intersections Inherit Bounds)

Corollary 31.3.8 (Intersections Inherit Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \cap B$, and every lower bound of A is a lower bound of $A \cap B$.* *Go to proof.*

Remark (Standard quantified statement).

$$A \cap B \subseteq A.$$

Remark (Predicate reading).

$$\mathrm{UB}(u, A) \implies \mathrm{UB}(u, A \cap B),$$

and similarly for lower bounds.

Remark (Negated quantified statement). A failure would be a failure of subset inheritance for $A \cap B \subseteq A$.

Remark (Negation predicate reading).

$$\mathrm{UB}(u, A) \wedge \neg\, \mathrm{UB}(u, A \cap B)$$

is contradictory, and the same holds for lower bounds.

Remark (Failure modes). Intersection can lose elements, but it cannot introduce an element outside the parent set A.

Remark (Failure mode decomposition). Apply subset inheritance to $C = A \cap B$.

Remark (Dependencies).

Subsets Preserve Upper Bounds; Subsets Preserve Lower Bounds.

Corollary (Differences Inherit Bounds)

Corollary 31.3.9 (Differences Inherit Bounds). *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \setminus B$, and every lower bound of A is a lower bound of $A \setminus B$.* *Go to proof.*

Remark (Standard quantified statement).

$$A \setminus B \subseteq A.$$

Remark (Predicate reading).

$$\mathrm{UB}(u, A) \implies \mathrm{UB}(u, A \setminus B),$$

and similarly for lower bounds.

Remark (Negated quantified statement). A failure would be a failure of subset inheritance for $A \setminus B \subseteq A$.

Remark (Negation predicate reading).

$$\mathrm{UB}(u, A) \wedge \neg\, \mathrm{UB}(u, A \setminus B)$$

is contradictory, and the same holds for lower bounds.

Remark (Failure modes). Difference can remove elements from A, but it cannot create a new element outside A.

Remark (Failure mode decomposition). Apply subset inheritance to $C = A \setminus B$.

Remark (Dependencies).

Subsets Preserve Upper Bounds; Subsets Preserve Lower Bounds.

Corollary (Complements Inherit Ambient Bounds)

Corollary 31.3.10 (Complements Inherit Ambient Bounds). *Let $A \subseteq E \subseteq \mathbb{R}$, and take complements relative to E. Every upper bound of E is an upper bound of $E \setminus A$, and every lower bound of E is a lower bound of $E \setminus A$.* *Go to proof.*

Remark (Standard quantified statement).

$$E \setminus A \subseteq E.$$

Remark (Predicate reading).

$$\mathrm{UB}(u, E) \Longrightarrow \mathrm{UB}(u, E \setminus A), \qquad \mathrm{LB}(\ell, E) \Longrightarrow \mathrm{LB}(\ell, E \setminus A).$$

Remark (Negated quantified statement).

$$\mathrm{UB}(u, E) \wedge \neg\, \mathrm{UB}(u, E \setminus A)$$

or

$$\mathrm{LB}(\ell, E) \wedge \neg\, \mathrm{LB}(\ell, E \setminus A).$$

Remark (Negation predicate reading). Both negated configurations contradict subset inheritance because $E \setminus A \subseteq E$.

Remark (Failure modes). The complement does not inherit bounds from A; it inherits them from the ambient set E.

Remark (Failure mode decomposition). Every point of $E \setminus A$ is a point of E, so every ambient bound of E applies to the complement.

Remark (Dependencies).

Subsets Preserve Upper Bounds; Subsets Preserve Lower Bounds.

31.3.2 Lattice Operations and Bounds

Remark (Orientation). In this section, *lattice* refers to the max–min structure already present in the usual order on $\mathbb{R}$. For two real numbers, their join is their maximum and their meet is their minimum:

$$x \vee y = \max\{x, y\}, \qquad x \wedge y = \min\{x, y\}.$$

Thus lattice operations are not a new order relation; they are the operations created by the order. The set-level constructions below apply those pairwise operations to all pairs of points from two nonempty sets.

Definition (Pointwise Maximum and Minimum Sets)

Definition 31.3.11 (Pointwise Maximum and Minimum Sets)**.** Let $A, B \subseteq \mathbb{R}$ be nonempty. Define

$$\max(A, B) := \{\max\{a, b\} : a \in A,\ b \in B\},$$

and

$$\min(A, B) := \{\min\{a, b\} : a \in A,\ b \in B\}.$$

Remark (Standard quantified statement).

$$x \in \max(A, B) \iff (\exists a \in A)(\exists b \in B)\big(x = \max\{a, b\}\big),$$

and similarly for $\min(A, B)$ with $\min\{a, b\}$.

Remark (Definition predicate reading).

$$\text{PointwiseMaxSet}(A, B) = \max(A, B), \qquad \text{PointwiseMinSet}(A, B) = \min(A, B).$$

Remark (Negated quantified statement).

$$x \notin \max(A, B) \iff (\forall a \in A)(\forall b \in B)\big(x \neq \max\{a, b\}\big),$$

and similarly for $\min(A, B)$.

Remark (Negation predicate reading).

$$\neg\,\text{PointwiseMaxSet}(A, B) = C \quad \text{means} \quad (\exists x)\big[x \in C \,\triangle\, \max(A, B)\big],$$

and similarly for PointwiseMinSet.

Remark (Failure modes). The definition fails only if the image set omits a pairwise maximum or minimum, or if it includes a value not produced by one of the pairwise rules.

Remark (Failure mode decomposition). Every membership claim decomposes into two input witnesses and one order-based operation: either the larger input or the smaller input is selected.

Remark (Dependencies).

Maximum; Minimum.

Theorem (Supremum of the Pointwise Maximum Set)

Theorem 31.3.12 (Supremum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \max(A, B) = \max\{\sup A, \sup B\}.$$

Go to proof.

Remark (Standard quantified statement).

$$A, B \neq \varnothing, \quad A, B \text{ bounded above} \quad \Longrightarrow \quad \sup\{\max\{a, b\} : a \in A,\ b \in B\} = \max\{\sup A, \sup B\}. \blacksquare$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \Longrightarrow \mathrm{Supremum}(\max\{s_A, s_B\}, \max(A, B)).$$

Remark (Negated quantified statement).

$$\sup \max(A, B) \neq \max\{\sup A, \sup B\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \wedge \neg\, \mathrm{Supremum}(\max\{s_A, s_B\}, \max(A, B)).$$

Remark (Failure modes). The formula could fail only if the maximum of the two suprema failed to bound all pairwise maxima, or if it were not the least such bound.

Remark (Failure mode decomposition). Every pairwise maximum is bounded by $\max\{\sup A, \sup B\}$ Sharpness comes from holding one input fixed in the nonempty other set and letting the relevant set approach its own supremum.

Remark (Dependencies).

Supremum; Pointwise Maximum and Minimum Sets.

Theorem (Infimum of the Pointwise Maximum Set)

Theorem 31.3.13 (Infimum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \max(A, B) = \max\{\inf A, \inf B\}.$$

Go to proof.

Remark (Standard quantified statement).

$$A, B \neq \varnothing, \quad A, B \text{ bounded below} \quad \Longrightarrow \quad \inf\{\max\{a, b\} : a \in A,\ b \in B\} = \max\{\inf A, \inf B\}.$$

Remark (Predicate reading).

$$\mathrm{Infimum}(t_A, A) \wedge \mathrm{Infimum}(t_B, B) \Longrightarrow \mathrm{Infimum}(\max\{t_A, t_B\}, \max(A, B)).$$

Remark (Negated quantified statement).

$$\inf \max(A, B) \neq \max\{\inf A, \inf B\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(t_A, A) \wedge \mathrm{Infimum}(t_B, B) \wedge \neg\, \mathrm{Infimum}(\max\{t_A, t_B\}, \max(A, B)).$$

Remark (Failure modes)*.* The formula could fail only if the larger of the two lower thresholds were not a lower bound for all pairwise maxima, or if pairwise maxima could not approach that threshold from above.

Remark (Failure mode decomposition)*.* Every pairwise maximum is at least both lower thresholds, hence at least their maximum. Approximation of the two infima gives pairwise maxima arbitrarily close to the larger lower threshold.

Remark (Dependencies)*.*

Infimum; Pointwise Maximum and Minimum Sets.

Theorem (Supremum of the Pointwise Minimum Set)

Theorem 31.3.14 (Supremum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \min(A, B) = \min\{\sup A, \sup B\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$A, B \neq \varnothing, \quad A, B \text{ bounded above} \quad \Longrightarrow \quad \sup\{\min\{a, b\} : a \in A,\ b \in B\} = \min\{\sup A, \sup B\}. \blacksquare$$

Remark (Predicate reading)*.*

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \Longrightarrow \text{Supremum}(\min\{s_A, s_B\}, \min(A, B)).$$

Remark (Negated quantified statement)*.*

$$\sup \min(A, B) \neq \min\{\sup A, \sup B\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading)*.*

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge \neg\, \text{Supremum}(\min\{s_A, s_B\}, \min(A, B)).$$

Remark (Failure modes)*.* The formula could fail only if the smaller supremum failed to bound all pairwise minima, or if that upper bound were not sharp.

Remark (Failure mode decomposition)*.* Every pairwise minimum is below both suprema, hence below their minimum. Sharpness comes from taking both inputs near their own suprema.

Remark (Dependencies)*.*

Supremum; Pointwise Maximum and Minimum Sets.

Theorem (Infimum of the Pointwise Minimum Set)

Theorem 31.3.15 (Infimum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \min(A, B) = \min\{\inf A, \inf B\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$A, B \neq \varnothing, \quad A, B \text{ bounded below} \quad \Longrightarrow \quad \inf\{\min\{a, b\} : a \in A,\ b \in B\} = \min\{\inf A, \inf B\}. \blacksquare$$

Remark (Predicate reading)*.*

$$\mathrm{Infimum}(t_A, A) \wedge \mathrm{Infimum}(t_B, B) \Longrightarrow \mathrm{Infimum}(\min\{t_A, t_B\}, \min(A, B)).$$

Remark (Negated quantified statement)*.*

$$\inf \min(A, B) \neq \min\{\inf A, \inf B\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading)*.*

$$\mathrm{Infimum}(t_A, A) \wedge \mathrm{Infimum}(t_B, B) \wedge \neg\, \mathrm{Infimum}(\min\{t_A, t_B\}, \min(A, B)).$$

Remark (Failure modes)*.* The formula could fail only if the smaller infimum failed to bound all pairwise minima from below, or if that lower bound were not sharp.

Remark (Failure mode decomposition)*.* Every pairwise minimum is at least the smaller of the two infima. Sharpness comes from holding one set nonempty and approaching the smaller lower threshold.

Remark (Dependencies)*.*

Infimum; Pointwise Maximum and Minimum Sets.

31.3.3 Images, Inverse Images, and Extrema

Remark (Orientation)*.* One cannot usually move a supremum or infimum through an arbitrary function:

$$\sup f(A) \neq f(\sup A)$$

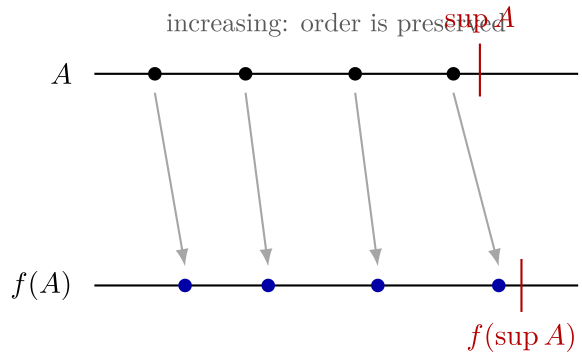
in general. Equality is an order-and-continuity statement. Monotone functions preserve or reverse the relevant inequalities, and continuity at the endpoint turns the endpoint bound into a sharp bound for the image. The inverse function statements are the same principle applied to f^{-1}.

Remark (Geometric meaning of the hypotheses). Think of a nonempty bounded set $A \subseteq \mathbb{R}$ as a cloud of points on the real line. The number $\sup A$ is the right-hand wall of that cloud: it may or may not belong to A, but points of A press arbitrarily close to it from the left. Applying a function f moves every point of the cloud and produces the image cloud

$$f(A) = \{f(a) : a \in A\}.$$

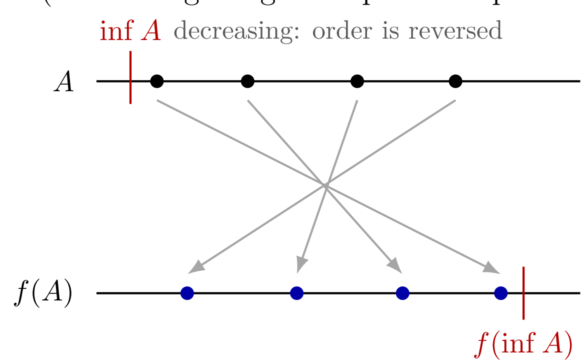
The question $\sup f(A) = f(\sup A)$ asks whether the function carries the right-hand wall of the original cloud to the right-hand wall of the image cloud.

Example (Increasing images keep the right edge on the right).



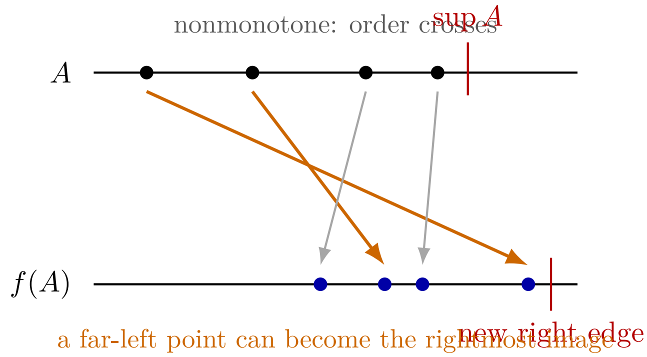
When f is increasing, order is preserved. A point that starts farther to the right cannot land to the left of a point that starts farther to the left. Thus the right edge of the image cloud is controlled by the right edge of the original cloud, provided continuity prevents a jump at the endpoint.

Example (Decreasing images swap the endpoint roles).



When f is decreasing, order is reversed. The left edge of the original cloud becomes the right edge of the image cloud, and the right edge of the original cloud becomes the left edge of the image cloud. This is why decreasing image theorems exchange suprema and infima.

Example (Nonmonotone images can cross).



If f is not monotone on the relevant domain, the arrows can cross. Then the right edge of $f(A)$ need not come from $\sup A$ or from $\inf A$ alone in a uniform way. The image edge may be produced by an interior point, by the left branch of a piecewise monotone function, or by the right branch. In that situation the set must be split into pieces where f is monotone, and the image pieces must be recombined using the algebra of suprema over unions.

Example (Why $x \mapsto x^2$ does not fit either global theorem). The function $f(x) = x^2$ is not increasing on all of $\mathbb{R}$. Indeed, $-2 \leq 1$, but

$$(-2)^2 = 4 > 1 = 1^2.$$

It is also not decreasing on all of $\mathbb{R}$. Indeed, $1 \leq 2$, but

$$1^2 = 1 < 4 = 2^2.$$

Thus neither the increasing image theorem nor the decreasing image theorem can be applied globally to $x \mapsto x^2$ on $\mathbb{R}$.

Let

$$A = (-1, 2).$$

Then $\inf A = -1$ and $\sup A = 2$. The image under the square map is

$$f(A) = \{x^2 : -1 < x < 2\} = [0, 4),$$

so

$$\sup f(A) = 4.$$

The increasing-theorem expression gives

$$f(\sup A) = f(2) = 4,$$

which happens to agree in this example. But the agreement is not justified by global increasingness, because x^2 is not globally increasing.

The decreasing-theorem expression gives

$$f(\inf A) = f(-1) = 1,$$

and this does not agree:

$$\sup f(A) = 4 \neq 1 = f(\inf A).$$

The failure occurs because the square map folds the real line at 0. It is decreasing on $(-\infty, 0]$ and increasing on $[0, \infty)$, so a set that straddles 0 must be analyzed branch by branch. For a nonempty bounded set $A \subseteq \mathbb{R}$ with finite endpoints $\alpha = \inf A$ and $\beta = \sup A$, the largest square is controlled by whichever endpoint is farther from 0:

$$\sup\{x^2 : x \in A\} = \max\{\alpha^2, \beta^2\}.$$

Theorem (Increasing Image Preserves Suprema)

Theorem 31.3.16 (Increasing Image Preserves Suprema)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be increasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\sup f(A) = f(\sup A), \qquad f(A) = \{f(a) : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad s = \sup A \in I, \quad f \text{ increasing and continuous at } s \Longrightarrow \sup\{f(a) : a \in A\} = f(s).$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Increasing}(f, I) \wedge \text{ContinuousAt}(f, s) \Longrightarrow \text{Supremum}(f(s), f(A)).$$

Remark (Negated quantified statement).

$$\sup f(A) \neq f(\sup A)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Increasing}(f, I) \wedge \text{ContinuousAt}(f, s) \wedge \neg\,\text{Supremum}(f(s), f(A)).$$

Remark (Failure modes). The equality can fail if f is not increasing, if $\sup A$ is not in the domain, or if the endpoint value of f is not approached by values $f(a)$ with $a \in A$.

Remark (Failure mode decomposition). Increasingness gives $f(a) \leq f(s)$ for every $a \in A$. The defining approximation property of the supremum gives points of A arbitrarily close to s from below, and continuity at s makes their images approach $f(s)$.

Remark (Dependencies).

Supremum; Increasing Function; Continuity at a Point.

Theorem (Increasing Image Preserves Infima)

Theorem 31.3.17 (Increasing Image Preserves Infima)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be increasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\inf f(A) = f(\inf A).$$

Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad t = \inf A \in I, \quad f \text{ increasing and continuous at } t \Longrightarrow \inf\{f(a) : a \in A\} = f(t).$$

Remark (Predicate reading).

$$\mathrm{Infimum}(t, A) \wedge \mathrm{Increasing}(f, I) \wedge \mathrm{ContinuousAt}(f, t) \Longrightarrow \mathrm{Infimum}(f(t), f(A)).$$

Remark (Negated quantified statement).

$$\inf f(A) \neq f(\inf A)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(t, A) \wedge \mathrm{Increasing}(f, I) \wedge \mathrm{ContinuousAt}(f, t) \wedge \neg\, \mathrm{Infimum}(f(t), f(A)).$$

Remark (Failure modes). The equality can fail if f is not increasing, if $\inf A$ is not in the domain, or if the endpoint value is not reached as a limit of image values.

Remark (Failure mode decomposition). Increasingness gives $f(t) \leq f(a)$ for every $a \in A$. Approximation of the infimum from above and continuity at t make this lower bound sharp.

Remark (Dependencies).

Infimum; Increasing Function; Continuity at a Point.

Theorem (Decreasing Image Sends Infimum to Supremum)

Theorem 31.3.18 (Decreasing Image Sends Infimum to Supremum). *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\sup f(A) = f(\inf A).$$

Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad t = \inf A \in I, \quad f \text{ decreasing and continuous at } t \Longrightarrow \sup\{f(a) : a \in A\} = f(t).$$

Remark (Predicate reading).

$$\mathrm{Infimum}(t, A) \wedge \mathrm{Decreasing}(f, I) \wedge \mathrm{ContinuousAt}(f, t) \Longrightarrow \mathrm{Supremum}(f(t), f(A)).$$

Remark (Negated quantified statement).

$$\sup f(A) \neq f(\inf A)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Decreasing}(f, I) \wedge \text{ContinuousAt}(f, t) \wedge \neg\,\text{Supremum}(f(t), f(A)).$$

Remark (Failure modes). The equality can fail if f is not decreasing, if $\inf A$ is outside the domain, or if f is not continuous at the lower endpoint.

Remark (Failure mode decomposition). Decreasingness gives $f(a) \le f(t)$ for every $a \in A$. Approximation of t from above and continuity at t make this upper bound sharp.

Remark (Dependencies).
Infimum; Supremum; Decreasing Function; Continuity at a Point.

Theorem (Decreasing Image Sends Supremum to Infimum)

Theorem 31.3.19 (Decreasing Image Sends Supremum to Infimum). *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\inf f(A) = f(\sup A).$$

Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad s = \sup A \in I, \quad f \text{ decreasing and continuous at } s \Longrightarrow \inf\{f(a) : a \in A\} = f(s).$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Decreasing}(f, I) \wedge \text{ContinuousAt}(f, s) \Longrightarrow \text{Infimum}(f(s), f(A)).$$

Remark (Negated quantified statement).

$$\inf f(A) \neq f(\sup A)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Decreasing}(f, I) \wedge \text{ContinuousAt}(f, s) \wedge \neg\,\text{Infimum}(f(s), f(A)).$$

Remark (Failure modes). The equality can fail if f is not decreasing, if $\sup A$ is outside the domain, or if f is not continuous at the upper endpoint.

Remark (Failure mode decomposition). Decreasingness gives $f(s) \le f(a)$ for every $a \in A$. Approximation of s from below and continuity at s make this lower bound sharp.

Remark (Dependencies).
Supremum; Infimum; Decreasing Function; Continuity at a Point.

Theorem (Increasing Inverse Preserves Suprema)

Theorem 31.3.20 (Increasing Inverse Preserves Suprema)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if* $\sup B \in J$*, then*

$$\sup f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$B \neq \varnothing, \quad \sup B \in J, \quad f^{-1} \text{ increasing and continuous at } \sup B \Longrightarrow \sup f^{-1}(B) = f^{-1}(\sup B).$$

Remark (Predicate reading)*.*

$$\text{Supremum}(u, B) \wedge \text{Increasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, u) \Longrightarrow \text{Supremum}(f^{-1}(u), f^{-1}(B)).$$

Remark (Negated quantified statement)*.*

$$\sup f^{-1}(B) \neq f^{-1}(\sup B)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading)*.*

$$\text{Supremum}(u, B) \wedge \text{Increasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, u) \wedge \neg\, \text{Supremum}(f^{-1}(u), f^{-1}(B)).$$

Remark (Failure modes)*.* The formula can fail if the inverse is not increasing, if $\sup B$ is outside the codomain set J, or if the inverse is not continuous at $\sup B$.

Remark (Failure mode decomposition)*.* Apply the increasing image supremum theorem to $g = f^{-1}$ and the set B.

Remark (Dependencies)*.*

Increasing Image Preserves Suprema.

Theorem (Increasing Inverse Preserves Infima)

Theorem 31.3.21 (Increasing Inverse Preserves Infima)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if* $\inf B \in J$*, then*

$$\inf f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$B \neq \varnothing, \quad \inf B \in J, \quad f^{-1} \text{ increasing and continuous at } \inf B \Longrightarrow \inf f^{-1}(B) = f^{-1}(\inf B).$$

Remark (Predicate reading).

$$\mathrm{Infimum}(v, B) \land \mathrm{Increasing}(f^{-1}, J) \land \mathrm{ContinuousAt}(f^{-1}, v) \implies \mathrm{Infimum}(f^{-1}(v), f^{-1}(B)).$$

Remark (Negated quantified statement).

$$\inf f^{-1}(B) \neq f^{-1}(\inf B)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(v, B) \land \mathrm{Increasing}(f^{-1}, J) \land \mathrm{ContinuousAt}(f^{-1}, v) \land \neg\,\mathrm{Infimum}(f^{-1}(v), f^{-1}(B)).$$

Remark (Failure modes). The formula can fail if the inverse is not increasing, if $\inf B$ is outside the codomain set J, or if the inverse is not continuous at $\inf B$.

Remark (Failure mode decomposition). Apply the increasing image infimum theorem to $g = f^{-1}$ and the set B.

Remark (Dependencies).

Increasing Image Preserves Infima.

Theorem (Decreasing Inverse Sends Infimum to Supremum)

Theorem 31.3.22 (Decreasing Inverse Sends Infimum to Supremum)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Remark (Standard quantified statement).

$$B \neq \varnothing, \quad \inf B \in J, \quad f^{-1} \text{ decreasing and continuous at } \inf B \implies \sup f^{-1}(B) = f^{-1}(\inf B).$$

Remark (Predicate reading).

$$\mathrm{Infimum}(v, B) \land \mathrm{Decreasing}(f^{-1}, J) \land \mathrm{ContinuousAt}(f^{-1}, v) \implies \mathrm{Supremum}(f^{-1}(v), f^{-1}(B)).$$

Remark (Negated quantified statement).

$$\sup f^{-1}(B) \neq f^{-1}(\inf B)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(v, B) \land \mathrm{Decreasing}(f^{-1}, J) \land \mathrm{ContinuousAt}(f^{-1}, v) \land \neg\,\mathrm{Supremum}(f^{-1}(v), f^{-1}(B)).$$

Remark (Failure modes). The formula can fail if the inverse is not decreasing, if $\inf B$ is outside J, or if the inverse is not continuous at $\inf B$.

Remark (Failure mode decomposition). Apply the decreasing image infimum-to-supremum theorem to $g = f^{-1}$ and B.

Remark (Dependencies).

Decreasing Image Sends Infimum to Supremum.

Theorem (Decreasing Inverse Sends Supremum to Infimum)

Theorem 31.3.23 (Decreasing Inverse Sends Supremum to Infimum)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

Remark (Standard quantified statement).

$$B \neq \varnothing, \quad \sup B \in J, \quad f^{-1} \text{ decreasing and continuous at } \sup B \Longrightarrow \inf f^{-1}(B) = f^{-1}(\sup B).$$

Remark (Predicate reading).

$$\mathrm{Supremum}(u, B) \wedge \mathrm{Decreasing}(f^{-1}, J) \wedge \mathrm{ContinuousAt}(f^{-1}, u) \Longrightarrow \mathrm{Infimum}(f^{-1}(u), f^{-1}(B)).$$

Remark (Negated quantified statement).

$$\inf f^{-1}(B) \neq f^{-1}(\sup B)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(u, B) \wedge \mathrm{Decreasing}(f^{-1}, J) \wedge \mathrm{ContinuousAt}(f^{-1}, u) \wedge \neg\, \mathrm{Infimum}(f^{-1}(u), f^{-1}(B)).$$

Remark (Failure modes). The formula can fail if the inverse is not decreasing, if $\sup B$ is outside J, or if the inverse is not continuous at $\sup B$.

Remark (Failure mode decomposition). Apply the decreasing image supremum-to-infimum theorem to $g = f^{-1}$ and B.

Remark (Dependencies).

Decreasing Image Sends Supremum to Infimum.

31.3.4 Dilation and Displacement

Remark (Orientation). Dilation and displacement are affine operations on sets. Translation preserves the order of all comparisons, positive dilation preserves order, and negative dilation reverses order.

Definition (Displacement, Dilation, and Reflection)

Definition 31.3.24 (Displacement, Dilation, and Reflection)**.** Let $A \subseteq \mathbb{R}$ and let $c, \lambda \in \mathbb{R}$. Define

$$A + c := \{a + c : a \in A\}, \qquad A - c := \{a - c : a \in A\}, \qquad \lambda A := \{\lambda a : a \in A\}.$$

The set $-A := (-1)A$ is the *reflection* of A through the origin.

Remark (Standard quantified statement)*.*

$$x \in A + c \iff (\exists a \in A)(x = a + c), \qquad x \in \lambda A \iff (\exists a \in A)(x = \lambda a).$$

Remark (Definition predicate reading)*.*

$$\mathrm{Translate}(A, c) = A + c, \qquad \mathrm{Dilate}(A, \lambda) = \lambda A, \qquad \mathrm{Reflect}(A) = -A.$$

Remark (Negated quantified statement)*.*

$$x \notin A + c \iff (\forall a \in A)(x \neq a + c), \qquad x \notin \lambda A \iff (\forall a \in A)(x \neq \lambda a).$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{Translate}(A, c) = B \quad \text{means} \quad (\exists x)\big[x \in B \triangle (A + c)\big],$$

and similarly for dilation and reflection.

Remark (Failure modes)*.* The definition fails only if the image set omits an actual transformed point or contains a point that is not produced by transforming an element of A.

Remark (Failure mode decomposition)*.* Every membership claim in a transformed set decomposes into an original element and the transformation rule that produced the new point.

Remark (Interpretation)*.* Displacement slides the set, dilation stretches or collapses it, and reflection is the special dilation that reverses the real line.

Remark (Dependencies)*.*

Subset; Multiplication on $\mathbb{R}$; Addition on $\mathbb{R}$.

Proposition (Translation Preserves Upper Bounds)

Proposition 31.3.25 (Translation Preserves Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If u is an upper bound of A, then $u + c$ is an upper bound of $A + c$.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$(\forall a \in A)(a \leq u) \implies (\forall y \in A + c)(y \leq u + c).$$

Remark (Predicate reading)*.*

$$\mathrm{UB}(u, A) \implies \mathrm{UB}(u + c, A + c).$$

Remark (Negated quantified statement).

$$(\forall a \in A)(a \le u), \qquad (\exists y \in A + c)(u + c < y).$$

Remark (Negation predicate reading).

$$\mathrm{UB}(u, A) \wedge \neg\, \mathrm{UB}(u + c, A + c).$$

Remark (Failure modes). A failure would require a translated point above the translated upper bound.

Remark (Failure mode decomposition). Write $y = a + c$ with $a \in A$. Then $a \le u$ gives $a + c \le u + c$.

Remark (Dependencies).

Upper Bound; Displacement, Dilation, and Reflection.

Proposition (Translation Preserves Lower Bounds)

Proposition 31.3.26 (Translation Preserves Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $\ell + c$ is a lower bound of $A + c$.* *Go to proof.*

Remark (Standard quantified statement).

$$(\forall a \in A)(\ell \le a) \implies (\forall y \in A + c)(\ell + c \le y).$$

Remark (Predicate reading).

$$\mathrm{LB}(\ell, A) \implies \mathrm{LB}(\ell + c, A + c).$$

Remark (Negated quantified statement).

$$(\forall a \in A)(\ell \le a), \qquad (\exists y \in A + c)(y < \ell + c).$$

Remark (Negation predicate reading).

$$\mathrm{LB}(\ell, A) \wedge \neg\, \mathrm{LB}(\ell + c, A + c).$$

Remark (Failure modes). A failure would require a translated point below the translated lower bound.

Remark (Failure mode decomposition). Write $y = a + c$ with $a \in A$. Then $\ell \le a$ gives $\ell + c \le a + c$.

Remark (Dependencies).

Lower Bound; Displacement, Dilation, and Reflection.

Proposition (Positive Dilation Preserves Upper Bounds)

Proposition 31.3.27 (Positive Dilation Preserves Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If u is an upper bound of A, then λu is an upper bound of λA.* *Go to proof.*

Remark (Standard quantified statement).

$$\lambda > 0 \land (\forall a \in A)(a \leq u) \implies (\forall y \in \lambda A)(y \leq \lambda u).$$

Remark (Predicate reading).

$$\lambda > 0 \land \mathrm{UB}(u, A) \implies \mathrm{UB}(\lambda u, \lambda A).$$

Remark (Negated quantified statement).

$$\lambda > 0, \qquad (\forall a \in A)(a \leq u), \qquad (\exists y \in \lambda A)(\lambda u < y).$$

Remark (Negation predicate reading).

$$\lambda > 0 \land \mathrm{UB}(u, A) \land \neg\, \mathrm{UB}(\lambda u, \lambda A).$$

Remark (Failure modes). A failure would require positive multiplication to reverse an inequality.

Remark (Failure mode decomposition). Write $y = \lambda a$ with $a \in A$. Since $a \leq u$ and $\lambda > 0$, $\lambda a \leq \lambda u$.

Remark (Dependencies).

Upper Bound; Displacement, Dilation, and Reflection.

Proposition (Positive Dilation Preserves Lower Bounds)

Proposition 31.3.28 (Positive Dilation Preserves Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If ℓ is a lower bound of A, then $\lambda\ell$ is a lower bound of λA.* *Go to proof.*

Remark (Standard quantified statement).

$$\lambda > 0 \land (\forall a \in A)(\ell \leq a) \implies (\forall y \in \lambda A)(\lambda\ell \leq y).$$

Remark (Predicate reading).

$$\lambda > 0 \land \mathrm{LB}(\ell, A) \implies \mathrm{LB}(\lambda\ell, \lambda A).$$

Remark (Negated quantified statement).

$$\lambda > 0, \qquad (\forall a \in A)(\ell \leq a), \qquad (\exists y \in \lambda A)(y < \lambda\ell).$$

Remark (Negation predicate reading).

$$\lambda > 0 \land \mathrm{LB}(\ell, A) \land \neg\, \mathrm{LB}(\lambda\ell, \lambda A).$$

Remark (Failure modes). A failure would require a positively dilated point below the positively dilated lower bound.

Remark (Failure mode decomposition). Write $y = \lambda a$ with $a \in A$. Since $\ell \leq a$ and $\lambda > 0$, $\lambda\ell \leq \lambda a$.

Remark (Dependencies).

Lower Bound; Displacement, Dilation, and Reflection.

Proposition (Negative Dilation Sends Lower Bounds to Upper Bounds)

Proposition 31.3.29 (Negative Dilation Sends Lower Bounds to Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If ℓ is a lower bound of A, then $\lambda\ell$ is an upper bound of λA.* *Go to proof.*

Remark (Standard quantified statement).

$$\lambda < 0 \wedge (\forall a \in A)(\ell \leq a) \implies (\forall y \in \lambda A)(y \leq \lambda\ell).$$

Remark (Predicate reading).

$$\lambda < 0 \wedge \mathrm{LB}(\ell, A) \implies \mathrm{UB}(\lambda\ell, \lambda A).$$

Remark (Negated quantified statement).

$$\lambda < 0, \qquad (\forall a \in A)(\ell \leq a), \qquad (\exists y \in \lambda A)(\lambda\ell < y).$$

Remark (Negation predicate reading).

$$\lambda < 0 \wedge \mathrm{LB}(\ell, A) \wedge \neg\, \mathrm{UB}(\lambda\ell, \lambda A).$$

Remark (Failure modes). A failure would require negative multiplication not to reverse the lower-bound inequality.

Remark (Failure mode decomposition). Write $y = \lambda a$ with $a \in A$. Since $\ell \leq a$ and $\lambda < 0$, $\lambda a \leq \lambda\ell$.

Remark (Dependencies).

Lower Bound; Upper Bound; Displacement, Dilation, and Reflection.

Proposition (Negative Dilation Sends Upper Bounds to Lower Bounds)

Proposition 31.3.30 (Negative Dilation Sends Upper Bounds to Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If u is an upper bound of A, then λu is a lower bound of λA.* *Go to proof.*

Remark (Standard quantified statement).

$$\lambda < 0 \wedge (\forall a \in A)(a \leq u) \implies (\forall y \in \lambda A)(\lambda u \leq y).$$

Remark (Predicate reading).

$$\lambda < 0 \wedge \mathrm{UB}(u, A) \implies \mathrm{LB}(\lambda u, \lambda A).$$

Remark (Negated quantified statement).

$$\lambda < 0, \qquad (\forall a \in A)(a \le u), \qquad (\exists y \in \lambda A)(y < \lambda u).$$

Remark (Negation predicate reading).

$$\lambda < 0 \wedge \mathrm{UB}(u, A) \wedge \neg\, \mathrm{LB}(\lambda u, \lambda A).$$

Remark (Failure modes). A failure would require negative multiplication not to reverse the upper-bound inequality.

Remark (Failure mode decomposition). Write $y = \lambda a$ with $a \in A$. Since $a \le u$ and $\lambda < 0$, $\lambda u \le \lambda a$.

Remark (Dependencies).

Upper Bound; Lower Bound; Displacement, Dilation, and Reflection.

Corollary (Reflection Swaps Upper and Lower Bounds)

Corollary 31.3.31 (Reflection Swaps Upper and Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $-\ell$ is an upper bound of $-A$. If u is an upper bound of A, then $-u$ is a lower bound of $-A$. Go to proof.*

Remark (Standard quantified statement).

$$(\forall a \in A)(\ell \le a) \implies (\forall y \in -A)(y \le -\ell),$$

and

$$(\forall a \in A)(a \le u) \implies (\forall y \in -A)(-u \le y).$$

Remark (Predicate reading).

$$\mathrm{LB}(\ell, A) \implies \mathrm{UB}(-\ell, -A), \qquad \mathrm{UB}(u, A) \implies \mathrm{LB}(-u, -A).$$

Remark (Negated quantified statement). A failure is a failure of negative dilation at the scalar $\lambda = -1$.

Remark (Negation predicate reading).

$$\mathrm{LB}(\ell, A) \wedge \neg\, \mathrm{UB}(-\ell, -A)$$

or

$$\mathrm{UB}(u, A) \wedge \neg\, \mathrm{LB}(-u, -A).$$

Remark (Failure modes). Reflection reverses the real line, so lower information becomes upper information and upper information becomes lower information.

Remark (Failure mode decomposition). Apply negative dilation with $\lambda = -1$.

Remark (Dependencies).

Negative Dilation Sends Lower Bounds to Upper Bounds; Negative Dilation Sends Upper Bounds to Lower Bounds.

31.3.5 Algebra of Suprema and Infima

Current Algebraic Properties of Supremum and Infimum

Remark (Section boundary)*.* Basic bound, supremum, infimum, maximum, and minimum facts now live in Upper and Lower Bounds, Suprema and Infima, and Maxima and Minima. This section begins with algebraic transformations of those extremal values.

Definition (Set Arithmetic Images)

Definition 31.3.32 (Set Arithmetic Images)**.** Let $A, B \subseteq \mathbb{R}$. Define

$$A + B := \{a + b : a \in A,\ b \in B\}, \qquad A - B := \{a - b : a \in A,\ b \in B\},$$

$$AB := \{ab : a \in A,\ b \in B\}, \qquad A/B := \{a/b : a \in A,\ b \in B,\ b \neq 0\}.$$

Remark (Standard quantified statement)*.*

$$x \in A + B \iff (\exists a \in A)(\exists b \in B)(x = a + b),$$

and similarly for difference, product, and quotient with the quotient condition $b \neq 0$.

Remark (Definition predicate reading)*.*

$$\mathrm{SetSum}(A, B) = A+B, \quad \mathrm{SetDifference}(A, B) = A-B, \quad \mathrm{SetProduct}(A, B) = AB, \quad \mathrm{SetQuotient}(A, B) =$$

Remark (Negated quantified statement)*.*

$$x \notin A + B \iff (\forall a \in A)(\forall b \in B)(x \neq a + b),$$

with the analogous readings for the other arithmetic images.

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{SetProduct}(A, B) = C \quad \text{means} \quad (\exists x)\big[x \in C \triangle AB\big],$$

and similarly for sum, difference, and quotient.

Remark (Failure modes)*.* An arithmetic-image definition fails only if it omits a value produced by the rule or includes a value not produced by the rule.

Remark (Failure mode decomposition)*.* Every membership claim decomposes into witnesses from the input sets and the arithmetic operation that produced the image point.

Remark (Dependencies)*.*

Subset; Addition on $\mathbb{R}$; Multiplication on $\mathbb{R}$.

Definition (Reciprocal Image)

Definition 31.3.33 (Reciprocal Image)**.** Let $A \subseteq \mathbb{R}$ and suppose $0 \notin A$. The reciprocal image of A is

$$A^{-1} := \{1/a : a \in A\}.$$

Remark (Standard quantified statement).

$$x \in A^{-1} \iff (\exists a \in A)(x = 1/a).$$

Remark (Definition predicate reading).

$$\text{ReciprocalImage}(A) = A^{-1}.$$

Remark (Negated quantified statement).

$$x \notin A^{-1} \iff (\forall a \in A)(x \neq 1/a).$$

Remark (Negation predicate reading).

$$\neg\,\text{ReciprocalImage}(A) = B \quad \text{means} \quad (\exists x)\big[x \in B \triangle A^{-1}\big].$$

Remark (Failure modes). The reciprocal image is undefined if zero is allowed as an input value. Once zero is excluded, the only definition failures are omitted or spurious reciprocal values.

Remark (Failure mode decomposition). Every element of A^{-1} has the form $1/a$ for a nonzero $a \in A$.

Remark (Dependencies).

Subset; Multiplication on $\mathbb{R}$.

Theorem (Translation Invariance of the Supremum)

Theorem 31.3.34 (Translation Invariance of the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A + c) = \sup A + c, \qquad A + c = \{a + c : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$.
$\sup\{a + c : a \in A\} = \sup A + c.$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \implies \text{Supremum}(s + c, A + c).$$

Remark (Negated quantified statement).

$$\sup(A + c) \neq \sup A + c$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \neg\, \text{Supremum}(s + c, A + c)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if translation either failed to preserve upper bounds or failed to preserve leastness. Adding the same real number to every comparison preserves the order, so neither failure mode can occur.

Remark (Failure mode decomposition).

$$\forall a \in A\ (a \leq s) \iff \forall a \in A\ (a + c \leq s + c),$$
$$\forall u\, [\text{UpperBound}(u, A) \implies s \leq u] \iff \forall v\, [\text{UpperBound}(v, A + c) \implies s + c \leq v].$$

Remark (Interpretation). Translation shifts the entire set and all of its ceilings by the same amount. The least ceiling therefore shifts by exactly that amount.

Remark (Dependencies).

Upper Bound; Supremum.

Theorem (Translation Invariance of the Infimum)

Theorem 31.3.35 (Translation Invariance of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $c \in \mathbb{R}$. Then*

$$\inf(A + c) = \inf A + c.$$

Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad A \text{ bounded below} \quad \implies \quad \inf\{a + c : a \in A\} = \inf A + c.$$

Remark (Predicate reading).

$$\text{Infimum}(t, A) \implies \text{Infimum}(t + c, A + c).$$

Remark (Negated quantified statement).

$$\inf(A + c) \neq \inf A + c$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \neg\, \text{Infimum}(t + c, A + c).$$

Remark (Failure modes). The formula could fail only if translation failed to preserve lower bounds or failed to preserve greatestness among lower bounds.

Remark (Failure mode decomposition).

$$(\forall a \in A)(t \leq a) \iff (\forall a \in A)(t + c \leq a + c).$$

Remark (Interpretation). Translation shifts every floor by the same amount, so the greatest floor shifts by that amount.

Remark (Dependencies).

Lower Bound; Infimum; Set Arithmetic Images.

Theorem (Scalar Multiplication by a Positive Scalar)

Theorem 31.3.36 (Scalar Multiplication by a Positive Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded above, and let } k > 0.\\ &\sup\{ka : a \in A\} = k \sup A.\end{aligned}$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge k > 0 \Longrightarrow \text{Supremum}(ks, kA).$$

Remark (Negated quantified statement).

$$k > 0, \quad \sup(kA) \neq k \sup A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge k > 0 \wedge \neg\,\text{Supremum}(ks, kA)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if multiplying by a positive scalar failed to preserve upper bounds or failed to preserve leastness. Positive multiplication preserves the order, so both properties scale exactly.

Remark (Failure mode decomposition).

$$\begin{gathered}\forall a \in A\ (a \leq s) \iff \forall a \in A\ (ka \leq ks),\\ \text{UpperBound}(u, A) \iff \text{UpperBound}(ku, kA).\end{gathered}$$

Remark (Interpretation). Positive scaling stretches the set without reversing order. The least ceiling stretches by the same positive factor.

Remark (Dependencies).

Upper Bound; Supremum.

Theorem (Positive Scalar Multiplication of the Infimum)

Theorem 31.3.37 (Positive Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k > 0$. Then*

$$\inf(kA) = k \inf A.$$

Go to proof.

Remark (Standard quantified statement).

$$k > 0, \quad A \neq \varnothing, \quad A \text{ bounded below} \quad \Longrightarrow \quad \inf\{ka : a \in A\} = k \inf A.$$

Remark (Predicate reading).

$$\text{Infimum}(t, A) \wedge k > 0 \Longrightarrow \text{Infimum}(kt, kA).$$

Remark (Negated quantified statement).

$$k > 0, \quad \inf(kA) \neq k \inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge k > 0 \wedge \neg\,\text{Infimum}(kt, kA).$$

Remark (Failure modes). The formula could fail only if positive multiplication failed to preserve lower bounds or greatestness among lower bounds.

Remark (Failure mode decomposition).

$$(\forall a \in A)(t \leq a) \Longleftrightarrow (\forall a \in A)(kt \leq ka) \qquad (k > 0).$$

Remark (Interpretation). Positive scaling stretches the floor of the set by the same factor.

Remark (Dependencies).

Lower Bound; Infimum; Displacement, Dilation, and Reflection.

Theorem (Scalar Multiplication by a Negative Scalar)

Theorem 31.3.38 (Scalar Multiplication by a Negative Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded below, and let } k < 0.\\ &\sup\{ka : a \in A\} = k \inf A.\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{Infimum}(t, A) \wedge k < 0 \Longrightarrow \mathrm{Supremum}(kt, kA).$$

Remark (Negated quantified statement).

$$k < 0, \quad \sup(kA) \neq k \inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(t, A) \wedge k < 0 \wedge \neg\,\mathrm{Supremum}(kt, kA)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if negative multiplication did not reverse lower bounds into upper bounds, or if the extremal value lost its leastness after reversal. Multiplication by a negative scalar reverses order exactly, so the infimum of A becomes the supremum of kA.

Remark (Failure mode decomposition).

$$\begin{aligned}\forall a \in A\ (t \le a) &\Longleftrightarrow \forall a \in A\ (ka \le kt),\\ \mathrm{LowerBound}(v, A) &\Longleftrightarrow \mathrm{UpperBound}(kv, kA) \qquad (k < 0).\end{aligned}$$

Remark (Interpretation). Negative scaling reflects the set and changes floors into ceilings. The lowest point threshold of A becomes the highest point threshold of the reflected and scaled set.

Remark (Dependencies).

Lower Bound; Upper Bound; Infimum; Supremum.

Theorem (Negative Scalar Multiplication of the Infimum)

Theorem 31.3.39 (Negative Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k < 0$. Then*

$$\inf(kA) = k \sup A.$$

Go to proof.

Remark (Standard quantified statement).

$$k < 0, \quad A \neq \varnothing, \quad A \text{ bounded above} \quad \Longrightarrow \quad \inf\{ka : a \in A\} = k \sup A.$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \wedge k < 0 \Longrightarrow \mathrm{Infimum}(ks, kA).$$

Remark (Negated quantified statement).

$$k < 0, \quad \inf(kA) \neq k \sup A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge k < 0 \wedge \neg\,\mathrm{Infimum}(ks, kA).$$

Remark (Failure modes). The formula could fail only if negative multiplication failed to reverse upper bounds into lower bounds or failed to preserve the extremal threshold after reversal.

Remark (Failure mode decomposition).

$$(\forall a \in A)(a \leq s) \Longleftrightarrow (\forall a \in A)(ks \leq ka) \qquad (k < 0).$$

Remark (Interpretation). Negative scaling reflects the set, so the ceiling of A becomes the floor of kA.

Remark (Dependencies).

Upper Bound; Lower Bound; Supremum; Infimum.

Theorem (Negation Exchanges Supremum and Infimum)

Theorem 31.3.40 (Negation Exchanges Supremum and Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below. Then $-A = \{-a : a \in A\}$ is bounded above and*

$$\sup(-A) = -\inf A.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded below.}\\ &\sup\{-a : a \in A\} = -\inf A.\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{Infimum}(t, A) \Longrightarrow \mathrm{Supremum}(-t, -A).$$

Remark (Negated quantified statement).

$$\sup(-A) \neq -\inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \neg\, \text{Supremum}(-t, -A)$$

is a contradictory configuration.

Remark (Failure modes). The theorem could fail only if negation did not convert lower bounds into upper bounds or did not preserve the extremal threshold after reversing order. The map $x \mapsto -x$ reverses all inequalities, so the exchange is exact.

Remark (Failure mode decomposition).

$$\forall a \in A\ (t \leq a) \iff \forall a \in A\ (-a \leq -t),$$
$$\text{LowerBound}(v, A) \iff \text{UpperBound}(-v, -A).$$

Remark (Interpretation). Negation reflects the real line through the origin. Floors become ceilings, so the infimum of the original set becomes the negative of the supremum of the reflected set.

Remark (Dependencies).

Lower Bound; Upper Bound; Infimum; Supremum.

Theorem (Supremum of a Sum Set)

Theorem 31.3.41 (Supremum of a Sum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup(A + B) = \sup A + \sup B, \qquad A + B = \{a + b : a \in A,\ b \in B\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded above.}$$
$$\sup\{a + b : a \in A,\ b \in B\} = \sup A + \sup B.$$

Remark (Predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies \text{Supremum}(s_A + s_B, A + B).$$

Remark (Negated quantified statement).

$$\sup(A + B) \neq \sup A + \sup B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge \neg\, \text{Supremum}(s_A + s_B, A + B)$$

is a contradictory configuration.

Remark (Failure modes). The equality could fail only if the sum of the two suprema did not bound the sum set, or if it were not the least such bound. The first part follows from adding the two upper-bound inequalities; the second follows by approximating each supremum from within its set.

Remark (Failure mode decomposition).

$$a \le \sup A, \quad b \le \sup B \Longrightarrow a + b \le \sup A + \sup B,$$
$$\varepsilon > 0 \Longrightarrow \exists a \in A,\ b \in B : \sup A + \sup B - \varepsilon < a + b.$$

Remark (Interpretation). The highest limiting value of all pairwise sums is obtained by combining values that approach the separate ceilings of A and B.

Remark (Dependencies).

Supremum; Epsilon Characterization of Supremum.

Theorem (Infimum of a Sum Set)

Theorem 31.3.42 (Infimum of a Sum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf(A + B) = \inf A + \inf B.$$

Go to proof.

Remark (Standard quantified statement).

$$A, B \neq \varnothing, \quad A, B \text{ bounded below} \quad \Longrightarrow \quad \inf\{a + b : a \in A,\ b \in B\} = \inf A + \inf B.$$

Remark (Predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \Longrightarrow \text{Infimum}(t_A + t_B, A + B).$$

Remark (Negated quantified statement).

$$\inf(A + B) \neq \inf A + \inf B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \neg\,\text{Infimum}(t_A + t_B, A + B).$$

Remark (Failure modes). The equality could fail only if the sum of the two infima failed to be a lower bound of the sum set or failed to be the greatest such lower bound.

Remark (Failure mode decomposition).

$$t_A \le a, \quad t_B \le b \Longrightarrow t_A + t_B \le a + b,$$

and sharpness follows by choosing elements near each infimum.

Remark (Interpretation). The lowest limiting value of pairwise sums is obtained by combining values near the separate floors of the two sets.

Remark (Dependencies).

Infimum; Set Arithmetic Images.

Theorem (Supremum of a Difference Set)

Theorem 31.3.43 (Supremum of a Difference Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \qquad A - B = \{a - b : a \in A,\ b \in B\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A, B \subseteq \mathbb{R}$ be nonempty, A bounded above, and B bounded below.
$\sup\{a - b : a \in A,\ b \in B\} = \sup A - \inf B$.

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \Longrightarrow \text{Supremum}(s - t, A - B).$$

Remark (Negated quantified statement).

$$\sup(A - B) \neq \sup A - \inf B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \wedge \neg\, \text{Supremum}(s - t, A - B)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if subtracting elements of B did not convert the lower extremal behavior of B into an upper contribution. Since $a - b = a + (-b)$, this is the sum theorem combined with the negation exchange.

Remark (Failure mode decomposition).

$$A - B = A + (-B), \qquad \sup(-B) = -\inf B, \qquad \sup(A + (-B)) = \sup A + \sup(-B).$$

Remark (Interpretation). The largest possible difference comes from making the first term as large as possible and the subtracted term as small as possible.

Remark (Dependencies).

Supremum; Infimum; Negation Exchanges Supremum and Infimum; Supremum of a Sum Set.

Theorem (Infimum of a Difference Set)

Theorem 31.3.44 (Infimum of a Difference Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded below and B bounded above. Then*

$$\inf(A - B) = \inf A - \sup B.$$

Go to proof.

Remark (Standard quantified statement).

$$A, B \neq \varnothing, \quad A \text{ bounded below}, \quad B \text{ bounded above} \quad \Longrightarrow \quad \inf\{a-b : a \in A,\ b \in B\} = \inf A - \sup B. \blacksquare$$

Remark (Predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Supremum}(s, B) \Longrightarrow \text{Infimum}(t - s, A - B).$$

Remark (Negated quantified statement).

$$\inf(A - B) \neq \inf A - \sup B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Supremum}(s, B) \wedge \neg\,\text{Infimum}(t - s, A - B).$$

Remark (Failure modes). The formula could fail only if subtracting B failed to convert its upper extremal behavior into a lower contribution.

Remark (Failure mode decomposition).

$$A - B = A + (-B), \qquad \inf(-B) = -\sup B, \qquad \inf(A + (-B)) = \inf A + \inf(-B).$$

Remark (Interpretation). The smallest possible difference comes from making the first term as small as possible and the subtracted term as large as possible.

Remark (Dependencies).

Infimum; Supremum; Infimum of a Sum Set; Negative Scalar Multiplication of the Infimum.

Theorem (Supremum of a Dilation)

Theorem 31.3.45 (Supremum of a Dilation)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \qquad \lambda A = \{\lambda a : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded, and let } \lambda \in \mathbb{R}.\\ &\sup\{\lambda a : a \in A\} = \max\{\lambda \sup A, \lambda \inf A\}.\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \wedge \mathrm{Infimum}(t, A) \Longrightarrow \mathrm{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A).$$

Remark (Negated quantified statement).

$$\sup(\lambda A) \neq \max\{\lambda \sup A, \lambda \inf A\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge \mathrm{Infimum}(t, A) \wedge \neg\,\mathrm{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A)$$

is a contradictory configuration.

Remark (Failure modes). The formula separates into the cases $\lambda > 0$, $\lambda = 0$, and $\lambda < 0$. Positive dilation scales the supremum, zero dilation collapses the set to $\{0\}$, and negative dilation turns the infimum into the supremum.

Remark (Failure mode decomposition).

$$\sup(\lambda A) = \begin{cases} \lambda \sup A, & \lambda > 0, \\ 0, & \lambda = 0, \\ \lambda \inf A, & \lambda < 0. \end{cases}$$

Remark (Interpretation). A dilation either preserves the order, collapses all points to zero, or reverses the order. The displayed maximum packages those three cases into one formula.

Remark (Dependencies).

Supremum; Infimum; Maximum; Scalar Multiplication by a Positive Scalar; Scalar Multiplication by a Negative Scalar.

Theorem (Supremum of the Absolute-Value Image)

Theorem 31.3.46 (Supremum of the Absolute-Value Image)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup|A| = \max\{|\sup A|, |\inf A|\}, \qquad |A| = \{|a| : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded.}\\ &\sup\{|a| : a \in A\} = \max\{|\sup A|, |\inf A|\}.\end{aligned}$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \Longrightarrow \text{Supremum}(\max\{|s|, |t|\}, |A|).$$

Remark (Negated quantified statement).

$$\sup |A| \neq \max\{|\sup A|, |\inf A|\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \wedge \neg\, \text{Supremum}(\max\{|s|, |t|\}, |A|)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if the farthest absolute-value size were not achieved at one of the two extremal thresholds. Since every $a \in A$ lies between $\inf A$ and $\sup A$, its distance from zero is bounded by the larger endpoint magnitude.

Remark (Failure mode decomposition).

$$\inf A \leq a \leq \sup A \Longrightarrow |a| \leq \max\{|\sup A|, |\inf A|\}.$$

Remark (Interpretation). The absolute-value image measures distance from the origin. For a bounded subset of the real line, the largest possible distance is controlled by whichever extremal endpoint lies farther from zero.

Remark (Dependencies).

Supremum; Infimum; Maximum; Every Element Lies Below the Supremum.

Theorem (Supremum of a Reciprocal Set)

Theorem 31.3.47 (Supremum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\sup(A^{-1}) = \frac{1}{\inf A}.$$

Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A] \quad \Longrightarrow \quad \sup\{1/a : a \in A\} = (\inf A)^{-1}.$$

Remark (Predicate reading).

$$\text{Infimum}(m, A) \wedge \text{Supremum}(M, A) \wedge 0 \notin [m, M] \Longrightarrow \text{Supremum}(m^{-1}, A^{-1}).$$

Remark (Negated quantified statement).

$$A \neq \varnothing, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A], \quad \sup(A^{-1}) \neq (\inf A)^{-1}.$$

Remark (Negation predicate reading).

$$\text{Infimum}(m, A) \wedge 0 \notin [m, \sup A] \wedge \neg\, \text{Supremum}(m^{-1}, A^{-1})$$

is contradictory under the stated hypotheses.

Remark (Failure modes). The formula fails without a nonzero gap from zero: reciprocal images may be unbounded or undefined. With the gap, reciprocal is order-reversing on the whole interval containing A, so the floor of A becomes the ceiling of A^{-1}.

Remark (Failure mode decomposition). Every $a \in A$ lies between $\inf A$ and $\sup A$, and the interval does not cross zero. The reciprocal map reverses comparisons there, so $1/a \leq 1/\inf A$. Leastness follows by applying the same reversal to values of A approaching $\inf A$.

Remark (Interpretation). Taking reciprocals turns small nonzero magnitudes into large magnitudes. The least value threshold of the original set therefore controls the largest value threshold of the reciprocal set.

Remark (Dependencies).

Bounded Set; Infimum; Supremum.

Theorem (Infimum of a Reciprocal Set)

Theorem 31.3.48 (Infimum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\inf(A^{-1}) = \frac{1}{\sup A}.$$

Go to proof.

Remark (Standard quantified statement).

$$A \neq \varnothing, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A] \quad \Longrightarrow \quad \inf\{1/a : a \in A\} = (\sup A)^{-1}.$$

Remark (Predicate reading).

$$\text{Infimum}(m, A) \wedge \text{Supremum}(M, A) \wedge 0 \notin [m, M] \Longrightarrow \text{Infimum}(M^{-1}, A^{-1}).$$

Remark (Negated quantified statement).

$$A \neq \varnothing, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A], \quad \inf(A^{-1}) \neq (\sup A)^{-1}.$$

Remark (Negation predicate reading).

$$\text{Supremum}(M, A) \wedge 0 \notin [\inf A, M] \wedge \neg\, \text{Infimum}(M^{-1}, A^{-1})$$

is contradictory under the stated hypotheses.

Remark (Failure modes). The formula fails without excluding zero from the closed extremal interval. Once zero is excluded, reciprocal reverses order and the ceiling of A becomes the floor of A^{-1}.

Remark (Failure mode decomposition). Every reciprocal $1/a$ lies above $1/\sup A$ because reciprocal reverses order on the nonzero interval containing A. Greatestness of $\sup A$ gives elements of A arbitrarily close to that ceiling, which makes the lower bound sharp.

Remark (Interpretation). For nonzero bounded sets that stay on one side of zero, reciprocal swaps the two extremal thresholds.

Remark (Dependencies).

Bounded Set; Infimum; Supremum; Supremum of a Reciprocal Set.

Theorem (Supremum of a Product Set)

Theorem 31.3.49 (Supremum of a Product Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup(AB) = \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Remark (Standard quantified statement).

$$A, B \neq \varnothing, \quad A, B \text{ bounded} \quad \Longrightarrow \quad \sup\{ab : a \in A,\ b \in B\}$$

equals the maximum of the four endpoint products.

Remark (Predicate reading).

$$\mathrm{Infimum}(\alpha, A) \wedge \mathrm{Supremum}(\beta, A) \wedge \mathrm{Infimum}(\gamma, B) \wedge \mathrm{Supremum}(\delta, B) \Longrightarrow \mathrm{Supremum}(\max\{\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta\}, A$$

Remark (Negated quantified statement).

$$\sup(AB) \neq \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading). The product endpoint maximum is not the supremum of AB.

Remark (Failure modes). A failure would require a product above all four endpoint products or would require the endpoint maximum not to be least.

Remark (Failure mode decomposition). Every pair (a, b) lies in the rectangle $[\inf A, \sup A] \times [\inf B, \sup B]$. A bilinear product on a rectangle has its extreme values at corners, and elements can approximate the relevant corners from within the two sets.

Remark (Dependencies).

Set Arithmetic Images; Infimum; Supremum; Maximum.

Theorem (Infimum of a Product Set)

Theorem 31.3.50 (Infimum of a Product Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf(AB) = \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$A, B \neq \varnothing, \quad A, B \text{ bounded} \quad \Longrightarrow \quad \inf\{ab : a \in A,\ b \in B\}$$

equals the minimum of the four endpoint products.

Remark (Predicate reading)*.*

$$\text{Infimum}(\alpha, A) \wedge \text{Supremum}(\beta, A) \wedge \text{Infimum}(\gamma, B) \wedge \text{Supremum}(\delta, B) \Longrightarrow \text{Infimum}(\min\{\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta\}, AB)$$

Remark (Negated quantified statement)*.*

$$\inf(AB) \neq \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading)*.* The product endpoint minimum is not the infimum of AB.

Remark (Failure modes)*.* A failure would require a product below all four endpoint products or would require the endpoint minimum not to be greatest.

Remark (Failure mode decomposition)*.* Reduce to the same endpoint analysis as the supremum product formula, replacing the corner maximum by the corner minimum.

Remark (Dependencies)*.*

Set Arithmetic Images; Infimum; Supremum; Minimum; Supremum of a Product Set.

Theorem (Supremum of a Quotient Set)

Theorem 31.3.51 (Supremum of a Quotient Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\sup(A/B) = \max\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$A, B \neq \varnothing, \quad A, B \text{ bounded}, \quad 0 \notin [\inf B, \sup B]$$

implies that the supremum of $\{a/b : a \in A,\ b \in B\}$ is the maximum of the four endpoint quotients.

Remark (Predicate reading).

$$\operatorname{Supremum}\left(\max\left\{\frac{\inf A}{\inf B},\frac{\inf A}{\sup B},\frac{\sup A}{\inf B},\frac{\sup A}{\sup B}\right\},A/B\right).$$

Remark (Negated quantified statement). The displayed endpoint maximum is not the supremum of A/B.

Remark (Negation predicate reading).

$$\neg\operatorname{Supremum}(\text{endpoint quotient maximum}, A/B).$$

Remark (Failure modes). A failure would require the quotient set to see values not controlled by the reciprocal endpoint interval for B.

Remark (Failure mode decomposition). Use $A/B = AB^{-1}$, the reciprocal endpoint formulas for B^{-1}, and the product endpoint formula.

Remark (Dependencies).

Set Arithmetic Images; Reciprocal Image; Supremum of a Reciprocal Set; Infimum of a Reciprocal Set; Supremum of a Product Set.

Theorem (Infimum of a Quotient Set)

Theorem 31.3.52 (Infimum of a Quotient Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\inf(A/B) = \min\left\{\frac{\inf A}{\inf B},\frac{\inf A}{\sup B},\frac{\sup A}{\inf B},\frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Remark (Standard quantified statement).

$$A, B \neq \varnothing, \quad A, B \text{ bounded}, \quad 0 \notin [\inf B, \sup B]$$

implies that the infimum of $\{a/b : a \in A,\ b \in B\}$ is the minimum of the four endpoint quotients.

Remark (Predicate reading).

$$\operatorname{Infimum}\left(\min\left\{\frac{\inf A}{\inf B},\frac{\inf A}{\sup B},\frac{\sup A}{\inf B},\frac{\sup A}{\sup B}\right\},A/B\right).$$

Remark (Negated quantified statement). The displayed endpoint minimum is not the infimum of A/B.

Remark (Negation predicate reading).

$$\neg\operatorname{Infimum}(\text{endpoint quotient minimum}, A/B).$$

Remark (Failure modes). A failure would require a quotient below all endpoint quotients or would require the endpoint minimum not to be greatest among lower bounds.

Remark (Failure mode decomposition). Use $A/B = AB^{-1}$ and combine the reciprocal endpoint formulas with the product endpoint infimum formula.

Remark (Dependencies).

Set Arithmetic Images; Reciprocal Image; Supremum of a Reciprocal Set; Infimum of a Reciprocal Set; Infimum of a Product Set.

Proposition (Bounds for Suprema and Infima Under Set Addition)

Proposition 31.3.53 (Bounds for Suprema and Infima Under Set Addition)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A+B) \qquad \textit{and} \qquad \sup(A+B) \leq \sup A + \sup B.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded.}\\ &\inf A + \inf B \leq \inf(A+B) \quad \wedge \quad \sup(A+B) \leq \sup A + \sup B.\end{aligned}$$

Remark (Predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies t_A + t_B \leq \inf(A+B) \leq \sup(A+$$

Remark (Negated quantified statement).

$$\inf(A+B) < \inf A + \inf B \quad \text{or} \quad \sup A + \sup B < \sup(A+B)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\big[\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \inf(A+B) < t_A + t_B\big] \vee \big[\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge s_A + s_B <$$

is a contradictory configuration.

Remark (Failure modes). The lower inequality could fail only if a sum fell below the sum of the two lower thresholds. The upper inequality could fail only if a sum rose above the sum of the two upper thresholds.

Remark (Failure mode decomposition).

$$\inf A \leq a \leq \sup A, \quad \inf B \leq b \leq \sup B \implies \inf A + \inf B \leq a + b \leq \sup A + \sup B.$$

Remark (Interpretation). Every element of the sum set is built from one element of A and one element of B. Adding the individual lower and upper controls gives immediate lower and upper controls for every such sum.

Remark (Dependencies).

Infimum; Supremum; Supremum of a Sum Set.

Proposition (Upper Bounds Depend on the Ambient Order)

Proposition 31.3.54 (Upper Bounds Depend on the Ambient Order). *Let $(S, \leq_S)$ and $(T, \leq_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order. Go to proof.*

Remark (Standard quantified statement).

$$u \text{ is an upper bound of } A \text{ in } S \iff u \in S \text{ and } \forall x \in A\ (x \leq_S u),$$
$$u \text{ is an upper bound of } A \text{ in } T \iff u \in T \text{ and } \forall x \in A\ (x \leq_T u).$$

Remark (Predicate reading).

$$\mathrm{UpperBound}_S(u, A) \iff u \in S \wedge \forall x \in A\ (x \leq_S u).$$

Remark (Negated quantified statement).

$$\text{The phrase "}u\text{ is an upper bound of }A\text{"}$$

is not well-determined until the ambient ordered set and its order relation are fixed.

Remark (Negation predicate reading).

$$\neg\,\mathrm{WellDetermined}(\mathrm{UpperBound}(u, A))$$

when the ambient order is omitted.

Remark (Failure modes). Ambiguity occurs when the same set A is considered inside different ordered ambient sets or under different order relations. The candidate bound may belong to one ambient set and not another, or the relevant comparison relation may change.

Remark (Failure mode decomposition).

$$\mathrm{UpperBound}_S(u, A) \text{ and } \mathrm{UpperBound}_T(u, A)$$

are separate assertions unless $S = T$ and $\leq_S = \leq_T$ on the elements being compared.

Remark (Interpretation). An upper bound is not only about the subset; it is also about where the subset lives. Changing the ambient order changes the universe of possible bounds and the comparison used to test them.

Remark (Dependencies).

Upper Bound.

Proposition (Suprema Depend on the Ambient Set)

Proposition 31.3.55 (Suprema Depend on the Ambient Set). *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S, if it exists, need not agree with the supremum of A relative to S'. Go to proof.*

Remark (Standard quantified statement).

$$\sup_S A \text{ and } \sup_{S'} A$$

are ambient-relative objects and may differ in existence or value.

Remark (Predicate reading).

$$\mathrm{Supremum}_S(s, A) \quad \text{and} \quad \mathrm{Supremum}_{S'}(s, A)$$

are different assertions unless the ambient upper-bound sets agree.

Remark (Negated quantified statement).

$$\sup_S A = \sup_{S'} A$$

is not guaranteed by $A \subseteq S \subseteq S'$ alone.

Remark (Negation predicate reading).

$$\mathrm{Supremum}_{S'}(s, A) \wedge \neg\, \mathrm{Supremum}_S(s, A)$$

is possible when the larger ambient set contains the least upper bound but the smaller one does not.

Remark (Failure modes). The ambient-relative supremum can fail to transfer because the candidate least upper bound may not belong to the smaller ambient set, or because the smaller ambient set may have a different collection of upper bounds.

Remark (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S \Longrightarrow s \text{ cannot be the supremum of } A \text{ relative to } S.$$

Remark (Interpretation). Suprema are least upper bounds inside a chosen universe. Enlarging or shrinking that universe can create or remove the point that should serve as the least ceiling.

Remark (Dependencies).

Upper Bound; Supremum.

Theorem (Ambient Existence of Supremum)

Theorem 31.3.56 (Ambient Existence of Supremum)**.** *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S. Go to proof.*

Remark (Standard quantified statement).

$$\exists S \subseteq S' \; \exists A \subseteq S : \quad \sup_{S'} A \text{ exists} \quad \text{and} \quad \sup_S A \text{ does not exist.}$$

Remark (Predicate reading).

$$\exists S \subseteq S' \; \exists A \subseteq S \; \exists s \in S' : \text{Supremum}_{S'}(s, A) \wedge \neg \exists t \in S \; \text{Supremum}_{S}(t, A).$$

Remark (Negated quantified statement).

Every ambient enlargement preserves and reflects existence of suprema.

This assertion is false.

Remark (Negation predicate reading).

$$\forall S \subseteq S' \; \forall A \subseteq S : \left[\exists s \in S' \; \text{Supremum}_{S'}(s, A) \implies \exists t \in S \; \text{Supremum}_{S}(t, A)\right]$$

is false.

Remark (Failure modes). The transfer from S' to S fails when the least upper bound exists in the larger ambient set but is missing from the smaller one. The smaller ambient set may contain upper bounds without containing a least upper bound.

Remark (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S, \quad \text{and no } t \in S \text{ is least among the upper bounds of } A \text{ in } S.$$

Remark (Interpretation). Completeness is an ambient property. A larger ordered universe can fill a gap that remains absent in a smaller ordered universe.

Remark (Dependencies).

Upper Bound; Supremum; Suprema Depend on the Ambient Set.

Lemma (Rational Gap Example for Suprema)

Lemma 31.3.57 (Rational Gap Example for Suprema). *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$. *Go to proof.*

Remark (Standard quantified statement).

$$A = \{q \in \mathbb{Q} : q^2 < 2\} \quad \implies \quad \sup_{\mathbb{Q}} A \text{ does not exist.}$$

Remark (Predicate reading).

$$\neg \exists s \in \mathbb{Q} \; \text{Supremum}_{\mathbb{Q}}(s, A), \qquad A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Remark (Negated quantified statement).

$$\exists s \in \mathbb{Q} \; (s = \sup_{\mathbb{Q}} A)$$

is false for this set.

Remark (Negation predicate reading).

$$\exists s \in \mathbb{Q}\ \mathrm{Supremum}_{\mathbb{Q}}(s, A)$$

is false.

Remark (Failure modes). A rational candidate below $\sqrt{2}$ fails to be an upper bound. A rational candidate above $\sqrt{2}$ is an upper bound but is not least, because another rational upper bound lies between it and $\sqrt{2}$.

Remark (Failure mode decomposition).

$$\begin{cases} s^2 < 2 & \Longrightarrow s \text{ is not an upper bound of } A, \\ s^2 > 2 & \Longrightarrow s \text{ is not the least upper bound of } A. \end{cases} \qquad (s \in \mathbb{Q})$$

Remark (Interpretation). The set wants $\sqrt{2}$ as its least ceiling, but that number is not rational. The rational ambient set therefore has a visible gap where the supremum should be.

Remark (Dependencies).

Upper Bound; Supremum; Ambient Existence of Supremum.

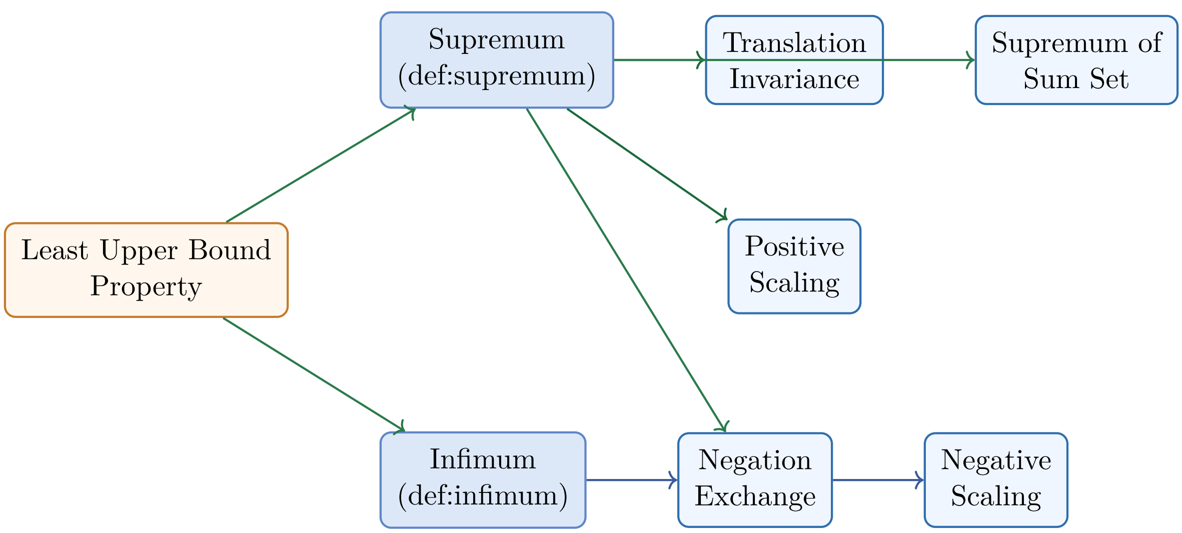
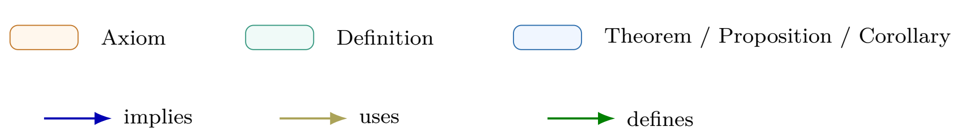


Figure 31.9: Bound Algebra Cluster Dependency Map. The map originates from the 'Least Upper Bound Property' (Axiom) and branches into the dual properties of the continuum. The Negation Exchange serves as the structural bridge between supremum and infimum algebra.



Chapter 32

Functions

SERIES	VOLUME III	BOOK I	CHAPTER 4
From Cantor to Ito	Classical Analysis	Bounds, Sequences, and Series	**Functions**

32.1 Algebra of Functions

Algebra of Functions — Quick Reference

Statement	Role
Injective composition	Injectivity is preserved by composition
Surjective composition	Surjectivity is preserved by composition
Bijective composition	Bijections compose to bijections
Inverse bijection	Bijections have bijective inverses
Preimage algebra	Preimages preserve unions and intersections exactly

Algebra of Functions

Remark (Intervals on $\mathbb{R}$). Every open interval (a, b) consists entirely of cluster points; $[a, b]$ is closed and bounded. Natural domains for functions are typically intervals or unions of intervals.

Proposition

Proposition 32.1.1 (Composition of Injections Is Injective)**.** *Let $f : A \to B$ and let $g : B \to C$. If f is injective and g is injective, then $g \circ f : A \to C$ is injective.*

Go to proof.

Remark (Standard quantified statement).

$$\Big(\forall a_1, a_2 \in A\ (f(a_1) = f(a_2) \Rightarrow a_1 = a_2)\Big) \wedge \Big(\forall b_1, b_2 \in B\ (g(b_1) = g(b_2) \Rightarrow b_1 = b_2)\Big)$$
$$\Rightarrow \forall a_1, a_2 \in A\ (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2).$$

Remark (Definition predicate reading).

$$\text{Injective}(f) \wedge \text{Injective}(g) \implies \text{Injective}(g \circ f).$$

Remark (Negated quantified statement).

$$\exists a_1, a_2 \in A\ \big(a_1 \neq a_2\ \wedge\ g(f(a_1)) = g(f(a_2))\big).$$

Remark (Negation predicate reading).

$$\neg\ \text{Injective}(g \circ f).$$

Remark (Failure modes). The composite fails to be injective exactly when two distinct inputs of A are sent to the same output in C after applying both functions.

Remark (Failure mode decomposition).

$$\exists a_1, a_2 \in A\ \big(a_1 \neq a_2\ \wedge\ g(f(a_1)) = g(f(a_2))\big).$$

Remark (Interpretation). Injectivity survives composition: equality after $g \circ f$ can be pulled back first through g and then through f.

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 32.1.2 (Composition of Surjections Is Surjective)**.** Surjective$(f) \wedge$ Surjective$(g) \Rightarrow$ Surjective$(g \circ f)$.

Go to proof.

Remark (Standard quantified statement).

$$\text{Surjective}(f) \wedge \text{Surjective}(g)\ \Rightarrow\ \forall c \in C,\ \exists a \in A : g(f(a)) = c.$$

Remark (Definition predicate reading).

$$\text{Surjective}(g \circ f, C)\ :\equiv\ \forall c \in C,\ \exists a \in A : (g \circ f)(a) = c.$$

Remark (Dependencies).

- Composition

- Function

Corollary

Corollary 32.1.3 (Composition of Bijections Is Bijective)**.** *Let $f : A \to B$ and let $g : B \to C$. If f is bijective and g is bijective, then $g \circ f : A \to C$ is bijective.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\big(\forall a_1, a_2 \in A\ (f(a_1) = f(a_2) \Rightarrow a_1 = a_2)\big) \wedge \big(\forall b \in B\ \exists a \in A\ (f(a) = b)\big)\Big]\\ &\quad \wedge \Big[\big(\forall b_1, b_2 \in B\ (g(b_1) = g(b_2) \Rightarrow b_1 = b_2)\big) \wedge \big(\forall c \in C\ \exists b \in B\ (g(b) = c)\big)\Big]\\ &\quad \Rightarrow \Big[\big(\forall a_1, a_2 \in A\ (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2)\big) \wedge \big(\forall c \in C\ \exists a \in A\ (g(f(a)) = c)\big)\Big].\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Bijective}(f) \wedge \mathrm{Bijective}(g) \implies \mathrm{Bijective}(g \circ f).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists a_1, a_2 \in A\ \big(a_1 \neq a_2\ \wedge\ g(f(a_1)) = g(f(a_2))\big)\\ &\quad \vee\ \exists c \in C\ \forall a \in A\ \big(g(f(a)) \neq c\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Bijective}(g \circ f).$$

Remark (Failure modes). A composite bijection can fail only by losing injectivity or by missing some target value in C.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{Bijective}(g \circ f) \iff \neg\,\mathrm{Injective}(g \circ f)\ \vee\ \neg\,\mathrm{Surjective}(g \circ f).$$

Remark (Interpretation). A bijection is an injective and surjective function. Since both of those properties are preserved by composition, bijectivity is preserved too.

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 32.1.4 (Inverse of a Bijection Is a Bijection)**.** $\mathrm{Bijective}(f) \Rightarrow \mathrm{Bijective}(f^{-1})$.
Go to proof.

Remark (Standard quantified statement).

$$\text{Bijective}(f) \;\Rightarrow\; \text{InverseFunction}(f, f^{-1}),\; f^{-1} \circ f = \text{id}_A,\; f \circ f^{-1} = \text{id}_B.$$

Remark (Definition predicate reading).

$$\text{InverseFunction}(f, f^{-1}) \;:\equiv\; f^{-1} \circ f = \text{id}_A \;\wedge\; f \circ f^{-1} = \text{id}_B.$$

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 32.1.5 (Preimage Distributes over Union and Intersection)**.** *Let $f : A \to B$, $S, T \subseteq B$. Then:*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

(iii) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} (i) \quad & x \in f^{-1}(S \cup T) \iff f(x) \in S \vee f(x) \in T \iff x \in f^{-1}(S) \cup f^{-1}(T). \\ (ii) \quad & x \in f^{-1}(S \cap T) \iff f(x) \in S \wedge f(x) \in T \iff x \in f^{-1}(S) \cap f^{-1}(T). \end{aligned}$$

Remark (Negated quantified statement).

$$x \notin f^{-1}(S) \iff f(x) \notin S \iff x \in f^{-1}(S^c).$$

Remark (Interpretation). Preimage is a complete lattice homomorphism: it preserves all set operations including complement. Contrast with image: $f(A \cap B) \subseteq f(A) \cap f(B)$ with strict containment possible when f is not injective. Preimage identities are load-bearing in topology (preimages of open sets are open) and measure theory (preimages of measurable sets are measurable).

Remark (Dependencies).

- Function

32.2 Real-Valued Functions

Real-Valued Functions — Quick Reference

Concept	Meaning
Pointwise sum	Algebraic addition of function values
Pointwise product	Product fg, distinct from composition
Pointwise quotient	Quotient where denominator is nonzero
Bounded function	A single bound controls all values on A
Supremum on a set	Least upper bound of $f(A)$
Maximum point	Point where the supremum is attained
Monotone function	Increasing or decreasing behavior on a set
Constant function	Same value at every input

Real-Valued Functions

Definition (Pointwise Sum of Functions)

Definition 32.2.1 (Pointwise Sum of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise sum $f + g : X \to \mathbb{R}$ is the function defined by

$$(f + g)(x) = f(x) + g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((f + g)(x) = f(x) + g(x)\big).$$

Remark (Interpretation). The sum is computed output by output; at each input, add the two function values at that same input.

Remark (Dependencies).

- Function
- Subset
- Addition on $\mathbb{R}$

Definition (Pointwise Difference of Functions)

Definition 32.2.2 (Pointwise Difference of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise difference $f - g : X \to \mathbb{R}$ is the function defined by

$$(f - g)(x) = f(x) - g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((f - g)(x) = f(x) - g(x)\big).$$

Remark (Interpretation). The difference subtracts the output of g from the output of f at each input separately.

Remark (Dependencies).

- Function
- Subset
- Subtraction on $\mathbb{R}$

Definition (Pointwise Product of Functions)

Definition 32.2.3 (Pointwise Product of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise product $fg : X \to \mathbb{R}$ is the function defined by

$$(fg)(x) = f(x)g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((fg)(x) = f(x)g(x)\big).$$

Remark (Interpretation). The product is not composition: it multiplies the two real outputs at the same input.

Remark (Dependencies).

- Function
- Subset
- Multiplication on $\mathbb{R}$

Definition (Pointwise Scalar Multiple of a Function)

Definition 32.2.4 (Pointwise Scalar Multiple of a Function)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $c \in \mathbb{R}$. The pointwise scalar multiple $cf : X \to \mathbb{R}$ is the function defined by

$$(cf)(x) = cf(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((cf)(x) = cf(x)\big).$$

Remark (Interpretation). Scalar multiplication stretches every output of the function by the same real scalar.

Remark (Dependencies).

- Function

- Subset
- Multiplication on $\mathbb{R}$

Definition (Pointwise Absolute Value of a Function)

Definition 32.2.5 (Pointwise Absolute Value of a Function)**.** Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$. The pointwise absolute value $|f| : X \to \mathbb{R}$ is the function defined by

$$|f|(x) = |f(x)| \qquad (x \in X).$$

Remark (Standard quantified statement)*.*

$$\forall x \in X \ \big(|f|(x) = |f(x)|\big).$$

Remark (Interpretation)*.* The absolute value operation applies the ordinary real absolute value to each output.

Remark (Dependencies)*.*

- Function
- Subset
- Absolute Value on $\mathbb{R}$

Definition (Pointwise Maximum of Two Functions)

Definition 32.2.6 (Pointwise Maximum of Two Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise maximum $\max(f, g) : X \to \mathbb{R}$ is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \qquad (x \in X).$$

Remark (Standard quantified statement)*.*

$$\forall x \in X \ \big(\max(f, g)(x) = \max\{f(x), g(x)\}\big).$$

Remark (Interpretation)*.* At each input, the new function keeps the larger of the two output values.

Remark (Dependencies)*.*

- Function
- Subset
- Maximum

Definition (Pointwise Quotient of Functions)

Definition 32.2.7 (Pointwise Quotient of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. Suppose $g(x) \neq 0$ for every $x \in X$. The pointwise quotient $f/g : X \to \mathbb{R}$ is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\big(\forall x \in X \ (g(x) \neq 0)\big) \implies \forall x \in X \ \big((f/g)(x) = f(x)/g(x)\big).$$

Remark (Interpretation). The quotient is defined pointwise only when the denominator function never vanishes on the domain.

Remark (Dependencies).

- Function
- Subset
- Division on $\mathbb{R}$

Definition (Linear Combination of Real-Valued Functions)

Definition 32.2.8 (Linear Combination of Real-Valued Functions)**.** Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha f + \beta g : X \to \mathbb{R}$ is the function defined by

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x)\big).$$

Remark (Interpretation). A linear combination is built only from scalar multiples and sums. It is the shared algebraic pattern behind sum rules, constant-multiple rules, and finite linear-combination rules for functions.

Remark (Dependencies).

- Pointwise Scalar Multiple of a Function
- Pointwise Sum of Functions

Definition (Real-Linear Rule)

Definition 32.2.9 (Real-Linear Rule)**.** Let $\mathcal{C}$ be a class of real-valued functions closed under real linear combinations. A rule T on $\mathcal{C}$ is real-linear if, whenever $f, g \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

whenever both sides are defined.

Remark (Standard quantified statement).

$$f, g \in \mathcal{C} \implies T(\alpha f + \beta g) = \alpha T(f) + \beta T(g).$$

Remark (Interpretation). Linearity records exactly the behavior of preserving addition and scalar multiplication. Nonlinear operations such as products, quotients, absolute values, and order comparisons require their own dependency information.

Remark (Dependencies).

- Function
- Linear Combination of Real-Valued Functions

Definition (Bounded Above Function)

Definition 32.2.10 (Bounded Above Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded above on A if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \qquad (x \in A).$$

Remark (Standard quantified statement).

$$\exists M \in \mathbb{R}\ \forall x \in A\ \big(f(x) \leq M\big).$$

Remark (Negated quantified statement).

$$\forall M \in \mathbb{R}\ \exists x \in A\ \big(f(x) > M\big).$$

Remark (Failure modes). No proposed real ceiling works: whenever a candidate upper bound is chosen, some input has an output larger than that candidate.

Remark (Failure mode decomposition).

$$\forall M \in \mathbb{R}\ \exists x \in A\ \big(f(x) > M\big).$$

Remark (Interpretation). Bounded above means all output values lie below one common real ceiling.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$
- Bounded Above

Definition (Bounded Below Function)

Definition 32.2.11 (Bounded Below Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded below on A if there exists $m \in \mathbb{R}$ such that

$$m \le f(x) \qquad (x \in A).$$

Remark (Standard quantified statement)*.*

$$\exists m \in \mathbb{R}\; \forall x \in A\; \big(m \le f(x)\big).$$

Remark (Negated quantified statement)*.*

$$\forall m \in \mathbb{R}\; \exists x \in A\; \big(f(x) < m\big).$$

Remark (Failure modes)*.* No proposed real floor works: whenever a candidate lower bound is chosen, some input has an output smaller than that candidate.

Remark (Failure mode decomposition)*.*

$$\forall m \in \mathbb{R}\; \exists x \in A\; \big(f(x) < m\big).$$

Remark (Interpretation)*.* Bounded below means all output values lie above one common real floor.

Remark (Dependencies)*.*

- Function
- Subset
- Order on $\mathbb{R}$
- Bounded Below

Definition (Bounded Function)

Definition 32.2.12 (Bounded Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded on A if there exists $B > 0$ such that

$$|f(x)| \le B \qquad (x \in A).$$

Remark (Standard quantified statement)*.*

$$\exists B > 0\; \forall x \in A\; \big(|f(x)| \le B\big).$$

Remark (Definition predicate reading)*.*

$$\mathrm{BoundedOn}(f, A) \;:=\; \exists B > 0\; \forall x \in A\; \big(|f(x)| \le B\big).$$

Remark (Negated quantified statement).

$$\forall B > 0\ \exists x \in A\ \big(|f(x)| > B\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f, A) \iff \forall B > 0\ \exists x \in A\ \big(|f(x)| > B\big).$$

Remark (Failure modes). Every symmetric interval around 0 fails to contain all output values of f on A.

Remark (Failure mode decomposition).

$$\forall B > 0\ \exists x \in A\ \big(f(x) > B\ \lor\ f(x) < -B\big).$$

Remark (Interpretation). A bounded function has all of its values trapped inside some symmetric interval $[-B, B]$. Being bounded above alone is not enough; the function $f(x) = -x$ on $\mathbb{R}$ is bounded above by 0 but is not bounded.

Remark (Dependencies).

- Bounded Above Function
- Bounded Below Function

Definition (Supremum of a Function on a Set)

Definition 32.2.13 (Supremum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The supremum of f on A is

$$\sup_{x \in A} f(x) = \sup f(A),$$

provided the supremum of $f(A)$ exists.

Remark (Standard quantified statement).

$$s = \sup_{x \in A} f(x) \iff \big(\forall x \in A\ (f(x) \le s)\big)\ \land\ \big(\forall u \in \mathbb{R}\ ((\forall x \in A\ (f(x) \le u)) \Rightarrow s \le u)\big).$$

Remark (Interpretation). The supremum is the least real ceiling for the output set $f(A)$; it need not be an output value of f.

Remark (Dependencies).

- Function
- Subset
- Image Set
- Supremum

Definition (Infimum of a Function on a Set)

Definition 32.2.14 (Infimum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The infimum of f on A is

$$\inf_{x \in A} f(x) = \inf f(A),$$

provided the infimum of $f(A)$ exists.

Remark (Standard quantified statement).

$$t = \inf_{x \in A} f(x) \iff \big(\forall x \in A\ (t \le f(x))\big) \wedge \big(\forall \ell \in \mathbb{R}\ ((\forall x \in A\ (\ell \le f(x))) \Rightarrow \ell \le t)\big).$$

Remark (Interpretation). The infimum is the greatest real floor for the output set $f(A)$; it need not be an output value of f.

Remark (Dependencies).

- Function
- Subset
- Image Set
- Infimum

Definition (Maximum Point of a Function)

Definition 32.2.15 (Maximum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x^* \in A$ is a maximum point of f on A if

$$f(x) \le f(x^*) \qquad (x \in A).$$

Remark (Standard quantified statement).

$$x^* \in A \ \wedge\ \forall x \in A\ \big(f(x) \le f(x^*)\big).$$

Remark (Negated quantified statement).

$$x^* \notin A \ \vee\ \exists x \in A\ \big(f(x) > f(x^*)\big).$$

Remark (Failure modes). A proposed maximum point fails either by not belonging to the domain or by being beaten by another output value.

Remark (Failure mode decomposition).

$$x^* \notin A \ \vee\ \exists x \in A\ \big(f(x) > f(x^*)\big).$$

Remark (Interpretation). A maximum point is an input where the largest output value is attained.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$
- Maximum

Definition (Minimum Point of a Function)

Definition 32.2.16 (Minimum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x_* \in A$ is a minimum point of f on A if

$$f(x_*) \leq f(x) \qquad (x \in A).$$

Remark (Standard quantified statement).

$$x_* \in A \ \wedge \ \forall x \in A \ \big(f(x_*) \leq f(x)\big).$$

Remark (Negated quantified statement).

$$x_* \notin A \ \vee \ \exists x \in A \ \big(f(x) < f(x_*)\big).$$

Remark (Failure modes). A proposed minimum point fails either by not belonging to the domain or by being undercut by another output value.

Remark (Failure mode decomposition).

$$x_* \notin A \ \vee \ \exists x \in A \ \big(f(x) < f(x_*)\big).$$

Remark (Interpretation). A minimum point is an input where the smallest output value is attained.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$
- Minimum

Definition (Increasing Function)

Definition 32.2.17 (Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is increasing on A if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \ \big(x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) > f(x_2)\big).$$

Remark (Failure modes). The function fails to be increasing if some later input has a smaller output than an earlier input.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) > f(x_2)\big).$$

Remark (Interpretation). As the input moves to the right in A, the output is not allowed to go down.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Strictly Increasing Function)

Definition 32.2.18 (Strictly Increasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly increasing on A if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \ \big(x_1 < x_2 \Rightarrow f(x_1) < f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) \geq f(x_2)\big).$$

Remark (Failure modes). Strict increase fails if some later input has an output that is no larger than the earlier output.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) \geq f(x_2)\big).$$

Remark (Interpretation). Strict increase requires every genuine move to the right in the domain to produce a genuine increase in output.

Remark (Dependencies).

- Function
- Subset
- Strict Order on $\mathbb{R}$

Definition (Decreasing Function)

Definition 32.2.19 (Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is decreasing on A if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement)*.*

$$\forall x_1, x_2 \in A \ \big(x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)\big).$$

Remark (Negated quantified statement)*.*

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) < f(x_2)\big).$$

Remark (Failure modes)*.* The function fails to be decreasing if some later input has a larger output than an earlier input.

Remark (Failure mode decomposition)*.*

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) < f(x_2)\big).$$

Remark (Interpretation)*.* As the input moves to the right in A, the output is not allowed to go up.

Remark (Dependencies)*.*

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Strictly Decreasing Function)

Definition 32.2.20 (Strictly Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly decreasing on A if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement)*.*

$$\forall x_1, x_2 \in A \ \big(x_1 < x_2 \Rightarrow f(x_1) > f(x_2)\big).$$

Remark (Negated quantified statement)*.*

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) \leq f(x_2)\big).$$

Remark (Failure modes). Strict decrease fails if some later input has an output that is no smaller than the earlier output.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ \big(x_1 < x_2 \ \wedge \ f(x_1) \leq f(x_2)\big).$$

Remark (Interpretation). Strict decrease requires every genuine move to the right in the domain to produce a genuine decrease in output.

Remark (Dependencies).

- Function
- Subset
- Strict Order on $\mathbb{R}$

Definition (Monotone Function)

Definition 32.2.21 (Monotone Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is monotone on A if f is increasing on A or decreasing on A.

Remark (Standard quantified statement).

$$\big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))\big) \ \vee \ \big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2))\big).$$

Remark (Definition predicate reading).

$$\mathrm{MonotoneFunction}(f, A) \ :\Leftrightarrow \ \big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))\big) \vee \big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \geq$$

Remark (Negated quantified statement).

$$\big(\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) > f(x_2))\big) \ \wedge \ \big(\exists y_1, y_2 \in A \ (y_1 < y_2 \wedge f(y_1) < f(y_2))\big).$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{MonotoneFunction}(f, A)$$

Remark (Failure modes). A nonmonotone function has evidence of both kinds of reversal: one pair where the outputs go down as inputs increase, and another pair where the outputs go up as inputs increase.

Remark (Failure mode decomposition).

$$\begin{aligned} &\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) > f(x_2)) \\ &\qquad \wedge \ \exists y_1, y_2 \in A \ (y_1 < y_2 \wedge f(y_1) < f(y_2)). \end{aligned}$$

Remark (Interpretation). A monotone function moves consistently in one direction: nondecreasing or nonincreasing.

Remark (Dependencies).

- Increasing Function
- Decreasing Function

Definition (Constant Function)

Definition 32.2.22 (Constant Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is constant on A if

$$f(x_1) = f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \; \big(f(x_1) = f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \; \big(f(x_1) \neq f(x_2)\big).$$

Remark (Failure modes). Constancy fails as soon as two inputs in the domain have different outputs.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \; \big(f(x_1) \neq f(x_2)\big).$$

Remark (Interpretation). A constant function assigns the same real value to every input in its domain.

Remark (Dependencies).

- Function
- Subset

32.3 Subsets of $\mathbb{R}$

Subsets of $\mathbb{R}$ — Quick Reference

Concept	Meaning
Epsilon neighbourhood	Unpunctured condition $\lvert x - c\rvert < r$
Deleted neighbourhood	Punctured condition $0 < \lvert x - c\rvert < \delta$
Cluster point	Every neighbourhood contains a distinct point of X
Adherent point	Every neighbourhood contains some point of X
Closure	Points adherent to X
Closed set	Set containing its closure or sequential limits
Bounded set	Set contained in a large interval
Heine–Borel	Closed and bounded iff sequentially compact in $\mathbb{R}$

Subsets of $\mathbb{R}$

Remark (Terminological note). Cluster point = limit point = accumulation point. Bartle and Lebl use *cluster point*; Rudin uses *limit point*; Tao uses both. Adherent point (Tao) is the weaker notion that includes isolated points.

Definition (ε-Neighbourhood)

Definition 32.3.1 (ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **ε-neighbourhood** of x is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in V_\varepsilon(x) \iff |y - x| < \varepsilon.$$

Remark (Interpretation). The ε-neighbourhood is the open interval of radius ε centered at x.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Deleted ε-Neighbourhood)

Definition 32.3.2 (Deleted ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **deleted ε-neighbourhood** of x is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in V_\varepsilon^*(x) \iff 0 < |y - x| < \varepsilon.$$

Remark (Interpretation). The deleted neighbourhood removes the base point. It is the punctured input region used in limit definitions.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Cluster Point)

Definition 32.3.3 (Cluster Point)**.** Let $X \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. The point x is a **cluster point** of X if

$$\forall \varepsilon > 0,\ \exists\, y \in X \setminus \{x\} :\ |y - x| < \varepsilon.$$

Remark (Standard quantified statement).

$$x \text{ is a cluster point of } X \iff \forall \varepsilon > 0,\ \exists\, y \in X \setminus \{x\} :\ |y - x| < \varepsilon.$$

Remark (Definition predicate reading).

$$x \text{ is a cluster point of } X \ :\equiv \ \forall \varepsilon > 0,\ V_\varepsilon^*(x) \cap X \neq \emptyset.$$

Remark (Negated quantified statement).

$$\neg\, x \text{ is a cluster point of } X \iff \exists \varepsilon > 0,\ \forall\, y \in X \setminus \{x\} :\ \neg |y - x| < \varepsilon.$$

Remark (Negation predicate reading).

$$\neg\, x \text{ is a cluster point of } X \ :\equiv \ \exists \varepsilon > 0 :\ V_\varepsilon^*(x) \cap X = \emptyset.$$

Remark (Interpretation). A cluster point is approached by X via distinct witnesses arbitrarily close. A cluster point need not belong to X: 0 is a cluster point of $(0, 1)$ but $0 \notin (0, 1)$. The condition $\lim_{x \to c} f(x) = L$ requires $\mathrm{ClusterPoint}(c, \mathrm{dom}(f))$; without it the punctured ball condition is vacuous.

Remark (Dependencies).

- Deleted ε-Neighbourhood

Theorem

Theorem 32.3.4 (Sequential Characterization of Cluster Points)**.** *c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \to c$.*

Go to proof.

Remark (Standard quantified statement).

$$c \text{ is a cluster point of } A \iff \exists (a_n) \subseteq A \setminus \{c\} :\ a_n \to c.$$

Remark (Interpretation). The forward direction: use $\varepsilon = 1/n$ to build the witness sequence. The reverse direction is immediate since the sequence witnesses the ball condition for every ε. This bridges the ε-δ definition with the sequential language.

Remark (Dependencies).

- Cluster Point

Definition (Adherent Point)

Definition 32.3.5 (Adherent Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **adherent point** of X if

$$\forall \varepsilon > 0\ \exists y \in X\ (|y - x| < \varepsilon).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \big(V_\varepsilon(x) \cap X \neq \emptyset\big).$$

Remark (Definition predicate reading).

$$\mathrm{AdherentPoint}(x, X) \;:=\; \forall \varepsilon > 0 \; \exists y \in X \; (|y - x| < \varepsilon).$$

Remark (Negated quantified statement).

$$\exists \varepsilon > 0 \; \forall y \in X \; (|y - x| \geq \varepsilon).$$

Remark (Interpretation). An adherent point can be witnessed by points of X arbitrarily close to x, including possibly x itself.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Isolated Point)

Definition 32.3.6 (Isolated Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in X$. The point x is an **isolated point** of X if

$$\exists \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X = \{x\}\big).$$

Remark (Standard quantified statement).

$$\exists \varepsilon > 0 \; \forall y \in X \; \big(|y - x| < \varepsilon \Rightarrow y = x\big).$$

Remark (Definition predicate reading).

$$\mathrm{IsolatedPoint}(x, X) \;:=\; x \in X \land \exists \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X = \{x\}\big).$$

Remark (Negated quantified statement).

$$\forall \varepsilon > 0 \; \exists y \in X \setminus \{x\} \; (|y - x| < \varepsilon).$$

Remark (Interpretation). An isolated point belongs to the set but has a neighbourhood containing no other points of the set.

Remark (Dependencies).

- Adherent Point
- Cluster Point

Definition (Interior Point)

Definition 32.3.7 (Interior Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **interior point** of X if

$$\exists \varepsilon > 0 \; \big(V_\varepsilon(x) \subseteq X\big).$$

Remark (Standard quantified statement).

$$\exists \varepsilon > 0 \; \forall y \in \mathbb{R} \; \big(|y - x| < \varepsilon \Rightarrow y \in X\big).$$

Remark (Definition predicate reading).

$$\text{InteriorPoint}(x, X) \;:=\; \exists \varepsilon > 0\ \big(V_\varepsilon(x) \subseteq X\big).$$

Remark (Interpretation). An interior point has a full open neighbourhood contained in the set.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Boundary Point)

Definition 32.3.8 (Boundary Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is a **boundary point** of X if

$$\forall \varepsilon > 0\ \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Definition predicate reading).

$$\text{BoundaryPoint}(x, X) \;:=\; \forall \varepsilon > 0\ \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Interpretation). A boundary point sees both the set and its complement in every neighbourhood.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Interior)

Definition 32.3.9 (Interior)**.** Let $X \subseteq \mathbb{R}$. The **interior** of X is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0\ V_\varepsilon(x) \subseteq X\}.$$

Remark (Standard quantified statement).

$$x \in X^\circ \iff \exists \varepsilon > 0\ V_\varepsilon(x) \subseteq X.$$

Remark (Interpretation). The interior collects exactly the points around which the set contains a whole neighbourhood.

Remark (Dependencies).

- Interior Point

Definition (Boundary)

Definition 32.3.10 (Boundary)**.** Let $X \subseteq \mathbb{R}$. The **boundary** of X is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0\ (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

Remark (Standard quantified statement)*.*

$$x \in \partial X \iff \forall \varepsilon > 0\ (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset).$$

Remark (Interpretation)*.* The boundary records the points where every neighbourhood touches both inside and outside of the set.

Remark (Dependencies)*.*

- Boundary Point

Definition (Closure)

Definition 32.3.11 (Closure)**.**

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0\ V_\varepsilon(x) \cap X \neq \emptyset\}.$$

Remark (Standard quantified statement)*.*

$$x \in \overline{X} \iff x \text{ is an adherent point of } X \iff \forall \varepsilon > 0 : V_\varepsilon(x) \cap X \neq \emptyset.$$

Remark (Interpretation)*.* The closure adds all adherent points of X, so it fills in the limiting points that can be reached by points of X.

Remark (Canonical examples)*.* $\overline{(a,b)} = [a,b]$; $\overline{\mathbb{Q}} = \mathbb{R}$; $\overline{\mathbb{Z}} = \mathbb{Z}$; $\overline{\{1/n\}} = \{1/n\} \cup \{0\}$.

Remark (Dependencies)*.*

- Adherent Point

Lemma

Lemma 32.3.12 (Elementary Properties of Closures)**.** *Let $X, Y \subseteq \mathbb{R}$. Then: (i) $X \subseteq \overline{X}$; (ii) $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; (iii) $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; (iv) $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\overline{X \cup Y} = \overline{X} \cup \overline{Y}; \quad \overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}.$$

Remark (Failure of equality in (iii))*.* Take $X = (0,1)$, $Y = (1,2)$: $\overline{X \cap Y} = \emptyset$ but $\overline{X} \cap \overline{Y} = \{1\}$.

Remark (Interpretation)*.* Closure preserves inclusions and finite unions, but intersection can lose limit points that are approached separately by the two sets.

Remark (Dependencies).

- Closure

Definition (Closed Subset of $\mathbb{R}$)

Definition 32.3.13 (Closed Subset of $\mathbb{R}$)**.** E is closed $\iff$ $\overline{E} = E$.

Remark (Standard quantified statement).

$$E \text{ is closed} \iff \forall x \in \mathbb{R} : \ x \text{ is an adherent point of } E \Rightarrow x \in E.$$

Remark (Definition predicate reading).

$$E \text{ is closed} \ :\equiv \ \forall x : \ x \text{ is an adherent point of } E \Rightarrow x \in E.$$

Remark (Negated quantified statement).

$$\neg\, E \text{ is closed} \iff \exists x \notin E : \ x \text{ is an adherent point of } E.$$

Remark (Negation predicate reading).

$$\neg\, E \text{ is closed} \ :\equiv \ \exists x \notin E : \ \forall \varepsilon > 0 : V_\varepsilon(x) \cap E \neq \emptyset.$$

Remark (Interpretation). E is closed iff it contains its own boundary. Examples: $[a, b]$ closed; (a, b) not closed; $\mathbb{R}$ and $\emptyset$ both closed; $\mathbb{Q}$ not closed.

Remark (Dependencies).

- Closure

Corollary

Corollary 32.3.14 (Closed iff Sequentially Closed)**.** *X is closed* $\iff$ $\forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X.$

Go to proof.

Remark (Standard quantified statement).

$$X \text{ is closed} \iff \forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X.$$

Remark (Interpretation). A closed set cannot be escaped by convergent sequences. Every adherent point has a witness sequence; closedness forces the limit into the set.

Remark (Dependencies).

- Closed Subset of $\mathbb{R}$
- Sequential Characterization of Cluster Points

Corollary

Corollary 32.3.15 (Every Point of an Interval Is a Cluster Point). *If I is an interval, then every $x \in I$ is a cluster point of I.*

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \forall \varepsilon > 0 \ \exists y \in I \setminus \{x\} \ (|y - x| < \varepsilon).$$

Remark (Interpretation). This justifies intervals as natural domains for function limits. $\lim_{x \to c} f(x) = L$ requires $\mathrm{ClusterPoint}(c, \mathrm{dom}(f))$; on any interval this is free.

Remark (Dependencies).

- Cluster Point

Definition (Bounded Subset of $\mathbb{R}$)

Definition 32.3.16 (Bounded Subset of $\mathbb{R}$). X is bounded $\iff$ $\exists M > 0, \ \forall x \in X : |x| \leq M$.

Remark (Standard quantified statement).

$$\exists M > 0 \ \forall x \in X \ (|x| \leq M).$$

Remark (Negated quantified statement).

$$\forall M > 0 \ \exists x \in X \ (|x| > M).$$

Remark (Interpretation). A bounded subset of $\mathbb{R}$ fits inside some symmetric interval $[-M, M]$.

Remark (Dependencies).

- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem

Theorem 32.3.17 (Heine–Borel Theorem for $\mathbb{R}$). *X is closed $\wedge$ X is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Go to proof.

Remark (Standard quantified statement).

$$X \text{ is closed} \wedge X \text{ is bounded} \iff \forall (a_n) \subseteq X, \ \exists (a_{n_k}), \ \exists L \in X : a_{n_k} \to L.$$

Remark (Definition predicate reading).

$$\text{SequentiallyCompact}(X) \;:\equiv\; X \text{ is closed} \wedge X \text{ is bounded}.$$

Remark (Negated quantified statement).

$$\neg\big(X \text{ is closed} \wedge X \text{ is bounded}\big) \iff \exists (a_n) \subseteq X: \text{ no subsequence converges to a limit in } X.$$

Remark (Failure modes). The compactness conclusion fails when either closedness or boundedness fails.

Remark (Failure mode decomposition). Either X is bounded fails — a sequence escapes to $\pm\infty$ — or X is closed fails — a convergent subsequence exists but its limit lies outside X.

Remark (Interpretation). Closed prevents limits from escaping through boundary gaps; bounded prevents escape to infinity. Both are required. Forward: bounded gives Bolzano–Weierstrass; closed keeps the limit in X. Reverse: negate either hypothesis to construct a sequence with no convergent subsequence in X.

Remark (Dependencies).

- Closed Subset of $\mathbb{R}$
- Bounded Subset of $\mathbb{R}$

Definition (True Near a Point)

Definition 32.3.18 (True Near a Point)**.** Property $P(f(x))$ is **true near** x_0 if $\exists \delta > 0,\ \forall x : 0 < |x - x_0| < \delta) \Rightarrow P(f(x))$.

Remark (Standard quantified statement).

$$P(f(x)) \text{ true near } x_0 \iff \exists \delta > 0,\ \forall x : 0 < |x - x_0| < \delta \Rightarrow P(f(x)).$$

Remark (Interpretation). $\lim_{x \to x_0} f(x) = L$ states: for every $\varepsilon > 0$, the property $|f(x) - L| < \varepsilon$ is true near x_0.

Remark (Dependencies).

- ε-Neighbourhood

Chapter 33

Discrete Calculus

SERIES	VOLUME III	BOOK I	CHAPTER 5
From Cantor to Ito	Classical Analysis	Bounds, Sequences, and Series	Discrete Calculus

33.1 Motivation

Motivation

Classical calculus studies functions defined on continuous domains, typically functions $f : \mathbb{R} \to \mathbb{R}$. Its central tools are the derivative and the integral:

- The derivative measures *instantaneous change*.
- The integral accumulates *total change*.

However, many important mathematical objects are inherently *discrete*. Examples include

- sequences a_n,
- recurrence relations,
- sums of integers or combinatorial quantities,
- discrete probability distributions,
- algorithms and computational processes.

In these contexts the domain is typically the natural numbers

$$f : \mathbb{Z} \to \mathbb{R}$$

rather than the real line.

To study change in such settings, calculus is replaced by its discrete counterpart.

Discrete Analogs of Calculus

The central idea of *discrete calculus* is that the roles of derivative and integral are played by two simple operations.

Continuous vs. Discrete Calculus

Continuous Calculus	Discrete Calculus
Derivative $\dfrac{d}{dx}f(x)$	Forward difference $\Delta f(n) = f(n+1) - f(n)$
Integral $\displaystyle\int f(x)\,dx$	Summation $\sum f(n)$

The forward difference operator measures change between consecutive values of a function.

$$\Delta f(n) = f(n+1) - f(n).$$

This quantity plays the same role for sequences that the derivative plays for functions of a real variable.

Likewise, summation serves as the discrete analogue of integration.

$$\sum_{k=a}^{b} f(k)$$

accumulates the values of a function across a finite discrete interval.

Telescoping Phenomena

One of the most important identities in discrete mathematics arises when differences are summed.

$$\sum_{k=a}^{b-1} \bigl(f(k+1) - f(k)\bigr) = f(b) - f(a).$$

The intermediate terms cancel, leaving only the endpoints. Such sums are called *telescoping sums*.

This identity is the discrete counterpart of the *Fundamental Theorem of Calculus*.

Why Discrete Calculus Matters

Discrete calculus appears throughout mathematics:

- in combinatorics and enumerative formulas,
- in recurrence relations and generating functions,
- in probability theory and stochastic processes,

- in numerical analysis and finite difference methods,
- in computer science and algorithm analysis.

For students of analysis, discrete calculus is particularly valuable because it reveals the structural skeleton behind many identities involving sums and sequences.

Many techniques used to estimate limits of sequences or series ultimately rely on discrete analogues of differential identities.

A Structural View

Continuous calculus studies how functions change along the real line.

Discrete calculus studies how functions change along the integers.

Despite the difference in setting, the two theories mirror each other remarkably closely. Understanding this parallel provides powerful intuition and useful techniques for manipulating sums, sequences, and recurrence relations.

In the sections that follow we develop the basic tools of discrete calculus, beginning with the forward difference operator and the algebra of finite differences.

Structural Correspondence Between Continuous and Discrete Calculus

$\frac{d}{dx}f(x)$ Derivative $\longrightarrow$ $\Delta f(n) = f(n+1) - f(n)$ Forward Difference

$\int f(x)\,dx$ Integral $\longrightarrow$ $\sum f(n)$ Summation

$\frac{d}{dx}x^n = nx^{n-1}$ Power Rule $\longrightarrow$ $\Delta n^{\underline{k}} = k\,n^{\underline{k-1}}$ Falling Factorial Rule

Fundamental Theorem of Calculus $\longrightarrow$ Telescoping Sum Identity

33.2 Foundations

33.2.1 Foundations

Definition (Discrete Function)

A *discrete function* is a function

$$f : \mathbb{Z} \to \mathbb{R}$$

(or more generally $f : \mathbb{N} \to \mathbb{R}$). It assigns a real value to each integer input, producing a sequence $f(0)$, $f(1)$, $f(2)$, ...

Remark (Fully quantified form). $\forall n \in \mathbb{Z},\ f(n) \in \mathbb{R}$. The domain is the integers (or natural numbers); the codomain is the real line.

Remark (Discrete vs. continuous). In discrete calculus the independent variable moves in *integer steps* rather than continuously. Change is therefore measured between successive integer inputs rather than through infinitesimal limits. The fundamental operator of discrete calculus is the *difference operator*

$$\Delta f(n) = f(n+1) - f(n),$$

which plays a role analogous to the derivative in continuous calculus.

Remark (Sequences as discrete functions). A discrete function is simply another way of viewing a *sequence*. A sequence $a_0, a_1, a_2, \ldots$ is identified with

$$f : \mathbb{N} \to \mathbb{R}, \qquad f(n) = a_n.$$

Several equivalent notations appear in the literature: $f(n)$, a_n, $(a_n)_{n\in\mathbb{N}}$, $\{a_n\}_{n=0}^{\infty}$. All describe the same object. The functional notation $f(n)$ is preferred in discrete calculus because the difference operator acts on it much as a derivative acts on functions in continuous calculus.

33.2.2 Summation Notation

Definition (Summation)

Let f be a function defined on the integers. For integers $a \le b$, the expression

$$\sum_{k=a}^{b} f(k)$$

denotes the sum $f(a) + f(a+1) + f(a+2) + \cdots + f(b)$. The symbol $\sum$ (Greek capital sigma) represents summation; k is the *index of summation*; a and b are the *lower* and *upper limits*; $f(k)$ is the *summand*.

Remark (Standard quantified statement). $\displaystyle\sum_{k=a}^{b} f(k) \ := \ f(a) + f(a+1) + \cdots + f(b)$, where $a, b \in \mathbb{Z}$, $a \le b$, and $f : \mathbb{Z} \to \mathbb{R}$.

Remark (Interpretation). Sigma notation compresses a finite list of additions into one indexed symbol.

Remark (Dummy variable). The index of summation is a *dummy variable*: $\sum_{k=a}^{b} f(k) = \sum_{i=a}^{b} f(i)$. Its name carries no information about the value of the sum.

Basic Properties of Summation

Remark (Basic properties of summation). For functions f and g and constants c,

$$\sum_{k=m}^{n}(f(k)+g(k)) = \sum_{k=m}^{n} f(k) + \sum_{k=m}^{n} g(k),$$

$$\sum_{k=m}^{n} c\, f(k) = c\sum_{k=m}^{n} f(k).$$

These express the *linearity* of summation.

Splitting Sums

Remark (Splitting sums). A sum over an interval may be divided into smaller pieces:

$$\sum_{k=a}^{c} f(k) = \sum_{k=a}^{b} f(k) + \sum_{k=b+1}^{c} f(k) \qquad \text{whenever } a \le b < c.$$

Changing the Index

Remark (Changing the index). It is often useful to shift the summation index. For example,

$$\sum_{k=0}^{n} f(k) = \sum_{j=1}^{n+1} f(j-1).$$

Such substitutions frequently simplify expressions or align sums with known formulas.

Empty Sums

Remark (Empty sums). If the upper bound is smaller than the lower bound, the sum contains no terms. By convention,

$$\sum_{k=a}^{b} f(k) = 0 \qquad \text{when } a > b.$$

This convention simplifies many algebraic identities.

Nested Summations

Remark (Nested summations). Summations may be nested:

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij} = \sum_{i=1}^{n}\big(a_{i1} + a_{i2} + \cdots + a_{im}\big).$$

The inner sum is evaluated first. When the limits are independent, the order of summation may be exchanged:

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij} = \sum_{j=1}^{m}\sum_{i=1}^{n} a_{ij}.$$

Remark (Analogy with integration). Nested sums behave like iterated integrals: exchanging summation order corresponds to Fubini's theorem in the discrete setting.

Common Examples

Remark (Common examples).

$$\sum_{k=1}^{n} 1 = n, \qquad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

These formulas appear throughout discrete mathematics and analysis.

33.2.3 The Binomial Theorem

Theorem 33.2.1 (Binomial Theorem)**.** *For any integer $n \geq 0$ and real numbers x, y,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k,$$

where the binomial coefficients *are*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Remark (Standard quantified statement). $\forall n \in \mathbb{N},\ \forall x, y \in \mathbb{R} : (x+y)^n = \displaystyle\sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$

Remark (Interpretation). The theorem expands a power of a sum by counting how many terms choose each possible number of y-factors.

Remark (Combinatorial meaning). The binomial coefficients count the number of ways to choose k factors of y from the n factors in the product $(x+y)(x+y)\cdots(x+y)$. Thus $\binom{n}{k}$ is the number of ways to select k objects from a set of n, and the theorem describes how a power of a sum expands into a sum of products.

Basic Properties of Binomial Coefficients

Remark (Basic properties of binomial coefficients).

$$\binom{n}{0} = 1, \qquad \binom{n}{n} = 1, \qquad \binom{n}{k} = \binom{n}{n-k}.$$

They also satisfy the *Pascal recursion*:

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}.$$

This recursion produces Pascal's triangle, in which each entry is the sum of the two entries directly above it.

$$\begin{array}{ccccccccccc}
 & & & & & \binom{0}{0} & & & & & \\
 & & & & \binom{1}{0} & & \binom{1}{1} & & & & \\
 & & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & & \\
 & & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & & \\
 & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} & \\
\binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5}
\end{array}$$

Remark (Pascal's triangle). Each entry $\binom{n}{k}$ is obtained from the sum of the two entries above it: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. These coefficients appear throughout discrete calculus, especially in formulas for higher finite differences and in Newton's forward difference expansion.

Connection with Discrete Calculus

Remark (Connection with discrete calculus). The binomial theorem plays a central role in discrete calculus because difference operators naturally produce binomial coefficients.

Proposition 33.2.2 (Higher difference formula)**.** *For any function f,*

$$\Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k).$$

Go to proof.

Remark (Standard quantified statement). $\forall n \in \mathbb{N},\ \forall f : \mathbb{Z} \to \mathbb{R},\ \forall x \in \mathbb{Z} :\ \ \Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k).$

Remark (Interpretation). The n-th forward difference is a signed binomial combination of $n+1$ successive function values.

Remark (Proof idea). Applying the difference operator $\Delta = E - I$ (where E is the shift operator) repeatedly gives $\Delta^n = (E-I)^n$. Expanding by the binomial theorem yields the coefficients $(-1)^{n-k}\binom{n}{k}$, and $E^k f(x) = f(x+k)$. This derivation is carried out in full in the Difference Operators section.

Remark (Dependencies).

- Binomial Theorem.
- Forward Difference Operator.
- Higher Differences.

Explicit Examples

Remark (Explicit examples). Applying the difference operator repeatedly to a function f yields

$$\begin{aligned}\Delta f(x) &= f(x+1) - f(x),\\ \Delta^2 f(x) &= f(x+2) - 2f(x+1) + f(x),\\ \Delta^3 f(x) &= f(x+3) - 3f(x+2) + 3f(x+1) - f(x).\end{aligned}$$

The coefficients $1, 2, 1$ and $1, 3, 3, 1$ are precisely the rows of Pascal's triangle, confirming that binomial coefficients determine the pattern of coefficients in higher-order finite differences.

33.3 Algebra of Summations

Summation Algebra — Quick Reference

Linearity	$\sum(af+bg) = a\sum f + b\sum g$
Splitting	$\sum_a^c = \sum_a^b + \sum_{b+1}^c$
Index shift	$\sum_{k=a}^b f(k) = \sum_{k=a+1}^{b+1} f(k-1)$
Constant sum	$\sum_{k=a}^b c = (b-a+1)c$
Empty sum	$\sum_{k=a}^b f(k) = 0$ when $a > b$
Nested sums	$\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$ (independent limits)

Remark (Orientation). The difference operator and telescoping identity provide the background for the basic algebraic rules used to manipulate summations. These identities allow sums to be combined, split, and reindexed while preserving their value.

Linearity

Remark (Linearity). Summation is a linear operator. For functions f and g and constants a and b,

$$\sum_{k=m}^{n}(a\,f(k) + b\,g(k)) = a\sum_{k=m}^{n} f(k) + b\sum_{k=m}^{n} g(k).$$

This allows constants to be factored out of a sum and sums of expressions to be separated.

Example. $\sum_{k=1}^{n}(2k+3) = 2\sum_{k=1}^{n} k + 3\sum_{k=1}^{n} 1.$

Splitting a Sum

Remark (Splitting a sum). A sum over an interval may be divided into pieces:

$$\sum_{k=a}^{c} f(k) = \sum_{k=a}^{b} f(k) + \sum_{k=b+1}^{c} f(k) \qquad \text{whenever } a \leq b < c.$$

Example. $\sum_{k=1}^{10} f(k) = \sum_{k=1}^{5} f(k) + \sum_{k=6}^{10} f(k).$

Index Renaming

Remark (Index renaming). The index of summation is a *dummy variable*:

$$\sum_{k=a}^{b} f(k) = \sum_{i=a}^{b} f(i).$$

This is useful when manipulating expressions containing several sums, since the index variable can be renamed to avoid conflicts.

Index Shifting

Remark (Index shifting). It is often convenient to shift the summation index:

$$\sum_{k=a}^{b} f(k) = \sum_{k=a+1}^{b+1} f(k-1).$$

Index shifting allows sums to be aligned so that terms cancel or combine properly. It appears frequently in proofs involving recurrence relations and discrete calculus identities.

Example. $\sum_{k=0}^{n} f(k+1) = \sum_{k=1}^{n+1} f(k).$

Constant Sum

Remark (Constant sum). If the summand does not depend on the index, the sum counts terms:

$$\sum_{k=a}^{b} c = (b - a + 1)\, c.$$

The quantity $b - a + 1$ is the number of integers between a and b, inclusive.

Example. $\displaystyle\sum_{k=3}^{7} 5 = 5 \cdot 5 = 25.$

Empty Sum

Remark (Empty sum). If the upper limit is smaller than the lower limit, the sum contains no terms. By convention,

$$\sum_{k=a}^{b} f(k) = 0 \qquad \text{whenever } a > b.$$

This convention simplifies many algebraic identities and allows formulas to remain valid without special cases.

Nested Sums

Remark (Nested sums). Summations may be nested:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} = \sum_{i=1}^{n} \Big(a_{i1} + a_{i2} + \cdots + a_{im} \Big).$$

When the limits are independent, the order may be exchanged:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} = \sum_{j=1}^{m} \sum_{i=1}^{n} a_{ij}.$$

Remark (Analogy with double integrals). Exchanging summation order is the discrete analogue of Fubini's theorem for iterated integrals.

Common Summation Identities

Remark (Common summation identities). Many sums that appear in discrete mathematics and discrete calculus have closed-form expressions. The following identities are among the most frequently encountered.

Common Summation Identities

$$\sum_{k=1}^{n} 1 = n \qquad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$\sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r} \quad (r \neq 1) \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Remark (Role in discrete calculus). Polynomial summation formulas play an important role when working with finite differences and Newton series, while geometric sums arise naturally in recurrence relations and generating functions. Many of these identities can be derived systematically using the falling factorial basis and the Fundamental Theorem of Finite Calculus introduced later in this chapter.

33.4 Difference Operators

33.4.1 Difference Operators

Discrete calculus studies functions defined on the integers and the way their values change from one integer to the next. The central tool for measuring this change is the *difference operator*, which plays a role analogous to the derivative in continuous calculus.

Motivation

In ordinary calculus the derivative measures the rate of change of a function:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

In the discrete setting the independent variable changes in steps of size 1. Instead of taking a limit, we measure the difference between successive values of the function directly.

Forward Difference

Definition 33.4.1 (Forward Difference Operator)**.** Let $f : \mathbb{Z} \to \mathbb{R}$. The *forward difference operator* Δ is defined by

$$\Delta f(n) = f(n+1) - f(n).$$

The function Δf measures the change in f when the input increases by one unit.

Remark (Fully quantified form). $\forall f : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z} :\ \Delta f(n) := f(n+1) - f(n).$

Remark (Interpretation). The forward difference records the one-step change of a discrete function.

Remark (Analogy with differentiation). The operator Δ is the discrete analogue of the derivative. While the derivative measures infinitesimal change through a limit, the forward difference measures change between successive integer inputs exactly, with no limit required.

Example

Consider the function $f(n) = n^2$. Applying the difference operator gives

$$\begin{aligned}\Delta f(n) &= f(n+1) - f(n) \\ &= (n+1)^2 - n^2 \\ &= 2n + 1.\end{aligned}$$

Thus the forward difference of the quadratic function n^2 is the linear function $2n + 1$.

Remark (Degree reduction). This mirrors a familiar fact from ordinary calculus: $\frac{d}{dx}x^2 = 2x$. Just as derivatives reduce the degree of a polynomial by one, finite differences also lower the degree of polynomial sequences. Note, however, that $\Delta(n^2) = 2n + 1$ rather than $2n$; ordinary powers do not behave as cleanly under Δ as they do under differentiation. This motivates introducing the *falling factorial* basis later in this chapter.

33.4.2 Higher Differences

Definition 33.4.2 (Higher Differences). The *second difference* of f is defined by

$$\Delta^2 f(n) = \Delta(\Delta f(n)).$$

More generally, the k-th difference $\Delta^k f(n)$ denotes the k-fold application of the difference operator, defined recursively by

$$\Delta^0 f = f, \qquad \Delta^k f = \Delta(\Delta^{k-1} f).$$

Remark (Fully quantified form). $\forall k \in \mathbb{N},\ \forall f : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z} :\ \Delta^k f(n) := (\underbrace{\Delta \circ \cdots \circ \Delta}_{k})\, f(n)$

Remark (Interpretation). Higher differences measure repeated one-step change, just as higher derivatives measure repeated infinitesimal change.

Remark (Dependencies).

- Forward Difference Operator.

Example

Let $f(n) = n^2$. Then

$$\begin{aligned}\Delta f(n) &= 2n + 1, \\ \Delta^2 f(n) &= \Delta(2n+1) = 2(n+1) + 1 - (2n+1) = 2, \\ \Delta^3 f(n) &= \Delta(2) = 0.\end{aligned}$$

Remark (Analogy with higher derivatives). Higher differences behave analogously to higher derivatives. For polynomial functions of degree d, the d-th difference is constant and all higher differences are zero — just as the d-th derivative of a degree-d polynomial is constant and the next derivative vanishes.

33.4.3 Properties of the Difference Operator

The forward difference operator satisfies algebraic properties that closely mirror the familiar rules of differentiation.

Linearity

Proposition 33.4.3 (Linearity of Δ)**.** *For functions $f, g : \mathbb{Z} \to \mathbb{R}$ and constants $a, b \in \mathbb{R}$,*

$$\Delta(af + bg) = a\,\Delta f + b\,\Delta g.$$

Remark (Fully quantified form)*.* $\forall f, g : \mathbb{Z} \to \mathbb{R},\ \forall a, b \in \mathbb{R},\ \forall n \in \mathbb{Z} :\ \ \Delta(af + bg)(n) = a\,\Delta f(n) + b\,\Delta g(n)$.

Remark (Interpretation)*.* Taking a forward difference respects pointwise linear combinations.

Remark (Dependencies)*.*

- Forward Difference Operator.

Proof. By definition of the difference operator,

$$\begin{aligned}\Delta(af + bg)(n) &= (af + bg)(n+1) - (af + bg)(n)\\ &= af(n+1) + bg(n+1) - af(n) - bg(n)\\ &= a(f(n+1) - f(n)) + b(g(n+1) - g(n))\\ &= a\,\Delta f(n) + b\,\Delta g(n). \quad \square\end{aligned}$$

Constant Rule

Proposition 33.4.4 (Constant rule)**.** *If c is a constant function, then $\Delta c = 0$.*

Remark (Fully quantified form)*.* $\forall c \in \mathbb{R},\ \forall n \in \mathbb{Z} :\ \ \Delta c(n) = 0$.

Remark (Interpretation)*.* A constant discrete function has no one-step change.

Remark (Dependencies)*.*

- Forward Difference Operator.

Proof. $\Delta c(n) = c - c = 0$. □

Remark (Analogy)*.* This mirrors $\frac{d}{dx}c = 0$ in continuous calculus.

The Shift Operator and Operator Algebra

Definition 33.4.5 (Shift Operator)**.** The *shift operator* E and the *identity operator* I are defined by

$$Ef(n) = f(n+1), \qquad If(n) = f(n).$$

Remark (Fully quantified form)*.* $\forall f : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z} :\ Ef(n) = f(n+1)$ and $If(n) = f(n)$.

Remark (Interpretation). The shift operator advances the input by one step, while the identity operator leaves the input unchanged.

Theorem 33.4.6 ($\Delta = E - I$)**.**

$$\Delta = E - I.$$

Remark (Fully quantified form). $\forall f : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z} :\ \Delta f(n) = (E - I)f(n) = f(n+1) - f(n)$.

Remark (Interpretation). The forward difference is the shifted value minus the original value.

Remark (Dependencies).

- Forward Difference Operator.
- Shift Operator.

Proof. Applying the operator $E - I$ to a function f gives

$$(E - I)f(n) = Ef(n) - If(n) = f(n+1) - f(n) = \Delta f(n).$$ □

Remark (Higher differences via the binomial theorem). Expressing $\Delta = E - I$ reveals that higher differences can be computed using ordinary algebra. Applying the binomial theorem to $\Delta^n = (E - I)^n$ gives

$$\Delta^n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} E^k.$$

Since $E^k f(n) = f(n+k)$, this yields

$$\Delta^n f(n) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(n+k).$$

This explains why binomial coefficients and Pascal's triangle appear naturally throughout discrete calculus: they are a direct consequence of the algebraic identity $\Delta = E - I$.

33.4.4 Difference Tables and Polynomial Detection

One of the most useful tools in discrete calculus is the *difference table*. By repeatedly applying the difference operator to a sequence, one can detect whether the sequence arises from a polynomial function and determine its degree.

Definition 33.4.7 (Difference Table)**.** Given a sequence $f(0), f(1), f(2), \ldots$, the *difference table* is constructed by listing the successive values of the function and then computing their forward differences $\Delta f(n) = f(n+1) - f(n)$, then $\Delta^2 f$, $\Delta^3 f$, and so on.

Remark (Interpretation). A difference table organizes repeated forward differences into rows so that eventual constant or zero rows become visible.

Remark (Dependencies).

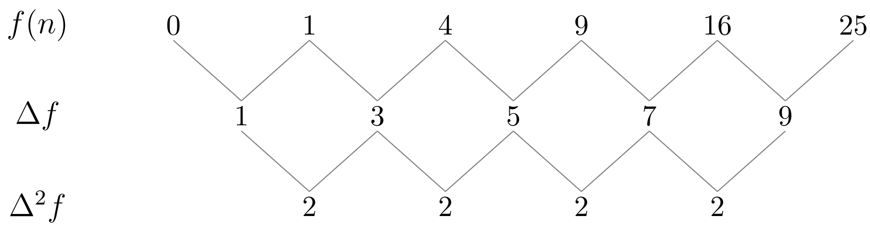
- Forward Difference Operator.
- Higher Differences.

Example

For $f(n) = n^2$:

n	$f(n)$	$\Delta f(n)$	$\Delta^2 f(n)$
0	0	1	
1	1	3	2
2	4	5	2
3	9	7	2
4	16	9	2
5	25		

The second differences are constant (all equal to 2), and the third differences are zero.



Theorem 33.4.8 (Polynomial detection)**.** *Let $f(n)$ be a sequence generated by a polynomial of degree d. Then $\Delta^d f(n)$ is constant, and $\Delta^{d+1} f(n) = 0$.*

Proof. Write $f(n) = a_d n^d + a_{d-1} n^{d-1} + \cdots + a_0$ with $a_d \neq 0$. We show by induction on d that Δf is a polynomial of degree $d-1$ with leading coefficient $d\, a_d$.

Base case ($d = 0$): a constant polynomial has $\Delta f = 0$. ✓

Inductive step: Assume the result holds for all polynomials of degree less than d. Then

$$\Delta f(n) = f(n+1) - f(n) = a_d\big((n+1)^d - n^d\big) + (\text{lower-degree terms}).$$

Expanding $(n+1)^d - n^d = d\, n^{d-1} + \binom{d}{2} n^{d-2} + \cdots$, the leading term is $d\, a_d\, n^{d-1}$. Thus Δf is a polynomial of degree $d-1$ with leading coefficient $d\, a_d \neq 0$.

By induction, applying Δ a further $d-1$ times yields the constant $\Delta^d f(n) = d!\, a_d$, and one further application gives 0. □

Remark (Fully quantified form). $\forall d \in \mathbb{N}$, $\forall f : \mathbb{Z} \to \mathbb{R}$ polynomial of degree d:
$\exists c \in \mathbb{R} : \forall n \in \mathbb{Z},\ \Delta^d f(n) = c$, and $\forall n \in \mathbb{Z},\ \Delta^{d+1} f(n) = 0$.

Remark (Interpretation). Repeated differences reveal polynomial degree: a degree d polynomial has constant d-th differences and vanishing next differences.

Remark (Consequence). Difference tables provide a practical method for identifying polynomial sequences and determining their degree. This observation leads naturally to the *falling factorial basis*: a family of polynomials that interact particularly cleanly with the difference operator, forming the foundation for Newton's discrete analogue of the Taylor expansion.

Remark (Dependencies).

- Forward Difference Operator.
- Higher Differences.
- Difference Table.
- Binomial Theorem.

33.5 Telescoping Sums

33.5.1 Telescoping Sums

One of the most useful patterns in summation is the phenomenon of *telescoping*. In a telescoping sum, consecutive terms cancel with one another, leaving only a small number of terms from the beginning and end of the expression.

A Simple Example

Consider the sum

$$(2-1)+(3-2)+(4-3)+(5-4).$$

Expanding gives

$$2-1+3-2+4-3+5-4.$$

Most terms cancel in adjacent pairs:

$$\cancel{2}-1+\cancel{3}-\cancel{2}+\cancel{4}-\cancel{3}+5-\cancel{4}.$$

After cancellation, only the first and last terms remain: $5-1$. Thus

$$(2-1)+(3-2)+(4-3)+(5-4)=5-1.$$

Definition (Telescoping Sum)

A sum is called a *telescoping sum* if consecutive terms cancel when the sum is expanded, leaving only a few remaining terms — typically the first and last.

General Form

The general telescoping structure is

$$\sum_{k=a}^{b-1}(f(k+1)-f(k)).$$

Writing out the terms explicitly:

$$\begin{aligned}&(f(a+1)-f(a))\\&+(f(a+2)-f(a+1))\\&+(f(a+3)-f(a+2))\\&\quad\vdots\\&+(f(b)-f(b-1)).\end{aligned}$$

All intermediate terms cancel in pairs, leaving only $f(b)-f(a)$.

Remark (Why "telescoping"). The name comes from an old collapsible telescope: each section slides into the next. The intermediate terms in the sum "slide together" and vanish, collapsing the entire sum down to its endpoints.

The Telescoping Identity

Theorem (Telescoping Identity)

For any function $f:\mathbb{Z}\to\mathbb{R}$ and integers $a\le b$,

$$\sum_{k=a}^{b-1}(f(k+1)-f(k))=f(b)-f(a).$$

Remark (Fully quantified form). $\forall f:\mathbb{Z}\to\mathbb{R},\ \forall a,b\in\mathbb{Z},\ a\le b:\ \sum_{k=a}^{b-1}(f(k+1)-f(k))=f(b)-f(a)$.

Proof. Expanding the sum gives

$$\begin{aligned}(f(a+1)-f(a))+(f(a+2)-f(a+1))&\\+\cdots+(f(b)-f(b-1)).&\end{aligned}$$

Every intermediate term $f(a+1),f(a+2),\ldots,f(b-1)$ appears once with a positive sign and once with a negative sign, so they all cancel. The remaining terms are $f(b)$ and $-f(a)$, giving $f(b)-f(a)$. □

Cancellation Pattern

The cancellation can be seen explicitly:

$$
\begin{array}{cc}
f(a+1) - f(a) & -f(a) \\
f(a+2) - f(a+1) & \cancel{f(a+1)} - \cancel{f(a+1)} \\
f(a+3) - f(a+2) & \cancel{f(a+2)} - \cancel{f(a+2)} \\
\vdots & \vdots \\
f(b) - f(b-1) & f(b)
\end{array}
$$

Remark (Intuition). Each intermediate term appears twice in the expanded sum: once with a positive sign and once with a negative sign. These pairs cancel, leaving only the first and last terms.

Operator Form

Using the forward difference operator $\Delta f(n) = f(n+1) - f(n)$, the telescoping identity takes the compact form

$$\sum_{k=a}^{b-1} \Delta f(k) = f(b) - f(a).$$

Remark (Discrete Fundamental Theorem of Calculus). This identity is the discrete analogue of the Fundamental Theorem of Calculus:

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

It reveals a deep relationship between the difference operator and summation: summation is the inverse of differencing, just as integration is the inverse of differentiation. This connection is made precise in the next section through the concept of the *discrete antiderivative*.

33.6 Discrete Antiderivatives

33.6.1 Discrete Antiderivatives

The telescoping identity established in the previous section,

$$\sum_{k=a}^{b-1} \Delta f(k) = f(b) - f(a),$$

reveals that summation acts as an inverse operation to the difference operator, much like integration acts as an inverse to differentiation in ordinary calculus. This motivates the following definition.

Definition 33.6.1 (Discrete Antiderivative)**.** Let $f : \mathbb{Z} \to \mathbb{R}$ be a discrete function. A function $F : \mathbb{Z} \to \mathbb{R}$ is called a *discrete antiderivative* (or *discrete indefinite sum*) of f if

$$\Delta F(n) = f(n),$$

that is, $F(n+1) - F(n) = f(n)$ for all $n \in \mathbb{Z}$.

Remark (Standard quantified statement)*.* $\forall f, F : \mathbb{Z} \to \mathbb{R} : \ F$ is a discrete antiderivative of $f \iff \forall n \in \mathbb{Z},\ F(n+1) - F(n) = f(n)$.

Remark (Interpretation)*.* A discrete antiderivative is a function whose one-step increments reproduce the given summand.

Remark (Analogy)*.* A discrete antiderivative plays the same role in discrete calculus that an antiderivative plays in ordinary calculus: it reverses the fundamental differentiation-like operator of the theory. Instead of reversing differentiation, it reverses the forward difference Δ.

Remark (Dependencies)*.*

- Forward Difference Operator.

Examples

Example 1. Let $f(n) = 1$. Take $F(n) = n$. Then

$$\Delta F(n) = (n+1) - n = 1 = f(n). \checkmark$$

Example 2. Let $f(n) = n$. Take $F(n) = \frac{n(n-1)}{2}$. Then

$$\Delta F(n) = \frac{(n+1)n}{2} - \frac{n(n-1)}{2} = \frac{n(n+1-n+1)}{2} = n = f(n). \checkmark$$

Remark (Connection to falling factorials)*.* Both examples above are instances of a general pattern: the discrete antiderivative of the falling factorial $n^{\underline{k}}$ is $\frac{1}{k+1} n^{\underline{k+1}}$, exactly mirroring the antiderivative rule for ordinary powers. This is developed in the Polynomial Basis section.

Constant of Integration

Discrete antiderivatives are not unique.

Proposition 33.6.2 (Constant of integration)**.** *If F is a discrete antiderivative of f, then so is $F + C$ for any constant $C \in \mathbb{R}$.*

Remark (Fully quantified form)*.* $\forall C \in \mathbb{R} : \ \Delta F = f \Rightarrow \Delta(F + C) = f$.

Remark (Interpretation). Adding a constant changes the level of a discrete antiderivative but not its one-step increments.

Remark (Dependencies).

- Discrete Antiderivative.
- Forward Difference Operator.

Proof.

$$\Delta(F + C)(n) = (F + C)(n + 1) - (F + C)(n) = F(n + 1) - F(n) = \Delta F(n) = f(n). \quad \square$$

Fundamental Theorem of Finite Calculus

Theorem (Fundamental Theorem of Finite Calculus)

Let F be a discrete antiderivative of f, so that $\Delta F(n) = f(n)$ for all n. Then

$$\sum_{k=a}^{b-1} f(k) = F(b) - F(a).$$

Remark (Fully quantified form). $\forall f, F : \mathbb{Z} \to \mathbb{R},\ \forall a, b \in \mathbb{Z},\ a \le b :\ (\forall n,\ \Delta F(n) = f(n)) \Rightarrow \sum_{k=a}^{b-1} f(k) = F(b) - F(a).$

Proof. Since $f(k) = \Delta F(k)$ for all k, substituting gives

$$\sum_{k=a}^{b-1} f(k) = \sum_{k=a}^{b-1} \Delta F(k).$$

By the Telescoping Identity (Theorem 33.5.1),

$$\sum_{k=a}^{b-1} \Delta F(k) = F(b) - F(a). \quad \square$$

Remark (Analogy with continuous calculus). This theorem is the discrete analogue of the Fundamental Theorem of Calculus,

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

In continuous calculus, differentiation and integration are inverse operations. In discrete calculus, the analogous inverse pair is the difference operator Δ and summation. Any sum can be evaluated by finding a discrete antiderivative of the summand — just as any definite integral can be evaluated by finding an antiderivative.

Remark (Practical consequence). The theorem reduces the problem of computing $\sum_{k=a}^{b-1} f(k)$ to the algebraic problem of finding F with $\Delta F = f$. Combined with the falling factorial rules developed in the next section, this gives a systematic method for evaluating a wide family of sums in closed form.

33.7 Polynomial Basis

33.7.1 Polynomial Basis: Falling Factorials

In ordinary calculus, polynomial functions are naturally expressed using the monomial powers $1, x, x^2, x^3, \ldots$ because the derivative acts simply on them: $\frac{d}{dx}x^n = nx^{n-1}$.

In discrete calculus, however, the difference operator does not behave as cleanly on ordinary powers. For example,

$$\Delta(n^2) = (n+1)^2 - n^2 = 2n + 1.$$

Although the degree decreases by one, the result contains an extra constant term. For this reason it is more natural to work with a different family of polynomials — the *falling factorials* — which interact with Δ exactly as monomials interact with $\frac{d}{dx}$.

Definition (Falling Factorial)

For a nonnegative integer k, the *falling factorial* (or *falling power*) is defined by

$$x^{\underline{k}} = x(x-1)(x-2)\cdots(x-k+1) \qquad (k \text{ factors}).$$

By convention, $x^{\underline{0}} = 1$.

Remark (Fully quantified form). $\forall k \in \mathbb{N},\ \forall x \in \mathbb{R} :\ x^{\underline{k}} := \prod_{j=0}^{k-1}(x-j)$, with $x^{\underline{0}} := 1$.

Remark (Relation to factorials). When $x = n \in \mathbb{N}$ and $k \le n$, the falling factorial equals $\frac{n!}{(n-k)!}$. In particular, $n^{\underline{n}} = n!$. Thus falling factorials generalise factorials to arbitrary inputs.

The first few falling factorials are:

$$\begin{aligned} x^{\underline{0}} &= 1, \\ x^{\underline{1}} &= x, \\ x^{\underline{2}} &= x(x-1), \\ x^{\underline{3}} &= x(x-1)(x-2). \end{aligned}$$

The Difference Rule for Falling Factorials

Theorem (Difference Rule for Falling Factorials)

For $k \ge 1$,

$$\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}.$$

Remark (Fully quantified form). $\forall k \in \mathbb{N},\ k \ge 1,\ \forall x \in \mathbb{R} :\ (x+1)^{\underline{k}} - x^{\underline{k}} = k\, x^{\underline{k-1}}$.

Proof. Observe that

$$(x+1)^{\underline{k}} = (x+1)\, x\, (x-1)\cdots(x-k+2).$$

Therefore

$$\Delta\, x^{\underline{k}} = (x+1)^{\underline{k}} - x^{\underline{k}} = x(x-1)\cdots(x-k+2)\big[(x+1)-(x-k+1)\big].$$

The bracket simplifies:

$$(x+1)-(x-k+1) = k.$$

The remaining product $x(x-1)\cdots(x-k+2)$ is exactly $x^{\underline{k-1}}$. Thus

$$\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}. \qquad \square$$

Remark (Mirror of the power rule). This formula exactly mirrors the continuous power rule $\frac{d}{dx}x^k = kx^{k-1}$. Because of this clean behaviour under Δ, falling factorials form the natural polynomial basis for discrete calculus.

Remark (Antiderivative rule). The theorem also yields the discrete antiderivative rule:

$$\Delta\left(\frac{x^{\underline{k+1}}}{k+1}\right) = x^{\underline{k}}.$$

Thus $\frac{x^{\underline{k+1}}}{k+1}$ is a discrete antiderivative of $x^{\underline{k}}$, mirroring the continuous rule $\int x^k\, dx = \frac{x^{k+1}}{k+1}$.

Summary Table

$f(x)$	$\Delta f(x)$
$x^{\underline{1}} = x$	1
$x^{\underline{2}} = x(x-1)$	$2x$
$x^{\underline{3}} = x(x-1)(x-2)$	$3x(x-1)$
$x^{\underline{4}} = x(x-1)(x-2)(x-3)$	$4x(x-1)(x-2)$

Polynomial Representation in the Falling Factorial Basis

Every polynomial can be written as a linear combination of falling factorials:

$$f(x) = c_0 + c_1\, x + c_2\, x^{\underline{2}} + c_3\, x^{\underline{3}} + \cdots + c_n\, x^{\underline{n}}.$$

Remark (Role in Newton's expansion). This representation plays the same role in discrete calculus that the Taylor expansion plays in continuous calculus. The coefficients c_k are determined by the successive finite differences of f evaluated at 0: $c_k = \Delta^k f(0)/k!$. This leads directly to Newton's forward difference expansion, developed in the Expansions section.

Consequences of the Difference Rule

The difference rule $\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}$ (Theorem 33.7.1) has several important consequences.

Higher Differences of Falling Factorials

Applying the rule repeatedly gives:

$$\Delta^m x^{\underline{k}} = \begin{cases} \dfrac{k!}{(k-m)!} x^{\underline{k-m}} & \text{if } m \le k, \\ 0 & \text{if } m > k. \end{cases}$$

In particular, evaluating at $x = 0$:

$$\Delta^m x^{\underline{k}}\Big|_{x=0} = \begin{cases} k! & \text{if } m = k, \\ 0 & \text{if } m \neq k. \end{cases}$$

Remark (Basis orthogonality)*.* This vanishing property means the falling factorials behave like a "triangular" basis: Δ^m kills all $x^{\underline{k}}$ with $k < m$, and returns a scalar multiple for $k = m$. This is what makes the coefficient extraction in Newton's expansion work cleanly.

Connection to Binomial Coefficients

The higher difference formula from the Binomial Theorem section (Proposition 33.2.2),

$$\Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k),$$

together with the difference rule, gives an explicit formula for the coefficients of the falling factorial expansion:

$$c_k = \frac{\Delta^k f(0)}{k!}.$$

Remark (Pascal's triangle revisited)*.* The coefficients appearing in higher finite differences follow exactly the pattern of Pascal's triangle — a direct consequence of the algebraic identity $\Delta = E - I$ and the binomial theorem.

Worked Example: Expanding n^2

We find the falling factorial expansion of $f(n) = n^2$.

Computing the difference table at $n = 0$:

$$f(0) = 0, \quad \Delta f(0) = 1, \quad \Delta^2 f(0) = 2, \quad \Delta^3 f(0) = 0.$$

The coefficients are $c_k = \Delta^k f(0)/k!$:

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = \frac{2}{2!} = 1, \quad c_3 = 0.$$

Therefore:

$$n^2 = 0 + n^{\underline{1}} + n^{\underline{2}} = n + n(n-1) = n^2 - n + n = n^2. \checkmark$$

Remark (Practical value)*.* Once a function is expressed in the falling factorial basis, both differencing and antidifferencing become purely mechanical, following the rule $\Delta x^{\underline{k}} = k\, x^{\underline{k-1}}$ and its inverse. This makes the Fundamental Theorem of Finite Calculus immediately applicable to compute sums like $\sum_{k=0}^{n-1} k^2$ without clever tricks.

33.8 Calculus Rules

33.8.1 Calculus Rules

With the difference operator and the polynomial basis in place, we can now develop the full suite of differentiation-like rules for discrete calculus. The linearity of Δ was established in the Difference Operators section (Proposition 33.4.3). Here we add the *product rule* — the discrete counterpart of the Leibniz rule $(fg)' = f'g + fg'$.

Product Rule

Theorem (Discrete Product Rule)

For functions $f, g : \mathbb{Z} \to \mathbb{R}$,

$$\Delta(fg)(n) = f(n)\,\Delta g(n) + g(n+1)\,\Delta f(n).$$

Remark (Fully quantified form). $\forall f, g : \mathbb{Z} \to \mathbb{R},\ \forall n \in \mathbb{Z} :\ (f(n+1)g(n+1)) - (f(n)g(n)) = f(n)(g(n+1) - g(n)) + g(n+1)(f(n+1) - f(n))$.

Proof. By definition of Δ,

$$\Delta(fg)(n) = f(n+1)g(n+1) - f(n)g(n).$$

Add and subtract $f(n)g(n+1)$:

$$= f(n+1)g(n+1) - f(n)g(n+1) + f(n)g(n+1) - f(n)g(n).$$

Grouping the first two and last two terms:

$$= g(n+1)\big(f(n+1) - f(n)\big) + f(n)\big(g(n+1) - g(n)\big) = g(n+1)\,\Delta f(n) + f(n)\,\Delta g(n). \quad \square$$

Remark (Comparison with the continuous product rule). In continuous calculus, $(fg)' = f'g + fg'$ is symmetric in f and g. In discrete calculus the rule is slightly asymmetric: one factor is evaluated at n and the other at $n+1$. This asymmetry is unavoidable and is a characteristic feature of the discrete setting.

Remark (Consequence — summation by parts). The product rule leads directly to the summation by parts formula: summing both sides of $\Delta(fg) = f\,\Delta g + g(E)\,\Delta f$ over $k = a$ to $b - 1$ and applying the Fundamental Theorem gives the formula in the next section.

Summation by Parts

Summation by parts is the discrete analogue of integration by parts. It is derived from the product rule exactly as the continuous version is derived from the Leibniz rule.

Theorem (Summation by Parts)

For functions $f, g : \mathbb{Z} \to \mathbb{R}$ and integers $a \le b$,

$$\sum_{k=a}^{b-1} f(k)\,\Delta g(k) = \Big[f(k)g(k)\Big]_{k=a}^{k=b} - \sum_{k=a}^{b-1} g(k+1)\,\Delta f(k).$$

That is,

$$\sum_{k=a}^{b-1} f(k)\,\Delta g(k) = f(b)g(b) - f(a)g(a) - \sum_{k=a}^{b-1} g(k+1)\,\Delta f(k).$$

Remark (Fully quantified form). $\forall f, g : \mathbb{Z} \to \mathbb{R},\ \forall a, b \in \mathbb{Z},\ a \le b : \displaystyle\sum_{k=a}^{b-1} f(k)\Delta g(k) = f(b)g(b) - f(a)g(a) - \sum_{k=a}^{b-1} g(k+1)\Delta f(k).$

Proof. By the Discrete Product Rule (Theorem 33.8.1),

$$\Delta(fg)(k) = f(k)\,\Delta g(k) + g(k+1)\,\Delta f(k).$$

Rearranging,

$$f(k)\,\Delta g(k) = \Delta(fg)(k) - g(k+1)\,\Delta f(k).$$

Sum both sides from $k = a$ to $k = b-1$:

$$\sum_{k=a}^{b-1} f(k)\,\Delta g(k) = \sum_{k=a}^{b-1} \Delta(fg)(k) - \sum_{k=a}^{b-1} g(k+1)\,\Delta f(k).$$

Applying the Fundamental Theorem of Finite Calculus (Theorem 33.6.1) to the first sum on the right gives

$$\sum_{k=a}^{b-1} \Delta(fg)(k) = f(b)g(b) - f(a)g(a),$$

which completes the proof. □

Remark (Analogy with integration by parts). The continuous integration by parts formula,

$$\int_a^b f\,dg = [fg]_a^b - \int_a^b g\,df,$$

is recovered from summation by parts by replacing $\sum$ with $\int$ and Δ with d. The asymmetry in the discrete formula — $g(k+1)$ rather than $g(k)$ in the remaining sum — reflects the shift inherent in the product rule.

Remark (Applications). Summation by parts is useful for evaluating sums involving products, particularly when one factor is easy to sum (as a falling factorial) and the other is easy to difference. Classic applications include evaluating $\sum kr^k$ (where $f(k) = k$, $\Delta g(k) = r^k$ leads to a geometric-type sum) and establishing Abel's summation formula.

33.9 Expansions

33.9.1 Newton's Forward Difference Expansion

In continuous calculus, smooth functions can be expanded as power series called Taylor series:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}\, x^k = f(0) + f'(0)\, x + \frac{f''(0)}{2!}\, x^2 + \cdots .$$

The coefficients are determined by successive derivatives evaluated at 0, and the basis functions are ordinary powers x^k.

Discrete calculus has a precise analogue: the *Newton forward difference expansion*, in which derivatives are replaced by finite differences and ordinary powers are replaced by falling factorials.

Theorem (Newton Forward Difference Expansion)

Let $f : \mathbb{Z} \to \mathbb{R}$. Then f can be expressed as

$$f(x) = \sum_{k=0}^{n} \frac{\Delta^k f(0)}{k!}\, x^{\underline{k}} = f(0) + \Delta f(0)\, x + \frac{\Delta^2 f(0)}{2!}\, x^{\underline{2}} + \frac{\Delta^3 f(0)}{3!}\, x^{\underline{3}} + \cdots ,$$

where $x^{\underline{k}} = x(x-1)\cdots(x-k+1)$ is the falling factorial. For a polynomial of degree n, the expansion is exact (the series terminates at $k = n$).

Remark (Fully quantified form). $\forall f : \mathbb{Z} \to \mathbb{R}$ polynomial of degree $\le n$, $\forall x \in \mathbb{Z} :\ f(x) = \sum_{k=0}^{n} \binom{x}{k} \Delta^k f(0)$, where $\binom{x}{k} := x^{\underline{k}}/k!$.

Structural Parallel with the Taylor Series

The Newton expansion mirrors the Taylor series at every point. The correspondence is exact:

Taylor Series (continuous)	**Newton Expansion (discrete)**
Coefficients $f^{(k)}(0)/k!$	Coefficients $\Delta^k f(0)/k!$
Basis functions x^k	Basis functions $x^{\underline{k}}$
Derivative $f^{(k)}$	Difference Δ^k
Convergence required in general	Exact for polynomials

Remark (Why falling factorials play the role of powers). In the Taylor series, powers x^k are the natural basis because the derivative satisfies $\frac{d}{dx}x^k = kx^{k-1}$ and $\frac{d^m}{dx^m}x^k\big|_{x=0} = k!$ when $m = k$, and 0 otherwise. The falling factorials satisfy the same two properties under Δ: $\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}$ and $\Delta^m\, x^{\underline{k}}\big|_{x=0} = k!$ when $m = k$, else 0. This "triangular" property is exactly what allows the coefficients $c_k = \Delta^k f(0)/k!$ to be read off independently, level by level.

Proof Sketch

We construct the expansion term by term, matching the values of f at successive integers.

The constant term $f(0)$ matches f at $x = 0$.

Adding the linear term $\Delta f(0) \cdot x$ matches f at $x = 0$ and $x = 1$.

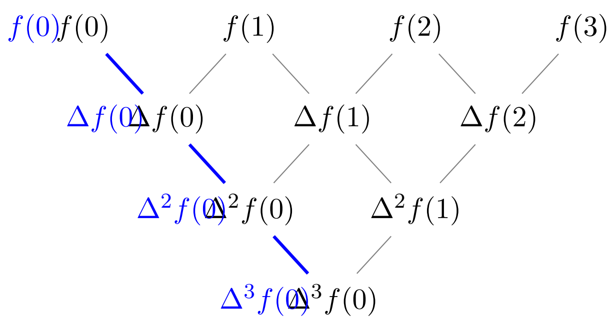
Adding the quadratic term $\frac{\Delta^2 f(0)}{2!} x^{\underline{2}}$ corrects the value at $x = 2$ while leaving earlier matches intact, because $x^{\underline{2}} = x(x-1)$ vanishes at $x = 0$ and $x = 1$.

At each stage, $x^{\underline{k}}$ vanishes at $x = 0, 1, \ldots, k-1$, so the k-th term corrects only the value at $x = k$. The coefficient $\Delta^k f(0)/k!$ achieves this correction by the triangular property noted above.

Reading Coefficients from the Difference Table

The coefficients in the Newton expansion are read from the *left edge* of the difference table — the "Newton diagonal":

$$f(0), \quad \Delta f(0), \quad \Delta^2 f(0), \quad \Delta^3 f(0), \ldots$$



Each level of the difference table contributes exactly one term to the falling-factorial expansion.

Worked Example: Expanding $f(n) = n^3$

We compute the difference table at $n = 0$:

n	$f(n)$	$\Delta f(n)$	$\Delta^2 f(n)$	$\Delta^3 f(n)$
0	0	1	6	6
1	1	7	12	
2	8	19		
3	27			

The coefficients are $c_k = \Delta^k f(0)/k!$:

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = \frac{6}{2} = 3, \quad c_3 = \frac{6}{6} = 1.$$

Therefore the Newton expansion is:

$$n^3 = n + 3\,n^{\underline{2}} + n^{\underline{3}} = n + 3n(n-1) + n(n-1)(n-2).$$

Verification: $n + 3n^2 - 3n + n^3 - 3n^2 + 2n = n^3$. ✓

Application: Computing $\sum_{k=0}^{n-1} k^2$

Using the expansion $k^2 = k^{\underline{1}} + k^{\underline{2}}$ and the Fundamental Theorem of Finite Calculus (Theorem 33.6.1),

$$\sum_{k=0}^{n-1} k^2 = \sum_{k=0}^{n-1} k^{\underline{1}} + \sum_{k=0}^{n-1} k^{\underline{2}}.$$

Since $\Delta\left(\frac{k^{\underline{2}}}{2}\right) = k^{\underline{1}}$ and $\Delta\left(\frac{k^{\underline{3}}}{3}\right) = k^{\underline{2}}$, the Fundamental Theorem gives:

$$\sum_{k=0}^{n-1} k^{\underline{1}} = \frac{n^{\underline{2}}}{2} - \frac{0^{\underline{2}}}{2} = \frac{n(n-1)}{2},$$

$$\sum_{k=0}^{n-1} k^{\underline{2}} = \frac{n^{\underline{3}}}{3} - \frac{0^{\underline{3}}}{3} = \frac{n(n-1)(n-2)}{3}.$$

Adding:

$$\sum_{k=0}^{n-1} k^2 = \frac{n(n-1)}{2} + \frac{n(n-1)(n-2)}{3} = \frac{n(n-1)}{6}\big[3 + 2(n-2)\big] = \frac{n(n-1)(2n-1)}{6}.$$

Remark (Check against the standard formula). The standard formula is $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$. The computed result $\sum_{k=0}^{n-1} k^2 = \frac{(n-1)n(2n-1)}{6}$ matches: substitute $n \mapsto n-1$ in the standard formula to get $\frac{(n-1)n(2n-1)}{6}$. ✓

This example illustrates the power of the method: the sum was evaluated by pure algebraic antidifferentiation, with no need for induction or clever manipulation.

33.10 Special Functions

33.10.1 Special Functions in Discrete Calculus

Just as continuous calculus has its distinguished special functions — the exponential e^{ax}, the trigonometric functions, and the logarithm — discrete calculus has analogous functions that behave simply under the difference operator.

The Discrete Exponential

In continuous calculus, the exponential function e^{ax} is its own derivative (up to a constant):

$$\frac{d}{dx} e^{ax} = a\, e^{ax}.$$

The analogous function in discrete calculus is the *discrete exponential* $(1+a)^n$.

Definition 33.10.1 (Discrete Exponential)**.** For a constant $a \neq -1$ and $n \in \mathbb{Z}$, the *discrete exponential* with base $(1+a)$ is the function

$$E_a(n) = (1+a)^n.$$

Remark (Fully quantified form). $\forall a \in \mathbb{R} \setminus \{-1\},\ \forall n \in \mathbb{Z} :\ E_a(n) := (1+a)^n.$

Remark (Interpretation). The discrete exponential is the geometric function whose ratio between successive inputs is $1+a$.

Proposition 33.10.2 (Discrete exponential rule)**.**

$$\Delta\,(1+a)^n = a\,(1+a)^n.$$

Remark (Fully quantified form). $\forall a \in \mathbb{R} \setminus \{-1\},\ \forall n \in \mathbb{Z} :\ (1+a)^{n+1} - (1+a)^n = a(1+a)^n.$

Remark (Interpretation). The forward difference of a discrete exponential is a scalar multiple of the same discrete exponential.

Proof.

$$\Delta\,(1+a)^n = (1+a)^{n+1} - (1+a)^n = (1+a)^n\big[(1+a)-1\big] = a(1+a)^n. \qquad \square$$

Remark (Analogy). The continuous exponential satisfies $\frac{d}{dx}e^{ax} = ae^{ax}$. The discrete exponential satisfies $\Delta(1+a)^n = a(1+a)^n$. The base e in continuous calculus is replaced by $(1+a)$ in discrete calculus. When a is small, $(1+a) \approx e^a$, recovering the continuous limit.

Remark (Antiderivative). From the proposition, a discrete antiderivative of $a(1+a)^n$ is $(1+a)^n$ itself. Equivalently, a discrete antiderivative of $(1+a)^n$ is $\frac{(1+a)^n}{a}$. Combined with the Fundamental Theorem of Finite Calculus, this gives the geometric series formula:

$$\sum_{k=0}^{n-1}(1+a)^k = \frac{(1+a)^n - 1}{a} \qquad (a \neq 0).$$

Remark (Dependencies).

- Discrete Exponential.
- Forward Difference Operator.

33.10.2 The Function 2^n in Discrete Calculus

Among all discrete exponentials, the case $a = 1$ — giving $(1+1)^n = 2^n$ — is perhaps the most important. It appears in combinatorics as the count of subsets, in probability as the sample space of fair coin sequences, in computer science as the natural measure of binary information, and in discrete calculus as a canonical test function. This section develops its properties systematically.

Differencing 2^n

Setting $a = 1$ in the discrete exponential rule (Proposition 33.10.2) gives immediately:

Proposition ($\Delta 2^n = 2^n$)

$$\Delta\, 2^n = 2^n.$$

Remark (Standard quantified statement). $\forall n \in \mathbb{Z} :\ 2^{n+1} - 2^n = 2^n$.

Proof.

$$\Delta\, 2^n = 2^{n+1} - 2^n = 2 \cdot 2^n - 2^n = (2-1) \cdot 2^n = 2^n. \qquad \square$$

Remark (Self-similarity). This says that 2^n is its own forward difference. In continuous calculus, the analogous property is $\frac{d}{dx}e^x = e^x$: the exponential is its own derivative. The function 2^n is the discrete analogue of e^x in the sense that $\Delta(2^n) = 2^n$. More precisely, $2^n = e^{n \ln 2}$, and the relationship $\Delta(2^n) = 2^n$ corresponds to the special case $a = 1$ of $\Delta(1+a)^n = a(1+a)^n$.

Antiderivative and Geometric Sum

Since $\Delta(2^n) = 2^n$, a discrete antiderivative of 2^n is 2^n itself (up to a constant).

Proposition (Antiderivative of 2^n)

A discrete antiderivative of 2^n is 2^n itself:

$$\Delta(2^n) = 2^n \qquad \text{so} \qquad \sum_{k=0}^{n-1} 2^k = 2^n - 1.$$

Remark (Standard quantified statement). $\forall n \in \mathbb{N} : \displaystyle\sum_{k=0}^{n-1} 2^k = 2^n - 1.$

Proof. Applying the Fundamental Theorem of Finite Calculus (Theorem 33.6.1) with $F(k) = 2^k$ and $f(k) = \Delta F(k) = 2^k$:

$$\sum_{k=0}^{n-1} 2^k = F(n) - F(0) = 2^n - 2^0 = 2^n - 1. \qquad \square$$

Remark (Verification by expansion). Expanding: $1 + 2 + 4 + 8 + \cdots + 2^{n-1}$. Multiply the partial sum $S = \sum_{k=0}^{n-1} 2^k$ by 2 to get $2S = \sum_{k=1}^{n} 2^k$. Subtracting: $2S - S = 2^n - 1$, so $S = 2^n - 1$. This is the classical derivation of the geometric series formula; the discrete calculus approach via antidifferentiation gives the same result mechanically.

Higher Differences of 2^n

Proposition 33.10.3 (Higher differences of 2^n)**.** *For all* $k \geq 0$,

$$\Delta^k 2^n = 2^n.$$

Remark (Standard quantified statement)*.* $\forall k \in \mathbb{N},\ \forall n \in \mathbb{Z}:\ \Delta^k(2^n) = 2^n$.

Remark (Interpretation)*.* Every forward-difference row for 2^n reproduces the same function.

Proof. By induction. The base case $k = 0$ is $\Delta^0(2^n) = 2^n$ by definition. For the inductive step, assume $\Delta^k(2^n) = 2^n$. Then

$$\Delta^{k+1}(2^n) = \Delta(\Delta^k(2^n)) = \Delta(2^n) = 2^n. \qquad \square$$

Remark (Consequence for difference tables)*.* Since every finite difference of 2^n is 2^n again, the difference table of 2^n consists entirely of the sequence 2^n at every level. This is the discrete analogue of the fact that the Taylor series of e^x has the same coefficient pattern at every derivative level. In particular, 2^n is *not* a polynomial — a polynomial of degree d would reach 0 at the $(d+1)$-th difference row. The non-termination of the difference table is the signature of a genuinely transcendental discrete function.

Remark (Dependencies)*.*

- Higher Differences.
- Discrete Exponential Rule.

2^n in Combinatorics and the Newton Expansion

The Binomial Theorem (Theorem 33.2.1) gives a direct connection: setting $x = y = 1$ yields

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

This counts the total number of subsets of an n-element set: there are $\binom{n}{k}$ subsets of size k, and the sum over all k gives 2^n.

More deeply, the Newton expansion (Theorem 33.9.1) of 2^n in the falling factorial basis is:

$$2^n = \sum_{k=0}^{n} \binom{n}{k} = \sum_{k=0}^{n} \frac{n^{\underline{k}}}{k!},$$

since $\Delta^k(2^n)\big|_{n=0} = 2^0 = 1$ for every k. This expresses 2^n as the sum of all normalised falling factorials up to degree n, and reveals why binomial coefficients count subsets: they are the coefficients of 2^n in its own Newton expansion.

Remark (Analogy with e^x)*.* In continuous calculus, $e^x = \sum_{k=0}^{\infty} x^k/k!$. The Newton expansion gives $2^n = \sum_{k=0}^{n} n^{\underline{k}}/k!$. Both express a canonical exponential as a sum of normalised basis functions, with all coefficients equal to 1. The analogy is exact: ordinary powers replace falling factorials, and the infinite series terminates at degree n in the discrete case because 2^n has degree exactly n in the falling factorial basis.

Discrete Differential Equation

In continuous calculus, the equation $f' = f$ (with $f(0) = 1$) has the unique solution $f(x) = e^x$. In discrete calculus, the analogous *difference equation* is:

$$\Delta F(n) = F(n), \qquad F(0) = 1.$$

That is, $F(n+1) - F(n) = F(n)$, so $F(n+1) = 2F(n)$. The unique solution to this initial value problem is $F(n) = 2^n$.

Proposition 33.10.4 (Discrete IVP)**.** *The unique function $F : \mathbb{N} \to \mathbb{R}$ satisfying*

$$F(n+1) = 2\,F(n), \qquad F(0) = 1$$

is $F(n) = 2^n$.

Remark (Standard quantified statement)*.* $\forall n \in \mathbb{N} : \ (F(n+1) = 2F(n) \text{ and } F(0) = 1) \Rightarrow F(n) = 2^n$.

Remark (Interpretation)*.* The recurrence $F(n+1) = 2F(n)$ with initial value 1 determines the discrete exponential 2^n uniquely.

Proof. Uniqueness: if G is another solution, then $G(0) = 1 = F(0)$, and inductively $G(n+1) = 2G(n) = 2F(n) = F(n+1)$, so $G = F$. Existence: verify directly that $2^{n+1} = 2 \cdot 2^n$ and $2^0 = 1$. ☐

Remark (Significance)*.* This characterises 2^n as the unique fixed point of "multiply by 2" starting from 1. The analogous characterisation of e^x is: unique solution to $f' = f$, $f(0) = 1$. Both are initial-value problems in their respective calculi, and both have closed-form solutions that serve as the canonical exponential of that calculus.

Remark (Dependencies)*.*

- Discrete Exponential.

Difference Formulas for Special Functions

The table below collects the difference formulas for the principal special functions of discrete calculus.

Remark (Difference formulas)*.*

Function	Forward Difference
$x^{\underline{k}}$ (falling factorial)	$k\,x^{\underline{k-1}}$
$(1+a)^n$ (discrete exponential)	$a\,(1+a)^n$
$\binom{x}{k} = x^{\underline{k}}/k!$	$\binom{x}{k-1}$
c (constant)	0
n	1
$n^{\underline{2}} = n(n-1)$	$2n$

Remark (Binomial coefficient as a special case). The formula $\Delta\binom{x}{k} = \binom{x}{k-1}$ follows immediately from the falling factorial rule $\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}$ by dividing through by $k!$. It says that the binomial coefficient $\binom{x}{k}$ is its own natural "unit" in discrete calculus, analogous to the monomial $x^k/k!$ in the Taylor expansion.

Antiderivative Table

By reversing the difference formulas, we obtain the corresponding discrete antiderivatives:

Remark (Discrete antiderivatives).

Function $f(n)$	**Antiderivative** $F(n)$ **with** $\Delta F = f$
$x^{\underline{k}}$	$\dfrac{x^{\underline{k+1}}}{k+1}$
$(1+a)^n$ $(a \neq 0)$	$\dfrac{(1+a)^n}{a}$
$\binom{x}{k}$	$\binom{x}{k+1}$
c (constant)	cn

Remark (Using the Fundamental Theorem). Each antiderivative in the table, combined with the Fundamental Theorem of Finite Calculus (Theorem 33.6.1), yields an immediate summation formula. For instance, from $\Delta\left(\frac{x^{\underline{k+1}}}{k+1}\right) = x^{\underline{k}}$:

$$\sum_{j=0}^{n-1} j^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}.$$

This is the discrete analogue of $\int_0^n x^k\, dx = \frac{n^{k+1}}{k+1}$.

Chapter 34

Sequences

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 6
Sequences

34.1 Sequence Definitions

34.1.1 Definitions

Definition (Sequence)

Definition 34.1.1 (Sequence)**.** Let X be a nonempty set. A **sequence** in X is a function

$$x : \mathbb{N} \to X.$$

The value $x(n)$ is written x_n and called the **n-th term** of the sequence. The sequence itself is denoted $(x_n)_{n\in\mathbb{N}}$, or simply (x_n) when the index set is clear.

Remark (Standard quantified statement).

$$\forall X \neq \varnothing\ \forall x : \mathbb{N} \to X\ \big(\mathrm{Sequence}(x, X) \iff x : \mathbb{N} \to X\big).$$

Remark (Definition predicate reading).

$$\mathrm{Sequence}(x_n, X)\ :=\ x : \mathbb{N} \to X.$$

Remark (Interpretation)*.* A sequence is not a set of terms but a function: the index carries independent weight. Two sequences (x_n) and (y_n) in X are equal precisely when $x_n = y_n$ for every $n \in \mathbb{N}$, so order and repetition are intrinsic to the object. The notational collapse from $x(n)$ to x_n is conventional suppression of function-application syntax; the underlying object remains a function $\mathbb{N} \to X$. When $X = \mathbb{R}$, the sequence is called a **real sequence**, which is the default ambient context for this chapter.

Remark (Dependencies).

- Function
- Natural Numbers

Definition (Real Sequence)

Definition 34.1.2 (Real Sequence)**.** A **real sequence** is a sequence whose image is contained in $\mathbb{R}$. Equivalently, a real sequence is a function

$$x : \mathbb{N} \to \mathbb{R}.$$

Remark (Standard quantified statement).

$$\mathrm{RealSequence}(x_n) \iff x : \mathbb{N} \to \mathbb{R}.$$

Remark (Definition predicate reading).

$$\mathrm{RealSequence}(x_n) \; := \; \mathrm{Sequence}(x_n, \mathbb{R}).$$

Remark (Interpretation). A real sequence is not a new kind of indexing object; it is the specialization of a sequence to the ambient space $\mathbb{R}$. Thus the generic function structure is inherited from sequences, while order, absolute value, and arithmetic enter through the real codomain.

Remark (Dependencies).

- Sequence
- Real Numbers

Example (Constant Sequence). Let $c \in \mathbb{R}$. The **constant sequence** is defined by

$$x_n := c \quad \forall n \in \mathbb{N}.$$

The constant sequence converges to c: given any $\varepsilon > 0$, take $N = 1$, and observe $|x_n - c| = 0 < \varepsilon$ for all $n \geq 1$. It is the degenerate case of convergence in which the tail is not merely eventually close to L but identically equal to it from the first term.

Example (The Sequence $(1/n)$). The sequence (x_n) defined by

$$x_n := \frac{1}{n} \quad \forall n \in \mathbb{N}$$

converges to 0. Given $\varepsilon > 0$, the Archimedean property supplies $N \in \mathbb{N}$ with $N > 1/\varepsilon$, and then for all $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

This is the canonical example of a sequence that approaches its limit monotonically from above, never reaching it. It also serves as the standard witness in squeeze arguments and in the proof that $\inf\{1/n : n \in \mathbb{N}\} = 0$.

Example (The Sequence (n)). The sequence (x_n) defined by

$$x_n := n \quad \forall n \in \mathbb{N}$$

is divergent. For any proposed limit $L \in \mathbb{R}$, take $\varepsilon = 1$; the Archimedean property supplies $n \in \mathbb{N}$ with $n > L + 1$, so $|x_n - L| = |n - L| > 1 = \varepsilon$ for infinitely many n. The sequence is unbounded, and every convergent sequence is bounded, so divergence also follows from that theorem once it is established. This is the canonical example of divergence to $+\infty$: the terms are not oscillating outside a neighborhood of L but growing without bound in a single direction.

34.1.2 Null and Constant Sequences

Null and Constant Sequences

Definition (Constant Sequence)

Definition 34.1.3 (Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **constant** exactly when there exists $c \in \mathbb{R}$ such that $x_n = c$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is constant} \iff \exists c \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n = c).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{ConstantSequence}(x_n, c) := \forall n \in \mathbb{N}\ (x_n = c).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not constant} \iff \forall c \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n \neq c).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \exists c \in \mathbb{R}\ \text{ConstantSequence}(x_n, c).$$

Remark (Failure modes). Every proposed constant value fails at some index.

Remark (Failure mode decomposition).

$$\forall c \in \mathbb{R} \qquad \underbrace{\exists n \in \mathbb{N}\ (x_n \neq c)}_{\text{a term differs from the proposed constant value}} .$$

Remark (Interpretation). A constant sequence has no genuine dependence on the index. The indexing is still part of the object, but every index is assigned the same real value.

Remark (Dependencies).

- Real Sequence

Definition (Null Sequence)

Definition 34.1.4 (Null Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **null sequence** exactly when for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for every $n \geq N$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is a null sequence} \iff \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, (|x_n| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{NullSequence}(x_n) :\equiv \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, (|x_n| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not a null sequence} \iff \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, (|x_n| \geq \varepsilon).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{NullSequence}(x_n).$$

Remark (Failure modes)*.* Some positive tolerance is missed infinitely far out in the sequence.

Remark (Failure mode decomposition).

$$\exists \varepsilon > 0\, \forall N \in \mathbb{N} \qquad \underbrace{\exists n \geq N\, (|x_n| \geq \varepsilon)}_{\text{the tail is not trapped inside the } \varepsilon\text{-band}} \qquad .$$

Remark (Interpretation)*.* A null sequence is a sequence whose tail is eventually contained in every symmetric neighborhood of zero. The first finitely many terms are irrelevant; only the eventual behavior matters.

Remark (Dependencies).

- Real Sequence

Theorem (Constant Sequence Convergence)

Theorem 34.1.5 (Constant Sequence Convergence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and } c \in \mathbb{R}.\\&\big(\forall n \in \mathbb{N}\, (x_n = c)\big) \implies \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, (|x_n - c| < \varepsilon).\end{aligned}$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \Longrightarrow \text{ConvergentSequence}(x_n, c).$$

Remark (Interpretation). A constant sequence is already equal to its proposed limit at every index. Every positive tolerance is therefore satisfied by any tail of the sequence.

Remark (Dependencies).

- Constant Sequence
- Convergent Sequence

Theorem (Zero Sequence is Null)

Theorem 34.1.6 (Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Go to proof.

Remark (Standard quantified statement).

Let (x_n) be a real sequence.

$$\big(\forall n \in \mathbb{N}\ (x_n = 0)\big) \Longrightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, 0) \Longrightarrow \text{NullSequence}(x_n).$$

Remark (Interpretation). The zero sequence satisfies every null-sequence tolerance immediately because the distance from each term to zero is exactly zero.

Remark (Dependencies).

- Null Sequence
- Constant Sequence

Theorem (Constant Null Sequence)

Theorem 34.1.7 (Constant Null Sequence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Remark (Standard quantified statement).

Let (x_n) be a real sequence and $c \in \mathbb{R}$.

$$\big(\forall n \in \mathbb{N}\ (x_n = c)\big) \Longrightarrow \Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon)\big] \Longleftrightarrow c = 0\Big).$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \big(\text{NullSequence}(x_n) \iff c = 0\big).$$

Remark (Interpretation). A sequence that never changes can approach zero only when its fixed value is already zero. Conversely, the fixed value zero gives the zero sequence.

Remark (Dependencies).

- Constant Sequence
- Null Sequence

Definition (Ultimately Constant Sequence)

Definition 34.1.8 (Ultimately Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **ultimately constant** exactly when there exist $N \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is ultimately constant} \iff \exists N \in \mathbb{N}\ \exists c \in \mathbb{R}\ \forall n \geq N\ (x_n = c).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) :\equiv \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n = c).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not ultimately constant} \iff \forall N \in \mathbb{N}\ \forall c \in \mathbb{R}\ \exists n \geq N\ (x_n \neq c).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \exists c \in \mathbb{R}\ \ \text{UltimatelyConstantSequence}(x_n, c).$$

Remark (Failure modes). No tail of the sequence becomes permanently equal to a single real value.

Remark (Failure mode decomposition).

$$\forall N \in \mathbb{N}\ \forall c \in \mathbb{R} \qquad \underbrace{\exists n \geq N\ (x_n \neq c)}_{\text{the proposed tail value fails later in the sequence}} \qquad .$$

Remark (Interpretation). An ultimately constant sequence may vary on an initial finite segment, but its tail becomes fixed. The concept isolates eventual behavior from finite initial behavior.

Remark (Dependencies).

- Real Sequence

Theorem (Difference from the Limit is Null)

Theorem 34.1.9 (Difference from the Limit is Null)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and } L \in \mathbb{R}.\\ &\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\big]\\ &\qquad \Longleftrightarrow \qquad \big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\big].\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \Longleftrightarrow \mathrm{NullSequence}(x_n - L).$$

Remark (Interpretation)*.* Subtracting the proposed limit translates the convergence problem to the special case of convergence to zero. Null sequences are therefore the local normal form for sequence convergence.

Remark (Dependencies).

- Convergent Sequence
- Null Sequence
- Pointwise Difference of Sequences

Theorem (Ultimately Constant Sequence Convergence)

Theorem 34.1.10 (Ultimately Constant Sequence Convergence)**.** *Let (x_n) be a real sequence. If there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$, then (x_n) converges to c.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\forall c \in \mathbb{R}\ \Big(\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = c)\big]\\ &\qquad \Longrightarrow \qquad \big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - c| < \varepsilon)\big]\Big).\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{UltimatelyConstantSequence}(x_n, c) \Longrightarrow \mathrm{ConvergentSequence}(x_n, c).$$

Remark (Interpretation). Once the tail is equal to a fixed value, all sufficiently late terms are already exactly at the proposed limit.

Remark (Dependencies).

- Ultimately Constant Sequence
- Convergent Sequence

Theorem (Constant Implies Ultimately Constant)

Theorem 34.1.11 (Constant Implies Ultimately Constant)**.** *Every constant real sequence is ultimately constant.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &\big[\exists c \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n = c)\big] \\ &\qquad \Longrightarrow \qquad \big[\exists N \in \mathbb{N}\ \exists c \in \mathbb{R}\ \forall n \geq N\ (x_n = c)\big]. \end{aligned}$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \Longrightarrow \text{UltimatelyConstantSequence}(x_n, c).$$

Remark (Interpretation). A constant sequence has already stabilized from the first index, so it is automatically stabilized on some tail.

Remark (Dependencies).

- Constant Sequence
- Ultimately Constant Sequence

Theorem (Ultimately Zero Sequence is Null)

Theorem 34.1.12 (Ultimately Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = 0)\big] \\ &\qquad \Longrightarrow \qquad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon). \end{aligned}$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, 0) \Longrightarrow \text{NullSequence}(x_n).$$

Remark (Interpretation). If the tail is identically zero, then every sufficiently late term lies inside every positive tolerance around zero.

Remark (Dependencies).

- Ultimately Constant Sequence
- Null Sequence

Theorem (Ultimately Constant Null Sequence)

Theorem 34.1.13 (Ultimately Constant Null Sequence)**.** *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\forall c \in \mathbb{R} \Big(\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = c)\big] \\ &\qquad \implies \qquad \Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon)\big] \iff c = 0 \Big)\Big).\end{aligned}$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) \implies \big(\text{NullSequence}(x_n) \iff c = 0\big).$$

Remark (Interpretation). Only the eventual value controls whether an ultimately constant sequence is null. A finite initial segment cannot prevent or create null behavior.

Remark (Dependencies).

- Ultimately Constant Sequence
- Null Sequence

Theorem (Tail Equality Preserves Convergence)

Theorem 34.1.14 (Tail Equality Preserves Convergence)**.** *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.}\\ &\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = y_n)\big] \\ &\qquad \implies \qquad \forall L \in \mathbb{R} \Big(\big[\forall \varepsilon > 0\ \exists N_x \in \mathbb{N}\ \forall n \geq N_x\ (|x_n - L| < \varepsilon)\big] \\ &\qquad\qquad \iff \qquad \big[\forall \varepsilon > 0\ \exists N_y \in \mathbb{N}\ \forall n \geq N_y\ (|y_n - L| < \varepsilon)\big]\Big).\end{aligned}$$

Remark (Predicate reading).

$$\text{EventuallyEqual}(x_n, y_n) \implies \forall L \in \mathbb{R}\ \big(\text{ConvergentSequence}(x_n, L) \iff \text{ConvergentSequence}(y_n, L)\big).$$

Remark (Interpretation). Convergence is a tail property. Changing finitely many initial terms cannot change whether a sequence converges to a specified real number.

Remark (Dependencies).

- Convergent Sequence

Definition (Sequence Bounded Above)

Definition 34.1.15 (Sequence Bounded Above)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded above** exactly when there exists $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is bounded above} \iff \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{BoundedAboveSequence}(x_n) :\equiv \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not bounded above} \iff \forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (M < x_n).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{BoundedAboveSequence}(x_n).$$

Remark (Failure modes). Every proposed upper bound is exceeded by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall M \in \mathbb{R} \qquad \underbrace{\exists n \in \mathbb{N}\ (M < x_n)}_{\text{a term rises above the proposed ceiling}} \qquad .$$

Remark (Interpretation). A bounded-above sequence has a finite ceiling. Failure means no real number serves as a ceiling for all terms.

Remark (Dependencies).

- Real Sequence

Definition (Sequence Bounded Below)

Definition 34.1.16 (Sequence Bounded Below)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded below** exactly when there exists $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is bounded below} \iff \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{BoundedBelowSequence}(x_n) :\equiv \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is not bounded below} \iff \forall m \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n < m).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{BoundedBelowSequence}(x_n).$$

Remark (Failure modes). Every proposed lower bound is undercut by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall m \in \mathbb{R} \qquad \underbrace{\exists n \in \mathbb{N}\ (x_n < m)}_{\text{a term falls below the proposed floor}} \qquad .$$

Remark (Interpretation). A bounded-below sequence has a finite floor. Failure means no real number serves as a floor for all terms.

Remark (Dependencies).

- Real Sequence

Definition (Bounded Sequence)

Definition 34.1.17 (Bounded Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded** exactly when there exists $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &(x_n) \text{ is bounded} \iff \exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{BoundedSequence}(x_n) :\equiv \exists M > 0\, \forall n \in \mathbb{N}\, (|x_n| \leq M).$$

Remark (Negated quantified statement).

$$\begin{array}{l}\text{Let } (x_n) \text{ be a real sequence.}\\ (x_n) \text{ is not bounded} \iff \forall M > 0\, \exists n \in \mathbb{N}\, (|x_n| > M).\end{array}$$

Remark (Negation predicate reading).

$$\neg\, \text{BoundedSequence}(x_n).$$

Remark (Failure modes). Every symmetric bound around zero is escaped by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall M > 0 \qquad \underbrace{\exists n \in \mathbb{N}\, (|x_n| > M)}_{\text{a term escapes the symmetric band of radius } M} \qquad .$$

Remark (Interpretation). A bounded sequence is trapped in a finite symmetric band around zero. This is equivalent to having both a finite ceiling and a finite floor.

Remark (Dependencies).

- Real Sequence

Theorem (Eventually Bounded Above Tail Formulation)

Theorem 34.1.18 (Eventually Bounded Above Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \geq N_0$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{array}{l}\text{Let } (x_n) \text{ be a real sequence.}\\ \big[\exists M \in \mathbb{R}\, \forall n \in \mathbb{N}\, (x_n \leq M)\big]\\ \qquad \iff \qquad \big[\exists N_0 \in \mathbb{N}\, \exists M \in \mathbb{R}\, \forall n \geq N_0\, (x_n \leq M)\big].\end{array}$$

Remark (Predicate reading).

$$\text{BoundedAboveSequence}(x_n) \iff \exists M \in \mathbb{R}\ \text{Eventually}\big(x_n,\ x_n \leq M\big).$$

Remark (Interpretation). One-sided boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a larger upper bound.

Remark (Dependencies).

- Sequence Bounded Above

Theorem (Eventually Bounded Below Tail Formulation)

Theorem 34.1.19 (Eventually Bounded Below Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \le x_n$ for every $n \ge N_0$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big]\\ &\qquad \iff \qquad \big[\exists N_0 \in \mathbb{N}\ \exists m \in \mathbb{R}\ \forall n \ge N_0\ (m \le x_n)\big].\end{aligned}$$

Remark (Predicate reading).

$$\text{BoundedBelowSequence}(x_n) \iff \exists m \in \mathbb{R}\ \text{Eventually}\big(x_n,\ m \le x_n\big).$$

Remark (Interpretation)*.* One-sided lower boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a lower bound.

Remark (Dependencies).

- Sequence Bounded Below

Theorem (Eventually Bounded Tail Formulation)

Theorem 34.1.20 (Eventually Bounded Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \le M$ for every $n \ge N_0$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\big[\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M)\big]\\ &\qquad \iff \qquad \big[\exists N_0 \in \mathbb{N}\ \exists M > 0\ \forall n \ge N_0\ (|x_n| \le M)\big].\end{aligned}$$

Remark (Predicate reading).

$$\text{BoundedSequence}(x_n) \iff \exists M > 0\ \text{Eventually}\big(x_n,\ |x_n| \le M\big).$$

Remark (Interpretation)*.* Boundedness is a tail property up to finitely many exceptions. A finite initial segment can be enclosed by increasing the bound.

Remark (Dependencies).

- Bounded Sequence

Theorem (Bounded Sequence If and Only If Bounded Above and Below)

Theorem 34.1.21 (Bounded Sequence If and Only If Bounded Above and Below)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\big[\exists K > 0\ \forall n \in \mathbb{N}\ (|x_n| \le K)\big]\\ &\qquad \iff \qquad \big[\exists m \in \mathbb{R}\ \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n \land x_n \le M)\big].\end{aligned}$$

Remark (Predicate reading).

$$\text{BoundedSequence}(x_n) \iff \big(\text{BoundedBelowSequence}(x_n) \land \text{BoundedAboveSequence}(x_n)\big).$$

Remark (Interpretation). An absolute-value bound gives a symmetric interval around zero. Having both a floor and a ceiling gives an interval that may not be symmetric, but it can be enlarged to a symmetric one.

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Absolute Bound Gives Upper and Lower Bounds)

Theorem 34.1.22 (Absolute Bound Gives Upper and Lower Bounds)**.** *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \le K$ for every $n \in \mathbb{N}$, then $-K \le x_n \le K$ for every $n \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\forall K > 0\ \big[\forall n \in \mathbb{N}\ (|x_n| \le K)\big]\\ &\qquad \implies \qquad \forall n \in \mathbb{N}\ (-K \le x_n \land x_n \le K).\end{aligned}$$

Remark (Interpretation). An absolute-value inequality simultaneously controls the positive and negative directions. The upper barrier is K, and the lower barrier is $-K$.

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Upper and Lower Bounds Give an Absolute Bound)

Theorem 34.1.23 (Upper and Lower Bounds Give an Absolute Bound)**.** *Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \le x_n \le M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \le K$ for every $n \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\big[\exists m \in \mathbb{R}\ \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n \wedge x_n \le M)\big]\\&\qquad \Longrightarrow \qquad \big[\exists K > 0\ \forall n \in \mathbb{N}\ (|x_n| \le K)\big].\end{aligned}$$

Remark (Interpretation). Two one-sided barriers can always be replaced by one symmetric barrier around zero. Any positive K at least as large as both $|m|$ and $|M|$ works.

Remark (Dependencies).

- Sequence Bounded Above
- Sequence Bounded Below
- Bounded Sequence

34.2 Examples and Counterexamples

Sequence Examples — Quick Reference

Example	**Role**
Constant sequence	Basic convergent model
Reciprocal sequence	Canonical null sequence
Alternating null sequence	Sign oscillation with convergence to 0
Oscillating sequence	Bounded nonconvergent model
Geometric sequence	Ratio-driven convergence and divergence
Bounded not convergent	Boundedness alone is insufficient
Small differences not Cauchy	Vanishing increments do not ensure Cauchy

Examples and Counterexamples

Examples provide test objects for the definitions. A good sequence example usually isolates one behavior: settling to a limit, oscillating between several values, escaping to infinity, or satisfying a tempting condition that is still too weak to force convergence.

Example (Constant Sequence). Let $c \in \mathbb{R}$ and define $x_n = c$ for every $n \in \mathbb{N}$. Then (x_n) converges to c.

Remark (Interpretation). A constant sequence is the baseline example of convergence: every term is already equal to the proposed limit.

Example (Reciprocal Sequence). Define $x_n = 1/n$ for every $n \in \mathbb{N}$. Then (x_n) is decreasing, bounded below by 0, and converges to 0.

Remark (Interpretation). The reciprocal sequence is the standard positive null sequence. It is also the model for estimates of the form "eventually smaller than any fixed positive tolerance."

Example (Alternating Null Sequence). Define

$$x_n = \frac{(-1)^n}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Then (x_n) is not monotone, but it converges to 0.

Remark (Interpretation). Oscillation in sign does not by itself prevent convergence. The amplitudes must be watched: here the oscillation collapses to zero.

Example (Oscillating Sequence). Define

$$x_n = (-1)^n \qquad \text{for every } n \in \mathbb{N}.$$

Then (x_n) is bounded and divergent. Its even-indexed subsequence converges to 1, and its odd-indexed subsequence converges to -1.

Remark (Interpretation). This is the canonical oscillator. The sequence never settles because two subsequences settle to different limits.

Example (Geometric Sequence). Let $r \in \mathbb{R}$ and define $x_n = r^n$ for every $n \in \mathbb{N}$. Then:

- if $|r| < 1$, then $x_n \to 0$;
- if $r = 1$, then $x_n \to 1$;
- if $r = -1$, then (x_n) oscillates and diverges;
- if $|r| > 1$, then (x_n) is unbounded.

Remark (Interpretation). The geometric sequence is the standard laboratory for rate, sign, and magnitude. The threshold $|r| = 1$ separates decay from growth.

Example (Bounded Does Not Imply Convergent). The sequence $x_n = (-1)^n$ is bounded, since $|x_n| \leq 1$ for every $n \in \mathbb{N}$, but it does not converge.

Remark (Interpretation). Boundedness prevents escape to infinity, but it does not prevent persistent oscillation.

Example (Vanishing Successive Differences Do Not Imply Cauchy). Let

$$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Then

$$x_{n+1} - x_n = \frac{1}{n+1} \to 0,$$

but (x_n) is not Cauchy.

Remark (Interpretation). Cauchy control requires all sufficiently late terms to be mutually close. Controlling only adjacent jumps is weaker.

34.3 Convergence

Definition (Convergence of a Sequence)

Definition 34.3.1 (Convergence of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$. The sequence (x_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(|x_n - L| < \varepsilon\big).$$

The value L is called the **limit** of (x_n), written $x_n \to L$ or $\lim_{n\to\infty} x_n = L$. A sequence that converges to some $L \in \mathbb{R}$ is called **convergent**; otherwise it is called **divergent**.

Remark (Standard quantified statement).

$$\begin{gathered}\forall (x_n) \subseteq \mathbb{R} \, \forall L \in \mathbb{R} \\ \Big(x_n \to L \iff \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(|x_n - L| < \varepsilon\big)\Big).\end{gathered}$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, L) \;:=\; \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \;\; \text{OutputTolerance}(x_n, L, \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{gathered}x_n \not\to L \iff \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \\ \big(|x_n - L| \geq \varepsilon\big).\end{gathered}$$

Remark (Negation predicate reading).

$$\text{NotConvergentTo}(x_n, L) \;:=\; \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \, \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Failure modes). Failure of convergence to a fixed L means that some positive tolerance cannot be controlled on any tail. The escaped terms may oscillate, drift monotonically away, or grow without bound, but all such behavior is captured by the same persistent-tolerance obstruction.

Remark (Failure mode decomposition). The negation decomposes into a single structural obstruction:

$$\underbrace{\exists \varepsilon > 0}_{\text{a fixed tolerance}} \quad \underbrace{\forall N \in \mathbb{N}}_{\text{no tail suffices}} \quad \underbrace{\exists n \geq N}_{\text{terms escape infinitely often}} \quad \underbrace{|x_n - L| \geq \varepsilon}_{\text{outside the target interval}} \quad .$$

The sequence fails to converge to L not by drifting away once, but by returning outside the ε-interval infinitely often regardless of how far along the tail one looks.

Remark (Interpretation). The quantifier order $\forall \varepsilon \, \exists N \, \forall n \geq N$ encodes a strict dependency: N is permitted to depend on ε, but not on n. This asymmetry is load-bearing. Reversing the outer quantifiers to $\exists N \, \forall \varepsilon$ would assert that a single index N controls all tolerances simultaneously, which is a strictly stronger and generally false condition. The condition $|x_n - L| < \varepsilon$ is the membership assertion $x_n \in (L - \varepsilon, \, L + \varepsilon)$; convergence is therefore the statement that the tail of the sequence is eventually contained in every ε-neighborhood of L.

Remark (Dependencies).

- Real Sequence

Definition (Convergence — Neighborhood Formulation)

Definition 34.3.2 (Convergence — Neighborhood Formulation)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The sequence (x_n) **converges** to L if for every $\varepsilon > 0$ the ε-neighborhood

$$V_\varepsilon(L) \ := \ (L - \varepsilon, \, L + \varepsilon)$$

contains all but finitely many terms of (x_n); that is,

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R} \, \forall L \in \mathbb{R} \\ &\quad \Big(x_n \to L \iff \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, L) \ := \ \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}x_n \not\to L \iff \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \\ \big(x_n \notin V_\varepsilon(L)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\text{NotConvergentTo}(x_n, L) \ := \ \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \, \big(x_n \notin V_\varepsilon(L)\big).$$

Remark (Failure modes). Failure of the neighborhood formulation is exactly failure to trap a tail in every neighborhood of L. Different examples may escape by oscillation, by monotone drift, or by unbounded growth, but the formal failure is always the existence of one neighborhood that every tail misses at least once.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon > 0}_{\text{a fixed neighborhood}} \quad \underbrace{\forall N \in \mathbb{N}}_{\text{no tail suffices}} \quad \underbrace{\exists n \geq N}_{\text{terms escape infinitely often}} \quad \underbrace{x_n \notin V_\varepsilon(L)}_{\text{outside every candidate neighborhood}} \quad .$$

The sequence fails to converge to L precisely when some neighborhood of L contains only finitely many terms: infinitely many terms lie permanently outside it.

Remark (Interpretation). The neighborhood formulation reframes convergence as a set-containment condition rather than an inequality condition. The assertion $x_n \in V_\varepsilon(L)$ is identical to $|x_n - L| < \varepsilon$; the two definitions are logically equivalent and the choice between them is one of language, not content. The neighborhood language is the natural bridge to the topological formulation: in a general topological space, open sets replace ε-neighborhoods and the definition reads identically with $V_\varepsilon(L)$ replaced by any open set U containing L. The phrase *all but finitely many terms* is the canonical prose compression of the quantifier block $\exists N \in \mathbb{N}\, \forall n \geq N$, and is standard in the literature.

Remark (Dependencies).

- Convergence of a Sequence
- ε-neighbourhood

Theorem (Equivalence of Convergence Formulations)

Theorem 34.3.3 (Equivalence of Convergence Formulations)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The following statements are equivalent.*

(a) $x_n \to L$.

(b) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(|x_n - L| < \varepsilon\big)$.

(c) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(L - \varepsilon < x_n < L + \varepsilon\big)$.

(d) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(x_n \in V_\varepsilon(L)\big)$.

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R} \\ \Big((a) \iff (b) \iff (c) \iff (d)\Big).$$

Remark (Theorem predicate reading).

EquivalentConvergenceForms(x_n, L) := ConvergentSequence$(x_n, L) \iff$ NeighborhoodConvergentSeque

Remark (Interpretation). The four conditions are the same geometric fact written in four notational registers. Condition (a) is the informal assertion. Condition (b) is the canonical ε-K definition: the absolute value formulation. Condition (c) makes the interval structure explicit by expanding $|x_n - L| < \varepsilon$ into its equivalent double inequality $L - \varepsilon < x_n < L + \varepsilon$; this is useful in proofs where upper and lower bounds are handled separately. Condition (d) is the neighborhood formulation: $x_n \in V_\varepsilon(L)$ is identical to conditions (b) and (c) but frames convergence as set membership, which is the natural language for the topological generalization. The index K plays the same role as N in the base definition Definition 34.3.1; the letter changes here to signal that K is the tail offset in the sense of Definition 34.4.1.

Remark (Dependencies).

- Convergence of a Sequence
- Convergence — Neighborhood Formulation

34.3.1 Limits

Theorem (Uniqueness of Limits)

Theorem 34.3.4 (Uniqueness of Limits)**.** *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \to L$ and $x_n \to K$, then $L = K$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall K \in \mathbb{R}\\ &\quad \Big(\big(x_n \to L \ \wedge\ x_n \to K \big) \implies L = K \Big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{UniqueSequentialLimit}(x_n, L, K) \ := \ \big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(x_n, K)\big) \implies L =$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists (x_n) \subseteq \mathbb{R}\ \exists L \in \mathbb{R}\ \exists K \in \mathbb{R}\\ &\quad \Big(\big(x_n \to L \ \wedge\ x_n \to K \big) \ \wedge\ L \neq K \Big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{UniqueSequentialLimit}(x_n, L, K) \ := \ \text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(x_n, K) \wedge L \neq K.$$

Remark (Failure modes). The only failure mode is the existence of one real sequence with two distinct real limits. In $\mathbb{R}$ this is impossible because distinct real numbers can be separated by disjoint neighborhoods.

Remark (Failure mode decomposition).

$$\underbrace{\exists (x_n) \subseteq \mathbb{R}}_{\text{some sequence}}\ \underbrace{x_n \to L\ \wedge\ x_n \to K}_{\text{converges to two values}}\ \ \underbrace{L \neq K}_{\text{distinct limits}}\ .$$

The theorem asserts this configuration is impossible in $\mathbb{R}$. The failure mode is not merely exotic: it occurs in non-Hausdorff topological spaces where distinct points cannot be separated by disjoint open sets. The proof that it cannot occur in $\mathbb{R}$ is precisely an argument that $\mathbb{R}$ has enough room to separate any two distinct points by disjoint ε-neighborhoods.

Remark (Interpretation). The theorem licenses the notation $\lim_{n\to\infty} x_n = L$: the definite article is justified precisely because the limit, when it exists, is unique. Without uniqueness the limit would be a relation rather than a function, and the notation would be ambiguous. The proof strategy is a separation argument: if $L \neq K$ then $|L - K| > 0$, and choosing $\varepsilon = |L - K|/2$ produces two disjoint neighborhoods that the tail of (x_n) cannot simultaneously occupy. The Hausdorff remark in the failure mode block is the forward pointer: when sequences are studied in general topological spaces, uniqueness of limits is equivalent to the Hausdorff separation axiom.

Remark (Dependencies).

- Convergence of a Sequence
- Triangle inequality

Theorem (Limit Preserves Eventual Order)

Theorem 34.3.5 (Limit Preserves Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \to L$ and $y_n \to M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N_0$, then $L \leq M$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall L, M \in \mathbb{R} \\
&\quad \Big(x_n \to L\ \wedge\ y_n \to M\ \wedge\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n \leq y_n) \Big) \\
&\quad \implies \quad L \leq M.
\end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{ConvergentSequence}(y_n, M) \wedge \mathrm{Eventually}(x_n,\ x_n \leq y_n)\big) \implies L \leq M.$$

Remark (Interpretation). An inequality that holds on a tail survives passage to the limit, but only as a weak inequality. Strict inequalities on the terms may collapse at the limit, as in $0 < x_n$ with $x_n \to 0$.

Remark (Dependencies).

- Convergence of a Sequence
- Uniqueness of Limits

Theorem (Strict Limit Separation Gives Eventual Order)

Theorem 34.3.6 (Strict Limit Separation Gives Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R} \\ &\quad \Big(x_n \to A \ \wedge \ y_n \to B \ \wedge \ A < B \Big) \\ &\quad \Longrightarrow \quad \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < y_n). \end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}(x_n, A) \wedge \mathrm{ConvergentSequence}(y_n, B) \wedge A < B\big) \Longrightarrow \mathrm{Eventually}(x_n,\ x_n < y_n).$$

Remark (Interpretation). If the limiting values have a positive gap, then sufficiently late terms inherit the same order. The proof uses the open interval between A and B: eventually (x_n) lies near A and (y_n) lies near B, with room left between the two tails.

Remark (Dependencies).

- Convergence of a Sequence
- Uniqueness of Limits

Theorem (Eventual Strict Comparison Preserves Weak Limit Order)

Theorem 34.3.7 (Eventual Strict Comparison Preserves Weak Limit Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R} \\ &\quad \Big(x_n \to A \ \wedge \ y_n \to B \ \wedge \ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < y_n) \Big) \Longrightarrow A \leq B, \\ &\quad \Big(x_n \to A \ \wedge \ y_n \to B \ \wedge \ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n > y_n) \Big) \Longrightarrow A \geq B. \end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\text{ConvergentSequence}(x_n, A) \land \text{ConvergentSequence}(y_n, B) \land \text{Eventually}(x_n,\ x_n < y_n)\big) \implies A \le B,$$
$$\big(\text{ConvergentSequence}(x_n, A) \land \text{ConvergentSequence}(y_n, B) \land \text{Eventually}(x_n,\ x_n > y_n)\big) \implies A \ge B.$$

Remark (Interpretation). Strict comparison of terms does not force strict comparison of limits. The strict inequality can collapse in the limit, so the conclusion is weak: for example $1/n > 0$ for every $n \in \mathbb{N}$, but both sides have limit 0.

Remark (Dependencies).

- Limit Preserves Eventual Order

Theorem (Constant Comparison for Sequence Limits)

Theorem 34.3.8 (Constant Comparison for Sequence Limits)**.** *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n \le B$ for every $n \ge N$, then $A \le B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \ge N$, then $A \le B$.

(c) If there exists $N \in \mathbb{N}$ such that $x_n \ge B$ for every $n \ge N$, then $A \ge B$.

(d) If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \ge N$, then $A \ge B$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R} \\
&\quad \Big(x_n \to A\ \land\ \exists N \in \mathbb{N}\ \forall n \ge N\ (x_n \le B)\Big) \implies A \le B, \\
&\quad \Big(x_n \to A\ \land\ \exists N \in \mathbb{N}\ \forall n \ge N\ (x_n < B)\Big) \implies A \le B, \\
&\quad \Big(x_n \to A\ \land\ \exists N \in \mathbb{N}\ \forall n \ge N\ (x_n \ge B)\Big) \implies A \ge B, \\
&\quad \Big(x_n \to A\ \land\ \exists N \in \mathbb{N}\ \forall n \ge N\ (x_n > B)\Big) \implies A \ge B.
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{ConvergentSequence}(x_n, A) \land \text{Eventually}(x_n,\ x_n \le B) \implies A \le B, \\
&\text{ConvergentSequence}(x_n, A) \land \text{Eventually}(x_n,\ x_n < B) \implies A \le B, \\
&\text{ConvergentSequence}(x_n, A) \land \text{Eventually}(x_n,\ x_n \ge B) \implies A \ge B, \\
&\text{ConvergentSequence}(x_n, A) \land \text{Eventually}(x_n,\ x_n > B) \implies A \ge B.
\end{aligned}$$

Remark (Interpretation). This is the constant version of the comparison theorem. A fixed real number B may be treated as the constant sequence with value B, so each conclusion is a direct application of eventual order preservation. Again, strict term inequalities produce only weak inequalities between limits.

Remark (Dependencies).

- Constant Sequence
- Limit Preserves Eventual Order
- Eventual Strict Comparison Preserves Weak Limit Order

Theorem (Constant Squeeze Theorem)

Theorem 34.3.9 (Constant Squeeze Theorem). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \leq x_n \leq L$ for every $n \geq N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R} \\
&\quad \Big(\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (L \leq x_n \leq L)\Big) \\
&\quad \implies \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{Eventually}(x_n,\ L \leq x_n \leq L) \implies \text{ConvergentSequence}(x_n, L).$$

Remark (Interpretation). This is the degenerate squeeze theorem: the lower and upper barriers are the same constant sequence. It says that eventual equality to a real number is enough for convergence to that real number.

Remark (Dependencies).

- Convergence of a Sequence
- Constant Sequence

Theorem (Sequence Squeeze Theorem)

Theorem 34.3.10 (Sequence Squeeze Theorem). *Let (a_n), (x_n), and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \to L$ and $b_n \to L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \leq x_n \leq b_n$ for every $n \geq N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (a_n), (x_n), (b_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R} \\
&\quad \Big(a_n \to L\ \wedge\ b_n \to L\ \wedge\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (a_n \leq x_n \leq b_n)\Big) \\
&\quad \implies \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).
\end{aligned}$$

Remark (Theorem predicate reading).

$\big(\text{ConvergentSequence}(a_n, L) \wedge \text{ConvergentSequence}(b_n, L) \wedge \text{Eventually}(x_n,\ a_n \le x_n \le b_n)\big) \implies \text{Convergent}$

Remark (Interpretation). The middle sequence is trapped between two tails that approach the same number. Once both barriers lie inside an ε-neighborhood of L, the trapped sequence must also lie inside that neighborhood.

Remark (Dependencies).

- Convergence of a Sequence
- Limit Preserves Eventual Order

Theorem (Absolute-Value Squeeze Theorem)

Theorem 34.3.11 (Absolute-Value Squeeze Theorem)**.** *Let (x_n) and (u_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $u_n \to 0$. If there exists $N_0 \in \mathbb{N}$ such that $|x_n - L| \le u_n$ for every $n \ge N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n), (u_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R} \\ &\quad \Big(u_n \to 0\ \wedge\ \exists N_0 \in \mathbb{N}\ \forall n \ge N_0\ (|x_n - L| \le u_n)\Big) \\ &\quad \implies \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \ge N\ \big(|x_n - L| < \varepsilon\big). \end{aligned}$$

Remark (Theorem predicate reading).

$\big(\text{ConvergentSequence}(u_n, 0) \wedge \text{Eventually}(x_n,\ |x_n - L| \le u_n)\big) \implies \text{ConvergentSequence}(x_n, L).$ ∎

Remark (Interpretation). This is the estimate form of the squeeze theorem. Instead of constructing two explicit bounding sequences, it bounds the error $|x_n - L|$ by a null sequence.

Remark (Dependencies).

- Convergence of a Sequence
- Null Sequence
- Sequence Squeeze Theorem

34.3.2 Algebra of Limits

Algebra of Convergent Sequences

The algebra of convergent sequences records that limits are compatible with the ordinary algebraic operations on real numbers. The pointwise operations below turn real sequences into new real sequences, and the theorem blocks that follow state how convergence passes through those operations. The reciprocal and quotient laws require nonzero denominator hypotheses, while the square-root law requires nonnegative terms.

Definition (Pointwise Sum of Sequences)

Definition 34.3.12 (Pointwise Sum of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise sum** of (x_n) and (y_n) is the real sequence $(x_n + y_n)$ defined by

$$(x + y)_n := x_n + y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement)*.*

$$\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.}$$
$$\forall n \in \mathbb{N} \ \big((x + y)_n = x_n + y_n\big).$$

Remark (Interpretation)*.* The sum sequence is formed term by term. No limiting process is involved in the definition; convergence enters only when the behavior of the resulting sequence is studied.

Remark (Dependencies)*.*

- Real Sequence

Definition (Pointwise Difference of Sequences)

Definition 34.3.13 (Pointwise Difference of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise difference** of (x_n) and (y_n) is the real sequence $(x_n - y_n)$ defined by

$$(x - y)_n := x_n - y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement)*.*

$$\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.}$$
$$\forall n \in \mathbb{N} \ \big((x - y)_n = x_n - y_n\big).$$

Remark (Interpretation)*.* The difference sequence subtracts corresponding terms. It is equivalently the sum of (x_n) with the pointwise negation of (y_n).

Remark (Dependencies)*.*

- Real Sequence
- Pointwise Sum of Sequences

Definition (Scalar Multiple of a Sequence)

Definition 34.3.14 (Scalar Multiple of a Sequence)**.** Let (x_n) be a real sequence, and let $\alpha \in \mathbb{R}$. The **scalar multiple** $\alpha(x_n)$ is the real sequence (αx_n) defined by

$$(\alpha x)_n := \alpha x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and } \alpha \in \mathbb{R}.\\&\forall n \in \mathbb{N}\ \big((\alpha x)_n = \alpha x_n\big).\end{aligned}$$

Remark (Interpretation). Scalar multiplication rescales every term by the same real number. It changes the vertical scale of the sequence but not its indexing.

Remark (Dependencies).

- Real Sequence

Definition (Linear Combination of Real Sequences)

Definition 34.3.15 (Linear Combination of Real Sequences)**.** Let (x_n) and (y_n) be real sequences, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha(x_n) + \beta(y_n)$ is the real sequence $(\alpha x_n + \beta y_n)$ defined by

$$(\alpha x + \beta y)_n := \alpha x_n + \beta y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n), (y_n) \text{ be real sequences and } \alpha, \beta \in \mathbb{R}.\\&\forall n \in \mathbb{N}\ \big((\alpha x + \beta y)_n = \alpha x_n + \beta y_n\big).\end{aligned}$$

Remark (Interpretation). A linear combination of sequences is built term by term from scalar multiples and sums. It is the common algebraic shape behind sequence limit rules and Cauchy closure under addition and scalar multiplication.

Remark (Dependencies).

- Scalar Multiple of a Sequence
- Pointwise Sum of Sequences

Definition (Pointwise Negation of a Sequence)

Definition 34.3.16 (Pointwise Negation of a Sequence)**.** Let (x_n) be a real sequence. The **pointwise negation** of (x_n) is the real sequence $(-x_n)$ defined by

$$(-x)_n := -x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\forall n \in \mathbb{N}\ \big((-x)_n = -x_n\big).\end{aligned}$$

Remark (Interpretation). Pointwise negation reflects the sequence across zero. It is the special scalar multiple corresponding to the scalar -1.

Remark (Dependencies).

- Scalar Multiple of a Sequence

Definition (Pointwise Product of Sequences)

Definition 34.3.17 (Pointwise Product of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise product** of (x_n) and (y_n) is the real sequence $(x_n y_n)$ defined by

$$(xy)_n := x_n y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.}$$
$$\forall n \in \mathbb{N} \ \big((xy)_n = x_n y_n\big).$$

Remark (Interpretation). The product sequence multiplies corresponding terms. The product law for limits is subtler than the sum law because one factor must be controlled while the other is approximated.

Remark (Dependencies).

- Real Sequence

Definition (Reciprocal Sequence)

Definition 34.3.18 (Reciprocal Sequence)**.** Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$. The **reciprocal sequence** of (x_n) is the real sequence $(1/x_n)$ defined by

$$\left(\frac{1}{x}\right)_n := \frac{1}{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\ (x_n \neq 0).$$
$$\forall n \in \mathbb{N} \ \left(\left(\frac{1}{x}\right)_n = \frac{1}{x_n}\right).$$

Remark (Interpretation). The reciprocal sequence is defined only when division by each term is legal. For limit theorems, a nonzero limit guarantees eventual nonzero behavior, but the sequence-level construction itself requires nonzero terms wherever it is defined.

Remark (Dependencies).

- Real Sequence

Definition (Pointwise Quotient of Sequences)

Definition 34.3.19 (Pointwise Quotient of Sequences)**.** Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$. The **pointwise quotient** of (x_n) by (y_n) is the real sequence (x_n/y_n) defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n), (y_n) \text{ be real sequences with } \forall n \in \mathbb{N}\ (y_n \neq 0).\\ &\forall n \in \mathbb{N}\ \left(\left(\frac{x}{y}\right)_n = \frac{x_n}{y_n}\right).\end{aligned}$$

Remark (Interpretation). The quotient sequence is the product of (x_n) with the reciprocal of (y_n). Its convergence law is therefore a product law plus a reciprocal law.

Remark (Dependencies).

- Pointwise Product of Sequences
- Reciprocal Sequence

Definition (Square Sequence)

Definition 34.3.20 (Square Sequence)**.** Let (x_n) be a real sequence. The **square sequence** of (x_n) is the real sequence (x_n^2) defined by

$$(x^2)_n := x_n^2 \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\ &\forall n \in \mathbb{N}\ \big((x^2)_n = x_n^2\big).\end{aligned}$$

Remark (Interpretation). The square sequence is the product of a sequence with itself. Its limit law is therefore a direct specialization of the product law.

Remark (Dependencies).

- Pointwise Product of Sequences

Definition (Absolute Value Sequence)

Definition 34.3.21 (Absolute Value Sequence)**.** Let (x_n) be a real sequence. The **absolute value sequence** of (x_n) is the real sequence $(|x_n|)$ defined by

$$(|x|)_n := |x_n| \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall n \in \mathbb{N} \; \big((|x|)_n = |x_n|\big).$$

Remark (Interpretation). The absolute value sequence records the magnitude of each term. It removes sign information while preserving distance from zero.

Remark (Dependencies).

- Real Sequence

Definition (Square Root Sequence)

Definition 34.3.22 (Square Root Sequence)**.** Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$. The **square root sequence** of (x_n) is the real sequence $(\sqrt{x_n})$ defined by

$$(\sqrt{x})_n := \sqrt{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N} \; (0 \leq x_n).$$
$$\forall n \in \mathbb{N} \; \big((\sqrt{x})_n = \sqrt{x_n}\big).$$

Remark (Interpretation). The square root sequence is defined term by term on the non-negative real terms. The nonnegativity hypothesis is part of the construction, not a proof artifact.

Remark (Dependencies).

- Real Sequence

Theorem (Limit of a Scalar Multiple)

Theorem 34.3.23 (Limit of a Scalar Multiple)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \to L$, then $\alpha x_n \to \alpha L$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L, \alpha \in \mathbb{R}.$$
$$\big[\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N \; (|x_n - L| < \varepsilon)\big]$$
$$\implies \qquad \big[\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N \; (|\alpha x_n - \alpha L| < \varepsilon)\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(\alpha x_n, \alpha L).$$

Remark (Interpretation). Multiplying every term by a fixed scalar multiplies the limiting value by the same scalar. The case $\alpha = 0$ collapses the sequence to the constant zero sequence.

Remark (Dependencies).

- Convergence of a Sequence
- Scalar Multiple of a Sequence

Theorem (Limit of a Sum)

Theorem 34.3.24 (Limit of a Sum). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n + y_n \to L + M$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}.$$
$$\big[x_n \to L \ \wedge \ y_n \to M\big] \implies \big[x_n + y_n \to L + M\big].$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \implies \text{ConvergentSequence}(x_n + y_n, L + M).$$

Remark (Interpretation). The sum law says that convergent sequences may be added before or after taking limits. The proof splits a target tolerance into two smaller tolerances, one for each summand.

Remark (Dependencies).

- Convergence of a Sequence
- Pointwise Sum of Sequences
- Triangle Inequality

Theorem (Limit of a Negation)

Theorem 34.3.25 (Limit of a Negation). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $-x_n \to -L$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[-x_n \to -L\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(-x_n, -L).$$

Remark (Interpretation). Negation is scalar multiplication by -1, so this theorem is the scalar multiple law in its most common special case.

Remark (Dependencies).

- Limit of a Scalar Multiple
- Pointwise Negation of a Sequence

Theorem (Limit of a Difference)

Theorem 34.3.26 (Limit of a Difference)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n - y_n \to L - M$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}.$$
$$\big[x_n \to L \;\wedge\; y_n \to M\big] \implies \big[x_n - y_n \to L - M\big].$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \implies \text{ConvergentSequence}(x_n - y_n, L - M).$$

Remark (Interpretation). The difference law follows by adding (x_n) to the negation of (y_n). It is therefore not a new analytic mechanism, but a useful named rule.

Remark (Dependencies).

- Limit of a Sum
- Limit of a Negation
- Pointwise Difference of Sequences

Theorem (Limit of a Product)

Theorem 34.3.27 (Limit of a Product)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n y_n \to LM$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}.$$
$$\big[x_n \to L \;\wedge\; y_n \to M\big] \implies \big[x_n y_n \to LM\big].$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \implies \text{ConvergentSequence}(x_n y_n, LM). \blacksquare$$

Remark (Interpretation). The product law uses both approximation and boundedness. A convergent sequence is eventually bounded, so one factor can be controlled while the other is made small.

Remark (Dependencies).

- Convergence of a Sequence
- Pointwise Product of Sequences
- Bounded Sequence

Theorem (Nonzero Limit is Eventually Nonzero)

Theorem 34.3.28 (Nonzero Limit is Eventually Nonzero)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}. \\ &\big[L \neq 0 \ \wedge \ x_n \to L\big] \implies \exists N \in \mathbb{N}\, \forall n \geq N\ (x_n \neq 0). \end{aligned}$$

Remark (Interpretation). A sequence converging to a nonzero number eventually stays away from zero. This is the denominator-control result needed for reciprocal and quotient rules.

Remark (Dependencies).

- Convergence of a Sequence

Theorem (Limit of a Reciprocal)

Theorem 34.3.29 (Limit of a Reciprocal)**.** *Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then $1/x_n \to 1/L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\ (x_n \neq 0), \text{ and let } L \neq 0. \\ &\big[x_n \to L\big] \implies \Big[\frac{1}{x_n} \to \frac{1}{L}\Big]. \end{aligned}$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}\left(\frac{1}{x_n}, \frac{1}{L}\right).$$

Remark (Interpretation). Reciprocation is continuous away from zero. The nonzero limit supplies a tail on which the denominators stay uniformly separated from zero.

Remark (Dependencies).

- Reciprocal Sequence
- Nonzero Limit is Eventually Nonzero

Theorem (Limit of a Quotient)

Theorem 34.3.30 (Limit of a Quotient)**.** *Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \to L$ and $y_n \to M$, then $x_n/y_n \to L/M$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences with } \forall n \in \mathbb{N}\ (y_n \neq 0), \text{ and let } M \neq 0.$$
$$\big[x_n \to L \ \wedge\ y_n \to M\big] \implies \Big[\frac{x_n}{y_n} \to \frac{L}{M}\Big].$$

Remark (Predicate reading).

$$\big(\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{ConvergentSequence}(y_n, M)\big) \implies \mathrm{ConvergentSequence}\left(\frac{x_n}{y_n}, \frac{L}{M}\right).$$

Remark (Interpretation). The quotient law is the product law applied to (x_n) and the reciprocal of (y_n). The condition $M \neq 0$ prevents the limiting denominator from collapsing.

Remark (Dependencies).

- Pointwise Quotient of Sequences
- Limit of a Product
- Limit of a Reciprocal

Theorem (Limit of a Square)

Theorem 34.3.31 (Limit of a Square)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $x_n^2 \to L^2$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[x_n^2 \to L^2\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(x_n^2, L^2).$$

Remark (Interpretation). Squaring preserves limits because it is multiplication of a sequence by itself.

Remark (Dependencies).

- Square Sequence
- Limit of a Product

Theorem (Limit of an Absolute Value)

Theorem 34.3.32 (Limit of an Absolute Value)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $|x_n| \to |L|$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[|x_n| \to |L|\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(|x_n|, |L|).$$

Remark (Interpretation). Taking absolute values cannot increase distance by more than the original distance: the magnitude function is continuous on $\mathbb{R}$.

Remark (Dependencies).

- Absolute Value Sequence
- Triangle Inequality

Theorem (Positive Limit is Eventually Positive)

Theorem 34.3.33 (Positive Limit is Eventually Positive)**.** *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L > 0.$$
$$\big[x_n \to L\big] \implies \exists N \in \mathbb{N}\ \forall n \geq N\ (0 < x_n).$$

Remark (Interpretation). Convergence to a positive limit eventually traps the sequence inside the positive half-line. This result is the sign-control counterpart of the eventual nonzero theorem.

Remark (Dependencies).

- Convergence of a Sequence

Theorem (Limit of a Square Root)

Theorem 34.3.34 (Limit of a Square Root)**.** *Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \to L$, then $0 \leq L$ and $\sqrt{x_n} \to \sqrt{L}$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\ (0 \leq x_n), \text{ and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[0 \leq L \ \wedge\ \sqrt{x_n} \to \sqrt{L}\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(\sqrt{x_n}, \sqrt{L}).$$

Remark (Interpretation). The square-root function is continuous on the nonnegative real line. Near zero the proof uses the order relation; away from zero it can be rationalized by multiplying by the conjugate.

Remark (Dependencies).

- Square Root Sequence
- Convergence of a Sequence

Theorem (Polynomial Sequence Limit)

Theorem 34.3.35 (Polynomial Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \to L$, then $p(x_n) \to p(L)$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence, } L \in \mathbb{R}, \text{ and } p \in \mathbb{R}[t].$$
$$\big[x_n \to L\big] \implies \big[p(x_n) \to p(L)\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(p(x_n), p(L)).$$

Remark (Interpretation). Polynomial limits are built from finitely many additions, scalar multiples, and products. The theorem packages those algebra laws into one reusable composition principle.

Remark (Dependencies).

- Limit of a Sum
- Limit of a Scalar Multiple
- Limit of a Product
- Limit of a Square

Theorem (Rational Sequence Limit)

Theorem 34.3.36 (Rational Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \to L$, then*

$$\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence, } L \in \mathbb{R}, \text{ and } p, q \in \mathbb{R}[t].\\ &\big[q(L) \neq 0 \wedge \forall n \in \mathbb{N}\ (q(x_n) \neq 0) \wedge x_n \to L\big]\\ &\qquad \Longrightarrow \qquad \left[\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}\right].\end{aligned}$$

Remark (Predicate reading)*.*

$$\text{ConvergentSequence}(x_n, L) \Longrightarrow \text{ConvergentSequence}\left(\frac{p(x_n)}{q(x_n)}, \frac{p(L)}{q(L)}\right).$$

Remark (Interpretation)*.* A rational expression is a quotient of two polynomial expressions. The condition $q(L) \neq 0$ keeps the limiting denominator legitimate, and the pointwise condition $q(x_n) \neq 0$ keeps every term of the quotient sequence defined.

Remark (Dependencies)*.*

- Polynomial Sequence Limit
- Limit of a Quotient
- Nonzero Limit is Eventually Nonzero

34.4 Tails

Definition (M-Tail of a Sequence)

Definition 34.4.1 (M-Tail of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $M \in \mathbb{N}$. The **M-tail** of (x_n) is the sequence

$$(x_n)_{n \geq M} \;:=\; \big(x_M,\, x_{M+1},\, x_{M+2},\, \ldots\big).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall M \in \mathbb{N}\ \Big((x_n)_{n \geq M} = (x_M, x_{M+1}, x_{M+2}, \ldots) \Big).$$

Remark (Definition predicate reading).

$$\text{TailSequence}(x_n, M, y_k)\ :=\ \forall k \in \mathbb{N}\ \big(y_k = x_{M+k-1} \big).$$

Remark (Interpretation). The M-tail discards the first $M-1$ terms and retains everything from index M onward. It is the formal object underlying the phrase *all but finitely many terms*: to say that all but finitely many terms of (x_n) satisfy a property P is precisely to say that some M-tail of (x_n) satisfies P termwise. The index M is an offset into the sequence, not a bound on the terms themselves; two sequences with identical M-tails share all asymptotic properties, differing only in finitely many initial values which convergence arguments discard entirely.

Remark (Dependencies).

- Real Sequence

Theorem (Convergence of a Tail)

Theorem 34.4.2 (Convergence of a Tail). *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m-tail $(x_{m+n})_{n \in \mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n \to \infty} x_{m+n}\ =\ \lim_{n \to \infty} x_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R}\ \forall m \in \mathbb{N} \\ &\quad \Big(\big(x_{m+n}\big)_{n \in \mathbb{N}} \text{ converges} \iff (x_n) \text{ converges} \Big) \\ &\quad \wedge\ \Big((x_n) \text{ converges} \implies \lim_{n \to \infty} x_{m+n} = \lim_{n \to \infty} x_n \Big). \end{aligned}$$

Remark (Theorem predicate reading).

$$\text{TailConvergenceInvariant}(x_n, m)\ :=\ \Big(\exists L \in \mathbb{R}\ \text{ConvergentSequence}(x_n, L) \Big) \iff \Big(\exists L \in \mathbb{R}\ \text{ConvergentSeq}$$

Remark (Contrapositive quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R}\ \forall m \in \mathbb{N} \\ &\quad \Big(\big(x_{m+n}\big)_{n \in \mathbb{N}} \text{ diverges} \iff (x_n) \text{ diverges} \Big). \end{aligned}$$

Remark (Contrapositive predicate reading).

$$\neg\, \text{TailConvergenceInvariant}(x_n, m)\ :=\ \big((x_n) \text{ diverges} \iff (x_{m+n})_{n \in \mathbb{N}} \text{ diverges} \big).$$

Remark (Interpretation). Convergence is a *tail property*: it depends only on the eventual behavior of the sequence, not on any finite initial segment. Discarding the first m terms neither creates nor destroys convergence, and when the limit exists it is preserved exactly. This theorem formalizes the informal principle that finitely many terms are irrelevant to asymptotic behavior. It is the rigorous justification for the common proof move of assuming without loss of generality that n is large enough for some auxiliary condition to hold: one simply passes to a suitable m-tail, applies the argument there, and concludes for the full sequence by this theorem. The contrapositive is equally useful: to show (x_n) diverges it suffices to show that some m-tail diverges.

Remark (Dependencies).

- Convergence of a Sequence
- M-Tail of a Sequence
- Uniqueness of Limits

Theorem (Convergence by Domination)

Theorem 34.4.3 (Convergence by Domination)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim_{n\to\infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \;\le\; c\, a_n \qquad \forall n \ge m,$$

then $\lim_{n\to\infty} x_n = L$.
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\; \forall L \in \mathbb{R}\; \forall (a_n) \subseteq (0,\infty)$$
$$\Big(\Big(\lim_{n\to\infty} a_n = 0 \;\wedge\; \exists c > 0\; \exists m \in \mathbb{N}\; \forall n \ge m\; \big(|x_n - L| \le c\, a_n\big)\Big) \implies \lim_{n\to\infty} x_n = L\Big).$$

Remark (Contrapositive quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\; \forall L \in \mathbb{R}\; \forall (a_n) \subseteq (0,\infty)$$
$$\Big(\lim_{n\to\infty} x_n \ne L \implies \lim_{n\to\infty} a_n \ne 0 \;\vee\; \forall c > 0\; \forall m \in \mathbb{N}\; \exists n \ge m\; \big(|x_n - L| > c\, a_n\big)\Big).$$

Remark (Contrapositive predicate reading).

$$\mathrm{NotDominatedByNullError}(x_n, L, a_n) \;:=\; \lim_{n\to\infty} a_n \ne 0 \vee \forall c > 0\, \forall m \in \mathbb{N}\, \exists n \ge m \big(|x_n - L| > c\, a_n\big).$$

Remark (Interpretation). The theorem is a *domination principle*: it is not necessary to verify the ε-N condition directly for (x_n). It suffices to find a null sequence (a_n) – one converging to zero – that dominates the error $|x_n - L|$ on some tail, up to a fixed scalar c. The scalar c

is irrelevant to the conclusion: since $a_n \to 0$, the product $c\, a_n \to 0$ regardless of the size of c, and any $\varepsilon > 0$ is eventually beaten by $c\, a_n$. This theorem is the rigorous engine behind the common proof pattern of bounding $|x_n - L|$ by a simpler expression and observing that the simpler expression tends to zero. The canonical instance is $a_n = 1/n$, where domination by c/n is sufficient to conclude convergence to L.

Remark (Dependencies).

- Convergence of a Sequence
- M-Tail of a Sequence

Theorem (Ratio Limit Less Than One Implies Null)

Theorem 34.4.4 (Ratio Limit Less Than One Implies Null)**.** *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n\to\infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n\to\infty} x_n = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq (0,\infty)\ \forall L \in \mathbb{R}$$
$$\Big(\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L \ \wedge\ L < 1\Big) \Longrightarrow \lim_{n\to\infty} x_n = 0.$$

Remark (Theorem predicate reading).

$$\big(\text{ConvergentSequence}\big(x_{n+1}/x_n, L\big) \wedge L < 1\big) \Longrightarrow \text{NullSequence}(x_n).$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq (0,\infty)\ \exists L \in \mathbb{R}$$
$$\Big(\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L \ \wedge\ L < 1 \ \wedge\ \lim_{n\to\infty} x_n \neq 0\Big).$$

Remark (Negation predicate reading).

$$\text{ConvergentSequence}\big(x_{n+1}/x_n, L\big) \wedge L < 1 \wedge \text{NotConvergentTo}(x_n, 0).$$

Remark (Failure modes). A failure would be a positive sequence whose consecutive ratios settle below one in the limit but whose terms do not tend to zero. The theorem says this cannot happen: once the ratio limit is less than one, some tail has ratios bounded by a fixed number $r < 1$, forcing geometric decay.

Remark (Failure mode decomposition).

$$\underbrace{(x_n) \subseteq (0,\infty)}_{\text{positive terms}} \quad \underbrace{\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L < 1}_{\text{eventual ratio gap}} \quad \underbrace{\lim_{n\to\infty} x_n \neq 0}_{\text{fails to be null}}.$$

The positive-term hypothesis makes each ratio meaningful. The strict gap $L < 1$ gives a tail on which $x_{n+1} \le r x_n$ for some $r < 1$, and repeated iteration then gives domination by a null geometric sequence.

Remark (Interpretation). This is the sequence form of the ratio test. The theorem does not merely say that the terms decrease eventually; it says they decrease at an eventually geometric rate. The strict inequality $L < 1$ is essential. When the ratio limit equals 1, the conclusion may hold, as with $1/n$, or fail, as with the constant sequence 1.

Remark (Dependencies).

- Convergence of a Sequence
- Null Sequence
- M-Tail of a Sequence
- Convergence by Domination

34.5 Monotonicity

Monotonicity of Sequences

Monotonicity is an order condition on successive terms of a sequence. It is independent of convergence by itself, but when combined with boundedness it becomes one of the fundamental convergence mechanisms in real analysis.

Definition (Increasing Sequence)

Definition 34.5.1 (Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **increasing** if

$$x_n \le x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is increasing} \iff \forall n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) :\equiv \forall n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not increasing} \iff \exists n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{MonotoneIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Interpretation). An increasing sequence may stay flat. The term "increasing" is therefore used here in the non-strict sense: each term is at least as large as the one before it.

Remark (Dependencies).

- Real Sequence

Definition (Decreasing Sequence)

Definition 34.5.2 (Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **decreasing** if

$$x_{n+1} \le x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is decreasing} \iff \forall n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Definition predicate reading).

$$\mathrm{MonotoneDecreasingSequence}(x_n) :\equiv \forall n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not decreasing} \iff \exists n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{MonotoneDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Interpretation). A decreasing sequence may stay flat. The order condition says the sequence never rises as the index advances.

Remark (Dependencies).

- Real Sequence

Definition (Strictly Increasing Sequence)

Definition 34.5.3 (Strictly Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly increasing** if

$$x_n < x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is strictly increasing} \iff \forall n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{StrictlyIncreasingSequence}(x_n) :\equiv \forall n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not strictly increasing} \iff \exists n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Negation predicate reading).

$$\neg\,\text{StrictlyIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Interpretation). Strict increase excludes equality between consecutive terms. Every strictly increasing sequence is increasing, but the converse need not hold.

Remark (Dependencies).

- Real Sequence
- Increasing Sequence

Definition (Strictly Decreasing Sequence)

Definition 34.5.4 (Strictly Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly decreasing** if

$$x_{n+1} < x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is strictly decreasing} \iff \forall n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Definition predicate reading).

$$\text{StrictlyDecreasingSequence}(x_n) :\equiv \forall n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not strictly decreasing} \iff \exists n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\text{StrictlyDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Interpretation). Strict decrease excludes equality between consecutive terms. Every strictly decreasing sequence is decreasing, but the converse need not hold.

Remark (Dependencies).

- Real Sequence

- Decreasing Sequence

Definition (Monotone Sequence)

Definition 34.5.5 (Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **monotone** if it is increasing or decreasing.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is monotone}\\ &\quad\iff \big[\forall n \in \mathbb{N}\ (x_n \leq x_{n+1})\big] \vee \big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big].\end{aligned}$$

Remark (Definition predicate reading).

$$\text{MonotoneSequence}(x_n) :\equiv \text{MonotoneIncreasingSequence}(x_n) \vee \text{MonotoneDecreasingSequence}(x_n).$$

Remark (Negated quantified statement).

$$\begin{aligned}&(x_n) \text{ is not monotone}\\ &\quad\iff \big[\exists n \in \mathbb{N}\ (x_{n+1} < x_n)\big] \wedge \big[\exists m \in \mathbb{N}\ (x_m < x_{m+1})\big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{MonotoneSequence}(x_n) \iff \neg\,\text{MonotoneIncreasingSequence}(x_n) \wedge \neg\,\text{MonotoneDecreasingSequence}(x_n).$$

Remark (Interpretation). Monotonicity means the sequence has one global direction. It may move upward, move downward, or stay constant, but it cannot reverse direction.

Remark (Dependencies).

- Increasing Sequence
- Decreasing Sequence

Definition (Eventually Increasing Sequence)

Definition 34.5.6 (Eventually Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually increasing** if there exists $N \in \mathbb{N}$ such that

$$x_n \leq x_{n+1} \qquad \text{for every } n \geq N.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is eventually increasing} \iff \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{EventuallyIncreasingSequence}(x_n) :\equiv \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not eventually increasing} \iff \forall N \in \mathbb{N}\, \exists n \geq N\ (x_{n+1} < x_n).$$

Remark (Negation predicate reading).

$$\neg\,\text{EventuallyIncreasingSequence}(x_n) \iff \forall N \in \mathbb{N}\, \exists n \geq N\ (x_{n+1} < x_n).$$

Remark (Interpretation). Eventual increase allows finitely many early reversals. Only the tail of the sequence must move in a nondecreasing direction.

Remark (Dependencies).

- M-Tail of a Sequence
- Increasing Sequence

Definition (Eventually Decreasing Sequence)

Definition 34.5.7 (Eventually Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually decreasing** if there exists $N \in \mathbb{N}$ such that

$$x_{n+1} \leq x_n \qquad \text{for every } n \geq N.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is eventually decreasing} \iff \exists N \in \mathbb{N}\, \forall n \geq N\ (x_{n+1} \leq x_n).$$

Remark (Definition predicate reading).

$$\text{EventuallyDecreasingSequence}(x_n) \coloneqq \exists N \in \mathbb{N}\, \forall n \geq N\ (x_{n+1} \leq x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not eventually decreasing} \iff \forall N \in \mathbb{N}\, \exists n \geq N\ (x_n < x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\text{EventuallyDecreasingSequence}(x_n) \iff \forall N \in \mathbb{N}\, \exists n \geq N\ (x_n < x_{n+1}).$$

Remark (Interpretation). Eventual decrease allows the initial terms to behave irregularly. The order condition begins only after some finite threshold.

Remark (Dependencies).

- M-Tail of a Sequence
- Decreasing Sequence

Definition (Eventually Monotone Sequence)

Definition 34.5.8 (Eventually Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually monotone** if it is eventually increasing or eventually decreasing.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is eventually monotone}\\ &\quad\iff \quad \big[\exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1})\big] \vee \big[\exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n)\big].\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{EventuallyMonotoneSequence}(x_n) :\!= \mathrm{EventuallyIncreasingSequence}(x_n) \vee \mathrm{EventuallyDecreasingSequence}($$

Remark (Negated quantified statement).

$$\begin{aligned}&(x_n) \text{ is not eventually monotone}\\ &\quad\iff \quad \big[\forall N \in \mathbb{N}\ \exists n \geq N\ (x_{n+1} < x_n)\big] \wedge \big[\forall N \in \mathbb{N}\ \exists n \geq N\ (x_n < x_{n+1})\big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{EventuallyMonotoneSequence}(x_n) \iff \neg\,\mathrm{EventuallyIncreasingSequence}(x_n) \wedge \neg\,\mathrm{EventuallyDecreasingSe}$$

Remark (Interpretation). Eventual monotonicity is the tail version of monotonicity. It is often the right hypothesis in applications, because finitely many early terms do not affect convergence.

Remark (Dependencies).

- Eventually Increasing Sequence
- Eventually Decreasing Sequence
- Convergence of a Tail

Theorem (Monotone Convergence Theorem)

Theorem 34.5.9 (Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and bounded above, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If (x_n) is decreasing and bounded below, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big] \Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ \Big(\lim_{n\to\infty} x_n = L \wedge L = \sup\{x_n : n \in \mathbb{N}\} \Big), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \wedge \big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big] \Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ \Big(\lim_{n\to\infty} x_n = L \wedge L = \inf\{x_n : n \in \mathbb{N}\} \Big).
\end{aligned}
$$

Remark (Theorem predicate reading).

$$
\begin{aligned}
&\mathrm{MonotoneIncreasingSequence}(x_n) \wedge \mathrm{BoundedAboveSequence}(x_n) \implies \mathrm{ConvergentSequence}\big(x_n, \sup\{x_n : n \\
&\mathrm{MonotoneDecreasingSequence}(x_n) \wedge \mathrm{BoundedBelowSequence}(x_n) \implies \mathrm{ConvergentSequence}\big(x_n, \inf\{x_n : n
\end{aligned}
$$

Remark (Negated quantified statement).

$$
\begin{aligned}
\exists (x_n) \subseteq \mathbb{R}\ \Big(&\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big] \\
&\wedge (x_n) \text{ does not converge} \Big), \\
\exists (x_n) \subseteq \mathbb{R}\ \Big(&\big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \wedge \big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big] \\
&\wedge (x_n) \text{ does not converge} \Big).
\end{aligned}
$$

Remark (Negation predicate reading).

$$
\begin{aligned}
&\mathrm{MonotoneIncreasingSequence}(x_n) \wedge \mathrm{BoundedAboveSequence}(x_n) \wedge \neg \exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L), \\
&\mathrm{MonotoneDecreasingSequence}(x_n) \wedge \mathrm{BoundedBelowSequence}(x_n) \wedge \neg \exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L).
\end{aligned}
$$

Remark (Failure modes). The theorem would fail only if a one-directional bounded sequence could avoid settling to a real limit. Completeness of $\mathbb{R}$ rules this out: an increasing bounded-above sequence is forced toward the least upper bound of its range, while a decreasing bounded-below sequence is forced toward the greatest lower bound of its range.

Remark (Failure mode decomposition).

$$
\underbrace{\forall n \in \mathbb{N}\ (x_n \le x_{n+1})}_{\text{one direction}} \quad \underbrace{\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)}_{\text{ceiling}} \quad \underbrace{(x_n) \text{ does not converge}}_{\text{forbidden by completeness}}.
$$

The decreasing case is dual: reverse the inequalities and replace the ceiling by a floor.

Remark (Interpretation). Boundedness prevents escape, and monotonicity prevents oscillation. Together they force convergence. The theorem is one of the standard forms in which the completeness of the real numbers becomes a sequence theorem.

Remark (Dependencies).

- Convergence of a Sequence
- Increasing Sequence
- Decreasing Sequence
- Sequence Bounded Above
- Sequence Bounded Below
- Supremum
- Infimum

Theorem (Strict Monotonicity Implies Monotonicity)

Theorem 34.5.10 (Strict Monotonicity Implies Monotonicity)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is strictly increasing, then (x_n) is increasing.

(b) If (x_n) is strictly decreasing, then (x_n) is decreasing.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad\Big(\big[\forall n \in \mathbb{N}\ (x_n < x_{n+1})\big] \implies \big[\forall n \in \mathbb{N}\ (x_n \leq x_{n+1})\big]\Big),\\ &\quad\Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} < x_n)\big] \implies \big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big]\Big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}&\text{StrictlyIncreasingSequence}(x_n) \implies \text{MonotoneIncreasingSequence}(x_n),\\ &\text{StrictlyDecreasingSequence}(x_n) \implies \text{MonotoneDecreasingSequence}(x_n).\end{aligned}$$

Remark (Interpretation). Strict monotonicity is stronger than non-strict monotonicity because every strict inequality is also a weak inequality.

Remark (Dependencies).

- Strictly Increasing Sequence
- Strictly Decreasing Sequence
- Increasing Sequence
- Decreasing Sequence

Theorem (Bounded Monotone Sequence Equivalences)

Theorem 34.5.11 (Bounded Monotone Sequence Equivalences)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.

(b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.

(c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \implies \Big(\exists L \in \mathbb{R}\ (x_n \to L) \iff \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\Big), \\
&\quad \big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \implies \Big(\exists L \in \mathbb{R}\ (x_n \to L) \iff \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\Big).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\mathrm{MonotoneIncreasingSequence}(x_n) \implies \Big(\exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{BoundedAboveSequence}$$

$$\mathrm{MonotoneDecreasingSequence}(x_n) \implies \Big(\exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{BoundedBelowSequence}$$

Remark (Interpretation). For monotone sequences, the only obstruction to real convergence is escape in the monotone direction. Boundedness removes that obstruction.

Remark (Dependencies).

- Monotone Convergence Theorem
- Bounded Sequence
- Convergence of a Sequence

Theorem (Eventually Monotone Convergence Theorem)

Theorem 34.5.12 (Eventually Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.

(b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.

(c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1}) \land \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M)\Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ (x_n \to L), \\
&\quad \Big(\exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n) \land \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n)\Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ (x_n \to L).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{EventuallyIncreasingSequence}(x_n) \land \text{BoundedAboveSequence}(x_n) \implies \exists L \in \mathbb{R}\ \text{ConvergentSequence}(x_n, L \\
&\text{EventuallyDecreasingSequence}(x_n) \land \text{BoundedBelowSequence}(x_n) \implies \exists L \in \mathbb{R}\ \text{ConvergentSequence}(x_n, L
\end{aligned}$$

Remark (Interpretation). This is the tail version of the Monotone Convergence Theorem. Once the tail is monotone and bounded in the right direction, finite initial behavior is irrelevant.

Remark (Dependencies).

- Eventually Increasing Sequence
- Eventually Decreasing Sequence
- Convergence of a Tail
- Monotone Convergence Theorem

Theorem (Unbounded Monotone Divergence)

Theorem 34.5.13 (Unbounded Monotone Divergence)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and is not bounded above, then $x_n \to +\infty$.

(b) If (x_n) is decreasing and is not bounded below, then $x_n \to -\infty$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \leq x_{n+1})\big] \land \big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (M < x_n)\big]\Big) \\
&\quad \implies \quad \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ (M < x_n), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big] \land \big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n < M)\big]\Big) \\
&\quad \implies \quad \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < M).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) \wedge \neg\,\text{BoundedAboveSequence}(x_n) \implies \text{DivergesToPositiveInfinity}(x_n),$$
$$\text{MonotoneDecreasingSequence}(x_n) \wedge \neg\,\text{BoundedBelowSequence}(x_n) \implies \text{DivergesToNegativeInfinity}(x_n).$$

Remark (Interpretation). For a monotone sequence, unboundedness has a direction. An increasing sequence that has no upper bound eventually exceeds every real number; the decreasing case is the order-dual statement.

Remark (Dependencies).

- Divergence to Positive Infinity
- Divergence to Negative Infinity
- Increasing Sequence
- Decreasing Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Algebraic Transformations Preserve Monotonicity)

Theorem 34.5.14 (Algebraic Transformations Preserve Monotonicity)**.** *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

(a) If (x_n) is increasing, then $(x_n + c)$ is increasing.

(b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.

(c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.

(d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.

(e) If (x_n) is increasing, then $(-x_n)$ is decreasing.

(f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall c, \alpha \in \mathbb{R} \\
&\quad \big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \implies \big[\forall n \in \mathbb{N}\ (x_n + c \le x_{n+1} + c)\big], \\
&\quad \big(\alpha > 0 \wedge \forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big) \implies \big[\forall n \in \mathbb{N}\ (\alpha x_n \le \alpha x_{n+1})\big], \\
&\quad \big(\alpha < 0 \wedge \forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big) \implies \big[\forall n \in \mathbb{N}\ (\alpha x_{n+1} \le \alpha x_n)\big].
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{MonotoneIncreasingSequence}(x_n) \implies \text{MonotoneIncreasingSequence}(x_n + c),\\
&\text{MonotoneIncreasingSequence}(x_n) \wedge \alpha > 0 \implies \text{MonotoneIncreasingSequence}(\alpha x_n),\\
&\text{MonotoneIncreasingSequence}(x_n) \wedge \alpha < 0 \implies \text{MonotoneDecreasingSequence}(\alpha x_n).
\end{aligned}$$

Remark (Interpretation). Translation preserves order, multiplication by a positive scalar preserves order, and multiplication by a negative scalar reverses order. Negation is the special case $\alpha = -1$.

Remark (Dependencies).

- Pointwise Sum of Sequences
- Scalar Multiple of a Sequence
- Pointwise Negation of a Sequence
- Increasing Sequence
- Decreasing Sequence

34.6 Subsequences

Subsequences

Subsequences record what remains after passing to an infinite ordered selection of indices. They are the basic tool for extracting structure from a sequence: limits, oscillation, boundedness, compactness, and failure of convergence are all detected by suitable subsequences.

Definition (Strictly Increasing Index Map)

Definition 34.6.1 (Strictly Increasing Index Map)**.** Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a function. The function σ is a **strictly increasing index map** if

$$k < \ell \implies \sigma(k) < \sigma(\ell) \qquad \text{for all } k, \ell \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\sigma : \mathbb{N} \to \mathbb{N} \text{ is strictly increasing} \iff \forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)).$$

Remark (Definition predicate reading).

$$\text{StrictlyIncreasing}(\sigma) := \forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)).$$

Remark (Negated quantified statement).

$$\sigma \text{ is not strictly increasing} \iff \exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \leq \sigma(k)).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{StrictlyIncreasing}(\sigma) \iff \exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \leq \sigma(k)).$$

Remark (Interpretation). The index map selects terms without repetition and without changing their order. This is the structural difference between a subsequence and an arbitrary rearrangement.

Remark (Dependencies).

- Function
- Natural Numbers
- Strict Order Successor Characterization

Definition (Subsequence)

Definition 34.6.2 (Subsequence)**.** Let (x_n) be a real sequence. A **subsequence** of (x_n) is a sequence $(x_{\sigma(k)})_{k\in\mathbb{N}}$, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a strictly increasing index map.

Remark (Standard quantified statement).

$$\begin{aligned} &(y_k) \text{ is a subsequence of } (x_n) \\ &\iff \quad \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N}\ (y_k = x_{\sigma(k)})\Big). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Subsequence}(y_k, x_n) :\equiv \exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N}\ (y_k = x_{\sigma(k)})\big).$$

Remark (Negated quantified statement).

$$\begin{aligned} &(y_k) \text{ is not a subsequence of } (x_n) \\ &\iff \quad \forall \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \leq \sigma(k)) \vee \exists k \in \mathbb{N}\ (y_k \neq x_{\sigma(k)})\Big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Subsequence}(y_k, x_n).$$

Remark (Interpretation). A subsequence keeps the original chronological order but may skip terms. The new index k counts the selected terms, while $\sigma(k)$ records where each selected term came from in the original sequence.

Remark (Dependencies).

- Real Sequence
- Strictly Increasing Index Map

Definition (Subsequential Limit)

Definition 34.6.3 (Subsequential Limit)**.** Let (x_n) be a real sequence and let $L \in \mathbb{R}$. The real number L is a **subsequential limit** of (x_n) if there exists a subsequence $(x_{\sigma(k)})$ such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned} &L \text{ is a subsequential limit of } (x_n) \\ \iff\quad &\exists \sigma : \mathbb{N} \to \mathbb{N} \left(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k\to\infty} x_{\sigma(k)} = L\right). \end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{SubsequentialLimit}(x_n, L) :\equiv \exists \sigma \left(\mathrm{StrictlyIncreasing}(\sigma) \wedge \mathrm{ConvergentSequence}(x_{\sigma(k)}, L)\right).$$

Remark (Interpretation)*.* Subsequential limits detect limiting behavior along an infinite selection of terms. A sequence may have no ordinary limit but still have one or more subsequential limits.

Remark (Dependencies)*.*

- Subsequence
- Convergence of a Sequence

Definition (Has Convergent Subsequence)

Definition 34.6.4 (Has Convergent Subsequence)**.** Let (x_n) be a real sequence. The sequence (x_n) **has a convergent subsequence** if there exist a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ and a real number L such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned} &(x_n) \text{ has a convergent subsequence} \\ \iff\quad &\exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R} \left(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k\to\infty} x_{\sigma(k)} = L\right). \end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{HasConvergentSubsequence}(x_n) :\equiv \exists L \in \mathbb{R}\ \mathrm{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation)*.* This property asks for at least one convergent infinite selection. It does not assert convergence of the whole sequence and does not require uniqueness of the subsequential limit.

Remark (Dependencies).

- Subsequential Limit

Theorem (Subsequence Indices Dominate the Identity)

Theorem 34.6.5 (Subsequence Indices Dominate the Identity). *Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a strictly increasing index map. Then*

$$k \leq \sigma(k) \qquad \text{for every } k \in \mathbb{N}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \sigma : \mathbb{N} \to \mathbb{N} \left(\left[\forall k, \ell \in \mathbb{N} \; (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \right] \Longrightarrow \left[\forall k \in \mathbb{N} \; (k \leq \sigma(k)) \right] \right).$$

Remark (Theorem predicate reading).

$$\text{StrictlyIncreasing}(\sigma) \Longrightarrow \forall k \in \mathbb{N} \; (k \leq \sigma(k)).$$

Remark (Interpretation). A subsequence cannot outrun the original indexing backwards. By the time the subsequence has selected its kth term, it must have reached at least the kth original position.

Remark (Dependencies).

- Strictly Increasing Index Map
- Well-Ordering Principle

Theorem (Subsequences Preserve Limits)

Theorem 34.6.6 (Subsequences Preserve Limits). *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then every subsequence of (x_n) converges to L.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \; \forall L \in \mathbb{R} \; \forall \sigma : \mathbb{N} \to \mathbb{N}$$
$$\left(x_n \to L \wedge \forall k, \ell \in \mathbb{N} \; (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \right) \Longrightarrow \lim_{k \to \infty} x_{\sigma(k)} = L.$$

Remark (Theorem predicate reading).

$$\text{ConvergentSequence}(x_n, L) \wedge \text{StrictlyIncreasing}(\sigma) \Longrightarrow \text{ConvergentSequence}(x_{\sigma(k)}, L).$$

Remark (Interpretation). Subsequences cannot escape a limit already possessed by the whole sequence. The domination of subsequence indices by the identity transfers every tail estimate for (x_n) to a tail estimate for $(x_{\sigma(k)})$.

Remark (Dependencies).

- Subsequence
- Convergence of a Sequence
- Subsequence Indices Dominate the Identity

Theorem (Subsequential Limit of a Convergent Sequence)

Theorem 34.6.7 (Subsequential Limit of a Convergent Sequence)**.** *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then L is the only possible subsequential limit of (x_n).*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, K \in \mathbb{R}$$
$$\Big(x_n \to L \wedge K \text{ is a subsequential limit of } (x_n)\Big) \Longrightarrow K = L.$$

Remark (Theorem predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K) \Longrightarrow K = L.$$

Remark (Interpretation). A convergent sequence has no hidden alternative limiting behavior. Every convergent subsequence inherits the same limit, and uniqueness of limits then forces equality.

Remark (Dependencies).

- Subsequences Preserve Limits
- Uniqueness of Limits

Theorem (Divergence by Two Subsequential Limits)

Theorem 34.6.8 (Divergence by Two Subsequential Limits)**.** *Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, K \in \mathbb{R}$$
$$\Big(L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n) \wedge L \neq K\Big)$$
$$\Longrightarrow \quad (x_n) \text{ does not converge.}$$

Remark (Theorem predicate reading).

$$\mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K) \wedge L \neq K \Longrightarrow \neg \exists A \in \mathbb{R}\ \mathrm{ConvergentSequence}(x_n, A).$$

Remark (Interpretation). Two different subsequential limits reveal persistent oscillation or splitting. If the whole sequence had a limit, every subsequence would have to inherit that same limit.

Remark (Dependencies).

- Subsequential Limit
- Subsequential Limit of a Convergent Sequence

Theorem (Boundedness Passes to Subsequences)

Theorem 34.6.9 (Boundedness Passes to Subsequences). *Every subsequence of a bounded real sequence is bounded.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma : \mathbb{N} \to \mathbb{N}\\&\quad \Big(\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M) \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\Big)\\&\quad \Longrightarrow \quad \exists M > 0\ \forall k \in \mathbb{N}\ (|x_{\sigma(k)}| \le M).\end{aligned}$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \wedge \mathrm{StrictlyIncreasing}(\sigma) \Longrightarrow \mathrm{BoundedSequence}(x_{\sigma(k)}).$$

Remark (Interpretation). A subsequence cannot create new values; it only selects from the original sequence. Any global bound for the original sequence is automatically a bound for the selection.

Remark (Dependencies).

- Bounded Sequence
- Subsequence

Theorem (Monotonicity Passes to Subsequences)

Theorem 34.6.10 (Monotonicity Passes to Subsequences). *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma : \mathbb{N} \to \mathbb{N}\\&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\big]\Big)\\&\quad \Longrightarrow \quad \forall k \in \mathbb{N}\ (x_{\sigma(k)} \le x_{\sigma(k+1)}).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{MonotoneIncreasingSequence}(x_{\sigma(k)}).$$

Remark (Interpretation). Selecting terms in the original order preserves the order pattern already present in the sequence. This is why the index map must be strictly increasing.

Remark (Dependencies).

- Increasing Sequence
- Decreasing Sequence
- Subsequence

Theorem (Subsequence of a Subsequence)

Theorem 34.6.11 (Subsequence of a Subsequence)**.** *Every subsequence of a subsequence of (x_n) is a subsequence of (x_n).*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma, \tau : \mathbb{N} \to \mathbb{N} \\ &\quad \Big(\big[\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell))\big] \wedge \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \tau(k) < \tau(\ell))\big] \Big) \\ &\quad \implies \quad (x_{\sigma(\tau(k))})_{k \in \mathbb{N}} \text{ is a subsequence of } (x_n). \end{aligned}$$

Remark (Theorem predicate reading).

$$\text{StrictlyIncreasing}(\sigma) \wedge \text{StrictlyIncreasing}(\tau) \implies \text{Subsequence}(x_{\sigma(\tau(k))}, x_n).$$

Remark (Interpretation). Passing to subsequences is closed under iteration. The composed index map $\sigma \circ \tau$ is still strictly increasing.

Remark (Dependencies).

- Subsequence
- Composition

Theorem (Eventual Properties Pass to Subsequences)

Theorem 34.6.12 (Eventual Properties Pass to Subsequences)**.** *Let $P(n)$ be a property of indices. If there exists $N \in \mathbb{N}$ such that $P(n)$ holds for every $n \geq N$, then for every strictly increasing index map σ there exists $K \in \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \geq K$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big(\exists N \in \mathbb{N}\, \forall n \geq N\ P(n) \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\Big)$$
$$\Longrightarrow \quad \exists K \in \mathbb{N}\, \forall k \geq K\ P(\sigma(k)).$$

Remark (Theorem predicate reading).

$$\text{Eventually}(x_n, P) \wedge \text{StrictlyIncreasing}(\sigma) \Longrightarrow \text{Eventually}(x_{\sigma(k)}, P).$$

Remark (Interpretation). Subsequences eventually enter every tail of the original sequence. Any property that holds on a tail of the original therefore holds on a tail of the subsequence.

Remark (Dependencies).

- Subsequence Indices Dominate the Identity
- Subsequence

Theorem (Frequent Properties Yield Subsequences)

Theorem 34.6.13 (Frequent Properties Yield Subsequences)**.** *Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \geq N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall N \in \mathbb{N}\, \exists n \geq N\ P(n)\Big] \Longrightarrow \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N}\ P(\sigma(k))\Big).$$

Remark (Theorem predicate reading).

$$\text{Frequently}(x_n, P) \Longrightarrow \exists \sigma\ \big(\text{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N}\ P(\sigma(k))\big).$$

Remark (Interpretation). An infinitely recurring property can be threaded into a subsequence. The construction chooses one later witness at each step.

Remark (Dependencies).

- Well-Ordering Principle
- Subsequence

Theorem (Subsequential Limits Respect Bounds)

Theorem 34.6.14 (Subsequential Limits Respect Bounds)**.** *Let (x_n) be a real sequence and let L be a subsequential limit of (x_n). If $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then $m \leq L \leq M$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, m, M \in \mathbb{R}$$
$$\Big(L \text{ is a subsequential limit of } (x_n) \wedge \forall n \in \mathbb{N}\ (m \leq x_n \leq M)\Big) \Longrightarrow m \leq L \leq M.$$

Remark (Theorem predicate reading).

$$\mathrm{SubsequentialLimit}(x_n, L) \wedge \forall n \in \mathbb{N}\ (m \leq x_n \leq M) \Longrightarrow m \leq L \leq M.$$

Remark (Interpretation). Bounds pass to subsequences, and limits preserve eventual inequalities. Hence subsequential limits remain inside any closed interval that contains all terms.

Remark (Dependencies).

- Boundedness Passes to Subsequences
- Constant Comparison for Sequence Limits

Theorem (Squeeze Passes to Subsequences)

Theorem 34.6.15 (Squeeze Passes to Subsequences)**.** *Let (a_n), (x_n), and (b_n) be real sequences. If $a_n \leq x_n \leq b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that*

$$a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)} \qquad \text{for every } k \geq K.$$

Go to proof.

Remark (Standard quantified statement).

$$\Big(\exists N \in \mathbb{N}\ \forall n \geq N\ (a_n \leq x_n \leq b_n) \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\Big)$$
$$\Longrightarrow \quad \exists K \in \mathbb{N}\ \forall k \geq K\ (a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)}).$$

Remark (Theorem predicate reading).

$$\mathrm{Eventually}(x_n,\ a_n \leq x_n \leq b_n) \wedge \mathrm{StrictlyIncreasing}(\sigma) \Longrightarrow \mathrm{Eventually}(x_{\sigma(k)},\ a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)}).$$

Remark (Interpretation). Squeeze hypotheses are tail hypotheses, so they survive passage to subsequences.

Remark (Dependencies).

- Eventual Properties Pass to Subsequences
- Sequence Squeeze Theorem

Theorem (Monotone Subsequence Theorem)

Theorem 34.6.16 (Monotone Subsequence Theorem)**.** *Every real sequence has a monotone subsequence.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \big[\forall k \in \mathbb{N}\ (x_{\sigma(k)} \leq x_{\sigma(k+1)}) \vee \forall k \in \mathbb{N}\ (x_{\sigma(k+1)} \leq x$$

Remark (Theorem predicate reading).

$$\exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \mathrm{MonotoneSequence}(x_{\sigma(k)})\big).$$

Remark (Interpretation). Every sequence contains an infinite ordered pattern. The proof uses the peak dichotomy: either there are infinitely many terms larger than all later terms, or there is an infinite rising chain.

Remark (Dependencies).

- Monotone Sequence
- Subsequence
- Well-Ordering Principle

Theorem (Bolzano-Weierstrass Theorem for Sequences)

Theorem 34.6.17 (Bolzano-Weierstrass Theorem for Sequences)**.** *Every bounded real sequence has a convergent subsequence.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M) \Longrightarrow \exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R}\ \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \lim_{k \to \infty}$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \Longrightarrow \mathrm{HasConvergentSubsequence}(x_n).$$

Remark (Interpretation). Bolzano-Weierstrass is the compactness principle for bounded real sequences. Boundedness prevents escape, and the monotone subsequence theorem supplies an ordered subsequence to which the Monotone Convergence Theorem applies.

Remark (Dependencies).

- Bounded Sequence
- Monotone Subsequence Theorem
- Boundedness Passes to Subsequences
- Bounded Monotone Sequence Equivalences

Theorem (Sequential Compactness of Closed Bounded Intervals)

Theorem 34.6.18 (Sequential Compactness of Closed Bounded Intervals)**.** *Let $a, b \in \mathbb{R}$ with $a \leq b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\forall a, b \in \mathbb{R}\ \forall (x_n) \subseteq \mathbb{R}\\ &\quad \Big(a \leq b \wedge \forall n \in \mathbb{N}\ (a \leq x_n \leq b)\Big)\\ &\quad \Longrightarrow \quad \exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \lim_{k \to \infty} x_{\sigma(k)} = L \wedge a \leq L \leq b\Big).\end{aligned}$$

Remark (Theorem predicate reading)*.*

$$\big[\forall n \in \mathbb{N}\ (a \leq x_n \leq b)\big] \Longrightarrow \exists L \in [a, b]\ \text{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation)*.* Closed bounded intervals are sequentially compact: a sequence trapped in the interval has a convergent subsequence, and closedness keeps the subsequential limit inside the interval.

Remark (Dependencies)*.*

- Bolzano-Weierstrass Theorem for Sequences
- Subsequential Limits Respect Bounds

34.7 Divergence

Divergence — Quick Reference

Concept	Meaning
Divergent sequence	Sequence that does not converge in $\mathbb{R}$
Diverges to $+\infty$	Eventually exceeds every real bound
Diverges to $-\infty$	Eventually lies below every real bound
Oscillatory sequence	Divergence by persistent oscillation
Infinite divergence	Divergence to infinity implies real divergence
Two subsequential limits	Distinct subsequential limits force divergence
Bounded divergence	Bounded divergence yields multiple subsequential limits

Divergence

Divergence is not a single behavior. A sequence can fail to converge because it escapes upward, escapes downward, oscillates between incompatible limiting patterns, or remains bounded while refusing to settle. The purpose of this section is to separate these failure modes and connect them to subsequences.

Definition (Divergent Sequence)

Definition 34.7.1 (Divergent Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **divergent** if there is no $L \in \mathbb{R}$ such that $x_n \to L$.

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is divergent} \\ &\iff \quad \forall L \in \mathbb{R}\, \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(|x_n - L| \geq \varepsilon\big). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{DivergentSequence}(x_n) :\equiv \forall L \in \mathbb{R}\, \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not divergent} \iff \exists L \in \mathbb{R}\, \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(|x_n - L| < \varepsilon\big).$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{DivergentSequence}(x_n) \iff \exists L \in \mathbb{R}\; \mathrm{ConvergentSequence}(x_n, L).$$

Remark (Interpretation)*.* Divergence is the formal negation of real convergence. It says that every candidate limit fails at some fixed tolerance on every tail.

Remark (Dependencies).

- Real Sequence
- Convergence of a Sequence

Definition (Divergence to Positive Infinity)

Definition 34.7.2 (Divergence to Positive Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $+\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Remark (Standard quantified statement).

$$x_n \to +\infty \iff \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToPositiveInfinity}(x_n) :\equiv \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Remark (Negated quantified statement).

$$x_n \not\to +\infty \iff \exists M \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(x_n \leq M\big).$$

Remark (Interpretation). Divergence to $+\infty$ means every horizontal threshold is eventually left below the sequence. It is not convergence to a real number; it is escape from the real line in the positive direction.

Remark (Dependencies).

- Real Sequence

Definition (Divergence to Negative Infinity)

Definition 34.7.3 (Divergence to Negative Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $-\infty$ if

$$\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n < M\big).$$

Remark (Standard quantified statement).

$$x_n \to -\infty \iff \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n < M\big).$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToNegativeInfinity}(x_n) :\equiv \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n < M\big).$$

Remark (Negated quantified statement).

$$x_n \not\to -\infty \iff \exists M \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(M \leq x_n\big).$$

Remark (Interpretation). Divergence to $-\infty$ is escape past every lower threshold. It is the order-dual of divergence to $+\infty$.

Remark (Dependencies).

- Real Sequence

Definition (Oscillatory Sequence)

Definition 34.7.4 (Oscillatory Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **oscillatory** if it is divergent, does not diverge to $+\infty$, and does not diverge to $-\infty$.

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is oscillatory} \\ \iff\quad &\Big[\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big)\Big] \\ &\wedge \Big[\exists M \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(x_n \leq M\big)\Big] \\ &\wedge \Big[\exists M \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(M \leq x_n\big)\Big]. \end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned}\mathrm{OscillatorySequence}(x_n) \coloneqq\ & \mathrm{DivergentSequence}(x_n)\\ & \wedge\ \neg\,\mathrm{DivergesToPositiveInfinity}(x_n)\\ & \wedge\ \neg\,\mathrm{DivergesToNegativeInfinity}(x_n).\end{aligned}$$

Remark (Interpretation). Oscillation means failure to settle without one-directional escape. The sequence may remain bounded, as $(-1)^n$ does, or it may be unbounded while still repeatedly returning in incompatible directions.

Remark (Dependencies).

- Divergent Sequence
- Divergence to Positive Infinity
- Divergence to Negative Infinity

Theorem (Divergence to Infinity Implies Real Divergence)

Theorem 34.7.5 (Divergence to Infinity Implies Real Divergence)**.** *Let (x_n) be a real sequence. If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) is divergent.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(M < x_n\big)\Big]\\ &\quad \vee \Big[\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n < M\big)\Big]\\ &\qquad \Longrightarrow \forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToPositiveInfinity}(x_n) \vee \mathrm{DivergesToNegativeInfinity}(x_n) \Longrightarrow \mathrm{DivergentSequence}(x_n).$$

Remark (Interpretation). Infinity divergence is a stronger, directional failure of real convergence. A sequence escaping past every real threshold cannot approach a finite real limit.

Remark (Dependencies).

- Divergent Sequence
- Divergence to Positive Infinity
- Divergence to Negative Infinity

Theorem (Two Subsequential Limits Force Divergence)

Theorem 34.7.6 (Two Subsequential Limits Force Divergence)**.** *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \neq K$, then (x_n) is divergent.*

Go to proof.

Remark (Standard quantified statement).

$$\exists L, K \in \mathbb{R} \Big(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)\Big)$$
$$\implies \forall A \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - A| \geq \varepsilon\big).$$

Remark (Definition predicate reading).

$$\exists L, K \in \mathbb{R}\ \big(L \neq K \wedge \mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K)\big)$$
$$\implies \mathrm{DivergentSequence}(x_n).$$

Remark (Interpretation). Convergence forces all subsequences to inherit the same limit. Two different subsequential limits therefore witness an irreparable conflict in the tail behavior of the original sequence.

Remark (Dependencies).

- Subsequential Limit
- Subsequences Preserve Limits
- Uniqueness of Limits
- Divergence by Two Subsequential Limits

Theorem (Unbounded Above Sequence Has a Positive-Infinity Subsequence)

Theorem 34.7.7 (Unbounded Above Sequence Has a Positive-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to +\infty$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ \big(M < x_n\big)\Big]$$
$$\implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall M \in \mathbb{R}\ \exists K \in \mathbb{N}\ \forall k \geq K\ \big(M < x_{\sigma(k)}\big)\Big).$$

Remark (Definition predicate reading).

$$\neg\,\mathrm{BoundedAboveSequence}(x_n) \implies \exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \mathrm{DivergesToPositiveInfinity}(x_{\sigma(k)})\big).$$

Remark (Interpretation). Unboundedness above may be sparse. This theorem says the high values can still be selected in increasing index order to form a subsequence that escapes to $+\infty$.

Remark (Dependencies).

- Sequence Bounded Above

- Subsequence
- Divergence to Positive Infinity

Theorem (Unbounded Below Sequence Has a Negative-Infinity Subsequence)

Theorem 34.7.8 (Unbounded Below Sequence Has a Negative-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to -\infty$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ \big(x_n < M\big)\Big]$$
$$\implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall M \in \mathbb{R}\ \exists K \in \mathbb{N}\ \forall k \geq K\ \big(x_{\sigma(k)} < M\big)\Big).$$

Remark (Definition predicate reading).

$$\neg\,\mathrm{BoundedBelowSequence}(x_n) \implies \exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \mathrm{DivergesToNegativeInfinity}(x_{\sigma(k)})\big).$$

Remark (Interpretation). Unboundedness below can be converted into a selected tail that escapes past every lower threshold. The proof is the order-reversed version of the positive-infinity subsequence construction.

Remark (Dependencies).

- Sequence Bounded Below
- Subsequence
- Divergence to Negative Infinity

Theorem (Bounded Divergence Produces Two Subsequential Limits)

Theorem 34.7.9 (Bounded Divergence Produces Two Subsequential Limits)**.** *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \neq K$ such that L and K are subsequential limits of (x_n).*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\exists M > 0\ \forall n \in \mathbb{N}\ \big(|x_n| \leq M\big)\Big] \wedge \Big[\forall A \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - A| \geq \varepsilon\big)\Big]$$
$$\implies \exists L, K \in \mathbb{R}\ \Big(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)\Big).$$

Remark (Definition predicate reading).

$$\mathrm{BoundedSequence}(x_n) \wedge \mathrm{DivergentSequence}(x_n) \implies \exists L, K \in \mathbb{R}\ \big(L \neq K \wedge \mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{Subse}$$

Remark (Interpretation). Bounded divergence cannot be explained by escape to infinity. In the real line, compactness forces convergent subsequences, and divergence forces at least two incompatible subsequential limits.

Remark (Dependencies).

- Bounded Sequence
- Divergent Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences
- Subsequences Preserve Limits

34.8 Liminf and Limsup

Liminf–Limsup — Quick Reference

Concept	Meaning
Tail supremum sequence	Supremum of each tail
Tail infimum sequence	Infimum of each tail
Limit superior	Limit of tail suprema
Limit inferior	Limit of tail infima
Liminf below limsup	Lower envelope never exceeds upper envelope
Convergence criterion	Convergence iff liminf equals limsup
Largest subsequential limit	Limsup is the top cluster value
Smallest subsequential limit	Liminf is the bottom cluster value
Oscillation criterion	Gap detects nonconvergence
Eventual-order squeeze	Squeeze estimates transfer to limsup and liminf

Liminf and Limsup

The limit superior and limit inferior measure the eventual upper and lower envelopes of a bounded sequence. Instead of asking whether all late terms settle to one number, they ask how high and how low the tails continue to reach. This makes them natural tools for detecting convergence, oscillation, and subsequential limits.

Definition (Tail Supremum Sequence)

Definition 34.8.1 (Tail Supremum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail supremum sequence** of (x_n) is the real sequence (s_n) defined by

$$s_n := \sup\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \ \big(s_n = \sup\{x_k : k \geq n\}\big).$$

Remark (Interpretation). The term s_n records the least ceiling of the n-th tail. As n grows, earlier terms are discarded, so the tail has fewer opportunities to reach large values.

Remark (Dependencies).

- Bounded Sequence
- Supremum
- M-Tail of a Sequence

Definition (Tail Infimum Sequence)

Definition 34.8.2 (Tail Infimum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail infimum sequence** of (x_n) is the real sequence (i_n) defined by

$$i_n := \inf\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \ \big(i_n = \inf\{x_k : k \geq n\}\big).$$

Remark (Interpretation). The term i_n records the greatest floor of the n-th tail. As earlier terms are discarded, the tail loses opportunities to reach small values.

Remark (Dependencies).

- Bounded Sequence
- Infimum
- M-Tail of a Sequence

Definition (Limit Superior of a Sequence)

Definition 34.8.3 (Limit Superior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. The **limit superior** of (x_n) is

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} s_n.$$

Remark (Standard quantified statement).

$$\limsup_{n\to\infty} x_n = L \iff \forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n \geq N \ \big(|s_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\mathrm{LimsupSequence}(x_n, L) := \mathrm{ConvergentSequence}(s_n, L).$$

Remark (Interpretation). The limit superior is the eventual upper envelope of the sequence. It is the number to which the ceilings of the tails descend.

Remark (Dependencies).

- Tail Supremum Sequence
- Convergence of a Sequence

Definition (Limit Inferior of a Sequence)

Definition 34.8.4 (Limit Inferior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. The **limit inferior** of (x_n) is

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} i_n.$$

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n = L \iff \forall\varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|i_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{LiminfSequence}(x_n, L) \coloneqq \text{ConvergentSequence}(i_n, L).$$

Remark (Interpretation). The limit inferior is the eventual lower envelope of the sequence. It is the number to which the floors of the tails ascend.

Remark (Dependencies).

- Tail Infimum Sequence
- Convergence of a Sequence

Theorem (Tail Suprema Are Decreasing)

Theorem 34.8.5 (Tail Suprema Are Decreasing)**.** *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*
Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N}\ \big(s_{n+1} \leq s_n\big).$$

Remark (Definition predicate reading).

$$\text{MonotoneDecreasingSequence}(s_n).$$

Remark (Interpretation). The $(n+1)$-st tail is contained in the n-th tail. Passing from a set to a smaller subset cannot raise its supremum.

Remark (Dependencies).

- Tail Supremum Sequence
- Supremum Is Monotone Under Inclusion

Theorem (Tail Infima Are Increasing)

Theorem 34.8.6 (Tail Infima Are Increasing)**.** *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*

Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(i_n \le i_{n+1}\right).$$

Remark (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(i_n).$$

Remark (Interpretation). The $(n+1)$-st tail is contained in the n-th tail. Passing from a set to a smaller subset cannot lower its infimum.

Remark (Dependencies).

- Tail Infimum Sequence
- Infimum

Theorem (Liminf Is Below Limsup)

Theorem 34.8.7 (Liminf Is Below Limsup)**.** *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n \le \limsup_{n\to\infty} x_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n \le \limsup_{n\to\infty} x_n.$$

Remark (Interpretation). Every tail infimum lies below every corresponding tail supremum. Passing to the limits preserves this order.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Limit Superior of a Sequence

- Constant Comparison for Sequence Limits

Theorem (Convergence iff Liminf Equals Limsup)

Theorem 34.8.8 (Convergence iff Liminf Equals Limsup)**.** *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \to L$ if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$x_n \to L \quad \iff \quad \liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Remark (Definition predicate reading)*.*

$$\text{ConvergentSequence}(x_n, L) \iff \big(\text{LiminfSequence}(x_n, L) \wedge \text{LimsupSequence}(x_n, L)\big).$$

Remark (Interpretation)*.* Convergence occurs exactly when the eventual lower envelope and eventual upper envelope collapse to the same number.

Remark (Dependencies)*.*

- Limit Inferior of a Sequence
- Limit Superior of a Sequence
- Convergence of a Sequence
- Sequence Squeeze Theorem

Theorem (Limsup Is the Largest Subsequential Limit)

Theorem 34.8.9 (Limsup Is the Largest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n\to\infty} x_n$. Then S is a subsequential limit of (x_n), and every subsequential limit of (x_n) is less than or equal to S.*
Go to proof.

Remark (Standard quantified statement)*.*

$$S = \limsup_{n\to\infty} x_n \implies \Big[S \text{ is a subsequential limit of } (x_n) \wedge \forall L \in \mathbb{R}\ \big(L \text{ is a subsequential limit of } (x_n) \implies L \le S\big)\Big].$$

Remark (Definition predicate reading)*.*

$$\text{LimsupSequence}(x_n, S) \implies \big(\text{SubsequentialLimit}(x_n, S) \wedge \forall L \in \mathbb{R}\ (\text{SubsequentialLimit}(x_n, L) \implies L \le S)\big).$$

Remark (Interpretation). The limit superior is not merely an upper bound on subsequential behavior. It is actually attained as a subsequential limit, and no subsequential limit lies above it.

Remark (Dependencies).

- Limit Superior of a Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences

Theorem (Liminf Is the Smallest Subsequential Limit)

Theorem 34.8.10 (Liminf Is the Smallest Subsequential Limit). *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n\to\infty} x_n$. Then I is a subsequential limit of (x_n), and every subsequential limit of (x_n) is greater than or equal to I.*

Go to proof.

Remark (Standard quantified statement).

$$I = \liminf_{n\to\infty} x_n \implies \Big[I \text{ is a subsequential limit of } (x_n) \land \forall L \in \mathbb{R}\ \big(L \text{ is a subsequential limit of } (x_n) \implies I \le L\big)\Big].$$

Remark (Definition predicate reading).

$$\mathrm{LiminfSequence}(x_n, I) \implies \big(\mathrm{SubsequentialLimit}(x_n, I) \land \forall L \in \mathbb{R}\ (\mathrm{SubsequentialLimit}(x_n, L) \implies I \le L)\big).$$

Remark (Interpretation). The limit inferior is attained as a subsequential limit and no subsequential limit lies below it.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences

Theorem (Oscillation Criterion via Liminf and Limsup)

Theorem 34.8.11 (Oscillation Criterion via Liminf and Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Go to proof.

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n \iff \exists L, K \in \mathbb{R} \left(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)\right).$$

Remark (Definition predicate reading).

$$\exists I, S \in \mathbb{R} \left(\mathrm{LiminfSequence}(x_n, I) \wedge \mathrm{LimsupSequence}(x_n, S) \wedge I < S\right) \iff \exists L, K \in \mathbb{R} \left(L \neq K \wedge \mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K)\right).$$

Remark (Interpretation). A strict gap between the lower and upper envelopes is exactly persistent oscillation in bounded sequences. The sequence keeps returning near at least two incompatible limiting levels.

Remark (Dependencies).

- Limsup Is the Largest Subsequential Limit
- Liminf Is the Smallest Subsequential Limit
- Two Subsequential Limits Force Divergence

Theorem (Limsup Comparison Under Eventual Order)

Theorem 34.8.12 (Limsup Comparison Under Eventual Order)**.** *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\exists N \in \mathbb{N}\, \forall n \geq N\ (x_n \leq y_n) \implies \limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

Remark (Interpretation). Eventual order of sequences passes to eventual order of their upper tail envelopes. Taking the limiting ceiling preserves that comparison.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Preserves Eventual Order

Theorem (Liminf Comparison Under Eventual Order)

Theorem 34.8.13 (Liminf Comparison Under Eventual Order)**.** *Let* (x_n) *and* (y_n) *be bounded real sequences. If there exists* $N \in \mathbb{N}$ *such that* $x_n \le y_n$ *for every* $n \ge N$*, then*

$$\liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\exists N \in \mathbb{N}\ \forall n \ge N\ (x_n \le y_n) \\ &\implies \quad \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n. \end{aligned}$$

Remark (Interpretation). Eventual order also passes to the lower tail envelopes. If one sequence is eventually below another, its eventual floor cannot sit above the other's eventual floor.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Limit Preserves Eventual Order

Theorem (Limsup Squeeze Under Eventual Order)

Theorem 34.8.14 (Limsup Squeeze Under Eventual Order)**.** *Let* (a_n)*,* (x_n)*, and* (b_n) *be bounded real sequences. If there exists* $N \in \mathbb{N}$ *such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\exists N \in \mathbb{N}\ \forall n \ge N\ (a_n \le x_n \le b_n) \\ &\implies \quad \limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n. \end{aligned}$$

Remark (Interpretation). If one sequence is eventually trapped between two others, then its eventual upper envelope is trapped between their eventual upper envelopes.

Remark (Dependencies).

- Limsup Comparison Under Eventual Order

Theorem (Liminf Squeeze Under Eventual Order)

Theorem 34.8.15 (Liminf Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\exists N \in \mathbb{N}\ \forall n \ge N\ (a_n \le x_n \le b_n)$$
$$\implies \quad \liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Remark (Interpretation). The lower-envelope version of the squeeze principle says that eventual two-sided order traps the eventual floors as well as the eventual ceilings.

Remark (Dependencies).

- Liminf Comparison Under Eventual Order

Remark (Illustrative cases). For the alternating sequence $x_n = (-1)^n$, the tail suprema are constantly 1 and the tail infima are constantly -1, so

$$\limsup_{n\to\infty}(-1)^n = 1, \qquad \liminf_{n\to\infty}(-1)^n = -1.$$

For any convergent sequence $x_n \to L$, both envelopes collapse to L.

34.9 Order-Theoretic Limit Constructions

Order-Theoretic Limits — Quick Reference

Statement	Role
Increasing sequence limit	Limit realized as a supremum
Decreasing sequence limit	Limit realized as an infimum
Tail extremals converge	Tail suprema and infima are monotone
Bounded sequence limsup/liminf	Bounded sequences have asymptotic envelopes

Order-Theoretic Limit Constructions

Completeness lets order data produce limits. An increasing sequence bounded above converges to the least ceiling of its range; a decreasing sequence bounded below converges to the greatest floor of its range. Applying these same ideas to tail suprema and tail infima produces limsup and liminf for every bounded sequence.

Theorem (Increasing Sequence Limit as Supremum)

Theorem 34.9.1 (Increasing Sequence Limit as Supremum)**.** *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left(\left[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\right] \wedge \left[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\right] \right.$$
$$\left. \implies \lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\} \right).$$

Remark (Theorem predicate reading).

$$\mathrm{MonotoneIncreasingSequence}(x_n) \wedge \mathrm{BoundedAboveSequence}(x_n) \implies \mathrm{ConvergentSequence}\big(x_n, \sup\{x_n : n \in$$

Remark (Interpretation). The supremum is the only possible destination: all terms lie below it, and the defining approximation property of the supremum forces the sequence eventually above every lower tolerance.

Remark (Dependencies).

- Increasing Sequence
- Sequence Bounded Above
- Supremum
- Monotone Convergence Theorem

Theorem (Decreasing Sequence Limit as Infimum)

Theorem 34.9.2 (Decreasing Sequence Limit as Infimum)**.** *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left(\left[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\right] \wedge \left[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\right] \right.$$
$$\left. \implies \lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\} \right).$$

Remark (Theorem predicate reading).

$$\text{MonotoneDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \Longrightarrow \text{ConvergentSequence}\big(x_n, \inf\{x_n : n \in$$

Remark (Interpretation). The infimum is the decreasing analogue of the supremum target: the sequence falls toward its greatest lower barrier.

Remark (Dependencies).

- Decreasing Sequence
- Sequence Bounded Below
- Infimum
- Monotone Convergence Theorem

Theorem (Tail Suprema and Infima Converge)

Theorem 34.9.3 (Tail Suprema and Infima Converge)**.** *Let (x_n) be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \qquad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) and (i_n) converge.

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \ \Big((x_n) \text{ is bounded} \Longrightarrow \\ \exists S, I \in \mathbb{R} \left[\lim_{n\to\infty} \sup\{x_k : k \geq n\} = S \wedge \lim_{n\to\infty} \inf\{x_k : k \geq n\} = I \right] \Big).$$

Remark (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \Longrightarrow \exists S, I \in \mathbb{R}\ \big(\text{ConvergentSequence}(\sup\{x_k : k \geq n\}, S) \wedge \text{ConvergentSequence}(\inf\{$$

Remark (Interpretation). Tail suprema decrease because later tails contain fewer terms; tail infima increase for the same reason. Boundedness supplies the opposite bound needed for monotone convergence.

Remark (Dependencies).

- Tail Supremum Sequence
- Tail Infimum Sequence
- Tail Suprema Are Decreasing
- Tail Infima Are Increasing

- Monotone Convergence Theorem

Theorem (Bounded Sequence Has Limsup and Liminf)

Theorem 34.9.4 (Bounded Sequence Has Limsup and Liminf)**.** *Every bounded real sequence has a limit superior and a limit inferior.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left((x_n) \text{ is bounded} \implies \exists S, I \in \mathbb{R} \left(S = \limsup_{n\to\infty} x_n \wedge I = \liminf_{n\to\infty} x_n \right) \right).$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \implies \exists S, I \in \mathbb{R} \left(\mathrm{LimsupSequence}(x_n, S) \wedge \mathrm{LiminfSequence}(x_n, I) \right).$$

Remark (Interpretation). For bounded sequences, limsup and liminf are not extra assumptions. They are guaranteed order-theoretic limits of the tail envelopes.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Inferior of a Sequence
- Tail Suprema and Infima Converge

34.10 Cluster Values of Sequences

Cluster Values — Quick Reference

Concept	Meaning
Cluster value	Value approached along a subsequence
Cluster equals subsequential limit	Converts frequent approach to subsequences
Bounded sequences have cluster values	Compactness produces accumulation
Extremal cluster values	Limsup and liminf are the outer cluster values

Cluster Values of Sequences

A convergent sequence has only one limiting value. A bounded divergent sequence may still have values that it approaches along selected subsequences. Cluster values name this repeated limiting behavior without requiring the entire sequence to converge.

Definition (Cluster Value of a Sequence)

Definition 34.10.1 (Cluster Value of a Sequence)**.** Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The number L is a **cluster value** of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that

$$|x_n - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$L \text{ is a cluster value of } (x_n) \iff \forall \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{SequenceClusterValue}(x_n, L) :\equiv \forall \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$L \text{ is not a cluster value of } (x_n) \iff \exists \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Interpretation). No matter how far out in the sequence one looks, some later term returns inside every neighborhood of L.

Remark (Dependencies).

- Real Sequence

Theorem (Cluster Values Are Subsequential Limits)

Theorem 34.10.2 (Cluster Values Are Subsequential Limits)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n).*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \Big(L \text{ is a cluster value of } (x_n) \iff L \text{ is a subsequential limit of } (x_n)\Big).$$

Remark (Theorem predicate reading).

$$\text{SequenceClusterValue}(x_n, L) \iff \text{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation). The neighborhood formulation and the subsequence formulation describe the same phenomenon: infinitely many opportunities to get arbitrarily close to L.

Remark (Dependencies).

- Cluster Value of a Sequence
- Subsequential Limit

- Frequent Properties Yield Subsequences

Theorem (Bounded Sequences Have Cluster Values)

Theorem 34.10.3 (Bounded Sequences Have Cluster Values)**.** *Every bounded real sequence has at least one cluster value.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left(\left[\exists M > 0 \, \forall n \in \mathbb{N} \, (|x_n| \leq M) \right] \Longrightarrow \exists L \in \mathbb{R} \, \big(L \text{ is a cluster value of } (x_n)\big) \right).$$

Remark (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \Longrightarrow \exists L \in \mathbb{R} \;\; \text{SequenceClusterValue}(x_n, L).$$

Remark (Interpretation). Boundedness traps the sequence in a finite interval, and Bolzano-Weierstrass extracts a convergent subsequence from that trap.

Remark (Dependencies).

- Bounded Sequence
- Bolzano-Weierstrass Theorem for Sequences
- Cluster Values Are Subsequential Limits

Theorem (Limsup and Liminf Are Extremal Cluster Values)

Theorem 34.10.4 (Limsup and Liminf Are Extremal Cluster Values)**.** *Let* (x_n) *be a bounded real sequence. Then* $\limsup_{n\to\infty} x_n$ *is the largest cluster value of* (x_n)*, and* $\liminf_{n\to\infty} x_n$ *is the smallest cluster value of* (x_n)*.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R} \Big((x_n) \text{ is bounded} \Longrightarrow \\ &\quad \Big[\limsup_{n\to\infty} x_n \text{ is a cluster value of } (x_n) \wedge \forall L \in \mathbb{R} \, \big(L \text{ is a cluster value of } (x_n) \Longrightarrow L \leq \limsup_{n\to\infty} x_n\big)\Big] \\ &\quad \wedge \Big[\liminf_{n\to\infty} x_n \text{ is a cluster value of } (x_n) \wedge \forall L \in \mathbb{R} \, \big(L \text{ is a cluster value of } (x_n) \Longrightarrow \liminf_{n\to\infty} x_n \leq L\big)\Big]\Big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}&\text{BoundedSequence}(x_n) \Longrightarrow \\ &\quad \text{SequenceClusterValue}\big(x_n, \limsup_{n\to\infty} x_n\big) \wedge \forall L \in \mathbb{R} \, \big(\text{SequenceClusterValue}(x_n, L) \Longrightarrow L \leq \limsup_{n\to\infty} x_n\big), \\ &\text{BoundedSequence}(x_n) \Longrightarrow \\ &\quad \text{SequenceClusterValue}\big(x_n, \liminf_{n\to\infty} x_n\big) \wedge \forall L \in \mathbb{R} \, \big(\text{SequenceClusterValue}(x_n, L) \Longrightarrow \liminf_{n\to\infty} x_n \leq L\big).\end{aligned}$$

Remark (Interpretation). Limsup and liminf describe the top and bottom edges of the sequence's subsequential behavior.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Inferior of a Sequence
- Limsup Is the Largest Subsequential Limit
- Liminf Is the Smallest Subsequential Limit
- Cluster Values Are Subsequential Limits

34.11 Cauchy

Cauchy Sequences

Cauchy sequences measure internal stabilization. Instead of comparing the terms to an already named limit, the Cauchy condition compares sufficiently late terms to each other. Over the real numbers this internal criterion is equivalent to convergence, and that equivalence is one of the central sequence forms of completeness.

Definition (Cauchy Sequence)

Definition 34.11.1 (Cauchy Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **Cauchy sequence** if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is Cauchy} \\ &\quad \iff \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ \big(|x_m - x_n| < \varepsilon\big). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) :\equiv \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned} &(x_n) \text{ is not Cauchy} \\ &\quad \iff \quad \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists m, n \geq N\ \big(|x_m - x_n| \geq \varepsilon\big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{CauchySequence}(x_n) \iff \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists m, n \geq N\ \big(|x_m - x_n| \geq \varepsilon\big).$$

Remark (Interpretation). The Cauchy condition says that the tails of the sequence eventually have arbitrarily small internal spread. It does not name a candidate limit; it only asserts that late terms become mutually close.

Remark (Dependencies).

- Real Sequence

Theorem (Convergent Sequences Are Cauchy)

Theorem 34.11.2 (Convergent Sequences Are Cauchy)**.** *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad \Big(\big(\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon) \big)\\ &\qquad \Longrightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon) \Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists L \in \mathbb{R}\ \operatorname{ConvergentSequence}(x_n, L) \Longrightarrow \operatorname{CauchySequence}(x_n) \Big).$$

Remark (Interpretation). Once every sufficiently late term is close to the same limit, any two sufficiently late terms are close to each other. This theorem is the easy half of the Cauchy criterion.

Remark (Dependencies).

- Convergence of a Sequence
- Cauchy Sequence

Theorem (Cauchy Sequences Are Bounded)

Theorem 34.11.3 (Cauchy Sequences Are Bounded)**.** *Every Cauchy real sequence is bounded.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad \Big(\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\\ &\qquad \Longrightarrow \exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M) \Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R} \left(\mathrm{CauchySequence}(x_n) \implies \mathrm{BoundedSequence}(x_n)\right).$$

Remark (Interpretation). A Cauchy sequence has one tightly controlled tail. The finitely many terms before that tail can be absorbed into a single larger bound.

Remark (Dependencies).

- Cauchy Sequence
- Bounded Sequence

Theorem (Cauchy Sequence with a Convergent Subsequence)

Theorem 34.11.4 (Cauchy Sequence with a Convergent Subsequence)**.** *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R} \\
&\quad \Big(\left[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\right] \\
&\qquad \wedge \left[\exists \sigma : \mathbb{N} \to \mathbb{N}\ \left(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall k \geq K\ (|x_{\sigma(k)} - L| < \varepsilon)\right)\right] \\
&\qquad \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\Big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R} \\
&\quad \Big(\mathrm{CauchySequence}(x_n) \wedge \mathrm{SubsequentialLimit}(x_n, L) \implies \mathrm{ConvergentSequence}(x_n, L)\Big).
\end{aligned}$$

Remark (Interpretation). The Cauchy condition prevents the rest of the tail from wandering away from any convergent subsequence. A single subsequential limit therefore determines the limit of the whole sequence.

Remark (Dependencies).

- Cauchy Sequence
- Subsequence
- Subsequential Limit
- Convergence of a Sequence

Theorem (Cauchy Criterion for Real Sequences)

Theorem 34.11.5 (Cauchy Criterion for Real Sequences)**.** *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad\Big(\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\\ &\qquad \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists L \in \mathbb{R}\ \text{ConvergentSequence}(x_n, L) \iff \text{CauchySequence}(x_n)\Big).$$

Remark (Interpretation). This theorem is the sequence form of completeness of $\mathbb{R}$. In an incomplete ambient space, a sequence may be internally stable without converging inside that space.

Remark (Dependencies).

- Convergent Sequences Are Cauchy
- Cauchy Sequences Are Bounded
- Bolzano-Weierstrass Theorem for Sequences
- Cauchy Sequence with a Convergent Subsequence

Theorem (Cauchy Criterion via Tails)

Theorem 34.11.6 (Cauchy Criterion via Tails)**.** *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N-tail has pairwise term distances less than ε.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is Cauchy}\\ &\quad\iff \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall p, q \in \mathbb{N}\ \big(|x_{N+p} - x_{N+q}| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall p, q \in \mathbb{N}\ \big(|x_{N+p} - x_{N+q}| < \varepsilon\big).$$

Remark (Interpretation). The Cauchy condition is a tail property. Reindexing a sufficiently late tail does not change the substance of the condition; it only changes the names of the indices.

Remark (Dependencies).

- Cauchy Sequence
- M-Tail of a Sequence

Theorem (Cauchy Tail Diameter Criterion)

Theorem 34.11.7 (Cauchy Tail Diameter Criterion)**.** *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is Cauchy} \\ \iff \quad &\forall \varepsilon > 0 \; \exists N_0 \in \mathbb{N} \; \forall N \geq N_0 \\ &\forall m, n \geq N \; \big(|x_m - x_n| < \varepsilon\big). \end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0 \; \exists N_0 \in \mathbb{N} \; \forall N \geq N_0 \; \forall m, n \geq N \; \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Interpretation). The tail-diameter form says that once one tail is small, every later tail is small as well. It records the monotone nature of tail control without introducing additional notation for diameters.

Remark (Dependencies).

- Cauchy Sequence
- M-Tail of a Sequence

Theorem (Cauchy Sequences Have Vanishing Successive Differences)

Theorem 34.11.8 (Cauchy Sequences Have Vanishing Successive Differences)**.** *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \to 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\\ \Big(\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\\ \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_{n+1} - x_n| < \varepsilon)\Big).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{NullSequence}\big(|x_{n+1} - x_n|\big).$$

Remark (Interpretation). Cauchy control applies to every pair of late indices, so it applies in particular to consecutive late indices. The converse is false in general: small successive steps do not prevent accumulated drift.

Remark (Dependencies).

- Cauchy Sequence
- Null Sequence

Theorem (Scalar Multiples of Cauchy Sequences)

Theorem 34.11.9 (Scalar Multiples of Cauchy Sequences)**.** *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall \alpha \in \mathbb{R}\\ \Big((x_n) \text{ is Cauchy} \implies (\alpha x_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{CauchySequence}(\alpha x_n).$$

Remark (Interpretation). Multiplying all terms by one fixed real number rescales all pairwise tail distances by the same fixed factor.

Remark (Dependencies).

- Cauchy Sequence
- Scalar Multiple of a Sequence

Theorem (Sums of Cauchy Sequences)

Theorem 34.11.10 (Sums of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n + y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (x_n + y_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \implies \mathrm{CauchySequence}(x_n + y_n).$$

Remark (Interpretation). The triangle inequality splits the tail distance of the sum into the two tail distances already controlled by the original sequences.

Remark (Dependencies).

- Cauchy Sequence
- Pointwise Sum of Sequences

Theorem (Differences of Cauchy Sequences)

Theorem 34.11.11 (Differences of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (x_n - y_n) \text{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \implies \mathrm{CauchySequence}(x_n - y_n).$$

Remark (Interpretation). The difference theorem is the sum theorem applied after negating one sequence. It is often the form used to compare two approximating sequences.

Remark (Dependencies).

- Sums of Cauchy Sequences
- Scalar Multiples of Cauchy Sequences
- Pointwise Difference of Sequences

Theorem (Linear Combinations of Cauchy Sequences)

Theorem 34.11.12 (Linear Combinations of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}\ \forall \alpha \in \mathbb{R}\ \forall \beta \in \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \Longrightarrow (\alpha x_n + \beta y_n) \text{ is Cauchy} \Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \Longrightarrow \mathrm{CauchySequence}(\alpha x_n + \beta y_n).$$

Remark (Interpretation). The Cauchy real sequences are closed under finite real linear combinations. This is the additive vector-space part of Cauchy algebra.

Remark (Dependencies).

- Linear Combination of Real Sequences
- Scalar Multiples of Cauchy Sequences
- Sums of Cauchy Sequences

Theorem (Products of Cauchy Sequences)

Theorem 34.11.13 (Products of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \Longrightarrow (x_n y_n) \text{ is Cauchy} \Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \Longrightarrow \mathrm{CauchySequence}(x_n y_n).$$

Remark (Interpretation). Products require boundedness in addition to tail closeness. The boundedness is supplied automatically because every Cauchy real sequence is bounded.

Remark (Dependencies).

- Cauchy Sequences Are Bounded
- Pointwise Product of Sequences

Theorem (Reciprocals of Cauchy Sequences)

Theorem 34.11.14 (Reciprocals of Cauchy Sequences)**.** *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|x_n| \geq c)$$
$$\implies (1/x_n) \text{ is Cauchy} \Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|x_n| \geq c) \implies \mathrm{CauchySequence}(1/x_n).$$

Remark (Interpretation). The lower bound away from zero prevents division from amplifying small tail differences without control. Without it, reciprocals need not preserve Cauchy behavior.

Remark (Dependencies).

- Cauchy Sequence
- Reciprocal Sequence

Theorem (Quotients of Cauchy Sequences)

Theorem 34.11.15 (Quotients of Cauchy Sequences)**.** *Let (x_n) and (y_n) be Cauchy real sequences. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|y_n| \geq c$ for every $n \geq N_0$, then (x_n/y_n) is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|y_n| \geq c)$$
$$\implies (x_n/y_n) \text{ is Cauchy} \Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \wedge \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|y_n| \geq c) \implies \mathrm{CauchySequence}(x_n/$$

Remark (Interpretation). The quotient theorem is the product theorem combined with the reciprocal theorem for a denominator eventually bounded away from zero.

Remark (Dependencies).

- Products of Cauchy Sequences
- Reciprocals of Cauchy Sequences
- Pointwise Quotient of Sequences

Theorem (Absolute Values of Cauchy Sequences)

Theorem 34.11.16 (Absolute Values of Cauchy Sequences)**.** *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left((x_n) \text{ is Cauchy} \Longrightarrow (|x_n|) \text{ is Cauchy} \right).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \Longrightarrow \text{CauchySequence}(|x_n|).$$

Remark (Interpretation). The reverse triangle inequality bounds the distance between absolute values by the original distance between terms.

Remark (Dependencies).

- Cauchy Sequence
- Absolute Value of a Sequence

34.12 Applications of Sequences

Sequence Applications — Quick Reference

Construction	Role
Newton sequence for $\sqrt{2}$	Iterative approximation model
Newton approximation	Converges to $\sqrt{2}$
Factorial partial sums	Series approximants for e
Approximation of e	Factorial sums converge to e
Compound-interest sequence	Classical exponential approximants
Compound-interest approximation	Compound interest converges to e
Decimal truncation sequence	Decimal approximants from below
Decimal truncations	Truncations converge to the target real

Applications of Sequences

The convergence theorems for sequences become numerical tools once the sequence is computable. Newton iteration turns an equation into a convergent recursive approximation; factorial partial sums and compound-interest terms approximate the Euler number; decimal truncations approximate arbitrary real numbers by finite decimals.

Definition (Newton Sequence for $\sqrt{2}$)

Definition 34.12.1 (Newton Sequence for $\sqrt{2}$)**.** The **Newton sequence for** $\sqrt{2}$ is the real sequence (x_n) defined by

$$x_1 := \frac{3}{2}, \qquad x_{n+1} := \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \quad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned} & x_1 = \frac{3}{2} \\ \wedge \quad & \forall n \in \mathbb{N} \left(x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)\right). \end{aligned}$$

Remark (Interpretation)*.* The recursion is Newton's method applied to the equation $x^2 - 2 = 0$. Each new term averages the current estimate with the correction $2/x_n$.

Remark (Dependencies)*.*

- Real Sequence

Theorem (Newton Approximation of $\sqrt{2}$)

Theorem 34.12.2 (Newton Approximation of $\sqrt{2}$)**.** *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(|x_n - \sqrt{2}| < \varepsilon\big).$$

Remark (Definition predicate reading)*.*

$$\text{ConvergentSequence}(x_n, \sqrt{2}).$$

Remark (Interpretation)*.* The Newton sequence is decreasing after the first term and bounded below by $\sqrt{2}$, so monotone convergence supplies a limit. Passing the recursion to the limit identifies the limit as the positive solution of $L^2 = 2$.

Remark (Dependencies)*.*

- Newton Sequence for $\sqrt{2}$
- Monotone Convergence Theorem

- Rational Sequence Limit
- Square Root Sequence

Definition (Factorial Partial Sums)

Definition 34.12.3 (Factorial Partial Sums)**.** The **factorial partial-sum sequence** is the real sequence (s_n) defined by

$$s_n := \sum_{k=0}^{n} \frac{1}{k!} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(s_n = \sum_{k=0}^{n} \frac{1}{k!} \right).$$

Remark (Interpretation). The sequence adds the reciprocal factorial terms one at a time. The terms decrease very rapidly, so the partial sums stabilize quickly.

Remark (Dependencies).

- Real Sequence

Theorem (Factorial Partial Sums Approximate e)

Theorem 34.12.4 (Factorial Partial Sums Approximate e)**.** *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e:*

$$\lim_{n\to\infty} s_n = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|s_n - e| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\mathrm{ConvergentSequence}(s_n, e).$$

Remark (Interpretation). This is the sequence form of the factorial-series definition of e. In practice the approximation is efficient because the error after n terms is controlled by a rapidly shrinking factorial tail.

Remark (Dependencies).

- Factorial Partial Sums

- The Number e
- Cauchy Criterion for Real Sequences

Definition (Compound-Interest Sequence)

Definition 34.12.5 (Compound-Interest Sequence)**.** The **compound-interest sequence** is the real sequence (c_n) defined by

$$c_n := \left(1 + \frac{1}{n}\right)^n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(c_n = \left(1 + \frac{1}{n}\right)^n \right).$$

Remark (Interpretation). The term c_n is the value of one unit of principal after one unit of time when interest is compounded n times at nominal rate 1.

Remark (Dependencies).

- Real Sequence

Theorem (Compound-Interest Approximation of e)

Theorem 34.12.6 (Compound-Interest Approximation of e)**.** *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n\to\infty} c_n = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(|c_n - e| < \varepsilon \big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(c_n, e).$$

Remark (Interpretation). The compound-interest approximation gives a second natural sequence converging to the same constant as the factorial partial sums. It is slower numerically, but it explains why e appears in continuous growth.

Remark (Dependencies).

- Compound-Interest Sequence
- The Number e

- Sequence Squeeze Theorem

Definition (Decimal Truncation Sequence)

Definition 34.12.7 (Decimal Truncation Sequence)**.** Let $\alpha \in \mathbb{R}$. The **decimal truncation sequence** of α is the real sequence (d_n) defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(d_n = \frac{\lfloor 10^n \alpha \rfloor}{10^n} \right).$$

Remark (Interpretation). The number d_n is obtained by cutting α off after n decimal places. Truncation is one-sided: it does not round to the nearest decimal.

Remark (Dependencies).

- Real Sequence

Theorem (Decimal Truncations Converge)

Theorem 34.12.8 (Decimal Truncations Converge)**.** *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α. Then*

$$\lim_{n\to\infty} d_n = \alpha.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N \, \big(|d_n - \alpha| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(d_n, \alpha).$$

Remark (Interpretation). Decimal truncations converge because the truncation error is always less than 10^{-n}. This is one of the most concrete examples of convergence: finite decimal data can approximate a real number to arbitrary accuracy.

Remark (Dependencies).

- Decimal Truncation Sequence
- Sequence Squeeze Theorem

Chapter 35

Series

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK I
Bounds, Sequences, and Series

CHAPTER 7
Series

35.1 Finite Series

Remark (Exposition). Tao introduces finite series by first fixing a finite real sequence indexed by consecutive integers and then defining its sum recursively Tao [2006]. The source bundles these conventions into one definition; here they are split into atomic definitions.

Definition (Finite Real Sequence on an Integer Interval)

Definition 35.1.1 (Finite Real Sequence on an Integer Interval)**.** Let $m, n \in \mathbb{Z}$ with $m \leq n$. A finite real sequence indexed from m to n is an assignment of a real number a_i to each integer i satisfying $m \leq i \leq n$. We denote such a sequence by $(a_i)_{i=m}^{n}$.

Remark (Interpretation). The indices form the integer interval $\{m, m+1, \ldots, n\}$. The notation $(a_i)_{i=m}^{n}$ records both the terms and the finite range of indices on which the terms are defined.

Remark (Dependencies).

- Real Sequence

Definition (Empty Finite Sum)

Definition 35.1.2 (Empty Finite Sum)**.** Let $m, n \in \mathbb{Z}$. If $n < m$, the finite sum from m to n is defined by

$$\sum_{i=m}^{n} a_i := 0.$$

Remark (Interpretation). When the upper index lies below the lower index, there are no terms to add. The sum is therefore assigned the additive identity 0.

Remark (Dependencies).

- Finite Real Sequence on an Integer Interval

Definition (Finite Sum)

Definition 35.1.3 (Finite Sum)**.** Let $m, n \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant integer indices. The notation

$$\sum_{i=m}^{n} a_i$$

denotes the finite sum, or finite series, of the terms indexed from m to n.

Remark (Standard quantified statement).

$$\sum_{i=m}^{n} a_i \quad \text{denotes the finite sum of the terms } a_i \text{ indexed by } m \leq i \leq n.$$

Remark (Interpretation). This definition fixes the notation and name of the object being defined. The value of the object is then determined by the empty-sum convention and the recursive rule below.

Remark (Dependencies).

- Finite Real Sequence on an Integer Interval

Definition (Recursive Finite Sum)

Definition 35.1.4 (Recursive Finite Sum)**.** Let $m \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant indices. The finite sum, also called a finite series, is defined recursively from the empty finite sum by the rule

$$\sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^{n} a_i \right) + a_{n+1} \qquad \text{whenever } n \geq m-1.$$

Remark (Standard quantified statement).

$$\forall m, n \in \mathbb{Z}, \quad n \geq m-1 \Longrightarrow \sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^{n} a_i \right) + a_{n+1}.$$

Remark (Interpretation). The recursive rule says that extending the upper endpoint by one index extends the sum by adding exactly the new last term. Together with the empty-sum convention, this determines every finite sum.

Remark (Examples). For a sequence indexed from m, the first few cases are

$$\sum_{i=m}^{m-2} a_i = 0, \qquad \sum_{i=m}^{m-1} a_i = 0, \qquad \sum_{i=m}^{m} a_i = a_m,$$

and

$$\sum_{i=m}^{m+1} a_i = a_m + a_{m+1}, \qquad \sum_{i=m}^{m+2} a_i = a_m + a_{m+1} + a_{m+2}.$$

Remark (Dependencies).

- Empty Finite Sum
- Finite Sum
- Finite Real Sequence on an Integer Interval

Lemma (Splitting a Finite Sum)

Lemma 35.1.5 (Splitting a Finite Sum). *Let $m \le n < p$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \le i \le p$. Then*

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i.$$

Go to proof.

Remark (Interpretation). A finite sum over one integer interval may be split into two adjacent finite sums without changing its value.

Remark (Dependencies).

- Recursive Finite Sum

Definition (Summation over a Finite Set)

Definition 35.1.6 (Summation over a Finite Set). Let X be a finite set with n elements, where $n \in \mathbb{N}$, and let $f : X \to \mathbb{R}$. Choose a bijection

$$g : \{i \in \mathbb{N} : 1 \le i \le n\} \to X.$$

The finite sum of f over X is defined by

$$\sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i)).$$

Remark (Standard quantified statement).

$$X = \{g(1), \dots, g(n)\} \quad \Longrightarrow \quad \sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i)).$$

Remark (Interpretation). The definition computes a sum over an unordered finite set by first listing its elements through a bijection from $\{1, \ldots, n\}$. The notation $\sum_{x \in X} f(x)$ records the set being summed over; the chosen listing is auxiliary.

Remark (Dependencies).

- Finite Sum

Proposition (Finite Summations Are Well-Defined)

Proposition 35.1.7 (Finite Summations Are Well-Defined). *Let X be a finite set with n elements, where $n \in \mathbb{N}$, let $f : X \to \mathbb{R}$, and let*

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X \qquad \text{and} \qquad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X$$

be bijections. Then

$$\sum_{i=1}^{n} f(g(i)) = \sum_{i=1}^{n} f(h(i)).$$

Go to proof.

Remark (Interpretation). The value of a finite summation over a finite set does not depend on which bijection is used to list the elements of that set. Tao notes that the corresponding issue for summations over infinite sets is more complicated.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Empty Finite Set Summation)

Proposition 35.1.8 (Empty Finite Set Summation). *If X is empty, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

Go to proof.

Remark (Interpretation). The finite summation over the empty set has value 0.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Singleton Finite Set Summation)

Proposition 35.1.9 (Singleton Finite Set Summation). *If X consists of a single element, $X = \{x_0\}$, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

Go to proof.

Remark (Interpretation). Summing a function over a one-element set gives the value of the function at that element.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Substitution for Finite Set Summation)

Proposition 35.1.10 (Substitution for Finite Set Summation)**.** *If X is a finite set, $f : X \to \mathbb{R}$ is a function, and $g : Y \to X$ is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

Go to proof.

Remark (Interpretation). A finite summation over X may be computed by summing over any set bijective with X, after composing the summand with the bijection.

Remark (Dependencies).

- Summation over a Finite Set
- Finite Summations Are Well-Defined

Proposition (Integer Interval as a Finite Set)

Proposition 35.1.11 (Integer Interval as a Finite Set)**.** *Let $n \le m$ be integers, and let*

$$X := \{i \in \mathbb{Z} : n \le i \le m\}.$$

If a_i is a real number assigned to each integer $i \in X$, then

$$\sum_{i=n}^{m} a_i = \sum_{i \in X} a_i.$$

Go to proof.

Remark (Interpretation). Summing over a consecutive integer interval agrees with summing over that same interval regarded as a finite set.

Remark (Dependencies).

- Finite Sum
- Summation over a Finite Set

Proposition (Disjoint Union Finite Set Summation)

Proposition 35.1.12 (Disjoint Union Finite Set Summation). *Let X and Y be disjoint finite sets, so $X \cap Y = \emptyset$, and let $f : X \cup Y \to \mathbb{R}$ be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left(\sum_{x \in X} f(x) \right) + \left(\sum_{y \in Y} f(y) \right).$$

Go to proof.

Remark (Interpretation). A sum over a disjoint union splits into the sum over each component set.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Addition Rule for Finite Set Summation)

Proposition 35.1.13 (Addition Rule for Finite Set Summation). *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

Go to proof.

Remark (Interpretation). Finite set summation distributes over pointwise addition.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Scalar Multiplication Rule for Finite Set Summation)

Proposition 35.1.14 (Scalar Multiplication Rule for Finite Set Summation). *Let X be a finite set, let $f : X \to \mathbb{R}$ be a function, and let c be a real number. Then*

$$\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x).$$

Go to proof.

Remark (Interpretation). A constant scalar may be pulled outside a finite set summation.

Remark (Dependencies).

- Summation over a Finite Set

Proposition (Monotonicity for Finite Set Summation)

Proposition 35.1.15 (Monotonicity for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions such that $f(x) \leq g(x)$ for all $x \in X$. Then*

$$\sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\big(\forall x \in X,\ f(x) \leq g(x)\big) \implies \sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

Remark (Interpretation)*.* Pointwise order between two functions on a finite set gives the same order between their finite sums.

Remark (Dependencies)*.*

- Summation over a Finite Set

Proposition (Triangle Inequality for Finite Set Summation)

Proposition 35.1.16 (Triangle Inequality for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ be a function. Then*

$$\left| \sum_{x \in X} f(x) \right| \leq \sum_{x \in X} |f(x)|.$$

Go to proof.

Remark (Interpretation)*.* The absolute value of a finite set summation is bounded by the corresponding finite set summation of absolute values.

Remark (Dependencies)*.*

- Summation over a Finite Set

Lemma (Iterated Summation over a Product)

Lemma 35.1.17 (Iterated Summation over a Product)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x,y) \in X \times Y} f(x, y).$$

Go to proof.

Remark (Interpretation). Summing first over Y for each $x \in X$, and then summing those values over X, gives the same value as summing over the product set $X \times Y$.

Remark (Dependencies).

- Summation over a Finite Set

Corollary (Fubini's Theorem for Finite Series)

Corollary 35.1.18 (Fubini's Theorem for Finite Series)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x\in X}\left(\sum_{y\in Y} f(x,y)\right) = \sum_{(x,y)\in X\times Y} f(x,y) = \sum_{(y,x)\in Y\times X} f(x,y) = \sum_{y\in Y}\left(\sum_{x\in X} f(x,y)\right).$$

Go to proof.

Remark (Standard quantified statement).

$$\sum_{x\in X}\left(\sum_{y\in Y} f(x,y)\right) = \sum_{y\in Y}\left(\sum_{x\in X} f(x,y)\right).$$

Remark (Interpretation). For a function on a finite product set, the order of summation over the two finite factors may be interchanged.

Remark (Dependencies).

- Iterated Summation over a Product
- Summation over a Finite Set

Lemma (Reindexing a Finite Sum)

Lemma 35.1.19 (Reindexing a Finite Sum)**.** *Let $m \le n$ be integers, let k be an integer, and let a_i be a real number assigned to each integer i satisfying $m \le i \le n$. Then*

$$\sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

Go to proof.

Remark (Interpretation). Shifting every index by the same integer changes the labels of the summation but not the ordered list of terms being summed.

Remark (Dependencies).

- Recursive Finite Sum

Lemma (Finite Sum of Sums)

Lemma 35.1.20 (Finite Sum of Sums)**.** *Let $m \le n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \le i \le n$. Then*

$$\sum_{i=m}^{n}(a_i + b_i) = \left(\sum_{i=m}^{n} a_i\right) + \left(\sum_{i=m}^{n} b_i\right).$$

Go to proof.

Remark (Interpretation). Finite summation distributes over termwise addition.

Remark (Dependencies).

- Recursive Finite Sum

Lemma (Finite Sum of Scalar Multiples)

Lemma 35.1.21 (Finite Sum of Scalar Multiples)**.** *Let $m \le n$ be integers, let a_i be a real number assigned to each integer i satisfying $m \le i \le n$, and let $c \in \mathbb{R}$. Then*

$$\sum_{i=m}^{n}(ca_i) = c\left(\sum_{i=m}^{n} a_i\right).$$

Go to proof.

Remark (Interpretation). A common scalar factor may be pulled outside a finite sum.

Remark (Dependencies).

- Recursive Finite Sum

Lemma (Triangle Inequality for Finite Series)

Lemma 35.1.22 (Triangle Inequality for Finite Series)**.** *Let $m \le n$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \le i \le n$. Then*

$$\left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n}|a_i|.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall m, n \in \mathbb{Z}, \quad m \le n \Longrightarrow \left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n}|a_i|.$$

Remark (Interpretation). The absolute value of a finite series is bounded by the corresponding finite series of absolute values.

Remark (Dependencies).

- Recursive Finite Sum

Lemma (Comparison Test for Finite Series)

Lemma 35.1.23 (Comparison Test for Finite Series)**.** *Let $m \leq n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \leq i \leq n$. Suppose that $a_i \leq b_i$ for all i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} a_i \leq \sum_{i=m}^{n} b_i.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \, (m \leq i \leq n \implies a_i \leq b_i) \implies \sum_{i=m}^{n} a_i \leq \sum_{i=m}^{n} b_i.$$

Remark (Interpretation)*.* Termwise comparison of two finite series gives comparison of their finite sums.

Remark (Dependencies).

- Recursive Finite Sum

35.2 Infinite Series

Definition (Formal Infinite Series)

Definition 35.2.1 (Formal Infinite Series)**.** A formal infinite series is an expression of the form

$$\sum_{n=m}^{\infty} a_n,$$

where m is an integer, and a_n is a real number for every integer $n \geq m$. This series is sometimes written as

$$a_m + a_{m+1} + a_{m+2} + \cdots .$$

Remark (Standard quantified statement).

$$\sum_{n=m}^{\infty} a_n \quad \text{denotes a formal infinite series with } m \in \mathbb{Z} \text{ and } a_n \in \mathbb{R} \text{ for every } n \geq m.$$

Remark (Interpretation)*.* The notation records an infinite list of real terms beginning at the integer index m. At this stage it is a formal expression rather than a claim that a value has already been assigned to the series.

Remark (Dependencies).

- Real Sequence

Chapter 36

Function Sequences

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BOOK I
Bounds, Sequences, and Series

CHAPTER 8
Function Sequences

Functions, Continuity, and Differentiation

Chapter 37

Elementary Functions

37.1 Hyperbolic Functions

Hyperbolic Functions — Quick Reference

Concept	**Meaning**
Hyperbolic functions	sinh and cosh from even/odd exponential parts
Hyperbolic Pythagorean identity	$\cosh^2 x - \sinh^2 x = 1$
Addition formulas	Hyperbolic angle-addition identities
Tanh limits	$\tanh x \to \pm 1$ as $x \to \pm\infty$
Inverse hyperbolic functions	Logarithmic inverse formulas

Hyperbolic Functions

Definition (Hyperbolic Functions)

Definition 37.1.1 (Hyperbolic Functions)**.** For all $x \in \mathbb{R}$:

$$\cosh x \;:=\; \frac{e^x + e^{-x}}{2}, \qquad \sinh x \;:=\; \frac{e^x - e^{-x}}{2}, \qquad \tanh x \;:=\; \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\coth x := \cosh x/\sinh x$.

Remark (Standard quantified statement)*.*

$$\forall x \in \mathbb{R}: \ \cosh x = \frac{e^x + e^{-x}}{2} \geq 1, \ \sinh x = \frac{e^x - e^{-x}}{2}, \ e^x = \cosh x + \sinh x.$$

Remark (Definition predicate reading).

$$\text{HyperbolicCosh}(x,y) \;:\equiv\; y = \frac{e^x + e^{-x}}{2}. \qquad \text{HyperbolicSinh}(x,y) \;:\equiv\; y = \frac{e^x - e^{-x}}{2}.$$

$$\text{HyperbolicTanh}(x,y) \;:\equiv\; \cosh x \neq 0 \;\wedge\; y = \frac{\sinh x}{\cosh x}.$$

Remark (Definition predicate reading).

$$\text{HyperbolicPythagorean}(x) \;:\equiv\; \text{HyperbolicCosh}(x,c) \wedge \text{HyperbolicSinh}(x,s) \wedge c^2 - s^2 = 1.$$

Remark (Negated quantified statement).

$$\neg\, \text{HyperbolicTanh}(x,y) \iff y \neq \frac{\sinh x}{\cosh x}.$$

Note $\cosh x > 0$ for all x, so HyperbolicTanh is defined on all of $\mathbb{R}$.

Remark (Contrast with trigonometric functions).

Trigonometric	**Hyperbolic**
$\cos^2 x + \sin^2 x = 1$	$\cosh^2 x - \sinh^2 x = 1$
$(\cos x)' = -\sin x$	$(\cosh x)' = \sinh x$
$(\sin x)' = \cos x$	$(\sinh x)' = \cosh x$
$e^{ix} = \cos x + i \sin x$	$e^x = \cosh x + \sinh x$
Parametrises unit circle	Parametrises unit hyperbola

The minus sign in $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$ reflects the substitution $x \mapsto ix$ in Euler's formula.

Proposition (Hyperbolic Pythagorean Identity)

Proposition 37.1.2 (Hyperbolic Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; \cosh^2 x - \sinh^2 x = 1.$$

Remark (Negated quantified statement).

$$\neg\,(\forall x : \cosh^2 x - \sinh^2 x = 1) \iff \exists x \in \mathbb{R} : \cosh^2 x - \sinh^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

Remark (Proof). Direct computation:

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4} = \frac{4}{4} = 1.$$

Remark (Dependencies).

- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition (Addition Formulas for Hyperbolic Functions)

Proposition 37.1.3 (Addition Formulas for Hyperbolic Functions)**.** *For all $x, y \in \mathbb{R}$:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

Remark (Definition predicate reading).

$$\text{HyperbolicAddition}(x, y) \ :\equiv \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sinh) \iff \exists x, y \in \mathbb{R} : \ \sinh(x+y) \neq \sinh x \cosh y + \cosh x \sinh y.$$

The negation has no witness.

Remark (Proof strategy). Substitute the definitions and expand, or note the formulas follow from $e^{x+y} = e^x e^y$ by taking even and odd parts.

Remark (Comparison with trigonometric addition formulas). The sinh formula matches $\sin(x+y)$ exactly. The cosh formula differs from $\cos(x+y)$ only in the sign of the last term: $+\sinh x \sinh y$ versus $-\sin x \sin y$. The sign difference is a direct consequence of $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$.

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition (Limits of $\tanh$)

Proposition 37.1.4 (Limits of tanh)**.**

$$\lim_{x \to +\infty} \tanh x = 1, \qquad \lim_{x \to -\infty} \tanh x = -1.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ \wedge\ \text{LimitAtInfinity}(\tanh x,\ -\infty,\ -1).$$

Remark (Definition predicate reading).

$$\text{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ :\equiv\ \forall\varepsilon>0,\ \exists M\in\mathbb{R},\ \forall x:\ x>M\Rightarrow \text{OutputTolerance}(\tanh x,1,\varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to+\infty}\tanh x=1\right)\iff \exists\varepsilon_0>0,\ \forall M>0,\ \exists x>M:\ |\tanh x-1|\ge\varepsilon_0.$$

Remark (Proof strategy). Write $\tanh x=(1-e^{-2x})/(1+e^{-2x})$. As $x\to+\infty$, $e^{-2x}\to 0$, giving $\tanh x\to 1$. The limit at $-\infty$ follows by oddness: $\tanh(-x)=-\tanh(x)$.

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 37.1.5 (Inverse Hyperbolic Functions)**.** Since sinh is strictly increasing and bijective $\mathbb{R}\to\mathbb{R}$, its inverse arcsinh is defined on all of $\mathbb{R}$. Since cosh is strictly increasing on $[0,\infty)$ with range $[1,\infty)$, its inverse arccosh is defined on $[1,\infty)$. Since tanh is strictly increasing with range $(-1,1)$, arctanh is defined on $(-1,1)$.
Explicit logarithmic formulas:

$$\operatorname{arcsinh} x=\ln\left(x+\sqrt{x^2+1}\right),\quad x\in\mathbb{R}.$$
$$\operatorname{arccosh} x=\ln\left(x+\sqrt{x^2-1}\right),\quad x\ge 1.$$
$$\operatorname{arctanh} x=\tfrac{1}{2}\ln\left(\frac{1+x}{1-x}\right),\quad |x|<1.$$

Remark (Standard quantified statement).

$$\forall x\in\mathbb{R}:\ \sinh(\operatorname{arcsinh} x)=x\ \wedge\ \operatorname{arcsinh}(\sinh x)=x.$$

$$\forall x\ge 1:\ \cosh(\operatorname{arccosh} x)=x.\qquad \forall|x|<1:\ \tanh(\operatorname{arctanh} x)=x.$$

Remark (Definition predicate reading).

$$\text{Arcsinh}(y,x)\ :\equiv\ x\in\mathbb{R}\ \wedge\ \sinh x=y\ :\equiv\ x=\ln\left(y+\sqrt{y^2+1}\right).$$

$$\text{Arctanh}(y,x)\ :\equiv\ |y|<1\ \wedge\ \tanh x=y\ :\equiv\ x=\tfrac{1}{2}\ln\left(\frac{1+y}{1-y}\right).$$

Remark (Negated quantified statement).

$$\neg\ \mathrm{Arctanh}(y, x) \iff |y| \geq 1 \vee \tanh x \neq y.$$

Remark (Derivation of arcsinh). Set $y = \sinh x = (e^x - e^{-x})/2$ and let $u = e^x > 0$. Then $y = (u - 1/u)/2$, so $u^2 - 2yu - 1 = 0$. The positive root is $u = y + \sqrt{y^2 + 1}$, giving $x = \ln(y + \sqrt{y^2 + 1})$.

Remark (Derivatives). $(\operatorname{arcsinh} x)' = 1/\sqrt{x^2 + 1}$; $(\operatorname{arccosh} x)' = 1/\sqrt{x^2 - 1}$ for $x > 1$; $(\operatorname{arctanh} x)' = 1/(1 - x^2)$ for $|x| < 1$. These follow from the inverse function theorem applied to sinh, cosh, and tanh respectively.

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Limit at Infinity
- Limit of a Quotient

Remark (Dependencies).

- Hyperbolic Functions
- Natural Logarithm
- Continuous Inverse Theorem
- Absolute Value on $\mathbb{R}$

37.2 The Logarithm

Logarithmic Functions — Quick Reference

Concept	**Meaning**
Natural logarithm	Inverse of the exponential on $(0, \infty)$
Fundamental logarithmic limit	$\ln(1 + x)/x \to 1$
Power dominates log	Logarithmic growth is below every positive power

The Logarithm

Definition (Natural Logarithm)

Definition 37.2.1 (Natural Logarithm)**.** The **natural logarithm** $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x \;:=\; \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x \;:=\; \int_1^x \frac{1}{t}\,dt.$$

Remark (Standard quantified statement)*.*

$$\forall x > 0 :\; e^{\ln x} = x. \qquad \forall y \in \mathbb{R} :\; \ln(e^y) = y.$$

Remark (Definition predicate reading)*.*

$$\mathrm{NaturalLog}(x, y) \;:\equiv\; x > 0 \;\wedge\; e^y = x \;:\equiv\; x > 0 \;\wedge\; y = \int_1^x \frac{1}{t}\,dt.$$

Remark (Definition predicate reading)*.*

$$\mathrm{FunctionLimit}\left(\frac{\ln(1+x)}{x},\, 0,\, 1\right) \;:\equiv\; \forall \varepsilon > 0,\; \exists \delta > 0,\; \forall x :\; \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\Bigg(\frac{\ln$$

Remark (Negated quantified statement)*.*

$$\neg\, \mathrm{NaturalLog}(x, y) \iff x \le 0 \;\vee\; e^y \ne x.$$

Remark (Two definitions, same function)*.* The inverse-of-exponential definition requires e^x to be established first. The integral definition is self-contained and is preferred when building analysis from scratch. Both yield the same function.

Remark (Key properties)*.* (i) $\ln 1 = 0$; $\ln e = 1$.

(ii) **Functional equation:** $\ln(xy) = \ln x + \ln y$ for all $x, y > 0$.

(iii) $\ln(x/y) = \ln x - \ln y$; $\ln(x^r) = r \ln x$ for $r \in \mathbb{R}$.

(iv) $\ln$ is strictly increasing on $(0, \infty)$.

(v) $\ln x \to +\infty$ as $x \to \infty$; $\ln x \to -\infty$ as $x \to 0^+$.

(vi) $(\ln x)' = 1/x$ for all $x > 0$.

Remark (General logarithm)*.* For $a > 0$, $a \ne 1$, the **logarithm base** a is $\log_a x := \ln x / \ln a$. The common logarithm is $\log_{10}$; the binary logarithm is $\log_2$.

Proposition (Fundamental Logarithmic Limit)

Proposition 37.2.2 (Fundamental Logarithmic Limit)**.**

$$\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall\varepsilon > 0,\ \exists\delta > 0:\ 0 < |x| < \delta \Rightarrow \left|\frac{\ln(1+x)}{x} - 1\right| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1\right) \iff \exists\varepsilon_0 > 0,\ \forall\delta > 0,\ \exists x:\ 0 < |x| < \delta\ \wedge\ \left|\frac{\ln(1+x)}{x} - 1\right| \geq \varepsilon_0.$$

Remark (Equivalence with exponential limit). The fundamental exponential limit $\lim(e^x - 1)/x = 1$ and the fundamental logarithmic limit $\lim\ln(1+x)/x = 1$ are equivalent: each follows from the other via the substitution $x \leftrightarrow e^x - 1$. Both express the first-order Taylor approximation of the respective function at 0.

Remark (Proof strategy). From the series: $\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ (valid for $|x| < 1$). Dividing by x gives $1 - x/2 + x^2/3 - \cdots \to 1$ as $x \to 0$ by continuity of power series.

Remark (Dependencies).

- Exponential Function
- Inverse Bijection
- Continuous Inverse Theorem

Proposition (Every Power Dominates the Logarithm)

Proposition 37.2.3 (Every Power Dominates the Logarithm)**.** *For every $r > 0$:*

$$\lim_{x\to\infty}\frac{\ln x}{x^r} = 0 \qquad \textit{and} \qquad \lim_{x\to 0^+} x^r \ln x = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0:\ \lim_{x\to\infty} x^{-r}\ln x = 0\ \wedge\ \lim_{x\to 0^+} x^r \ln x = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^{-r}\ln x,\ +\infty,\ 0)\ :\equiv\ \forall\varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x:\ x > M \Rightarrow \mathrm{OutputTolerance}(x^{-r}\ln x, 0, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty}\frac{\ln x}{x^r}=0\right) \iff \exists\varepsilon_0>0,\ \forall M>0,\ \exists x>M:\ x^{-r}|\ln x|\geq\varepsilon_0.$$

Remark (Proof strategy). For the first: substitute $x=e^t$ to get $t/e^{rt}\to 0$ by exponential dominance. For the second: substitute $x=1/t$ and reduce to the first: $x^r\ln x=(-\ln t)/t^r\to 0$.

Remark (Significance). These limits establish the position of $\ln x$ in the growth hierarchy: $\ln x$ is dominated by every positive power. Combined with the exponential dominance result, the full ordering is $\ln x \ll x^r \ll e^x$ as $x\to\infty$ for any $r>0$. The second limit says that near 0^+, the positive power x^r wins over the logarithmic blow-up $|\ln x|\to\infty$. Used in integrability arguments near 0.

Remark (Dependencies).

- Natural Logarithm
- Fundamental Exponential Limit
- Function Limit

Remark (Dependencies).

- Natural Logarithm
- Power Function
- Exponential Dominates Every Power
- Limit at Infinity

37.3 Power Functions and the Exponential

Power and Exponential Functions — Quick Reference

Concept	Meaning
Power function	Real powers x^r on positive inputs
The number e	Series and compound-interest limit definitions
Exponential function	Power series with $e^{x+y}=e^xe^y$
Fundamental exponential limit	$(e^x-1)/x\to 1$
Compound-interest limit	$(1+1/n)^n\to e$
Growth hierarchy	e^x dominates powers at infinity

Power Functions and the Exponential

Definition (Power Function)

Definition 37.3.1 (Power Function)**.** Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r \ := \ e^{r \ln x}.$$

Remark (Standard quantified statement)*.*

$$\forall r \in \mathbb{R},\ \forall x > 0 :\ x^r = e^{r \ln x}.$$

For $r \in \mathbb{N}$ this reduces to the r-fold product $x \cdot x \cdots x$.

Remark (Definition predicate reading)*.*

$$\mathrm{RealPower}(x, r, y) \ :\equiv \ x > 0 \ \wedge \ y = e^{r \ln x}.$$

Remark (Laws of exponents)*.* For $x, y > 0$ and $r, s \in \mathbb{R}$:

$$x^r \cdot x^s = x^{r+s}, \quad (x^r)^s = x^{rs}, \quad (xy)^r = x^r y^r, \quad x^0 = 1, \quad x^1 = x.$$

Each follows from the corresponding property of the exponential via the definition $x^r = e^{r \ln x}$.

Remark (Interpretation)*.* Defining $x^r := e^{r \ln x}$ unifies all three cases — integer, rational, irrational exponents — into a single formula and makes differentiability of x^r immediate from that of e^x and $\ln x$ via the chain rule: $(x^r)' = r x^{r-1}$.

Definition (The Number e)

Definition 37.3.2 (The Number e)**.** The **Euler number** is

$$e \ := \ \sum_{n=0}^{\infty} \frac{1}{n!} \ = \ \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \ \approx \ 2.71828\ldots$$

Both expressions define the same real number.

Remark (Standard quantified statement)*.*

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \ \wedge \ e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

Remark (Interpretation)*.* The series $\sum 1/n!$ converges rapidly by comparison with a geometric series. The limit $(1 + 1/n)^n$ arises from continuous compound interest and is proved equal to the series sum via the fundamental logarithmic limit. Both representations are in common use.

Remark (Dependencies)*.*

- Integer Powers in a Field

- Exponential Function
- Natural Logarithm
- Multiplication on $\mathbb{R}$

Remark (Dependencies).

- Sequence
- Convergent Sequence

Definition (Exponential Function)

Definition 37.3.3 (Exponential Function)**.** The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is defined by the absolutely convergent power series

$$e^x \ := \ \exp(x) \ := \ \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in \mathbb{R}.$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \text{convergent absolutely since } \sum \frac{|x|^n}{n!} < \infty.$$

Remark (Definition predicate reading).

$$\text{Exponential}(x, y) \ :\equiv \ y = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Remark (Key properties).

(i) $e^0 = 1$; $e^1 = e$.

(ii) **Functional equation:** $e^{x+y} = e^x \cdot e^y$ for all $x, y \in \mathbb{R}$.

(iii) $e^x > 0$ for all x; $e^{-x} = 1/e^x$.

(iv) e^x is strictly increasing and bijective $\mathbb{R} \to (0, \infty)$.

(v) $(e^x)' = e^x$ — the exponential is its own derivative.

Remark (General exponential). For $a > 0$, $a \neq 1$: $a^x := e^{x \ln a}$. Strictly increasing if $a > 1$, strictly decreasing if $0 < a < 1$. $(a^x)' = a^x \ln a$. Only e^x is its own derivative.

Proposition (Fundamental Exponential Limit)

Proposition 37.3.4 (Fundamental Exponential Limit)**.**

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{e^x - 1}{x} - 1 \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{e^x - 1}{x}, 0, 1\right) :\equiv \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\left(\frac{e^x -}{x}\right.$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{e^x - 1}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \ \wedge \ \left| \frac{e^x - 1}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Interpretation). This is the statement $(e^x)'\big|_{x=0} = 1$. The series gives $(e^x - 1)/x = 1 + x/2! + x^2/3! + \cdots \to 1$ as $x \to 0$ by continuity of power series at their center. Combined with the functional equation, this implies $(e^x)' = e^x$ everywhere.

Remark (Dependencies).

- The Number e
- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition (Compound Interest Limit)

Proposition 37.3.5 (Compound Interest Limit)**.**

$$\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists M > 0 :\ x > M \Rightarrow \left| \left(1 + \frac{1}{x}\right)^x - e \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{LimitAtInfinity}\left(\left(1 + \tfrac{1}{x}\right)^x, +\infty, e\right) :\equiv \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \text{OutputTolerance}\left(\left(1 + \tfrac{1}{x}\right)^x, e, \varepsilon\right)$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to \infty} \left(1 + \tfrac{1}{x}\right)^x = e \right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ \left| \left(1 + \frac{1}{x}\right)^x - e \right| \geq \varepsilon_0.$$

Remark (Proof strategy). Write $(1+1/x)^x = e^{x\ln(1+1/x)}$. Set $t = 1/x \to 0^+$ and use the fundamental logarithmic limit $\ln(1+t)/t \to 1$. Then $x\ln(1+1/x) = \ln(1+t)/t \to 1$, so $(1+1/x)^x = e^{x\ln(1+1/x)} \to e^1 = e$ by continuity of the exponential.

Remark (Dependencies).

- Exponential Function
- Function Limit
- Derivative at a Point

Proposition (Exponential Dominates Every Power)

Proposition 37.3.6 (Exponential Dominates Every Power)**.** *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{x^r}{e^x} = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0 : \lim_{x\to\infty} x^r e^{-x} = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^r e^{-x},\ +\infty,\ 0)\ :\equiv\ \forall\varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x:\ x > M \Rightarrow \mathrm{OutputTolerance}(x^r e^{-x}, 0, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty} \frac{x^r}{e^x} = 0\right) \iff \exists\varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M:\ x^r e^{-x} \geq \varepsilon_0.$$

Remark (Proof strategy). Pick integer $n > r$. The series bound $e^x \geq x^n/n!$ gives $x^r/e^x \leq n!\,x^{r-n} \to 0$ since $r - n < 0$.

Remark (Growth hierarchy). For any $r > 0$:

$$\ln x \ \ll\ x^r \ \ll\ e^x \qquad \text{as } x \to \infty.$$

The exponential eventually dominates every polynomial and every power function. This ordering is used constantly in estimates, dominated convergence arguments, and comparison tests.

Remark (Dependencies).

- The Number e
- Exponential Function

- Natural Logarithm
- Fundamental Logarithmic Limit
- Limit at Infinity

Remark (Dependencies).

- Power Function
- Exponential Function
- Limit at Infinity
- Squeeze Theorem for Function Limits

37.4 Trigonometric Functions

Trigonometric Functions — Quick Reference

Concept	Meaning
Sine and cosine	Defined analytically by power series
Pythagorean identity	$\sin^2 x + \cos^2 x = 1$
Addition formulas	Angle addition identities from exponentials
Other trig functions	Tangent, cotangent, secant, cosecant
Inverse trig functions	Inverses on monotone restricted domains
Fundamental trig limit	$\sin x/x \to 1$ as $x \to 0$
Second trig limit	$(1 - \cos x)/x^2 \to 1/2$

Trigonometric Functions

Definition (Sine and Cosine)

Definition 37.4.1 (Sine and Cosine via Power Series)**.** The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\sin x \; := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \; = \; x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x \; := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \; = \; 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

both absolutely convergent by the ratio test.

Remark (Definition predicate reading).

$$\mathrm{Sine}(x, y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}. \qquad \mathrm{Cosine}(x, y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

Remark (Key properties). (i) $\sin 0 = 0$; $\cos 0 = 1$.

(ii) $\sin(-x) = -\sin x$ (odd); $\cos(-x) = \cos x$ (even).

(iii) $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$.

(iv) $|\sin x| \le 1$; $|\cos x| \le 1$ for all x.

(v) $\sin^2 x + \cos^2 x = 1$ (Pythagorean identity).

(vi) Periodic: $\sin(x + 2\pi) = \sin x$; $\cos(x + 2\pi) = \cos x$.

Remark (Relationship to e^{ix}). Euler's formula: $e^{ix} = \cos x + i \sin x$, where e^{ix} is defined by the complex exponential series. This gives $\cos x = \mathrm{Re}(e^{ix})$ and $\sin x = \mathrm{Im}(e^{ix})$. Euler's identity $e^{i\pi} + 1 = 0$ follows immediately.

Proposition (Pythagorean Identity)

Proposition 37.4.2 (Pythagorean Identity). *For all $x \in \mathbb{R}$:*

$$\sin^2 x + \cos^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; \sin^2 x + \cos^2 x = 1.$$

Remark (Definition predicate reading).

$$\mathrm{PythagoreanIdentity}(x) \;:\equiv\; \mathrm{Sine}(x, s) \;\wedge\; \mathrm{Cosine}(x, c) \;\wedge\; s^2 + c^2 = 1.$$

Remark (Negated quantified statement).

$$\neg\left(\forall x : \sin^2 x + \cos^2 x = 1\right) \iff \exists x \in \mathbb{R} : \sin^2 x + \cos^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

Remark (Proof strategy). Define $f(x) := \sin^2 x + \cos^2 x$. Then $f'(x) = 2 \sin x \cos x + 2 \cos x(-\sin x) = 0$, so f is constant. Since $f(0) = 0 + 1 = 1$, the identity follows.

Remark (Dependencies).

- Real Numbers

- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition (Addition Formulas)

Proposition 37.4.3 (Addition Formulas)**.** *For all $x, y \in \mathbb{R}$:*

$$\sin(x + y) = \sin x \cos y + \cos x \sin y, \qquad \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \ \sin(x + y) = \sin x \cos y + \cos x \sin y \ \wedge \ \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

Remark (Definition predicate reading).

$$\mathrm{TrigAddition}(x, y) \ :\equiv \ \sin(x+y) = \sin x \cos y + \cos x \sin y \wedge \cos(x+y) = \cos x \cos y - \sin x \sin y. \blacksquare$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sin) \iff \exists x, y \in \mathbb{R} : \ \sin(x + y) \neq \sin x \cos y + \cos x \sin y.$$

The negation has no witness.

Remark (Proof strategy). From the Cauchy product: $e^{i(x+y)} = e^{ix}e^{iy}$. Taking real and imaginary parts yields both addition formulas simultaneously.

Remark (Derived identities).

$$\begin{aligned}
\sin(2x) &= 2 \sin x \cos x \\
\cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 \\
\sin^2 x &= \tfrac{1}{2}(1 - \cos 2x) \\
\cos^2 x &= \tfrac{1}{2}(1 + \cos 2x)
\end{aligned}$$

Remark (Dependencies).

- Sine and Cosine
- Derivative at a Point
- Zero Derivative Implies Constant

Definition

Definition 37.4.4 (Tangent, Cotangent, Secant, Cosecant)**.**

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R},\ \cos x \neq 0 :\ \tan x = \frac{\sin x}{\cos x}.$$

Remark (Definition predicate reading).

$$\mathrm{Tangent}(x, y)\ :\equiv\ \cos x \neq 0\ \wedge\ y = \frac{\sin x}{\cos x}.$$

Remark (Pythagorean identities for tan). Dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$: $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$: $\cot^2 x + 1 = \csc^2 x$.

Remark (Dependencies).

- Sine and Cosine
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 37.4.5 (Inverse Trigonometric Functions).

- $\arcsin : [-1, 1] \to [-\pi/2, \pi/2]$ is the inverse of $\sin\big|_{[-\pi/2,\pi/2]}$.
- $\arccos : [-1, 1] \to [0, \pi]$ is the inverse of $\cos\big|_{[0,\pi]}$.
- $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is the inverse of $\tan\big|_{(-\pi/2,\pi/2)}$.

Remark (Standard quantified statement).

$$\forall y \in [-1, 1] :\ \sin(\arcsin y) = y. \qquad \forall x \in [-\pi/2, \pi/2] :\ \arcsin(\sin x) = x.$$

Remark (Definition predicate reading).

$$\mathrm{Arctan}(y, x)\ :\equiv\ x \in (-\pi/2, \pi/2)\ \wedge\ \tan x = y.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{\sin x}{x},\ 0,\ 1\right)\ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\left(\frac{\sin x}{x},\right.$$

Remark (Negated quantified statement).

$$\neg\ \mathrm{Arctan}(y, x)\ \iff\ x \notin (-\pi/2, \pi/2)\ \vee\ \tan x \neq y.$$

Remark (Key limits of arctan).

$$\lim_{x\to+\infty} \arctan x = \frac{\pi}{2}, \qquad \lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}.$$

arctan is bounded, strictly increasing, and continuous on $\mathbb{R}$; the Monotone Limit Theorem gives the limits as the supremum and infimum of its range.

Remark (Dependencies).

- Sine and Cosine
- $\mathbb{R}$ Is a Field

Theorem (Fundamental Trigonometric Limit)

Theorem 37.4.6 (Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{\sin x}{x} - 1 \right| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x\to 0} \frac{\sin x}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \ \wedge \ \left| \frac{\sin x}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Proof strategy — series). From the series: $\sin x/x = 1 - x^2/3! + x^4/5! - \cdots \to 1$ as $x \to 0$ by continuity of power series at their center.

Remark (Proof strategy — geometric squeeze). For $0 < x < \pi/2$, area comparison of two triangles and a circular sector gives $\cos x \leq \sin x/x \leq 1$. Since $\cos x \to 1$ as $x \to 0^+$, the Squeeze Theorem gives $\sin x/x \to 1$. Even symmetry extends this to $x \to 0^-$.

Remark (Significance). This is the statement $(\sin x)'\big|_{x=0} = 1$, which implies $(\sin x)' = \cos x$ everywhere. It underlies all trigonometric differentiation.

Remark (Dependencies).

- Sine and Cosine
- Tangent, Cotangent, Secant, Cosecant
- Continuous Inverse Theorem
- Closed Interval

Proposition (Second Fundamental Trigonometric Limit)

Proposition 37.4.7 (Second Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{1 - \cos x}{x^2},\ 0,\ \tfrac{1}{2}\right)\ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\Big(1$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta\ \wedge\ \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| \geq \varepsilon_0.$$

Remark (Proof strategy). Use the half-angle identity $1 - \cos x = 2\sin^2(x/2)$:

$$\frac{1 - \cos x}{x^2} = \frac{2\sin^2(x/2)}{x^2} = \frac{1}{2}\left(\frac{\sin(x/2)}{x/2}\right)^2 \to \frac{1}{2} \cdot 1^2 = \frac{1}{2}.$$

Remark (Dependencies).

- Sine and Cosine
- Function Limit
- Squeeze Theorem for Function Limits

Remark (Dependencies).

- Sine and Cosine
- Trigonometric Addition Formulas
- Fundamental Trigonometric Limit
- Function Limit

Chapter 38

Continuity

SERIES
From Cantor to Ito

VOLUME III
Classical Analysis

BOOK II
Functions, Continuity, and Differentiation

CHAPTER 9
Continuity

38.1 Limits

One-Sided Limits — Quick Reference

Concept/Statement	Meaning/Role
Right-hand limit	Approach the point from inputs larger than c.
Left-hand limit	Approach the point from inputs smaller than c.
Right sequential criterion	Test right limits by right-approaching sequences.
Left sequential criterion	Test left limits by left-approaching sequences.
Matching criterion	Two-sided limits exist exactly when side limits agree.

Limits of Functions — Quick Reference

Concept/Statement	Meaning/Role
Function limit	Punctured input control near a cluster point.
Sequential limit	All approaching sequences have the same output limit.
Neighbourhood limit	Neighbourhood formulation of the same condition.
Uniqueness	A function has at most one limit at a point.
Sequential criterion	Sequence convergence is the main working test.
Local boundedness	Finite limits bound the function near the point.
Squeeze theorem	Ordered comparison transfers limits.
Locality	Only behaviour near the approach point matters.

Algebra of Limits — Quick Reference

Concept/Statement	Meaning/Role
Sum rule	Limits add pointwise.
Difference rule	Limits subtract pointwise.
Scalar multiple rule	Scalars pass through limits.
Product rule	Products of finite limits multiply.
Quotient rule	Division is valid when the denominator limit is nonzero.

Limits of Functions

Definition (ε-Closeness of a Function to a Value)

Definition 38.1.1 (ε-Closeness of a Function to a Value)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function f is **ε-close to L** if $|f(x) - L| \le \varepsilon$ for all $x \in X$.

Remark (Standard quantified statement).

$$f\ \varepsilon\text{-close to } L \iff \forall x \in X\ \big(|f(x) - L| \le \varepsilon\big).$$

Remark (Interpretation). A global condition: every output lies within ε of L, regardless of where in X the input lies. This is Tao's scaffolding toward the limit definition, which localises the same output condition to a punctured neighbourhood of c.

Definition (Local δ-Closeness Near a Point)

Definition 38.1.2 (Local δ-Closeness Near a Point)**.** Let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, let x_0 be adherent to X, and let $\varepsilon, \delta > 0$. The function f is **ε-close to L near x_0 within δ** if

$$\forall x \in X\ \big(|x - x_0| < \delta \Rightarrow |f(x) - L| \le \varepsilon\big).$$

Remark (Standard quantified statement).

$$\forall x \in X :\ |x - x_0| < \delta\ \Rightarrow\ |f(x) - L| \le \varepsilon.$$

Remark (Interpretation). A local condition: only outputs from the δ-ball around x_0 need lie within ε of L. The input condition here is Within — unpunctured. The limit definition replaces this with InputTolerance (punctured) and asserts that for every ε such a δ exists.

Remark (Dependencies).

- ε-Closeness of a Function to a Value

Definition (Limit of a Function)

Definition 38.1.3 (Limit of a Function)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. A real number L is the **limit of f at** c, written $\lim_{x\to c} f(x) = L$, if

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A \; 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Go to uniqueness proof.

Remark (Standard quantified statement)*.*

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A \; 0 < |x - c| < \delta \;\; \Rightarrow \;\; |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading)*.* Define

$$\begin{aligned} \mathrm{OutputTol}(\varepsilon) &\iff \varepsilon > 0, \\ \mathrm{InputTol}(\delta) &\iff \delta > 0, \\ \mathrm{PuncturedNbhd}(x, c, \delta) &\iff 0 < |x - c| < \delta, \\ \mathrm{OutputNbhd}(y, L, \varepsilon) &\iff |y - L| < \varepsilon. \end{aligned}$$

Then

$$\mathrm{FunctionLimit}(f, A, c, L) \iff \forall \varepsilon \Big(\mathrm{OutputTol}(\varepsilon) \Rightarrow \exists \delta \left(\mathrm{InputTol}(\delta) \wedge \forall x \in A \left[\mathrm{PuncturedNbhd}(x, c, \delta) \Rightarrow \mathrm{O}\right.\right.$$

Remark (Negated quantified statement)*.*

$$\exists \varepsilon_0 > 0 \; \forall \delta > 0 \; \exists x \in A \; 0 < |x - c| < \delta \;\; \wedge \;\; |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading)*.*

$$\begin{aligned} \neg\, \mathrm{FunctionLimit}(f, A, c, L) &\iff \exists \varepsilon_0 \Big(\mathrm{OutputTol}(\varepsilon_0) \wedge \forall \delta \left(\mathrm{InputTol}(\delta) \Rightarrow \exists x \in A \left[\mathrm{PuncturedNbhd}(x, c, \delta)\right.\right. \\ &\iff \exists \varepsilon_0 > 0 \; \forall \delta > 0 \; \exists x \in A \; \big(0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big). \end{aligned}$$

Remark (Failure modes)*.*

- The proposed value L is the wrong target: the function approaches some value $M \neq L$ near c.
- The function does not settle near any single value as x approaches c through A.
- The outputs repeatedly stay at least a fixed distance from L no matter how tightly the inputs are restricted around c.

Remark (Failure mode decomposition)*.*

$$\neg\, \mathrm{FunctionLimit}(f, A, c, L) \iff \exists \varepsilon_0 > 0 \; \forall \delta > 0 \; \exists x \in A \; \Big(\underbrace{0 < |x - c| < \delta}_{\text{arbitrarily close to } c} \wedge \underbrace{|f(x) - L| \geq \varepsilon_0}_{\text{fixed output gap from } L}\Big).$$

Remark (Interpretation). The quantifier structure is asymmetric by design. The output tolerance ε is a *demand*: the caller specifies how precise the output must be. The input tolerance δ is a *response*: the definition asserts one always exists. The input condition is punctured (InputTolerance, not Within) because f may not be defined at c, and because continuity — not limits — is the concept that includes c itself. This is the deepest structural difference between the two definitions. Four critical features: (i) f need not be defined at c; (ii) c must be a cluster point of A else the condition is vacuously satisfied by every L; (iii) δ may depend on ε; (iv) the condition is about all x near c simultaneously, not just one.

Remark (Dependencies).

- Cluster Point

Definition (Sequential Definition of Limit)

Definition 38.1.4 (Sequential Definition of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, the sequence $(f(x_n))$ converges to L.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \Rightarrow f(x_n) \to L\big).$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}_{\text{seq}}(f, c, L) \ :\equiv \ \forall (x_n) \subseteq A \backslash \{c\} : \ \text{ConvergentSequence}(x_n, c) \ \Rightarrow \ \text{ConvergentSequence}(f(x_n),$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \ \wedge \ f(x_n) \not\to L\big).$$

Remark (Interpretation). The sequence (x_n) must lie in $A \setminus \{c\}$: the term $x_n = c$ is excluded, mirroring the $0 <$ in the ε-δ form. The sequential form is the primary tool for both proving limits (via the bridge principle) and disproving them (exhibit one bad sequence). The ε-δ form is better for constructing explicit δ functions.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Definition (Neighbourhood Form of Limit)

Definition 38.1.5 (Neighbourhood Form of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **neighbourhood sense** holds if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a deleted δ-neighbourhood $V^*_\delta(c)$ such that

$$x \in V^*_\delta(c) \cap A \quad \Longrightarrow \quad f(x) \in V_\varepsilon(L).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in V_\delta^*(c) \cap A \; f(x) \in V_\varepsilon(L).$$

Remark (Interpretation). The neighbourhood form packages the same local limit condition in terms of sets around the input point and output value. The domain is still restricted to A, and the neighbourhood of c is deleted so the value of f at c itself is irrelevant.

Remark (Dependencies).

- Limit of a Function
- ε-Neighbourhood
- Deleted ε-Neighbourhood

Proposition

Proposition 38.1.6 (Neighbourhood Form of the Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x \to c} f(x) = L$ if and only if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a δ-neighbourhood $V_\delta(c)$ such that $x \in V_\delta^*(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A \; 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in V_\delta^*(c) \cap A \; f(x) \in V_\varepsilon(L)\Big].$$

Remark (Interpretation). The equivalence is purely notational: $V_\delta^*(c) \cap A$ is exactly the set of $x \in A$ satisfying $0 < |x - c| < \delta$, and $f(x) \in V_\varepsilon(L)$ is exactly $\mathrm{OutputTolerance}(f(x), L, \varepsilon)$. The neighbourhood form makes the geometric content explicit: the image of every punctured δ-slice of A must fit inside the ε-ball around L. This is the form of continuity used in metric space topology.

Remark (Dependencies).

- Limit of a Function
- Neighbourhood Form of Limit
- ε-Neighbourhood
- Deleted ε-Neighbourhood

Theorem

Theorem 38.1.7 (Uniqueness of Limits of Functions)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then f has at most one limit at c.*

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \ \wedge \ \lim_{x\to c} f(x) = L' \ \Rightarrow \ L = L'.$$

Remark (Failure modes). Failure of uniqueness would mean that two distinct real numbers both satisfy the limit definition at the same cluster point.

Remark (Failure mode decomposition). Non-uniqueness would require two values $L \neq L'$ each satisfying the limit definition. The $\varepsilon = |L-L'|/2$ argument shows any x within $\min(\delta_1, \delta_2)$ of c satisfies both $|f(x)-L| < \varepsilon$ and $|f(x)-L'| < \varepsilon$, but then $|L-L'| \leq |L-f(x)|+|f(x)-L'| < 2\varepsilon = |L-L'|$, a contradiction. The cluster point condition is essential: it guarantees such an x exists.

Remark (Interpretation). The limit, when it exists, is uniquely determined by the local behavior of f near c. Uniqueness is what licenses writing $\lim_{x\to c} f(x) = L$ as an equation rather than an assertion.

Remark (Dependencies).

- Limit of a Function

Proposition

Proposition 38.1.8 (Sequential Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Negated quantified statement).

$$\neg \lim_{x\to c} f(x) = L \iff \exists (x_n) \subseteq A \setminus \{c\} :\ x_n \to c \ \wedge\ f(x_n) \not\to L.$$

Remark (Interpretation). The Sequential Criterion is the engine of the algebra of limits. Once it is in hand:

$$\text{algebra of sequence limits} \xRightarrow{\text{Sequential Criterion}} \text{algebra of function limits.}$$

Every sequence result — sum, product, quotient, squeeze — transfers to function limits by feeding a sequence into the criterion and reading the result back. The negation is equally powerful: to disprove $\lim_{x\to c} f(x) = L$, exhibit one sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$ and $f(x_n) \not\to L$.

Remark (Dependencies).

- Limit of a Function
- Sequential Definition of Limit

Corollary

Corollary 38.1.9 (Three Equivalent Forms of the Function Limit). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The ε-δ form of $\lim_{x\to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]\\ \iff &\Big[\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in V_\delta^*(c) \cap A\ f(x) \in V_\varepsilon(L)\Big]\\ \iff &\Big[\forall(x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].\end{aligned}$$

Remark (Interpretation). All three encodings capture the same mathematical condition. The ε-δ form is best for proving explicit bounds. The neighbourhood form is best for geometric reasoning and metric space generalisation. The sequential form is best for importing sequence results and constructing counterexamples.

Remark (Dependencies).

- Limit of a Function
- Neighbourhood Form of Limit
- Sequential Definition of Limit
- Neighbourhood Form of the Limit
- Sequential Criterion for Function Limits

Theorem

Theorem 38.1.10 (Limit Implies Local Boundedness). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \le M$ for all $x \in A$ with $0 < |x - c| < \delta$.*
Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \ \Rightarrow\ \exists\delta > 0\ \exists M > 0\ \forall x \in A\ \big(0 < |x - c| < \delta \Rightarrow |f(x)| \le M\big).$$

Remark (Proof sketch). Apply the definition with $\varepsilon = 1$: get $\delta > 0$ such that $|f(x) - L| < 1$ holds for all x satisfying $0 < |x - c| < \delta$. The triangle inequality then gives $|f(x)| \le |f(x) - L| + |L| < 1 + |L|$. Set $M = 1 + |L|$.

Remark (Interpretation). A finite limit does more than control the output near L: it prevents the function from growing without bound near c. Local boundedness is a coarser statement than limit existence, but it is what is needed for the product and quotient rules.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 38.1.11 (Bounded Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $a, b \in \mathbb{R}$. If $a \leq f(x) \leq b$ for all $x \in A$ with $x \neq c$, and $\lim_{x \to c} f(x) = L$, then $a \leq L \leq b$.*

Go to proof.

Remark (Standard quantified statement).

$$\big(\forall x \in A \setminus \{c\} : \; a \leq f(x) \leq b\big) \;\wedge\; \lim_{x \to c} f(x) = L \;\Rightarrow\; a \leq L \leq b.$$

Remark (Interpretation). If every nearby value of f is trapped in $[a, b]$, the limit cannot escape outside $[a, b]$. This is the function-limit analogue of the sequence result that limits preserve non-strict inequalities. The proof applies the Sequential Criterion and that sequence result directly.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 38.1.12 (Squeeze Theorem for Function Limits). *Let $f, g, h : A \to \mathbb{R}$ and let c be a cluster point of A. Suppose $\lim_{x \to c} f(x) = L = \lim_{x \to c} g(x) = L$, and suppose there exists $\delta_0 > 0$ such that*

$$\forall x \in A, \; 0 < |x - c| < \delta_0 \;\Rightarrow\; f(x) \leq h(x) \leq g(x).$$

Then $\lim_{x \to c} h(x) = L$.

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x \to c} f(x) = L = \lim_{x \to c} g(x) = L \;\wedge\; f \leq h \leq g \text{ near } c \;\Rightarrow\; \lim_{x \to c} h(x) = L.$$

Remark (Interpretation). The proof routes entirely through the Sequential Criterion: convert to sequences, apply the sequential squeeze theorem, convert back. No new ε-δ argument is needed. The condition $f \leq h \leq g$ is only required near c, not globally.

Remark (Dependencies).

- Limit of a Function

- Bounded Limits

Theorem

Theorem 38.1.13 (Limits Are Local)**.** *Let $E \subseteq X \subseteq \mathbb{R}$, let x_0 be a cluster point of E, let $f : X \to \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x \to x_0;\, x \in E} f(x) = L \iff \lim_{x \to x_0;\, x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\lim_{x \to x_0;\, x \in E} f(x) = L \iff \lim_{x \to x_0;\, x \in E \cap (x_0 - \delta,\, x_0 + \delta)} f(x) = L.$$

Remark (Interpretation)*.* The existence and value of FunctionLimit(f, x_0, L) depend only on the behavior of f within any fixed neighbourhood of x_0. Points outside that neighbourhood are irrelevant. This justifies focusing all limit arguments on small balls around c without any loss of generality.

Remark (Dependencies)*.*

- Limit of a Function

Proposition

Proposition 38.1.14 (Limit of Absolute Value)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x \to c} f(x) = L$, then $\lim_{x \to c} |f(x)| = |L|$.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\lim_{x \to c} f(x) = L \;\Rightarrow\; \lim_{x \to c} |f(x)| = |L|.$$

Remark (Negated quantified statement)*.*

$$\lim_{x \to c} |f(x)| = |L| \;\not\Rightarrow\; \lim_{x \to c} f(x) = L.$$

Remark (Failure modes)*.* The absolute values may converge while the signs of $f(x)$ oscillate or approach the opposite sign from the proposed limiting value.

Remark (Proof strategy)*.* The reverse triangle inequality $\big||f(x)| - |L|\big| \leq |f(x) - L|$ feeds the ε-δ definition directly: the same δ that delivers $|f(x) - L| < \varepsilon$ also delivers $\big||f(x)| - |L|\big| < \varepsilon$.

Remark (Interpretation)*.* Taking absolute values can collapse sign information, so convergence of f forces convergence of $|f|$, but convergence of $|f|$ alone does not recover the original limiting sign.

Remark (Dependencies)*.*

- Limit of a Function

Theorem

Theorem 38.1.15 (Composition of Limits)**.** *Let $f : A \to B$, let $g : B \to \mathbb{R}$, let c_1 be a cluster point of A, and let $c_2 \in \mathbb{R}$ be a cluster point of B. Suppose $\lim_{x\to c_1} f(x) = c_2$ and $\lim_{y\to c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x\to c_1} g(f(x)) = L$.*

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c_1} f(x) = c_2 \;\wedge\; \lim_{y\to c_2} g(y) = L \;\wedge\; \big(c_2 \in B \Rightarrow g(c_2) = L\big) \;\Rightarrow\; \lim_{x\to c_1} g(f(x)) = L.$$

Remark (Failure modes). The composition conclusion can fail if the outer function is not controlled at the intermediate value reached by the inner limit.

Remark (Failure mode decomposition). The hypothesis $g(c_2) = L$ is essential. Without it the composition can fail: take $f \equiv c_2$ (constant) and $g(c_2) \neq L$. Then $g(f(x)) = g(c_2) \neq L$ for all x, so $\lim_{x\to c_1} g(f(x)) = L$ fails even though both component limits exist. The failure is that $f(x) = c_2$ is excluded from the punctured condition on g, but f hits c_2 constantly.

Remark (Interpretation). Composition of limits requires care precisely because the input condition is punctured. The extra hypothesis $g(c_2) = L$ bridges the gap: when $f(x) = c_2$, the output condition $g(f(x)) = g(c_2) = L$ is satisfied directly. This hypothesis is equivalent to requiring g to be continuous at c_2 (within B).

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 38.1.16 (Monotone Limit Theorem)**.** *Let $a < b$ and let $f : (a, b) \to \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing f and every $x_0 \in (a, b)$:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

Go to proof.

Remark (Standard quantified statement).

$$f \text{ is monotone on } (a, b) \wedge \exists B > 0\; \forall x \in (a, b)\; (|f(x)| \le B) \;\Rightarrow\; \forall x_0 \in (a, b) : \lim_{x\to x_0^-} f(x) = f(x_0^-) \wedge \lim_{x\to x_0^+} f($$

Remark (Interpretation). Boundedness guarantees the supremum and infimum used as candidate limits exist. Monotonicity forces convergence toward them from each side: for increasing f, the values $f(x)$ for $x < x_0$ form a bounded increasing net approaching their supremum. No oscillation is possible. The theorem establishes that monotone functions can only have jump discontinuities — never oscillatory ones.

Remark (Dependencies).

- Monotone Function
- Bounded Function
- Left-Hand Limit
- Right-Hand Limit

Proposition

Proposition 38.1.17 (Cauchy Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall\ \mathrm{OutputTolerance}(\varepsilon),\ \exists\ \mathrm{InputTolerance}(\delta),\ \forall x, y \in A:\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta\ \Rightarrow\ |f(x)-f(y)$$

Go to proof.

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \Big(\lim_{x\to c} f(x) = L\Big) \iff \forall\varepsilon > 0,\ \exists\delta > 0,\ \forall x, y \in A:\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta \Rightarrow |f(x)-f(y)| <$$

Remark (Interpretation). The Cauchy criterion tests limit existence without knowing L. Two outputs from any sufficiently small punctured ball must lie within ε of each other — the outputs cluster even if their common limit is unknown. This is the function-limit analogue of the Cauchy completeness criterion for sequences, and relies on the completeness of $\mathbb{R}$ in exactly the same way.

Remark (Dependencies).

- Limit of a Function

Algebra of Limits

Theorem

Theorem 38.1.18 (Sum Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f+g)(x) = L + M.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\Big[\forall\varepsilon > 0\ \exists\delta_f > 0\ \forall x \in A\ 0 < |x-c| < \delta_f \Rightarrow |f(x) - L| < \varepsilon\Big]\\
&\quad\wedge \Big[\forall\varepsilon > 0\ \exists\delta_g > 0\ \forall x \in A\ 0 < |x-c| < \delta_g \Rightarrow |g(x) - M| < \varepsilon\Big]\\
&\quad\Rightarrow \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |(f+g)(x) - (L+M)| < \varepsilon.
\end{aligned}$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f + g, c, L + M).$$

Remark (Interpretation). Limits respect addition: sufficiently close values of f and g add to values of $f + g$ close to the sum of the two limits.

Remark (Dependencies).

- Limit of a Function
- Pointwise Sum of Functions

Theorem

Theorem 38.1.19 (Difference Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f - g)(x) = L - M.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right]$$
$$\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |(f - g)(x) - (L - M)| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f - g, c, L - M).$$

Remark (Interpretation). The difference rule is the sum rule applied to $f + (-g)$.

Remark (Dependencies).

- Limit of a Function
- Pointwise Difference of Functions

Theorem

Theorem 38.1.20 (Scalar Multiple Rule for Function Limits)**.** *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then*

$$\lim_{x\to c}(\alpha f)(x) = \alpha L.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |\alpha f(x) - \alpha L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \implies \text{FunctionLimit}(\alpha f, c, \alpha L).$$

Remark (Interpretation). Multiplying a function by a fixed scalar multiplies its limiting value by that same scalar.

Remark (Dependencies).

- Limit of a Function
- Pointwise Scalar Multiple of a Function

Theorem

Theorem 38.1.21 (Product Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c} (fg)(x) = LM.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right] \\ \Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x)g(x) - LM| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(fg, c, LM).$$

Remark (Interpretation). The product rule combines closeness of f to L with local boundedness of g near c.

Remark (Dependencies).

- Limit of a Function
- Pointwise Product of Functions

Theorem

Theorem 38.1.22 (Quotient Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right] \wedge M \neq 0$$
$$\Rightarrow \exists\delta_0 > 0\ \forall x \in A\ 0 < |x-c| < \delta_0 \Rightarrow g(x) \neq 0,$$
$$\text{and}\quad \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow \left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f,c,L) \wedge \mathrm{FunctionLimit}(g,c,M) \wedge M \neq 0 \implies \mathrm{FunctionLimit}(f/g,c,L/M).$$

Remark (Interpretation). A nonzero limiting denominator stays away from zero near the limit point, so division becomes locally legitimate and the quotient limit follows.

Remark (Dependencies).

- Limit of a Function
- Pointwise Quotient of Functions

One-Sided Limits

Definition (Right-Hand Limit)

Definition 38.1.23 (Right-Hand Limit)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. A real number L is the **right-hand limit** of f at c if
$$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ c < x < c + \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{RightLimit}(f,c,L) \ :=\ \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists\varepsilon_0 > 0\ \forall\delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg\ \mathrm{RightLimit}(f,c,L).$$

Remark (Failure modes). A proposed right-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the right.

Remark (Failure mode decomposition).

$$\exists\varepsilon_0 > 0\ \forall\delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the right of c are tested. The value of f at c, if it is defined, is irrelevant to the right-hand limit.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Definition (Left-Hand Limit)

Definition 38.1.24 (Left-Hand Limit)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. A real number L is the **left-hand limit** of f at c if

$$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ c - \delta < x < c \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{LeftLimit}(f, c, L)\ :=\ \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists\varepsilon_0 > 0\ \forall\delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg\ \text{LeftLimit}(f, c, L).$$

Remark (Failure modes). A proposed left-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the left.

Remark (Failure mode decomposition).

$$\exists\varepsilon_0 > 0\ \forall\delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the left of c are tested. This is the mirror image of the right-hand limit definition.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Theorem

Theorem 38.1.25 (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x\to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (c, \infty)\ x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Right-hand limits can be tested by sequences approaching c from above. Every such sequence must force the corresponding output sequence toward L.

Remark (Dependencies).

- Right-Hand Limit
- Sequential Definition of Limit

Corollary

Corollary 38.1.26 (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x\to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (-\infty, c)\ x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Left-hand limits can be tested by sequences approaching c from below. The criterion is the left-sided analogue of the right-hand sequential criterion.

Remark (Dependencies).

- Left-Hand Limit

- Sequential Definition of Limit

Theorem

Theorem 38.1.27 (Two-Sided Limit via One-Sided Limits)**.** *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x \to c} f(x) = L$ if and only if both $\lim_{x \to c^+} f(x) = L$ and $\lim_{x \to c^-} f(x) = L$.*
Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]\\ &\quad\iff \Big[\forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in A\ 0 < x - c < \delta_+ \Rightarrow |f(x) - L| < \varepsilon\Big]\\ &\quad\quad \wedge \Big[\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in A\ 0 < c - x < \delta_- \Rightarrow |f(x) - L| < \varepsilon\Big].\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{FunctionLimit}(f, c, L) \iff \mathrm{RightLimit}(f, c, L) \wedge \mathrm{LeftLimit}(f, c, L).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < |x - c| < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\\ &\quad\iff \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big]\\ &\quad\quad \vee \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big].\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{FunctionLimit}(f, c, L) \iff \neg\,\mathrm{RightLimit}(f, c, L) \vee \neg\,\mathrm{LeftLimit}(f, c, L).$$

Remark (Failure modes)*.* A two-sided limit fails precisely when the right-hand limit fails, the left-hand limit fails, or the two one-sided behaviours do not meet at the same value.

Remark (Failure mode decomposition)*.*

$$\neg\,\mathrm{FunctionLimit}(f, c, L) \iff \neg\,\mathrm{RightLimit}(f, c, L) \vee \neg\,\mathrm{LeftLimit}(f, c, L).$$

Remark (Interpretation)*.* A two-sided limit is exactly the agreement of the two one-sided tests. Both sides must approach the same value.

Remark (Dependencies)*.*

- Limit of a Function
- Right-Hand Limit
- Left-Hand Limit

38.2 Point Continuity

Oscillation — Quick Reference

Concept/Statement	Meaning/Role
Oscillation on a set	Diameter of the image over a set.
Oscillation at a point	Limiting local oscillation near the point.
Zero oscillation criterion	Continuity is exactly vanishing point oscillation.
Discontinuity set	Discontinuities are detected by positive oscillation.

Discontinuity — Quick Reference

Concept/Statement	Meaning/Role
Point of discontinuity	Negation of continuity at a point.
Sequential discontinuity	A bad sequence witnesses failure.
Neighbourhood discontinuity	Neighbourhood form of the same failure.
Equivalence lemma	The discontinuity formulations agree.
Types of discontinuity	Removable, jump, and essential behavior.

Continuity — Quick Reference

Concept/Statement	Meaning/Role
Relative neighbourhood	Neighbourhoods are read inside the domain.
Point continuity	The unpunctured ε-δ condition.
Neighbourhood continuity	Images of nearby inputs stay near the value.
Sequential continuity	Convergent input sequences preserve limits.
Oscillation criterion	Continuity is equivalent to zero local oscillation.

Continuity

Definition (Relative Neighborhood)

Definition 38.2.1 (Relative Neighborhood)**.** Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of a in E if and only if there exists $\delta > 0$ such that $U = E \cap (a-\delta, a+\delta)$. Equivalently, every neighborhood of a in E is a set of the form $U_E(a) := E \cap (a-\delta, a+\delta)$ for some $\delta > 0$.

Remark (Standard quantified statement).

$$\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.$$
$$U \text{ is a neighborhood of } a \text{ in } E \iff \exists \delta > 0 \ \big(U = E \cap (a-\delta, a+\delta)\big).$$

Remark (Definition predicate reading).

$$\text{RelativeNeighborhood}(U, E, a) := \exists \delta > 0 \ \big(U = E \cap (a-\delta, a+\delta)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.\\ &U \text{ fails to be a neighborhood of } a \text{ in } E \iff \forall \delta > 0\ \big(U \neq E \cap (a-\delta, a+\delta)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{RelativeNeighborhood}(U, E, a) \iff \forall \delta > 0\ \big(U \neq E \cap (a-\delta, a+\delta)\big).$$

Remark (Failure modes). Failure occurs precisely when no radius δ makes U coincide with the relative interval slice: for every $\delta > 0$ either U contains a point outside $E \cap (a-\delta, a+\delta)$ or the slice contains a point missing from U.

Remark (Failure mode decomposition).

$$\forall \delta > 0\ \underbrace{\big(\exists x \in U :\ x \notin E \cap (a-\delta, a+\delta)\big)}_{\text{extraneous point}} \vee \underbrace{\big(\exists y \in E \cap (a-\delta, a+\delta) :\ y \notin U\big)}_{\text{missing point}}.$$

Remark (Interpretation). Relative neighborhoods capture the local behavior of E around a by intersecting E with an ordinary open interval centered at a. The construction restricts the ambient notion of openness to the geometry of E, so it is suited to relative topologies. Failure simply means that U never matches the precise slice of E at any scale, typically because U omits some nearby points of E or includes points that fall outside E near a.

Remark (Dependencies).

- Subset
- Epsilon Neighbourhood
- Order on $\mathbb{R}$

Definition (Continuity at a Point)

Definition 38.2.2 (Continuity at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is **continuous at** c if and only if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A : \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{ContinuousAt}(f, A, c) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is not continuous at } c \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f, A, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).$$

Remark (Failure modes). Continuity at c fails precisely when there exists a persistent output gap $\varepsilon_0 > 0$ such that no matter how small the chosen neighborhood of c is, a point of A within that neighborhood keeps the function values at least ε_0 away from $f(c)$. The single failure branch records this inescapable oscillation near c.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{persistent gap}}\quad \underbrace{\forall \delta > 0}_{\text{every neighborhood}}\quad \underbrace{\exists x \in A}_{\text{nearby witness}}\ :\ |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0.$$

Remark (Interpretation). Continuity at c means that the output $f(x)$ can be kept arbitrarily close to $f(c)$ by taking x sufficiently close to c while staying in the domain A. The negated form highlights that a discontinuity arises only when there is a fixed positive separation in output that persists no matter how tightly one zooms in on c, so nearby inputs cannot force the outputs to settle near $f(c)$. This captures the geometric intuition that the graph of a continuous function has no abrupt jumps or breaks at c.

Remark (Dependencies).

- Definition: Limit of a Function

Definition (Continuity at a Point — Neighbourhood Form)

Definition 38.2.3 (Continuity at a Point — Neighbourhood Form)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is continuous at c if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0 :\\ &\qquad f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, c) :\equiv \forall \varepsilon > 0\ \exists \delta > 0 :\ f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A.\\ &\neg\big(f \text{ is continuous at } c\big) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 :\\ &\qquad f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0).\end{aligned}$$

Remark (Negation predicate reading).

$$\text{DiscontinuousAt}(f,c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 :\ f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0).$$

Remark (Failure modes). Failure occurs precisely when some fixed output tolerance cannot be matched by any neighborhood about c. This manifests by finding points of A arbitrarily close to c whose images either exceed the upper endpoint $f(c)+\varepsilon_0$ or fall below the lower endpoint $f(c)-\varepsilon_0$.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \geq f(c)+\varepsilon_0\Big)}_{\text{overshoot}} \vee \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \leq f(c)-\varepsilon_0}_{\text{undershoot}}$$

Remark (Interpretation). Continuity at c demands that every output tolerance around $f(c)$ has a matching neighborhood around c whose image remains confined to the tolerance band. The negation says that one tolerance band resists every neighborhood choice, so some sequence of points from A approaches c while their images either spike above or dip below $f(c)$ by at least a fixed amount. Geometrically, arbitrarily tight vertical strips around the graph above c must be achievable by shrinking the horizontal window; failure means the graph keeps escaping that strip no matter how closely the input variable hugs c.

Definition (Continuity at a Point — Sequential Form)

Definition 38.2.4 (Continuity at a Point — Sequential Form)**.** Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f\colon A \to \mathbb{R}$. The function f is continuous at c if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in A, the convergence $\lim_{n\to\infty} x_n = c$ implies $\lim_{n\to\infty} f(x_n) = f(c)$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.\\ &f \text{ is continuous at } c \iff \forall (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) = f(c)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{ContinuousAt}(f,c) :\!= \forall x_n\ \Big(\text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \Rightarrow \text{ConvergentSequence}(f(x_n), f$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.\\ &f \text{ is not continuous at } c \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A\ \Big(\lim_{n\to\infty} x_n = c \wedge \lim_{n\to\infty} f(x_n) \neq f(c)\Big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f,c) \iff \exists x_n\ \Big(\mathrm{Sequence}(x_n,A) \wedge \mathrm{ConvergentSequence}(x_n,c) \wedge \neg\,\mathrm{ConvergentSequence}(f(x_n$$

Remark (Failure modes). Continuity at c fails sequentially when there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in A with $x_n \to c$ but $f(x_n)$ does not converge to $f(c)$. This manifests in two distinct ways: either $f(x_n)$ converges to some limit different from $f(c)$, or $f(x_n)$ fails to converge entirely (oscillating without settling or diverging to infinity).

Remark (Failure mode decomposition).

$$\neg(f \text{ continuous at } c) \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A :\ \underbrace{\lim_{n\to\infty} x_n = c}_{\text{valid approach}} \wedge \underbrace{\lim_{n\to\infty} f(x_n) \neq f(c)}_{\text{image sequence fails}}.$$

Remark (Interpretation). The sequential characterization asserts that information about f near c is completely encoded by its action on limits of sequences taken within the domain. Any approach to c through admissible points must transport convergent input sequences to convergent output sequences with the expected limit; conversely, a single pathological sequence suffices to expose a discontinuity. Geometrically, this captures the idea that f cannot introduce jumps or oscillations that persist along every infinitesimal path toward c.

Remark (Dependencies).

Definition 38.2.2.

Characterization of Discontinuity

Definition (Point of Discontinuity)

Definition 38.2.5 (Point of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $a \in A$. The function f is discontinuous at a if and only if f is not continuous at a.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ a \in A$.

$$f \text{ is discontinuous at } a \iff \neg\Big(\forall\varepsilon>0\ \exists\delta>0\ \forall x\in A\ \big(|x-a|<\delta \Rightarrow |f(x)-f(a)|<\varepsilon\big)\Big).$$

Remark (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f,a) :\!= \neg\,\mathrm{ContinuousAt}(f,a).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ a \in A$.

$$f \text{ is discontinuous at } a \iff \exists\varepsilon>0\ \forall\delta>0\ \exists x\in A\ \big(|x-a|<\delta \wedge |f(x)-f(a)| \geq \varepsilon\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f,a) \iff \mathrm{ContinuousAt}(f,a) \iff \forall\varepsilon>0\ \exists\delta>0\ \forall x\in A\ \big(|x-a|<\delta \Rightarrow |f(x)-f(a)| <$$

Remark (Failure modes). Discontinuity arises either because the nearby values of f do not approach any single real number as $x \to a$, or because a well-defined limit exists but disagrees with the actual value $f(a)$.

Remark (Failure mode decomposition).

$$\underbrace{\neg\exists L \in \mathbb{R}\ \operatorname{FunctionLimit}(f, A, a, L)}_{\text{no limit exists}} \vee \underbrace{\big(\exists L \in \mathbb{R}\ \operatorname{FunctionLimit}(f, A, a, L) \wedge L \neq f(a)\big)}_{\text{limit-value mismatch}}.$$

Remark (Interpretation). A point of discontinuity is detected by the breakdown of the usual epsilon-delta control. Understanding removable, jump, and essential cases guides how to remedy or classify singular behavior in proofs.

Remark (Dependencies).

Definition 38.2.2.

Definition (Sequential Characterization of Discontinuity)

Definition 38.2.6 (Sequential Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq A$ such that $x_n \to c$ while $f(x_n) \not\to f(c)$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists (x_n) \subseteq A\ \big[x_n \to c\ \wedge\ f(x_n) \not\to f(c)\big].$$

Remark (Definition predicate reading).

$$\operatorname{DiscontinuousAt}(f, c) := \exists (x_n) \subseteq A :\ x_n \to c\ \wedge\ \neg\big(f(x_n) \to f(c)\big).$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c).$$

Remark (Negation predicate reading).

$$\neg \operatorname{DiscontinuousAt}(f, c) \iff \operatorname{ContinuousAt}(f, c) \iff \forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c).$$ ∎

Remark (Failure modes). The characterization fails — meaning f is NOT sequentially discontinuous — when every sequence (x_n) in A converging to c has $f(x_n) \to f(c)$. No sequential witness exists.

Remark (Failure mode decomposition).

$$\neg \operatorname{DiscontinuousAt}(f, c) \iff \underbrace{\forall (x_n) \subseteq A :\ x_n \to c\ \Rightarrow\ f(x_n) \to f(c)}_{\text{every approach preserves the limit}}.$$

Remark (Interpretation). A discontinuity at c can be certified by a single sequence approaching c whose images refuse to settle at $f(c)$.

Remark (Dependencies).

Definition 38.2.4; Definition 38.2.5.

Definition (Neighbourhood Characterization of Discontinuity)

Definition 38.2.7 (Neighbourhood Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a neighborhood V of $f(c)$ such that for every neighborhood U of c relative to A there exists $x \in U \cap A$ with $f(x) \notin V$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A\ \big(f(x) \notin V\big).$$

Remark (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f, c) \coloneqq \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V.$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \mathrm{ContinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V$$

Remark (Failure modes)*.* The neighbourhood discontinuity characterization fails — meaning f is actually continuous — when every output neighbourhood V of $f(c)$ can be matched by some input neighbourhood U such that f maps all of $U \cap A$ into V. No perpetually-missed output neighbourhood exists.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{DiscontinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c)\ \underbrace{\exists U \text{ nbhd of } c}_{\text{some input window}} :\ \underbrace{\forall x \in U \cap A :\ f(x) \in V}_{\text{all outputs inside tolerance}}.$$

Remark (Interpretation)*.* Discontinuity means a fixed output neighborhood V is perpetually missed no matter how tightly the input is restricted around c.

Remark (Dependencies).

Definition 38.2.1.

Lemma (Equivalent Formulations of Discontinuity at a Point)

Lemma 38.2.8 (Equivalent Formulations of Discontinuity at a Point)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The following are equivalent:*

1. *c is a point of discontinuity in the ε-δ sense.*
2. *c is a sequential point of discontinuity.*
3. *c is a neighbourhood point of discontinuity.*

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&(1)\ \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A:\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\
\iff &(2)\ \exists (x_n) \subseteq A:\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\
\iff &(3)\ \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A:\ f(x) \notin V.
\end{aligned}
$$

Remark (Predicate reading).

$$
\begin{aligned}
\mathrm{DiscontinuousAt}(f, c) &\iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A:\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\
&\iff \exists (x_n) \subseteq A:\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\
&\iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A:\ f(x) \notin V.
\end{aligned}
$$

Remark (Failure modes). The equivalence cannot fail for functions at limit points of A: the proof that all three characterizations are equivalent is constructive in both directions. In the degenerate case where c is not a limit point of A, the punctured-neighbourhood conditions are vacuously satisfied by every L, and the equivalence becomes trivial rather than informative.

Remark (Interpretation). The three characterizations of discontinuity are mutually convertible. Analysts select whichever viewpoint simplifies a particular argument.

Remark (Dependencies).

- Point of Discontinuity
- Sequential Characterization of Discontinuity
- Neighbourhood Characterization of Discontinuity

Definition (Types of Discontinuity)

Definition 38.2.9 (Types of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $x_0 \in A$ with f discontinuous at x_0.

(i) *Removable* iff $\exists L \in \mathbb{R}:\ \lim_{x \to x_0} f(x) = L$ and $L \neq f(x_0)$.

(ii) *Jump* iff $\exists L_-, L_+ \in \mathbb{R}:\ \lim_{x \to x_0^-} f(x) = L_-$, $\lim_{x \to x_0^+} f(x) = L_+$, and $L_- \neq L_+$.

(iii) *Essential* iff it is not removable.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\text{Removable} \iff \exists L \in \mathbb{R}\ \Big(\lim_{x \to x_0} f(x) = L \wedge L \neq f(x_0)\Big). \\
&\text{Jump} \iff \exists L_-, L_+ \in \mathbb{R}\ \Big(\lim_{x \to x_0^-} f(x) = L_- \wedge \lim_{x \to x_0^+} f(x) = L_+ \wedge L_- \neq L_+\Big). \\
&\text{Essential} \iff \neg \exists L \in \mathbb{R}\ \Big(\lim_{x \to x_0} f(x) = L \wedge L \neq f(x_0)\Big).
\end{aligned}
$$

Remark (Definition predicate reading).

$$
\begin{aligned}
\mathrm{RemovableDiscontinuity}(f, x_0) &:= \exists L \in \mathbb{R}\ \big(\mathrm{FunctionLimit}(f, A, x_0, L) \wedge L \neq f(x_0)\big). \\
\mathrm{JumpDiscontinuity}(f, x_0) &:= \exists L_-, L_+ \in \mathbb{R}\ \big(\mathrm{LeftLimit}(f, A, x_0, L_-) \wedge \mathrm{RightLimit}(f, A, x_0, L_+) \wedge L_- \neq \\
\mathrm{EssentialDiscontinuity}(f, x_0) &:= \neg\, \mathrm{RemovableDiscontinuity}(f, x_0).
\end{aligned}
$$

Remark (Negated quantified statement).

$$\neg\text{Removable} \iff \forall L \in \mathbb{R}\ \big(\lim_{x\to x_0} f(x) \neq L \lor L = f(x_0)\big),$$
$$\neg\text{Jump} \iff \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x\to x_0^-} f(x) \neq L_- \lor \lim_{x\to x_0^+} f(x) \neq L_+ \lor L_- = L_+\big),$$
$$\neg\text{Essential} \iff \text{Removable}.$$

Remark (Negation predicate reading).

$$\neg\,\text{RemovableDiscontinuity}(f, x_0) \iff \forall L \in \mathbb{R}\ \big(\neg\,\text{FunctionLimit}(f, A, x_0, L) \lor L = f(x_0)\big).$$
$$\neg\,\text{JumpDiscontinuity}(f, x_0) \iff \forall L_-, L_+ \in \mathbb{R}\ \big(\neg\,\text{LeftLimit}(f, A, x_0, L_-) \lor \neg\,\text{RightLimit}(f, A, x_0, L_+)$$
$$\neg\,\text{EssentialDiscontinuity}(f, x_0) \iff \text{RemovableDiscontinuity}(f, x_0).$$

Remark (Failure modes).

- Removable discontinuity fails when no bilateral limit exists at x_0, or when the bilateral limit exists but equals $f(x_0)$ (making the function actually continuous).
- Jump discontinuity fails when at least one of the one-sided limits does not exist, or when both one-sided limits exist but are equal (making it potentially removable or continuous).
- Essential discontinuity fails when the discontinuity is removable (the bilateral limit exists and differs from $f(x_0)$).

Remark (Failure mode decomposition).

$$\neg\,\text{RemovableDiscontinuity}(f, x_0) \iff \underbrace{\neg\exists L \in \mathbb{R}\ \lim_{x\to x_0} f(x) = L}_{\text{no bilateral limit exists}} \lor \underbrace{\lim_{x\to x_0} f(x) = f(x_0)}_{\text{limit equals function value}}.$$
$$\neg\,\text{JumpDiscontinuity}(f, x_0) \iff \underbrace{\neg\exists L_- \in \mathbb{R}\ \lim_{x\to x_0^-} f(x) = L_-}_{\text{no left limit}} \lor \underbrace{\neg\exists L_+ \in \mathbb{R}\ \lim_{x\to x_0^+} f(x) = L_+}_{\text{no right limit}}$$
$$\lor \underbrace{\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x)}_{\text{one-sided limits agree}}.$$

Remark (Interpretation). Removable discontinuities can be healed by redefining $f(x_0)$. Jump discontinuities provide well-defined one-sided limits useful in integration theory. Essential discontinuities are pathological and require more global tools.

Remark (Dependencies).

- Point of Discontinuity
- Limit of a Function
- Left-Hand Limit
- Right-Hand Limit

Oscillation

Definition (Oscillation on a Set)

Definition 38.2.10 (Oscillation on a Set)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $E \subseteq A$. The oscillation of f on E is the extended real number

$$\omega(f; E) \coloneqq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\} = \operatorname{diam}(f(E)).$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0, \infty].\\ &\omega(f; E) = \omega \iff \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\operatorname{OscillationOn}(f, E, \omega) \coloneqq \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0, \infty].\\ &\omega(f; E) \neq \omega \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg \operatorname{OscillationOn}(f, E, \omega) \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Failure modes)*.* The proposed value ω fails either because it is not an upper bound for pairwise differences of function values on E, or because it is an upper bound but not the least such upper bound.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists x_1, x_2 \in E :\ |f(x_1) - f(x_2)| > \omega}_{\text{not an upper bound}} \vee \underbrace{\exists \eta < \omega\ \forall x_1, x_2 \in E :\ |f(x_1) - f(x_2)| \le \eta}_{\text{not least}}.$$

Remark (Interpretation)*.* Oscillation on a set records the spread of the image — the diameter of $f(E)$. Small oscillation means all values of f on E lie close together. Zero oscillation means f is constant on E.

Remark (Dependencies)*.*

- Function
- Subset
- Supremum
- Absolute Value on $\mathbb{R}$

Definition (Oscillation at a Point)

Definition 38.2.11 (Oscillation at a Point)**.** Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \varnothing$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of f at x_0 is the extended real number

$$\omega(f; x_0) = \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big),$$

with $\omega(f; A)$ denoting the oscillation of f on the set A. Because the function $\delta \mapsto \omega\big(f;\, U_\delta(x_0) \cap E\big)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty].\\
&\big[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing)\ \wedge\ \omega(f; x_0) = \omega\big]\\
&\iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall r\ (0 < r < \delta \Rightarrow |\omega(f; U_r(x_0) \cap E) - \omega| < \varepsilon).
\end{aligned}$$

Remark (Definition predicate reading).

$$\operatorname{OscillationAt}(f, x_0, \omega) := \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big) = \omega, \quad U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}.$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty].\\
&\neg\big[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing)\ \wedge\ \omega(f; x_0) = \omega\big]\\
&\iff \Big[\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing\Big]\\
&\qquad \vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0)\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg \operatorname{OscillationAt}(f, x_0, \omega) \iff \Big[\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing\Big] \vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E$$

Remark (Failure modes). The definition fails in two distinct ways: either the point x_0 has a punctured neighborhood missing E, so there are no arbitrarily small samples to measure, or the local oscillation values near x_0 refuse to stabilize, preventing the limit from existing.

Remark (Failure mode decomposition).

$$\neg \operatorname{OscillationAt}(f, x_0, \omega) = \underbrace{\exists \delta_0 > 0 :\ U_{\delta_0}(x_0) \cap E = \varnothing}_{\text{no nearby sample points}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E)}_{\text{oscillation limit does not exist}}$$

Remark (Interpretation). Oscillation at x_0 measures the largest possible jump in function values that survives after one zooms in arbitrarily tightly around x_0. The shrinking neighborhoods $U_r(x_0) \cap E$ supply finer and finer samples, and the oscillation on each such set records the spread of f there. If these spreads settle to a single number, that number captures the local variability; it is zero precisely when f becomes arbitrarily close to a single value, i.e., when f is continuous at x_0. Failure occurs either because x_0 has no nearby domain points or because the spreads fluctuate indefinitely, reflecting wild local behavior.

Remark (Dependencies).

Definition 38.2.10

Theorem (Continuity Characterised by Oscillation)

Theorem 38.2.12 (Continuity Characterised by Oscillation)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f;c) = 0$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A.\\
&\big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\big)\\
&\iff\\
&\left(\inf_{\delta>0} \sup\big\{|f(x) - f(y)| : x, y \in A,\ |x-c| < \delta,\ |y-c| < \delta\big\} = 0\right).
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousAt}(f,c) \iff \mathrm{OscillationAt}(f,c,0).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A.$

$\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ (|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x) - f(y)| : x, y \in A,\ |x-c| < \delta,$

$\vee$

$\Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ (|x-c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x) - f(y)| : x, y \in A,\ |x-c| <$

Remark (Negation predicate reading).

$$\neg\big(\mathrm{ContinuousAt}(f,c) \iff \mathrm{OscillationAt}(f,c,0)\big).$$

Remark (Failure modes). The equivalence fails precisely in two mutually exclusive ways: either f remains continuous at c while the oscillation retains a positive lower bound, or $\omega(f;c)$ vanishes even though f does not satisfy the ε-δ continuity condition at c.

Remark (Failure mode decomposition).

$$\underbrace{\mathrm{ContinuousAt}(f,c) \wedge \neg\,\mathrm{OscillationAt}(f,c,0)}_{\text{Positive oscillation despite continuity}} \vee \underbrace{\neg\,\mathrm{ContinuousAt}(f,c) \wedge \mathrm{OscillationAt}(f,c,0)}_{\text{Zero oscillation without continuity}}.$$

Remark (Interpretation). The theorem states that measuring the oscillation of f near c captures exactly the same local regularity as the classical ε-δ definition of continuity. If oscillation collapses to zero, every pair of nearby function values becomes arbitrarily close, forcing $f(x)$ to approach $f(c)$. Conversely, any discontinuity manifests as a nonzero oscillation, because nearby points can produce separated outputs. The standard failure therefore arises from detecting a residual jump or oscillatory behavior in arbitrarily small neighborhoods of c.

Remark (Dependencies).

Definition 38.2.2, Definition 38.2.11.

Proposition

Proposition 38.2.13 (Discontinuity Set via Oscillation)**.** *Let* $f : E \to \mathbb{R}$*. Then*

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f;x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f;x) \geq 1/n\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } f : E \to \mathbb{R}.\\
&\forall x \in E : \ x \in D(f) \iff \omega(f;x) > 0 \iff \exists n \in \mathbb{N} : \ \omega(f;x) \geq \tfrac{1}{n}.
\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}
x \in D(f) &\iff \text{DiscontinuousAt}(f,x)\\
&\iff \exists \omega \ (\text{OscillationAt}(f,x,\omega) \wedge \omega > 0)\\
&\iff \exists n \in \mathbb{N} \ \exists \omega \ (\text{OscillationAt}(f,x,\omega) \wedge \omega \geq \tfrac{1}{n}).
\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } f : E \to \mathbb{R}.\\
&\neg\Big[\forall x \in E \ \big(x \in D(f) \iff \omega(f;x) > 0 \iff \exists n \in \mathbb{N} \ \omega(f;x) \geq \tfrac{1}{n}\big)\Big]\\
&\iff \exists x \in E \ \Big((x \in D(f) \wedge \omega(f;x) = 0) \vee (x \notin D(f) \wedge \omega(f;x) > 0)\\
&\qquad \vee (\omega(f;x) > 0 \wedge \forall n \in \mathbb{N} \ \omega(f;x) < \tfrac{1}{n})\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
\exists x \in E \ \Big(&\exists \omega \ (\text{DiscontinuousAt}(f,x) \wedge \text{OscillationAt}(f,x,\omega) \wedge \omega = 0) \vee\\
&\exists \omega \ (\neg \text{DiscontinuousAt}(f,x) \wedge \text{OscillationAt}(f,x,\omega) \wedge \omega > 0) \vee\\
&\exists \omega \ (\text{OscillationAt}(f,x,\omega) \wedge \omega > 0 \wedge \forall n \in \mathbb{N} \ \omega < \tfrac{1}{n})\Big).
\end{aligned}$$

Remark (Failure modes). The proposition fails precisely when some $x \in E$ is marked as discontinuous although its oscillation vanishes, or when x is declared continuous despite a positive oscillation, or when a positive oscillation never exceeds any reciprocal integer, contradicting the Archimedean step that converts $\omega(f;x) > 0$ into the countable union description.

Remark (Failure mode decomposition).

$$\exists x \in E \begin{cases} x \in D(f) \wedge \omega(f;x) = 0 & \text{(Type I)}, \\ x \notin D(f) \wedge \omega(f;x) > 0 & \text{(Type II)}, \\ \omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\ \omega(f;x) < \frac{1}{n} & \text{(Type III)}. \end{cases}$$

Remark (Interpretation). Oscillation detects discontinuity quantitatively: every discontinuity forces $\omega(f;x)$ to stay above some positive threshold, and conversely the vanishing of oscillation characterises continuity. Because each set $\{x \in E : \omega(f;x) \geq 1/n\}$ is closed, the discontinuity set $D(f)$ is an F_σ, a countable union of closed sets. This structural description is the gateway to measure-theoretic criteria such as the Lebesgue characterisation of Riemann integrability, where the size of $D(f)$ controls integrability.

Remark (Dependencies).

Definition 38.2.5; Definition 38.2.11.

38.3 Global Theorems

Global Theorems — Quick Reference

Concept/Statement	Meaning/Role
Bounded on a set	The output set is bounded in $\mathbb{R}$.
Boundedness theorem	Continuous functions on $[a,b]$ are bounded.
Absolute extrema	Global maximum and minimum values on a set.
Extreme value theorem	Continuous functions on $[a,b]$ attain extrema.
Intermediate value theorem	Continuous functions do not skip values.
Image interval theorem	$f([a,b])$ is a closed bounded interval.
Darboux property	Interval values are preserved without gaps.

Intermediate Value Theorem

Definition (Bounded on a Set)

Definition 38.3.1 (Bounded on a Set)**.** Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The function f is bounded on A if and only if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.$$
$$f \text{ is bounded on } A \iff \exists M > 0\ \forall x \in A\ (|f(x)| \leq M).$$

Remark (Definition predicate reading).

$$\text{BoundedOn}(f, A) \coloneqq \exists M > 0\ \forall x \in A\ (|f(x)| \leq M).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \wedge \forall x \in [a,b]\ (f(x) \neq c) \Longrightarrow \Big[\big(\forall x \in [a,b]\ f(x) > c\big) \vee \big(\forall x \in [a,b]\ f(x) < c\big)\Big].$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.$$
$$\neg(f \text{ bounded on } A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f,A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).$$

Remark (Failure modes). Failure occurs precisely when every proposed positive bound M is exceeded by some value of $|f(x)|$ with $x \in A$.

Remark (Failure mode decomposition).

$$\underbrace{\forall M > 0}_{\text{no admissible bound}}\quad \underbrace{\exists x \in A\ (|f(x)| > M)}_{\text{value exceeding } M}.$$

Remark (Interpretation). The definition packages the informal idea that the values of f stay within a fixed vertical strip over A. The witness M serves as a uniform ceiling for the magnitudes $|f(x)|$, so once such an M is known the graph of f never escapes the band $[-M,M]$. Failure means that the outputs of f grow without limit along some sequence of points inside A, so no finite strip can contain the entire image. Boundedness on a set is the foundational regularity hypothesis that underlies the boundedness and extreme value theorems for continuous functions on compact domains.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Boundedness Theorem)

Theorem 38.3.2 (Boundedness Theorem)**.** *Let $a < b$ and let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a,b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a < b \text{ and } f : [a,b] \to \mathbb{R}.$$
$$\Big(\forall c \in [a,b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a,b]\big(|x-c| < \delta \Rightarrow |f(x)-f(c)| < \varepsilon\big)\Big)$$
$$\Longrightarrow \exists M > 0\ \forall x \in [a,b]\ (|f(x)| \leq M).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \Longrightarrow \mathrm{BoundedOn}(f,[a,b]).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a<b \text{ and } f:[a,b]\to\mathbb{R}.\\
&\Big(\forall c\in[a,b]\ \forall\varepsilon>0\ \exists\delta>0\ \forall x\in[a,b]\big(|x-c|<\delta\Rightarrow|f(x)-f(c)|<\varepsilon\big)\Big)\\
&\qquad\land\Big(\forall M>0\ \exists x\in[a,b]\ (|f(x)|>M)\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \land \neg\,\mathrm{BoundedOn}(f,[a,b]).$$

Remark (Failure modes). Each component hypothesis is indispensable: without closedness (for instance $f(x)=1/x$ on $(0,1]$) continuity does not prevent divergence near the missing endpoint; without bounded domain (for example $f(x)=x$ on $[0,\infty)$) linear growth forces unbounded values; without continuity (take $f(0)=0$ and $f(x)=1/x$ for $x\in(0,1]$) the discontinuity at 0 again creates arbitrarily large magnitudes.

Remark (Failure mode decomposition).

$$\neg\big[\mathrm{ClosedInterval}(a,b)\land\mathrm{BoundedSet}([a,b])\land\mathrm{ContinuousOn}(f,[a,b])\big] = \underbrace{\neg\,\mathrm{ClosedInterval}(a,b)}_{\text{missing endpoint}} \lor \underbrace{\neg\,\mathrm{Bounde}}_{\text{unbounde}}$$

Remark (Interpretation). Continuity ties nearby inputs to nearby outputs, and the compact interval $[a,b]$ can be covered by finitely many neighborhoods in which the oscillation of f is controlled; taking the maximal oscillation among this finite cover supplies a global bound. Failure typically arises when the domain lacks compactness: an omitted endpoint allows the function to escape control near the hole, an unbounded interval lets values diverge along the ray, and a point of discontinuity can spike the function without upsetting nearby behavior elsewhere.

Remark (Dependencies).

Definition 38.2.2; Definition 38.3.1.

Definition (Absolute Maximum and Absolute Minimum)

Definition 38.3.3 (Absolute Maximum and Absolute Minimum)**.** Let $A\subseteq\mathbb{R}$ and let $f\colon A\to\mathbb{R}$. A point $x^*\in A$ is an absolute maximum of f on A if and only if $f(x^*)\geq f(x)$ for every $x\in A$. A point $x_*\in A$ is an absolute minimum of f on A if and only if $f(x_*)\leq f(x)$ for every $x\in A$.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A\subseteq\mathbb{R},\ f\colon A\to\mathbb{R},\ x^*,x_*\in A.\\
&x^* \text{ is an absolute maximum of } f \text{ on } A \Longleftrightarrow \forall x\in A\ (f(x)\leq f(x^*)).\\
&x_* \text{ is an absolute minimum of } f \text{ on } A \Longleftrightarrow \forall x\in A\ (f(x_*)\leq f(x)).
\end{aligned}$$

Remark (Definition predicate reading).

$$(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A)) \iff (x^* \in A) \wedge \forall x \in A\ (f(x) \le f(x^*)),$$
$$(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A)) \iff (x_* \in A) \wedge \forall x \in A\ (f(x_*) \le f(x)).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x^*, x_* \in \mathbb{R}.\\
&x^* \text{ fails to be an absolute maximum of } f \text{ on } A\\
&\iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big).\\
&x_* \text{ fails to be an absolute minimum of } f \text{ on } A\\
&\iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\big[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))\big] \iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big),$$
$$\neg\big[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))\big] \iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).$$

Remark (Failure modes). An alleged absolute maximum can fail because the proposed point lies outside the domain A, or because some $x \in A$ produces a strictly larger function value. An alleged absolute minimum fails for the analogous reasons, either by leaving the domain or by admitting a point in A with a strictly smaller value.

Remark (Failure mode decomposition).

$$\neg\big[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))\big] = \underbrace{(x^* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x) > f(x^*))\big)}_{\text{better value}},$$
$$\neg\big[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))\big] = \underbrace{(x_* \notin A)}_{\text{external point}} \vee \underbrace{\big(\exists x \in A\ (f(x_*) > f(x))\big)}_{\text{better value}}.$$

Remark (Interpretation). An absolute maximum identifies the highest value that f attains on its domain A, together with a point of attainment, and the absolute minimum identifies the lowest attained value. These extrema describe the global ceiling and floor of the graph restricted to A, so they are the endpoints of the vertical range. Failure occurs either when the candidate point is not part of the domain or when another domain point produces a superior value, reflecting that global optimization is sensitive to both membership and comparative size. Geometrically the absolute maximum sits at the tallest point of the graph above A, while the absolute minimum sits at the lowest point.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

- Maximum
- Minimum

Theorem (Extreme Value Theorem)

Theorem 38.3.4 (Extreme Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \le f(x) \le f(x_M)$ for every $x \in [a, b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}$$
$$\exists x_m, x_M \in [a, b]\ \forall x \in [a, b]\ (f(x_m) \le f(x) \le f(x_M)).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \implies \exists x_m, x_M \in [a, b]\ \big(\mathrm{Maximum}(f(x_M), f([a, b])) \wedge \mathrm{Minimum}(f(x_m), f([a, b]))\big).$$

Remark (Failure modes). The theorem can fail if the compact interval hypothesis is lost, or if the function is not continuous on the whole interval. Without compactness an extremal value may exist only as a limiting value, and without continuity the graph may jump over its supremum or infimum.

Remark (Failure mode decomposition).

$$\underbrace{\neg(a < b)}_{\text{invalid interval}} \vee \underbrace{\neg\big(f : [a, b] \to \mathbb{R}\big)}_{\text{wrong function type}} \vee \underbrace{\exists c \in [a, b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \ge \varepsilon\big)}_{\text{loss of continuity}}.$$

Remark (Interpretation). Continuity on a closed and bounded interval forces the image of the interval to inherit the compact features of the domain. The function cannot escape to infinity or oscillate without settling, so it must reach both its highest and lowest values somewhere in the interval. When the hypotheses fail, the typical obstruction is the loss of compactness: on an open interval the function might drift off without achieving its supremum, and without continuity it may jump over candidate extrema.

Remark (Dependencies).

Uses Definition 38.3.3 and Theorem 38.3.2.

Theorem (Location of Roots)

Theorem 38.3.5 (Location of Roots)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a, b)$ such that $f(c) = 0$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } f : [a, b] \to \mathbb{R} \text{ continuous.}$$
$$\big(f(a) < 0 < f(b) \vee f(a) > 0 > f(b)\big) \Rightarrow \exists c \in (a, b)\ \big(f(c) = 0\big).$$

Remark (Failure modes). The root conclusion can fail if f is not continuous on $[a, b]$, or if the endpoint values do not have opposite signs.

Remark (Failure mode decomposition).

$$\underbrace{\exists c \in [a,b]\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in [a,b]\ \big(|x-c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}} \vee \underbrace{\big(f(a)f(b) \geq 0\big)}_{\text{no sign change}}.$$

Remark (Interpretation). Continuity forces the image of the interval $[a, b]$ to contain every intermediate value between $f(a)$ and $f(b)$. A sign change therefore compels the graph to cross the horizontal axis, producing a root strictly inside the interval. Failure occurs when the endpoint values share the same sign or when f is not continuous, because then the image need not sweep across zero. Geometrically, the theorem asserts that a continuous curve connecting opposite sides of the x-axis must intersect the axis somewhere in between.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

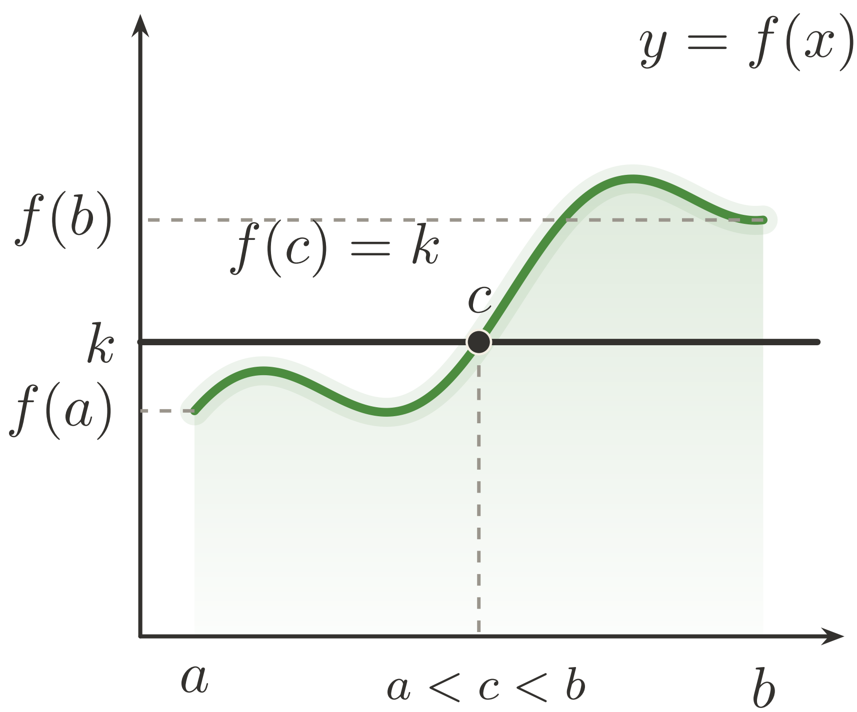


Figure 38.1: Intermediate Value Theorem: a continuous path hits every intermediate height.

Theorem (Bolzano's Intermediate Value Theorem)

Theorem 38.3.6 (Bolzano's Intermediate Value Theorem). *Let $I \subseteq \mathbb{R}$ be an interval, let $f\colon I \to \mathbb{R}$ be continuous on I, and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that $\min\{a, b\} < c < \max\{a, b\}$ and $f(c) = k$. Go to proof.*

Remark (Standard quantified statement).

$$\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall a, b \in I\ \forall k \in \mathbb{R}\ \Big[\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in$$

Remark (Failure modes). The conclusion can fail only if the continuity-on-an-interval hypothesis breaks down, or if the proposed level k is not strictly between the two attained endpoint values.

Remark (Failure mode decomposition).

$$\begin{aligned}&\underbrace{\exists x, y \in I\ \exists z \in \mathbb{R}\ \big(x < z < y \wedge z \notin I\big)}_{\text{not an interval}}\\ \vee\ &\underbrace{\exists c \in I\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}}\\ \vee\ &\underbrace{\neg\big(f(a) < k < f(b)\big)}_{\text{level not between endpoint values}}.\end{aligned}$$

Remark (Interpretation). A continuous real-valued function on an interval cannot skip any intermediate scalar between two attained values, so the graph must cross every horizontal level that lies strictly between $f(a)$ and $f(b)$. The conclusion provides an interior point c whose image equals the prescribed level k, ensuring root-finding for $f - k$. If the hypothesis fails, either f is not continuous or the level k does not lie between the endpoint values, and no such interior point need exist.

Remark (Dependencies).

- Function
- Subset
- Continuity at a Point
- Order on $\mathbb{R}$
- Maximum
- Minimum

Theorem (Image of Closed Bounded Interval Is Closed Bounded Interval)

Theorem 38.3.7 (Image of Closed Bounded Interval Is Closed Bounded Interval). *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then $f([a, b]) = [\min_{[a,b]} f, \max_{[a,b]} f]$. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \forall f : [a, b] \to \mathbb{R} :$$
$$\Big(a < b \wedge \forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big) \Longrightarrow f([a, b]) = [\min_{[a,b]} f$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \Longrightarrow f([a,b]) = [\min_{[a,b]} f, \max_{[a,b]} f].$$

Remark (Interpretation). The statement combines the extreme value theorem, which supplies both a minimum and a maximum on $[a,b]$, with the intermediate value theorem, which fills in every intermediate value between them. Consequently the image $f([a,b])$ is a closed interval with endpoints given by the absolute extremal values of f on $[a,b]$, showing that continuous functions map compact intervals onto compact intervals without introducing gaps.

Remark (Dependencies).

Extreme Value Theorem, Bolzano's Intermediate Value Theorem

Theorem (Preservation of Intervals)

Theorem 38.3.8 (Preservation of Intervals)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous. Then $f(I)$ is an interval.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\, \forall f : I \to \mathbb{R} :\\
&\quad \Big[\big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)\\
&\quad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\Big] \Longrightarrow \forall y_1, y_2 \in f(I)\ \forall y \in \mathbb{R}\ \big(y_1 <
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, I) \Longrightarrow \mathrm{Interval}(f(I)).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\exists I \subseteq \mathbb{R}\ \exists f : I \to \mathbb{R}\ \exists y_1, y_2 \in f(I)\ \exists y \in \mathbb{R} :\\
&\quad \big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)\\
&\quad \wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\\
&\quad \wedge y_1 < y < y_2 \wedge y \notin f(I).
\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{Interval}(I) \wedge \mathrm{ContinuousOn}(f, I) \wedge \neg\,\mathrm{Interval}(f(I)).$$

Remark (Interpretation). Continuous functions cannot tear gaps open inside their images of intervals. Once the function achieves two output values, continuity and the Intermediate Value Theorem force every intermediate value to occur as well, so the image remains an interval even when I is unbounded or half-open. Failure occurs precisely when continuity breaks, because discontinuities can skip intermediate heights and leave disconnected image sets.

Remark (Dependencies).

thm:bolzano-intermediate-value

Theorem (Darboux Property)

Theorem 38.3.9 (Darboux Property)**.** *Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$ or $f(x) < c$ for all $x \in [a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \Rightarrow \Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

Remark (Negated quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \Big(\exists u \in [a, b]\ f(u) \leq c\Big) \wedge \Big(\exists v \in [a, b]\ f(v) \geq c\Big).$$

Remark (Negation predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \neg\Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

Remark (Contrapositive quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\exists x \in [a, b]\ f(x) \leq c\Big) \wedge \Big(\exists y \in [a, b]\ f(y) \geq c\Big) \Rightarrow \exists z \in [a, b]\ f(z) = c.$$

Remark (Contrapositive predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \text{MeetsBothSides}(f, [a, b], c) \Longrightarrow \exists z \in [a, b]\ (f(z) = c).$$

Remark (Interpretation). A continuous function on a compact interval cannot oscillate across a level c without actually attaining that level. Stated contrapositive fashion, if c is never taken then the entire graph lies strictly above or strictly below the horizontal line y=c. The only way this guarantee can fail is when the function produces values on both sides of c, in which case continuity forces a crossing point. Geometrically the conclusion describes the image of a connected set under a continuous map: it is an interval, so omitting c forces the image to stay in one of the open half-lines determined by c.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

38.4 Uniform Continuity

Uniform Continuity — Quick Reference

Concept/Statement	Meaning/Role
Uniform continuity	One $\delta(\varepsilon)$ works for all pairs in the domain.
Sequential test	Close inputs with separated outputs witness failure.
Heine–Cantor	Continuity on $[a, b]$ upgrades to uniform continuity.
Cauchy preservation	Uniform continuity preserves Cauchy sequences.
Lipschitz	A global slope bound gives a uniform modulus.
Lipschitz $\Rightarrow$ uniform	Lipschitz control implies uniform continuity.
Bi-Lipschitz maps	Distances are controlled above and below.

Uniform Continuity

Definition (Uniform Continuity)

Definition 38.4.1 (Uniform Continuity)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is uniformly continuous on A if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

Remark (Standard quantified statement).

$$\begin{aligned}&(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R})\\ &\qquad \Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{UniformlyContinuous}(f, A) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R})\\ &\qquad \vee \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{UniformlyContinuous}(f, A) \iff \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).$$

Remark (Failure modes). The definition fails exactly when a single positive tolerance ε_0 exposes pairs of inputs arbitrarily close together whose function values remain at least ε_0 apart, so no universal δ works for that ε_0.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{UniformlyContinuous}(f, A) = \underbrace{\exists \varepsilon_0 > 0}_{\text{frozen tolerance}} \wedge \underbrace{\forall \delta > 0}_{\text{no global control}} \wedge \underbrace{\exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon_0)}_{\text{bad witness pair}}.$$

Remark (Interpretation). Uniform continuity demands that a single proximity scale δ answers any prescribed accuracy ε simultaneously across the whole domain. The negation shows what goes wrong: a fixed output tolerance resists all choices of δ because pairs of nearby inputs can always be found that stretch the function values too far apart. Geometrically, the graph of a uniformly continuous function can be traversed with a ruler of length δ horizontally and ε vertically without ever overstepping the vertical tolerance, whereas a nonuniformly continuous function exhibits spikes that defeat every fixed ruler length.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Algebra of Uniform Continuity on a General Domain)

Theorem 38.4.2 (Algebra of Uniform Continuity on a General Domain)**.** *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on A, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A.*

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product fg is not uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\Big[\big(\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\big)$$
$$\wedge \big(\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\big)\Big] \Rightarrow \Big[\forall \varepsilon > 0\ \exists \delta_1 > 0\ \forall x, y \in A\ (|x -$$
$$\wedge \forall \varepsilon > 0\ \exists \delta_2 > 0\ \forall x, y \in A\ (|x - y| < \delta_2 \Rightarrow |(f - g)(x) - (f - g)(y)| < \varepsilon)$$
$$\wedge \forall \varepsilon > 0\ \exists \delta_3 > 0\ \forall x, y \in A\ (|x - y| < \delta_3 \Rightarrow |cf(x) - cf(y)| < \varepsilon)\Big],$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\ \Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]$$
$$\wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\Big]$$
$$\wedge \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ \big(|x - y| < \delta \wedge |f(x)g(x) - f(y)g(y)| \geq \varepsilon_0\big).$$

Remark (Predicate reading).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$
$$\big(\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A)\big) \Longrightarrow \big(\mathrm{UniformlyContinuous}(f + g, A) \wedge \mathrm{U}$$
$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\quad \mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \neg\, \mathrm{UniformlyContin}$$

Remark (Interpretation). Uniform continuity is stable under linear combinations on any common domain, so sums, differences, and scalar multiples inherit the same global control on oscillation from the original functions. The proof decomposes the linear expressions into epsilon-delta bounds that add and subtract without upsetting the uniform response. In contrast, products can amplify local variation when one factor lacks a bounded range, so even uniformly continuous multiplicands can produce a product that fails the global tolerance requirement. The theorem explains both the algebraic closure that analysts routinely exploit and the standard obstruction, guiding expectations when building uniformly continuous functions from simpler pieces.

Remark (Dependencies).

- Uniform Continuity
- Linear Combination of Real-Valued Functions
- Pointwise Difference of Functions

Theorem (Algebra of Uniform Continuity on a Bounded Domain)

Theorem 38.4.3 (Algebra of Uniform Continuity on a Bounded Domain)**.** *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A.

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R} \text{ be bounded and } f, g : A \to \mathbb{R}.\\
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)g(x) - f(y)g(y)| < \varepsilon),\\
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big]\\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big]\\
&\wedge \big[\exists k > 0\ \forall x \in A\ (|g(x)| \geq k)\big]\\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)/g(x) - f(y)/g(y)| < \varepsilon).
\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}
&\text{Assume } A \subseteq \mathbb{R} \text{ is bounded and } f, g : A \to \mathbb{R}.\\
&\text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A)\\
&\qquad \Longrightarrow \text{UniformlyContinuous}(fg, A),\\
&\text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A) \wedge \exists k > 0\ \forall x \in A\ (|g(x)| \geq k)\\
&\qquad \Longrightarrow \text{UniformlyContinuous}(f/g, A).
\end{aligned}$$

Remark (Failure modes). The product conclusion can fail only when one of the hypotheses used to control the product is missing: the domain is not bounded, f is not uniformly continuous on A, or g is not uniformly continuous on A. The quotient conclusion can also fail if the denominator is not bounded away from zero on A.

Remark (Failure mode decomposition).

$$\underbrace{\neg\,\mathrm{BoundedSet}(A)}_{\text{unbounded domain}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(f,A)}_{\text{first factor not uniformly continuous}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(g,A)}_{\text{second factor not uniformly continuous}}$$
$$\vee \underbrace{\neg\exists k>0\ \forall x\in A:\ |g(x)|\geq k}_{\text{denominator not bounded away from zero}}.$$

Remark (Interpretation). Bounded subsets of $\mathbb{R}$ prevent the product of two uniformly continuous functions from developing large oscillations, so the shared modulus of continuity for f and g can be combined to control fg. The quotient is uniformly continuous whenever g never approaches zero, because the bounded-away-from-zero hypothesis allows inversion to act as another uniformly continuous operation that preserves the common modulus after scaling. Failure of boundedness or a loss of a positive lower bound for $|g|$ are the only ways these closure properties can break, and in either case the oscillation of the algebraic combination can no longer be controlled uniformly across the entire set.

Remark (Dependencies).

Definition 38.4.1.

Proposition (Sequential Characterization of Uniform Continuity)

Proposition 38.4.4 (Sequential Characterization of Uniform Continuity)**.** *Let $A\subseteq\mathbb{R}$ and let $f:A\to\mathbb{R}$. Then f is uniformly continuous on A if and only if for every pair of sequences (x_n) and (y_n) in A with $\lim_{n\to\infty}(x_n-y_n)=0$ we have $\lim_{n\to\infty}(f(x_n)-f(y_n))=0$. Go to proof.*

Remark (Standard quantified statement).

$$\Big[\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in A\ \big(|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon\big)\Big] \iff \forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to 0\Rightarrow|f(x_n$$

Remark (Predicate reading).

$$\mathrm{UniformlyContinuous}(f,A)\iff\forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to0\Rightarrow|f(x_n)-f(y_n)|\to0\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists A\subseteq\mathbb{R}\ \exists f:A\to\mathbb{R}:\\&\Big([\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in A\ (|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon)]\\&\qquad\wedge\exists x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to0\wedge\neg(|f(x_n)-f(y_n)|\to0)\big)\Big)\\&\vee\Big([\exists\varepsilon_0>0\ \forall\delta>0\ \exists x,y\in A\ (|x-y|<\delta\wedge|f(x)-f(y)|\geq\varepsilon_0)]\wedge\forall x_n,y_n:\mathbb{N}\to A\ (|x_n-y_n|\to0\Rightarrow|\end{aligned}$$

Remark (Negation predicate reading).

$$\exists A \subseteq \mathbb{R}\ \exists f : A \to \mathbb{R} :$$
$$\Big(\mathrm{UniformlyContinuous}(f, A) \wedge \exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)\Big)$$
$$\vee \Big(\neg\,\mathrm{UniformlyContinuous}(f, A) \wedge \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)\Big).$$

Remark (Failure modes). The equivalence can fail in exactly two ways: either f is uniformly continuous while some pair of asymptotically coincident sequences in A yields outputs that do not converge together, or every such pair of sequences does converge together while f nonetheless fails to be uniformly continuous on A.

Remark (Failure mode decomposition).

$$\underbrace{\mathrm{UniformlyContinuous}(f, A)}_{\text{uniform continuity holds}} \wedge \underbrace{\exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)}_{\text{sequential test fails}}$$
$$\vee\ \underbrace{\neg\,\mathrm{UniformlyContinuous}(f, A)}_{\text{uniform continuity fails}} \wedge \underbrace{\forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)}_{\text{sequential test holds}}.$$

Remark (Interpretation). Uniform continuity prevents the function from pulling together inputs that are asymptotically identical while letting their images drift apart. The sequential formulation packages this requirement into a tail test that ignores explicit epsilon-delta bookkeeping: whenever two points of A are eventually arbitrarily close, their function values must also be eventually arbitrarily close. Proving uniform continuity via sequences is often simpler, and conversely, producing two sequences that collide in A but whose images do not collide gives a concrete witness that uniform continuity fails. The two failure modes correspond to breaking one direction or the other of the claimed equivalence.

Remark (Dependencies).

Definition 38.4.1

Theorem (Heine–Cantor Theorem)

Theorem 38.4.5 (Heine–Cantor Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is uniformly continuous on $[a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}.$$
$$\Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big]$$
$$\Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in [a, b]\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \Rightarrow \mathrm{UniformlyContinuous}(f, [a, b]).$$

Remark (Negated quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a,b] \to \mathbb{R}.$$
$$\Big[\forall c \in [a,b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a,b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big]$$
$$\wedge \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in [a,b]\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big].$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f, [a,b]) \wedge \neg\, \mathrm{UniformlyContinuous}(f, [a,b]).$$

Remark (Interpretation). Continuity on the compact interval $[a,b]$ forces a single modulus of continuity that works simultaneously at every point, so oscillations of f cannot sharpen when inspecting smaller neighborhoods. Proofs usually argue by contradiction: assuming the negated form produces sequences of points whose distances shrink while their images stay apart, contradicting compactness-driven convergence. The theorem shows that closed bounded domains prevent the pathological behavior that obstructs uniform control on open or unbounded sets.

Remark (Dependencies).
Definition 38.2.2; Definition 38.4.1.

Proposition (Uniform Continuity Preserves Cauchy Sequences)

Proposition 38.4.6 (Uniform Continuity Preserves Cauchy Sequences)**.** *Let $S \subseteq \mathbb{R}$, let $f : S \to \mathbb{R}$, and let (x_n) be a sequence in S. If f is uniformly continuous on S and (x_n) is a Cauchy sequence in S, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } S \subseteq \mathbb{R},\ f : S \to \mathbb{R},\ (x_n)_{n \in \mathbb{N}} \subseteq S.$$
$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in S\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]$$
$$\wedge \Big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_n - x_m| < \varepsilon)\Big]$$
$$\Rightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|f(x_n) - f(x_m)| < \varepsilon).$$

Remark (Predicate reading).

$$\mathrm{UniformlyContinuous}(f, S) \wedge \mathrm{CauchySequence}(x_n) \ \Rightarrow\ \mathrm{CauchySequence}(f(x_n)).$$

Remark (Interpretation). Uniform continuity controls the change in $f(x)$ solely through the distance between inputs, so any Cauchy sequence in the domain is sent to a sequence whose tails remain uniformly close. Ordinary pointwise continuity cannot guarantee this because the necessary input tolerance could shrink along the sequence, so uniformity is exactly the additional strength needed to transport the Cauchy property. The standard failure mode occurs when f is merely continuous and the sequence approaches a boundary point where continuity loses uniform control, producing large oscillations in the image sequence. Geometrically, uniform continuity supplies a single modulus of continuity that works for the entire path traced by (x_n), ensuring that eventually all image terms lie in an arbitrarily small common interval.

Remark (Dependencies).

Definition 38.4.1; Definition 34.11.1.

Definition (Lipschitz Condition)

Definition 38.4.7 (Lipschitz Condition)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $K > 0$. The function f satisfies the Lipschitz condition on A with constant K if and only if $|f(x) - f(u)| \leq K|x - u|$ for every $x, u \in A$.

Remark (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \wedge (K > 0)$$
$$\Longrightarrow \forall x, u \in A \, \big(|f(x) - f(u)| \leq K|x - u|\big).$$

Remark (Definition predicate reading).

$$\mathrm{Lipschitz}(f, A, K) \coloneqq (K > 0) \wedge \forall x, u \in A \, \big(|f(x) - f(u)| \leq K|x - u|\big).$$

Remark (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \leq 0)$$
$$\vee \, \exists x, u \in A \, \big(|f(x) - f(u)| > K|x - u|\big).$$

Remark (Negation predicate reading).

$$\neg \, \mathrm{Lipschitz}(f, A, K) \iff \neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \leq 0) \vee \exists x, u \in A \, \big(|f(x) - f(u)| > K|x - u|\big).$$

Remark (Failure modes). The Lipschitz condition fails when the ambient data are malformed, when the proposed constant is not positive, or when there exists a pair of points in A whose images separate faster than K times the separation of the points themselves.

Remark (Failure mode decomposition).

$$\underbrace{\neg(A \subseteq \mathbb{R})}_{\text{bad domain}} \vee \underbrace{\neg(f : A \to \mathbb{R})}_{\text{bad function type}} \vee \underbrace{K \leq 0}_{\text{bad constant}} \vee \underbrace{\exists x, u \in A : \; |f(x) - f(u)| > K|x - u|}_{\text{excessive separation of outputs}}.$$

Remark (Interpretation). The Lipschitz condition requires a single constant K that simultaneously bounds the rate at which f can stretch distances across the entire set A. It encapsulates a global slope cap: every secant line through the graph has absolute slope at most K. Failure occurs once a single pair of points forces a steeper secant, signaling that no uniform control constant works on the whole domain. Geometrically, the graph of a Lipschitz function lies inside a cone of opening determined by K, so nearby points cannot be blown apart faster than K times their original spacing.

Remark (Dependencies).

- Function

Proposition (Lipschitz Implies Uniform Continuity)

Proposition 38.4.8 (Lipschitz Implies Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then f is uniformly continuous on A. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R}, \text{ and suppose } \exists K \geq 0\ \forall x, y \in A\ (|f(x) - f(y)| \leq K|x - y|).$$
$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).$$

Remark (Predicate reading).

$$\big[\exists K \geq 0\ \text{ Lipschitz}(f, A, K)\big] \Rightarrow \text{UniformlyContinuous}(f, A).$$

Remark (Contrapositive quantified statement)*.* If f fails to be uniformly continuous on A, then f is not Lipschitz on A with any constant.

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R}.$$
$$\Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big]$$
$$\Longrightarrow \forall K \geq 0\ \exists x, y \in A\ (|f(x) - f(y)| > K|x - y|).$$

Remark (Contrapositive predicate reading).

$$\neg\, \text{UniformlyContinuous}(f, A) \implies \neg\exists K \geq 0 :\ \text{Lipschitz}(f, A, K).$$

Remark (Interpretation)*.* A Lipschitz bound controls the variation of f by the distance between inputs with a single constant K, so the same K yields a usable $\delta = \varepsilon/K$ for every pair of points at once, proving uniform continuity. When $K = 0$ the function is constant and trivially uniform. Failure of uniform continuity would force two points arbitrarily close together whose images stay a fixed distance apart, contradicting the Lipschitz inequality.

Remark (Strict implication note)*.* The implication Lipschitz $\Rightarrow$ uniform continuity is strict. The canonical counterexample is $f(x) = \sqrt{x}$ on $[0, 1]$, which is uniformly continuous by the Heine–Cantor theorem but satisfies $|f(x) - f(0)|/|x - 0| = 1/\sqrt{x} \to \infty$ as $x \to 0^+$, so no Lipschitz constant exists at the origin. This distinction is relevant to GPU Lipschitz-constant estimation: uniform continuity of a simulation function does not supply a computable K for the Lipschitz condition; only an explicit Lipschitz bound does.

Remark (Dependencies).

Definition 38.4.7, Definition 38.4.1.

Theorem (Algebra of Lipschitz Functions)

Theorem 38.4.9 (Algebra of Lipschitz Functions)**.** *Let $A \subseteq \mathbb{R}$ and let $f, g : A \to \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.*

(a) For every $c \in \mathbb{R}$, the function cf is Lipschitz on A with constant $|c|K_f$.

(b) The functions $f + g$ and $f - g$ are Lipschitz on A with constant $K_f + K_g$.

(c) If $B \subseteq \mathbb{R}$, $h : B \to \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$.

(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then fg is Lipschitz on A with constant $M_g K_f + M_f K_g$.

(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if g is Lipschitz on A with constant K_g, then f/g is Lipschitz on A with constant $K_f/k + M_f K_g/k^2$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f, g : A \to \mathbb{R},\ K_f, K_g \geq 0,\\
&\forall x, y \in A:\ |f(x) - f(y)| \leq K_f|x - y| \text{ and } |g(x) - g(y)| \leq K_g|x - y|.\\
&\text{(a) } \forall c \in \mathbb{R},\ \forall x, y \in A:\ |cf(x) - cf(y)| \leq |c|K_f|x - y|.\\
&\text{(b) } \forall x, y \in A:\ |(f \pm g)(x) - (f \pm g)(y)| \leq (K_f + K_g)|x - y|.\\
&\text{(c) If } B \subseteq \mathbb{R},\ f(A) \subseteq B,\ h : B \to \mathbb{R},\ \forall u, v \in B:\ |h(u) - h(v)| \leq K_h|u - v|,\\
&\qquad \text{then } \forall x, y \in A:\ |h(f(x)) - h(f(y))| \leq K_h K_f|x - y|.\\
&\text{(d) If } |f(x)| \leq M_f,\ |g(x)| \leq M_g\ \forall x \in A,\ \text{then } \forall x, y \in A:\\
&\qquad |f(x)g(x) - f(y)g(y)| \leq (M_g K_f + M_f K_g)|x - y|.\\
&\text{(e) If } |f(x)| \leq M_f,\ |g(x)| \geq k > 0\ \forall x \in A,\ \text{then } \forall x, y \in A:\\
&\qquad \left|\frac{f(x)}{g(x)} - \frac{f(y)}{g(y)}\right| \leq \left(\frac{K_f}{k} + \frac{M_f K_g}{k^2}\right)|x - y|.
\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}
&\text{Lipschitz}(f, A, K_f) \wedge \text{Lipschitz}(g, A, K_g)\\
&\implies \Big[\forall c \in \mathbb{R} : \text{Lipschitz}(cf, A, |c|K_f)\Big]\\
&\qquad \wedge \text{Lipschitz}(f + g, A, K_f + K_g) \wedge \text{Lipschitz}(f - g, A, K_f + K_g)\\
&\qquad \wedge \Big(\text{Lipschitz}(h, B, K_h) \wedge f(A) \subseteq B \Rightarrow \text{Lipschitz}(h \circ f, A, K_h K_f)\Big)\\
&\qquad \wedge \Big[(|f| \leq M_f) \wedge (|g| \leq M_g) \Rightarrow \text{Lipschitz}(fg, A, M_g K_f + M_f K_g)\Big]\\
&\qquad \wedge \left[(|f| \leq M_f) \wedge (|g| \geq k > 0) \Rightarrow \text{Lipschitz}\left(\frac{f}{g}, A, \frac{K_f}{k} + \frac{M_f K_g}{k^2}\right)\right].
\end{aligned}$$

Remark (Interpretation). Lipschitz functions are stable under the standard algebraic operations: scaling magnifies the constant proportionally, addition and subtraction add the errors, and composition multiplies constants because distortions accumulate multiplicatively. Products and quotients require uniform control of magnitudes; bounds on the factors prevent growth in the coefficients that accompany the derivatives of the product rule heuristic, while a positive lower bound on the denominator guarantees the quotient does not amplify oscillations uncontrollably. Together, these clauses justify building rich classes of Lipschitz maps from simpler ones without losing quantitative control.

Remark (Dependencies).

Definition 38.4.7, Definition 38.3.1

Definition (Bi-Lipschitz Map)

Definition 38.4.10 (Bi-Lipschitz Map)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The map f is bi-Lipschitz on A if and only if there exist constants $0 < \alpha \leq K$ such that for every $x, u \in A$,

$$\alpha|x-u| \leq |f(x) - f(u)| \leq K|x-u|.$$

Any such pair (α, K) is called a bi-Lipschitz constant pair for f on A.

Remark (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R})$$
$$\implies \exists \alpha, K \in \mathbb{R} \left(0 < \alpha \leq K \wedge \forall x, u \in A : \ \alpha|x-u| \leq |f(x) - f(u)| \leq K|x-u|\right).$$

Remark (Definition predicate reading).

$$\operatorname{BiLipschitz}(f, A, \alpha, K) \coloneqq (0 < \alpha \leq K) \wedge \forall x, u \in A \ \left(\alpha|x-u| \leq |f(x) - f(u)| \leq K|x-u|\right).$$

Remark (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R})$$
$$\vee \forall \alpha, K \in \mathbb{R} \Big(0 < \alpha \leq K \Rightarrow \exists x, u \in A : \ |f(x) - f(u)| < \alpha|x-u| \vee |f(x) -$$

Remark (Negation predicate reading).

$$\neg \operatorname{BiLipschitz}(f, A, \alpha, K) \iff \neg(0 < \alpha \leq K) \vee \exists x, u \in A \ \big(|f(x) - f(u)| < \alpha|x-u| \vee |f(x) - f(u)| > K|x -$$

Remark (Failure modes). The property fails either because f collapses some pair of points faster than the allowed lower constant α, or because it stretches some pair more than the upper constant K permits. Both distortions prevent a uniform two-sided control of distances.

Remark (Failure mode decomposition).

$$\exists x, u \in A : \quad \underbrace{|f(x) - f(u)| < \alpha|x-u|}_{\text{lower-distortion failure}} \quad \text{or} \quad \underbrace{|f(x) - f(u)| > K|x-u|}_{\text{upper-distortion failure}}.$$

Remark (Interpretation). A bi-Lipschitz map distorts every distance in A by factors lying between α and K, so it is simultaneously Lipschitz and has a Lipschitz inverse defined on its image. This two-sided control guarantees injectivity and preserves geometric features such as relative spacing up to uniform scaling, while failures arise precisely from excessive collapsing or stretching of some pair of points.

Remark (Dependencies).

- Function

Theorem (Inverse of a Bi-Lipschitz Map Is Lipschitz)

Theorem 38.4.11 (Inverse of a Bi-Lipschitz Map Is Lipschitz)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x-y| \le |f(x)-f(y)| \le K|x-y|$ for all $x, y \in A$. Then f is injective, the inverse mapping $f^{-1} : f(A) \to A$ is well defined, and*

$$\frac{1}{K}|u-v| \le |f^{-1}(u) - f^{-1}(v)| \le \frac{1}{\alpha}|u-v| \quad \text{for all } u, v \in f(A).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ \alpha, K > 0.\\
&\Big[\forall x, y \in A\ \big(\alpha|x-y| \le |f(x)-f(y)| \le K|x-y|\big)\Big]\\
&\implies \Big[\forall x, y \in A\ (f(x) = f(y) \Rightarrow x = y)\\
&\qquad \wedge \exists g : f(A) \to A\ \big(\forall u \in f(A)\ (f(g(u)) = u)\big)\\
&\qquad \wedge \forall u, v \in f(A)\ \Big(\frac{1}{K}|u-v| \le |g(u)-g(v)| \le \frac{1}{\alpha}|u-v|\Big)\Big].
\end{aligned}$$

Remark (Predicate reading).

$$\text{BiLipschitz}(f, A, \alpha, K) \Rightarrow \big(\text{Injective}(f) \wedge \text{BiLipschitz}(f^{-1}, f(A), 1/K, 1/\alpha)\big).$$

Remark (Interpretation). A bi-Lipschitz bound simultaneously controls the expansion and contraction of f, so no two points of A can share the same image and the inverse is automatically a well-defined function on $f(A)$. The inequalities for f^{-1} follow by applying the original bi-Lipschitz bounds to pairs of preimages, showing that reversing the mapping preserves the same type of two-sided Lipschitz control with the reciprocals of the original constants. Failure would require collapsing two distinct points or losing the two-sided bounds, either of which contradicts the defining hypothesis.

Remark (Dependencies).

Definition 38.4.7; Definition 38.4.10.

38.5 Monotone Functions

Monotone Functions — Quick Reference

Concept/Statement	Meaning/Role
One-sided limits	Monotone functions have finite or infinite side limits.
Continuity criterion	Continuity means both side limits equal the value.
Jump	The size of the monotone discontinuity.
First-kind gaps	Monotone discontinuities are jumps.
Countability	Only countably many jumps can occur.
Continuous inverse	Strict monotonicity gives a continuous inverse.

Limits Superior and Inferior — Quick Reference

Concept/Statement	Meaning/Role
Function $\limsup$ and $\liminf$	Tail suprema become punctured-neighbourhood suprema.
Order relation	Always $\limsup \geq \liminf$.
Limit criterion	A finite limit exists exactly when they agree.

Monotone Functions and Continuity

Theorem (One-Sided Limits of Monotone Functions)

Theorem 38.5.1 (One-Sided Limits of Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let c be interior to I, and let $f : I \to \mathbb{R}$ be increasing. Define*

$$L_- := \sup\{f(x) : x \in I,\ x < c\}, \qquad L_+ := \inf\{f(x) : x \in I,\ c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \to c^-} f(x) = L_-, \qquad \lim_{x \to c^+} f(x) = L_+.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&I \subseteq \mathbb{R} \wedge \big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge f : I \to \mathbb{R} \\
&\wedge \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \leq f(y))\big) \wedge c \in I \wedge \exists a, b \in I\ (a < c < b) \\
&\wedge L_- = \sup\{f(x) : x \in I,\ x < c\} \wedge L_+ = \inf\{f(x) : x \in I,\ c < x\} \\
&\Longrightarrow \\
&\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in I\ \big((0 < c - x < \delta_- \wedge x < c) \Rightarrow |f(x) - L_-| < \varepsilon\big) \\
&\wedge \forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in I\ \big((0 < x - c < \delta_+ \wedge c < x) \Rightarrow |f(x) - L_+| < \varepsilon\big).
\end{aligned}$$

Remark (Predicate reading).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ increasing, and c interior to I.

$$\mathrm{LeftLimit}\Big(f, c, \sup\{f(x) : x \in I,\ x < c\}\Big) \wedge \mathrm{RightLimit}\Big(f, c, \inf\{f(x) : x \in I,\ x > c\}\Big).$$

Remark (Interpretation). For an increasing function, the values just to the left of c form a monotone increasing set bounded above by any right neighborhood, so their supremum matches the left-hand limit. Symmetrically, the values just to the right form a monotone increasing tail bounded below, making the infimum coincide with the right-hand limit. The theorem shows that monotonicity alone guarantees both one-sided limits exist and equals these extremal values. Failure occurs when c is not an interior point or f is not monotone, because the comparison sets may be empty or oscillatory, preventing the suprema or infima from governing the boundary behavior.

Remark (Dependencies).

- Completeness of $\mathbb{R}$
- Monotone Function
- Supremum
- Infimum

Corollary (Continuity Criterion for Monotone Functions)

Corollary 38.5.2 (Continuity Criterion for Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone increasing, and let c be an interior point of I. Then f is continuous at c if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x\to c^-} f(x)$ and $f(c^+) = \lim_{x\to c^+} f(x)$. Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) = f(c) = \lim_{x\to c^+} f(x)\right).$$

Remark (Predicate reading).

$$\mathrm{ContinuousAt}(f, c) \iff \big(\mathrm{LeftLimit}(f, c, f(c)) \wedge \mathrm{RightLimit}(f, c, f(c))\big).$$

Remark (Negated quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$\neg\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x\to c^-} f(x) < f(c)\right) \vee \left(f(c) < \lim_{x\to c^+} f(x)\right).$$

Remark (Negation predicate reading).

$$\neg\,\text{ContinuousAt}(f,c) \iff \big(\exists L^-\,[\text{LeftLimit}(f,c,L^-) \wedge L^- < f(c)]\big) \vee \big(\exists L^+\,[\text{RightLimit}(f,c,L^+) \wedge f(c) < L$$

Remark (Failure modes). Continuity can fail only through a jump: either the graph arrives at c from the left strictly below $f(c)$, or it departs to the right strictly above $f(c)$. Both jumps reflect the inability of the one-sided limits to match the function value at the interior point.

Remark (Failure mode decomposition).

$$\neg\,\text{ContinuousAt}(f,c) = \underbrace{\big(\exists L^-\,[\text{LeftLimit}(f,c,L^-) \wedge L^- < f(c)]\big)}_{\text{left jump at } c} \vee \underbrace{\big(\exists L^+\,[\text{RightLimit}(f,c,L^+) \wedge f(c) < L^+]}_{\text{right jump at } c}$$

Remark (Interpretation). A monotone graph possesses finite one-sided limits at every interior point. Continuity at c demands that these approach values exactly meet the function value, so the graph neither overshoots nor undershoots when passing through c. Any failure manifests as a jump discontinuity: the graph either leaps upward at c or remains below the right-hand approach. This criterion is indispensable when classifying discontinuities of monotone functions, since it converts continuity into a simple equality check between the function value and the adjacent plateau levels.

Remark (Dependencies).

Definition 38.2.2; Theorem 38.5.1.

Definition (Jump of a Function)

Definition 38.5.3 (Jump of a Function)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be increasing, and let c be an interior point of I. If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of f at c is

$$j_f(c) := f(c^+) - f(c^-).$$

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R} \\
&\Big[\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \exists a, b \in I\ (a < c < b) \\
&\qquad \wedge \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\lim_{x \to c^-} f(x) = L_- \wedge \lim_{x \to c^+} f(x) = L_+\big)\Big] \\
&\qquad \Rightarrow \big(j = j_f(c) \iff j = f(c^+) - f(c^-)\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\text{JumpOfFunction}(f,I,c,j) := \text{Interval}(I) \wedge \text{MonotoneIncreasing}(f,I) \wedge \text{InteriorPoint}(c,I) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\text{Le}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R} : \\
&\neg\big(j = j_f(c)\big) \iff \neg\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \\
&\qquad \vee \neg\big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \vee \neg\big(\exists a, b \in I\ (a < c < b)\big) \\
&\qquad \vee \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x \to c^-} f(x) \ne L_- \vee \lim_{x \to c^+} f(x) \ne L_+ \vee j \ne L_+ - L_-\big).
\end{aligned}$$

Remark (Negation predicate reading).

$\neg\,\mathrm{JumpOfFunction}(f, I, c, j) \iff \neg\,\mathrm{Interval}(I) \vee \neg\,\mathrm{MonotoneIncreasing}(f, I) \vee \neg\,\mathrm{InteriorPoint}(c, I) \vee \forall L_-, L$

Remark (Failure modes). A proposed jump value fails when a required one-sided limit does not exist, or when both one-sided limits exist but the proposed value is not their difference.

Remark (Failure mode decomposition).

$$\underbrace{\neg\big(f(c^-)\ \text{exists}\big)}_{\text{missing left limit}} \vee \underbrace{\neg\big(f(c^+)\ \text{exists}\big)}_{\text{missing right limit}} \vee \underbrace{j \neq f(c^+) - f(c^-)}_{\text{incorrect jump magnitude}}.$$

Remark (Interpretation). For a monotone function the one-sided limits exist at every interior point, and their difference measures the vertical gap in the graph at that point. A zero jump means the two limiting heights agree; a positive jump records a discontinuity of first kind.

Remark (Dependencies).

Theorem 38.5.1.

Proposition (Discontinuities of a Monotone Function Are of First Kind)

Proposition 38.5.4 (Discontinuities of a Monotone Function Are of First Kind). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a. If f is discontinuous at a, then both one-sided limits $\lim_{x\uparrow a} f(x)$ and $\lim_{x\downarrow a} f(x)$ exist in $\mathbb{R}$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } I \subseteq \mathbb{R} \text{ be an interval, } f : I \to \mathbb{R} \text{ monotone, and } a \in I \text{ interior to } I.$$

$$\big[f \text{ is discontinuous at } a\big] \implies \exists L^-, L^+ \in \mathbb{R} : \begin{cases} \lim\limits_{x\uparrow a} f(x) = L^-, \\ \lim\limits_{x\downarrow a} f(x) = L^+. \end{cases}$$

Remark (Predicate reading).

$\mathrm{MonotoneFunction}(f, I) \wedge \mathrm{DiscontinuousAt}(f, a) \implies \exists L^-, L^+ \in \mathbb{R}\ \big(\mathrm{LeftLimit}(f, a, L^-) \wedge \mathrm{RightLimit}(f, a, L$

Remark (Interpretation). A monotone graph can jump only by settling on distinct finite one-sided limits; it cannot oscillate wildly or blow up to infinity at a point of discontinuity. Thus every discontinuity of a monotone function is a jump discontinuity of the first kind, which explains why such functions are easy to integrate and why their exceptional sets are well behaved. Geometrically, the graph approaches a finite height from the left and another finite height from the right before making the jump.

Remark (Dependencies).

Definition 38.2.5

Corollary (Jump Intervals Are Disjoint)

Corollary 38.5.5 (Jump Intervals Are Disjoint). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of f, then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint.Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in I \ \Big[a \neq b \wedge \big(\exists \varepsilon_a > 0 \ \forall \delta > 0 \ \exists x \in I \ (|x-a| < \delta \wedge |f(x)-f(a)| \geq \varepsilon_a)\big) \wedge \big(\exists \varepsilon_b > 0 \ \forall \delta > 0 \ \exists y \in I \ (|y-b| <$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f, I) \wedge \begin{cases} a, b \in I, \\ a \neq b, \\ \text{DiscontinuousAt}(f, a), \\ \text{DiscontinuousAt}(f, b) \end{cases} \implies (f(a^-), f(a^+)) \cap (f(b^-), f(b^+)) = \varnothing.$$

Remark (Interpretation). The corollary states that a nondecreasing function assigns disjoint open value intervals to distinct jump discontinuities. Because the function never reverses direction, the lower one-sided limit at the later point cannot drop below the upper one-sided limit at the earlier point, so the corresponding ranges cannot overlap. This distinction of jump intervals ensures that the cumulative vertical jumps of a monotone function stay separated, preventing the graph from revisiting the same band of values via different discontinuities.

Remark (Dependencies).

Definition 38.5.3; Proposition 38.5.4

Theorem (Discontinuities of a Monotone Function Are Countable)

Theorem 38.5.6 (Discontinuities of a Monotone Function Are Countable)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } I \subseteq \mathbb{R} \text{ be an interval and } f : I \to \mathbb{R} \text{ be monotone.} \\ &\#\{c \in I : f \text{ fails to be continuous at } c\} \leq \aleph_0. \end{aligned}$$

Remark (Predicate reading).

$$\text{MonotoneFunction}(f, I) \implies \#\{c \in I : \text{DiscontinuousAt}(f, c)\} \leq \aleph_0.$$

Remark (Interpretation). Every discontinuity of a monotone function is a jump discontinuity with a nondegenerate vertical gap $(f(c^-), f(c^+))$. Distinct discontinuities yield pairwise disjoint gaps, and each such interval contains a rational number. Associating to every discontinuity the rational number inside its gap produces an injection of the discontinuity set into $\mathbb{Q}$, forcing the set to be countable. The only way the conclusion could fail would be the existence of uncountably many disjoint jump intervals, which is impossible because the rationals are countable.

Remark (Dependencies).

Theorem 38.5.1 and Corollary 38.5.5.

Corollary (Monotone Function Is Continuous on a Dense Set)

Corollary 38.5.7 (Monotone Function Is Continuous on a Dense Set)**.** *Let $f : [a, b] \to \mathbb{R}$ be monotone. Then the set $\{x \in [a, b] : f$ is continuous at $x\}$ is dense in $[a, b]$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } f : [a, b] \to \mathbb{R} \text{ be monotone.}\\&\forall u, v \in [a, b]\ (u < v \Rightarrow \exists c \in (u, v) \text{ such that } f \text{ is continuous at } c).\end{aligned}$$

Remark (Predicate reading)*.*

$$\mathrm{MonotoneFunction}(f, [a, b]) \Longrightarrow \mathrm{DenseSubset}\big(\{x \in [a, b] : \mathrm{ContinuousAt}(f, x)\}, [a, b]\big).$$

Remark (Negated quantified statement)*.*

$$\exists u, v \in [a, b]\ \Big(u < v \wedge \forall c \in (u, v)\ \exists \varepsilon_c > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_c\big)\Big).$$

Remark (Negation predicate reading)*.*

$$\exists u, v \in [a, b]\ \Big(u < v \wedge \forall c \in (u, v)\ \neg\, \mathrm{ContinuousAt}(f, c)\Big).$$

Remark (Interpretation)*.* A monotone function on a closed interval can only jump at most countably many times, so the points of discontinuity leave room inside every subinterval for a continuity point. Thus, although the function may fail to be continuous at isolated or countably many locations, continuity still occurs throughout any interval one inspects. This property ensures that monotone functions retain many features of continuous functions when arguments depend only on dense sets of continuity points.

Remark (Dependencies)*.*

Discontinuities of a Monotone Function Are Countable

Proposition (Continuous Injective Is Strictly Monotone)

Proposition 38.5.8 (Continuous Injective Is Strictly Monotone)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is injective if and only if f is strictly monotone on $[a, b]$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous on } [a, b].\\&\big[\forall x, y \in [a, b]\ (f(x) = f(y) \Rightarrow x = y)\big] \iff \big[\forall x, y \in [a, b]\ (x < y \Rightarrow f(x) < f(y))\big]\\&\qquad\qquad \vee \big[\forall x, y \in [a, b]\ (x < y \Rightarrow f(y) < f(x))\big].\end{aligned}$$

Remark (Predicate reading)*.*

$$\mathrm{ContinuousOn}(f, [a, b]) \Longrightarrow \big(\mathrm{Injective}(f) \iff \mathrm{StrictlyMonotone}(f, [a, b])\big).$$

Remark (Interpretation). A continuous function on a closed interval cannot reverse direction without repeating a value, so injectivity enforces a single increasing or decreasing trend across the entire domain. Conversely any strictly monotone map is automatically injective. The result emphasizes that topological constraints supplied by continuity interact with order-theoretic injectivity to force global monotonicity; failure occurs precisely when the graph turns back on itself, producing repeated values. Geometrically the continuous injective graph must cross each horizontal line at most once, which is only possible when the graph rises everywhere or falls everywhere.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

Theorem (Continuous Inverse Theorem)

Theorem 38.5.9 (Continuous Inverse Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Then the inverse function f^{-1} : $f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$. Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$.

$$\Big[(\forall x, y \in I)(x < y \Rightarrow f(x) < f(y)) \wedge \forall c \in I\; \forall \varepsilon > 0\; \exists \delta > 0\; \forall x \in I\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big]$$

$$\Longrightarrow \Big[\exists g : f(I) \to I \text{ such that } (\forall x \in I)\, g(f(x)) = x \wedge (\forall y \in f(I))\, f(g(y)) = y$$

$$\wedge\, (\forall u, v \in f(I))(u < v \Rightarrow g(u) < g(v)) \wedge \forall y \in f(I)\; \forall \varepsilon > 0\; \exists \eta > 0\; \forall u \in f(I)\big(|u - y| < \eta \Rightarrow |g(u) - g$$

Remark (Interpretation). A continuous strictly monotone function on an interval never reverses the order of points, so it is automatically bijective onto its image. This theorem records that the inverse map retains the same order orientation and inherits continuity, reflecting the fact that no sudden jumps can occur when the original graph crosses each horizontal line exactly once. Failure would require either loss of monotonicity or a discontinuity, either of which would create a flat spot or a jump preventing the existence of a continuous inverse. Geometrically, the graph of f^{-1} is the reflection of the graph of f across the diagonal line $y = x$, so the smoothness and strict increase or decrease are visibly preserved by symmetry.

Remark (Dependencies).

- Bolzano's Intermediate Value Theorem
- Strictly Increasing Function
- Strictly Decreasing Function
- Continuity at a Point

Limits Superior and Inferior of Functions

Definition (Limits Superior and Inferior of a Function)

Definition 38.5.10 (Limits Superior and Inferior of a Function)**.** Let $f : E \to \mathbb{R}$ and let x_0 be an accumulation point of E. For each $\delta > 0$ set

$$U(\delta) = \{\, f(x) : x \in E,\ 0 < |x - x_0| < \delta \,\}.$$

The extended real numbers $\limsup_{x\to x_0} f(x)$ and $\liminf_{x\to x_0} f(x)$ are defined by

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta), \qquad \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

Remark (Standard quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$
$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta),$$
$$\liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta).$$

Remark (Definition predicate reading)*.*

$$\mathrm{FunctionLimsup}(L, f, E, x_0) :\!= L = \inf_{\delta>0} \sup U(\delta),$$
$$\mathrm{FunctionLiminf}(l, f, E, x_0) :\!= l = \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

Remark (Negated quantified statement)*.*

Let $f : E \to \mathbb{R},\ x_0$ an accumulation point of $E,$ and $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}.$
$$\neg\Big[\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta) \ \wedge\ \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta)\Big]$$
$$\iff \limsup_{x\to x_0} f(x) \neq \inf_{\delta>0} \sup U(\delta) \vee \liminf_{x\to x_0} f(x) \neq \sup_{\delta>0} \inf U(\delta).$$

Remark (Negation predicate reading)*.*

$$\neg\big(\mathrm{FunctionLimsup}(L, f, E, x_0) \wedge \mathrm{FunctionLiminf}(l, f, E, x_0)\big)$$
$$\iff L \neq \inf_{\delta>0} \sup U(\delta) \vee l \neq \sup_{\delta>0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ 0 < |x - x_0| < \delta\}$.

Remark (Failure modes)*.* The definition can fail only in structurally distinct ways: punctured neighborhoods might be empty despite the accumulation-point hypothesis, or the upper envelopes may blow up while a finite limsup is asserted, or the lower envelopes may dive to negative infinity while a finite liminf is asserted.

Remark (Failure mode decomposition).

$$\underbrace{\exists\delta > 0 :\ U(\delta) = \varnothing}_{\text{missing punctured samples}} \vee \underbrace{\left(\inf_{\delta>0} \sup U(\delta) = +\infty \wedge L < +\infty\right)}_{\text{upper envelope blow-up}} \vee \underbrace{\left(\sup_{\delta>0} \inf U(\delta) = -\infty \wedge l > -\infty\right)}_{\text{lower envelope collapse}}.$$

Remark (Interpretation). The limit superior records the smallest height that eventually dominates all nearby function values, while the limit inferior records the largest depth that eventually underestimates all nearby values. Taking suprema within punctured neighborhoods captures every oscillation that can be witnessed arbitrarily close to x_0, and the outer infimum or supremum squeezes these oscillations to the sharpest possible bounds. Failure occurs only if the neighborhood sampling breaks down or if the oscillations race off to infinite extremal heights, signaling that no finite envelope can control the function near the accumulation point.

Remark (Dependencies).

- Function
- Supremum
- Infimum
- Epsilon Neighbourhood

Proposition

Proposition 38.5.11 ($\limsup \geq \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let x_0 be a limit point of A, and let $f : A \to \mathbb{R}$. Then*

$$\limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{R}\ \forall f : A \to \mathbb{R}\ \forall x_0 \in \mathbb{R} :$$
$$\Big(\forall \eta > 0\ \exists x \in A \setminus \{x_0\}\ (|x - x_0| < \eta)\Big) \Longrightarrow \limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

Remark (Interpretation). The limsup records the largest cluster value that f attains near x_0, while the liminf records the smallest cluster value. Because every subsequential limit contributing to the liminf also contributes to the limsup, the supremum of all cluster values cannot lie below their infimum. The only way equality holds is when all cluster values coincide, i.e. when the ordinary limit exists.

Remark (Dependencies).

Definition (Limits Superior and Inferior of a Function).

Proposition

Proposition 38.5.12 (Limit Exists iff $\limsup = \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals L if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$. Go to proof.*

Remark (Standard quantified statement)*.*

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ (\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A \setminus \{x_0\}\ (0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon))\\ \iff \quad \left(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\right).\end{gathered}$$

Remark (Predicate reading)*.*

$$\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\big).$$

Remark (Negated quantified statement)*.*

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)\Big]\\ \iff \quad \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).\end{gathered}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).$$

Remark (Failure modes)*.* The criterion fails only when at least one side of the equivalence breaks: either the function fails to approach L in the epsilon-delta sense, or one of the two extremal envelopes near x_0 does not equal L.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)}_{\text{limit condition fails}} \vee \underbrace{\limsup_{x\to x_0} f(x) \neq L}_{\text{upper envelope mismatch}} \vee \underbrace{\liminf_{x\to x_0} f(x) \neq}_{\text{lower envelope misn}}$$

Remark (Interpretation)*.* The two-sided limit of a real-valued function at a finite limit point exists precisely when the largest subsequential accumulation value and the smallest subsequential accumulation value coincide, in which case they both equal the limit. If the limsup exceeds the liminf, as happens for $f(x) = \sin(1/x)$ near $x_0 = 0$, the function oscillates between distinct cluster heights and therefore has no finite limit. The theorem isolates the limit as the unique height compatible with every oscillatory envelope and shows that agreeing envelopes are the exact signature of convergence in this local setting.

Remark (Dependencies)*.*

Definition 38.5.10

38.6 Limits at Infinity

Limits at Infinity — Quick Reference

Concept/Statement	Meaning/Role
Infinite adherent points	Unboundedness makes $\pm\infty$ available as approach points.
Limit at infinity	Replace $0 < \lvert x - c\rvert < \delta$ by sufficiently large $\lvert x\rvert$.
Sequential criterion	Limits at infinity are tested along escaping sequences.
Limit laws	Algebraic limit rules carry over unchanged.

Limits at Infinity

Definition (Infinite Adherent Points)

Definition 38.6.1 (Infinite Adherent Points)**.** Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

Remark (Standard quantified statement).

$$\begin{aligned} +\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\, \exists x \in X\ (x > M), \\ -\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R}\, \exists x \in X\ (x < M). \end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned} \mathrm{PlusInfAdherent}(X) &:= \forall M \in \mathbb{R}\, \exists x \in X\ (x > M), \\ \mathrm{MinusInfAdherent}(X) &:= \forall M \in \mathbb{R}\, \exists x \in X\ (x < M). \end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned} \neg(+\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\, \forall x \in X\ (x \le M), \\ \neg(-\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R}\, \forall x \in X\ (x \ge M). \end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned} \neg\,\mathrm{PlusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\, \forall x \in X\ (x \le M), \\ \neg\,\mathrm{MinusInfAdherent}(X) &\iff \exists M \in \mathbb{R}\, \forall x \in X\ (x \ge M). \end{aligned}$$

Remark (Failure modes). Infinite adherence fails at $+\infty$ exactly when X is bounded above, and fails at $-\infty$ exactly when X is bounded below.

Remark (Failure mode decomposition).

$$\underbrace{\exists M \in \mathbb{R}\, \forall x \in X\ (x \le M)}_{\text{finite upper bound}} \qquad \underbrace{\exists M \in \mathbb{R}\, \forall x \in X\ (x \ge M)}_{\text{finite lower bound}}.$$

Remark (Interpretation). Both finite and infinite adherence are instances of adherence in $\overline{\mathbb{R}}$: finite adherence uses balls; infinite adherence uses rays.

Definition (Limit at Infinity)

Definition 38.6.2 (Limit at Infinity)**.** Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of f at $+\infty$ equals L if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ f : X \to \mathbb{R},\ L \in \mathbb{R}.\\ &\lim_{x\to+\infty} f(x) = L \iff \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\\ &\qquad (x > M \Rightarrow |f(x) - L| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(f, +\infty, L) :\equiv \forall \varepsilon > 0\ \exists M \in \mathbb{R}\ \forall x \in X\ (x > M \Rightarrow |f(x) - L| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),\ f : X \to \mathbb{R},\ L \in \mathbb{R}.\\ &\neg\big(\lim_{x\to+\infty} f(x) = L\big) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\\ &\qquad (x > M \wedge |f(x) - L| \geq \varepsilon_0).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{LimitAtInfinity}(f, +\infty, L) \iff \exists \varepsilon_0 > 0\ \forall M \in \mathbb{R}\ \exists x \in X\ (x > M \wedge |f(x) - L| \geq \varepsilon_0).$$

Remark (Failure modes)*.* The definition fails only when there is a fixed tolerance ε_0 that the function cannot satisfy on any tail of X, meaning every attempt to choose M leaves some point $x > M$ with $|f(x) - L| \geq \varepsilon_0$.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{fixed tolerance}}\ \underbrace{\forall M \in \mathbb{R}}_{\text{no tail succeeds}}\ \underbrace{\exists x \in X\ (x > M \wedge |f(x) - L| \geq \varepsilon_0)}_{\text{far-point violation}}.$$

Remark (Interpretation)*.* The limit at $+\infty$ demands that eventually the function values stay within any prescribed accuracy of L. Intuitively, once x is large enough, the graph of f must lie in the horizontal band of height ε around L. Failure occurs when some positive tolerance is violated infinitely often by points escaping to $+\infty$. This definition allows us to treat unbounded domains using the same epsilon control that governs finite limits, capturing the long-range behavior of f with respect to the ideal point at infinity.

Remark (Dependencies).

Definition 38.6.1.

38.7 Approximation

Approximation — Quick Reference

Concept/Statement	Meaning/Role
Step approximation	Uniform approximation by step functions.
Piecewise linear function	Linear pieces joined across a partition.
PL approximation	Uniform approximation by polygonal functions.
Weierstrass	Uniform polynomial approximation on compact intervals.
Bernstein polynomial	Constructive polynomial approximants on $[0, 1]$.
Bernstein convergence	Bernstein polynomials converge uniformly.

Approximation of Continuous Functions

Theorem (Step Function Approximation)

Theorem 38.7.1 (Step Function Approximation)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a, b] \to \mathbb{R}$ such that*

$$\sup_{x\in[a,b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous and let } \varepsilon > 0.$$
$$\exists s_\varepsilon : [a, b] \to \mathbb{R} \text{ step function } \forall x \in [a, b] \ (|f(x) - s_\varepsilon(x)| < \varepsilon).$$

Remark (Predicate reading)*.*

$$\text{StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) := \text{StepFunction}(s_\varepsilon, [a, b]) \wedge \forall x \in [a, b] \ (|f(x) - s_\varepsilon(x)| < \varepsilon).$$

Remark (Negated quantified statement)*.*

$$\text{Let } f : [a, b] \to \mathbb{R} \text{ be continuous and let } \varepsilon > 0.$$
$$\forall s : [a, b] \to \mathbb{R} \text{ step function } \exists x \in [a, b] \ (|f(x) - s(x)| \geq \varepsilon).$$

Remark (Negation predicate reading)*.*

$$\neg\exists s_\varepsilon \ \text{StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) \iff \forall s : [a, b] \to \mathbb{R} \ \Big(\text{StepFunction}(s, [a, b]) \Rightarrow \exists x \in [a, b] \ (|f(x) - s$$

Remark (Failure modes)*.* The statement fails precisely when a continuous f and tolerance ε are fixed but every step function leaves an error of at least ε at some point in $[a, b]$.

Remark (Failure mode decomposition)*.*

$$\underbrace{\forall s : [a, b] \to \mathbb{R} \text{ step function}}_{\text{all approximants}} \ \underbrace{\exists x \in [a, b] \ (|f(x) - s(x)| \geq \varepsilon)}_{\text{witnessed large deviation}}.$$

Remark (Interpretation). Uniform continuity on $[a, b]$ guarantees that f does not oscillate rapidly at small scales, so subdividing the interval finely enough allows a step function to track f within any prescribed uniform error. The typical construction samples f at one point of each subinterval and keeps that value constant across the piece. Failure would mean that no matter how the interval is partitioned, some subinterval forces an error at least ε, contradicting uniform continuity. The result is a foundational approximation fact used to show that continuous functions on compact intervals are Riemann integrable and to connect continuous functions with simple combinatorial models.

Remark (Dependencies).

Heine–Cantor Theorem.

Definition (Piecewise Linear Function)

Definition 38.7.2 (Piecewise Linear Function)**.** Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a, b] \to \mathbb{R}$. The function g is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \cdots < x_m = b$ such that $g|_{[x_{k-1}, x_k]}$ is affine for every $k \in \{1, \ldots, m\}$.

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}. \\ &g \text{ is piecewise linear} \\ &\iff \exists m \in \mathbb{N} \Big(m \geq 1 \land \exists x_0, \ldots, x_m \in [a, b] \; \big[a = x_0 < x_1 < \cdots < x_m = b \\ &\qquad\qquad \land \forall k \in \{1, \ldots, m\} \; g|_{[x_{k-1}, x_k]} \text{ is affine} \big] \Big). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{PiecewiseLinear}(g, [a, b]) :\equiv \exists m \in \mathbb{N} \Big(m \geq 1 \land \exists x_0, \ldots, x_m \in [a, b] \; \big[a = x_0 < \cdots < x_m = b \land \forall k \in \{1, \ldots, m\}$$

Remark (Negated quantified statement).

$$\begin{aligned} &\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and } g : [a, b] \to \mathbb{R}. \\ &g \text{ is not piecewise linear} \\ &\iff \forall m \in \mathbb{N} \Big(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b] \; \big[a = x_0 < \cdots < x_m = b \\ &\qquad\qquad \Rightarrow \exists k \in \{1, \ldots, m\} \; g|_{[x_{k-1}, x_k]} \text{ is not affine} \big] \Big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{PiecewiseLinear}(g, [a, b]) \iff \forall m \in \mathbb{N} \Big(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a, b] \; \big[a = x_0 < \cdots < x_m = b \Rightarrow \exists k \in \{1$$

Remark (Failure modes). A proposed piecewise-linear structure can fail because no finite ordered partition of $[a, b]$ is available, or because for every such partition at least one subinterval has a restriction of g that is not affine.

Remark (Failure mode decomposition).

$$\underbrace{\forall m \in \mathbb{N}\ (m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a,b]\ (a = x_0 < \cdots < x_m = b \Rightarrow \cdots))}_{\text{no valid finite affine subdivision}} \vee \underbrace{\exists k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is not}}_{\text{non-affine segment}}$$

Remark (Interpretation). A piecewise linear function on $[a, b]$ has a graph made from finitely many straight line segments. The definition permits different affine rules on different subintervals, but requires a finite ordered list of nodes that covers the whole interval.

Remark (Dependencies).

- Function
- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

Theorem

Theorem 38.7.3 (Piecewise Linear Approximation)**.** *Let $f\colon [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon\colon [a, b] \to \mathbb{R}$ such that $\sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } f\colon [a, b] \to \mathbb{R},\ f \text{ continuous},\ \varepsilon > 0.\\
&\exists \text{ continuous piecewise linear } \ell_\varepsilon\colon [a, b] \to \mathbb{R} : \sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon.
\end{aligned}$$

Remark (Interpretation). Every continuous function on a compact interval can be uniformly approximated by a polygonal curve matching finitely many sampled values of the original graph. The construction refines the interval into a sufficiently fine partition guaranteed by compactness, then connects the sampled values with straight segments, producing a uniformly small error over the entire domain. Failure would require a uniform oscillation exceeding ε on some subinterval, which cannot occur once the partition is fine enough. Geometrically, the theorem states that continuous graphs admit arbitrarily close polyline shadows, providing a bridge from rough step approximations to smooth polynomial approximations.

Remark (Dependencies).

- Heine–Cantor Theorem
- Piecewise Linear Function
- Continuity at a Point

Theorem (Weierstrass Approximation Theorem)

Theorem 38.7.4 (Weierstrass Approximation Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}\\&\forall \varepsilon > 0 \; \exists p \in \mathbb{R}[x] \; (\|f - p\|_{\infty,[a,b]} < \varepsilon).\end{aligned}$$

Remark (Predicate reading)*.*

$$\mathrm{ContinuousOn}(f, [a, b]) \Longrightarrow \forall \varepsilon > 0 \; \exists p \in \mathbb{R}[x] \; \big(\|f - p\|_{\infty,[a,b]} < \varepsilon\big).$$

Remark (Interpretation)*.* Every continuous function on a closed interval can be uniformly approximated by polynomials, so polynomials are dense in the space of continuous functions under the supremum norm. The result holds because convolutions with Bernstein polynomials preserve continuity and converge uniformly to the original function. Failure would mean there exists a uniform tolerance that no polynomial can match, contradicting the approximation scheme built from Bernstein polynomials. Geometrically, the theorem states that the graph of a continuous function can be matched arbitrarily well by the graph of a polynomial, so polynomial graphs form a flexible mesh that can shadow any continuous curve on a compact interval.

Remark (Dependencies)*.*

Depends on thm:bernstein-approximation.

Definition (Bernstein Polynomial)

Definition 38.7.5 (Bernstein Polynomial)**.** Let $n \in \mathbb{N}$ and let $f : [0, 1] \to \mathbb{R}$. A function $B_n : [0, 1] \to \mathbb{R}$ is called the nth Bernstein polynomial of f if and only if, for every $x \in [0, 1]$,

$$B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1 - x)^{n-k}.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } n \in \mathbb{N}, \; f : [0, 1] \to \mathbb{R}, \; B_n : [0, 1] \to \mathbb{R}.\\&B_n \text{ is the } n\text{th Bernstein polynomial of } f\\&\Longleftrightarrow \forall x \in [0, 1] \; \left(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1 - x)^{n-k}\right).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{BernsteinPolynomial}(B_n, f, n) :\equiv \Big(B_n : [0, 1] \to \mathbb{R}\Big) \wedge \forall x \in [0, 1] \; \left(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1 - x)^{n-k}\right).$$

Remark (Interpretation). The nth Bernstein polynomial samples f at the uniformly spaced nodes k/n, weights those samples by the binomial probabilities of order n, and produces for each x an average that depends polynomially on x. This construction smooths f into a polynomial that agrees with f at the endpoints and approximates f uniformly as n grows, so describing it explicitly is essential for subsequent approximation theorems. Failure arises only when the candidate polynomial disagrees with the binomial averaging recipe at some point of the unit interval, reflecting an incorrect coefficient or evaluation in the polynomial expression.

Remark (Dependencies).

- Function
- Natural Numbers
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{R}$

Theorem (Bernstein Approximation Theorem)

Theorem 38.7.6 (Bernstein Approximation Theorem). *Let $f : [0,1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n denote the nth Bernstein polynomial of f. Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$. Go to proof.*

Remark (Standard quantified statement).

Let $f : [0,1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n be the nth Bernstein polynomial of f.
$\forall \varepsilon > 0\ \exists n_\varepsilon \in \mathbb{N}\ \forall n \geq n_\varepsilon : \ \|f - B_n\|_\infty < \varepsilon.$

Remark (Interpretation). The theorem asserts that the Bernstein polynomials of a continuous function on $[0,1]$ converge uniformly to the original function, so the entire graph of f can be approximated to any prescribed sup-norm accuracy by explicit degree-n polynomials once n is large enough. The proof works by showing that the Bernstein operators preserve positivity and linear growth while averaging values of f, which forces the uniform error to vanish as $n \to \infty$. This matters because it gives an elementary constructive route from continuous functions to polynomials, strengthening the Weierstrass approximation theorem with a concrete approximating sequence. The approximation can fail only when f is not continuous on $[0,1]$, in which case the Bernstein polynomials may smooth out jumps and no longer capture the original function in the sup norm. Geometrically, each B_n is a convex combination of the samples $f(k/n)$, so the polygons formed by these samples become finer and eventually hug the graph of f everywhere on the interval.

Remark (Dependencies).

Definition (Continuity at a Point), Definition (Bernstein Polynomial).

38.8 Gauge

Gauges and Tagged Partitions — Quick Reference

Concept/Statement	Meaning/Role
Interval partition	Finite subdivision of a compact interval.
Tagged partition	Each subinterval carries a selected tag.
Gauge	A position-dependent positive tolerance.
δ-fine partition	Each subinterval lies within its tag's gauge window.
Cousin's theorem	Every gauge on $[a, b]$ admits a fine tagged partition.
Mesh	The largest subinterval length.
Common refinement	A partition refining two given partitions.

Gauges and Tagged Partitions

Definition (Partition)

Definition 38.8.1 (Partition)**.** Let $a < b$ and set $I = [a, b]$. A finite ordered set $P = \{x_0, \ldots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of I if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \cdots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \ldots, n$ subdivide I.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \ldots, x_n\} \subseteq I.\\ &P \text{ is a partition of } I\\ &\iff \big(x_0 = a \wedge x_n = b \wedge \forall i \in \{1, \ldots, n\}\ (x_{i-1} < x_i)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{PartitionOfInterval}(P, I) :\equiv \exists n \in \mathbb{N}\ \Big(n \geq 1 \wedge P = \{x_0, \ldots, x_n\} \subseteq I \wedge x_0 = \min I \wedge x_n = \max I \wedge \forall i \in \{1, \ldots,$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \ldots, x_n\} \subseteq I.\\ &P \text{ is not a partition of } I\\ &\iff \big((x_0 \neq a) \vee (x_n \neq b) \vee \exists i \in \{1, \ldots, n\}\ (x_{i-1} \geq x_i)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{PartitionOfInterval}(P, I) \iff \forall n \in \mathbb{N}\ \Big(n \geq 1 \Rightarrow \big(P \neq \{x_0, \ldots, x_n\} \vee P \nsubseteq I \vee x_0 \neq \min I \vee x_n \neq \max I \vee \exists i$$

Remark (Failure modes). The definition fails precisely in three ways: the left endpoint is omitted, the right endpoint is omitted, or the sequence of nodes ceases to be strictly increasing, leading to repeated or out-of-order points.

Remark (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint missing}} \quad \vee \quad \underbrace{x_n \neq b}_{\text{right endpoint missing}} \quad \vee \underbrace{\exists i \in \{1, \ldots, n\} \ (x_{i-1} \geq x_i)}_{\text{nodes not strictly increasing}}.$$

Remark (Interpretation). A partition records the finite list of nodes that slice a closed interval into adjacent subintervals. The strict ordering ensures that every interior point appears exactly once in this linear traversal, so the induced subintervals $[x_{i-1}, x_i]$ tile $[a, b]$ without overlap or gaps. Computations of Riemann sums, Darboux sums, and gauge integrals use partitions as the combinatorial scaffolding that supports sampling and measurement. Failure arises when the endpoints are not captured or when the nodes regress or stagnate, which would leave gaps or redundancies in the subdivision.

Remark (Dependencies).

- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

Definition (Tagged Partition)

Definition 38.8.2 (Tagged Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ is a tagged partition of $[a, b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^{n}$ of $[a, b]$ with $a = x_0 < x_1 < \cdots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \ldots, n$. Equivalently, a tagged partition is exactly a partition P of $[a, b]$ together with one tag chosen inside every subinterval of P.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}. \\
&\dot{\mathcal{P}} \text{ is a tagged partition of } [a, b] \\
&\iff \big(a = x_0 < \cdots < x_n = b \wedge \forall i \in \{1, \ldots, n\} \ (t_i \in [x_{i-1}, x_i])\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) :\!= \Big(\mathrm{Partition}(P, [a, b]) \wedge \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n} \wedge \forall i \ (t_i \in [x_{i-1}, x_i])\Big),$$

where $\mathrm{Partition}(P, [a, b])$ abbreviates $a = x_0 < \cdots < x_n = b$ for $P = \{[x_{i-1}, x_i]\}_{i=1}^{n}$.

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}. \\
&\dot{\mathcal{P}} \text{ is not a tagged partition of } [a, b] \\
&\iff \Big(x_0 \neq a \vee x_n \neq b \vee \exists j \in \{1, \ldots, n\} \ (x_{j-1} \geq x_j) \vee \exists i \in \{1, \ldots, n\} \ (t_i \notin [x_{i-1}, x_i])\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{TaggedPartition}(\dot{\mathcal{P}},[a,b]) \iff (x_0 \neq a) \vee (x_n \neq b) \vee \Big(\exists j\ (x_{j-1} \geq x_j)\Big) \vee \Big(\exists i\ (t_i \notin [x_{i-1},x_i])\Big).$$

Remark (Failure modes). A list fails to be a tagged partition either because the subintervals do not form a valid partition (the mesh is disordered or has gaps) or because at least one chosen tag lies outside its designated subinterval.

Remark (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint defect}} \quad \vee \quad \underbrace{x_n \neq b}_{\text{right endpoint defect}} \quad \vee \; \underbrace{\exists j\ (x_{j-1} \geq x_j)}_{\text{ordering defect}} \;\vee\; \underbrace{\exists i\ (t_i \notin [x_{i-1},x_i])}_{\text{tag misplaced}}.$$

Remark (Interpretation). A tagged partition augments the usual subdivision of a closed interval by selecting a representative sampling point inside each component subinterval. These tags serve as evaluation points for Riemann sums and gauge integrals. The construction encapsulates both the geometric slicing of the interval and the analytic data needed for sample values. Failure occurs when the slices no longer tile the interval in order or when a sample point is chosen outside its slice, breaking the correspondence between tags and subintervals.

Remark (Dependencies).

Definition 38.8.1.

Definition (Gauge)

Definition 38.8.3 (Gauge)**.** Let $a,b \in \mathbb{R}$ with $a<b$ and set $I=[a,b]$. A gauge on I is a function $\delta : I \to (0,\infty)$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a,b \in \mathbb{R} \text{ with } a<b, \text{ put } I=[a,b], \text{ and let } \delta : I \to \mathbb{R}.\\ &\delta \text{ is a gauge on } I \iff \forall x \in I\ (\delta(x)>0).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Gauge}(\delta, I) \coloneqq \big(\delta : I \to \mathbb{R}\big) \wedge \forall x \in I\ (\delta(x) > 0).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } a,b \in \mathbb{R} \text{ with } a<b, \text{ put } I=[a,b], \text{ and let } \delta : I \to \mathbb{R}.\\ &\delta \text{ fails to be a gauge on } I \iff \exists x \in I\ (\delta(x) \leq 0).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Gauge}(\delta, I) \iff \neg\big(\delta : I \to \mathbb{R}\big) \vee \exists x \in I\ (\delta(x) \leq 0).$$

Remark (Failure modes). A purported gauge fails either because it is not a function from I to $\mathbb{R}$, or because it takes on a nonpositive value somewhere in I.

Remark (Failure mode decomposition).

$$\underbrace{\neg\big(\delta : I \to \mathbb{R}\big)}_{\text{wrong function type}} \;\vee\; \underbrace{\exists x \in I\ (\delta(x) \leq 0)}_{\text{nonpositive gauge value}}$$

Remark (Interpretation). A gauge assigns to each point of the closed interval a positive radius, encoding how tightly a tagged subinterval must cluster around that point in fine partition arguments. The sole obstruction is positivity: if any value is zero or negative, the construction cannot deliver legitimate local scales, so the gauge structure breaks down.

Remark (Dependencies).

- Function
- Closed Interval
- Order on $\mathbb{R}$

Definition (δ-Fine Partition)

Definition 38.8.4 (δ-Fine Partition)**.** Let $[a, b]$ be a closed interval, let $\delta : [a, b] \to (0, \infty)$ be a gauge, and let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ be a tagged partition of $[a, b]$. The tagged partition $\dot{\mathcal{P}}$ is δ-fine if and only if for every $i \in \{1, \ldots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

Remark (Standard quantified statement).

Let $[a, b]$ be a closed interval, $\delta : [a, b] \to (0, \infty)$, and $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ a tagged partition of $[a, b]$. $\dot{\mathcal{P}}$ is δ-fine $\iff \forall i \in \{1, \ldots, n\}\ \forall x \in [x_{i-1}, x_i]\ (|x - t_i| < \delta(t_i))$.

Remark (Definition predicate reading).

$$\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) :\equiv \forall i \in \{1, \ldots, n\}\ \forall x \in [x_{i-1}, x_i]\ (|x - t_i| < \delta(t_i)).$$

Remark (Negated quantified statement).

$$\exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]\ \big(|x - t_i| \geq \delta(t_i)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) \iff \exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]\ (|x - t_i| \geq \delta(t_i)).$$

Remark (Failure modes). A tagged partition fails to be δ-fine precisely when some subinterval contains a point whose distance from its tag reaches or exceeds the available radius $\delta(t_i)$, so the corresponding subinterval is not contained in the prescribed open neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]}_{\text{choose offending subinterval and point}}\ \underbrace{(|x - t_i| \geq \delta(t_i))}_{\text{distance not controlled by } \delta}.$$

Remark (Interpretation). A δ-fine tagged partition refines the interval so that each subinterval sits entirely inside the open ball centered at its tag with radius dictated by the gauge. This ensures that when summing local contributions, every evaluation point remains within a tolerance chosen for that location, a key requirement for gauge-based integral constructions. Failure occurs when a subinterval is too wide for the tolerance prescribed at its tag, so some point of the subinterval lies on or outside the allowed neighborhood.

Remark (Dependencies).

Definition 38.8.3; Definition 38.8.2

Lemma

Lemma 38.8.5 (Every Point Is Covered by a Tag). *Let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ that is δ-fine on $[a, b]$, and let $x \in [a, b]$. Then there exists $i \in \{1, \dots, n\}$ such that $|x - t_i| < \delta(t_i)$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a < b,\ \delta : [a, b] \to (0, \infty),\ \text{and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n \text{ be } \delta\text{-fine on } [a, b].\\ &\forall x \in [a, b]\ \exists i \in \{1, \dots, n\}\ (|x - t_i| < \delta(t_i)).\end{aligned}$$

Remark (Interpretation). A δ-fine tagged partition matches each subinterval to a tag that lies inside the interval and records the gauge radius allowed at that tag. The lemma asserts that every point of the underlying interval falls within the gauge ball centered at one of the tags, so the gauge control at that tag can be applied to the point. This coverage property is the mechanism that transfers local control encoded by the gauge to arbitrary points of the interval.

Remark (Dependencies).

Definition 38.8.2, Definition 38.8.4.

Theorem (Cousin's Theorem)

Theorem 38.8.6 (Cousin's Theorem). *Every gauge $\delta : [a, b] \to (0, \infty)$ on a closed interval $[a, b]$ admits a δ-fine tagged partition of $[a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } \delta : [a, b] \to (0, \infty).\\ &\exists m \in \mathbb{N}\ \exists x_0, \dots, x_m\ \exists t_1, \dots, t_m\\ &\qquad \big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j \in \{1, \dots, m\}\ (t_j \in [x_{j-1}, x_j])\\ &\qquad \wedge \forall j \in \{1, \dots, m\}\ [x_{j-1}, x_j] \subseteq (t_j - \delta(t_j), t_j + \delta(t_j))\big).\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}\forall \delta : [a, b] \to (0, \infty) \quad \exists \dot{\mathcal{P}}\ \Big(&\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b])\\ &\wedge \mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)\Big).\end{aligned}$$

Remark (Negated quantified statement).

$$
\begin{aligned}
&\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } \delta : [a,b] \to (0,\infty).\\
&\neg\Big(\text{there exists a } \delta\text{-fine tagged partition of } [a,b]\Big)\\
&\iff \forall m \in \mathbb{N}\ \forall x_0, \ldots, x_m\ \forall t_1, \ldots, t_m\\
&\qquad \Big[\big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j\ (t_j \in [x_{j-1}, x_j])\big)\\
&\qquad \Rightarrow \exists j \in \{1, \ldots, m\}\ [x_{j-1}, x_j] \nsubseteq (t_j - \delta(t_j), t_j + \delta(t_j))\Big].
\end{aligned}
$$

Remark (Negation predicate reading).

$$\exists \delta : [a,b] \to (0,\infty)\ \forall \dot{\mathcal{P}}\ \Big(\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a,b]) \Rightarrow \neg\, \mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)\Big).$$

Remark (Failure modes). Failure would mean that a positive gauge δ defeats every tagged partition: no matter how the interval is subdivided and tagged, at least one subinterval is not contained in the gauge neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\forall \dot{\mathcal{P}}\ \ \mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a,b])}_{\text{all tagged partitions}} \quad \Rightarrow \quad \underbrace{\neg\, \mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)}_{\text{some interval escapes its gauge ball}} \quad .$$

Remark (Interpretation). The theorem is the gauge analogue of compactness for a closed interval. Although the allowed radius may vary from point to point, finitely many tagged subintervals can still be chosen so that each lies inside the neighborhood dictated by its own tag. The standard bisection proof uses the Nested Interval Property to rule out a gauge that defeats all finite tagged partitions.

Remark (Dependencies).

Definition 38.8.3, Definition 38.8.2, Definition 38.8.4.

Remark (Gauge proofs of the four classical theorems). Each proof follows the same pattern: define a gauge δ from the continuity hypothesis, invoke Cousin's theorem for a δ-fine partition, apply the Covering Lemma to reduce to finitely many tags, derive the global conclusion.

Boundedness. At each t, continuity gives $\delta(t)$ with $\mathrm{Within}(x, t, \delta(t)) \Rightarrow |f(x) - f(t)| \leq 1$. A δ-fine partition with tags $t_1, \ldots, t_n$ gives $K = \max_i |f(t_i)|$; the Covering Lemma yields $|f(x)| \leq 1 + K$ for all x.

EVT. Let $M = \sup f(I)$; suppose $f(x) < M$ everywhere. At each t, take $\delta(t)$ with $\mathrm{Within}(x, t, \delta(t)) \Rightarrow f(x) < \frac{1}{2}(M + f(t))$. A δ-fine partition produces $\widetilde{M} = \frac{1}{2}(M + \max_i f(t_i)) < M$ bounding f everywhere — contradiction.

Location of Roots. Assume $f(t) \neq 0$ for all t. At each t, take $\delta(t)$ preserving $\mathrm{sgn}(f(t))$. A δ-fine partition forces consistent sign from $f(a) < 0$ to $f(b) > 0$ — contradiction.

Heine–Cantor. Given $\varepsilon > 0$, at each t let $\delta(t)$ satisfy $\mathrm{Within}(x, t, 2\delta(t)) \Rightarrow |f(x) - f(t)| \leq \varepsilon/2$. A δ-fine partition yields $\delta_\varepsilon = \min_i \delta(t_i) > 0$. For $|x - u| < \delta_\varepsilon$: find t_i covering x; then $|u - t_i| \leq |u - x| + |x - t_i| < 2\delta(t_i)$; so $|f(x) - f(u)| \leq \varepsilon$.

Tagged Partitions and Mesh

Definition (Mesh of a Partition)

Definition 38.8.7 (Mesh of a Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. A real number m is the *mesh* (or *norm*) of P if and only if

$$m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

The mesh of P is denoted by $\|P\|$ and is the unique m satisfying this equivalence.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \ldots, x_n\} \text{ be a partition of } [a, b].\\ &m \text{ is the mesh of } P \iff m = \max_{1 \le i \le n} (x_i - x_{i-1}).\end{aligned}$$

Remark (Definition predicate reading)*.*

$$\mathrm{NormOfPartition}(P, m) :\!= m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

Remark (Negated quantified statement)*.*

$$\begin{aligned}&\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and let } P = \{x_0, \ldots, x_n\} \text{ be a partition of } [a, b].\\ &\neg\big(m \text{ is the mesh of } P\big) \iff \big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big) \lor \big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big).\end{aligned}$$

Remark (Negation predicate reading)*.*

$$\neg\,\mathrm{NormOfPartition}(P, m) \iff \big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big) \lor \big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big).$$

Remark (Failure modes)*.* Failure occurs either when m underestimates the largest subinterval of P or when m overshoots that largest subinterval.

Remark (Failure mode decomposition)*.*

$$\neg\,\mathrm{NormOfPartition}(P, m) = \underbrace{\big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big)}_{\text{Underestimate}} \lor \underbrace{\big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big)}_{\text{Overestimate}}.$$

Remark (Interpretation). The mesh of a finite partition records the length of the widest subinterval and thus measures the resolution of the partition. Refining a partition strictly decreases or preserves this value, so a small mesh signals uniformly short subintervals, a critical input for Riemann and gauge integral estimates. Failure arises when a proposed number does not coincide with the actual maximum subinterval length, either because some subinterval is longer than the proposal or because all subintervals are shorter.

Remark (Dependencies)*.*

Definition 38.8.1.

Definition (Refinement)

Definition 38.8.8 (Refinement)**.** Let $[a, b] \subseteq \mathbb{R}$ and let P and Q be partitions of $[a, b]$. Then Q is a refinement of P if and only if every point of P is also a point of Q; equivalently, $P \subseteq Q$.

Remark (Standard quantified statement)*.*

$$\forall x \ (x \in P \Rightarrow x \in Q).$$

Remark (Definition predicate reading)*.*

$$\text{Refinement}(Q, P) :\equiv P \subseteq Q.$$

Remark (Negated quantified statement)*.*

$$\exists x \ (x \in P \wedge x \notin Q).$$

Remark (Negation predicate reading)*.*

$$\neg \text{Refinement}(Q, P) \iff \exists x \in P \ (x \notin Q).$$

Remark (Failure modes)*.* Failure occurs precisely when some point listed in P fails to appear in Q, meaning Q omits at least one of the partition points it was supposed to inherit from P.

Remark (Failure mode decomposition)*.*

$$\underbrace{\exists x \in P \ (x \notin Q)}_{\text{point of } P \text{ omitted by } Q}.$$

Remark (Interpretation)*.* Refinement formalizes the idea of subdividing an interval more finely without losing any previously chosen partition points. The condition $P \subseteq Q$ guarantees that every subinterval defined by P is still represented when one passes to Q, allowing comparisons of sums or integrals computed on different meshes. Failure means that Q discards at least one point of P, so information tied to that breakpoint would be lost when transitioning between partitions.

Remark (Dependencies)*.*

Definition 38.8.1.

Definition (Common Refinement)

Definition 38.8.9 (Common Refinement)**.** Let $a < b$ and let P and Q be partitions of $[a, b]$. Their common refinement is the partition R obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a, b]$.

Remark (Standard quantified statement)*.*

Let $a < b$ in $\mathbb{R}$, and let P, Q be partitions of $[a, b]$.

$\forall R$ partition of $[a, b]$: R is the common refinement of P and $Q \iff R = P \cup Q$.

Remark (Definition predicate reading).

$$\mathrm{CommonRefinement}(R, P, Q) :\!= (R = P \cup Q).$$

Remark (Negated quantified statement).

$$\begin{aligned} &a < b \wedge P, Q \subseteq [a, b] \wedge R \subseteq [a, b] \\ &\implies \neg\big(\forall x \in \mathbb{R}\ (x \in R \iff x \in P \cup Q)\big) \\ &\iff \big(\exists x \in R\ (x \notin P \cup Q)\big) \vee \big(\exists x \in P \cup Q\ (x \notin R)\big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{CommonRefinement}(R, P, Q) \iff R \neq P \cup Q.$$

Remark (Failure modes). The claim that a partition R is the common refinement of P and Q fails exactly when R either contains a node not present in $P \cup Q$ or omits a node that $P \cup Q$ contributes. These are the only two structural ways in which R can differ from $P \cup Q$.

Remark (Failure mode decomposition).

$$R \neq P \cup Q \iff \underbrace{\exists x \in R \setminus (P \cup Q)}_{\text{extra node}} \vee \underbrace{\exists x \in (P \cup Q) \setminus R}_{\text{missing node}}.$$

Remark (Interpretation). Forming the common refinement of two partitions simply collects every partition point that either partition already uses, producing the unique partition whose nodes are exactly those existing ones. No new nodes are created beyond those already in P or Q, and none are discarded; the construction merely merges the lists so that later comparisons, such as evaluating tagged sums on both partitions simultaneously, can be performed without further adjustment.

Remark (Dependencies).

Definition 38.8.1

Chapter 39

Differentiation

SERIES	VOLUME III	BOOK II	CHAPTER 10
From Cantor to Ito	Classical Analysis	Functions, Continuity, and Differentiation	Differentiation

39.1 The Tangent Problem

Secant and Tangent — Quick Reference

Concept	Meaning
Secant line	Line through two points of a graph
Tangent line	Limiting local line determined by the derivative

Remark (Section guide). Before the derivative is a formula it is a question: what straight line best matches a curve at a single point? The route in is indirect: choose a second point, draw the computable secant line, and let that point slide toward the first. If the secants settle toward a limiting line, that line is the tangent and its slope is the derivative.

Definition (Secant Line)

Definition 39.1.1 (Secant Line)**.** Let $f : A \to \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A\ (x_1 \neq x_2 \Rightarrow \text{SecantLine}(f, x_1, x_2) \text{ has slope } \tfrac{f(x_2)-f(x_1)}{x_2-x_1}).$$

Remark (Interpretation). With increments $\Delta x := x_2 - x_1$ and $\Delta y := f(x_2) - f(x_1)$, the secant slope is $\Delta y/\Delta x$ — the average rate of change of f over the interval $[x_1, x_2]$. This is an exact algebraic identity; no limit is involved. As $x_2 \to x_1$, if the secant slope converges, its limit is the derivative $f'(x_1)$ — the instantaneous rate of change. Geometrically, the secant line rotates about the fixed point $(x_1, f(x_1))$ as x_2 approaches x_1, and the limit position is the tangent line.

Definition (Difference Quotient)

Definition 39.1.2 (Difference Quotient)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and let $x \in A$ with $x \neq c$. The *difference quotient* of f from c to x is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

Remark (Interpretation)*.* The difference quotient is the slope of the secant line through $(c, f(c))$ and $(x, f(x))$. It is the quantity whose limiting value, when it exists as $x \to c$, becomes the derivative.

Remark (Dependencies)*.*

- Secant Line

Definition (Tangent Line)

Definition 39.1.3 (Tangent Line)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c. The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

Remark (Standard quantified statement)*.*

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \text{LinearApproximationAt}(f, c) = f(c) + f'(c)(x - c).$$

Remark (Interpretation)*.* The tangent line is the limit of the family of secant lines through $(c, f(c))$ and $(x, f(x))$ as $x \to c$. More precisely: the slope $m_{\text{sec}}(c, x) = (f(x) - f(c))/(x - c)$ converges to $f'(c)$, so the tangent line is the line whose slope is that limit. It is also the unique linear function $\ell(x) = f(c) + f'(c)(x - c)$ satisfying $\ell(c) = f(c)$ and $\ell'(c) = f'(c)$ — the best first-order approximation to f near c. The error $f(x) - \ell(x) = o(x - c)$ as $x \to c$, meaning it vanishes faster than the displacement.

Remark (Dependencies)*.*

- Definition: Secant Line
- Definition: Derivative at a Point

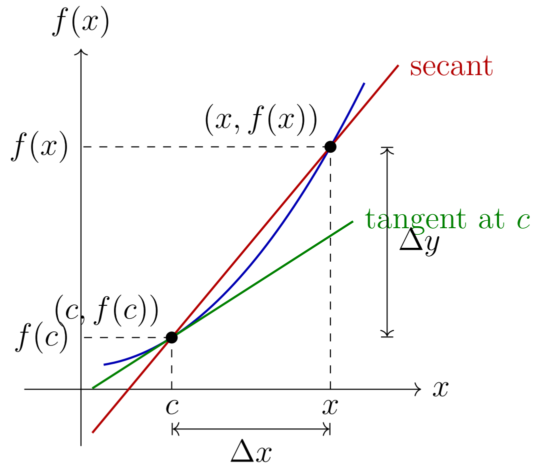


Figure 39.1: The secant line through $(c, f(c))$ and $(x, f(x))$ has slope $\Delta y/\Delta x$. As $x \to c$, the secant line rotates toward the tangent line at $(c, f(c))$, whose slope is $f'(c)$.

39.2 The Derivative

Derivative Definition — Quick Reference

Concept	**Meaning**
Derivative at a point	Epsilon–delta limiting secant slope
Example: derivative of x^2	The definition in a removable-singularity computation
Topological derivative	Neighbourhood formulation of the same limit
Sequential derivative	Sequence test for differentiability
Equivalent definitions	Epsilon–delta, topological, and sequential forms agree
h-form	Difference quotient written with $x = c + h$
Differentiability implies continuity	Differentiability is stronger than continuity
Example: one corner	Continuous but not differentiable at one point
Example: sawtooth	Continuous with infinitely many isolated corners
Example: Weierstrass function	Continuous and nowhere differentiable
Uniqueness	The derivative value is single-valued
Identity derivative	Base case $D(x) = 1$
Constant derivative	Base case $D(C) = 0$

Remark (Section guide). The derivative is the number the secant slopes approach, but the word "limit" has three useful faces. The epsilon–delta form is the contract; the topological form strips it to neighbourhoods; the sequential form is the disproving tool, since one bad approach sequence is enough.

Definition (Derivative at a Point)

Definition 39.2.1 (Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ be a real-valued function and $c \in \operatorname{int}(D)$. We say f is *differentiable at* c with *derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D\Big(0 < |x - c| < \delta \implies \left|\frac{f(x) - f(c)}{x - c} - L\right| < \varepsilon\Big).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D\Big(0 < |x - c| < \delta \implies \left|\frac{f(x) - f(c)}{x - c} - L\right| < \varepsilon\Big).$$

Remark (Definition predicate reading).

$$\text{DerivativeAt}(f, c, L) \iff \text{FunctionLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\, D,\, c,\, L\right).$$

The predicate DerivativeAt(f, c, L) asserts that L is the value of the limit of the difference quotient of f at c.

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in D\Big(0 < |x - c| < \delta\ \wedge\ \left|\frac{f(x) - f(c)}{x - c} - L\right| \geq \varepsilon\Big).$$

Remark (Negation predicate reading).

$$\neg\,\text{DerivativeAt}(f, c, L) \iff \neg\,\text{FunctionLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\, D,\, c,\, L\right).$$

Remark (Failure modes).

- The difference quotient has no finite limit at c (e.g. $|f(x) - f(c)|/|x - c| \to \infty$).
- The difference quotient oscillates without settling near any single value.
- The proposed value L is the wrong candidate.

Remark (Interpretation). The derivative is a limit of slopes of secant lines. The ε-δ form demands that the difference quotient can be made arbitrarily close to L by taking x sufficiently close to c. The input condition $0 < |x - c|$ is punctured: $x = c$ is excluded because $(f(c) - f(c))/(c - c)$ is $0/0$.

Remark (Dependencies).

- Limit of a Function

Example (Derivative of x^2 from the Definition). Let $f(x) = x^2$ and fix $c \in \mathbb{R}$. For $x \neq c$,

$$\frac{f(x) - f(c)}{x - c} = \frac{x^2 - c^2}{x - c} = x + c.$$

The singularity has been removed by exact algebra, so the limit as $x \to c$ is $2c$. Hence $f'(c) = 2c$. Read through Carathéodory, this is the factor $\varphi(x) = x + c$, continuous at c with $\varphi(c) = 2c$.

Definition (Topological Definition of Derivative at a Point)

Definition 39.2.2 (Topological Definition of Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ and $c \in \operatorname{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L, there exists a neighbourhood U of c such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Standard quantified statement).

$$\forall V \text{ neighbourhood of } L \; \exists U \text{ neighbourhood of } c \; \forall x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \forall V \ni L \; \exists U \ni c \; \forall x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \in V.$$

Remark (Negated quantified statement).

$$\exists V \text{ neighbourhood of } L \; \forall U \text{ neighbourhood of } c \; \exists x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Negation predicate reading).

$$\neg \, \mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \exists V \ni L \; \forall U \ni c \; \exists x \in (U \setminus \{c\}) \cap D : \; \frac{f(x) - f(c)}{x - c} \notin V.$$

Remark (Failure modes)*.* The topological derivative fails when a fixed output neighbourhood of L cannot be entered by the difference quotient no matter how tightly the input is restricted around c. Geometric reading: no punctured neighbourhood around c maps all secant slopes into any prescribed window around L.

Remark (Interpretation)*.* The topological form replaces ε-balls with abstract neighbourhoods, making the definition coordinate-free and suited for generalisation to metric spaces and manifolds.

Remark (Dependencies).

- Function
- Relative Neighborhood
- Subtraction on $\mathbb{R}$
- Division on $\mathbb{R}$

Definition (Sequential Definition of Derivative at a Point)

Definition 39.2.3 (Sequential Definition of Derivative at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. We say that f is differentiable at c with derivative L if

$$\forall (x_n) \subseteq A \setminus \{c\} \; \big(x_n \to c \Rightarrow \tfrac{f(x_n) - f(c)}{x_n - c} \to L \big).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \; \Rightarrow \; \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{seq}}(f, c, L) \iff \forall (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \; \Rightarrow \; \frac{f(x_n) - f(c)}{x_n - c} \to L.$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \; \wedge \; \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{DerivativeAt}_{\mathrm{seq}}(f, c, L) \iff \exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \; \wedge \; \frac{f(x_n) - f(c)}{x_n - c} \not\to L.$$

Remark (Failure modes). Failure occurs when a single approach sequence $(x_n) \to c$ exists along which the difference quotients either: (i) diverge to infinity (vertical tangent); (ii) oscillate without settling; (iii) converge but to a value other than L (wrong slope candidate). A single such sequence certifies non-differentiability.

Remark (Failure mode decomposition).

$$\underbrace{x_n \to c}_{\text{approach condition}} \quad \wedge \quad \underbrace{\frac{f(x_n) - f(c)}{x_n - c} \not\to L}_{\text{quotient failure}}.$$

The two canonical witness types are: two subsequences with different limiting difference quotients (e.g., $f(x) = |x|$ at $c = 0$), or quotients growing without bound.

Remark (Interpretation). The sequential form is the primary tool for disproving differentiability: exhibit a single sequence $x_n \to c$ along which the difference quotient fails to converge to any single value.

Remark (Dependencies).

- Sequence
- Convergent Sequence
- Derivative at a Point
- Limit of a Function

Theorem (Equivalence of Definitions of the Derivative at a Point)

Theorem 39.2.4 (Equivalence of Definitions of the Derivative at a Point)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$ be a limit point of A. The following are equivalent:*

1. *f is differentiable at c in the ε-δ sense with derivative L.*
2. *f is differentiable at c in the topological sense with derivative L.*
3. *f is differentiable at c in the sequential sense with derivative L.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{(i)} \iff \text{(ii)} \iff \text{(iii)}.$$

Remark (Predicate reading)*.*

$$\mathrm{DerivativeAt}(f, c, L) \iff \mathrm{DerivativeAt}_{\mathrm{top}}(f, c, L) \iff \mathrm{DerivativeAt}_{\mathrm{seq}}(f, c, L).$$

All three predicates name the same underlying condition on the difference quotient of f at c.

Remark (Negated quantified statement)*.*

$$\begin{aligned} \neg\text{(i)} &\iff \neg\text{(ii)} \iff \neg\text{(iii)} \\ &\iff \exists (x_n) \subseteq A \setminus \{c\} : \; x_n \to c \;\wedge\; \frac{f(x_n) - f(c)}{x_n - c} \not\to L. \end{aligned}$$

Remark (Failure modes)*.* The equivalence cannot produce contradictory verdicts: if c is a limit point of A, all three forms agree. Degenerate case: if c is not a limit point of A, the punctured-ball and sequence conditions are vacuously satisfied by every candidate L simultaneously, making the equivalence trivially true for all L rather than characterizing a unique derivative.

Remark (Interpretation)*.* The three forms are notational restatements of the same underlying limit condition on the difference quotient. Analysts choose whichever form is most convenient for the proof at hand.

Remark (Dependencies)*.*

- Derivative at a Point
- Topological Definition
- Sequential Definition

Proposition (Derivative: h-Form Equivalence)

Proposition 39.2.5 (Derivative: h-Form Equivalence)**.** *Let $f : A \to \mathbb{R}$ and $c \in \operatorname{int}(A)$. Then f is differentiable at c with derivative L if and only if*

$$\lim_{h\to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{DerivativeAt}(f, c, L) \iff \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Remark (Negated quantified statement)*.*

$$\neg\,\text{DerivativeAt}(f, c, L) \iff \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} \neq L.$$

Remark (Failure modes)*.* The h-form fails in two branches: (i) The incremental ratio $(f(c+h) - f(c))/h$ has no finite limit as $h \to 0$ (corner, cusp, or vertical tangent at c). (ii) The ratio converges to a finite value different from L (proposed L is the wrong slope).

Remark (Interpretation)*.* The h-form substitutes $x = c + h$ in the difference quotient. It is the standard notation in calculus and makes the incremental structure of the derivative explicit.

Remark (Dependencies)*.*

- Derivative at a Point

Remark (Discrete and continuous change)*.* The forward difference operator

$$\Delta f(n) = f(n+1) - f(n)$$

is the discrete analogue of the derivative. The h-form above is its continuous descendant: replace the fixed step 1 by a shrinking step h, divide by h, and pass to the limit.

The kinship matters, but so does the divergence. The discrete product rule has an unavoidable shift:

$$\Delta(fg)(n) = (\Delta f)(n)g(n) + f(n+1)(\Delta g)(n).$$

The continuous product rule loses this shift because $f(x+h) \to f(x)$ as $h \to 0$. Likewise, ordinary powers are not the clean basis for Δ; falling factorials are:

$$\Delta\, x^{\underline{k}} = k\, x^{\underline{k-1}}.$$

Passing to the limit is therefore not just a sharpening of a quotient. It restores symmetries that the discrete operator must carry as shifts.

39.2.1 First Consequences

Theorem (Differentiable Implies Continuous)

Theorem 39.2.6 (Differentiable Implies Continuous)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, then f is continuous at c.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\mathrm{DerivableAt}(f, c) \;\Rightarrow\; \mathrm{ContinuousAt}(f, c).$$

Remark (Predicate reading)*.*

$$\mathrm{DifferentiableAt}(f, c) \implies \mathrm{ContinuousAt}(f, c).$$

Remark (Failure modes)*.* The implication can only fail in the reverse direction: continuity at c does not force differentiability at c.

Remark (Failure mode decomposition)*.* The converse fails: $f(x) = |x|$ is continuous at 0 but not differentiable there.

Remark (Contrapositive quantified statement)*.*

$$\neg\mathrm{ContinuousAt}(f, c) \;\Rightarrow\; \neg\mathrm{DifferentiableAt}(f, c).$$

Remark (Contrapositive predicate reading)*.*

$$\neg\, \mathrm{ContinuousAt}(f, c) \implies \neg\, \mathrm{DifferentiableAt}(f, c).$$

Remark (Interpretation)*.* Differentiability is strictly stronger than continuity: the existence of a well-defined tangent slope forces the function values to coalesce at $f(c)$ as $x \to c$.

Remark (Dependencies)*.*

- Derivative at a Point
- Continuity at a Point

Remark (Converse failure)*.* The theorem runs one way only. Differentiability forces continuity, but continuity does not force differentiability. The negation of the converse is

$$\exists f\ \exists c :\ \mathrm{ContinuousAt}(f, c) \wedge \neg\, \mathrm{DifferentiableAt}(f, c).$$

The examples below escalate the failure: one corner, infinitely many isolated corners, and no differentiability anywhere.

Example (One Corner: $|x|$)*.* The function $f(x) = |x|$ is continuous on $\mathbb{R}$, but it is not differentiable at 0. Its one-sided derivatives are -1 and $+1$, so the two one-sided slopes do not agree. Thus a single corner is enough to separate continuity from differentiability.

Example (Finitely Many Corners: A Sawtooth). On a compact interval, a triangular sawtooth made from finitely many line segments can be continuous while failing to be differentiable at each corner. For example,

$$s(x) = \begin{cases} x, & 0 \le x \le 1, \\ 2 - x, & 1 \le x \le 2, \end{cases}$$

is continuous on $[0, 2]$, but the left derivative at 1 is 1 and the right derivative is -1. Adding more teeth gives any finite number of corner failures without breaking continuity.

Example (The Weierstrass Function). For $0 < a < 1$ and an odd integer b satisfying $ab > 1 + \frac{3}{2}\pi$, the Weierstrass function

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

is continuous on $\mathbb{R}$, because the series converges uniformly by the Weierstrass M-test. Classically, under the displayed condition, it is differentiable at no point. Thus continuity may hold everywhere while differentiability holds nowhere.

Remark (Proof status). The continuity of W follows from uniform convergence once the function-series machinery is available. The nowhere-differentiability proof is a classical result of Weierstrass and is deferred to the function-series development; it is used here only as a counterexample to the converse of Differentiable Implies Continuous.

Remark (Forward flag). Brownian motion has sample paths that are almost surely continuous everywhere and differentiable nowhere. This is the stochastic analogue of the Weierstrass phenomenon and one reason ordinary calculus must be replaced by the Itô integral in stochastic analysis.

Theorem (Uniqueness of the Derivative)

Theorem 39.2.7 (Uniqueness of the Derivative). *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, its derivative is unique.*

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \ \wedge \ \mathrm{DerivativeAt}(f, c, L') \ \Rightarrow \ L = L'.$$

Remark (Predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \ \wedge \ \mathrm{DerivativeAt}(f, c, L') \implies L = L'.$$

Remark (Failure modes). The uniqueness proof assumes $L \ne L'$ and takes $\varepsilon = |L - L'|/2 > 0$. The definition supplies δ making the difference quotient within ε of both L and L' simultaneously, forcing $|L - L'| < 2\varepsilon = |L - L'|$, a contradiction. The cluster-point hypothesis on c is essential: without it, the punctured-ball condition is vacuous and the limit is trivially satisfied by every value.

Remark (Contrapositive quantified statement).

$$L \neq L' \;\Rightarrow\; \neg\big(\mathrm{DerivativeAt}(f,c,L) \;\wedge\; \mathrm{DerivativeAt}(f,c,L')\big).$$

Two distinct candidates cannot both be the derivative of f at c.

Remark (Contrapositive predicate reading).

$$L \neq L' \implies \neg\,\mathrm{DerivativeAt}(f,c,L) \;\vee\; \neg\,\mathrm{DerivativeAt}(f,c,L').$$

Remark (Interpretation). Uniqueness licenses writing $f'(c)$ as a definite value rather than a set of candidates.

Remark (Dependencies).

- Derivative at a Point

Proposition (Derivative of the Identity)

Proposition 39.2.8 (Derivative of the Identity). *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall c \in \mathbb{R} : \; \frac{d}{dx}[x]\Big|_{x=c} = 1.$$

Remark (Interpretation). The identity function has slope 1 everywhere. It is the base case for computing derivatives of polynomials via the algebra of derivatives.

Remark (Dependencies).

- Derivative at a Point
- Function

Proposition (Derivative of a Constant)

Proposition 39.2.9 (Derivative of a Constant). *Let $k \in \mathbb{R}$ and let $f(x) = k$ for all $x \in \mathbb{R}$. Then $f'(c) = 0$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall k \in \mathbb{R}\, \forall c \in \mathbb{R} : \; \frac{d}{dx}[k]\Big|_{x=c} = 0.$$

Remark (Interpretation). A constant function has zero change across every secant line, so its instantaneous rate of change is zero everywhere. Together with the derivative of the identity, this is one of the base cases for the algebraic computation rules.

Remark (Dependencies).

- Derivative at a Point

39.3 One-Sided Derivatives

One-Sided Derivatives — Quick Reference

Concept	Meaning
Left-hand derivative	Limiting slope approached from the left
Right-hand derivative	Limiting slope approached from the right
Two-sided criterion	Differentiability iff both one-sided derivatives agree
Example: $\lvert x\rvert$ at 0	A continuous corner with unequal one-sided slopes

Remark (Section guide). A corner is a place where a curve has a left answer and a right answer and refuses to choose. One-sided derivatives interrogate each side alone, and their agreement criterion is the fastest way to certify many failures of differentiability.

Definition (Left-Hand Derivative)

Definition 39.3.1 (Left-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I. We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x\to c^-} \frac{f(x)-f(c)}{x-c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x\to c^-} \frac{f(x)-f(c)}{x-c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \Big(-(\delta) < x - c < 0 \implies \left| \frac{f(x)-f(c)}{x-c} - L \right| < \varepsilon \Big).$$

In this case $f'_-(c) = L$.

Remark (Definition predicate reading).

$$\mathrm{LeftDerivativeAt}(f, c, L) \iff \mathrm{LeftLimit}\left(\lambda x.\ \frac{f(x)-f(c)}{x-c},\ c,\ L \right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \Big(-\delta < x - c < 0\ \wedge\ \left| \frac{f(x)-f(c)}{x-c} - L \right| \geq \varepsilon \Big).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R}\ \neg\, \mathrm{LeftDerivativeAt}(f, c, L).$$

No finite L is the left-hand limit of the difference quotient.

Remark (Interpretation). The left-hand derivative $f'_-(c)$ is the limit of the difference quotient as x approaches c *strictly from the left*. It measures the instantaneous rate of change of f at c as seen from the left side. The right-hand derivative $f'_+(c)$ is the symmetric object from the right. At interior points of an interval, both one-sided derivatives are defined, and the full derivative $f'(c)$ exists if and only if they agree. At endpoints, only the inward one-sided derivative is accessible, and it serves as the boundary derivative. The canonical example: for $f(x) = |x|$ at $c = 0$, $f'_-(0) = -1$ and $f'_+(0) = +1$; since these disagree, $f'(0)$ does not exist.

Remark (Dependencies).

- Definition: Derivative at a Point

Definition (Right-Hand Derivative)

Definition 39.3.2 (Right-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I. We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \Big(0 < x - c < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

In this case $f'_+(c) = L$.

Remark (Definition predicate reading).

$$\text{RightDerivativeAt}(f, c, L) \iff \text{RightLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\ c,\ L\right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \Big(0 < x - c < \delta\ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \Big).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R}\ \neg\, \text{RightDerivativeAt}(f, c, L).$$

Remark (Interpretation). The right-hand derivative $f'_+(c)$ is the instantaneous rate of change seen from inputs larger than c. It is the endpoint derivative available at the left endpoint of a closed interval, and it is one half of the two-sided derivative test at interior points.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Left-Hand Derivative

Theorem

Theorem 39.3.3 (Differentiability and One-Sided Derivatives)**.** *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$. Then f is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f, c) \iff \Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L)\Big) \text{ for some } L \in \mathbb{R}.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{DifferentiableAt}(f, c) \iff \neg\Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L)\Big) \text{ for all } L \in \mathbb{R}.$$

That is: either a one-sided derivative fails to exist, or both exist but are unequal.

Remark (Interpretation). This theorem decomposes differentiability at an interior point into two one-sided conditions. It is the standard tool for proving non-differentiability: to show f is not differentiable at c, compute $f'_-(c)$ and $f'_+(c)$ and observe they disagree. For $f(x) = |x|$ at $c = 0$: $f'_-(0) = -1 \neq +1 = f'_+(0)$, so f is not differentiable at 0. The theorem also provides the correct definition of differentiability on a closed interval $[a, b]$: differentiable on the open interior (a, b) in the full sense, plus $f'_+(a)$ and $f'_-(b)$ both exist at the endpoints.

Remark (Dependencies).

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Definition: Derivative at a Point

Example ($|x|$ Has No Derivative at 0). For $f(x) = |x|$ at $c = 0$, the one-sided difference quotients are

$$\frac{|x| - 0}{x - 0} = \frac{|x|}{x} = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

Thus $f'_-(0) = -1$ and $f'_+(0) = 1$. The two values disagree, so the two-sided derivative does not exist at 0, even though f is continuous there.

39.4 Carathéodory and the Chain Rule

Carathéodory and Chain Rule — Quick Reference

Concept	Role
Carathéodory factorization	Continuous slope factor for differentiability
Carathéodory characterization	Differentiability iff the slope factor extends continuously
Chain Rule	Derivative of a composition
Leibniz Rule	n-th derivative of a product
Faa di Bruno	n-th derivative of a composition
Faa di Bruno at $n = 2$	Coefficient check for the second derivative
Faa di Bruno at $n = 3$	Coefficient check for the third derivative

39.4.1 Carathéodory Characterization

Remark (Section guide). Carathéodory replaces division by factorisation: $f(x) - f(c) = (x - c)\varphi(x)$, with φ continuous at c. Differentiability becomes continuity of the slope factor, and continuity composes. That is why the chain rule becomes clean rather than a fight with vanishing denominators.

Definition (Carathéodory Factorization)

Definition 39.4.1 (Carathéodory Factorization)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Remark (Standard quantified statement).

$$\exists \phi : I \to \mathbb{R}\ \big(\text{ContinuousAt}(\phi, c)\ \wedge\ \forall x \in I\ [f(x) = f(c) + (x - c)\phi(x)]\big).$$

Remark (Definition predicate reading). The existence of a Carathéodory factorization at c is equivalent to DifferentiableAt(f, c) by the characterization theorem below. The factorizing function is explicit:

$$\phi(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

Differentiability of f at c is precisely the statement that this piecewise-defined ϕ is continuous at c.

Remark (Interpretation). The Carathéodory factorization replaces the difference quotient — a function with a removable discontinuity at c — with a function ϕ that is genuinely continuous at c. The factorization $f(x) - f(c) = (x - c)\phi(x)$ holds everywhere on I, including at

$x = c$ where both sides equal zero. The derivative is recovered as $\phi(c)$ without ever dividing by zero. This reframing converts analytical limit conditions into topological continuity conditions, which compose and factor more cleanly. It is the preferred tool for proving the product rule, quotient rule, and chain rule, because continuity of a product or composition is easier to establish than the existence of a limit of a quotient.

Carathéodory vs. Lipschitz. Both control the increment $f(x) - f(c)$ by the factor $(x - c)$, but differently. The Carathéodory factorization requires the slope function $\phi(x)$ to extend *continuously* to c; it controls the *behaviour* of the slope, not just its size. The Lipschitz condition $|f(x)-f(c)| \leq L|x-c|$ only bounds the size of the increment. Carathéodory implies Lipschitz: if ϕ is continuous at c it is bounded near c, so $|f(x)-f(c)| = |\phi(x)||x-c| \leq K|x-c|$ for some K near c. The converse fails: $f(x) = |x|$ is Lipschitz on $\mathbb{R}$ with constant 1 but the slope function takes values ± 1 on either side of 0 with no continuous extension.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 39.4.2 (Carathéodory Characterization of Differentiability)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

(i) f is differentiable at c.

(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \textit{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. *Go to proof. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} \text{DifferentiableAt}(f, c) \iff \; & \exists \phi : I \to \mathbb{R} \; \big(\text{ContinuousAt}(\phi, c) \\ & \wedge \; \forall x \in I \; [f(x) = f(c) + (x - c)\phi(x)]\big). \end{aligned}$$

In this case $\phi(c) = f'(c)$.

Remark (Interpretation). This theorem establishes that differentiability and the existence of a Carathéodory factorization are exactly the same condition. The proof is a two-direction construction: in the forward direction, define ϕ as the difference quotient extended by $f'(c)$ at c and verify continuity from the definition of the derivative; in the backward direction, recover the difference quotient from ϕ and read off its limit as $\phi(c)$. The theorem's power is that it converts every future differentiability argument into a continuity argument: to show $g \circ f$ is differentiable, find a continuous Carathéodory function for it; composing the Carathéodory functions of f and g yields one automatically.

Remark (Dependencies).

- Definition: Carathéodory Factorization
- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 39.4.3 (Chain Rule). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c, and*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, f(c)) \implies (g \circ f)'(c) = g'(f(c))f'(c).$$

Remark (Interpretation). The derivative of a composition is the derivative of the outside function evaluated at the inside value, multiplied by the derivative of the inside function.

Remark (Interpretation). Let ϕ_f and ϕ_g be the Carathéodory slope factors for f at c and g at $f(c)$. Then

$$g(f(x)) - g(f(c)) = \phi_g(f(x))(f(x) - f(c)) = \phi_g(f(x))\phi_f(x)(x - c).$$

The new slope factor is

$$\phi_{g\circ f}(x) = \phi_g(f(x))\phi_f(x),$$

which is continuous at c and has value $g'(f(c))f'(c)$. This proof never divides by $f(x) - f(c)$, so it avoids the failure in the informal $\Delta g/\Delta f \cdot \Delta f/\Delta x$ argument when $f(x) = f(c)$ near c.

Remark (Dependencies).

- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Differentiable Implies Continuous
- Theorem: Composition of Limits
- Theorem: Product Rule for Function Limits

Theorem (Leibniz Rule)

Theorem 39.4.4 (Leibniz Rule). *Let $I \subseteq \mathbb{R}$ be an interval, let $x \in I$, and suppose that $f, g : I \to \mathbb{R}$ are n-times differentiable on a neighbourhood of x. Then fg is n-times differentiable at x, and*

$$(fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x)g^{(n-k)}(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Diff}^n(f,x) \wedge \mathrm{Diff}^n(g,x) \Longrightarrow (fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Remark (Interpretation). The binomial pattern is not an accident. Each of the n differentiations lands on either f or g, and $\binom{n}{k}$ counts the ways to send exactly k differentiations to f. At $n = 1$, this is the ordinary product rule.

Remark (Dependencies).

- Theorem: Product Rule
- Definition: Higher Derivatives

Theorem (Faa di Bruno's Formula)

Theorem 39.4.5 (Faa di Bruno's Formula)**.** *Let $I, J \subseteq \mathbb{R}$ be intervals, let $x \in I$, suppose $g : I \to J$ is n-times differentiable on a neighbourhood of x, and suppose $f : J \to \mathbb{R}$ is n-times differentiable on a neighbourhood of $g(x)$. Then $f \circ g$ is n-times differentiable at x, and*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^{n} m k_m = n} \frac{n!}{\prod_{m=1}^{n} k_m!\,(m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^{n} \left(g^{(m)}(x)\right)^{k_m},$$

where $k_m \geq 0$ and $k = \sum_{m=1}^{n} k_m$. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Diff}^n(g,x) \wedge \mathrm{Diff}^n(f,g(x)) \Longrightarrow (f \circ g)^{(n)}(x) = \sum_{\sum m k_m = n} \frac{n!}{\prod_m k_m!\,(m!)^{k_m}} f^{(k)}(g(x)) \prod_m \left(g^{(m)}(x)\right)^{k_m}.$$

Remark (Interpretation). The chain rule gives the first derivative of a composition. Repeated differentiation is deeper: product-rule terms and chain-rule terms interlace, and the bookkeeping is governed by integer partitions. Faa di Bruno's formula is the all-orders chain rule.

Remark (Hypotheses). Smoothness is more than this formula needs. The n-th order statement requires n derivatives of g near x and n derivatives of f near $g(x)$. At $n = 1$, the formula is exactly the chain rule.

Remark (Structural reading). The tuple $(k_1, \ldots, k_n)$ records an integer partition of n: k_m is the number of parts of size m. The coefficient

$$\frac{n!}{\prod_m k_m!\,(m!)^{k_m}}$$

counts set partitions with that block-size profile. The outer derivative $f^{(k)}$ is differentiated once per block, while each block of size m contributes one factor $g^{(m)}$.

Remark (Normalization). The formula uses the bare-product normalization: the coefficient contains $(m!)^{k_m}$, and the product contains $\big(g^{(m)}(x)\big)^{k_m}$. Equivalently, one may put $g^{(m)}(x)/m!$ inside the product and remove $(m!)^{k_m}$ from the denominator. Mixing these two conventions gives the wrong coefficients.

Remark (Cross-volume callback). The constraint $\sum_m mk_m = n$ is precisely the multiplicity form of an integer partition of n. Thus the number of summands is $p(n)$, the partition function from the discrete and number-system material, now appearing as bookkeeping for iterated differentiation.

Remark (Scope). This theorem is enrichment, not a critical dependency for the integration, measure, or manifold arcs. It is included because it is the complete all-orders chain rule and a useful callback to partitions.

Remark (Dependencies).

- Theorem: Chain Rule
- Theorem: Leibniz Rule
- Definition: Higher Derivatives
- Definition: Class C^k

Example (Faa di Bruno at $n = 2$). The partitions of 2 are represented by $(k_1, k_2) = (2, 0)$ and $(0, 1)$. The first gives $f''(g)(g')^2$, and the second gives $f'(g)g''$. Hence

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g''.$$

The coefficient on $f'(g)g''$ is 1, which is the quickest check that the normalization has not been mixed.

Example (Faa di Bruno at $n = 3$). The partitions of 3 produce the three terms

$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The middle coefficient 3 counts the three ways to choose which differentiation contributes the second-derivative block of g.

39.5 The Shape Questions

Shape Questions — Quick Reference

Concept	Meaning
Increasing at a point	Local right-up, left-down behavior
Decreasing at a point	Local right-down, left-up behavior
Relative minimum	Locally smallest function value
Relative maximum	Locally largest function value
Relative extremum	Relative minimum or maximum
Convex function	Graph lies below its chords
Concave function	Graph lies above its chords
Inflection point	Local change in bending behavior

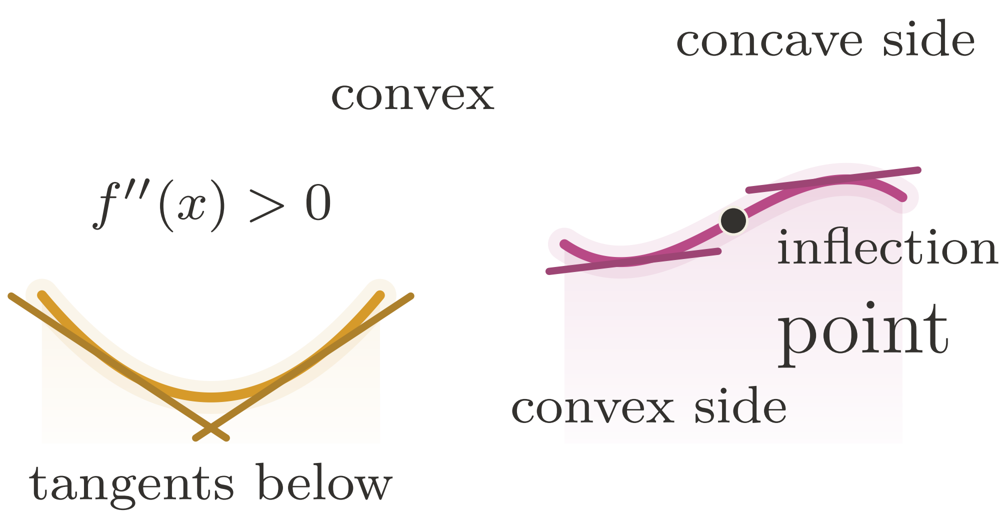


Figure 39.2: Convexity, concavity, and the change of bending at an inflection point.

Definition (Convex Function)

Definition 39.5.1 (Convex Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *convex* on I if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Remark (Standard quantified statement)*.*

$$\forall x, y \in I \ \forall \lambda \in [0, 1] : \quad f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Remark (Interpretation)*.* Convexity is defined by chords, not derivatives. The graph lies below the line segment joining any two of its points. Derivative tests are later tools for detecting this geometric condition, not the definition of the condition.

Remark (Dependencies).

- Closed Interval

Definition (Concave Function)

Definition 39.5.2 (Concave Function). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *concave* on I if $-f$ is convex on I.

Remark (Standard quantified statement).

$$\mathrm{ConcaveOn}(f, I) \iff \mathrm{ConvexOn}(-f, I).$$

Remark (Interpretation). Concavity is the downward-bending counterpart of convexity. Equivalently, the graph of a concave function lies above each chord joining two of its points.

Remark (Dependencies).

- Convex Function

Definition (Inflection Point)

Definition 39.5.3 (Inflection Point). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be an interior point. The point c is an *inflection point* of f if f changes convexity at c: there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that f is convex on one side of c and concave on the other side.

Remark (Standard quantified statement).

$$\mathrm{InflectionPoint}(f, c) \iff f \text{ changes convexity at } c.$$

Remark (Interpretation). An inflection point is a change in the direction of bending. It is not merely a point where $f''(c) = 0$; that equation is only a necessary condition under additional hypotheses.

Remark (Dependencies).

- Convex Function
- Concave Function

39.5.1 Interior Extrema

Interior Extrema — Quick Reference

Statement	Role
Relative minimum	Local lower comparison point
Relative maximum	Local upper comparison point
Relative extremum	Umbrella term for local optima
Interior Extremum Theorem	Interior differentiable extremum has zero derivative
Necessary condition	Extrema occur at critical behavior
Critical-point corollary	Derivative zero or derivative undefined

Definition (Relative Minimum)

Definition 39.5.4 (Relative Minimum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \leq f(x) \qquad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(c) < f(x) \qquad \text{for all } x \in \big(A \cap (c - \delta, c + \delta)\big) \setminus \{c\},$$

then f has a *strict relative minimum* at c.

Remark (Standard quantified statement)*.*

$$\exists \delta > 0 \ \forall x \in A \ \big(|x - c| < \delta \implies f(c) \leq f(x)\big).$$

Remark (Definition predicate reading)*.*

$$\mathrm{LocalMinimum}(f, c) \iff \exists \delta > 0 \ \forall x \in A \ \big(\mathrm{Within}(x, c, \delta) \implies f(c) \leq f(x)\big).$$

Remark (Interpretation)*.* f has a relative minimum at c if $f(c)$ is the smallest value of f in some open neighbourhood of c within the domain. The word *relative* (or *local*) signals that we are comparing $f(c)$ only to nearby values, not to all values of f globally. The strict variant excludes the degenerate case where nearby points share the minimum value. The symmetric notion — relative maximum — reverses the inequality. Together they constitute a relative extremum. These are purely neighbourhood conditions; no derivative appears in the definitions.

Remark (Dependencies)*.*

- Function
- Subset
- Open Interval
- Order on $\mathbb{R}$

Definition (Relative Maximum)

Definition 39.5.5 (Relative Maximum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \leq f(c) \qquad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(x) < f(c) \qquad \text{for all } x \in \big(A \cap (c - \delta, c + \delta)\big) \setminus \{c\},$$

then f has a *strict relative maximum* at c.

Remark (Standard quantified statement).

$$\exists \delta > 0 \; \forall x \in A \; \big(|x - c| < \delta \implies f(x) \le f(c)\big).$$

Remark (Definition predicate reading).

$$\mathrm{LocalMaximum}(f, c) \iff \exists \delta > 0 \; \forall x \in A \; \big(\mathrm{Within}(x, c, \delta) \implies f(x) \le f(c)\big).$$

Remark (Interpretation). A relative maximum is a point whose function value is at least as large as all nearby function values in the domain.

Remark (Dependencies).

- Definition: Relative Minimum

Definition (Relative Extremum)

Definition 39.5.6 (Relative Extremum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

Remark (Definition predicate reading).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

Remark (Interpretation). Relative extrema are the local turning candidates of a graph: nearby values lie all on one side of $f(c)$.

Remark (Dependencies).

- Definition: Relative Minimum
- Definition: Relative Maximum

Theorem

Theorem 39.5.7 (Interior Extremum Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c, then*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \land \mathrm{DifferentiableAt}(f, c) \Rightarrow \mathrm{DerivativeAt}(f, c, 0).$$

Remark (Contrapositive quantified statement).

$$\text{DerivativeAt}(f, c, L) \ \wedge \ L \neq 0 \ \Rightarrow \ \neg\,\text{LocalExtremum}(f, c).$$

If $f'(c)$ exists and is nonzero, then c is not a relative extremum.

Remark (Contrapositive predicate reading).

$$\text{DerivativeAt}(f, c, L) \ \wedge \ L \neq 0 \ \Rightarrow \ \neg\,\text{LocalMinimum}(f, c) \ \wedge \ \neg\,\text{LocalMaximum}(f, c).$$

Remark (Interpretation). The theorem gives a necessary condition for interior extrema under differentiability: the derivative must vanish. The geometric picture is immediate — at a local max or min, the tangent line must be horizontal, since the function is neither consistently increasing nor consistently decreasing. The proof is a sign argument: at a relative maximum, the difference quotient is non-positive for $x > c$ and non-negative for $x < c$; since the derivative is a single limit, it is squeezed to zero. The converse fails: $f'(c) = 0$ does not imply an extremum ($f(x) = x^3$ at $c = 0$). Critical points — where $f'(c) = 0$ or $f'(c)$ does not exist — are candidates for extrema, not guarantees.

Remark (Dependencies).

- Definition: Relative Extremum
- Definition: Derivative at a Point
- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Theorem: Differentiability and One-Sided Derivatives

Theorem

Theorem 39.5.8 (Necessary Condition for an Interior Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \ \Rightarrow \ \neg\,\text{DifferentiableAt}(f, c) \ \vee \ \text{DerivativeAt}(f, c, 0).$$

Remark (Interpretation). This is a weakening of the Interior Extremum Theorem that drops the differentiability hypothesis. Without assuming $f'(c)$ exists, the conclusion is a disjunction: either the derivative fails to exist at c, or it exists and equals zero. This is the correct form for identifying *critical points* — points that are candidates for extrema. The set of critical points is the union of points where f' is zero and points where f' does not exist.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem

Corollary

Corollary 39.5.9 (Necessary Condition for an Extremum)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\mathrm{LocalExtremum}(f, c) \;\Rightarrow\; \mathrm{DerivativeAt}(f, c, 0) \;\vee\; \neg\,\mathrm{DifferentiableAt}(f, c).$$

Remark (Interpretation)*.* An extremum can occur only at a critical point: either the derivative vanishes or the derivative fails to exist.

Remark (Dependencies)*.*

- Theorem: Necessary Condition for an Interior Extremum

39.6 The Mean Value Theorems

Mean Value Theorems — Quick Reference

Statement	**Role**
Rolle's Theorem	Equal endpoint values force a horizontal tangent
Mean Value Theorem	Some tangent slope equals the secant slope
Cauchy Mean Value Theorem	Compares two functions' rates of change
Derivative bound implies Lipschitz	Turns a slope bound into a global modulus

Remark (Section guide)*.* Rolle's theorem says that a differentiable curve returning to the same height must level off somewhere between. The Mean Value Theorem tilts that picture: between any two points, some interior tangent runs parallel to the chord. Cauchy's version compares two functions at once and later powers L'Hôpital's rule.

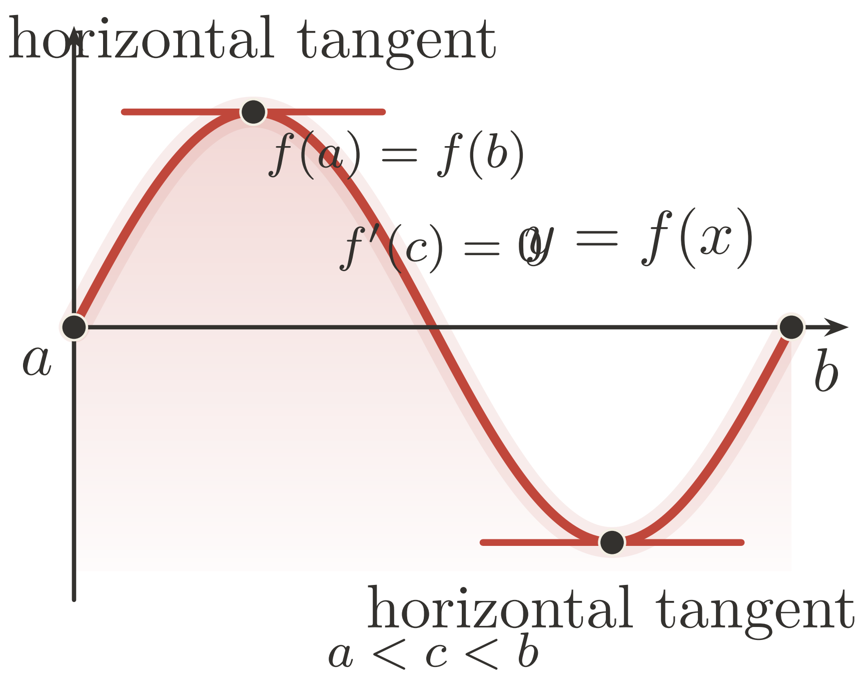


Figure 39.3: Rolle's Theorem: equal endpoint heights force a horizontal tangent.

Theorem (Rolle's Theorem)

Theorem 39.6.1 (Rolle's Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \ \wedge\ \forall x \in (a, b)\ \mathrm{DifferentiableAt}(f, x) \ \wedge\ f(a) = f(b)\\&\quad \Rightarrow\ \exists c \in (a, b)\ \mathrm{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \ \wedge\ \mathrm{DifferentiableOn}(f, (a, b)) \ \wedge\ f(a) = f(b)\\&\quad \Longrightarrow\ \exists c \in (a, b)\ :\ \mathrm{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \mathrm{sgn}(x - 1/2)$ on $[0, 1]$. Then $f(0) = f(1) = -1$, but f has a jump discontinuity at $x = 1/2$. The derivative does not exist anywhere, so there is no horizontal tangent.
- **Differentiability fails:** Let $f(x) = |x - 1/2|$ on $[0, 1]$. Then $f(0) = f(1) = 1/2$, and f is continuous, but f has a corner at $x = 1/2$ where the derivative is undefined. For all $c \neq 1/2$, we have $f'(c) = \pm 1 \neq 0$.

- **Equal endpoints fail:** Let $f(x) = x$ on $[0, 1]$. Then f is continuous and differentiable with $f'(x) = 1 > 0$ everywhere, but $f(0) = 0 \neq 1 = f(1)$.

Remark (Contrapositive quantified statement).

$$\big(\forall c \in (a,b) \ : \ f'(c) \neq 0\big) \ \Rightarrow \ f \text{ not continuous on } [a,b] \vee f \text{ not differentiable on } (a,b) \vee f(a) \neq f(b). \blacksquare$$

Remark (Contrapositive predicate reading).

$$\begin{aligned}&\big(\forall c \in (a,b) \ : \ \neg\,\mathrm{DerivativeAt}(f,c,0)\big)\\ &\quad \Longrightarrow \neg\,\mathrm{ContinuousOn}(f,[a,b]) \ \vee \ \neg\,\mathrm{DifferentiableOn}(f,(a,b)) \ \vee \ f(a) \neq f(b).\end{aligned}$$

Remark (Interpretation). If a differentiable function returns to the same value at both endpoints of an interval, the derivative must vanish somewhere in between. Geometrically: if the curve starts and ends at the same height, at some interior point the tangent line is horizontal. The proof combines two ingredients — the Extreme Value Theorem guarantees that a continuous function on a closed bounded interval attains its maximum and minimum, and the Interior Extremum Theorem guarantees that at an interior extremum the derivative is zero. If both extrema occur at the endpoints then f is constant and $f' \equiv 0$ throughout. Rolle's Theorem is the special case of the MVT where the secant slope is zero.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem
- Theorem: Extreme Value Theorem
- Definition: Derivative at a Point

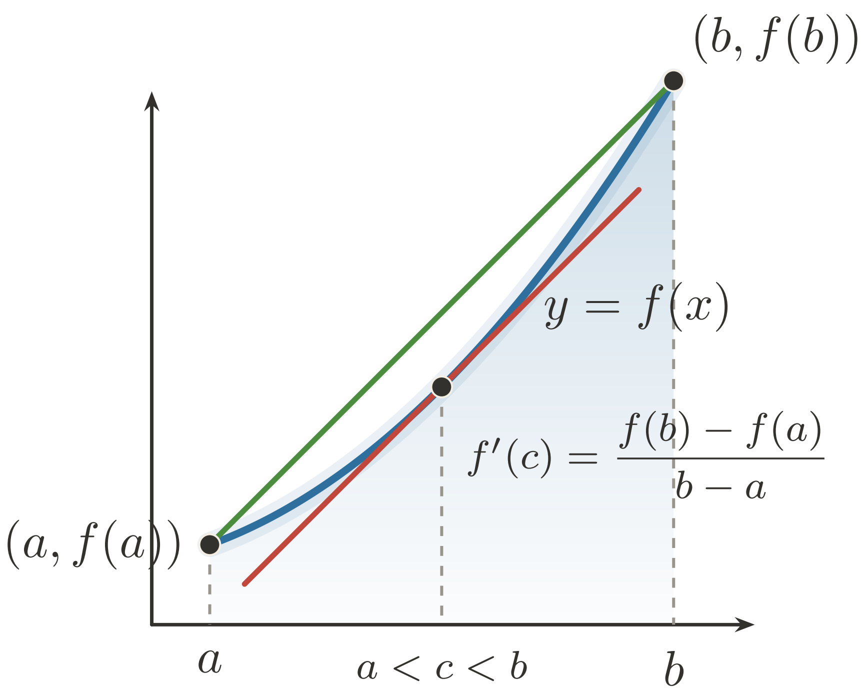


Figure 39.4: Mean Value Theorem: an interior tangent matches the secant slope.

Theorem (Mean Value Theorem)

Theorem 39.6.2 (Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \;\wedge\; \forall x \in (a, b)\;\mathrm{DifferentiableAt}(f, x)\\ &\Rightarrow\; \exists c \in (a, b)\;\mathrm{DerivativeAt}\Big(f,\; c,\; \tfrac{f(b)-f(a)}{b-a}\Big).\end{aligned}$$

Remark (Predicate reading)*.*

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \;\wedge\; \mathrm{DifferentiableOn}(f, (a, b))\\ &\implies \exists c \in (a, b)\;:\;\mathrm{DerivativeAt}\Big(f,\; c,\; \tfrac{f(b)-f(a)}{b-a}\Big).\end{aligned}$$

Remark (Failure modes)*.* Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$. Then f has a jump discontinuity at $x = 1/2$, and the secant slope is $\frac{f(1)-f(0)}{1-0} = 1$. But $f'(x) = 0$ wherever the derivative exists, so no interior point achieves the average rate of change.
- **Differentiability fails:** Let $f(x) = |x|$ on $[-1, 1]$. Then f is continuous and the secant slope is $\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$. But $f'(x) = \pm 1$ everywhere f is differentiable (i.e., for $x \neq 0$), so no interior point has derivative equal to the secant slope.

The MVT is an existence result. It guarantees some c exists but gives no formula for it.

Remark (Contrapositive quantified statement)*.*

$$\left(\forall c \in (a, b)\;:\; f'(c) \neq \frac{f(b) - f(a)}{b - a}\right) \;\Rightarrow\; f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b).$$

Remark (Contrapositive predicate reading)*.*

$$\begin{aligned}&\Big(\forall c \in (a, b)\;:\;\neg\,\mathrm{DerivativeAt}\Big(f,\; c,\; \tfrac{f(b)-f(a)}{b-a}\Big)\Big)\\ &\implies \neg\,\mathrm{ContinuousOn}(f, [a, b]) \;\vee\; \neg\,\mathrm{DifferentiableOn}(f, (a, b)).\end{aligned}$$

Remark (Interpretation)*.* The MVT asserts that the instantaneous rate of change equals the average rate of change at some interior point. Geometrically: the tangent line at some point c is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$. The proof reduces to Rolle's Theorem by subtracting the secant: define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, which satisfies $g(a) = g(b) = f(a)$, then apply Rolle's. The theorem is an existence result, not a construction: it guarantees some c exists but gives no formula for it. Its immediate corollaries — the constant function corollary, the monotonicity corollary, and the Lipschitz bound corollary — each flow from the MVT by a one-line argument and are the theorem's primary applications.

Remark (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point

Theorem (Cauchy Mean Value Theorem)

Theorem 39.6.3 (Cauchy Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$\big(f(b) - f(a)\big)g'(c) = \big(g(b) - g(a)\big)f'(c).$$

If, in addition, $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\mathrm{ContinuousOn}(f, [a, b]) \wedge \mathrm{ContinuousOn}(g, [a, b]) \wedge \mathrm{DifferentiableOn}(f, (a, b)) \wedge \mathrm{DifferentiableOn}(g, (a, b)) \\ &\quad \implies \exists c \in (a, b)\ \big[(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)\big].\end{aligned}$$

Remark (Predicate reading). $\mathrm{CauchyMVT}(f, g, [a, b])$ asserts the existence of an interior point where the velocity vector $(g'(c), f'(c))$ is parallel to the chord vector $(g(b)-g(a), f(b)-f(a))$. Taking $g(x) = x$ recovers the ordinary Mean Value Theorem.

Remark (Negated quantified statement).

$$\forall c \in (a, b) :\ \big(f(b) - f(a)\big)g'(c) \neq \big(g(b) - g(a)\big)f'(c).$$

Remark (Failure modes). The continuity and differentiability hypotheses fail for the same reasons they fail in the ordinary MVT: jumps break endpoint-to-interior control, and corners leave no derivative to match. The quotient form has the additional nonvanishing hypothesis $g' \neq 0$ on (a, b); without it the cross-multiplied conclusion may still hold, but the displayed quotient can be undefined.

Remark (Contrapositive quantified statement).

$$\begin{aligned}&\big(\forall c \in (a, b) :\ (f(b) - f(a))g'(c) \neq (g(b) - g(a))f'(c)\big) \\ &\quad \implies \neg\,\mathrm{ContinuousOn}(f, [a, b]) \vee \neg\,\mathrm{ContinuousOn}(g, [a, b]) \vee \neg\,\mathrm{DifferentiableOn}(f, (a, b)) \vee \neg\,\mathrm{Differentia}\end{aligned}$$

Remark (Contrapositive predicate reading). If no interior point makes the derivative pair $(g'(c), f'(c))$ parallel to the chord pair $(g(b)-g(a), f(b)-f(a))$, then at least one of the four regularity hypotheses fails: continuity of f or g on $[a,b]$, or differentiability of f or g on (a,b).

Remark (Interpretation). This is the parametric Mean Value Theorem. For the planar curve $t \mapsto (g(t), f(t))$, some interior tangent is parallel to the chord joining its endpoints. It compares two rates of change rather than comparing one function to the horizontal axis. This is the form that later underlies L'Hopital-type ratio arguments and the Cauchy form of Taylor's remainder.

Remark (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point
- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

Corollary (Derivative Bound Implies Lipschitz)

Corollary 39.6.4 (Derivative Bound Implies Lipschitz). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. If there exists $M \geq 0$ such that*

$$|f'(x)| \leq M \qquad \text{for every } x \in I,$$

then f is Lipschitz on I with constant M:

$$|f(x) - f(y)| \leq M|x-y| \qquad \text{for every } x, y \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\big(\exists M \geq 0\ \forall x \in I : \ |f'(x)| \leq M\big) \implies \forall x, y \in I : \ |f(x) - f(y)| \leq M|x-y|.$$

Remark (Predicate reading).

$$\mathrm{BoundedDerivativeOn}(f, I, M) \implies \mathrm{Lipschitz}(f, I, M).$$

Remark (Failure modes). The derivative bound must hold throughout the interval. A function such as $f(x) = \sqrt{x}$ on $[0,1]$ is continuous and uniformly continuous, but its derivative is unbounded near 0, so no global Lipschitz constant follows from this test. The interval hypothesis is also essential: across a disconnected gap, the MVT cannot compare the two components.

Remark (Interpretation). This corollary converts a pointwise slope bound into a global modulus of continuity. It is the precise bridge from derivative estimates to the Lipschitz condition: controlling the size of f' controls every secant slope on the interval.

Remark (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Lipschitz Condition
- Definition: Derivative at a Point

39.7 Reading the Graph from f' and f''

39.7.1 First-Order Shape

First-Order Shape — Quick Reference

Statement	Role
Zero derivative	Vanishing derivative forces constant behavior
Equal derivatives	Functions differ by a constant
Nonnegative derivative	Detects nondecreasing behavior
Nonpositive derivative	Detects nonincreasing behavior
Positive derivative	Local increasing behavior
Negative derivative	Local decreasing behavior
First derivative test: max	Sign change detects a local maximum
First derivative test: min	Sign change detects a local minimum

Remark (Section guide). The Mean Value Theorem is now available, so the harvest can begin. Vanishing derivatives force constancy, derivative signs force monotonicity, and second derivatives detect bending when enough regularity is present. Darboux and the smoothness hierarchy show that derivatives carry hidden rigidity even when they are not continuous.

Definition (Increasing at a Point)

Definition 39.7.1 (Increasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *increasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c-\delta, c) \quad \text{and} \quad y \in A \cap (c, c+\delta),$$

we have $f(x) \leq f(c) \leq f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0\ \forall x \in A \cap (c-\delta, c)\ \forall y \in A \cap (c, c+\delta)\ \big(f(x) \leq f(c) \leq f(y)\big).$$

Remark (Interpretation). f is increasing at c if c lies between the values of f on either side in some punctured neighbourhood: f is smaller to the left and larger to the right (weakly). This is a strictly local notion — it says nothing about f outside the δ-neighbourhood of c. A function can be increasing at c without being increasing (globally or on any interval). The non-strict inequalities permit the function to be flat on one or both sides of c, distinguishing this from the notion of a strict local increase.

Remark (Dependencies).

- Definition: Strictly Increasing Function (functions chapter)

Definition (Decreasing at a Point)

Definition 39.7.2 (Decreasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *decreasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \geq f(c) \geq f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0 \, \forall x \in A \cap (c - \delta, c) \, \forall y \in A \cap (c, c + \delta) \, \big(f(x) \geq f(c) \geq f(y)\big).$$

Remark (Interpretation). The function is decreasing at c when nearby values to the left sit above $f(c)$ and nearby values to the right sit below it.

Remark (Dependencies).

- Definition: Increasing at a Point

Theorem (Zero Derivative Implies Constant)

Theorem 39.7.3 (Zero Derivative Implies Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = 0 \qquad \textit{for all } x \in (a, b),$$

then f is constant on I. Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b) \ \text{DerivativeAt}(f, x, 0) \ \Rightarrow \ \exists C \in \mathbb{R} \, \forall x \in I \ (f(x) = C).$$

Remark (Contrapositive quantified statement).

$$\neg\big(\exists C \in \mathbb{R} \, \forall x \in I \ (f(x) = C)\big) \ \Rightarrow \ \exists x \in (a, b) \, \neg \, \text{DerivativeAt}(f, x, 0).$$

If f is not constant on I, then f' is not identically zero on (a, b).

Remark (Contrapositive predicate reading).

$$\neg \, \text{ConstantOn}(f, I) \ \Rightarrow \ \exists x \in (a, b) \ (f'(x) \neq 0).$$

Remark (Interpretation). The MVT is the engine here: if $f'(x) = 0$ everywhere on (a, b), then for any two points $x_1 < x_2$ in I, the MVT supplies a c with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$, so $f(x_1) = f(x_2)$. Since this holds for all pairs, f is constant. The theorem is the foundation of antiderivative uniqueness: two antiderivatives of the same function differ by a constant. The contrapositive is used in ODE uniqueness arguments — if a solution is not identically the zero solution, its derivative is not identically zero somewhere.

Remark (Dependencies).

- Theorem: Mean Value Theorem

Corollary

Corollary 39.7.4 (Equal Derivatives Imply Difference Is Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, and let $f, g : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = g'(x) \qquad \textit{for all } x \in (a, b),$$

then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$. *Go to proof.*

Remark (Standard quantified statement).

$$\forall x \in (a, b)\ (f'(x) = g'(x)) \ \Rightarrow\ \exists C \in \mathbb{R}\ \forall x \in I\ (f(x) = g(x) + C).$$

Remark (Contrapositive quantified statement).

$$\forall C \in \mathbb{R}\ \exists x \in I\ (f(x) \neq g(x) + C) \ \Rightarrow\ \exists x \in (a, b)\ (f'(x) \neq g'(x)).$$

If $f - g$ is not a constant function, then f' and g' disagree somewhere.

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{ConstantOn}(f - g,\ I) \ \Rightarrow\ \exists x \in (a, b)\ (f'(x) \neq g'(x)).$$

Remark (Interpretation). This is the corollary that makes antiderivatives well-defined up to a constant: any two antiderivatives of the same function on an interval differ by a constant. The proof is one line — apply the previous theorem to $h = f - g$, whose derivative is $(f - g)' = f' - g' = 0$.

Remark (Dependencies).

- Theorem: Zero Derivative Implies Constant

Theorem (Nondecreasing Iff Nonnegative Derivative)

Theorem 39.7.5 (Nondecreasing Iff Nonnegative Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nondecreasing on I if and only if*

$$f'(x) \geq 0 \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{NondecreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \geq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\,\text{NondecreasingOn}(f, I).$$

If the derivative is negative at some point, the function is not nondecreasing.

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\,\text{NondecreasingOn}(f, I).$$

Remark (Interpretation). The derivative is the instantaneous rate of change; requiring it to be nonnegative everywhere exactly characterises functions that never decrease. The forward direction (f nondecreasing $\Rightarrow f' \geq 0$) follows from the definition: the difference quotient is nonnegative when f is nondecreasing, so the limit is nonnegative. The reverse direction ($f' \geq 0 \Rightarrow$ nondecreasing) is a consequence of the MVT: for $x < y$, the MVT gives $f(y) - f(x) = f'(c)(y - x) \geq 0$.

Note carefully: $f' \geq 0$ characterises *nondecreasing* (weak), not *strictly increasing* (strict). Strict monotonicity requires $f' > 0$ at all but isolated points. The function $f(x) = x^3$ satisfies $f'(0) = 0$ but is strictly increasing; $f' \geq 0$ with equality only at isolated points is sufficient for strict increase, but $f' > 0$ everywhere is not necessary.

Remark (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Derivative at a Point

Theorem

Theorem 39.7.6 (Nonincreasing Iff Nonpositive Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nonincreasing on I if and only if*

$$f'(x) \leq 0 \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{NonincreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \leq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) > 0) \ \Rightarrow\ \neg\,\text{NonincreasingOn}(f, I).$$

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) > 0) \ \Rightarrow\ \neg\,\text{NonincreasingOn}(f, I).$$

Remark (Interpretation). Nonpositive derivative everywhere is exactly the infinitesimal signature of a function that never increases.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

Lemma

Lemma 39.7.7 (Positive Derivative Implies Locally Increasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \textit{for all } x \in I \textit{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \textit{for all } x \in I \textit{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \wedge L > 0 \;\Rightarrow\; \exists \delta > 0\, \forall x \in I\, \big(c{-}\delta < x < c \Rightarrow f(x) < f(c)\big) \wedge \big(c < x < c{+}\delta \Rightarrow f(x) >$$

Remark (Contrapositive quantified statement).

$$\forall \delta > 0\, \exists x \in I\, \big((c{-}\delta < x < c \wedge f(x) \geq f(c)) \vee (c < x < c{+}\delta \wedge f(x) \leq f(c))\big) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Contrapositive predicate reading).

$$\neg\, \mathrm{LocallyIncreasingAt}(f, c) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Interpretation). If the instantaneous rate of change is strictly positive at c, the function is genuinely climbing through c: it is strictly below $f(c)$ immediately to the left and strictly above immediately to the right. The proof is a direct ε-δ argument: take $\varepsilon = f'(c)/2 > 0$; the definition of the derivative supplies δ such that the difference quotient is within ε of $f'(c)$, forcing it to be positive, which then forces the correct sign on $f(x) - f(c)$. This lemma is the key ingredient in the proof of the First Derivative Test.

Remark (Dependencies).

- Definition: Derivative at a Point

Lemma

Lemma 39.7.8 (Negative Derivative Implies Locally Decreasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \textit{for all } x \in I \textit{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \textit{for all } x \in I \textit{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L < 0 \;\Rightarrow\; \exists \delta > 0\, \forall x \in I\, \big(c-\delta < x < c \Rightarrow f(x) > f(c)\big) \wedge \big(c < x < c+\delta \Rightarrow f(x) <$$

Remark (Contrapositive quantified statement).

$$\neg\, \text{LocallyDecreasingAt}(f, c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Contrapositive predicate reading).

$$\neg\, \text{LocallyDecreasingAt}(f, c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Interpretation). If the derivative at c is strictly negative, nearby values move downward as the input passes through c from left to right.

Remark (Dependencies).

- Lemma: Positive Derivative Implies Locally Increasing

Theorem (First Derivative Test: Relative Maximum)

Theorem 39.7.9 (First Derivative Test: Relative Maximum). *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c - \delta, c), \qquad f'(x) \leq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative maximum at c. *Go to proof.*

Remark (Standard quantified statement).

$$\exists \delta > 0 \left[(c - \delta, c + \delta) \subseteq I \;\wedge\; \forall x \in (c - \delta, c)\, (f'(x) \geq 0) \;\wedge\; \forall x \in (c, c + \delta)\, (f'(x) \leq 0)\right]$$
$$\Rightarrow\; \text{LocalMaximum}(f, c).$$

Remark (Interpretation). If the function is nondecreasing immediately to the left of c and nonincreasing immediately to the right, then c is a local maximum — the function climbs to c and then falls away. The proof integrates: by the nondecreasing characterisation applied on $(c - \delta, c)$, $f(x) \leq f(c)$ for $x < c$; by the nonincreasing characterisation on $(c, c + \delta)$, $f(x) \leq f(c)$ for $x > c$. The sign condition on f' replaces the Interior Extremum Theorem's converse, which does not hold in general; this is a sufficient condition, not a necessary one. The test is applicable even when $f'(c) = 0$ or $f'(c)$ does not exist.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Definition: Relative Maximum

Theorem

Theorem 39.7.10 (First Derivative Test: Relative Minimum)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \leq 0 \quad \text{for all } x \in (c - \delta, c), \qquad f'(x) \geq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative minimum at c. *Go to proof.*

Remark (Standard quantified statement).

$$\exists \delta > 0 \left[(c - \delta, c + \delta) \subseteq I \ \wedge\ \forall x \in (c - \delta, c)\ (f'(x) \leq 0)\ \wedge\ \forall x \in (c, c + \delta)\ (f'(x) \geq 0)\right] \\ \Rightarrow \ \mathrm{LocalMinimum}(f, c).$$

Remark (Interpretation). If the derivative changes from nonpositive to nonnegative at c, the graph falls into c and then rises away from it, so c is a local minimum.

Remark (Dependencies).

- Theorem: First Derivative Test: Relative Maximum
- Definition: Relative Minimum

39.7.2 Second-Order Shape

Second-Order Shape — Quick Reference

Concept	Role
Second derivative	Derivative of the derivative
Higher derivatives	Iterated derivatives $f^{(n)}$
Convexity test	$f'' \geq 0$ detects convexity
Concavity test	$f'' \leq 0$ detects concavity
Second derivative test	Classifies critical points
Inflection condition	Continuous f'' changes sign through zero

39.7.3 Higher-Order Derivatives

Definition (Second Derivative)

Definition 39.7.11 (Second Derivative)**.** Let f be differentiable on a neighbourhood of c. If f' is differentiable at c, the *second derivative* of f at c is

$$f''(c) := (f')'(c).$$

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f', c, f''(c)) \quad \Longleftrightarrow \quad f''(c) = (f')'(c).$$

Remark (Interpretation). The first derivative measures instantaneous slope. The second derivative measures how that slope itself changes.

Remark (Dependencies).

- Derivative at a Point

Definition (Higher Derivatives)

Definition 39.7.12 (Higher Derivatives). Define $f^{(0)} := f$ and $f^{(1)} := f'$. For $n \geq 2$, if $f^{(n-1)}$ is differentiable at c, define

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

Remark (Standard quantified statement).

$$\forall n \geq 2 : \quad f^{(n)}(c) = (f^{(n-1)})'(c)$$

whenever the derivative on the right exists.

Remark (Interpretation). Higher derivatives are obtained by iterating the derivative operator. Their existence is not automatic: each stage requires differentiability of the previous derivative.

Remark (Dependencies).

- Second Derivative

39.7.4 Second-Order Shape

Theorem (Second-Derivative Convexity Test)

Theorem 39.7.13 (Second-Derivative Convexity Test). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*

$$f''(x) \geq 0 \qquad \text{for all } x \in I,$$

then f is convex on I. *Go to proof.*

Remark (Standard quantified statement).

$$\big(\forall x \in I,\ f''(x) \geq 0\big) \Longrightarrow \mathrm{ConvexOn}(f, I).$$

Remark (Interpretation). Nonnegative second derivative means the slopes do not decrease. This is the derivative-based signature of convex bending.

Remark (Dependencies).

- Second Derivative
- Convex Function

Theorem (Second-Derivative Concavity Test)

Theorem 39.7.14 (Second-Derivative Concavity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*

$$f''(x) \le 0 \qquad \text{for all } x \in I,$$

then f is concave on I. Go to proof.

Remark (Standard quantified statement)*.*

$$\big(\forall x \in I,\ f''(x) \le 0\big) \implies \mathrm{ConcaveOn}(f, I).$$

Remark (Interpretation)*.* Nonpositive second derivative means the slopes do not increase. This is the derivative-based signature of concave bending.

Remark (Interpretation)*.* The second-derivative tests are narrower than the chord definitions of convexity and concavity. The function $|x|$ is convex, but it is not twice differentiable at 0.

Remark (Dependencies)*.*

- Second Derivative
- Concave Function

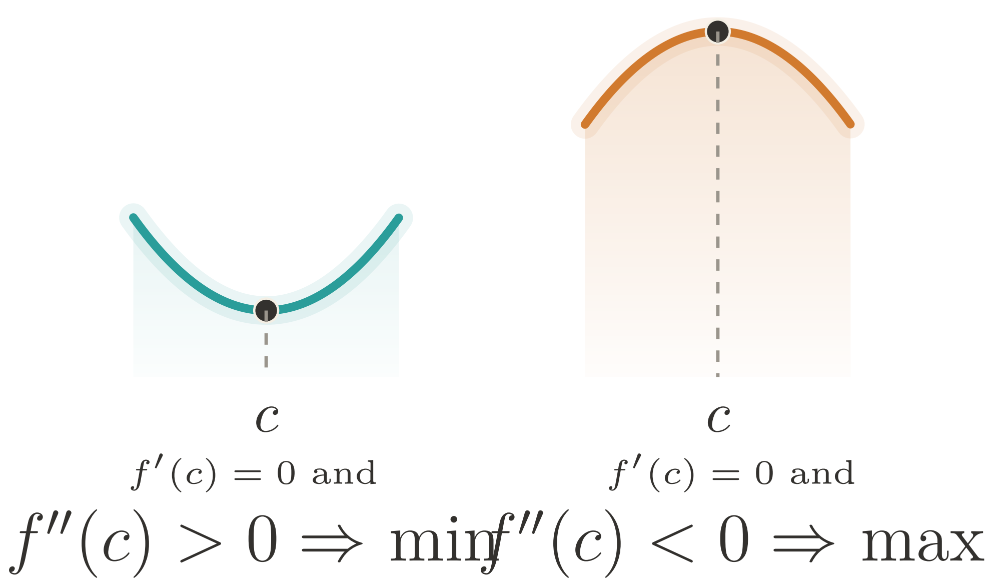


Figure 39.5: Second derivative test: the sign of $f''(c)$ decides the local shape at a critical point.

Theorem (Second Derivative Test)

Theorem 39.7.15 (Second Derivative Test)**.** *Let f be twice differentiable in a neighbourhood of c, and suppose $f'(c) = 0$.*

- *If $f''(c) > 0$, then f has a strict relative minimum at c.*
- *If $f''(c) < 0$, then f has a strict relative maximum at c.*

Go to proof.

Remark (Standard quantified statement).

$$f'(c) = 0 \wedge f''(c) > 0 \Rightarrow \text{StrictRelativeMinimum}(f, c),$$

and

$$f'(c) = 0 \wedge f''(c) < 0 \Rightarrow \text{StrictRelativeMaximum}(f, c).$$

Remark (Interpretation). At a critical point, positive second derivative means the graph bends upward; negative second derivative means it bends downward.

Remark (Dependencies).

- Second Derivative
- Relative Minimum
- Relative Maximum

Proposition (Inflection Point Necessary Condition)

Proposition 39.7.16 (Inflection Point Necessary Condition)**.** *Let f have an inflection point at c. If f'' exists on a neighbourhood of c and is continuous at c, then*

$$f''(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{InflectionPoint}(f, c) \wedge \text{ContinuousAt}(f'', c) \implies f''(c) = 0.$$

Remark (Interpretation). An inflection point is a change in bending. When the second derivative is continuous, such a change can pass through c only by passing through zero. The converse can fail: $f''(c) = 0$ does not by itself guarantee an inflection point.

Remark (Dependencies).

- Inflection Point
- Second Derivative
- Continuity at a Point

39.7.5 Darboux's Theorem

Darboux — Quick Reference

Statement	Role
Darboux's Theorem	Derivatives have the intermediate value property

Theorem (Darboux's Theorem)

Theorem 39.7.17 (Darboux's Theorem)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on I. If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = k.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall k \; \big(\min(f'(a), f'(b)) < k < \max(f'(a), f'(b)) \;\Rightarrow\; \exists c \in (a, b) \; (f'(c) = k)\big).$$

Remark (Interpretation). The Intermediate Value Theorem says continuous functions have the intermediate value property. Darboux's theorem says derivatives also have that property even when the derivative is not continuous. Thus a derivative cannot have a jump discontinuity. The proof uses differentiability of f to construct an auxiliary function $g(x) = f(x) - kx$ and then applies the Extreme Value Theorem and the Interior Extremum Theorem.

Remark (Dependencies).

- Theorem: Extreme Value Theorem
- Theorem: Interior Extremum Theorem
- Definition: Derivative at a Point

39.7.6 Smoothness Classes

Smoothness Classes — Quick Reference

Concept	Meaning
Class C^1	Differentiable with continuous first derivative
Class C^k	k derivatives with continuous top derivative
Class C^∞	Continuous derivatives of every order
Class C^ω	Equal to local Taylor series
Smoothness tower	Strict hierarchy $C^\omega \subsetneq C^\infty \subsetneq \cdots$
Class $C^{1,1}$	C^1 with Lipschitz first derivative
Placement of $C^{1,1}$	Refines the gap between C^1 and C^2
Bounded f''	Bounded second derivative gives $C^{1,1}$
Example: differentiable not C^1	Oscillatory derivative discontinuity at 0

Definition (Continuously Differentiable; Class C^1)

Definition 39.7.18 (Continuously Differentiable; Class C^1)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *continuously differentiable* on I, written $f \in C^1(I)$, if f is differentiable at every point of I and the derivative $f' : I \to \mathbb{R}$ is continuous on I.

Remark (Standard quantified statement).

$$f \in C^1(I) \iff \big(\forall x \in I : \text{DifferentiableAt}(f, x)\big) \wedge \text{ContinuousOn}(f', I).$$

Remark (Interpretation). The condition $f \in C^1(I)$ is strictly stronger than differentiability. A function may be differentiable everywhere while its derivative fails to be continuous; Darboux's Theorem still constrains that failure, because derivatives cannot jump. Thus the gap between differentiability and C^1 is oscillatory rather than jump-like.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Continuity at a Point

Example (Differentiable Everywhere, but Not C^1). Let

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

At the origin,

$$f'(0) = \lim_{x \to 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \to 0} x \sin(1/x) = 0$$

by squeezing. For $x \neq 0$,

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

The first term tends to 0, but $\cos(1/x)$ oscillates, so f' is not continuous at 0. Thus f is differentiable on all of $\mathbb{R}$, but $f \notin C^1(\mathbb{R})$. The failure is oscillatory, not a jump, as Darboux's theorem requires.

Definition (Class C^k)

Definition 39.7.19 (Class C^k)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $k \in \mathbb{N}$, and let $f : I \to \mathbb{R}$. The function f is of *class* C^k on I, written $f \in C^k(I)$, if the derivatives $f^{(0)}, f^{(1)}, \ldots, f^{(k)}$ exist on I and $f^{(k)}$ is continuous on I. By convention, $C^0(I)$ is the class of continuous functions on I.

Remark (Standard quantified statement).

$$f \in C^k(I) \iff \big(\forall j \in \{0, 1, \ldots, k\} : f^{(j)} \text{ exists on } I\big) \wedge \text{ContinuousOn}\big(f^{(k)}, I\big).$$

Remark (Predicate reading).

$$\mathrm{ClassC}(f, I, k).$$

For a fixed interval I, the classes are nested:

$$f \in C^{k+1}(I) \Longrightarrow f \in C^k(I).$$

Remark (Negated quantified statement).

$$f \notin C^k(I) \Longleftrightarrow \exists x \in I : \; \big[f^{(k)} \text{ fails to exist at } x\big] \vee \big[f^{(k)} \text{ exists on } I \wedge \neg \, \mathrm{ContinuousAt}(f^{(k)}, x)\big].$$

Remark (Negation predicate reading). $\neg\,\mathrm{ClassC}(f, I, k)$: somewhere the k-th derivative either does not exist or exists but is not continuous. A single point where the top derivative is discontinuous certifies $f \notin C^k(I)$.

Remark (Interpretation). The notation $C^k(I)$ records k layers of differentiability together with continuity at the top layer. This is the regularity scale used to state how many times a theorem may differentiate a function without leaving the controlled setting.

Remark (Dependencies).

- Definition: Higher Derivatives
- Definition: Continuously Differentiable; Class C^1

Definition (Smooth Function; Class C^∞)

Definition 39.7.20 (Smooth Function; Class C^∞)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *smooth*, or of *class* C^∞, written $f \in C^\infty(I)$, if $f \in C^k(I)$ for every $k \in \mathbb{N}$. Equivalently,

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

Remark (Standard quantified statement).

$$f \in C^\infty(I) \Longleftrightarrow \forall k \in \mathbb{N} : \; f \in C^k(I).$$

Remark (Negated quantified statement).

$$f \notin C^\infty(I) \Longleftrightarrow \exists k \in \mathbb{N} : \; f \notin C^k(I).$$

A single order at which differentiability or continuity breaks demotes the function out of C^∞.

Remark (Interpretation). Smoothness is the regularity language imported by differential geometry. Smooth manifolds are assembled from C^∞ transition maps, and the tangent-map construction seeded by the differential uses this vocabulary.

Remark (Dependencies).

- Definition: Class C^k

Definition (Real-Analytic Function; Class C^ω)

Definition 39.7.21 (Real-Analytic Function; Class C^ω)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *real-analytic*, or of *class* C^ω, written $f \in C^\omega(I)$, if for every $a \in I$ there exists $r > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \qquad \text{for every } x \in (a-r, a+r) \cap I.$$

Remark (Standard quantified statement).

$$f \in C^\omega(I) \iff \forall a \in I\ \exists r > 0\ \forall x \in (a-r, a+r) \cap I :\ f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Remark (Interpretation). Analytic regularity means that the Taylor series does more than approximate the function: near each point, the series equals the function. The infinite Taylor series and its convergence are developed in Taylor's construction and error analysis, which supplies the series machinery behind this definition.

Remark (Dependencies).

- Definition: Smooth Function; Class C^∞
- Definition: Taylor Polynomial at a Point

Theorem (The Smoothness Tower)

Theorem 39.7.22 (The Smoothness Tower)**.** *For every nondegenerate interval I, the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

hold, and each displayed inclusion is strict. *Go to proof.*

Remark (Standard quantified statement).

$$\forall k \in \mathbb{N} :\ C^{k+1}(I) \subsetneq C^k(I), \qquad C^\omega(I) \subsetneq C^\infty(I).$$

Remark (Failure modes). The inclusions do not fail; only their strictness needs witnesses.

- $C^k \setminus C^{k+1}$: $\varphi_k(x) = x^k|x|$, whose k-th derivative is $(k+1)!\,|x|$, continuous but not differentiable at 0.
- $C^\infty \setminus C^\omega$: the flat function e^{-1/x^2}, smooth with all derivatives zero at 0, so its Maclaurin series is zero while the function is not.

These separators prove every displayed containment is a genuine drop.

Remark (Interpretation). The inclusions are definitional; strictness is witnessed by examples. On any nondegenerate interval, translated and rescaled copies of $x \mapsto x^k|x|$ separate C^k from C^{k+1}. Translated copies of the flat function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

separate C^∞ from C^ω: all derivatives at the flat point vanish, so the Taylor series is identically zero, while the function is positive on one side of that point.

Remark (Dependencies).

- Definition: Class C^k
- Definition: Smooth Function; Class C^∞
- Definition: Real-Analytic Function; Class C^ω
- Theorem: Darboux's Theorem

Remark (Section guide). Between C^1 and C^2 there is a regularity floor worth naming. A C^1 function has a continuous derivative; a C^2 function has a derivative that is itself differentiable. Many functions used in analysis live in the gap: their first derivative is continuous and controlled, but a classical second derivative may fail to exist at a few points. The control is a Lipschitz bound on f', and the class is written $C^{1,1}$.

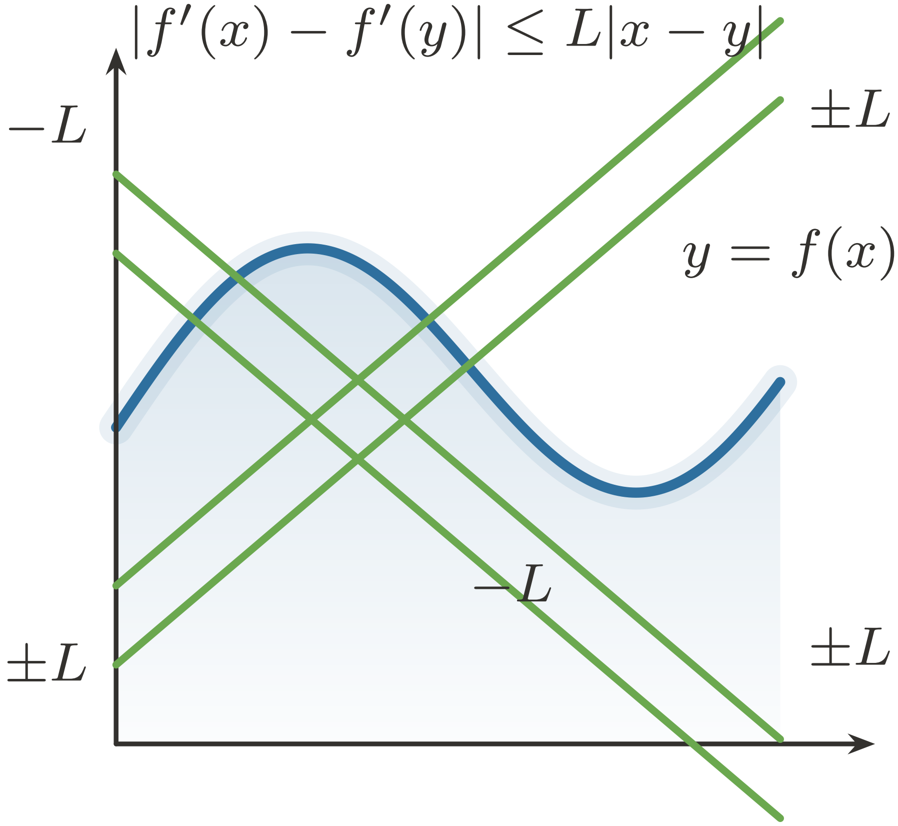


Figure 39.6: A Lipschitz derivative changes at a uniformly bounded rate.

Definition (Lipschitz-Derivative Class $C^{1,1}$)

Definition 39.7.23 (Lipschitz-Derivative Class $C^{1,1}$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is of *class* $C^{1,1}$ on I, written $f \in C^{1,1}(I)$, if $f \in C^1(I)$ and f' is Lipschitz on I: there exists $L \ge 0$ such that

$$|f'(x) - f'(y)| \le L|x-y| \qquad \text{for all } x, y \in I.$$

Remark (Standard quantified statement)*.*

$$f \in C^{1,1}(I) \iff f \in C^1(I) \wedge \exists L \ge 0\ \forall x, y \in I:\ |f'(x) - f'(y)| \le L|x-y|.$$

Remark (Definition predicate reading)*.*

$$\mathrm{ClassC11}(f, I) \iff \mathrm{ClassC}(f, I, 1) \wedge \exists L \ge 0 : \mathrm{Lipschitz}(f', I, L).$$

The smallest admissible L is the Lipschitz constant of f':

$$\sup_{x \ne y} \frac{|f'(x) - f'(y)|}{|x-y|}.$$

Remark (Negated quantified statement)*.*

$$f \notin C^{1,1}(I) \iff f \notin C^1(I) \vee \big(\forall L \ge 0\ \exists x, y \in I:\ |f'(x) - f'(y)| > L|x-y|\big).$$

Remark (Interpretation). $C^{1,1}$ says that f' behaves as if $|f''| \le L$, without requiring f'' to exist everywhere. It is the exact regularity tracked by many first-order estimates: the derivative is not merely continuous, but changes at a bounded rate.

Remark (Dependencies).

- Definition: Continuously Differentiable; Class C^1
- Corollary: Derivative Bound Implies Lipschitz

Theorem (Placement of $C^{1,1}$)

Theorem 39.7.24 (Placement of $C^{1,1}$). *For every nondegenerate interval I,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

and $C^{1,1}(I)$ is not contained in $C^2(I)$. If K is a compact interval, then

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

and the first inclusion is strict. *Go to proof.*

Remark (Standard quantified statement).

$$\big(f \in C^{1,1}(I) \Rightarrow f \in C^1(I)\big), \qquad \big(K \text{ compact} \wedge f \in C^2(K) \Rightarrow f \in C^{1,1}(K)\big),$$

with the displayed containments strict.

Remark (Failure modes). The global inclusion $C^2(I) \subseteq C^{1,1}(I)$ is false on arbitrary intervals: $f(x) = x^3$ is $C^2(\mathbb{R})$, but $f'(x) = 3x^2$ is not Lipschitz on $\mathbb{R}$. Strictness is witnessed by two elementary functions.

- $C^{1,1} \setminus C^2$: $f(x) = x|x|$ has $f'(x) = 2|x|$, which is Lipschitz, but $f''(0)$ does not exist.
- $C^1 \setminus C^{1,1}$: $f(x) = |x|^{4/3}$ has continuous first derivative, but f' has unbounded difference quotients at 0.

Remark (Interpretation). $C^{1,1}$ is a genuine intermediate regularity condition, but it is not the same as C^2. A bounded second derivative forces the class; a Lipschitz first derivative may have corners; and a continuous first derivative may still vary too sharply to be Lipschitz.

Remark (Dependencies).

- Definition: Lipschitz-Derivative Class $C^{1,1}$
- Theorem: The Smoothness Tower
- Corollary: Derivative Bound Implies Lipschitz

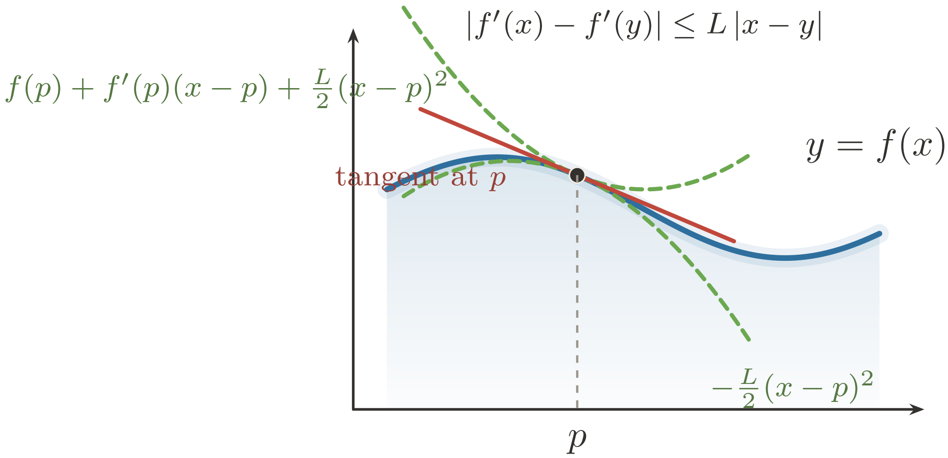


Figure 39.7: $C^{1,1}$ control gives quadratic upper and lower bounds around a tangent line.

Corollary (Bounded Second Derivative Implies $C^{1,1}$)

Corollary 39.7.25 (Bounded Second Derivative Implies $C^{1,1}$)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f \in C^2(I)$. If $|f''(x)| \le M$ for all $x \in I$, then $f \in C^{1,1}(I)$, and f' is Lipschitz with constant M. Go to proof.*

Remark (Standard quantified statement).

$$\big(f \in C^2(I) \wedge \forall x \in I : \ |f''(x)| \le M\big) \Rightarrow f \in C^{1,1}(I) \wedge \text{Lipschitz}(f', I, M).$$

Remark (Interpretation). This is Derivative Bound Implies Lipschitz applied one level up, to f' in place of f. The converse requires almost-everywhere language and is therefore deferred.

Remark (Dependencies).

- Corollary: Derivative Bound Implies Lipschitz
- Definition: Class C^k
- Definition: Lipschitz-Derivative Class $C^{1,1}$

Remark (Forward flag). A Lipschitz derivative is differentiable at most points, but making that sentence precise requires measure zero. In one variable this passes through bounded variation; in several variables it becomes Rademacher's theorem. Those results belong to the integration and measure-theoretic development, not to this chapter.

39.8 The Algebra of Derivatives

Algebra of Derivatives — Quick Reference

Rule	Role
Constant multiple	Scalars pass through differentiation
Sum	Derivative distributes over addition
Product	Differentiates pointwise products
Quotient	Differentiates quotients where denominator is nonzero
Finite sum	Iterates the sum rule over finitely many terms
Extended product	Iterates the product rule over finitely many factors
Integer power	Differentiates powers built from products
Finite linear combination	Packages linearity in finite form
Inverse Function Theorem	Produces a C^1 local inverse
Inverse derivative	Gives the reciprocal derivative formula
L'Hôpital $0/0$	Evaluates indeterminate zero ratios
L'Hôpital ∞/∞	Evaluates indeterminate divergent ratios

39.8.1 From Characterization to Computation

Remark (Section guide). With Carathéodory in hand, computing derivatives becomes bookkeeping with continuous slope factors. The basic rules record one structural fact: differentiation is a linear operator that also obeys a product law.

Remark (Interpretation). Carathéodory's characterization turns differentiability into continuity of a slope factor. If

$$f(x) - f(c) = (x-c)\phi_f(x), \qquad g(x) - g(c) = (x-c)\phi_g(x),$$

with ϕ_f and ϕ_g continuous at c, then the standard algebraic rules for derivatives follow from the algebra of continuous functions. The recurring move is no longer to manipulate a punctured quotient directly, but to construct a continuous factor whose value at c is the derivative.

Remark (Dependencies).

- Carathéodory Characterization of Differentiability
- Algebra of Limits

Theorem (Constant Multiple Rule)

Theorem 39.8.1 (Constant Multiple Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $\alpha \in \mathbb{R}$. If f is differentiable at c, then αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \text{DerivativeAt}(\alpha f,\, c,\, \alpha f'(c)).$$

Remark (Interpretation). Multiplying a function by a constant scales its derivative by the same constant. The proof is one line: the scalar α factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits passes it through the limit. Differentiation is therefore homogeneous of degree one with respect to scalar multiplication.

Remark (Dependencies).

- Definition: Derivative at a Point
- Constant Multiple Rule for Limits

Theorem (Sum Rule)

Theorem 39.8.2 (Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then $f + g$ is differentiable at c, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\wedge\; \text{DifferentiableAt}(g, c) \;\Rightarrow\; \text{DerivativeAt}(f + g,\, c,\, f'(c) + g'(c)).$$

Remark (Interpretation). The derivative of a sum is the sum of the derivatives. The proof is immediate: the difference quotient of $f + g$ splits linearly into the sum of the individual difference quotients, and the Sum Rule for Limits distributes the limit across the sum. Together with the Constant Multiple Rule, this establishes that differentiation is a linear operation: $(\alpha f + \beta g)' = \alpha f' + \beta g'$.

Remark (Dependencies).

- Definition: Derivative at a Point
- Sum Rule for Limits

Theorem (Product Rule)

Theorem 39.8.3 (Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \;\Rightarrow\; \text{DerivativeAt}(fg,\ c,\ f'(c)g(c) + f(c)g'(c)).$$

Remark (Interpretation). The derivative of a product is not simply the product of the derivatives. The correct formula adds two terms: the rate of change of f weighted by the current value of g, plus the rate of change of g weighted by the current value of f. The proof inserts the auxiliary term $f(x)g(c)$ to decouple the increments of f and g, isolating the individual difference quotients. Continuity of f at c (which follows from differentiability) is used to evaluate $\lim_{x\to c} f(x) = f(c)$ in one of the factors.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits

Theorem (Quotient Rule)

Theorem 39.8.4 (Quotient Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \wedge g(c) \neq 0 \;\Rightarrow\; \text{DerivativeAt}\left(\frac{f}{g},\ c,\ \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}\right).$$

Remark (Interpretation). The quotient rule is the product rule applied to $f \cdot g^{-1}$, but stated directly. The hypothesis $g(c) \neq 0$ ensures the quotient is defined near c — continuity of g (from its differentiability) and the quotient rule for limits guarantee $g(x) \neq 0$ in a deleted neighbourhood of c, so the difference quotient of f/g is well-defined throughout. The add-and-subtract technique with $f(c)g(c)$ decouples the increments of f and g in the numerator, exactly as in the product rule proof.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits
- Quotient Rule for Limits

39.8.2 Closure

Corollary

Corollary 39.8.5 (Finite Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable at c, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \in \{1, \ldots, n\}\ \text{DifferentiableAt}(f_i, c) \ \Rightarrow\ \text{DerivativeAt}\big(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)\big).$$

Remark (Interpretation). Finite sums and scalar weights do not introduce a new kind of differentiation rule; they package repeated use of the Sum Rule and Constant Multiple Rule.

Remark (Dependencies).

- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

Corollary

Corollary 39.8.6 (Extended Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\prod_{i=1}^{n} f_i$ is differentiable at c, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(c) = \sum_{k=1}^{n} f_k'(c) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \text{DifferentiableAt}(f_i, c) \ \Rightarrow\ \text{DerivativeAt}\Big(\textstyle\prod_i f_i,\ c,\ \sum_k f_k'(c) \prod_{i \neq k} f_i(c)\Big).$$

Remark (Interpretation). In a finite product, each term takes one turn being differentiated while all other factors are held at their values.

Remark (Dependencies).

- Theorem: Product Rule

Corollary

Corollary 39.8.7 (Power Rule for Integer Powers of a Function)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $n \in \mathbb{N}$. If f is differentiable at c, then f^n is differentiable at c, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\mathrm{DifferentiableAt}(f, c) \ \wedge \ n \in \mathbb{N} \ \Rightarrow \ \mathrm{DerivativeAt}(f^n,\, c,\, n(f(c))^{n-1} f'(c)).$$

Remark (Interpretation)*.* The integer power rule is the extended product rule applied to n identical factors.

Remark (Dependencies)*.*

- Corollary: Extended Product Rule

Theorem

Theorem 39.8.8 (Finite Linear Combination Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i \right)' (c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall i \ \mathrm{DifferentiableAt}(f_i, c) \ \Rightarrow \ \mathrm{DerivativeAt}\big(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)\big).$$

Remark (Interpretation)*.* The Finite Linear Combination Rule is the full statement that differentiation is a *linear operator*: it commutes with scalar multiplication and addition simultaneously, for any finite collection of differentiable functions. This is not a separate theorem from the Constant Multiple and Sum Rules — it is their inductive closure. Stated in operator language: if $\mathcal{D}(I)$ denotes the vector space of functions differentiable at c, then $D_c : \mathcal{D}(I) \to \mathbb{R}$ defined by $D_c(f) = f'(c)$ is a linear functional. The Finite Linear Combination Rule says $D_c(\sum_i \alpha_i f_i) = \sum_i \alpha_i D_c(f_i)$.

Remark (Dependencies)*.*

- Real-Linear Rule
- Linear Combination of Real-Valued Functions
- Theorem: Constant Multiple Rule
- Theorem: Sum Rule

39.8.3 Global Forms on an Interval

Theorem

Theorem 39.8.9 (Constant Multiple Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I, then αf is differentiable on I, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \text{DifferentiableAt}(f, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(\alpha f,\ x,\ \alpha f'(x)).$$

Remark (Interpretation)*.* The pointwise constant-multiple rule becomes an interval statement by applying it at every point of the interval.

Remark (Dependencies).

- Theorem: Constant Multiple Rule

Theorem

Theorem 39.8.10 (Sum Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then $f + g$ is differentiable on I, and*

$$(f + g)'(x) = f'(x) + g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \big(\text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x)\big) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(f+g,\ x,\ f'(x)+g'(x)).$$

Remark (Interpretation)*.* The derivative of a sum is computed point by point across the whole interval.

Remark (Dependencies).

- Theorem: Sum Rule

Theorem

Theorem 39.8.11 (Product Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then fg is differentiable on I, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$\forall x \in I\ \big(\text{DifferentiableAt}(f,x) \wedge \text{DifferentiableAt}(g,x)\big) \ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}(fg,\ x,\ f'(x)g(x)+f(x)g'($

Remark (Interpretation). The interval product rule records the product rule as an identity of derivative functions.

Remark (Dependencies).

- Theorem: Product Rule

Theorem

Theorem 39.8.12 (Quotient Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$\forall x \in I\ \big(\text{DifferentiableAt}(f,x) \wedge \text{DifferentiableAt}(g,x) \wedge g(x) \neq 0\big) \ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}\Big(\frac{f}{g},\ x,\ \frac{f'(x)g(x)-f}{(g(x))}$

Remark (Interpretation). The quotient rule on an interval is the pointwise quotient rule with the nonvanishing denominator hypothesis required at every point.

Remark (Dependencies).

- Theorem: Quotient Rule

Corollary

Corollary 39.8.13 (Finite Sum Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then $\sum_{i=1}^n \alpha_i f_i$ is differentiable on I, and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \forall x \in I\ \ \text{DifferentiableAt}(f_i, x) \ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x))\,.$$

Remark (Interpretation). Finite linear combinations commute with differentiation as functions on the whole interval.

Remark (Dependencies).

- Corollary: Finite Sum Rule

Corollary

Corollary 39.8.14 (Extended Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \ldots, f_n : I \to \mathbb{R}$. If each f_i is differentiable on I, then $\prod_{i=1}^{n} f_i$ is differentiable on I, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(x) = \sum_{k=1}^{n} f_k'(x) \prod_{\substack{i=1 \\ i\neq k}}^{n} f_i(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \forall x \in I\ \ \text{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}\Big(\prod_i f_i,\ x,\ \sum_k f_k'(x) \prod_{i\neq k} f_i(x)\Big).$$

Remark (Interpretation). The product rule for many factors holds uniformly point by point on the interval.

Remark (Dependencies).

- Corollary: Extended Product Rule

Corollary

Corollary 39.8.15 (Power Rule for Integer Powers of a Function on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I, then f^n is differentiable on I, and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I\ \ \text{DifferentiableAt}(f, x)\ \wedge\ n \in \mathbb{N}\ \Rightarrow\ \forall x \in I\ \ \text{DerivativeAt}(f^n,\ x,\ n(f(x))^{n-1} f'(x)).$$

Remark (Interpretation). The interval power rule says that the pointwise integer-power formula assembles into an equality of derivative functions.

Remark (Dependencies).

- Corollary: Power Rule for Integer Powers of a Function

Theorem

Theorem 39.8.16 (Finite Linear Combination Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \forall x \in I\ \operatorname{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \operatorname{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x))\,.$$

Remark (Interpretation). The pointwise statements above — each proved at a single point c — extend to global statements when the hypotheses hold at every point of I. If f is differentiable on I, then $f' : I \to \mathbb{R}$ is itself a function and differentiation acts as a linear operator on the vector space $\mathcal{D}(I)$ of functions differentiable on I:

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

These equalities hold as functions on I. The Finite Linear Combination Rule in its global form is precisely the statement that $D : \mathcal{D}(I) \to \mathcal{F}(I)$ defined by $D(f) = f'$ is a linear map between function spaces, where $\mathcal{F}(I)$ denotes functions on I. The interval variants are not separate theorems — they are the pointwise results collected into function notation by applying the pointwise result at each $x \in I$.

Remark (Dependencies).

- Real-Linear Rule
- Linear Combination of Real-Valued Functions
- Theorem: Finite Linear Combination Rule

When f is strictly monotone and differentiable on I with $f'(x) \neq 0$ for all $x \in I$, the inverse $g : J \to I$ is differentiable on $J = f(I)$ and

$$g' = \frac{1}{f' \circ g},$$

as functions on J. This is the global form of the Inverse Function Theorem for derivatives and is treated in the inverse function section.

39.8.4 Inverting the Operation

Theorem (Inverse Function Theorem, One Variable)

Theorem 39.8.17 (Inverse Function Theorem, One Variable)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$ be of class C^1, and let $c \in I$ satisfy $f'(c) \neq 0$. Then there exist open intervals $U \subseteq I$ and $V \subseteq \mathbb{R}$, with $c \in U$ and $f(c) \in V$, such that:*

1. *$f|_U : U \to V$ is a bijection;*
2. *the inverse $g = (f|_U)^{-1} : V \to U$ is of class C^1;*
3. *for every $y \in V$,*
$$g'(y) = \frac{1}{f'(g(y))}.$$

Go to proof.

Remark (Standard quantified statement).

$$f \in C^1(I) \wedge c \in I \wedge f'(c) \neq 0$$
$$\implies \exists U \ni c\, \exists V \ni f(c)\, \big[\text{Bijective}(f|_U, U, V) \wedge (f|_U)^{-1} \in C^1(V) \wedge \forall y \in V :\ ((f|_U)^{-1})'(y) = 1/f'((f|_U)^{-1}$$

Remark (Predicate reading). LocallyInvertible(f, c) holds with a C^1 local inverse. The condition $f'(c) \neq 0$ upgrades local set-theoretic invertibility to local analytic invertibility: the inverse exists as a differentiable function and has controlled derivative.

Remark (Failure modes). The hypotheses fail in different ways.

- If $f'(c) = 0$, the inverse may exist without being differentiable. The map $f(x) = x^3$ is globally bijective, but its inverse $x \mapsto x^{1/3}$ is not differentiable at 0.
- If f' is not continuous at c, the sign of f' need not persist near c. Mere differentiability with $f'(c) \neq 0$ is therefore not enough to force local monotonicity or local invertibility.

Remark (Contrapositive quantified statement).

$$\neg\, \text{LocallyInvertible}_{C^1}(f, c) \implies f \notin C^1(\text{near } c) \ \vee\ f'(c) = 0.$$

Remark (Contrapositive predicate reading). If f has no C^1 local inverse at c, then either f fails to be continuously differentiable near c, or its derivative vanishes there. The witnesses $x \mapsto x^3$ and $x \mapsto x + 2x^2 \sin(1/x)$ isolate the two failure mechanisms.

Remark (Interpretation). This theorem is the existence result behind the inverse-derivative formula. A continuous nonzero derivative keeps its sign on a small interval; constant sign forces strict monotonicity; strict monotonicity gives a local inverse; and the Carathéodory characterization upgrades that inverse to a differentiable map with reciprocal slope. In several variables, the condition $f'(c) \neq 0$ becomes invertibility of the differential.

Remark (Dependencies).

- Definition: Continuously Differentiable; Class C^1
- Theorem: Monotonicity from the Sign of the Derivative
- Theorem: Continuous Inverse Theorem
- Theorem: Carathéodory Characterization of Differentiability

Corollary (Derivative of the Inverse Function)

Corollary 39.8.18 (Derivative of the Inverse Function)**.** *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse g satisfies*

$$g' = \frac{1}{f' \circ g}$$

on V. More globally, if $f : I \to J$ is a C^1 bijection with $f'(x) \neq 0$ for every $x \in I$, then $f^{-1} : J \to I$ is C^1 and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \qquad \text{for every } y \in J.$$

Go to proof.

Remark (Standard quantified statement).

$$g = (f|_U)^{-1} \Longrightarrow \forall y \in V : \; g'(y) = \frac{1}{f'(g(y))}.$$

Remark (Interpretation). The formula says that undoing a differentiable change of variable reverses the local stretch factor. If f multiplies small increments near x by $f'(x)$, then its inverse multiplies corresponding increments near $f(x)$ by $1/f'(x)$.

Remark (Dependencies).

- Theorem: Inverse Function Theorem, One Variable
- Theorem: Chain Rule

39.8.5 Indeterminate Forms: L'Hôpital's Rule

Theorem (L'Hôpital's Rule, $0/0$ Form)

Theorem 39.8.19 (L'Hôpital's Rule, 0/0 Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0 \qquad \textit{and} \qquad \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.

Remark (Standard quantified statement)*.*

$$\left(\lim_{x\to a^+} f(x) = 0 \wedge \lim_{x\to a^+} g(x) = 0 \wedge \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L\right) \Longrightarrow \lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

Remark (Failure modes)*.* The rule is a one-way implication and is routinely misapplied.

- *Form not indeterminate.* If f/g is not genuinely $0/0$, differentiating numerator and denominator can answer a different problem.
- $\lim f'/g'$ *fails to exist.* With $f(x) = x + \sin x$ and $g(x) = x$ as $x \to +\infty$, the quotient $f/g \to 1$, but $f'/g' = 1 + \cos x$ oscillates. Absence of $\lim f'/g'$ says nothing by itself about $\lim f/g$.
- g' *vanishes near the target.* The denominator derivative must stay nonzero so the Cauchy Mean Value Theorem applies and the derivative ratio is defined.

Remark (Interpretation)*.* The $0/0$ form is the Cauchy Mean Value Theorem read as a limit law. On a shrinking interval, a ratio of changes in f and g is represented by a ratio of derivatives at an interior point. If the derivative ratio settles, the original quotient is forced to settle to the same value.

Remark (Dependencies)*.*

- Theorem: Cauchy Mean Value Theorem
- Definition: Limit of a Function
- Theorem: Squeeze Theorem for Function Limits

Theorem (L'Hôpital's Rule, ∞/∞ Form)

Theorem 39.8.20 (L'Hôpital's Rule, ∞/∞ Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x \to a^+} |g(x)| = +\infty \qquad \textit{and} \qquad \lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

No separate hypothesis on $\lim_{x \to a^+} f(x)$ is required. *Go to proof.*

Remark (Standard quantified statement).

$$\left(\lim_{x \to a^+} |g(x)| = +\infty \wedge \lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L \right) \Longrightarrow \lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

Remark (Failure modes). L'Hôpital's Rule is a one-way implication and requires an indeterminate form. If the quotient is not of type $0/0$ or ∞/∞, differentiating numerator and denominator can change the problem. The derivative ratio must also have a limit: for $f(x) = x + \sin x$ and $g(x) = x$ as $x \to +\infty$, the quotient f/g tends to 1, but $f'/g' = 1 + \cos x$ has no limit.

Remark (Interpretation). The ∞/∞ form compares two functions whose magnitudes escape. The Cauchy Mean Value Theorem controls the quotient after subtracting fixed endpoint values, and the divergent denominator makes those fixed values negligible.

Remark (Dependencies).

- Theorem: Cauchy Mean Value Theorem
- Definition: Limit of a Function
- Theorem: Squeeze Theorem for Function Limits

39.9 Taylor: Construction and Its Error

Taylor Expansion — Quick Reference

Concept	Meaning
Taylor polynomial	Polynomial forced by derivative data at a point
Taylor remainder	Exact error after subtracting the Taylor polynomial
Maclaurin polynomial	Taylor polynomial centered at 0
Lagrange remainder	Error controlled by an intermediate derivative value
Peano remainder	Local asymptotic error smaller than the retained order
Example: $(\sin x - x)/x^3$	Taylor and L'Hôpital compared on one limit
Flat function	Smooth function whose Taylor series misses it

Taylor's theorem is the derivative's most ambitious claim: local behaviour to finite order is encoded by derivatives at one point, with an error term that can be estimated. The first-order case is the seed of the closing reframe, where the derivative stops being only a number $f'(c)$ and becomes the linear map $h \mapsto f'(c)h$.

Definition (Taylor Polynomial at a Point)

Definition 39.9.1 (Taylor Polynomial at a Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \leq k \leq n$. The **Taylor polynomial of order** n for f at a is

$$T_{n,a}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{Let } I \subseteq \mathbb{R},\ a \in I,\ n \in \mathbb{N},\ f : I \to \mathbb{R}.\\ &\text{If } f^{(k)}(a) \text{ exists for every } 0 \leq k \leq n, \text{ then}\\ &\qquad \forall x \in I\ \left(T_{n,a}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \right).\end{aligned}$$

Remark (Interpretation)*.* The Taylor polynomial is the unique polynomial of degree at most n whose first n derivatives at a agree with those of f. Its coefficients are not chosen by fitting sample points; they are dictated by derivative data at one point.

Remark (Dependencies)*.*

- Derivative at a Point
- Power Rule for Integer Powers of a Function
- Finite Sum Rule

Definition (Taylor Remainder)

Definition 39.9.2 (Taylor Remainder)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order** n at a is the function $R_{n,a}f : I \to \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

Remark (Standard quantified statement)*.*

$$\forall x \in I\ \big(R_{n,a}f(x) = f(x) - T_{n,a}f(x)\big).$$

Remark (Interpretation)*.* The remainder is the exact error made by replacing f with its Taylor polynomial. Taylor's theorem is valuable because it gives usable formulas or estimates for this error.

Remark (Dependencies).

- Taylor Polynomial at a Point

Definition (Maclaurin Polynomial)

Definition 39.9.3 (Maclaurin Polynomial)**.** Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \le k \le n$. The **Maclaurin polynomial of order** n for f is

$$T_{n,0}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k.$$

Remark (Standard quantified statement).

$$\forall x \in I \left(T_{n,0}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k \right).$$

Remark (Interpretation). Maclaurin polynomials are Taylor polynomials centered at the origin. They are not a separate approximation theory; they are the centered-at-zero case of the same construction.

Remark (Dependencies).

- Taylor Polynomial at a Point

Theorem (Taylor's Theorem with Lagrange Remainder)

Theorem 39.9.4 (Taylor's Theorem with Lagrange Remainder)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a,b] \to \mathbb{R}$. Suppose that $f, f', \ldots, f^{(n)}$ are continuous on $[a,b]$ and that $f^{(n+1)}$ exists on (a,b). Then, for every $x \in (a,b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a,b)\ \exists c \in (a,x) \left(f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \right).$$

Remark (Interpretation). Taylor's theorem says that the error after truncating at degree n has the same shape as the next Taylor term, but with the next derivative evaluated at some intermediate point rather than at the center. The point c is generally not known explicitly; the theorem is most often used to estimate the size of the remainder.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Mean Value Theorem
- Finite Sum Rule
- Power Rule for Integer Powers of a Function

Corollary (Taylor Expansion with Peano Remainder)

Corollary 39.9.5 (Taylor Expansion with Peano Remainder)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order n in a neighbourhood of a. If $f^{(n)}$ is continuous at a, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big) \qquad \text{as } x \to a.$$

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x \to a} \frac{f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}{(x-a)^n} = 0.$$

Remark (Interpretation). The Peano form is a local asymptotic statement. It says that the Taylor polynomial captures all terms through order n, and the remaining error is smaller than $(x-a)^n$ near the center.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Limit of a Function
- Derivative at a Point

Corollary (First-Order Peano Remainder)

Corollary 39.9.6 (First-Order Peano Remainder)**.** *Let f be differentiable at c. Then, as $h \to 0$,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \;\Rightarrow\; \lim_{h \to 0} \frac{f(c+h) - f(c) - f'(c)h}{h} = 0.$$

Remark (Interpretation). The first-order Peano formula is the derivative rewritten as an approximation theorem. It says that the affine approximation $f(c) + f'(c)h$ captures all first-order change, and the remaining error is smaller than h itself.

Remark (Dependencies).

- Derivative at a Point

Example ($(\sin x - x)/x^3$ Two Ways). Taylor expansion gives

$$\sin x = x - \frac{x^3}{6} + o(x^3),$$

so

$$\frac{\sin x - x}{x^3} = \frac{-x^3/6 + o(x^3)}{x^3} \longrightarrow -\frac{1}{6}.$$

L'Hôpital's rule reaches the same value by three differentiations:

$$\lim_{x \to 0} \frac{\sin x - x}{x^3} = \lim_{x \to 0} \frac{\cos x - 1}{3x^2} = \lim_{x \to 0} \frac{-\sin x}{6x} = \lim_{x \to 0} \frac{-\cos x}{6} = -\frac{1}{6}.$$

Taylor exposes the reason for the answer: the limit is the cubic coefficient.

Proposition (The Flat Function: Smooth but Not Analytic)

Proposition 39.9.7 (The Flat Function: Smooth but Not Analytic)**.** *Define $f : \mathbb{R} \to \mathbb{R}$ by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then:

1. $f \in C^\infty(\mathbb{R})$*;*
2. $f^{(n)}(0) = 0$ *for every* $n \geq 0$*;*
3. *the Maclaurin series of* f *is identically* 0*, so* $f \notin C^\omega$ *on any neighbourhood of* 0*.*

In particular, $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \varnothing$*.* *Go to proof.*

Remark (Standard quantified statement).

$$\big(\forall n \in \mathbb{N} : f^{(n)}(0) = 0\big) \wedge \big(\forall r > 0 \;\exists x : \; 0 < |x| < r \wedge f(x) \neq 0\big) \Longrightarrow f \in C^\infty \setminus C^\omega.$$

Remark (Interpretation). The Taylor series sees only derivative data at the center, and all of that data is zero for the flat function at 0. The function is nevertheless positive away from 0. This separates smoothness from analyticity and supplies the strictness witness referenced by the smoothness tower.

Remark (Dependencies).

- Definition: Higher Derivatives
- Definition: Smooth Function; Class C^∞
- Definition: Real-Analytic Function; Class C^ω
- Definition: Maclaurin Polynomial
- Theorem: The Smoothness Tower

39.9.1 The Differential

Remark (Interpretation). The derivative value $f'(c) \in \mathbb{R}$ is a number. The differential df_c is a linear map from increments to first-order changes. In one dimension these two objects are represented by the same scalar, but they play different roles.

Definition (Differentiability by a Differential)

Definition 39.9.8 (Differentiability by a Differential)**.** Let f be defined on a neighbourhood of $c \in \mathbb{R}$. The function f is *differentiable at c in the differential sense* if there exists a linear map $L : \mathbb{R} \to \mathbb{R}$ such that

$$f(c+h) - f(c) = L(h) + o(h) \qquad \text{as } h \to 0.$$

In that case the *differential* of f at c is $df_c := L$.

Remark (Standard quantified statement).

$$\exists L : \mathbb{R} \to \mathbb{R} \text{ linear such that } f(c+h) - f(c) = L(h) + o(h).$$

Remark (Interpretation). The differential is the linear part of the increment. The function itself is approximated affinely by $f(c) + df_c(x - c)$, while df_c alone is the linear map acting on the displacement.

Remark (Dependencies).

- First-Order Peano Remainder

Theorem (Differential and Derivative Agree)

Theorem 39.9.9 (Differential and Derivative Agree)**.** *Let f be defined on a neighbourhood of $c \in \mathbb{R}$. Then f is differentiable at c in the ordinary derivative sense if and only if it is differentiable at c in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \exists df_c \text{ with } df_c(h) = f'(c)h.$$

Remark (Interpretation). This theorem is the bridge from the slope number to the linear map. The one-dimensional derivative is multiplication by the scalar $f'(c)$.

Remark (Dependencies).

- Derivative at a Point
- Differentiability by a Differential

Theorem (Uniqueness of the Differential)

Theorem 39.9.10 (Uniqueness of the Differential). *If $L_1, L_2 : \mathbb{R} \to \mathbb{R}$ are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \qquad \text{as } h \to 0,$$

then $L_1 = L_2$. *Go to proof.*

Remark (Standard quantified statement).

$$\big(L_1(h) - L_2(h) = o(h)\big) \implies L_1 = L_2.$$

Remark (Interpretation). The differential is not a choice of linear approximation among many. If a first-order linear part exists, it is forced.

Remark (Dependencies).

- Differentiability by a Differential

Theorem (Differential Continuity Criterion)

Theorem 39.9.11 (Differential Continuity Criterion). *If f is differentiable at c in the differential sense, then f is continuous at c.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{DifferentiableByDifferentialAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

Remark (Interpretation). From $f(c+h) - f(c) = df_c(h) + o(h)$, both terms on the right tend to zero as $h \to 0$. Continuity is therefore immediate in the differential language.

Remark (Dependencies).

- Differentiability by a Differential

Theorem (Chain Rule for Differentials)

Theorem 39.9.12 (Chain Rule for Differentials). *If f is differentiable at c and g is differentiable at $f(c)$, then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Go to proof.

Remark (Standard quantified statement).

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Remark (Interpretation). In one dimension this says that composing two scalar multiplications multiplies their scalars. In several variables the same formula becomes matrix multiplication.

Remark (Dependencies).

- Chain Rule
- Differential and Derivative Agree

Theorem (Linearity of the Differential)

Theorem 39.9.13 (Linearity of the Differential)**.** *If f and g are differentiable at c and $\alpha, \beta \in \mathbb{R}$, then*

$$d(\alpha f + \beta g)_c = \alpha\, df_c + \beta\, dg_c.$$

Go to proof.

Remark (Standard quantified statement).

$$d(\alpha f + \beta g)_c = \alpha\, df_c + \beta\, dg_c.$$

Remark (Interpretation). The differential packages the linearity of differentiation as equality of linear maps.

Remark (Interpretation). On a manifold, the same construction becomes a tangent map

$$T_c f : T_c M \to T_{f(c)} N.$$

The one-dimensional differential is the prototype: replace real increments by tangent vectors and replace scalar multiplication by a linear map between tangent spaces.

Remark (Dependencies).

- Finite Linear Combination Rule
- Differential and Derivative Agree

Integration

Chapter 40

Integration

SERIES	VOLUME III	BOOK IV	CHAPTER 11
From Cantor to Ito	Classical Analysis	Integration	Integration

40.1 Partitions and Tags

Remark (Reframing note)*.* This chapter is the narrative version of the reference list. The reference orders the definitions correctly, but the real subject is the sequence of defects and patches:

$$\begin{aligned}\text{geometric problem} &\to \text{Cauchy} \to \text{Riemann}\\ &\to \text{Darboux} \to \text{Stieltjes}\\ &\to \text{Henstock–Kurzweil and McShane}\\ &\to \text{Lebesgue.}\end{aligned}$$

Each integral answers a concrete complaint about the one before it. The point is not to memorize several integrals, but to understand why each one had to be invented.

The Geometric Problem

Forget integrals for a moment. The physical question is always the same:

> Given a pointwise rate f, how much total quantity accumulates from a to b?

A speedometer reads $f(t)$; the accumulated quantity is distance. A wire has linear density $f(x)$; the accumulated quantity is mass. A curve $y = f(x)$ over $[a, b]$ asks for signed area. The only available strategy is:

1. Chop $[a, b]$ into thin slabs.
2. Pretend f is constant on each slab.
3. Add the resulting rectangles.

4. Refine the chopping and ask whether the sums settle.

Every construction below makes two choices: which height to use in a slab, and what it means for all these rectangle sums to settle to one number.

Remark (Construction comparison). The constructions differ by the height rule, the ruler, or the fineness rule:

Construction	Height rule	Ruler / fineness
Cauchy	$f(x_{i-1})$, left endpoint	Δx_i
Riemann	$f(\xi_i)$, arbitrary tag	Δx_i
Darboux	$\inf_I f,\ \sup_I f$ on each slab	Δx_i
Riemann–Stieltjes	$f(\xi_i)$, tagged height	$\Delta \alpha_i$
Henstock–Kurzweil	$f(\xi_i),\ \xi_i \in I_i$	$\Delta x_i,\ \delta(\xi_i)$-fine, tag tethered
McShane	$f(t_i),\ t_i$ may float	$\Delta x_i,\ \delta(t_i)$-fine, tag untethered

Henstock–Kurzweil does not change the sampled height or the ordinary ruler; it changes the admissible chopping by replacing a global mesh bound with a local gauge condition. McShane keeps the same gauge idea but frees the tag from the slab, which pulls the theory back toward Lebesgue's absolute notion of size.

Definition (Partition)

Definition 40.1.1 (Partition of a Compact Interval)**.** A *partition* of $[a, b]$ is a finite increasing list

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Equivalently,

$$\begin{aligned}\text{Partition}(P, a, b) \iff {} & \exists n \in \mathbb{N}\, \exists x_0, \ldots, x_n \in \mathbb{R} \\ & \big(P = \{x_0, \ldots, x_n\} \wedge x_0 = a \wedge x_n = b\big) \\ & \wedge \forall i \,\big(1 \le i \le n \Rightarrow x_{i-1} < x_i\big).\end{aligned}$$

Definition (Subinterval Width)

Definition 40.1.2 (Subinterval Width)**.** The i-th slab has width

$$\Delta x_i = x_i - x_{i-1}.$$

Definition (Mesh)

Definition 40.1.3 (Mesh of a Partition)**.** The *mesh* or *norm* of P is the width of its fattest slab:

$$\|P\| = \max_{1 \le i \le n} \Delta x_i.$$

Remark (Geometric reading). The mesh is the resolution dial. The phrase "as the partition is refined" will usually mean that $\|P\| \to 0$: no slab is allowed to remain fat.

Definition (Tagged Partition)

Definition 40.1.4 (Tagged Partition)**.** A *tagged partition* is a partition $P = \{x_0, \dots, x_n\}$ together with points $\xi_i \in [x_{i-1}, x_i]$. The tag is the point where the height $f(\xi_i)$ is sampled.

Definition (Refinement)

Definition 40.1.5 (Refinement of a Partition)**.** A partition P' *refines* P when $P \subseteq P'$. Refinement adds cut-points; it does not move or delete existing cut-points.

Lemma (Common Refinement)

Lemma 40.1.6 (Common Refinement of Two Partitions)**.** *Any two partitions P and P' of $[a, b]$ have a common refinement: order the finite set $P \cup P'$ increasingly.*

Remark (Why the lemma is load-bearing). This small fact is the reason independently chosen partitions can be compared. Whenever a proof says "refine them together," it is using this lemma. It is the structural engine behind Cauchy's existence proof and Darboux's squeeze.

40.2 The Cauchy Integral

Geometric Intuition

Cauchy makes the simplest possible height choice: always sample the left edge of each slab. The result is a staircase of left-anchored rectangles. As the slabs thin out, the question is whether those staircase areas settle.

Remark (Engineering reading). Cauchy's construction is the seed. It answers the geometric problem for the case where the curve is tame enough that left-edge staircases forget their initial sampling bias as the mesh shrinks. The construction is deliberately primitive: it fixes a height rule first and asks for convergence second.

Definition (Cauchy Sum)

Definition 40.2.1 (Left-Endpoint Cauchy Sum)**.** For $f : [a, b] \to \mathbb{R}$ and $P = \{a = x_0 < \cdots < x_n = b\}$, define

$$C(f, P) = \sum_{i=1}^{n} f(x_{i-1})\,\Delta x_i.$$

Definition (Cauchy Integral)

Definition 40.2.2 (Cauchy Integral)**.** The function f is *Cauchy-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall P\ \big(\|P\| < \delta \Rightarrow |C(f, P) - L| < \varepsilon\big).$$

The value L is denoted $\int_a^b f(x)\,dx$.

Proposition (Integral of a Constant)

Proposition 40.2.3 (Cauchy Integral of a Constant). *If $f(x) = c$ on $[a, b]$, then $\int_a^b c\,dx = c(b-a)$.*

Theorem (Linearity)

Theorem 40.2.4 (Linearity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem (Monotonicity)

Theorem 40.2.5 (Monotonicity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\,dx \le \int_a^b g\,dx.$$

Corollary (Bounds)

Corollary 40.2.6 (Bounds for the Cauchy Integral). *If f is continuous on $[a, b]$, $m = \min_{[a,b]} f$, and $M = \max_{[a,b]} f$, then*

$$m(b-a) \le \int_a^b f\,dx \le M(b-a).$$

Theorem (Triangle Inequality)

Theorem 40.2.7 (Triangle Inequality for the Cauchy Integral). *If f and $|f|$ are Cauchy-integrable on $[a, b]$, then*

$$\left| \int_a^b f\,dx \right| \le \int_a^b |f|\,dx.$$

Theorem (Interval Additivity)

Theorem 40.2.8 (Interval Additivity for the Cauchy Integral). *If $c \in [a, b]$ and f is Cauchy-integrable on the relevant intervals, then*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Definition (Oscillation on an Interval)

Definition 40.2.9 (Oscillation on an Interval)**.** For bounded f on an interval I, define

$$\omega_f(I) = \sup_I f - \inf_I f.$$

Oscillation is the error controller: if $\omega_f(I)$ is small, every sample height on I is close to every other sample height.

Remark (Oscillation as the visible error)*.* Figure 40.1 shows the same quantity geometrically: the Darboux gap $U(f,P) - L(f,P)$ is the sum of local oscillations $\omega_f(I_i)$ multiplied by the slab widths Δx_i. The trapezoid and Simpson panels are numerical side branches; the load-bearing panel here is the oscillation identity.

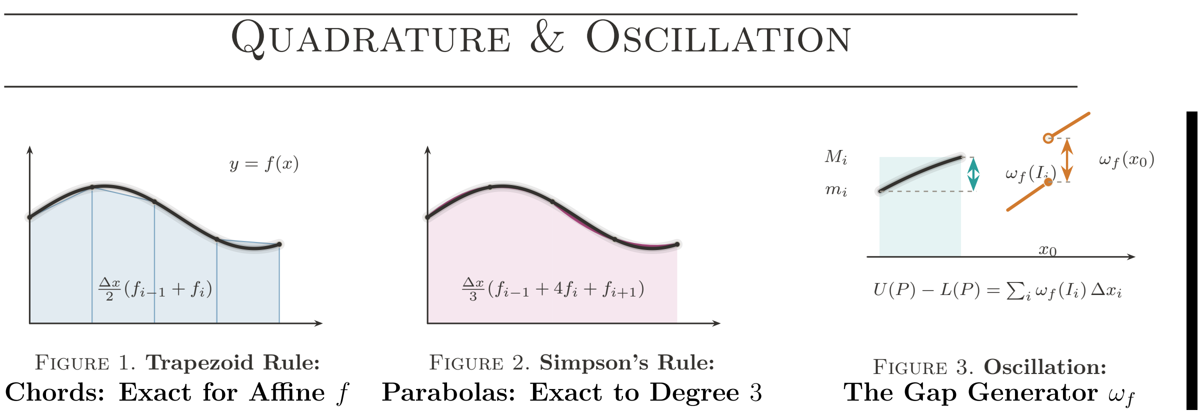


Figure 40.1: Quadrature and oscillation

Theorem (Continuous Implies Cauchy-Integrable)

Theorem 40.2.10 (Continuous Functions Are Cauchy-Integrable)**.** *Every continuous function $f : [a,b] \to \mathbb{R}$ is Cauchy-integrable.*

Remark (Proof skeleton)*.* Continuity on the compact interval gives uniform continuity. Hence short slabs have small oscillation. Given two fine partitions, pass to their common refinement and compare sums slab by slab; each re-sampling error is bounded by $\omega_f(I)$ times the slab width. The sums form a Cauchy net, and completeness of $\mathbb{R}$ supplies the limit.

Remark (What continuity buys)*.* Uniform continuity is the load-bearing ingredient. It converts pointwise control into one global scale: if the slabs are all shorter than that scale, then every slab has small oscillation. Without this global scale, Cauchy's left-endpoint definition has no mechanism for proving that unrelated partitions settle to the same number.

Remark (A first discontinuous edge case)*.* A single jump should still have an area. If

$$f(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c, \end{cases}$$

on $[a, b]$, the only dangerous slab is the one crossing c. Its total possible error is bounded by its width, and that width tends to zero as the mesh tends to zero. This example is not hard; the point is that Cauchy's definition gives no internal test explaining why it is harmless. The proof above proves continuity is sufficient, not that controlled discontinuities are allowed.

Theorem (Tag Independence)

Theorem 40.2.11 (Tag Independence for Continuous Functions)**.** *If f is continuous on $[a, b]$, then every choice of tags gives the same limit as the left-endpoint sums as $\|P\| \to 0$.*

Remark (Bug report). Cauchy has two defects. First, the left endpoint is privileged by convention; tag independence is a theorem only after continuity has already been assumed. Second, the existence proof gives no usable criterion for discontinuous functions. The patch request is to stop privileging the left endpoint and make tag independence part of the definition.

Remark (Patch request). The next definition should not say "sample on the left." It should say: sample anywhere, even adversarially, and require all fine enough choices to agree. Riemann is exactly Cauchy's tag-independence theorem promoted into the definition of integrability.

40.3 The Riemann Integral

Geometric Intuition

Riemann promotes Cauchy's tag-independence theorem into a definition. A function is integrable when every sufficiently fine staircase, even with adversarial tags, is forced near the same number.

Remark (Engineering reading). The sampling convention is no longer chosen by the author of the definition. It is chosen by a universal quantifier. A Riemann-integrable function is one whose accumulated area survives every possible choice of sample heights once the slabs are fine enough.

Definition (Riemann Sum)

Definition 40.3.1 (Riemann Sum)**.** For a tagged partition (P, ξ), define

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

Definition (Riemann Integral)

Definition 40.3.2 (Riemann Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Riemann-integrable* with value L if

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall (P, \xi) \; \big(\|P\| < \delta \Rightarrow |S(f, P, \xi) - L| < \varepsilon\big).$$

Remark (The one symbol that changed everything). Cauchy quantifies over choppings. Riemann quantifies over choppings and every possible choice of heights:

$$\forall P \qquad \rightsquigarrow \qquad \forall (P, \xi).$$

That extra quantifier removes the arbitrary left endpoint and turns tag independence into the price of admission.

Remark (Payoff). The definition is no longer tied to continuity. It can admit bounded functions with discontinuities, because the question is not whether f is continuous but whether all fine tagged sums are forced into one narrow numerical corridor. The Cauchy criterion below makes that statement intrinsic: the value need not be known in advance.

Remark (The ruler function example). Thomae's function, also called the ruler function, is

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms,} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous at every rational and continuous at every irrational, so its discontinuities are dense. Nevertheless its Riemann integral on any compact interval is 0. The reason is that the rational points where T can be large form only a finite set above any fixed height threshold. Fine partitions can isolate those finitely many tall spikes, while all remaining slabs have uniformly small height. This is exactly the kind of function Cauchy's continuity proof could not diagnose, but Riemann's tag-independent definition can admit.

Theorem (Continuous Implies Riemann-Integrable)

Theorem 40.3.3 (Continuous Functions Are Riemann-Integrable). *Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.*

Theorem (Linearity)

Theorem 40.3.4 (Linearity of the Riemann Integral). *If f and g are Riemann-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\, dx = \alpha \int_a^b f\, dx + \beta \int_a^b g\, dx.$$

Theorem (Monotonicity)

Theorem 40.3.5 (Monotonicity of the Riemann Integral). *If f and g are Riemann-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\, dx \le \int_a^b g\, dx.$$

Theorem (Triangle Inequality)

Theorem 40.3.6 (Triangle Inequality for the Riemann Integral)**.** *If f is Riemann-integrable on $[a,b]$, then $|f|$ is Riemann-integrable and*

$$\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx.$$

Theorem (Interval Additivity)

Theorem 40.3.7 (Interval Additivity for the Riemann Integral)**.** *If $c \in [a,b]$, then f is Riemann-integrable on $[a,b]$ if and only if it is Riemann-integrable on $[a,c]$ and $[c,b]$, and in that case*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Theorem (Cauchy Criterion)

Theorem 40.3.8 (Cauchy Criterion for Riemann Integrability)**.** *A bounded function is Riemann-integrable if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that any two tagged partitions (P,ξ) and (Q,η) with $\|P\|, \|Q\| < \delta$ have*

$$|S(f,P,\xi) - S(f,Q,\eta)| < \varepsilon.$$

Remark (Bug report). Riemann fixed the arbitrary tag, but the definition is operationally hostile. To verify integrability directly one must control all tagged partitions at once. The patch request is to remove tags from the verification problem: replace every possible sample height by the slab infimum and supremum, then squeeze.

Remark (Why this is hard to use). Riemann is logically clean but not mechanically friendly. Refining a partition does not make an arbitrary Riemann sum move monotonically, because every refinement lets the tags move again. There is no single upper or lower quantity improving under refinement. Darboux supplies exactly that missing monotone control.

40.4 The Darboux Integral

Geometric Intuition

Darboux stops chasing tags. On each slab, the adversary can do no better than the supremum and no worse than the infimum. So build the upper staircase from $\sup f$, the lower staircase from $\inf f$, and ask whether the gap can be squeezed to zero.

Remark (Engineering reading). Darboux is the usable face of Riemann integration. It does not change the integral; it changes the test. Instead of quantifying over every possible tag, it brackets all tags at once.

Definition (Darboux Sums)

Definition 40.4.1 (Lower and Upper Darboux Sums)**.** For bounded f on $[a, b]$, set

$$m_i = \inf_{[x_{i-1},x_i]} f, \qquad M_i = \sup_{[x_{i-1},x_i]} f,$$

and define

$$L(f, P) = \sum_i m_i \Delta x_i, \qquad U(f, P) = \sum_i M_i \Delta x_i.$$

Lemma (Monotonicity Under Refinement)

Lemma 40.4.2 (Darboux Sums Under Refinement)**.** *If P' refines P, then*

$$L(f, P) \le L(f, P') \le U(f, P') \le U(f, P).$$

Remark (Geometric reading). Thinner slabs give lower rectangles room to rise and upper rectangles room to fall. This monotonicity is exactly what Riemann sums lack when tags move adversarially.

Remark (One-point refinement proof idea). It is enough to add one cut-point c inside one slab and iterate. The infimum over the whole slab is less than or equal to the infimum over each piece, so the lower contribution rises. The supremum over the whole slab is greater than or equal to the supremum over each piece, so the upper contribution falls.

Definition (Darboux Integrability)

Definition 40.4.3 (Darboux Integrability)**.** The lower and upper integrals are

$$\underline{I} = \sup_P L(f, P), \qquad \overline{I} = \inf_P U(f, P).$$

The function f is *Darboux-integrable* if $\underline{I} = \overline{I}$.

Theorem (Darboux Criterion)

Theorem 40.4.4 (Darboux Criterion)**.** *A bounded function f is Darboux-integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that*

$$U(f, P) - L(f, P) < \varepsilon.$$

Remark (The squeeze in one plate). Figure 40.2 places the Cauchy/Riemann tag choices, the lower–upper Darboux bracket, the refinement squeeze, and the gauge panel on the same page. The Darboux row is the chapter's hinge: replacing moving tags by inf and sup turns integrability into the checkable $U - L$ squeeze.

Probes of the Integral

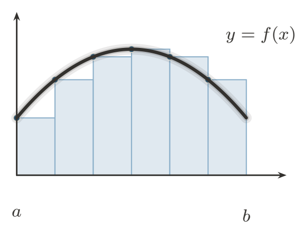


FIGURE 1. **Left Riemann Sum:**
Tag at the Left Endpoint

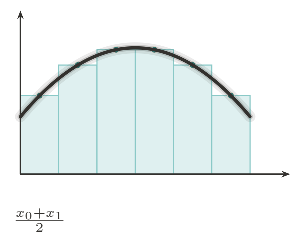


FIGURE 2. **Midpoint Rule:**
The Centered Tag

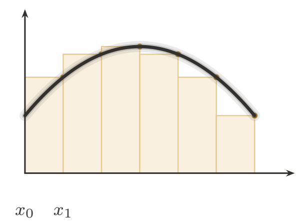


FIGURE 3. **Right Riemann Sum:**
Tag at the Right Endpoint

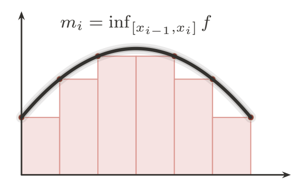


FIGURE 4. **Lower Darboux Sum:**
The Infimum Probe

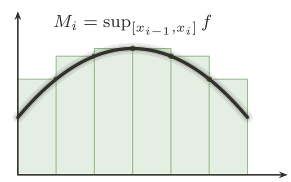


FIGURE 5. **Upper Darboux Sum:**
The Supremum Probe

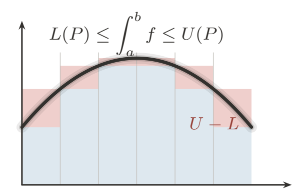


FIGURE 6. **The Squeeze:**
Integrability as a Closing Gap

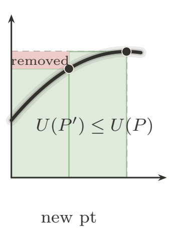


FIGURE 7. **Partition Refinement:**
Adding Points Tightens the Sums

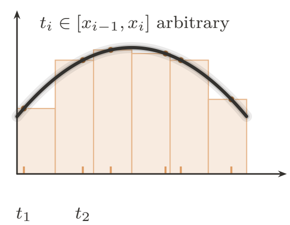


FIGURE 8. **Tagged Partition:**
The General Riemann Sum

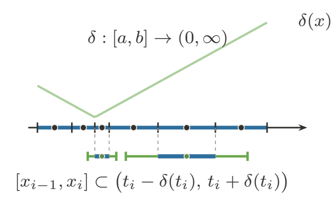


FIGURE 9. **The Gauge:**
δ-Fine Partitions (Henstock–Kurzweil)

Figure 40.2: Probes of the integral

Theorem (Riemann Equals Darboux)

Theorem 40.4.5 (Equivalence of Riemann and Darboux Integrability)**.** *For bounded functions on $[a, b]$, Riemann integrability and Darboux integrability are equivalent, and the integral values agree.*

Theorem (Continuous Functions Are Integrable)

Theorem 40.4.6 (Continuous Functions Are Darboux-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem (Monotone Functions Are Integrable)

Theorem 40.4.7 (Monotone Functions Are Darboux-Integrable)**.** *Every monotone function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Remark (Why this is a harvest theorem). Monotone functions may have infinitely many discontinuities. Darboux still handles them because their total vertical movement is finite. On each slab the oscillation is just the endpoint jump across that slab, and these jumps telescope over the partition. Equal-width partitions therefore force $U(f, P) - L(f, P)$ to zero.

Theorem (Finitely Many Discontinuities)

Theorem 40.4.8 (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable)**.** *If $f : [a, b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Darboux-integrable.*

Remark (Why finite bad sets are harmless). The Darboux gap is

$$U(f, P) - L(f, P) = \sum_i \omega_f([x_{i-1}, x_i])\Delta x_i.$$

A finite set of discontinuities can be trapped in intervals of arbitrarily small total width. On the complement, f is continuous on compact pieces, so its oscillation is uniformly small on fine slabs. Thus the bad slabs are short and the good slabs are flat.

Theorem (Algebra of Darboux-Integrable Functions)

Theorem 40.4.9 (Sums and Scalar Multiples of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then $\alpha f + \beta g$ is Darboux-integrable for all $\alpha, \beta \in \mathbb{R}$.*

Theorem (Products)

Theorem 40.4.10 (Products of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then fg is Darboux-integrable on $[a, b]$.*

Theorem (Absolute Values)

Theorem 40.4.11 (Absolute Values of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, then $|f|$ is Darboux-integrable on $[a, b]$.*

Theorem (Continuous Composition)

Theorem 40.4.12 (Continuous Images of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, φ is continuous on an interval containing $f([a, b])$, and f is bounded, then $\varphi \circ f$ is Darboux-integrable on $[a, b]$.*

Remark (Why the harvest is cheap). The Darboux criterion reduces closure questions to controlling the upper-lower gap. Boundedness controls coefficients and products, while continuity of φ converts small oscillation of f into small oscillation of $\varphi \circ f$. This is the point of switching from tags to a squeeze: theorems become estimates on $U(f,P) - L(f,P)$.

Remark (Bug report). Darboux is a better tool for the same class. It does not enlarge Riemann integrability. The remaining defects now fork. One can change the ruler Δx to $\Delta\alpha$, which leads to Riemann–Stieltjes. Or one can diagnose the size of the discontinuity set, which leads to Lebesgue and also to the gauge-integral cure.

Remark (Residual failures). The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ on $[0,1]$ has infimum 0 and supremum 1 on every slab, since rationals and irrationals are dense. Thus $L(f,P) = 0$ and $U(f,P) = 1$ for every partition, so the Darboux gap never closes. This is not a Darboux-specific failure; by equivalence, it is the failure of the whole Riemann family. The failure modes in Figure 40.3 separate the common pathologies: Dirichlet breaks the continuity-almost-everywhere condition, Thomae survives it, unbounded examples violate the boundedness hypothesis, and pointwise limits show why Riemann integration has weak limit behavior.

Where Riemann Fails

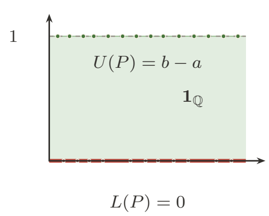


Figure 1. **Dirichlet $\mathbf{1}_{\mathbb{Q}}$: The Gap That Never Closes**

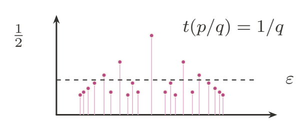


Figure 2. **Thomae's Function: Dense Jumps, Yet Integrable**

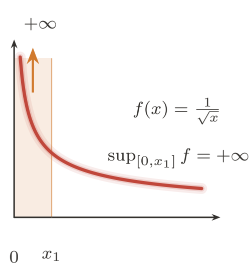


Figure 3. **Unbounded Integrand: Boundedness Is Not Optional**

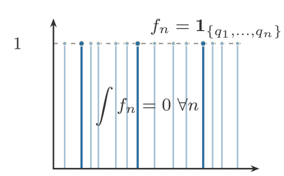


Figure 4. **Pointwise Limit:** $\lim\int \neq \int\lim$ **(Lebesgue's Cue)**

Figure 40.3: Failure modes of the Riemann family

Remark (What Darboux makes visible). For Thomae's function, Darboux sees the opposite picture. On every slab the infimum is 0, because every slab contains irrationals. The supremum can be large only near rationals with small denominator; above any fixed height threshold, there are only finitely many such rationals. A partition can isolate those finitely many tall rational spikes inside intervals of tiny total width, and the remaining upper rectangles are short. Thus the upper sum can be forced below any prescribed ε, while every lower sum is 0.

This is the practical gain: Riemann gives the correct definition, but Darboux gives the hand tool. Prove $U - L$ can be squeezed, and the Riemann value follows from the equivalence theorem.

Remark (The fork). Darboux exposes two separate defects:

Residual defect	Successor	Change
only ordinary length	Riemann–Stieltjes	$\Delta x_i \mapsto \Delta \alpha_i$
domain slabs see every bad set	Lebesgue / HK	measure bad sets or adapt fineness

Stieltjes is therefore a sideways generalization, not the final repair of Riemann's domain-partition limitation.

40.5 The Riemann–Stieltjes Integral

Geometric Intuition

Riemann and Darboux weigh each slab by ordinary length, Δx_i. Riemann–Stieltjes changes the ruler. A slab is weighed by how much an integrator α changes across it:

$$\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1}).$$

If $\alpha(x) = x$, ordinary Riemann integration returns. If α jumps, the integral sees a point mass.

Remark (Engineering reading). Stieltjes does not repair the discontinuity-set problem. It answers a different complaint: ordinary integration has only one ruler. By replacing Δx_i with $\Delta \alpha_i$, the integral can count regions with nonuniform weight and can encode discrete masses as jumps of α.

Definition (Total Variation)

Definition 40.5.1 (Total Variation)**.** The total variation of α on $[a, b]$ is

$$V_a^b(\alpha) = \sup_P \sum_i |\alpha(x_i) - \alpha(x_{i-1})|.$$

Definition (Bounded Variation)

Definition 40.5.2 (Bounded Variation)**.** The function α has *bounded variation* on $[a, b]$ if

$$V_a^b(\alpha) < \infty.$$

Geometrically, this says the ruler has finite total up-and-down travel.

Proposition (Monotone Implies BV)

Proposition 40.5.3 (Monotone Functions Have Bounded Variation)**.** *If α is monotone on $[a, b]$, then α has bounded variation.*

Remark (Why bounded variation is the honest ruler condition). If α is increasing, the increments telescope:

$$\sum_i \big(\alpha(x_i) - \alpha(x_{i-1})\big) = \alpha(b) - \alpha(a).$$

Bounded variation is the signed analogue of that finite ruler budget. It is what prevents the weighted sums from accumulating uncontrolled oscillatory ruler error.

Definition (Riemann–Stieltjes Integral)

Definition 40.5.4 (Riemann–Stieltjes Integral)**.** The Riemann–Stieltjes sum is

$$\sum_i f(\xi_i)\big(\alpha(x_i) - \alpha(x_{i-1})\big).$$

When these sums have a tag-independent limit as $\|P\| \to 0$, the limit is denoted $\int_a^b f\, d\alpha$.

Theorem (Continuous Integrand, BV Integrator)

Theorem 40.5.5 (Continuous Integrand and BV Integrator)**.** *If f is continuous on $[a, b]$ and α has bounded variation, then $\int_a^b f\, d\alpha$ exists.*

Theorem (Bilinearity)

Theorem 40.5.6 (Bilinearity of the Riemann–Stieltjes Integral)**.** *Whenever the displayed integrals exist,*

$$\int_a^b (\lambda f + \mu g)\, d\alpha = \lambda \int_a^b f\, d\alpha + \mu \int_a^b g\, d\alpha$$

and

$$\int_a^b f\, d(\lambda\alpha + \mu\beta) = \lambda \int_a^b f\, d\alpha + \mu \int_a^b f\, d\beta.$$

Theorem (Interval Additivity)

Theorem 40.5.7 (Interval Additivity for the Riemann–Stieltjes Integral)**.** *If $c \in [a, b]$ and the relevant Riemann–Stieltjes integrals exist, then*

$$\int_a^b f\, d\alpha = \int_a^c f\, d\alpha + \int_c^b f\, d\alpha.$$

Theorem (Integration by Parts)

Theorem 40.5.8 (Riemann–Stieltjes Integration by Parts). *If f and α are such that one of $\int_a^b f\,d\alpha$ and $\int_a^b \alpha\,df$ exists, then the other exists under the same Riemann–Stieltjes convention, and*

$$\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a).$$

Theorem (Smooth Integrator Reduction)

Theorem 40.5.9 (Reduction to Riemann Integration for a C^1 Integrator). *If $\alpha \in C^1([a,b])$ and f is Riemann-integrable on $[a,b]$, then*

$$\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx.$$

Theorem (Step Integrator)

Theorem 40.5.10 (Step Integrators Give Finite Sums). *If α is constant except for finitely many jumps at points c_k and f is continuous at each c_k, then*

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

Remark (Proof idea). The Cauchy common-refinement argument runs again. Uniform continuity makes $\omega_f(I)$ small on short slabs, and bounded variation controls $\sum_i |\Delta\alpha_i|$. The error has the form

$$\sum_i \omega_f(I_i)\,|\Delta\alpha_i|,$$

so small oscillation multiplied by finite total variation is small.

Remark (Sums as integrals). If α is a step function with jumps at points c_k, then the Stieltjes integral collapses to a weighted sum of the values $f(c_k)$ times the jumps. This is the conceptual doorway to probability: a cumulative distribution function is an integrator, and expectation is an integral against that integrator.

Remark (Step integrator calculation). Suppose α is constant except for finitely many jumps $\Delta\alpha(c_k) = \alpha(c_k+) - \alpha(c_k-)$. Any sufficiently fine partition has one tiny slab around each c_k carrying that jump, while all other slabs have $\Delta\alpha_i = 0$. If f is continuous at each c_k, then the tagged height on the k-th jump slab is forced toward $f(c_k)$, so

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

This is why a finite weighted sum is not a different species from an integral: it is a Stieltjes integral against a discrete ruler.

Remark (Shared-jump failure). The new pathology is real. On $[0, 1]$, let $c = 1/2$, let

$$\alpha(x) = \mathbf{1}_{[c,1]}(x), \qquad f(x) = \mathbf{1}_{[c,1]}(x).$$

The only nonzero $\Delta\alpha_i$ occurs on the slab containing c. If the tag for that slab is chosen left of c, its contribution is 0; if the tag is chosen at or right of c, its contribution is 1. This discrepancy does not shrink with the slab width, because the ruler has a whole unit of mass concentrated at the jump. Tag independence fails, so the Riemann–Stieltjes integral does not exist in this convention.

Remark (Bug report). Stieltjes is a sideways generalization, not the Lebesgue fix. Shared discontinuities of f and α can destroy tag independence, and the standard existence theorem leans on the finite ruler budget $V_a^b(\alpha) < \infty$. Brownian paths have infinite variation almost surely, so stochastic integration cannot be ordinary pathwise Riemann–Stieltjes integration.

Remark (Patch request). The next stochastic patch is not "make Stieltjes a little better." Brownian motion forces a different limiting mode: an L^2 or probabilistic limit rather than a pathwise Riemann–Stieltjes limit. That is the structural reason Ito integration is not just Stieltjes integration with a rougher integrator.

40.6 The Henstock–Kurzweil Integral

Where This Sits

Darboux exposed the Riemann disease before measure theory names it precisely: one uniform fineness parameter must serve the whole interval. There are two ways forward. Lebesgue will change what is measured. The gauge integrals keep the tagged-sum picture and change exactly one symbol: the constant fineness $\delta > 0$ becomes a positive function $\delta(x)$.

Remark (Sibling cure). Henstock–Kurzweil belongs here, before McShane and the final measure-zero diagnosis, because it is still visibly a Riemann-style tagged-sum integral. It patches the uniform-mesh defect without abandoning slabs, tags, or sums. McShane will modify the tag rule one more time, producing a gauge-sum version of Lebesgue integration.

Geometric Intuition

A gauge is adaptive mesh refinement written as analysis. It is tiny where the function needs tight local control and generous where the function is calm. A tagged slab is allowed only if it fits inside the local radius prescribed by its own tag.

Remark (The one-symbol upgrade). Riemann uses a positive number:

$$\exists \delta > 0.$$

Henstock–Kurzweil uses a positive function:

$$\exists \delta : [a, b] \to (0, \infty).$$

The mesh condition $\|P\| < \delta$ becomes the local condition that each slab is δ-fine at its tag.

Definition (Gauge)

Definition 40.6.1 (Gauge on an Interval)**.** A *gauge* on $[a, b]$ is a function

$$\delta : [a, b] \to (0, \infty).$$

Definition (Delta-Fine Tagged Partition)

Definition 40.6.2 (δ-Fine Tagged Partition)**.** A tagged partition (P, ξ) is *δ-fine* if

$$[x_{i-1}, x_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Definition (Henstock–Kurzweil Integral)

Definition 40.6.3 (Henstock–Kurzweil Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Henstock–Kurzweil integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a, b] \to (0, \infty)\ \forall (P, \xi)\ \big((P, \xi) \text{ is } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon\big).$$

The value is denoted (HK) $\int_a^b f$.

Lemma (Cousin)

Lemma 40.6.4 (Cousin's Lemma)**.** *For every gauge δ on $[a, b]$, there exists a δ-fine tagged partition of $[a, b]$.*

Remark (Non-vacuity). Cousin's Lemma prevents the definition from being empty. No matter how wildly the gauge varies, compactness of $[a, b]$ guarantees that at least one tagged partition respects all the local step-size demands.

Theorem (Riemann Sits Inside Henstock–Kurzweil)

Theorem 40.6.5 (Riemann Integrable Implies Henstock–Kurzweil Integrable)**.** *If f is Riemann-integrable on $[a, b]$, then f is Henstock–Kurzweil integrable on $[a, b]$, and the two integrals have the same value.*

Remark (Constant gauges). Riemann is the special case where the gauge is constant. If a Riemann proof hands over one mesh tolerance δ_0, the gauge proof uses $\delta(x) = \delta_0$ for every x.

Lemma (Straddle)

Lemma 40.6.6 (Straddle Lemma)**.** *If F is differentiable at ξ with $F'(\xi) = f(\xi)$, then for every $\varepsilon > 0$ there is $\delta(\xi) > 0$ such that whenever $u \le \xi \le v$, $u, v \in (\xi - \delta(\xi), \xi + \delta(\xi))$,*

$$|F(v) - F(u) - f(\xi)(v - u)| \le \varepsilon(v - u).$$

Theorem (FTC for the Gauge Integral)

Theorem 40.6.7 (Fundamental Theorem for the Henstock–Kurzweil Integral)**.** *If $F : [a,b] \to \mathbb{R}$ is differentiable at every point and $F' = f$, then f is Henstock–Kurzweil integrable and*

$$(\mathrm{HK}) \int_a^b f = F(b) - F(a).$$

Theorem (Continuous Functions Are Henstock–Kurzweil Integrable)

Theorem 40.6.8 (Continuous Functions Are Henstock–Kurzweil Integrable)**.** *Every continuous function $f : [a,b] \to \mathbb{R}$ is Henstock–Kurzweil integrable.*

Remark (What it buys)*.* On a compact interval,

$$\mathcal{R}[a,b] \subsetneq \mathcal{M}[a,b] = \mathcal{L}[a,b] \subsetneq \mathcal{HK}[a,b].$$

Figure 40.4 records this inclusion lattice and the standard witness at each frontier. The first strict inclusion is witnessed by functions like $\mathbf{1}_{\mathbb{Q}}$, which are Lebesgue-integrable but not Riemann integrable. The second is witnessed by derivatives such as that of $F(x) = x^2 \sin(1/x^2)$ with $F(0) = 0$: the derivative is HK-integrable by the gauge FTC even when its absolute value is not Lebesgue-integrable.

This inclusion statement anticipates the next volume's Lebesgue theory. The present chapter has not built Lebesgue integration yet; it has only isolated the Riemann-side machinery, the gauge machinery, and the reason a measure-theoretic successor is needed.

The Integration Lattice

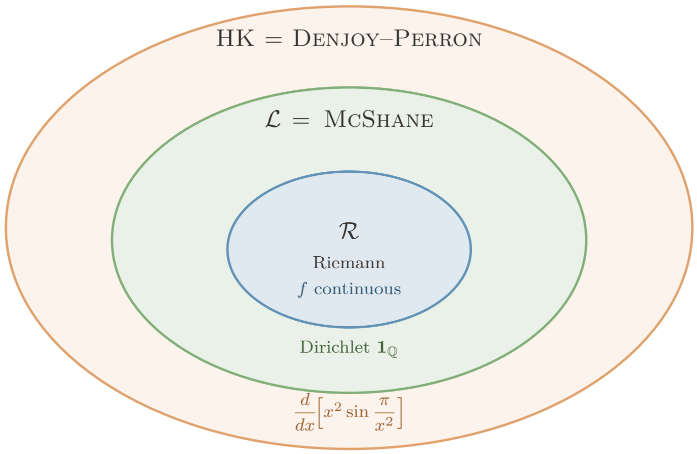


Figure. The Integration Lattice: each frontier is crossed by exactly one witness.

$\mathcal{R} \subsetneq \mathcal{L} = \text{McShane} \subsetneq \mathcal{HK}$, with the strict gaps witnessed by:

- $\mathcal{L} \setminus \mathcal{R}$: $\mathbf{1}_{\mathbb{Q}}$ — Lebesgue (and McShane) integrable, $\int = 0$; not Riemann.
- $\mathcal{HK} \setminus \mathcal{L}$: F' for $F(x) = x^2 \sin(\pi/x^2)$, $F(0) = 0$ — HK-integrable ($\int_0^1 F' = F(1) - F(0) = 0$) but $\int_0^1 |F'| = \infty$, so not Lebesgue.

Engine: $\mathcal{L}$ = McShane uses *free* tags (absolute integration); $\mathcal{HK}$ uses *tethered* tags, which preserve cancellation — the source of the outermost ring, and the reason HK integrates *every* everywhere-derivative.

Figure 40.4: The inclusion lattice for Riemann, Lebesgue–McShane, and Henstock–Kurzweil integration

Remark (Killer example). Let $F(x) = x^2 \sin(1/x^2)$ for $x \neq 0$, and set $F(0) = 0$. Then F is differentiable at 0, with $F'(0) = 0$, while for $x \neq 0$,

$$F'(x) = 2x \sin(1/x^2) - \frac{2}{x} \cos(1/x^2).$$

The oscillatory term has nonintegrable absolute size near 0, so this derivative is the standard witness that HK reaches beyond Lebesgue while the FTC remains exact.

Remark (HK's own bug report). Henstock–Kurzweil is not simply "the best integral." Lebesgue has stronger limit theorems, a complete L^1 normed-space home, and a clean abstract measure-space setting. HK is the jewel for the one-dimensional FTC and conditional accumulation; Lebesgue and Ito remain the engines for probability and stochastic integration.

Remark (Computational bridge). A gauge $\delta(x)$ is a spatially varying step-size field. Lipschitz control is the case where a constant gauge is enough; singular or non-Lipschitz regions demand a genuinely varying gauge. Cousin's Lemma is the compactness guarantee that a valid adaptive partition exists.

Remark (One-page summary).

Integral	Patch	Remaining defect
Cauchy	left rectangles settle for continuous f	privileged tag
Riemann	all tags must agree	hard to verify
Darboux	sup/inf squeeze	same class as Riemann
Stieltjes	change the ruler	needs BV; shared jumps
Henstock–Kurzweil	local gauge fineness	weaker limit machinery
McShane	free gauge tags	same class as Lebesgue
Lebesgue	measure level sets	abstract machinery required

The thread is the right-hand column: every integral is a direct response to a specific defect.

40.7 The McShane Integral

Where This Sits

McShane is the gauge integral that looks almost indistinguishable from Henstock–Kurzweil until one notices where the tags are allowed to live. HK keeps each tag inside its own slab. McShane lets a tag control a slab from outside it, as long as the slab lies inside the tag's gauge ball.

Remark (The tiny syntactic change). Henstock–Kurzweil says:

$$\xi_i \in [x_{i-1}, x_i] \quad \text{and} \quad [x_{i-1}, x_i] \subset (\xi_i - \delta(\xi_i), \xi_i + \delta(\xi_i)).$$

McShane drops only the first requirement. The interval still has to be locally small around its tag, but the tag need not lie in the interval it controls. That small freedom is decisive: it prevents the conditional cancellation that lets HK integrate every derivative.

Remark (HK versus McShane at a glance).

Integral	Gauge	Tag position	Class on $[a, b]$
Henstock–Kurzweil	$\delta(x)$	$\xi_i \in I_i$	$\mathcal{HK}$
McShane	$\delta(x)$	$\xi_i \in [a, b]$ and $I_i \subset B(\xi_i, \delta(\xi_i))$	$\mathcal{L}$

Both use a variable gauge. The difference is not the step-size field; it is whether the tag must sit inside the interval whose height it supplies. Figure 40.5 shows this tethered-versus-free distinction directly.

Two Ways to Tag

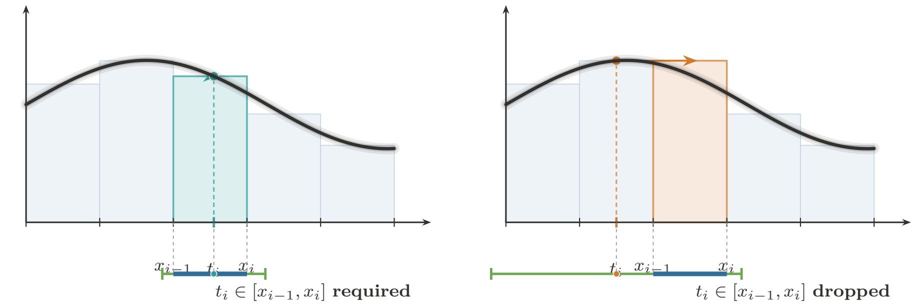


Figure 1. **Henstock–Kurzweil: Tag Tethered to Its Cell**

Figure 2. **McShane: The Tag Roams Free**

Same integrand, same gauge condition $I_i \subseteq (t_i - \delta(t_i),\, t_i + \delta(t_i))$. The *only* difference is whether the tag must sit in its own cell.

HK keeps the tether $\Rightarrow$ conditional convergence survives $\Rightarrow$ $\mathcal{HK}$ = Denjoy–Perron (nonabsolute). McShane cuts it $\Rightarrow$ only absolute integrals remain $\Rightarrow$ McShane $= \mathcal{L}^1$ (Lebesgue).

Figure 40.5: Henstock–Kurzweil versus McShane fineness

Geometric Intuition

In HK, a tag is a sample point inside the slab. In McShane, a tag is more like a local address that authorizes nearby slabs. The gauge still says how far the tag's influence reaches, but the rectangle height may be chosen from a point near the slab rather than inside it. This makes the integral less forgiving about sign-changing singular behavior and much closer to Lebesgue's absolute notion of size.

Definition (McShane-Fine Tagged Partition)

Definition 40.7.1 (McShane-Fine Tagged Partition)**.** Let $\delta : [a,b] \to (0,\infty)$ be a gauge. A finite tagged partition $\{([u_i,v_i],\xi_i)\}_{i=1}^n$ is *McShane δ-fine* if the intervals $[u_i,v_i]$ are non-overlapping, cover $[a,b]$, each $\xi_i \in [a,b]$, and

$$[u_i,v_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Unlike an HK tagged partition, the tag ξ_i is not required to lie in $[u_i,v_i]$.

Remark (Untethered tag). Figure 40.6 isolates the McShane move: a tag may sit outside the cell whose height it supplies, provided the cell lies inside the tag's gauge ball. The ruler is still Δx_i; the change is entirely in the admissible tag geometry.

The McShane Integral

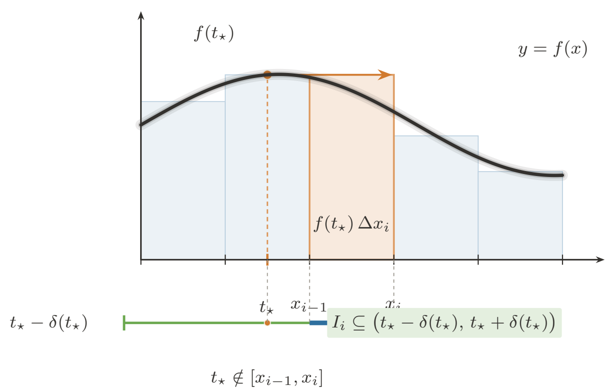


Figure. The McShane Integral — the untethered tag: a δ-fine partition whose tags roam free.

Rule (the only change from Henstock–Kurzweil). A tagged partition is *McShane δ-fine* iff $\forall i\ \big(I_i \subseteq (t_i - \delta(t_i),\, t_i + \delta(t_i))\big)$, with tags $t_i \in [a,b]$ *arbitrary* — the clause $t_i \in I_i$ is dropped. The sum is still $S = \sum_i f(t_i)\,\Delta x_i$, and

$$f \text{ McShane integrable} \iff f \in L^1[a,b], \qquad \text{with equal values.}$$

Where it sits: $\mathcal{R} \subsetneq \mathcal{L} = \text{McShane} \subsetneq \mathcal{HK}$ (Riemann $\subsetneq$ Lebesgue = McShane $\subsetneq$ Henstock–Kurzweil).

Figure 40.6: The McShane integral and the untethered tag

Definition (McShane Integral)

Definition 40.7.2 (McShane Integral)**.** A function $f : [a,b] \to \mathbb{R}$ is *McShane integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a,b] \to (0,\infty)\ \forall (P,\xi)\ \big((P,\xi) \text{ is McShane } \delta\text{-fine} \Rightarrow |S(f,P,\xi) - A| < \varepsilon\big),$$

where

$$S(f,P,\xi) = \sum_i f(\xi_i)(v_i - u_i).$$

Remark (Relation to HK)*.* Every McShane-fine partition is not necessarily HK-fine, because its tags may float outside their intervals. But every HK-fine partition is McShane-fine. Therefore McShane integrability is a stronger demand: the sums must survive all the usual gauge-fine tests and also the extra free-tag tests.

Theorem (McShane Sits Between Riemann and HK)

Theorem 40.7.3 (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable)**.** *On a compact interval,*

$$\mathcal{R}[a,b] \subseteq \mathcal{M}[a,b] \subseteq \mathcal{HK}[a,b],$$

and the integral values agree whenever a function belongs to the smaller class.

Theorem (McShane Equals Lebesgue)

Theorem 40.7.4 (McShane Integrability Equals Lebesgue Integrability)**.** *For real-valued functions on a compact interval,*

$$\mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b],$$

and the McShane and Lebesgue integral values agree.

Remark (The Lebesgue connection). For real-valued functions on a compact interval, the McShane integral recovers the Lebesgue integral. In practical terms,

$$\mathcal{M}[a,b] = \mathcal{L}[a,b]$$

with equal values. Thus McShane is not a second route beyond Lebesgue in the way HK is. It is a gauge-sum presentation of Lebesgue integration.

Remark (Comparison chain). Combining the Riemann inclusion, the McShane–Lebesgue identification, and the HK derivative witness gives

$$\mathcal{R}[a,b] \subsetneq \mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b] \subsetneq \mathcal{HK}[a,b].$$

The first strict inclusion is the old Riemann failure at dense null discontinuities; the second is the HK recovery of derivatives whose absolute size is not Lebesgue integrable.

Remark (Forward dependency). The proof that every Riemann-integrable function is McShane-integrable uses the measure-zero diagnosis in the next section: Riemann integrability means the discontinuity set is null. The exposition places McShane first because it belongs syntactically with the gauge integrals; the proof ledger records that it imports the Lebesgue criterion.

Remark (Why free tags force Lebesgue behavior). The HK derivative example

$$F'(x) = 2x\sin(1/x^2) - \frac{2}{x}\cos(1/x^2)$$

is conditionally integrable by HK because the tags remain inside the tiny slabs where the derivative is being locally linearized. McShane's free tags can sample nearby oscillatory peaks in a way that destroys this conditional balance. What survives that stronger test is exactly the Lebesgue-integrable, absolutely controlled part of the gauge world.

Remark (What it buys). McShane gives a tagged-sum construction of the Lebesgue integral without starting from simple functions or sigma-algebras. Conceptually, that is its job in this chapter: it shows that adaptive Riemann-style sums can reach the Lebesgue class, while HK shows that keeping tags inside slabs reaches beyond Lebesgue to conditionally integrable derivatives.

Remark (Bug report). McShane is elegant but not the terminal fix for this chapter's story. If one wants a tagged-sum definition of the Lebesgue integral, it is excellent. If one wants the perfect one-dimensional Fundamental Theorem of Calculus, HK is strictly stronger. And if one wants probability, stochastic processes, and abstract spaces, the next volume still needs measure theory explicitly.

Remark (Handoff to measure zero). McShane reveals that gauge machinery and Lebesgue size are compatible. The next section names the exact size condition already controlling the older Riemann–Darboux family: the discontinuity set must have measure zero.

40.8 Measure Zero and the Lebesgue Criterion

Geometric Intuition

Darboux integrability fails where upper and lower staircases refuse to meet. That refusal is local oscillation. Thus the exact Riemann question becomes: how large is the set where f is discontinuous?

Remark (Engineering reading). This final section is not another integral. It is the diagnostic scan of the entire Riemann–Darboux family, placed after Henstock–Kurzweil and McShane so the chapter ends by naming the exact obstruction that opens the Lebesgue volume. Once Darboux has made the gap visible, the remaining question is whether the bad part of the domain is small enough to cover by slabs of negligible total length.

Definition (Measure Zero)

Definition 40.8.1 (Measure Zero)**.** A set $E \subseteq \mathbb{R}$ has *measure zero* if for every $\varepsilon > 0$ there are open intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \qquad \sum_{k=1}^{\infty} |I_k| < \varepsilon.$$

Definition (Point Oscillation)

Definition 40.8.2 (Point Oscillation)**.** For bounded f on $[a, b]$, define

$$\omega_f(x) = \inf_{\delta>0} \omega_f\big((x-\delta, x+\delta) \cap [a, b]\big).$$

Then f is continuous at x exactly when $\omega_f(x) = 0$.

Remark (Oscillation levels). The sets

$$D_\eta = \{x : \omega_f(x) \geq \eta\}$$

separate the bad set by severity. The full discontinuity set is

$$D = \bigcup_{k=1}^{\infty} D_{1/k}.$$

The proof of the criterion works by showing that positive-size high oscillation forces a permanent Darboux gap, while null high-oscillation sets can be covered by intervals of tiny total length.

Theorem (Lebesgue Criterion)

Theorem 40.8.3 (Lebesgue Criterion for Riemann Integrability)**.** *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if its discontinuity set*

$$D = \{x \in [a, b] : \omega_f(x) > 0\}$$

has measure zero.

Remark (Terminal diagnosis). Riemann integration works precisely when the set of bad points is invisible to length. The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ is discontinuous everywhere, so $D = [0, 1]$, not a null set. The Lebesgue patch is to stop partitioning the domain into slabs and instead measure the sets where the function takes given ranges of values. Henstock–Kurzweil keeps tagged sums, but replaces uniform fineness by local fineness; McShane shows that a slightly stricter gauge rule recovers the Lebesgue class itself. The pathology plate in Figure 40.3 is the visual companion to this diagnosis.

Remark (Lebesgue handoff). The slogan is precise: Riemann partitions the domain, so every slab that meets a dense bad set still sees the bad behavior. Lebesgue partitions the range: it groups points by height and measures the set carrying each height. For $\mathbf{1}_{\mathbb{Q}}$, the height 1 lives on a null set and therefore contributes nothing.

Appendix: Pathologies of the Riemann Integral

Remark (Source placement). This appendix is the converted support note from `integration-failures.tex`. Its standalone theorem machinery and document preamble have been removed; the material is kept as commentary so it does not duplicate the formal Lebesgue criterion in Section 40.8.3.

Remark (The one theorem behind the failures). The Riemann failures in Figure 40.3 are not a bag of examples. They are tests against the two hypotheses in the Lebesgue criterion: boundedness, and a discontinuity set of measure zero. Dirichlet fails because its discontinuity set is all of $[0, 1]$. Unbounded examples fail before the criterion even begins. Pointwise limits expose a different weakness: the Riemann class is not stable under the limit operations that Lebesgue integration was built to support.

Remark (Thomae as the diagnostic example)*.* Thomae's function is the useful warning. Its discontinuities are dense, but they are countable and therefore null. Darboux can isolate the finitely many large spikes above any fixed threshold inside intervals of tiny total length; outside those intervals, the upper sums are uniformly small. Thus density of the bad set is not the obstruction. Size is.

Remark (Through-line)*.* The atlas failure note should be read as the bridge from Darboux to Measure Zero: $U(f,P) - L(f,P)$ is local oscillation times length, so the only bad oscillation that matters is bad oscillation carried by a set of positive length. That observation closes the Riemann family and points directly to Lebesgue's range-partition construction.

Appendix: The Gauge Integrals

Remark (Source placement)*.* This appendix is the converted support note from `gauge-integrals.tex`. The standalone preamble, local theorem declarations, and duplicate figure includes have been removed; the shared figures are the pinned figures already included in Sections 1.6 and 1.7.

Remark (The gauge framework)*.* The gauge integrals replace Riemann's single global mesh tolerance by a local step-size controller $\delta : [a,b] \to (0,\infty)$. Henstock–Kurzweil and McShane then differ by one geometric clause. HK requires the tag to sit inside the interval it controls. McShane drops that tether and allows the tag to range over $[a,b]$, provided the interval lies inside the tag's gauge ball. The contrast is visualized in Figures 40.5 and 40.6.

Remark (Free tags and Lebesgue)*.* McShane's free tags force absolute, Lebesgue-style behavior. Since a tag can authorize nearby intervals without lying inside them, the sums can test whether the mass assigned to height bands is genuinely controlled. On compact intervals this recovers the Lebesgue integral: $\mathcal{M}[a,b] = \mathcal{L}[a,b]$, with equal values.

Remark (Tethered tags and the FTC)*.* HK keeps the tag tethered to its interval, which preserves local cancellation for derivatives. That is why the HK integral has the exact one-dimensional FTC: if F is differentiable everywhere, then F' is HK-integrable and (HK) $\int_a^b F' = F(b) - F(a)$, even when F' is not Lebesgue integrable.

Remark (Lattice)*.* The resulting compact-interval lattice is

$$\mathcal{R}[a,b] \subsetneq \mathcal{L}[a,b] = \mathcal{M}[a,b] \subsetneq \mathcal{HK}[a,b],$$

as recorded in Figure 40.4. This appendix is kept after the main chapter because the book has not yet built the measure theory needed to prove the McShane–Lebesgue identification.

Part IV

Volume IV: Mathematical Spaces

Mathematical Spaces

Chapter 41

Mathematical Spaces

SERIES
From Cantor to Ito

VOLUME IV
Mathematical Spaces

BOOK I
Mathematical Spaces

CHAPTER 1
Mathematical Spaces

41.1 Spaces as Structured Arenas

Definition — Space

Definition 41.1.1 (Space)**.** A **space** is a nonempty set.

Remark (Standard quantified statement)*.*

$$X \text{ is a space} \iff X \neq \varnothing.$$

Remark (Interpretation)*.* At the base level, the word "space" records that there is an arena of points under discussion. No distance, topology, operation, or measure has yet been chosen. Later phrases such as metric space, topological space, and measure space add structure to this underlying nonempty set.

Remark (Historical note)*.* This convention appears as Definition 2.1 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Exposition)*.* A mathematical space is an arena in which a particular kind of mathematics can be performed. At the base there is a set of objects. Added structure then specifies the ground rules: which objects can be compared, which subsets are distinguished, which operations are allowed, which limits make sense, or which quantities can be measured. The same underlying set can support many different spaces because different structures make different questions meaningful.

Remark (Structural Pattern)*.* The general pattern is

$$\text{space} = \text{underlying set} + \text{chosen structure}.$$

A metric space chooses a distance function. A topological space chooses a collection of open sets. A measurable space chooses a σ-algebra of measurable sets, and a measure space adds a measure to that measurable structure. Algebraic spaces choose operations and laws. Each choice controls which statements are legitimate inside that setting.

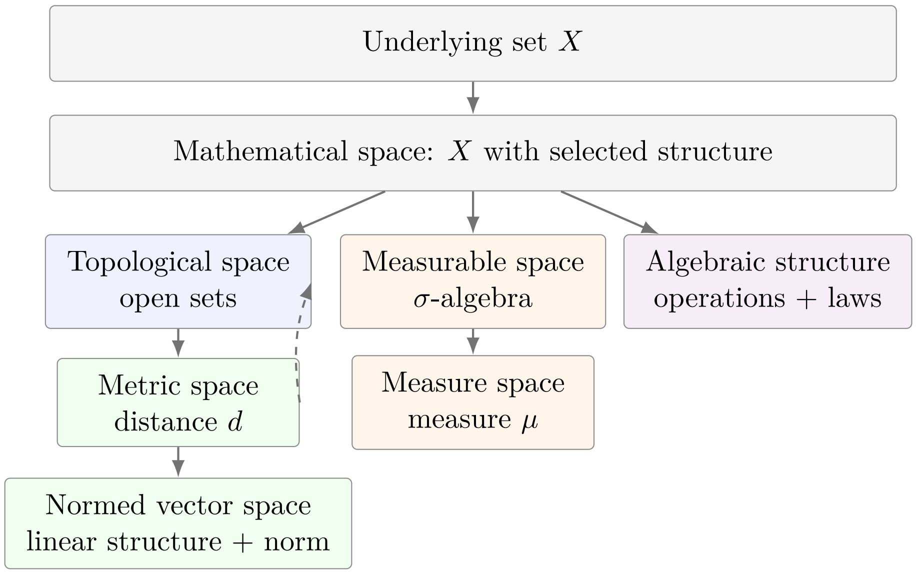
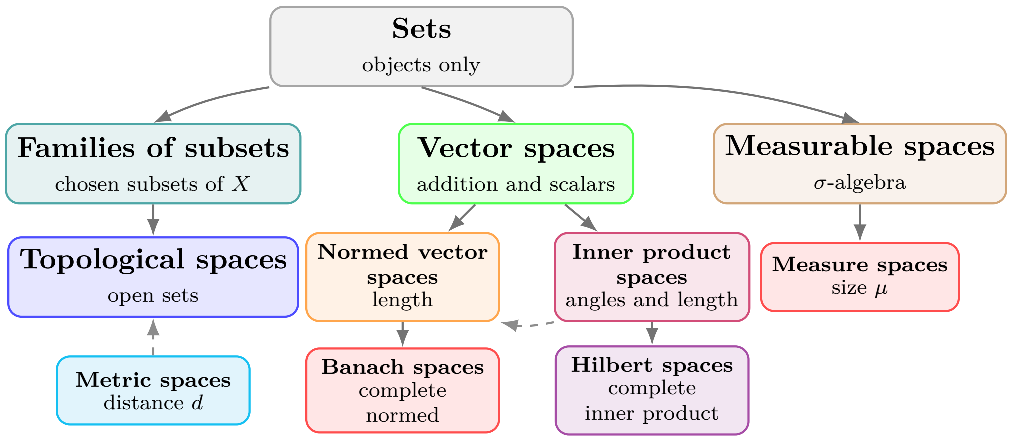


Figure 41.1: Mathematical spaces are built by adding structure to an underlying set. Some structures refine others: a metric induces a topology, while a measure requires measurable sets.

Remark (Interpretation). The nesting in Figure 41.1 should be read as an increase in available structure, not as a claim that every subject lies on a single chain. Metric spaces sit naturally over topology because every metric determines open balls and hence open sets. Measure spaces follow a different branch: they organize size rather than nearness. The common feature is the same: mathematical work begins only after the arena and its rules have been specified.

Remark (Exposition). The hierarchy in Figure 41.2 is a guide to structure, not a universal taxonomy. Some spaces are formed by adding data to a set. Others are formed by adding compatibility requirements to an already structured space. The following table records the construction pattern that will be used throughout this volume.



The branches record added structure. A topology selects open sets, a metric supplies distance and induces open sets, linear spaces add operations, and measure spaces add measurable sets together with size.

Figure 41.2: One common route from sets to structured spaces. A topology begins with selected open subsets of a set. A metric gives a distance and thereby generates open balls, so metric spaces naturally carry topologies. Linear, normed, inner product, Banach, and Hilbert spaces add further compatible structure.

Remark (Structure Table).

Space	**Comes from**	**Additional structure or rules**
Set	Primitive arena	A collection of objects; no nearness, size, or operation is specified.
Family of subsets	Set X	A selected collection $\mathcal{A} \subseteq \mathcal{P}(X)$.
Topological space	Set X	A topology $\mathcal{T}$: open sets containing $\varnothing$ and X, closed under arbitrary unions and finite intersections.
Metric space	Set X	A metric $d : X \times X \to [0, \infty)$ satisfying identity of indiscernibles, symmetry, and the triangle inequality; the metric induces open balls and hence a topology.
Vector space	Set V	Addition and scalar multiplication satisfying the vector space axioms over a chosen field.
Normed vector space	Vector space	A norm $\lVert\cdot\rVert$ measuring vector length, compatible with scalar multiplication and addition.
Inner product space	Vector space	An inner product $\langle \cdot, \cdot \rangle$ measuring angles and lengths; it induces a norm.
Banach space	Normed vector space	Completeness: every Cauchy sequence converges in the norm.
Hilbert space	Inner product space	Completeness with respect to the norm induced by the inner product.
Measurable space	Set X	A σ-algebra $\mathcal{M}$ of subsets regarded as measurable.
Measure space	Measurable space	A measure $\mu : \mathcal{M} \to [0, \infty]$ assigning size and satisfying countable additivity.

Remark (Structural Cards). The following cards use the structural presentation layer: each card records the inherited arena, the new data or laws, and the role of the resulting space. These are blueprint artifacts, not formal definitions; formal definitions remain authoritative when they are introduced.

Structural Card: Set

TAXONOMY

Primitive arena.

NEW AT

A nonempty domain X of objects.

AXIOMS / THEORY

No nearness, size, operation, or distinguished family of subsets is specified.

Structural Card: Family of Subsets

INHERITS

Set X.

NEW AT

A chosen collection $\mathcal{A} \subseteq \mathcal{P}(X)$.

TAXONOMY

Subset-selection structure. No closure law is imposed at this level.

Structural Card: Topological Space

INHERITS

Set X with a distinguished family of subsets.

NEW AT

A topology $\mathcal{T} \subseteq \mathcal{P}(X)$, read as the open subsets of X.

AXIOMS / THEORY

$\varnothing, X \in \mathcal{T}$; arbitrary unions of members of $\mathcal{T}$ lie in $\mathcal{T}$; finite intersections of members of $\mathcal{T}$ lie in $\mathcal{T}$.

TAXONOMY

Open-set structure for continuity, convergence, interior, closure, and neighborhood arguments.

Structural Card: Metric Space

INHERITS

Set X.

NEW AT

A distance function

$$d : X \times X \to [0, \infty).$$

AXIOMS / THEORY

Identity of indiscernibles, symmetry, and the triangle inequality.

USES-A

Open balls $B(x, r) = \{y \in X : d(x, y) < r\}$.

TAXONOMY

Distance structure. The metric induces a topology by declaring sets open when they contain a metric ball around each of their points.

Structural Card: Vector Space

INHERITS

Set V and a scalar field $\mathbb{F}$.

NEW AT

Vector addition $+ : V \times V \to V$, scalar multiplication $\cdot : \mathbb{F} \times V \to V$, and a zero vector $0 \in V$.

AXIOMS / THEORY

$(V, +, 0)$ is an abelian group; scalar multiplication is unital and associative; scalar multiplication distributes over vector addition and field addition.

TAXONOMY

Linear structure. The space supports linear combinations before any length, angle, or limit structure is chosen.

Structural Card: Normed Vector Space

INHERITS

Vector space V over $\mathbb{F}$.

NEW AT

A norm $\|\cdot\| : V \to [0, \infty)$.

AXIOMS / THEORY

$\|v\| = 0$ if and only if $v = 0$; $\|\alpha v\| = |\alpha| \, \|v\|$; $\|u + v\| \leq \|u\| + \|v\|$.

USES-A

The induced metric $d(u, v) = \|u - v\|$.

TAXONOMY

Linear metric structure. Algebraic operations and metric notions become compatible.

Structural Card: Inner Product Space

INHERITS

Real or complex vector space V.

NEW AT

An inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$.

AXIOMS / THEORY

Positive definiteness, conjugate symmetry, and linearity in the chosen argument.

USES-A

The induced norm $\|v\| = \sqrt{\langle v, v \rangle}$.

TAXONOMY

Linear geometry. Length, angle, and orthogonality are available, and the metric structure is derived from the inner product.

Structural Card: Banach Space

INHERITS

Normed vector space $(V, \|\cdot\|)$.

NEW AT

Completeness with respect to the norm-induced metric.

AXIOMS / THEORY

Every Cauchy sequence in V converges to a point of V in the norm.

TAXONOMY

Complete normed linear space. Approximation by Cauchy sequences remains inside the space.

Structural Card: Hilbert Space

INHERITS

Inner product space H.

NEW AT

Completeness with respect to the norm induced by the inner product.

AXIOMS / THEORY

Every Cauchy sequence in the induced norm converges to a point of H.

TAXONOMY

Complete inner product space. Orthogonality and projection arguments take place in a complete metric environment.

Structural Card: Measurable Space

INHERITS

Set X with a distinguished family of subsets.

NEW AT

A σ-algebra $\mathcal{M} \subseteq \mathcal{P}(X)$, read as the measurable subsets of X.

AXIOMS / THEORY

$X \in \mathcal{M}$; complements of measurable sets are measurable; countable unions of measurable sets are measurable.

TAXONOMY

Countably closed subset-selection structure. It specifies which subsets are legal inputs for size assignments.

Structural Card: Measure Space

INHERITS

Measurable space $(X, \mathcal{M})$.

NEW AT

A measure $\mu : \mathcal{M} \to [0, \infty]$.

AXIOMS / THEORY

$\mu(\varnothing) = 0$, and μ is countably additive on pairwise disjoint measurable sets:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

TAXONOMY

Quantitative measurable structure. The σ-algebra determines the admissible subsets, and the measure assigns their size.

Chapter 42

Metric Spaces

42.1 Metric Spaces

42.2 Definitions

Definition 42.2.1 (Metric Space)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ *(positive definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

A *metric space* is a pair (X, d), where X is a nonempty set and d is a metric on X.

Remark (Standard quantified statement)*.*

$$\begin{aligned} \mathrm{Metric}(X, d) \iff\ & X \neq \varnothing \ \wedge\ d : X \times X \to \mathbb{R}_{\geq 0} \\ & \wedge \forall x, y, z \in X\colon d(x, y) \geq 0 \wedge \big(d(x, y) = 0 \iff x = y\big) \\ & \wedge d(x, y) = d(y, x) \wedge d(x, y) \leq d(x, z) + d(z, y). \end{aligned}$$

Remark (Interpretation)*.* A metric turns a set into a distance arena. Positive definiteness says that distance is never negative and that zero distance identifies exactly one point. Symmetry says that distance does not depend on direction. The triangle inequality says that moving through an intermediate point cannot make the direct distance larger than the two-step route.

Remark (Historical note)*.* This convention appears as Definition 2.2 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Definition 42.2.2 (Pseudo-Metric)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *pseudo-metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, x) = 0$ *(positive semi-definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

Remark (Standard quantified statement)*.*

$$\begin{aligned} \mathrm{PseudoMetric}(X, d) \iff & X \neq \varnothing \ \wedge \ d : X \times X \to \mathbb{R}_{\geq 0} \\ & \wedge \forall x, y, z \in X \colon d(x, y) \geq 0 \wedge d(x, x) = 0 \\ & \wedge d(x, y) = d(y, x) \wedge d(x, y) \leq d(x, z) + d(z, y). \end{aligned}$$

Remark (Interpretation)*.* A pseudo-metric keeps the distance-like laws of nonnegativity, symmetry, and the triangle inequality, but it weakens positive definiteness. Distinct points may have distance 0. Thus a pseudo-metric measures separation only up to possible zero-distance identifications.

Remark (Historical note)*.* This convention appears as Definition 2.6 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Lemma

Lemma 42.2.3 (Metrics are Pseudo-Metrics)**.** *Every metric on a nonempty set X is a pseudo-metric on X. The converse is false: there exist pseudo-metrics that are not metrics.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\forall X \, \forall d \colon \ \mathrm{Metric}(X, d) \implies \mathrm{PseudoMetric}(X, d), \\ &\exists X \, \exists d \colon \ \mathrm{PseudoMetric}(X, d) \wedge \neg \, \mathrm{Metric}(X, d). \end{aligned}$$

Remark (Interpretation)*.* The implication is immediate because the metric axioms contain all pseudo-metric axioms. The only difference is the zero-distance law: a metric forces $d(x, y) = 0$ exactly when $x = y$, while a pseudo-metric only requires $d(x, x) = 0$.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Pseudo-Metric

42.3 Metrics

Remark (Exposition). A metric is the distance function itself. A metric space is the set together with that function. This section records standard metric functions and then states, one by one, that each function satisfies the metric axioms.

42.3.1 Norms Induce Metrics

Definition — Norm

Definition 42.3.1 (Norm)**.** Let V be a real vector space. A *norm* on V is a function

$$\|\cdot\| : V \to \mathbb{R}_{\geq 0}$$

such that for all $x, y \in V$ and all $\lambda \in \mathbb{R}$,

1. $\|x\| = 0$ if and only if $x = 0$ *(positive definiteness)*;
2. $\|\lambda x\| = |\lambda|\,\|x\|$ *(absolute homogeneity)*;
3. $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality)*.

Remark (Standard quantified statement).

$$\begin{aligned}\mathrm{Norm}(V, \|\cdot\|) \iff\ & \|\cdot\| : V \to \mathbb{R}_{\geq 0}\\ & \wedge \forall x \in V\colon \|x\| = 0 \iff x = 0\\ & \wedge \forall \lambda \in \mathbb{R}\ \forall x \in V\colon \|\lambda x\| = |\lambda|\,\|x\|\\ & \wedge \forall x, y \in V\colon \|x + y\| \leq \|x\| + \|y\|.\end{aligned}$$

Remark (Interpretation). A norm measures the size of one vector. The metric induced by a norm measures the size of the displacement vector $x - y$. Thus a norm is a length structure, and the induced metric is the corresponding distance structure.

Theorem

Theorem 42.3.2 (A Norm Induces a Metric)**.** *Let V be a real vector space and let $\|\cdot\|$ be a norm on V. Define*

$$d_{\|\cdot\|} : V \times V \to \mathbb{R}_{\geq 0}, \qquad d_{\|\cdot\|}(x, y) := \|x - y\|.$$

Then $d_{\|\cdot\|}$ is a metric on V.

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Norm}(V, \|\cdot\|) \Longrightarrow \mathrm{Metric}(V, d_{\|\cdot\|}), \qquad d_{\|\cdot\|}(x, y) = \|x - y\|.$$

Remark (Interpretation). The metric axioms are exactly the norm axioms applied to displacement vectors. Zero distance says the displacement $x - y$ has zero norm, hence is the zero vector. Symmetry follows from absolute homogeneity with scalar -1. The triangle inequality follows by writing $x - y = (x - z) + (z - y)$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Norm

Definition — Discrete Metric

Definition 42.3.3 (Discrete Metric)**.** Let X be a nonempty set. The *discrete metric* on X is the function

$$d_{\text{disc}} : X \times X \to \mathbb{R}_{\geq 0}$$

defined by

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Remark (Standard quantified statement).

$$\begin{aligned} &X \neq \varnothing, && d_{\text{disc}} : X \times X \to \mathbb{R}_{\geq 0}, \\ &\forall x, y \in X\colon && d_{\text{disc}}(x, y) = 0 \iff x = y, \\ &\forall x, y \in X\colon && x \neq y \iff d_{\text{disc}}(x, y) = 1. \end{aligned}$$

Remark (Interpretation). The discrete metric treats equality as the only closeness relation. Distinct points are all placed at the same positive distance from each other.

Proposition

Proposition 42.3.4 (The Discrete Distance is a Metric)**.** *For every nonempty set X, the discrete distance d_{disc} is a metric on X.*
Go to proof.

Remark (Standard quantified statement).

$$X \neq \varnothing \implies \text{Metric}(X, d_{\text{disc}}).$$

Remark (Interpretation). The only point at distance 0 from x is x itself. Symmetry is built into the two-case definition, and the triangle inequality is automatic because the only possible distances are 0 and 1.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Discrete Metric

Definition — Euclidean Distance on the Real Line

Definition 42.3.5 (Euclidean Distance on $\mathbb{R}$)**.** The *Euclidean distance* on $\mathbb{R}$ is the function

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}, \qquad d_{\mathbb{R}}(x, y) = |x - y|.$$

Remark (Standard quantified statement)*.*

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}, \qquad \forall x, y \in \mathbb{R}: \quad d_{\mathbb{R}}(x, y) = |x - y|.$$

Remark (Interpretation)*.* The real-line distance records absolute displacement between two real numbers. It forgets direction and keeps only separation.

Proposition

Proposition 42.3.6 (The Real-Line Euclidean Distance is a Metric)**.** *The function $d_{\mathbb{R}}(x, y) = |x - y|$ is a metric on $\mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\mathrm{Metric}(\mathbb{R}, d_{\mathbb{R}}).$$

Remark (Interpretation)*.* This proposition says that the usual absolute-value distance satisfies the abstract metric axioms.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}$

Definition — Euclidean Space

Definition 42.3.7 (Euclidean Space)**.** For $n \in \mathbb{N}$, the *n-dimensional Euclidean space* is $\mathbb{R}^n$ equipped with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Remark (Standard quantified statement)*.* For each $n \in \mathbb{N}$, $\mathrm{EuclideanSpace}(n)$ denotes $\mathbb{R}^n$ together with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Remark (Interpretation)*.* Euclidean space is the coordinate model of ordinary distance geometry.

Remark (Historical note)*.* This convention is the setting for the Euclidean examples in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Definition — Euclidean Distance on $\mathbb{R}^n$

Definition 42.3.8 (Euclidean Distance on $\mathbb{R}^n$)**.** For $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$, the *Euclidean distance* is the function

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_2(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Remark (Standard quantified statement).

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_2(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

Remark (Interpretation). The Euclidean distance measures the length of the coordinate difference vector $x - y$.

Theorem

Theorem 42.3.9 (Cauchy–Schwarz Inequality)**.** *For every $x, y \in \mathbb{R}^n$,*

$$|x \cdot y| \leq |x|\,|y|.$$

Equality holds if and only if x and y are linearly dependent.

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}^n\colon \quad |x \cdot y| \leq |x|\,|y|.$$

Remark (Interpretation). Cauchy–Schwarz controls the dot product by the lengths of the two vectors. It is the algebraic estimate behind Euclidean triangle inequalities.

Remark (Dependencies).

- Definition: Euclidean Space
- Definition: Euclidean Distance on $\mathbb{R}^n$

Corollary

Corollary 42.3.10 (Minkowski's Inequality)**.** *For every $x, y \in \mathbb{R}^n$,*

$$|x + y| \leq |x| + |y|.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}^n\colon \quad |x + y| \leq |x| + |y|.$$

Remark (Interpretation). Minkowski's inequality is the triangle inequality for Euclidean length.

Remark (Dependencies).

- Theorem: Cauchy–Schwarz Inequality
- Definition: Euclidean Distance on $\mathbb{R}^n$

Proposition

Proposition 42.3.11 (The Euclidean Distance on $\mathbb{R}^n$ is a Metric). *The function d_2 is a metric on $\mathbb{R}^n$.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}^n, d_2).$$

Remark (Interpretation). The Euclidean distance makes $\mathbb{R}^n$ into the standard finite dimensional metric space.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Minkowski's Inequality

Corollary

Corollary 42.3.12 (Euclidean Metric). *For every $x, y \in \mathbb{R}^n$, define*

$$d_2(x, y) := \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2}.$$

Then $(\mathbb{R}^n, d_2)$ is a metric space.

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}^n, d_2).$$

Remark (Interpretation). This records the standard Euclidean metric-space structure on $\mathbb{R}^n$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Minkowski's Inequality

Definition — Taxicab Metric

Definition 42.3.13 (Taxicab Metric). For $x, y \in \mathbb{R}^n$, the *taxicab metric* is the function

$$d_1(x, y) := \sum_{i=1}^{n} |x_i - y_i|.$$

Remark (Standard quantified statement).

$$d_1 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|.$$

Remark (Interpretation). Taxicab distance measures coordinate displacement additively, as if travel were constrained to coordinate-parallel streets.

Proposition

Proposition 42.3.14 (Taxicab Metric). *The function d_1 is a metric on $\mathbb{R}^n$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}^n, d_1).$$

Remark (Interpretation). The taxicab metric satisfies the metric axioms coordinate by coordinate, with the real absolute-value triangle inequality summed over all coordinates.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Taxicab Metric

Definition — p-Metric on $\mathbb{R}^n$

Definition 42.3.15 (p-Metric on $\mathbb{R}^n$). Let $1 \leq p < \infty$. For $x, y \in \mathbb{R}^n$, define

$$d_p(x, y) := \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}.$$

Remark (Standard quantified statement).

$$1 \leq p < \infty \implies d_p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}.$$

Remark (Interpretation). The p-metrics interpolate between taxicab distance at $p = 1$ and Euclidean distance at $p = 2$, and they form the standard finite-dimensional ℓ^p family.

Theorem

Theorem 42.3.16 (d_p is a Metric). *For every $1 \leq p < \infty$, the function d_p is a metric on $\mathbb{R}^n$.*
Go to proof.

Remark (Standard quantified statement).

$$1 \leq p < \infty \implies \mathrm{Metric}(\mathbb{R}^n, d_p).$$

Remark (Interpretation). The proof is the p-norm version of Minkowski's inequality. The theorem packages a whole family of metric structures on the same underlying set.

Remark (Dependencies).

- Definition: Metric Space
- Definition: p-Metric on $\mathbb{R}^n$

Definition — Complex Metric

Definition 42.3.17 (Complex Metric)**.** Identify $\mathbb{C}$ with $\mathbb{R}^2$. The *complex metric* is

$$d_{\mathbb{C}}(z, w) := |z - w|.$$

Remark (Standard quantified statement).

$$d_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{R}_{\geq 0}, \qquad d_{\mathbb{C}}(z, w) = |z - w|.$$

Remark (Interpretation). As a distance function, the complex metric is the Euclidean metric on $\mathbb{R}^2$ written in complex notation.

Proposition

Proposition 42.3.18 (The Complex Plane is a Metric Space)**.** *The pair* $(\mathbb{C}, d_{\mathbb{C}})$ *is a metric space.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{C}, d_{\mathbb{C}}).$$

Remark (Interpretation). The metric on $\mathbb{C}$ is inherited from the Euclidean plane.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Complex Metric
- Corollary: Euclidean Metric

Definition — Supremum Metric on Bounded Real Sequences

Definition 42.3.19 (Supremum Metric on Bounded Real Sequences)**.** Let $\ell^\infty(\mathbb{R})$ denote the set of all bounded real sequences:

$$\ell^\infty(\mathbb{R}) := \{x = (x_n)_{n\in\mathbb{N}} : \exists M \geq 0 \; \forall n \in \mathbb{N} \; |x_n| \leq M\}.$$

For $x, y \in \ell^\infty(\mathbb{R})$, define

$$d_\infty(x, y) := \sup_{n\in\mathbb{N}} |x_n - y_n|.$$

Remark (Standard quantified statement).

$$d_\infty : \ell^\infty(\mathbb{R}) \times \ell^\infty(\mathbb{R}) \to \mathbb{R}_{\geq 0}, \qquad d_\infty(x, y) = \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

Remark (Interpretation). The supremum metric measures the largest coordinatewise discrepancy between two bounded sequences.

Proposition

Proposition 42.3.20 (The Supremum Distance on Bounded Sequences is a Metric)**.** *The function d_∞ is a metric on $\ell^\infty(\mathbb{R})$.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\ell^\infty(\mathbb{R}), d_\infty).$$

Remark (Interpretation). Two bounded sequences are close in this metric when every coordinate is close by the same tolerance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Supremum Metric on Bounded Real Sequences

42.3.2 Point Functions and Pointlike Functions

Remark (Exposition). A metric is a two-variable distance function

$$d : X \times X \to \mathbb{R}_{\geq 0}.$$

The codomain $\mathbb{R}_{\geq 0}$ records only a nonnegative distance value; by itself, it does not remember which point of X served as the reference point. Point functions adapt the metric by freezing one input. For a fixed $z \in X$, the function $\delta_z : X \to \mathbb{R}_{\geq 0}$ sends x to the distance $d(z, x)$. Thus a point of the metric space can be represented by its distance profile across the whole space.

Definition — Point Function

Definition 42.3.21 (Point Function)**.** Let (X, d) be a metric space and let $z \in X$. The *point function at z* is the function

$$\delta_z : X \to \mathbb{R}_{\geq 0}, \qquad \delta_z(x) := d(z, x).$$

The set of all point functions on X is denoted by

$$\delta(X) := \{\, \delta_z : z \in X \,\}.$$

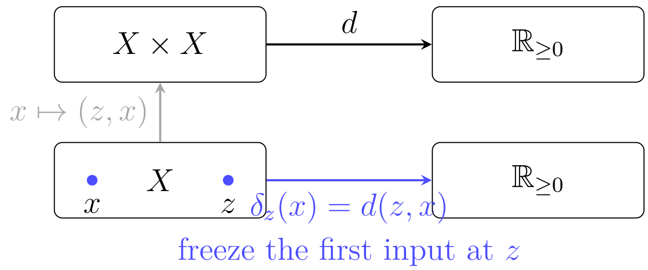


Figure 42.1: A metric $d : X \times X \to \mathbb{R}_{\geq 0}$ becomes the point function $\delta_z : X \to \mathbb{R}_{\geq 0}$ when the first input is fixed at z.

Remark (Standard quantified statement).

$$z \in X \implies \delta_z : X \to \mathbb{R}_{\geq 0} \quad \text{and} \quad \forall x \in X\colon \delta_z(x) = d(z, x).$$

Remark (Interpretation). The point function δ_z is the distance-from-z profile. It turns the fixed point z into a one-variable nonnegative real-valued function on the whole space.

Remark (Examples). In $\mathbb{R}$ with the usual metric $d(x, y) = |x - y|$, the point function at $z \in \mathbb{R}$ is

$$\delta_z(x) = d(z, x) = |x - z|.$$

For example, the point function at 2 is

$$\delta_2(x) = |x - 2|.$$

It has value 0 exactly at $x = 2$, and its graph is the familiar V-shape with vertex at $(2, 0)$. Thus the point 2 is encoded by the whole distance profile $x \mapsto |x - 2|$.

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 42.3.22 (Point Functions Identify Points)**.** *Let (X, d) be a metric space. The map*

$$X \to \delta(X), \qquad z \mapsto \delta_z$$

is a bijection.

Go to proof.

Remark (Standard quantified statement).

$$z \mapsto \delta_z \quad \text{is a bijection from } X \text{ onto } \delta(X).$$

Remark (Interpretation). No two different points have the same distance profile. The zero-distance law recovers z from δ_z, because $\delta_z(z) = 0$ and no other point has zero distance from z.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Point Function

Theorem

Theorem 42.3.23 (Point Function Inequalities). *Let (X, d) be a nonempty metric space and let $z \in X$. Then, for all $a, b \in X$,*

$$\delta_z(b) - \delta_z(a) \leq d(a, b) \leq \delta_z(b) + \delta_z(a),$$

and

$$\delta_z(z) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in X\colon \quad \delta_z(b) - \delta_z(a) \leq d(a, b) \leq \delta_z(b) + \delta_z(a), \qquad \delta_z(z) = 0.$$

Remark (Interpretation). Point functions mimic the metric because they are built from the metric. The right-hand inequality is the triangle inequality through z, while the left-hand inequality is the reverse triangle estimate.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Point Function

Definition — Pointlike Function

Definition 42.3.24 (Pointlike Function). Let (X, d) be a nonempty metric space. A function

$$u : X \to \mathbb{R}_{\geq 0}$$

is *pointlike* on X if, for all $a, b \in X$,

$$u(a) - u(b) \leq d(a, b) \leq u(a) + u(b).$$

Remark (Standard quantified statement).

$$u \text{ is pointlike} \quad \iff \quad u : X \to \mathbb{R}_{\geq 0} \ \wedge \ \forall a, b \in X\colon u(a) - u(b) \leq d(a, b) \leq u(a) + u(b).$$

Remark (Interpretation). A pointlike function satisfies the inequalities that every point function satisfies. It behaves like a distance-from-something profile, even before a point realizing that profile has been identified.

Remark (Dependencies).

- Definition: Metric Space
- Theorem: Point Function Inequalities

Theorem

Theorem 42.3.25 (Pointlike Functions with a Zero Are Point Functions)**.** *Let (X, d) be a metric space and let $u : X \to \mathbb{R}_{\geq 0}$. Then $u \in \delta(X)$ if and only if u is pointlike on X and*

$$0 \in u(X).$$

In that case, there is a unique $w \in X$ such that $u(w) = 0$, and

$$u = \delta_w.$$

Go to proof.

Remark (Standard quantified statement).

$$u \in \delta(X) \iff u \text{ is pointlike on } X \wedge 0 \in u(X).$$

Remark (Interpretation). The zero value anchors a pointlike function to an actual point of X. Once $u(w) = 0$, the pointlike inequalities force $u(a) = d(w, a)$ for every $a \in X$, so u is exactly the point function δ_w.

Remark (Dependencies).

- Definition: Point Function
- Definition: Pointlike Function

42.4 Examples of Metric Spaces

Remark (Exposition). The preceding metrics become metric spaces once their domains are paired with the verified distance functions. This section records the resulting spaces as objects in their own right.

Definition — Discrete Metric Space

Definition 42.4.1 (Discrete Metric Space)**.** Let X be a nonempty set. The *discrete metric space on X* is the metric space

$$(X, d_{\text{disc}}),$$

where d_{disc} is the discrete metric on X.

Remark (Standard quantified statement).

$$X \neq \varnothing \implies \text{MetricSpace}(X, d_{\text{disc}}).$$

Remark (Interpretation). A discrete metric space is a space where distinct points are all separated by the same positive distance. Its geometry records equality and inequality, but no finer notion of nearness.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Discrete Metric
- Proposition: The Discrete Distance is a Metric

Definition — Euclidean Metric Space

Definition 42.4.2 (Euclidean Metric Space)**.** For $n \in \mathbb{N}$, the *standard Euclidean metric space* is

$$(\mathbb{R}^n, d_2),$$

where d_2 is the Euclidean metric on $\mathbb{R}^n$.

Remark (Standard quantified statement).

$$n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_2).$$

Remark (Interpretation). The Euclidean metric space is the standard finite-dimensional distance model: points are coordinate vectors, and distance is ordinary straight-line length.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Definition — Taxicab Metric Space

Definition 42.4.3 (Taxicab Metric Space)**.** For $n \in \mathbb{N}$, the *taxicab metric space* is

$$(\mathbb{R}^n, d_1),$$

where d_1 is the taxicab metric on $\mathbb{R}^n$.

Remark (Standard quantified statement).

$$n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_1).$$

Remark (Interpretation). The taxicab metric space has the same underlying set as Euclidean space, but distance is measured by summing coordinatewise displacements.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Taxicab Metric
- Proposition: Taxicab Metric

Definition — p-Metric Space

Definition 42.4.4 (p-Metric Space)**.** Let $1 \leq p < \infty$ and let $n \in \mathbb{N}$. The *p-metric space* is

$$(\mathbb{R}^n, d_p),$$

where d_p is the p-metric on $\mathbb{R}^n$.

Remark (Standard quantified statement).

$$1 \leq p < \infty,\ n \in \mathbb{N} \implies \mathrm{MetricSpace}(\mathbb{R}^n, d_p).$$

Remark (Interpretation). The p-metric spaces form a family of finite-dimensional metric spaces on the same coordinate set, with different rules for turning coordinate displacements into distance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: p-Metric on $\mathbb{R}^n$
- Theorem: d_p is a Metric

Definition — Complex Metric Space

Definition 42.4.5 (Complex Metric Space)**.** The *complex metric space* is

$$(\mathbb{C}, d_{\mathbb{C}}),$$

where $d_{\mathbb{C}}$ is the usual Euclidean distance on the complex plane.

Remark (Standard quantified statement).

$$\mathrm{MetricSpace}(\mathbb{C}, d_{\mathbb{C}}).$$

Remark (Interpretation). The complex metric space treats complex numbers as points in the Euclidean plane, while retaining the notation and algebra of $\mathbb{C}$.

Remark (Dependencies).

- Definition: Metric Space

- Definition: Complex Metric
- Proposition: The Complex Plane is a Metric Space

Definition — Bounded Sequence Metric Space

Definition 42.4.6 (Bounded Sequence Metric Space)**.** The *bounded sequence metric space* is

$$(\ell^\infty(\mathbb{R}), d_\infty),$$

where $\ell^\infty(\mathbb{R})$ is the set of bounded real sequences and d_∞ is the supremum metric.

Remark (Standard quantified statement).

$$\mathrm{MetricSpace}(\ell^\infty(\mathbb{R}), d_\infty).$$

Remark (Interpretation). The bounded sequence metric space is an infinite-dimensional example. Its distance measures the largest coordinatewise discrepancy between two bounded real sequences.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Supremum Metric on Bounded Real Sequences
- Proposition: Supremum Metric on Bounded Sequences

42.5 Distances, Boundedness, and Balls

42.5.1 Distances and Boundedness

Remark (Exposition). Before metric topology begins, the metric already supplies quantitative measurements. It measures the distance between two points, the spread of a set, the distance from a point to a set, and the separation between two sets. These distance notions are the numerical layer beneath balls, neighborhoods, closedness, limit points, and convergence.

Definition — Diameter

Definition 42.5.1 (Diameter)**.** Let (X, d) be a metric space and let $A \subseteq X$ be nonempty. The *diameter* of A is

$$\mathrm{diam}(A) := \sup\{\, d(a, b) : a, b \in A \,\},$$

whenever this supremum is finite, and is $+\infty$ otherwise.

Remark (Standard quantified statement).

$$\mathrm{diam}(A) = \sup\{\, d(a, b) : a, b \in A \,\}.$$

Remark (Interpretation). The diameter records the largest distance needed to travel between two points of A. It measures internal spread rather than location.

Remark (Dependencies).

- Definition: Metric Space

Definition — Distance from a Point to a Set

Definition 42.5.2 (Distance from a Point to a Set)**.** Let (X, d) be a metric space, let $x \in X$, and let $A \subseteq X$ be nonempty. The *distance from x to A* is

$$d(x, A) := \inf\{\, d(x, a) : a \in A \,\}.$$

Remark (Standard quantified statement).

$$d(x, A) = \inf_{a \in A} d(x, a).$$

Remark (Interpretation). The number $d(x, A)$ measures how close x can get to A. The infimum may be achieved by a point of A, but it need not be achieved in every metric space.

Remark (Dependencies).

- Definition: Metric Space

Definition — Distance Between Sets

Definition 42.5.3 (Distance Between Sets)**.** Let (X, d) be a metric space, and let $A, B \subseteq X$ be nonempty. The *distance between A and B* is

$$d(A, B) := \inf\{\, d(a, b) : a \in A,\ b \in B \,\}.$$

Remark (Standard quantified statement).

$$d(A, B) = \inf_{a \in A,\ b \in B} d(a, b).$$

Remark (Interpretation). Set distance records the smallest separation that points of the two sets can approach. Distinct sets can have distance 0 if they touch in the limit.

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 42.5.4 (Distance to a Union)**.** *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ be nonempty. Then*

$$d(x, A \cup B) = \min\{d(x, A), d(x, B)\}.$$

Go to proof.

Remark (Standard quantified statement).

$$d(x, A \cup B) = \min\{d(x, A), d(x, B)\}.$$

Remark (Interpretation). To get close to a union, it is enough to get close to one of its pieces. The nearest piece determines the distance.

Remark (Dependencies).

- Definition: Distance from a Point to a Set

Theorem

Theorem 42.5.5 (Distance to an Intersection Bound). *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ have nonempty intersection. Then*

$$d(x, A \cap B) \geq \max\{d(x, A), d(x, B)\}.$$

Go to proof.

Remark (Standard quantified statement).

$$A \cap B \neq \varnothing \implies d(x, A \cap B) \geq \max\{d(x, A), d(x, B)\}.$$

Remark (Interpretation). An intersection has fewer candidate points than either set alone. Fewer points can only make the infimum distance larger, not smaller.

Remark (Dependencies).

- Definition: Distance from a Point to a Set

Definition — Metric Bounded Set

Definition 42.5.6 (Metric Bounded Set). Let (X, d) be a metric space and let $A \subseteq X$. The set A is *bounded* if there exist $x \in X$ and $R > 0$ such that

$$d(x, a) \leq R$$

for every $a \in A$.

Remark (Standard quantified statement).

$$A \text{ is bounded} \iff \exists x \in X\, \exists R > 0\, \forall a \in A\colon d(x, a) \leq R.$$

Remark (Interpretation). A bounded set has finite spread from at least one center. The center is auxiliary: boundedness records finite reach, not a preferred location. Important trap: an infinite set can be bounded; for example, $(0, 1) \subseteq \mathbb{R}$ is bounded in the usual metric.

Remark (Examples). In $\mathbb{R}$ with the usual metric, the intervals $(0, 1)$, $[0, 1]$, (a, b), and $[a, b]$ are bounded whenever $a < b$. The rays $(0, \infty)$ and $[0, \infty)$, and the whole line $\mathbb{R}$, are not bounded.

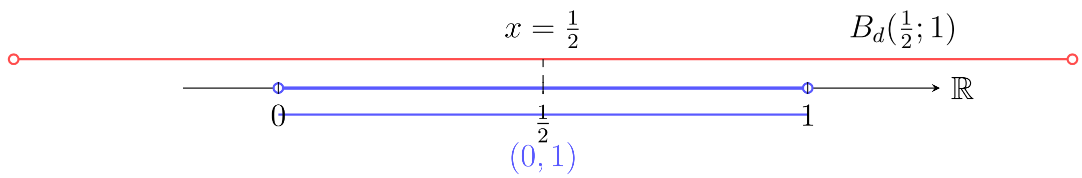
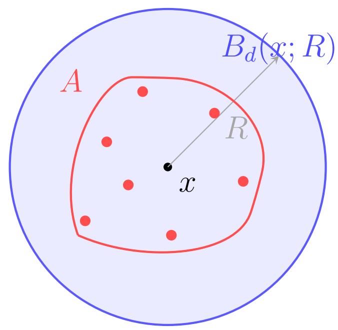


Figure 42.2: In $\mathbb{R}$, the interval $(0, 1)$ is bounded because every point of the interval is within distance 1 of $1/2$.



$A \subseteq B_d(x; R)$

Figure 42.3: An arbitrary subset A of a metric space is bounded when all of its points lie within one finite distance bound from a chosen center.

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 42.5.7 (Equivalent Boundedness Criterion)**.** *Let (X, d) be a metric space and let $A \subseteq X$. Then A is bounded if and only if there exist $x \in X$ and $R > 0$ such that*

$$d(x, a) < R$$

for every $a \in A$.

Remark (Standard quantified statement).

$$A \text{ is bounded} \iff \exists x \in X\, \exists R > 0\, \forall a \in A \colon d(x, a) < R.$$

Remark (Interpretation). Strict and non-strict radius bounds are equivalent after enlarging the radius. Both express the same finite-spread condition.

Remark (Dependencies).

- Definition: Metric Bounded Set

Theorem

Theorem 42.5.8 (Center Independence)**.** *Let (X, d) be a metric space. If $A \subseteq X$ and there exist $x \in X$ and $R > 0$ such that $d(x, a) \le R$ for every $a \in A$, then for every $y \in X$,*

$$d(y, a) \le R + d(x, y)$$

for every $a \in A$. Thus boundedness does not depend on the chosen center.

Remark (Standard quantified statement).

$$\big(\forall a \in A\colon d(x, a) \le R\big) \implies \forall y \in X\, \forall a \in A\colon d(y, a) \le R + d(x, y).$$

Remark (Interpretation). Changing centers costs at most the distance between the old center and the new center. The triangle inequality transports one finite bound to any other center.

Remark (Dependencies).

- Definition: Metric Bounded Set

Theorem

Theorem 42.5.9 (Subsets of Bounded Sets Are Bounded)**.** *Let (X, d) be a metric space. If $A \subseteq B \subseteq X$ and B is bounded, then A is bounded.*

Remark (Standard quantified statement).

$$A \subseteq B \subseteq X \ \wedge\ B \text{ is bounded} \implies A \text{ is bounded}.$$

Remark (Interpretation). A subset cannot spread out more than the set containing it. Any bound for B automatically bounds A.

Remark (Dependencies).

- Definition: Metric Bounded Set

Theorem

Theorem 42.5.10 (Finite Sets Are Bounded)**.** *Every finite subset of a metric space is bounded.*

Remark (Standard quantified statement).

$$A \subseteq X \ \wedge\ A \text{ is finite} \implies A \text{ is bounded}.$$

Remark (Interpretation). Only finitely many distances must be controlled. Choosing one center and a bound larger than all distances from that center captures the whole finite set.

Remark (Dependencies).

- Definition: Metric Bounded Set

Theorem

Theorem 42.5.11 (Finite Unions of Bounded Sets Are Bounded)**.** *Let (X,d) be a metric space, let $n \in \mathbb{N}$, and let $A_1, \ldots, A_n \subseteq X$. If $A_1, \ldots, A_n$ are bounded, then*

$$\bigcup_{k=1}^{n} A_k$$

is bounded.

Remark (Standard quantified statement)*.*

$$\big(\forall k \in \{1, \ldots, n\}\colon A_k \text{ is bounded}\big) \implies \bigcup_{k=1}^{n} A_k \text{ is bounded.}$$

Remark (Interpretation)*.* Each bounded set has finite spread. A finite union still has only finitely many centers and bounds to absorb into one sufficiently large bound.

Remark (Dependencies)*.*

- Definition: Metric Bounded Set
- Theorem: Center Independence

Theorem

Theorem 42.5.12 (Boundedness via Diameter)**.** *Let (X,d) be a metric space and let $A \subseteq X$ be nonempty. Then A is bounded if and only if there exists $C > 0$ such that*

$$d(a,b) \leq C$$

for all $a, b \in A$.

Remark (Standard quantified statement)*.*

$$A \neq \varnothing \implies \Big(A \text{ is bounded} \iff \exists C > 0\, \forall a, b \in A\colon d(a,b) \leq C\Big).$$

Remark (Interpretation)*.* For nonempty sets, boundedness can be measured internally: all pairwise distances must be uniformly bounded. This is the diameter viewpoint.

Remark (Dependencies)*.*

- Definition: Metric Bounded Set
- Definition: Diameter

Theorem

Theorem 42.5.13 (Closure of a Bounded Set Is Bounded)**.** *Let (X,d) be a metric space. If $A \subseteq X$ is bounded, then $\operatorname{cl}(A)$ is bounded.*

Remark (Standard quantified statement).

$$A \text{ is bounded} \Longrightarrow \text{cl}(A) \text{ is bounded.}$$

Remark (Interpretation). Passing to the closure may add limit points, but it does not create points arbitrarily far away. A slightly enlarged bound still captures the closure.

Remark (Dependencies).

- Definition: Metric Bounded Set
- Definition: Metric Closure

Theorem

Theorem 42.5.14 (Compact Sets Are Bounded). *Every compact subset of a metric space is bounded.*

Remark (Standard quantified statement).

$$K \subseteq X \ \wedge \ K \text{ is compact} \Longrightarrow K \text{ is bounded.}$$

Remark (Interpretation). The open balls centered at any fixed point cover the compact set as their radii grow. Compactness reduces this cover to finitely many radii, and the largest one bounds the set.

Remark (Dependencies).

- Definition: Metric Bounded Set

Theorem

Theorem 42.5.15 (Totally Bounded Sets Are Bounded). *Every totally bounded subset of a metric space is bounded.*

Remark (Standard quantified statement).

$$A \subseteq X \ \wedge \ A \text{ is totally bounded} \Longrightarrow A \text{ is bounded.}$$

Remark (Interpretation). A totally bounded set can be covered by finitely many balls of one fixed radius. A finite collection of bounded balls has finite overall spread, so the set is bounded.

Remark (Dependencies).

- Definition: Metric Bounded Set
- Theorem: Finite Unions of Bounded Sets Are Bounded

42.5.2 Balls and Local Neighborhoods

Definition — Open Ball

Definition 42.5.16 (Open Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *open ball* of radius r centered at x is

$$B_d(x; r) := \{\, y \in X : d(y, x) < r \,\}.$$

Remark (Standard quantified statement).

$$\forall y \in X\colon \quad y \in B_d(x; r) \iff d(y, x) < r.$$

Remark (Interpretation). An open ball collects exactly the points whose distance from the center is strictly less than the chosen radius. The strict inequality leaves the boundary points out.

Remark (Historical note). This notation follows Definition 2.15 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Definition — Closed Ball

Definition 42.5.17 (Closed Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *closed ball* of radius r centered at x is

$$\overline{B}_d(x; r) := \{\, y \in X : d(y, x) \leq r \,\}.$$

Remark (Standard quantified statement).

$$\forall y \in X\colon \quad y \in \overline{B}_d(x; r) \iff d(y, x) \leq r.$$

Remark (Interpretation). A closed ball uses the same center and radius as the corresponding open ball, but it includes the boundary points at distance exactly r.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Lemma

Lemma 42.5.18 (Key Local Ball Lemma)**.** *Let (X, d) be a metric space. If $y \in B_d(x; r)$, then*

$$B_d(y; r - d(x, y)) \subseteq B_d(x; r).$$

Remark (Standard quantified statement).

$$\forall x, y \in X\, \forall r > 0: \quad y \in B_d(x;r) \Longrightarrow B_d(y; r - d(x,y)) \subseteq B_d(x;r).$$

Remark (Interpretation). Every point inside an open ball has some positive-radius room that remains inside the original ball. The available radius is the unused distance from y to the boundary.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 42.5.19 (Ball Nesting). *Let (X,d) be a metric space, let $x \in X$, and let $0 < r \leq s$. Then*

$$B_d(x;r) \subseteq B_d(x;s).$$

If $r < s$, then also

$$\overline{B}_d(x;r) \subseteq B_d(x;s).$$

Remark (Standard quantified statement).

$$0 < r \leq s \Longrightarrow B_d(x;r) \subseteq B_d(x;s),$$

and

$$0 < r < s \Longrightarrow \overline{B}_d(x;r) \subseteq B_d(x;s).$$

Remark (Interpretation). Balls with the same center are ordered by radius. A strictly larger open ball even contains the closed ball of the smaller radius.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball

Theorem

Theorem 42.5.20 (Ball Containment by Center Distance). *Let (X,d) be a metric space, let $x, y \in X$, and let $r, s > 0$.*
If

$$d(x,y) + r \leq s,$$

then

$$B_d(x;r) \subseteq B_d(y;s).$$

If

$$d(x,y) + r < s,$$

then

$$\overline{B}_d(x;r) \subseteq B_d(y;s).$$

Remark (Standard quantified statement).

$$d(x,y)+r \le s \Longrightarrow B_d(x;r) \subseteq B_d(y;s),$$

and

$$d(x,y)+r < s \Longrightarrow \overline{B}_d(x;r) \subseteq B_d(y;s).$$

Remark (Interpretation). A ball centered at x sits inside a ball centered at y when the radius available at y is large enough to pay for both the center distance and the radius around x.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Closed Ball

Theorem

Theorem 42.5.21 (Intersecting Balls Imply Center-Distance Bound). *Let (X,d) be a metric space, let $x,y \in X$, and let $r,s>0$. If*

$$B_d(x;r) \cap B_d(y;s) \neq \varnothing,$$

then

$$d(x,y) < r+s.$$

Remark (Standard quantified statement).

$$B_d(x;r) \cap B_d(y;s) \neq \varnothing \Longrightarrow d(x,y) < r+s.$$

Remark (Interpretation). If two open balls overlap, a point in the overlap gives a two-step path from one center to the other. The triangle inequality bounds the center distance by the sum of the two radii.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 42.5.22 (Closed Balls Contain Open Balls). *Let (X,d) be a metric space. For every $x \in X$ and every $r>0$,*

$$B_d(x;r) \subseteq \overline{B}_d(x;r).$$

Remark (Standard quantified statement).

$$d(x,y) < r \Longrightarrow d(x,y) \le r.$$

Remark (Interpretation). The closed ball of radius r keeps all points of the open ball and adds the possible boundary points at distance exactly r.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball

42.6 Open and Closed Sets

Remark (Exposition). Open and closed sets describe how subsets sit inside a metric space. The primitive local objects are open balls: a set is open when every one of its points has a small ball that remains inside the set. Closed sets are then defined by open complements, and closure, interior, and boundary record the points forced by these local tests.

Definition — Metric Open Set

Definition 42.6.1 (Metric Open Set)**.** Let (X, d) be a metric space. A subset $E \subseteq X$ is *open in* (X, d) if either $E = \varnothing$ or, for every $x \in E$, there exists $\varepsilon > 0$ such that

$$B_d(x; \varepsilon) \subseteq E.$$

Remark (Standard quantified statement).

$$E \text{ is open in } (X, d) \iff E \subseteq X \ \wedge \ \big(E = \varnothing \ \vee \ \forall x \in E\, \exists \varepsilon > 0\colon B_d(x; \varepsilon) \subseteq E\big).$$

Remark (Interpretation). An open set contains room around each of its points. The amount of room may depend on the point, but every point of the set must have some positive-radius ball that stays inside the set.

Remark (Examples). In $\mathbb{R}$ with the usual metric, each interval (a, b) is open. The interval $[a, b]$ is not open when $a < b$, because the endpoints have no positive-radius open interval contained in $[a, b]$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 42.6.2 (Open Balls are Open)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r > 0$, the open ball $B_d(x; r)$ is open in (X, d).*
Go to proof.

Remark (Standard quantified statement).

$$\forall x \in X\, \forall r > 0\colon \quad B_d(x; r) \text{ is open in } (X, d).$$

Remark (Interpretation). Every metric ball contains room around each of its points. The triangle inequality is the mechanism that transfers room around the center to room around an arbitrary point inside the ball.

Remark (Historical note). This result corresponds to Theorem 3.4 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Metric Open Set
- Lemma: Key Local Ball Lemma

Theorem

Theorem 42.6.3 (Metric Open Set Closure Properties)**.** *Let X be a metric space. Then:*

1. *the empty set $\varnothing$ and the space X are open sets;*
2. *an arbitrary union of open subsets of X is open;*
3. *any finite intersection of open subsets of X is open.*

Go to proof.

Remark (Standard quantified statement). If X is a metric space, then

$$\varnothing \text{ is open in } X, \qquad X \text{ is open in } X,$$

$$\{U_\alpha\}_{\alpha \in A} \text{ open in } X \Longrightarrow \bigcup_{\alpha \in A} U_\alpha \text{ is open in } X,$$

and, for $n \in \mathbb{N}$,

$$U_1, \ldots, U_n \text{ open in } X \Longrightarrow \bigcap_{k=1}^{n} U_k \text{ is open in } X.$$

Remark (Interpretation). Metric open sets are stable under the same basic operations that define a topology: arbitrary unions and finite intersections. Thus every metric determines a topology by declaring its metric-open subsets to be open.

Remark (Historical note). This result corresponds to Theorem 3.2 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Open Set

Definition — Metric Closed Set

Definition 42.6.4 (Metric Closed Set)**.** Let (X, d) be a metric space. A subset $E \subseteq X$ is *closed in* (X, d) if its complement $E^c = X \setminus E$ is open in (X, d).

Remark (Standard quantified statement).

$$E \text{ is closed in } (X, d) \iff E \subseteq X \ \wedge \ E^c = X \setminus E \ \wedge \ E^c \text{ is open in } (X, d).$$

Remark (Interpretation). A closed set is defined by the openness of what lies outside it. In metric spaces this means every point outside E has some positive-radius ball that misses E.

Remark (Notation). When E is a subset of a fixed ambient space X, the symbol E^c denotes the complement of E in X:

$$E^c = X \setminus E.$$

Remark (Examples). In $\mathbb{R}$ with the usual metric, each interval $[a, b]$ is closed. The interval (a, b) is not closed when $a < b$, because it omits its endpoints.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Open Set

Theorem

Theorem 42.6.5 (Complement of a Closed Ball Is Open)**.** *Let (X, d) be a metric space. For every $x \in X$ and every $r \geq 0$,*

$$X \setminus \overline{B}_d(x; r) = \{\, y \in X : d(x, y) > r \,\}$$

is open.

Remark (Standard quantified statement).

$$X \setminus \overline{B}_d(x; r) \text{ is open in } (X, d).$$

Remark (Interpretation). The complement of a closed ball is the region strictly beyond its radius. Every point there has a positive margin from the boundary, so a small ball around it stays in the complement.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball
- Definition: Metric Open Set

Theorem

Theorem 42.6.6 (Closed Balls Are Closed). *Let (X, d) be a metric space. For every $x \in X$ and every $r \geq 0$, the closed ball*

$$\overline{B}_d(x; r) = \{ y \in X : d(x, y) \leq r \}$$

is closed.

Remark (Standard quantified statement).

$$\forall x \in X \, \forall r \geq 0\colon \quad \overline{B}_d(x; r) \text{ is closed in } (X, d).$$

Remark (Interpretation). Closed balls are closed because their complements are open. Points outside a closed ball have a positive margin beyond the radius, and that margin gives an open ball disjoint from the closed ball.

Remark (Dependencies).

- Definition: Closed Ball
- Definition: Metric Closed Set
- Theorem: Complement of a Closed Ball Is Open

Definition — Metric Closure

Definition 42.6.7 (Metric Closure). Let (X, d) be a metric space and let $E \subseteq X$. The *closure* of E, denoted $\mathrm{cl}(E)$, is

$$\mathrm{cl}(E) = \{x \in X : \forall \varepsilon > 0, \ B_d(x; \varepsilon) \cap E \neq \varnothing\}.$$

Remark (Standard quantified statement).

$$x \in \mathrm{cl}(E) \iff \forall \varepsilon > 0\colon B_d(x; \varepsilon) \cap E \neq \varnothing.$$

Remark (Interpretation). The closure of E contains the points of E and all points that cannot be separated from E by any positive-radius ball.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 42.6.8 (Closure of an Open Ball Is Contained in the Closed Ball). *Let (X, d) be a metric space. For every $x \in X$ and every $r > 0$,*

$$\mathrm{cl}\big(B_d(x; r)\big) \subseteq \overline{B}_d(x; r).$$

Remark (Standard quantified statement).

$$z \in \mathrm{cl}\big(B_d(x;r)\big) \Longrightarrow d(x,z) \le r.$$

Remark (Interpretation). Taking the closure of an open ball may add boundary points, but it cannot add points whose distance from the center is larger than the radius. Important trap: equality need not hold in every metric space,

$$\mathrm{cl}\big(B_d(x;r)\big) \ne \overline{B}_d(x;r),$$

and can occur. For example, in a discrete metric space, the closure of an open ball may be strictly smaller than the corresponding closed ball.

Remark (Dependencies).

- Definition: Open Ball
- Definition: Closed Ball
- Definition: Metric Closure

Definition — Dense Subset

Definition 42.6.9 (Dense Subset)**.** Let (X,d) be a metric space and let $A \subseteq X$. The set A is *dense in* X if

$$\mathrm{cl}(A) = X.$$

Remark (Standard quantified statement).

$$A \text{ is dense in } X \quad \Longleftrightarrow \quad \mathrm{cl}(A) = X.$$

Remark (Interpretation). A dense set reaches every point of the ambient space by approximation. Every point of X either belongs to A or is arbitrarily close to points of A.

Remark (Examples). In $\mathbb{R}$ with the usual metric, $\mathbb{Q}$ is dense in $\mathbb{R}$, and $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$.

Remark (Dependencies).

- Definition: Metric Closure

Definition — Metric Interior Point

Definition 42.6.10 (Metric Interior Point)**.** Let (X,d) be a metric space, let $E \subseteq X$, and let $x \in X$. The point x is an *interior point of* E if there exists $\delta > 0$ such that

$$B_d(x;\delta) \subseteq E.$$

Remark (Standard quantified statement).

$$x \text{ is an interior point of } E \quad \Longleftrightarrow \quad x \in X \ \wedge\ \exists \delta > 0\colon B_d(x;\delta) \subseteq E.$$

Remark (Interpretation). An interior point of E is a point with positive-radius room inside E. The entire ball around the point must remain in E, not merely the point itself.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Definition — Metric Interior

Definition 42.6.11 (Metric Interior)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *interior of* E is the set of all interior points of E. It is denoted by

$$E^\circ.$$

Remark (Standard quantified statement).

$$E^\circ = \{x \in X : \exists \delta > 0 \text{ such that } B_d(x; \delta) \subseteq E\}.$$

Remark (Interpretation). The interior E° collects exactly the points of E that have positive-radius room inside E.

Remark (Notation). The notation E° is read as the *interior of* E or E*-interior*.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Interior Point

Theorem

Theorem 42.6.12 (Interior is the Largest Open Subset)**.** *Let E be a subset of a metric space X. Then E° is the largest open subset of X contained in E. In other words:*

1. $E^\circ \subseteq E$*;*
2. E° *is open;*
3. $E^\circ \supseteq O$*, for every open set* $O \subseteq E$*.*

Go to proof.

Remark (Standard quantified statement).

$$E^\circ \subseteq E, \qquad E^\circ \text{ is open in } X, \qquad \forall O \subseteq E\colon O \text{ open in } X \Longrightarrow O \subseteq E^\circ.$$

Remark (Interpretation). The interior operation extracts the open part of a set. It keeps exactly the points with room inside E, and it contains every open subset already lying inside E.

Remark (Historical note). This result corresponds to Theorem 3.5 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Interior
- Definition: Metric Open Set
- Theorem: Open Balls are Open

Definition — Metric Boundary

Definition 42.6.13 (Metric Boundary)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *boundary* of E is

$$\partial E = \mathrm{cl}(E) \cap \mathrm{cl}(X \setminus E).$$

Remark (Standard quantified statement).

$$x \in \partial E \iff x \in \mathrm{cl}(E) \ \wedge \ x \in \mathrm{cl}(X \setminus E).$$

Remark (Interpretation). Boundary points are touched from both sides: every small ball around a boundary point meets E and also meets the complement of E.

Remark (Dependencies).

- Definition: Metric Closure

42.7 Limit Points and Isolated Points

Remark (Exposition). Limit points and isolated points separate two ways a subset can contain or approach a point. A limit point is repeatedly approached by other points of the set at every scale. An isolated point belongs to the set but has a small ball in which it is the only point of the set.

Definition — Metric Limit Point and Isolated Point

Definition 42.7.1 (Metric Limit Point and Isolated Point)**.** Let E be a subset of a metric space (X, d), and let $x \in X$.

1. The point x is a *limit point of* E if, for every $\varepsilon > 0$, the ball $B_d(x; \varepsilon)$ contains a point of $E \setminus \{x\}$.
2. The point x is an *isolated point of* E if $x \in E$ and x is not a limit point of E.

Remark (Standard quantified statement).

$$x \text{ is a limit point of } E \iff \forall \varepsilon > 0\colon B_d(x; \varepsilon) \cap (E \setminus \{x\}) \neq \varnothing.$$

$$x \text{ is an isolated point of } E \iff x \in E \ \wedge \ \exists \varepsilon > 0\colon B_d(x; \varepsilon) \cap E = \{x\}.$$

Remark (Interpretation). A limit point of E may lie in E or outside E; what matters is that points of E distinct from x occur arbitrarily close to x. An isolated point must belong to E, and some positive-radius ball around it contains no other point of E.

Remark (Notation). The set of limit points of E is denoted by E', read as E*-prime*.

Remark (Exposition). Limit points are also called *cluster points* or *accumulation points*. In later work with functions on subsets of metric spaces, limits of the form $\lim_{y \to x} f(y)$ are only meaningful when x is a limit point of the domain being approached.

Remark (Examples). In the usual metric on $\mathbb{R}$, if

$$E = (0,1) \cup \{2\},$$

then 2 is an isolated point of E, while $[0,1]$ is the set of limit points of E.

There are also metric spaces with infinite subsets that have no limit points. For example, if X is an infinite discrete metric space and $E = X$, then $E' = \varnothing$. Also, in $X = \mathbb{R} \setminus \mathbb{Q}$, the set

$$E = \{\sqrt{2}/n : n \in \mathbb{N}\}$$

has no limit point in X.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 42.7.2 (Closed Sets Contain Their Limit Points). *Let X be a metric space and let $E \subseteq X$. Then E is closed if and only if $E' \subseteq E$.*

Go to proof.

Remark (Standard quantified statement).

$$E \text{ is closed in } X \iff E' \subseteq E.$$

Remark (Interpretation). Closed sets contain all points that can be approached by other points of the set. Equivalently, no limit point of a closed set is missing from the set.

Remark (Historical note). This result corresponds to Theorem 3.10 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Closed Set
- Definition: Metric Limit Point and Isolated Point

Theorem

Theorem 42.7.3 (Sequential Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if there exists a sequence of distinct terms in E that converges to x.*

Go to proof.

Remark (Standard quantified statement).

$$x \in E' \iff \exists \{x_n\} \subseteq E\colon \big(\forall n \neq m,\ x_n \neq x_m\big) \wedge x_n \to x.$$

Remark (Interpretation). Limit points can be detected sequentially. A point is a limit point exactly when the set supplies infinitely many distinct approximating terms converging to that point.

Remark (Historical note). This result corresponds to Theorem 3.11 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Definition: Metric Sequential Convergence

Corollary

Corollary 42.7.4 (Neighborhood Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if every neighborhood of x contains infinitely many points of E.*

Go to proof.

Remark (Standard quantified statement).

$$x \in E' \iff \forall U\colon \big(U \text{ is a neighborhood of } x\big) \implies U \cap E \text{ is infinite.}$$

Remark (Interpretation). The neighborhood form strengthens the ball definition into a count statement: near a limit point, every neighborhood contains not just one nearby point of E, but infinitely many.

Remark (Historical note). This result corresponds to Corollary 3.12 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Neighborhood
- Theorem: Sequential Characterization of Limit Points

Theorem

Theorem 42.7.5 (Bolzano–Weierstrass for Bounded Infinite Subsets)**.** *Every bounded infinite subset of $\mathbb{R}^m$ has a limit point in $\mathbb{R}^m$.*

Go to proof.

Remark (Standard quantified statement).

$$E \subseteq \mathbb{R}^m, \quad E \text{ bounded}, \quad E \text{ infinite} \quad \Longrightarrow \quad \exists x \in \mathbb{R}^m \colon x \in E'.$$

Remark (Interpretation). In finite-dimensional Euclidean space, boundedness prevents an infinite set from spreading out completely. Some point of the ambient space must be approached by infinitely many points of the set.

Remark (Historical note). This result corresponds to Theorem 3.13 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Theorem: Bolzano–Weierstrass in $\mathbb{R}^m$
- Theorem: Sequential Characterization of Limit Points

Proposition

Proposition 42.7.6 (The Derived Set is Closed). *The set of limit points of any set is closed.*

Go to proof.

Remark (Standard quantified statement).

$$E \subseteq X \quad \Longrightarrow \quad E' \text{ is closed in } X.$$

Remark (Interpretation). Taking limit points produces a closed set. Once a point is approached by limit points of E, it is itself forced to be a limit point of E.

Remark (Historical note). This result corresponds to Proposition 3.15 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Theorem: Closed Sets Contain Their Limit Points

42.8 Sequential Convergence

Definition — Metric Sequential Convergence

Definition 42.8.1 (Metric Sequential Convergence). Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *convergent in* X if there exists $x_0 \in X$ such that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_0) < \varepsilon \qquad \text{for all } n \geq N.$$

In this case, $\{x_n\}$ converges to x_0, written

$$x_n \to x_0, \qquad x_0 = \lim_{n \to \infty} x_n.$$

Remark (Standard quantified statement).

$$x_n \to x_0 \iff x_0 \in X \ \wedge\ \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \in \mathbb{N}\colon \\ n \geq N \Longrightarrow d(x_n, x_0) < \varepsilon.$$

Remark (Interpretation). A sequence converges to x_0 when its tail eventually lies inside every metric tolerance around x_0. The metric supplies the meaning of "within ε" in the ambient space X.

Remark (Historical note). This convention follows Definition 2.20 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Definition — Metric Neighborhood

Definition 42.8.2 (Metric Neighborhood)**.** Let (X, d) be a metric space and let $x \in X$. A subset $U \subseteq X$ is a *neighborhood of* x if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq U.$$

Remark (Standard quantified statement).

$$U \text{ is a neighborhood of } x \iff U \subseteq X \ \wedge\ \exists \delta > 0\colon B_d(x; \delta) \subseteq U.$$

Remark (Interpretation). A metric neighborhood of x is any subset of X that contains at least one open ball centered at x. The set U may be larger than that ball; the essential condition is that x has some positive-radius room inside U.

Remark (Historical note). This convention follows Definition 2.21 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Definition — Epsilon Neighborhood

Definition 42.8.3 (Epsilon Neighborhood)**.** Let (X, d) be a metric space, let $x \in X$, and let $\varepsilon > 0$. The *ε-neighborhood of x* is the open ball

$$B_d(x; \varepsilon) = \{y \in X : d(y, x) < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in B_d(x; \varepsilon) \iff y \in X \ \wedge\ d(y, x) < \varepsilon.$$

Remark (Interpretation). An ε-neighborhood is the concrete neighborhood obtained by choosing a particular positive radius ε. It consists exactly of the points whose distance from x is less than that tolerance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Metric Neighborhood

Theorem

Theorem 42.8.4 (Metric Neighborhood Criterion for Sequential Convergence)**.** *Let (X,d) be a metric space, let $\{x_n\}$ be a sequence in X, and let $x \in X$. Then*

$$x_n \to x$$

if and only if every neighborhood of x contains all but finitely many terms of $\{x_n\}$.
Go to proof.

Remark (Standard quantified statement).

$$x_n \to x \quad \Longleftrightarrow \quad \forall U \subseteq X \colon \big(U \text{ is a neighborhood of } x\big) \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N \colon x_n \in U.$$

Remark (Interpretation). Sequential convergence can be tested by neighborhoods instead of numerical ε-bounds. A convergent sequence eventually remains inside every set that contains some open ball around the limit.

Remark (Historical note). This is the neighborhood characterization stated immediately after Definition 2.21 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Neighborhood

Definition — Metric Cauchy Sequence

Definition 42.8.5 (Metric Cauchy Sequence)**.** Let (X,d) be a metric space. A sequence $\{x_n\}$ in X is *Cauchy* if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \qquad \text{for all } n, m \geq N.$$

Remark (Standard quantified statement).

$$\{x_n\} \text{ is Cauchy} \quad \Longleftrightarrow \quad \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n, m \in \mathbb{N} \colon n, m \geq N \Longrightarrow d(x_n, x_m) < \varepsilon.$$

Remark (Interpretation). A Cauchy sequence is eventually internally stable: sufficiently late terms are close to one another, regardless of whether a limit point has already been identified in X.

Remark (Historical note). This convention follows Definition 2.22 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 42.8.6 (Metric Convergent Sequences Have Unique Limits). *In metric spaces, convergent sequences have unique limits.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq X \, \forall x, y \in X\colon \quad \big(x_n \to x \ \wedge \ x_n \to y\big) \Longrightarrow x = y.$$

Remark (Interpretation). In a metric space, a convergent sequence cannot settle at two different points. The positive definiteness of the metric turns zero distance between candidate limits into equality of the points.

Remark (Historical note). This result corresponds to Theorem 2.23 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Sequential Convergence

Theorem

Theorem 42.8.7 (Metric Cauchy Sequence with Convergent Subsequence). *Let $\{x_n\}$ be a Cauchy sequence in a metric space (X, d), let $x \in X$, and let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ such that*

$$\lim_{k \to \infty} x_{n_k} = x.$$

Then $x_n \to x$.

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq X \, \forall x \in X\colon \quad \Big(\{x_n\} \text{ is Cauchy} \\ \wedge \ \exists \{x_{n_k}\} \text{ subsequence of } \{x_n\}\colon x_{n_k} \to x\Big) \Longrightarrow x_n \to x.$$

Remark (Interpretation). For a Cauchy sequence, one convergent subsequence determines the behavior of the whole sequence. The Cauchy condition forces the full tail to follow any subsequence tail that has already converged.

Remark (Historical note). This result corresponds to Theorem 2.24 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence

Theorem

Theorem 42.8.8 (Coordinatewise Sequences in $\mathbb{R}^m$). *Let $\{x_n\}$ be a sequence in $\mathbb{R}^m$, and let $x_0 \in \mathbb{R}^m$. Write*

$$x_n = \left(x_n^{(1)}, \ldots, x_n^{(m)}\right) \qquad \text{for all } n \in \mathbb{N} \cup \{0\}.$$

Then:

1. $x_n \to x_0$ *if and only if* $x_n^{(j)} \to x_0^{(j)}$ *for every* $j = 1, \ldots, m$.
2. $\{x_n\}$ *is Cauchy if and only if* $\{x_n^{(j)}\}$ *is Cauchy for every* $j = 1, \ldots, m$.

Go to proof.

Remark (Standard quantified statement).

$$x_n \to x_0 \iff \forall j \in \{1, \ldots, m\}\colon x_n^{(j)} \to x_0^{(j)},$$
$$\{x_n\} \text{ is Cauchy} \iff \forall j \in \{1, \ldots, m\}\colon \{x_n^{(j)}\} \text{ is Cauchy}.$$

Remark (Interpretation). In finite-dimensional Euclidean space, convergence and Cauchy behavior can be checked one coordinate at a time. The Euclidean metric packages the finitely many coordinate estimates into one distance estimate.

Remark (Historical note). This result corresponds to Theorem 2.25 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Cauchy Sequence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Theorem

Theorem 42.8.9 (Euclidean Cauchy Sequences Converge)**.** *Every Cauchy sequence in* $\mathbb{R}^m$ *is convergent in* $\mathbb{R}^m$*.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq \mathbb{R}^m\colon \quad \{x_n\} \text{ is Cauchy} \Longrightarrow \exists x \in \mathbb{R}^m\colon x_n \to x.$$

Remark (Interpretation). $\mathbb{R}^m$ contains the limits of all of its Cauchy sequences. This is the finite-dimensional Euclidean completeness property.

Remark (Historical note). This result corresponds to Theorem 2.26 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence
- Theorem: Coordinatewise Sequences in $\mathbb{R}^m$

Theorem

Theorem 42.8.10 (Bolzano–Weierstrass in $\mathbb{R}^m$)**.** *Every bounded sequence in* $\mathbb{R}^m$ *contains a subsequence that converges in* $\mathbb{R}^m$*.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq \mathbb{R}^m\colon \quad \{x_n\} \text{ is bounded} \Longrightarrow \exists \{x_{n_k}\} \text{ subsequence } \exists x \in \mathbb{R}^m\colon x_{n_k} \to x.$$

Remark (Interpretation). Boundedness in finite-dimensional Euclidean space is strong enough to force some subsequence to settle to a Euclidean limit.

Remark (Historical note). This result corresponds to Theorem 2.27 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Chapter 43

Topology

SERIES
From Cantor to Ito

VOLUME IV
Mathematical Spaces

BOOK I
Mathematical Spaces

CHAPTER 3
Topology

43.1 Topology

43.2 Definitions

Definition — Topological Space

Definition 43.2.1 (Topological Space)**.** Let X be a set. A *topology* on X is a collection $\tau \subseteq \mathcal{P}(X)$ such that:

1. $\emptyset \in \tau$ and $X \in \tau$;
2. if $\{U_i\}_{i \in I}$ is any family of sets in τ, then $\bigcup_{i \in I} U_i \in \tau$;
3. if $U_1, \ldots, U_n \in \tau$, then $\bigcap_{k=1}^{n} U_k \in \tau$.

A *topological space* is a pair (X, τ), where X is a set and τ is a topology on X. The members of τ are called the *open sets* of the space.

Remark (Interpretation). A topology records which subsets are treated as open. The axioms say that openness includes the impossible and total regions, is preserved under arbitrary unions, and is preserved under finite intersections.

Remark (Examples). The same set may carry different topologies. If τ_1 and τ_2 are different topologies on the same set X, then (X, τ_1) and (X, τ_2) are different topological spaces because they generally declare different subsets of X to be open.

Remark (Dependencies).

- Definition: Metric Open Set

Definition — Discrete Topology

Definition 43.2.2 (Discrete Topology)**.** Let X be a set. The *discrete topology* on X is the topology

$$\tau_{\text{disc}} = \mathcal{P}(X).$$

Thus every subset of X is open in the topological space (X, τ_{disc}).

Remark (Standard quantified statement).

$$\tau_{\text{disc}} = \mathcal{P}(X), \qquad U \text{ is open in } (X, \tau_{\text{disc}}) \iff U \subseteq X.$$

Remark (Interpretation). The discrete topology makes the strongest possible declaration of openness: no subset is excluded. It is the topology with the most open sets on the underlying set X. Equivalently, it is the *finest* topology on X: every topology on X is contained in τ_{disc}.

Remark (Dependencies).

- Definition: Topological Space

Definition — Sierpinski Topology

Definition 43.2.3 (Sierpinski Topology)**.** Let $X = \{0, 1\}$. The *Sierpinski topology* on X is the topology

$$\tau_{\text{S}} = \{\emptyset, \{1\}, X\}.$$

The topological space (X, τ_{S}) is called the *Sierpinski space*.

Remark (Standard quantified statement).

$$U \text{ is open in } (X, \tau_{\text{S}}) \iff U = \emptyset \text{ or } U = \{1\} \text{ or } U = X.$$

Remark (Interpretation). The Sierpinski topology is the smallest nontrivial topology on a two-point set. It distinguishes the two points topologically: one singleton is open and the other singleton is not.

Remark (Dependencies).

- Definition: Topological Space

Definition — Cofinite Topology

Definition 43.2.4 (Cofinite Topology)**.** Let X be a set. The *cofinite topology* on X is the topology

$$\tau_{\text{cof}} = \{\, U \subseteq X : X \setminus U \text{ is finite} \,\} \cup \{\emptyset\}.$$

Thus a subset of X is open exactly when it is empty or its complement in the ambient set is finite.

Remark (Standard quantified statement).

$$U \text{ is open in } (X, \tau_{\text{cof}}) \iff U = \emptyset \text{ or } X \setminus U \text{ is finite}.$$

Remark (Interpretation). The cofinite topology treats almost all of X as open. A nonempty open set may omit only finitely many points, so on an infinite set any two nonempty open sets must intersect.

Remark (Dependencies).

- Definition: Topological Space

Definition — Trivial Topology

Definition 43.2.5 (Trivial Topology)**.** Let X be a set. The *trivial topology* on X is the topology

$$\tau_{\text{triv}} = \{\emptyset, X\}.$$

Thus only $\emptyset$ and X are open in the topological space (X, τ_{triv}).

Remark (Standard quantified statement).

$$\tau_{\text{triv}} = \{\emptyset, X\}, \qquad U \text{ is open in } (X, \tau_{\text{triv}}) \iff U = \emptyset \text{ or } U = X.$$

Remark (Interpretation). The trivial topology makes the weakest possible declaration of openness: only the sets required by the topology axioms are open. It is the topology with the fewest open sets on the underlying set X. Equivalently, it is the *coarsest* topology on X: it is contained in every topology on X.

Remark (Dependencies).

- Definition: Topological Space

43.3 Topological Space Subsets

Definition — Closed Subset of a Topological Space

Definition 43.3.1 (Closed Subset of a Topological Space)**.** Let (X, τ) be a topological space, and let $F \subseteq X$. The subset F is *closed in* (X, τ) if its complement

$$X \setminus F$$

is open in (X, τ).

Remark (Standard quantified statement).

$$F \text{ is closed in } (X, \tau) \iff F \subseteq X \text{ and } X \setminus F \in \tau.$$

Remark (Interpretation). Closedness is defined by openness of the complement. A subset is closed when everything outside it forms an open subset of the ambient topological space.

Remark (Examples). The empty set $\emptyset$ and the whole space X are both open and closed in every topological space (X, τ). The set X is open by the topology axioms and closed because $X \setminus X = \emptyset$ is open. The set $\emptyset$ is open by the topology axioms and closed because $X \setminus \emptyset = X$ is open.

Remark (Dependencies).

- Definition: Topological Space

Theorem

Theorem 43.3.2 (Closed-Set Characterization of Topologies). *Let X be a set, and let $\mathcal{C}$ be a collection of subsets of X. Suppose that:*

1. *$\emptyset \in \mathcal{C}$ and $X \in \mathcal{C}$;*
2. *the intersection of any family of elements of $\mathcal{C}$ belongs to $\mathcal{C}$;*
3. *the union of any two elements of $\mathcal{C}$ belongs to $\mathcal{C}$.*

Then there is a unique topology τ on X whose family of closed sets is exactly $\mathcal{C}$.
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\emptyset, X \in \mathcal{C} \wedge \forall \mathcal{A} \subseteq \mathcal{C} \colon \bigcap_{A \in \mathcal{A}} A \in \mathcal{C} \\
&\wedge \forall C_1, C_2 \in \mathcal{C} \colon C_1 \cup C_2 \in \mathcal{C} \\
&\implies \exists! \tau \colon \operatorname{Topology}(X, \tau) \wedge \mathcal{C} = \{F \subseteq X : X \setminus F \in \tau\}.
\end{aligned}$$

Remark (Interpretation). The topology axioms can be stated dually in terms of closed sets. Open sets are closed under arbitrary unions and finite intersections, while closed sets are closed under arbitrary intersections and finite unions.

Remark (Dependencies).

- Definition: Topological Space
- Definition: Closed Subset of a Topological Space

Definition — Finer and Coarser Topologies

Definition 43.3.3 (Finer and Coarser Topologies). Let τ_1 and τ_2 be topologies on a set X. The topology τ_1 is *finer than* τ_2 if

$$\tau_2 \subseteq \tau_1.$$

Equivalently, τ_2 is *coarser than* τ_1. The two topologies are *comparable* if one is finer than the other.

Remark (Standard quantified statement).

$$\tau_1 \text{ is finer than } \tau_2 \iff \tau_2 \subseteq \tau_1.$$

$$\tau_1 \text{ and } \tau_2 \text{ are comparable} \iff \tau_2 \subseteq \tau_1 \text{ or } \tau_1 \subseteq \tau_2.$$

Remark (Interpretation). A finer topology has at least as many open sets as the coarser topology. Thus it can distinguish points and subsets at least as sharply as the coarser topology.

Remark (Examples). If $\mathcal{F}_i$ is the family of closed subsets determined by τ_i, then

$$\tau_1 \text{ is finer than } \tau_2 \iff \mathcal{F}_2 \subseteq \mathcal{F}_1.$$

The discrete topology is the finest topology on a given set because every subset is open.

Remark (Dependencies).

- Definition: Topological Space
- Definition: Discrete Topology
- Definition: Closed Subset of a Topological Space

43.4 Basis

Definition — Basis for a Topology

Definition 43.4.1 (Basis for a Topology)**.** Let (X, τ) be a topological space. A collection $\mathcal{B} \subseteq \tau$ is a *basis for the topology* τ if every open set $O \in \tau$ is the union of some subcollection of $\mathcal{B}$. The elements of $\mathcal{B}$ are called *basic elements* or *basic open sets.*

Remark (Standard quantified statement).

$$\mathcal{B} \text{ is a basis for } \tau \iff \mathcal{B} \subseteq \tau \text{ and } \forall O \in \tau\, \exists \mathcal{A} \subseteq \mathcal{B} \colon O = \bigcup_{B \in \mathcal{A}} B.$$

Remark (Interpretation). A basis is a smaller supply of open sets from which all open sets can be assembled by unions. It records local building blocks for the topology.

Remark (Dependencies).

- Definition: Topological Space

Proposition

Proposition 43.4.2 (Pointwise Characterization of a Basis)**.** *Let (X, τ) be a topological space, and let $\mathcal{B} \subseteq \tau$. Then $\mathcal{B}$ is a basis for τ if and only if for every open set $O \in \tau$ and every $x \in O$, there is $B \in \mathcal{B}$ such that*

$$x \in B \subseteq O.$$

Consequently, a subset $A \subseteq X$ is open if and only if for each $x \in A$, there is $B \in \mathcal{B}$ such that

$$x \in B \subseteq A.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathcal{B} \text{ is a basis for } \tau \iff \forall O \in \tau\, \forall x \in O\, \exists B \in \mathcal{B} \colon x \in B \subseteq O.$$

$$A \in \tau \iff A \subseteq X \text{ and } \forall x \in A\, \exists B \in \mathcal{B} \colon x \in B \subseteq A.$$

Remark (Interpretation). The pointwise test says that a basis can be checked locally. Around every point of every open set, there must be a basic open set that still lies inside the original open set.

Remark (Dependencies).

- Definition: Topological Space
- Definition: Basis for a Topology

Chapter 44

Set Algebra

SERIES	VOLUME IV	BOOK I	CHAPTER 4
From Cantor to Ito	Mathematical Spaces	Mathematical Spaces	Set Algebra

44.1 Set Algebras

44.2 Systems of Sets

44.2.1 Systems of Sets

Comparison of Set Systems

Property	Semiring	Ring	Algebra	σ-algebra	Topology
Contains $\varnothing$	✓	✓	✓	✓	✓
Contains X	—	—	✓	✓	✓
Closed under $\cap$ (finite)	✓	✓	✓	✓	✓
Closed under $\cup$ (finite)	—	✓	✓	✓	✓
Closed under $\cup$ (countable)	—	—	—	✓	—
Closed under $\cup$ (arbitrary)	—	—	—	—	✓
Closed under complement	—	—	✓	✓	—
Difference $A \setminus B$	finite disj. union	✓	✓	✓	—

The definitions and theorems of the previous sections have been about individual sets and their operations. A second level of structure emerges when one asks: which *collections* of sets are closed under given operations? This question is not abstract generality for its own sake. The three subjects that depend most heavily on this chapter — topology, measure theory, and functional analysis — each begin by specifying a particular type of collection on a given set X. A topology specifies which subsets are "open." A σ-algebra specifies which subsets are "measurable." In both cases, the definition is precisely a list of closure conditions: which set operations applied to members of the collection must produce another member.

Mastering these four structures here, as purely set-theoretic objects, means that when topology and measure theory are encountered, all the foundational work is already done. The task will be to understand the *geometric* or *analytic* meaning of the axioms, not to struggle with the set-theoretic mechanics.

Definition (Semiring of Sets)

Definition 44.2.1 (Semiring of Sets)**.** A collection $\mathcal{S}$ of subsets of a set X is a *semiring* if:

(i) $\varnothing \in \mathcal{S}$;

(ii) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$;

(iii) for any $A, B \in \mathcal{S}$, the difference $A \setminus B$ is a *finite disjoint union* of members of $\mathcal{S}$: there exist pairwise disjoint $C_1, \ldots, C_n \in \mathcal{S}$ with $A \setminus B = C_1 \sqcup \cdots \sqcup C_n$.

Remark (Dependencies).

- Subset
- Empty Set
- Intersection
- Set Difference
- Union
- Pairwise Disjoint Family
- Finite and Infinite Sets

Example (Half-open intervals form a semiring). Let $X = \mathbb{R}$ and let $\mathcal{S}$ be the collection of all half-open intervals $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$, together with $\varnothing$. Then $\mathcal{S}$ is a semiring:

- $[a, b) \cap [c, d) = [\max(a, c), \min(b, d))$, which is again a half-open interval (or $\varnothing$);
- $[a, b) \setminus [c, d)$ is either $\varnothing$, $[a, c)$, $[d, b)$, or $[a, c) \sqcup [d, b)$ — in every case, a finite disjoint union of half-open intervals.

This semiring is the natural starting point for constructing Lebesgue measure: one first defines length on $\mathcal{S}$ (as $b - a$ for $[a, b)$), then extends to the generated ring and σ-algebra.

Remark (Why semirings appear in measure theory). Lebesgue measure is most naturally defined first on a semiring, because semirings have enough structure to state the countable additivity condition for a *premeasure*, yet are simple enough that explicit formulas for sets like $[a, b)$ are easy to write down. The general extension theorem (Carathéodory's theorem) then extends the premeasure to the full σ-algebra. This is why Kolmogorov & Fomin introduce semirings before algebras: they are the base case of the extension hierarchy.

Definition (Ring of Sets)

Definition 44.2.2 (Ring of Sets)**.** A collection $\mathcal{R}$ of subsets of X is a *ring of sets* if:

(i) $\varnothing \in \mathcal{R}$;

(ii) $A, B \in \mathcal{R}$ implies $A \cup B \in \mathcal{R}$;

(iii) $A, B \in \mathcal{R}$ implies $A \setminus B \in \mathcal{R}$.

Remark (Dependencies).

- Subset
- Empty Set
- Union
- Set Difference

Remark (Derived closure properties of a ring). A ring is closed under finitely many of each operation. Specifically, if $\mathcal{R}$ is a ring then:

- $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$ (intersection follows from set difference);
- $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{R}$ (symmetric difference follows from union and difference).

A ring is closed under all finite Boolean combinations of its elements.

Remark (Why "ring"). The name comes from algebra. Under the operations of symmetric difference $\triangle$ (playing the role of addition) and intersection $\cap$ (multiplication), a ring of sets is a commutative ring in the algebraic sense: $\triangle$ is commutative and associative with identity $\varnothing$, every element is its own inverse ($A \triangle A = \varnothing$), and $\cap$ distributes over $\triangle$.

Definition (Algebra of Sets)

Definition 44.2.3 (Algebra of Sets)**.** A collection $\mathcal{A}$ of subsets of X is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

(i) $X \in \mathcal{A}$;

(ii) $A \in \mathcal{A}$ implies $A^c = X \setminus A \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$.

Remark (Dependencies).

- Subset
- Complement
- Union

Remark (Derived properties of an algebra). An algebra is a ring that also contains X. Since $\varnothing = X^c$ and $X \in \mathcal{A}$, every algebra contains $\varnothing$. Closure under finite intersection follows: $A \cap B = (A^c \cup B^c)^c \in \mathcal{A}$ by De Morgan. An algebra is therefore closed under all finite Boolean operations: finite unions, finite intersections, complements, and differences.

Definition (σ-Algebra)

Definition 44.2.4 (σ-Algebra)**.** A collection $\mathcal{M}$ of subsets of X is a σ*-algebra* (or σ*-field*) on X if:

(i) $X \in \mathcal{M}$;

(ii) $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$;

(iii) if $\{A_n\}_{n\in\mathbb{N}}$ is a countable family in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space.*

Remark (Dependencies).

- Subset
- Empty Set
- Complement
- Union
- Countable Set

Remark (Derived closure properties of a σ-algebra). Every σ-algebra is an algebra (take the union of a finite family by padding with $\varnothing$). But a σ-algebra is additionally closed under countable intersections: by De Morgan,

$$\bigcap_{n=1}^{\infty} A_n \;=\; \Big(\bigcup_{n=1}^{\infty} A_n^c\Big)^c \;\in\; \mathcal{M}.$$

The σ in "σ-algebra" is notation for "countable" (from the German *Summe*), signalling the step up from finite to countable closure.

Remark (Why countable and not arbitrary). A σ-algebra is closed under countable unions but not arbitrary ones. This is not a deficiency — it is the right condition for measure theory. Measure is defined to be countably additive: if $\{A_n\}$ is a countable disjoint family, then $\mu\big(\bigcup A_n\big) = \sum \mu(A_n)$. Extending to uncountable unions would force every set to have either measure zero or infinite measure (a useless theory). The countability restriction is what makes the theory nontrivial.

By contrast, a topology (defined below) is closed under *arbitrary* unions but only *finite* intersections. This is a different trade-off: open sets are defined by local conditions, and any union of open sets is open because being open is a pointwise property. Countable intersections of open sets are *not* required to be open; those are a separate class called G_δ sets.

Definition (Topology)

Definition 44.2.5 (Topology)**.** A *topology* on a set X is a collection τ of subsets of X satisfying:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha \in I}$ is any family of sets in τ (indexed by any set I), then $\bigcup_{\alpha \in I} U_\alpha \in \tau$;

(iii) if $U_1, \ldots, U_n \in \tau$ (finitely many), then $U_1 \cap \cdots \cap U_n \in \tau$.

The pair (X, τ) is a *topological space*. Members of τ are called *open sets*.

Remark (Dependencies).

- Subset
- Empty Set
- Indexed Family of Sets
- Indexed Union
- Indexed Intersection
- Finite and Infinite Sets

Remark (Why arbitrary unions but only finite intersections)*.* A topology formalizes the idea of "nearness" without a distance function. The condition that arbitrary unions of open sets are open reflects the following: if a point x is near some set, and that set is a union of smaller sets, then x is near one of those smaller sets. This is a purely local condition and works for any union.

Finite intersections must be open because: if x lies in the interior of U_1 and also in the interior of U_2, it lies in the interior of $U_1 \cap U_2$. This argument works for finitely many sets (take the smallest neighborhood). For a countably infinite intersection, the "smallest neighborhood" might shrink to nothing — consider $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$ in $\mathbb{R}$, which is not open. So infinite intersections of open sets are not required to be open.

Theorem (Intersection of σ-algebras is a σ-algebra)

Theorem 44.2.6 (Intersection of σ-algebras is a σ-algebra)**.** *Let $\{\mathcal{M}_\alpha\}_{\alpha\in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} := \bigcap_{\alpha\in I} \mathcal{M}_\alpha$$

is a σ-algebra on X. Go to proof.

Remark (Standard quantified statement). $\big(\forall \alpha \in I,\ \mathcal{M}_\alpha$ is a σ-algebra on $X\big) \to \bigcap_{\alpha\in I}\mathcal{M}_\alpha$ is a σ-algebra Explicitly: $X \in \bigcap \mathcal{M}_\alpha$; $A \in \bigcap \mathcal{M}_\alpha \to A^c \in \bigcap \mathcal{M}_\alpha$; $\forall n,\ A_n \in \bigcap \mathcal{M}_\alpha \to \bigcup_n A_n \in \bigcap \mathcal{M}_\alpha$.

Remark (Dependencies).

- σ-Algebra
- Indexed Family of Sets
- Intersection

Remark (Proof). Check each axiom: (i) $X \in \mathcal{M}_\alpha$ for all α, so $X \in \bigcap_\alpha \mathcal{M}_\alpha$. (ii) If $A \in \mathcal{M}$ then $A \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}$. (iii) If $\{A_n\} \subseteq \mathcal{M}$ then $\{A_n\} \subseteq \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}$. All three conditions are verified. The same argument works for algebras, rings, and topologies (with appropriate closure conditions).

Definition (Generated σ-Algebra)

Definition 44.2.7 (Generated σ-Algebra)**.** Let $\mathcal{F}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{F}$*, written $\sigma(\mathcal{F})$, is the smallest σ-algebra on X that contains $\mathcal{F}$:

$$\sigma(\mathcal{F}) := \bigcap \big\{ \mathcal{M} \ \big|\ \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \big\}.$$

The collection $\mathcal{F}$ is called a *generating family* for $\sigma(\mathcal{F})$.

Remark (Dependencies).

- σ-Algebra
- Subset
- Intersection
- Intersection of σ-Algebras

Remark (Why the definition is valid). The set of σ-algebras on X containing $\mathcal{F}$ is nonempty: $\mathcal{P}(X)$ is a σ-algebra containing every subset of X. Theorem 44.2.6 guarantees their intersection is again a σ-algebra. So $\sigma(\mathcal{F})$ is always well-defined. The same construction works for generated algebras, generated rings, and (with a small modification for topologies) generated topologies.

Remark (How to think about generated structures). $\sigma(\mathcal{F})$ contains $\mathcal{F}$, and then contains everything that must be present in any σ-algebra that contains $\mathcal{F}$: complements of elements of $\mathcal{F}$, countable unions of elements of $\mathcal{F}$, countable unions of complements of elements of $\mathcal{F}$, and so on — iterating indefinitely. The generated σ-algebra is the "closure" of $\mathcal{F}$ under all σ-algebraic operations. In practice, one specifies $\mathcal{F}$ as a convenient collection (e.g., intervals) and then works with the full σ-algebra by knowing which operations are allowed.

Definition (Borel σ-Algebra)

Definition 44.2.8 (Borel σ-Algebra)**.** Let (X, τ) be a topological space. The *Borel σ-algebra* on X, written $\mathcal{B}(X)$, is the σ-algebra generated by the open sets:

$$\mathcal{B}(X) \; := \; \sigma(\tau).$$

Members of $\mathcal{B}(X)$ are called *Borel sets.*

Remark (Dependencies).

- Topology
- Generated σ-Algebra

Example (F_σ and G_δ sets)*.* In a topological space X:

- A G_δ *set* is a countable intersection of open sets: $A = \bigcap_{n=1}^{\infty} U_n$ with each U_n open.
- An F_σ *set* is a countable union of closed sets: $A = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed.

These names come from German: G for *Gebiet* (open set), δ for *Durchschnitt* (intersection); F for *Ferme* (closed), σ for *Summe* (union). Both G_δ and F_σ sets are Borel sets. They are dual: A is G_δ if and only if A^c is F_σ.

These classes appear in analysis and measure theory as the natural "next level" above open and closed sets. In $\mathbb{R}$: the set of continuity points of a monotone function is a G_δ set; the rational numbers $\mathbb{Q}$ are an F_σ set; the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are a G_δ set.

Remark (Transition)*.* The four structures — semiring, ring, algebra, σ-algebra — and the topology form a hierarchy of increasing closure strength. Measure theory builds from the bottom (semiring, ring) upward; topology is its own branch with a different closure pattern. The Borel σ-algebra bridges the two: it is generated by the topology, so it captures all sets describable by topological operations, and it is the standard "default" σ-algebra in any analytic context. Whenever a later chapter says "let f be a measurable function" without specifying the σ-algebra, it almost certainly means measurability with respect to the Borel σ-algebra.

44.3 The Power Set and Characteristic Functions

44.3.1 The Power Set and Characteristic Functions

Chapter 45

Algebras of Sets

45.1 Set Operations and Hierarchy

45.2 Boolean Algebra of Sets

45.3 Rings of Sets

Bridge — Rings of Sets and Boolean Rings

Concept family: the set-theoretic operations behind the algebraic Boolean-ring model.

Remark (Dictionary).

Set operation	Ring operation
$A \triangle B$	$A + B$
$A \cap B$	$A \cdot B$
$\varnothing$	0
X	1

Under this dictionary, a ring of sets corresponds to a Boolean rng, and an algebra of sets corresponds to a Boolean ring with $1 = X$.

Remark (Structure theory lives in Volume VI). For the algebraic structure theory – idempotence implies characteristic 2, which implies $-p = p$, which implies commutativity, together with Stone representation and the finite-order consequence that no Boolean ring has order 3 – see Volume VI, Section 1.9, "Boolean Rings."

45.4 Sigma Algebras

45.5 Borel Sets

Chapter 46

Measure Spaces

Part V

Volume V: Modern Analysis

Measure Theory

Chapter 47

Measure Theory

SERIES
From Cantor to Ito

VOLUME V
Modern Analysis

BOOK I
Measure Theory

CHAPTER 1
Measure Theory

47.1 Measure Theory Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Probability

Chapter 48

Probability Theory

SERIES
From Cantor to Ito

VOLUME V
Modern Analysis

BOOK II
Probability

CHAPTER 2
Probability Theory

48.1 Probability Theory Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Functional Analysis

Chapter 49

Functional Analysis

SERIES
From Cantor to Ito
VOLUME V
Modern Analysis
BOOK III
Functional Analysis
CHAPTER 3
Functional Analysis

49.1 Functional Analysis Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Complex Analysis

Chapter 50

Complex Analysis

SERIES
From Cantor to Ito

VOLUME V
Modern Analysis

BOOK IV
Complex Analysis

CHAPTER 4
Complex Analysis

50.1 Complex Analysis Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Part VI

Volume VI: Algebra

Algebra

Chapter 51

Introduction to Algebraic Structures

[

Where You Are in the Journey]

Propositional Logic $\to$ Predicate Calculus $\to$ Sets & Functions $\to$ Proof Techniques $\to$ Real Analysis $\to$ **Intro to Algebraic Structures** $\to$ Linear Algebra $\to$ Topology $\to$ $\cdots$

This chapter fixes the vocabulary of algebraic structure first, then builds the standard one-law and two-law structure tower by adding one axiom at a time.

51.1 What Is a Mathematical Structure?

51.1.1 Algebraic Structures

Remark (Geometric picture). Before any specific algebra, fix what kind of thing we are studying: a set of elements together with stipulated ways to combine them, compare them, and name some of them. The axioms say which stipulations we keep. A group, a ring, and a field are the same general idea with different stipulations.

Definition (Algebraic Structure)

Definition 51.1.1 (Algebraic Structure)**.** An **algebraic structure** of signature $\mathcal{L}$ is a pair

$$\mathcal{A} = (A, I)$$

where A is a nonempty carrier set and I interprets each symbol of $\mathcal{L}$ on A: each n-ary function symbol as an operation $A^n \to A$, each n-ary relation symbol as a subset of A^n, and each constant symbol as an element of A. The structure is algebraic when $\mathcal{L}$ has no relation symbols beyond equality.

Remark (Standard quantified statement).

$$\mathrm{AlgStructure}_{\mathcal{L}}(\mathcal{A}) \iff \mathcal{A} = (A, I),\ A \neq \varnothing, \text{ and } I \text{ interprets every symbol of } \mathcal{L} \text{ on } A.$$

Remark (Interpretation). The signature is the type; the structure is the instance. “Group” is a signature with a binary operation, an identity constant, and an inverse operation; a particular group interprets these on a carrier.

Remark (Bourbaki anchor). Species of structure, *Theory of Sets*, Chapter IV, Section 1.

Remark (Dependencies).

- Carrier Set
- First-Order Signature

Definition (First-Order Signature)

Definition 51.1.2 (First-Order Signature)**.** A **first-order signature**

$$\mathcal{L} = (\mathrm{Const}_{\mathcal{L}}, \mathrm{Func}_{\mathcal{L}}, \mathrm{Rel}_{\mathcal{L}}, \mathrm{arity})$$

lists constant symbols, function symbols, and relation symbols, with each function or relation symbol carrying a declared arity.

Remark (Dependencies).

- Nullary Operation
- Unary Operation
- Binary Operation

Definition ($\mathcal{L}$-Structure)

Definition 51.1.3 ($\mathcal{L}$-Structure)**.** Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** is a nonempty carrier set A together with an interpretation of every symbol of $\mathcal{L}$ on A.

Remark (Dependencies).

- First-Order Signature
- Carrier Set

51.1.2 Laws of Composition

Definition (Binary Operation / Law of Composition)

Definition 51.1.4 (Binary Operation / Law of Composition)**.** Let A be a set. A **binary operation** on A (Bourbaki: an internal law of composition) is a function

$$\star : A \times A \to A.$$

The image of (a, b) is written $a \star b$.

Remark (Standard quantified statement).

$$\mathrm{BinOp}(\star, A) \iff \forall a, b \in A,\ a \star b \in A.$$

Remark (Interpretation). The codomain A is closure: combining two elements never leaves A.

Remark (Bourbaki anchor). "Law of composition," *Algebra I*, Section 1, Definition 1. "Binary operation" is the model-theoretic alias used in this volume.

Remark (Dependencies).

- Function (Volume I recall)
- Carrier Set

Definition (n-ary Operation)

Definition 51.1.5 (n-ary Operation)**.** An **n-ary operation** on A is a function $f : A^n \to A$, with $n \geq 0$. It is unary if $n = 1$, binary if $n = 2$, and nullary if $n = 0$.

Remark (Standard quantified statement).

$$\mathrm{Op}_n(f, A) \iff f : A^n \to A.$$

Definition (Unary Operation)

Definition 51.1.6 (Unary Operation)**.** A **unary operation** on A is a function $f : A \to A$.

Definition (Nullary Operation / Constant)

Definition 51.1.7 (Nullary Operation / Constant)**.** A **nullary operation** on A is the selection of a distinguished element $c \in A$; equivalently, it is an operation $A^0 \to A$.

Remark (Forward reference: external laws). An external law or action $\Omega \times A \to A$, whose elements of Ω are operators, is introduced with actions when modules and vector spaces are reached.

51.1.3 Distinguished Elements

Remark (Geometric picture). Some elements have a privileged role under a law: one that does nothing and one that swallows everything. Defining the one-sided versions first lets the two-sided collapse become a theorem rather than an assumption.

Definition 51.1.8 (Left Identity)**.** An element $e_L \in A$ is a **left identity** for $\star$ if $e_L \star a = a$ for all $a \in A$.

Definition 51.1.9 (Right Identity)**.** An element $e_R \in A$ is a **right identity** for $\star$ if $a \star e_R = a$ for all $a \in A$.

Definition 51.1.10 (Identity / Neutral Element)**.** An element $e \in A$ is a **two-sided identity** (or neutral element) for $\star$ if it is both a left and a right identity:

$$e \star a = a \star e = a \quad \text{for all } a \in A.$$

Remark (Standard quantified statement).

$$\mathrm{Identity}(e, \star, A) \iff e \in A \text{ and } \forall a \in A,\ e \star a = a \star e = a.$$

Proposition 51.1.11 (Identity Collapse and Uniqueness)**.** *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique.* *Go to proof.*

Remark (Proof structure). Evaluate $e_L \star e_R$ two ways.

Definition 51.1.12 (Left Absorbing Element)**.** An element $z_L \in A$ is **left absorbing** for $\star$ if $z_L \star a = z_L$ for all $a \in A$.

Definition 51.1.13 (Right Absorbing Element)**.** An element $z_R \in A$ is **right absorbing** for $\star$ if $a \star z_R = z_R$ for all $a \in A$.

Definition 51.1.14 (Absorbing Element)**.** An element $z \in A$ is **absorbing** (a zero element) for $\star$ if it is both left absorbing and right absorbing.

Remark (Standard quantified statement).

$$\mathrm{Absorbing}(z, \star, A) \iff z \in A \text{ and } \forall a \in A,\ z \star a = a \star z = z.$$

Proposition 51.1.15 (Uniqueness of the Absorbing Element)**.** *A two-sided absorbing element, if it exists, is unique.* *Go to proof.*

51.1.4 Inverses

Remark (Geometric picture). An inverse undoes an element, returning the identity. Undoing from the left and undoing from the right can a priori differ; associativity forces them to agree.

Definition 51.1.16 (Left Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $b \in A$ is a **left inverse** of a if $b \star a = e$.

Definition 51.1.17 (Right Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $c \in A$ is a **right inverse** of a if $a \star c = e$.

Definition 51.1.18 (Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $a^{-1} \in A$ is an **inverse** of a if

$$a \star a^{-1} = a^{-1} \star a = e.$$

Remark (Standard quantified statement)*.*

$$\text{Inverse}(b, a, \star, e) \iff a \star b = b \star a = e.$$

Definition 51.1.19 (Invertible Element)**.** An element a is **invertible** (a unit) if it has a two-sided inverse.

Remark (Standard quantified statement)*.*

$$\text{Invertible}(a) \iff \exists b,\ a \star b = b \star a = e.$$

Proposition 51.1.20 (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique.* *Go to proof.*

Remark (Proof structure)*.* Compute $b \star a \star c$ two ways; associativity is essential.

Definition 51.1.21 (Additive Inverse)**.** When $\star = +$ with identity 0, the inverse of a is written $-a$ and is called the **additive inverse** or negative.

Definition 51.1.22 (Multiplicative Inverse)**.** When $\star = \cdot$ with identity 1, the inverse of a is written a^{-1} and is called the **multiplicative inverse**.

Proposition 51.1.23 (Units of a Monoid Form a Group)**.** *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group.* *Go to proof.*

51.1.5 Properties of Laws

Definition 51.1.24 (Associativity)**.** A binary operation $\star$ on A is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in A.$$

Remark (Standard quantified statement)*.*

$$\text{Assoc}(\star, A) \iff \forall a, b, c \in A,\ (a \star b) \star c = a \star (b \star c).$$

Theorem 51.1.25 (General Associativity)**.** *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization.* *Go to proof.*

Remark (Proof structure)*.* Induct on n, reducing every bracketing to the right-nested one.

Definition 51.1.26 (Powers / Iterated Operation)**.** For associative $\star$ with identity e, define $a^0 := e$ and $a^{n+1} := a^n \star a$. In additive notation, define $0a := 0$ and $(n+1)a := na + a$.

Definition 51.1.27 (Commutativity)**.** A binary operation $\star$ on A is **commutative** if $a \star b = b \star a$ for all $a, b \in A$.

Definition 51.1.28 (Left Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **left distributive** over addition if

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

for all $a, b, c \in A$.

Definition 51.1.29 (Right Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **right distributive** over addition if

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

for all $a, b, c \in A$.

Definition 51.1.30 (Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **distributive** over addition if it is both left distributive and right distributive.

Remark (Interpretation). Distributivity is the bridge between the two laws of a ring.

Definition 51.1.31 (Idempotent Operation)**.** A binary operation $\star$ on A is **idempotent** if $a \star a = a$ for all $a \in A$.

51.1.6 Regularity Properties

Remark (Geometric picture). A cancellable element is one you can divide out of an equation even when no inverse exists. Where cancellation fails, the witnesses are zero divisors: the obstruction the ring tower is built to remove.

Definition 51.1.32 (Left Cancellable Element)**.** An element $a \in A$ is **left cancellable** if

$$a \star x = a \star y \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 51.1.33 (Right Cancellable Element)**.** An element $a \in A$ is **right cancellable** if

$$x \star a = y \star a \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 51.1.34 (Cancellable Element)**.** An element $a \in A$ is **cancellable** (regular) if it is both left cancellable and right cancellable.

Proposition 51.1.35 (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable. Go to proof.*

Remark (Proof structure). Left-multiply $a \star x = a \star y$ by a^{-1} and use associativity.

Definition 51.1.36 (Zero Divisor)**.** In a ring R, a nonzero element a is a **zero divisor** if there is a nonzero b with $a \cdot b = 0$ or $b \cdot a = 0$.

Remark (Standard quantified statement).

$$\mathrm{ZeroDivisor}(a, R) \Longleftrightarrow a \neq 0 \text{ and } \exists b \neq 0,\ a \cdot b = 0 \text{ or } b \cdot a = 0.$$

Remark (Carried fact). In any ring, 0 is absorbing for multiplication. This is Proposition 51.1.53, derived from distributivity rather than assumed as an axiom.

51.1.7 One Internal Law

Remark (Structure tower rule). Each structure below is the previous structure plus one axiom. Closure is absorbed into the phrase "binary operation."

Definition 51.1.37 (Magma)**.** A **magma** is a pair $(A, \star)$ where $A \neq \varnothing$ and $\star$ is a binary operation on A.

Definition 51.1.38 (Semigroup)**.** A **semigroup** is a magma whose operation is associative.

Definition 51.1.39 (Monoid)**.** A **monoid** is a semigroup with an identity element.

Definition 51.1.40 (Group)**.** A **group** is a monoid in which every element has an inverse.

Definition 51.1.41 (Abelian Group)**.** An **abelian group** is a group whose operation is commutative.

Definition 51.1.42 (Order of a Group)**.** The **order** of a group G, denoted $|G|$, is the cardinality of its carrier. The group is finite if $|G|$ is finite.

Proposition 51.1.43 (Uniqueness of the Identity in a Group)**.** *The identity of a group is unique.* *Go to proof.*

Remark (Dependency). This specializes Proposition 51.1.11.

Proposition 51.1.44 (Uniqueness of Inverses in a Group)**.** *Each element of a group has a unique inverse.* *Go to proof.*

Remark (Dependency). This specializes Proposition 51.1.20.

Proposition 51.1.45 (Cancellation Laws in a Group)**.** *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.* *Go to proof.*

Proposition 51.1.46 (Socks–Shoes)**.** *In a group, $(ab)^{-1} = b^{-1}a^{-1}$.* *Go to proof.*

51.1.8 Two Internal Laws

Remark (Convention). Throughout this volume, **ring** means unital ring. A non-unital object with ring-like addition, multiplication, associativity, and distributivity is called a **rng**.

Definition 51.1.47 (Rng)**.** A **rng** is a triple $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$. No multiplicative identity is required.

Definition 51.1.48 (Ring)**.** A **ring** is a rng with a multiplicative identity 1; equivalently, $(R, \cdot, 1)$ is a monoid. Thus the axioms are: additive abelian group, multiplicative associativity, distributivity, and unity.

Definition 51.1.49 (Commutative Ring)**.** A **commutative ring** is a ring whose multiplication is commutative.

Definition 51.1.50 (Integral Domain)**.** An **integral domain** is a commutative ring with $0 \neq 1$ and no zero divisors.

Definition 51.1.51 (Division Ring / Skew Field)**.** A **division ring** (or skew field) is a ring with $0 \neq 1$ in which every nonzero element is invertible under multiplication. Multiplication is not required to be commutative.

Definition 51.1.52 (Field)**.** A **field** is a commutative division ring; equivalently, it is a commutative ring with $0 \neq 1$ in which $(F \setminus \{0\}, \cdot)$ is a group.

Proposition 51.1.53 (Multiplication by Zero)**.** *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a. Go to proof.*

Remark (Proof structure)*.* Distribute $a(0 + 0)$ and cancel additively in $(R, +)$.

Proposition 51.1.54 (Multiplication by Additive Inverse)**.** *In a ring,*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

In a ring, $(-1) \cdot a = -a$. Go to proof.

Proposition 51.1.55 (Cancellation in Integral Domains)**.** *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$. Go to proof.*

Proposition 51.1.56 (Every Field is an Integral Domain)**.** *Every field is an integral domain. Go to proof.*

Proposition 51.1.57 (Finite Integral Domain is a Field)**.** *Every finite integral domain is a field. Go to proof.*

51.1.9 The Structural Lattice

Magma	Rng
$\Downarrow$ add associativity	$\Downarrow$ add unity
Semigroup	Ring
$\Downarrow$ add identity	$\Downarrow$ add multiplicative commutativity
Monoid	Commutative Ring
$\Downarrow$ add inverses	$\Downarrow$ add $0 \neq 1$ and no zero divisors
Group	Integral Domain
$\Downarrow$ add commutativity	$\Downarrow$ finite, or add inverses directly
Abelian Group	Field

Remark (Proof-stub inventory)*.* The one-proof-per-file pipeline for this chapter contains: identity-collapse, absorbing-unique, inverse-collapse, units-form-group, general-associativity, invertible-implies-cancellable, group-identity-unique, group-inverse-unique, group-cancellation, socks-shoes, ring-mult-zero, ring-mult-neg, domain-cancellation, field-is-domain, and finite-domain-is-field.

51.2 Relations

51.3 Operations

51.4 Laws of Operations

51.5 Functions

51.6 Distinguished Elements

51.7 Algebraic Structures

51.8 Groups

51.9 Rings

51.10 Fields

51.11 Number Systems as Structures

51.12 Number Systems as Models

Chapter 52

Abstract Algebra

52.1 Induction

52.1.1 Induction

Definitions and Theorems

52.2 Integers

52.2.1 Integers

Basic Definitions and Theorems

52.3 Modular Arithmetic

52.3.1 Modular Arithmetic

Basic Definitions and Theorems

Chapter 53

Algebraic Geometry

SERIES	VOLUME VI	BOOK I	CHAPTER 3
From Cantor to Ito	Algebra	Algebra	Algebraic Geometry

53.1 Algebraic Geometry

53.1.1 Preliminary Definitions

Definition 53.1.1 (Ring)**.** A set R with two binary operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.
2. **Associativity of multiplication:**
$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$
3. **Multiplicative identity:** There exists $1 \in R$ such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$
4. **Distributivity:**
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Remark (Standard quantified statement)*.* A ring is a set equipped with addition and multiplication satisfying the additive-group, multiplicative-associativity, multiplicative-identity, and distributive laws.

Remark (Interpretation)*.* A ring generalizes a field by dropping the requirement that nonzero elements have multiplicative inverses, and by not requiring multiplication to be commutative. When multiplication is commutative, we call R a *commutative ring*.

Definition 53.1.2 (Commutative Ring)**.** A ring R is *commutative* if

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Remark (Standard quantified statement). A ring R is commutative when multiplication in R is commutative.

Remark (Structural Summary). The hierarchy of algebraic structures encountered so far is:

Structure	Additive Group	Multiplicative Structure
Ring	Abelian	Associative, with identity
Commutative Ring	Abelian	Associative, commutative, with identity
Field	Abelian	Abelian on nonzero elements

Every field is a commutative ring, but not every commutative ring is a field.

Polynomial Rings

Definition 53.1.3 (Polynomial)**.** Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression

$$f = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \ldots, a_n \in R$, and x is a formal symbol called an *indeterminate*.

Remark (Standard quantified statement). A polynomial over R is a finite formal sum $\sum_i a_i x^i$ with coefficients $a_i \in R$.

Remark (Interpretation). The indeterminate x is not a variable ranging over values in R. It is a formal placeholder that encodes the coefficient sequence. Two polynomials are equal if and only if all their coefficients are equal.

Definition 53.1.4 (Degree)**.** Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R. If $a_n \neq 0$, then the *degree* of f is

$$\deg(f) := n.$$

The coefficient a_n is called the *leading coefficient* of f. A polynomial with leading coefficient 1 is called *monic*.

Remark (Standard quantified statement). For a nonzero polynomial $f = a_n x^n + \cdots + a_0$ with leading coefficient $a_n \neq 0$, the degree of f is n.

Remark (Interpretation). The zero polynomial $f = 0$ has no leading coefficient. Its degree is left undefined, or assigned $-\infty$ by convention to preserve the identity $\deg(fg) = \deg(f) + \deg(g)$.

Definition 53.1.5 (Polynomial Ring)**.** Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x, denoted $R[x]$, is the set of all polynomials in x with coefficients in R, equipped with addition and multiplication defined by:

$$\left(\sum_i a_i x^i\right) + \left(\sum_i b_i x^i\right) := \sum_i (a_i + b_i) x^i,$$

$$\left(\sum_i a_i x^i\right) \cdot \left(\sum_j b_j x^j\right) := \sum_k \left(\sum_{i+j=k} a_i b_j\right) x^k.$$

Remark (Standard quantified statement)*.* The set $R[x]$ is the collection of finite formal sums in x with coefficients in R, equipped with coefficientwise addition and Cauchy-product multiplication.

Example (Polynomial arithmetic over the integers)*.* Let $R = \mathbb{Z}$ and consider

$$f = 2x^2 + 3x + 1, \qquad g = x + 4 \quad \in \mathbb{Z}[x].$$

Then:

$$f + g = 2x^2 + 4x + 5,$$
$$f \cdot g = 2x^3 + 11x^2 + 13x + 4.$$

Remark (Structural Summary)*.* With these operations, $R[x]$ is itself a commutative ring. The original ring R embeds into $R[x]$ as the constant polynomials.

Property	Holds in $R[x]$?
Commutative ring	Always
Field	Only in degenerate cases
R embeds in $R[x]$	Always

Polynomial Rings in Several Variables

Definition 53.1.6 (Polynomial Ring in n Variables)**.** Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R, denoted

$$R[x_1, x_2, \ldots, x_n],$$

is defined inductively by

$$R[x_1, \ldots, x_n] := R[x_1, \ldots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_\alpha a_\alpha \, x^\alpha,$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_\alpha \in R$, and $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Remark (Standard quantified statement). The ring $R[x_1, \ldots, x_n]$ consists of finite formal sums $\sum_\alpha a_\alpha x^\alpha$ indexed by multi-indices.

Definition 53.1.7 (Monomial)**.** A *monomial* in $R[x_1, \ldots, x_n]$ is a polynomial of the form

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Remark (Standard quantified statement). A monomial in $R[x_1, \ldots, x_n]$ is a single multi-index power x^α.

Definition 53.1.8 (Total Degree)**.** The *total degree* of the monomial x^α is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \ldots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Remark (Standard quantified statement). The total degree of x^α is $|\alpha| = \alpha_1 + \cdots + \alpha_n$.

Example (Example). In $\mathbb{R}[x, y, z]$, the polynomial

$$f = 3x^2y + xy^2z - 5z^3$$

has three terms with total degrees 3, 4, and 3 respectively. Hence $\deg(f) = 4$.

Remark (Relevance to Algebraic Geometry). Polynomial rings in several variables are the foundational algebraic object of algebraic geometry. The geometric objects studied in subsequent sections — affine varieties, ideals, coordinate rings — are all defined in terms of $k[x_1, \ldots, x_n]$ for a field k. The interplay between the algebra of $k[x_1, \ldots, x_n]$ and the geometry of its zero sets is the central theme of Clader–Ross.

Ideals

Definition 53.1.9 (Ideal)**.** Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Remark (Standard quantified statement). An ideal of R is an additive subgroup of R closed under multiplication by arbitrary elements of R.

Definition 53.1.10 (Generated Ideal)**.** Let R be a commutative ring and let $f_1, \ldots, f_m \in R$. The *ideal generated by* $f_1, \ldots, f_m$ is

$$\langle f_1, \ldots, f_m \rangle := \left\{ \sum_{i=1}^{m} r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated*.

Remark (Standard quantified statement). The ideal generated by $f_1, \ldots, f_m$ is the set of all finite R-linear combinations of those generators.

Definition 53.1.11 (Finitely Generated Ideal)**.** An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \ldots, f_m \in R$ such that $I = \langle f_1, \ldots, f_m \rangle$.

Remark (Standard quantified statement). An ideal I is finitely generated when $I = \langle f_1, \ldots, f_m \rangle$ for some finite list $f_1, \ldots, f_m$ of elements of R.

Definition 53.1.12 (Noetherian Ring)**.** A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Remark (Standard quantified statement). A commutative ring R is Noetherian when every ideal $I \subseteq R$ is finitely generated.

Theorem 53.1.13 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*

In particular, $k[x_1, \ldots, x_n]$ is Noetherian for any field k. *Go to proof.*

Remark (Standard quantified statement). If R is Noetherian, then $R[x]$ is Noetherian.

Remark (Interpretation). The Hilbert Basis Theorem guarantees that every ideal in $k[x_1, \ldots, x_n]$ is finitely generated. This is a foundational finiteness result: it means every algebraic variety can be cut out by finitely many polynomial equations.

Quotient Rings

Definition 53.1.14 (Quotient Ring)**.** Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$

$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Remark (Standard quantified statement). For an ideal I of R, the quotient R/I is the set of additive cosets of I with operations induced from R.

Remark (Interpretation). The quotient ring R/I is a commutative ring. Elements of R/I are equivalence classes under the relation $a \sim b \iff a - b \in I$.

Example (A complex quotient). In $\mathbb{R}[x]$, consider the ideal $I = \langle x^2 + 1 \rangle$. The quotient ring

$$\mathbb{R}[x]/\langle x^2 + 1 \rangle \;\cong\; \mathbb{C},$$

since in the quotient, x satisfies $x^2 = -1$, which is precisely the defining relation of $i \in \mathbb{C}$.

Remark (Structural Position). Quotient rings of polynomial rings are the coordinate rings of affine varieties, which are studied in depth in the next chapter. The passage

$$k[x_1, \ldots, x_n] \;\longrightarrow\; k[x_1, \ldots, x_n]/I$$

is the algebraic encoding of restricting from all of $\mathbb{A}^n$ to the variety $V(I)$.

Linear Algebra

Chapter 54

Linear Algebra

SERIES
From Cantor to Ito

VOLUME VI
Algebra

BOOK II
Linear Algebra

CHAPTER 4
Linear Algebra

54.1 Vector Space Preliminaries

Remark (Toolkit — Vector Space Preliminaries)*.* **Concept family:** The concrete foundations for $\mathbf{F}^n$ before the abstract vector space axioms are introduced.

Atomic items in this section:

- Definition: List
- Definition: $\mathbf{F}^n$
- Definition: Addition in $\mathbf{F}^n$
- Theorem: Addition Is Commutative in $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$
- Definition: Additive Inverse in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Key predicate:

$$\mathrm{ListEquality}((x_1,\ldots,x_n),(y_1,\ldots,y_m)) \;:\equiv\; n=m \;\wedge\; \forall k\le n\;(x_k=y_k).$$

Definition (List)

Definition 54.1.1 (List)**.** Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Remark (Standard quantified statement).

$$\text{List}(L) \iff \exists n \in \mathbb{N} \cup \{0\} : \big(\text{Length}(L) = n \wedge L = (x_1, x_2, \ldots, x_n)\big)$$

Remark (Definition predicate reading).

$$\text{List}(L) \iff \exists n \in \mathbb{N}_0, \exists (x_1, \ldots, x_n) : L = (x_1, \ldots, x_n)$$

Remark (Interpretation). A list is the primary mechanism for discussing ordered finite sets of vectors or scalars. Unlike a set, the structural integrity of a list depends on both the identity and the position of its members.

- **Order Sensitivity**: The lists (a, b) and (b, a) are distinct unless $a = b$.
- **Finiteness**: By Axler's convention, a list must have a finite number of elements. Infinite ordered collections are categorized as sequences.
- **Empty List**: A list of length $n = 0$ is called the empty list, denoted by $()$.

Remark (Dependencies).

- Function
- Whole Numbers

Definition ($\mathbf{F}^n$)

Definition 54.1.2 ($\mathbf{F}^n$). Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Remark (Standard quantified statement).

$$\begin{aligned} &x \in \mathbf{F}^n \\ &\iff \quad \exists x_1, \ldots, x_n \in \mathbf{F} \text{ such that } x = (x_1, \ldots, x_n). \end{aligned}$$

Remark (Definition predicate reading).

$$\text{Fn}(\mathbf{F}, n) \;:\equiv\; \mathbf{F}^n = \{(x_1, \ldots, x_n) : x_k \in \mathbf{F}\}.$$

Remark (Interpretation). An element of $\mathbf{F}^n$ is an ordered list of n scalars from $\mathbf{F}$. The order matters, and the list is completely specified by its coordinates.

Remark (Dependencies).

- Definition: List

Definition (Addition in $\mathbf{F}^n$)

Definition 54.1.3 (Addition in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\forall x, y \in \mathbf{F}^n, \\ &\quad x = (x_1, \ldots, x_n) \wedge y = (y_1, \ldots, y_n) \;\Rightarrow\; x + y = (x_1 + y_1, \ldots, x_n + y_n). \end{aligned}$$

Remark (Interpretation)*.* Addition in $\mathbf{F}^n$ is defined coordinatewise: add the first coordinates, then the second coordinates, and so on through the nth coordinates.

Remark (Dependencies)*.*

- Definition: $\mathbf{F}^n$

Theorem (Addition Is Commutative in $\mathbf{F}^n$)

Theorem 54.1.4 (Addition Is Commutative in $\mathbf{F}^n$)**.** *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall x, y \in \mathbf{F}^n,\; x + y = y + x.$$

Remark (Interpretation)*.* Because addition in $\mathbf{F}^n$ is coordinatewise and scalar addition in $\mathbf{F}$ is commutative, vector addition in $\mathbf{F}^n$ is commutative.

Remark (Dependencies)*.*

- Definition: Addition in $\mathbf{F}^n$

Definition (Zero Vector in $\mathbf{F}^n$)

Definition 54.1.5 (Zero Vector in $\mathbf{F}^n$)**.** The symbol 0 denotes the list of length n all of whose coordinates are 0:

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Standard quantified statement)*.*

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Interpretation). The zero vector is the vector whose every coordinate is the scalar 0 in $\mathbf{F}$. It is the additive identity for $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Definition (Additive Inverse in $\mathbf{F}^n$)

Definition 54.1.6 (Additive Inverse in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \dots, -x_n).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\forall x \in \mathbf{F}^n,\\ &\quad x = (x_1, \dots, x_n) \Rightarrow -x = (-x_1, \dots, -x_n).\end{aligned}$$

Remark (Interpretation). The additive inverse in $\mathbf{F}^n$ is formed coordinatewise by taking the additive inverse of each coordinate in $\mathbf{F}$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$

Definition (Scalar Multiplication in $\mathbf{F}^n$)

Definition 54.1.7 (Scalar Multiplication in $\mathbf{F}^n$)**.** Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \dots, \lambda x_n).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\forall \lambda \in \mathbf{F}\ \forall x \in \mathbf{F}^n,\\ &\quad x = (x_1, \dots, x_n) \Rightarrow \lambda x = (\lambda x_1, \dots, \lambda x_n).\end{aligned}$$

Remark (Interpretation). Scalar multiplication in $\mathbf{F}^n$ is defined coordinatewise: multiply each coordinate of the vector by the scalar λ.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Remark (Toolkit — Vector Spaces). **Concept family:** The abstract vector space axioms and their immediate consequences.

Atomic items in this section:

- Definition: Vector Space (10 axioms)
- Definition: Vector and Point
- Definition: Real Vector Space
- Definition: Complex Vector Space
- Theorem: Unique Additive Identity
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$
- Theorem: $a0 = 0$
- Theorem: $(-1)v = -v$

Key predicate:

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot).$$

The 10 axioms are closure, commutativity, associativity, additive identity, additive inverse, scalar closure, both distributive laws, scalar associativity, and the scalar identity.

Definition (Vector Space)

Definition 54.1.8 (Vector Space)**.** A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \to V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

(i) $u + v \in V$;

(ii) $u + v = v + u$;

(iii) $(u + v) + w = u + (v + w)$;

(iv) there exists $0 \in V$ such that $v + 0 = v$;

(v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;

(vi) $av \in V$;

(vii) $a(u + v) = au + av$;

(viii) $(a + b)v = av + bv$;

(ix) $a(bv) = (ab)v$;

(x) $1v = v$.

Remark (Standard quantified statement).

$$\begin{aligned}
&V \text{ is a vector space over } \mathbf{F} \\
&\quad \iff \quad V \text{ is closed under addition and scalar multiplication,} \\
&\qquad \text{addition on } V \text{ is commutative and associative,} \\
&\qquad V \text{ has an additive identity and additive inverses,} \\
&\qquad \text{and scalar multiplication satisfies the two distributive laws,} \\
&\qquad \text{the associative law for scalar multiplication, and the identity law.}
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot)$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot) \iff \text{at least one of the ten axioms fails for some } u, v, w \in V,\ a, b \in \mathbf{F}.$$ ∎

Remark (Interpretation). A vector space is a set whose elements can be added together and multiplied by scalars, with these operations satisfying the natural algebraic rules familiar from $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$ (motivating example)
- Definition: Addition in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Definition (Vector and Point)

Definition 54.1.9 (Vector and Point)**.** An element of a vector space V is called a **vector**.
When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Remark (Standard quantified statement).

$$v \in V \implies v \text{ is a vector},$$
$$x \in \mathbf{F}^n \implies x \text{ is a point of } \mathbf{F}^n.$$

Remark (Interpretation). The word *vector* emphasizes the algebraic role of an element of V. In $\mathbf{F}^n$, the same object can also be viewed geometrically as a point with coordinates in $\mathbf{F}$.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Real Vector Space)

Definition 54.1.10 (Real Vector Space)**.** A **real vector space** is a vector space over $\mathbb{R}$.

Remark (Standard quantified statement).

$$V \text{ is a real vector space} \iff V \text{ is a vector space over } \mathbb{R}.$$

Remark (Definition predicate reading).

$$\mathrm{RealVectorSpace}(V) :\equiv \mathrm{VectorSpace}(V, \mathbb{R}, +, \cdot).$$

Remark (Interpretation). In a real vector space, the scalars used in scalar multiplication are real numbers.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Complex Vector Space)

Definition 54.1.11 (Complex Vector Space)**.** A **complex vector space** is a vector space over $\mathbb{C}$.

Remark (Standard quantified statement)*.*

$$V \text{ is a complex vector space} \iff V \text{ is a vector space over } \mathbb{C}.$$

Remark (Definition predicate reading)*.*

$$\mathrm{ComplexVectorSpace}(V) \;:\equiv\; \mathrm{VectorSpace}(V, \mathbb{C}, +, \cdot).$$

Remark (Interpretation)*.* In a complex vector space, the scalars used in scalar multiplication are complex numbers.

Remark (Notation)*.* For each positive integer m, the notation $\mathbf{F}^m$ denotes the vector space of all lists of length m of elements of $\mathbf{F}$.

The notation $\mathbf{F}^\infty$ denotes the set of all infinite lists

$$(x_1, x_2, x_3, \dots)$$

with each $x_j \in \mathbf{F}$, equipped with coordinatewise addition and coordinatewise scalar multiplication.

Remark (Dependencies)*.*

- Definition: Vector Space

Remark (Coordinates in Vector Spaces)*.* A pointwise proof is available when vectors are given explicitly by coordinates, as in $\mathbf{F}^n$, because the operations are defined coordinatewise. In an arbitrary vector space, vectors need not come with coordinates, so proofs must use the vector space axioms rather than coordinate calculations.

Theorem (Unique Additive Identity)

Theorem 54.1.12 (Unique Additive Identity)**.** *A vector space has a unique additive identity.* *Go to proof.*

Remark (Standard quantified statement)*.*

$$\begin{aligned}&\text{If } 0, 0' \in V \text{ each satisfy}\\ &\qquad \forall v \in V,\ v + 0 = v \quad \text{and} \quad \forall v \in V,\ v + 0' = v,\\ &\text{then } 0 = 0'.\end{aligned}$$

Remark (Interpretation)*.* Although the vector space axioms assert the existence of an additive identity, they force that identity to be unique.

Remark (Dependencies)*.*

- Definition: Vector Space

Theorem (Unique Additive Inverse)

Theorem 54.1.13 (Unique Additive Inverse)**.** *Every vector in a vector space has a unique additive inverse. Go to proof.*

Remark (Standard quantified statement)*.*

$$\forall v \in V,\ \text{if } w, u \in V \text{ satisfy } v + w = 0 \text{ and } v + u = 0,$$
$$\text{then } w = u.$$

Remark (Interpretation)*.* The vector space axioms guarantee that each vector has an additive inverse, and that inverse is determined uniquely by the requirement that the sum be 0.

Remark (Notation)*.* If $v \in V$, then the unique additive inverse of v is denoted by $-v$.
If $w, v \in V$, then

$$w - v \ := \ w + (-v).$$

Remark (Dependencies)*.*

- Definition: Vector Space
- Theorem: Unique Additive Identity

Theorem ($0v = 0$)

Theorem 54.1.14 ($0v = 0$)**.** *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall v \in V,\ 0v = 0.$$

Remark (Interpretation)*.* Multiplying any vector by the scalar 0 gives the zero vector. The 0 on the left is the scalar zero in $\mathbf{F}$; the 0 on the right is the zero vector in V.

Remark (Dependencies)*.*

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem ($a0 = 0$)

Theorem 54.1.15 ($a0 = 0$)**.** *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbf{F},\ a0 = 0.$$

Remark (Interpretation). Every scalar sends the zero vector to the zero vector. The 0 on the right of a is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem ($(-1)v = -v$)

Theorem 54.1.16 ($(-1)v = -v$)**.** *For every $v \in V$,*

$$(-1)v = -v.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ (-1)v = -v.$$

Remark (Interpretation). Scalar multiplication by -1 produces the additive inverse of a vector. This justifies the notation $-v$ for the additive inverse: it is literally (-1) times v.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$

Remark (Toolkit — Vector Subspaces). **Concept family:** Subspaces, their sums, and direct sums.

Atomic items in this section:

- Definition: Subspace
- Theorem: Conditions for a Subspace
- Definition: Sum of Subspaces
- Theorem: Sum of Subspaces Is the Smallest Containing Subspace
- Definition: Direct Sum
- Theorem: Condition for a Direct Sum

- Theorem: Direct Sum of Two Subspaces

Key predicates:

Subspace(U, V), ClosedUnderAddition(U), ClosedUnderScalarMultiplication$(U, \mathbf{F})$, DirectSu

Key test: U is a subspace $\iff$ $0 \in U$, U closed under $+$, U closed under scalar multiplication.

Definition (Subspace)

Definition 54.1.17 (Subspace)**.** Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\iff \quad U \subseteq V \text{ and } U \text{ is a vector space over } \mathbf{F} \\ &\quad \text{under the operations inherited from } V. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Subspace}(U, V) \;:\equiv\; U \subseteq V \;\wedge\; \mathrm{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Subspace}(U, V) \iff U \not\subseteq V \;\vee\; \neg\,\mathrm{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Interpretation). A subspace is a subset of a vector space that is closed under the vector-space operations and therefore forms a vector space in its own right.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Conditions for a Subspace)

Theorem 54.1.18 (Conditions for a Subspace)**.** *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$;

(ii) for all $u, v \in U$, one has $u + v \in U$;

(iii) for all $a \in \mathbf{F}$ and all $u \in U$, one has $au \in U$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\quad\iff \quad 0 \in U \\ &\qquad\quad \wedge\ \text{ClosedUnderAddition}(U) \\ &\qquad\quad \wedge\ \text{ClosedUnderScalarMultiplication}(U, \mathbf{F}). \end{aligned}$$

Remark (Failure modes). U fails to be a subspace via exactly one of three routes: it omits the zero vector, it contains two vectors whose sum falls outside U, or it contains a vector and a scalar whose product falls outside U.

Remark (Contrapositive quantified statement).

$$\neg\ \text{Subspace}(U, V) \iff 0 \notin U \vee \neg\ \text{ClosedUnderAddition}(U) \vee \neg\ \text{ClosedUnderScalarMultiplication}(U, \mathbf{F}).$$

Remark (Interpretation). To check that a subset is a subspace, it suffices to verify the three conditions. The remaining seven vector space axioms are inherited automatically from V since U uses the same operations. The three conditions are minimal: condition (i) ensures U is non-empty; (ii) and (iii) ensure the inherited operations stay within U.

Remark (Dependencies).

- Definition: Subspace
- Definition: Vector Space

Definition (Sum of Subspaces)

Definition 54.1.19 (Sum of Subspaces)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. The **sum** of $U_1, \ldots, U_m$ is the set

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \ldots, u_m \in U_m\}.$$

Remark (Standard quantified statement).

$$\begin{aligned} &v \in U_1 + \cdots + U_m \\ &\quad\iff \quad \exists u_1 \in U_1 \ \cdots\ \exists u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m. \end{aligned}$$

Remark (Definition predicate reading).

$$\text{SumOfSubspaces}(W, \{U_1, \ldots, U_m\}, V) \ :\equiv\ W = U_1 + \cdots + U_m.$$

Remark (Interpretation). The sum of subspaces consists of all vectors that can be built by adding together one vector from each subspace. Vectors from different subspaces interact; the representation $v = u_1 + \cdots + u_m$ need not be unique.

Remark (Dependencies).

- Definition: Subspace

Theorem (Sum of Subspaces Is the Smallest Containing Subspace)

Theorem 54.1.20 (Sum of Subspaces Is the Smallest Containing Subspace)**.** *Let V be a vector space over* $\mathbf{F}$*, and let $U_1, \ldots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \ldots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \ldots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\begin{aligned}
&\text{Subspace}(U_1 + \cdots + U_m,\, V),\\
&U_j \subseteq U_1 + \cdots + U_m \quad \text{for } j = 1, \ldots, m,\\
&\forall W \subseteq V\ \Big[\big(W \text{ is a subspace of } V \wedge U_1 \subseteq W \wedge \cdots \wedge U_m \subseteq W\big)\\
&\qquad\qquad \Rightarrow\ U_1 + \cdots + U_m \subseteq W\Big].
\end{aligned}$$

Remark (Interpretation)*.* The sum is the smallest subspace containing all the given subspaces: it is their join in the lattice of subspaces of V. Any subspace containing all of $U_1, \ldots, U_m$ must contain every sum $u_1 + \cdots + u_m$, so it contains the sum subspace.

Remark (Dependencies)*.*

- Definition: Sum of Subspaces
- Theorem: Conditions for a Subspace

Definition (Direct Sum)

Definition 54.1.21 (Direct Sum)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V such that

$$V = U_1 + \cdots + U_m.$$

Then V is called the **direct sum** of $U_1, \ldots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \cdots + u_m, \qquad u_1 \in U_1, \ldots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \cdots \oplus U_m.$$

Remark (Standard quantified statement)*.*

$$\begin{aligned}
&V = U_1 \oplus \cdots \oplus U_m\\
&\quad \iff \quad V = U_1 + \cdots + U_m\\
&\qquad \wedge\ \forall v \in V\ \exists!\, u_1 \in U_1\ \cdots\ \exists!\, u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m.
\end{aligned}$$

Remark (Definition predicate reading).

$$\text{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \;:\equiv\; V = U_1 \oplus \cdots \oplus U_m.$$

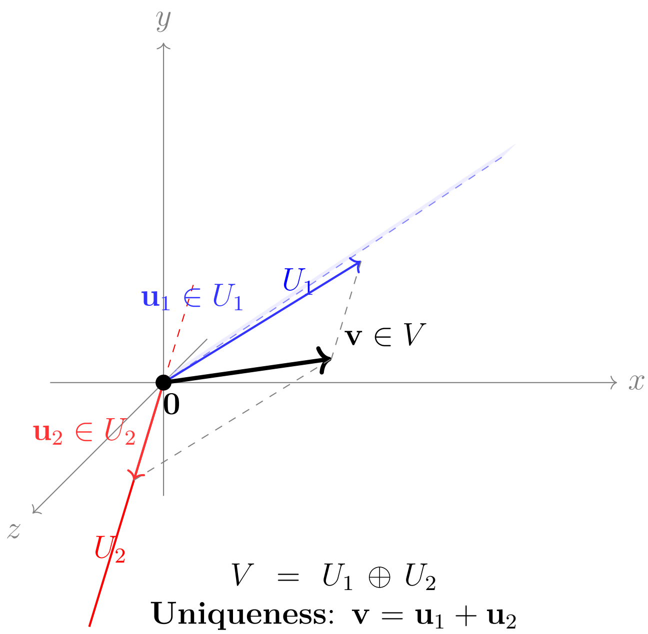
Remark (Negated quantified statement).

$$\neg\, \text{DirectSum}(V, \{U_1, \ldots, U_m\}, \mathbf{F}) \iff \exists v \in V\ \exists \text{ two distinct decompositions of } v.$$

Remark (Interpretation). A direct sum is a sum in which every vector has a unique decomposition into pieces coming from the given subspaces. The uniqueness condition is the structural rigidity that makes direct sums so useful — it means the subspaces contribute independently, with no redundancy.

Remark (Dependencies).

- Definition: Sum of Subspaces



$V \;=\; U_1 \oplus U_2$
Uniqueness: $\mathbf{v} = \mathbf{u}_1 + \mathbf{u}_2$
Trivial Intersection:
$U_1 \cap U_2 \;=\; \{\mathbf{0}\}$

Theorem (Condition for a Direct Sum)

Theorem 54.1.22 (Condition for a Direct Sum)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$*;*

(ii) if $u_1 \in U_1, \ldots, u_m \in U_m$ *and*

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} & V = U_1 \oplus \cdots \oplus U_m \\ & \quad \iff \quad \forall u_1 \in U_1 \ \cdots \ \forall u_m \in U_m \\ & \qquad \big(u_1 + \cdots + u_m = 0 \Rightarrow u_1 = \cdots = u_m = 0\big). \end{aligned}$$

Remark (Interpretation). A sum is direct exactly when the zero vector has only the trivial decomposition as a sum of vectors from the given subspaces. This is the practical test: rather than verifying uniqueness for every $v \in V$, it suffices to check uniqueness at $v = 0$.

Remark (Dependencies).

- Definition: Direct Sum
- Theorem: Unique Additive Identity

Theorem (Direct Sum of Two Subspaces)

Theorem 54.1.23 (Direct Sum of Two Subspaces)**.** *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} & U + W = U \oplus W \\ & \quad \iff \quad U \cap W = \{0\}. \end{aligned}$$

Remark (Contrapositive quantified statement).

$$U \cap W \neq \{0\} \iff U + W \neq U \oplus W.$$

If the two subspaces share a nonzero vector, the sum is not direct.

Remark (Interpretation). For two subspaces, directness is equivalent to saying they overlap only in the zero vector. This gives a simple geometric test: draw both subspaces and check whether their intersection is just the origin. The result does not generalise to three or more subspaces — it is possible for $U_1 \cap U_2$, $U_1 \cap U_3$, and $U_2 \cap U_3$ to all equal $\{0\}$ yet $U_1 + U_2 + U_3$ still fail to be a direct sum.

Remark (Dependencies).

- Theorem: Condition for a Direct Sum
- Definition: Direct Sum

Lattice and Order Theory

Chapter 55

Lattice Order

SERIES
From Cantor to Ito

VOLUME VI
Algebra

BOOK III
Lattice and Order Theory

CHAPTER 5
Lattice Order

55.1 Breadcrumb

Why Lattice Theory Gets Its Own Home

Remark (Why lattice theory gets its own home). Lattice theory begins with a simple observation: in many ordered structures, two elements have a natural least upper bound and a natural greatest lower bound. Once those operations exist everywhere, order starts behaving like algebra.

Remark (Where the first pass already lives). The first introduction to lattices is already in the lattice section of the order notes. That is the right place for the first encounter, because it grows directly out of partial orders, suprema, and infima.

Remark (Why return later). Once lattices stop being examples and start becoming the main character, the story gets richer. A fuller lattice-theory chapter will naturally gather topics such as:

- distributive, modular, and Boolean lattices;
- complete lattices and closure operators;
- sublattices, ideals, and filters;
- Galois connections and adjoint situations;
- fixed-point theorems such as Knaster–Tarski;
- the lattice viewpoint behind topologies, sigma-algebras, and domain theory.

Remark (Why this matters for the rest of the curriculum)*.* This chapter is a breadcrumb because lattice order quietly reappears all over the place. Power sets form Boolean lattices, sigma-algebras behave like closure-stable lattice structures, open sets form a join-rich order, and fixed-point methods show up in logic, measure, computation, and analysis. Giving lattice order its own future home makes those later echoes easier to recognize.

55.2 Ordered Sets

55.2.1 Ordered Sets

Remark (Toolkit: Ordered Sets)*.* Order is a relation, not a property. A set carries no intrinsic ordering; an order is an additional structure imposed on the set by specifying which pairs of elements stand in the ordering relation. The same underlying set can support many distinct orders simultaneously. The atomic notions developed here are:

- the **partial order** — the minimal framework in which order makes sense;
- the **strict partial order** — the asymmetric companion to the partial order, and the native language of open-ball conditions and limits;
- the **total order** — the specialisation in which every pair of elements is comparable.

Definition (Partial Order)

Definition 55.2.1 (Partial Order)**.** A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \Longrightarrow a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \Longrightarrow a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset*.

Remark (Standard quantified statement)*.*

$$\mathrm{PartialOrder}(S,\leq) \;\coloneqq\; \big(\forall a \in S,\ a \leq a\big) \wedge \big(\forall a,b \in S,\ a \leq b \wedge b \leq a \Rightarrow a = b\big) \wedge \big(\forall a,b,c \in S,\ a \leq b \wedge b \leq c$$

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{PartialOrder}(S,\leq) \;\coloneqq\; \big(\exists a \in S,\ a \not\leq a\big) \vee \big(\exists a,b \in S,\ a \leq b \wedge b \leq a \wedge a \neq b\big) \vee \big(\exists a,b,c \in S,\ a \leq b \wedge b \leq c \wedge$$

Remark (Interpretation)*.* Order is not an intrinsic quality of a set's elements — it is an external binary relation. The three axioms together enforce that the relation be a consistent, cycle-free, directed comparison. Reflexivity says every element is at least as large as itself. Anti-symmetry is the formal engine behind squeeze arguments: if $a \leq b$ and $b \leq a$ then a and b cannot be distinct. Transitivity ensures that the ordering chains: comparisons can be composed without breaking consistency. The same set of objects can carry many distinct partial orders simultaneously — standard, reverse, or lexicographic — each defining a different relational geometry.

Remark (Dependencies).

- Partial Order

Definition

Definition 55.2.2 (Strict Partial Order)**.** A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \Longrightarrow \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \Longrightarrow a < c$.

Remark (Standard quantified statement).

$$\mathrm{StrictPartialOrder}(S, <) \;:=\; \big(\forall a \in S,\ \neg(a < a)\big) \wedge \big(\forall a, b \in S,\ a < b \Rightarrow \neg(b < a)\big) \wedge \big(\forall a, b, c \in S,\ a < b \wedge b$$

Remark (Interpretation). The strict order is the companion of the non-strict order, eliminating reflexivity and replacing anti-symmetry with the stronger asymmetry. Every strict partial order determines a partial order by $a \leq b \iff (a < b \vee a = b)$, and every partial order determines a strict partial order by $a < b \iff (a \leq b \wedge a \neq b)$. In metric spaces and analysis, strict orders are the natural language of open conditions: open balls, open intervals, and limit definitions all use $<$ rather than $\leq$ to exclude boundary contact.

Remark (Dependencies).

- Partial Order

Definition (Total Order)

Definition 55.2.3 (Total Order)**.** A partial order $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \qquad a = b, \qquad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

Remark (Standard quantified statement).

$$\mathrm{TotalOrder}(S, \leq) \;:=\; \mathrm{PartialOrder}(S, \leq) \;\wedge\; \forall a, b \in S,\ (a < b) \oplus (a = b) \oplus (b < a),$$

where $\oplus$ denotes exclusive or. Equivalently, comparability suffices: $\forall a, b \in S,\ a \leq b \vee b \leq a$.

Remark (Interpretation). A partial order permits elements to be incomparable: it is possible that neither $a \leq b$ nor $b \leq a$. A total order removes this possibility entirely — every pair of elements is forced to be comparable. The rational numbers $\mathbb{Q}$ and real numbers $\mathbb{R}$ under their standard orderings are totally ordered fields. By contrast, the power set $\mathcal{P}(X)$ of any set X with more than one element, ordered by inclusion, is typically not a total order: the singletons $\{a\}$ and $\{b\}$ with $a \neq b$ are incomparable.

Remark (Dependencies).

Partial order, strict partial order.

55.2.2 Chains and Antichains

Remark (Toolkit: Chains and Antichains). A poset can contain subsets whose elements are either completely aligned or entirely incomparable. These two extremes are formalised as chains and antichains. They measure the "linearity" and "width" of the order structure respectively.

Definition (Chain)

Definition 55.2.4 (Chain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \ \vee \ y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a total order on C.

Remark (Standard quantified statement).

$$\operatorname{Chain}(C, S, \leq) \ \coloneqq \ C \subseteq S \ \wedge \ \forall x, y \in C, \ x \leq y \vee y \leq x.$$

Remark (Negated quantified statement).

$$\neg \operatorname{Chain}(C, S, \leq) \ \coloneqq \ \exists x, y \in C, \ x \not\leq y \wedge y \not\leq x.$$

Remark (Interpretation). A chain is a subset in which the partial order behaves as a total order: no two elements are left incomparable. The entire real line $\mathbb{R}$ under its standard ordering is itself a chain. Any finite totally ordered set — a ranking, a sequence of nested intervals, a linearly ordered list of stages — is a chain. Chains model the "ladder" structure of order: there is a clear direction from smaller to larger, and the path from any element to any other is unambiguous.

Remark (Dependencies).

Partial order, total order.

Definition (Antichain)

Definition 55.2.5 (Antichain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \Longrightarrow \neg(x \leq y) \wedge \neg(y \leq x).$$

Remark (Standard quantified statement).

$$\operatorname{Antichain}(A, S, \leq) \;\coloneqq\; A \subseteq S \;\wedge\; \forall x, y \in A,\; x \neq y \Rightarrow x \not\leq y \wedge y \not\leq x.$$

Remark (Negated quantified statement).

$$\neg \operatorname{Antichain}(A, S, \leq) \;\coloneqq\; \exists x, y \in A,\; x \neq y \wedge (x \leq y \vee y \leq x).$$

Remark (Interpretation). An antichain is a subset in which the partial order makes no comparisons at all: every pair of distinct elements is incomparable. In the power set $\mathcal{P}(X)$ ordered by inclusion, the collection of all subsets of a fixed cardinality k forms an antichain — no subset of size k contains another. Antichains measure the "width" of a poset: Dilworth's theorem states that the minimum number of chains needed to cover a finite poset equals the maximum size of an antichain. In higher-dimensional structures such as product orders, large antichains arise naturally and are the primary tool for understanding the lateral spread of the order.

Remark (Dependencies).

Partial order.

55.2.3 Partial and Total Maps

Remark (Toolkit: Partial and Total Maps). Before studying maps that preserve order structure, the foundational function types are recorded: a partial map need not be defined everywhere, while a total map is defined on every input. These are the base layer upon which the order-theoretic notions rest.

Definition (Partial Map)

Definition 55.2.6 (Partial Map)**.** Let A and B be sets. A *partial map* $f : A \rightharpoonup B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A,\; \forall y_1, y_2 \in B, \quad (x, y_1) \in f \;\wedge\; (x, y_2) \in f \implies y_1 = y_2.$$

Remark (Standard quantified statement).

$$\operatorname{PartialMap}(f, A, B) \;\coloneqq\; f \subseteq A \times B \wedge \forall x \in A,\; \forall y_1, y_2 \in B,\; (x, y_1) \in f \wedge (x, y_2) \in f \Rightarrow y_1 = y_2.$$

Remark (Interpretation). A partial map assigns to each element of A at most one element of B. Its domain of definition $\operatorname{dom}(f) = \{x \in A : \exists y \in B,\, (x, y) \in f\}$ may be a proper subset of A. Partial maps arise in order theory when a function is well-defined only on elements satisfying some order-theoretic condition — for example, on elements that admit an upper bound.

Definition

Definition 55.2.7 (Total Map)**.** A partial map $f : A \rightharpoonup B$ is a *total map* if it is defined on every element of A:

$$\forall x \in A,\; \exists y \in B, \quad (x, y) \in f.$$

A total map is written $f : A \to B$ and is precisely a function in the usual sense.

Remark (Standard quantified statement).

$$\mathrm{TotalMap}(f, A, B) \;\coloneqq\; \mathrm{PartialMap}(f, A, B) \;\wedge\; \forall x \in A,\; \exists y \in B,\; (x, y) \in f.$$

Remark (Dependencies).

Partial map, Function.

55.2.4 Maps Between Ordered Sets

Remark (Toolkit: Maps Between Ordered Sets). The structure-preserving maps between posets form a hierarchy of increasing strength. An order-preserving map carries the direction of the order forward. An order-embedding carries it in both directions. An order-isomorphism is a surjective order-embedding — a perfect structural identification between two posets.

Definition (Order-Preserving Map)

Definition 55.2.8 (Order-Preserving Map)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, \quad x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Remark (Standard quantified statement).

$$\mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\; x \leq_P y \Rightarrow \varphi(x) \leq_Q \varphi(y).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \exists x, y \in P,\; x \leq_P y \wedge \varphi(x) \not\leq_Q \varphi(y).$$

Remark (Interpretation). An order-preserving map carries the direction of the order from P into Q: whenever x is at or below y in P, $\varphi(x)$ is at or below $\varphi(y)$ in Q. The implication is one-directional — order-preserving maps need not reflect the order back. Two order-preserving maps compose to an order-preserving map, making order-preserving maps the morphisms of the category of posets.

Remark (Dependencies).

Partial order.

Definition (Order-Embedding)

Definition 55.2.9 (Order-Embedding)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is an *order-embedding* if

$$\forall x, y \in P, \quad x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \;:=\; \forall x, y \in P,\; x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

Remark (Interpretation). An order-embedding both preserves and reflects the order: the order structure of P is faithfully copied into Q without distortion. Unlike mere preservation, reflection means that order-relations between images in Q genuinely originate from order-relations in P. Every order-embedding is injective (Lemma 55.2.11) and identifies P order-isomorphically with its image in Q.

Remark (Dependencies).

Order-preserving map.

Definition (Order-Isomorphism)

Definition 55.2.10 (Order-Isomorphism)**.** An order-embedding $\varphi : P \to Q$ that is surjective is an *order-isomorphism.* When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \;:=\; \mathrm{OrderEmbedding}(\varphi, P, Q) \;\wedge\; \forall q \in Q,\; \exists p \in P,\; \varphi(p) = q.$$

Remark (Interpretation). An order-isomorphism is a bijection that perfectly translates the order of P into Q and back. Two order-isomorphic posets are order-theoretically identical: any statement about P expressed purely in terms of $\leq_P$ translates to an equally valid statement about Q under φ. Order-isomorphism is the correct notion of equality for posets.

Remark (Dependencies).

Order-embedding.

Lemma

Lemma 55.2.11 (Every Order-Embedding Is Injective)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.* *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \implies \forall x, y \in P,\; \varphi(x) = \varphi(y) \Rightarrow x = y.$$

Remark (Interpretation). Injectivity follows automatically from the biconditional in the embedding definition. If $\varphi(x) = \varphi(y)$, the image is simultaneously $\leq$ and $\geq$ itself in Q; order-reflection pulls this back to P, and anti-symmetry then forces $x = y$.

Remark (Dependencies).

Order-embedding, anti-symmetry of partial orders.

55.2.5 The Covering Relation

Remark (Toolkit: The Covering Relation). The covering relation identifies the irreducible steps of a partial order: y covers x when $x < y$ and there is nothing strictly between them. In a finite poset the entire order is determined by its covering relation, which is encoded exactly by the Hasse diagram.

Definition (Covering Relation)

Definition 55.2.12 (Covering Relation)**.** Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x, written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y \;:=\; x < y \;\wedge\; \neg\big(\exists z \in P,\; x < z < y\big).$$

Remark (Standard quantified statement).

$$x \prec y \;:=\; x < y \;\wedge\; \forall z \in P,\; x < z \leq y \Longrightarrow z = y.$$

Remark (Negated quantified statement).

$$\neg(x \prec y) \;:=\; x \not< y \;\vee\; \exists z \in P,\; x < z < y.$$

Remark (Interpretation). The covering relation captures the direct steps of the order — the comparisons that cannot be decomposed further. In a finite poset every strict comparison $x < y$ can be resolved into a finite chain of covers $x = x_0 \prec x_1 \prec \cdots \prec x_n = y$, so the covering relation alone determines the full order. The Hasse diagram of a finite poset is exactly the directed graph of the covering relation. In a dense order such as $\mathbb{Q}$ or $\mathbb{R}$, no element covers any other: between any two elements there is always a third.

Remark (Dependencies).

Partial order, strict partial order.

Lemma

Lemma 55.2.13 (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.*
3. *For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \iff \big(\forall x, y \in P,\; x <_P y \Leftrightarrow \varphi(x) <_Q \varphi(y)\big) \iff \big(\forall x, y \in P,\; x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y)\big)$$

Remark (Interpretation). In a finite poset the order relation, the strict order, and the covering relation each determine the others completely. This lemma shows that an order-isomorphism can be detected at any one of these three levels. The covering characterisation is the most useful in practice: since Hasse diagrams encode exactly the covering relation, two finite posets are order-isomorphic precisely when their diagrams are isomorphic as directed graphs.

Remark (Dependencies).

Order-isomorphism, covering relation.

Proposition

Proposition 55.2.14 (Hasse Diagram Characterization for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements.* *Go to proof.*

Remark (Standard quantified statement).

$$P \cong Q \iff \exists \varphi : P \xrightarrow{\sim} Q,\ \forall x, y \in P,\ x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y).$$

Remark (Interpretation). The Hasse diagram is a complete invariant of the order structure of a finite poset. Checking order-isomorphism between finite posets reduces to a graph isomorphism problem on the covering graphs — two finite posets with non-isomorphic Hasse diagrams cannot be order-isomorphic.

Remark (Dependencies).

Characterizations of order-isomorphisms for finite posets, covering relation.

55.2.6 Duality

Remark (Toolkit: Duality). Every partially ordered set has a dual obtained by reversing all order relations. Any statement that holds in every poset therefore holds in every dual poset, which means its order-reversed version also holds universally. This is the Duality Principle — the formal engine behind phrases like "by duality, the corresponding result for infima follows."

Definition (Dual Ordered Set)

Definition 55.2.15 (Dual Ordered Set). Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P, denoted P^{op}, is the set P equipped with the reversed order:

$$x \leq_{P^{\mathrm{op}}} y \iff y \leq_P x.$$

Remark (Standard quantified statement).

$$x \leq_{P^{\mathrm{op}}} y \coloneqq y \leq_P x.$$

Remark (Interpretation). Dualising a poset exchanges the roles of upper and lower: every upper bound becomes a lower bound, every supremum becomes an infimum, and every maximal element becomes a minimal element. In a finite poset the Hasse diagram of P^{op} is obtained from that of P by turning it upside down. The dual of a total order is again a total order.

Remark (Dependencies).

Partial order.

Proposition

Proposition 55.2.16 (Duality Principle). *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.* *Go to proof.*

Remark (Standard quantified statement). Every universally valid statement about partially ordered sets has a universally valid order-dual statement.

Remark (Interpretation). The Duality Principle eliminates the need to re-prove results whose dual is already established. Any theorem about upper bounds, suprema, maximal elements, or top elements automatically yields a theorem about lower bounds, infima, minimal elements, or bottom elements by applying the principle to P^{op}.

Remark (Dependencies).

Dual ordered set.

55.2.7 Extremal Elements

Remark (Toolkit: Extremal Elements). A poset may contain elements that sit at the boundary of the order. There are two distinct senses of "extreme": an element can be extreme *locally* (nothing in the poset exceeds it, though it may be incomparable with some elements) or extreme *globally* (it dominates or is dominated by every element). The definitions below make these distinctions precise, proceeding from global extremes of the whole poset, through local extremes of a subset, to the proposition that finite posets always contain the local kind.

Definition (Bottom Element)

Definition 55.2.17 (Bottom Element). Let $(P, \leq)$ be a partially ordered set. An element $\bot \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \bot \leq x.$$

If a bottom element exists it is unique; it is denoted $\bot_P$ or simply $\bot$.

Remark (Standard quantified statement).

$$\mathrm{Bottom}(\bot, P, \leq) \;:=\; \bot \in P \;\wedge\; \forall x \in P,\ \bot \leq x.$$

Remark (Interpretation). A bottom element is below every element of the poset without exception. In $(\mathcal{P}(X), \subseteq)$ the bottom element is $\varnothing$. The natural numbers have a bottom element; the integers do not. An antichain with more than one element has no bottom element, since no member is below any other.

Remark (Dependencies).

Partial order.

Definition (Top Element)

Definition 55.2.18 (Top Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Remark (Standard quantified statement).

$$\mathrm{Top}(\top, P, \leq) \;\coloneqq\; \top \in P \;\wedge\; \forall x \in P,\; x \leq \top.$$

Remark (Interpretation). Dual to the bottom element via the Duality Principle. In $(\mathcal{P}(X), \subseteq)$ the top element is X itself.

Remark (Dependencies).

Bottom element, duality principle.

Definition

Definition 55.2.19 (Maximal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in* Q if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\mathrm{Max}(Q)$.

Remark (Standard quantified statement).

$$\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; a \leq x \Rightarrow a = x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; \exists x \in Q,\; a < x,$$

that is, some element of Q strictly exceeds a.

Remark (Interpretation). A maximal element asserts only that nothing in Q strictly dominates it. It need not be comparable with every element of Q, so a poset can have multiple maximal elements simultaneously. In domain theory, maximal elements correspond to fully-specified or total objects.

Remark (Dependencies).

Partial order.

Definition

Definition 55.2.20 (Minimal Element)**.** Let $(P, \le)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in* Q if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \le a \implies x = a.$$

Remark (Standard quantified statement)*.*

$$\text{Minimal}(a, Q, \le) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; x \le a \Rightarrow x = a.$$

Remark (Interpretation)*.* Dual to maximal element via the Duality Principle.

Remark (Dependencies)*.*

Maximal element, duality principle.

Definition

Definition 55.2.21 (Greatest Element)**.** Let $(P, \le)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q:

$$\forall x \in Q, \quad x \le m.$$

If it exists it is unique; it is denoted $\max Q$.

Remark (Standard quantified statement)*.*

$$\text{Greatest}(m, Q, \le) \;\coloneqq\; m \in Q \;\wedge\; \forall x \in Q,\; x \le m.$$

Remark (Interpretation)*.* A greatest element is a maximal element that is additionally comparable with every element of Q. When $\max Q$ exists, $\text{Max}(Q) = \{\max Q\}$, but the converse fails: a unique maximal element need not dominate every element.

Remark (Dependencies)*.*

Maximal element.

Definition

Definition 55.2.22 (Least Element)**.** Let $(P, \le)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q:

$$\forall x \in Q, \quad m \le x.$$

If it exists it is unique; it is denoted $\min Q$.

Remark (Standard quantified statement)*.*

$$\text{Least}(m, Q, \le) \;\coloneqq\; m \in Q \;\wedge\; \forall x \in Q,\; m \le x.$$

Remark (Interpretation)*.* Dual to greatest element.

Remark (Dependencies).

Greatest element, duality principle.

Proposition

Proposition 55.2.23 (Every Nonempty Finite Poset Has a Maximal Element)**.** *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \mathrm{Max}(P)$ with $x \leq y$.*
Go to proof.

Remark (Standard quantified statement).

$$P \text{ finite}, \ Q \subseteq P, \ Q \neq \varnothing \implies \mathrm{Max}(Q) \neq \varnothing.$$

$$\forall x \in P, \ \exists y \in \mathrm{Max}(P), \ x \leq y.$$

Remark (Interpretation). Finiteness is essential. The collection of all finite subsets of $\mathbb{N}$, ordered by inclusion, is infinite and has no maximal element. In a finite poset one can always ascend from any element to a ceiling. This property is the finite shadow of Zorn's Lemma and underlies ascending chain arguments throughout algebra and order theory.

Remark (Dependencies).

Maximal element, partial order.

55.2.8 Lifting and Co-Lifting

Remark (Toolkit: Lifting and Co-Lifting). A poset that lacks a bottom or top element can be given one by adjoining a new universal bound. Lifting adjoins a bottom; co-lifting adjoins a top. These constructions are dual to each other and are used throughout domain theory, where a bottom element represents an undefined state.

Definition (Lifting)

Definition 55.2.24 (Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp := P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \qquad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new bottom element $\perp$ adjoined.

Remark (Standard quantified statement).

$$P_\perp = P \cup \{\perp\}, \quad \perp \notin P, \quad \forall x \in P_\perp, \ \perp \leq x, \quad \forall x, y \in P, \ x \leq_{P_\perp} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Lifting adjoins a universal lower bound to a poset that may not have one. In domain theory $\perp$ represents an undefined, divergent, or unspecified state. Any set S viewed as an antichain can be lifted to the *flat poset* $S_\perp$, in which all elements of S are incomparable above $\perp$.

Remark (Dependencies).

Partial order, Union, bottom element.

Definition

Definition 55.2.25 (Co-Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \qquad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new top element $\top$ adjoined.

Remark (Standard quantified statement).

$$P^\top = P \cup \{\top\}, \quad \top \notin P, \quad \forall x \in P^\top,\ x \leq \top, \quad \forall x, y \in P,\ x \leq_{P^\top} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Dual to lifting via the Duality Principle: co-lifting adjoins a universal upper bound. The co-lifting of an antichain produces a flat poset in which all elements are incomparable below $\top$.

Remark (Dependencies).

Lifting, top element, duality principle.

55.2.9 Consistency

Remark (Toolkit: Consistency). Two elements of a poset are consistent when they share a common upper bound — when they can be jointly extended to some element above both of them. In lattice-theoretic settings this is automatic; in general posets it is a genuine condition.

Definition

Definition 55.2.26 (Consistency)**.** Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \ \wedge \ y \leq z.$$

Remark (Standard quantified statement).

$$\mathrm{Consistent}(x, y, S, \leq) \ :=\ \exists z \in S,\ x \leq z \wedge y \leq z.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Consistent}(x, y, S, \leq) \ :=\ \forall z \in S,\ x \not\leq z \vee y \not\leq z.$$

Remark (Interpretation). Consistency asserts that x and y are compatible — there is some element of S above both. In a lattice where binary joins always exist, consistency is automatic: take $z = x \vee y$. In a general poset the condition is nontrivial and may fail. In domain theory, consistency captures the idea that two partial computations describe information that can be combined without contradiction.

Remark (Dependencies).

Partial order.

55.2.10 Down-Sets and Up-Sets

Remark (Toolkit: Down-Sets and Up-Sets). A down-set is a subset of a poset that is closed downward: if an element belongs to it, everything below that element belongs to it too. An up-set is the dual notion — closed upward. Down-sets are the order ideals of lattice theory; up-sets are the order filters. The collection of all down-sets of a poset, ordered by inclusion, is itself an important poset.

Definition (Down-Set)

Definition 55.2.27 (Down-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q,\ \forall y \in P, \quad y \leq x \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{DownSet}(Q, P, \leq) \ :=\ Q \subseteq P \ \wedge\ \forall x \in Q,\ \forall y \in P,\ y \leq x \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{DownSet}(Q, P, \leq) \ :=\ \exists x \in Q,\ \exists y \in P,\ y \leq x \wedge y \notin Q.$$

Remark (Interpretation). A down-set is a downward-closed region of the poset: once an element is included, the entire lower section below it is forced in. Down-sets generalise downward-closed intervals. The collection of all down-sets of P, denoted $\mathcal{O}(P)$ and ordered by inclusion, is itself a poset and in the finite case a distributive lattice. Birkhoff's representation theorem states that every finite distributive lattice arises this way.

Remark (Dependencies).

Partial order.

Definition (Up-Set)

Definition 55.2.28 (Up-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q,\ \forall y \in P, \quad x \leq y \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{UpSet}(Q, P, \leq) \ :=\ Q \subseteq P \ \wedge\ \forall x \in Q,\ \forall y \in P,\ x \leq y \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{UpSet}(Q, P, \leq) \ :=\ \exists x \in Q,\ \exists y \in P,\ x \leq y \wedge y \notin Q.$$

Remark (Interpretation). Dual to down-set via the Duality Principle: the up-sets of P are exactly the down-sets of P^{op}. In topology, filters are special up-sets closed under finite intersection and satisfying a directedness condition. The complement of a down-set is an up-set, and vice versa.

Remark (Dependencies).

Down-set, duality principle.

55.2.11 Families of Sets

Remark (Toolkit: Families of Sets). The power set of any set is naturally ordered by inclusion. Every collection of subsets — whether arising from algebra, topology, or logic — inherits this order. The key observation is that subsets correspond exactly to predicates via their characteristic functions, and that this correspondence is an order-isomorphism.

Remark (Families as ordered sets). Let X be a set. The *power set* $\mathcal{P}(X)$, consisting of all subsets of X, is ordered by inclusion: for $A, B \in \mathcal{P}(X)$,

$$A \leq B \iff A \subseteq B.$$

Any subcollection of $\mathcal{P}(X)$ inherits this order. Such a collection is called a *family of sets*.

Families of sets arise naturally when X carries additional structure. If X is a group G, the set $\mathrm{Sub}(G)$ of all subgroups and the set $\mathrm{NSub}(G)$ of all normal subgroups are each ordered by inclusion. If X is a vector space V, the set $\mathrm{Sub}(V)$ of all subspaces is ordered by inclusion. If X is a ring R, the sets of all subrings and of all ideals of R are similarly ordered. If $(X, \mathcal{T})$ is a topological space, the collections of open sets, closed sets, and clopen sets are each families of sets ordered by inclusion.

A particularly important example is the family $\mathcal{O}(P)$ of all down-sets of a partially ordered set P, ordered by inclusion.

Remark (Predicate formulation). The ordered set $(\mathcal{P}(X), \subseteq)$ admits an equivalent formulation in terms of predicates. A *predicate* on X is a function

$$p : X \to \{T, F\},$$

where T and F denote truth values. Let $\mathcal{P}^*(X)$ denote the set of all predicates on X, ordered by implication:

$$p \leq q \iff \{x \in X : p(x) = T\} \subseteq \{x \in X : q(x) = T\}.$$

Define the map

$$\Phi : \mathcal{P}^*(X) \to \mathcal{P}(X), \qquad \Phi(p) \coloneqq \{x \in X : p(x) = T\}.$$

Proposition

Proposition 55.2.29 (Predicates and Power Sets Are Order-Isomorphic). *The map* $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ *is an order-isomorphism. Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\Phi, \mathcal{P}^*(X), \mathcal{P}(X)).$$

Remark (Interpretation). Every subset of X can be represented as the truth set of a predicate, and every predicate determines a subset. The correspondence is bijective and order-preserving in both directions: $p \leq q$ if and only if the truth set of p is contained in the truth set of q, which is exactly the inclusion order on $\mathcal{P}(X)$. This identification is fundamental in logic, where sets and their characteristic predicates are treated as interchangeable, and in computer science, where sets are often implemented via their membership tests.

Remark (Dependencies).

Order-isomorphism, partial order.

Part VII

Volume VII: Advanced Logic

Category Theory

Chapter 56

Category Theory

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VOLUME VII
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BOOK I
Category Theory

CHAPTER 1
Category Theory

56.1 Category Theory

Remark (Planned content). Active mathematical content has not yet been added.

Model Theory

Chapter 57

Model Theory

57.1 Languages and Structures

L-Structures Toolkit — Quick Reference

Concept	Symbol / Notation
First-order language	$\mathcal{L}$
L-structure	$\mathcal{M} = (M, \mathcal{L}^{\mathcal{M}})$
Domain / universe	$\lvert\mathcal{M}\rvert$ or M
Interpretation of f	$f^{\mathcal{M}} : M^n \to M$
Interpretation of R	$R^{\mathcal{M}} \subseteq M^n$
Interpretation of c	$c^{\mathcal{M}} \in M$
Variable assignment	$s : \mathrm{Var} \to M$
Term evaluation	$s(t) \in M$
Satisfaction	$\mathcal{M} \models \varphi[s]$

First-Order Languages

Definition (First-Order Language $\mathcal{L}$)

Definition 57.1.1 (First-Order Language $\mathcal{L}$)**.** A **first-order language** $\mathcal{L}$ consists of three (possibly empty) sets of non-logical symbols:

- a set $\mathcal{R}$ of **relation symbols** (also called *predicate symbols*), each assigned a fixed arity $n \geq 1$;
- a set $\mathcal{F}$ of **function symbols**, each assigned a fixed **arity** $n \geq 0$ (0-ary function symbols are *constant symbols*);
- a set $\mathcal{C}$ of **constant symbols** (equivalently, 0-ary function symbols — listed separately for emphasis).

Together with the logical symbols common to all first-order languages (variables, connectives $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$, quantifiers $\forall, \exists$, equality $=$, parentheses), the triple $(\mathcal{F}, \mathcal{R}, \mathcal{C})$ determines $\mathcal{L}$.

Remark (Dependencies).

- Arity
- Relation Symbol
- Function Symbol
- Constant Symbol
- Variable
- Logical Connective

Remark (Fully quantified form). $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ where $\mathrm{ar}(f) \in \mathbb{N}$ for each $f \in \mathcal{F}$ and $\mathrm{ar}(R) \in \mathbb{N}_{\geq 1}$ for each $R \in \mathcal{R}$.

Remark (Intuition). A language is a *vocabulary*: it names the operations, relations, and distinguished elements that can appear in sentences, without yet saying what those names mean. The meaning is supplied by a structure.

Remark (Source note). Some sources (e.g. Marker §1.1, Hodges) absorb constant symbols into 0-ary function symbols. Others (e.g. Chang–Keisler) list them separately. The content is equivalent; only the bookkeeping differs.

Remark (Consequence). The set of $\mathcal{L}$-terms and the set of $\mathcal{L}$-formulas are both defined inductively from $\mathcal{L}$; see term evaluation below.

Example (Standard examples of first-order languages).

1. **Language of ordered fields** $\mathcal{L}_{\mathrm{of}}$: function symbols $\{+, \cdot, -\}$ of arities $2, 2, 1$; relation symbol $\{<\}$ of arity 2; constant symbols $\{0, 1\}$. The real field $(\mathbb{R}, +, \cdot, -, <, 0, 1)$ is an $\mathcal{L}_{\mathrm{of}}$-structure.

2. **Language of graphs** $\mathcal{L}_{\mathrm{gr}}$: no function symbols, no constants; one binary relation symbol $\{E\}$ (for "edge").

3. **Language of groups** $\mathcal{L}_{\mathrm{gp}}$: function symbols $\{\cdot, ^{-1}\}$ of arities $2, 1$; constant symbol $\{e\}$.

4. **Pure equality language** $\mathcal{L}_{=}$: no non-logical symbols at all. Structures are arbitrary non-empty sets.

L-Structures

Definition ($\mathcal{L}$-Structure)

Definition 57.1.2 ($\mathcal{L}$-Structure)**.** Let $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ be a first-order language. An **$\mathcal{L}$-structure** $\mathcal{M}$ consists of:

1. a non-empty set M called the **domain** (or **universe**) of $\mathcal{M}$, written $|\mathcal{M}|$;
2. for each n-ary relation symbol $R \in \mathcal{R}$, a relation $R^{\mathcal{M}} \subseteq M^n$;
3. for each n-ary function symbol $f \in \mathcal{F}$, a function $f^{\mathcal{M}} : M^n \to M$;
4. for each constant symbol $c \in \mathcal{C}$, a distinguished element $c^{\mathcal{M}} \in M$.

The superscript notation $f^{\mathcal{M}}, R^{\mathcal{M}}, c^{\mathcal{M}}$ denotes the **interpretation** of each symbol in $\mathcal{M}$.

Remark (Dependencies).

- First-Order Language
- Arity
- Relation Symbol
- Function Symbol
- Constant Symbol
- Relation
- Function
- Cartesian Product
- Subset

Remark (Fully quantified form). $\mathcal{M}$ is an $\mathcal{L}$-structure iff $M \neq \emptyset$, and $\forall f \in \mathcal{F}_{(n)},\ f^{\mathcal{M}} : M^n \to M$, and $\forall R \in \mathcal{R}_{(n)},\ R^{\mathcal{M}} \subseteq M^n$, and $\forall c \in \mathcal{C},\ c^{\mathcal{M}} \in M$.

Remark (Intuition). A structure is an $\mathcal{L}$-vocabulary *made concrete*: each symbol is assigned an actual mathematical object of the correct type. The domain provides the raw material; the interpretations give the symbols their meaning inside that domain.

Remark (Common error). The domain M must be *non-empty*. First-order logic with an empty domain is pathological: $\forall x\, \varphi$ would be vacuously true and $\exists x\, \varphi$ vacuously false for every φ, breaking the duality of the quantifiers.

Remark (Source note). Some sources write $\mathfrak{M}$ (Fraktur) for structures and reserve M for the domain; others use $\mathcal{M}$ throughout and write $|\mathcal{M}|$ for the domain. The latter convention is used here. Marker uses $\mathcal{M}$; Enderton uses $\mathfrak{A}$.

Example (L-structures for familiar mathematical objects).

1. $(\mathbb{R}, +^{\mathbb{R}}, \cdot^{\mathbb{R}}, -^{\mathbb{R}}, <^{\mathbb{R}}, 0^{\mathbb{R}}, 1^{\mathbb{R}})$ is an $\mathcal{L}_{\text{of}}$-structure with domain $M = \mathbb{R}$.

2. $(\mathbb{Q}, +^{\mathbb{Q}}, \cdot^{\mathbb{Q}}, -^{\mathbb{Q}}, <^{\mathbb{Q}}, 0^{\mathbb{Q}}, 1^{\mathbb{Q}})$ is another $\mathcal{L}_{\text{of}}$-structure with domain $M = \mathbb{Q}$.

3. The complete graph K_3 on $\{1, 2, 3\}$ is an $\mathcal{L}_{\text{gr}}$-structure with $E^{K_3} = \{(1,2), (2,1), (1,3), (3,1), (2,3), (3,$

Two different structures can share the same language — the language only fixes the *type* of the interpretations, not their specific content.

Variable Assignments

Definition (Variable Assignment)

Definition 57.1.3 (Variable Assignment)**.** Let $\mathcal{M}$ be an $\mathcal{L}$-structure with domain M, and let $\mathrm{Var} = \{v_1, v_2, v_3, \ldots\}$ be a fixed countable set of first-order variables.
A **variable assignment** (or **environment**) for $\mathcal{M}$ is a function $s : \mathrm{Var} \to M$.
For a variable assignment s, a variable v_i, and an element $a \in M$, write $s[v_i \mapsto a]$ for the assignment that agrees with s on all variables except v_i, where it returns a:

$$s[v_i \mapsto a](v_j) = \begin{cases} a & \text{if } j = i, \\ s(v_j) & \text{if } j \neq i. \end{cases}$$

Remark (Dependencies).

- $\mathcal{L}$-Structure
- Variable
- Function
- Countable

Remark (Fully quantified form). $s : \mathrm{Var} \to M$, and $s[v_i \mapsto a](v_j) = a$ if $j = i$, else $s(v_j)$.

Remark (Intuition). A variable assignment gives temporary values to free variables before evaluating a formula. The $s[v_i \mapsto a]$ notation is the standard tool for handling quantifier rules: to check $\exists v_i\, \varphi$, test whether $\mathcal{M} \models \varphi[s[v_i \mapsto a]]$ for some $a \in M$.

Term Evaluation

Definition (Term Evaluation)

Definition 57.1.4 (Term Evaluation)**.** Let $\mathcal{M}$ be an $\mathcal{L}$-structure and $s : \mathrm{Var} \to M$ a variable assignment. The **evaluation** $s(t) \in M$ of an $\mathcal{L}$-term t under s is defined inductively:

1. If $t = v_i$ is a variable, then $s(t) = s(v_i)$.
2. If $t = c$ is a constant symbol, then $s(t) = c^{\mathcal{M}}$.
3. If $t = f(t_1, \dots, t_n)$ for an n-ary function symbol f, then $s(t) = f^{\mathcal{M}}(s(t_1), \dots, s(t_n))$.

Remark (Dependencies).

- $\mathcal{L}$-Structure
- Variable Assignment
- Term
- Variable
- Arity
- Function Symbol
- Constant Symbol

Remark (Fully quantified form). $s : \mathrm{Var} \to M$, and $s(t) \in M$ for every $\mathcal{L}$-term t. Formally: $s(\cdot)$ is the unique function on terms satisfying clauses (1)–(3).

Remark (Intuition). Term evaluation is the "computation" step: given concrete values for all variables, every term reduces to an element of M. The inductive definition mirrors the inductive definition of terms themselves.

Satisfaction [Stub]

Definition (Satisfaction, $\mathcal{M} \models \varphi[s]$) — Stub

Let $\mathcal{M}$ be an $\mathcal{L}$-structure, s a variable assignment, and φ an $\mathcal{L}$-formula. The **satisfaction relation** $\mathcal{M} \models \varphi[s]$ (read: "$\mathcal{M}$ satisfies φ under s") is defined inductively on the structure of φ:

[To be filled in — clauses for atomic formulas, negation, conjunction, disjunction, implication, existential quantifier, universal quantifier.]

Remark (Fully quantified form). *[Stub — to be completed.]*

Remark (Intuition). Satisfaction is the semantic notion of truth: $\mathcal{M} \models \varphi[s]$ means φ is true in $\mathcal{M}$ when each free variable v_i takes the value $s(v_i)$. The definition proceeds by induction on formula complexity, grounding out on atomic formulas evaluated via term evaluation.

Elementary Equivalence and Substructures [Stub]

Proof Theory

Chapter 58

Proof Theory

58.1 Proof Theory

Remark (Planned content). Active mathematical content has not yet been added.

Type Theory

Chapter 59

Type Theory

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59.1 Type Theory

Remark (Planned content). Active mathematical content has not yet been added.

Lambda Calculus

Chapter 60

Lambda Calculus

60.1 Lambda Terms

Section Toolkit — Quick Reference

Concept	One-line summary
λ-term	Variable x, abstraction $\lambda x.M$, application MN
α-equivalence	$\lambda x.M =_\alpha \lambda y.M[y/x]$ when y fresh
β-reduction	$(\lambda x.M)N \to_\beta M[N/x]$
Normal form	Term with no β-redex; may not exist (non-termination)
Church boolean **T**	$\lambda x.\lambda y.x$
Church boolean **F**	$\lambda x.\lambda y.y$
Church numeral $\overline{n}$	$\lambda f.\lambda x.f^n(x)$ (f applied n times to x)
Type $\tau \to \sigma$	Function type: input of type τ, output of type σ
Typing judgment	$\Gamma \vdash M : \tau$ reads "M has type τ in context Γ"
Curry–Howard	Propositions are types; proofs are terms; $A \to B$ is implication

Untyped Lambda Calculus

Background References

- **First-order logic:** Enderton [1972], Barwise [1977a].
- **General topology:** Kelly [1955].
- **Set theory:** Halmos [1960], van Dalen et al. [1978].
- **Recursion theory:** Rogers [1967].
- **Category theory:** Arbib and Manes [1975], MacLane [1972].

Definition — Lambda Terms

Definition 60.1.1 (Lambda Terms)**.** The set Λ of λ-terms is defined inductively as follows:

1. Every variable x is a λ-term.
2. If M is a λ-term and x is a variable, then
$$\lambda x.\, M$$
is a λ-term.
3. If M and N are λ-terms, then
$$MN$$
is a λ-term.

Remark (Standard quantified statement)*.*

$M \in \Lambda \iff M$ is generated from variables by finitely many applications of $\lambda x.(\,\cdot\,)$ and $(\,\cdot\,)(\,\cdot\,)$. ∎

Remark (Interpretation)*.* The λ-terms are the smallest set closed under three constructions: variables, abstraction (function formation), and application (function evaluation). This inductive definition plays the same role as recursive definitions in analysis or algebra: every λ-term is built from variables by finitely many applications of abstraction and application.

Remark (Structural View)*.* Every λ-term has a tree structure:

- variables are leaves,
- abstractions $\lambda x.\, M$ are unary nodes,
- applications MN are binary nodes.

Thus Λ is a freely generated syntactic algebra.

Part VIII

Volume VIII: Applied and Computational Mathematics

Applied Methods

Chapter 61

Calculus

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Applied Methods

CHAPTER 3
Calculus

61.1 Limits

61.1.1 Limits

Definitions

Definition (Intuitive Limit Idea)**.** The limit $\lim_{x\to a} f(x) = L$ says that the values of $f(x)$ can be made close to L by taking x close to a, without requiring $f(a)$ itself to be defined or equal to L.

Definition (Epsilon-Delta Reference Definition)**.** The statement $\lim_{x\to a} f(x) = L$ means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

In Volume VI this definition is mainly a reference point for interpreting limit notation; detailed proofs belong in Real Analysis.

Definition (One-Sided Limits)**.** The right-hand limit $\lim_{x\to a^+} f(x)$ tracks values with $x > a$. The left-hand limit $\lim_{x\to a^-} f(x)$ tracks values with $x < a$. The two-sided limit exists exactly when both one-sided limits exist and agree.

Definition (Limits at Infinity)**.** The notation $\lim_{x\to\infty} f(x) = L$ describes the value approached by $f(x)$ as x becomes arbitrarily large. The notation $\lim_{x\to a} f(x) = \infty$ describes unbounded growth near a.

Definition (Indeterminate Forms)**.** Forms such as $0/0$, ∞/∞, $0\cdot\infty$, $\infty-\infty$, 1^∞, 0^0, and ∞^0 do not determine a limit by inspection. They signal that algebraic or analytic rewriting is needed.

Definition (Asymptote)**.** A horizontal, vertical, or oblique asymptote records limiting behavior of a graph near infinity or near a point where the function grows without bound.

Computational Methods

Situation	First method to try
Polynomial or rational expression	Direct substitution, then factoring
Radicals	Rationalize with the conjugate
Common factor 0/0	Cancel only after factoring
Trigonometric small-angle form	Use standard trigonometric limits
Bounded expression times vanishing expression	Squeeze theorem
Large rational functions	Compare leading powers

Example (Factoring a Removable Form). For

$$\lim_{x\to 3}\frac{x^2-9}{x-3},$$

factor $x^2-9=(x-3)(x+3)$, cancel the common nonzero factor for $x\neq 3$, and evaluate $x+3$ at $x=3$. The limit is 6.

Theorem Reference

Theorem (Limit Laws)**.** *When the relevant limits exist, sums, differences, scalar multiples, products, and quotients of functions have limits computed by applying the same algebraic operation to the limiting values. Quotients require a nonzero denominator limit.*

Theorem (Squeeze Theorem)**.** *If $g(x)\le f(x)\le h(x)$ near a and*

$$\lim_{x\to a} g(x)=\lim_{x\to a} h(x)=L,$$

then $\lim_{x\to a} f(x)=L$.

Limits Exercises

Status: stubbed.

Exercise sets for limits will be routed here.

61.2 Continuity

61.2.1 Continuity

Definitions

Definition (Continuity at a Point)**.** A function f is continuous at a when $f(a)$ is defined, $\lim_{x\to a} f(x)$ exists, and

$$\lim_{x\to a} f(x)=f(a).$$

Definition (Continuity on an Interval)**.** A function is continuous on an interval when it is continuous at each interior point and has the appropriate one-sided continuity at any endpoint included in the interval.

Definition (Common Discontinuities)**.** A removable discontinuity is a hole, a jump discontinuity has unequal one-sided limiting values, and an infinite discontinuity occurs when function values become unbounded near the point.

Computational Methods

Task	Practical check
Polynomial continuity	Continuous everywhere
Rational continuity	Exclude zeros of the denominator
Piecewise continuity	Check each piece and the junction values
Root/log continuity	Check the natural domain
Classify a discontinuity	Compare one-sided behavior and function value

Example (Piecewise Junction). For a piecewise function at $x = a$, compute $f(a)$, $\lim_{x\to a^-} f(x)$, and $\lim_{x\to a^+} f(x)$. Continuity requires all three quantities to exist and agree.

Theorem Reference

Theorem (Algebra of Continuous Functions)**.** *Sums, differences, scalar multiples, products, quotients with nonzero denominator, and compositions of continuous functions are continuous wherever the resulting expressions are defined.*

Theorem (Intermediate Value Theorem)**.** *If f is continuous on $[a, b]$ and N lies between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that $f(c) = N$.*

Continuity Exercises

Status: stubbed.

Exercise sets for continuity will be routed here.

61.3 Derivatives

61.3.1 Derivatives

Definition

Definition (Derivative as a Difference Quotient Limit)**.** The derivative of f at a is

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h},$$

when this limit exists.

Remark (Slope and Rate of Change). Geometrically, $f'(a)$ is the slope of the tangent line to $y = f(x)$ at $x = a$. In applications, it is the instantaneous rate of change of the output with respect to the input.

Example (Power from the Definition). For $f(x) = x^2$,

$$\frac{f(a+h) - f(a)}{h} = \frac{(a+h)^2 - a^2}{h} = 2a + h,$$

so $f'(a) = 2a$. This is a short computational derivation, not a theorem proof vault entry.

Derivative Algebra

Theorem (Constant, Sum, Difference, and Scalar Multiple Rules)**.** *For differentiable functions f and g and a constant c,*

$$(c)' = 0, \qquad (f+g)' = f' + g', \qquad (f-g)' = f' - g', \qquad (cf)' = cf'.$$

Theorem (Product Rule)**.** *If f and g are differentiable at x, then*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Theorem (Quotient Rule)**.** *If f and g are differentiable at x and $g(x) \neq 0$, then*

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}.$$

Theorem (Chain Rule)**.** *If g is differentiable at x and f is differentiable at $g(x)$, then*

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

Standard Derivative Table

Function	Derivative
x^n	nx^{n-1}
$\sqrt{x}$	$1/(2\sqrt{x})$
$1/x$	$-1/x^2$
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \ln a)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\arcsin x$	$1/\sqrt{1-x^2}$
$\arctan x$	$1/(1+x^2)$

Chain-Rule Patterns

Outer form	Pattern
$(u(x))^n$	$n(u(x))^{n-1}u'(x)$
$e^{u(x)}$	$e^{u(x)}u'(x)$
$\ln(u(x))$	$u'(x)/u(x)$
$\sin(u(x))$	$\cos(u(x))u'(x)$
$\sqrt{u(x)}$	$u'(x)/(2\sqrt{u(x)})$

Remark (Method)*.* When differentiating a composition, name the inside expression u, apply the derivative rule to the outside function, and multiply by u'.

Implicit Differentiation

Remark (Method)*.* Differentiate both sides with respect to x, treating y as a function of x. Every derivative of an expression involving y picks up a factor of dy/dx. Then solve algebraically for dy/dx.

Example (Circle)*.* From $x^2 + y^2 = 1$, differentiate to get

$$2x + 2y\frac{dy}{dx} = 0, \qquad \frac{dy}{dx} = -\frac{x}{y}.$$

Logarithmic Differentiation

Remark (Method)*.* Use logarithmic differentiation when products, quotients, powers with variable exponents, or many repeated factors make direct differentiation unwieldy. Take logarithms, expand using log laws, differentiate implicitly, and multiply by the original function.

Example (Variable Power)*.* If $y = x^x$ for $x > 0$, then

$$\ln y = x \ln x, \qquad \frac{y'}{y} = \ln x + 1, \qquad y' = x^x(\ln x + 1).$$

Applications

Application	Derivative role
Tangent lines	Slope at a point
Linear approximation	$f(a) + f'(a)(x - a)$
Critical points	Solve $f'(x) = 0$ or undefined
Curve sketching	Use f' and f'' sign information
Optimization	Find and compare candidates
Related rates	Differentiate a relation with respect to time

Theorem (Mean Value Theorem)**.** *If f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem (L'Hopital's Rule)**.** *For suitable differentiable functions producing an indeterminate form* $0/0$ *or* ∞/∞*,*

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

when the derivative quotient limit exists under the hypotheses of the rule.

Exercise-vault records are organized by mutable topic paths. Stable exercise set and item identifiers are canonical and are recorded in `exercise-ledger.yaml`.

Derivatives

Status: active.

Derivative Rules

Exercise Set: Derivative Rules **Stable set ID:** `set-derivative-rules-001`. **Mode:** computation. **Topic:** derivatives. **Status:** memorialized.

1. **Stable item ID:** `ex-deriv-rules-001`.

   Exercise. Find the derivative of

   $$y = 7x^5 - 4x^3 + 9x - 2.$$

   Solution. By the power rule and linearity,

   $$\frac{dy}{dx} = 35x^4 - 12x^2 + 9.$$

   Verification. The result is correct.

2. **Stable item ID:** `ex-deriv-rules-002`.

   Exercise. Find the derivative of

   $$f(x) = x^2 \sin x.$$

   Solution. By the product rule,

   $$f'(x) = 2x \sin x + x^2 \cos x.$$

   Verification. The result is corrected. The problem was initially copied as a quotient, but the memorialized exercise is the product $x^2 \sin x$.

   Rework note. Preserve the correction in the ledger so later refactors do not recover the discarded quotient transcription.

3. **Stable item ID:** `ex-deriv-rules-003`.

   Exercise. Find the derivative of

$$y = \frac{x^2+1}{x-3}.$$

   Solution. By the quotient rule,

$$\frac{dy}{dx} = \frac{(x-3)(2x)-(x^2+1)}{(x-3)^2} = \frac{x^2-6x-1}{(x-3)^2}.$$

   Verification. The result is correct, with the final simplification added.

Mixed Derivative Practice

Exercise Set: Mixed Derivative Practice **Stable set ID:** `set-derivative-mixed-001`. **Mode:** computation. **Topic:** derivatives. **Status:** generated placeholder.

This exercise set is reserved for future mixed derivative practice. No solved exercises have been memorialized here yet.

61.4 Integrals

61.4.1 Integrals

Definitions

Definition (Antiderivative)**.** An antiderivative of f on an interval is a function F such that $F' = f$ on that interval.

Definition (Definite Integral)**.** The definite integral $\int_a^b f(x)\,dx$ measures signed accumulation of f over $[a,b]$. Computationally, it is evaluated by antiderivatives whenever the Fundamental Theorem of Calculus applies.

Theorem (Fundamental Theorem of Calculus)**.** *If $F' = f$ on an interval containing a and b, then*

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Integral Algebra

Rule	Formula
Linearity	$\int (af+bg) = a\int f + b\int g$
Interval additivity	$\int_a^b f + \int_b^c f = \int_a^c f$
Reversal	$\int_a^b f = -\int_b^a f$
Zero width	$\int_a^a f = 0$
Constant multiple	$\int_a^b c\,dx = c(b-a)$

Techniques

Technique	Use when
Substitution	The integrand contains a composition and its derivative
Integration by parts	The integrand is a product with a simpler derivative
Trigonometric integrals	Powers of trig functions can be reduced
Trigonometric substitution	Radicals match $a^2 - x^2$, $a^2 + x^2$, or $x^2 - a^2$
Partial fractions	A rational function has a factorable denominator
Improper integrals	The interval is infinite or the integrand is unbounded

Remark (Method). Before choosing a technique, simplify algebraically, check for a direct antiderivative, and look for an inside function whose derivative is also present.

Applications

Application	Typical integral
Area	$\int_a^b (\text{top} - \text{bottom})\,dx$
Volume by slices	$\int_a^b A(x)\,dx$
Volume by shells	$\int 2\pi(\text{radius})(\text{height})\,dx$
Arc length	$\int_a^b \sqrt{1 + (y')^2}\,dx$
Surface area	$\int 2\pi(\text{radius})\,ds$

Integrals Exercises

Status: stubbed.

Exercise sets for integrals will be routed here.

61.5 Sequences and Series

61.5.1 Sequences and Series

Sequences

Definition (Sequence)**.** A sequence is a function from positive integers to numbers, usually written (a_n).

Definition (Convergence of a Sequence)**.** The sequence (a_n) converges to L when a_n approaches L as $n \to \infty$. Computationally, this is evaluated using algebraic simplification, standard limits, squeezing, monotonicity, and comparison.

Series

Definition (Series and Partial Sums)**.** The series $\sum_{n=1}^{\infty} a_n$ is studied through its partial sums

$$s_N = \sum_{n=1}^{N} a_n.$$

The series converges when the sequence (s_N) converges.

Series	Convergence pattern
Geometric $\sum ar^n$	Converges for $\lvert r\rvert < 1$
p-series $\sum 1/n^p$	Converges for $p > 1$

Convergence Tests

Test	Best used when
Term test	Terms do not tend to zero
Comparison	Positive terms resemble a known benchmark
Limit comparison	Ratios to a benchmark have a finite positive limit
Integral test	Terms come from a positive decreasing function
Alternating series test	Signs alternate and magnitudes decrease to zero
Ratio test	Factorials or exponentials appear
Root test	nth powers appear

Power Series

Definition (Power Series)**.** A power series centered at a has the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n.$$

Its convergence is organized by its radius and interval of convergence.

Definition (Taylor and Maclaurin Series)**.** The Taylor series of f centered at a is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The Maclaurin series is the special case $a = 0$.

Sequences and Series Exercises

Status: stubbed.

Exercise sets for sequences and series will be routed here.

61.6 Multivariable Calculus

61.6.1 Multivariable Calculus

Vectors and Curves

Definition (Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$)**.** Vectors encode magnitude and direction. In $\mathbb{R}^2$ and $\mathbb{R}^3$, they are manipulated through component arithmetic, dot products, cross products, projections, and parametrized lines and planes.

Definition (Vector-Valued Function)**.** A vector-valued function $\mathbf{r}(t)$ traces a curve. Its derivative $\mathbf{r}'(t)$ is the velocity vector and points in the tangent direction when nonzero.

Partial Derivatives

Definition (Partial Derivative)**.** For a function $f(x, y)$, the partial derivative f_x differentiates with respect to x while holding y constant. The derivative f_y does the same with respect to y.

Remark (Several-Variable Chain Rule). The chain rule in several variables tracks every path by which a variable can affect the output. In computations, draw the dependency diagram before writing the derivative formula.

Definition (Tangent Plane)**.** For $z = f(x, y)$, the tangent plane at (a, b) is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

when the required differentiability hypotheses hold.

Gradients and Directional Derivatives

Definition (Gradient)**.** The gradient of $f(x, y, z)$ is

$$\nabla f = \langle f_x, f_y, f_z \rangle.$$

It points in the direction of steepest increase.

Definition (Directional Derivative)**.** For a unit vector $\mathbf{u}$,

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}.$$

Remark (Lagrange Multipliers). To optimize f subject to a constraint $g = c$, solve

$$\nabla f = \lambda \nabla g, \qquad g = c,$$

then compare candidate values.

Multiple Integrals

Definition (Double and Triple Integrals)**.** Double and triple integrals accumulate a quantity over a region in the plane or in space:

$$\iint_R f\,dA, \qquad \iiint_E f\,dV.$$

Coordinate system	Common use
Polar	Circular regions in the plane
Cylindrical	Circular symmetry around an axis
Spherical	Radial symmetry around a point

Vector Calculus Theorems

Definition (Line and Surface Integrals)**.** Line integrals accumulate along curves. Surface integrals accumulate over surfaces, often measuring flux through the surface.

Theorem (Green's Theorem)**.** *For suitable planar regions and vector fields, a circulation integral around the boundary equals a double integral of curl over the region.*

Theorem (Stokes' Theorem)**.** *For suitable oriented surfaces, a circulation integral around the boundary equals a surface integral of curl over the surface.*

Theorem (Divergence Theorem)**.** *For suitable solid regions, the flux across the boundary equals the triple integral of divergence over the region.*

Multivariable Calculus Exercises

Status: stubbed.

Exercise sets for multivariable calculus will be routed here.

Chapter 62

Geometric Modeling

62.1 Curves and Surfaces

Parametric Curves and Surfaces — Quick Reference

Curve	$C(u) : [a, b] \to \mathbb{R}^n$; vector-valued, one parameter.
Velocity	$C'(u)$; tangent direction and speed of traversal.
Unit tangent	$\hat{T}(u) = C'(u)/\|C'(u)\|$ at a regular point.
Surface	$S(u, v) : R \to \mathbb{R}^3$; vector-valued, two parameters.
Partial derivatives	$S_u = \partial S/\partial u,\ S_v = \partial S/\partial v$; tangent to isoparametric lines.
Unit normal	$N = (S_u \times S_v)/\|S_u \times S_v\|$ at a regular point.
Tangent plane	Spanned by S_u and S_v at the base point.
Directional deriv.	$D_{\mathbf{v}} S = \nabla S \cdot \hat{\mathbf{v}}$; rate of change in direction $\mathbf{v}$.

62.1.1 Parametric Curves and Surfaces

Parametric Curves

Definition (Parametric Curve)

A *parametric curve* in $\mathbb{R}^n$ is a vector-valued function

$$C : [a,b] \to \mathbb{R}^n, \qquad C(u) = \big(x_1(u),\, x_2(u),\, \dots,\, x_n(u)\big),$$

where each coordinate function $x_i(u)$ is a real-valued function of the scalar parameter u. The interval $[a,b]$ is typically normalised to $[0,1]$.

Definition (Velocity, Regularity, Unit Tangent)

The *velocity vector* (or *tangent vector*) at u is the derivative

$$C'(u) = \big(x_1'(u),\, x_2'(u),\, \dots,\, x_n'(u)\big).$$

The curve is *regular* at u if $C'(u) \neq \mathbf{0}$. At a regular point, the *unit tangent vector* is

$$\hat{T}(u) = \frac{C'(u)}{\|C'(u)\|}.$$

The scalar $\|C'(u)\|$ is the *speed* of traversal.

Remark (Properties of parametric curves). Parametric curves carry structure that implicit curves do not:

- **Direction.** The parameter increases from a to b, giving the curve a natural orientation. Reversing the parametrisation reverses the direction.
- **Velocity and speed.** $C'(u)$ measures both the direction of motion and how fast the curve is being traversed. A curve can be *reparametrised* (change of variable $u = \phi(t)$) without changing its shape; the velocity changes but the trace does not.
- **Endpoint interpolation.** The start and end points are $C(a)$ and $C(b)$, accessible directly.
- **Boundedness.** The curve is defined on a closed interval, so it represents a bounded segment, not an infinite line.
- **Cusps.** Where $C'(u) = \mathbf{0}$, the curve may have a cusp (velocity drops to zero, direction may reverse). These are degenerate points excluded by the regularity condition.

Parametric Surfaces

Definition (Parametric Surface)

A *parametric surface* in $\mathbb{R}^3$ is a vector-valued function of two parameters,

$$S : R \longrightarrow \mathbb{R}^3, \qquad S(u,v) = \big(x(u,v),\ y(u,v),\ z(u,v)\big),$$

where $R \subseteq \mathbb{R}^2$ is the *parameter domain* (typically the unit square $[0,1]^2$).

Partial Derivatives and Isoparametric Curves

Definition (Partial Derivative Vectors)

The *partial derivative vectors* of S at (u,v) are

$$S_u(u,v) = \frac{\partial S}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right), \qquad S_v(u,v) = \frac{\partial S}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right).$$

S_u is the tangent vector to the surface in the u-direction: it is the velocity of the curve obtained by fixing v and varying u. S_v is the corresponding tangent vector in the v-direction.

Definition (Isoparametric Curves)

Fixing one parameter on $S(u,v)$ produces a curve on the surface:

- *u-curve* at $v = v_0$: $C_{v_0}(u) = S(u,\ v_0)$, tangent $= S_u(u, v_0)$.
- *v-curve* at $u = u_0$: $C_{u_0}(v) = S(u_0,\ v)$, tangent $= S_v(u_0, v)$.

These *isoparametric curves* (isocurves) sweep the surface and form its parameter grid. The two families intersect at every surface point $S(u_0, v_0)$.

Directional Derivative

Definition (Directional Derivative on a Surface)

The *directional derivative* of S at (u_0, v_0) in the direction $\mathbf{d} = (d_u, d_v)$ (a unit vector in parameter space) is

$$D_{\mathbf{d}}\, S = d_u\, S_u(u_0, v_0) + d_v\, S_v(u_0, v_0).$$

It measures the rate of change of position on the surface as the parameter moves in direction $\mathbf{d}$ in the uv-plane. Every tangent vector to the surface at (u_0, v_0) is a directional derivative for some choice of $\mathbf{d}$.

Unit Normal and Tangent Plane

Definition (Regularity, Unit Normal, Tangent Plane)

The surface is *regular* at (u_0, v_0) if $S_u \times S_v \neq \mathbf{0}$ there.
At a regular point, the *unit normal vector* is

$$N(u_0, v_0) = \frac{S_u(u_0, v_0) \;\times\; S_v(u_0, v_0)}{\|S_u(u_0, v_0) \;\times\; S_v(u_0, v_0)\|}.$$

The *tangent plane* at $S(u_0, v_0)$ is the plane through $S(u_0, v_0)$ spanned by S_u and S_v:

$$\Pi \;=\; \left\{\, S(u_0, v_0) + \alpha\, S_u + \beta\, S_v \;\middle|\; \alpha, \beta \in \mathbb{R} \,\right\}.$$

Its equation in terms of position $\mathbf{p}$ is

$$N \cdot \big(\mathbf{p} - S(u_0, v_0)\big) = 0.$$

Remark (Geometric meaning of $S_u \times S_v$). The magnitude $\|S_u \times S_v\|$ equals the area of the parallelogram spanned by S_u and S_v, which is the *local area element* of the surface. Where $\|S_u \times S_v\| = 0$ the surface is degenerate (the two tangent directions are parallel or one vanishes), and the normal is not defined.

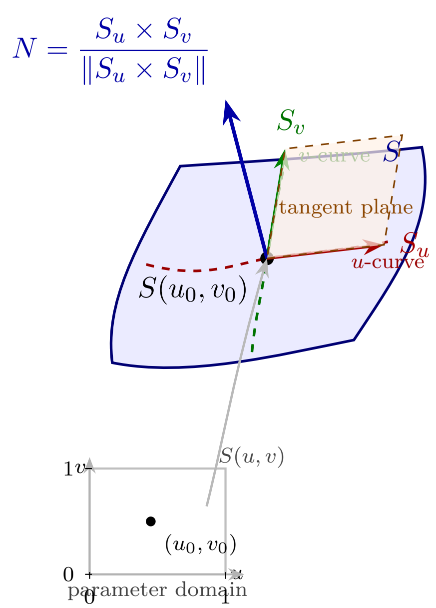

Figure: A parametric surface patch $S(u, v)$ with its parameter domain (lower left). At the point $S(u_0, v_0)$, the partial derivative vectors S_u (red) and S_v (green) are tangent to the isoparametric curves; together they span the tangent plane (orange). The unit normal $N = S_u \times S_v \,/\, \|S_u \times S_v\|$ (blue) is perpendicular to the tangent plane.

Status: Active

This section is actively expanding alongside work in Piegl & Tiller, *The NURBS Book*. Subsequent sections on Bézier curves, B-splines, NURBS, and continuity conditions follow.

62.2 Bezier Curves

Chapter 63

Computational Geometry

63.1 Convex Hulls

Toolkit: Convex Hulls

Ambient space	Finite point sets in $\mathbb{R}^2$.
Convexity	Line segments and convex combinations.
Certificates	Extreme points, supporting lines, and maximizing directions.
Order data	Distinct x-coordinates and slopes between ordered points.
Algorithmic shape	Build the upper hull by repeatedly choosing the next maximal slope.

Remark (Exposition). Computational geometry studies finite geometric data, such as point sets in the plane, using exact definitions and algorithms. Convex hulls are the first organizing example: they connect geometry, order and maximization, supporting lines, and algorithmic construction.

63.2 Convex Hulls

Convexity and Hulls

Convexity and Convex Hulls

This formulation makes precise the geometric idea of filling in all straight segments forced by a point set. Convex combinations give an algebraic language for the same operation, and the convex hull records the smallest set closed under that operation.

Definition

Definition 63.2.1 (Convex Set)**.** A set $S \subseteq \mathbb{R}^2$ is *convex* if for every pair of points $a, b \in S$ and every $\lambda \in [0,1]$,

$$\lambda a + (1-\lambda) b \in S.$$

Equivalently, S contains the line segment between any two of its points.

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Definition

Definition 63.2.2 (Convex Combination)**.** A *convex combination* of points $q_1, \ldots, q_n \in \mathbb{R}^2$ is a point of the form

$$\sum_{i=1}^{n} \lambda_i q_i, \qquad \lambda_i \geq 0 \text{ for all } i, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

The numbers λ_i are called the weights.

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Definition

Definition 63.2.3 (Convex Hull)**.** The *convex hull* of a set $P \subseteq \mathbb{R}^2$, written $\operatorname{conv}(P)$, is the set of all convex combinations of finitely many points of P:

$$\operatorname{conv}(P) = \left\{ \sum_{i=1}^{n} \lambda_i q_i \;\middle|\; n \geq 1,\ q_i \in P,\ \lambda_i \geq 0,\ \sum_{i=1}^{n} \lambda_i = 1 \right\}.$$

Equivalently, $\operatorname{conv}(P)$ is the smallest convex set containing P, i.e. the intersection of all convex sets that contain P.

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Lemma

Lemma 63.2.4 (Convex hull as the smallest convex set)**.** *Let $P \subseteq \mathbb{R}^2$. Then* conv(P) *is convex, contains P, and is contained in every convex set that contains P. Hence*

$$\operatorname{conv}(P) = \bigcap \{K \subseteq \mathbb{R}^2 : K \text{ is convex and } P \subseteq K\}.$$

Remark (Proof)*.* See the proof stub.

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Remark (Dependencies)*.*

- Convex Set
- Convex Combination
- Convex Hull

Supporting Functionals

Supporting Functionals and Vertices

This topic isolates the passage from geometry to optimization. A supporting line is a geometric certificate that a convex set lies on one side of a boundary, while a linear functional turns that certificate into a maximum.

Definition

Definition 63.2.5 (Extreme Point)**.** Let $S \subseteq \mathbb{R}^2$ be convex. A point $p \in S$ is an *extreme point* of S if whenever

$$p = \lambda a + (1-\lambda)b, \qquad a, b \in S, \quad \lambda \in (0,1),$$

it follows that $a = b = p$. For a finite point set P, the extreme points of conv(P) are called the vertices of the convex hull.

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Definition

Definition 63.2.6 (Supporting Line)**.** A line $\ell \subseteq \mathbb{R}^2$ is a *supporting line* of a convex set S at a point $p \in S$ if $p \in \ell$ and S lies entirely in one of the two closed half-planes bounded by ℓ. Equivalently, there exists $d \in \mathbb{R}^2 \setminus \{0\}$ such that

$$\ell = \{x \in \mathbb{R}^2 : \langle d, x\rangle = \langle d, p\rangle\}$$

and

$$\langle d, q\rangle \le \langle d, p\rangle \qquad \text{for all } q \in S.$$

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Lemma

Lemma 63.2.7 (Linear functionals preserve convex combinations)**.** *Let $d \in \mathbb{R}^2$, and let*

$$z = \sum_{i=1}^{n} \lambda_i q_i$$

be a convex combination of points $q_1, \ldots, q_n \in \mathbb{R}^2$. Then

$$\langle d, z\rangle = \sum_{i=1}^{n} \lambda_i \langle d, q_i\rangle.$$

Consequently, if $\langle d, q_i\rangle \le M$ for every i, then

$$\langle d, z\rangle \le M.$$

Remark (Proof)*.* See the proof stub.

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Theorem

Remark (Dependencies).

- Convex Combination
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Theorem 63.2.8 (Supporting functional characterization of hull vertices)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. A point $p \in P$ is a vertex of* $\operatorname{conv}(P)$ *if and only if there exists a direction $d \in \mathbb{R}^2 \setminus \{0\}$ such that*

$$\langle d, p \rangle > \langle d, q \rangle \qquad \textit{for all } q \in P \setminus \{p\}.$$

Equivalently, p is the unique point of P attaining

$$\max_{q \in P} \langle d, q \rangle.$$

Thus every hull vertex is a maximum of P under some linear functional.

Remark (Proof). See the proof stub.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Remark (Finite maximum form). The maximum is used here because P is finite. This is the finite-set version of the same supporting-functional intuition that appears in supremum language for more general sets.

Remark (Dependencies).

- Convex Combination
- Convex Hull
- Extreme Point
- Supporting Line
- Linear Functionals Preserve Convex Combinations

Upper Hull Structure

Monotone Slopes and Greedy Construction

This topic converts the supporting-line description into an ordered planar construction. Under the general-position assumptions, each relevant maximum is unique, and the upper hull can be recognized by the strict monotonicity of successive slopes.

Assumption 63.2.9 (General Position for the Hull Theorems)**.** Let $P \subset \mathbb{R}^2$ be a finite set in general position: the points have distinct x-coordinates and no three points are collinear. For $p \in \mathbb{R}^2$, write $x(p)$ and $y(p)$ for its coordinates.

Theorem

Theorem 63.2.10 (Monotone slope characterization of the upper hull)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. Let*

$$v_0, v_1, \ldots, v_m$$

be the vertices of the upper hull of P, listed in increasing x-coordinate, and define

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \qquad k = 0, 1, \ldots, m-1.$$

Then

$$s_0 > s_1 > \cdots > s_{m-1}.$$

Conversely, if points $w_0, \ldots, w_m$ satisfy

$$x(w_0) < x(w_1) < \cdots < x(w_m)$$

and their consecutive slopes are strictly decreasing, then every w_k is a vertex of

$$\operatorname{conv}(\{w_0, \ldots, w_m\}).$$

The lower hull is characterized similarly, with strictly increasing consecutive slopes.

Remark (Proof)*.* See the proof stub.

Remark (Standard quantified statement)*.* The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Theorem

Remark (Dependencies).

- Convex Hull
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- Supporting Functional Characterization of Hull Vertices

Theorem 63.2.11 (Greedy slope construction of the upper hull). *Let $P \subset \mathbb{R}^2$ be finite and in general position. Define*

$$v_0 = \arg\min_{p \in P} x(p).$$

Given v_k with

$$x(v_k) < \max_{p \in P} x(p),$$

define

$$v_{k+1} = \arg\max_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

Terminate once

$$v_k = \arg\max_{p \in P} x(p).$$

Then the points

$$v_0, v_1, \ldots, v_m$$

produced by this construction are exactly the vertices of the upper hull of P, and the slopes encountered along the way are strictly decreasing.

Remark (Proof). See the proof stub.

Remark (Standard quantified statement). The statement applies to the objects named in the hypotheses and asserts the corresponding containment, extremal, or construction property stated there.

Remark (Uniqueness of choices). The distinct x-coordinates make the initial and terminal x-extreme points unique. The no-three-collinear condition makes the slope maximizer in the greedy step unique.

Remark (Proof roadmap). The proof ladder is:

convex combinations → linear functionals → supporting lines → monotone slopes → greedy hull construc

The first two lemmas are algebraic. The monotone slope theorem is an analytic-geometry statement about three ordered points. The greedy construction theorem is the algorithmic consequence.

Remark (Dependencies).

- Convex Hull
- Supporting Line
- Linear Functionals Preserve Convex Combinations
- Supporting Functional Characterization of Hull Vertices
- Monotone slope characterization of the upper hull

Chapter 64

Computational Linear Algebra

64.1 Vectors

Vectors and Linear Combinations — Quick Reference

Vector in $\mathbb{R}^n$	An ordered list $(v_1, v_2, \ldots, v_n)$ of n real numbers.
Addition	$(v_1, \ldots, v_n) + (w_1, \ldots, w_n) = (v_1 + w_1, \ldots, v_n + w_n)$.
Scalar multiplication	$c(v_1, \ldots, v_n) = (cv_1, \ldots, cv_n)$.
Linear combination	$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$; scalars times vectors, summed.
Convex combination	Linear combination with $c_i \geq 0$ and $\sum c_i = 1$; result lies inside the convex hull.
Standard basis vectors	$\mathbf{e}_j \in \mathbb{R}^n$: all zeros except a 1 in position j. Every vector is a linear combination of $\mathbf{e}_1, \ldots, \mathbf{e}_n$.

Matrices — Quick Reference

Matrix	Rectangular array of numbers; $A \in \mathbb{R}^{m\times n}$ has m rows, n columns.
Matrix–vector	$(A\mathbf{x})_i = \sum_j A_{ij}x_j$; $A\mathbf{x}$ is a linear combination of columns of A.
Matrix–matrix	$(AB)_{ij} = \sum_k A_{ik}B_{kj}$; requires inner dimensions to match.
Transpose	$(A^T)_{ij} = A_{ji}$; rows become columns.
Identity	I_n: ones on diagonal, zeros elsewhere; $IA = AI = A$.
Column view	$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$ where $\mathbf{a}_j$ are the columns of A.

Lengths and Dot Products — Quick Reference

Dot product	$\mathbf{v}\cdot\mathbf{w} = v_1w_1 + v_2w_2 + \cdots + v_nw_n \in \mathbb{R}$.
Length (norm)	$\|\mathbf{v}\| = \sqrt{\mathbf{v}\cdot\mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}$.
Unit vector	$\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$; has length 1.
Angle	$\cos\theta = \dfrac{\mathbf{v}\cdot\mathbf{w}}{\|\mathbf{v}\|\,\|\mathbf{w}\|}$.
Orthogonality	$\mathbf{v}\perp\mathbf{w}$ iff $\mathbf{v}\cdot\mathbf{w} = 0$.
Cauchy–Schwarz	$\lvert\mathbf{v}\cdot\mathbf{w}\rvert \le \|\mathbf{v}\|\,\|\mathbf{w}\|$.

64.1.1 Vectors and Linear Combinations

Vectors

Definition (Vector in $\mathbb{R}^n$)

A *vector* in $\mathbb{R}^n$ is an ordered list of n real numbers, written $\mathbf{v} = (v_1, v_2, \ldots, v_n)$. The number n is the *dimension* of the vector.

Vector addition:

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1,\ v_2 + w_2,\ \ldots,\ v_n + w_n).$$

Scalar multiplication:

$$c\mathbf{v} = (cv_1,\ cv_2,\ \ldots,\ cv_n), \quad c \in \mathbb{R}.$$

Remark (Geometric meaning). In $\mathbb{R}^2$ and $\mathbb{R}^3$, a vector is an arrow indicating displacement. Its entries specify length and direction, but *not* location — the same vector can be placed anywhere in space. Addition corresponds to following one displacement and then another (head-to-tail rule). Scalar multiplication stretches or shrinks the arrow; negative scalars reverse its direction.

Theorem (Properties of Vector Operations)

For all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$:

1. $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ (commutativity)
2. $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$ (associativity)
3. $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$ (distributivity over vector addition)
4. $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$ (distributivity over scalar addition)
5. $c(d\mathbf{v}) = (cd)\mathbf{v}$ (compatibility)

Linear Combinations

Definition (Linear Combination)

A *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ is any vector of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k,$$

where $c_1, c_2, \ldots, c_k \in \mathbb{R}$ are called the *coefficients*.
A *convex combination* is a linear combination in which $c_i \geq 0$ for all i and $\sum_{i=1}^{k} c_i = 1$.

Remark (Why convex combinations matter for NURBS). Every point on a Bézier or B-spline curve is a convex combination of control points. The Bernstein basis functions $B_{i,n}(u)$ are non-negative and sum to 1, so $C(u) = \sum_{i=0}^{n} B_{i,n}(u)P_i$ is a convex combination of the control points $\{P_i\}$ at every parameter value u. This is the *convex hull property*: the curve lies entirely within the convex hull of its control polygon.

Definition (Standard Basis Vectors)

For $j = 1, 2, \ldots, n$, the *j-th standard basis vector* $\mathbf{e}_j \in \mathbb{R}^n$ has a 1 in position j and 0s everywhere else. Every vector $\mathbf{v} = (v_1, \ldots, v_n) \in \mathbb{R}^n$ can be written as

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n.$$

64.1.2 Lengths and Dot Products

The Dot Product

Definition (Dot Product)

The *dot product* (or *inner product*) of $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is

$$\mathbf{v} \cdot \mathbf{w} = v_1w_1 + v_2w_2 + \cdots + v_nw_n \ \in \ \mathbb{R}.$$

Theorem (Properties of the Dot Product)

For all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

1. $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$ (commutativity)
2. $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$ (distributivity)
3. $\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$ (scalar compatibility)
4. $\mathbf{v} \cdot \mathbf{v} \geq 0$, with equality iff $\mathbf{v} = \mathbf{0}$ (positive definiteness)

Length and Angle

Definition (Length and Unit Vector)

The *length* (or *norm*) of $\mathbf{v} \in \mathbb{R}^n$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

A *unit vector* has length 1. The *normalisation* of a nonzero vector $\mathbf{v}$ is $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$, which is the unit vector in the direction of $\mathbf{v}$.

Theorem (Cauchy–Schwarz Inequality)

For all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \, \|\mathbf{w}\|,$$

with equality if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel (one is a scalar multiple of the other).

Definition (Angle Between Vectors)

The *angle* $\theta \in [0, \pi]$ between nonzero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is defined by

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|}.$$

The vectors are *orthogonal* (written $\mathbf{v} \perp \mathbf{w}$) if $\theta = \pi/2$, equivalently if $\mathbf{v} \cdot \mathbf{w} = 0$.

Remark (Dot product and normals in NURBS). The unit normal to a parametric surface $S(u, v)$ is computed as $N = S_u \times S_v / \|S_u \times S_v\|$, which requires both the cross product and normalisation via the norm. The dot product also enters the convex hull argument: showing that a B-spline curve lies on a specific side of a hyperplane reduces to checking the sign of dot products of control points with the hyperplane's normal.

64.1.3 Introduction to Matrices

Matrices and Matrix–Vector Multiplication

Definition (Matrix)

An $m \times n$ *matrix* A is a rectangular array of real numbers with m rows and n columns:

$$A = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{pmatrix}.$$

The entry in row i, column j is denoted A_{ij} or $[A]_{ij}$. We write $A \in \mathbb{R}^{m\times n}$.

Definition (Matrix–Vector and Matrix–Matrix Multiplication)

If $A \in \mathbb{R}^{m\times n}$ and $\mathbf{x} \in \mathbb{R}^n$, then $A\mathbf{x} \in \mathbb{R}^m$ is defined by

$$(A\mathbf{x})_i = \sum_{j=1}^{n} A_{ij}x_j, \qquad i = 1, \ldots, m.$$

Equivalently, $A\mathbf{x}$ is the linear combination of the columns of A with coefficients from $\mathbf{x}$:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, their product $AB \in \mathbb{R}^{m\times p}$ has entries

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik}B_{kj}.$$

Remark (Matrix multiplication is not commutative). In general $AB \neq BA$, even when both products are defined. Order matters: applying transformation B first, then A, is written AB.

The Transpose

Definition (Transpose)

The *transpose* of $A \in \mathbb{R}^{m\times n}$ is the matrix $A^T \in \mathbb{R}^{n\times m}$ defined by $(A^T)_{ij} = A_{ji}$. Rows of A become columns of A^T. Key properties:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$
- $\mathbf{v}\cdot\mathbf{w} = \mathbf{v}^T\mathbf{w}$ (dot product as matrix multiplication)

Matrix Form of a Curve

Remark (NURBS connection: matrix form of curves and surfaces). The power basis curve $C(u) = \sum_{i=0}^{n} \mathbf{a}_i u^i$ can be written as

$$C(u) = [\mathbf{a}_0 \;\; \mathbf{a}_1 \;\; \cdots \;\; \mathbf{a}_n] \begin{pmatrix} 1 \\ u \\ \vdots \\ u^n \end{pmatrix} = A\,\mathbf{u}(u),$$

where A is the matrix of coefficient vectors and $\mathbf{u}(u)$ is the monomial vector. A Bézier curve has the equivalent matrix form $C(u) = [u^i]^T M [P_i]$, where M is the $(n+1)\times(n+1)$ change-of-basis matrix converting Bernstein basis to power basis. A tensor product surface takes the form $S(u,v) = [f_i(u)]^T [b_{i,j}][g_j(v)]$, a matrix sandwiched between two basis-function vectors. All of these representations rely on matrix–vector multiplication as defined above.

64.2 Solving Linear Equations

Matrix Operations — Quick Reference

Associativity	$A(BC) = (AB)C$.
Distributivity	$A(B + C) = AB + AC$; $(A + B)C = AC + BC$.
NOT commutative	$AB \neq BA$ in general.
Identity	$AI = IA = A$.
Transpose product	$(AB)^T = B^T A^T$.
Diagonal matrices	D_1D_2 is diagonal; DA scales rows; AD scales columns.

Inverse Matrices — Quick Reference

Inverse	A^{-1}: the unique matrix with $AA^{-1} = A^{-1}A = I$.
Existence	A^{-1} exists iff A is square and $\det(A) \neq 0$ (nonsingular).
2×2 **formula**	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
Product rule	$(AB)^{-1} = B^{-1}A^{-1}$ (order reverses).
Transpose rule	$(A^T)^{-1} = (A^{-1})^T$.
Change of basis	If M converts basis B_1 to B_2, then M^{-1} converts B_2 to B_1.

64.2.1 Rules for Matrix Operations

Algebraic Rules

Theorem (Rules for Matrix Multiplication)

When the dimensions are compatible:

1. $A(BC) = (AB)C$ (associativity)
2. $A(B + C) = AB + AC$ (left distributivity)
3. $(A + B)C = AC + BC$ (right distributivity)
4. $c(AB) = (cA)B = A(cB)$ for any scalar c
5. $(AB)^T = B^T A^T$ (transpose reverses order)

The following do **not** hold in general:

- $AB \neq BA$ (not commutative)
- $AB = AC \not\Rightarrow B = C$ (no cancellation in general)
- $AB = 0 \not\Rightarrow A = 0$ or $B = 0$

Diagonal Matrices

Definition (Diagonal Matrix)

A *diagonal matrix* $D \in \mathbb{R}^{n\times n}$ has $D_{ij} = 0$ for all $i \neq j$. Written $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$. Key computational facts:

- DA multiplies each *row* i of A by d_i.
- AD multiplies each *column* j of A by d_j.
- $D_1D_2 = \operatorname{diag}(d_1e_1, \ldots, d_ne_n)$ where e_j are the diagonal entries of D_2.

Block Matrices

Remark (Block matrix multiplication). Matrices can be partitioned into blocks, and multiplied block-by-block as if the blocks were scalar entries, provided the block sizes are compatible. If $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ and $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$, then

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix},$$

provided inner block dimensions match. The standard matrix of a linear transformation $[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)]$ is a block matrix with column vectors as blocks, and $[T]\mathbf{v} = v_1 T(\mathbf{e}_1) + \cdots + v_n T(\mathbf{e}_n)$ is block matrix–vector multiplication.

64.2.2 Inverse Matrices

Definition and Properties

Definition (Invertible Matrix)

A square matrix $A \in \mathbb{R}^{n\times n}$ is *invertible* (or *nonsingular*) if there exists a matrix $A^{-1} \in \mathbb{R}^{n\times n}$ such that

$$AA^{-1} = A^{-1}A = I_n.$$

The matrix A^{-1} is the *inverse* of A and is unique. A matrix with no inverse is called *singular*.

Theorem (Properties of Inverses)

When A and B are invertible $n \times n$ matrices:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$ (order reverses)
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(cA)^{-1} = \frac{1}{c}A^{-1}$ for $c \neq 0$

The 2×2 Case

Formula (2×2 Inverse)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad ad - bc \neq 0.$$

The quantity $ad - bc$ is the determinant. The matrix is singular when $ad = bc$.

Change of Basis via Inverse

Remark (NURBS connection: converting between Bernstein and power basis)*.* In Piegl & Tiller Exercise 1.14, the Bernstein basis functions of degree n can be written as $[B_{0,n}(u), \ldots, B_{n,n}(u)] = [1, u, \ldots, u^n]M$, where M is an $(n+1) \times (n+1)$ matrix. This means a Bézier curve has the matrix form

$$C(u) = [u^i]^T M[P_i],$$

where $[u^i]^T$ is the monomial row vector, $[P_i]$ is the column of control points, and M encodes the change from power basis to Bernstein basis.

The inverse M^{-1} converts the other direction: if you have a curve in power basis form with coefficients $[\mathbf{a}_i]$, then the equivalent Bézier control points are $[P_i] = M^{-1}[\mathbf{a}_i]$.

This is a direct application of the matrix inverse: M converts one basis to another, and M^{-1} inverts the conversion.

64.3 Vector Spaces

Basis and Dimension — Quick Reference

Linearly independent	No vector is a linear combination of the others; only $\mathbf{0} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$ has $c_i = 0$ for all i.
Span	$\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$: all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_k$.
Basis	A linearly independent set that spans the space.
Dimension	Number of vectors in any basis; all bases have the same size.
Change of basis	Transition matrix P: columns are the new basis vectors in terms of the old basis.

64.3.1 Independence, Basis, and Dimension

Linear Independence

Definition (Linear Independence)

Vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ are *linearly independent* if the only solution to

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

is $c_1 = c_2 = \cdots = c_k = 0$. Otherwise they are *linearly dependent*: at least one vector is a linear combination of the others.

Basis and Dimension

Definition (Basis)

A *basis* for a subspace $V \subseteq \mathbb{R}^n$ is a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ that is:

1. linearly independent, and
2. spans V (every vector in V is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$).

Theorem (Dimension is Well-Defined)

Any two bases for the same subspace V have the same number of vectors. This common number is called the *dimension* of V, written $\dim V$.

Remark. The standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$, so $\dim \mathbb{R}^n = n$. The key insight: every set of n linearly independent vectors in $\mathbb{R}^n$ is automatically a basis.

Change of Basis

Definition (Transition Matrix)

Let $\mathcal{B}_1 = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ and $\mathcal{B}_2 = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be two ordered bases for $\mathbb{R}^n$. The *transition matrix* P from $\mathcal{B}_1$ to $\mathcal{B}_2$ has as its i-th column the coordinates of $\mathbf{u}_i$ expressed in the basis $\mathcal{B}_2$. If $[\mathbf{x}]_{\mathcal{B}_1}$ denotes the coordinate vector of $\mathbf{x}$ in basis $\mathcal{B}_1$, then

$$[\mathbf{x}]_{\mathcal{B}_2} = P\,[\mathbf{x}]_{\mathcal{B}_1}.$$

The inverse P^{-1} converts in the opposite direction.

Remark (NURBS connection: Bernstein basis as a basis for polynomials). The Bernstein polynomials $\{B_{0,n}(u), B_{1,n}(u), \ldots, B_{n,n}(u)\}$ form a basis for the vector space of polynomials of degree $\leq n$. So does the monomial (power) basis $\{1, u, u^2, \ldots, u^n\}$. The matrix M from Piegl & Tiller Exercise 1.14 is precisely the transition matrix converting coordinates in the monomial basis to coordinates in the Bernstein basis. M^{-1} converts the other direction. This is why the Bézier curve $C(u) = [u^i]^T M[P_i]$ can be read as: "express the Bernstein basis functions in the monomial basis via M, then multiply by the control points."

64.4 Orthogonality

64.5 Determinants

Determinants and Cross Product — Quick Reference

2×2 **det**	$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$
3×3 **det**	Cofactor expansion along any row or column.
Cross product	$\mathbf{v} \times \mathbf{w}$: vector orthogonal to both $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^3$, computed via 3×3 determinant mnemonic.
Magnitude	$\lVert \mathbf{v} \times \mathbf{w} \rVert = \lVert \mathbf{v} \rVert \, \lVert \mathbf{w} \rVert \sin\theta$; equals area of parallelogram.
Direction	Right-hand rule; $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$.

64.5.1 Determinants and the Cross Product

The 3×3 Determinant

Definition (3×3 Determinant via Cofactor Expansion)

For $A \in \mathbb{R}^{3\times3}$, expanding along the first row:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

The signs alternate: $+, -, +$ along the first row.

The Cross Product

Definition (Cross Product)

For $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ in $\mathbb{R}^3$, the *cross product* is

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2w_3 - v_3w_2, \;\; v_3w_1 - v_1w_3, \;\; v_1w_2 - v_2w_1).$$

Theorem (Properties of the Cross Product)

For $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$ and $c \in \mathbb{R}$:

1. $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$ (anticommutativity)
2. $\mathbf{v} \times (\mathbf{w} + \mathbf{x}) = \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x}$ (distributivity)
3. $\mathbf{v} \times \mathbf{v} = \mathbf{0}$
4. $(c\mathbf{v}) \times \mathbf{w} = c(\mathbf{v} \times \mathbf{w})$
5. $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ and $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ (orthogonality)
6. $\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \, \|\mathbf{w}\| \sin\theta$ (magnitude = parallelogram area)

Remark (NURBS connection: surface normal vectors). The unit normal to a parametric surface at a point (u, v) is

$$N(u, v) = \frac{S_u \times S_v}{\|S_u \times S_v\|},$$

where S_u and S_v are the partial derivative vectors of the surface with respect to the parameters. The cross product gives the direction orthogonal to both tangent directions; dividing by the norm produces the unit normal. This appears in Piegl & Tiller §1.1 and is used throughout whenever normal vectors are needed (shading, offset surfaces, etc.).

Remark (Mnemonic for the cross product). Write the entries of $\mathbf{v}$ and $\mathbf{w}$ each twice in a 2×6 array. Ignore the first and last columns. Draw three "X" shapes between columns 2–3, 3–4, and 4–5. Each "X" contributes one entry of $\mathbf{v} \times \mathbf{w}$: add along forward diagonals, subtract along backward diagonals.

64.6 Eigenvalues

64.7 Singular Value Decomposition

64.8 Linear Transformations

Standard Matrix — Quick Reference

Standard matrix	$[T] = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \in \mathbb{R}^{m\times n}$.
How to build it	Evaluate T on each standard basis vector; place results as columns.
Uniqueness	$[T]$ is the *unique* matrix with $T(\mathbf{v}) = [T]\mathbf{v}$ for all $\mathbf{v}$.
Composition	$[S \circ T] = [S][T]$.
Change of basis	Same T, different bases $\Rightarrow$ different matrix; related by $B = P^{-1}AP$.

Linear Transformations — Quick Reference

Definition	$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w}$ and scalars c, d.
Origin fixed	Always $T(\mathbf{0}) = \mathbf{0}$; translation is *not* linear.
Standard matrix	$[T] = [T(\mathbf{e}_1) \mid \cdots \mid T(\mathbf{e}_n)]$; then $T(\mathbf{v}) = [T]\mathbf{v}$.
Composition	$S \circ T$ has matrix $[S][T]$.
Affine map	$\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$: linear part A plus translation $\mathbf{v}$.
Invariance	B-spline curves: apply affine map to control points only.

64.8.1 The Idea of a Linear Transformation

Definition

Definition (Linear Transformation)

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a *linear transformation* if

$$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w})$$

for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and all $c, d \in \mathbb{R}$.
Equivalently, T must satisfy both:

1. $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ (preserves addition), and
2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (preserves scalar multiplication).

Remark (The origin is always fixed). Every linear transformation satisfies $T(\mathbf{0}) = \mathbf{0}$. Therefore, *translation* (shifting by a nonzero vector) is not linear. The four geometric operations that *are* linear: rotation about the origin, scaling, reflection, and projection.

Geometric Catalog in $\mathbb{R}^2$

Standard Planar Transformations

Scaling	$\begin{pmatrix} c_1 & 0 \\ 0 & c_2 \end{pmatrix}$: scales x by c_1, y by c_2.
Rotation by θ	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$: rotates counter-clockwise.
Reflection ($y = x$)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: swaps x and y coordinates.
Projection (x-axis)	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$: drops the y component.

Remark (Composition is matrix multiplication). Applying T_1 then T_2 has matrix $[T_2][T_1]$. A sequence of geometric operations on a model (scale, then rotate, then translate) corresponds to multiplying their matrices together in that order.

Affine Transformations

Definition (Affine Transformation)

An *affine transformation* $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ has the form

$$\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v},$$

where $A \in \mathbb{R}^{n\times n}$ is the linear part and $\mathbf{v} \in \mathbb{R}^n$ is a translation vector. An affine transformation is a linear transformation when $\mathbf{v} = \mathbf{0}$.

Remark (Affine invariance of B-spline curves — NURBS Property P3.4). An affine transformation $\Phi(\mathbf{r}) = A\mathbf{r} + \mathbf{v}$ applied to a B-spline curve $C(u) = \sum_{i=0}^{n} N_{i,p}(u)P_i$ gives

$$\Phi(C(u)) = \sum_{i=0}^{n} N_{i,p}(u)\, \Phi(P_i).$$

The transformation can be applied to the curve by applying it to each control point individually. This works because the basis functions sum to 1 (partition of unity):

$$\Phi(C(u)) = A\Big(\sum N_{i,p}P_i\Big) + \mathbf{v} = \sum N_{i,p}(AP_i) + \underbrace{\Big(\sum N_{i,p}\Big)}_{=1}\mathbf{v} = \sum N_{i,p}(AP_i + \mathbf{v}).$$

This means rotating, translating, or scaling a B-spline curve costs only $n+1$ affine operations on control points, regardless of how many points are evaluated on the curve.

64.8.2 The Matrix of a Linear Transformation

The Standard Matrix

Theorem (Standard Matrix of a Linear Transformation)

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there exists a unique matrix $[T] \in \mathbb{R}^{m\times n}$ such that

$$T(\mathbf{v}) = [T]\mathbf{v} \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

The *standard matrix* $[T]$ is constructed by placing the images of the standard basis vectors as columns:

$$[T] = \big[\, T(\mathbf{e}_1) \;\big|\; T(\mathbf{e}_2) \;\big|\; \cdots \;\big|\; T(\mathbf{e}_n) \,\big].$$

Remark (Why this works). Since every $\mathbf{v} = v_1\mathbf{e}_1 + \cdots + v_n\mathbf{e}_n$, linearity gives

$$T(\mathbf{v}) = v_1T(\mathbf{e}_1) + \cdots + v_nT(\mathbf{e}_n) = [T]\mathbf{v}.$$

Knowing T on the basis vectors determines T on the entire space.

Matrix Under Change of Basis

Definition (Matrix Relative to a Basis)

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation and let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be an ordered basis. The *matrix of T relative to $\mathcal{B}$*, denoted $[T]_{\mathcal{B}}$, has as its j-th column the coordinates of $T(\mathbf{b}_j)$ expressed in the basis $\mathcal{B}$.
If P is the transition matrix whose columns are $\mathbf{b}_1, \ldots, \mathbf{b}_n$ expressed in the standard basis, then

$$[T]_{\mathcal{B}} = P^{-1}[T]P.$$

Two matrices related by $B = P^{-1}AP$ are called *similar*.

Remark (NURBS connection: Bézier curve in matrix form). The Bézier curve has the matrix representation

$$C(u) = [u^i]^T M[P_i],$$

where:

- $[u^i]^T = (1, u, u^2, \ldots, u^n)$ is the monomial basis vector,
- $[P_i]$ is the column vector of control points,
- M is the $(n + 1) \times (n + 1)$ matrix that encodes the Bernstein polynomials in the monomial basis.

The matrix M is the standard matrix of the linear transformation that converts from Bernstein basis coordinates to monomial basis coordinates. Setting $[\mathbf{a}_i] = M[P_i]$ gives the power basis representation of the same curve (so the Bézier coefficients $[P_i]$ are the coordinates of the curve in the Bernstein basis, and $[\mathbf{a}_i]$ are the coordinates in the monomial basis). Converting between the two is exactly the change-of-basis operation $P^{-1}AP$ applied to the coefficient vectors.

64.9 Complex Vectors and Matrices

64.10 Applications

64.11 Numerical Linear Algebra

64.12 Probability and Statistics

Chapter 65

Fourier Analysis

SERIES
From Turing to Carmack

VOLUME VIII
Applied and Computational Mathematics

BOOK II
Applied Methods

CHAPTER 7
Fourier Analysis

Chapter 66

Ordinary Differential Equations

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Applied and Computational Mathematics

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CHAPTER 8
Ordinary Differential Equations

66.1 Breadcrumb

Where This Chapter Lives

Calculus → **Ordinary Differential Equations** → Dynamical Systems / Numerical Methods / PDE

Why this chapter belongs here. Ordinary differential equations are where derivatives stop being only local rates and start governing whole families of evolving states. Once differentiation is in hand, equations such as

$$y'(t) = f(t, y(t))$$

become the natural language for motion, growth, decay, forcing, and feedback.

What this chapter will gather. The eventual ODE chapter is the right home for:

- first-order equations and separable models;
- linear equations and integrating factors;
- existence and uniqueness ideas;
- systems of ODEs and phase portraits;
- stability, equilibrium, and qualitative behavior;
- numerical methods such as Euler and Runge–Kutta schemes.

Why this is currently a breadcrumb. The surrounding curriculum is still building the supporting architecture: calculus, linear algebra, and numerical analysis. ODEs sit at the meeting point of all three, so this file marks the destination clearly even before the chapter is fully written.

66.2 Models

Definition

Definition 66.2.1 (Differential Equation)**.** A **differential equation** is an equation whose unknown is a function and whose terms involve one or more derivatives of that function.

For an unknown function $y = y(t)$, a differential equation may be written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0.$$

Here t is the independent variable, $y(t)$ is the unknown function, and

$$y', y'', \ldots, y^{(n)}$$

are derivatives of y.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). At the computational level, a differential equation describes how a quantity changes. Instead of giving the function $y(t)$ directly, it gives a relation involving the rate of change of $y(t)$, and possibly higher-order rates of change.

Example (A First Differential Equation). The equation

$$y' = 3y$$

is a differential equation because it relates the unknown function y to its derivative y'.

A solution is

$$y(t) = Ce^{3t},$$

where C is a constant. Indeed,

$$y'(t) = 3Ce^{3t} = 3y(t).$$

Definition

Definition 66.2.2 (Order of a Differential Equation)**.** The **order** of a differential equation is the highest derivative of the unknown function that appears in the equation.

For example, if an equation involving an unknown function $y = y(t)$ can be written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0$$

and $y^{(n)}$ is the highest derivative that appears, then the differential equation has **order** n.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Example (Orders of Differential Equations). The equation

$$y' = 3y$$

is first order, since the highest derivative appearing is y'.

The equation

$$y'' + 4y' + 5y = 0$$

is second order, since the highest derivative appearing is y''.

The equation

$$y^{(4)} - ty'' + y = 0$$

is fourth order, since the highest derivative appearing is $y^{(4)}$.

Definition

Definition 66.2.3 (Solution of a Differential Equation)**.** A **solution** of a differential equation is a function that satisfies the equation on a specified interval.
More precisely, if a differential equation is written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0,$$

then a solution on an interval I is a function $y : I \to \mathbb{R}$ such that y has all derivatives required by the equation and

$$F(t, y(t), y'(t), y''(t), \ldots, y^{(n)}(t)) = 0$$

for every $t \in I$.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). A solution is not a number; it is a function. The function solves the differential equation when substituting the function and its derivatives into the equation makes the equation true for every point in the interval under consideration.

Example (Checking a Solution). The function

$$y(t) = Ce^{3t}$$

is a solution of

$$y' = 3y$$

on $\mathbb{R}$, because

$$y'(t) = 3Ce^{3t} = 3y(t)$$

for every $t \in \mathbb{R}$.

Definition

Definition 66.2.4 (Autonomous Differential Equation)**.** An **autonomous differential equation** is a differential equation in which the independent variable does not appear explicitly.
For a first-order ordinary differential equation, an autonomous equation has the form

$$y' = f(y),$$

where the rate of change of y depends only on the current value of y, not directly on time t.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* An autonomous differential equation describes a system whose rule of evolution does not change with time. If the system is in the same state y at two different times, then the differential equation assigns the same rate of change at both times.

Example (Autonomous Equations)*.* The equation

$$y' = 3y$$

is autonomous, since the right-hand side depends only on y.

The equation

$$y' = t + y$$

is not autonomous, since the independent variable t appears explicitly.

Definition

Definition 66.2.5 (Non-Autonomous Differential Equation)**.** A **non-autonomous differential equation** is a differential equation in which the independent variable appears explicitly.
For a first-order ordinary differential equation, a non-autonomous equation has the form

$$y' = f(t, y),$$

where the rate of change of y may depend both on the current value of y and on the independent variable t.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* A non-autonomous differential equation describes a system whose rule of evolution may change over time. Even if the system has the same state y at two different times, the differential equation may assign different rates of change because the value of t may be different.

Example (Non-Autonomous Equations)*.* The equation

$$y' = t + y$$

is non-autonomous, since the independent variable t appears explicitly.

The equation

$$y' = 3y$$

is autonomous, not non-autonomous, since the right-hand side depends only on y.

Definition

Definition 66.2.6 (Linear Differential Equation)**.** A differential equation is called **linear** if the unknown function and its derivatives appear only to the first power and are not multiplied together.

An nth-order linear ordinary differential equation for an unknown function $y = y(t)$ has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

where the coefficient functions

$$a_0(t), a_1(t), \ldots, a_n(t)$$

and the forcing term $g(t)$ are given functions of the independent variable t.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). Linearity means that the equation is linear in the unknown function y and its derivatives. The coefficients may depend on t, but they may not depend on y or its derivatives. Thus terms such as y^2, yy', $\sin(y)$, and $(y')^2$ make an equation nonlinear.

Example (Linear and Nonlinear Equations)*.* The equation

$$y'' + 3ty' + 2y = \sin t$$

is linear.

The equation

$$y'' + y^2 = 0$$

is nonlinear because it contains the term y^2.

The equation

$$y' + yy' = t$$

is nonlinear because it contains the product yy'.

Ordinary differential equations are often used to model physical phenomena because many natural systems are governed by relationships between a quantity and its rate of change. In such models, the unknown function represents the state of the system, while the differential equation describes how that state evolves over time. For example, population models describe how a population changes, cooling models describe how temperature changes, motion models describe how position and velocity change, and circuit models describe how current or voltage changes. In each case, the differential equation does not usually give the quantity directly; instead, it gives a rule for how the quantity changes.

Definition

Definition 66.2.7 (Malthusian Population Growth Model)**.** The **Malthusian population growth model** assumes that the rate of change of a population is proportional to the current population.
If $P(t)$ denotes the population at time t, then the model is

$$P'(t) = kP(t),$$

where k is a constant called the **growth rate** or **proportionality constant**.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* The model says that population growth depends only on the current population size. If $k > 0$, the population grows exponentially. If $k < 0$, the population decays exponentially. If $k = 0$, the population remains constant.

Example (General Solution)*.* The general solution of

$$P'(t) = kP(t)$$

is

$$P(t) = Ce^{kt},$$

where C is a constant. If the initial population is $P(0) = P_0$, then

$$P(t) = P_0 e^{kt}.$$

Definition

Definition 66.2.8 (Decay)**.** A quantity is said to undergo **decay** if it decreases over time.
In a differential equation model, if $Q(t)$ denotes the quantity at time t, then decay is often modeled by an equation of the form

$$Q'(t) = -kQ(t),$$

where $k > 0$ is a constant.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* The equation

$$Q'(t) = -kQ(t)$$

says that the rate of change of the quantity is proportional to the current amount, but with a negative sign. Thus larger amounts decay faster in absolute size, while smaller amounts decay more slowly.

Example (Exponential Decay). The general solution of

$$Q'(t) = -kQ(t)$$

is

$$Q(t) = Ce^{-kt}.$$

If $Q(0) = Q_0$, then

$$Q(t) = Q_0e^{-kt}.$$

For $k > 0$, this function decreases toward 0 as t increases.

Definition

Definition 66.2.9 (Newton's Law of Cooling)**. Newton's Law of Cooling** states that the rate at which an object's temperature changes is proportional to the difference between the object's temperature and the surrounding ambient temperature.
If $T(t)$ denotes the temperature of the object at time t, and T_a denotes the ambient temperature, then the model is

$$T'(t) = -k\big(T(t) - T_a\big),$$

where $k > 0$ is a constant called the **cooling constant**.

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.
Remark (Interpretation). The quantity

$$T(t) - T_a$$

measures how far the object's temperature is from the ambient temperature. Newton's Law of Cooling says that the larger this temperature difference is, the faster the object cools or warms toward the ambient temperature.
Example (General Solution). The general solution of

$$T'(t) = -k\big(T(t) - T_a\big)$$

is

$$T(t) = T_a + Ce^{-kt},$$

where C is a constant. If $T(0) = T_0$, then

$$T(t) = T_a + (T_0 - T_a)e^{-kt}.$$

Definition

Definition 66.2.10 (General Solution)**.** A **general solution** of a differential equation is a family of solutions containing one or more arbitrary constants.
For an nth-order ordinary differential equation, the general solution typically contains n arbitrary constants:

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n).$$

Each choice of the constants $C_1, C_2, \ldots, C_n$ gives a particular solution of the differential equation.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* The general solution represents the full family of solutions produced by solving the differential equation. Initial conditions are then used to choose specific values of the arbitrary constants, producing a particular solution.

Example (General Solution)*.* The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t},$$

where C is an arbitrary constant. If an initial condition such as

$$y(0) = 5$$

is imposed, then $C = 5$, and the particular solution is

$$y(t) = 5e^{3t}.$$

Definition

Definition 66.2.11 (Family of Solutions)**.** A **family of solutions** of a differential equation is a collection of solution functions described by one or more parameters. For example, a family of solutions may be written in the form

$$y(t) = \Phi(t, C),$$

or more generally,

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n),$$

where $C, C_1, \ldots, C_n$ are parameters. Each choice of the parameters gives one particular solution.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* A family of solutions records many solutions at once. The parameters act as labels that distinguish one solution curve from another. When initial conditions are supplied, they usually determine the parameter values and select a single member of the family.

Example (A Family of Solutions)*.* The differential equation

$$y' = 3y$$

has the family of solutions

$$y(t) = Ce^{3t},$$

where $C \in \mathbb{R}$. Each value of C gives a different solution.

Definition

Definition 66.2.12 (Initial Condition)**.** An **initial condition** specifies the value of the unknown function at a particular value of the independent variable.
For a first-order differential equation involving an unknown function $y = y(t)$, an initial condition has the form

$$y(t_0) = y_0,$$

where t_0 is the initial time and y_0 is the initial value.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* An initial condition selects one solution from a family of solutions. If a differential equation has general solution

$$y(t) = \Phi(t, C),$$

then the condition

$$y(t_0) = y_0$$

is used to determine the constant C.

Example (Initial Condition)*.* The differential equation

$$y' = 3y$$

has general solution

$$y(t) = Ce^{3t}.$$

If the initial condition is

$$y(0) = 5,$$

then

$$5 = Ce^0 = C.$$

Thus the particular solution satisfying the initial condition is

$$y(t) = 5e^{3t}.$$

Definition

Definition 66.2.13 (Initial Value Problem)**.** An **initial value problem**, or **IVP**, is a differential equation together with one or more initial conditions.
For a first-order differential equation, an initial value problem has the form

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

The differential equation describes how the function changes, while the initial condition specifies the value of the solution at the initial time t_0.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation). An initial value problem asks for the particular solution of a differential equation that passes through a specified point. Geometrically, the condition

$$y(t_0) = y_0$$

means that the solution curve must pass through the point

$$(t_0, y_0).$$

Example (Initial Value Problem). Consider the initial value problem

$$y' = 3y, \qquad y(0) = 5.$$

The differential equation has general solution

$$y(t) = Ce^{3t}.$$

Using the initial condition,

$$5 = y(0) = Ce^0 = C.$$

Therefore the solution of the initial value problem is

$$y(t) = 5e^{3t}.$$

Definition

Definition 66.2.14 (Existence)**.** An initial value problem is said to have **existence** if there is at least one function that satisfies both the differential equation and the initial condition.
For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

existence means that there is a function $y(t)$, defined on some interval containing t_0, such that

$$y'(t) = f(t, y(t))$$

and

$$y(t_0) = y_0.$$

Remark (Standard quantified statement). This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Definition

Definition 66.2.15 (Uniqueness)**.** An initial value problem is said to have **uniqueness** if it has at most one solution on the interval under consideration.
For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

uniqueness means that if $y_1(t)$ and $y_2(t)$ are both solutions on the same interval I containing t_0, then

$$y_1(t) = y_2(t)$$

for every $t \in I$.

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Definition

Definition 66.2.16 (Interval of Existence)**.** An **interval of existence** for a solution of an initial value problem is an interval on which the solution is defined and satisfies the differential equation.
For an initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

an interval I is an interval of existence if $t_0 \in I$ and there is a solution $y : I \to \mathbb{R}$ such that

$$y'(t) = f(t, y(t))$$

for every $t \in I$, and

$$y(t_0) = y_0.$$

Remark (Standard quantified statement)*.* This definition fixes the named kind of differential-equation object by the conditions stated in the definition.

Remark (Interpretation)*.* Existence asks whether at least one solution passes through the initial point. Uniqueness asks whether only one solution passes through that point. The interval of existence records how long the solution is valid.

Example (Existence and Uniqueness)*.* The initial value problem

$$y' = 3y, \qquad y(0) = 5$$

has the solution

$$y(t) = 5e^{3t}.$$

This solution exists and is unique on $\mathbb{R}$.

The interval of existence is

$$(-\infty, \infty).$$

Chapter 67

Partial Differential Equations

67.1 Partial Differential Equations

Remark (Planned content). Active mathematical content has not yet been added.

Chapter 68

Computational Probability

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CHAPTER 10
Computational Probability

68.1 Computational Probability

Remark (Planned content). Active mathematical content has not yet been added.

Numerical Foundations

Chapter 69

Numerical Analysis

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BOOK I
Numerical Foundations

CHAPTER 1
Numerical Analysis

69.1 IEEE 754 Floating-Point Formats

Toolkit — Floating-Point Representation

This section develops the structural language of IEEE 754 binary floating-point representation. The key ideas are:

- every binary floating-point datum is organized into sign, exponent, and fraction fields,
- the exponent field determines whether the encoding is normalized, subnormal, zero, infinity, or NaN,
- the fraction field determines the significant bits of the represented value,
- each standard format is determined by its total bit width together with its exponent width, fraction width, and exponent bias.

IEEE 754 Floating-Point Formats

Definition - IEEE 754 Binary Floating-Point Datum

Definition 69.1.1 (IEEE 754 Binary Floating-Point Datum)**.** An *IEEE 754 binary floating-point datum* is a binary encoding partitioned into three fields:

- a *sign field* consisting of one bit,
- an *exponent field* consisting of a fixed number of bits,
- a *fraction field* consisting of a fixed number of bits.

Its interpretation depends on the exponent-field pattern and the fraction-field pattern.

Remark (Standard quantified statement)*.*

$$\text{IEEEBinaryFloatDatum}(x) \iff \text{HasSignField}(x) \wedge \text{HasExponentField}(x) \wedge \text{HasFractionField}(x). \blacksquare$$

Remark (Interpretation)*.* IEEE 754 does not interpret a floating-point bit string as a plain binary integer. Instead, the sign field determines sign, the exponent field determines scale and special cases, and the fraction field records the significant bits of the encoded value.

Definition - Normalized IEEE 754 Binary Number

Definition 69.1.2 (Normalized IEEE 754 Binary Number)**.** Let an IEEE 754 binary floating-point format have exponent bias bias. A finite encoded datum is called *normalized* if its exponent field is neither all zeros nor all ones.
If s is the sign bit, if e is the unsigned integer determined by the exponent field, and if f is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s(1.f)_2 2^{e-\text{bias}}.$$

Remark (Standard quantified statement)*.*

$$\text{NormalizedIEEEBinaryNumber}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldNeitherAllZerosNorA}$$

Remark (Interpretation)*.* In a normalized encoding, the leading binary digit of the significand is implicitly 1. This hidden leading bit is what gives normalized numbers one more bit of effective precision than the stored fraction field alone.

Definition - Subnormal IEEE 754 Binary Number

Definition 69.1.3 (Subnormal IEEE 754 Binary Number)**.** Let an IEEE 754 binary floating-point format have exponent bias bias. A finite encoded datum is called *subnormal* if its exponent field is all zeros and its fraction field is not all zeros.
If s is the sign bit and if f is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s(0.f)_2 2^{1-\text{bias}}.$$

Remark (Standard quantified statement).

$$\text{SubnormalIEEEBinaryNumber}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldAllZeros}(x) \wedge \neg\, \text{Fractio}$$

Remark (Interpretation). Subnormal numbers fill the gap between zero and the smallest positive normalized number. They preserve gradual underflow by allowing the leading significand bit to be 0 instead of the implicit 1 used for normalized numbers.

Definition - IEEE 754 Infinity Encoding

Definition 69.1.4 (IEEE 754 Infinity Encoding)**.** An IEEE 754 binary floating-point datum encodes *infinity* if its exponent field is all ones and its fraction field is all zeros. The sign bit determines whether the represented value is $+\infty$ or $-\infty$.

Remark (Standard quantified statement).

$$\text{IEEEInfinityEncoding}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldAllOnes}(x) \wedge \text{FractionFieldAllZe}$$

Remark (Interpretation). Infinity is a distinguished non-finite encoding used to represent overflow and certain limit-like arithmetic results. The sign field still matters, so IEEE 754 distinguishes positive infinity from negative infinity.

Definition - IEEE 754 NaN Encoding

Remark (Dependencies).

- IEEE 754 Binary Floating-Point Datum

Definition 69.1.5 (IEEE 754 NaN Encoding)**.** An IEEE 754 binary floating-point datum encodes *NaN* (Not a Number) if its exponent field is all ones and its fraction field is not all zeros.

Remark (Standard quantified statement).

$$\text{IEEENaNEncoding}(x) \iff \text{IEEEBinaryFloatDatum}(x) \wedge \text{ExponentFieldAllOnes}(x) \wedge \neg\, \text{FractionFieldAllZe}$$

Remark (Interpretation). A NaN encoding represents an undefined or invalid numerical result rather than a real number. For example, NaNs arise from operations such as invalid indeterminate forms or from the propagation of an already invalid datum through later computations.

Definition - IEEE 754 Binary16 Format

Definition 69.1.6 (IEEE 754 Binary16 Format)**.** The *IEEE 754 binary16 format* is the 16-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 5 exponent bits,
- 10 fraction bits,
- exponent bias 15.

For a normalized finite binary16 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-15}.$$

Remark (Standard quantified statement).

$\mathrm{Binary16}(x) \iff \mathrm{IEEEBinaryFloatDatum}(x) \wedge \mathrm{BitWidth}(x, 16) \wedge \mathrm{SignWidth}(x, 1) \wedge \mathrm{ExponentWidth}(x, 5) \wedge$

Remark (Interpretation)*.* Binary16 is commonly called *half precision.* It is useful when reduced storage and bandwidth matter more than wide dynamic range or high precision.

Remark (Bit layout).

15	14 … 10	9 … 0
s	exponent	fraction

Definition - IEEE 754 Binary32 Format

Definition 69.1.7 (IEEE 754 Binary32 Format)**.** The *IEEE 754 binary32 format* is the 32-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 8 exponent bits,
- 23 fraction bits,
- exponent bias 127.

For a normalized finite binary32 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-127}.$$

Remark (Standard quantified statement).

$\mathrm{Binary32}(x) \iff \mathrm{IEEEBinaryFloatDatum}(x) \wedge \mathrm{BitWidth}(x, 32) \wedge \mathrm{SignWidth}(x, 1) \wedge \mathrm{ExponentWidth}(x, 8) \wedge$

Remark (Interpretation)*.* Binary32 is commonly called *single precision.* It is the standard 32-bit floating-point format used broadly in scientific computing, computer graphics, simulation, and machine learning.

Remark (Bit layout).

31	30	23	22	0
s	exponent		fraction	

Definition - IEEE 754 Binary64 Format

Definition 69.1.8 (IEEE 754 Binary64 Format)**.** The *IEEE 754 binary64 format* is the 64-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 11 exponent bits,
- 52 fraction bits,
- exponent bias 1023.

For a normalized finite binary64 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-1023}.$$

Remark (Standard quantified statement).

$\mathrm{Binary64}(x) \iff \mathrm{IEEEBinaryFloatDatum}(x) \wedge \mathrm{BitWidth}(x, 64) \wedge \mathrm{SignWidth}(x, 1) \wedge \mathrm{ExponentWidth}(x, 11) \wedge$

Remark (Interpretation). Binary64 is commonly called *double precision*. It is the standard high-precision floating-point format for much of scientific and engineering computation.

Remark (Bit layout).

63	62	52	51	0
s	exponent		fraction	

Definition - IEEE 754 Binary128 Format

Definition 69.1.9 (IEEE 754 Binary128 Format)**.** The *IEEE 754 binary128 format* is the 128-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 15 exponent bits,
- 112 fraction bits,
- exponent bias 16383.

For a normalized finite binary128 datum, the represented value is

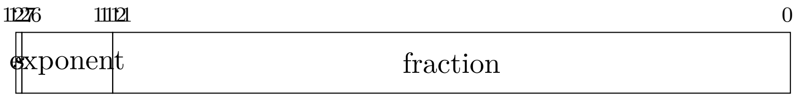
$$(-1)^s(1.f)_2 2^{e-16383}.$$

Remark (Standard quantified statement).

$\mathrm{Binary128}(x) \iff \mathrm{IEEEBinaryFloatDatum}(x) \wedge \mathrm{BitWidth}(x, 128) \wedge \mathrm{SignWidth}(x, 1) \wedge \mathrm{ExponentWidth}(x, 15$

Remark (Interpretation). Binary128 is commonly called *quadruple precision*. It provides substantially greater range and precision than binary64 at substantially greater storage and computational cost.

Remark (Bit layout).



Remark (Summary). The four standard IEEE 754 basic binary interchange formats considered here are:

$$\text{binary16 } (1, 5, 10), \qquad \text{binary32 } (1, 8, 23), \qquad \text{binary64 } (1, 11, 52), \qquad \text{binary128 } (1, 15, 112),$$

where each triple records

$$(\text{sign bits}, \ \text{exponent bits}, \ \text{fraction bits}).$$

69.2 Interval Arithmetic

Box (Interval) Arithmetic in $\mathbb{R}^n$ - Quick Reference

Operation	Rule
Box (product interval)	$[a, b] := \{x \in \mathbb{R}^n : a \le x \le b\}$
Containment	$[a, b] \subseteq [c, d] \iff c \le a \ \wedge \ b \le d$
Minkowski sum	$[a, b] \oplus [c, d] = [a + c, \ b + d]$
Difference	$[a, b] \ominus [c, d] = [a - d, \ b - c]$
Scalar mult. ($\lambda \ge 0$)	$\lambda[a, b] = [\lambda a, \ \lambda b]$
Scalar mult. ($\lambda < 0$)	$\lambda[a, b] = [\lambda b, \ \lambda a]$
Coordinatewise product	$[a, b] \odot [c, d] = \prod_i [\min S_i, \max S_i]$
Norm bound (Euclidean)	$\|x\|_2 \le \max\{\|a\|_2, \|b\|_2\}$ for $x \in [a, b]$ if $0 \le a \le b$

Interval Arithmetic - Quick Reference

Operation	Rule
Addition	$[a, b] + [c, d] = [a + c, \ b + d]$
Subtraction	$[a, b] - [c, d] = [a - d, \ b - c]$
Multiplication	$[a, b] \cdot [c, d]$ (case analysis)
Containment	$[a, b] \subseteq [c, d] \iff c \le a, b \le d$

69.2.1 Interval Arithmetic

Definition (Interval Addition)

Definition 69.2.1 (Interval Addition). For closed intervals $[a, b]$ and $[c, d]$, their *sum* is

$$[a, b] + [c, d] \ = \ [a + c, \ b + d].$$

Remark (Dependencies).

- Closed Interval
- Addition on $\mathbb{R}$

Definition (Interval Subtraction)

Definition 69.2.2 (Interval Subtraction)**.** For closed intervals $[a, b]$ and $[c, d]$, their *difference* is

$$[a, b] - [c, d] \;=\; [a - d,\; b - c].$$

Remark (Dependencies).

- Closed Interval
- Subtraction on $\mathbb{R}$

Definition (Interval Multiplication)

Definition 69.2.3 (Interval Multiplication)**.** For closed intervals $[a, b]$ and $[c, d]$, their *product* is

$$[a, b] \cdot [c, d] \;=\; \big[\min(ac, ad, bc, bd),\; \max(ac, ad, bc, bd)\big].$$

Remark (Dependencies).

- Closed Interval
- Multiplication on $\mathbb{R}$
- Order on $\mathbb{R}$

Definition (Interval Containment)

Definition 69.2.4 (Interval Containment)**.** $[a, b] \subseteq [c, d]$ if and only if $c \leq a$ and $b \leq d$.

Remark (Dependencies).

- Closed Interval
- Subset
- Order on $\mathbb{R}$

Remark (Motivation). Interval arithmetic provides a framework for rigorous numerical computation with guaranteed error bounds. A real number $x \in [a, b]$ means the computed value may lie anywhere in the interval; operations propagate these bounds.

Remark (Status). *This section is a stub. A full treatment – the sub-distributivity law, dependency problem, applications to root-finding, and connections to compactness – will be developed in a future revision.*

69.2.2 Interval Arithmetic in $\mathbb{R}^n$

Definition (Box / Product Interval)

Definition 69.2.5 (Box / Product Interval)**.** Let $a, b \in \mathbb{R}^n$ with $a \le b$ coordinatewise (i.e., $a_i \le b_i$ for all i). The *box* (or *interval vector*) determined by (a, b) is

$$[a, b] \;:=\; \{x \in \mathbb{R}^n : a \le x \le b\} \;=\; \prod_{i=1}^{n} [a_i, b_i].$$

Remark (Dependencies).

- Cartesian Product
- Closed Interval
- Order on $\mathbb{R}$

Definition (Box Containment)

Definition 69.2.6 (Box Containment)**.** For $a \le b$ and $c \le d$ in $\mathbb{R}^n$, we write $[a, b] \subseteq [c, d]$ iff

$$(\forall x \in \mathbb{R}^n)\ \big(x \in [a, b] \Rightarrow x \in [c, d]\big).$$

Equivalently (and most useful computationally),

$$[a, b] \subseteq [c, d] \quad \Longleftrightarrow \quad c \le a \ \wedge\ b \le d \qquad \text{(coordinatewise)}.$$

Remark (Dependencies).

- Box / Product Interval
- Subset
- Order on $\mathbb{R}$

Definition (Minkowski Addition of Boxes)

Definition 69.2.7 (Minkowski Addition of Boxes)**.** For boxes $X = [a, b]$ and $Y = [c, d]$ in $\mathbb{R}^n$, their *Minkowski sum* is

$$X \oplus Y \;:=\; \{x + y : x \in X,\ y \in Y\}.$$

For boxes, this closes in the class of boxes and satisfies the endpoint rule

$$[a, b] \oplus [c, d] \;=\; [a + c,\ b + d].$$

Remark (Dependencies).

- Box / Product Interval

- Addition on $\mathbb{R}$

Definition (Box Subtraction / Difference)

Definition 69.2.8 (Box Subtraction / Difference)**.** For boxes $X = [a, b]$ and $Y = [c, d]$ in $\mathbb{R}^n$, define

$$X \ominus Y \;:=\; \{x - y : x \in X,\; y \in Y\}.$$

Then

$$[a, b] \ominus [c, d] \;=\; [a - d,\; b - c].$$

Remark (Dependencies).

- Box / Product Interval
- Subtraction on $\mathbb{R}$

Definition (Scalar Multiplication of a Box)

Definition 69.2.9 (Scalar Multiplication of a Box)**.** For $\lambda \in \mathbb{R}$ and a box $[a, b] \subseteq \mathbb{R}^n$, define

$$\lambda[a, b] \;:=\; \{\lambda x : x \in [a, b]\}.$$

Then

$$\lambda[a, b] = \begin{cases} [\lambda a,\; \lambda b], & \lambda \geq 0, \\ [\lambda b,\; \lambda a], & \lambda < 0, \end{cases}$$

where the inequalities are coordinatewise.

Remark (Dependencies).

- Box / Product Interval
- Multiplication on $\mathbb{R}$
- Order on $\mathbb{R}$

Definition (Coordinatewise / Hadamard Product of Boxes)

Definition 69.2.10 (Coordinatewise / Hadamard Product of Boxes)**.** For $x, y \in \mathbb{R}^n$, write $(x \odot y)_i := x_i y_i$. For boxes $X = [a, b]$ and $Y = [c, d]$, define

$$X \odot Y \;:=\; \{x \odot y : x \in X,\; y \in Y\}.$$

Then $X \odot Y$ is again a box, computed coordinatewise:

$$[a, b] \odot [c, d] \;=\; \prod_{i=1}^{n} \Big[\min S_i,\; \max S_i\Big], \qquad S_i := \{a_i c_i,\; a_i d_i,\; b_i c_i,\; b_i d_i\}.$$

Remark (Dependencies).

- Box / Product Interval
- Multiplication on $\mathbb{R}$
- Order on $\mathbb{R}$

Remark (Interpretation: intervals as guaranteed enclosures). A box $X = [a, b] \subseteq \mathbb{R}^n$ represents a *set of possible vectors* (e.g., a computed vector plus uncertainty bounds). The operations above are *set operations* (Minkowski sum/difference, scalar image), so the result is a *guaranteed enclosure* of all possible true outcomes under the corresponding vector operation.

Remark (Norm bounds and why boxes are convenient). For many applications you also want quick bounds on a norm over a box. For example, if $0 \leq a \leq b$ coordinatewise, then for every $x \in [a, b]$ we have $0 \leq x \leq b$ and hence $\|x\|_2 \leq \|b\|_2$. More generally, bounding norms over a box can be reduced to checking finitely many "corners" when the objective is coordinatewise monotone.

Remark (Status). *This section is a stub. A full treatment – interval extensions of general functions $f : \mathbb{R}^n \to \mathbb{R}^m$, the dependency problem, sub-distributivity, natural vs. centered (affine) forms, and connections to compactness and continuity – will be developed in a future revision.*

69.3 Numerical Geometry

69.3.1 Week 1: Points, Vectors, and Floating-Point Geometry

Reading checklist.

- Atkinson, Chapter 1
- Solomon, Chapter 1 selectively
- Solomon, Chapter 2 selectively
- Vince, Chapter 7
- Existing Volume VII IEEE 754 notes
- Existing Volume VII interval arithmetic notes

Remark (Project placeholder). `lra-numerical-analysis/projects/point-vector-01`

Remark (Handwritten notes placeholder). TODO: Add handwritten-note extraction here after study.

69.3.2 Week 2: Linear Interpolation and Affine Sampling

Reading checklist.

- Salomon, Chapter 1 sections 1.1–1.3
- Salomon, Chapter 2 section 2.1
- Vince, Chapter 13
- Piegl–Tiller, Chapter 1 skim only
- Piegl–Tiller, Chapter 2 preview only

Remark (Project placeholder). `lra-numerical-analysis/projects/linear-interpolation-02`

Remark (Handwritten notes placeholder). TODO: Add handwritten-note extraction here after study.

69.3.3 Projects

- `point-vector-01`: planned
- `linear-interpolation-02`: planned
- `bezier-curve-03`: deferred
- `bspline-basis-04`: deferred
- `nurbs-curve-05`: deferred

69.3.4 Exercises

- Week 1 exercises: planned
- Week 2 exercises: planned
- TODO: Add exercises after handwritten notes are extracted.

Chapter 70

Numerical PDE

SERIES
From Turing to Carmack

VOLUME VIII
Applied and Computational Mathematics

BOOK I
Numerical Foundations

CHAPTER 2
Numerical PDE

70.1 Numerical PDE

Remark (Planned content). Active mathematical content has not yet been added.

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